



DeepLearning.AI

Math for Machine Learning

Probability and Statistics



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Math for Machine Learning

Probability and Statistics - Week 2

W2 Lesson 1

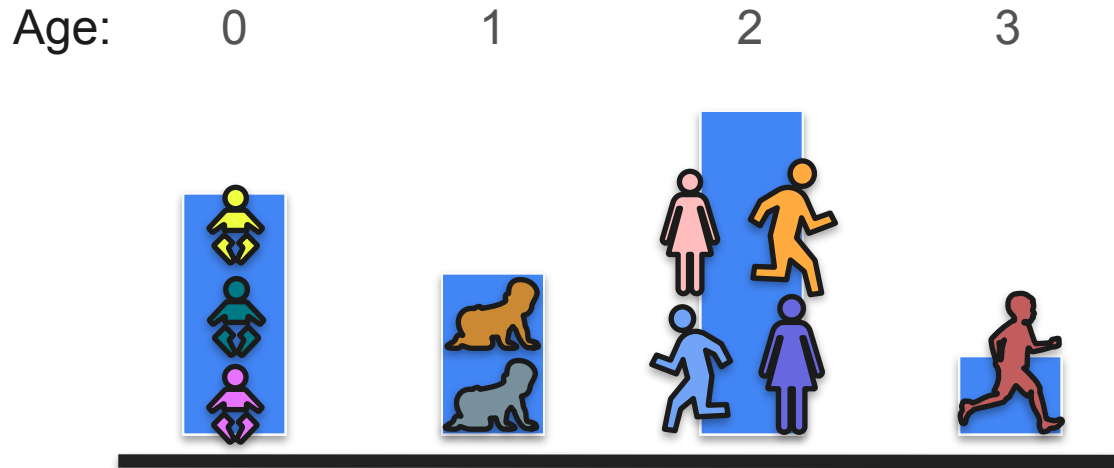


DeepLearning.AI

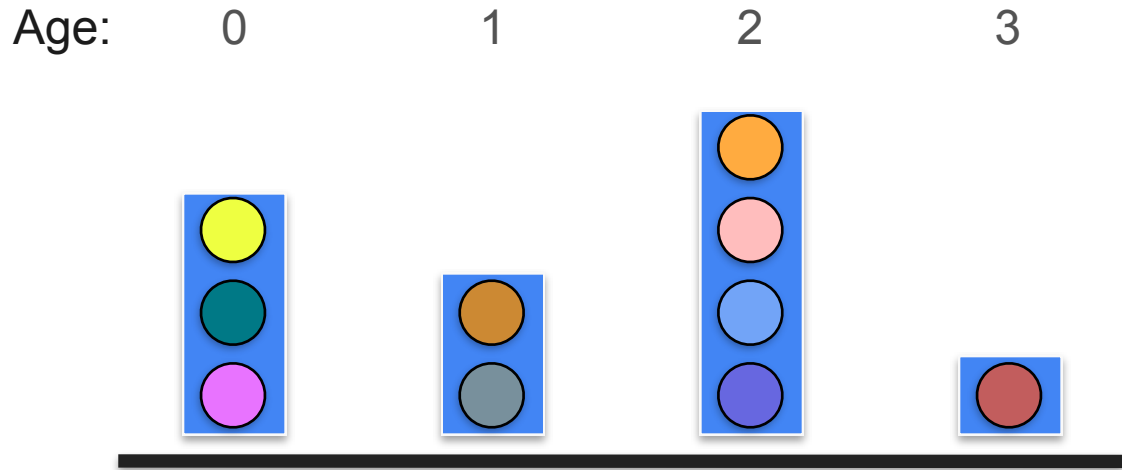
Describing Distributions

Measures of Central Tendency

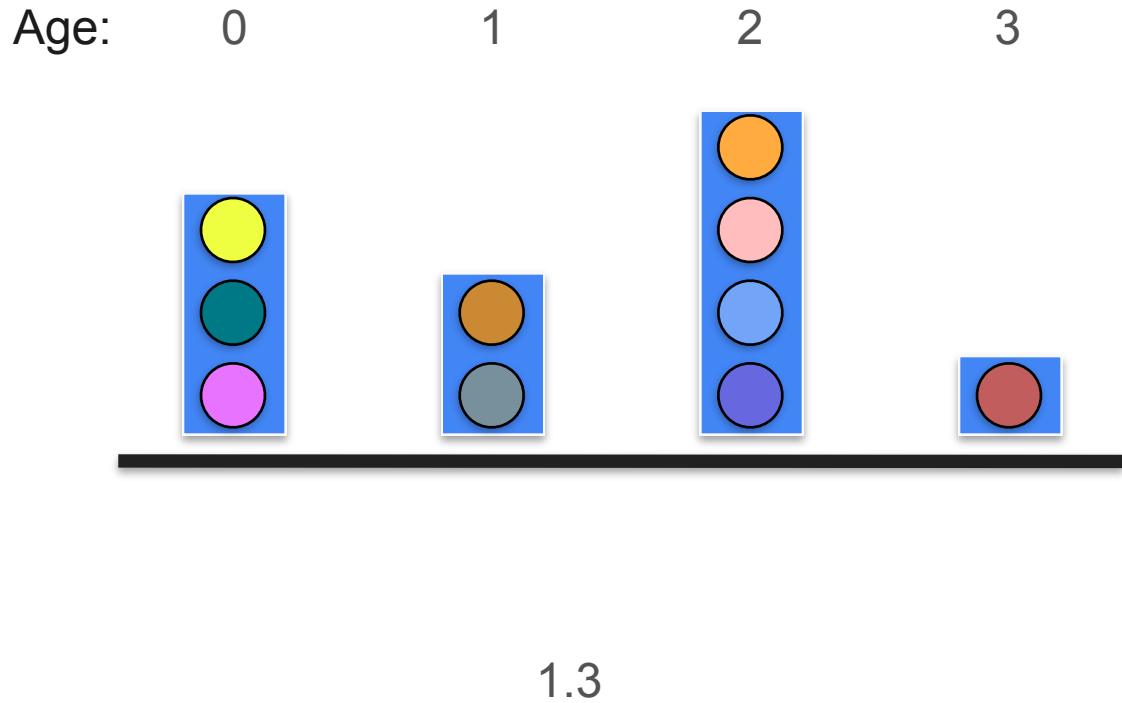
Mean: Example



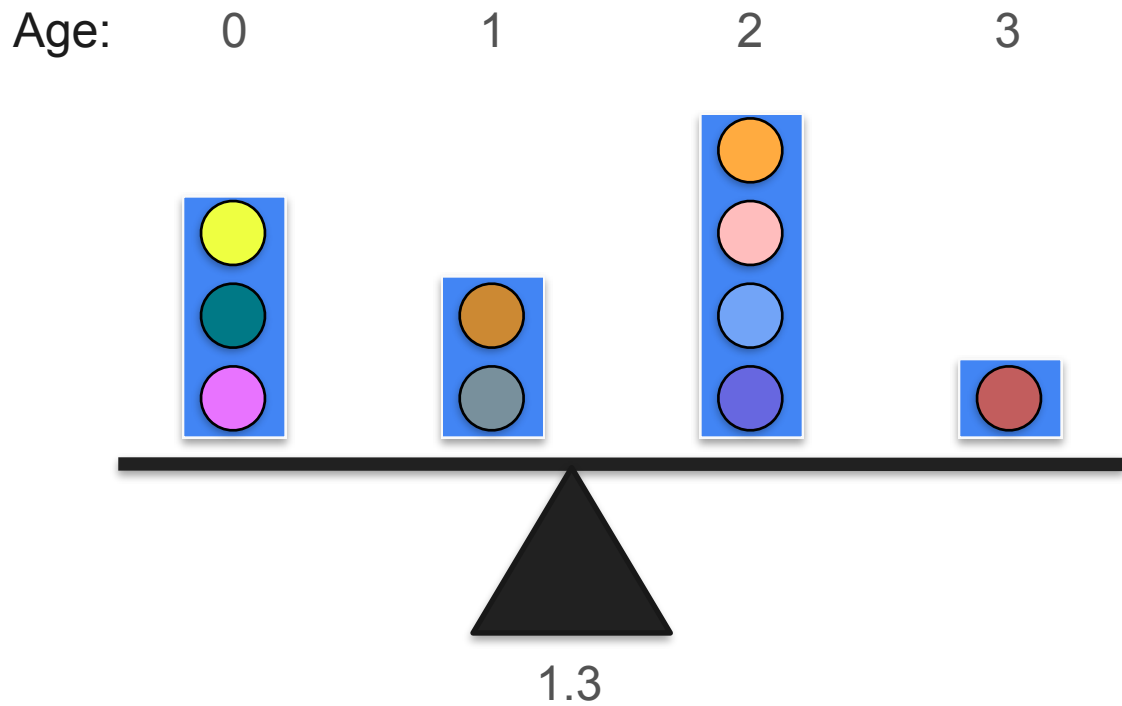
Mean: Example



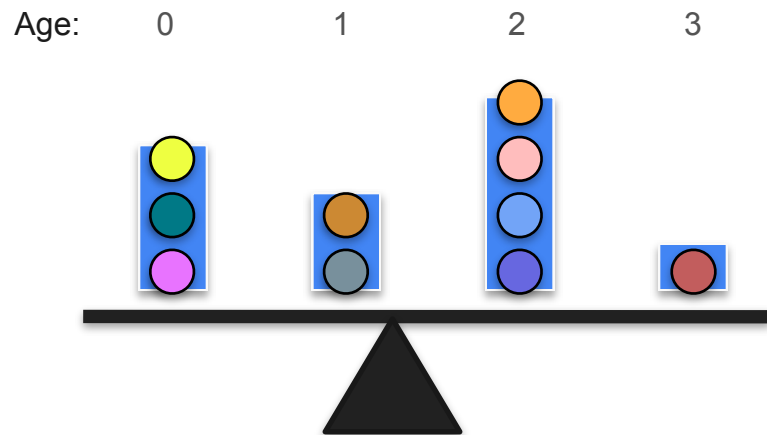
Mean: Example



Mean: Example

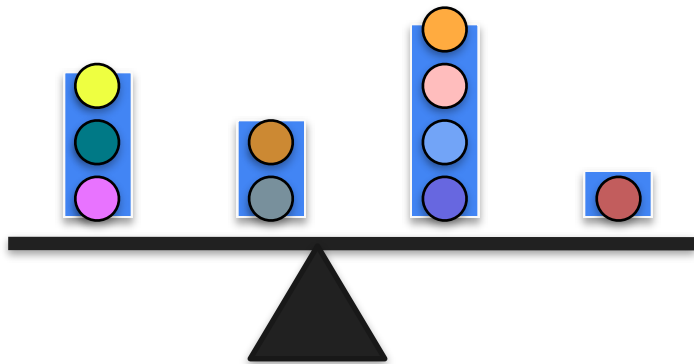


Mean: Example

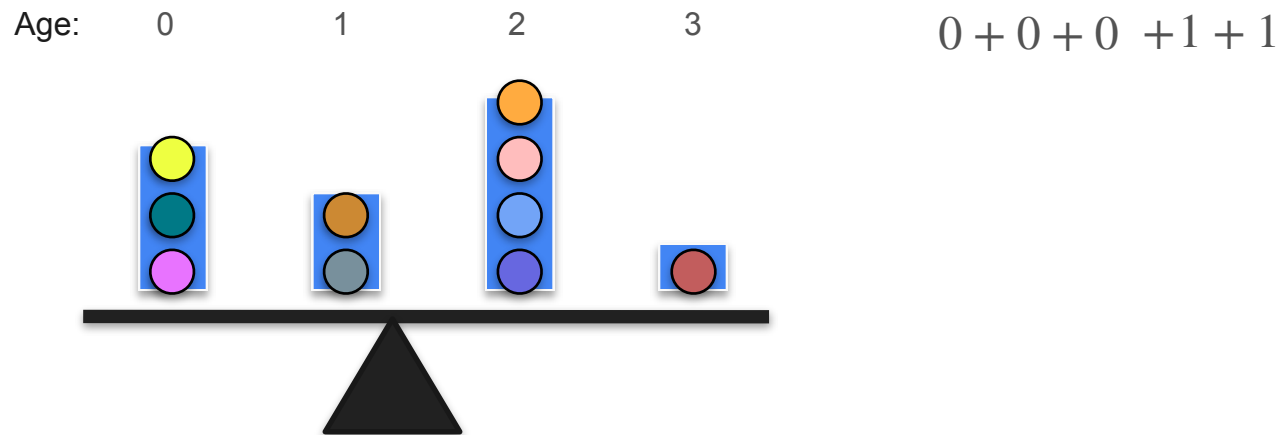


Mean: Example

Age: 0 1 2 3 $0 + 0 + 0$

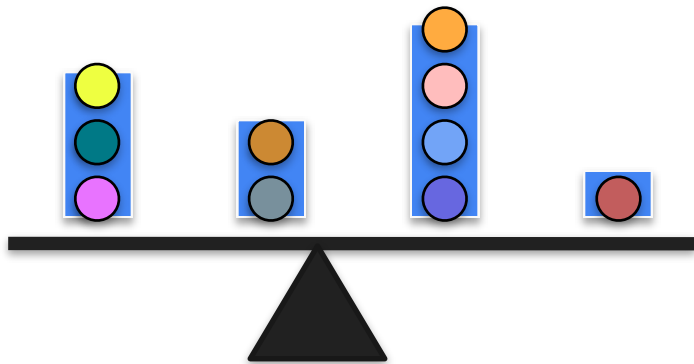


Mean: Example



Mean: Example

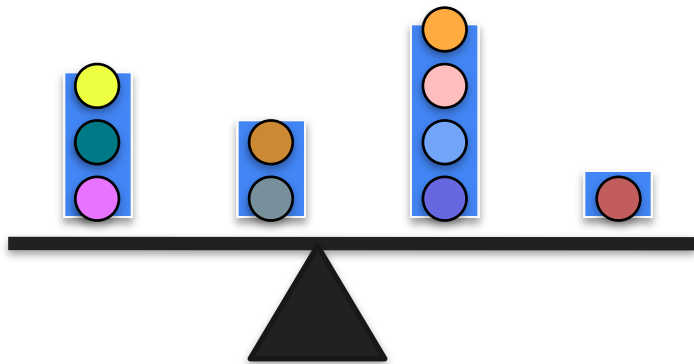
Age: 0 1 2 3



$$0 + 0 + 0 + 1 + 1 + 2 + 2 + 2 + 2$$

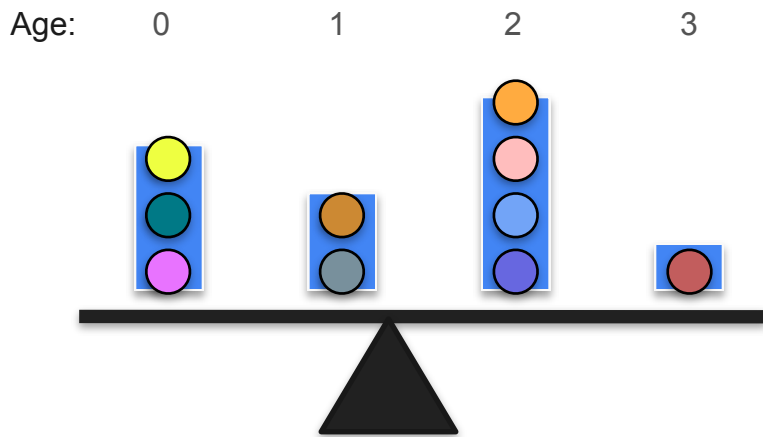
Mean: Example

Age: 0 1 2 3



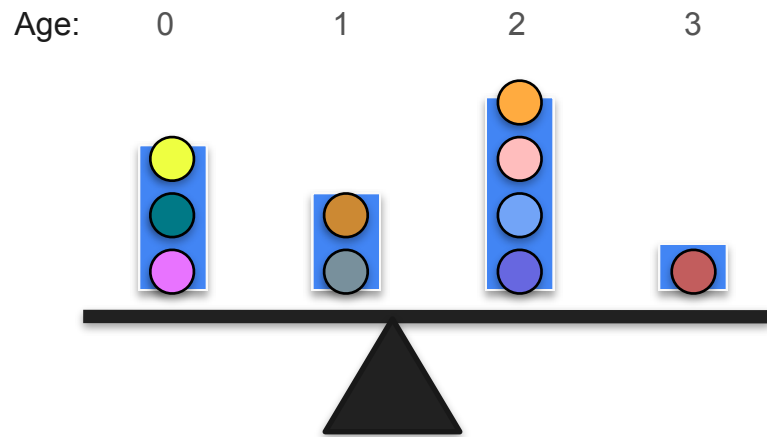
$$0 + 0 + 0 + 1 + 1 + 2 + 2 + 2 + 2 + 3$$

Mean: Example



$$\frac{0 + 0 + 0 + 1 + 1 + 2 + 2 + 2 + 2 + 3}{10}$$

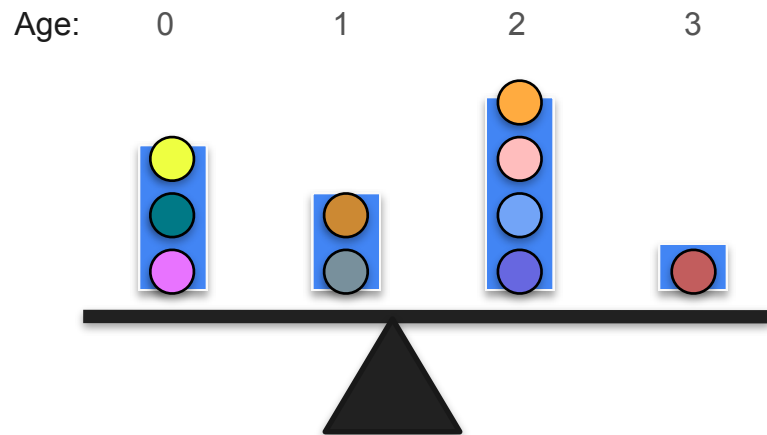
Mean: Example



$$\frac{0 + 0 + 0 + 1 + 1 + 2 + 2 + 2 + 2 + 3}{10}$$

$$= \frac{13}{10}$$

Mean: Example

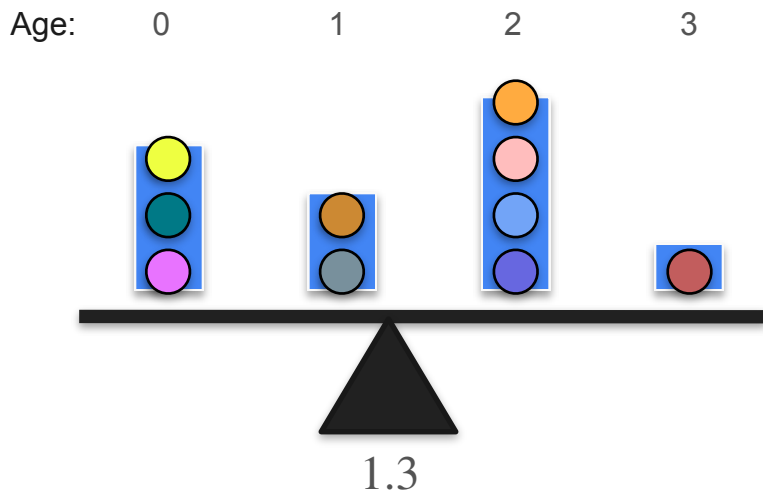


$$\frac{0 + 0 + 0 + 1 + 1 + 2 + 2 + 2 + 2 + 3}{10}$$

$$= \frac{13}{10}$$

$$= 1.3$$

Mean: Example

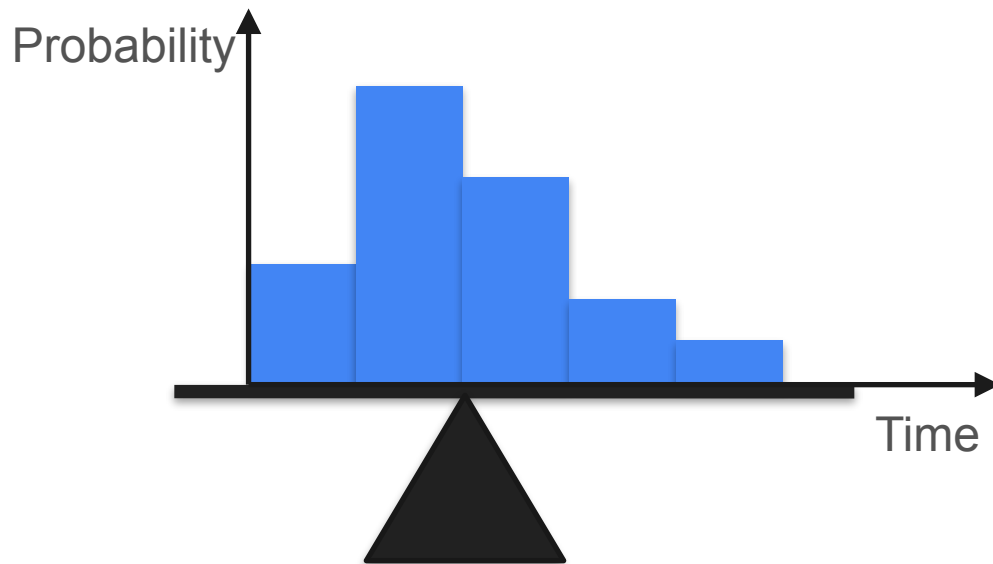
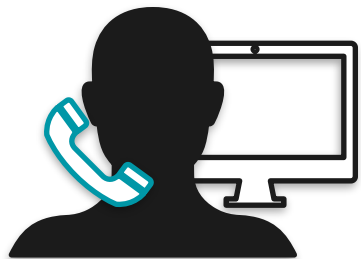


$$\frac{0 + 0 + 0 + 1 + 1 + 2 + 2 + 2 + 2 + 3}{10}$$

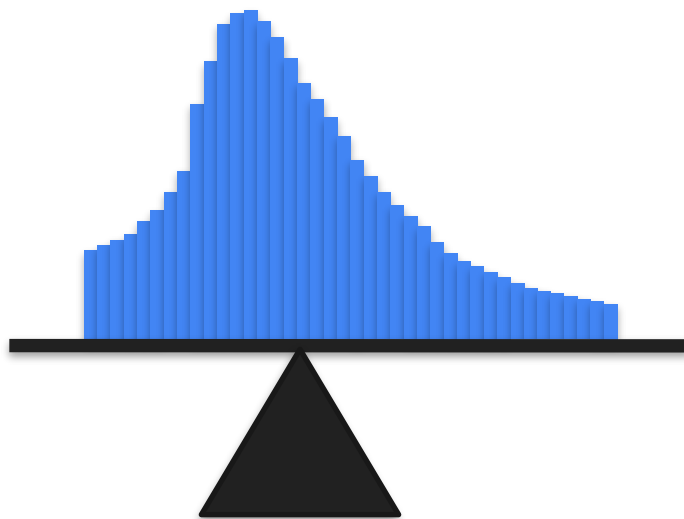
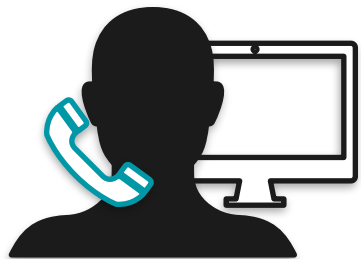
$$= \frac{13}{10}$$

$$= 1.3$$

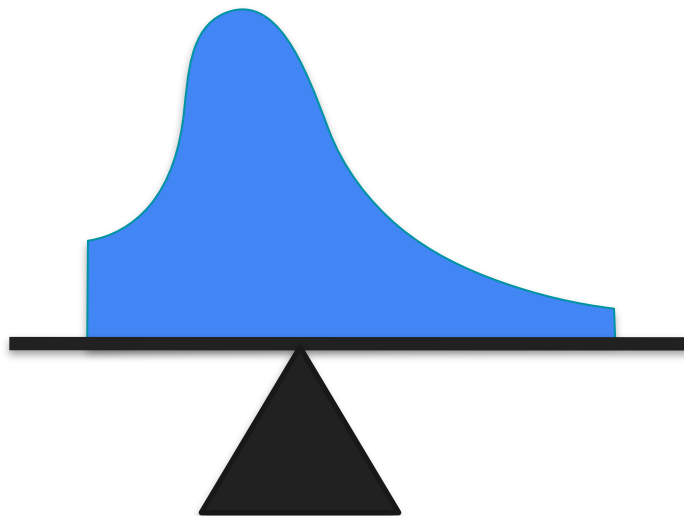
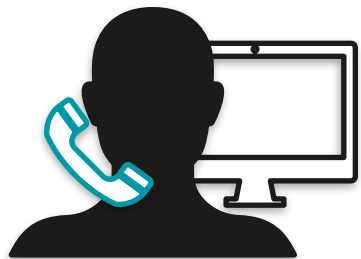
Mean



Mean



Mean



Median: Motivation

Median: Motivation

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- Why?

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- Why?
 1. The program was really good

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 3. One student made lots of money

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Michael Jordan

Outliers

Outliers

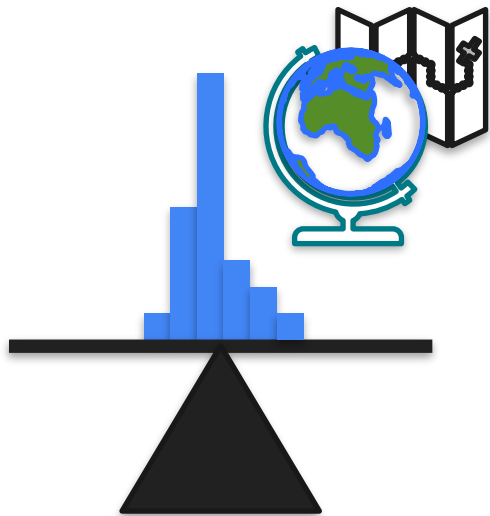


Outliers



Outliers

Graduates



Salary

Outliers

Graduates



Outliers

Graduates



Outliers

Graduates



Salary

Median

Graduates



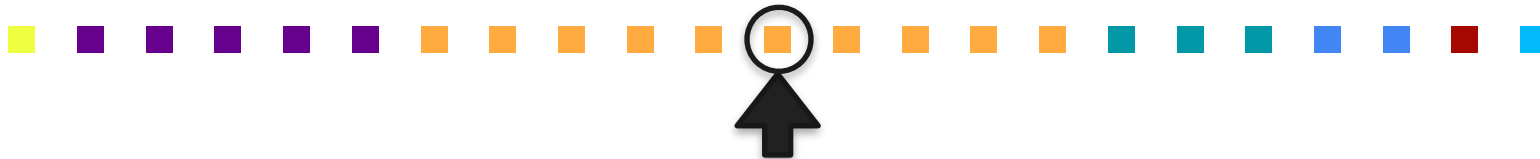
Salary

Median

Graduates



Salary



Graduates



Median

Salary



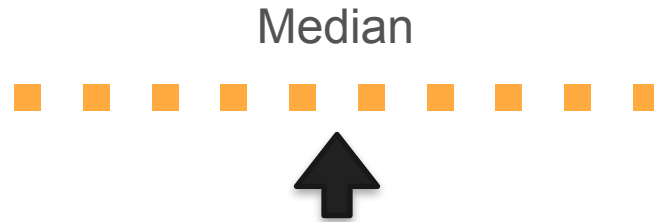
Median



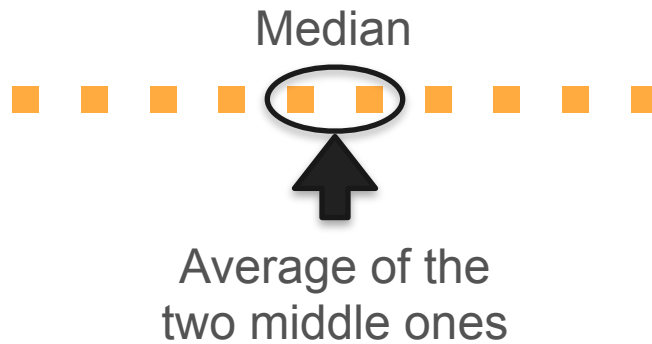
Median



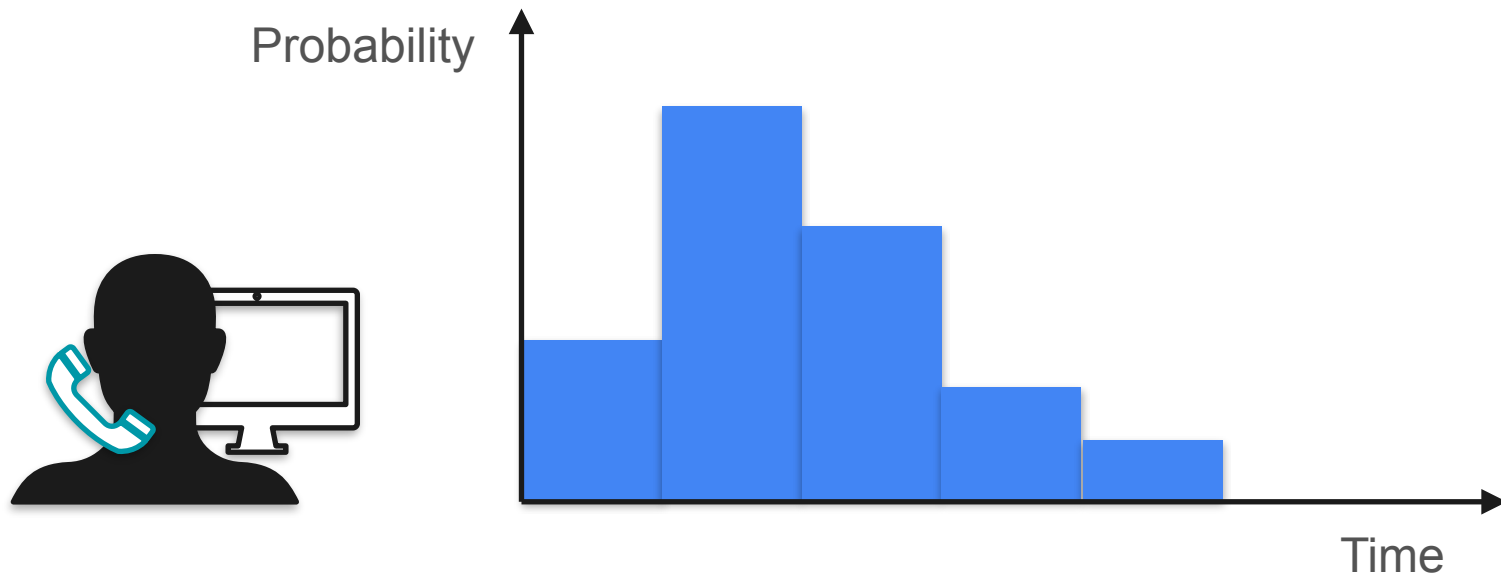
Median



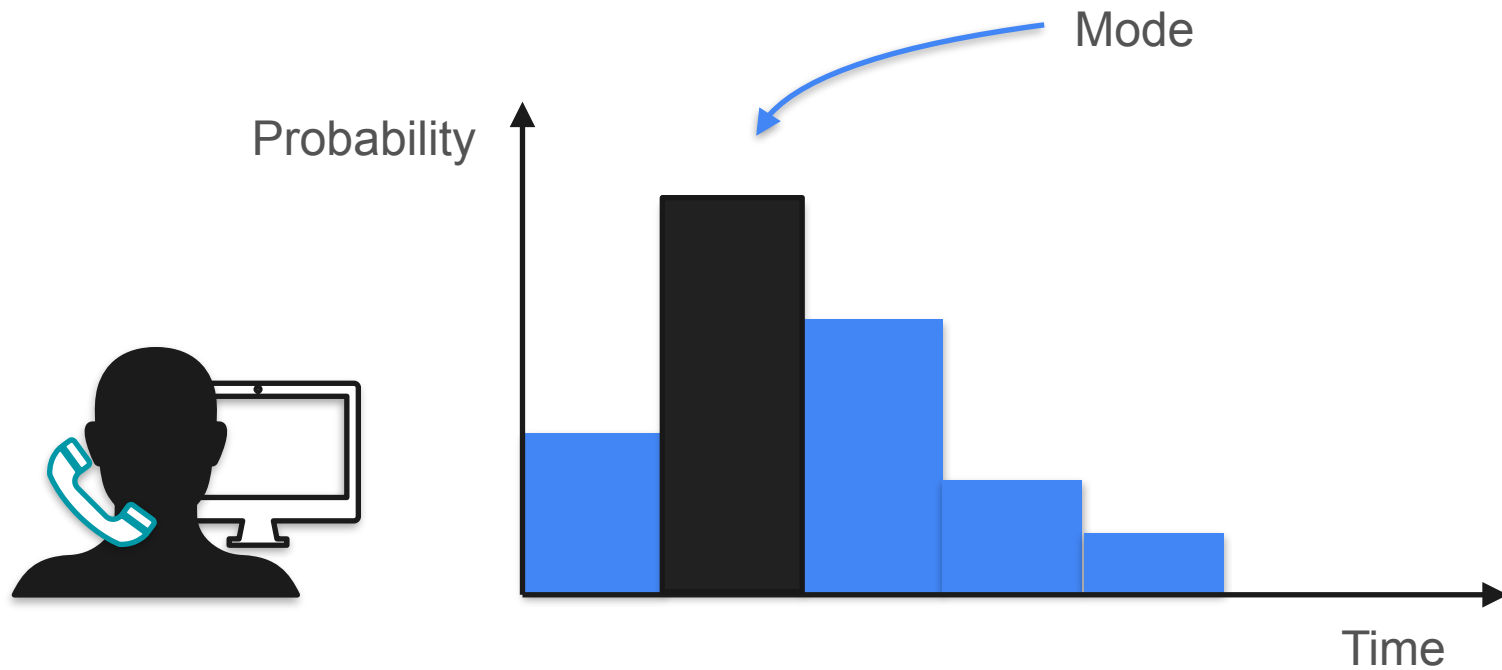
Median



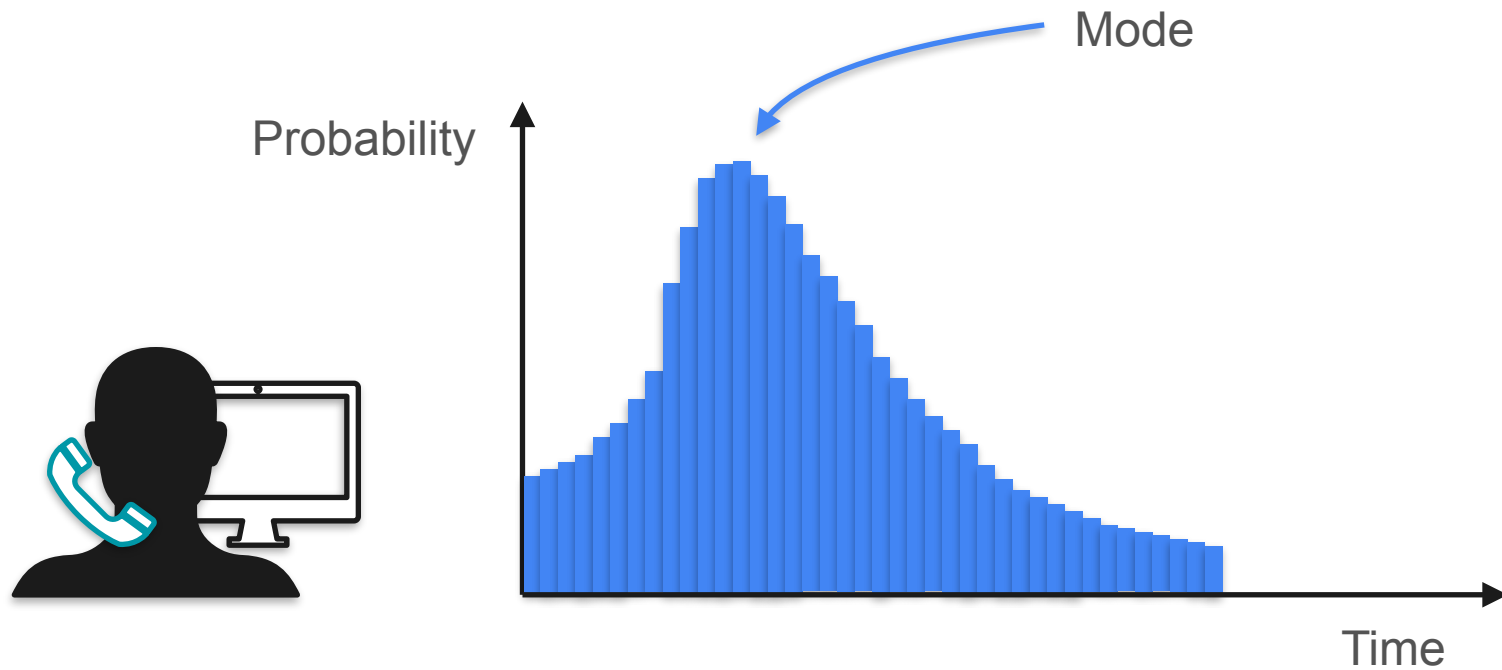
Mode



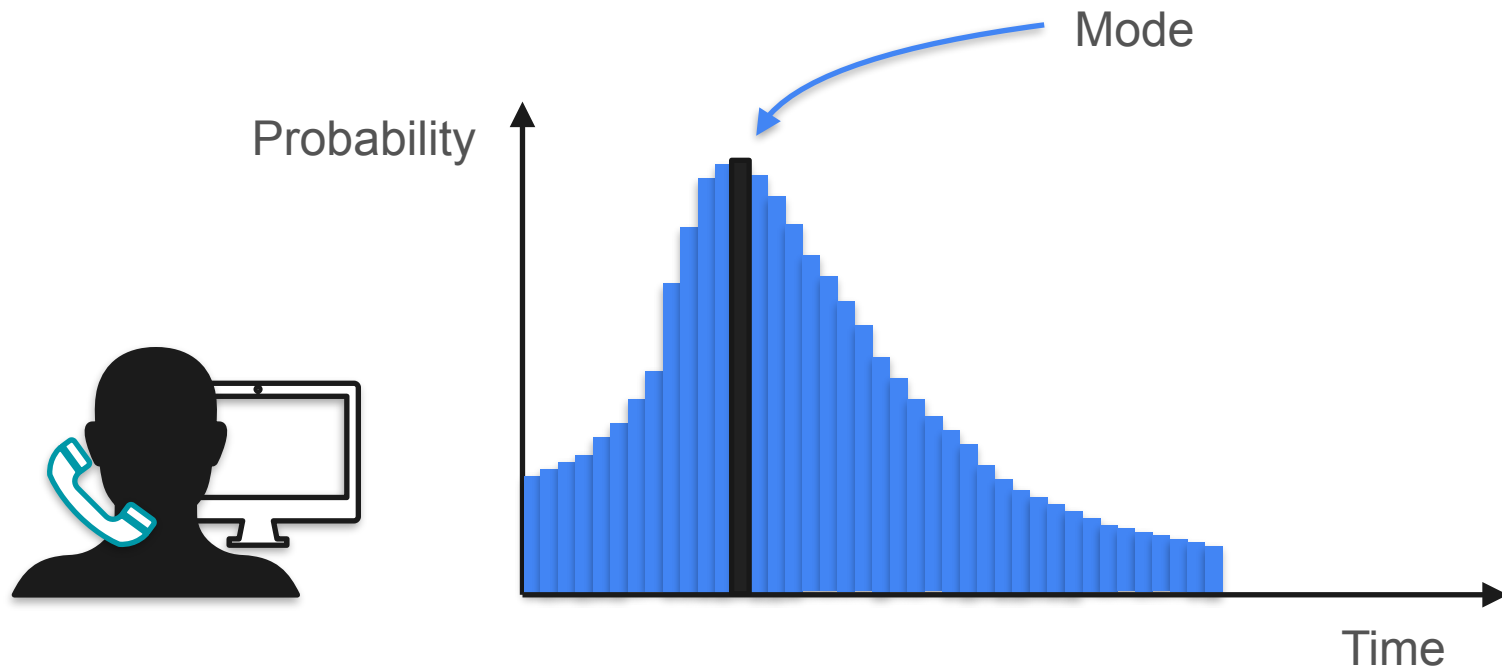
Mode



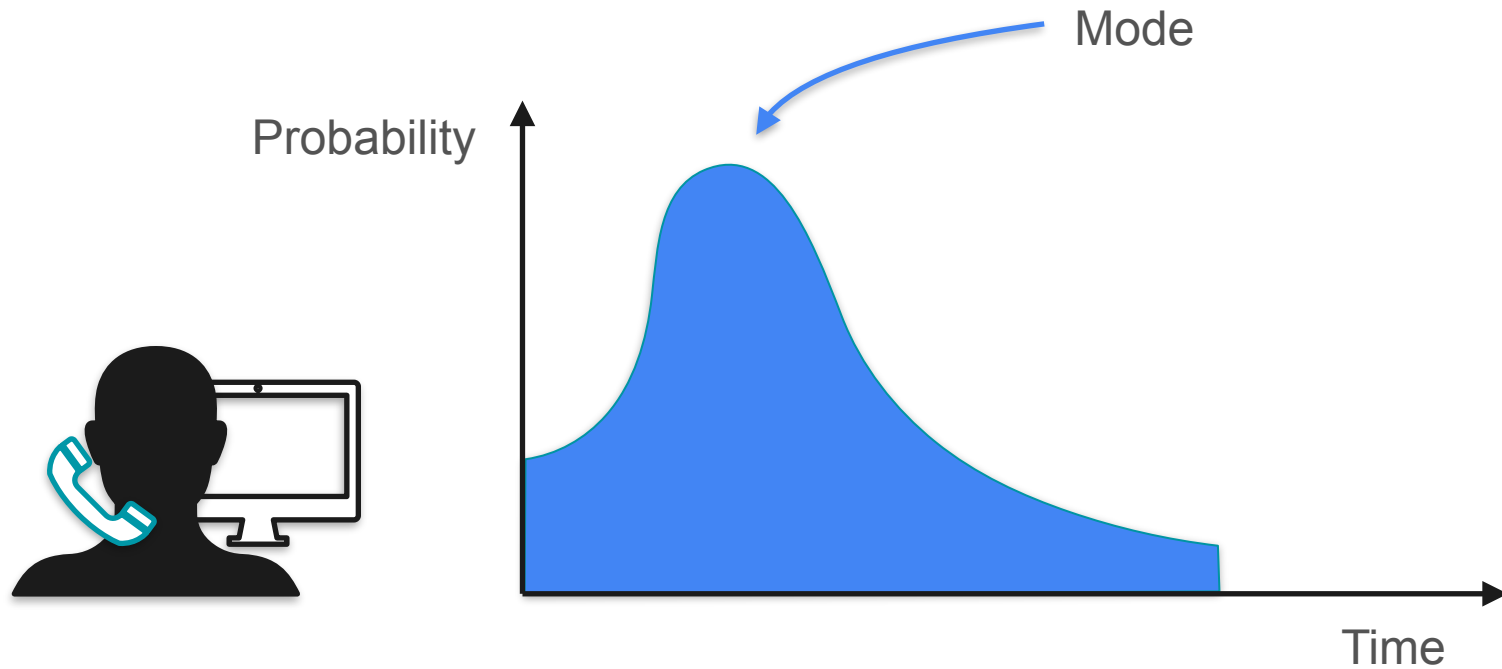
Mode



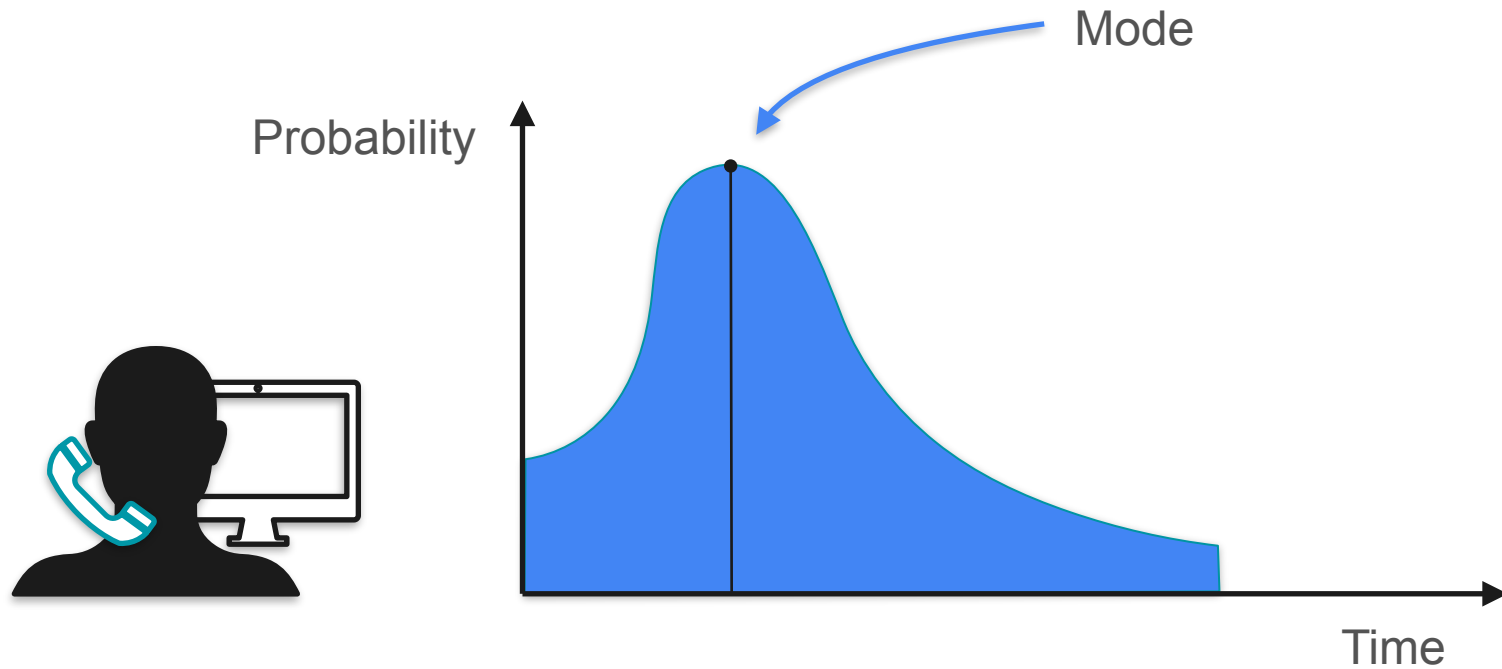
Mode



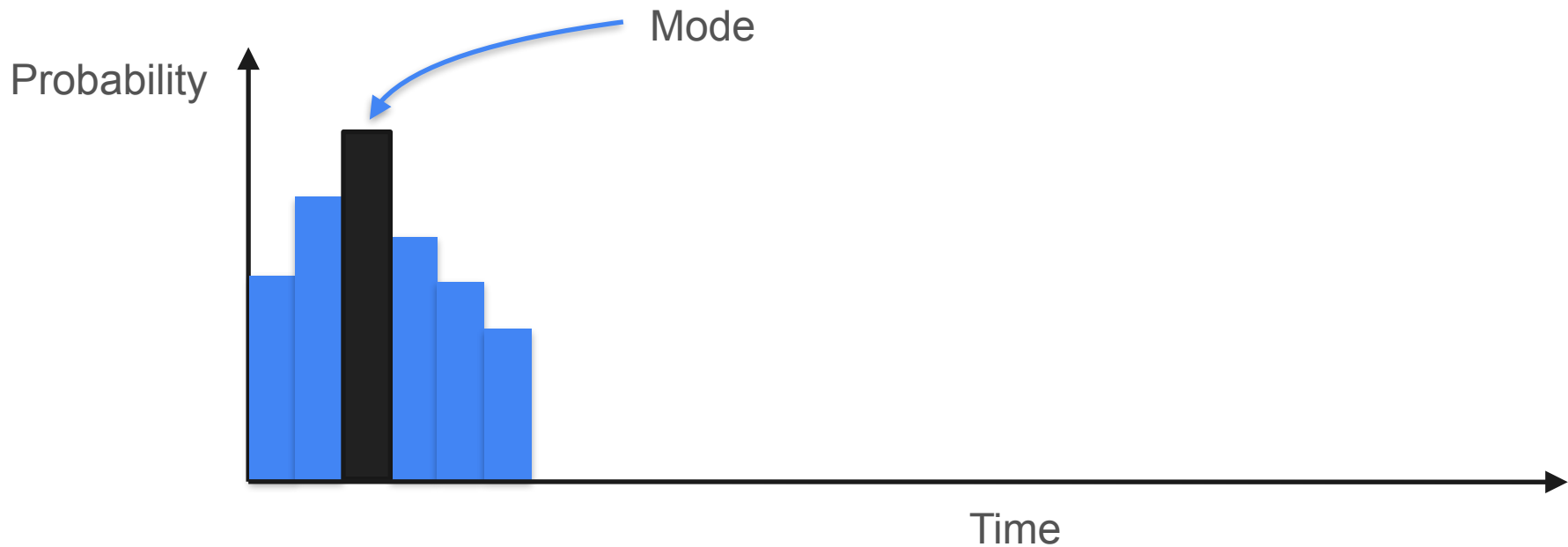
Mode



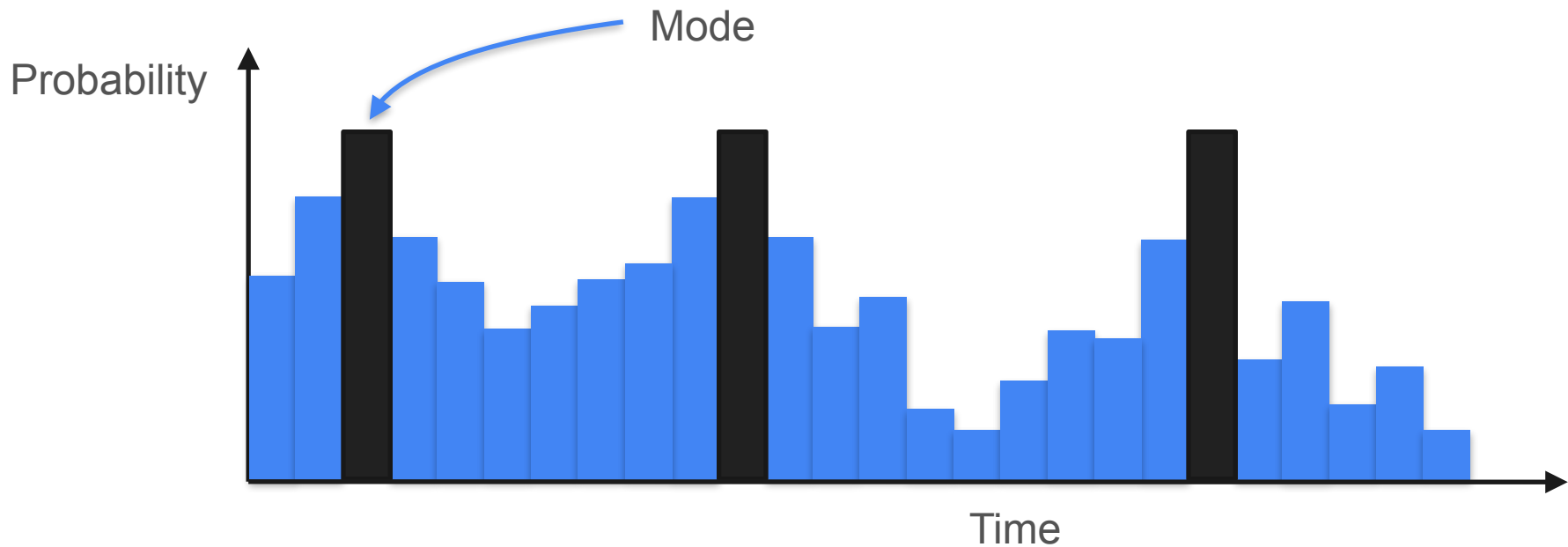
Mode



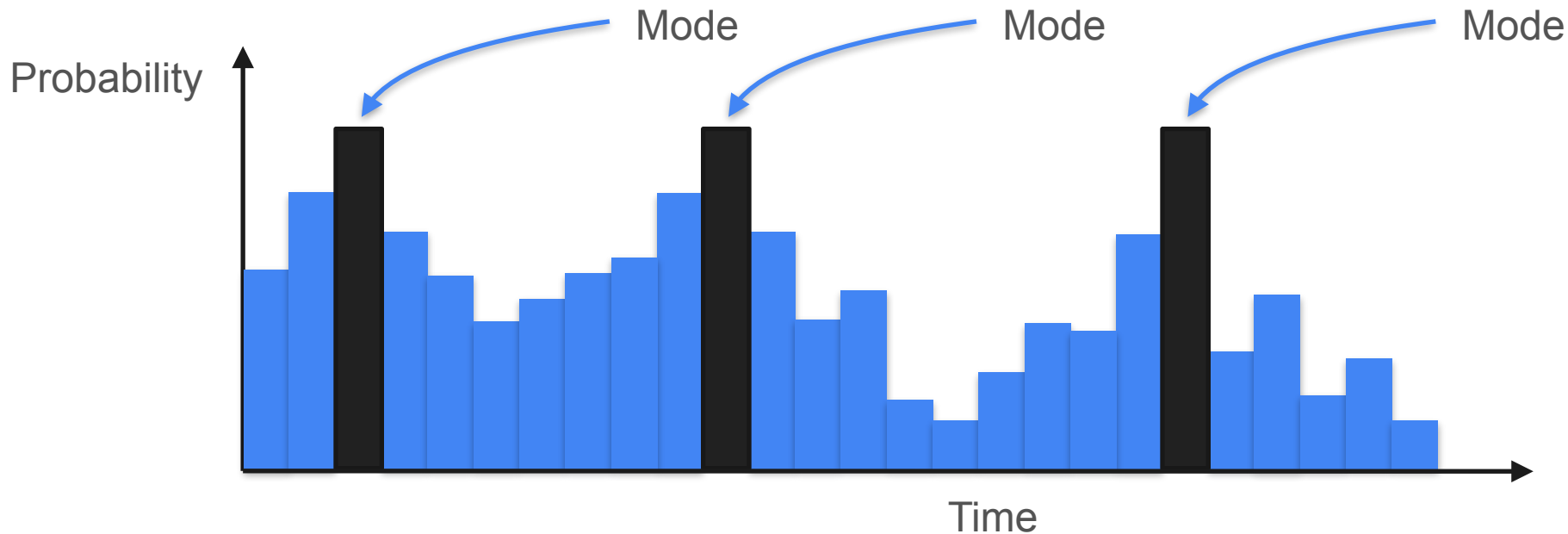
Mode: Multimodal Distribution



Mode: Multimodal Distribution

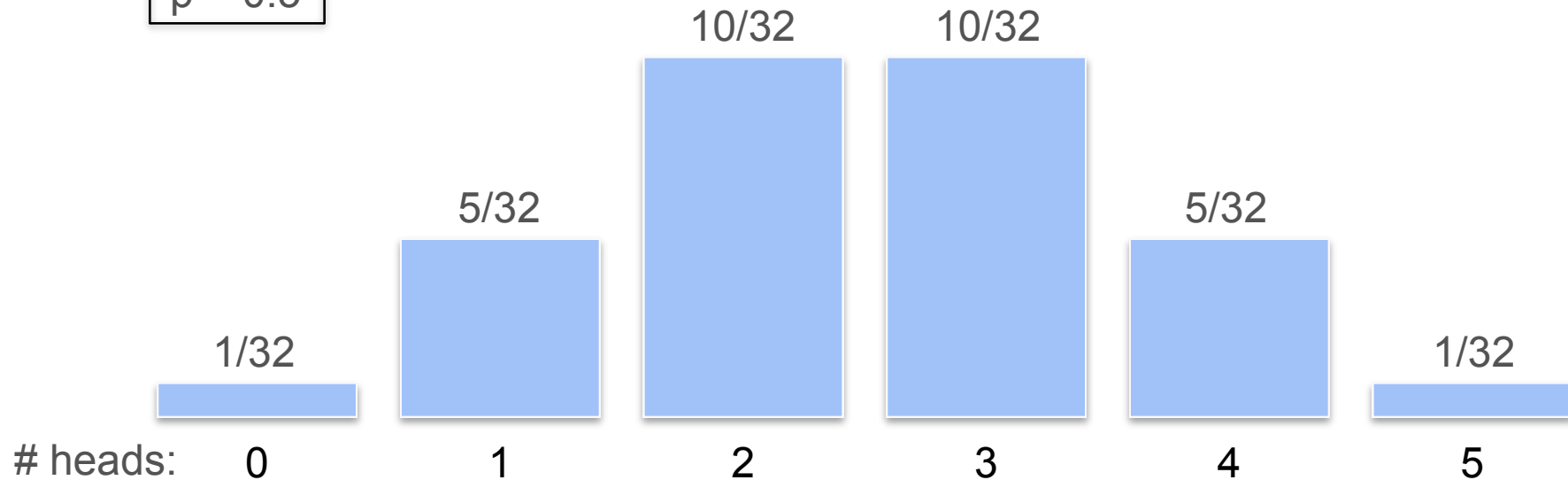


Mode: Multimodal Distribution



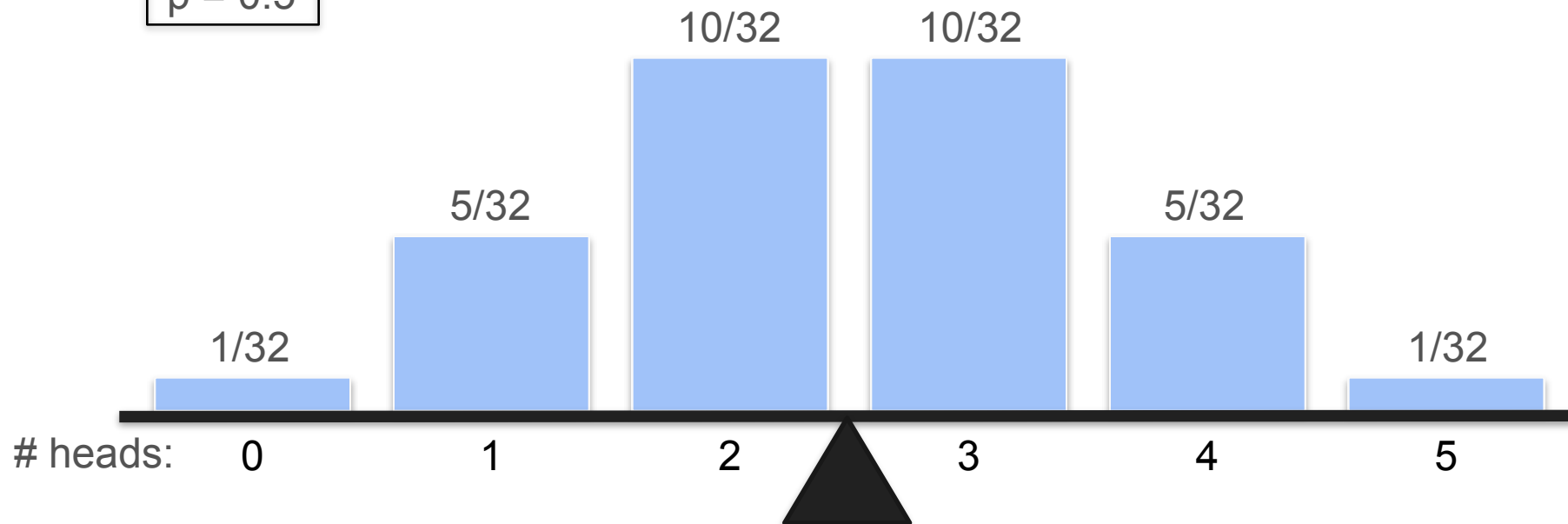
Mean, Median and Mode in Binomial Distribution

$n = 5$
 $p = 0.5$



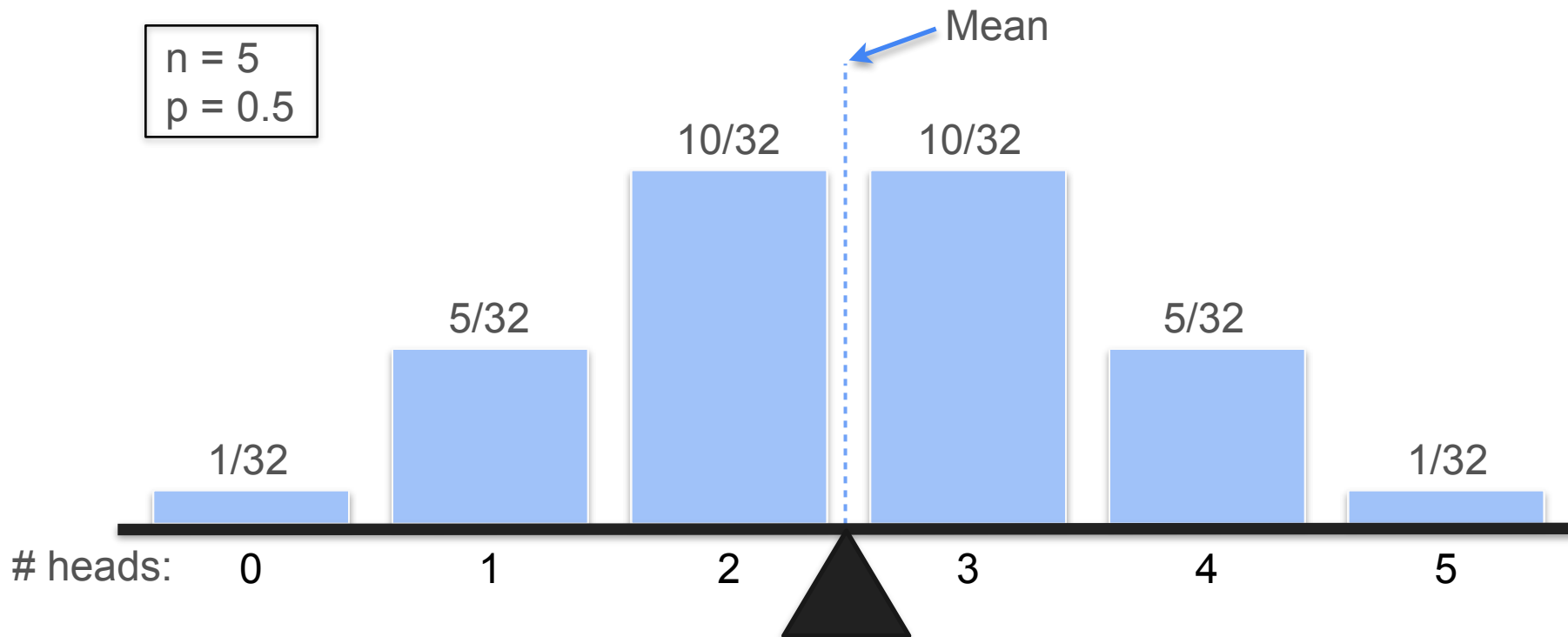
Mean, Median and Mode in Binomial Distribution

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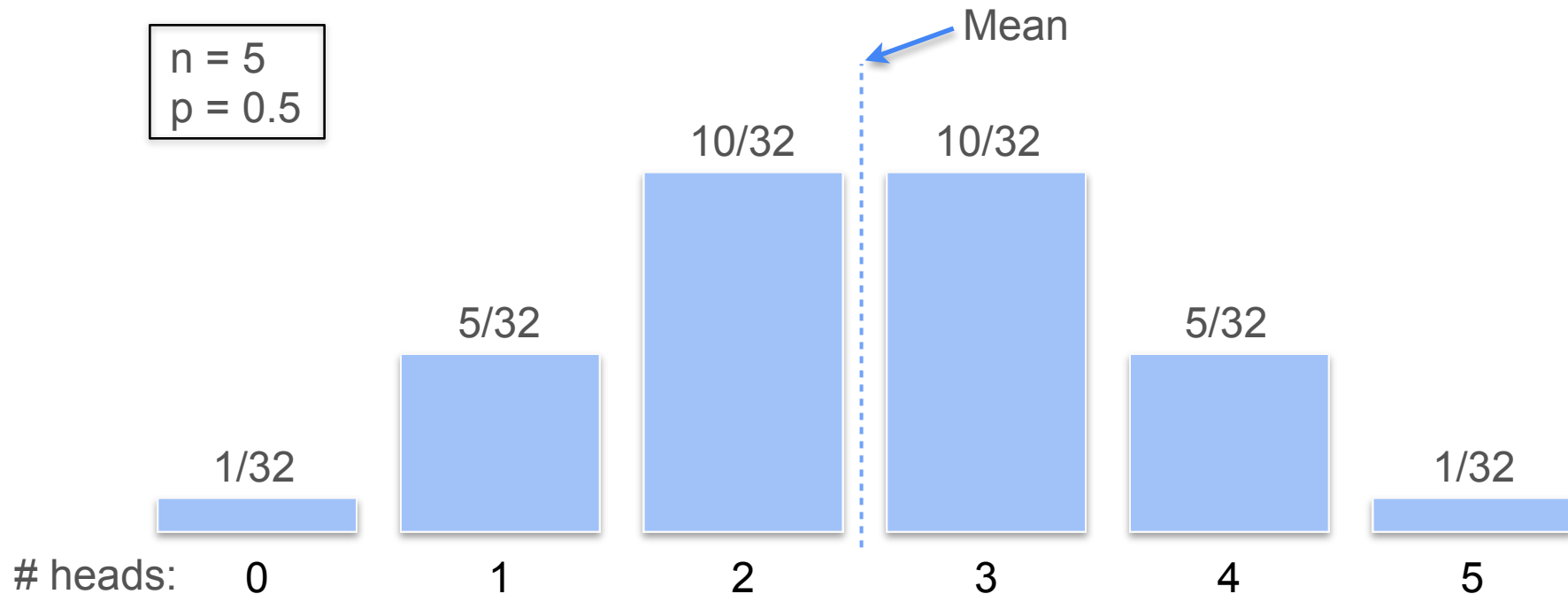
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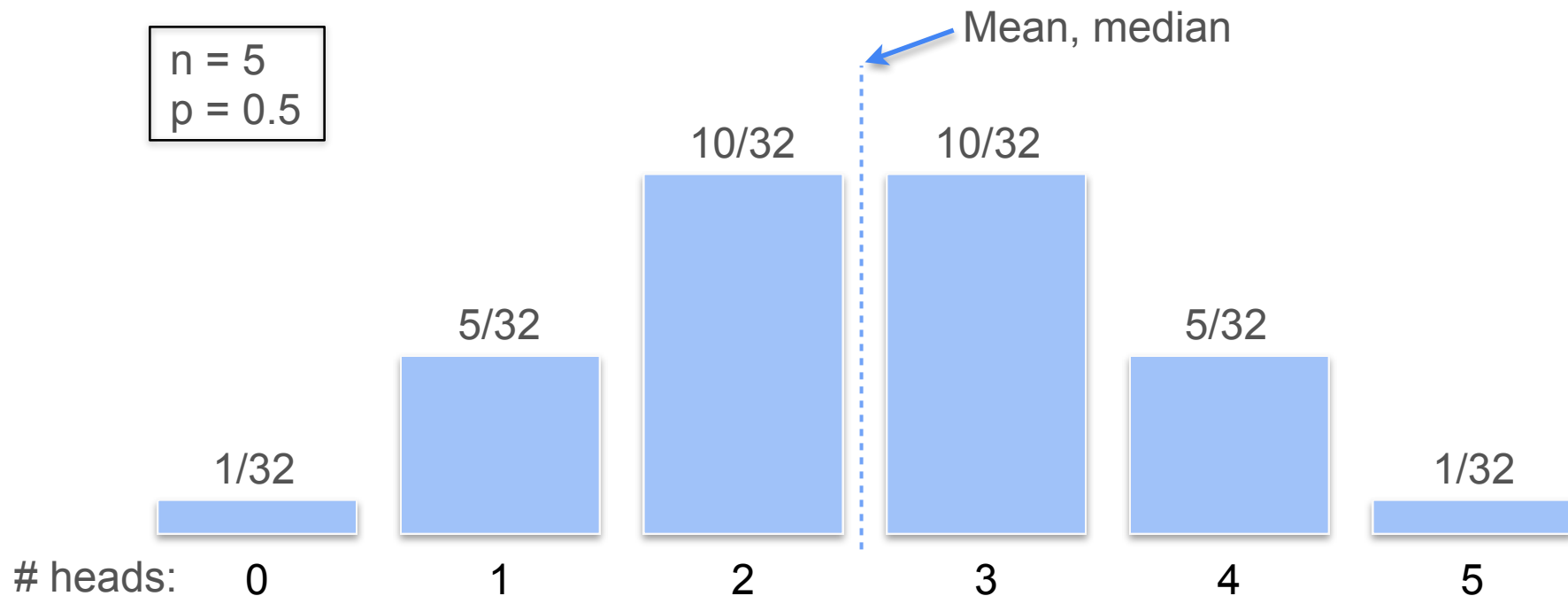
Mean, Median and Mode in Binomial Distribution

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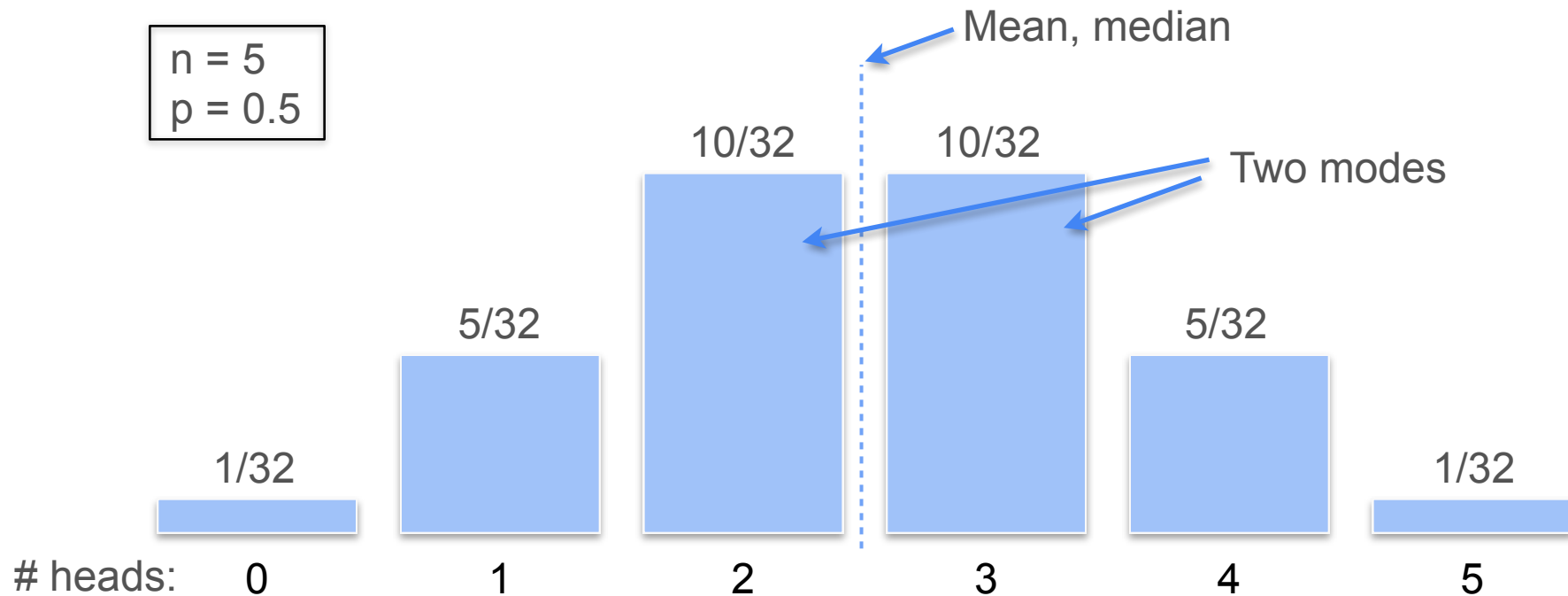
Mean, Median and Mode in Binomial Distribution

$$n = 5$$
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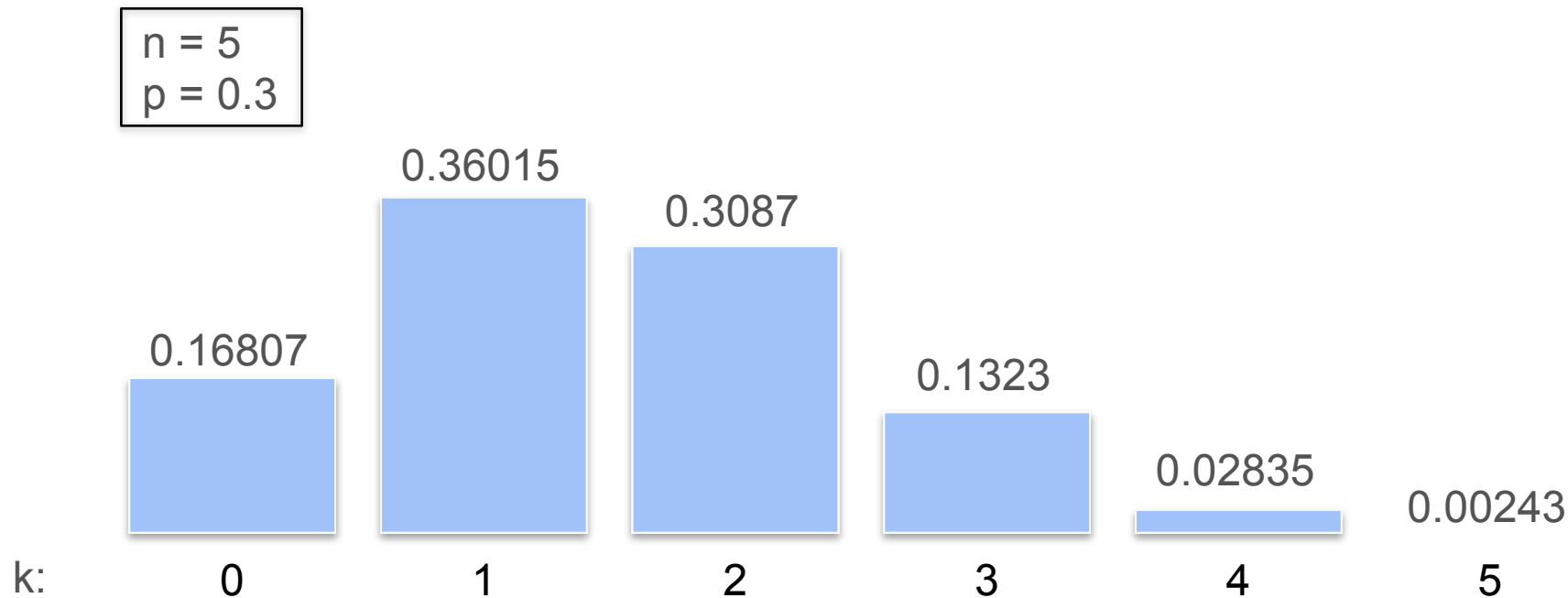


Mean, Median and Mode in Binomial Distribution

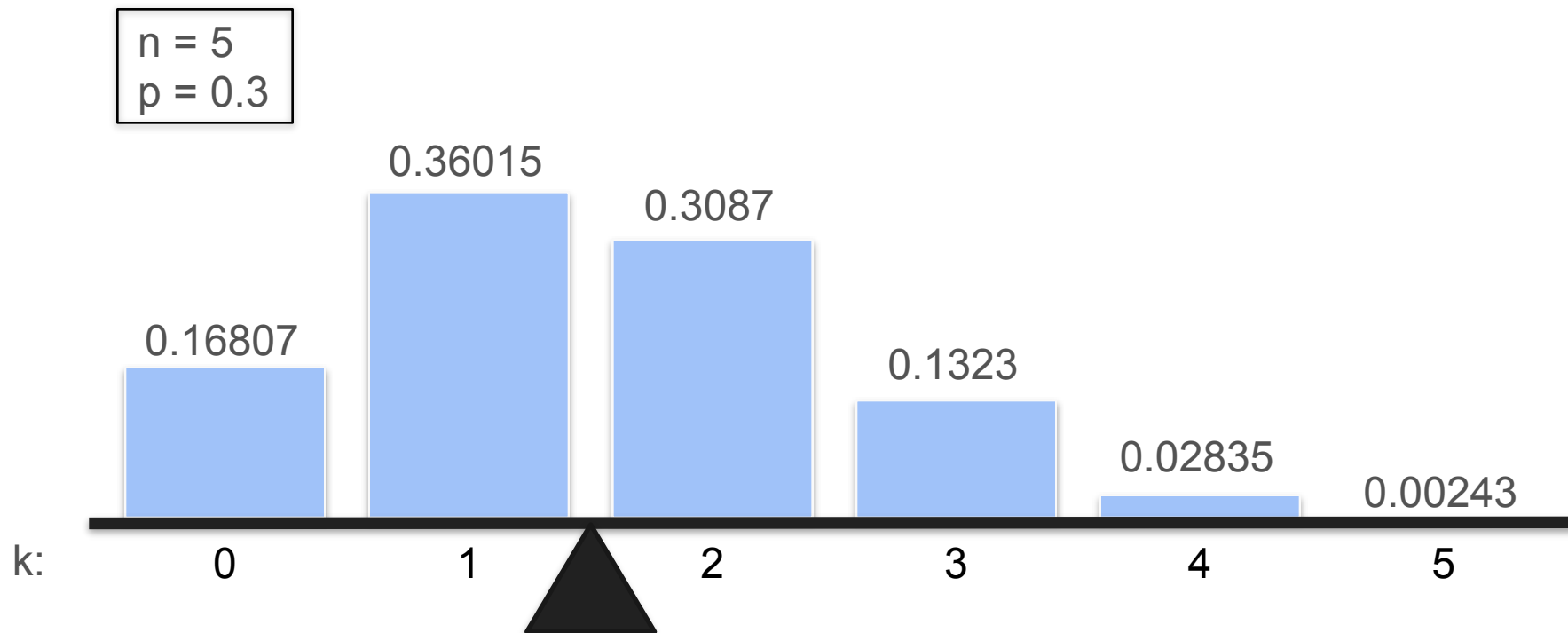
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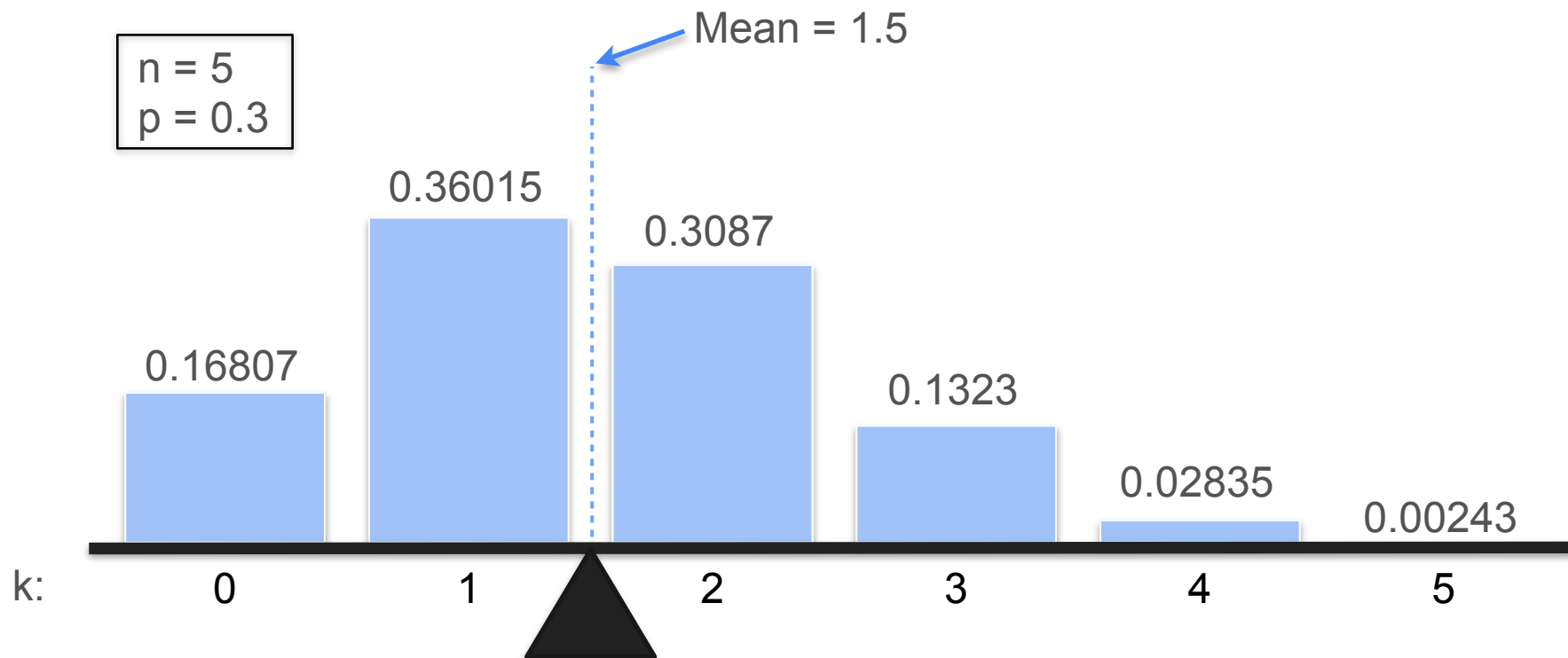
Mean, Median and Mode in Binomial Distribution



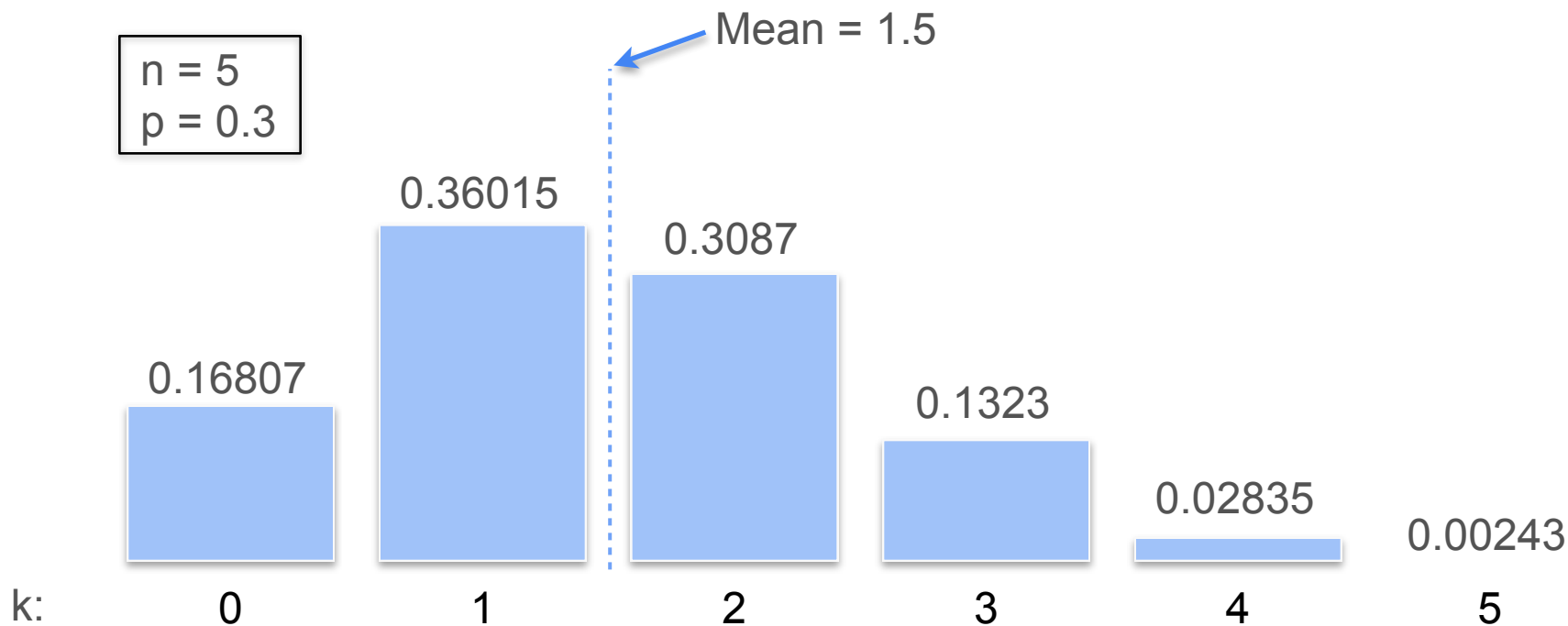
Mean, Median and Mode in Binomial Distribution



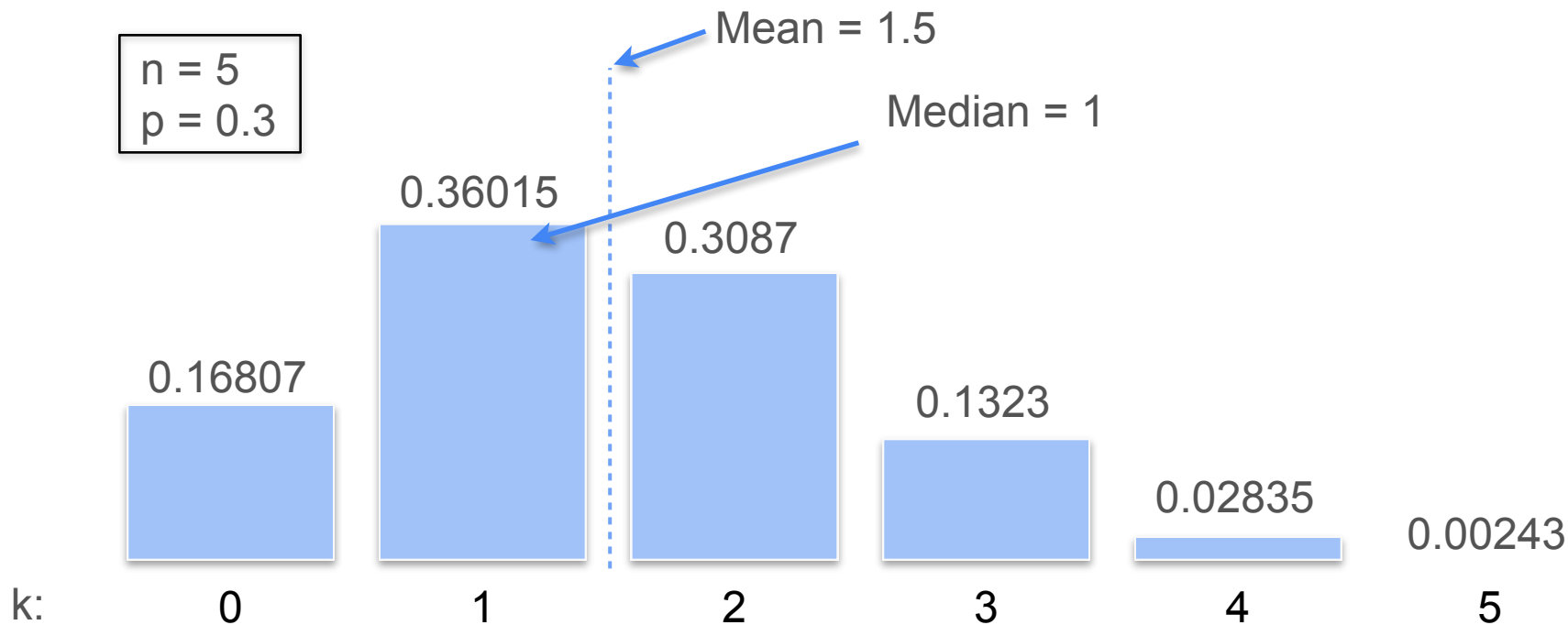
Mean, Median and Mode in Binomial Distribution



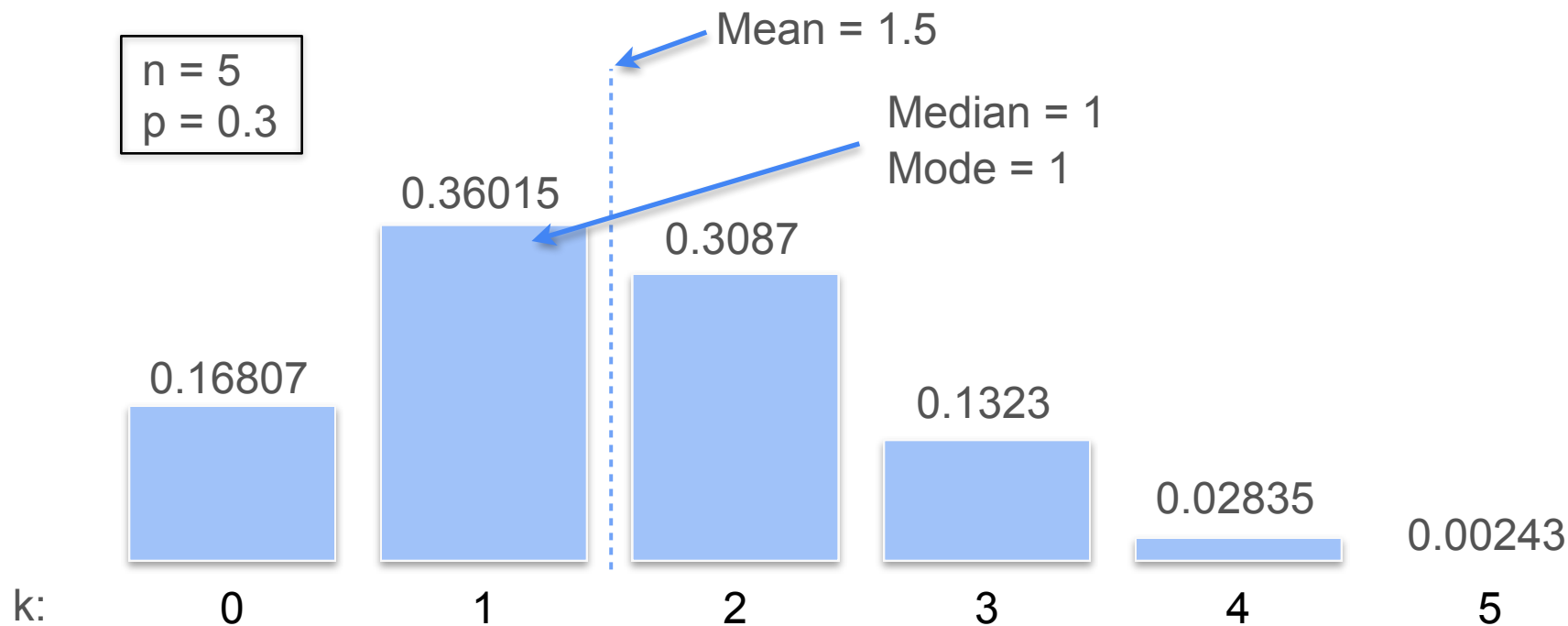
Mean, Median and Mode in Binomial Distribution



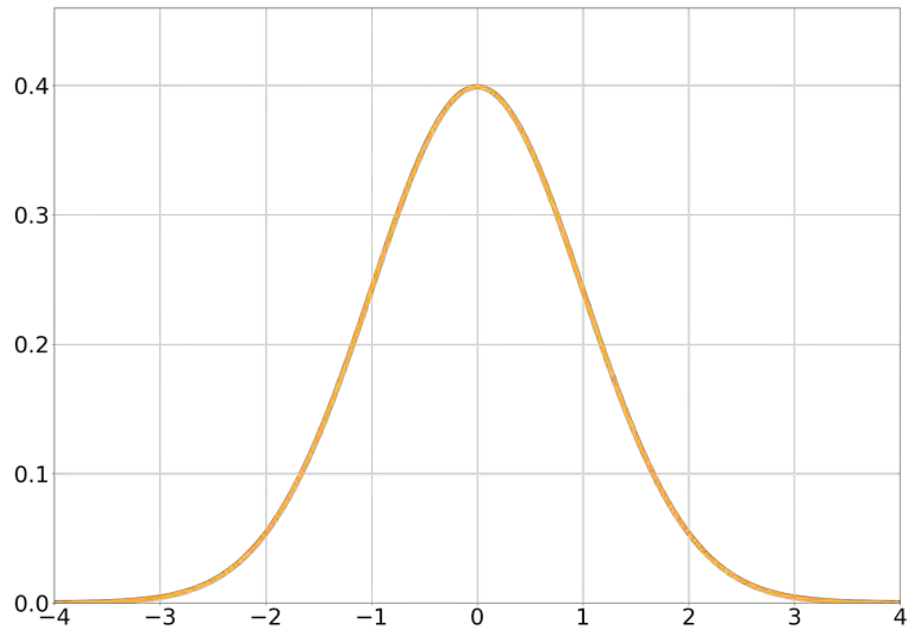
Mean, Median and Mode in Binomial Distribution



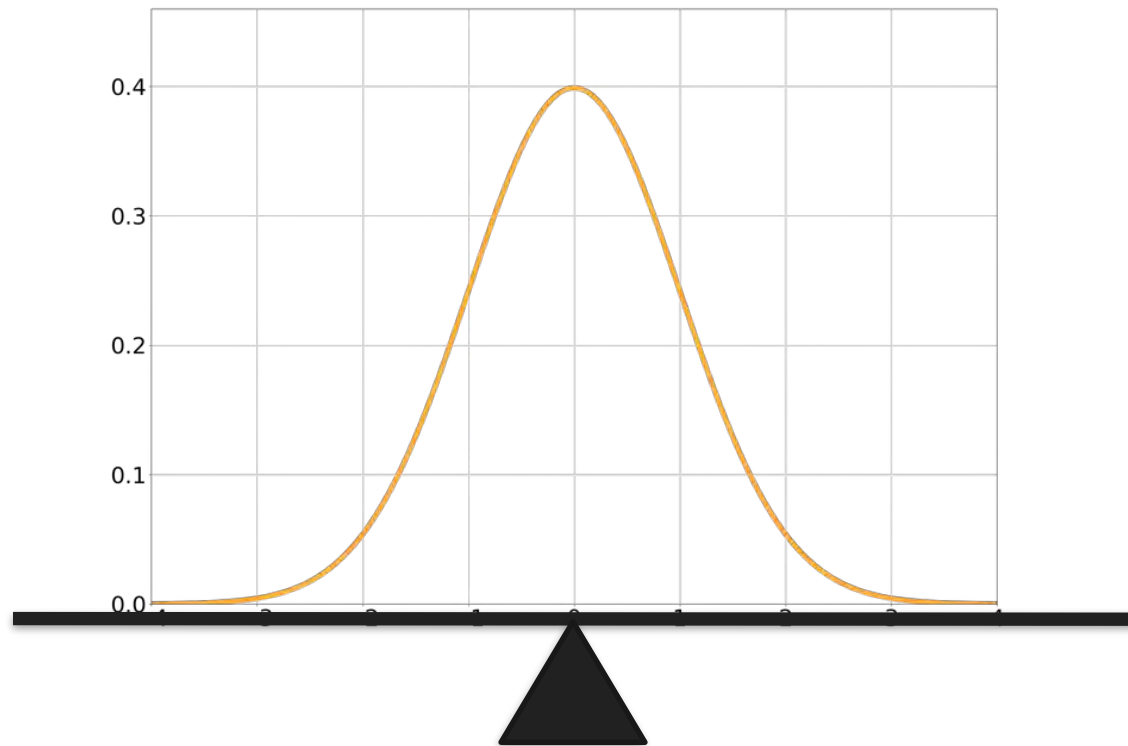
Mean, Median and Mode in Binomial Distribution



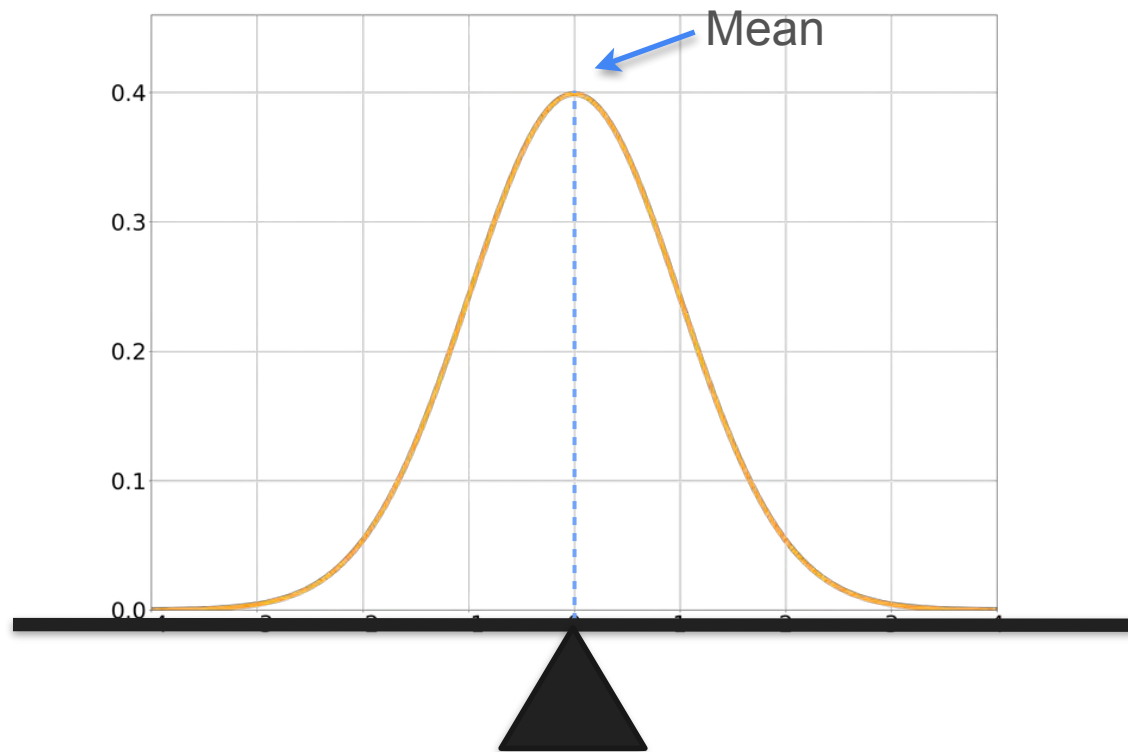
Mean, Median and Mode in Normal Distribution



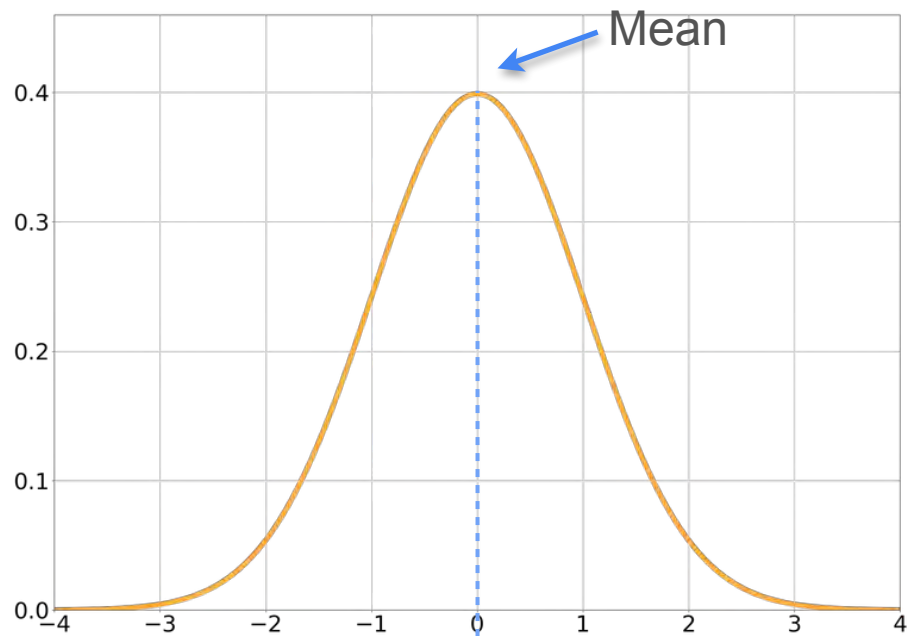
Mean, Median and Mode in Normal Distribution



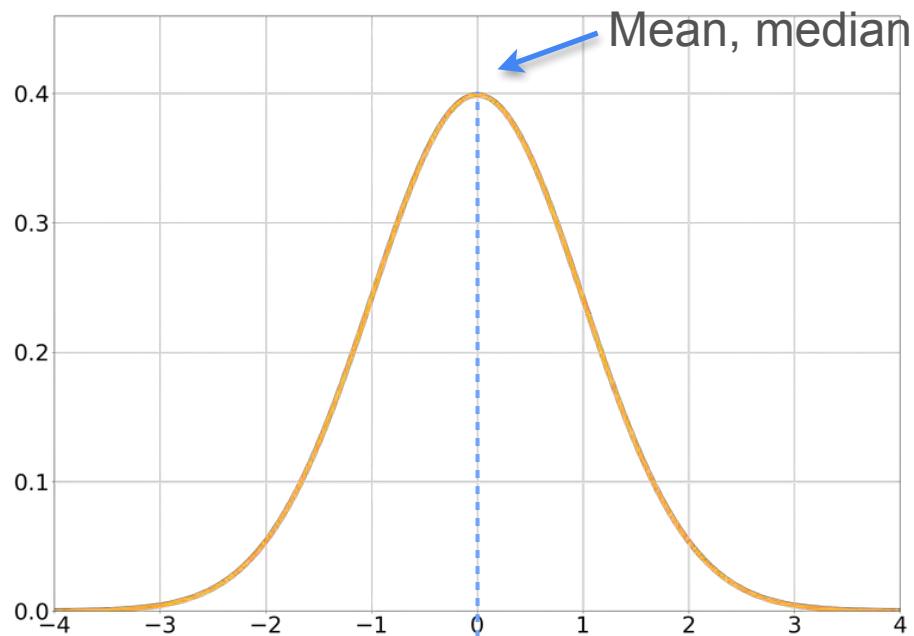
Mean, Median and Mode in Normal Distribution



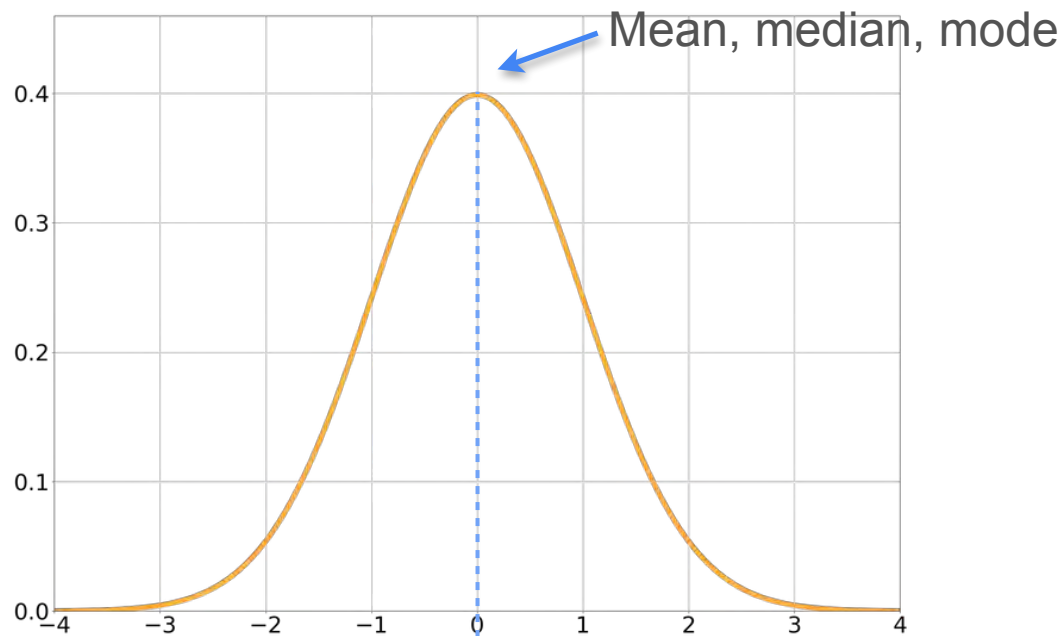
Mean, Median and Mode in Normal Distribution



Mean, Median and Mode in Normal Distribution



Mean, Median and Mode in Normal Distribution





DeepLearning.AI

Describing Distributions

Expected Value

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Game cost:

\$6

Do you play the game?

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Game cost:

\$4

Do you play the game?

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Game cost:

\$4

Do you play the game?

What is the maximum amount of money you would pay to play this game?

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Long term:

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Long term: $0.5 \cdot \$10$

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Long term: $0.5 \cdot \$10 + 0.5 \cdot \0

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Long term: $0.5 \cdot \$10 + 0.5 \cdot \$0 = \$5$

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Long term: $0.5 \cdot \$10 + 0.5 \cdot \$0 = \$5$  You expect to win \$5 on average
 $\mathbb{E}[X] = 5$

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Game cost:

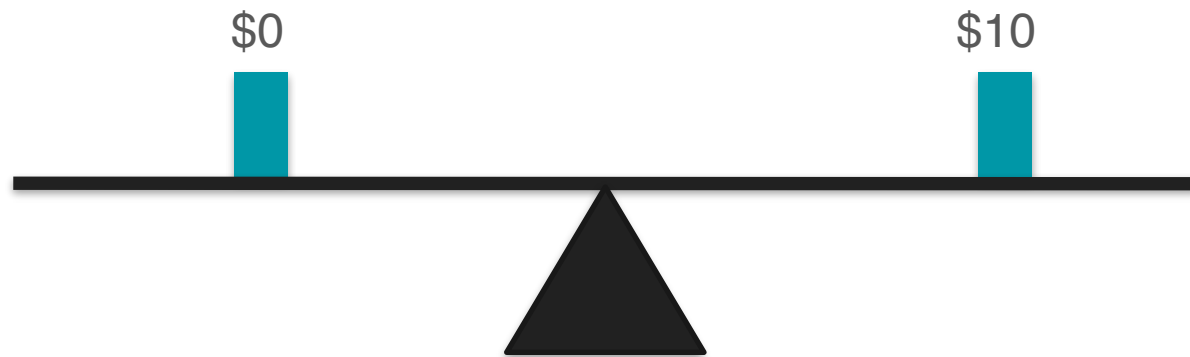
\$5

Long term: $0.5 \cdot \$10 + 0.5 \cdot \$0 = \$5$ $\Rightarrow$ You expect to win \$5 on average
 $\mathbb{E}[X] = 5$

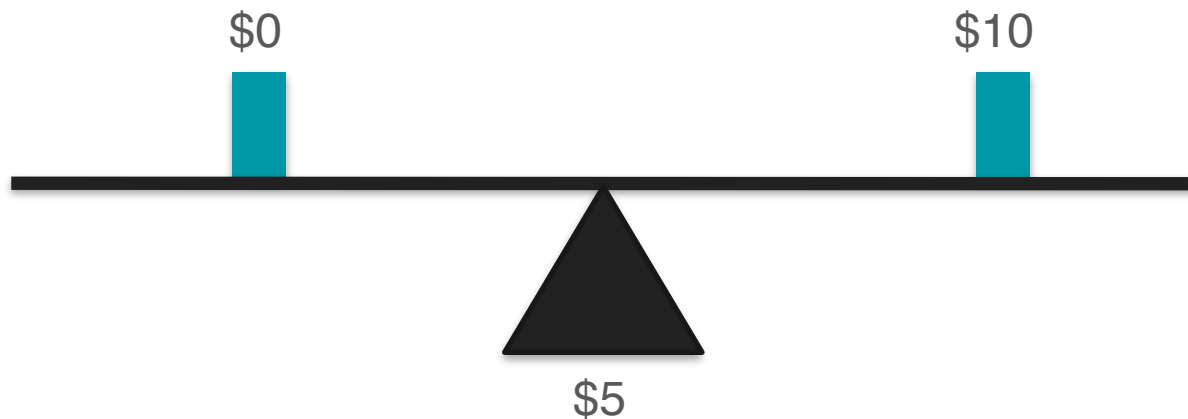
Expected Value: Motivation Example 1



Expected Value: Motivation Example 1



Expected Value: Motivation Example 1



Expected Value: Motivation Example 2

Play another game



Flip three coins. For each heads you win \$1

What is the maximum amount of money you would pay to play this game?

Expected Value: Motivation Example 2

Number of heads:

0

1

2

3



Expected Value: Motivation Example 2

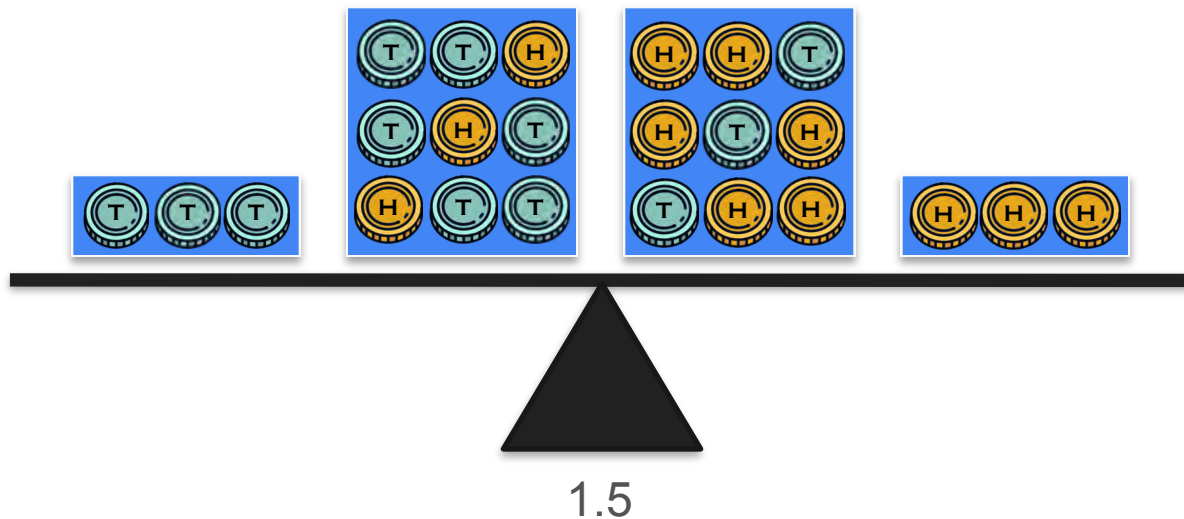
Number of heads:

0

1

2

3



Expected Value: Motivation Example 2

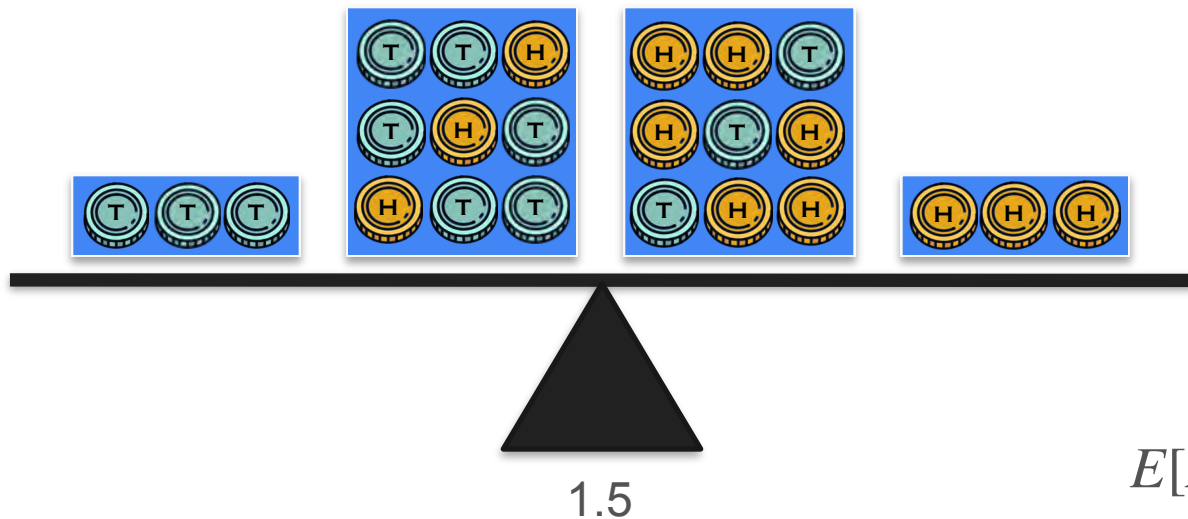
Number of heads:

0

1

2

3

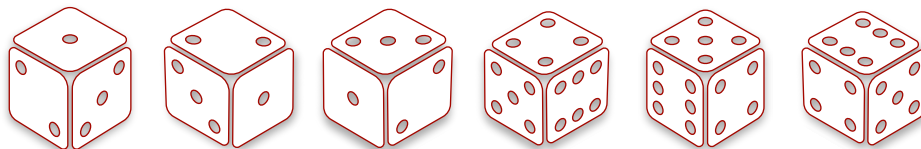


$$E[X] = 1.5$$

Expected Value: Motivation Example 3

Probability: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

Roll: 1 2 3 4 5 6

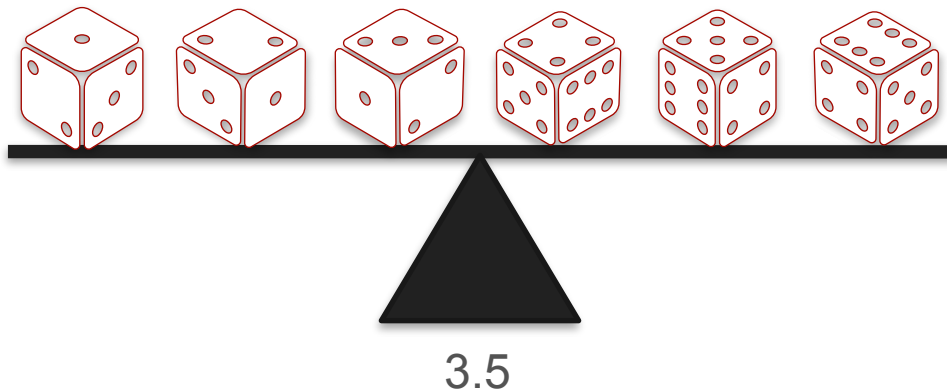


$$\frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

Expected Value: Motivation Example 3

Probability: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

Roll: 1 2 3 4 5 6



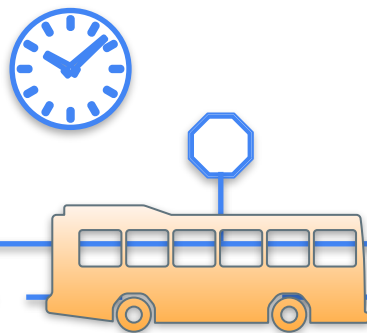
Expected Value



Expected Value



Expected Value



Expected Value

Waiting Time

15 min



Expected Value

Waiting Time

15 min



Expected Value

Waiting Time

15 min



Expected Value

Waiting Time

15 min



Expected Value

Waiting Time

15 min

32 min

58 min

1 min

47 min

14 min



Expected Value

Waiting Time

15 min

32 min

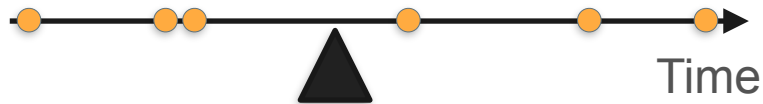
58 min

1 min

47 min

14 min

Average = 27.833



Expected Value

Waiting Time

15 min

32 min

58 min

1 min

47 min

14 min

37 min

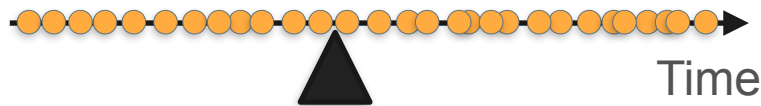
8 min

29 min

55 min

...

Average = 27.833



Expected Value

Waiting Time

15 min

32 min

58 min

1 min

47 min

14 min

37 min

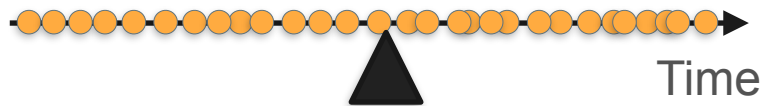
8 min

29 min

55 min

...

Average = 30



Expected Value

Waiting Time

15 min

32 min

58 min

1 min

47 min

14 min

37 min

8 min

29 min

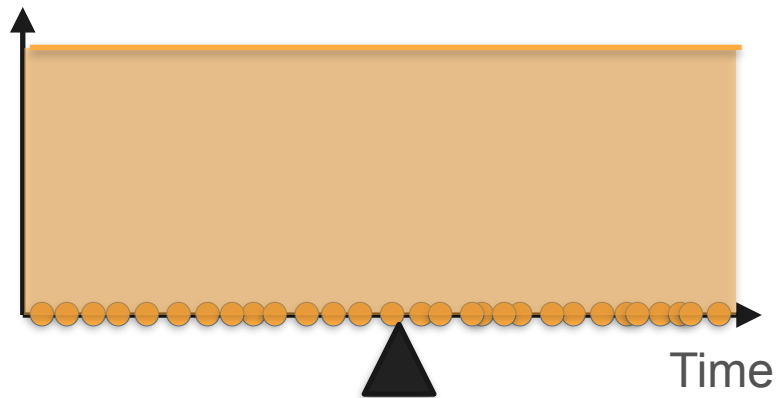
55 min

...

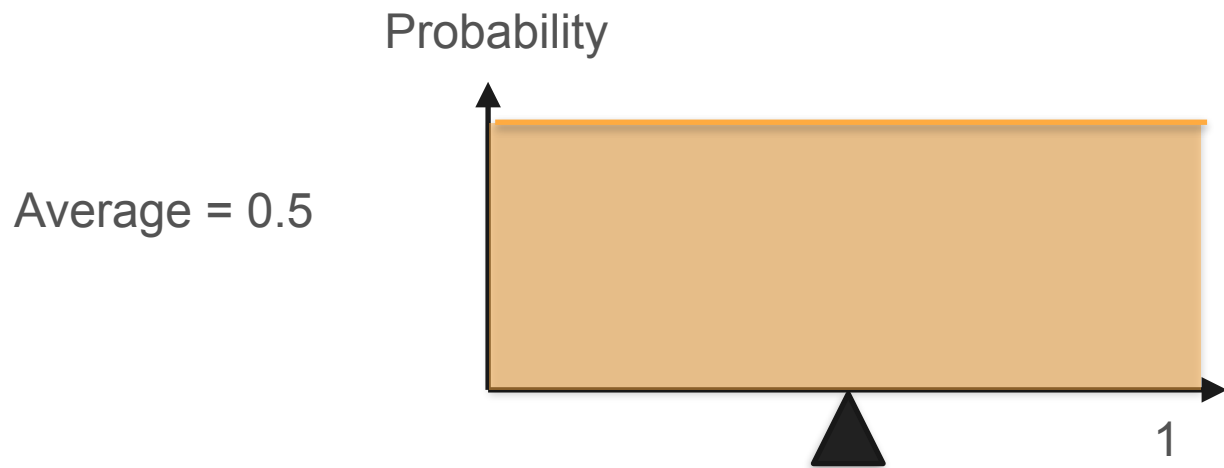
Average = 30



Probability

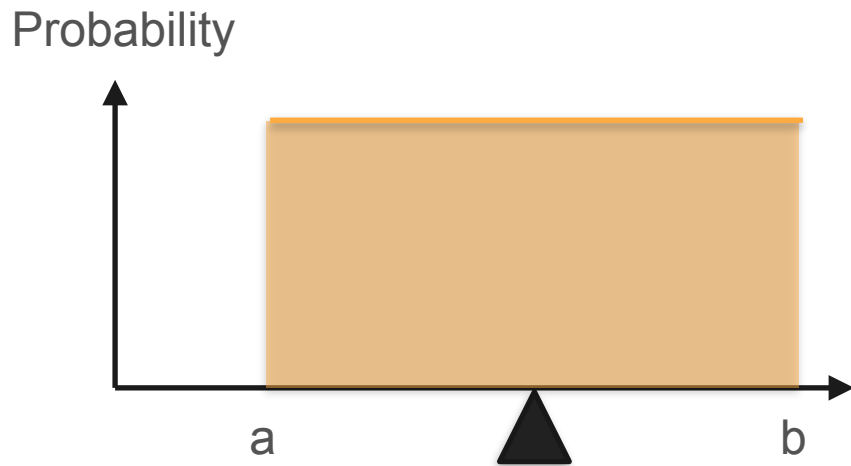


Expected Value: Uniform Distribution

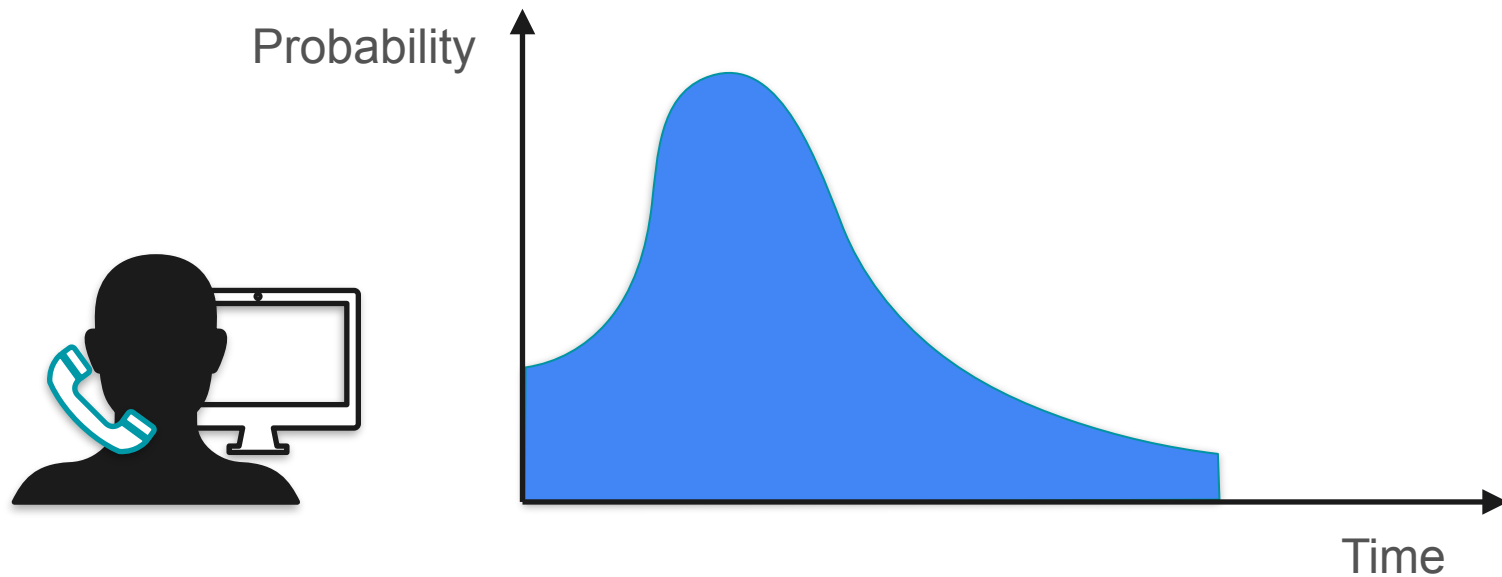


Expected Value: Uniform Distribution

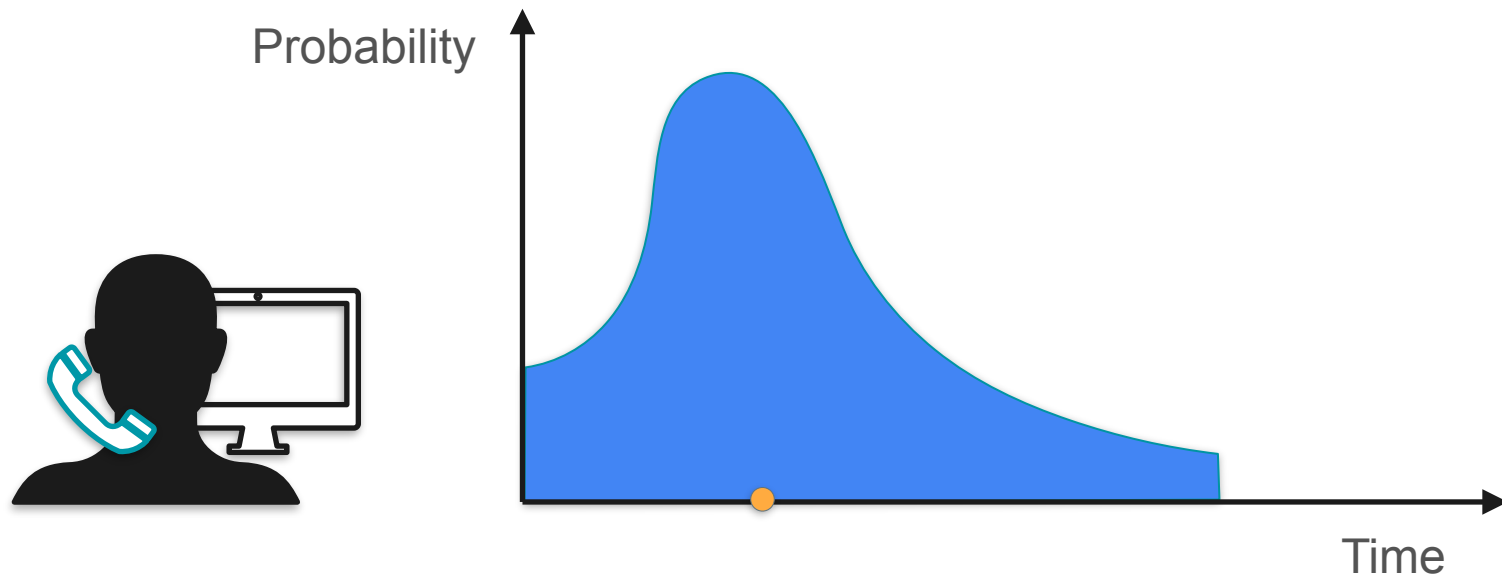
$$\text{Average} = \frac{a + b}{2}$$



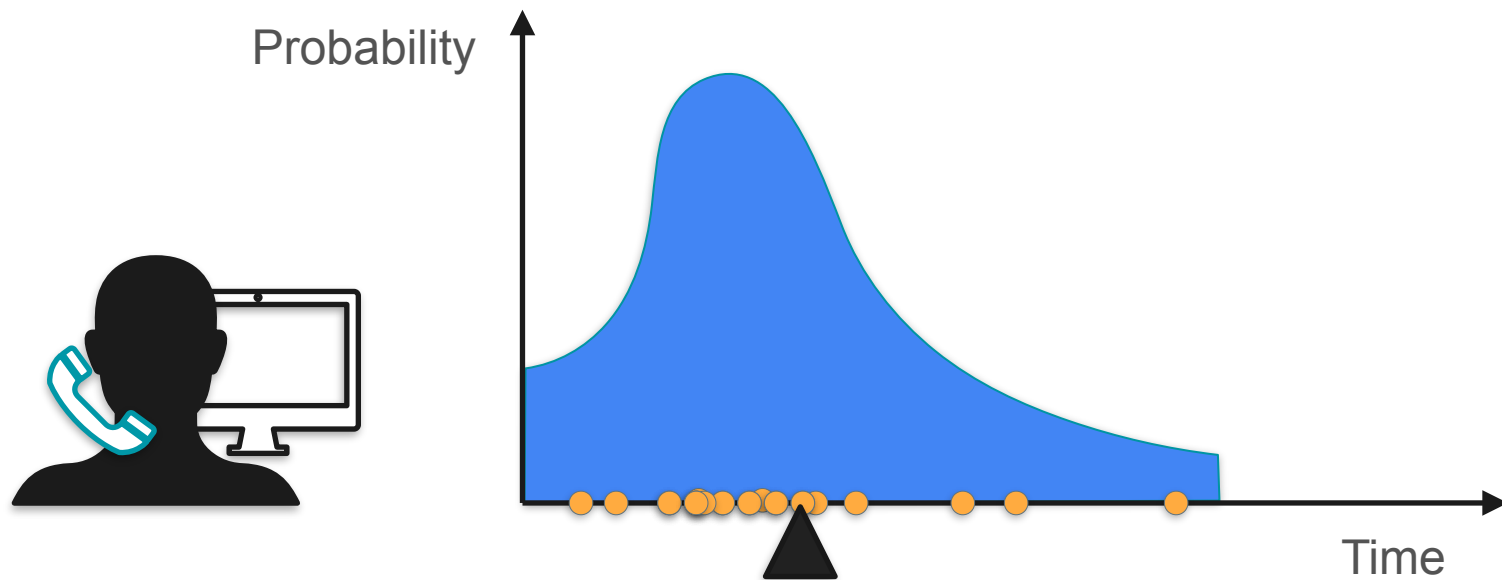
Expected Value



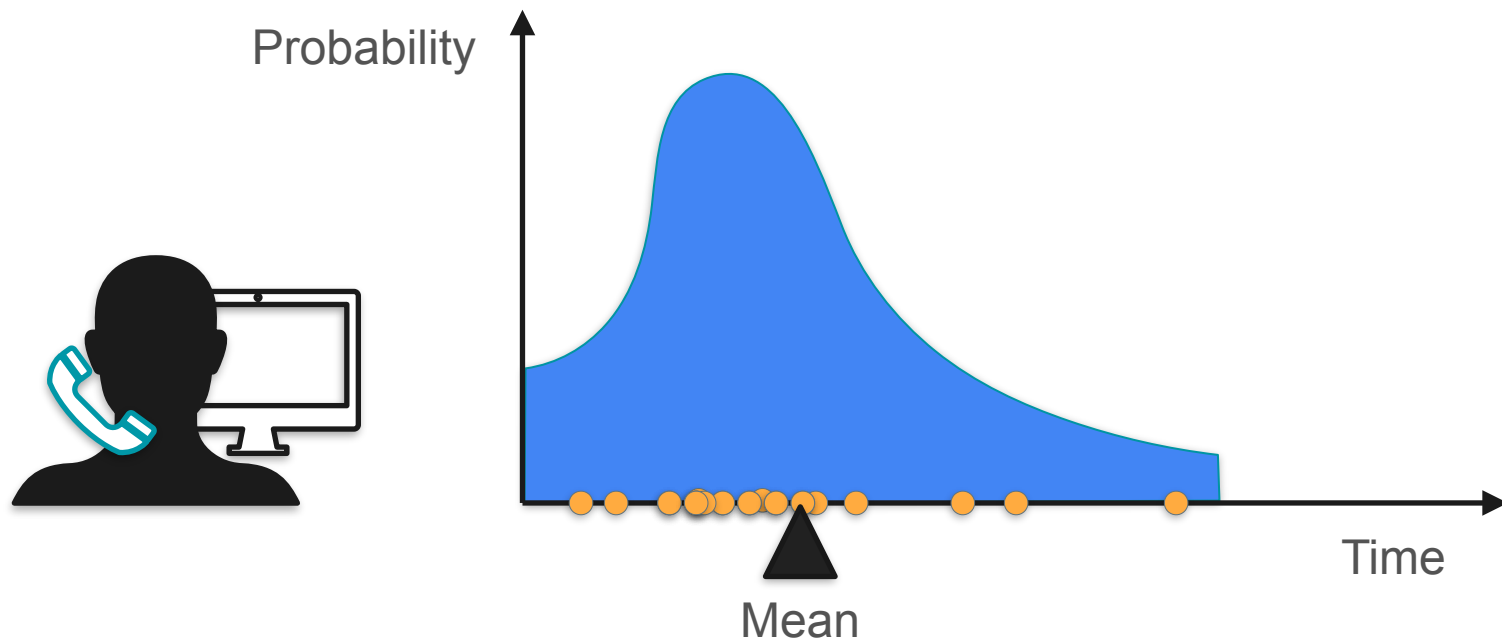
Expected Value



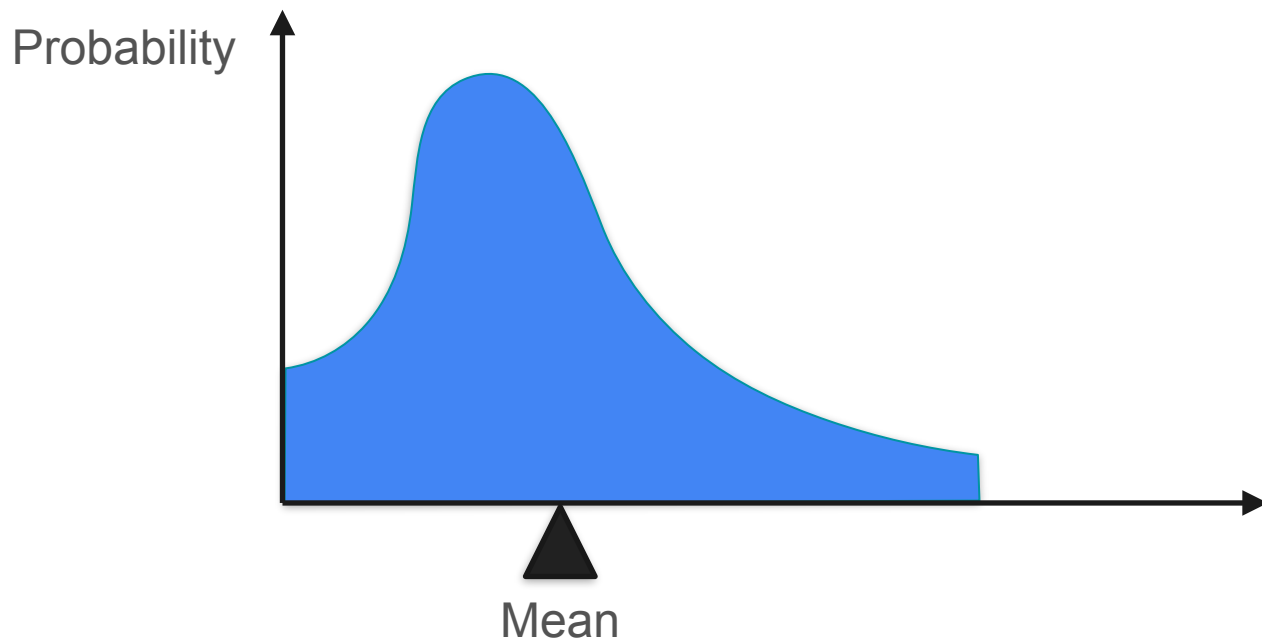
Expected Value



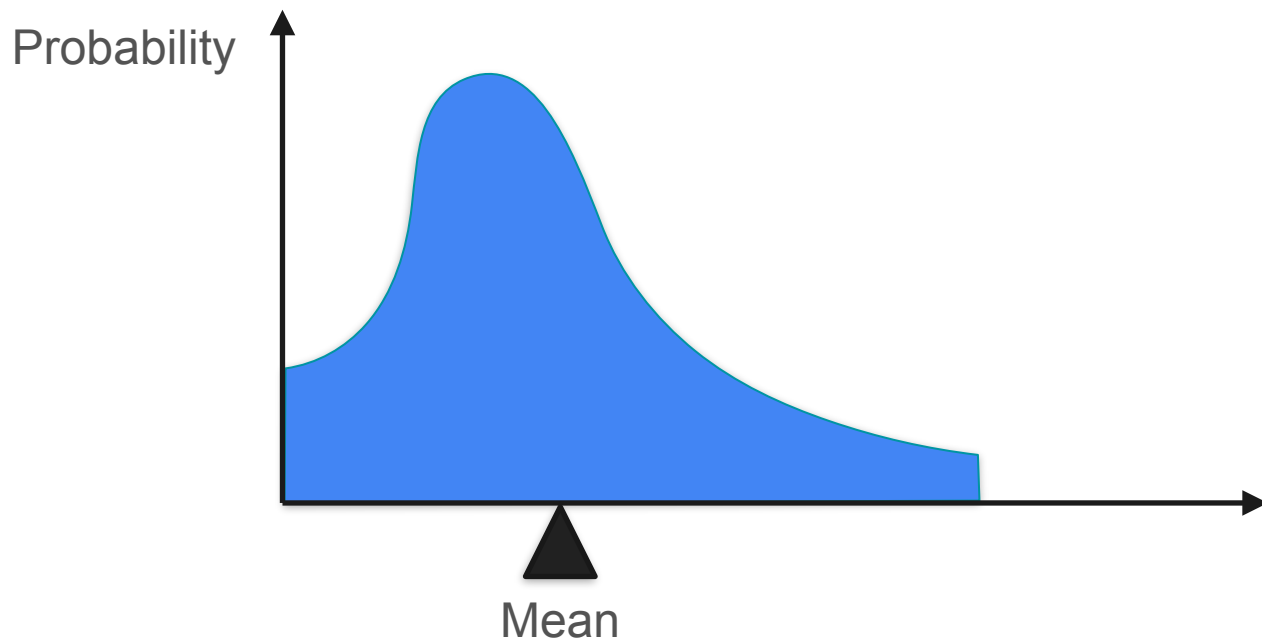
Expected Value



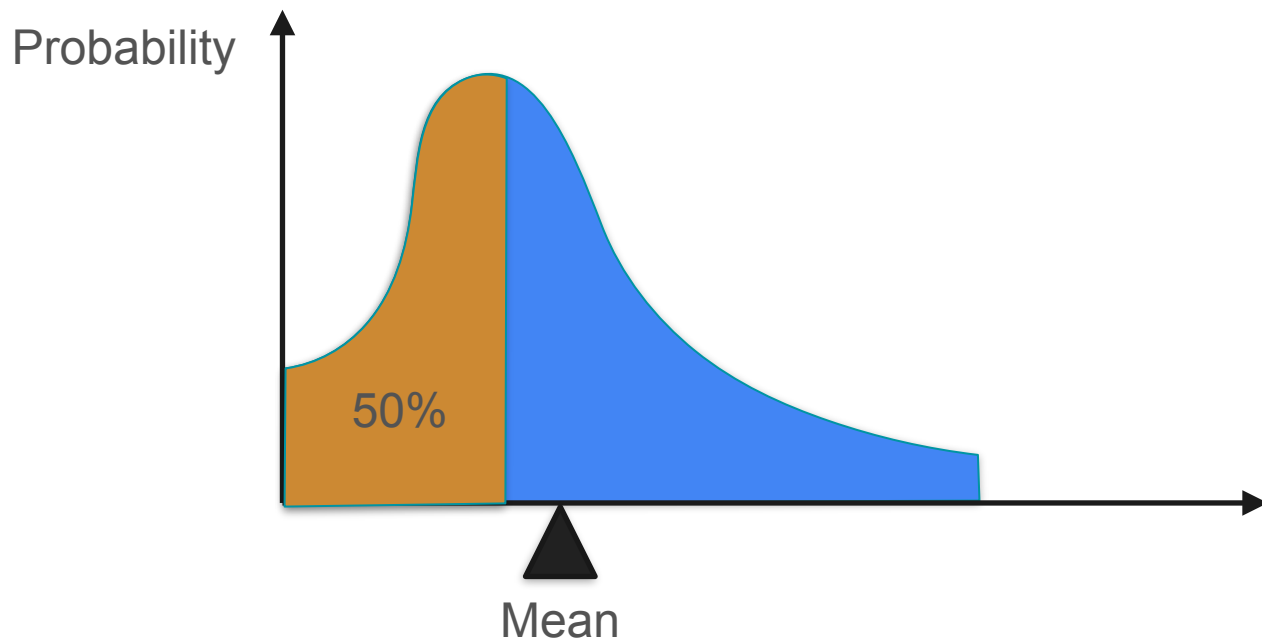
Expected Value: General Case



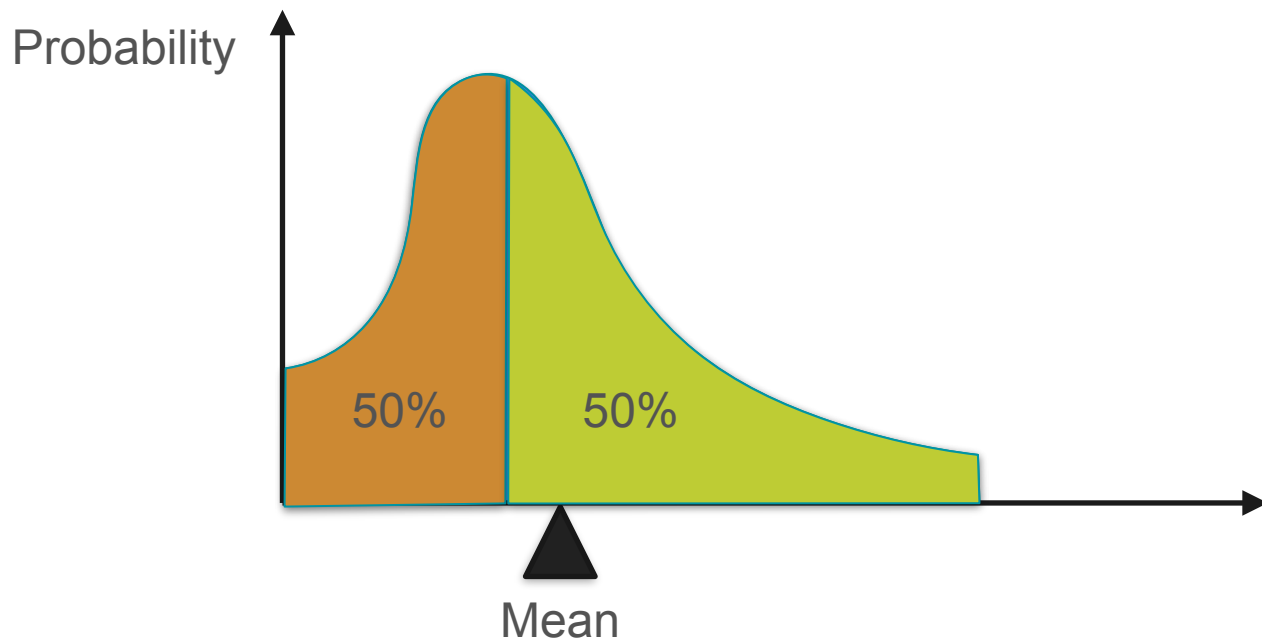
Expected Value: Common Misconception



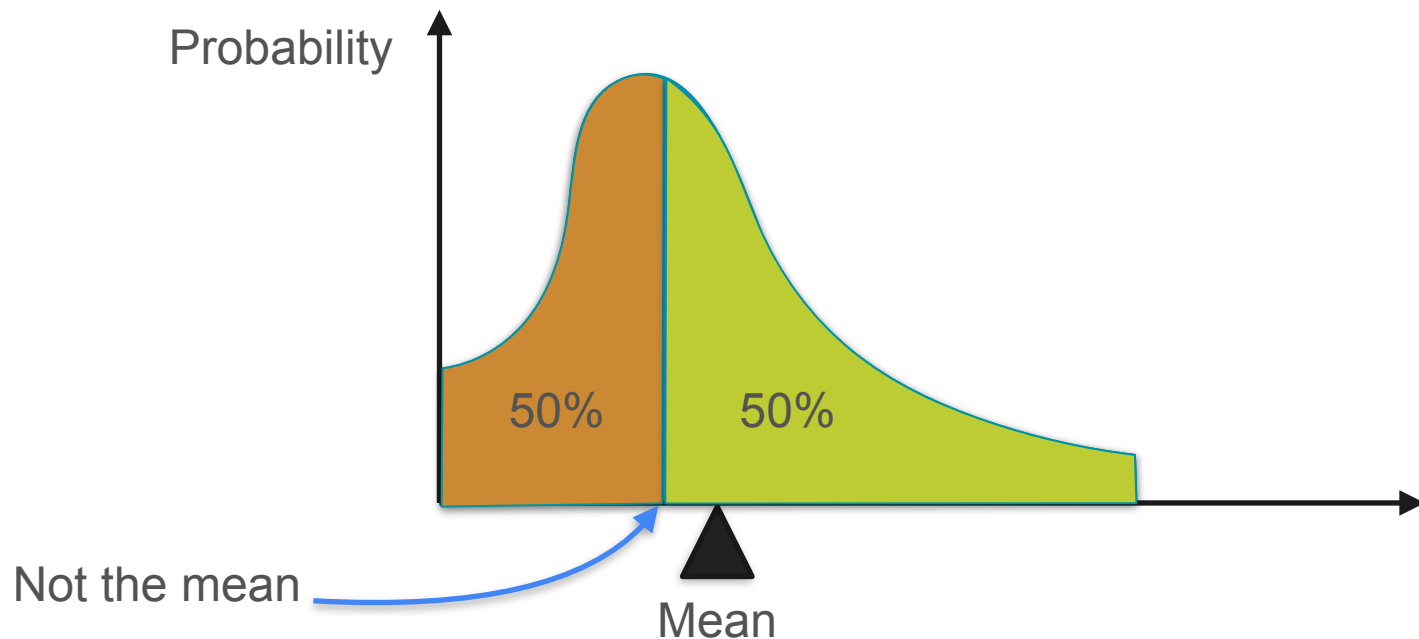
Expected Value: Common Misconception



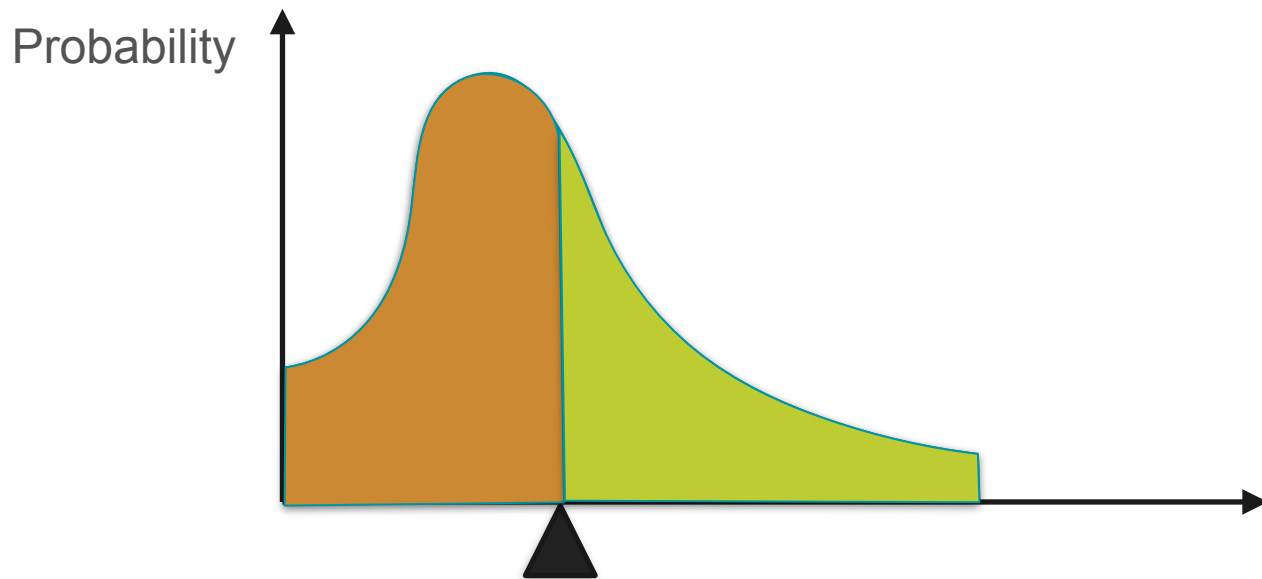
Expected Value: Common Misconception



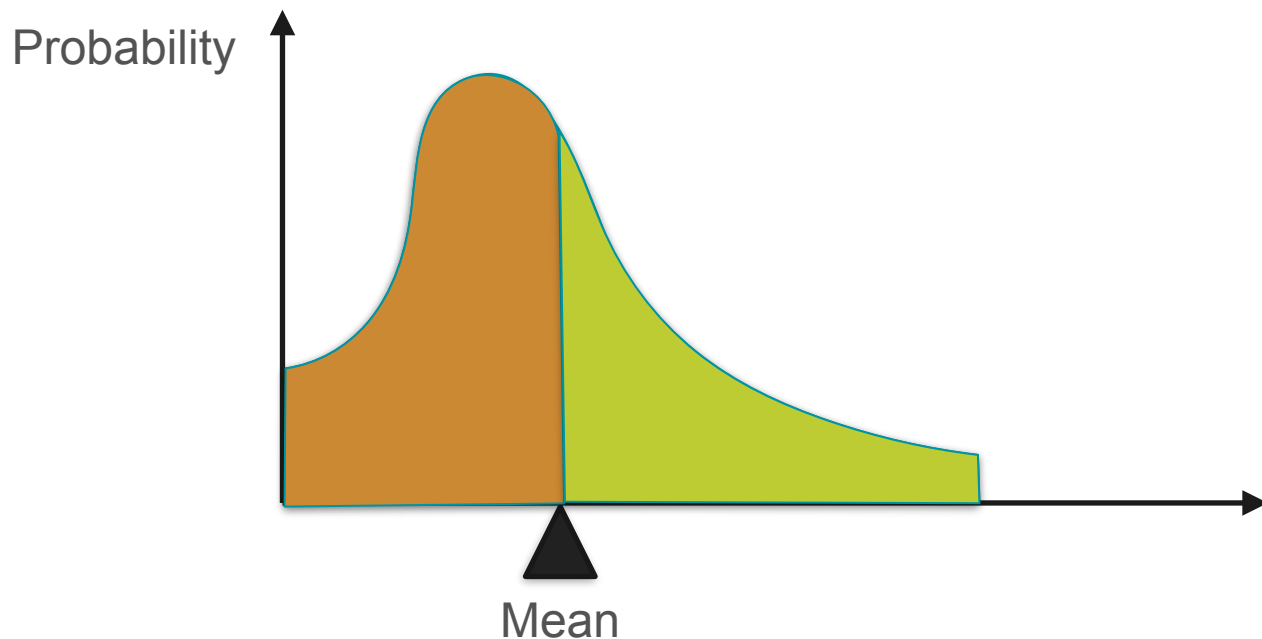
Expected Value: Common Misconception



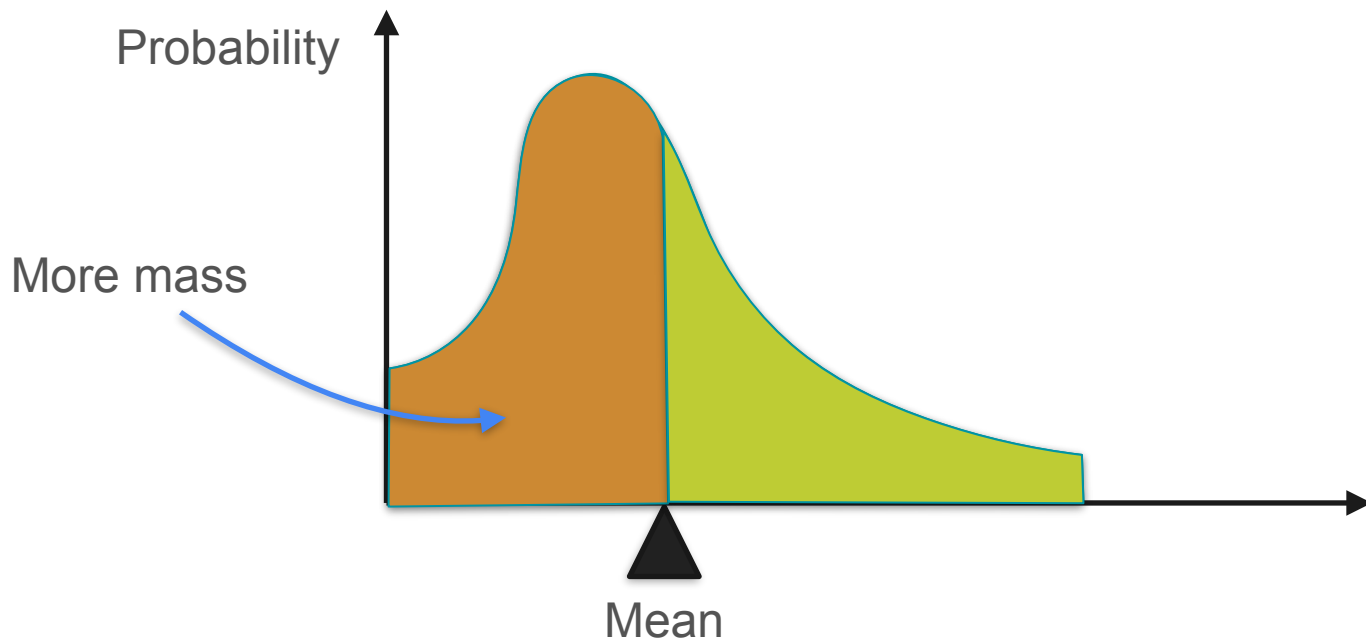
Expected Value: Common Misconception



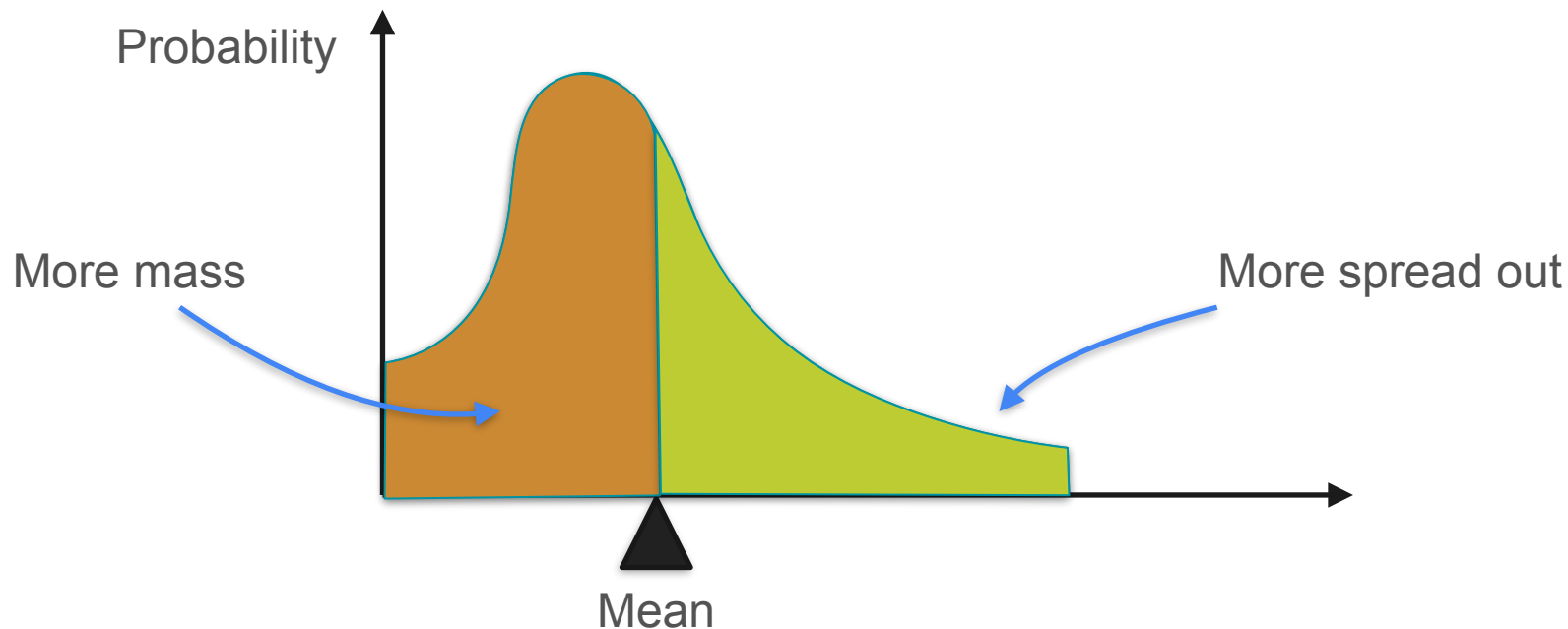
Expected Value: Common Misconception



Expected Value: Common Misconception



Expected Value: Common Misconception



Expected Value: Common Misconception

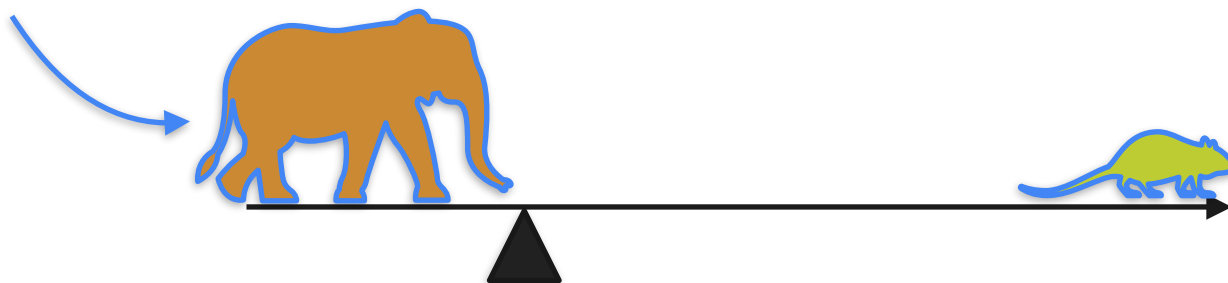


Expected Value: Common Misconception



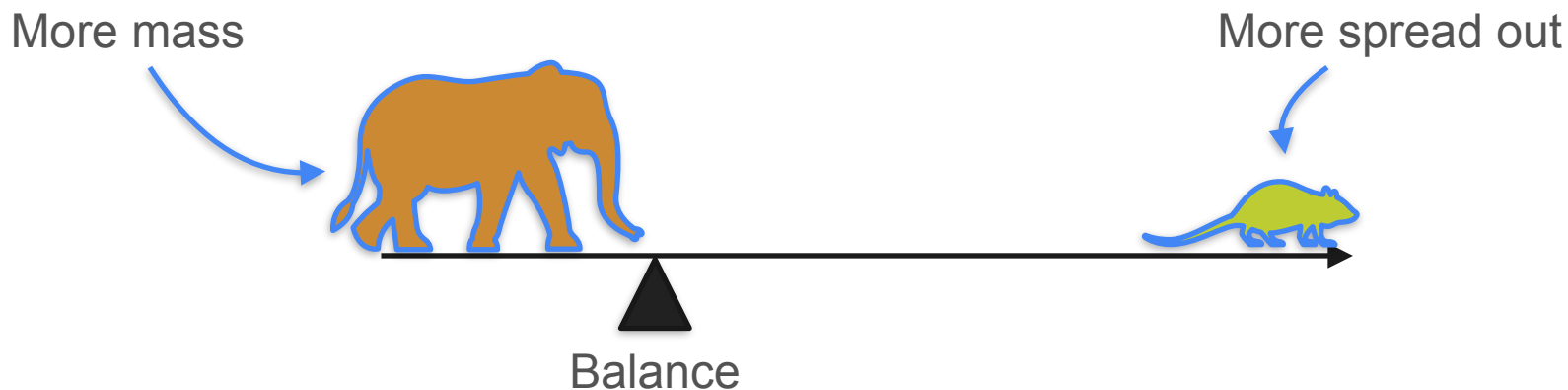
Expected Value: Common Misconception

More mass



Balance

Expected Value: Common Misconception



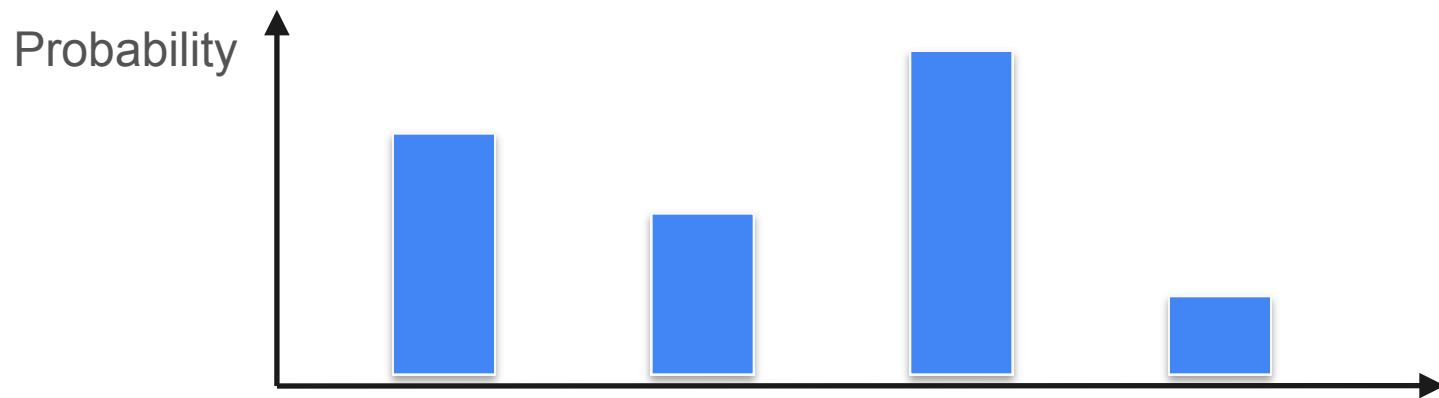


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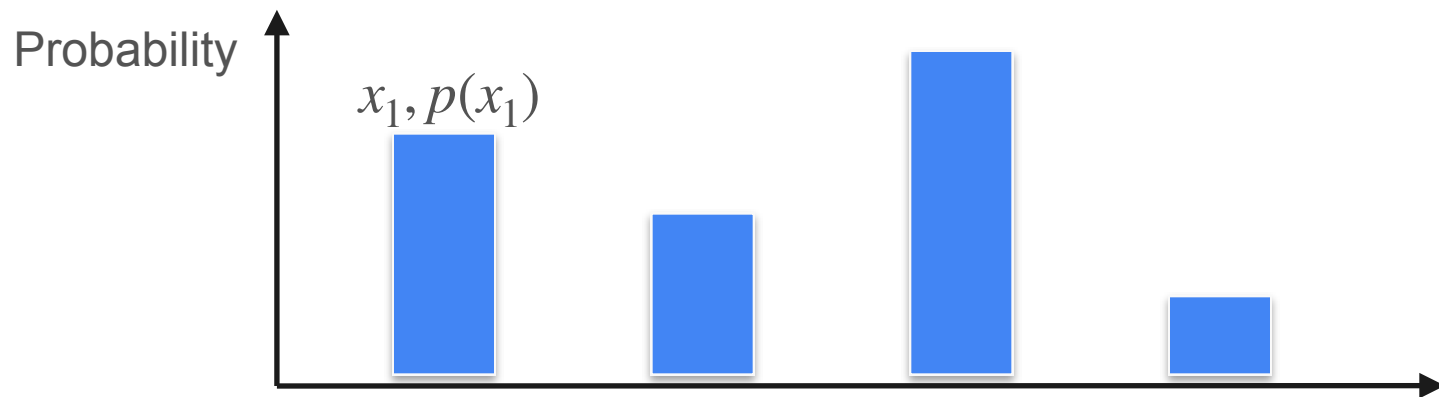
Describing Distributions

Expected value of a function

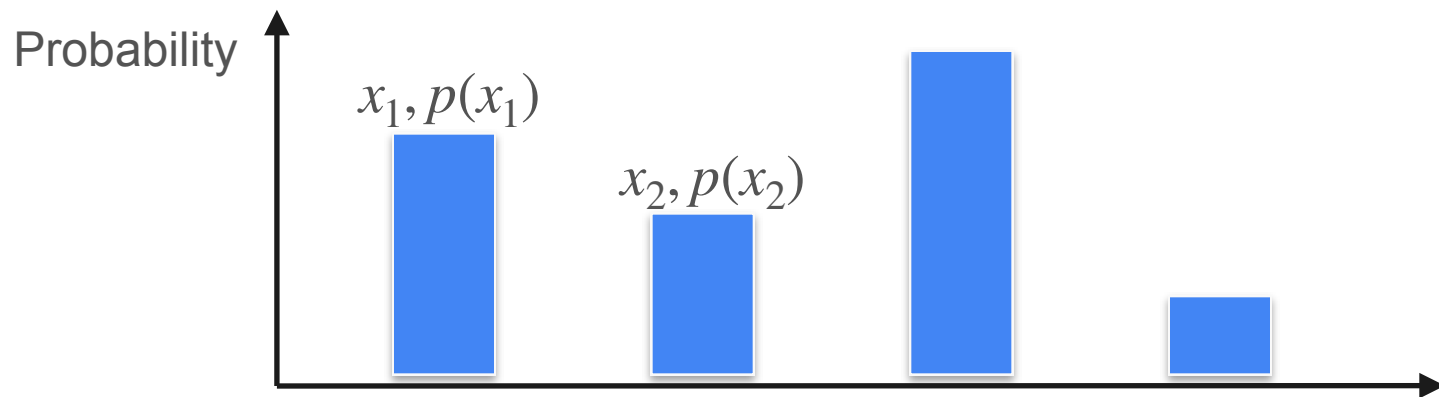
Expected Value of a Function



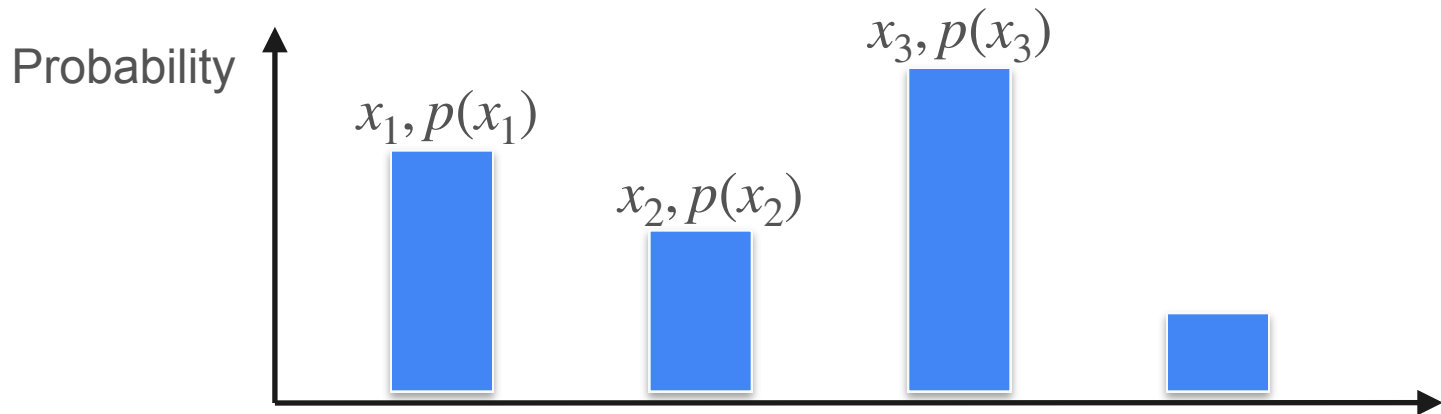
Expected Value of a Function



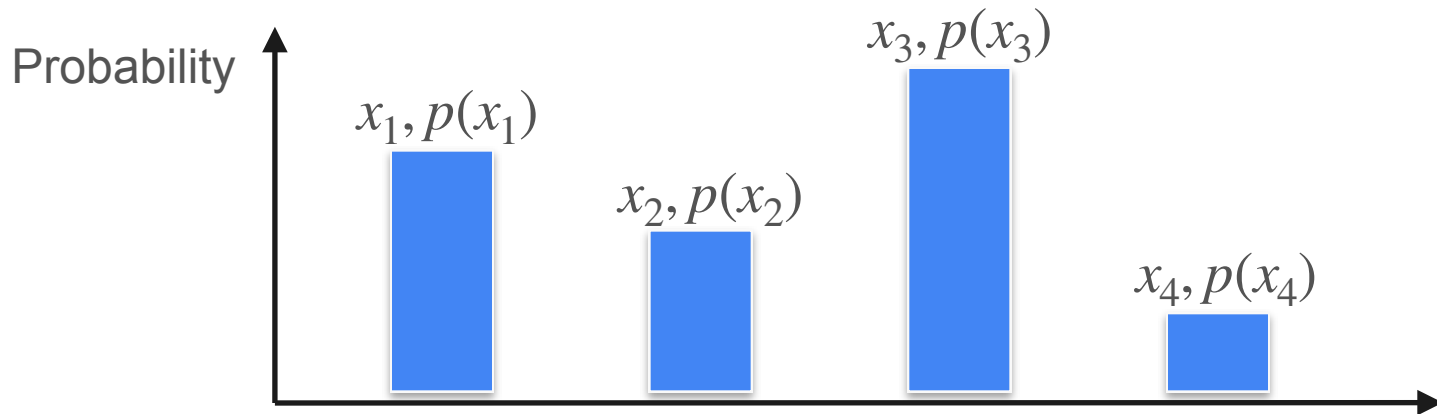
Expected Value of a Function



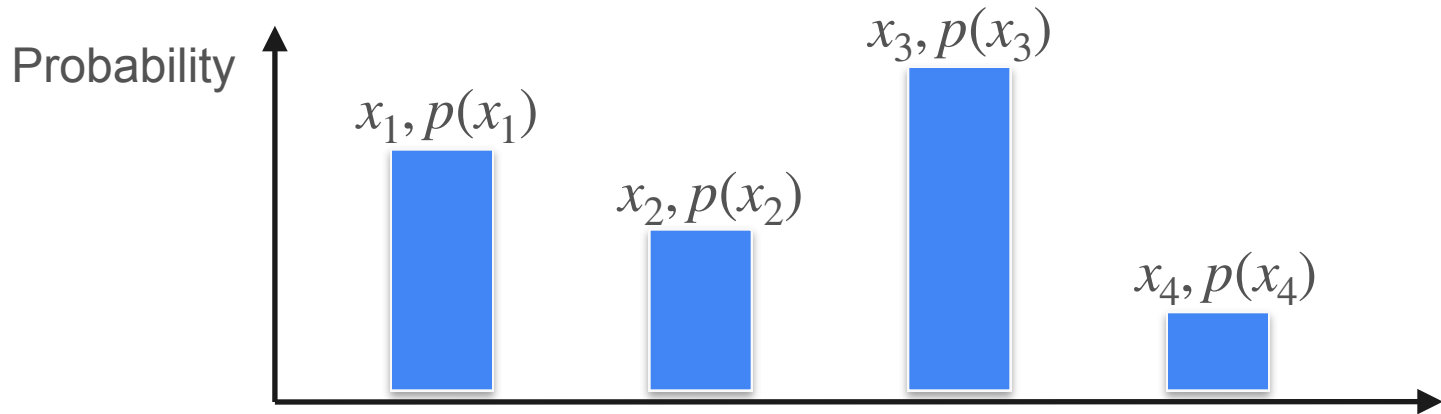
Expected Value of a Function



Expected Value of a Function

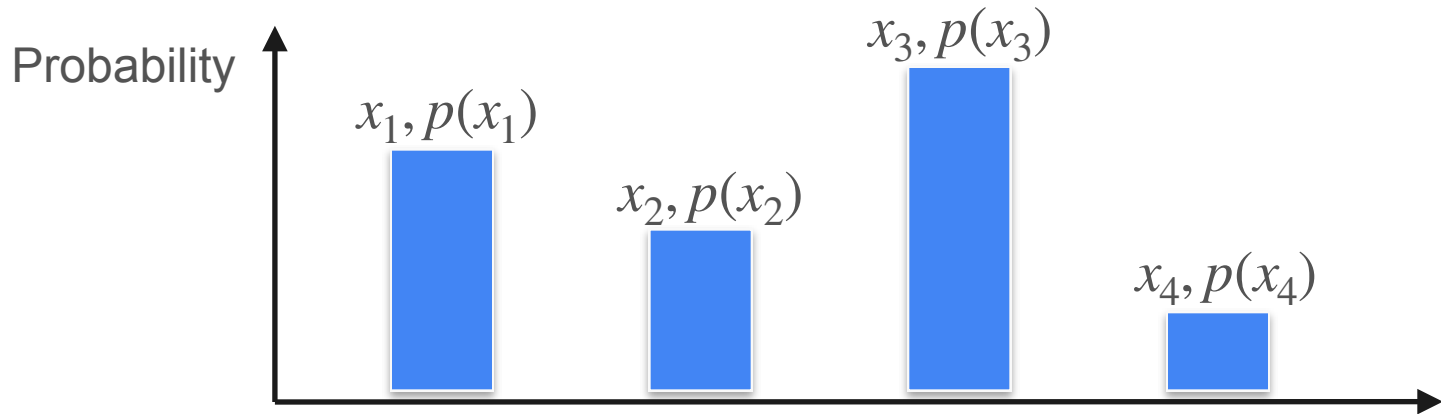


Expected Value of a Function



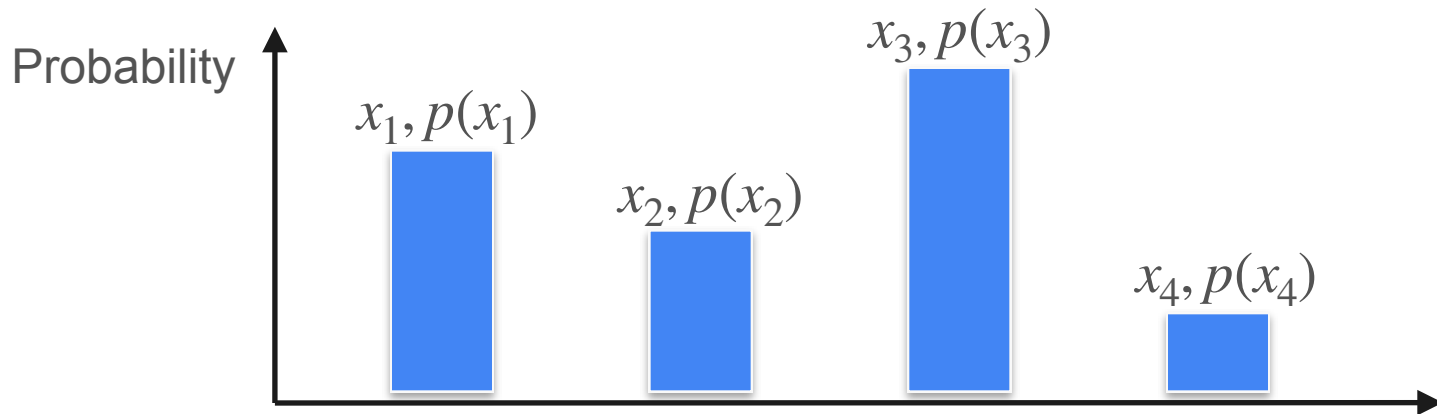
$$\mathbb{E}[X] =$$

Expected Value of a Function



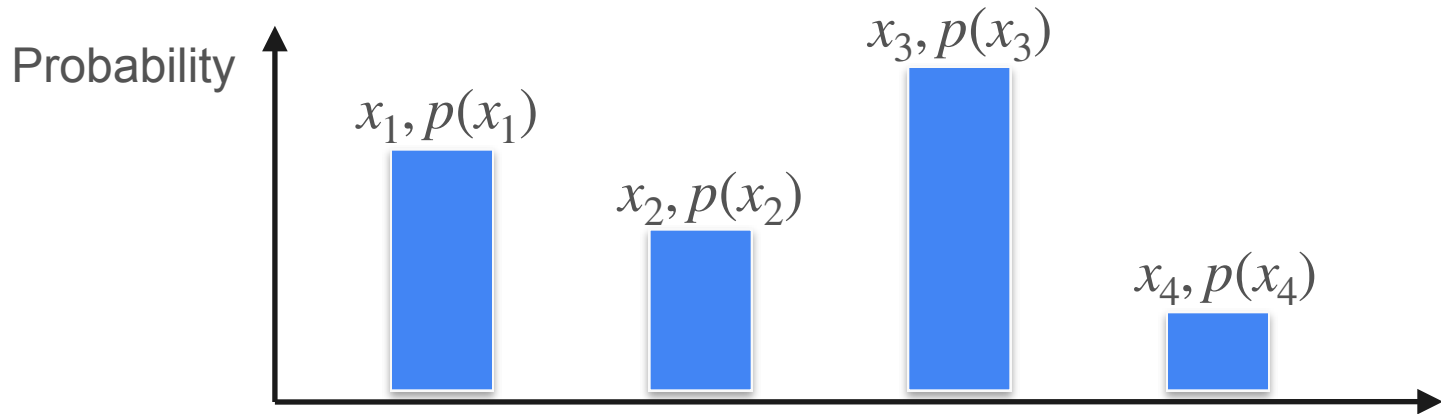
$$\mathbb{E}[X] = x_1 p(x_1)$$

Expected Value of a Function



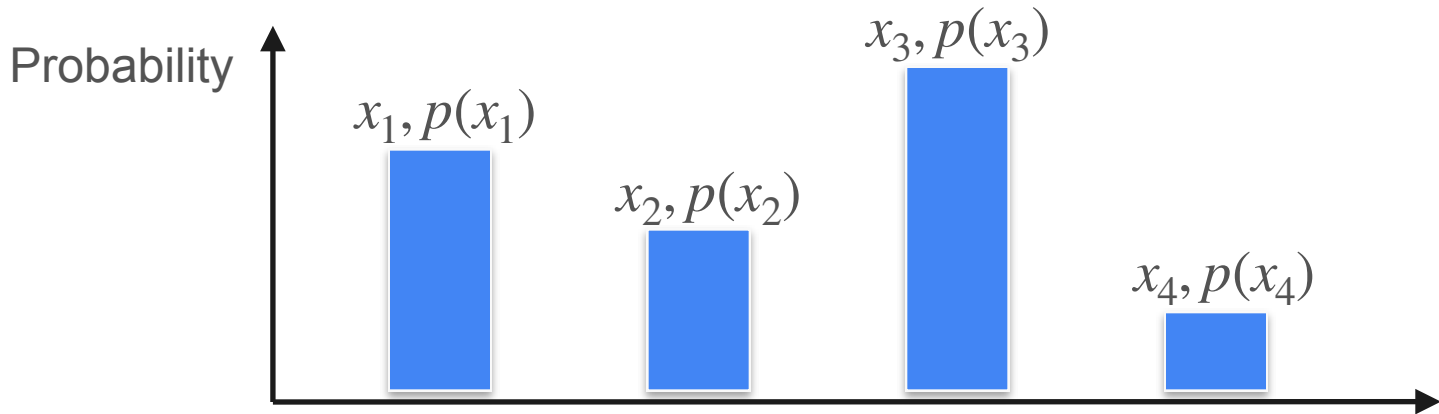
$$\mathbb{E}[X] = x_1p(x_1) + x_2p(x_2)$$

Expected Value of a Function



$$\mathbb{E}[X] = x_1p(x_1) + x_2p(x_2) + x_3p(x_3) + x_4p(x_4)$$

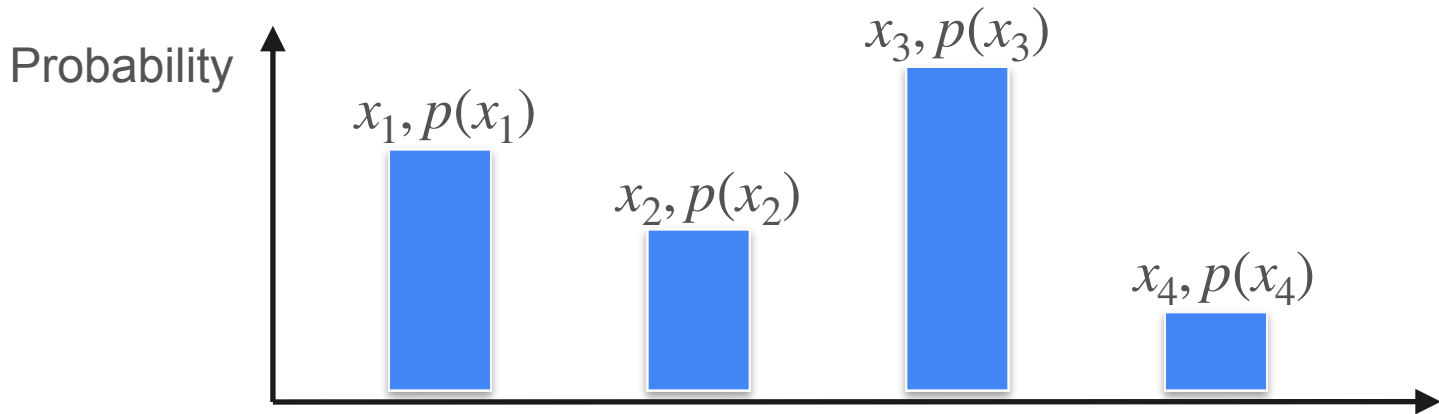
Expected Value of a Function



$$\mathbb{E}[X] = x_1p(x_1) + x_2p(x_2) + x_3p(x_3) + x_4p(x_4)$$

$$E[f(X)] =$$

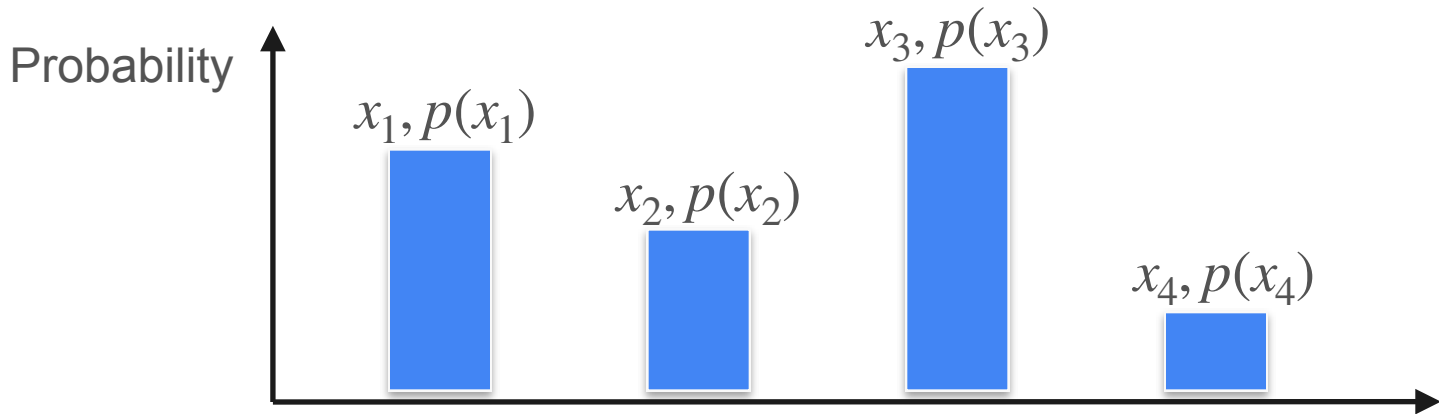
Expected Value of a Function



$$\mathbb{E}[X] = x_1p(x_1) + x_2p(x_2) + x_3p(x_3) + x_4p(x_4)$$

$$E[f(X)] = f(x_1)p(x_1)$$

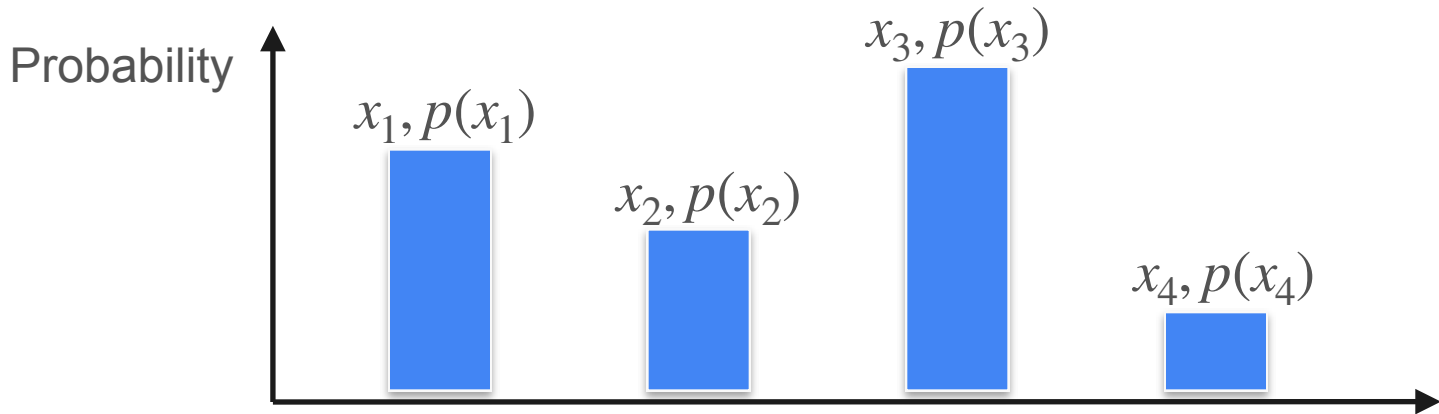
Expected Value of a Function



$$\mathbb{E}[X] = x_1p(x_1) + x_2p(x_2) + x_3p(x_3) + x_4p(x_4)$$

$$E[f(X)] = f(x_1)p(x_1) + f(x_2)p(x_2)$$

Expected Value of a Function



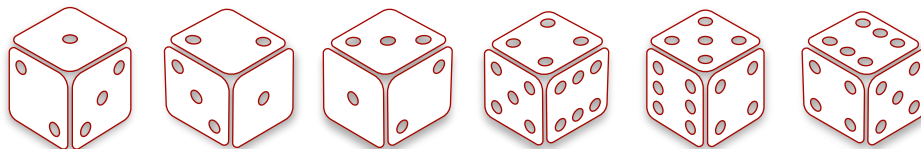
$$\mathbb{E}[X] = x_1p(x_1) + x_2p(x_2) + x_3p(x_3) + x_4p(x_4)$$

$$E[f(X)] = f(x_1)p(x_1) + f(x_2)p(x_2) + f(x_3)p(x_3) + f(x_4)p(x_4)$$

Expected Value of a Function

Probability: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

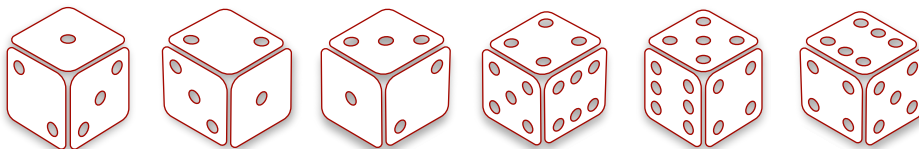
Roll: 1 2 3 4 5 6



Expected Value of a Function

Probability: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

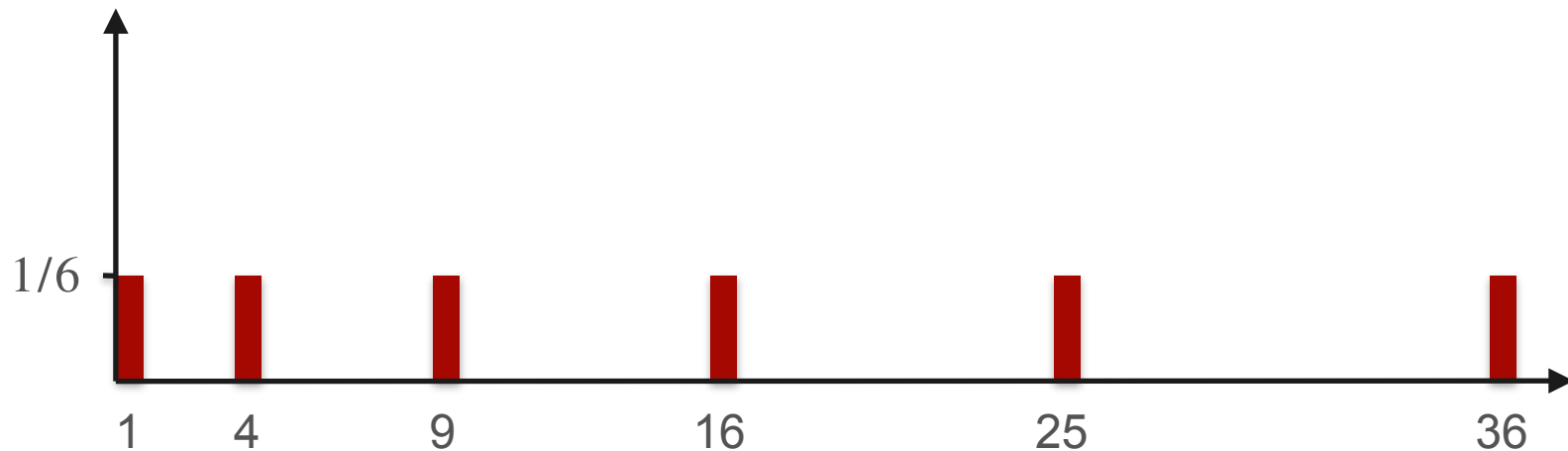
Roll: 1 2 3 4 5 6



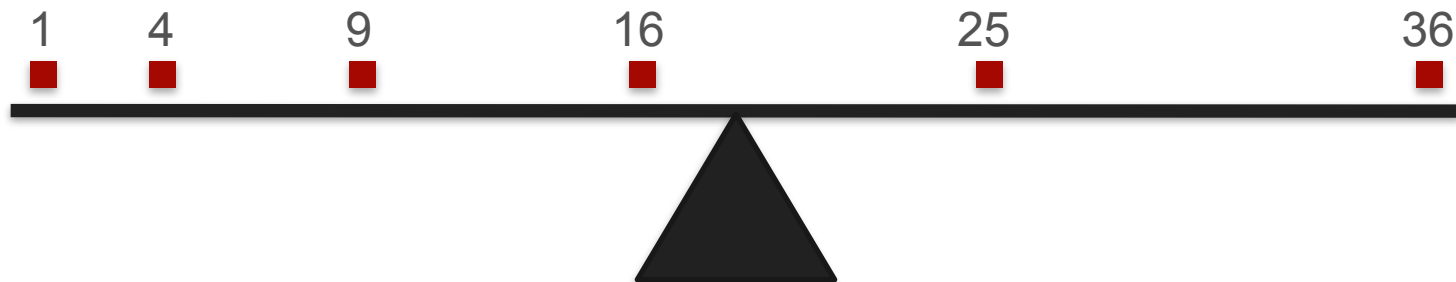
Square: 1 4 9 16 25 36

Expected Value of a Function

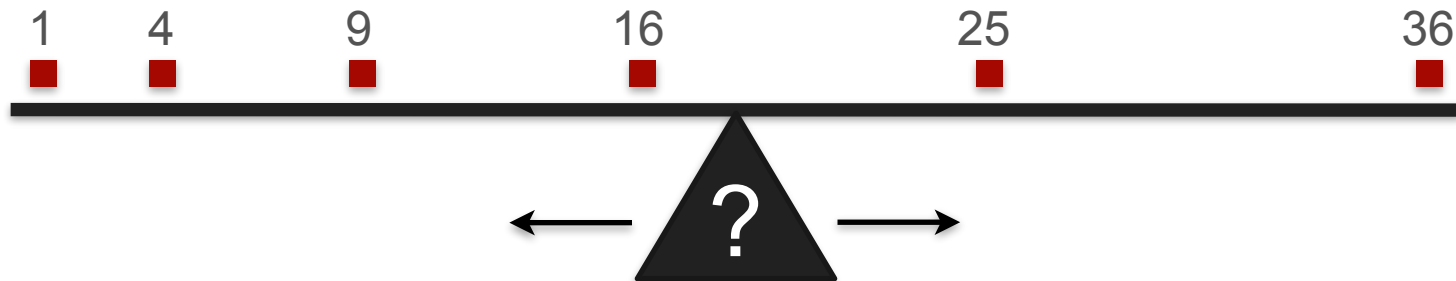
Probability



Expected Value of a Function

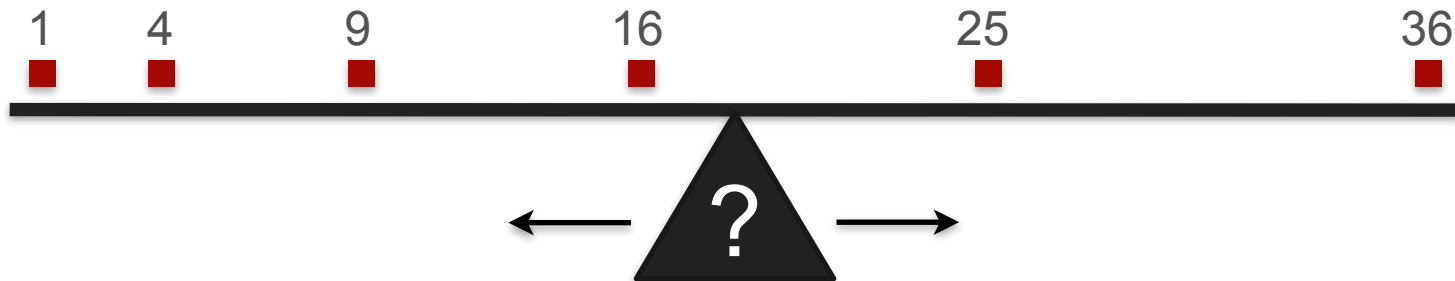


Expected Value of a Function



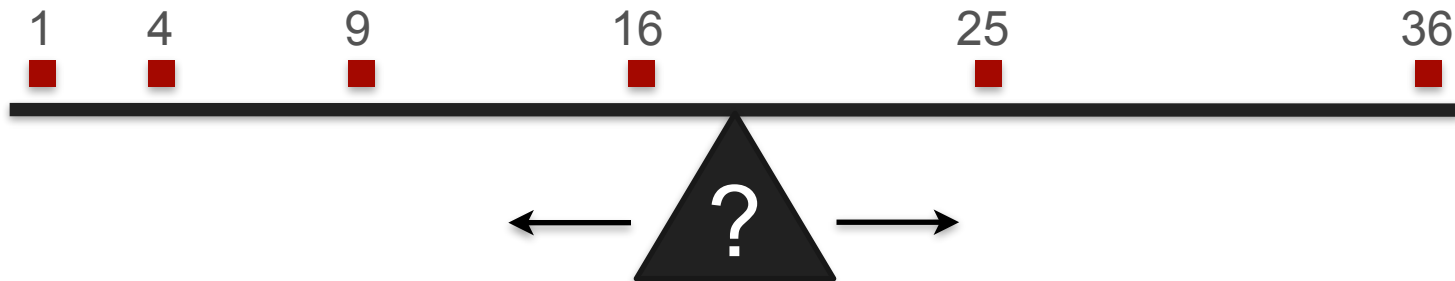
Expected Value of a Function

$$\frac{1 + 4 + 9 + 16 + 25 + 36}{6}$$

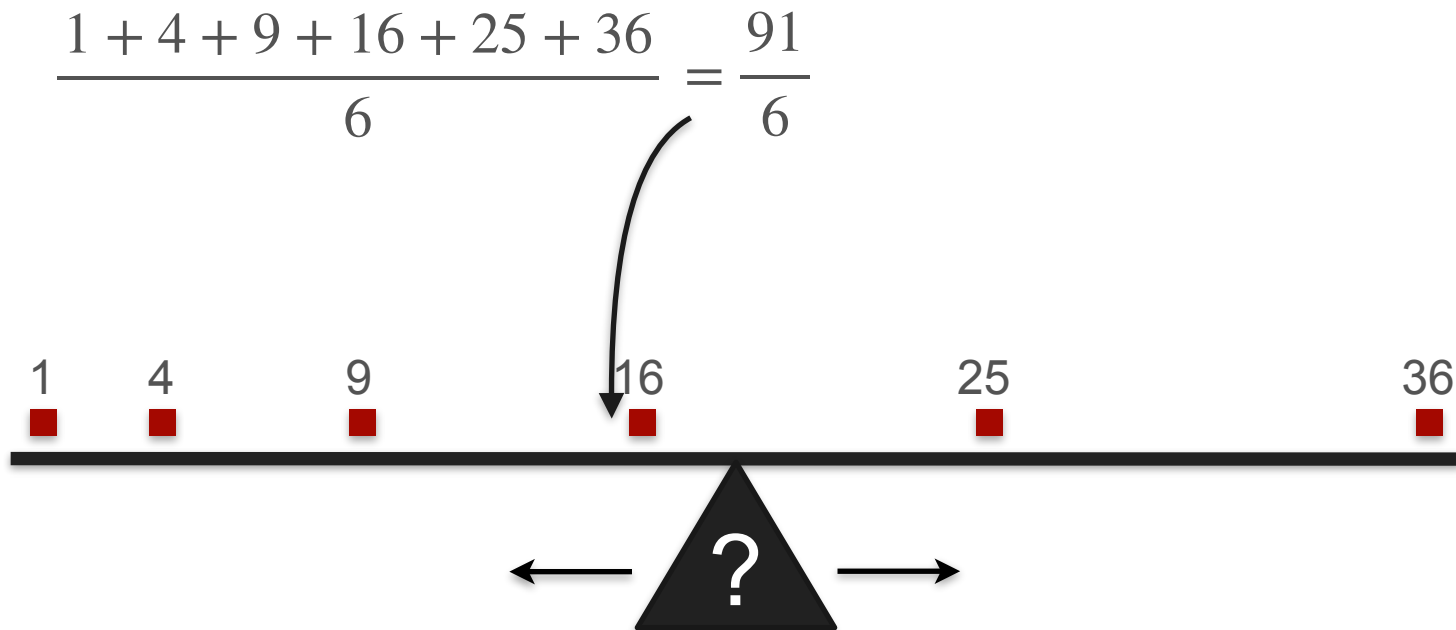


Expected Value of a Function

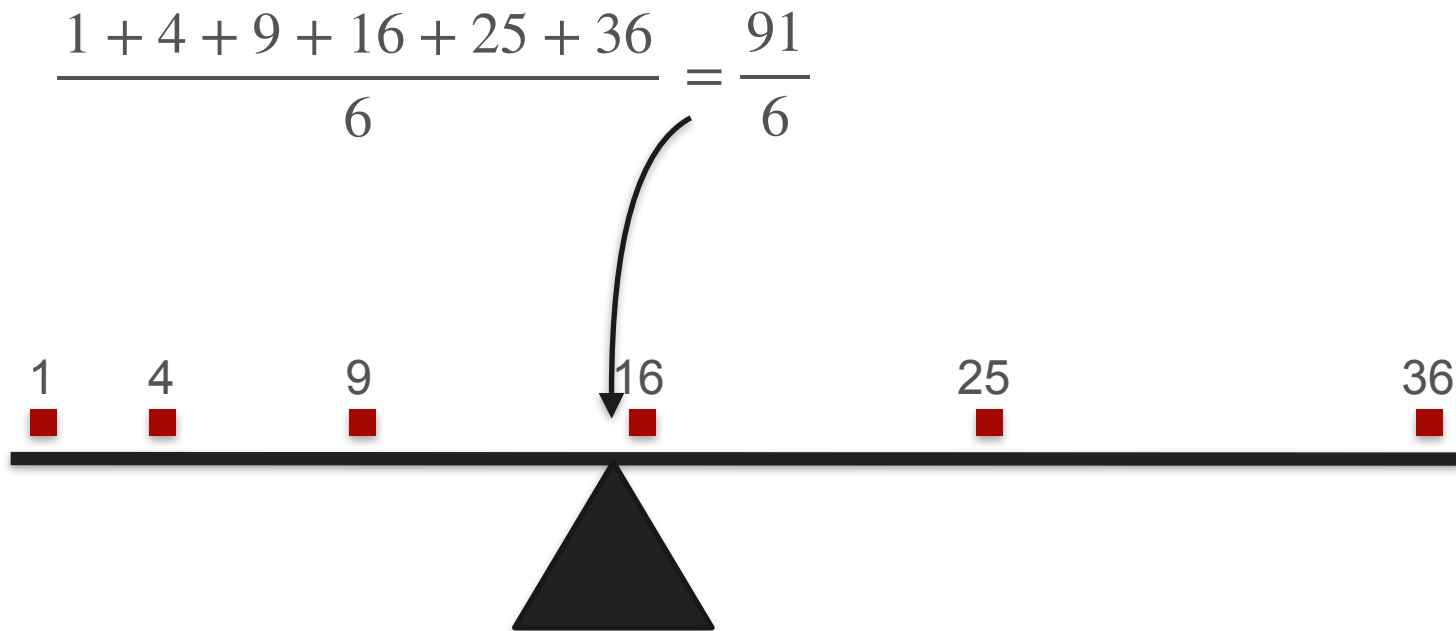
$$\frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6}$$



Expected Value of a Function

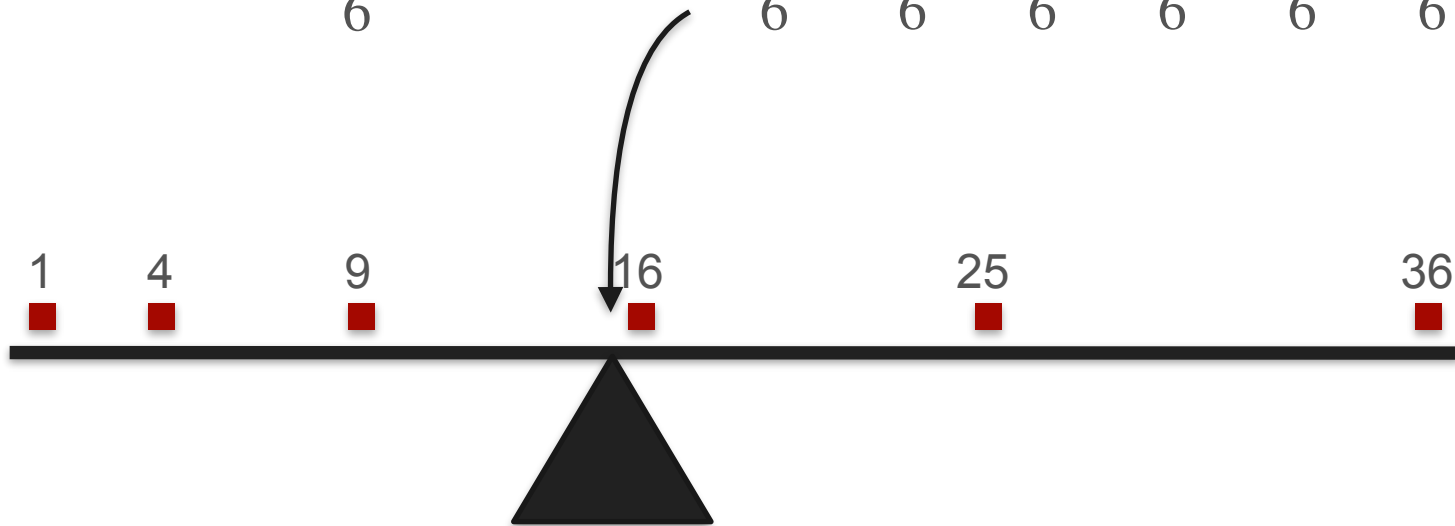


Expected Value of a Function



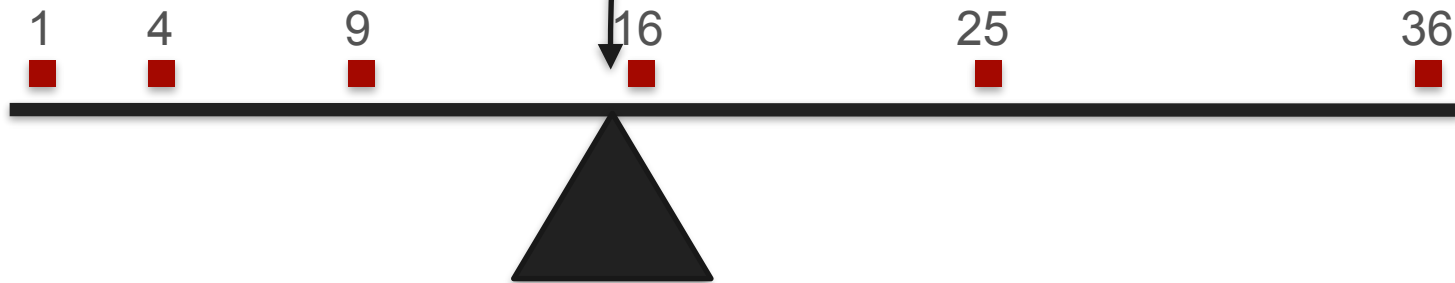
Expected Value of a Function

$$\frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6} = \frac{1^2}{6} + \frac{2^2}{6} + \frac{3^2}{6} + \frac{4^2}{6} + \frac{5^2}{6} + \frac{6^2}{6}$$



Expected Value of a Function

$$\frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6} = \frac{1^2}{6} + \frac{2^2}{6} + \frac{3^2}{6} + \frac{4^2}{6} + \frac{5^2}{6} + \frac{6^2}{6} \\ = \mathbb{E}[X^2]$$





DeepLearning.AI

Describing Distributions

Sum of expectations

Sum of Expectations

You play a game:

Sum of Expectations

You play a game:

Flip a coin. If heads you win \$ 1, else you win nothing.

Sum of Expectations

You play a game:

Flip a coin. If heads you win \$ 1, else you win nothing.



Win \$1



Win nothing

Sum of Expectations

You play a game:

Flip a coin. If heads you win \$ 1, else you win nothing.

Then roll a die. You win the amount you roll.



Win \$1



Win nothing

Sum of Expectations

You play a game:

Flip a coin. If heads you win \$ 1, else you win nothing.

Then roll a die. You win the amount you roll.



Win \$1

Win

\$1

\$2

\$3

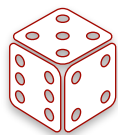
\$4

\$5

\$6



Win nothing



Sum of Expectations

You play a game:

Flip a coin. If heads you win \$ 1, else you win nothing.

Then roll a die. You win the amount you roll.



Win \$1

Win

\$1

\$2

\$3

\$4

\$5

\$6



Win nothing



What are your expected winnings for the game?

Sum of Expectations



Win \$1

Win

\$1

\$2

\$3

\$4

\$5

\$6



Win nothing



Sum of Expectations



Win \$1

Win

\$1

\$2

\$3

\$4

\$5

\$6



Win nothing



$$\mathbb{E}[X_{\text{coin}}] = \$0.5$$

Sum of Expectations



Win \$1

Win

\$1

\$2

\$3

\$4

\$5

\$6



Win nothing



$$\mathbb{E}[X_{\text{coin}}] = \$0.5$$

$$\mathbb{E}[X_{\text{dice}}] = \$3.5$$

Sum of Expectations



Win \$1

Win

\$1

\$2

\$3

\$4

\$5

\$6



Win nothing



$$\mathbb{E}[X_{\text{coin}}] = \$0.5$$

$$\mathbb{E}[X_{\text{dice}}] = \$3.5$$

$$\mathbb{E}[X] = \mathbb{E}[X_{\text{coin}}] + \mathbb{E}[X_{\text{dice}}] = \$0.5 + \$3.5 = \$4$$

Sum of Expectations



Win \$1

Win

\$1

\$2

\$3

\$4

\$5

\$6



Win nothing



$$\mathbb{E}[X_{coin}] = \$0.5$$

$$\mathbb{E}[X_{dice}] = \$3.5$$

$$\mathbb{E}[X] = \mathbb{E}[X_{coin}] + \mathbb{E}[X_{dice}] = \$0.5 + \$3.5 = \$4$$

$$\text{In general: } \mathbb{E}[X_1 + X_2] = \mathbb{E}[X_1] + \mathbb{E}[X_2]$$

Sum of Expectations



8 billion people

Sum of Expectations



8 billion people

Sum of Expectations



8 billion people

Sum of Expectations



Expected number of
correct assignments?



8 billion people

Sum of Expectations



1

Expected number of
correct assignments?



8 billion people

Sum of Expectations



1

Expected number of
correct assignments?

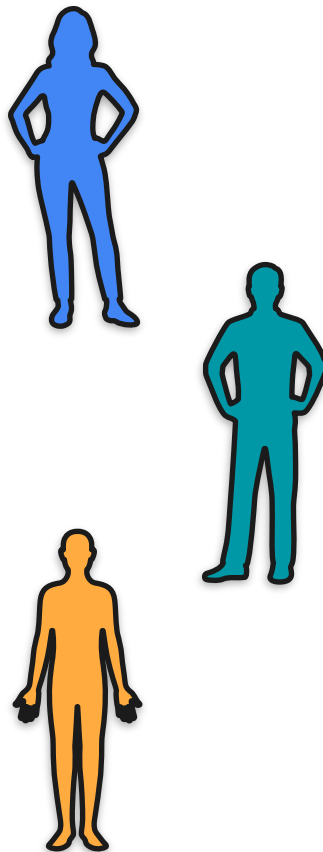
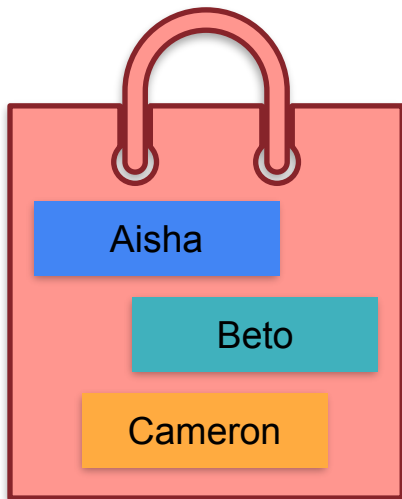


8 billion people

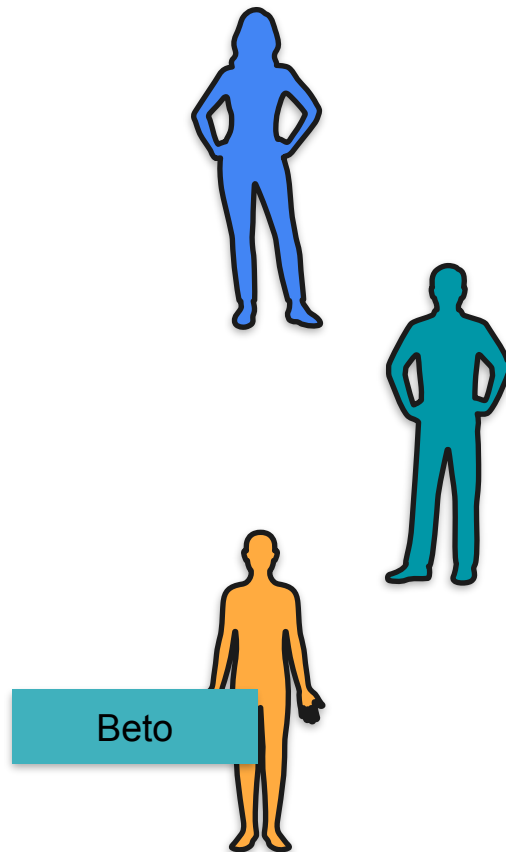
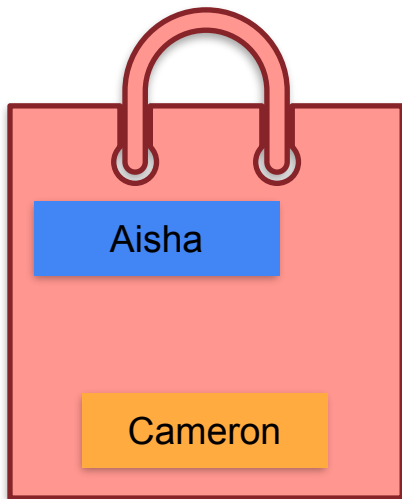


Bo Bleepityblop

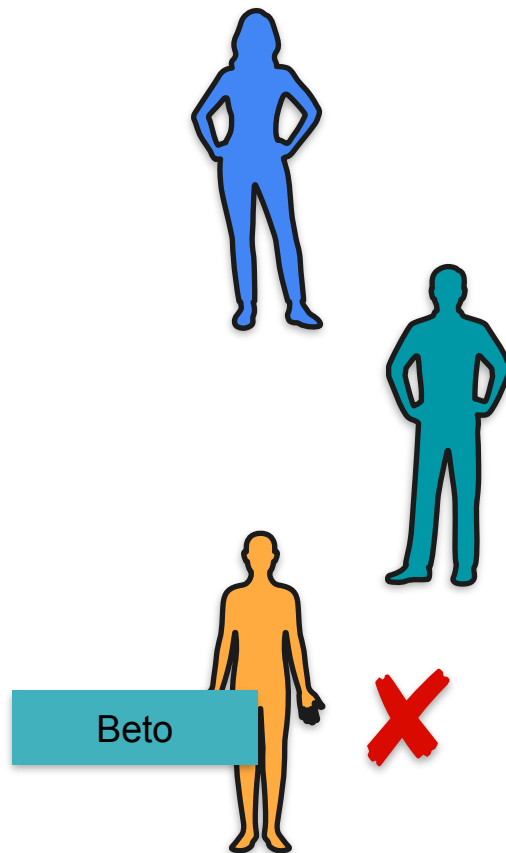
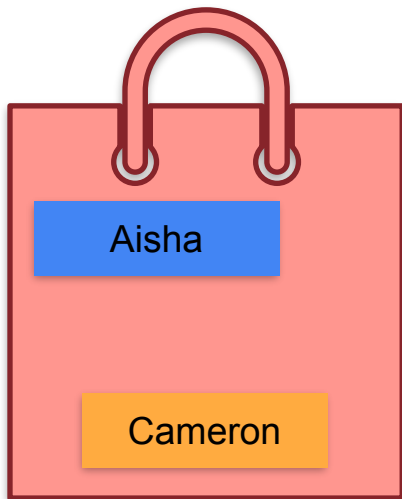
Sum of Expectations



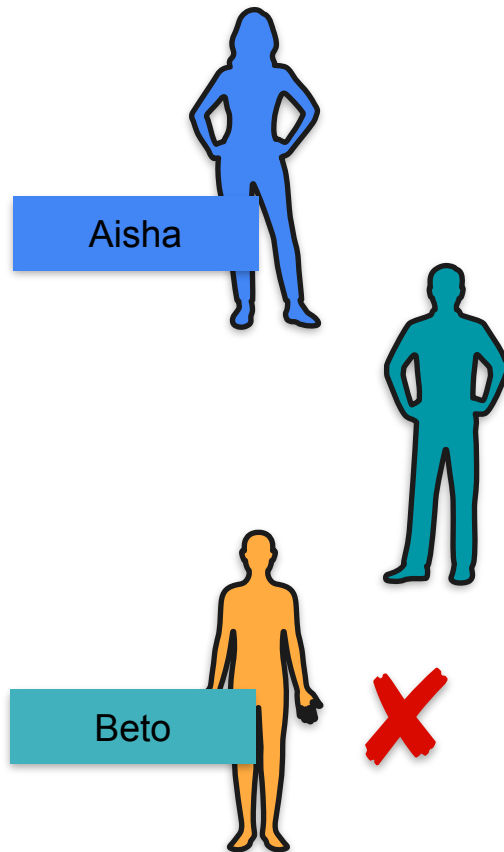
Sum of Expectations



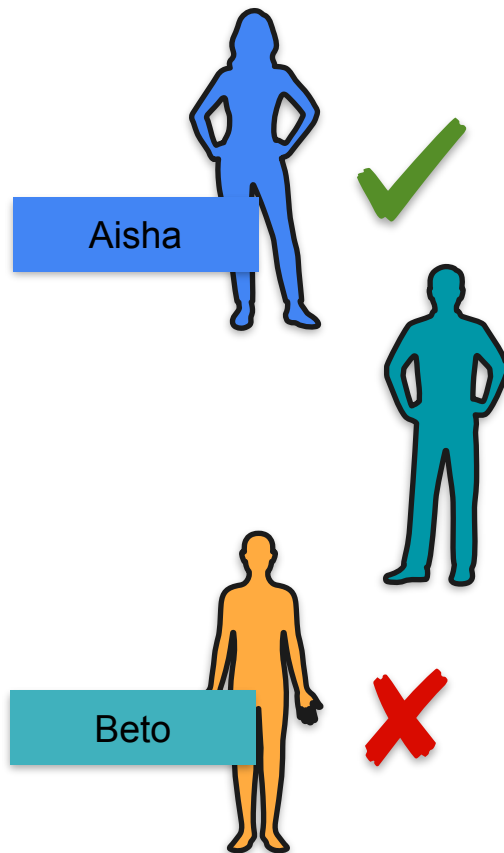
Sum of Expectations



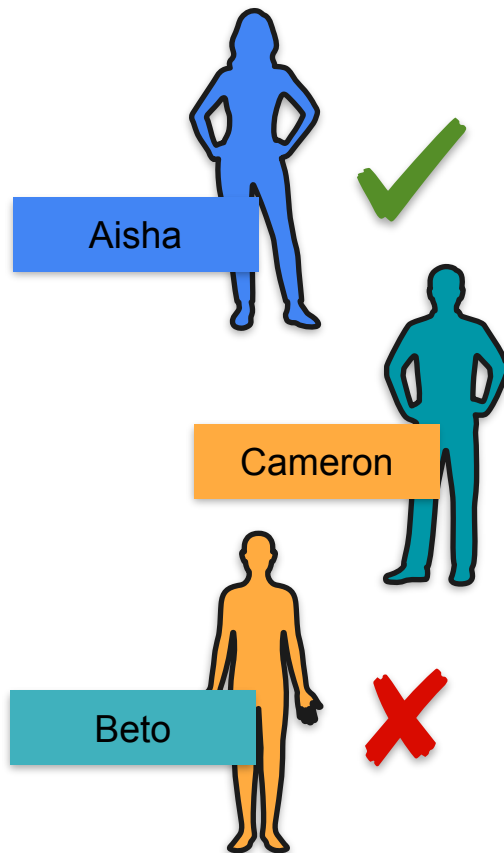
Sum of Expectations



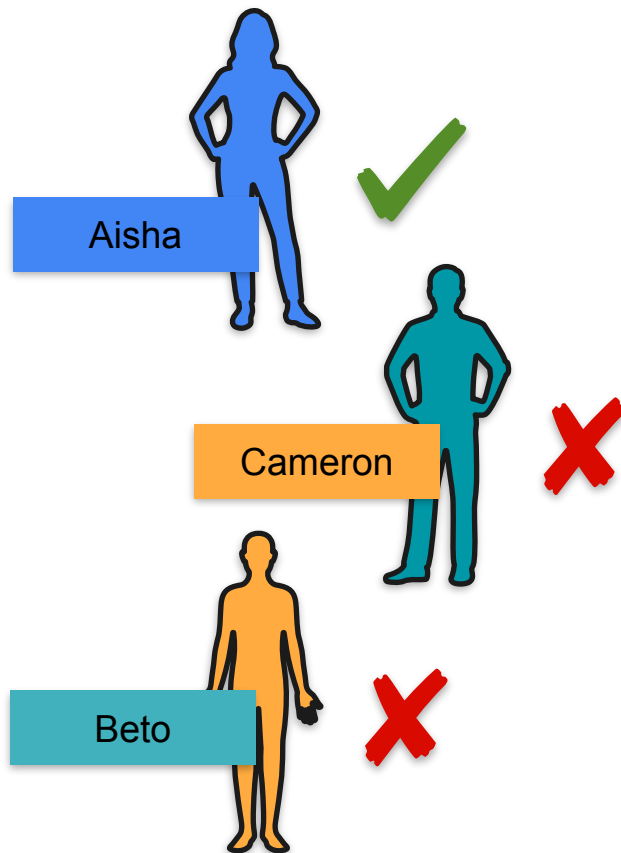
Sum of Expectations



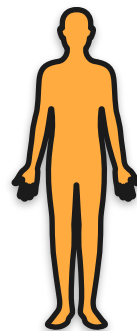
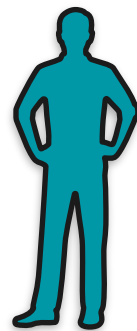
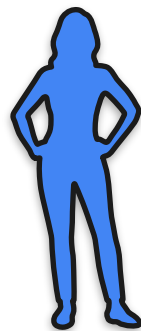
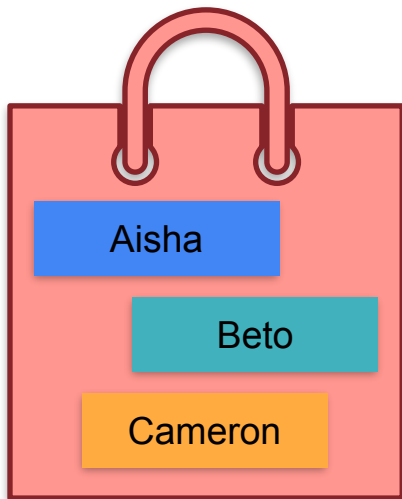
Sum of Expectations



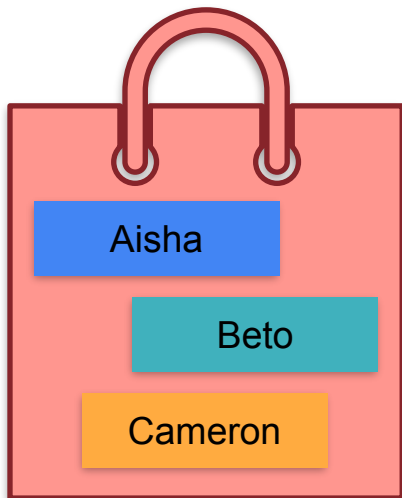
Sum of Expectations

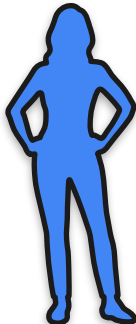
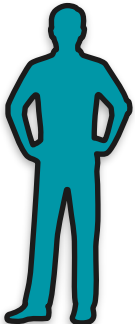
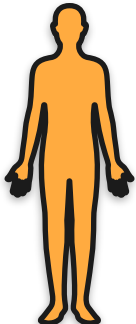


Sum of Expectations

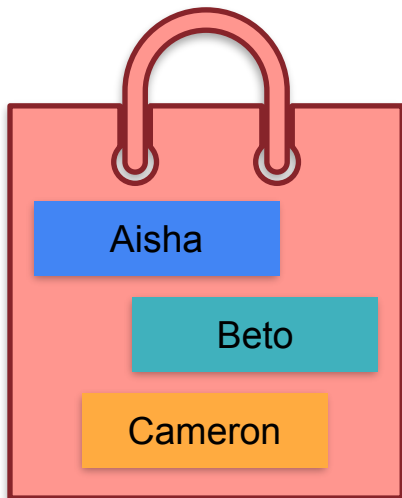


Sum of Expectations

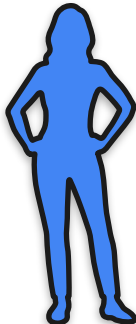
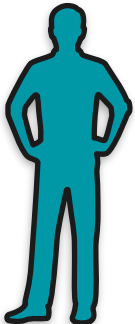
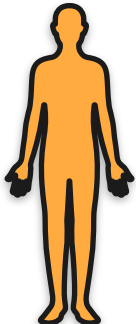


		
Aisha	Beto	Cameron
Aisha	Cameron	Beto
Beto	Aisha	Cameron
Beto	Cameron	Aisha
Cameron	Aisha	Beto
Cameron	Beto	Aisha

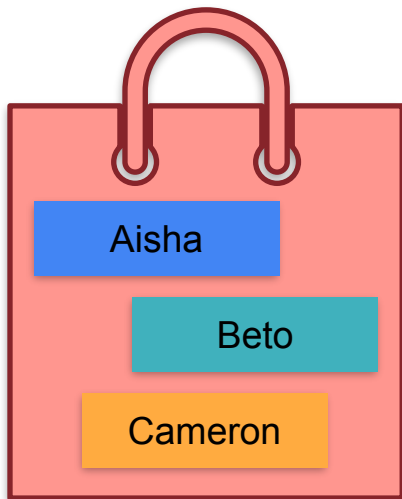
Sum of Expectations



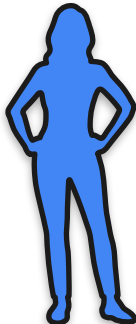
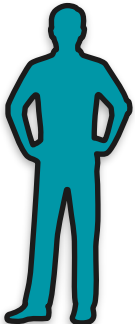
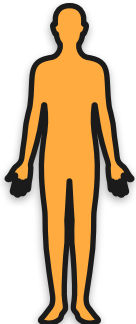
Correct

		
Aisha	Beto	Cameron
Aisha	Cameron	Beto
Beto	Aisha	Cameron
Beto	Cameron	Aisha
Cameron	Aisha	Beto
Cameron	Beto	Aisha

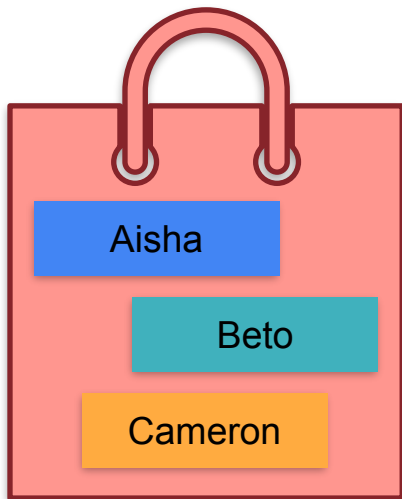
Sum of Expectations



Correct

		
Aisha	Beto	Cameron
Aisha	Cameron	Beto
Beto	Aisha	Cameron
Beto	Cameron	Aisha
Cameron	Aisha	Beto
Cameron	Beto	Aisha

Sum of Expectations



Correct



3

Aisha

Beto

Cameron

1

Aisha

Cameron

Beto

1

Beto

Aisha

Cameron

0

Beto

Cameron

Aisha

0

Cameron

Aisha

Beto

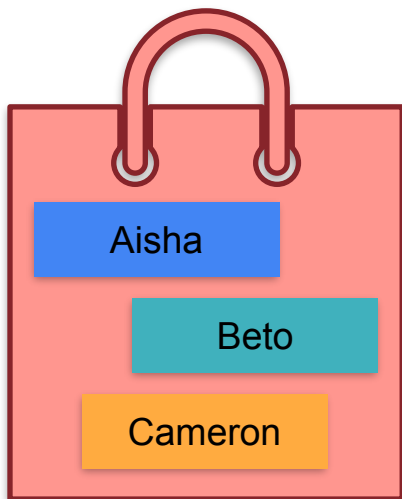
1

Cameron

Beto

Aisha

Sum of Expectations



Correct

6

3

1

1

0

0

1



Aisha



Beto



Cameron

Aisha

Cameron

Beto

Beto

Aisha

Cameron

Beto

Cameron

Aisha

Cameron

Aisha

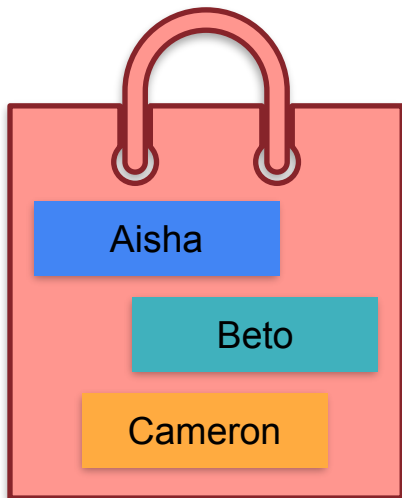
Beto

Cameron

Beto

Aisha

Sum of Expectations



Average
1

Correct

6

3

1

1

0

0

1



Aisha

Aisha

Beto

Beto

Cameron

Cameron



Beto

Cameron

Aisha

Cameron

Aisha

Beto



Cameron

Beto

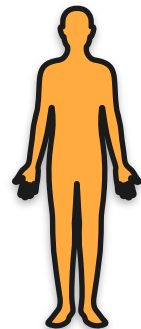
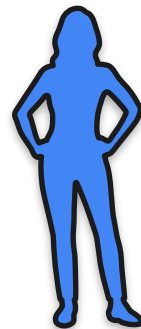
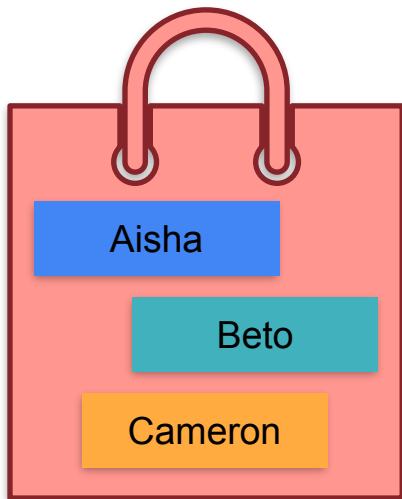
Cameron

Aisha

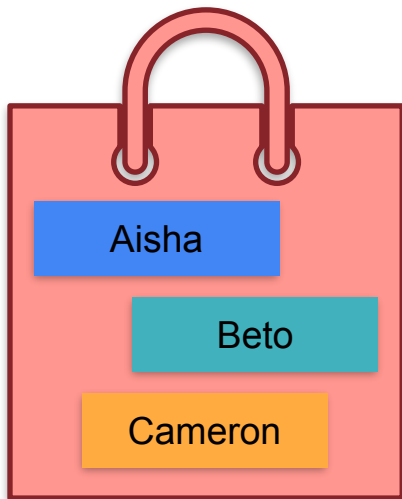
Beto

Aisha

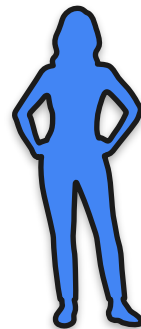
Sum of Expectations



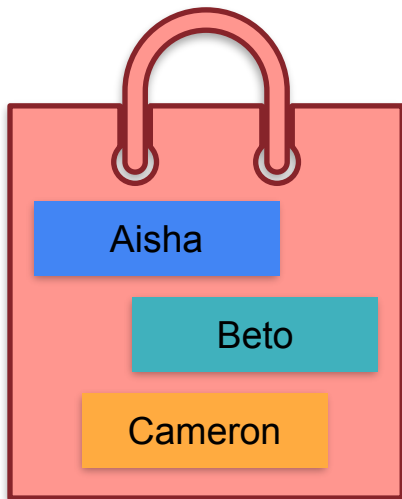
Sum of Expectations



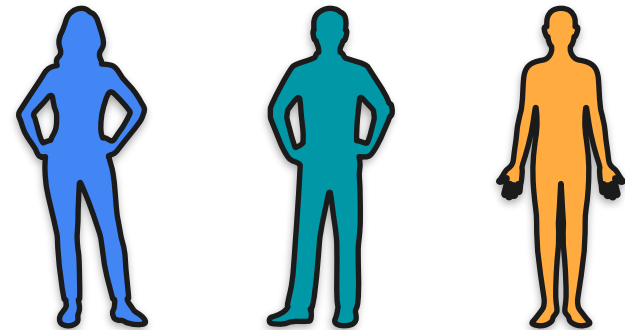
$\mathbb{E}[\text{Matches}]$



Sum of Expectations



$\mathbb{E}[\text{Matches}]$



$= \mathbb{E}[A]$

Sum of Expectations

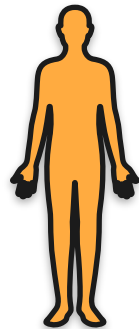
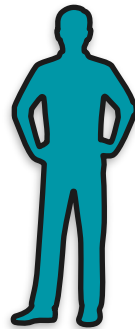
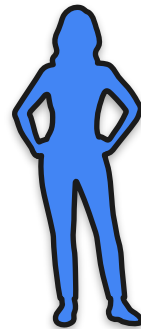


Aisha

Beto

Cameron

$\mathbb{E}[\text{Matches}]$



$= \mathbb{E}[A]$

Sum of Expectations



1/3

Aisha

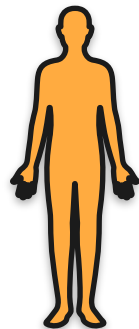
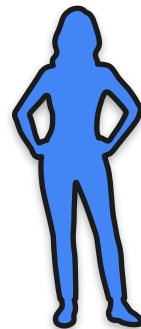
1/3

Beto

1/3

Cameron

$\mathbb{E}[\text{Matches}]$



$= \mathbb{E}[A]$

Sum of Expectations



1/3

Aisha

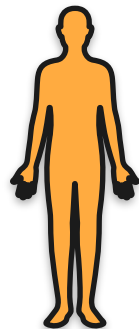
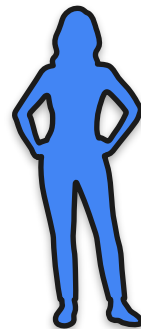
1/3

Beto

1/3

Cameron

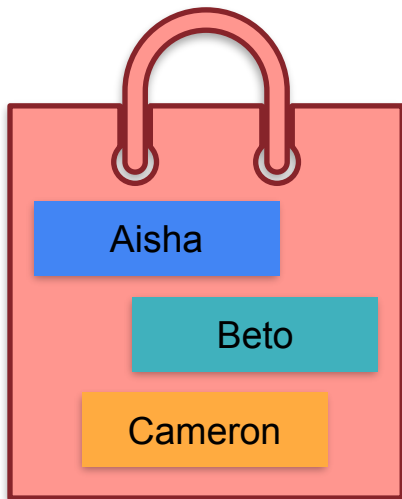
$\mathbb{E}[\text{Matches}]$



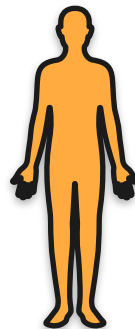
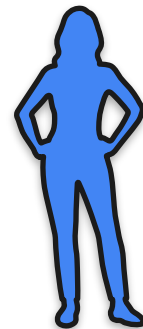
$= \mathbb{E}[A]$

$= 1/3$

Sum of Expectations



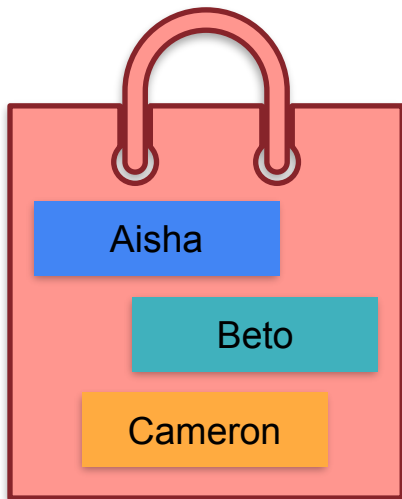
$\mathbb{E}[\text{Matches}]$



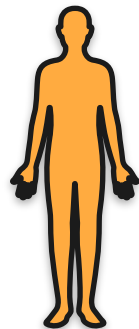
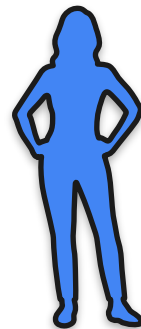
$= \mathbb{E}[A]$

$= 1/3$

Sum of Expectations



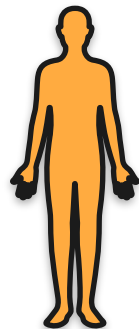
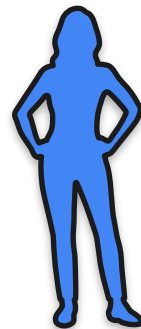
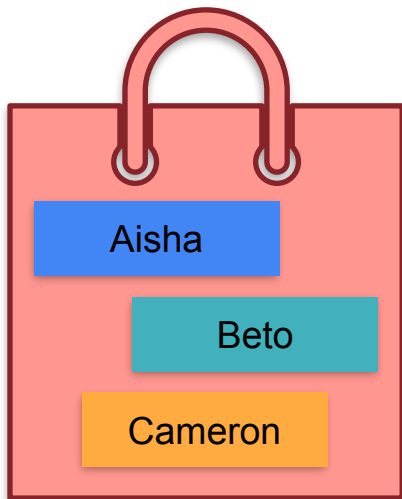
$\mathbb{E}[\text{Matches}]$



$$= \mathbb{E}[A] + \mathbb{E}[B]$$

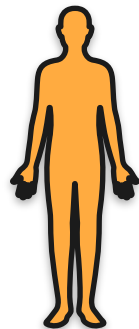
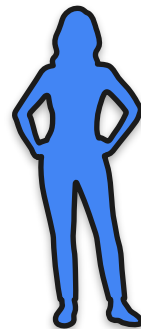
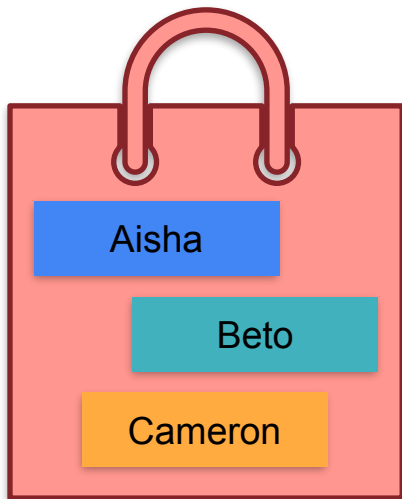
$$= 1/3 + 1/3$$

Sum of Expectations



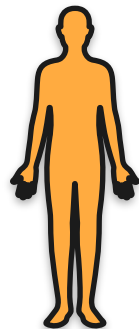
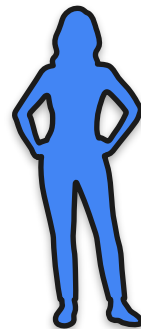
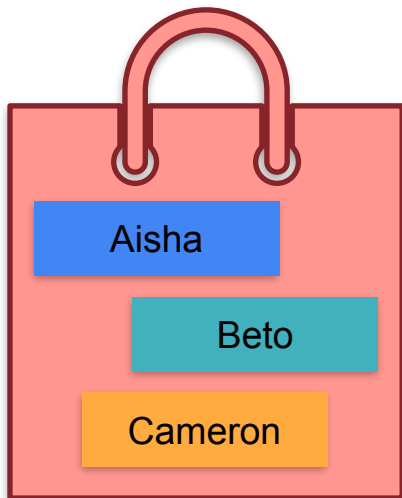
$$\begin{aligned}\mathbb{E}[\text{Matches}] &= \mathbb{E}[A] + \mathbb{E}[B] + \mathbb{E}[C] \\ &= 1/3 + 1/3 + 1/3\end{aligned}$$

Sum of Expectations



$$\begin{aligned}\mathbb{E}[\text{Matches}] &= \mathbb{E}[A] + \mathbb{E}[B] + \mathbb{E}[C] \\ &= 1/3 + 1/3 + 1/3 \\ &= 1\end{aligned}$$

Sum of Expectations



$$\begin{aligned}\mathbb{E}[\text{Matches}] &= \mathbb{E}[A] + \mathbb{E}[B] + \mathbb{E}[C] \\ &= 1/3 + 1/3 + 1/3 \\ &= 1\end{aligned}$$

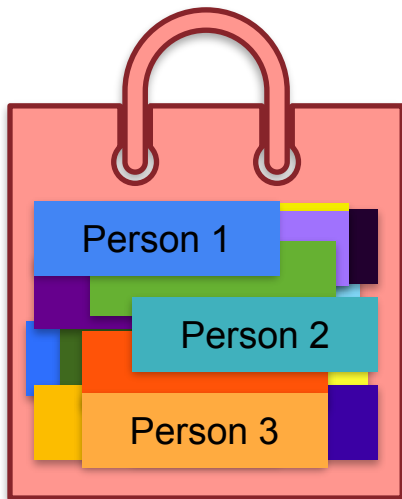
Average
1

Sum of Expectations



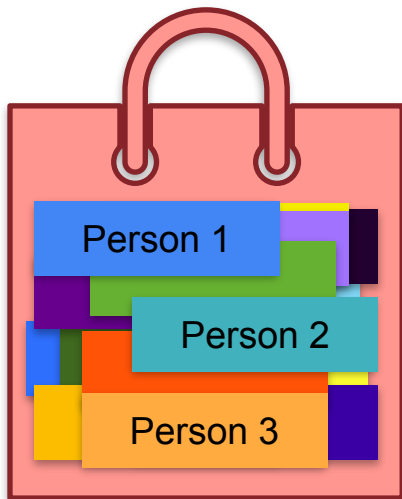
8 billion people

Sum of Expectations



8 billion people

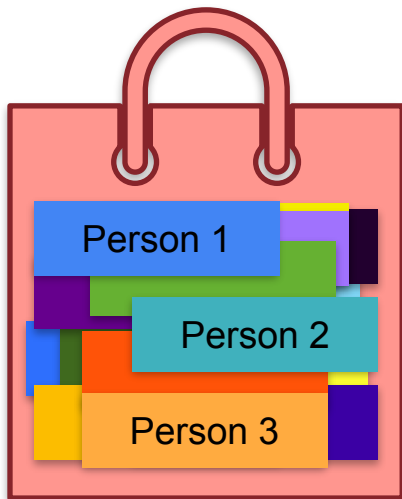
Sum of Expectations



8 billion people

Sum of Expectations

Expected number = ?



8 billion people

Sum of Expectations



Sum of Expectations



$$\mathbb{E} [\text{Matches}] = \mathbb{E} [X_{\text{person}_1}] + \mathbb{E} [X_{\text{person}_2}] + \dots + \mathbb{E} [X_{\text{person}_n}]$$

Sum of Expectations



n people (n = 8 billion)

$$\mathbb{E} [\text{Matches}] = \mathbb{E} [X_{\text{person}_1}] + \mathbb{E} [X_{\text{person}_2}] + \dots + \mathbb{E} [X_{\text{person}_n}]$$

Sum of Expectations



n people (n = 8 billion)

$$\mathbb{E}[\text{Matches}] = \mathbb{E}[X_{\text{person}_1}] + \mathbb{E}[X_{\text{person}_2}] + \dots + \mathbb{E}[X_{\text{person}_n}] = \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}$$

Sum of Expectations



n people (n = 8 billion)

$$\begin{aligned}\mathbb{E}[\text{Matches}] &= \mathbb{E}[X_{\text{person}_1}] + \mathbb{E}[X_{\text{person}_2}] + \dots + \mathbb{E}[X_{\text{person}_n}] \\ &= \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n} \\ &= n \cdot \frac{1}{n}\end{aligned}$$

Sum of Expectations



n people (n = 8 billion)

$$\begin{aligned}\mathbb{E}[\text{Matches}] &= \mathbb{E}[X_{\text{person}_1}] + \mathbb{E}[X_{\text{person}_2}] + \dots + \mathbb{E}[X_{\text{person}_n}] \\ &= \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n} \\ &= n \cdot \frac{1}{n} = 1\end{aligned}$$

Sum of Expectations



n people (n = 8 billion)

$$\mathbb{E} [\text{Matches}] = \mathbb{E} [X_{\text{person}_1}] + \mathbb{E} [X_{\text{person}_2}] + \dots + \mathbb{E} [X_{\text{person}_n}]$$

$$= \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}$$

In general:

$$\mathbb{E} [X_1 + X_2 + \dots + X_n] = \mathbb{E} [X_1] + \mathbb{E} [X_2] + \dots + \mathbb{E} [X_n]$$

$$= n \cdot \frac{1}{n} = 1$$



DeepLearning.AI

Describing Distributions

Variance

Variance Motivation: Fair Price To Play the Game

You play a game with a friend



You win 1 dollar



You lose 1 dollar

What is the fair amount of money to pay to play this game?

Variance Motivation: Fair Price To Play the Game

You play a game with a friend



You win 1 dollar



You lose 1 dollar

Game cost:

\$0

What is the fair amount of money to pay to play this game?

Variance Motivation: Fair Price To Play the Game

You play a game with a friend



You win 100 dollars



You lose 100 dollars

What is the fair amount of money to pay to play this game?

Variance Motivation: Fair Price To Play the Game

You play a game with a friend



You win 100 dollars



You lose 100 dollars

Game cost:

\$0

What is the fair amount of money to pay to play this game?

Variance Motivation: Fair Price To Play the Game

How do you tell these two games apart?

Game 1



You win 1 dollar



You lose 1 dollar

Game 2



You win 100 dollars



You lose 100 dollars

Variance Motivation: Fair Price To Play the Game

How do you tell these two games apart?

Game 1



You win 1 dollar



You lose 1 dollar

Game 2

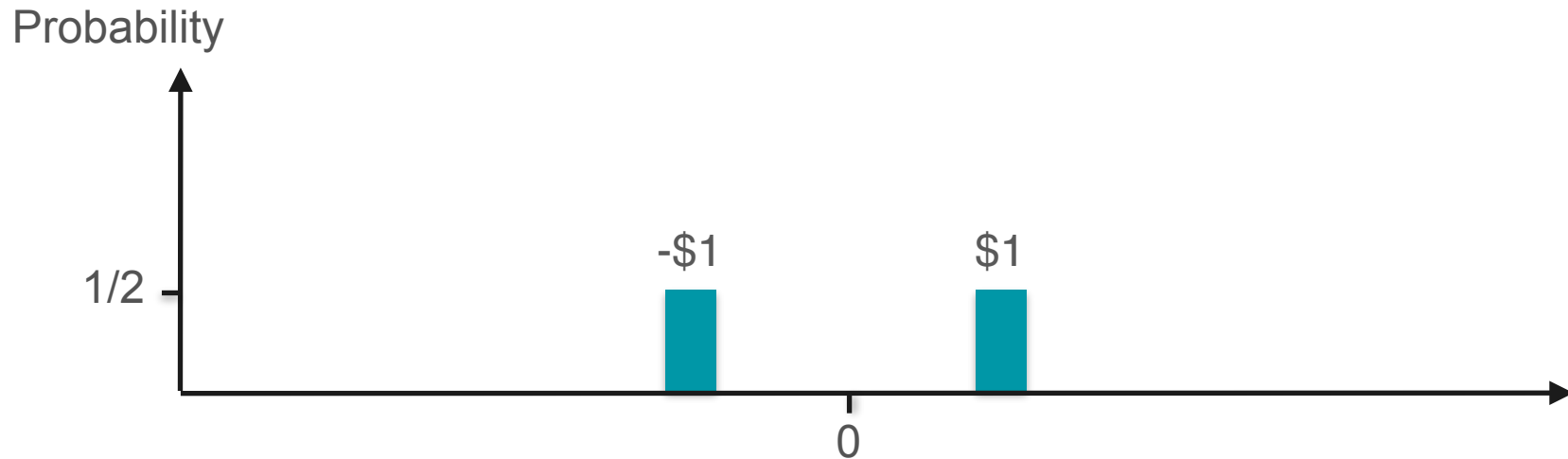
You win 100 dollars



You lose 100 dollars

Variance!

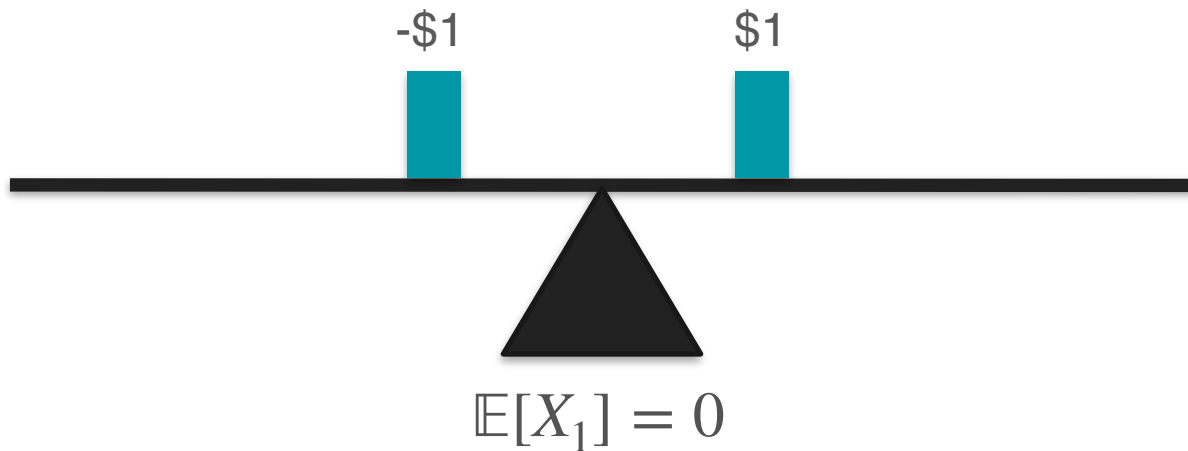
Variance Motivation: Measuring Spread



Variance Motivation: Measuring Spread

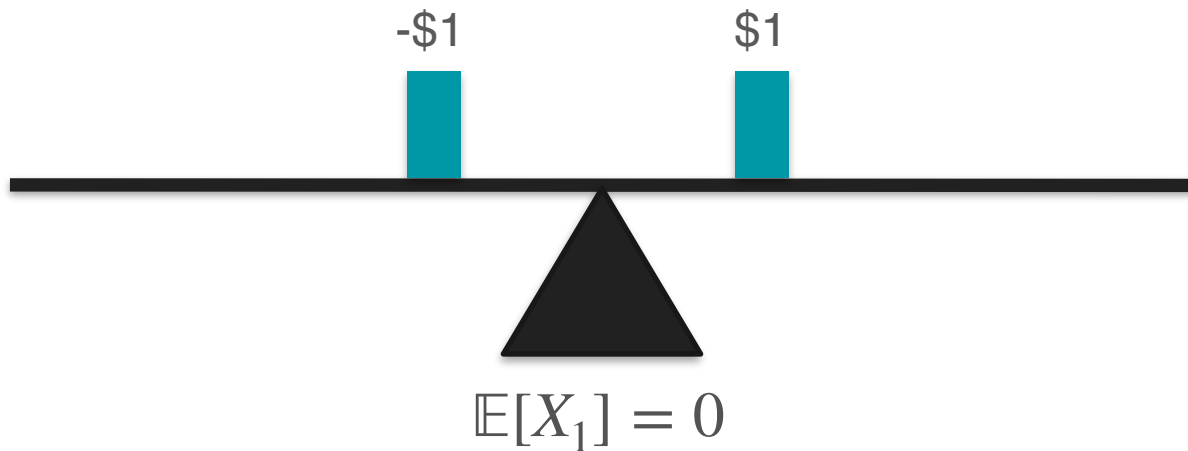


Variance Motivation: Measuring Spread

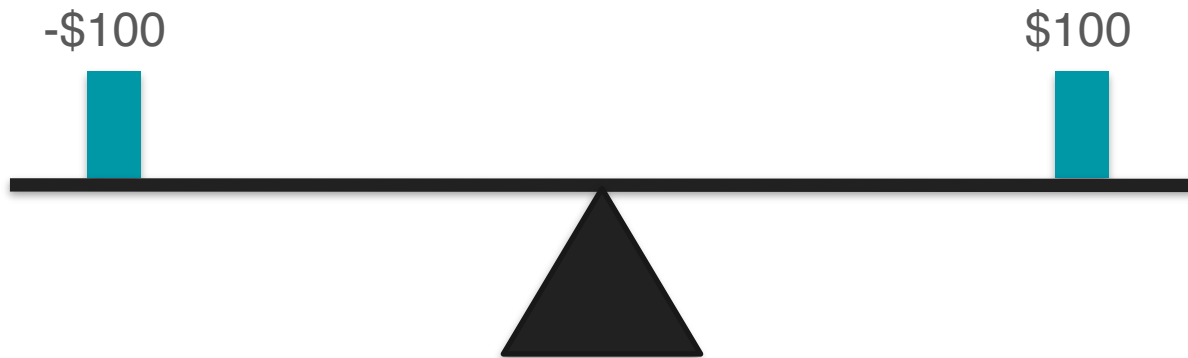


Variance Motivation: Measuring Spread

X_1 = expected amount of money gained in game 1

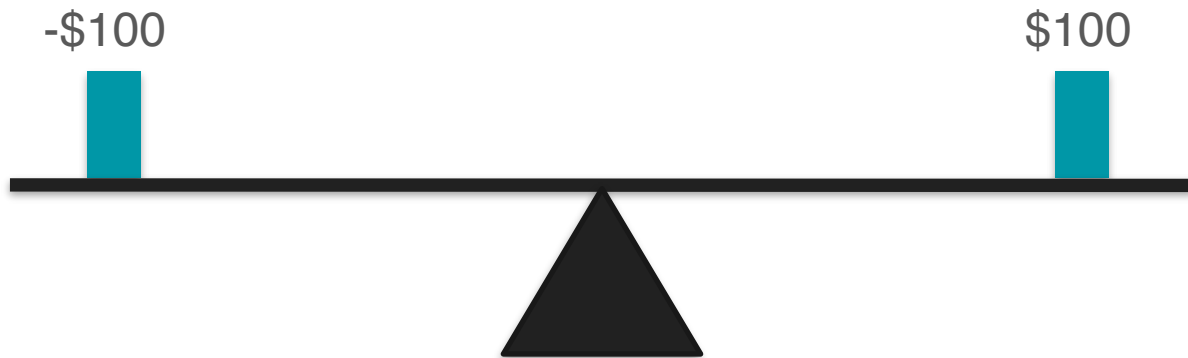


Variance Motivation: Measuring Spread



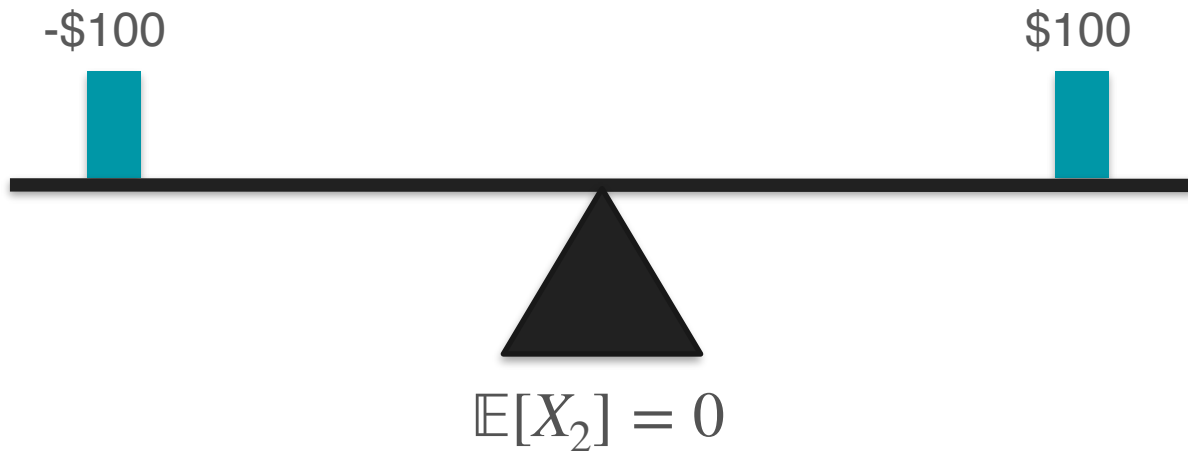
Variance Motivation: Measuring Spread

X_2 = expected amount of money gained in game 2

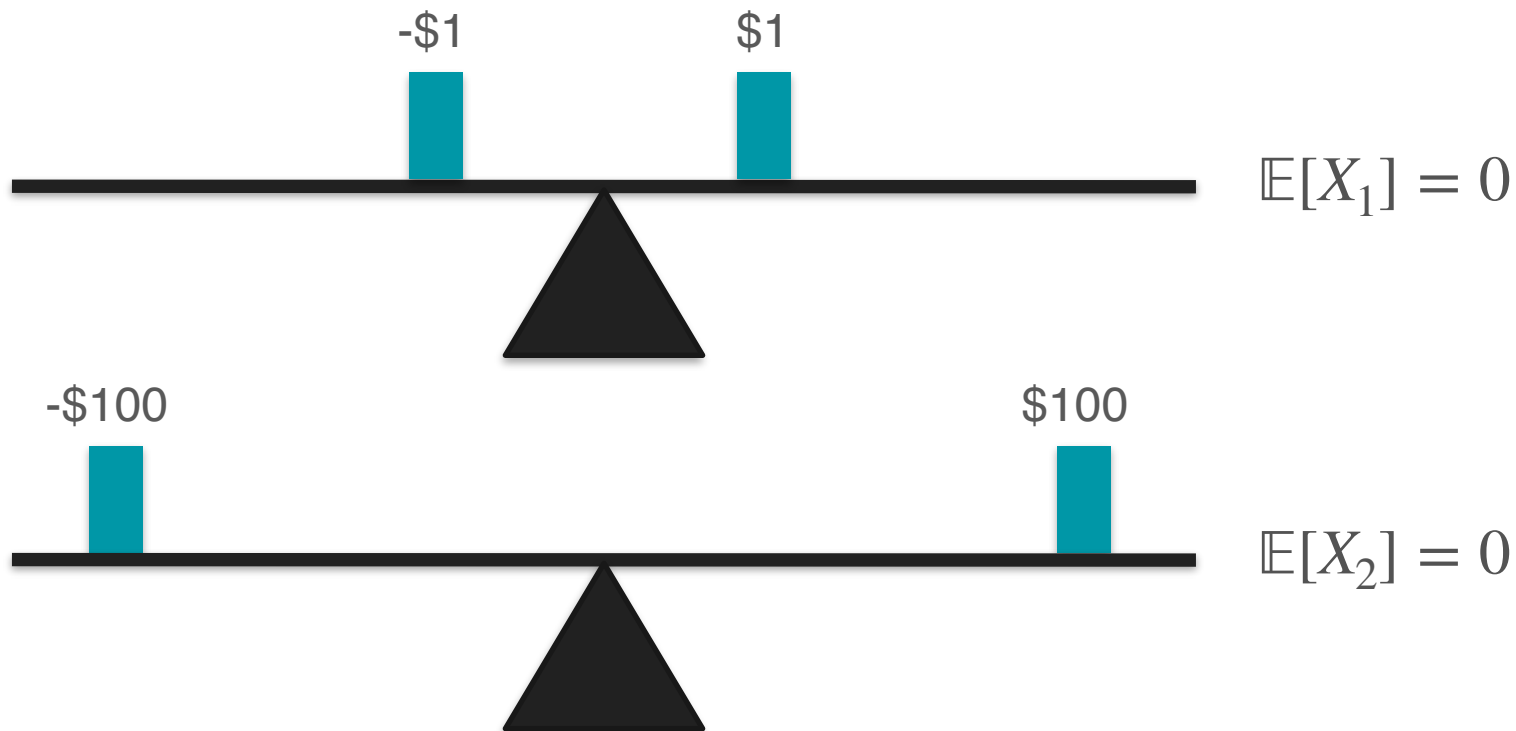


Variance Motivation: Measuring Spread

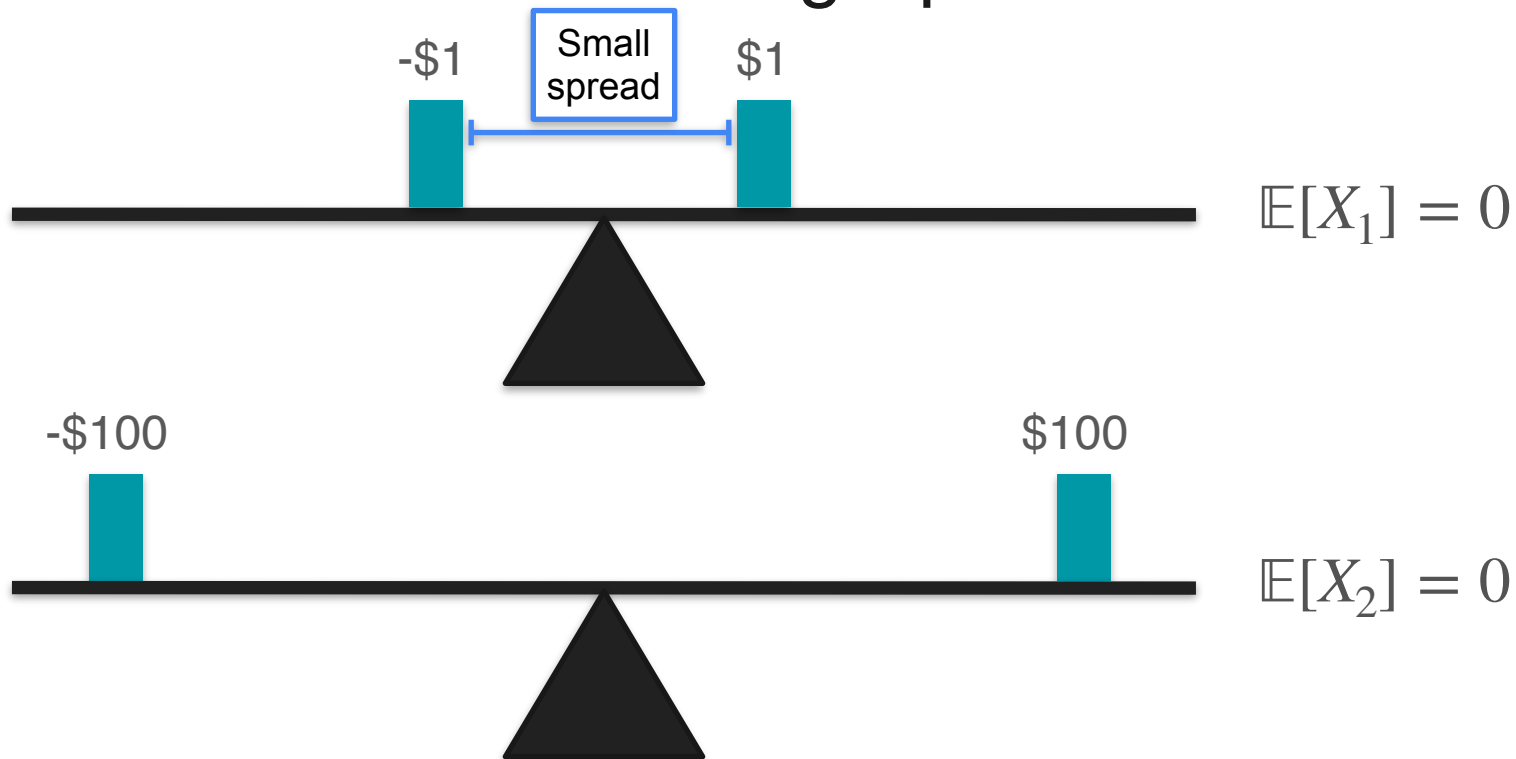
X_2 = expected amount of money gained in game 2



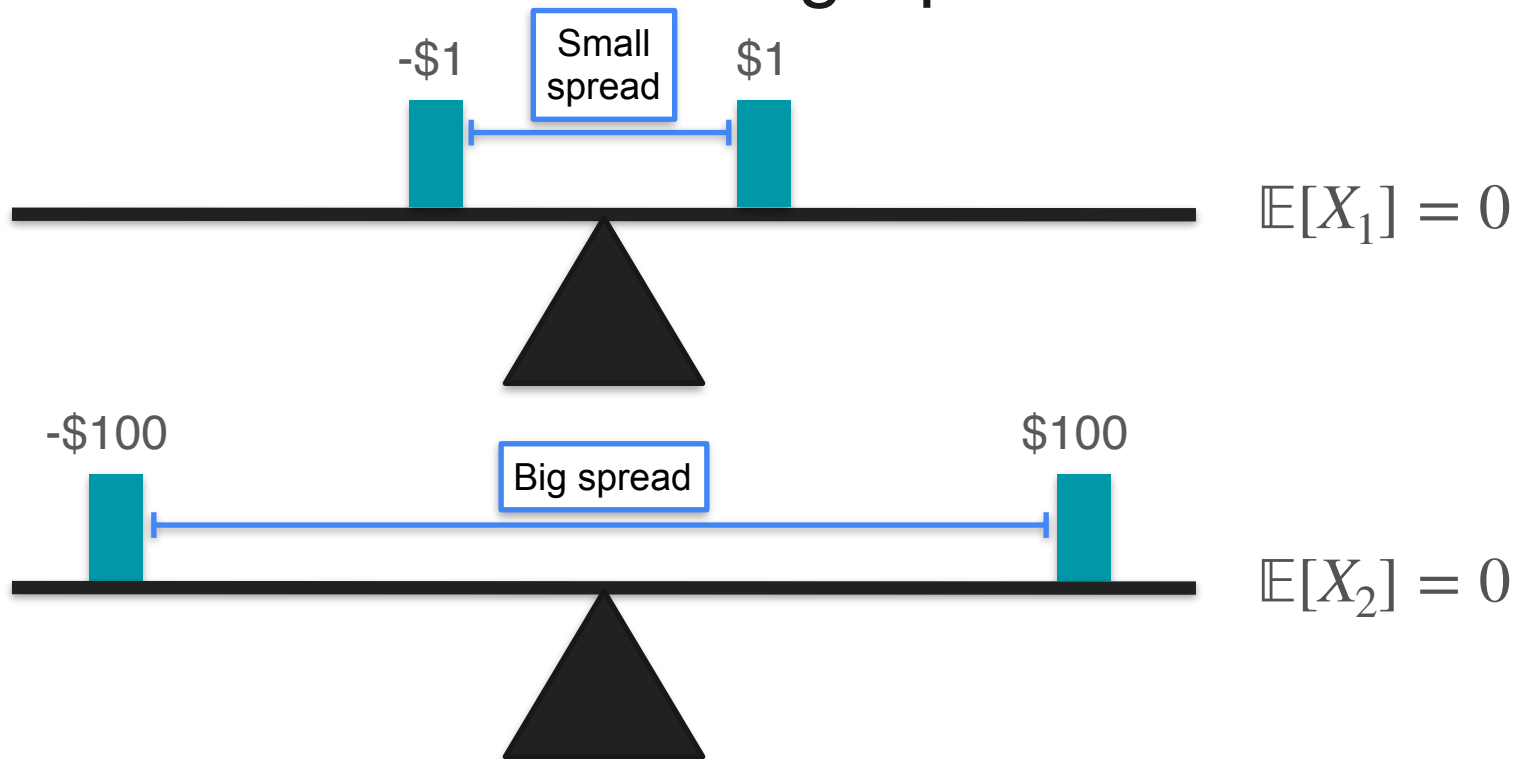
Variance Motivation: Measuring Spread



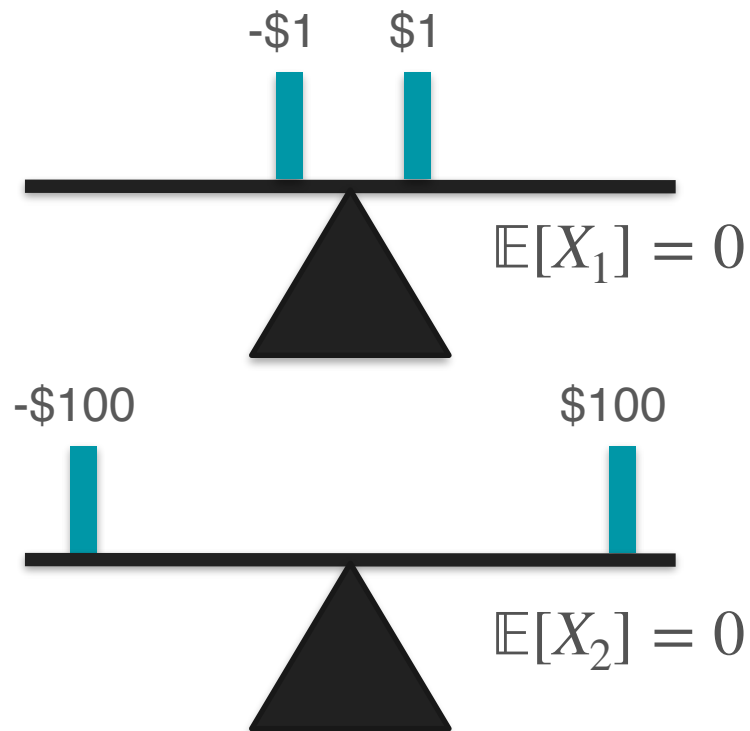
Variance Motivation: Measuring Spread



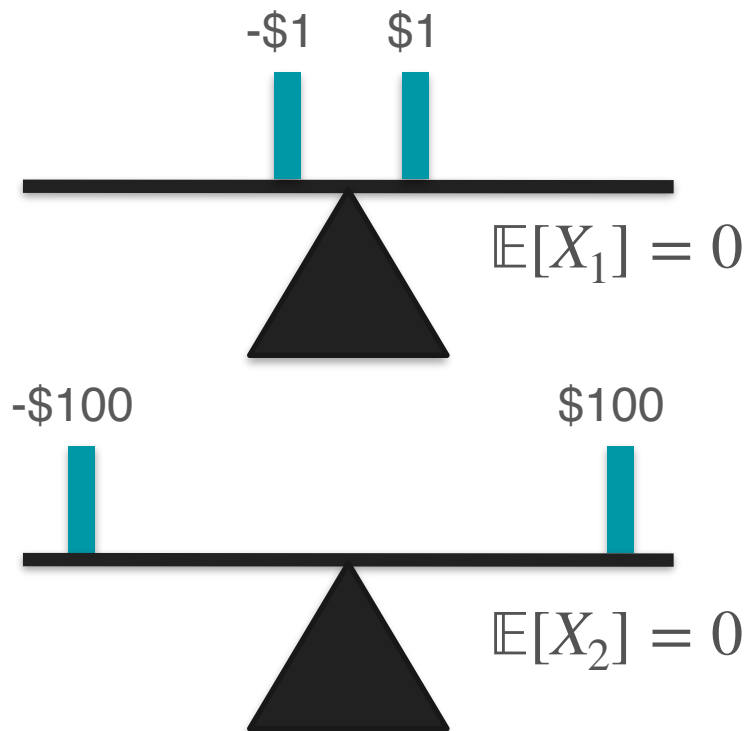
Variance Motivation: Measuring Spread



Variance Motivation: Measuring Spread



Variance Motivation: Measuring Spread



$$\mathbb{E}[X_1] = \frac{(1) + (-1)}{2} = 0$$

Variance Motivation: Measuring Spread

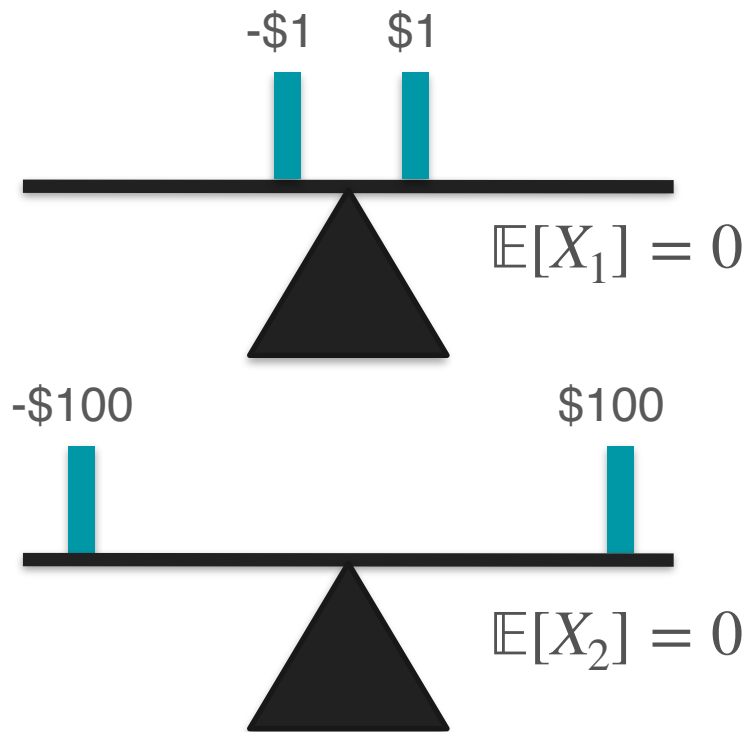


$$\mathbb{E}[X_1] = \frac{(1) + (-1)}{2} = 0$$



$$\mathbb{E}[X_2] = \frac{(100) + (-100)}{2} = 0$$

Variance Motivation: Measuring Spread

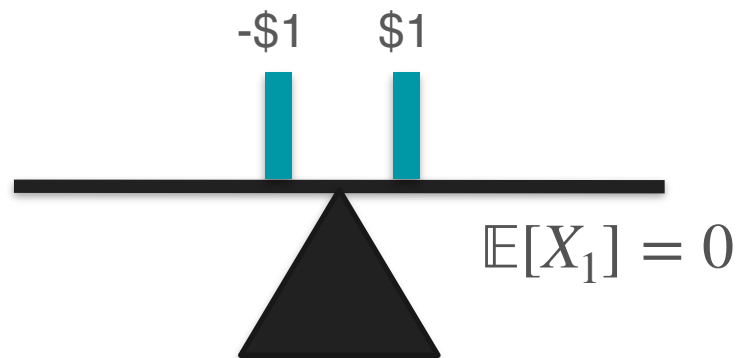


$$\mathbb{E}[X_1] = \frac{(1) + (-1)}{2} = 0$$

Turn into positive?

$$\mathbb{E}[X_2] = \frac{(100) + (-100)}{2} = 0$$

Variance Motivation: Measuring Spread



$$\mathbb{E}[X_1] = \frac{(1)^2 + (-1)^2}{2} = 0$$



$$\mathbb{E}[X_2] = \frac{(100)^2 + (-100)^2}{2} = 0$$

Variance Motivation: Measuring Spread



$$\mathbb{E}[X_1] = \frac{(1)^2 + (-1)^2}{2} = 0$$



$$\mathbb{E}[X_2] = \frac{(100)^2 + (-100)^2}{2} = 0$$

Variance Motivation: Measuring Spread



$$\mathbb{E}[X_1^2] = \frac{(1)^2 + (-1)^2}{2} = 0$$



$$\mathbb{E}[X_2^2] = \frac{(100)^2 + (-100)^2}{2} = 0$$

Variance Motivation: Measuring Spread



$$\mathbb{E}[X_1^2] = \frac{(1)^2 + (-1)^2}{2} = 1$$



$$\mathbb{E}[X_2^2] = \frac{(100)^2 + (-100)^2}{2} = 0$$

Variance Motivation: Measuring Spread

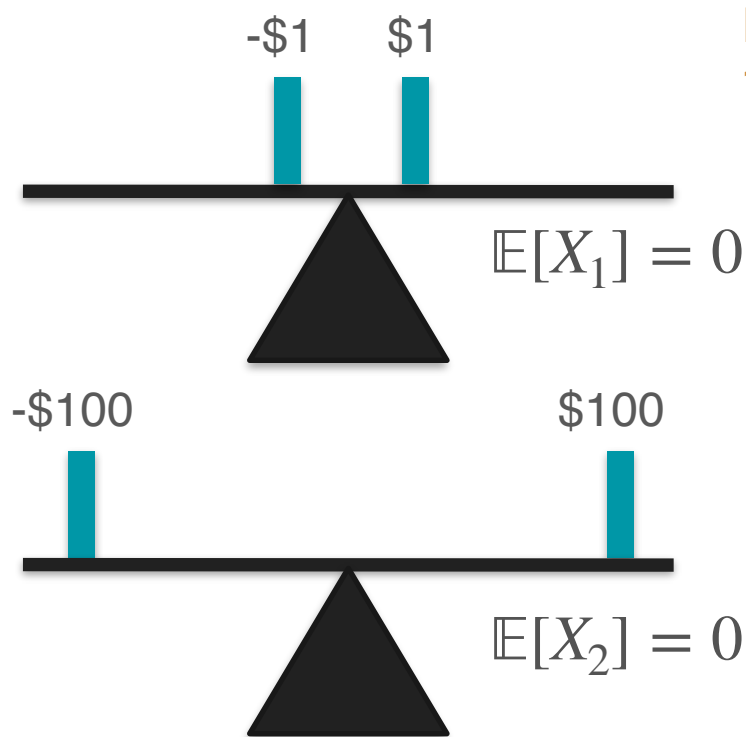


$$\mathbb{E}[X_1^2] = \frac{(1)^2 + (-1)^2}{2} = 1$$



$$\mathbb{E}[X_2^2] = \frac{(100)^2 + (-100)^2}{2} = 10,000$$

Variance Motivation: Measuring Spread

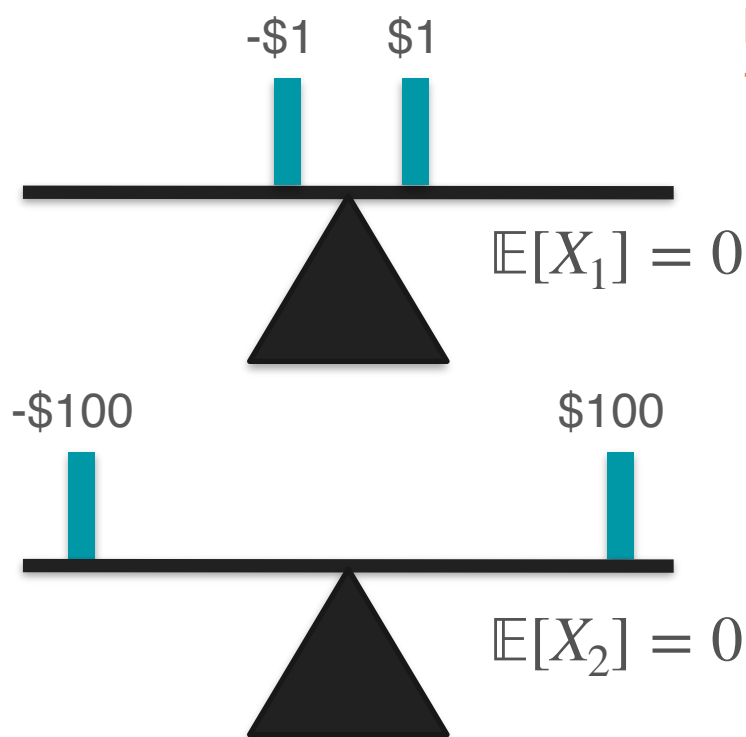


Key for telling game
1 and game 2 apart!

$$\mathbb{E}[X_1^2] = \frac{(1)^2 + (-1)^2}{2} = 1$$

$$\mathbb{E}[X_2^2] = \frac{(100)^2 + (-100)^2}{2} = 10,000$$

Variance Motivation: Measuring Spread



Key for telling game
1 and game 2 apart!

$$\mathbb{E}[X_1^2] = \frac{(1)^2 + (-1)^2}{2} = 1$$

$$\mathbb{E}[X_2^2] = \frac{(100)^2 + (-100)^2}{2} = 10,000$$

Measure of spread

Variance

$$\mathbb{E}[X^2]$$

Variance

$$\mathbb{E}[X^2]$$

Almost...

Variance Motivation: Centering With Mean

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

Game 2



You win 6 dollars



You win 4 dollars

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

Game 2



You win 6 dollars



You win 4 dollars

How risky are these two games in comparison?

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

Game 2



You win 6 dollars



You win 4 dollars

How risky are these two games in comparison?

Hint: Think of the spread

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

Game 2



You win 6 dollars



You win 4 dollars

They are equally risky

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

$$\mathbb{E}[X_1^2] = \frac{(-1)^2 + (1)^2}{2} = 1$$

Game 2



You win 6 dollars



You win 4 dollars

$$\mathbb{E}[X_2^2] = \frac{(4)^2 + (6)^2}{2} = 26$$

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

$$\mathbb{E}[X_1^2] = \frac{(-1)^2 + (1)^2}{2} = 1$$



Same risk?



$$\mathbb{E}[X_2^2] = \frac{(4)^2 + (6)^2}{2} = 26$$

Game 2



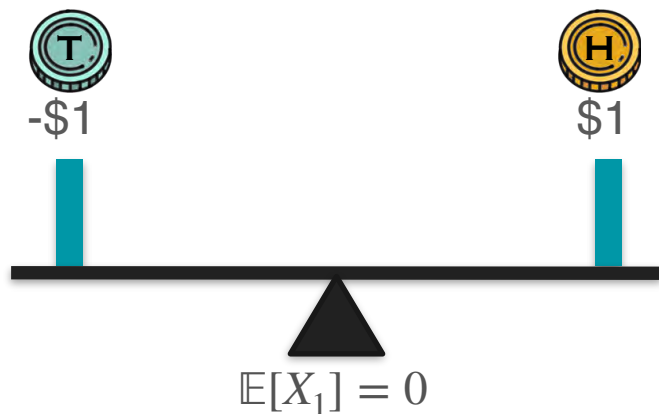
You win 6 dollars



You win 4 dollars

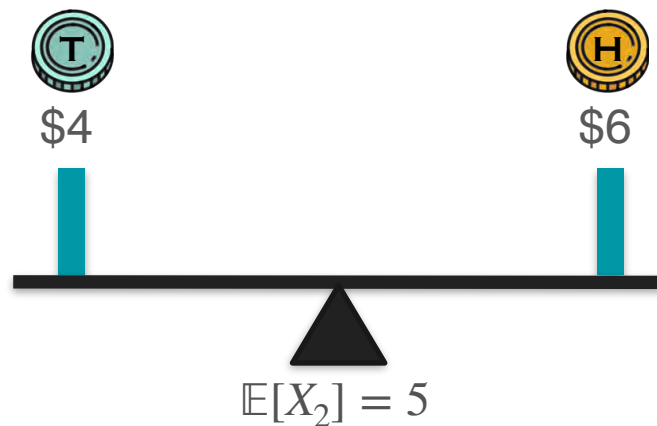
Variance Motivation: Centering With Mean

$$\mathbb{E}[X_1^2] = \frac{(-1)^2 + (1)^2}{2} = 1$$



Game 1

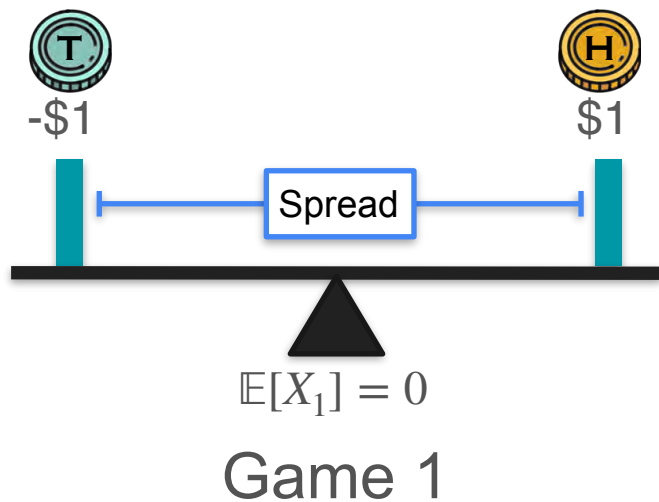
$$\mathbb{E}[X_2^2] = \frac{(4)^2 + (6)^2}{2} = 26$$



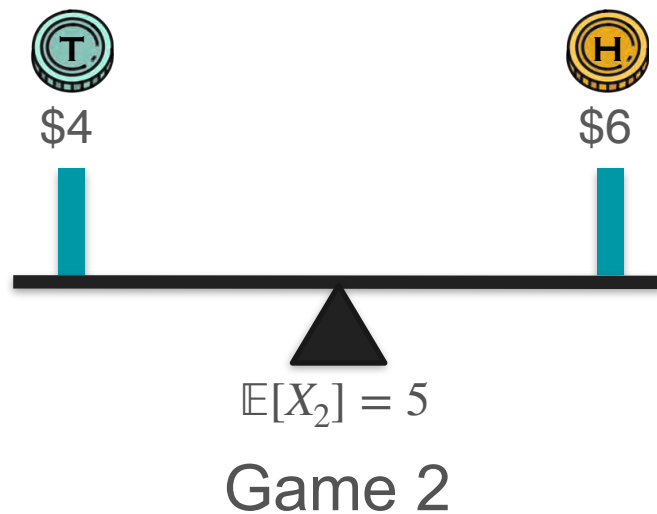
Game 2

Variance Motivation: Centering With Mean

$$\mathbb{E}[X_1^2] = \frac{(-1)^2 + (1)^2}{2} = 1$$

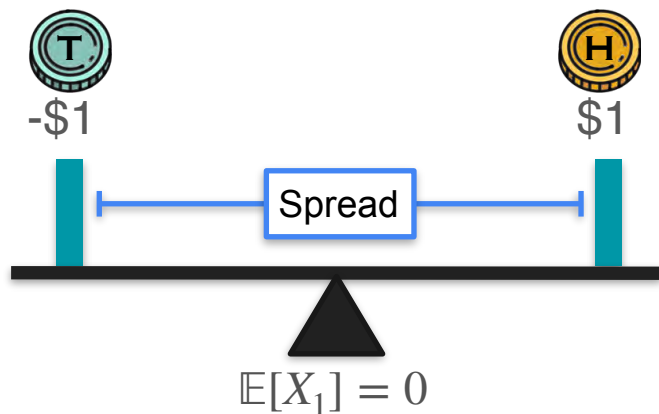


$$\mathbb{E}[X_2^2] = \frac{(4)^2 + (6)^2}{2} = 26$$



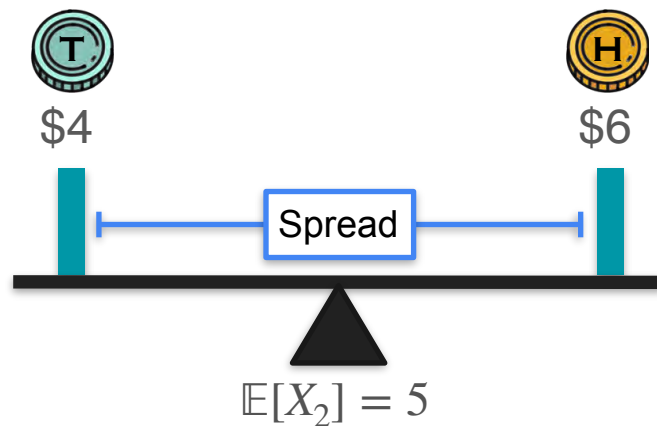
Variance Motivation: Centering With Mean

$$\mathbb{E}[X_1^2] = \frac{(-1)^2 + (1)^2}{2} = 1$$



Game 1

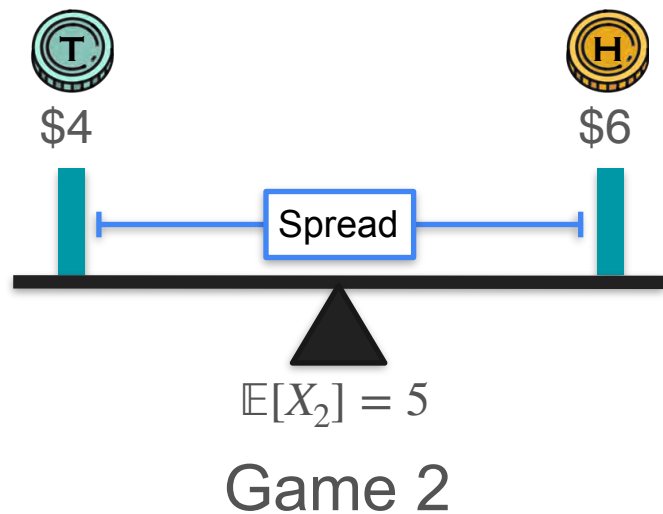
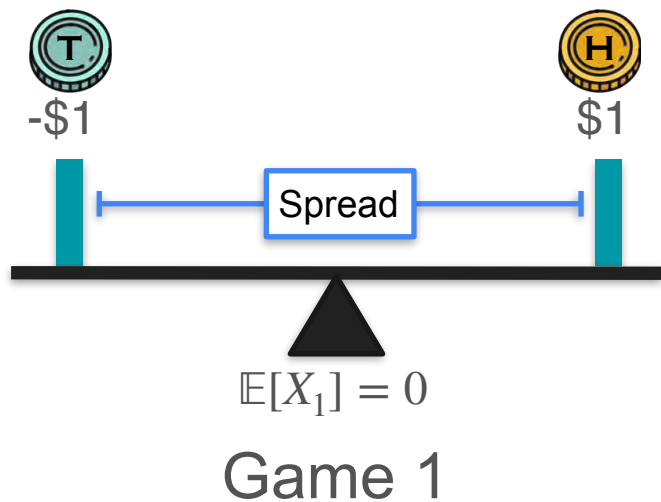
$$\mathbb{E}[X_2^2] = \frac{(4)^2 + (6)^2}{2} = 26$$



Game 2

Variance Motivation: Centering With Mean

$$\mathbb{E}[X_1^2] = \frac{(-1)^2 + (1)^2}{2} = 1 \quad \longleftarrow \text{Same spread?} \longrightarrow \quad \mathbb{E}[X_2^2] = \frac{(4)^2 + (6)^2}{2} = 26$$



Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$

1

1



-\$1



\$1

$$\mathbb{E}[X_1] = 0$$

Game 1

1

1



\$4



\$6

$$\mathbb{E}[X_2] = 5$$

Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$

1

1



-\$1



\$1

Spread

$$\mathbb{E}[X_1] = 0$$

Game 1

1

1



\$4



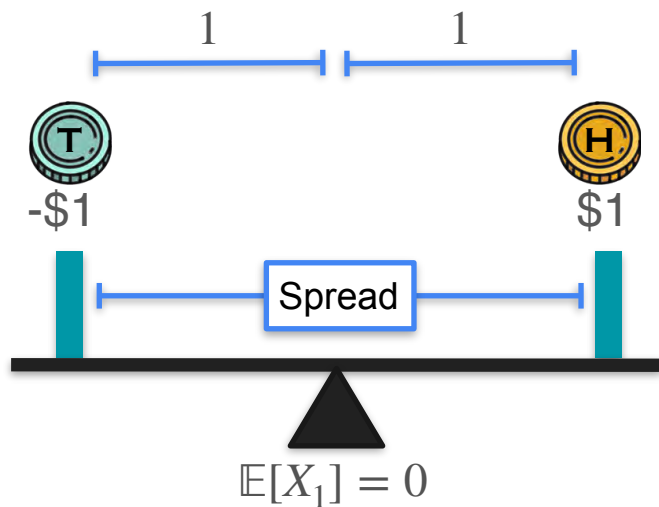
\$6

$$\mathbb{E}[X_2] = 5$$

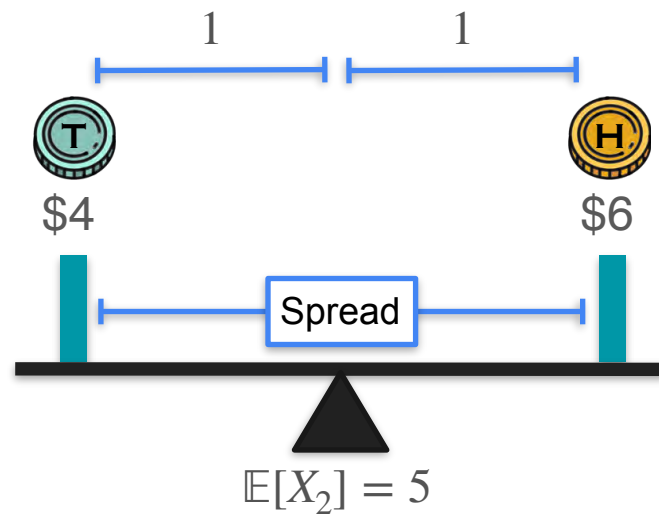
Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$



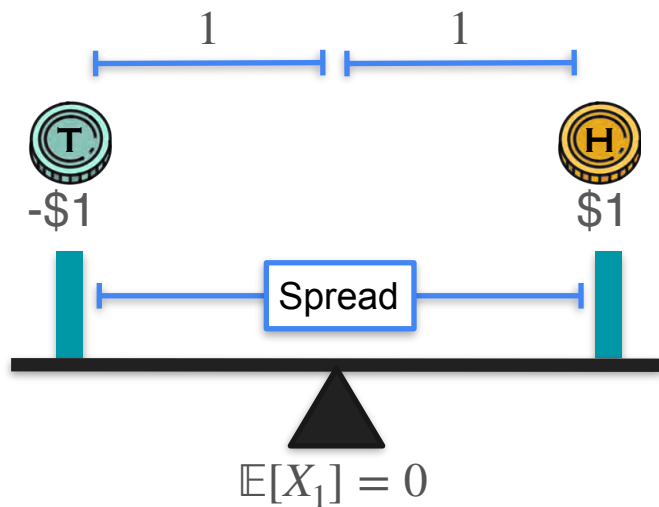
Game 1



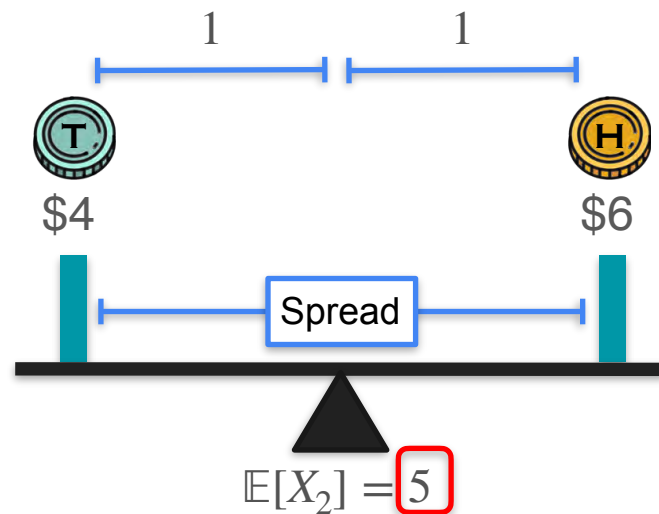
Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$



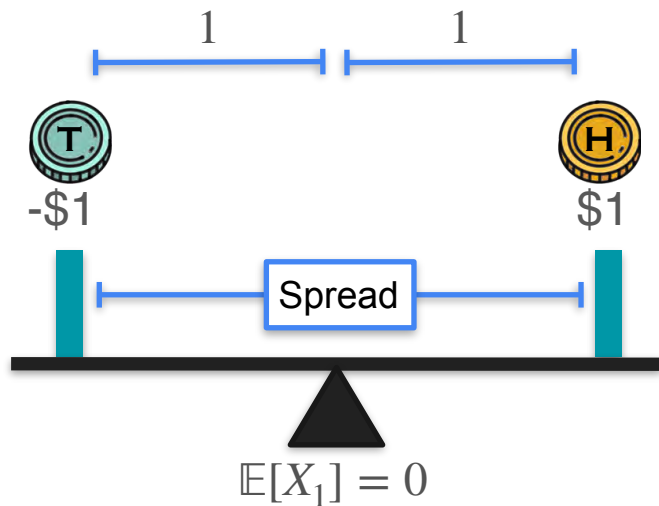
Game 1



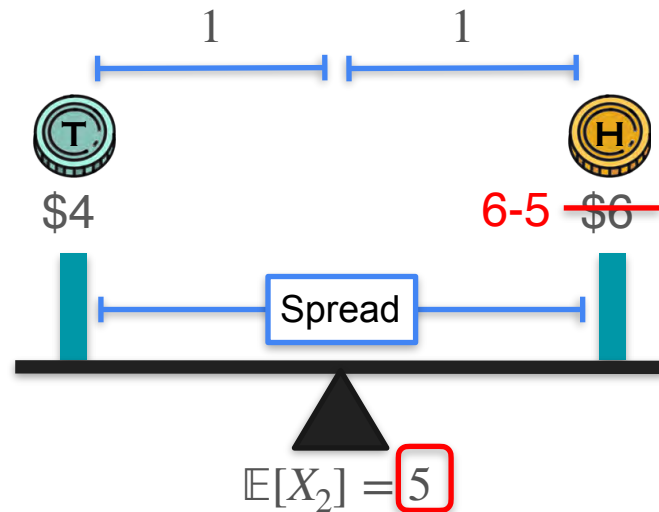
Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$



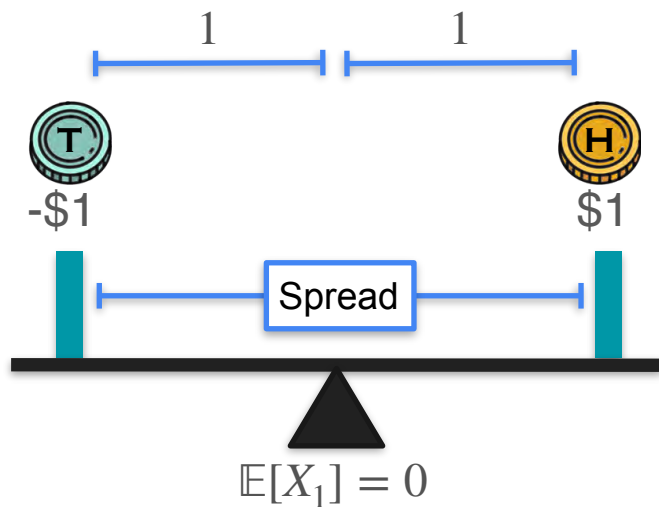
Game 1



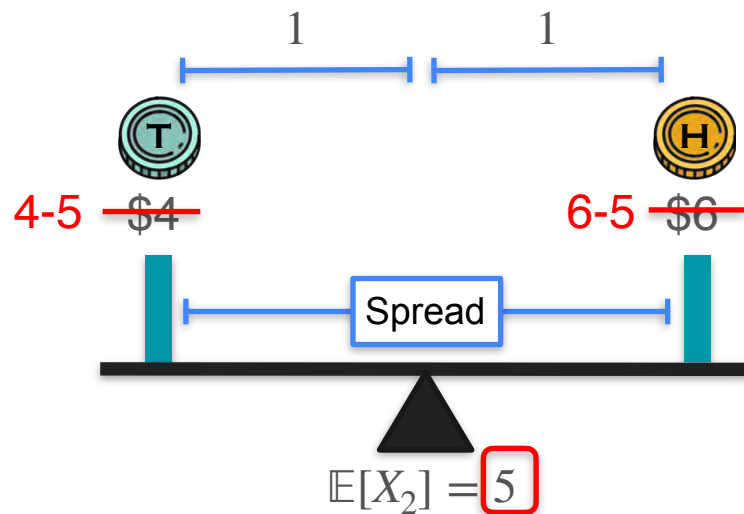
Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$



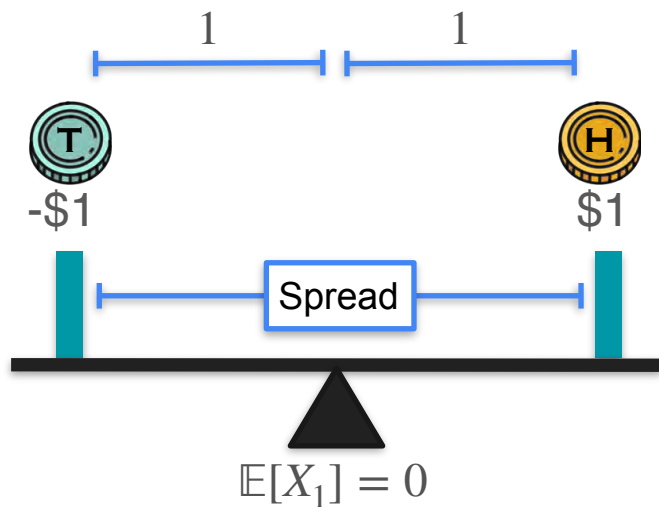
Game 1



Game 2

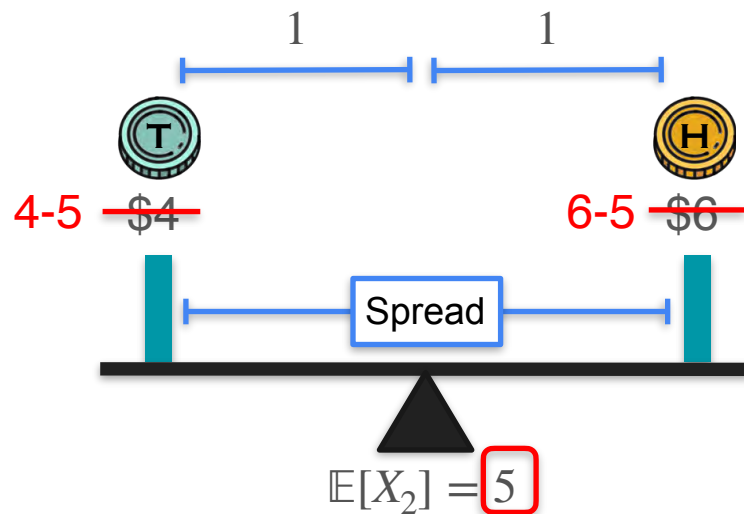
Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$



Game 1

$$\mathbb{E}[(X_2 - 5)^2]$$

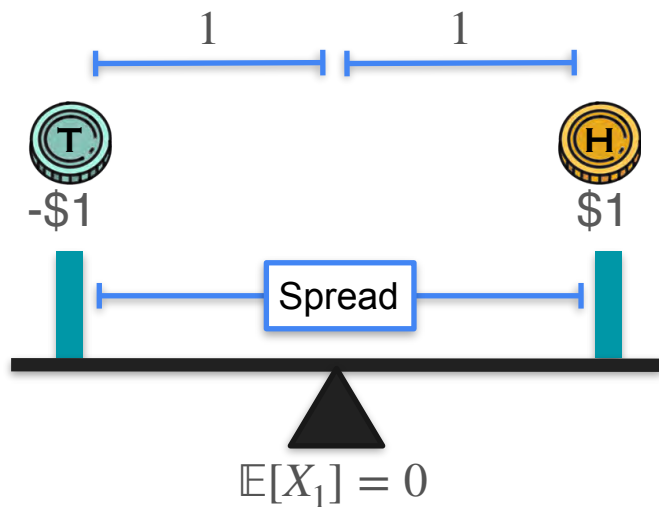


Game 2

Variance: Spread and Shift

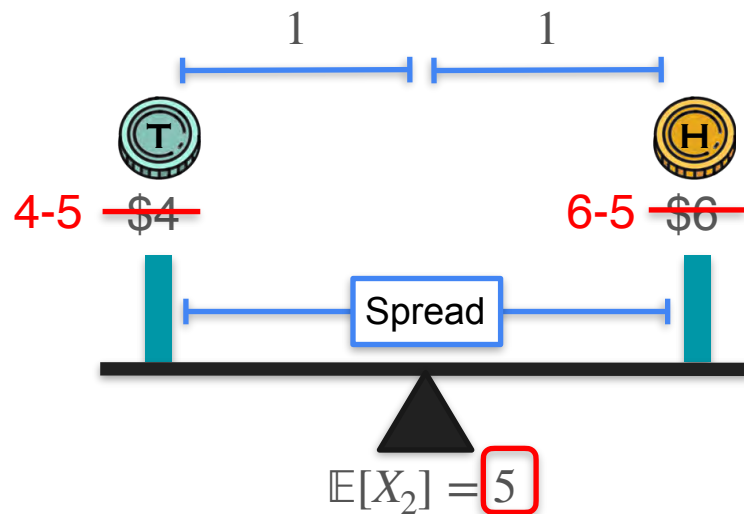
$$\mathbb{E}[X_1^2] = 1$$

Mean: μ



Game 1

$$\mathbb{E}[(X_2 - 5)^2]$$



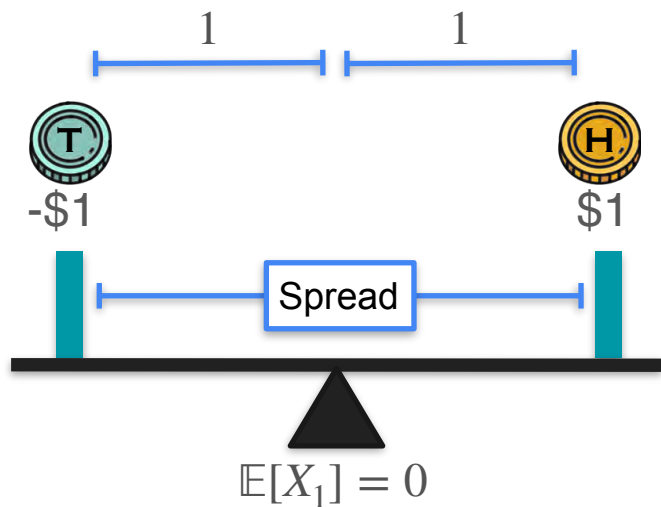
Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$

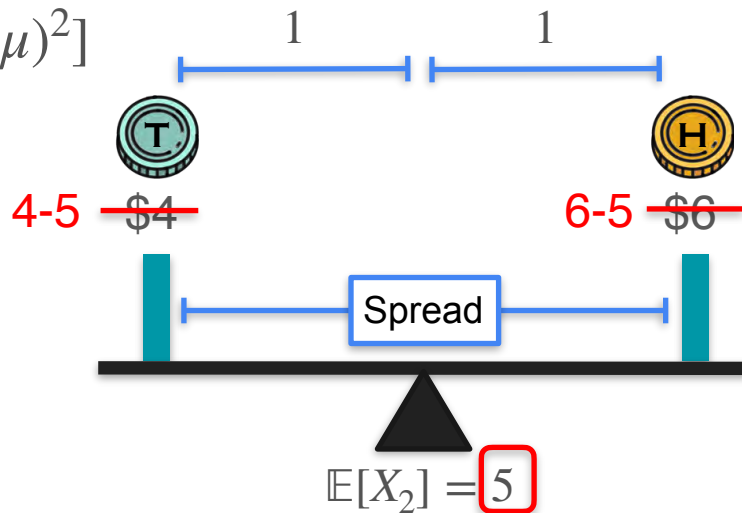
Mean: μ

$$\text{Variance: } \mathbb{E}[(X - \mu)^2]$$



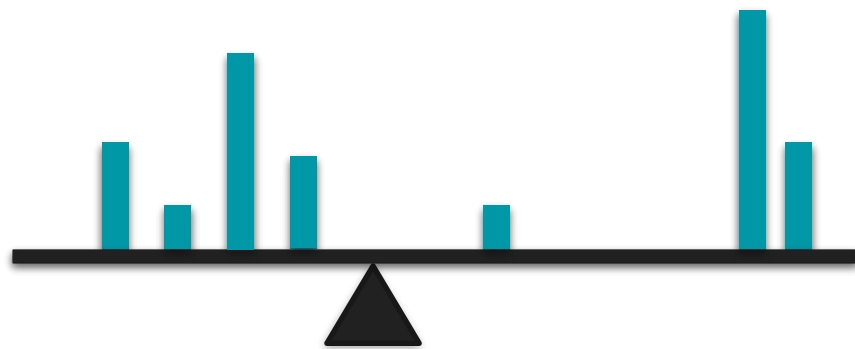
Game 1

$$\mathbb{E}[(X_2 - 5)^2]$$

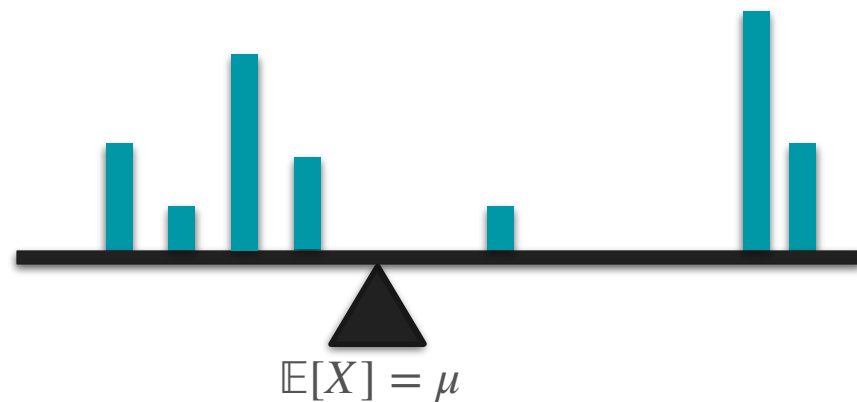


Game 2

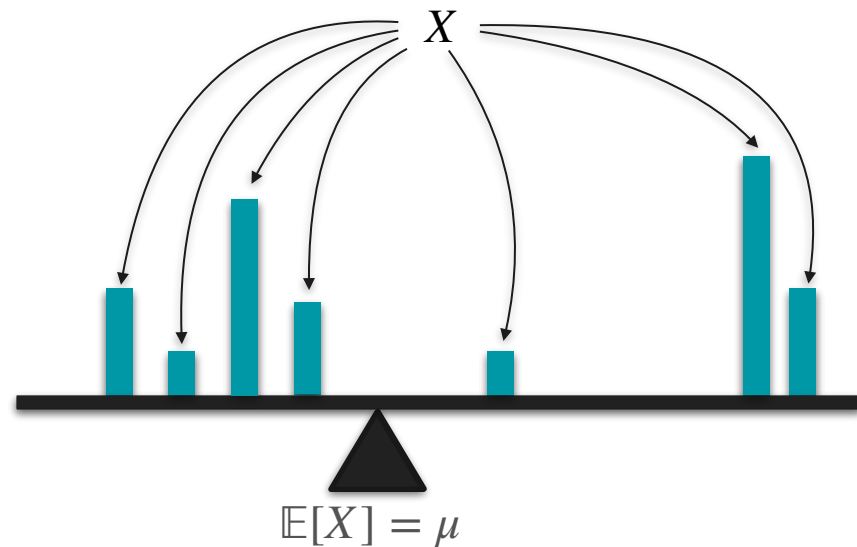
Variance Formula



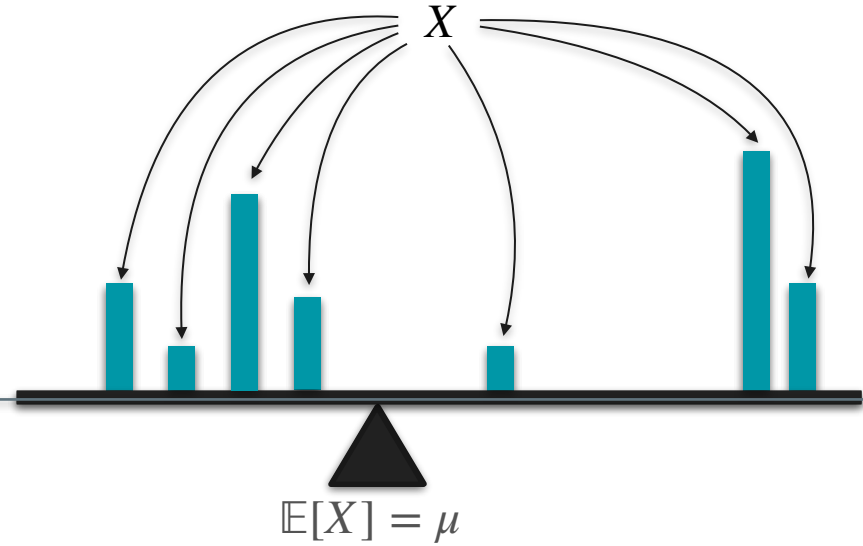
Variance Formula



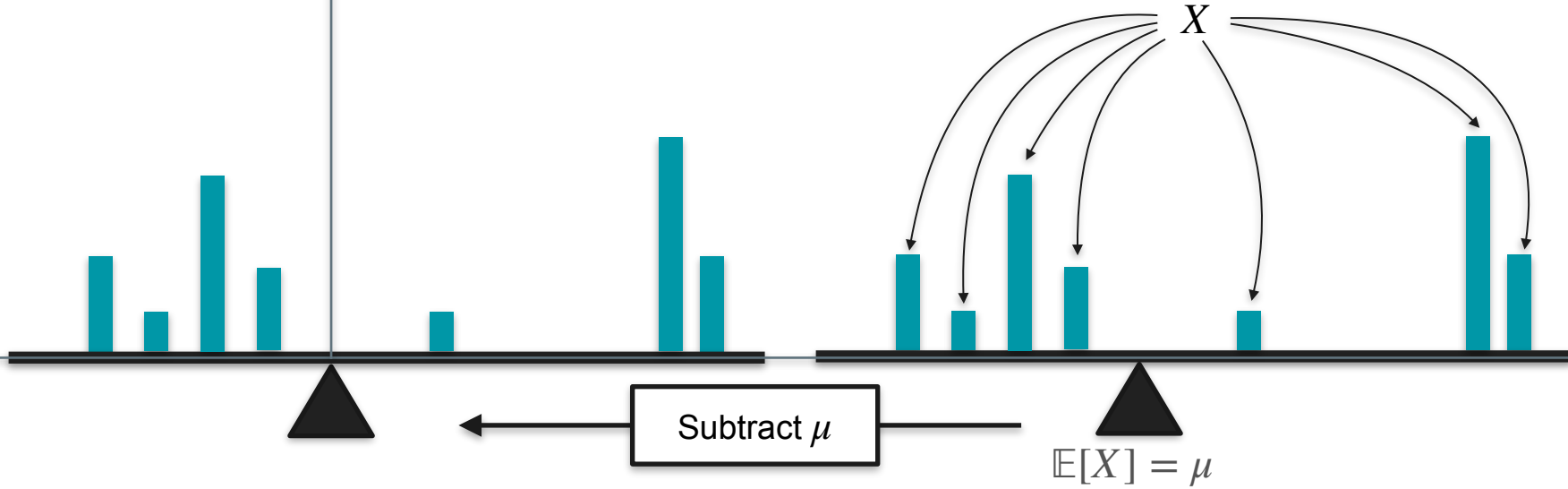
Variance Formula



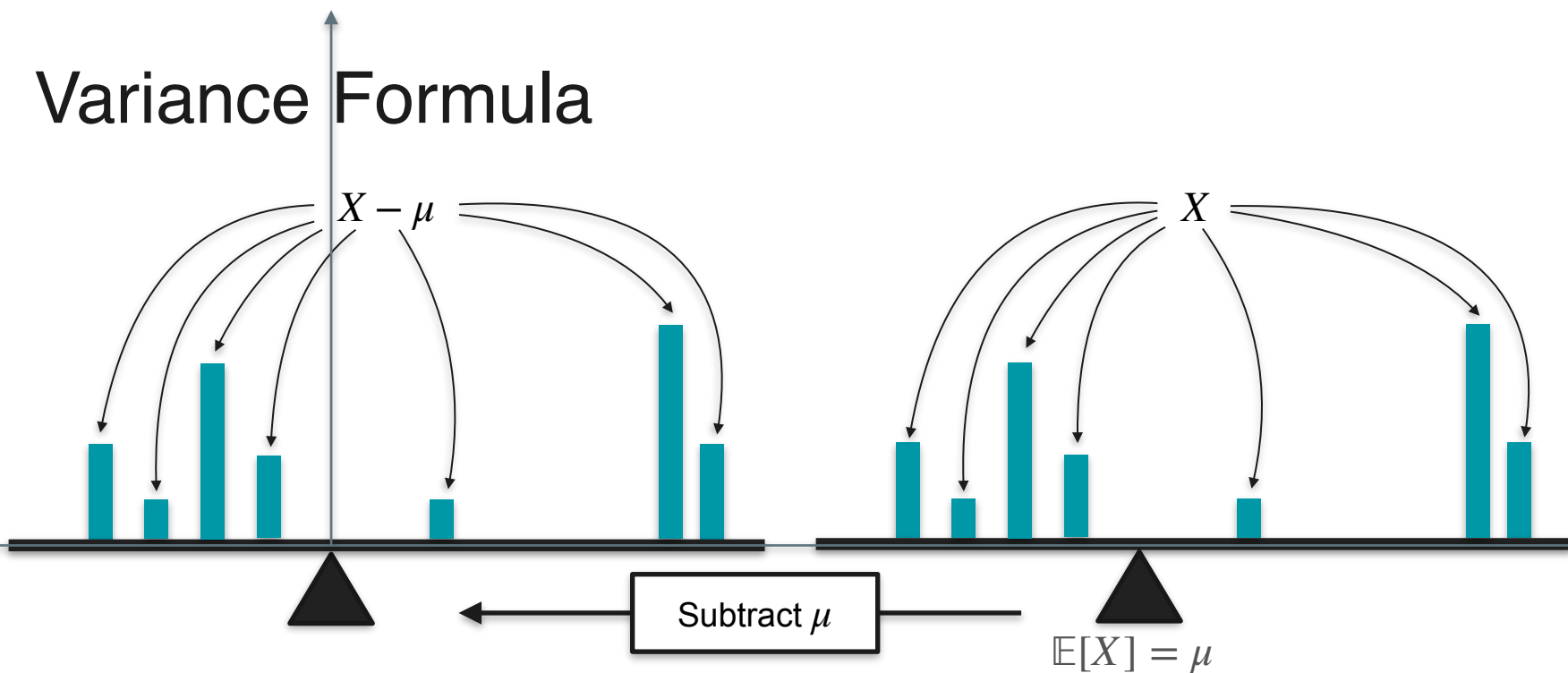
Variance Formula



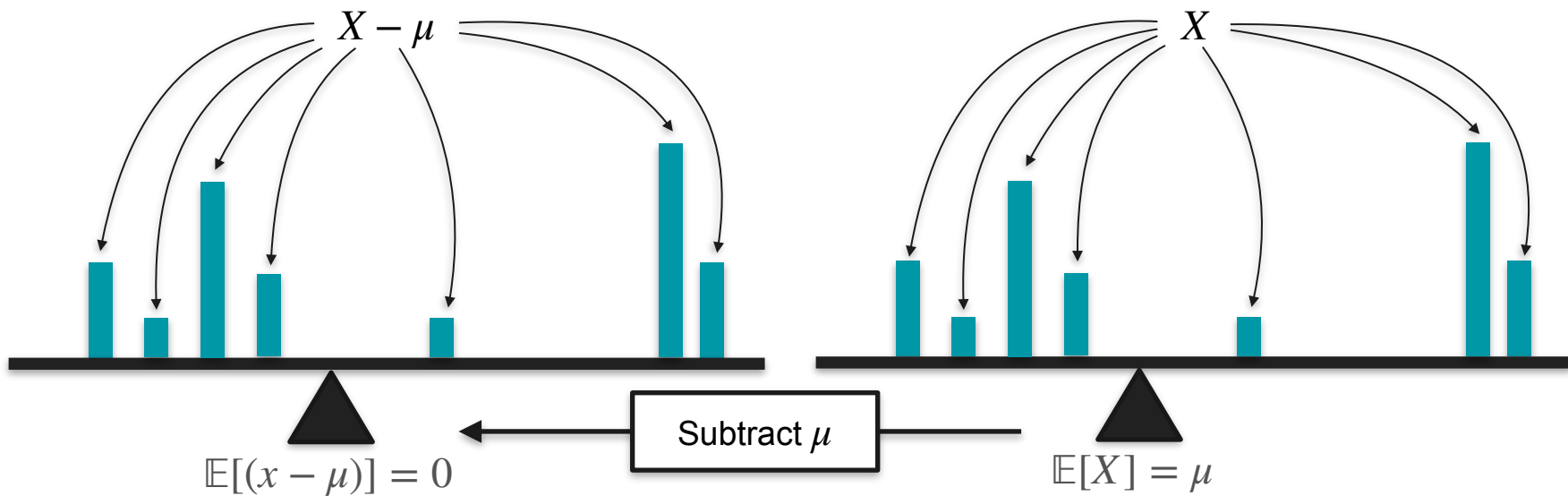
Variance Formula



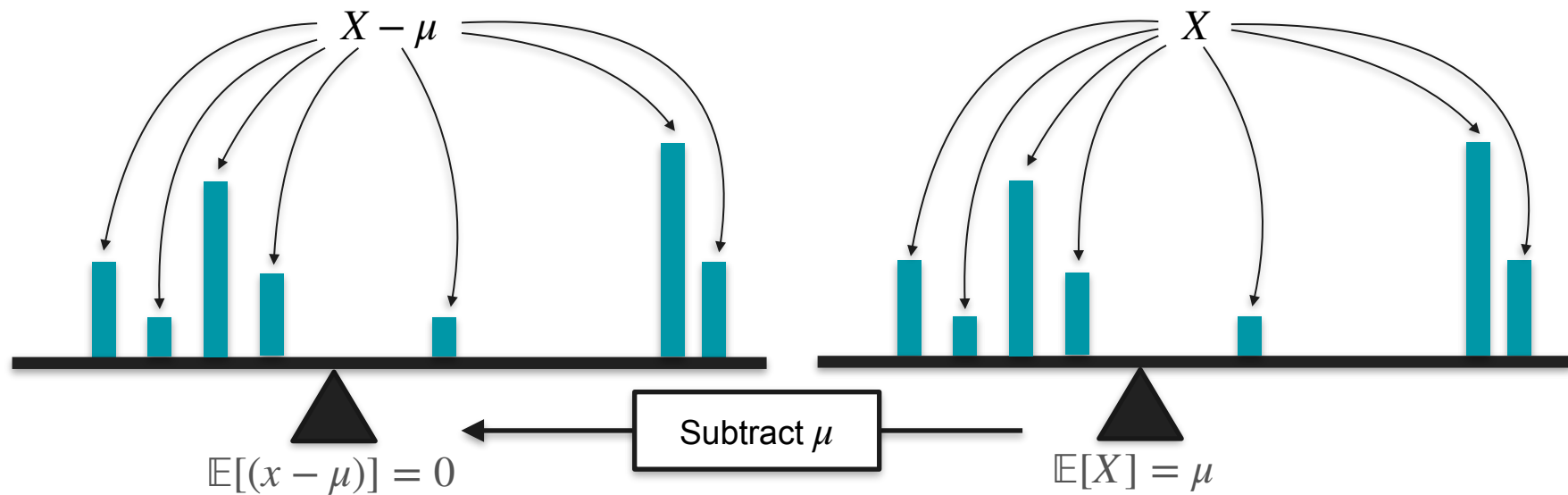
Variance Formula



Variance Formula



Variance Formula



$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$$

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Variance Formula

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \end{aligned}$$

$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

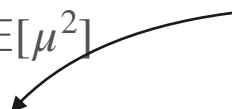
Variance Formula

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \quad \mathbb{E}[\text{constant} \cdot X] = \text{constant} \cdot \mathbb{E}[X] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \end{aligned}$$

Variance Formula

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \end{aligned}$$

$\mathbb{E}[\text{constant}] = \text{constant}$



Variance Formula

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\&= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \\&= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\&= \mathbb{E}[X^2] - 2\mathbb{E}[X] \cdot \mathbb{E}[X] + \mathbb{E}[X]^2\end{aligned}$$

$\mu = \mathbb{E}[X]$



Variance Formula

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\&= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \\&= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\&= \mathbb{E}[X^2] - 2\mathbb{E}[X] \cdot \mathbb{E}[X] + \mathbb{E}[X]^2 \\&= \mathbb{E}[X^2] - 2\mathbb{E}[X]^2 + \mathbb{E}[X]^2\end{aligned}$$

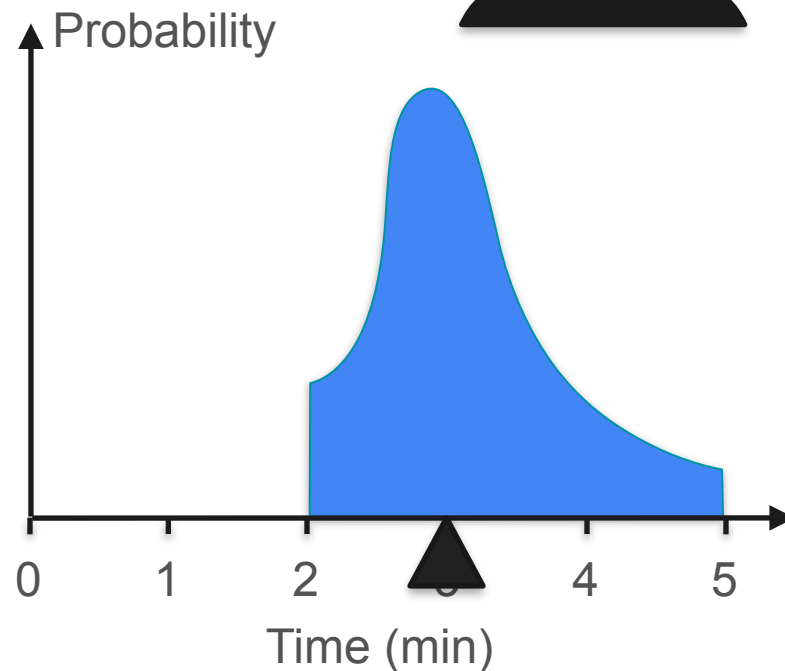
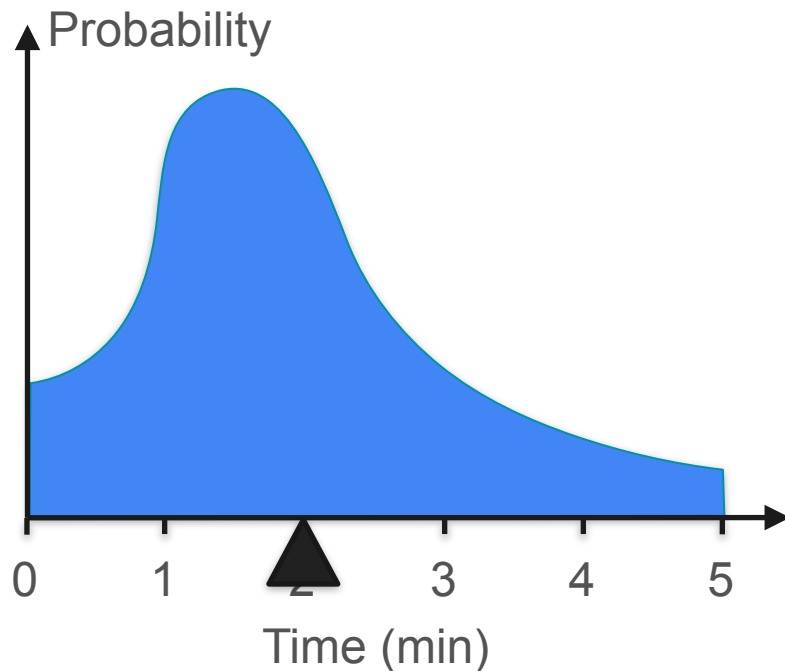
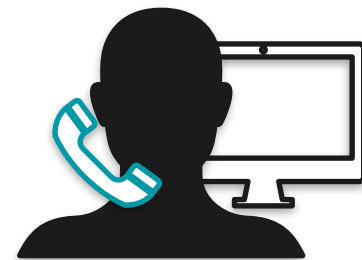
Variance Formula

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X] \cdot \mathbb{E}[X] + \mathbb{E}[X]^2 \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]^2 + \mathbb{E}[X]^2 \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \end{aligned}$$

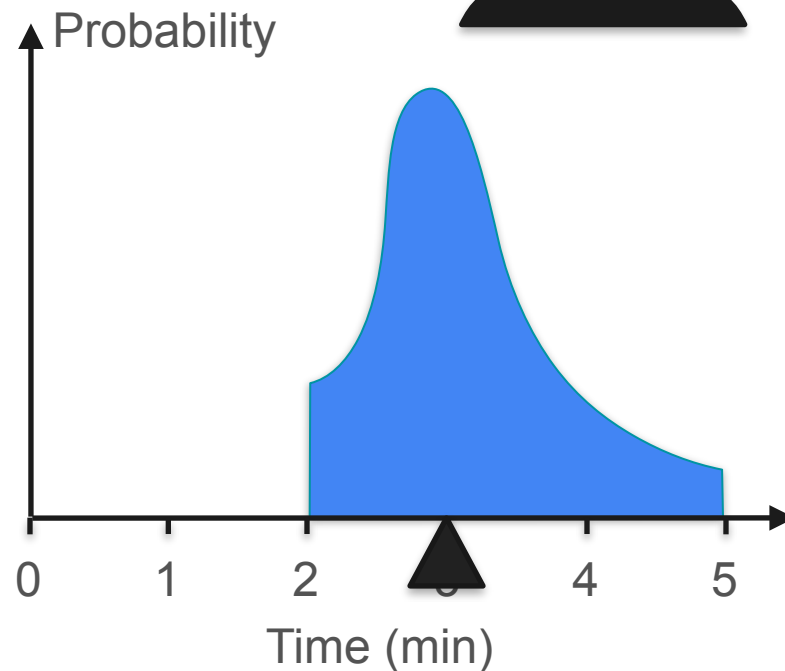
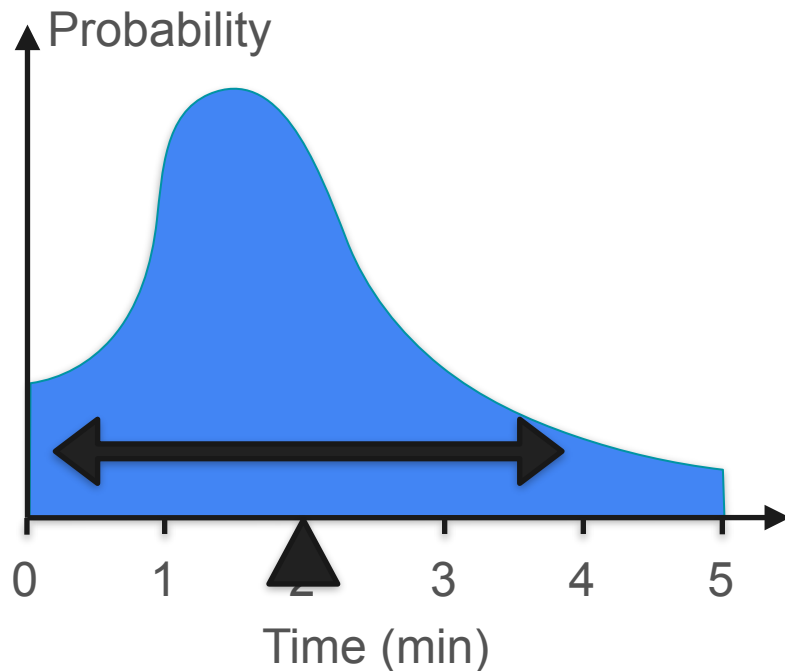
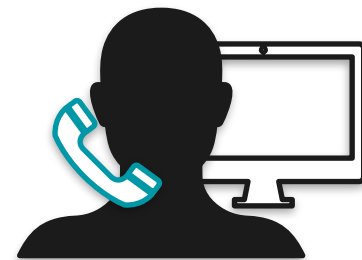
Variance Formula

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu)^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \end{aligned}$$

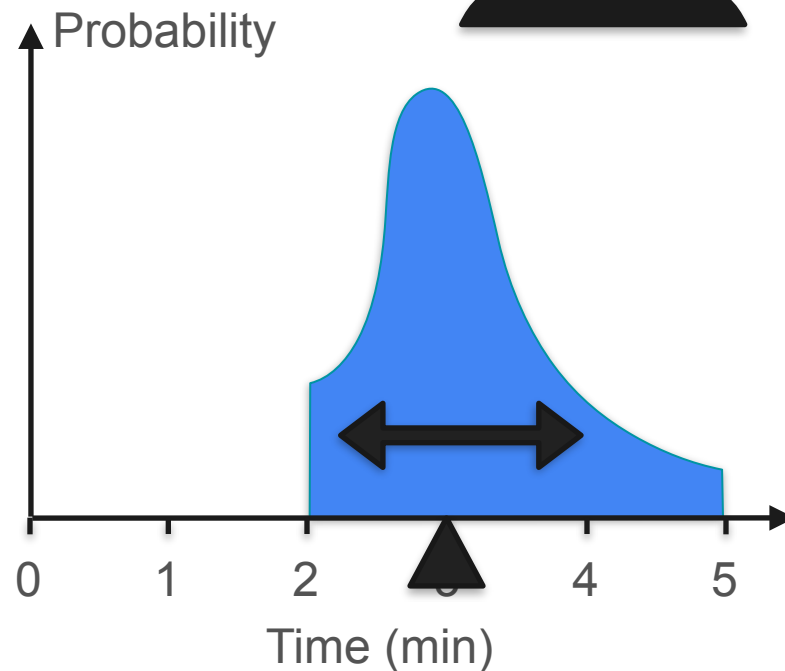
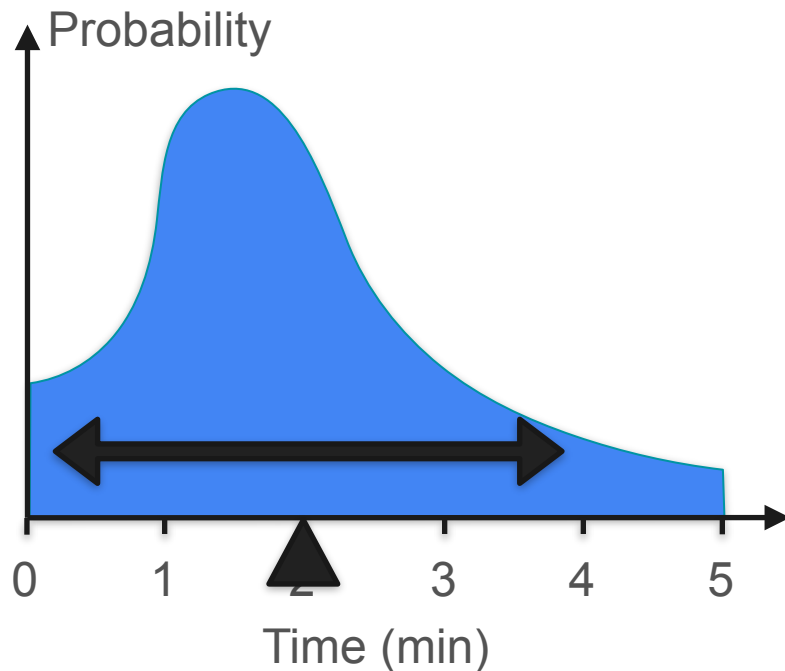
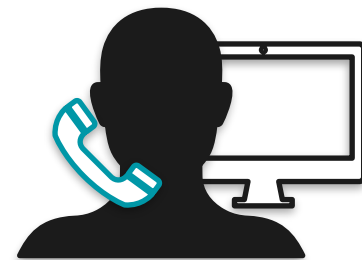
Variance for Continuous Distributions



Variance for Continuous Distributions

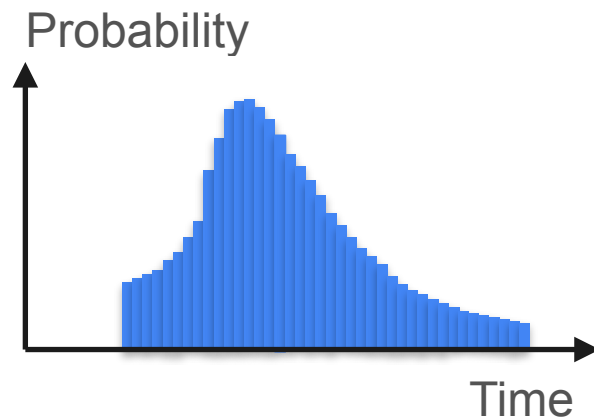


Variance for Continuous Distributions



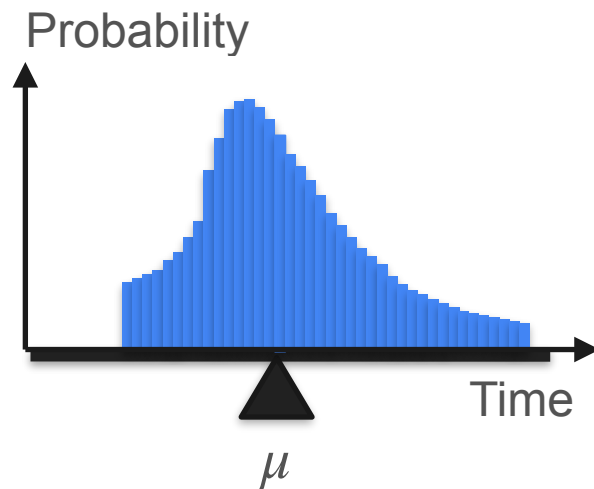
Variance

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$



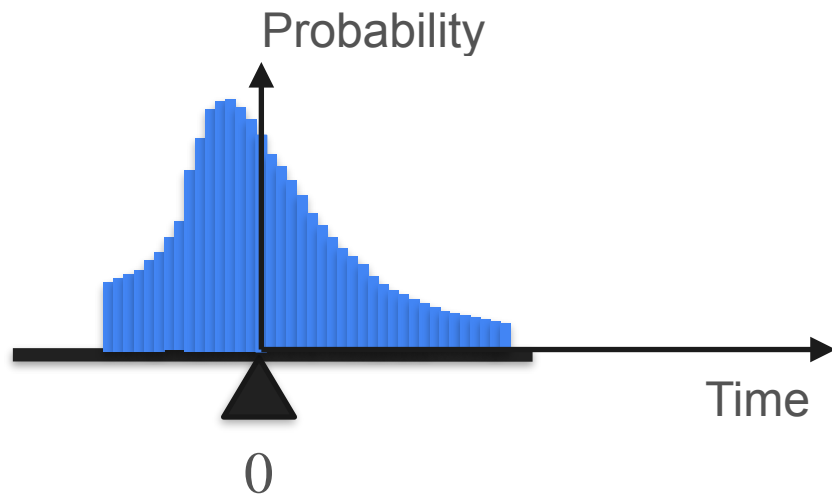
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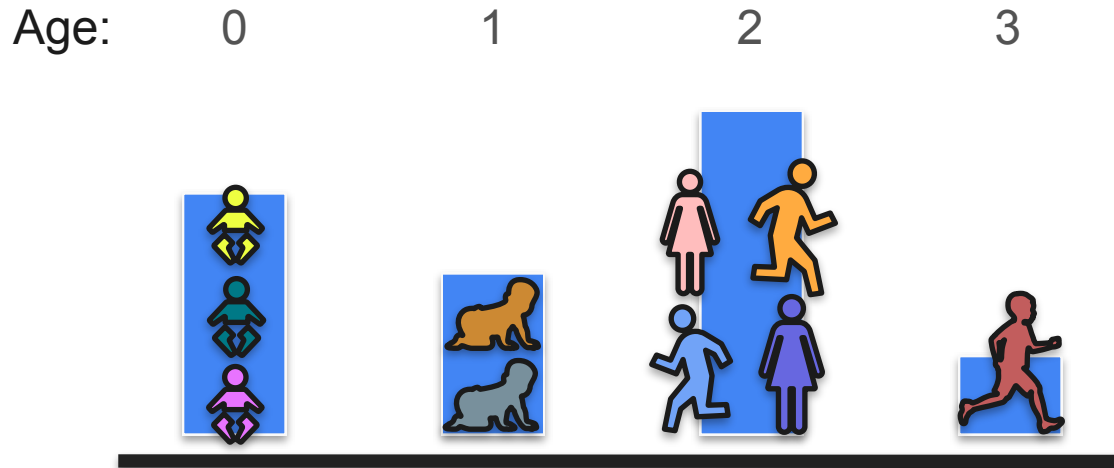


DeepLearning.AI

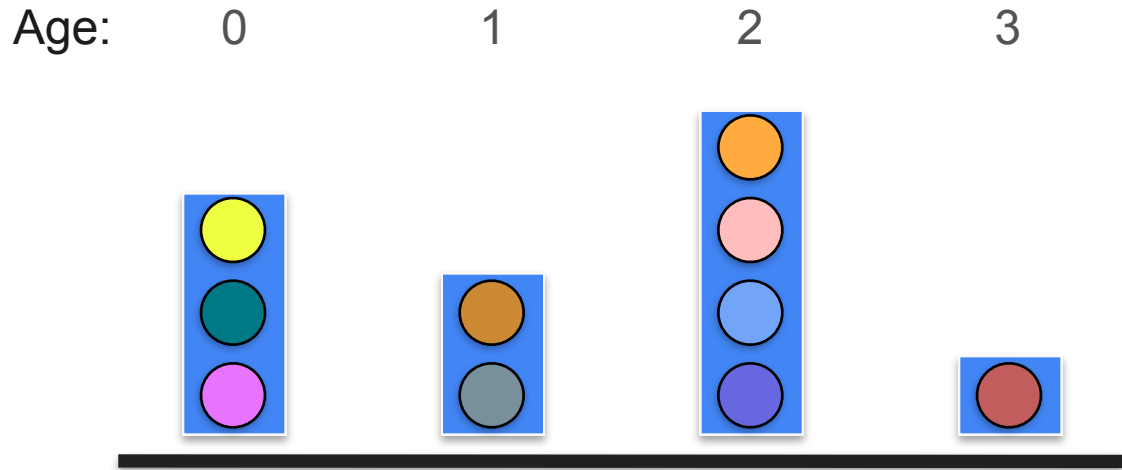
Describing Distributions

Measures of Central Tendency

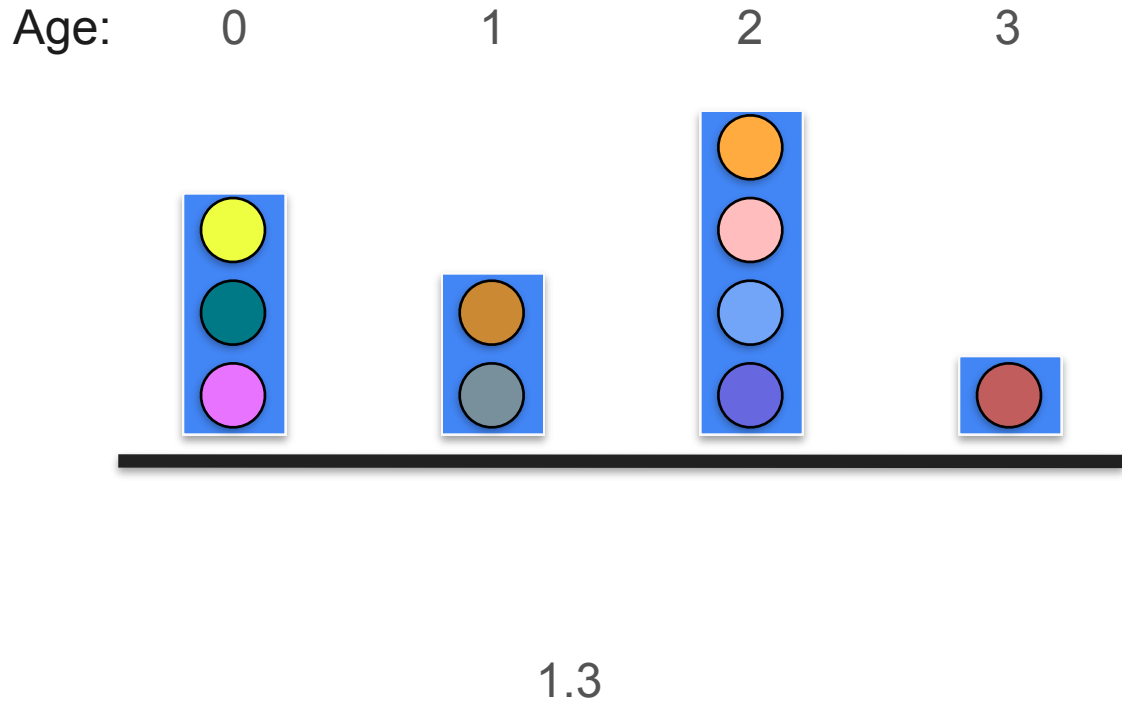
Mean: Example



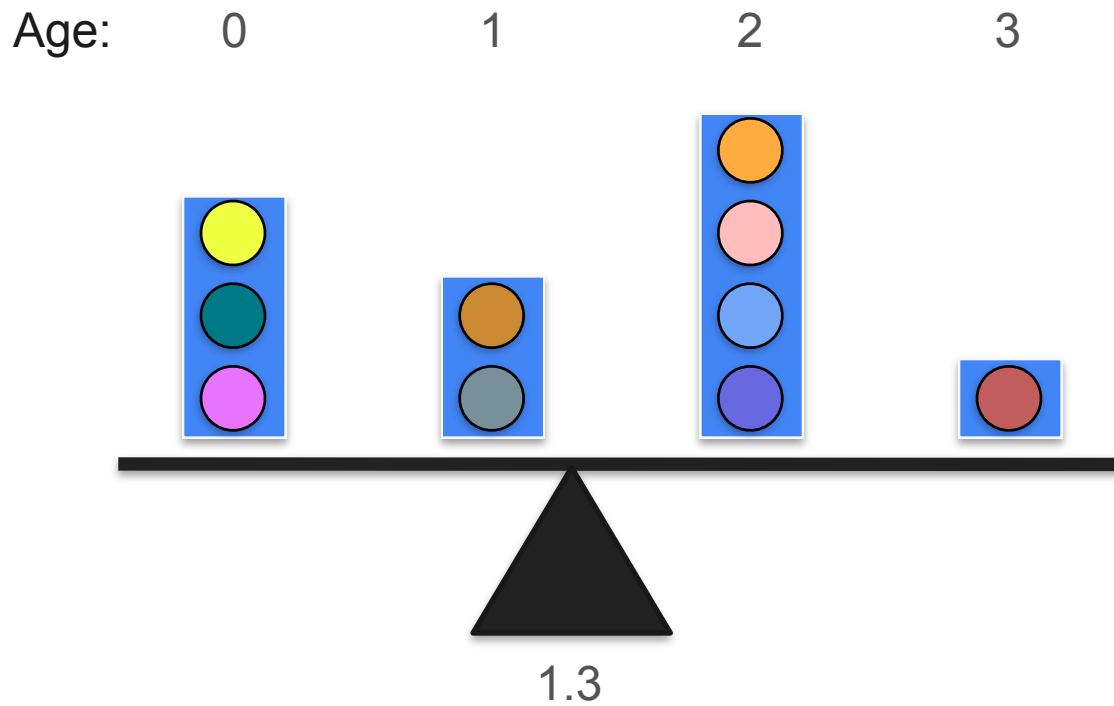
Mean: Example



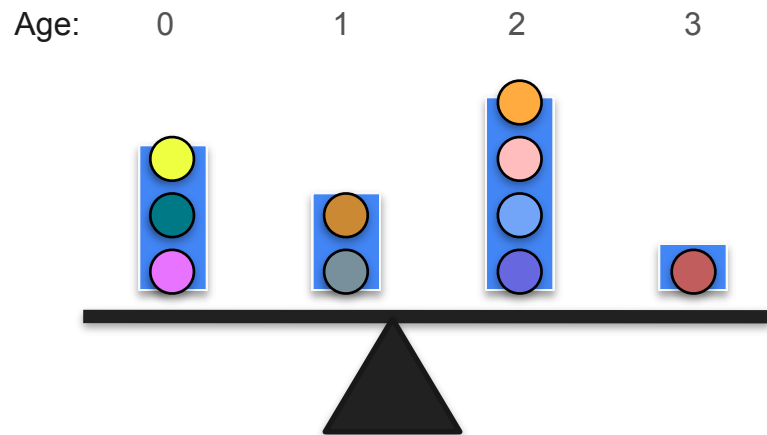
Mean: Example



Mean: Example

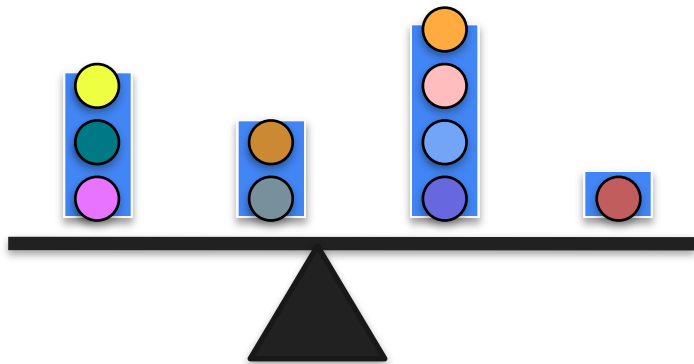


Mean: Example

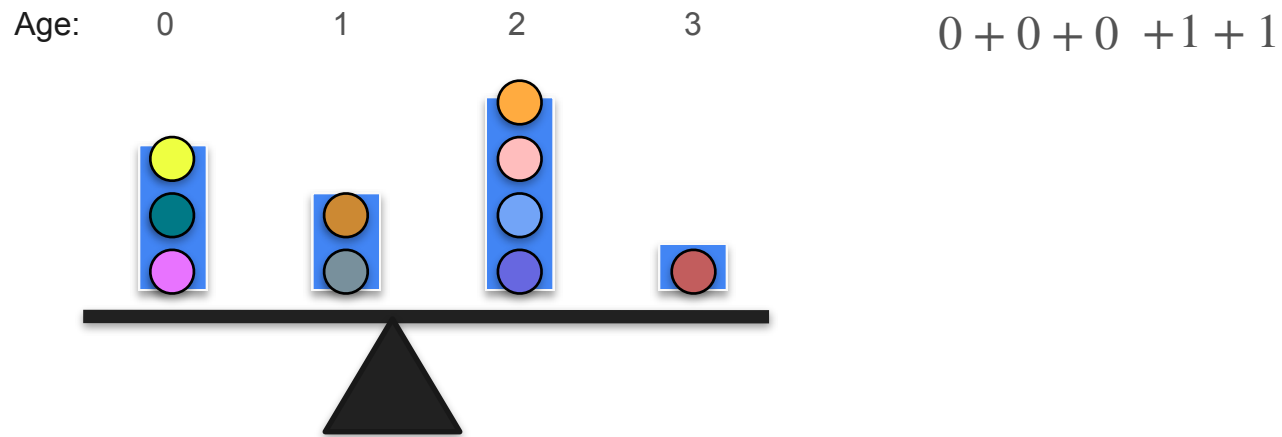


Mean: Example

Age: 0 1 2 3 $0 + 0 + 0$

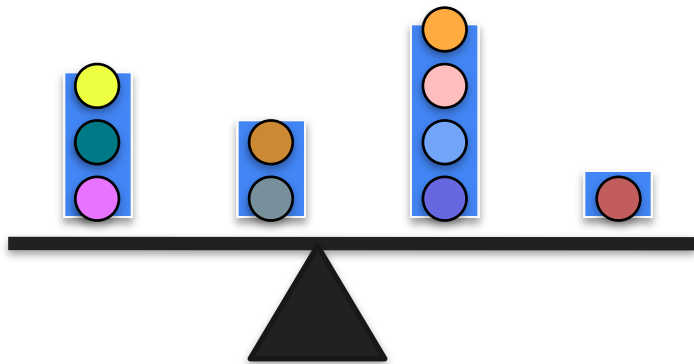


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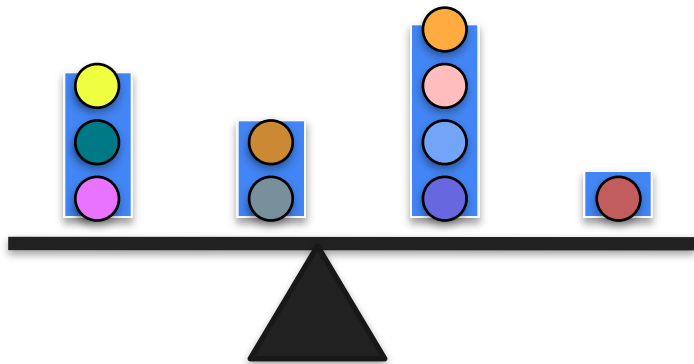
Age: 0 1 2 3



$$0 + 0 + 0 + 1 + 1 + 2 + 2 + 2 + 2$$

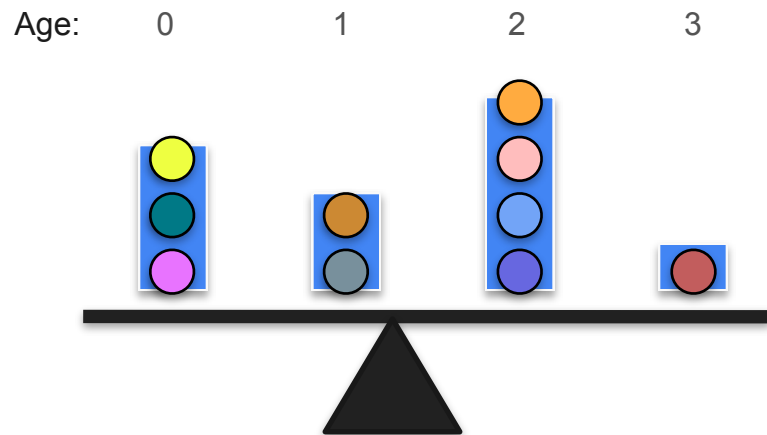
Mean: Example

Age: 0 1 2 3



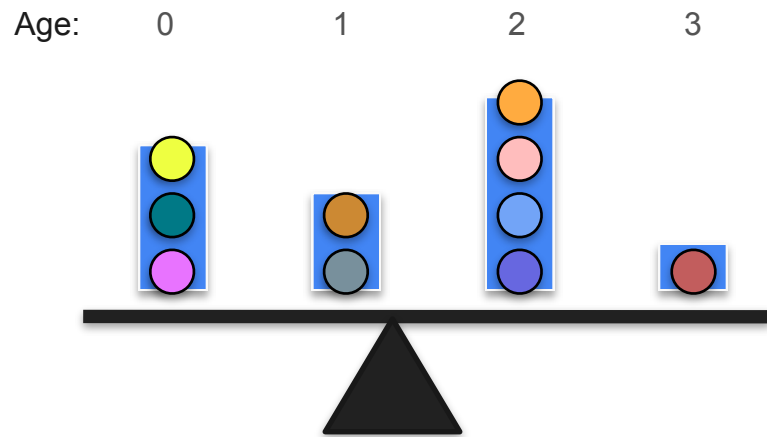
$$0 + 0 + 0 + 1 + 1 + 2 + 2 + 2 + 2 + 3$$

Mean: Example



$$\frac{0 + 0 + 0 + 1 + 1 + 2 + 2 + 2 + 2 + 3}{10}$$

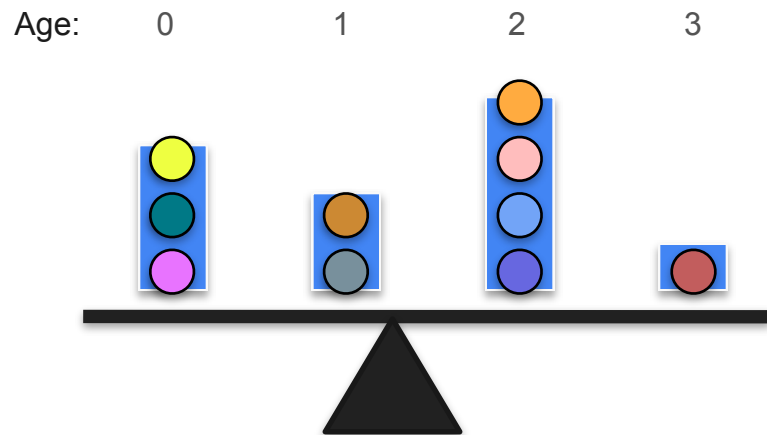
Mean: Example



$$\frac{0 + 0 + 0 + 1 + 1 + 2 + 2 + 2 + 2 + 3}{10}$$

$$= \frac{13}{10}$$

Mean: Example

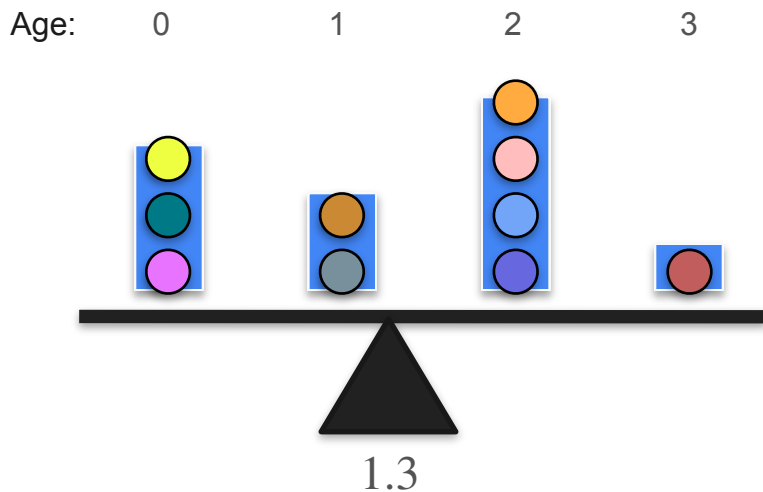


$$\frac{0 + 0 + 0 + 1 + 1 + 2 + 2 + 2 + 2 + 3}{10}$$

$$= \frac{13}{10}$$

$$= 1.3$$

Mean: Example

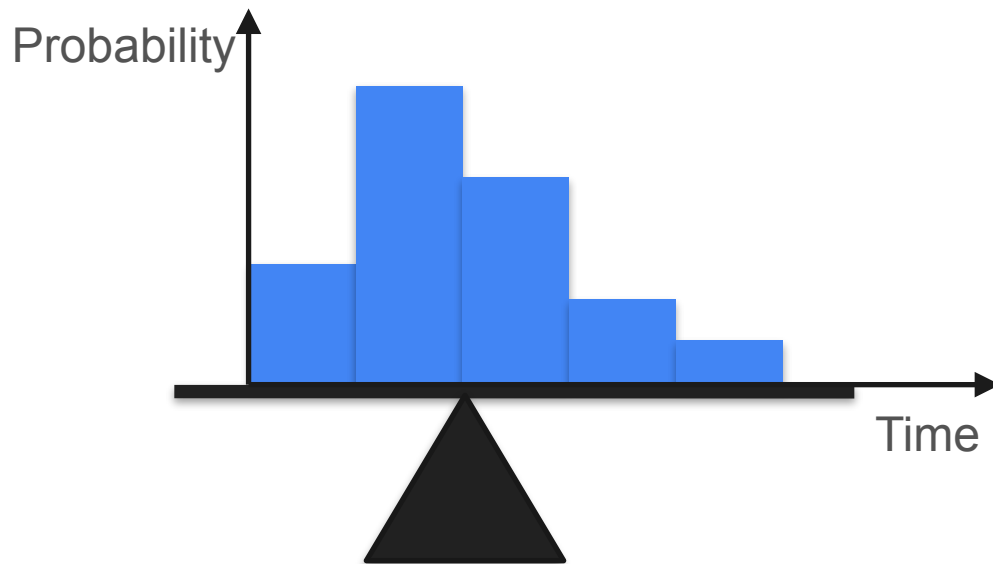
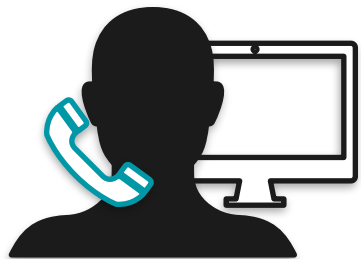


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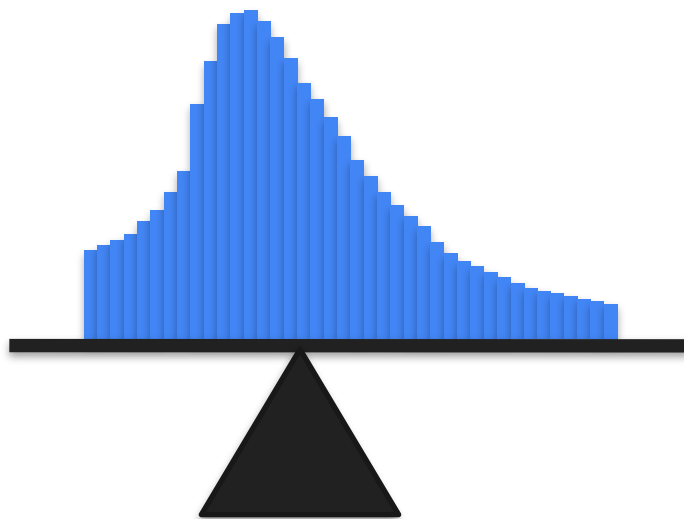
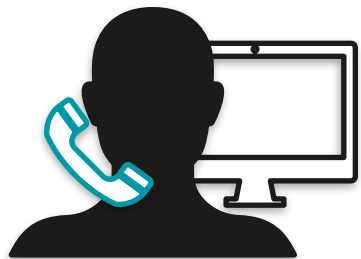
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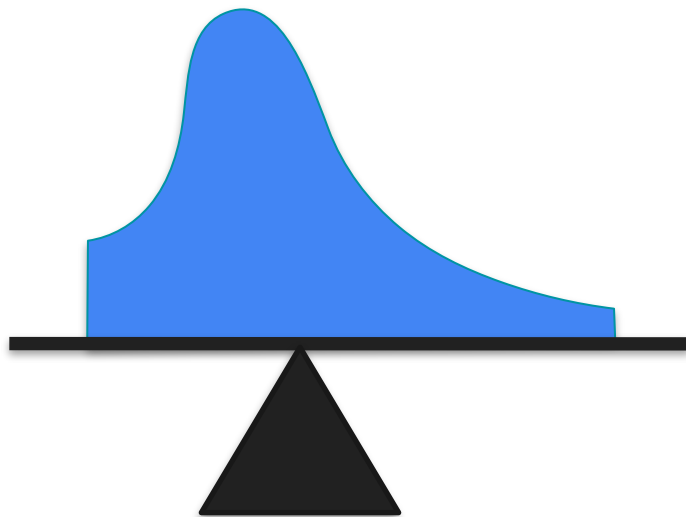
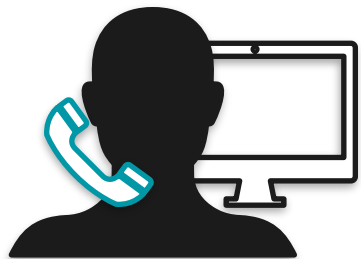
Mean



Mean



Mean



Median: Motivation

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Michael Jordan

Outliers

Outliers

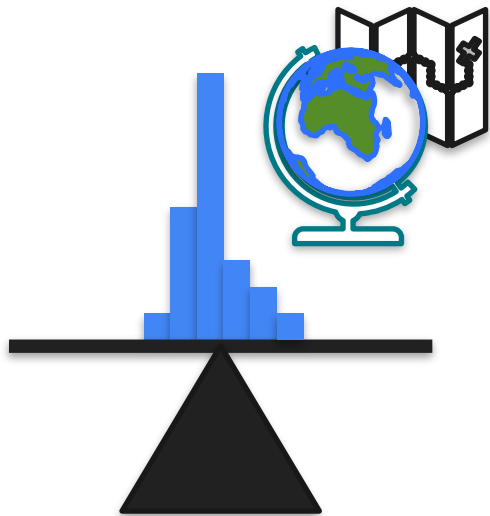


Outliers



Outliers

Graduates



Salary

Outliers

Graduates



Outliers

Graduates



Outliers



Median

Graduates



Salary

Median

Graduates



Salary

Graduates



Median

Salary

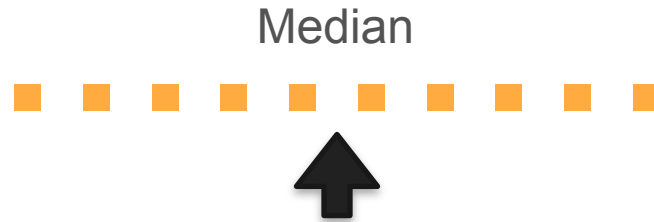
Median



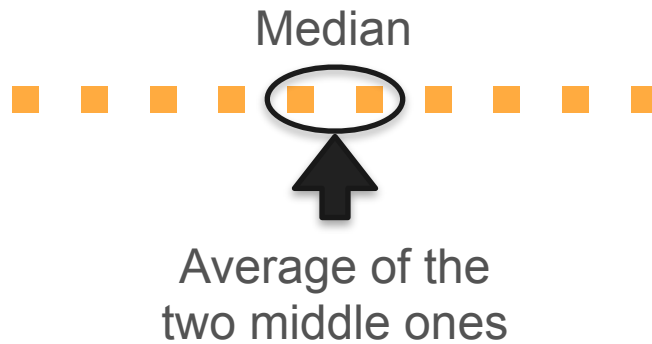
Median



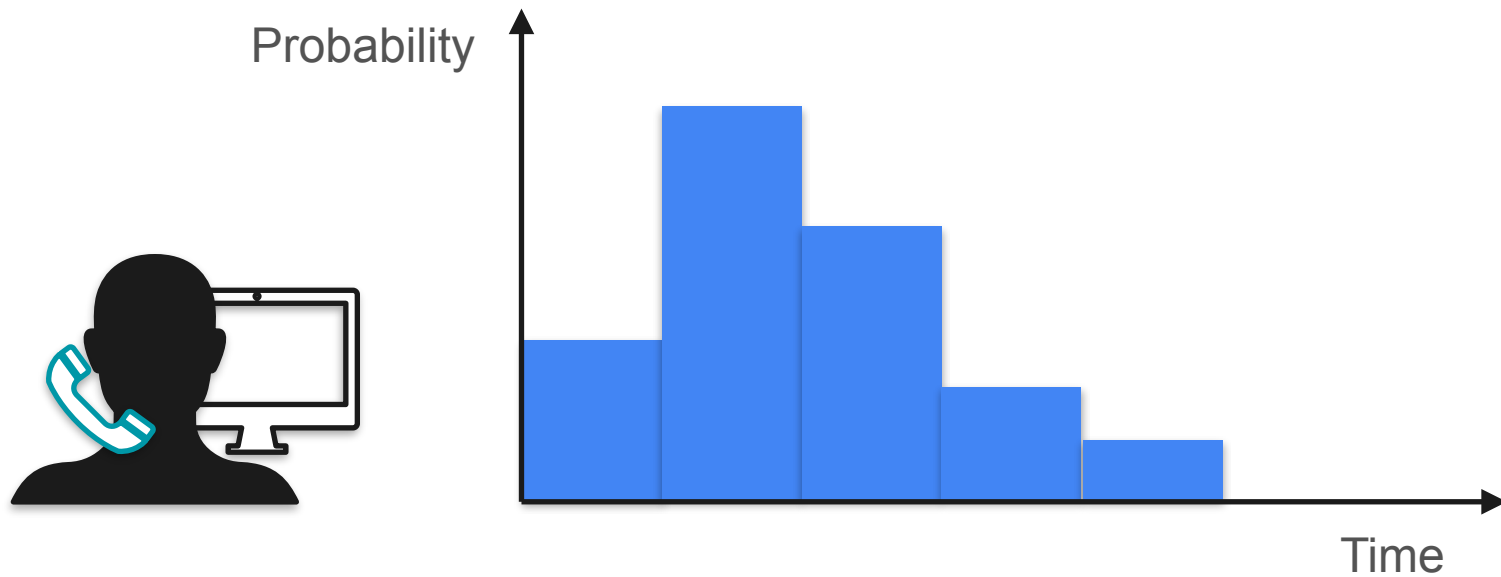
Median



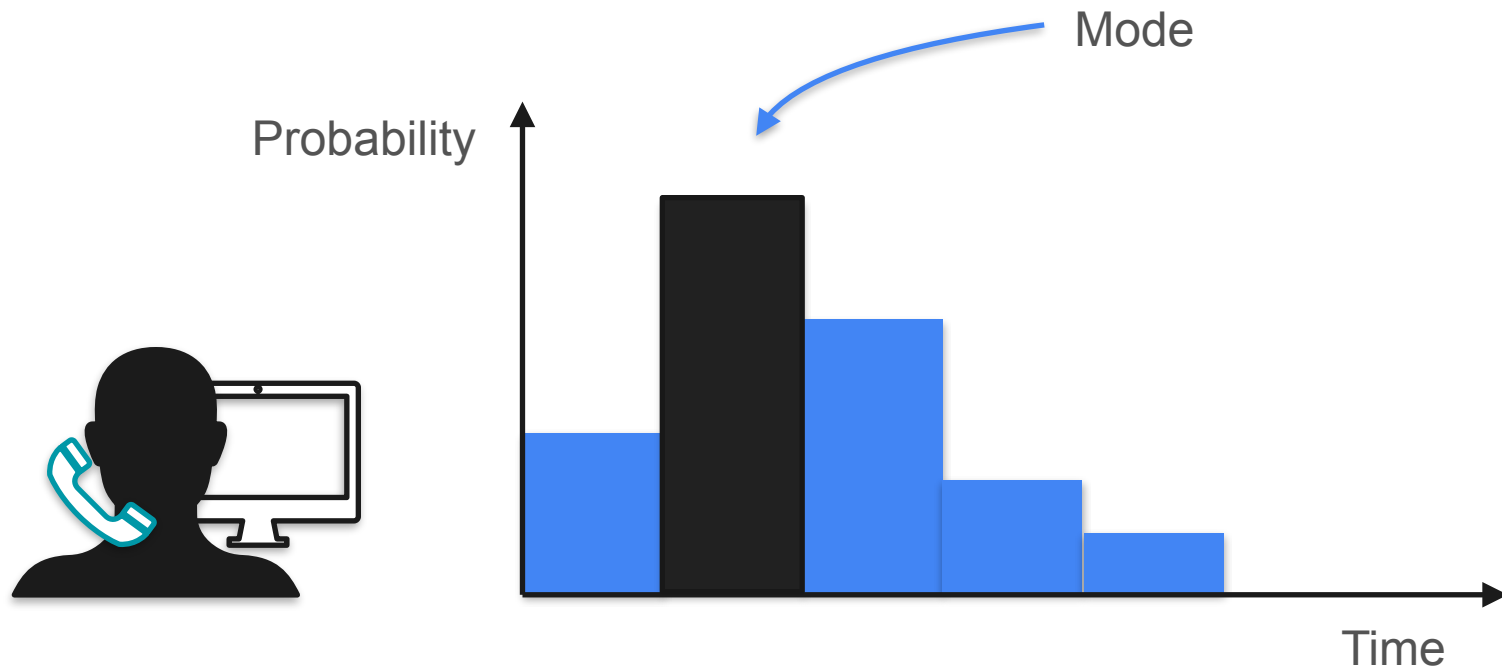
Median



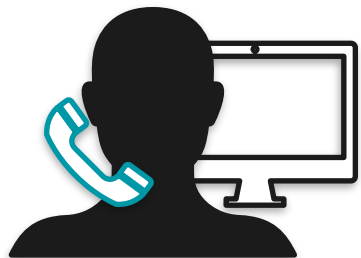
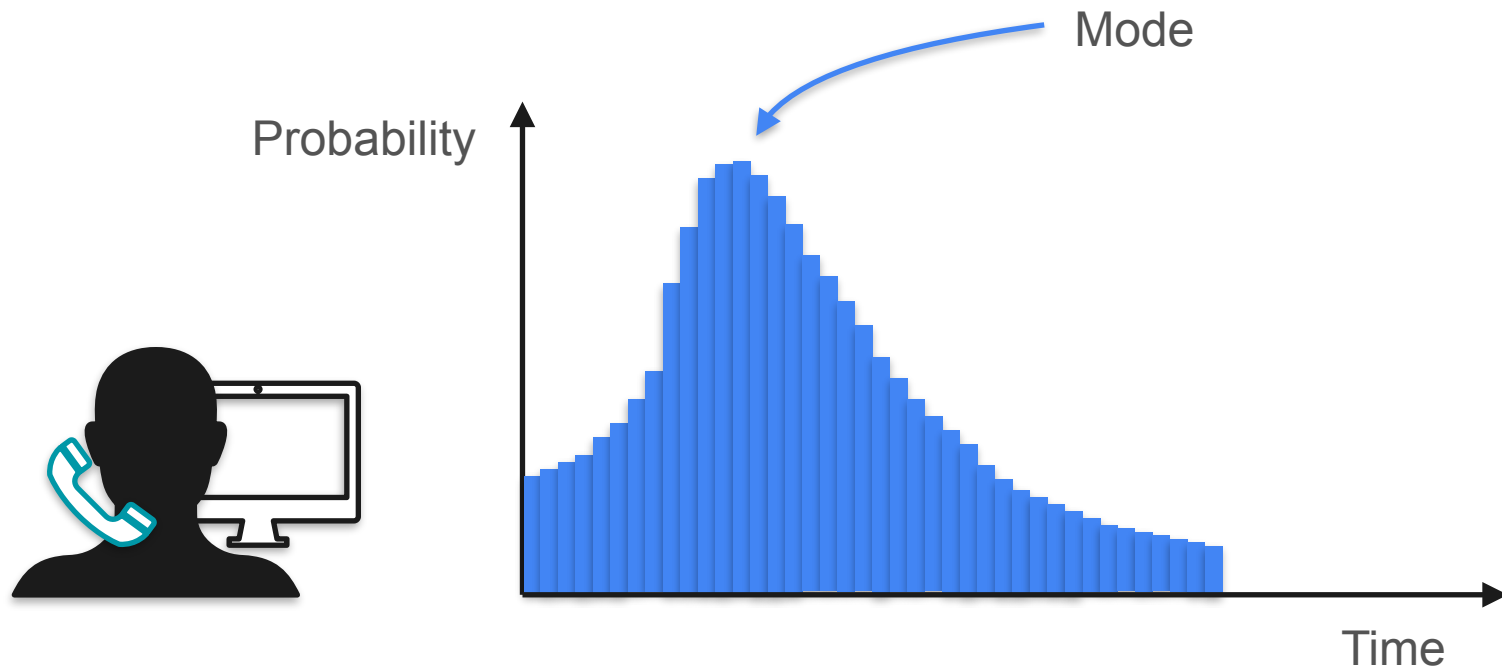
Mode



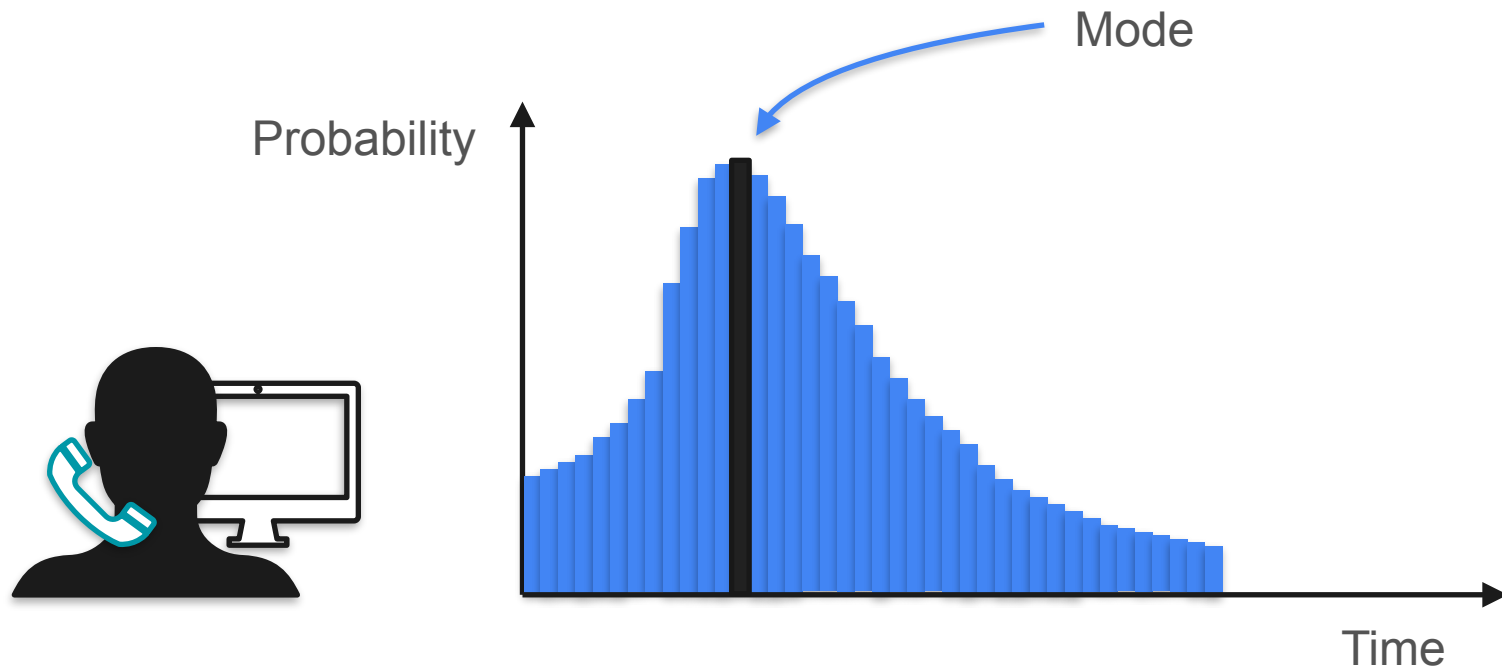
Mode



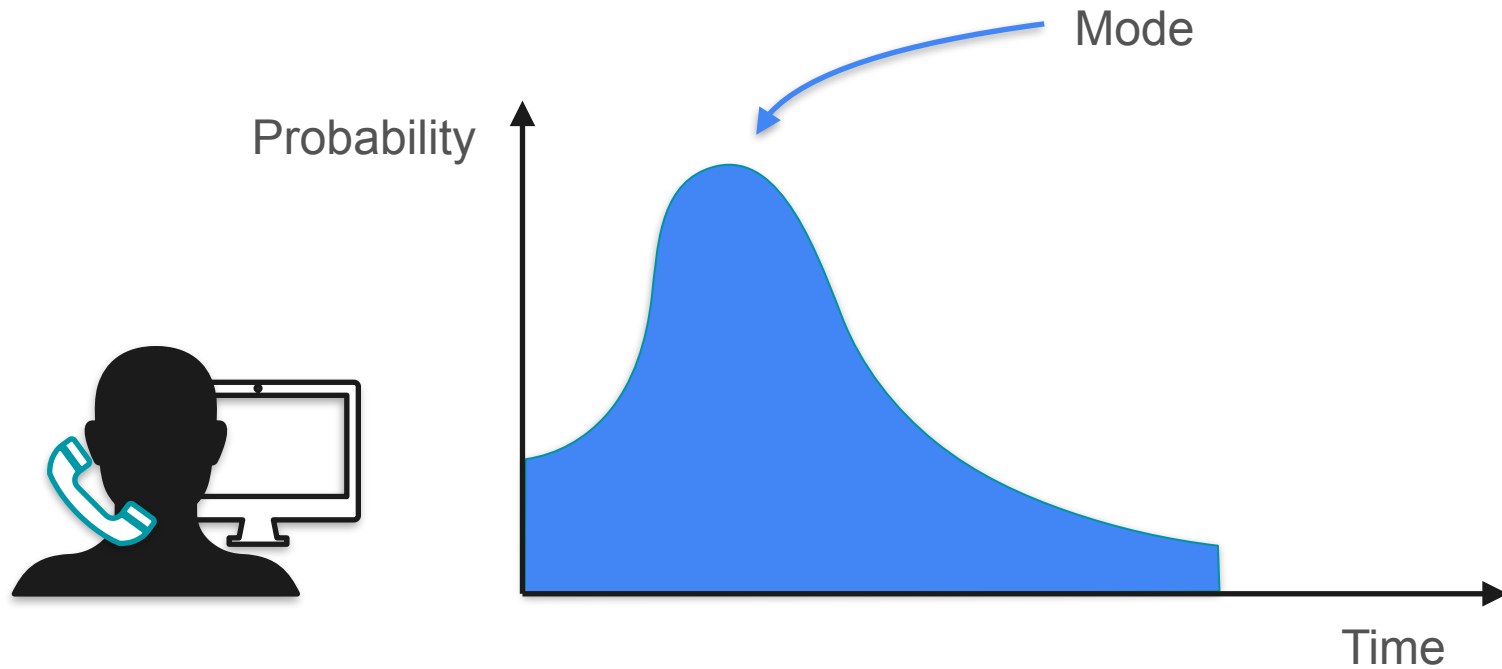
Mode



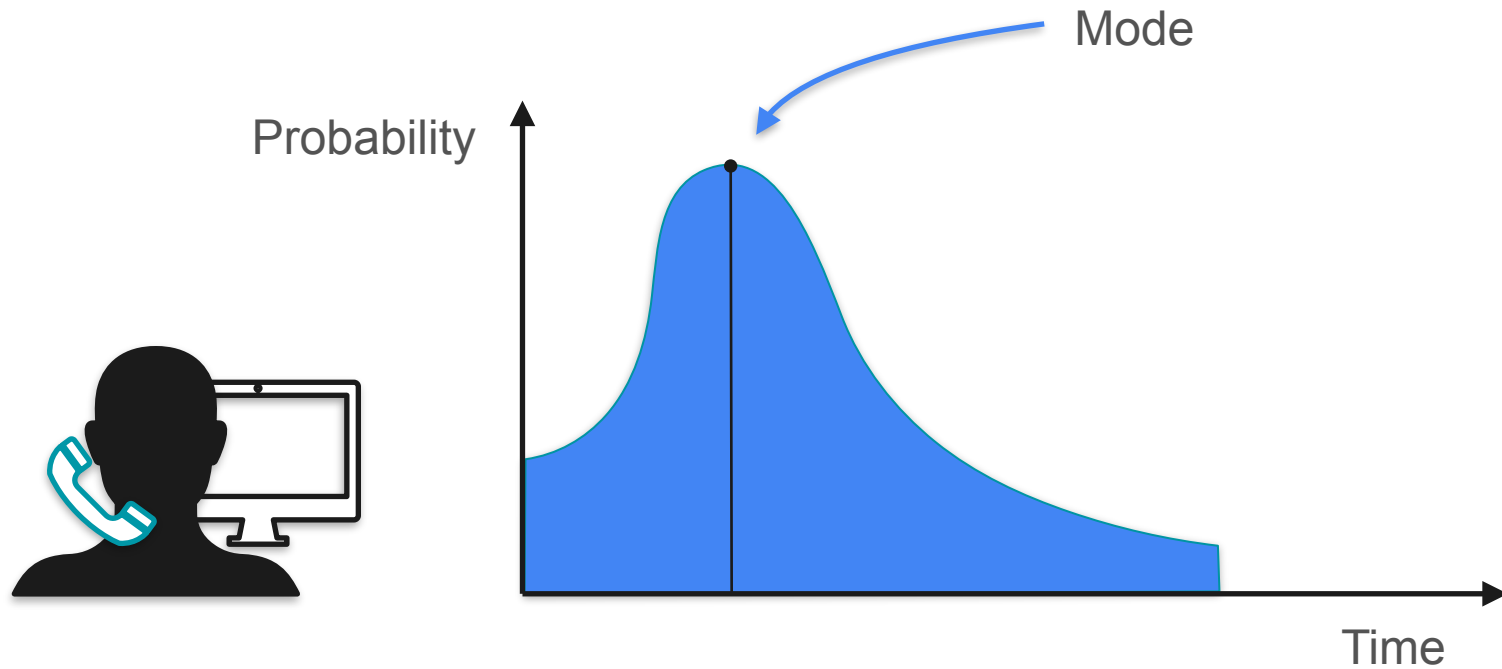
Mode



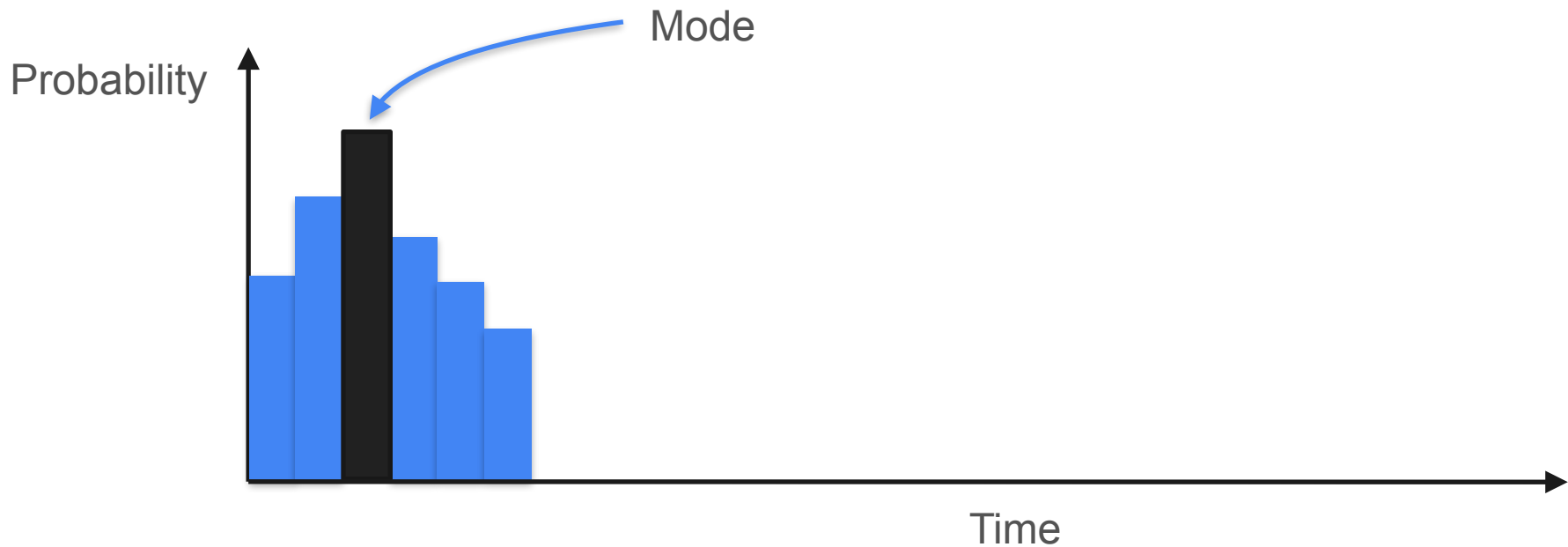
Mode



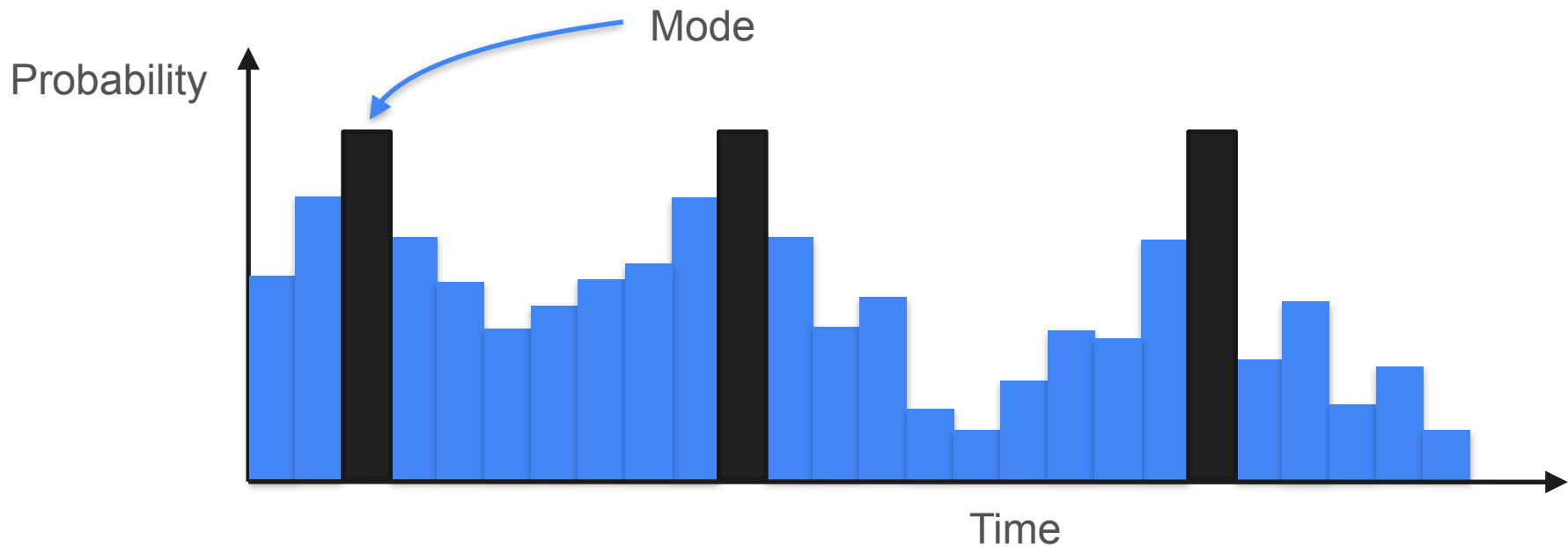
Mode



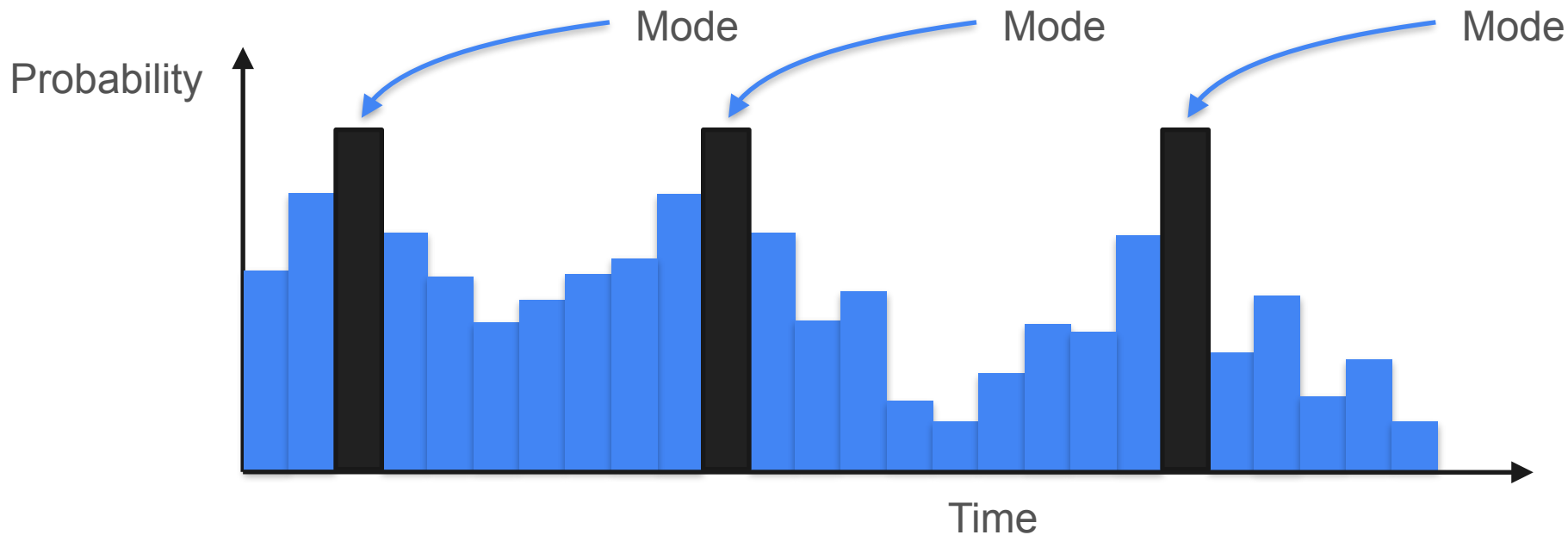
Mode: Multimodal Distribution



Mode: Multimodal Distribution

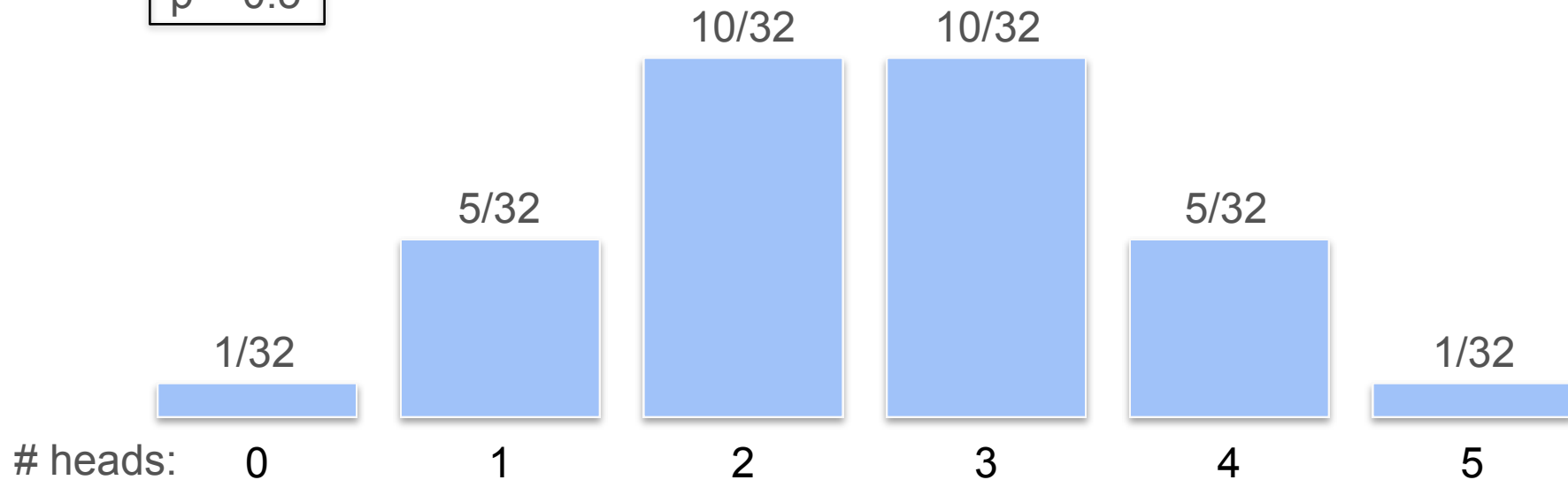


Mode: Multimodal Distribution



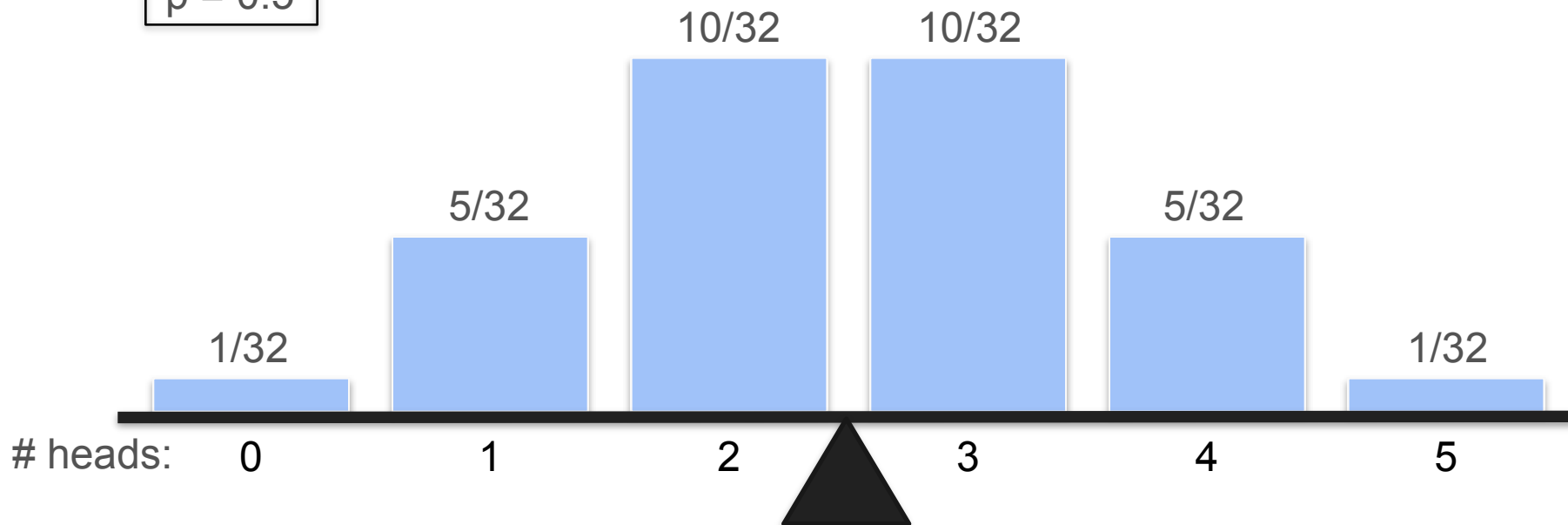
Mean, Median and Mode in Binomial Distribution

$n = 5$
 $p = 0.5$



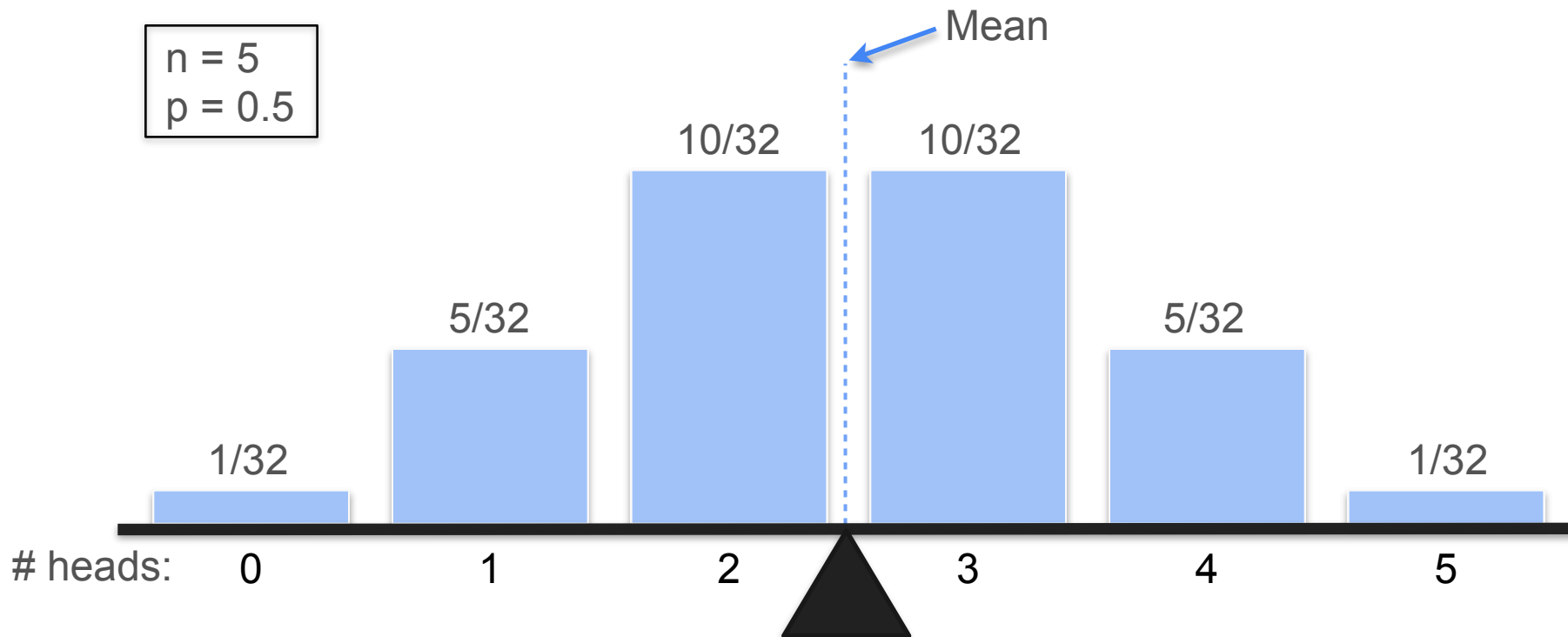
Mean, Median and Mode in Binomial Distribution

$n = 5$
 $p = 0.5$



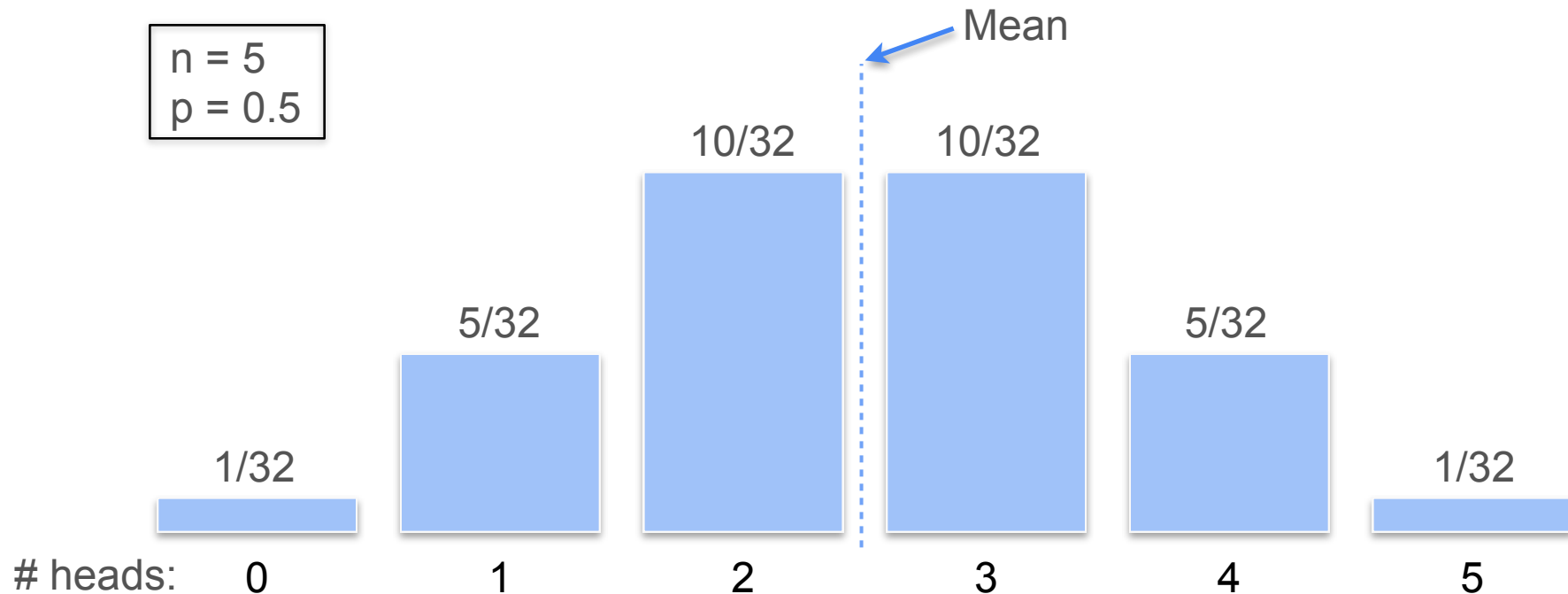
Mean, Median and Mode in Binomial Distribution

$$n = 5$$
$$p = 0.5$$



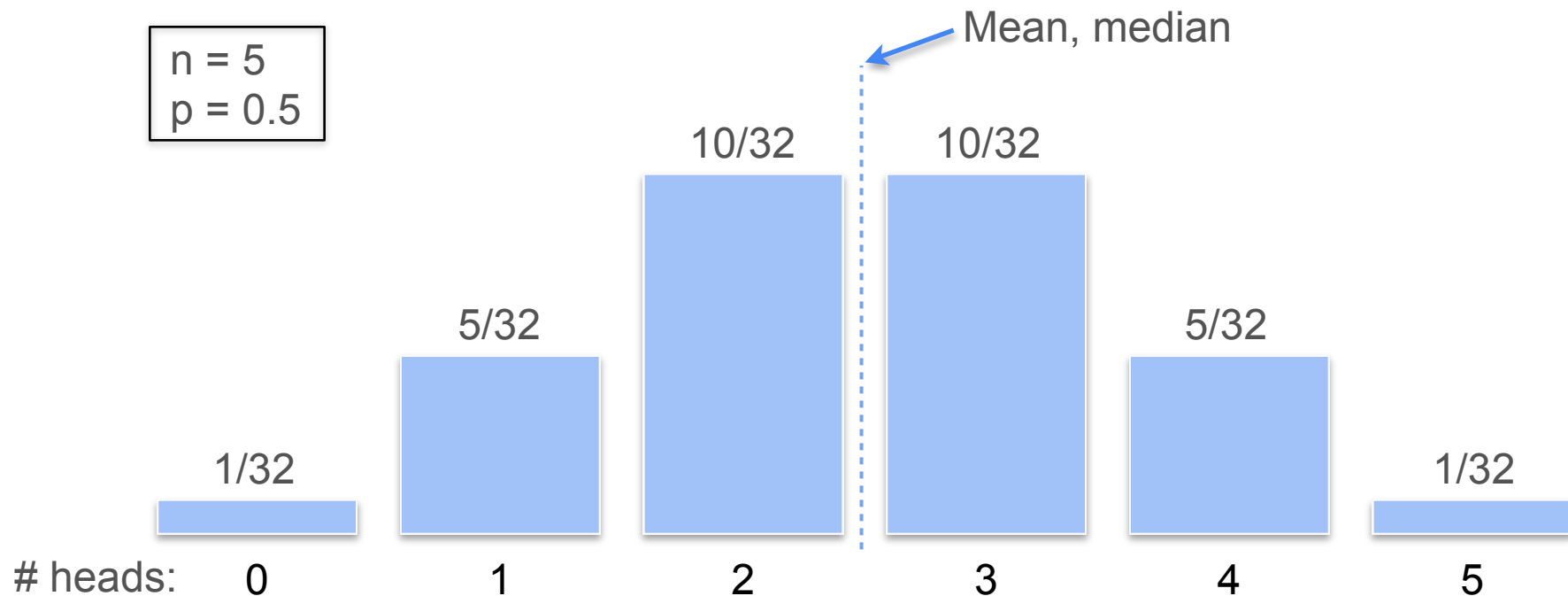
Mean, Median and Mode in Binomial Distribution

$$n = 5$$
$$p = 0.5$$



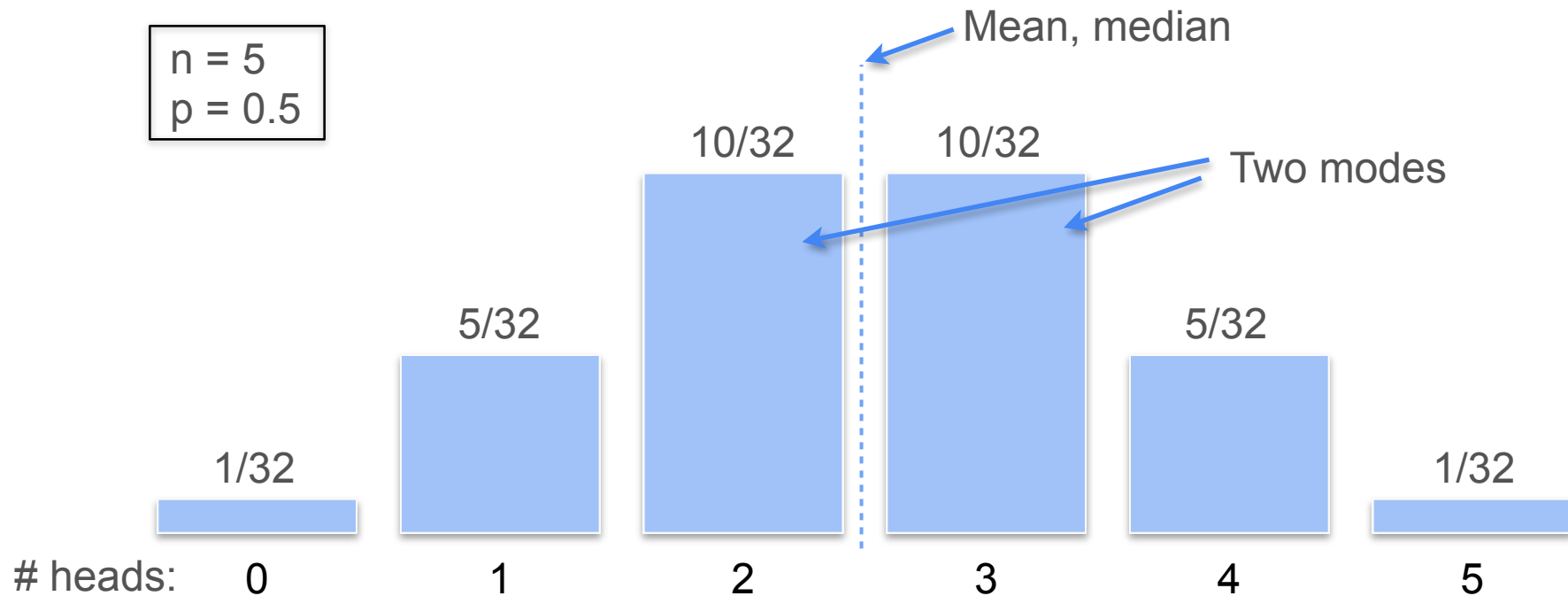
Mean, Median and Mode in Binomial Distribution

$$n = 5$$
$$p = 0.5$$

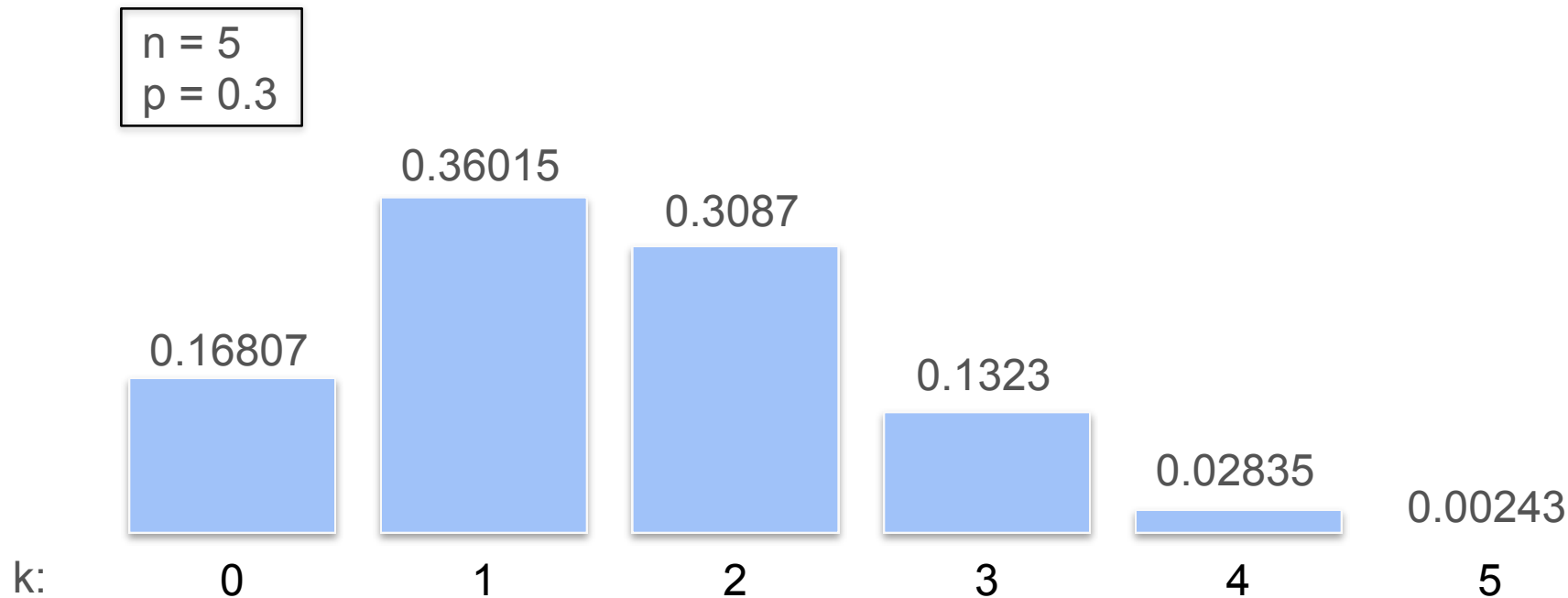


Mean, Median and Mode in Binomial Distribution

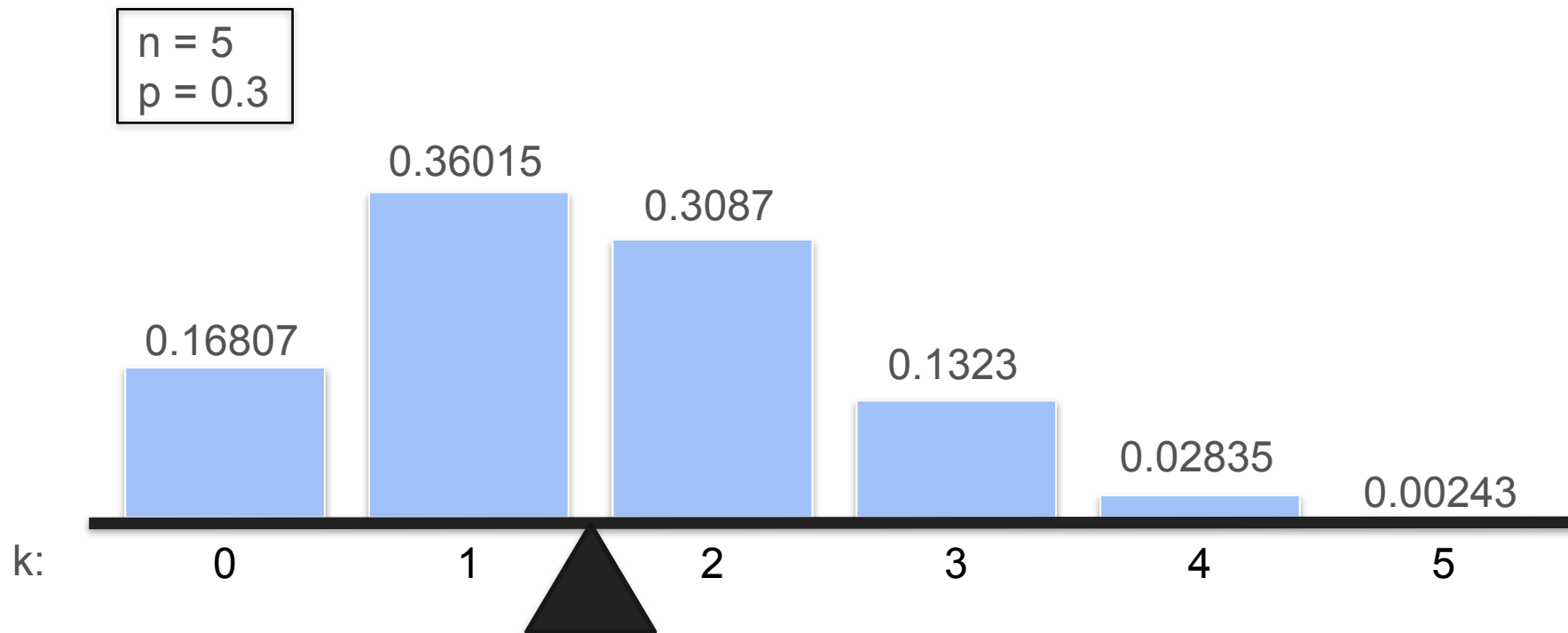
$$n = 5$$
$$p = 0.5$$



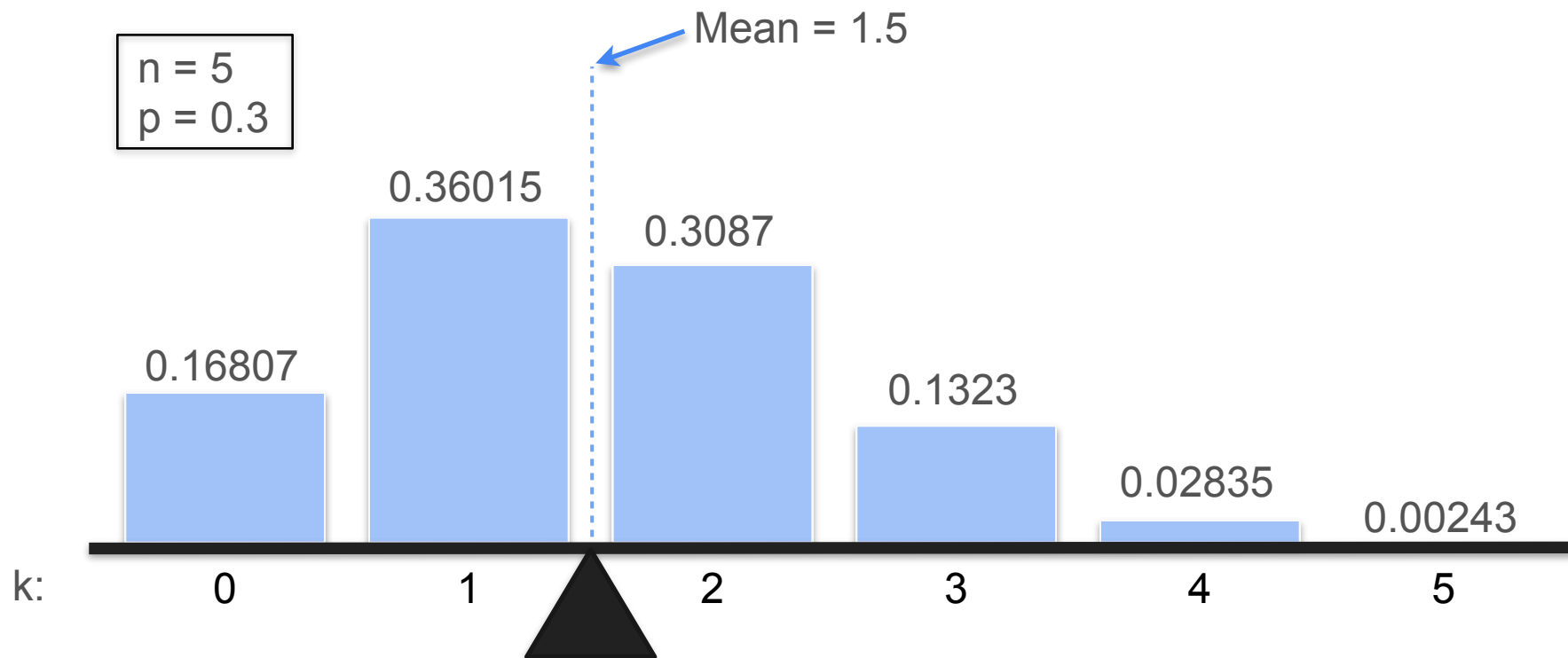
Mean, Median and Mode in Binomial Distribution



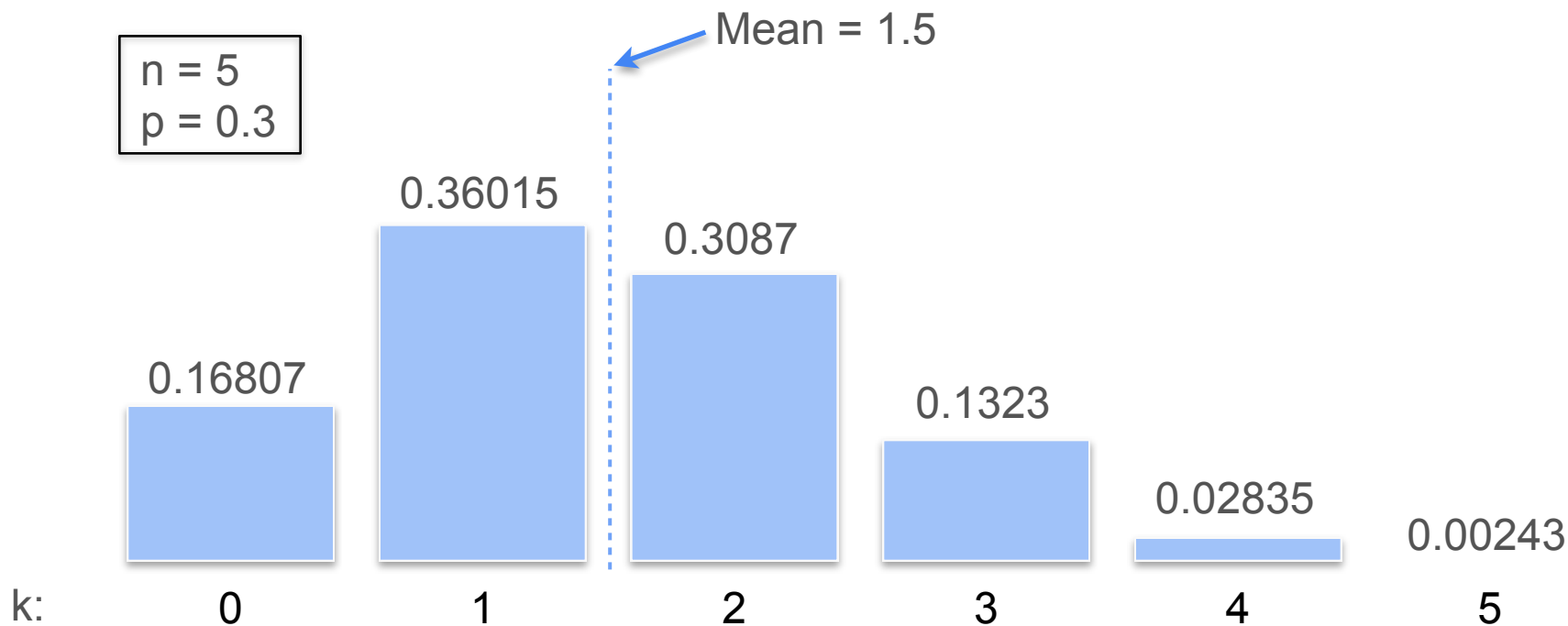
Mean, Median and Mode in Binomial Distribution



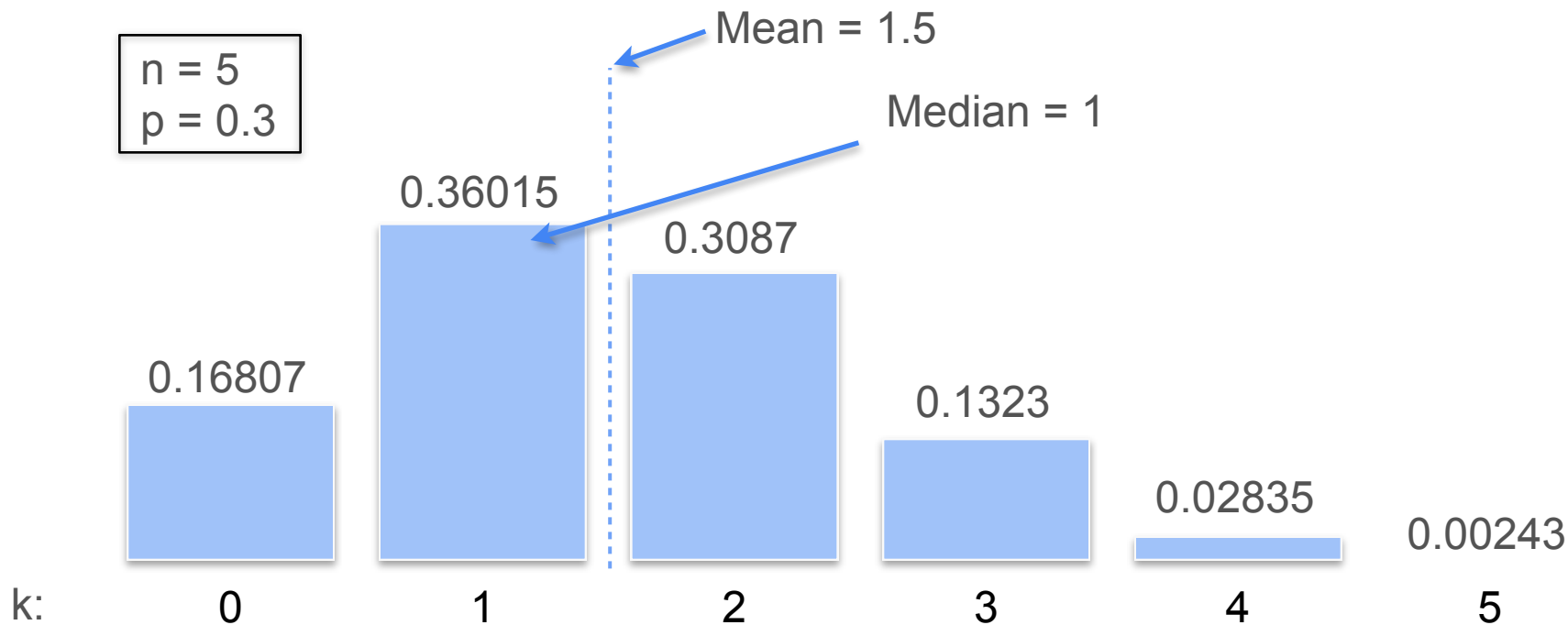
Mean, Median and Mode in Binomial Distribution



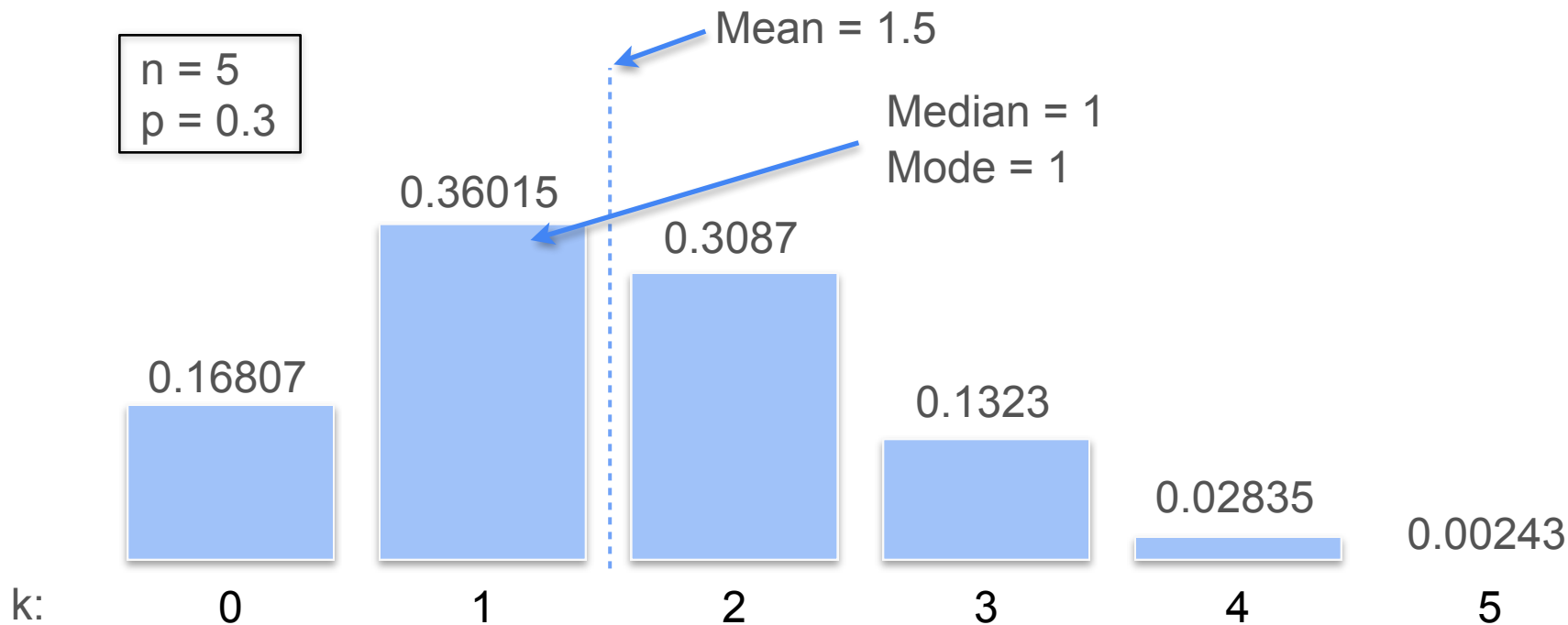
Mean, Median and Mode in Binomial Distribution



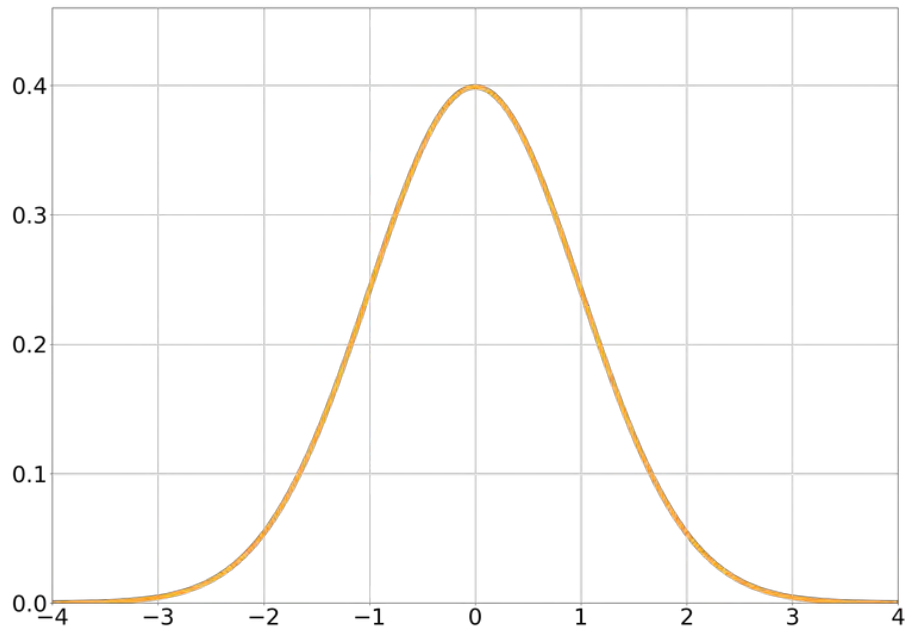
Mean, Median and Mode in Binomial Distribution



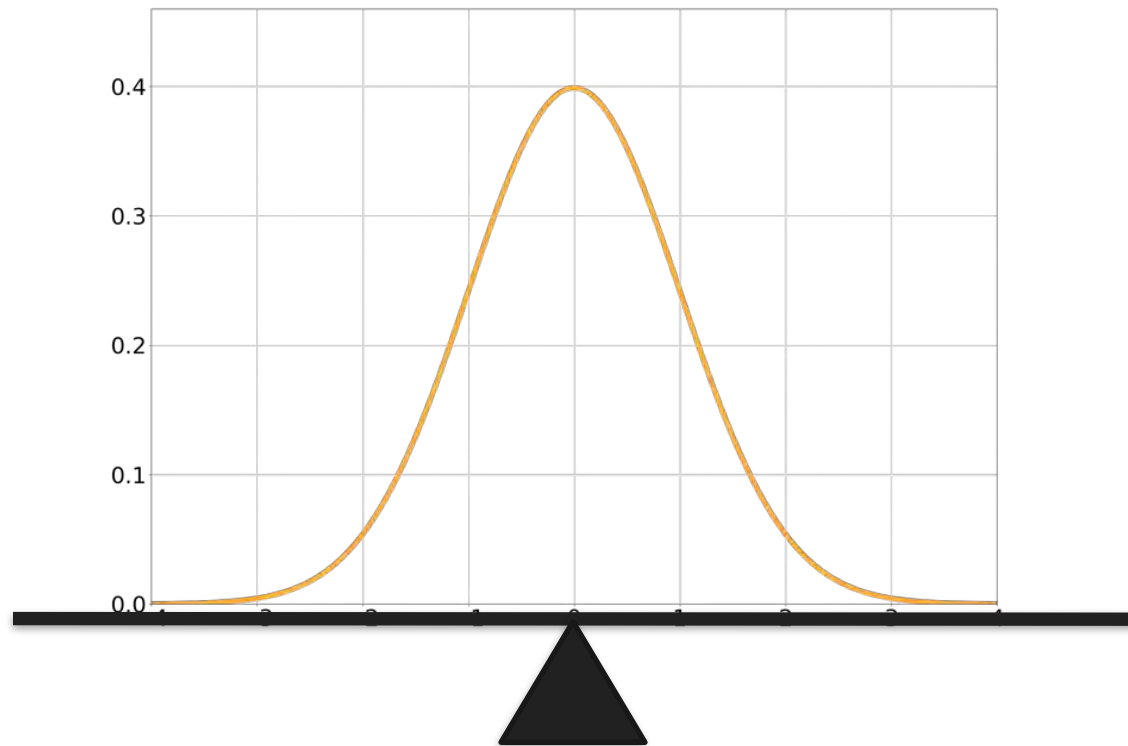
Mean, Median and Mode in Binomial Distribution



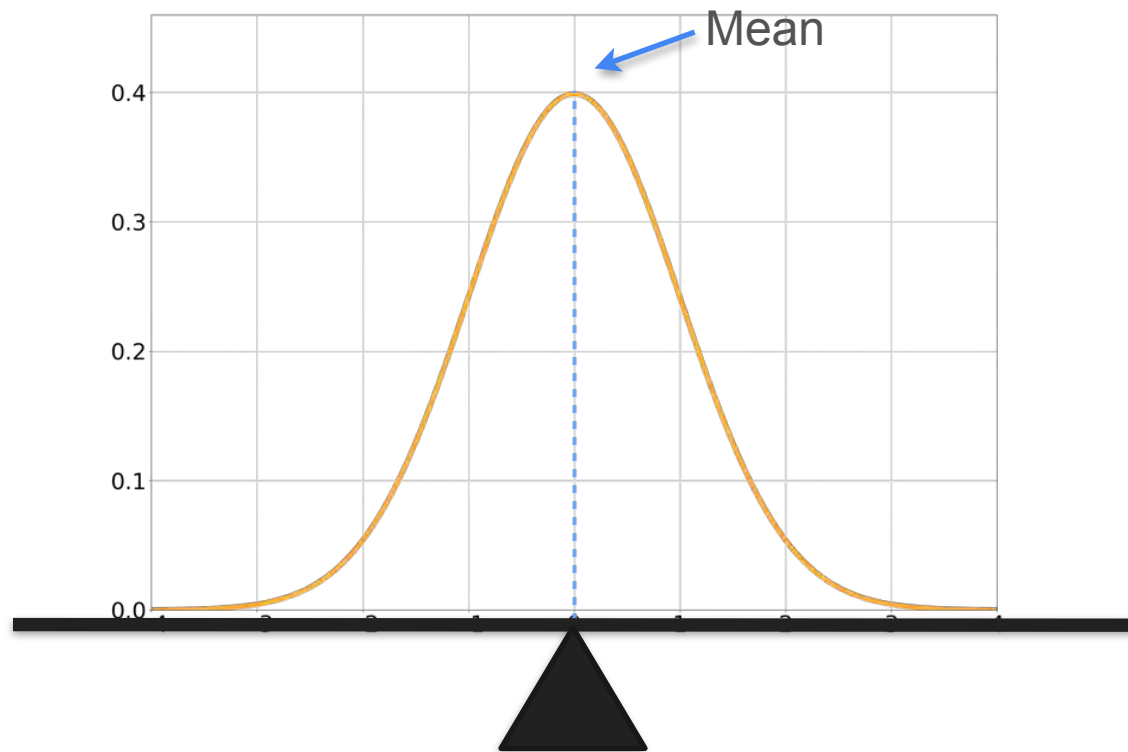
Mean, Median and Mode in Normal Distribution



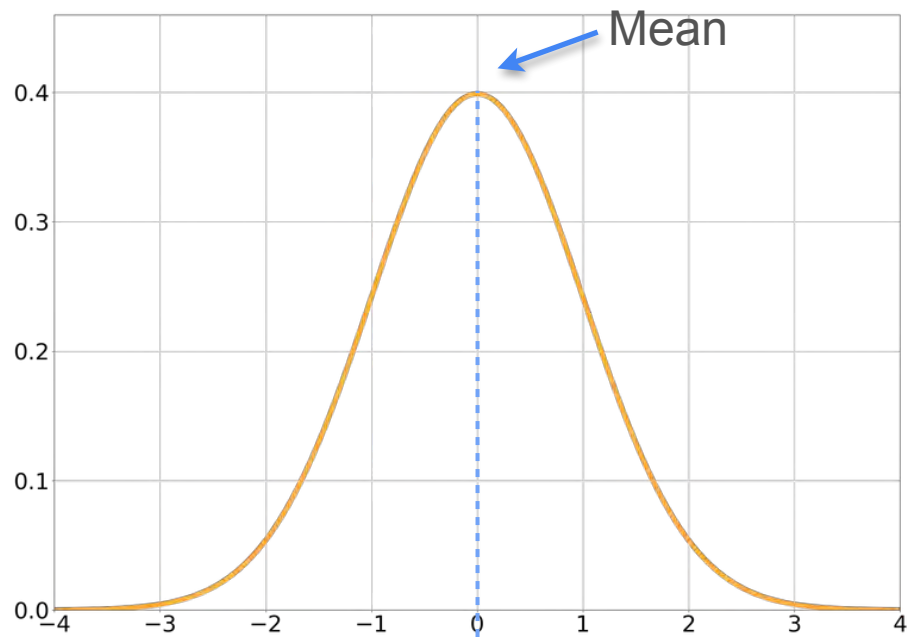
Mean, Median and Mode in Normal Distribution



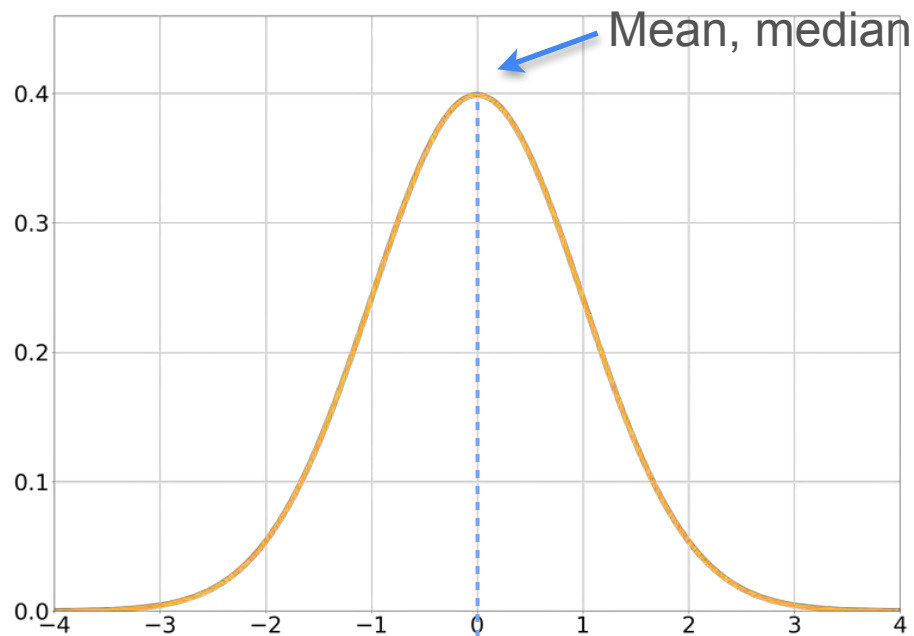
Mean, Median and Mode in Normal Distribution



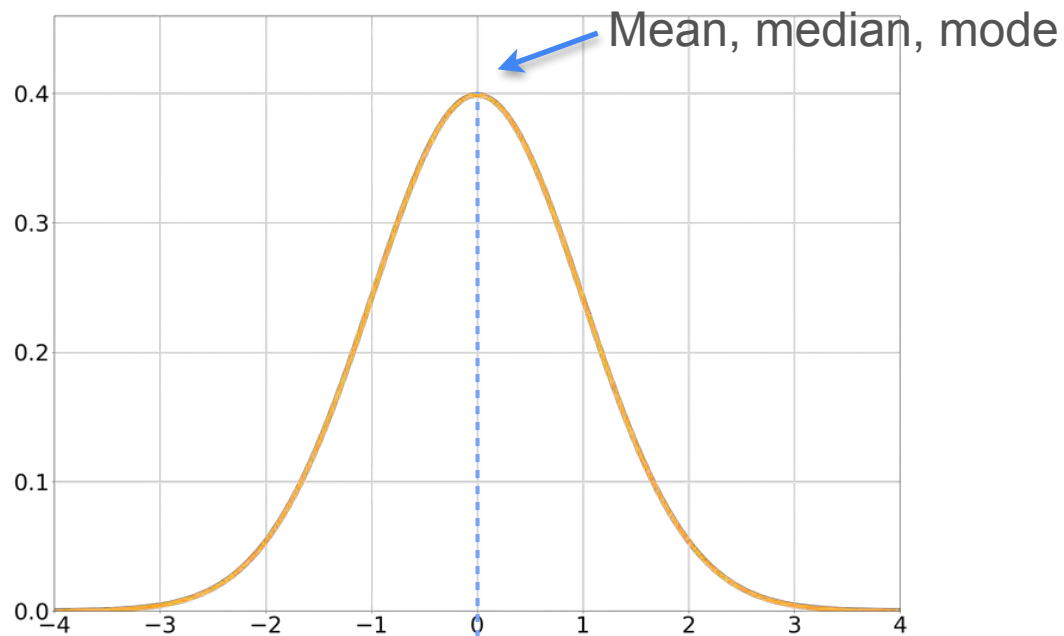
Mean, Median and Mode in Normal Distribution



Mean, Median and Mode in Normal Distribution



Mean, Median and Mode in Normal Distribution





DeepLearning.AI

Describing Distributions

Expected Value

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Game cost:

\$6

Do you play the game?

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Game cost:

\$4

Do you play the game?

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Game cost:

\$4

Do you play the game?

What is the maximum amount of money you would pay to play this game?

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Long term:

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Long term: $0.5 \cdot \$10$

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Long term: $0.5 \cdot \$10 + 0.5 \cdot \0

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Long term: $0.5 \cdot \$10 + 0.5 \cdot \$0 = \$5$

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Long term: $0.5 \cdot \$10 + 0.5 \cdot \$0 = \$5$  You expect to win \$5 on average
 $\mathbb{E}[X] = 5$

Expected Value: Motivation Example 1

You play a game with a friend



You win 10 dollars



You win nothing

Game cost:

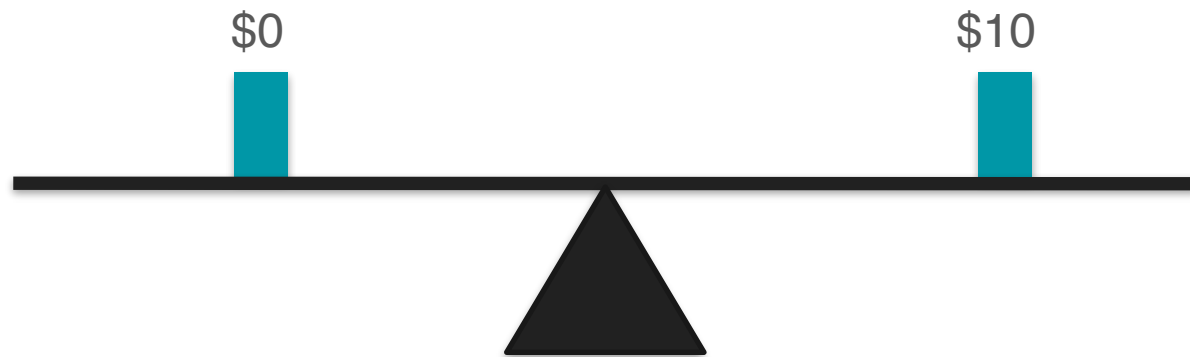
\$5

Long term: $0.5 \cdot \$10 + 0.5 \cdot \$0 = \$5$ $\Rightarrow$ You expect to win \$5 on average
 $\mathbb{E}[X] = 5$

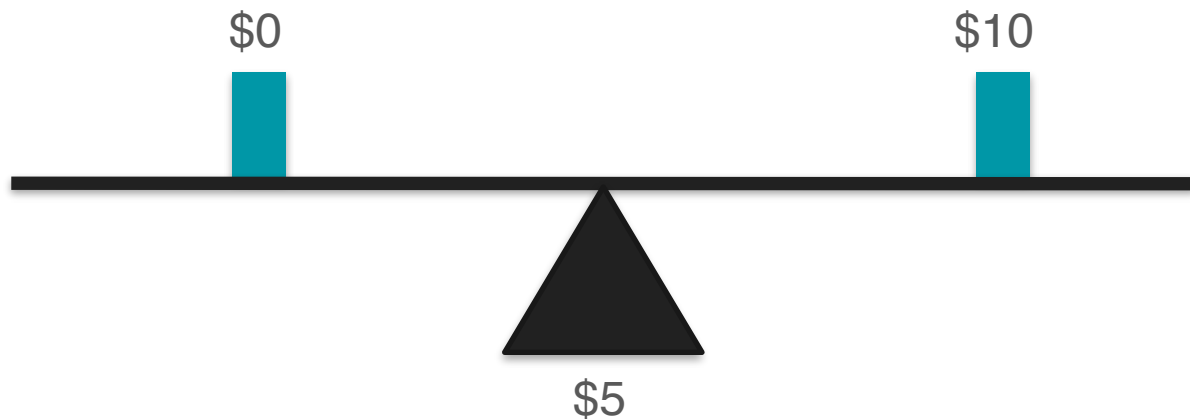
Expected Value: Motivation Example 1



Expected Value: Motivation Example 1



Expected Value: Motivation Example 1



Expected Value: Motivation Example 2

Play another game



Flip three coins. For each heads you win \$1

What is the maximum amount of money you would pay to play this game?

Expected Value: Motivation Example 2

Number of heads:

0

1

2

3



Expected Value: Motivation Example 2

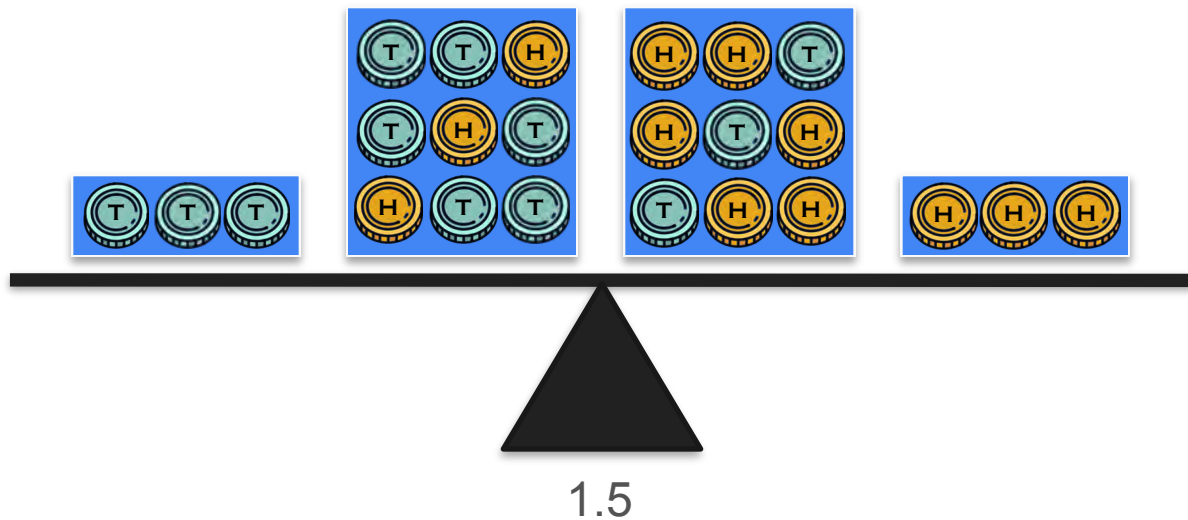
Number of heads:

0

1

2

3



Expected Value: Motivation Example 2

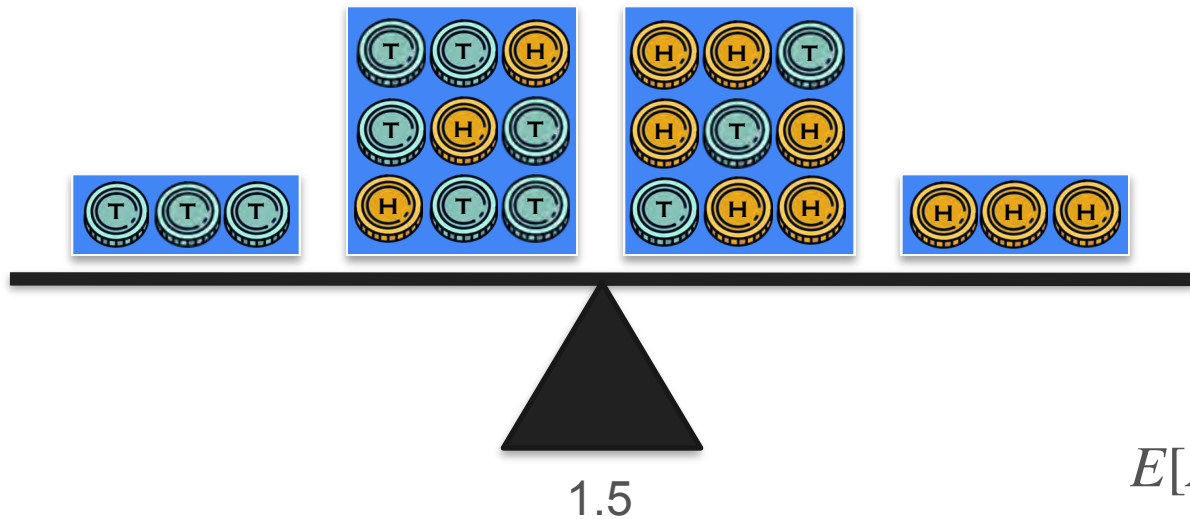
Number of heads:

0

1

2

3

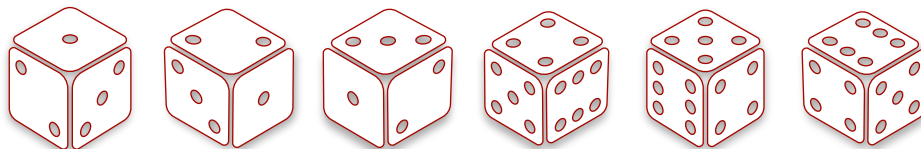


$$E[X] = 1.5$$

Expected Value: Motivation Example 3

Probability: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

Roll: 1 2 3 4 5 6

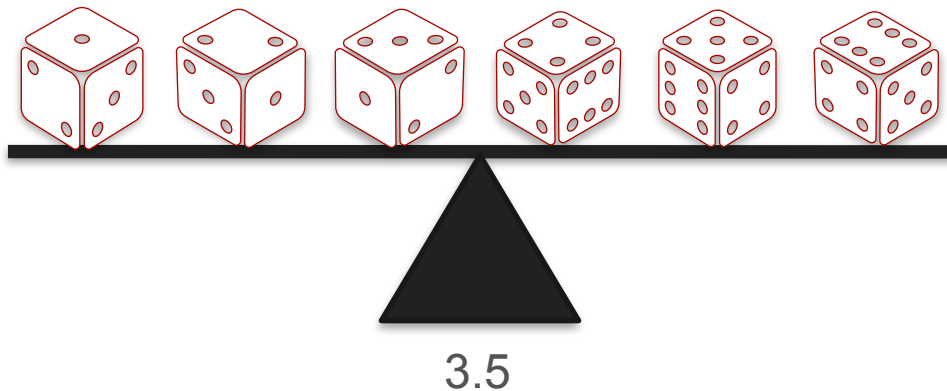


$$\frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

Expected Value: Motivation Example 3

Probability: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

Roll: 1 2 3 4 5 6



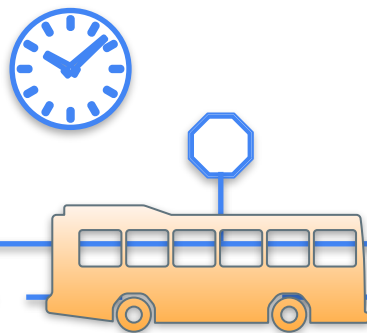
Expected Value



Expected Value



Expected Value



Expected Value

Waiting Time

15 min



Expected Value

Waiting Time

15 min



Expected Value

Waiting Time

15 min



Expected Value

Waiting Time

15 min



Expected Value

Waiting Time
15 min
32 min
58 min
1 min
47 min
14 min



Expected Value

Waiting Time

15 min

32 min

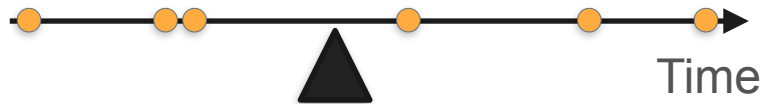
58 min

1 min

47 min

14 min

Average = 27.833



Expected Value

Waiting Time

15 min

32 min

58 min

1 min

47 min

14 min

37 min

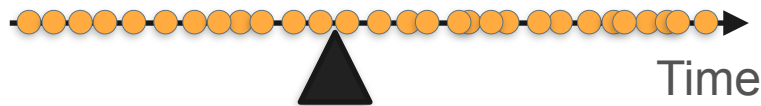
8 min

29 min

55 min

...

Average = 27.833



Expected Value

Waiting Time

15 min

32 min

58 min

1 min

47 min

14 min

37 min

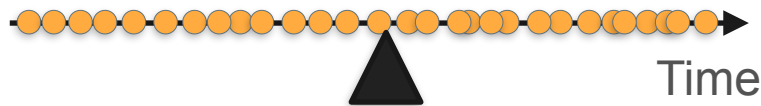
8 min

29 min

55 min

...

Average = 30



Expected Value

Waiting Time

15 min

32 min

58 min

1 min

47 min

14 min

37 min

8 min

29 min

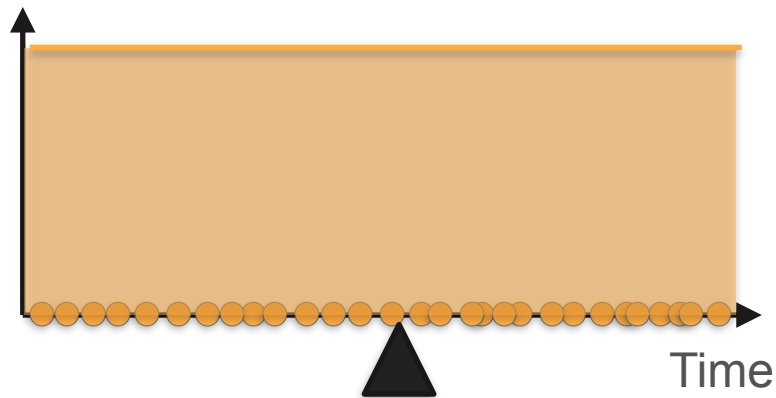
55 min

...

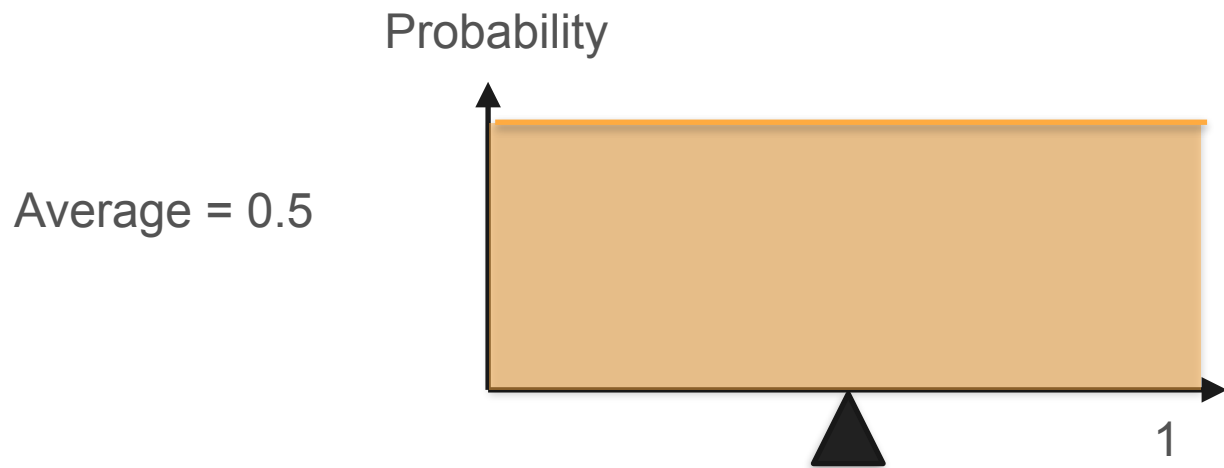
Average = 30



Probability

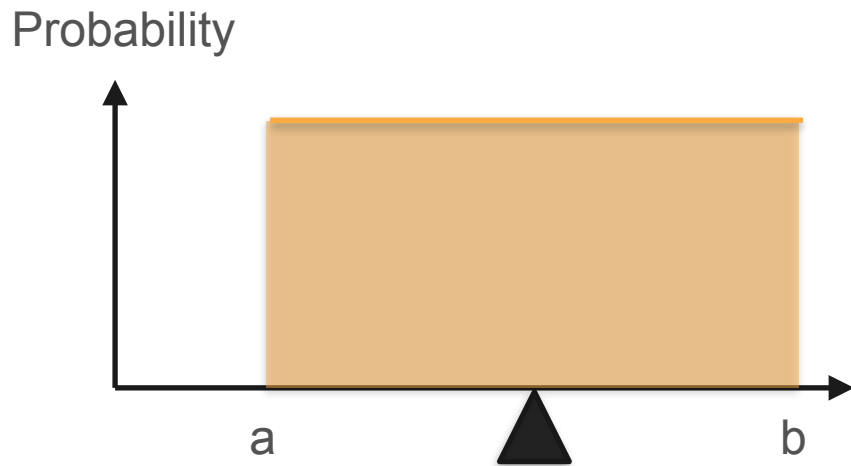


Expected Value: Uniform Distribution

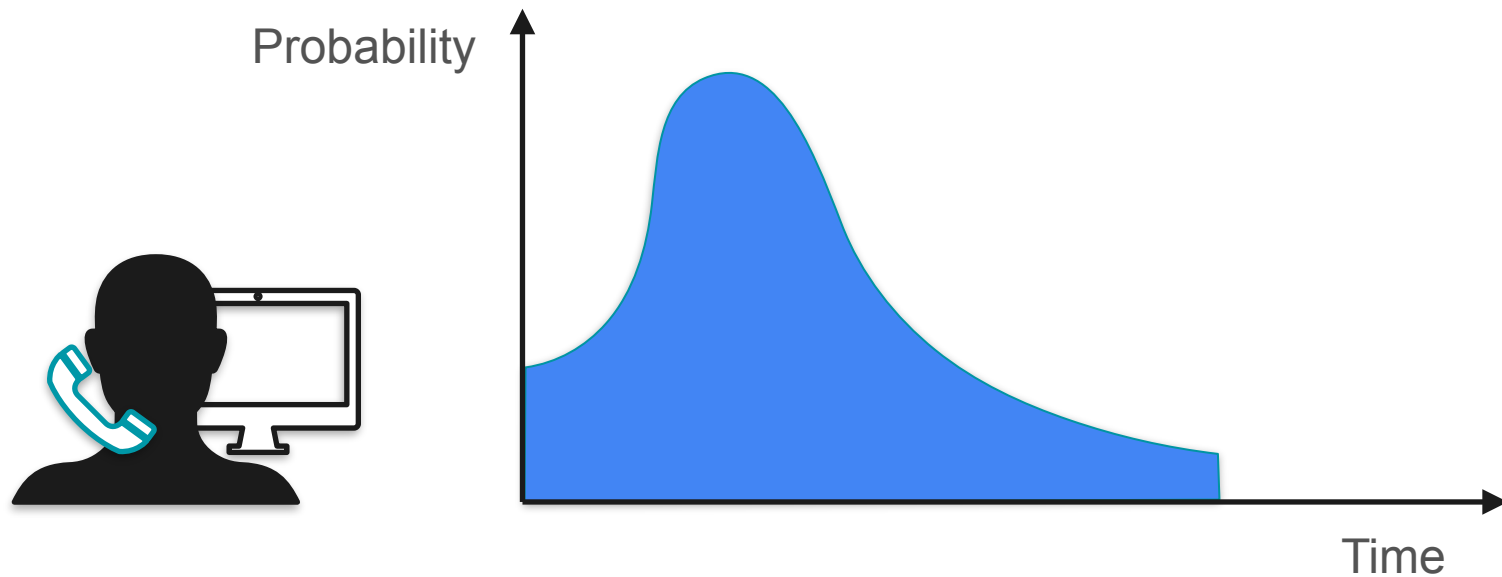


Expected Value: Uniform Distribution

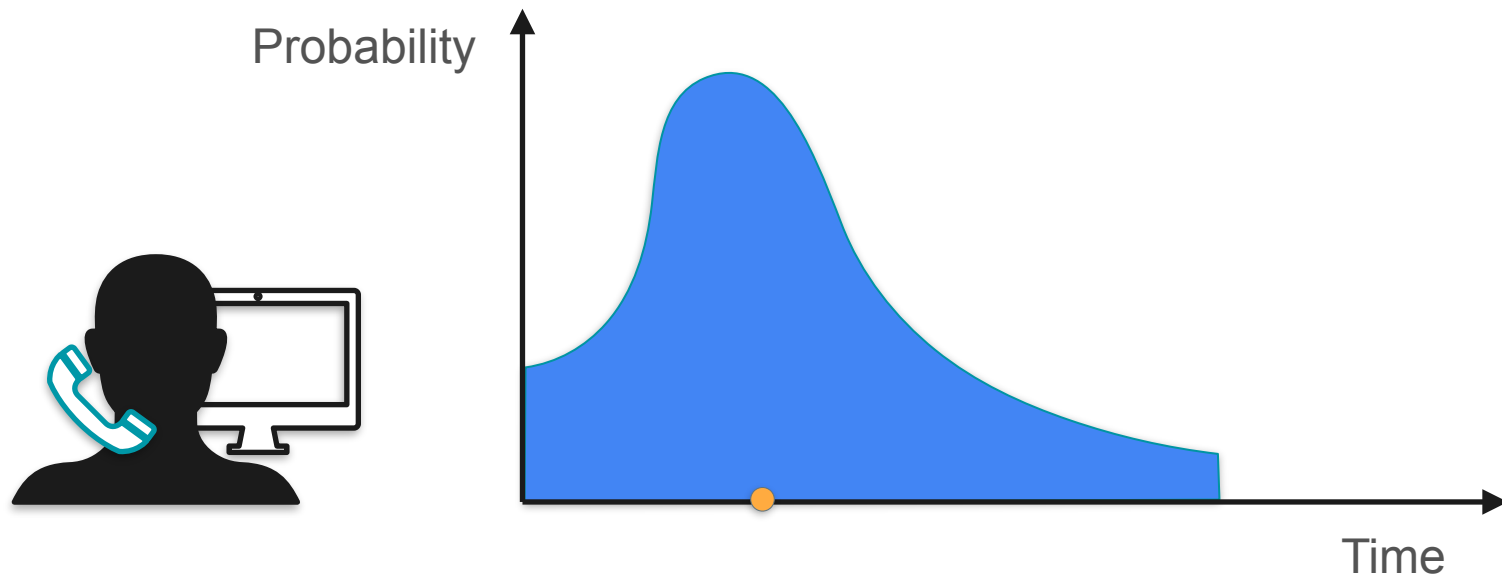
$$\text{Average} = \frac{a + b}{2}$$



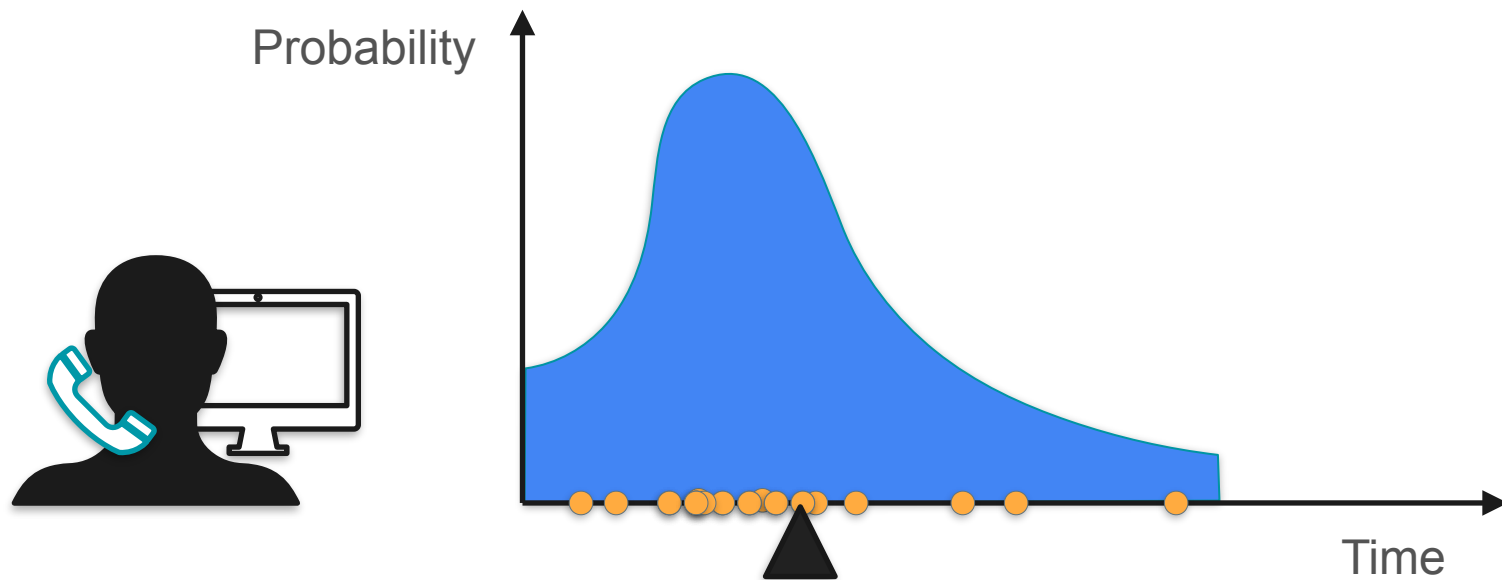
Expected Value



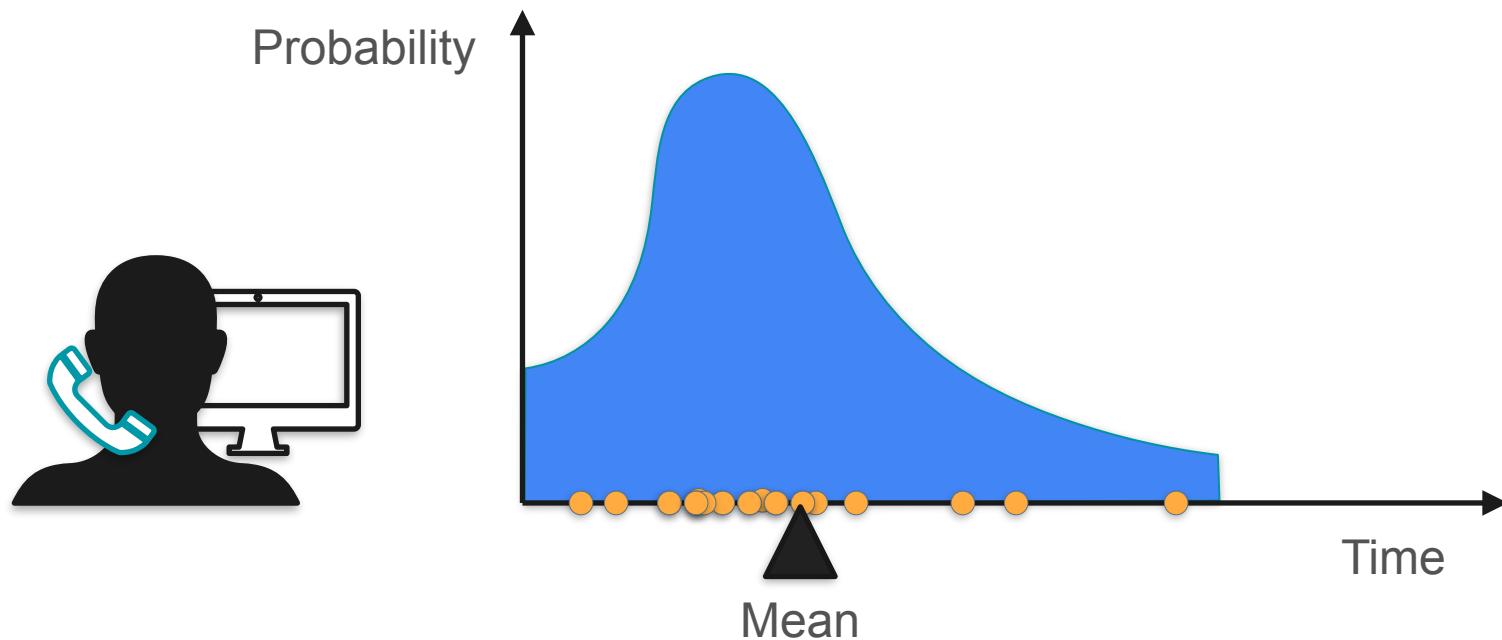
Expected Value



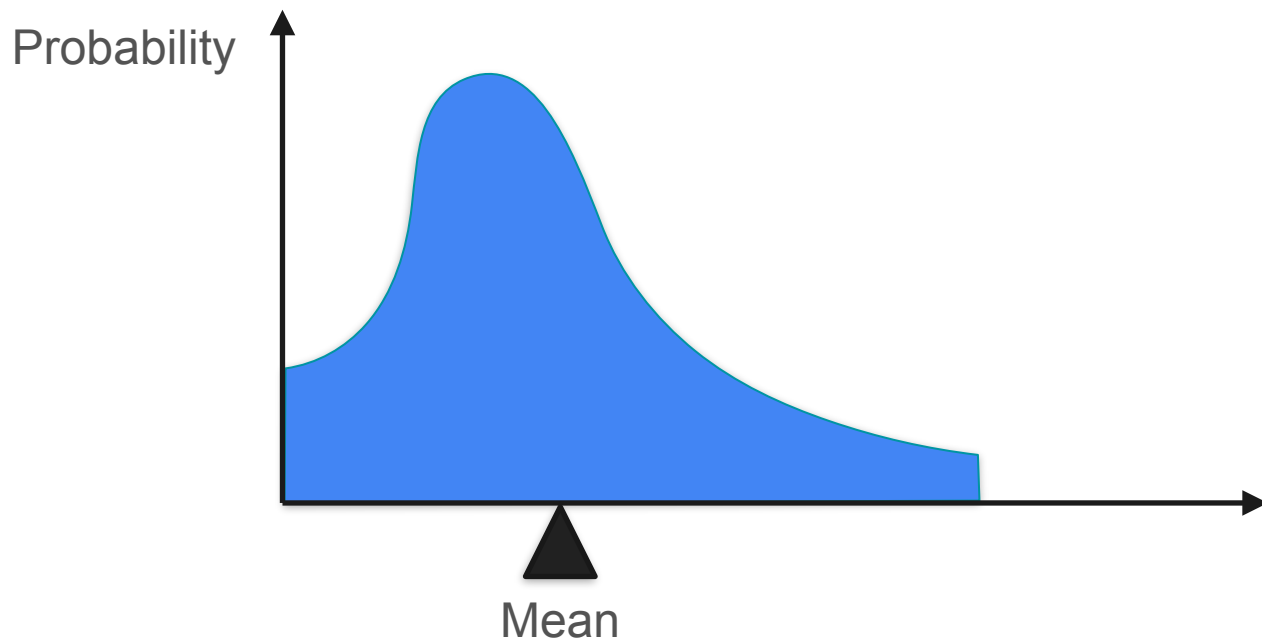
Expected Value



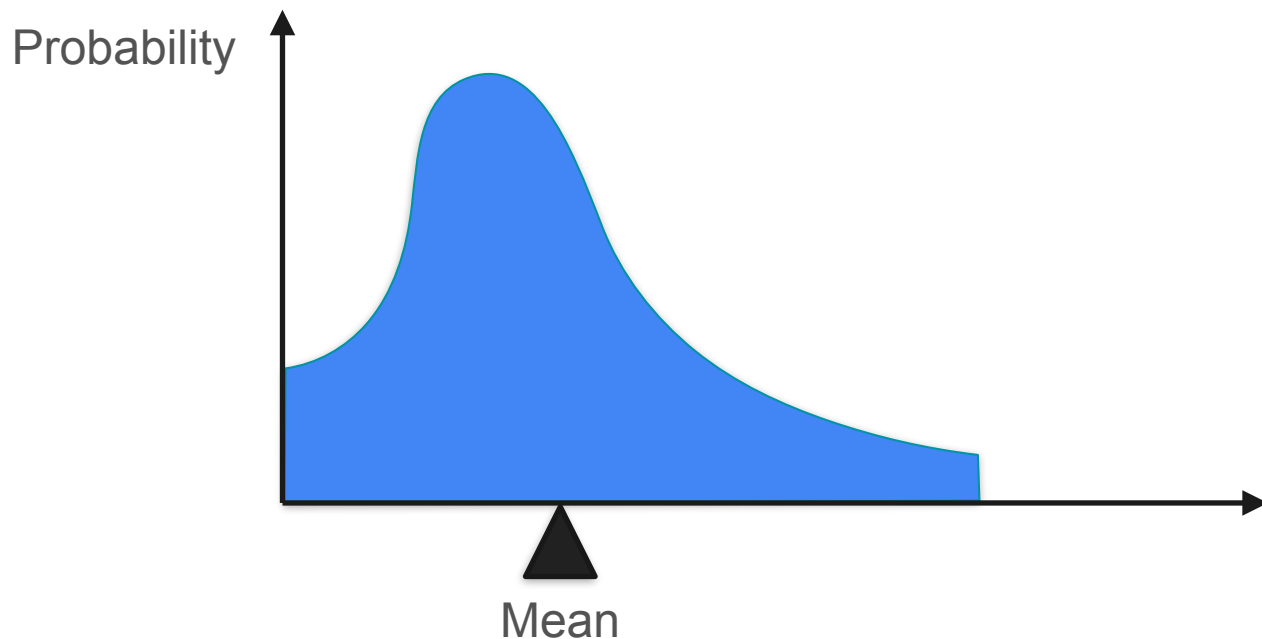
Expected Value



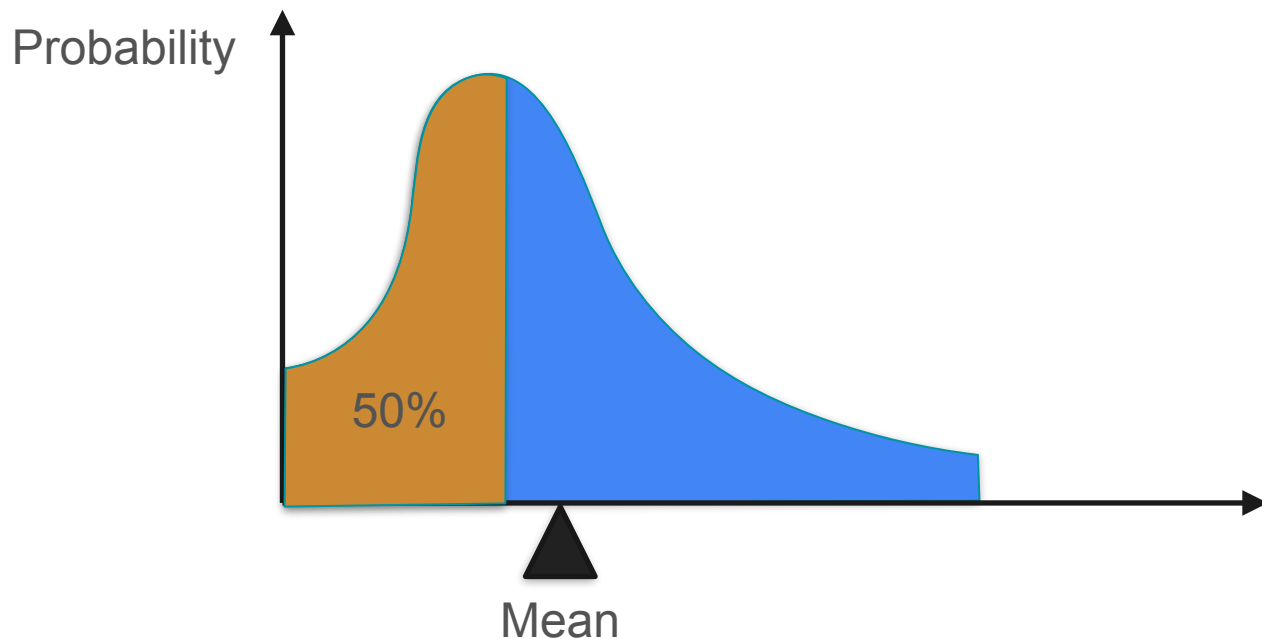
Expected Value: General Case



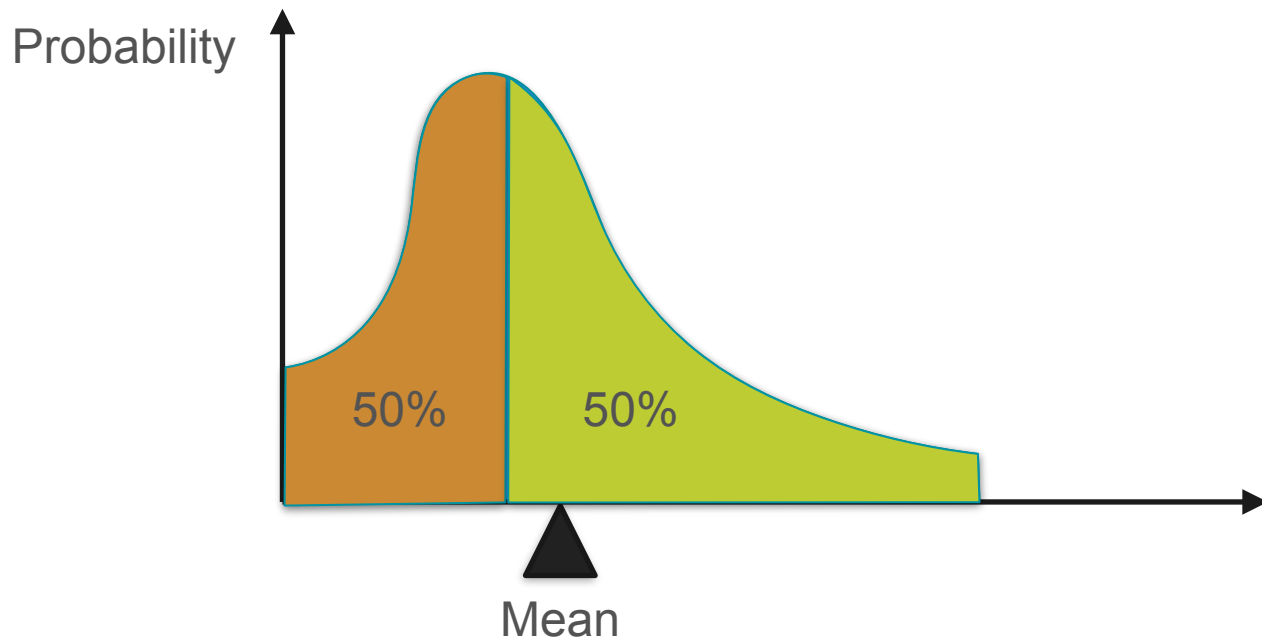
Expected Value: Common Misconception



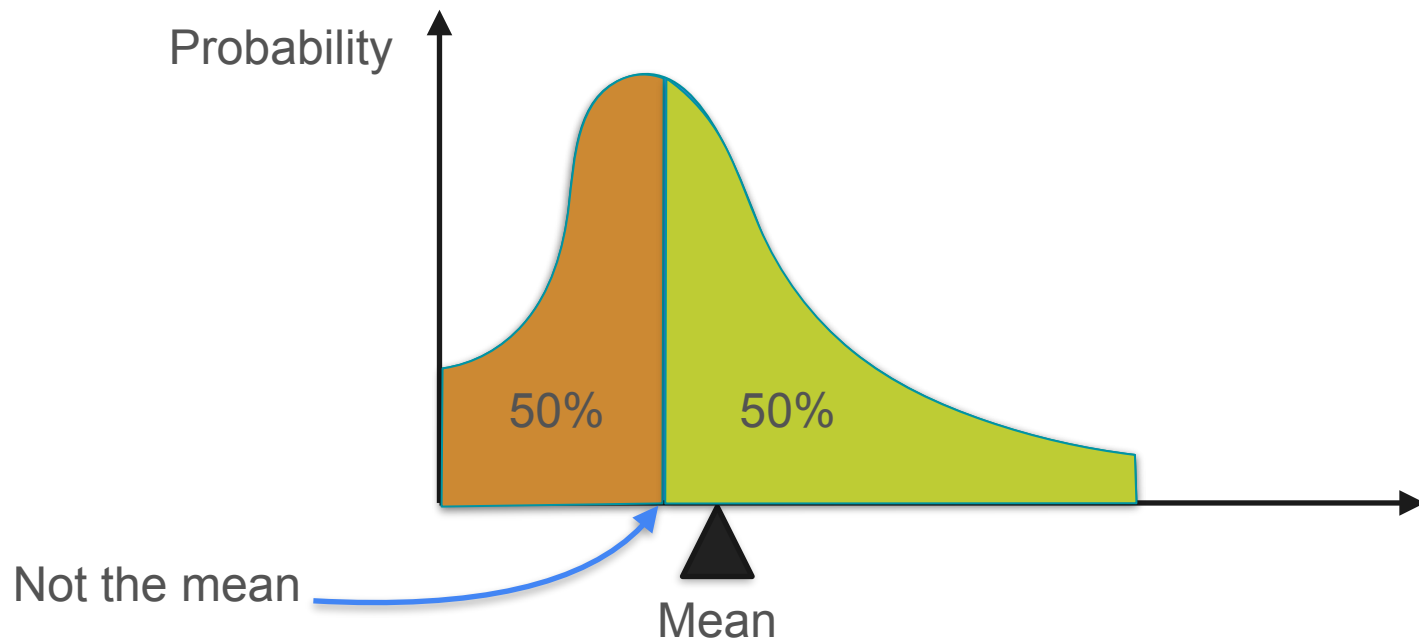
Expected Value: Common Misconception



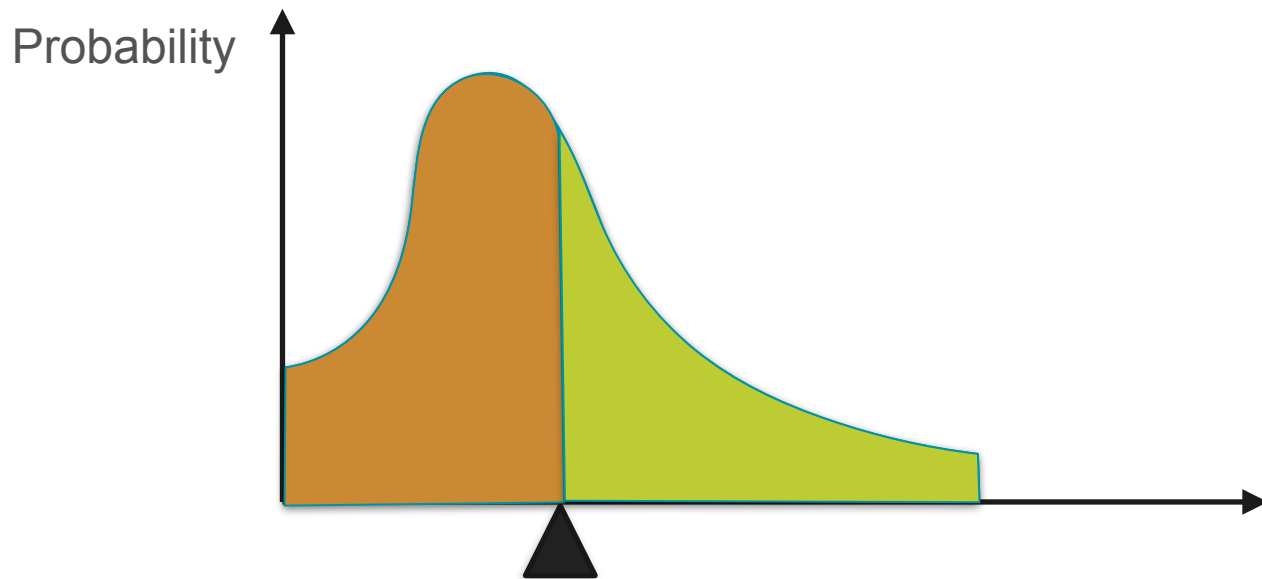
Expected Value: Common Misconception



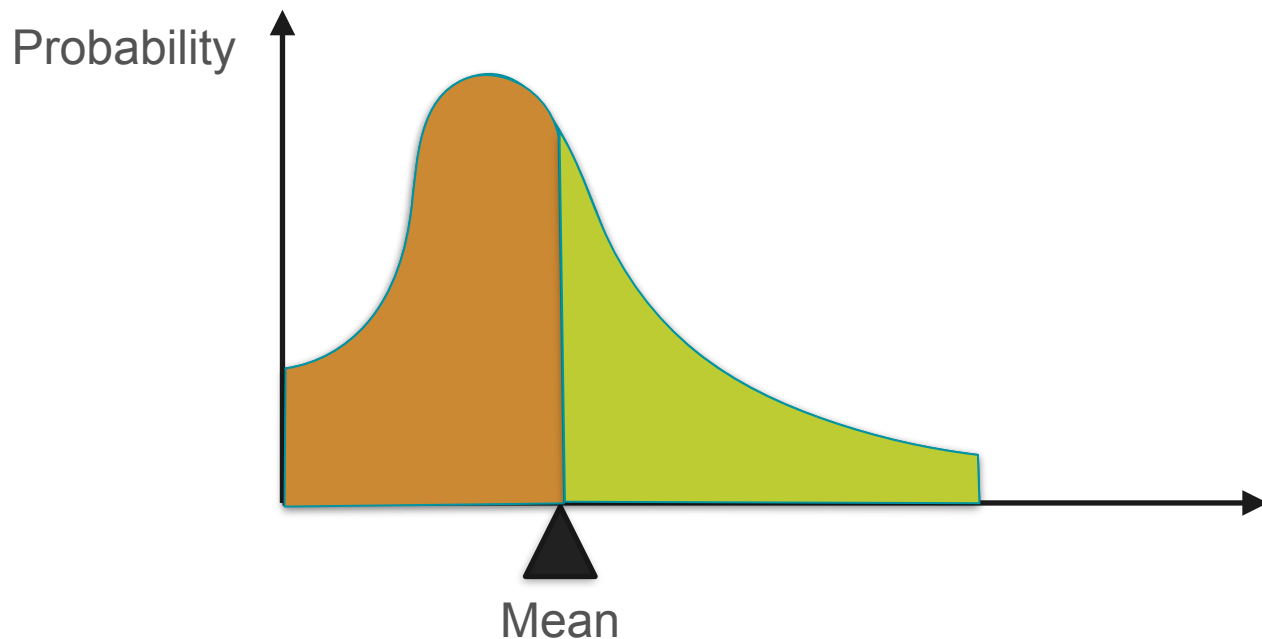
Expected Value: Common Misconception



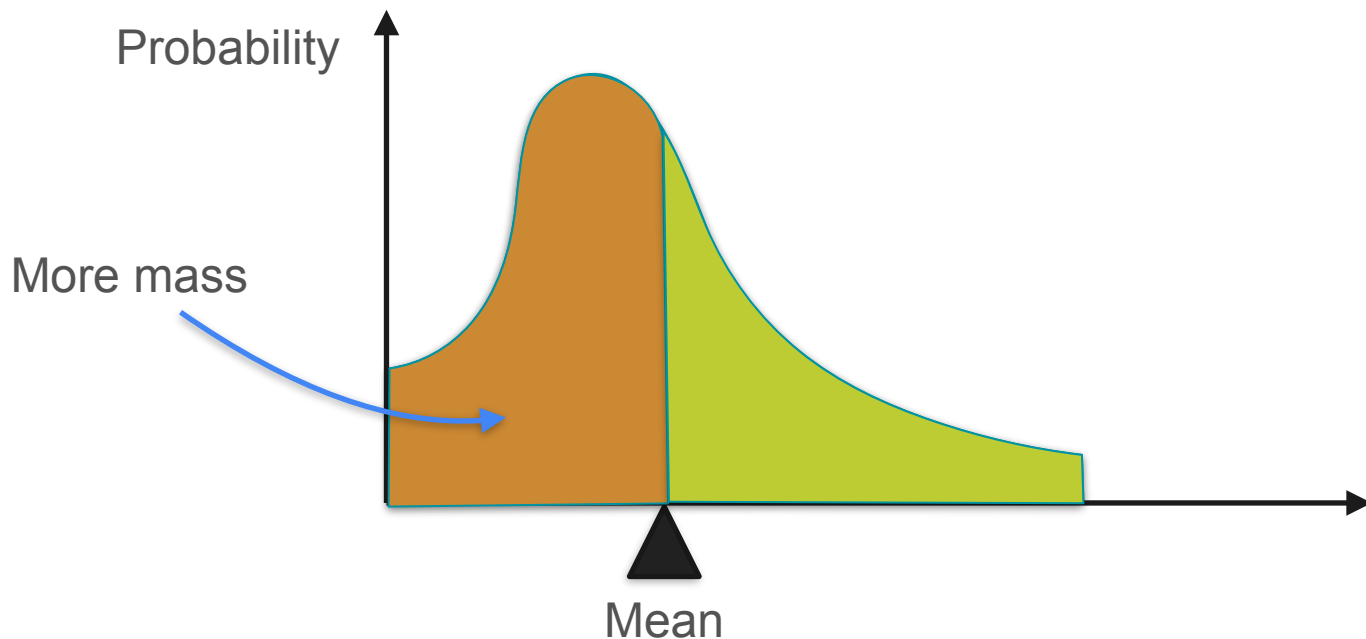
Expected Value: Common Misconception



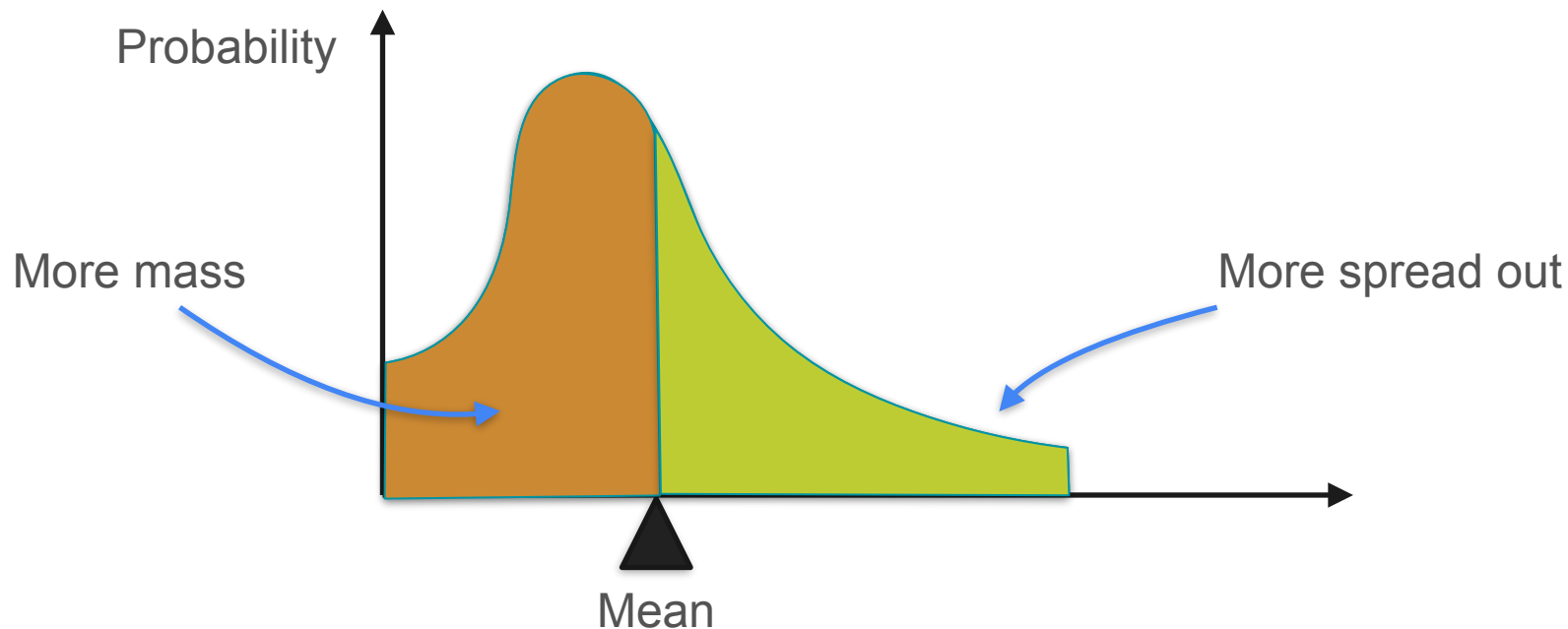
Expected Value: Common Misconception



Expected Value: Common Misconception



Expected Value: Common Misconception



Expected Value: Common Misconception

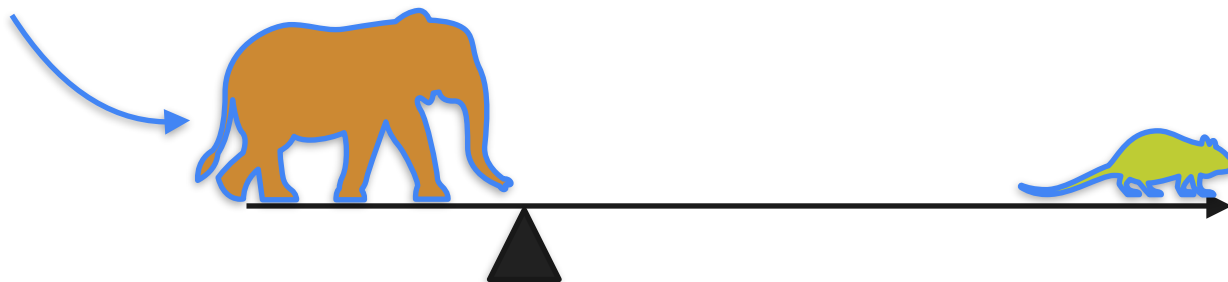


Expected Value: Common Misconception



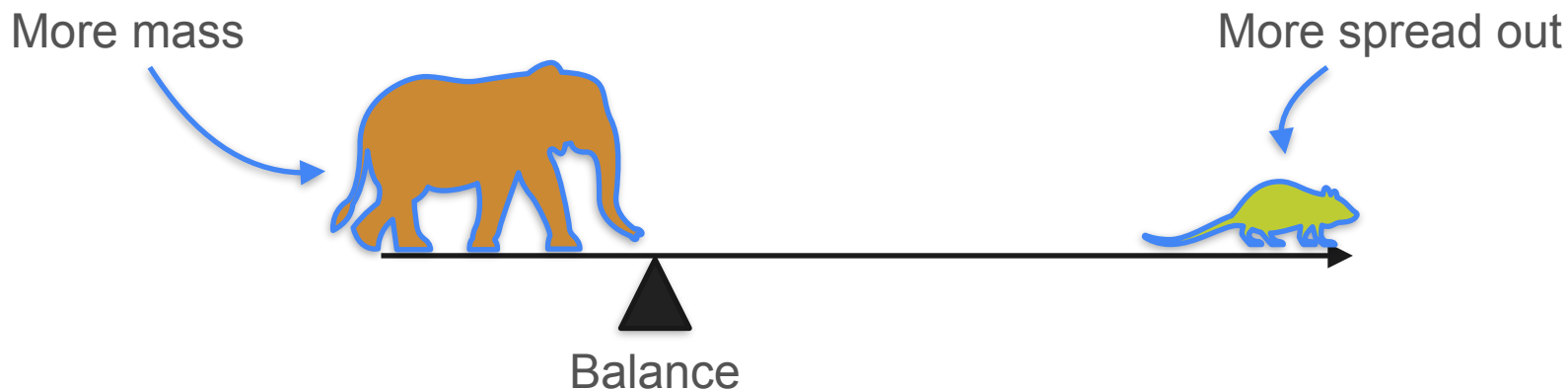
Expected Value: Common Misconception

More mass



Balance

Expected Value: Common Misconception



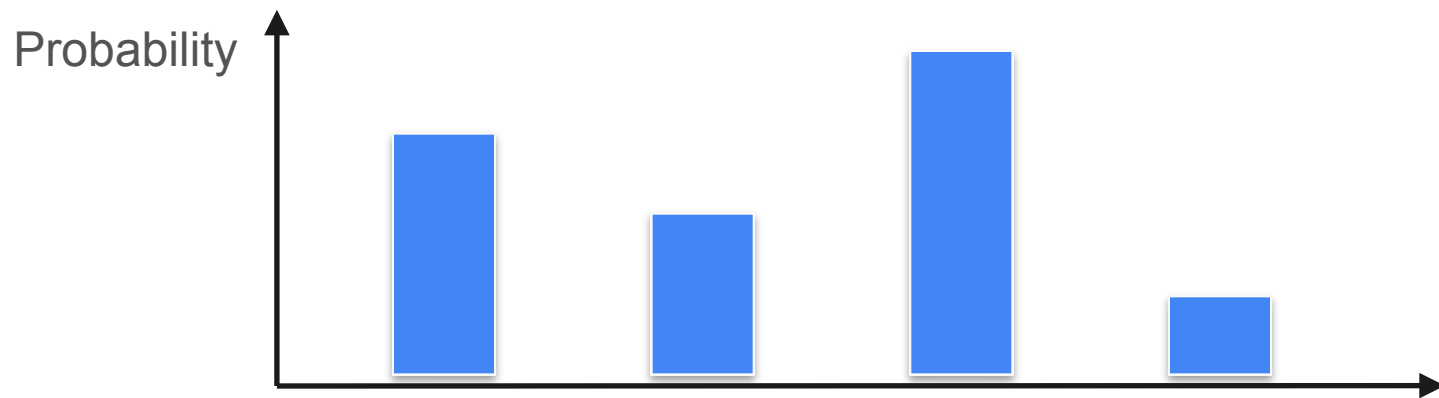


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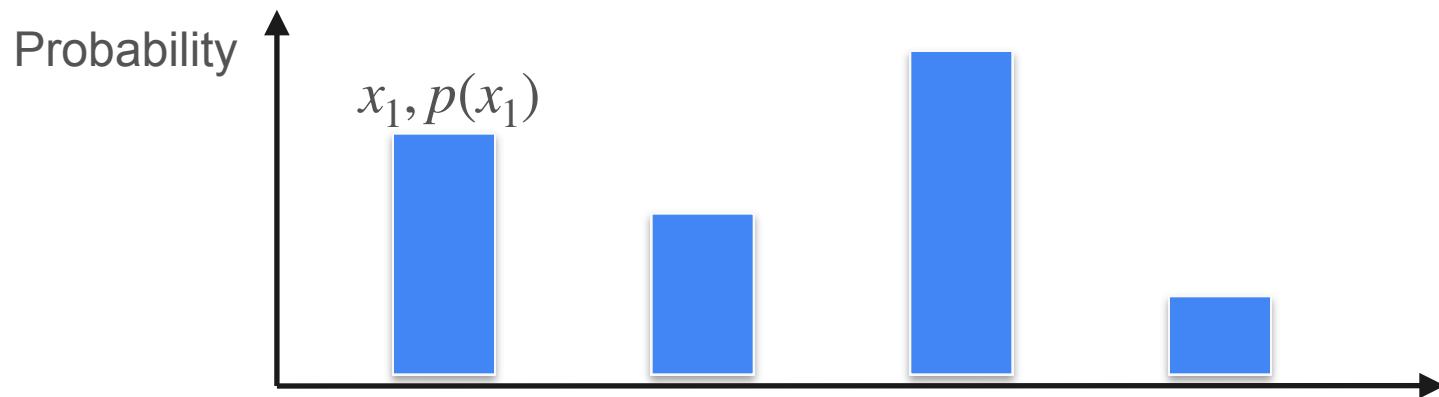
Describing Distributions

Expected value of a function

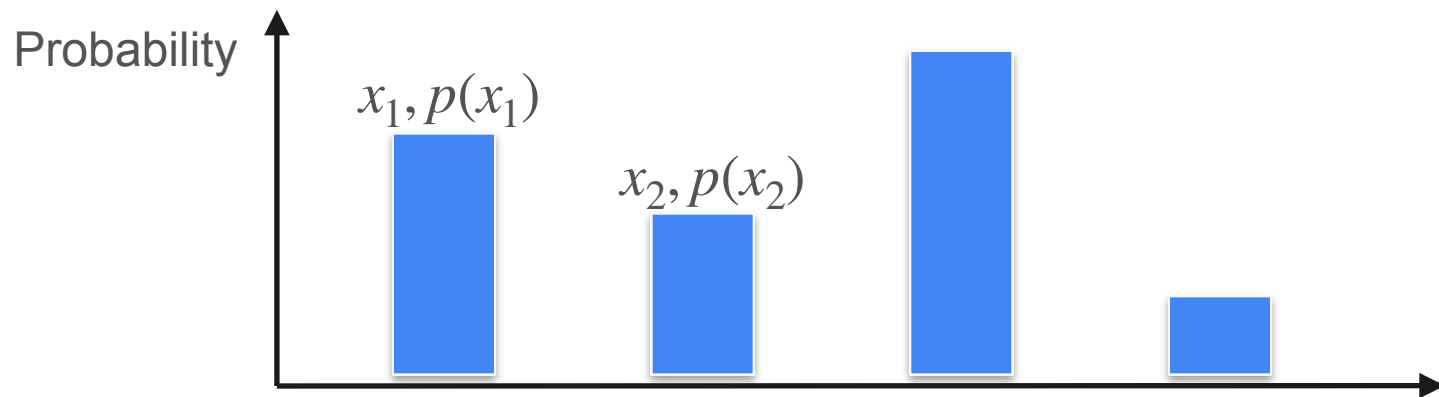
Expected Value of a Function



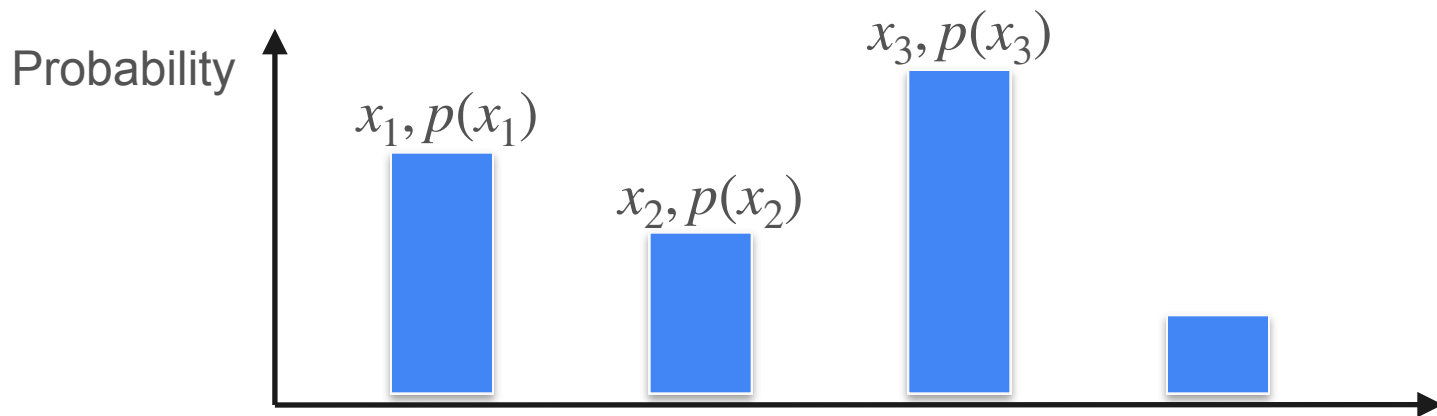
Expected Value of a Function



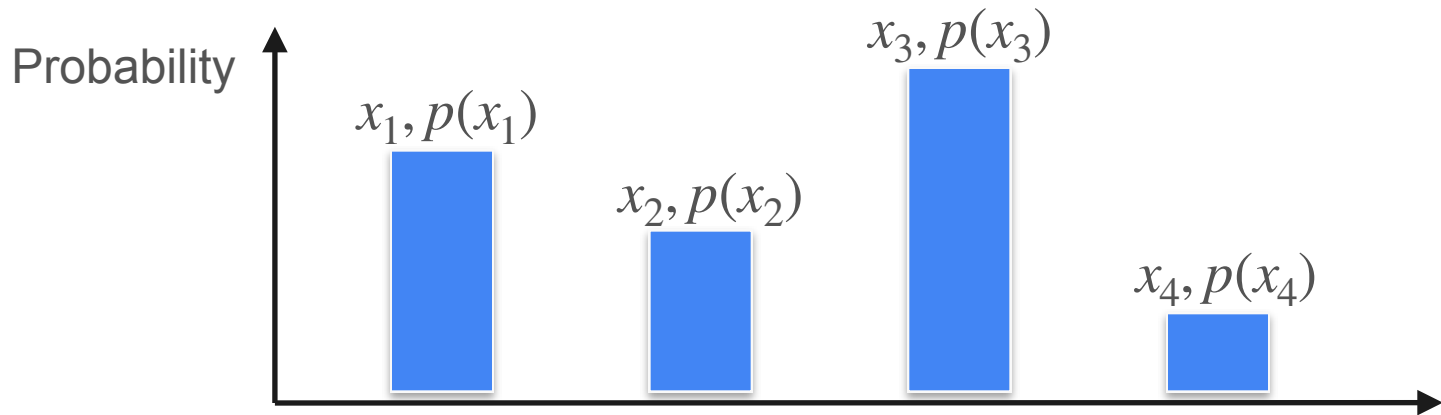
Expected Value of a Function



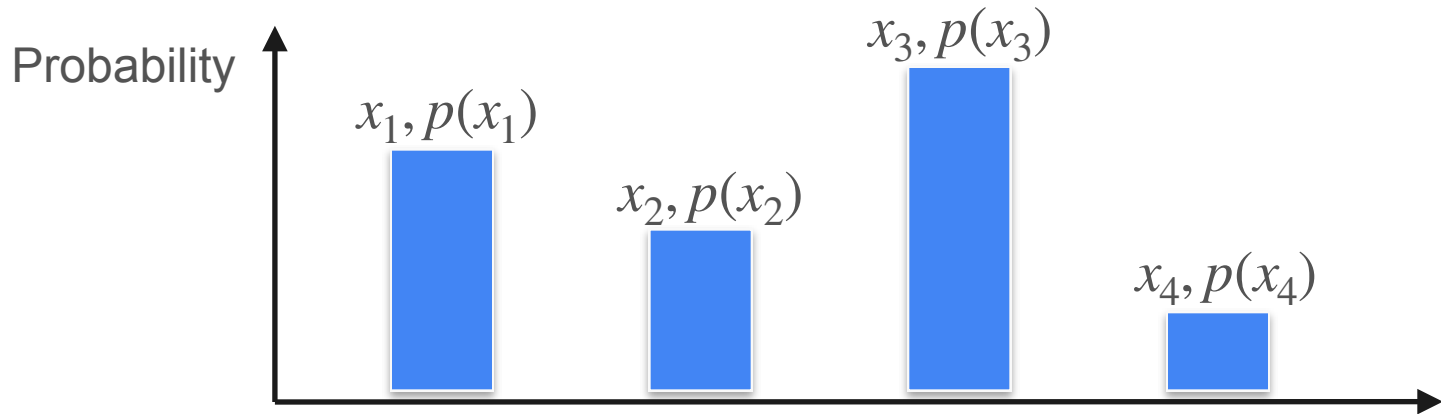
Expected Value of a Function



Expected Value of a Function

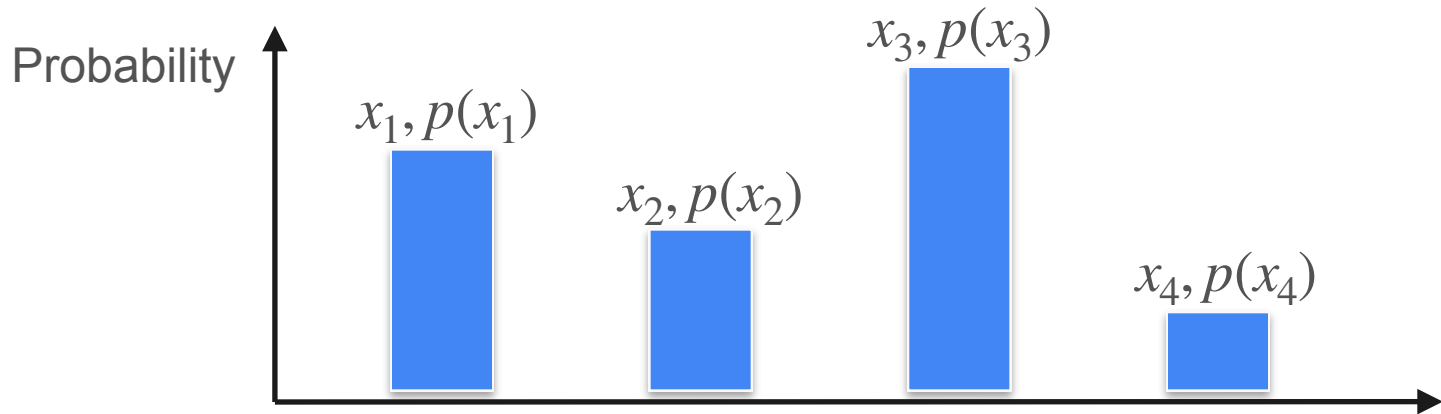


Expected Value of a Function



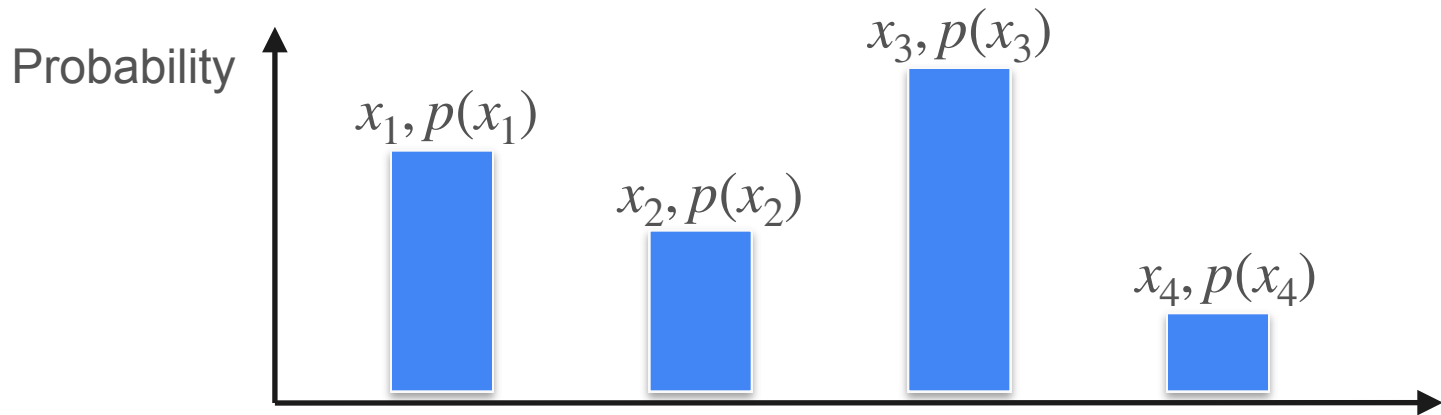
$$\mathbb{E}[X] =$$

Expected Value of a Function



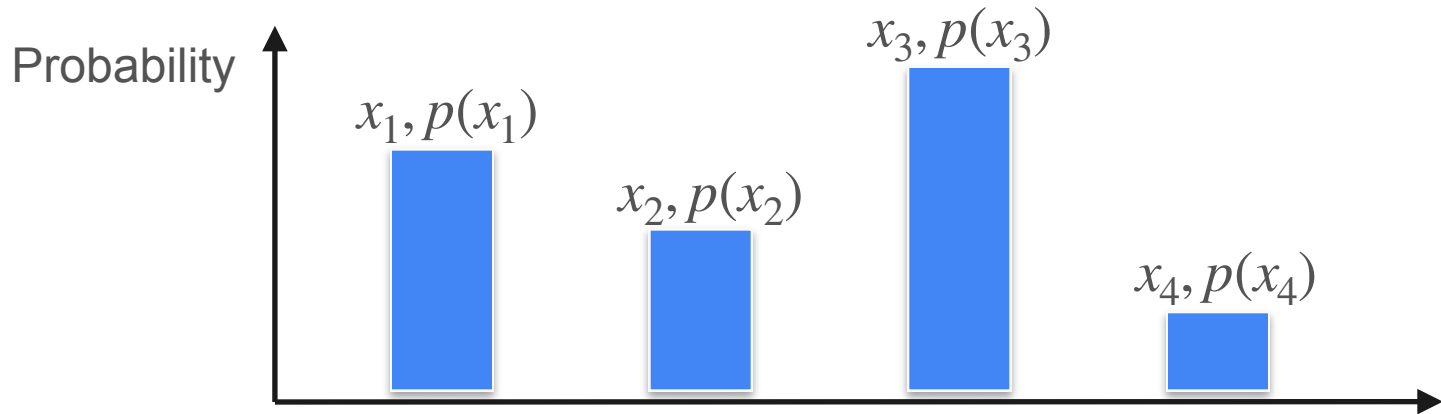
$$\mathbb{E}[X] = x_1 p(x_1)$$

Expected Value of a Function



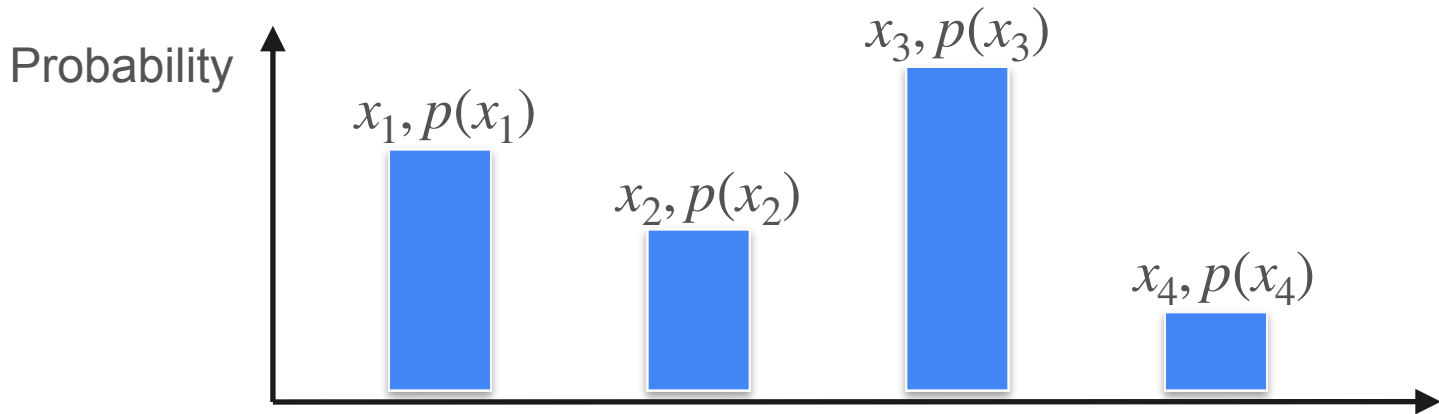
$$\mathbb{E}[X] = x_1p(x_1) + x_2p(x_2)$$

Expected Value of a Function



$$\mathbb{E}[X] = x_1p(x_1) + x_2p(x_2) + x_3p(x_3) + x_4p(x_4)$$

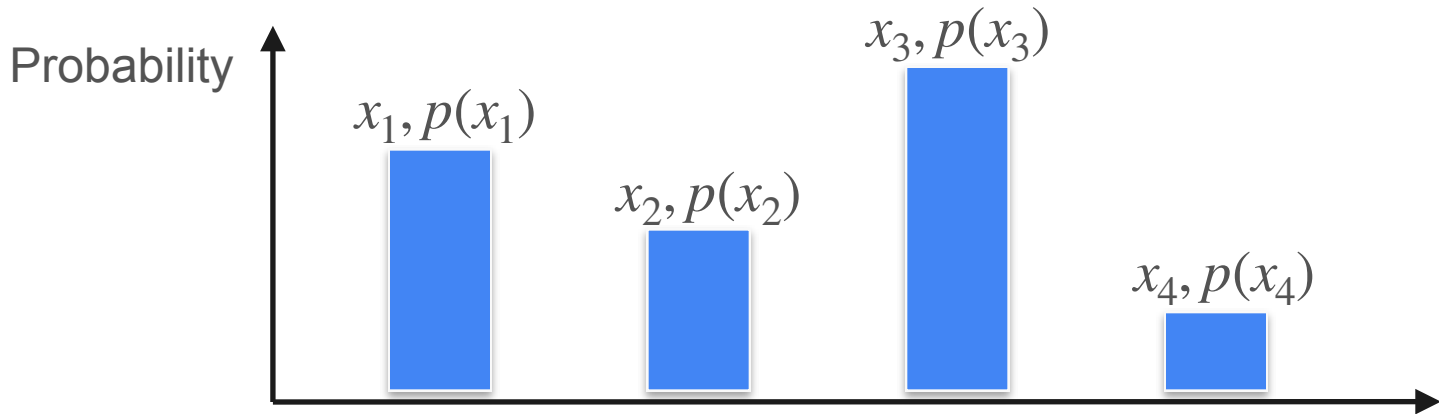
Expected Value of a Function



$$\mathbb{E}[X] = x_1p(x_1) + x_2p(x_2) + x_3p(x_3) + x_4p(x_4)$$

$$E[f(X)] =$$

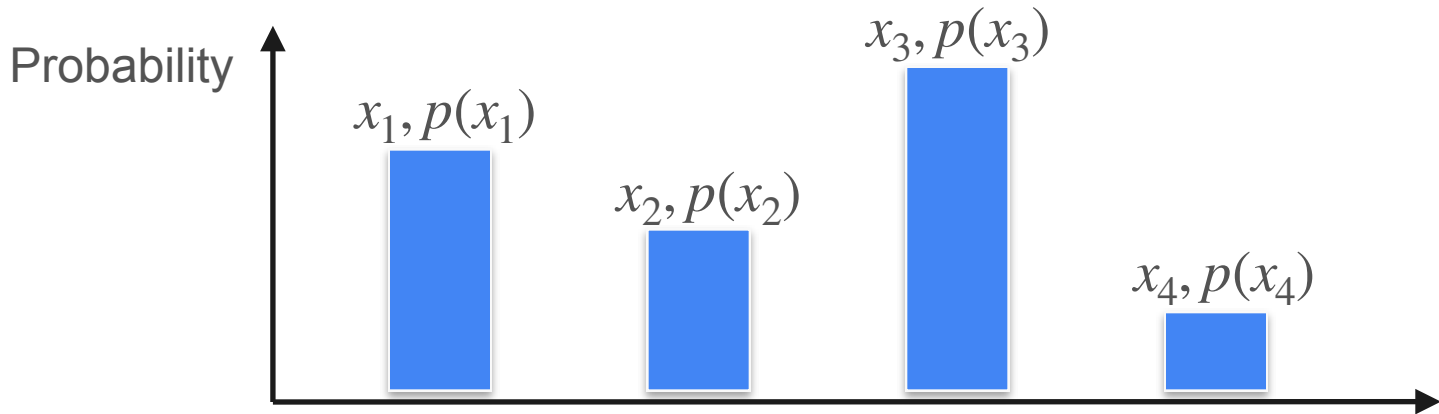
Expected Value of a Function



$$\mathbb{E}[X] = x_1p(x_1) + x_2p(x_2) + x_3p(x_3) + x_4p(x_4)$$

$$E[f(X)] = f(x_1)p(x_1)$$

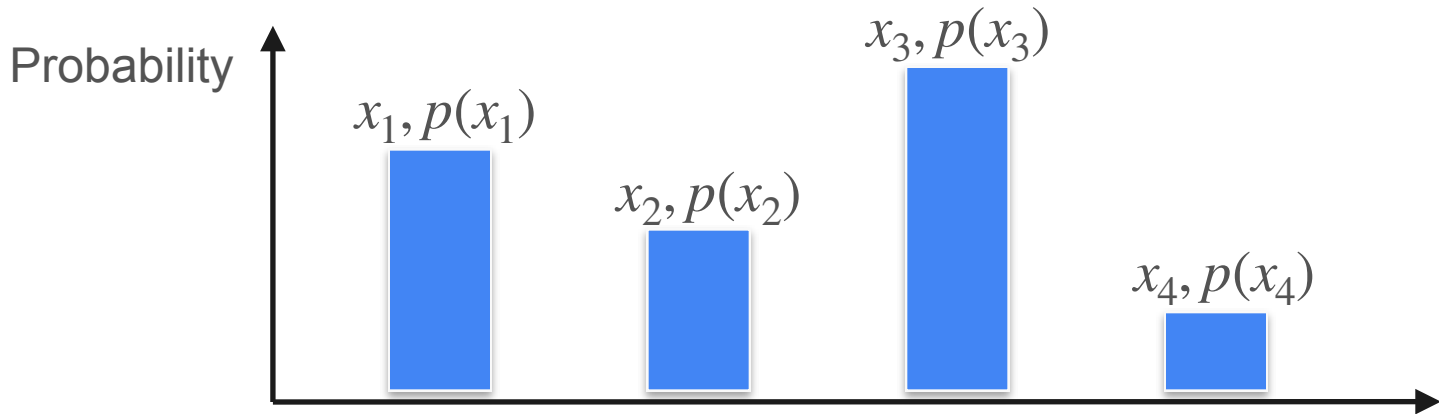
Expected Value of a Function



$$\mathbb{E}[X] = x_1p(x_1) + x_2p(x_2) + x_3p(x_3) + x_4p(x_4)$$

$$E[f(X)] = f(x_1)p(x_1) + f(x_2)p(x_2)$$

Expected Value of a Function



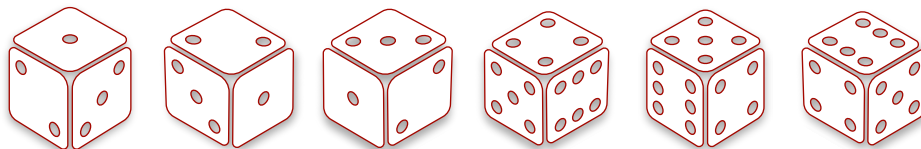
$$\mathbb{E}[X] = x_1p(x_1) + x_2p(x_2) + x_3p(x_3) + x_4p(x_4)$$

$$E[f(X)] = f(x_1)p(x_1) + f(x_2)p(x_2) + f(x_3)p(x_3) + f(x_4)p(x_4)$$

Expected Value of a Function

Probability: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

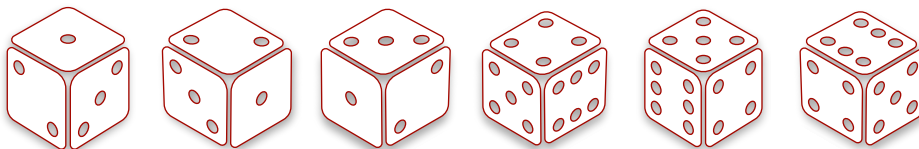
Roll: 1 2 3 4 5 6



Expected Value of a Function

Probability: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

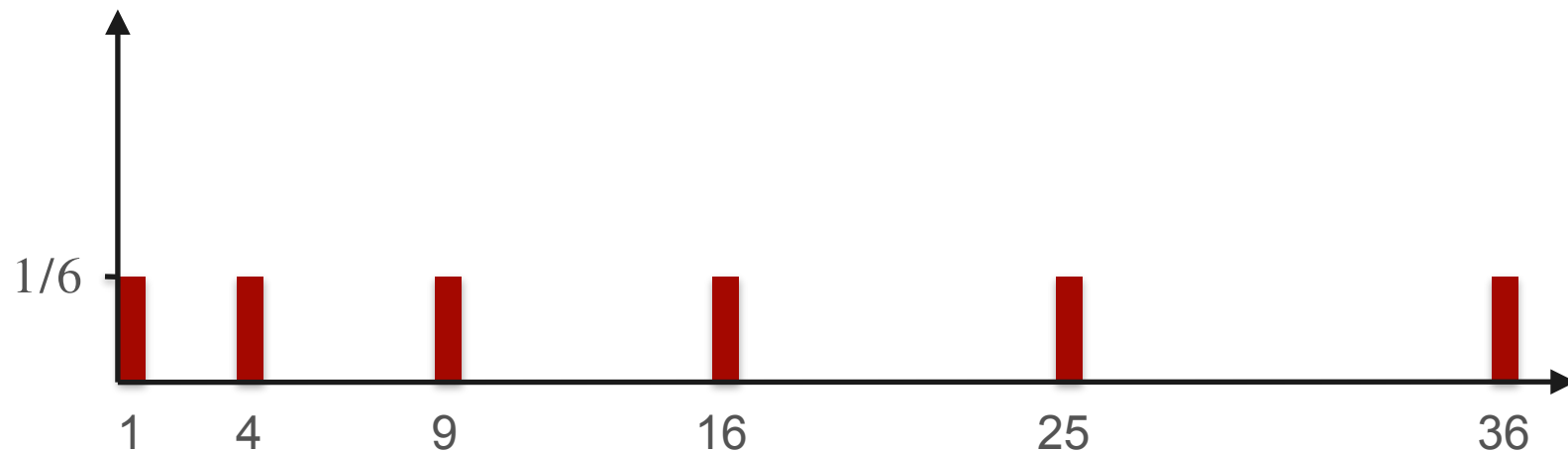
Roll: 1 2 3 4 5 6



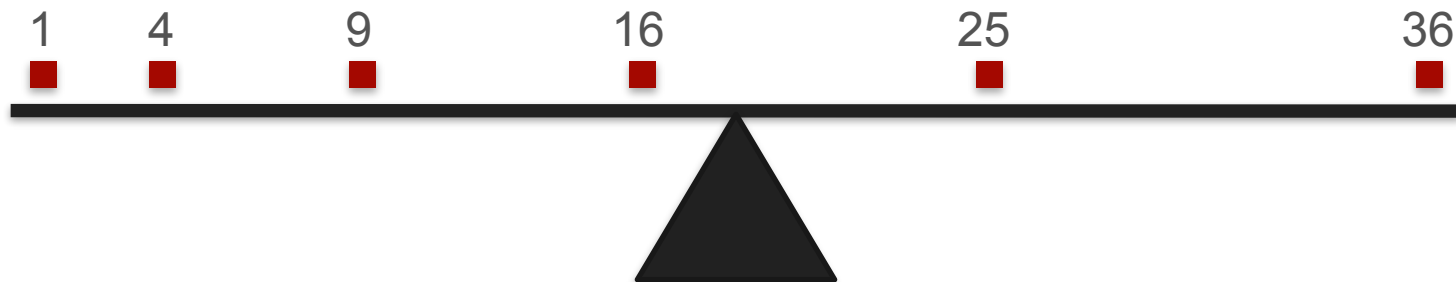
Square: 1 4 9 16 25 36

Expected Value of a Function

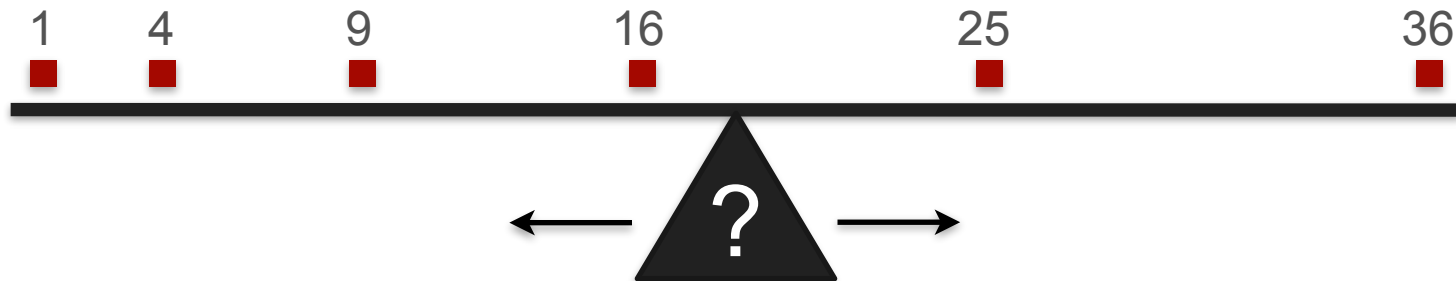
Probability



Expected Value of a Function

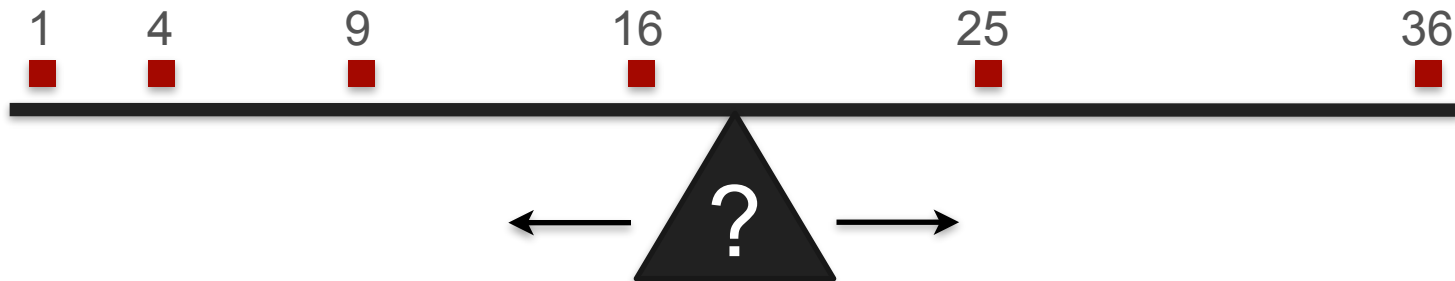


Expected Value of a Function



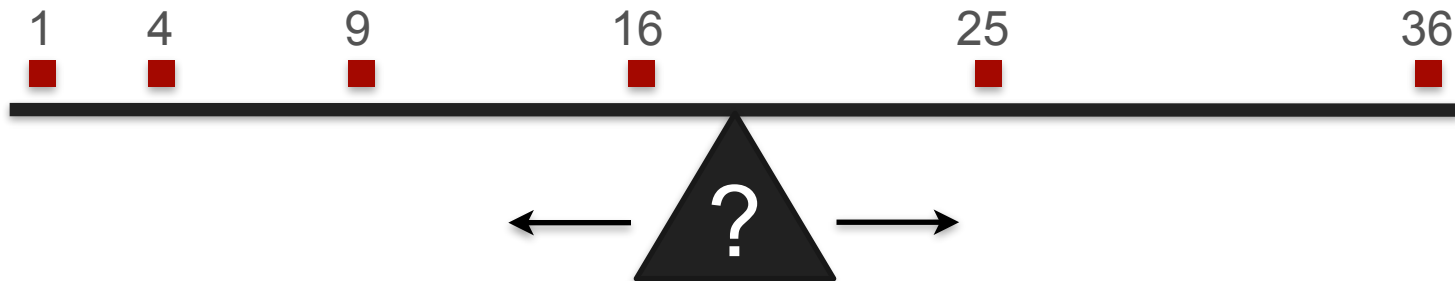
Expected Value of a Function

$$\frac{1 + 4 + 9 + 16 + 25 + 36}{6}$$

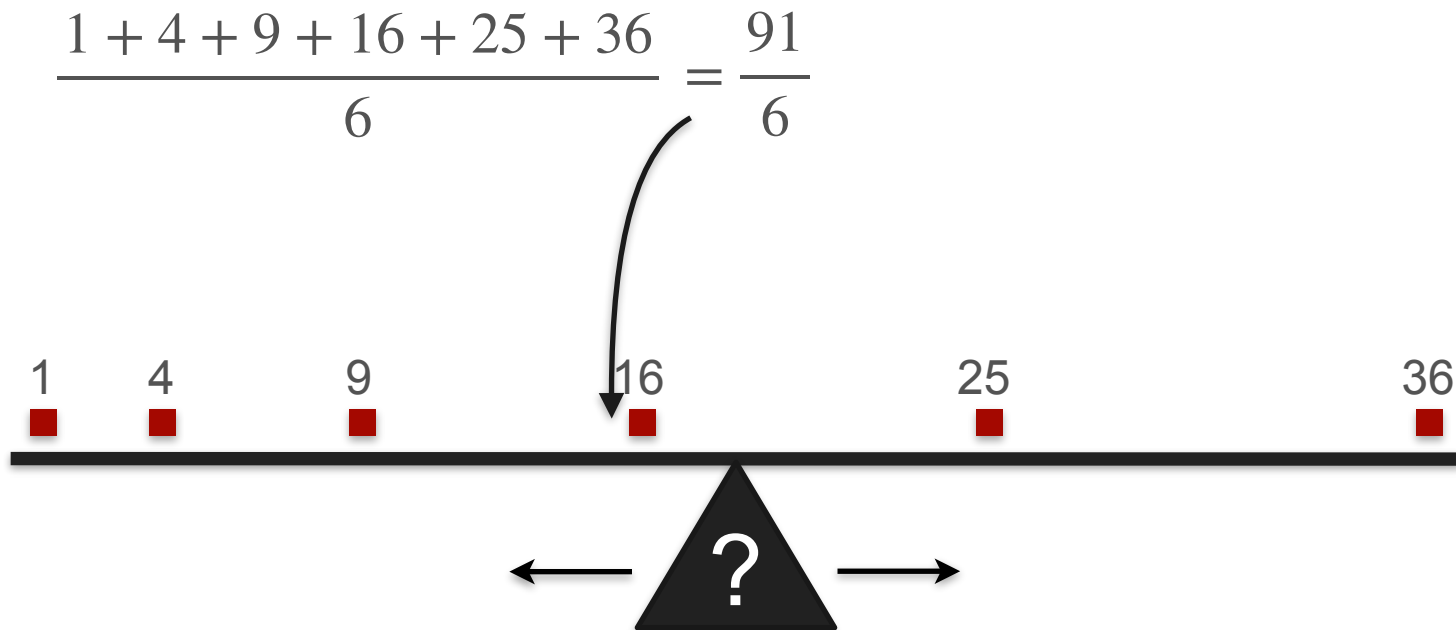


Expected Value of a Function

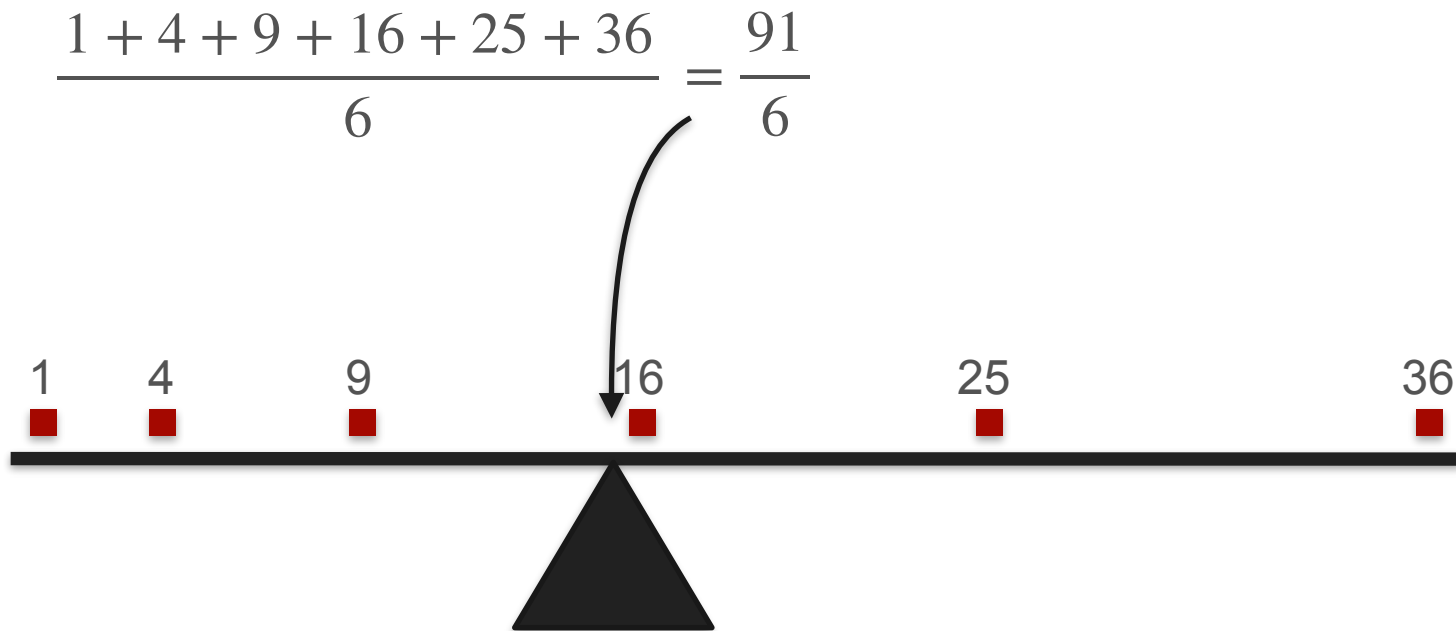
$$\frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6}$$



Expected Value of a Function

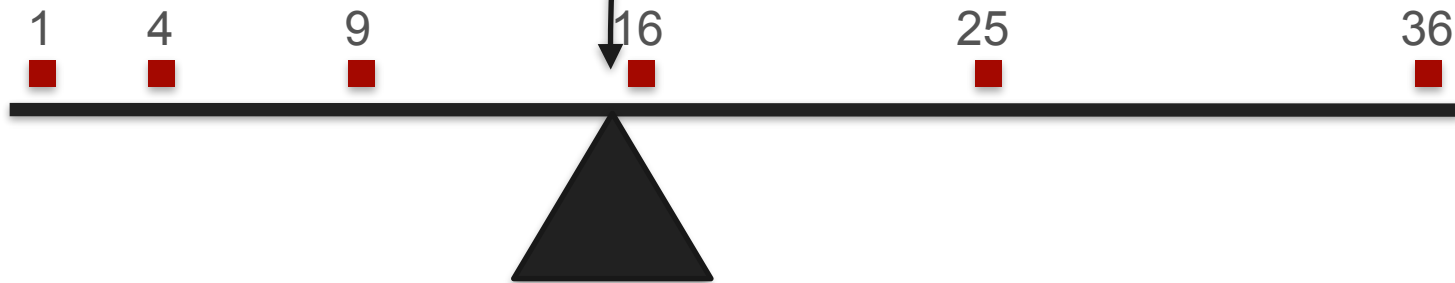


Expected Value of a Function



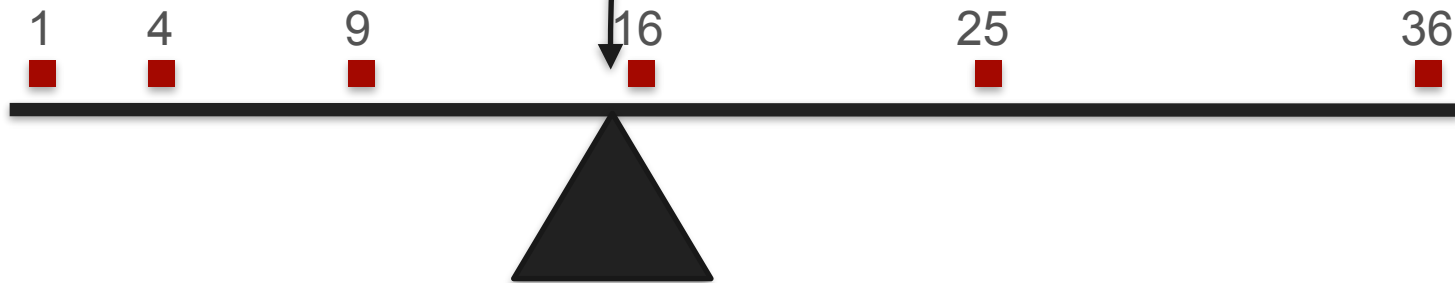
Expected Value of a Function

$$\frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6} = \frac{1^2}{6} + \frac{2^2}{6} + \frac{3^2}{6} + \frac{4^2}{6} + \frac{5^2}{6} + \frac{6^2}{6}$$



Expected Value of a Function

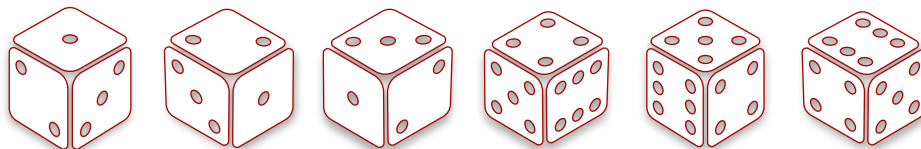
$$\frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6} = \frac{1^2}{6} + \frac{2^2}{6} + \frac{3^2}{6} + \frac{4^2}{6} + \frac{5^2}{6} + \frac{6^2}{6} \\ = \mathbb{E}[X^2]$$



Expectation of Linear Function

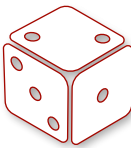
Probability: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

Roll: 1 2 3 4 5 6



Wins

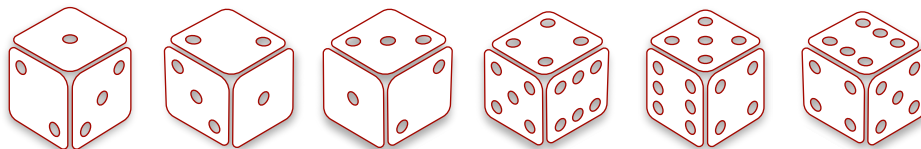
Expectation of Linear Function

Probability:	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$
Roll:	1	2	3	4	5	6
						
Double:	2	4	6	8	10	12
Wins	2 - 5	4 - 5	6 - 5	8 - 5	10 - 5	12 - 5

Expectation of Linear Function

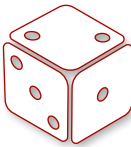
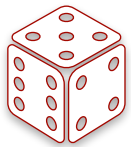
Probability: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

Roll: 1 2 3 4 5 6



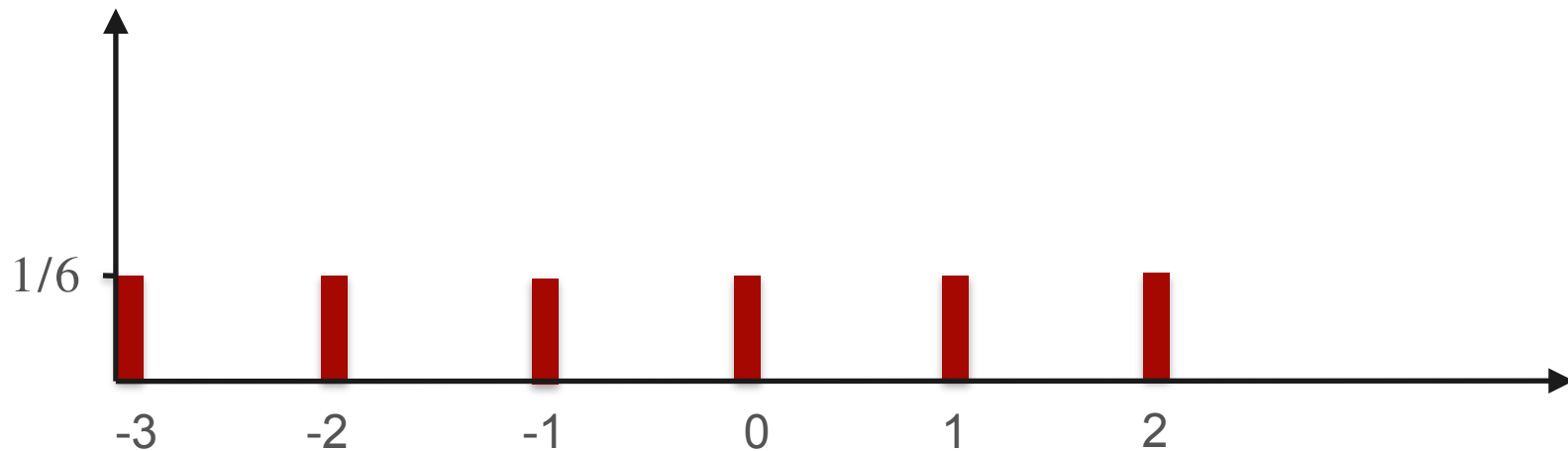
Wins

Expectation of Linear Function

Probability:	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$
Roll:	1	2	3	4	5	6
						
Double:	2	4	6	8	10	12
Wins	-3	-2	-1	0	1	2

Expected Value of a Function

Probability



Expected Value of a Function



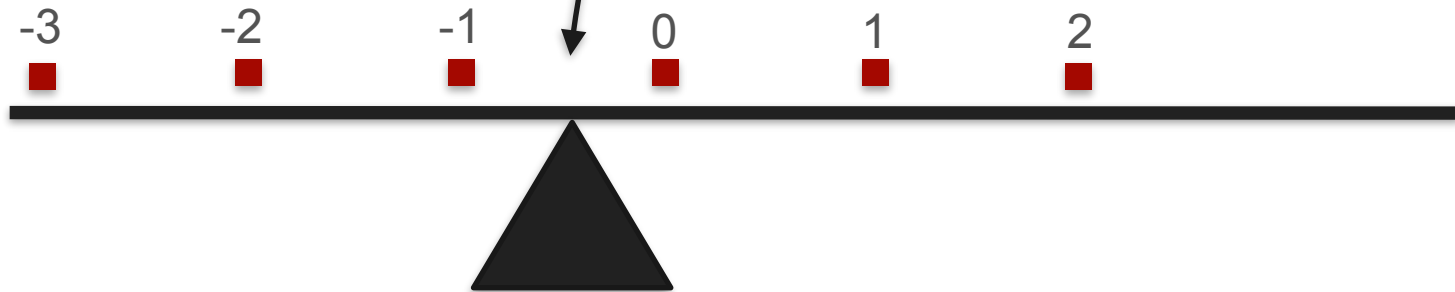
Expected Value of a Function

$$\frac{-3 + -2 + -1 + 0 + 1 + 2}{6}$$



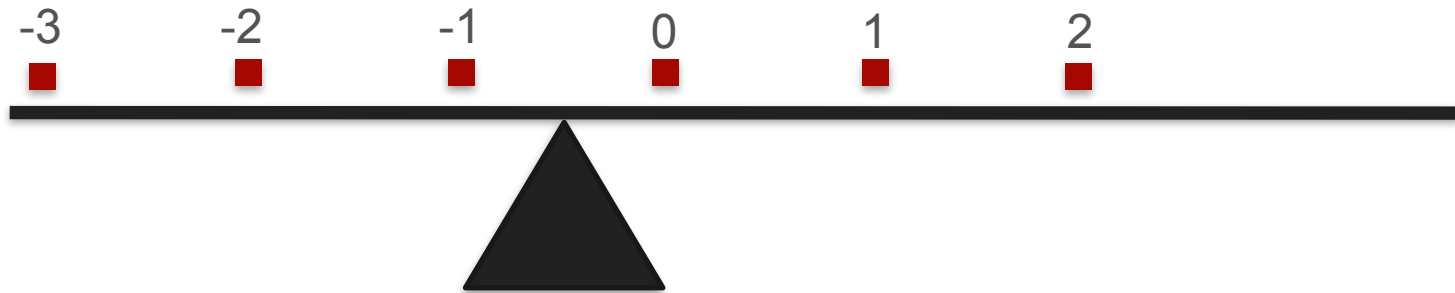
Expected Value of a Function

$$\frac{-3 + -2 + -1 + 0 + 1 + 2}{6} = \frac{-3}{6} = -0.5$$



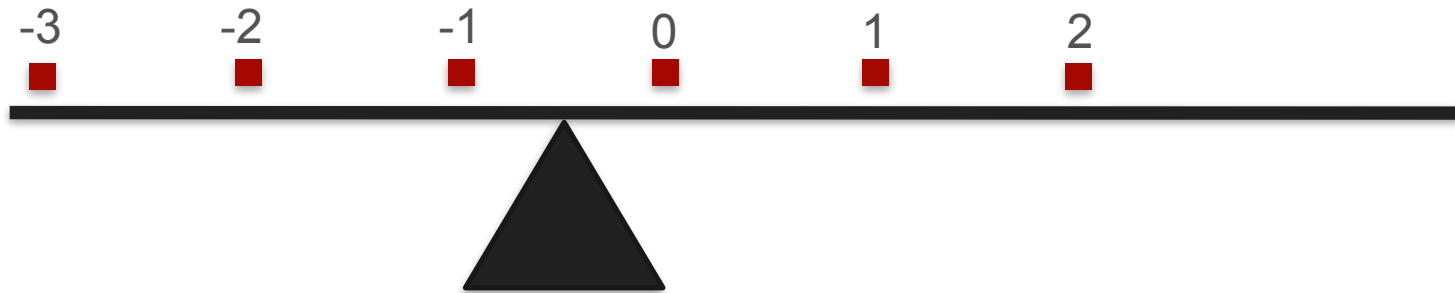
Expected Value of a Function

$$\frac{(2 \cdot 1 - 5) + (2 \cdot 2 - 5) + (2 \cdot 3 - 5) + (2 \cdot 4 - 5) + (2 \cdot 5 - 5) + (2 \cdot 6 - 5)}{6} = \frac{-3}{6} = -0.5$$



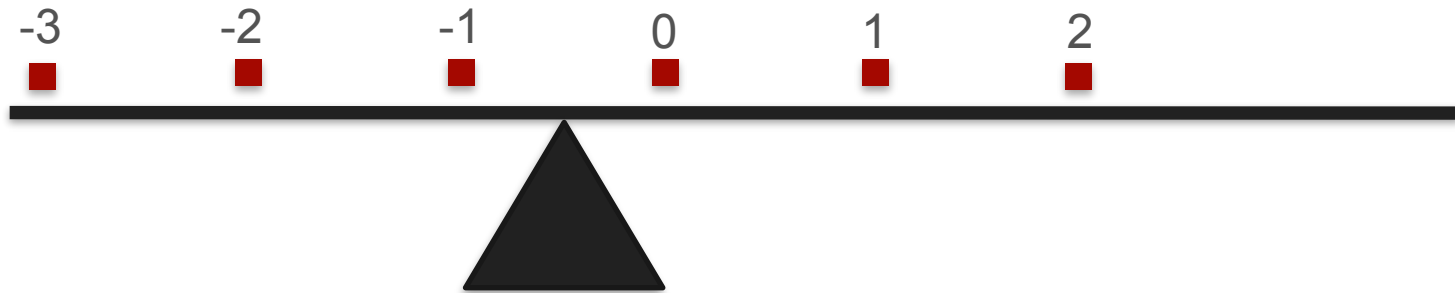
Expected Value of a Function

$$\frac{(2 \cdot 1) + (2 \cdot 2) + (2 \cdot 3) + (2 \cdot 4) + (2 \cdot 5) + (2 \cdot 6) + 6 \cdot (-5)}{6} = \frac{-3}{6} = -0.5$$



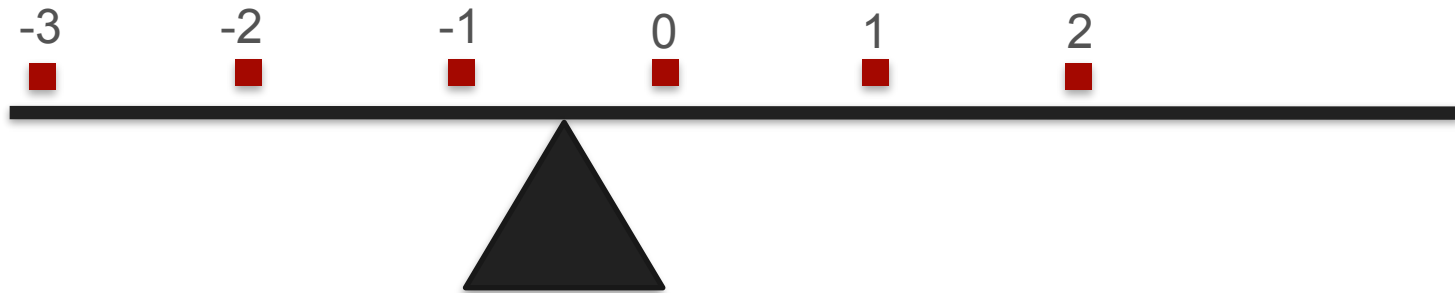
Expected Value of a Function

$$\frac{(2 \cdot 1) + (2 \cdot 2) + (2 \cdot 3) + (2 \cdot 4) + (2 \cdot 5) + (2 \cdot 6)}{6} + (-5) = \frac{-3}{6} = -0.5$$

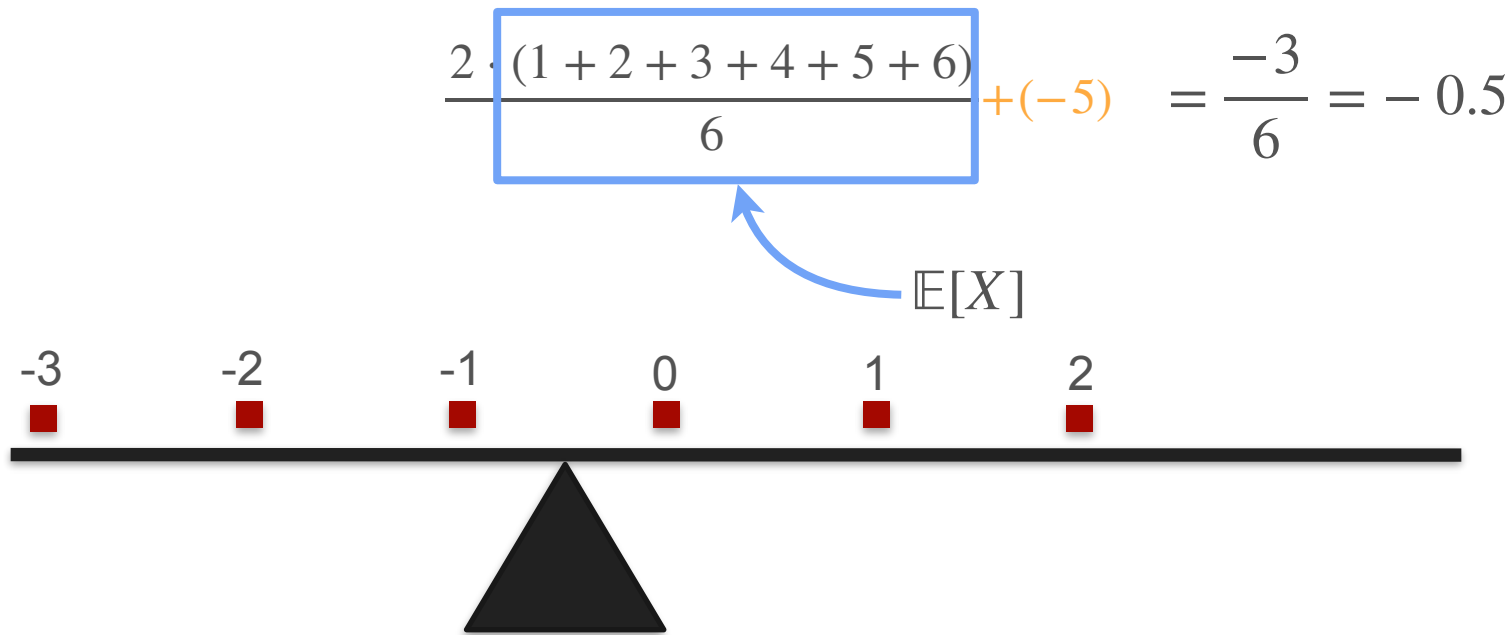


Expected Value of a Function

$$\frac{2 \cdot (1 + 2 + 3 + 4 + 5 + 6)}{6} + (-5) = \frac{-3}{6} = -0.5$$



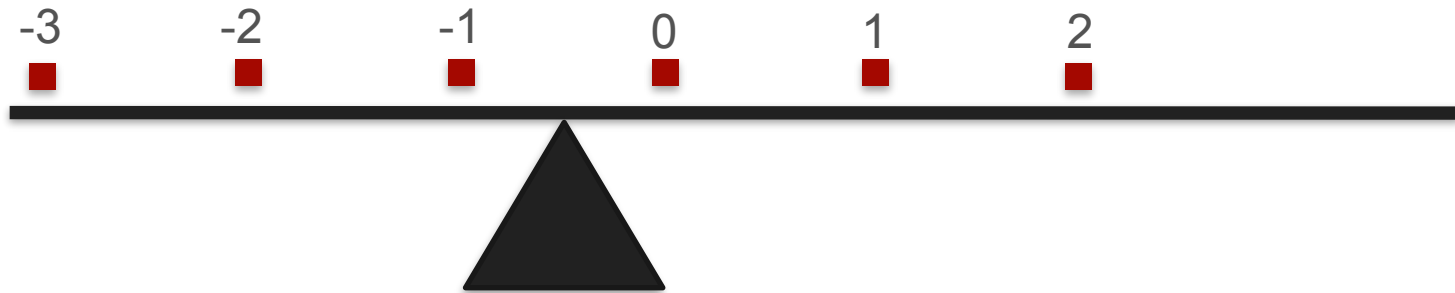
Expected Value of a Function



Expected Value of a Function

$$\mathbb{E}[2 \cdot X + (-5)] = \frac{2 \cdot (1 + 2 + 3 + 4 + 5 + 6)}{6} + (-5) = \frac{-3}{6} = -0.5$$

$\mathbb{E}[X]$



Expected Value of a Function

$$\mathbb{E}[2 \cdot X + (-5)] = \frac{2 \cdot (1 + 2 + 3 + 4 + 5 + 6)}{6} + (-5) = \frac{-3}{6} = -0.5$$

$\mathbb{E}[X]$

$= 2 \cdot \mathbb{E}[X] + (-5)$



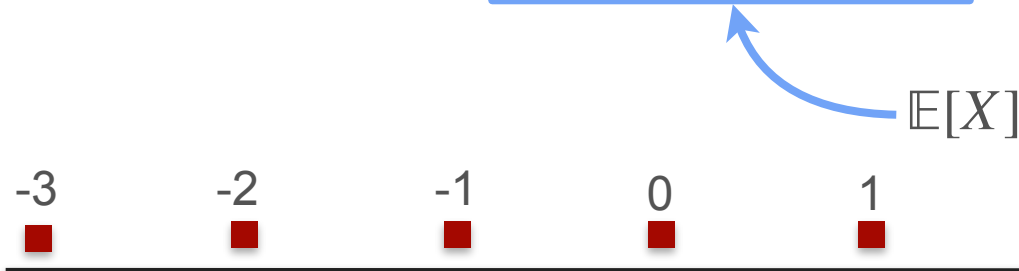
In general:

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$$

Expected Value of a Function

$$\mathbb{E}[2 \cdot X + (-5)] = \frac{2 \cdot (1 + 2 + 3 + 4 + 5 + 6)}{6} + (-5) = \frac{-3}{6} = -0.5$$

$= 2 \cdot \mathbb{E}[X] + (-5)$



In general:

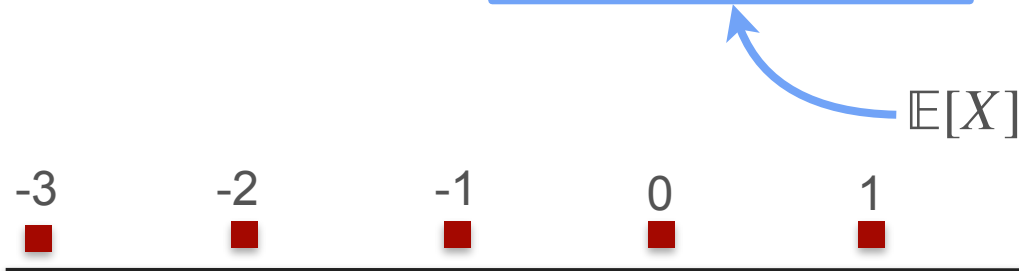
$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$$

$$\mathbb{E}[aX] = a\mathbb{E}[X]$$

Expected Value of a Function

$$\mathbb{E}[2 \cdot X + (-5)] = \frac{2 \cdot (1 + 2 + 3 + 4 + 5 + 6)}{6} + (-5) = \frac{-3}{6} = -0.5$$

$= 2 \cdot \mathbb{E}[X] + (-5)$



In general:

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$$

$$\mathbb{E}[aX] = a\mathbb{E}[X]$$

$$\mathbb{E}[b] = b$$



DeepLearning.AI

Describing Distributions

Sum of expectations

Sum of Expectation: Expected Winnings

You play a game:

Sum of Expectation: Expected Winnings

You play a game:

Flip a coin. If heads you win \$ 1, else you win nothing.

Sum of Expectation: Expected Winnings

You play a game:

Flip a coin. If heads you win \$ 1, else you win nothing.



Win \$1



Win nothing

Sum of Expectation: Expected Winnings

You play a game:

Flip a coin. If heads you win \$ 1, else you win nothing.

Then throw a dice. You win the amount you roll.



Win \$1



Win nothing

Sum of Expectation: Expected Winnings

You play a game:

Flip a coin. If heads you win \$ 1, else you win nothing.

Then throw a dice. You win the amount you roll.



Win \$1

Win

\$1

\$2

\$3

\$4

\$5

\$6



Win nothing



Sum of Expectation: Expected Winnings

You play a game:

Flip a coin. If heads you win \$ 1, else you win nothing.

Then throw a dice. You win the amount you roll.



Win \$1

Win

\$1

\$2

\$3

\$4

\$5

\$6



Win nothing



What are your expected winnings for the game?

Sum of Expectations: Expected Winnings



Win \$1

Win

\$1

\$2

\$3

\$4

\$5

\$6



Win nothing



Sum of Expectations: Expected Winnings



Win \$1

Win

\$1

\$2

\$3

\$4

\$5

\$6



Win nothing



$$\mathbb{E}[X_{\text{coin}}] = \$0.5$$

Sum of Expectations: Expected Winnings



Win \$1

Win

\$1

\$2

\$3

\$4

\$5

\$6



Win nothing



$$\mathbb{E}[X_{\text{coin}}] = \$0.5$$

$$\mathbb{E}[X_{\text{dice}}] = \$3.5$$

Sum of Expectations: Expected Winnings



Win \$1

Win

\$1

\$2

\$3

\$4

\$5

\$6



Win nothing



$$\mathbb{E}[X_{\text{coin}}] = \$0.5$$

$$\mathbb{E}[X_{\text{dice}}] = \$3.5$$

$$\mathbb{E}[X] = \mathbb{E}[X_{\text{coin}}] + \mathbb{E}[X_{\text{dice}}] = \$0.5 + \$3.5 = \$4$$

Sum of Expectations: Expected Winnings



Win \$1

Win

\$1

\$2

\$3

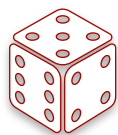
\$4

\$5

\$6



Win nothing



$$\mathbb{E}[X_{coin}] = \$0.5$$

$$\mathbb{E}[X_{dice}] = \$3.5$$

$$\mathbb{E}[X] = \mathbb{E}[X_{coin}] + \mathbb{E}[X_{dice}] = \$0.5 + \$3.5 = \$4$$

In general: $\mathbb{E}[X_1 + X_2] = \mathbb{E}[X_1] + \mathbb{E}[X_2]$

Sum of Expectations: Bag With Names



8 billion people

Sum of Expectations: Bag With Names



8 billion people

Sum of Expectations: Bag With Names



8 billion people

Sum of Expectations: Bag With Names



Expected number of
correct assignments?



8 billion people

Sum of Expectations: Bag With Names



1

Expected number of
correct assignments?



8 billion people

Sum of Expectations: Bag With Names



1

Expected number of
correct assignments?

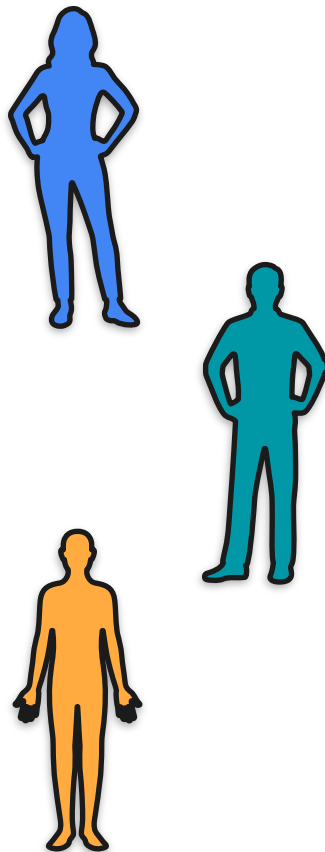
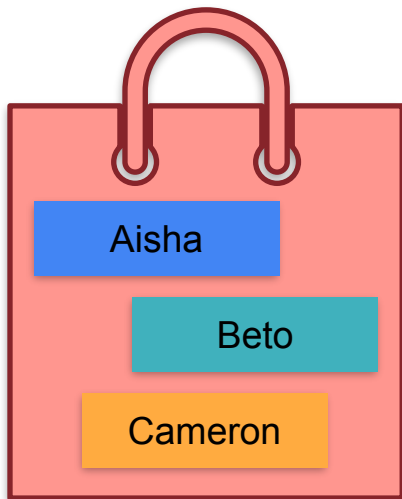


8 billion people

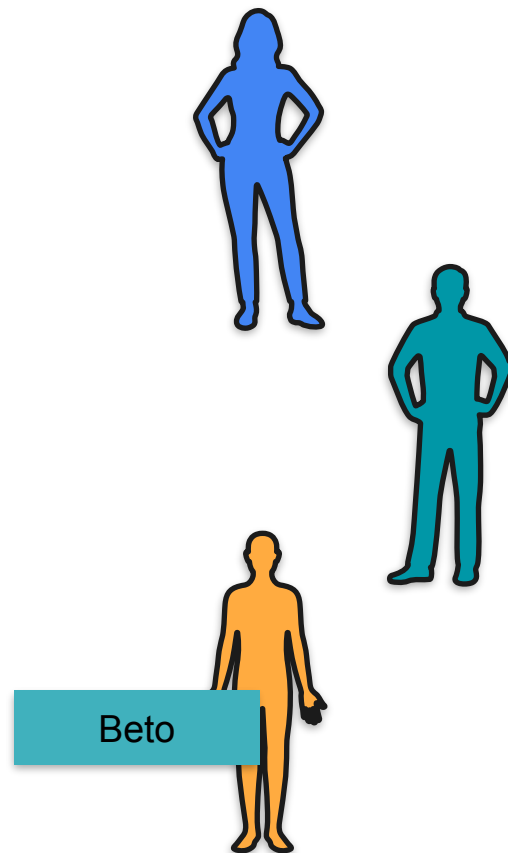
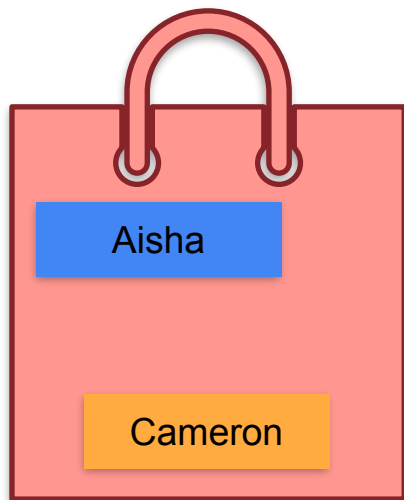


Bo Bleepityblop

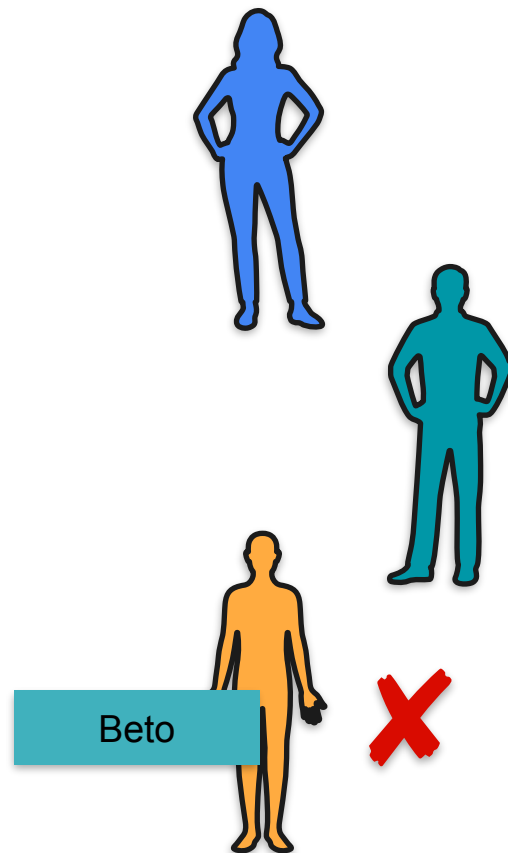
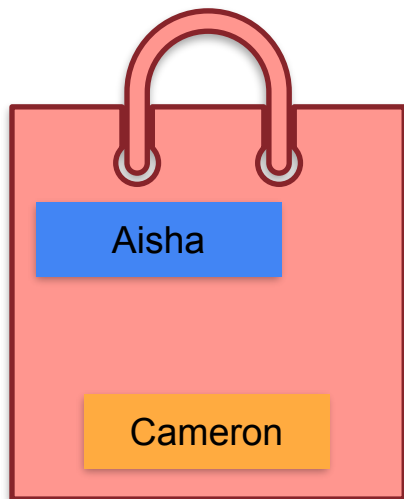
Bag With 3 Names



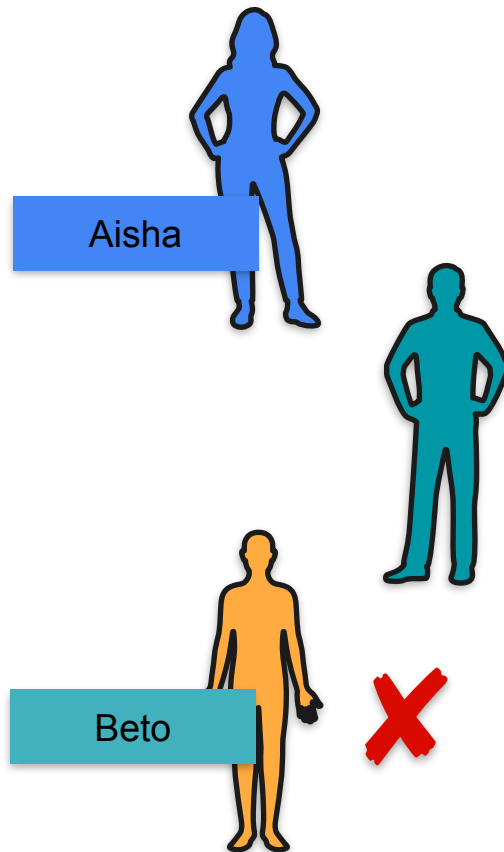
Bag With 3 Names



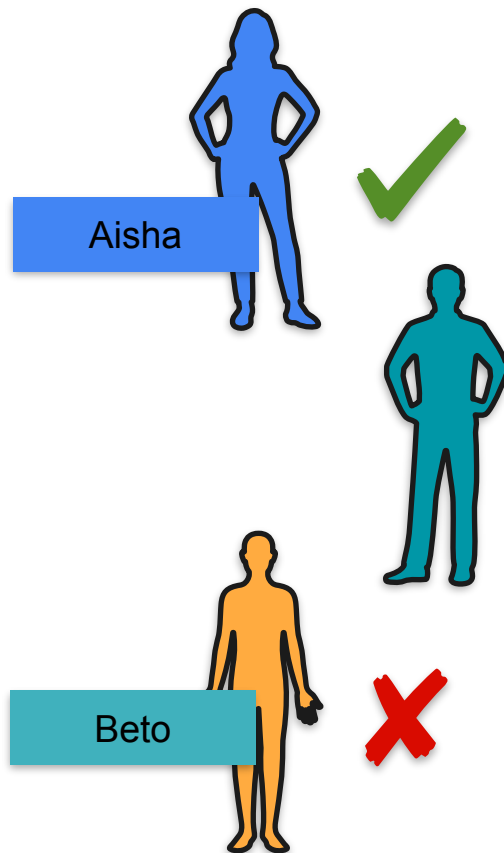
Bag With 3 Names



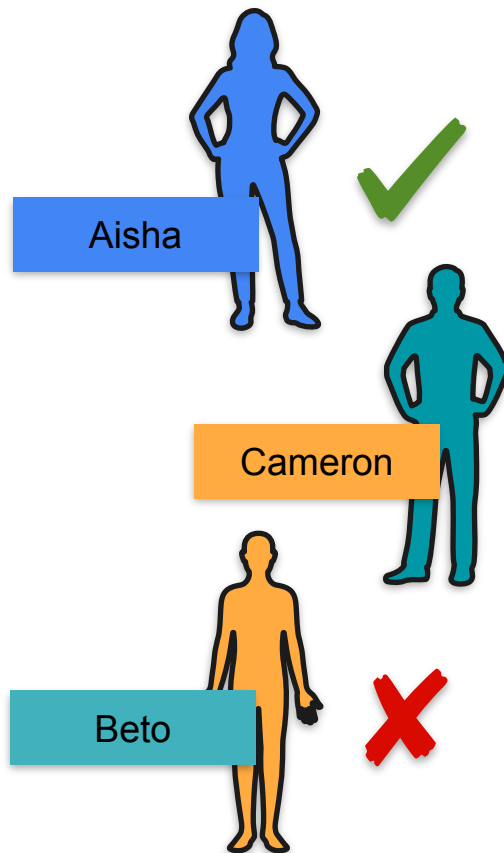
Bag With 3 Names



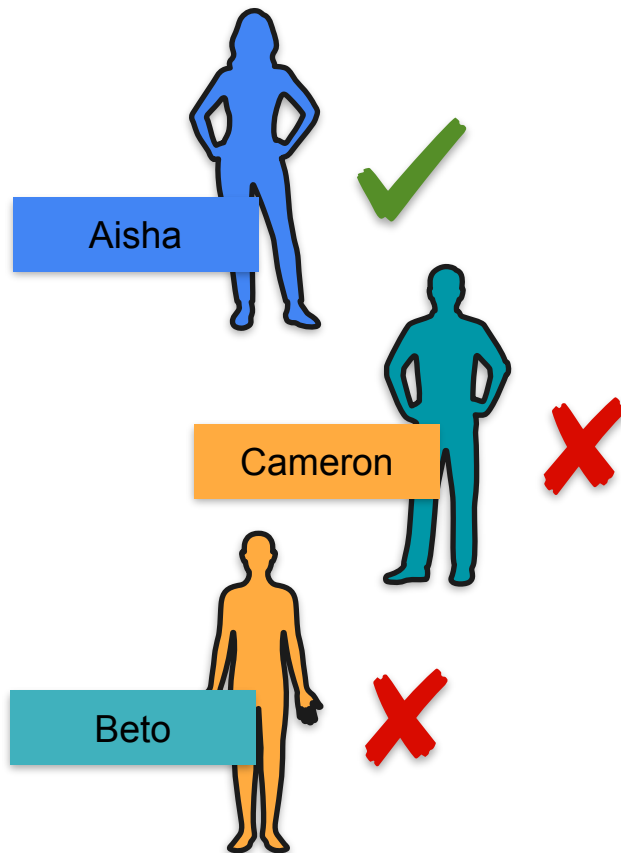
Bag With 3 Names



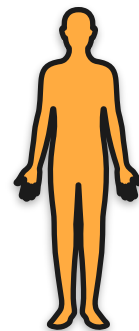
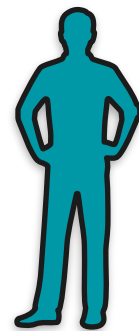
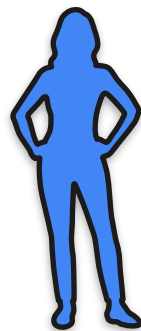
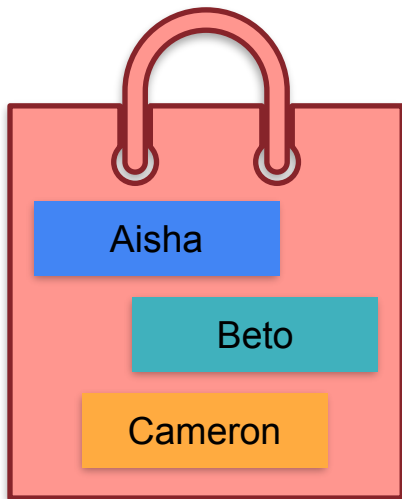
Bag With 3 Names



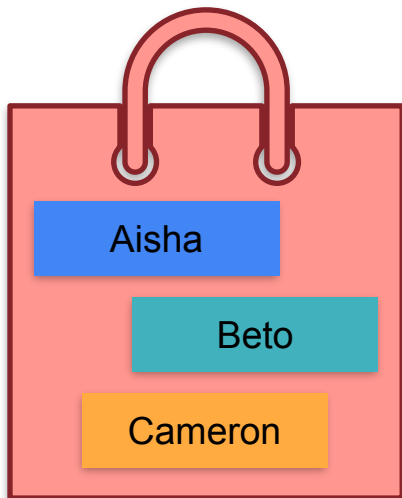
Bag With 3 Names



Bag With 3 Names



Bag With 3 Names



Aisha



Beto



Cameron

Aisha

Cameron

Beto

Beto

Aisha

Cameron

Beto

Cameron

Aisha

Cameron

Aisha

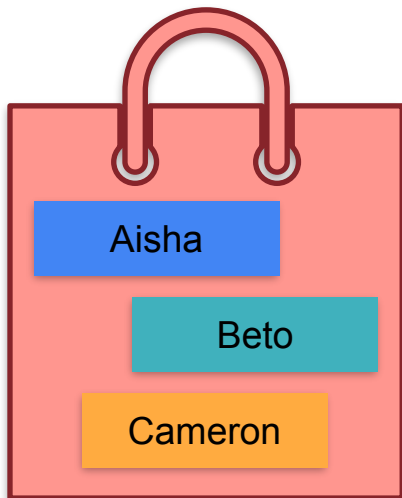
Beto

Cameron

Beto

Aisha

Bag With 3 Names



Correct



Aisha

Beto

Cameron

Aisha

Cameron

Beto

Beto

Aisha

Cameron

Beto

Cameron

Aisha

Cameron

Aisha

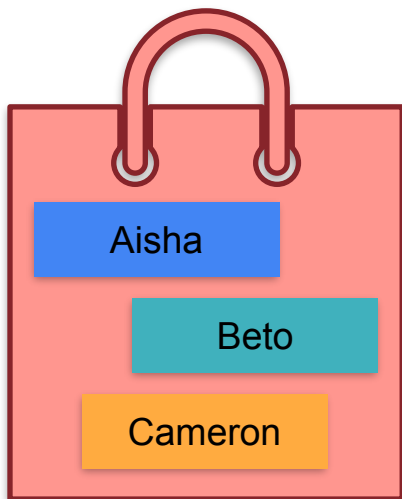
Beto

Cameron

Beto

Aisha

Bag With 3 Names



Correct



Aisha



Beto



Cameron

Aisha

Cameron

Beto

Beto

Aisha

Cameron

Beto

Cameron

Aisha

Cameron

Aisha

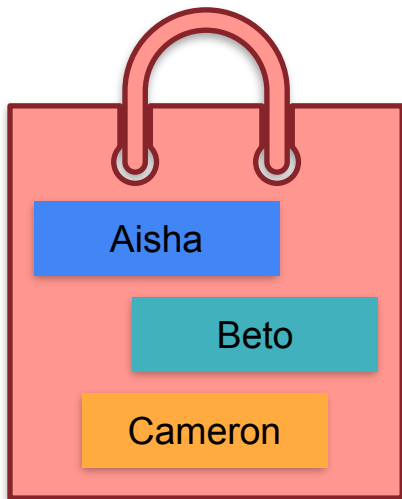
Beto

Cameron

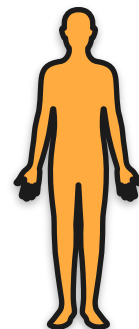
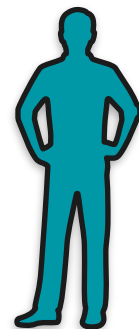
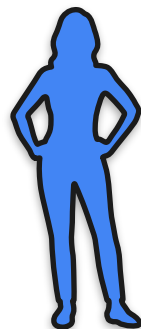
Beto

Aisha

Bag With 3 Names



Correct



3

Aisha

Beto

Cameron

1

Aisha

Cameron

Beto

1

Beto

Aisha

Cameron

0

Beto

Cameron

Aisha

0

Cameron

Aisha

Beto

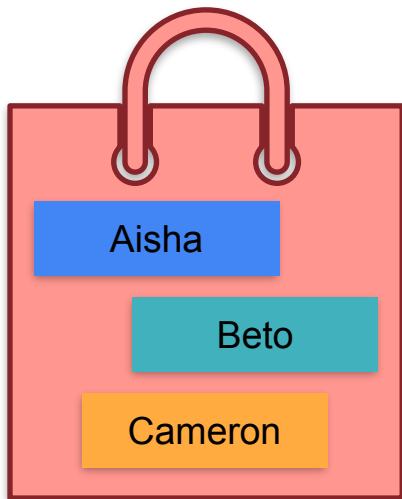
1

Cameron

Beto

Aisha

Bag With 3 Names



Correct

6

3

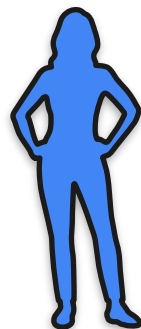
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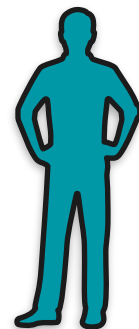
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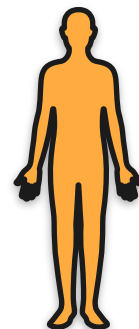
1



Aisha



Beto



Cameron

Aisha

Cameron

Beto

Beto

Aisha

Cameron

Beto

Cameron

Aisha

Cameron

Aisha

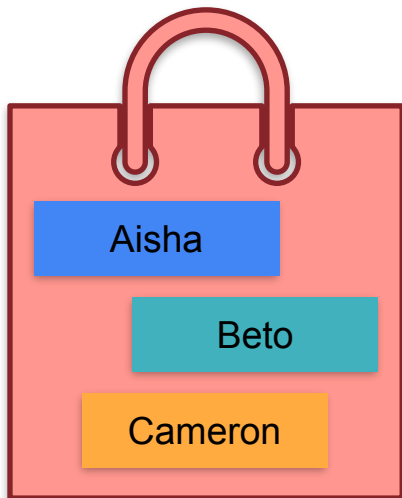
Beto

Cameron

Beto

Aisha

Bag With 3 Names



Average
1

Correct

6

3

1

1

0

0

1



Aisha

Aisha

Beto

Beto

Cameron

Cameron



Beto

Cameron

Aisha

Cameron

Aisha

Beto



Cameron

Beto

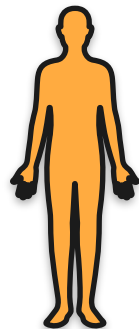
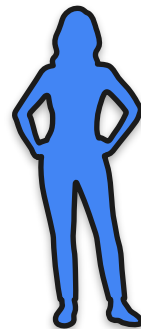
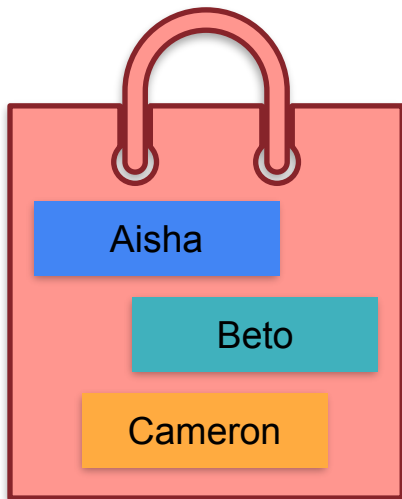
Cameron

Aisha

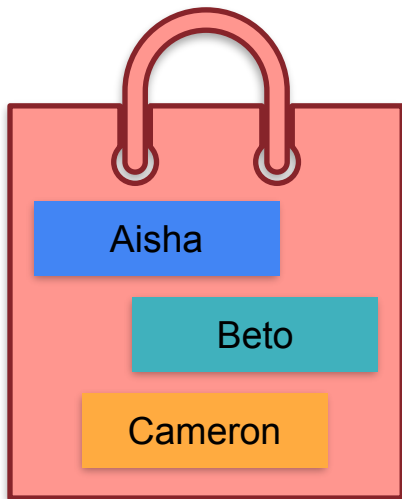
Beto

Aisha

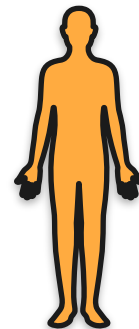
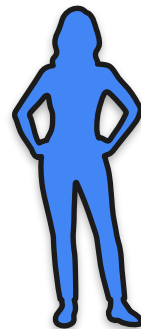
Adding Expectations



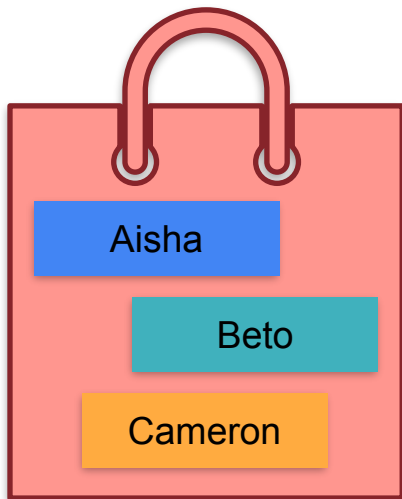
Adding Expectations



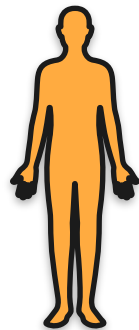
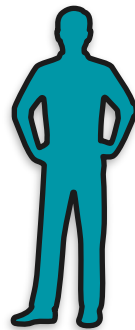
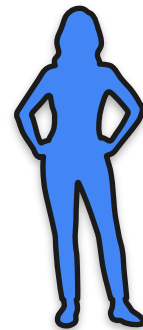
$\mathbb{E}[\text{Matches}]$



Adding Expectations



$\mathbb{E}[\text{Matches}]$



$= \mathbb{E}[A]$

Adding Expectations

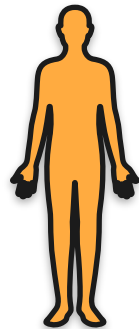
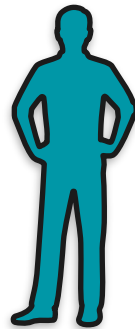
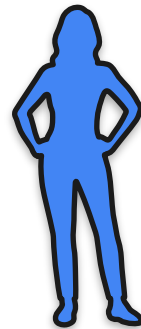


Aisha

Beto

Cameron

$\mathbb{E}[\text{Matches}]$



$= \mathbb{E}[A]$

Adding Expectations



1/3

Aisha

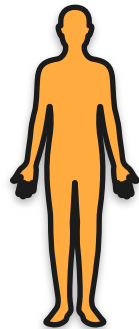
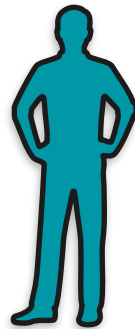
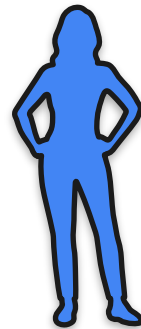
1/3

Beto

1/3

Cameron

$\mathbb{E}[\text{Matches}]$



$= \mathbb{E}[A]$

Adding Expectations



1/3

Aisha

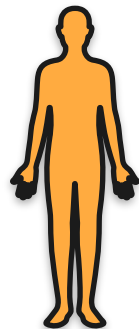
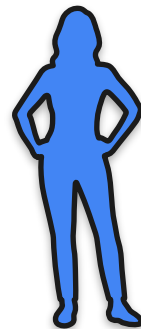
1/3

Beto

1/3

Cameron

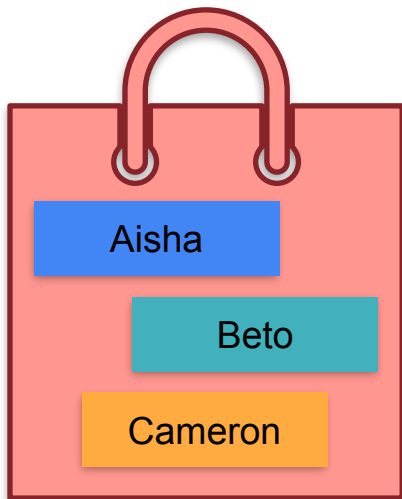
$\mathbb{E}[\text{Matches}]$



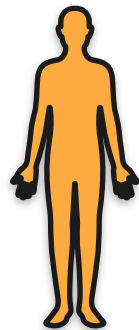
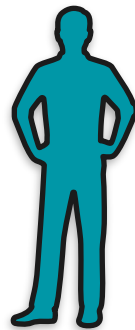
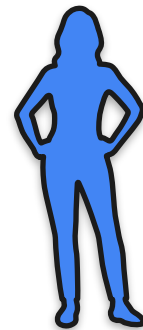
$= \mathbb{E}[A]$

$= 1/3$

Adding Expectations



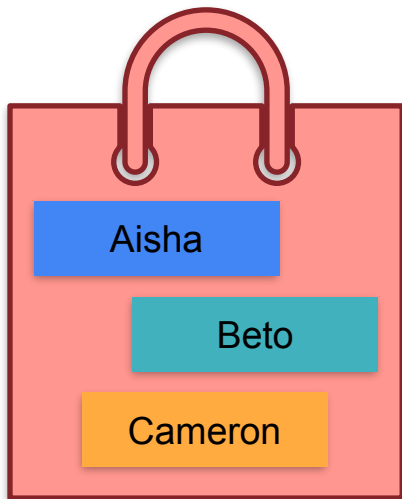
$\mathbb{E}[\text{Matches}]$



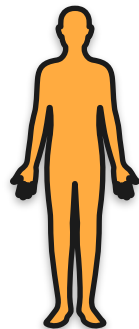
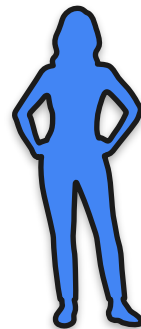
$= \mathbb{E}[A]$

$= 1/3$

Adding Expectations

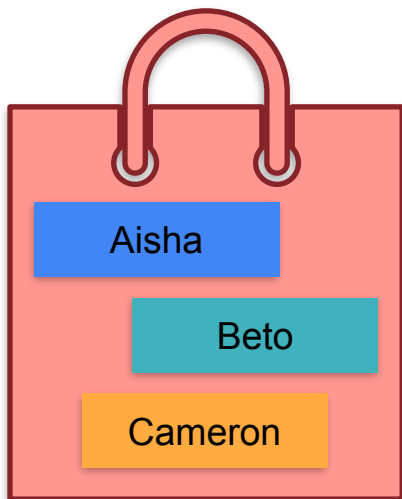


$\mathbb{E}[\text{Matches}]$

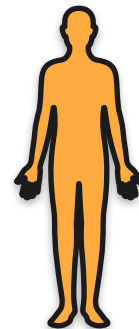
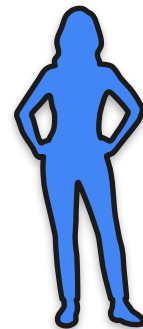


$$\begin{aligned}\mathbb{E}[\text{Matches}] &= \mathbb{E}[A] + \mathbb{E}[B] \\ &= 1/3 + 1/3\end{aligned}$$

Adding Expectations

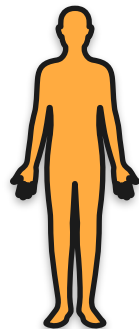
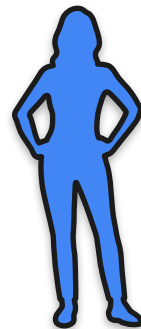
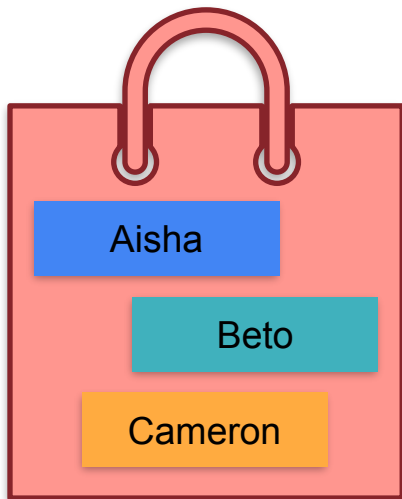


$\mathbb{E}[\text{Matches}]$



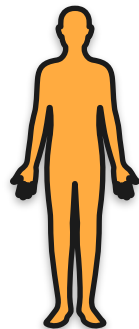
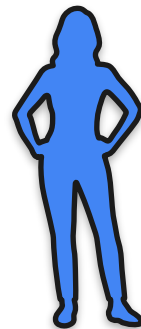
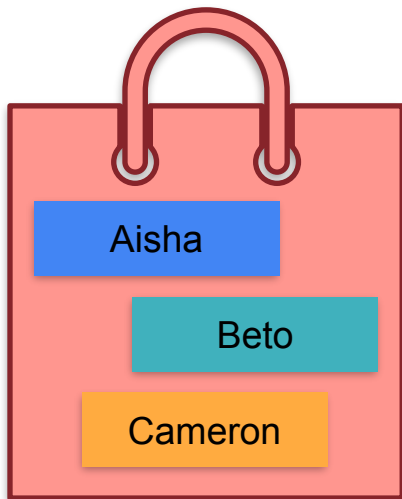
$$\begin{aligned}\mathbb{E}[\text{Matches}] &= \mathbb{E}[A] + \mathbb{E}[B] + \mathbb{E}[C] \\ &= 1/3 + 1/3 + 1/3\end{aligned}$$

Adding Expectations



$$\begin{aligned}\mathbb{E}[\text{Matches}] &= \mathbb{E}[A] + \mathbb{E}[B] + \mathbb{E}[C] \\ &= 1/3 + 1/3 + 1/3 \\ &= 1\end{aligned}$$

Adding Expectations



$$\begin{aligned}\mathbb{E}[\text{Matches}] &= \mathbb{E}[A] + \mathbb{E}[B] + \mathbb{E}[C] \\ &= 1/3 + 1/3 + 1/3 \\ &= 1\end{aligned}$$

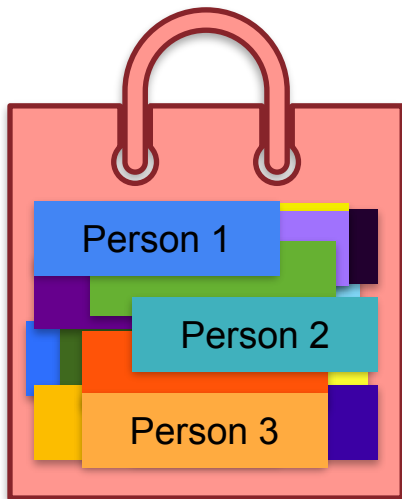
Average
1

Adding Expectations



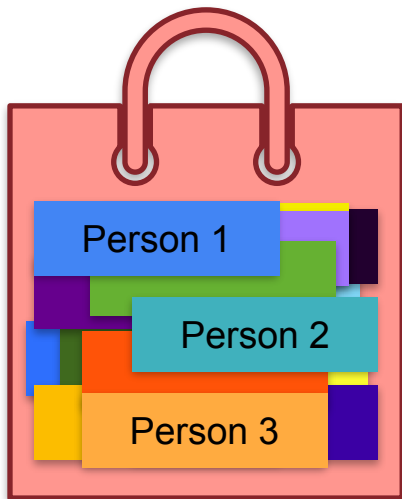
8 billion people

Adding Expectations



8 billion people

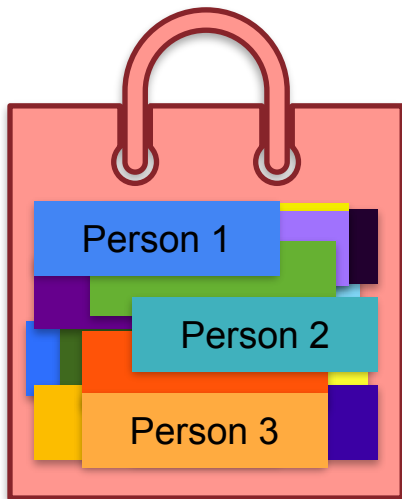
Adding Expectations



8 billion people

Adding Expectations

Expected number = ?



8 billion people

Adding Expectations



Adding Expectations



$$\mathbb{E} [\text{Matches}] = \mathbb{E} [X_{\text{person}_1}] + \mathbb{E} [X_{\text{person}_2}] + \dots + \mathbb{E} [X_{\text{person}_n}]$$

Adding Expectations



n people (n = 8 billion)

$$\mathbb{E} [\text{Matches}] = \mathbb{E} [X_{\text{person}_1}] + \mathbb{E} [X_{\text{person}_2}] + \dots + \mathbb{E} [X_{\text{person}_n}]$$

Adding Expectations



n people (n = 8 billion)

$$\mathbb{E} [\text{Matches}] = \mathbb{E} [X_{\text{person}_1}] + \mathbb{E} [X_{\text{person}_2}] + \dots + \mathbb{E} [X_{\text{person}_n}] = \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}$$

Adding Expectations



n people (n = 8 billion)

$$\begin{aligned}\mathbb{E}[\text{Matches}] &= \mathbb{E}[X_{\text{person}_1}] + \mathbb{E}[X_{\text{person}_2}] + \dots + \mathbb{E}[X_{\text{person}_n}] \\ &= \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n} \\ &= n \cdot \frac{1}{n}\end{aligned}$$

Adding Expectations



n people (n = 8 billion)

$$\begin{aligned}\mathbb{E}[\text{Matches}] &= \mathbb{E}[X_{\text{person}_1}] + \mathbb{E}[X_{\text{person}_2}] + \dots + \mathbb{E}[X_{\text{person}_n}] \\ &= \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n} \\ &= n \cdot \frac{1}{n} = 1\end{aligned}$$

Adding Expectations



n people (n = 8 billion)

$$\mathbb{E}[\text{Matches}] = \mathbb{E}[X_{\text{person}_1}] + \mathbb{E}[X_{\text{person}_2}] + \dots + \mathbb{E}[X_{\text{person}_n}] = \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}$$

In general:

$$\mathbb{E}[X_1 + X_2 + \dots + X_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \dots + \mathbb{E}[X_n]$$

$$= n \cdot \frac{1}{n} = 1$$



DeepLearning.AI

Describing Distributions

Variance

Variance Motivation: Fair Price To Play the Game

You play a game with a friend



You win 1 dollar



You lose 1 dollar

What is the fair amount of money to pay to play this game?

Variance Motivation: Fair Price To Play the Game

You play a game with a friend



You win 1 dollar



You lose 1 dollar

\$0

What is the fair amount of money to pay to play this game?

Variance Motivation: Fair Price To Play the Game

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You lose 1 dollar

What is the fair amount of money to pay to play this game?

Variance Motivation: Fair Price To Play the Game

You play a game with a friend

Game cost:



You win 1 dollar



You lose 1 dollar

What is the fair amount of money to pay to play this game?

Variance Motivation: Fair Price To Play the Game

You play a game with a friend



You win 1 dollar



You lose 1 dollar

Game cost:

\$0

What is the fair amount of money to pay to play this game?

Variance Motivation: Fair Price To Play the Game

You play a game with a friend



You win 100 dollars



You lose 100 dollars

What is the fair amount of money to pay to play this game?

Variance Motivation: Fair Price To Play the Game

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You win 100 dollars



You lose 100 dollars

\$0

What is the fair amount of money to pay to play this game?

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You play a game with a friend

Game cost:



You win 100 dollars



You lose 100 dollars

What is the fair amount of money to pay to play this game?

Variance Motivation: Fair Price To Play the Game

You play a game with a friend



You win 100 dollars



You lose 100 dollars

Game cost:

\$0

What is the fair amount of money to pay to play this game?

Variance Motivation: Fair Price To Play the Game

How do you tell these two games apart?

Game 1



You win 1 dollar



You lose 1 dollar

Game 2



You win 100 dollars



You lose 100 dollars

Variance Motivation: Fair Price To Play the Game

How do you tell these two games apart?

Game 1



You win 1 dollar



You lose 1 dollar

Game 2

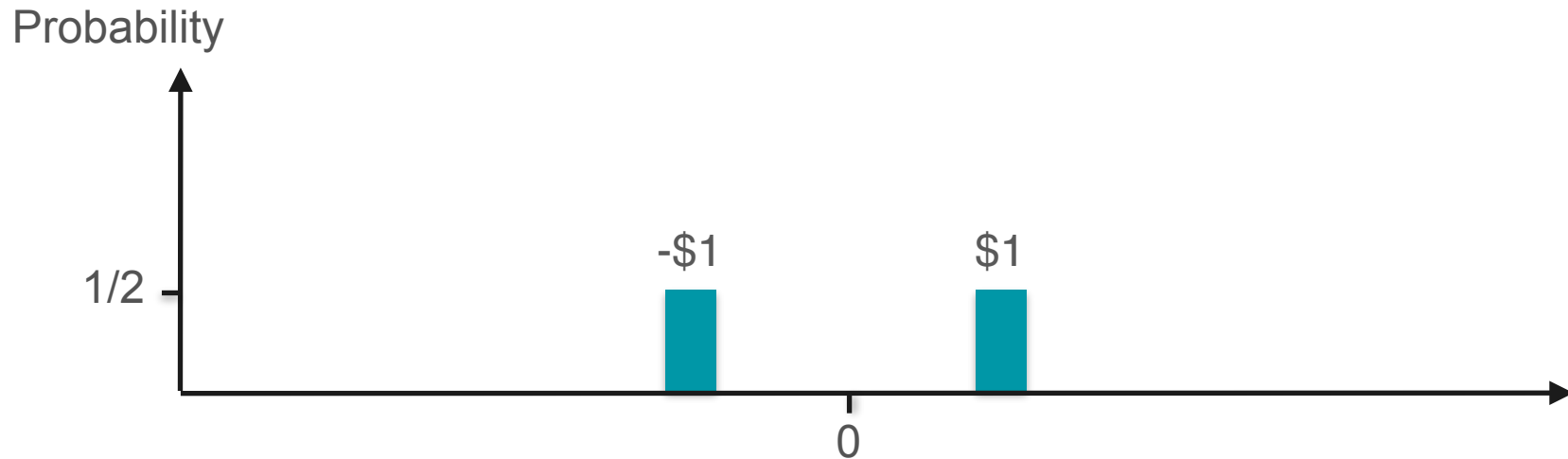
You win 100 dollars



You lose 100 dollars

Variance!

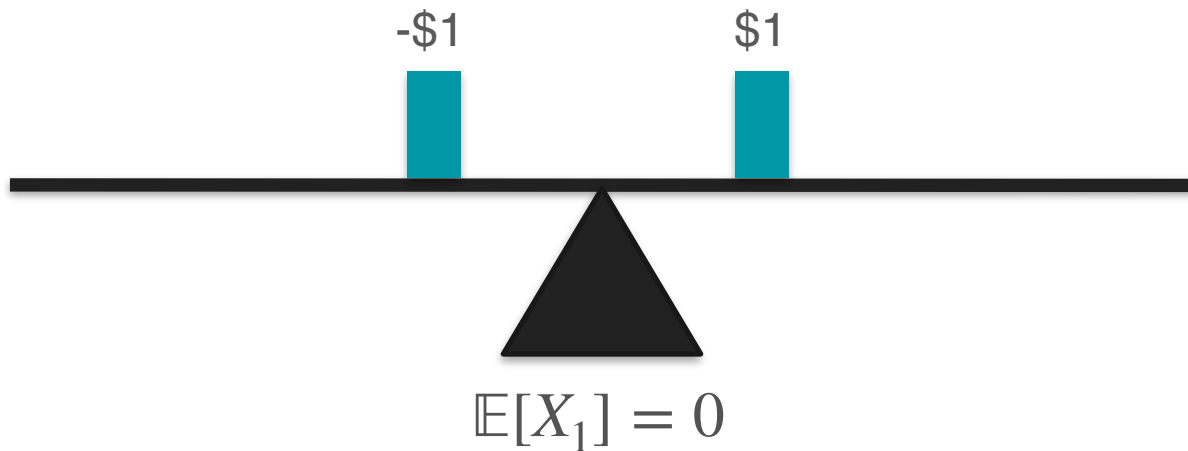
Variance Motivation: Measuring Spread



Variance Motivation: Measuring Spread

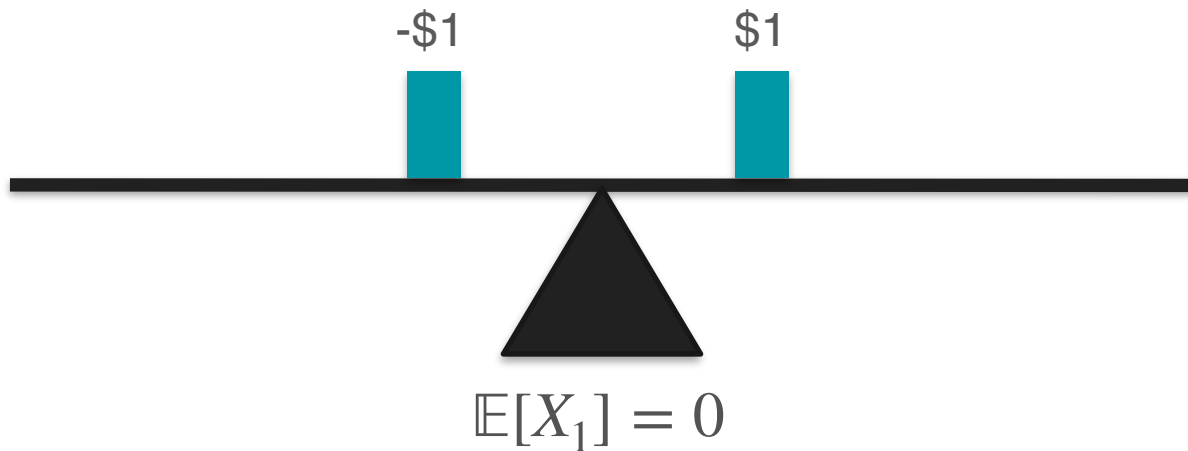


Variance Motivation: Measuring Spread

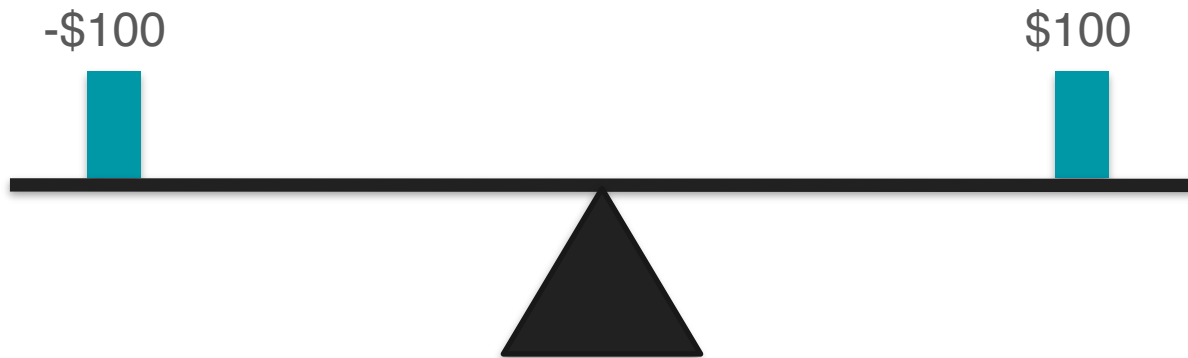


Variance Motivation: Measuring Spread

X_1 = expected amount of money gained in game 1

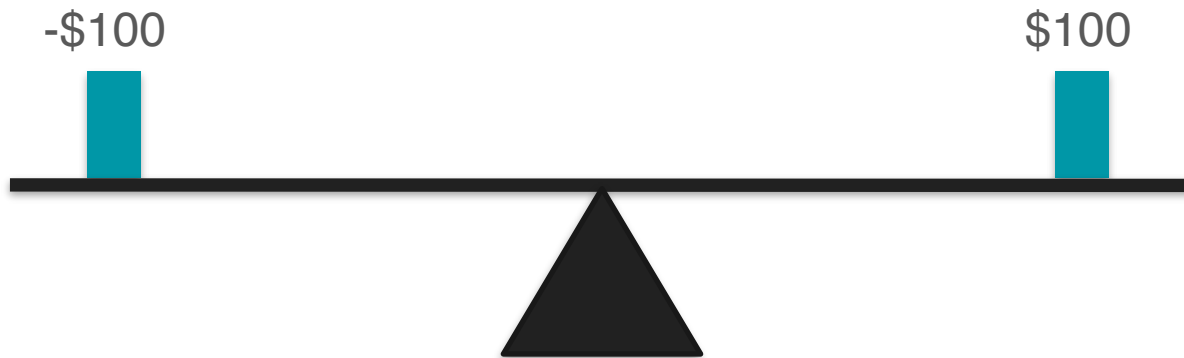


Variance Motivation: Measuring Spread



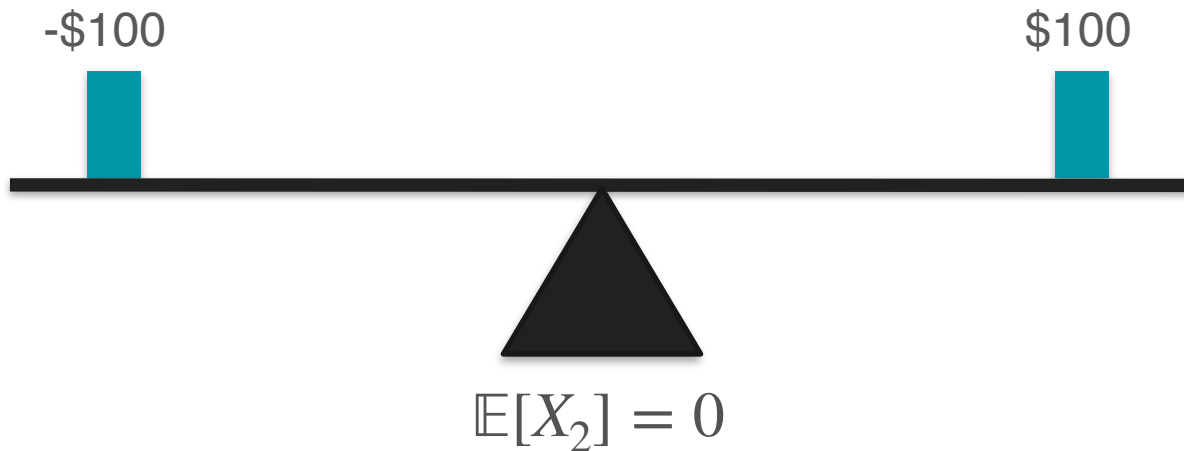
Variance Motivation: Measuring Spread

X_2 = expected amount of money gained in game 2

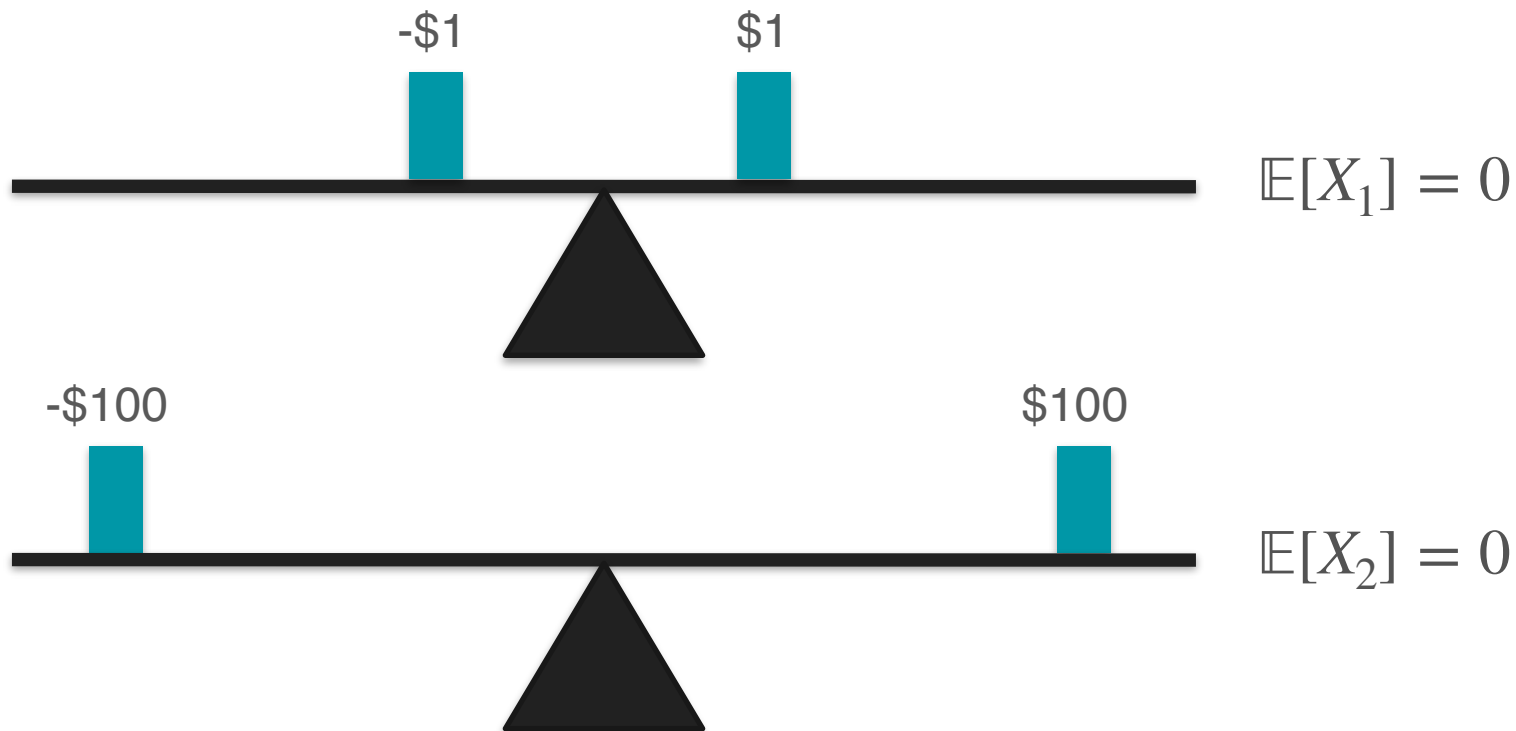


Variance Motivation: Measuring Spread

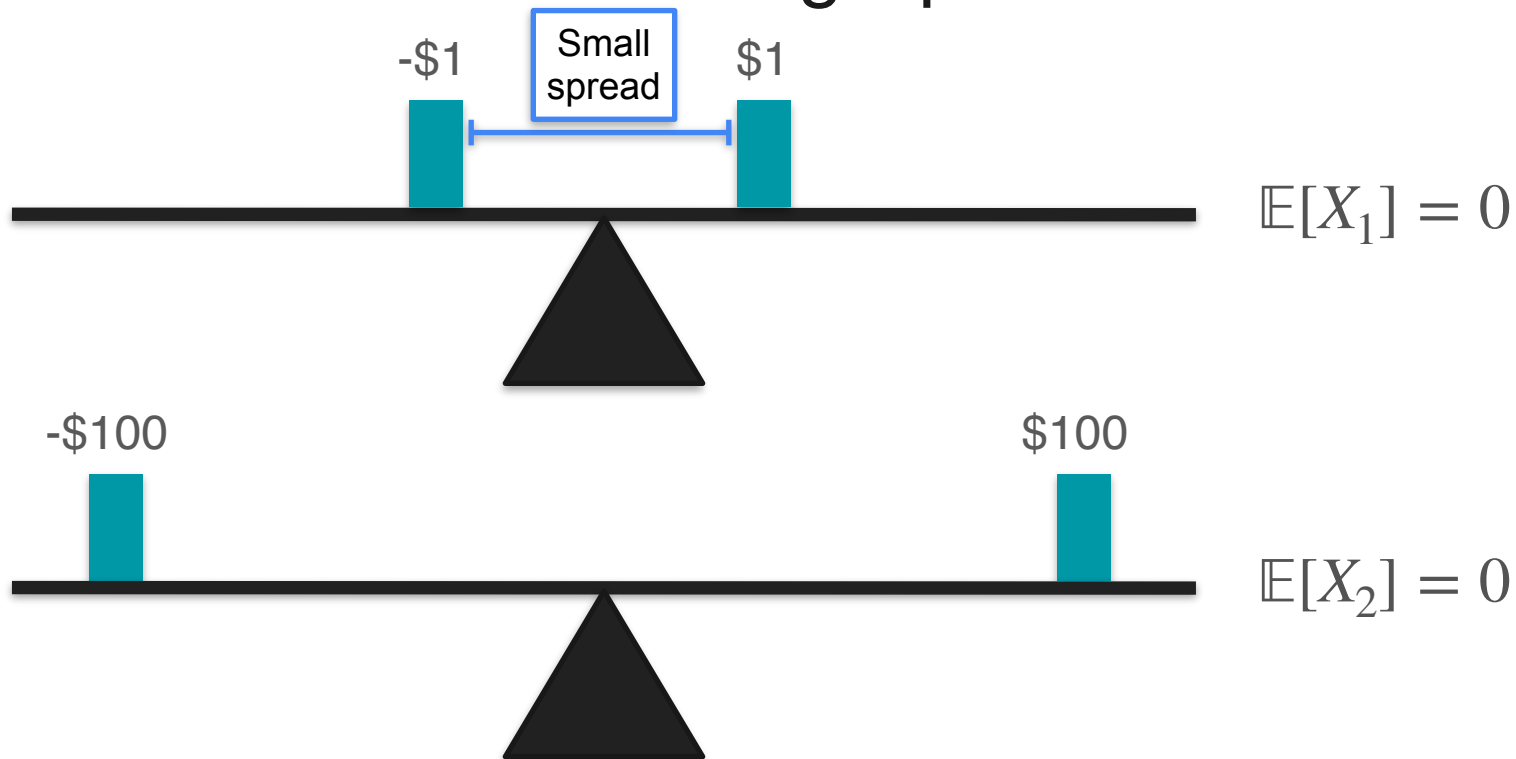
X_2 = expected amount of money gained in game 2



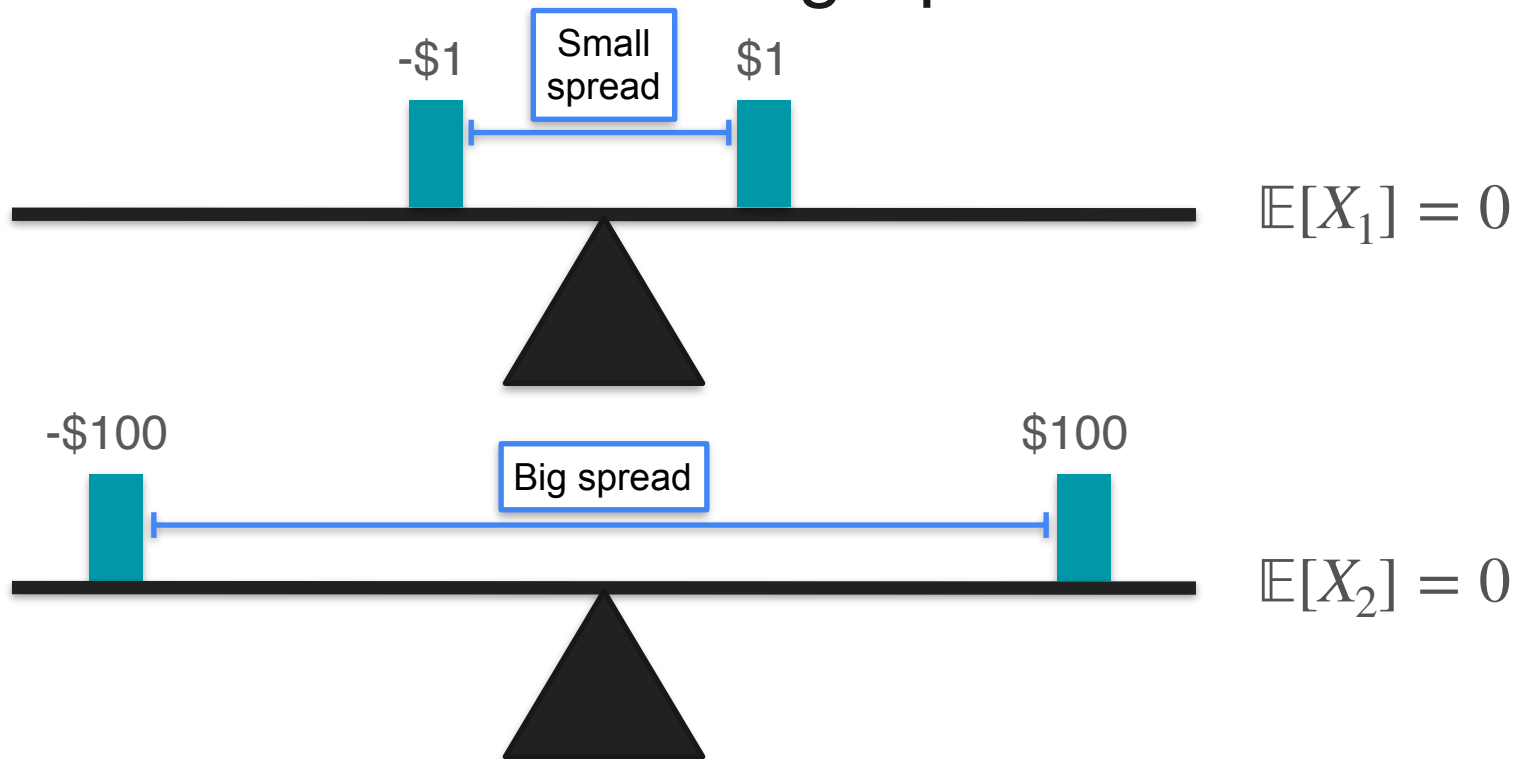
Variance Motivation: Measuring Spread



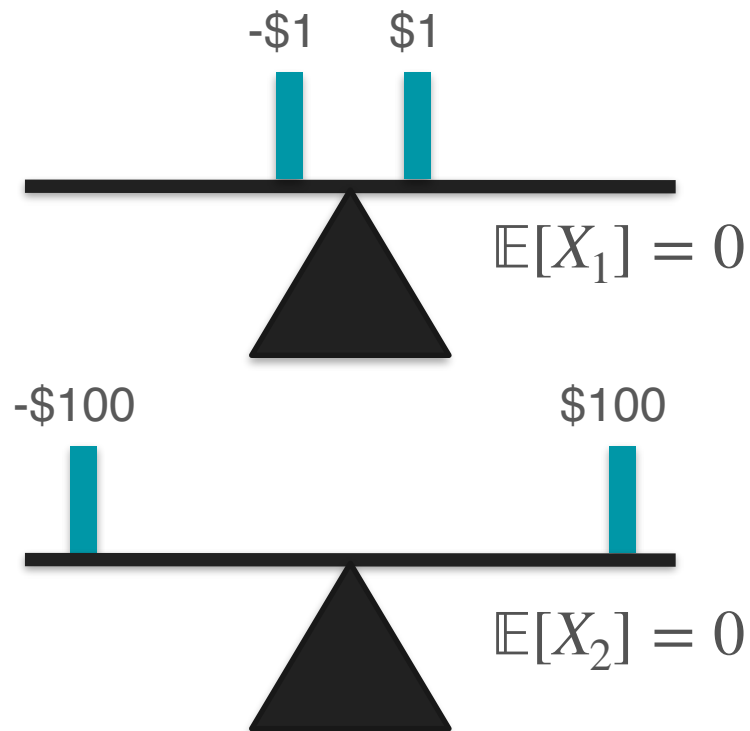
Variance Motivation: Measuring Spread



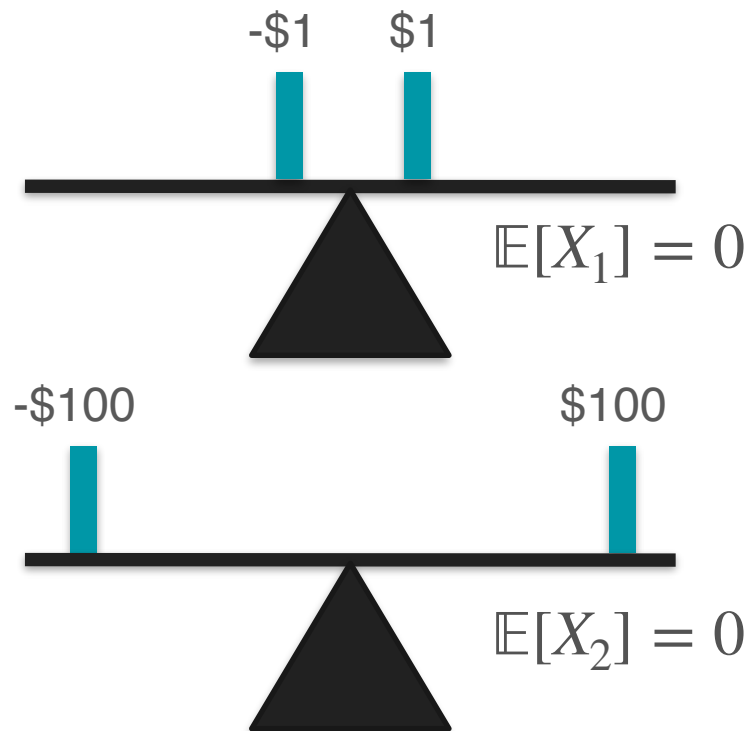
Variance Motivation: Measuring Spread



Variance Motivation: Measuring Spread

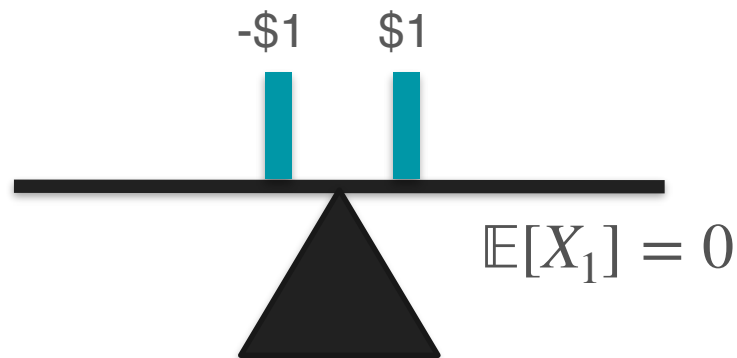


Variance Motivation: Measuring Spread



$$\mathbb{E}[X_1] = \frac{(1) + (-1)}{2} = 0$$

Variance Motivation: Measuring Spread

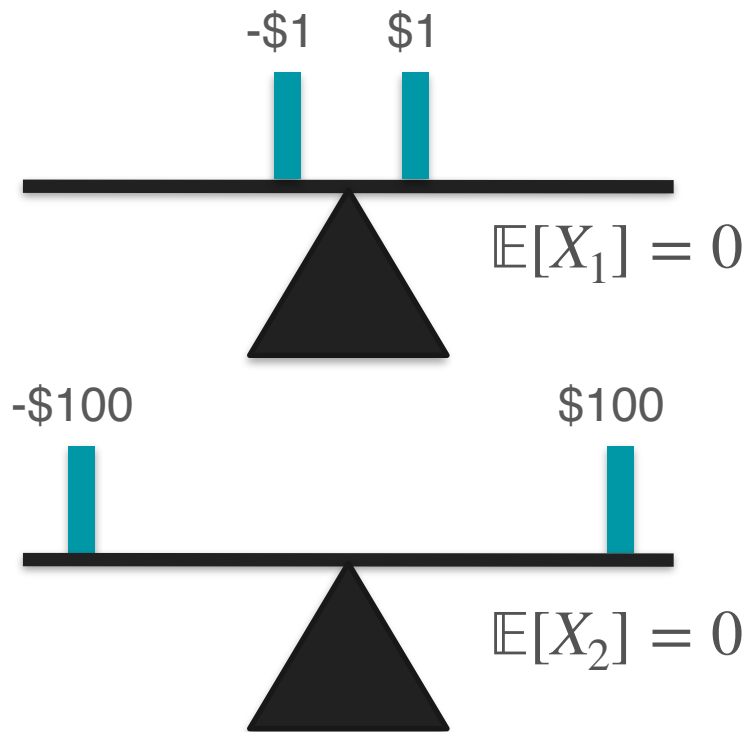


$$\mathbb{E}[X_1] = \frac{(1) + (-1)}{2} = 0$$



$$\mathbb{E}[X_2] = \frac{(100) + (-100)}{2} = 0$$

Variance Motivation: Measuring Spread

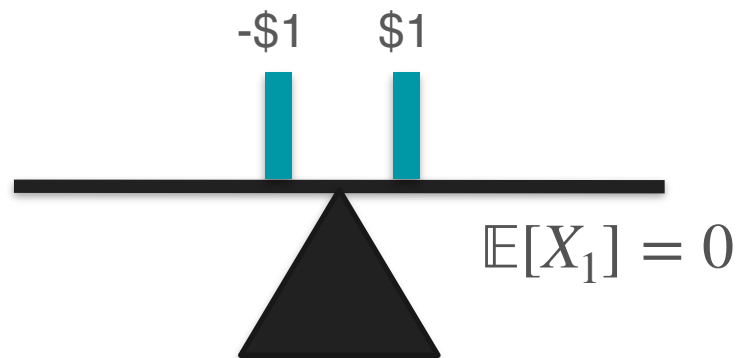


$$\mathbb{E}[X_1] = \frac{(1) + (-1)}{2} = 0$$

Turn into positive?

$$\mathbb{E}[X_2] = \frac{(100) + (-100)}{2} = 0$$

Variance Motivation: Measuring Spread



$$\mathbb{E}[X_1] = \frac{(1)^2 + (-1)^2}{2} = 0$$



$$\mathbb{E}[X_2] = \frac{(100)^2 + (-100)^2}{2} = 0$$

Variance Motivation: Measuring Spread



$$\mathbb{E}[X_1] = \frac{(1)^2 + (-1)^2}{2} = 0$$



$$\mathbb{E}[X_2] = \frac{(100)^2 + (-100)^2}{2} = 0$$

Variance Motivation: Measuring Spread



$$\mathbb{E}[X_1^2] = \frac{(1)^2 + (-1)^2}{2} = 0$$



$$\mathbb{E}[X_2^2] = \frac{(100)^2 + (-100)^2}{2} = 0$$

Variance Motivation: Measuring Spread



$$\mathbb{E}[X_1^2] = \frac{(1)^2 + (-1)^2}{2} = 1$$



$$\mathbb{E}[X_2^2] = \frac{(100)^2 + (-100)^2}{2} = 0$$

Variance Motivation: Measuring Spread

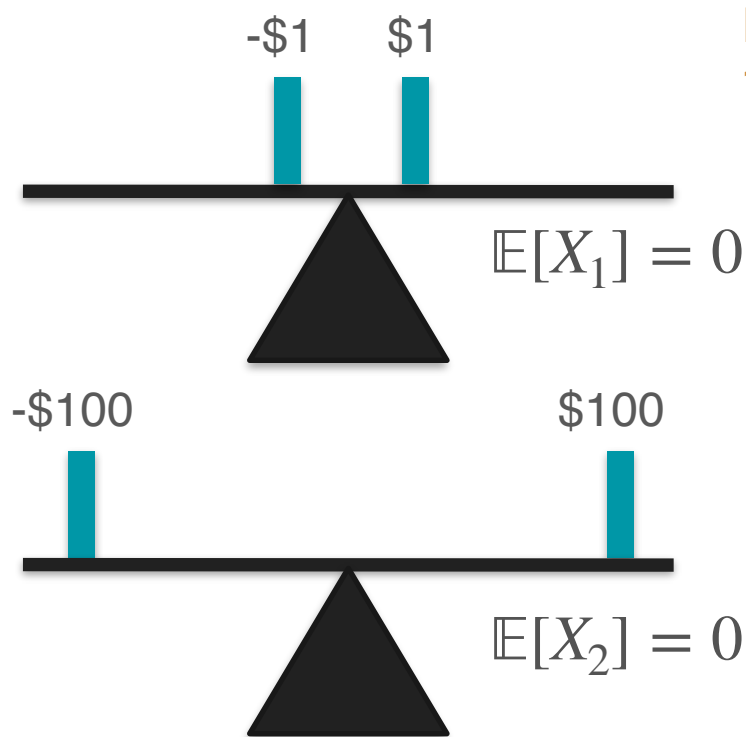


$$\mathbb{E}[X_1^2] = \frac{(1)^2 + (-1)^2}{2} = 1$$



$$\mathbb{E}[X_2^2] = \frac{(100)^2 + (-100)^2}{2} = 10,000$$

Variance Motivation: Measuring Spread

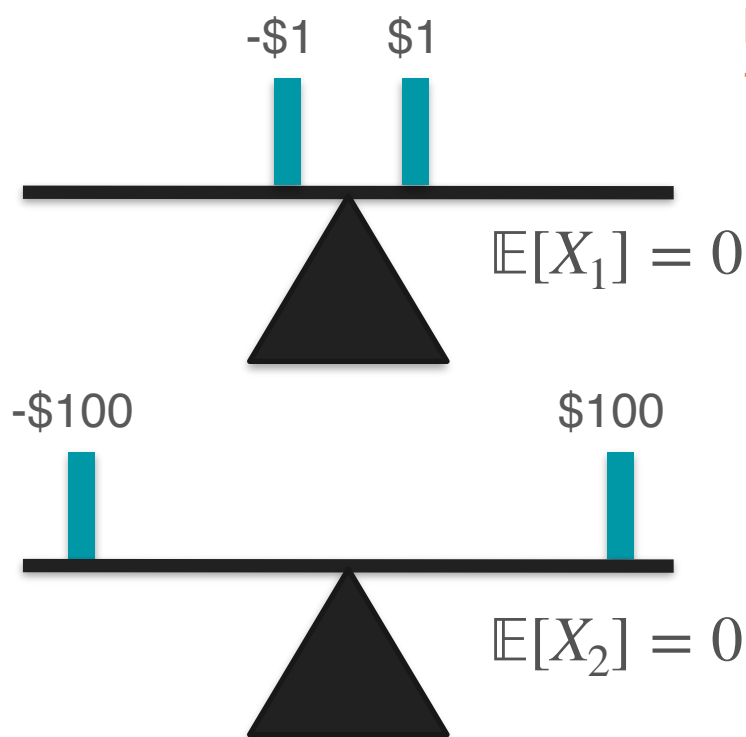


Key for telling game
1 and game 2 apart!

$$\mathbb{E}[X_1^2] = \frac{(1)^2 + (-1)^2}{2} = 1$$

$$\mathbb{E}[X_2^2] = \frac{(100)^2 + (-100)^2}{2} = 10,000$$

Variance Motivation: Measuring Spread



Key for telling game
1 and game 2 apart!

$$\mathbb{E}[X_1^2] = \frac{(1)^2 + (-1)^2}{2} = 1$$

$$\mathbb{E}[X_2^2] = \frac{(100)^2 + (-100)^2}{2} = 10,000$$

Measure of spread

Variance Motivation: Measuring Spread

$$\mathbb{E}[X^2]$$

Variance Motivation: Measuring Spread

$$\mathbb{E}[X^2]$$

Almost...

Variance Motivation: Centering With Mean

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

Game 2



You win 6 dollars



You win 4 dollars

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

Game 2



You win 6 dollars



You win 4 dollars

How risky are these two games in comparison?

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

Game 2



You win 6 dollars



You win 4 dollars

How risky are these two games in comparison?

Hint: Think of the spread

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

Game 2



You win 6 dollars



You win 4 dollars

They are equally risky

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

$$\mathbb{E}[X_1]^2 = \frac{(-1)^2 + (1)^2}{2} = 1$$

Game 2



You win 6 dollars



You win 4 dollars

$$\mathbb{E}[X_2]^2 = \frac{(4)^2 + (6)^2}{2} = 26$$

Variance Motivation: Centering With Mean

Game 1



You win 1 dollar



You lose 1 dollar

$$\mathbb{E}[X_1]^2 = \frac{(-1)^2 + (1)^2}{2} = 1$$



Same risk?



$$\mathbb{E}[X_2]^2 = \frac{(4)^2 + (6)^2}{2} = 26$$

Game 2



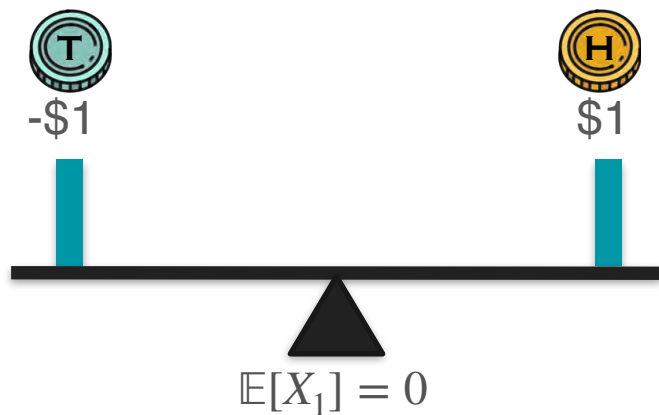
You win 6 dollars



You win 4 dollars

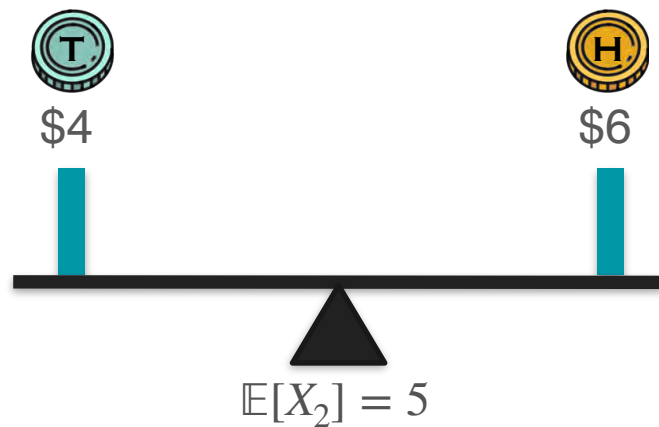
Variance Motivation: Centering With Mean

$$\mathbb{E}[X_1]^2 = \frac{(-1)^2 + (1)^2}{2} = 1$$



Game 1

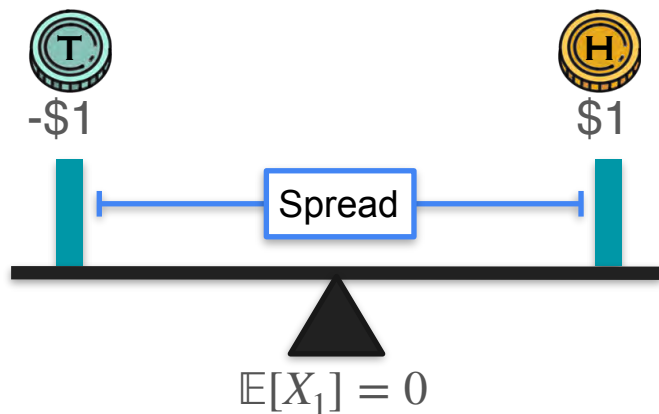
$$\mathbb{E}[X_2]^2 = \frac{(4)^2 + (6)^2}{2} = 26$$



Game 2

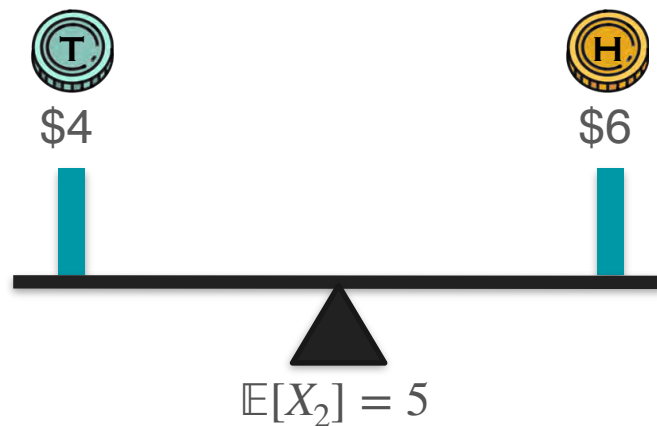
Variance Motivation: Centering With Mean

$$\mathbb{E}[X_1]^2 = \frac{(-1)^2 + (1)^2}{2} = 1$$



Game 1

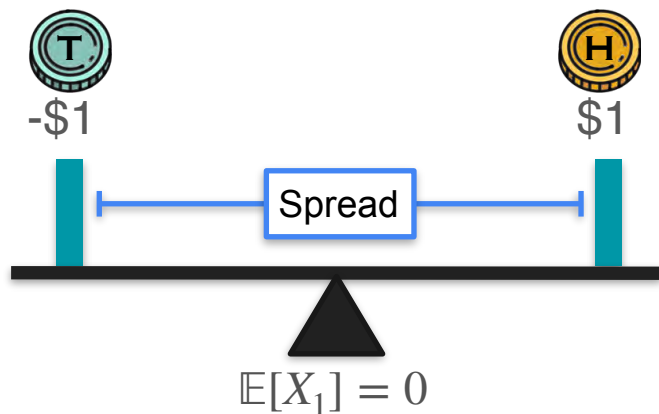
$$\mathbb{E}[X_2]^2 = \frac{(4)^2 + (6)^2}{2} = 26$$



Game 2

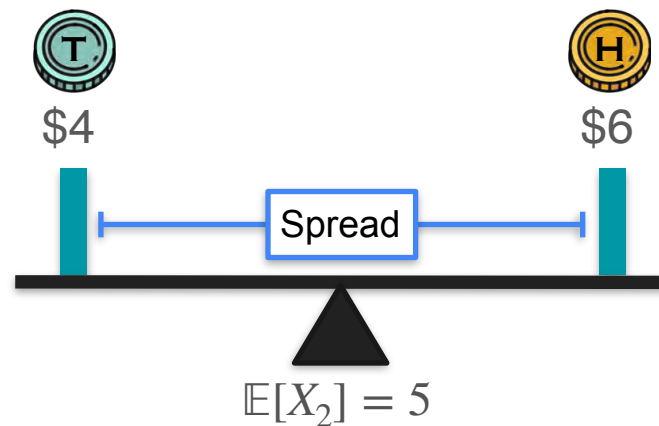
Variance Motivation: Centering With Mean

$$\mathbb{E}[X_1]^2 = \frac{(-1)^2 + (1)^2}{2} = 1$$



Game 1

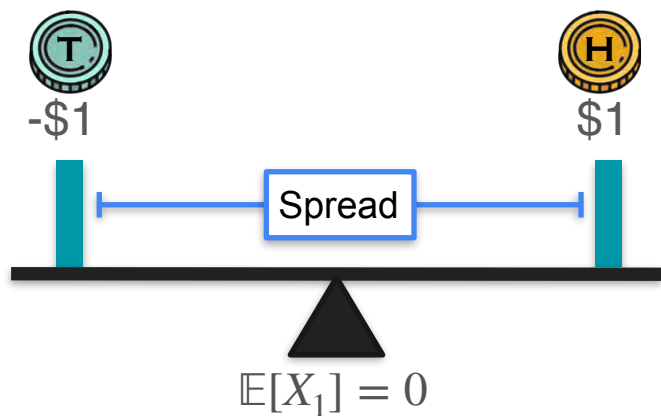
$$\mathbb{E}[X_2]^2 = \frac{(4)^2 + (6)^2}{2} = 26$$



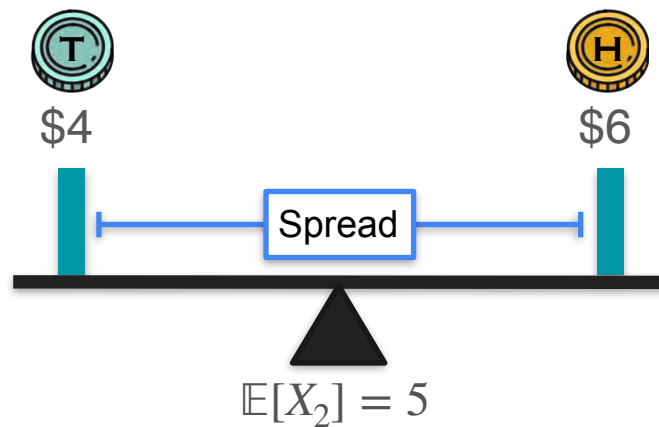
Game 2

Variance Motivation: Centering With Mean

$$\mathbb{E}[X_1]^2 = \frac{(-1)^2 + (1)^2}{2} = 1 \quad \longleftarrow \text{Same spread?} \longrightarrow \quad \mathbb{E}[X_2]^2 = \frac{(4)^2 + (6)^2}{2} = 26$$



Game 1



Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$

1

1



-\$1



\$1

$$\mathbb{E}[X_1] = 0$$

Game 1

1

1



\$4



\$6

$$\mathbb{E}[X_2] = 5$$

Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$

1

1



-\$1



\$1

Spread

$$\mathbb{E}[X_1] = 0$$

Game 1

1

1



\$4



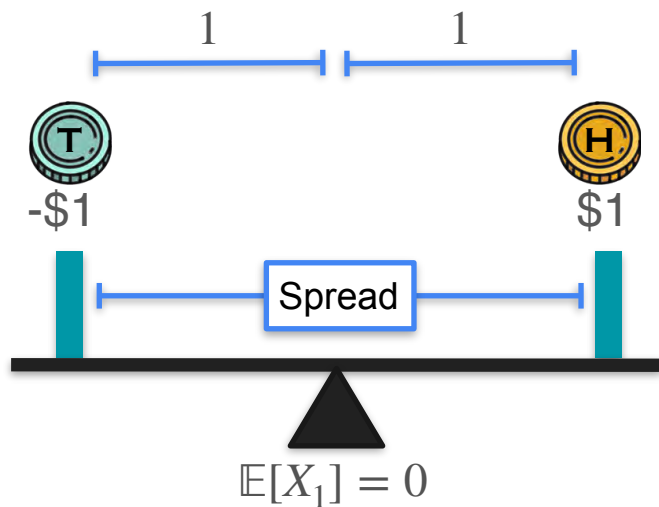
\$6

$$\mathbb{E}[X_2] = 5$$

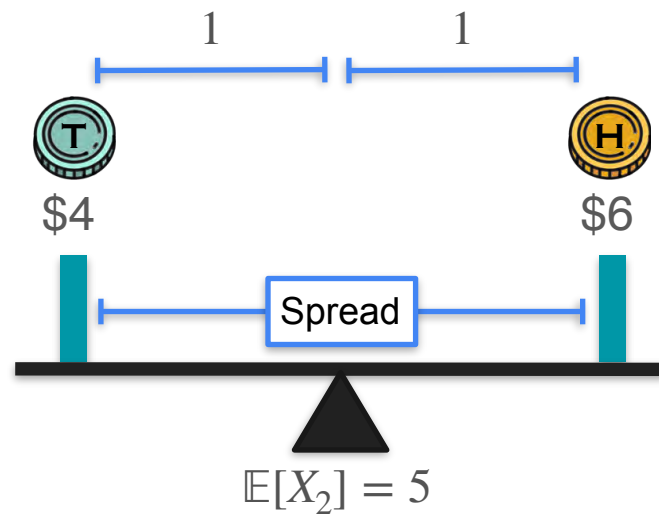
Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$



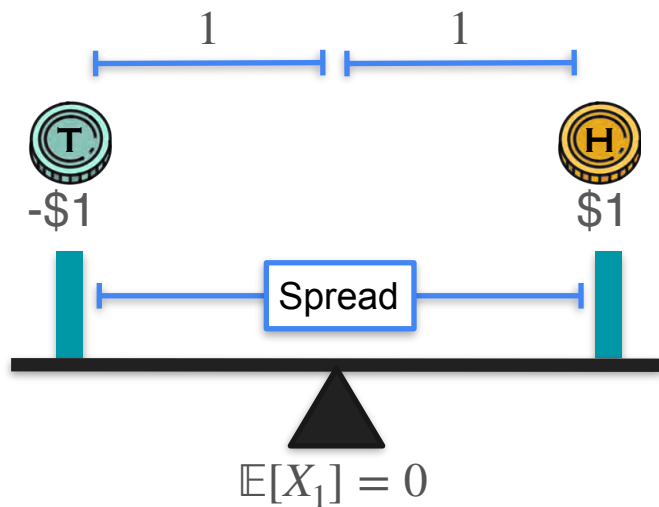
Game 1



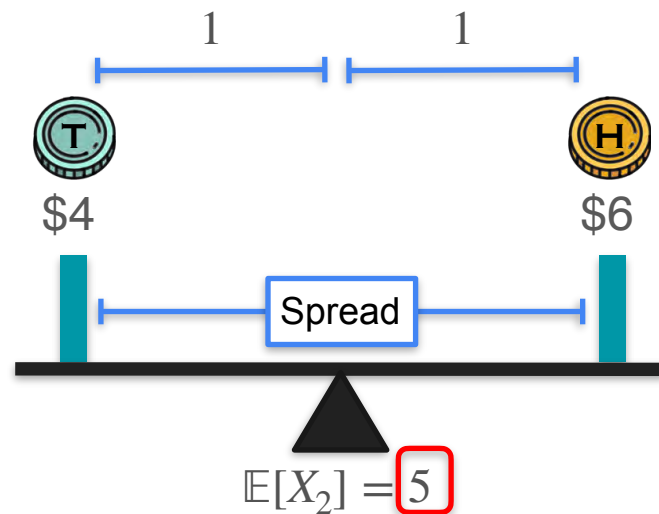
Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$



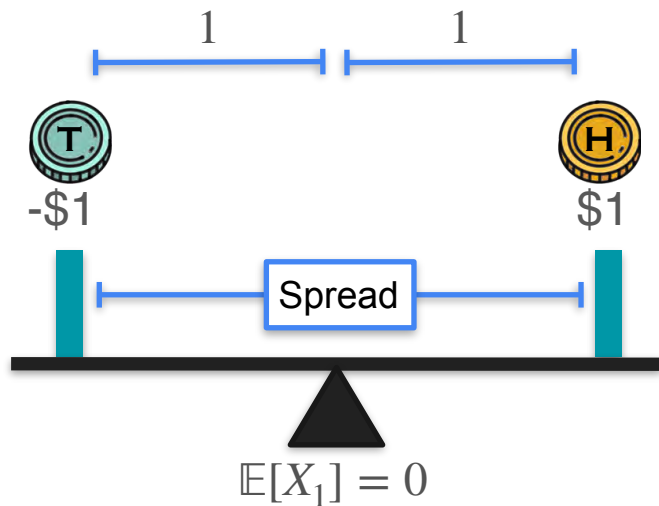
Game 1



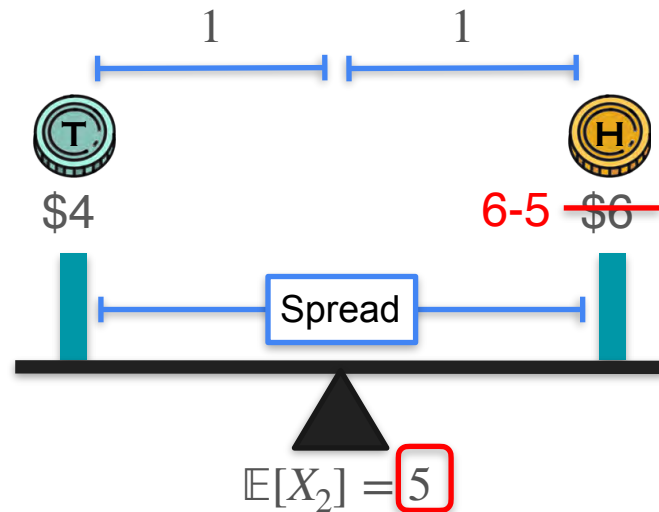
Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$



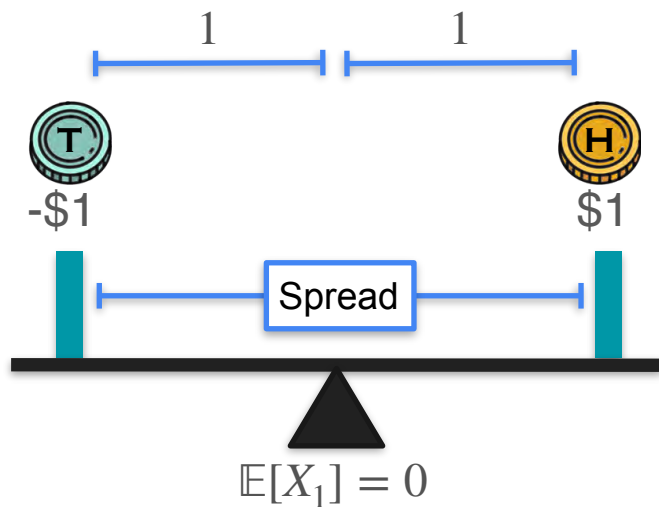
Game 1



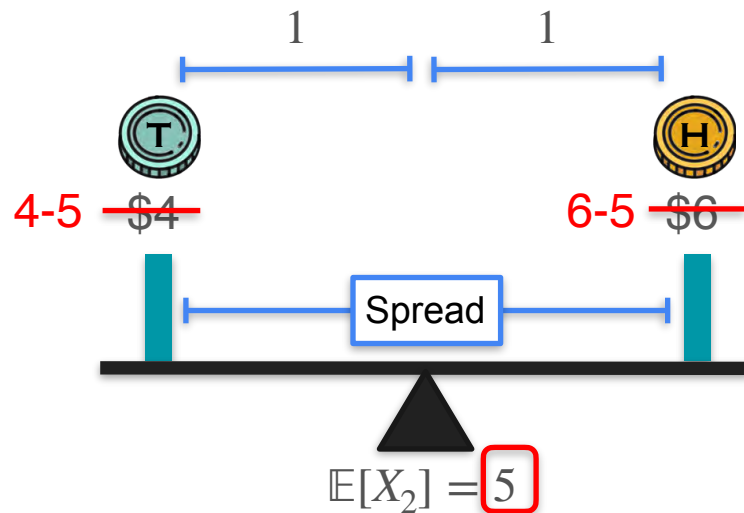
Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$



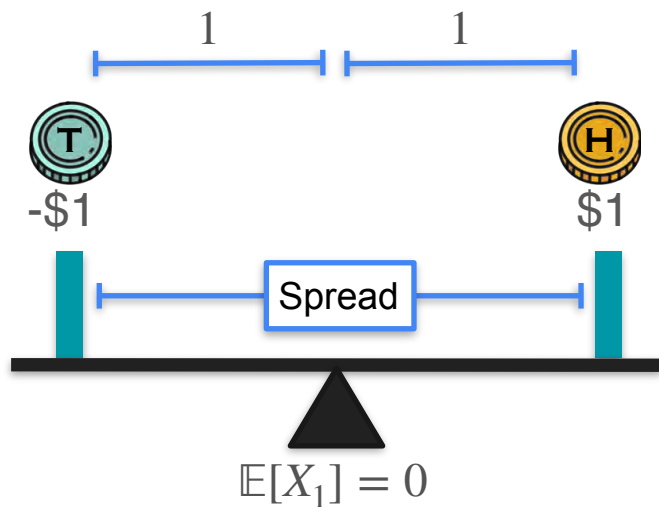
Game 1



Game 2

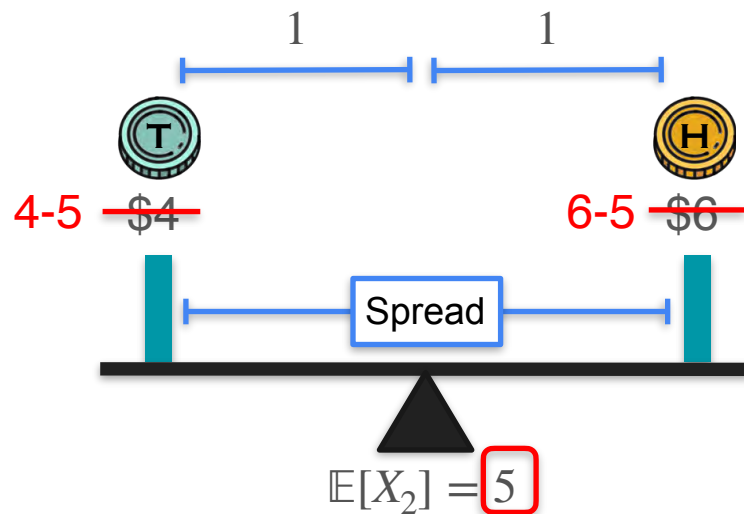
Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$



Game 1

$$\mathbb{E}[(X_2 - 5)^2]$$

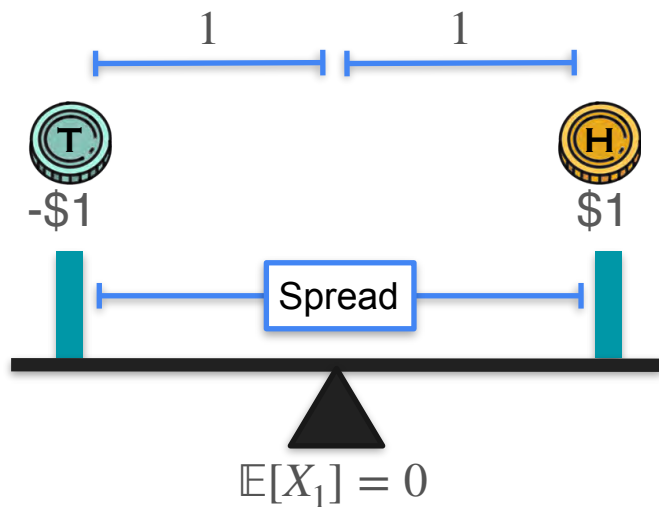


Game 2

Variance: Spread and Shift

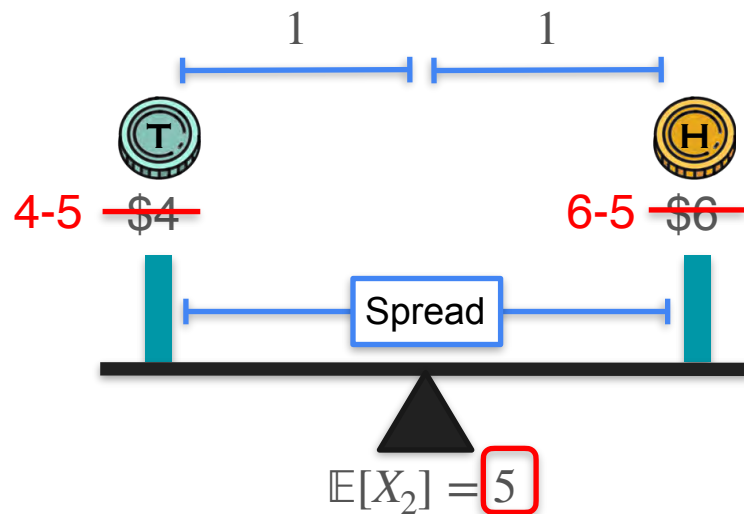
$$\mathbb{E}[X_1^2] = 1$$

Mean: μ



Game 1

$$\mathbb{E}[(X_2 - 5)^2]$$



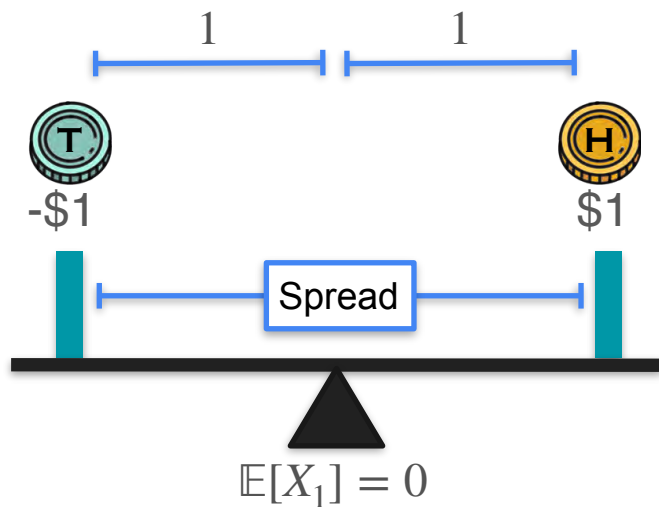
Game 2

Variance: Spread and Shift

$$\mathbb{E}[X_1^2] = 1$$

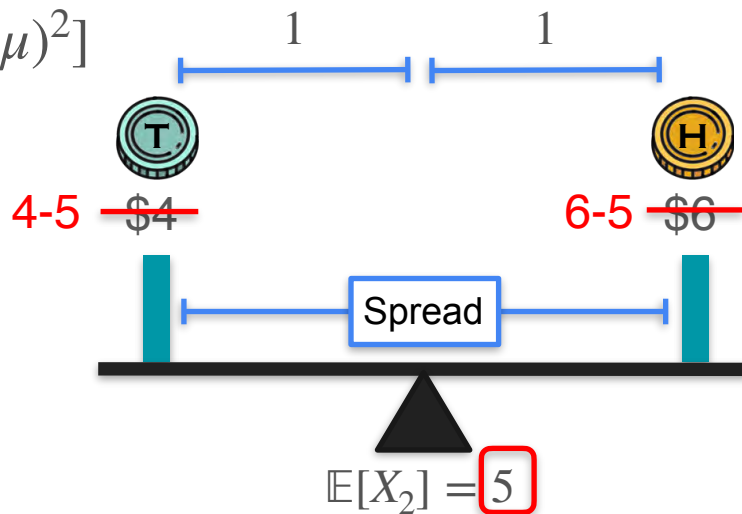
Mean: μ

$$\text{Variance: } \mathbb{E}[(X - \mu)^2]$$



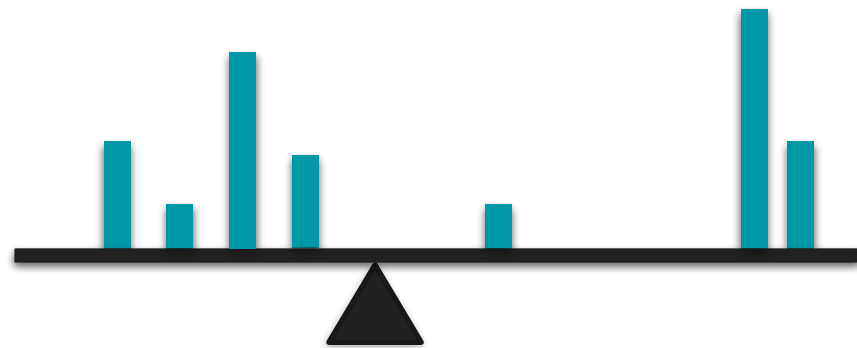
Game 1

$$\mathbb{E}[(X_2 - 5)^2]$$

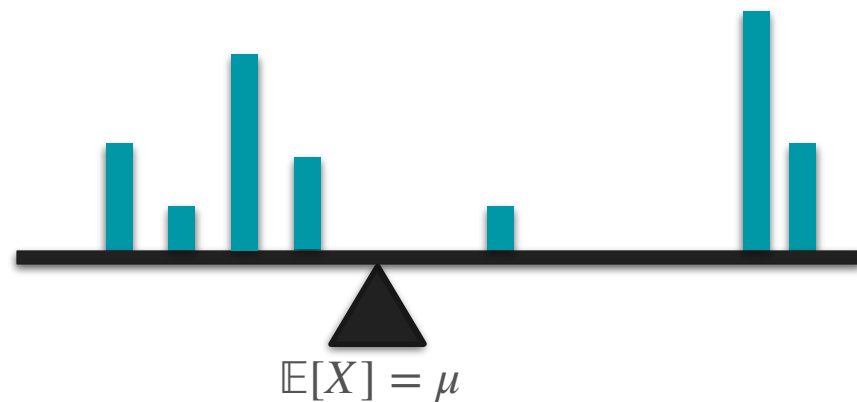


Game 2

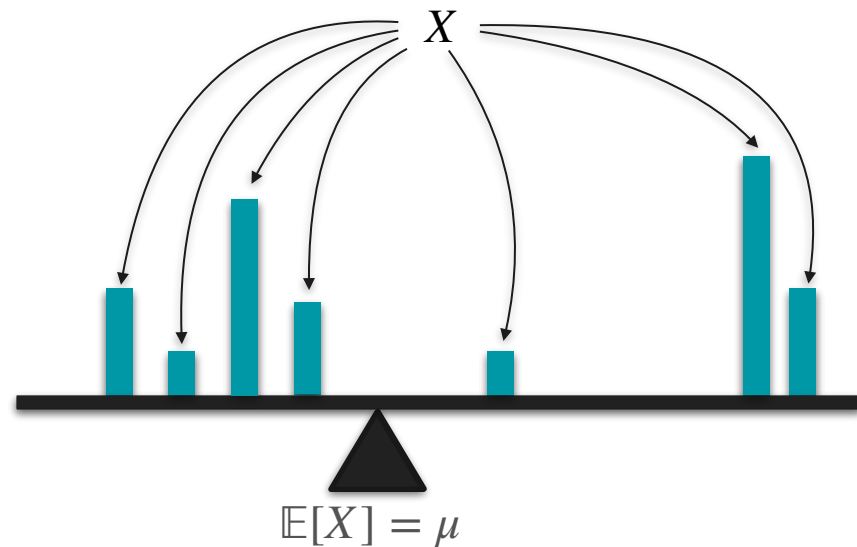
Variance Formula



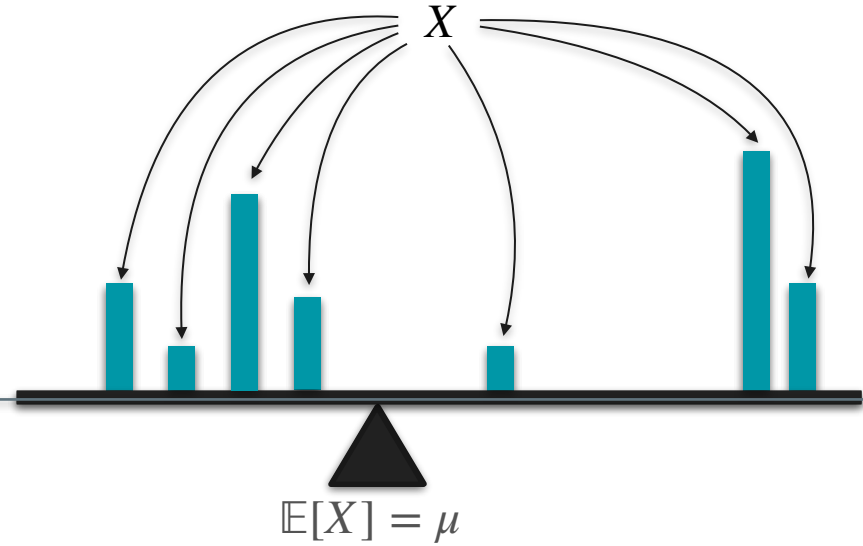
Variance Formula



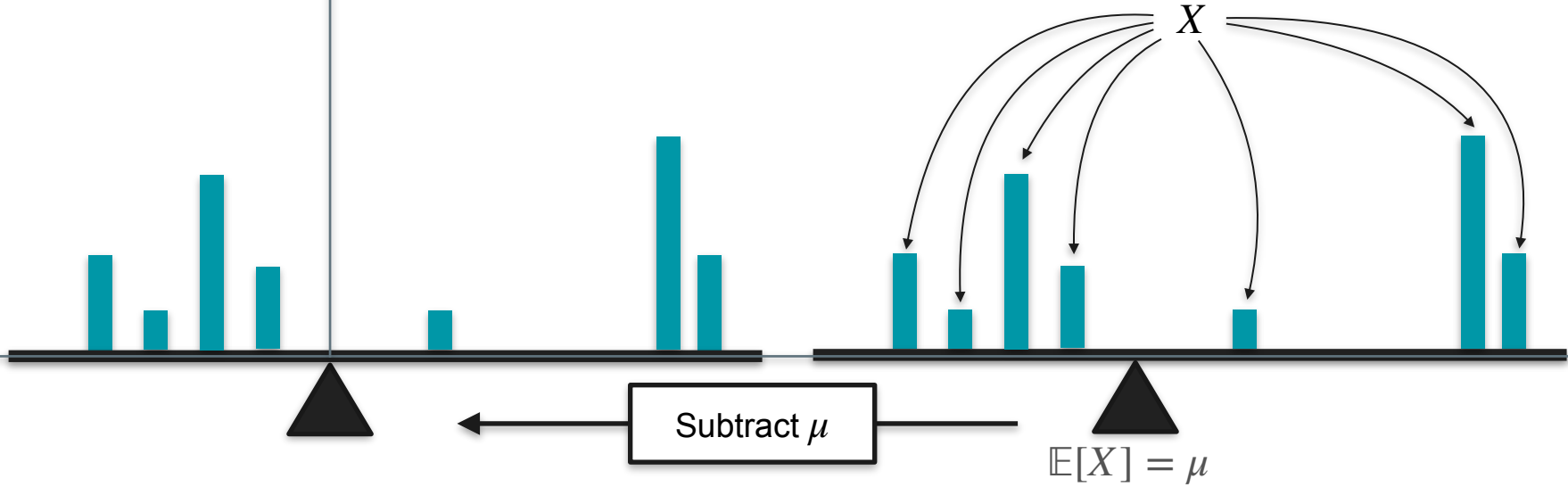
Variance Formula



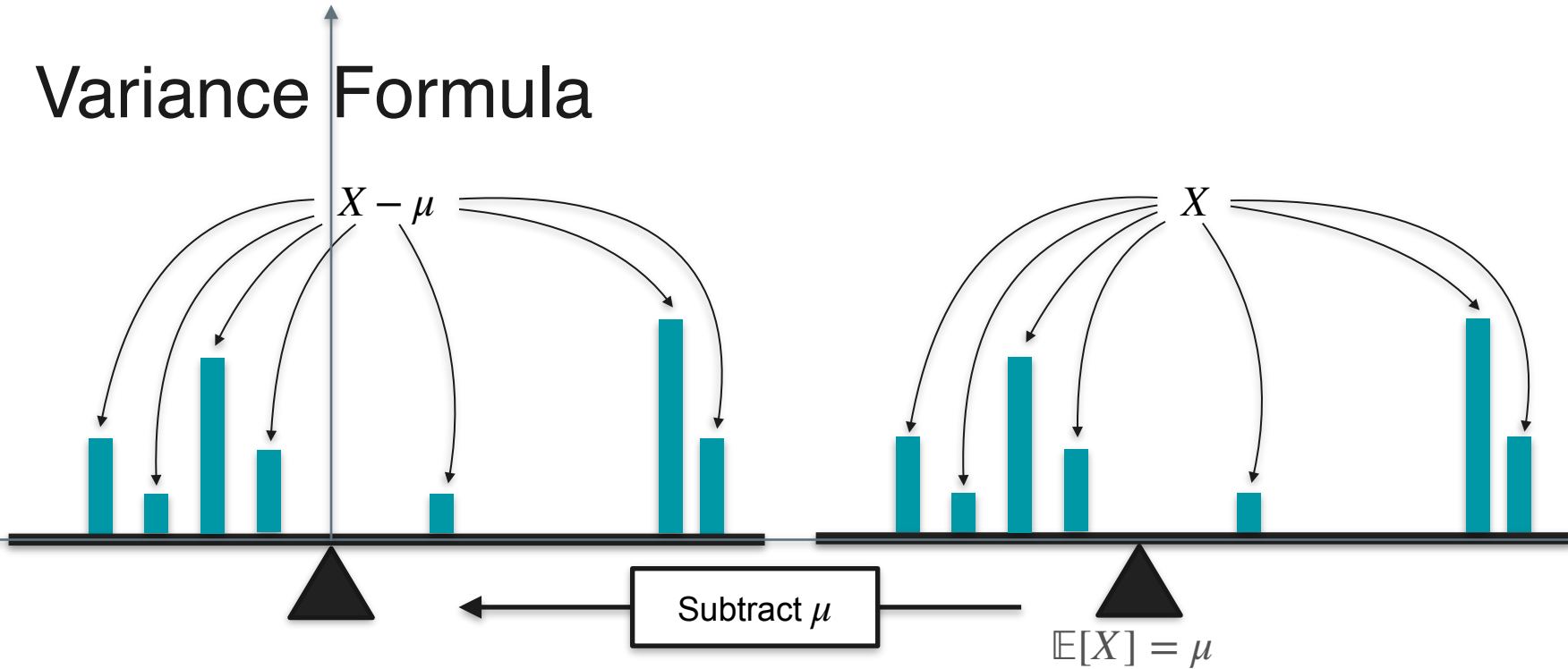
Variance Formula



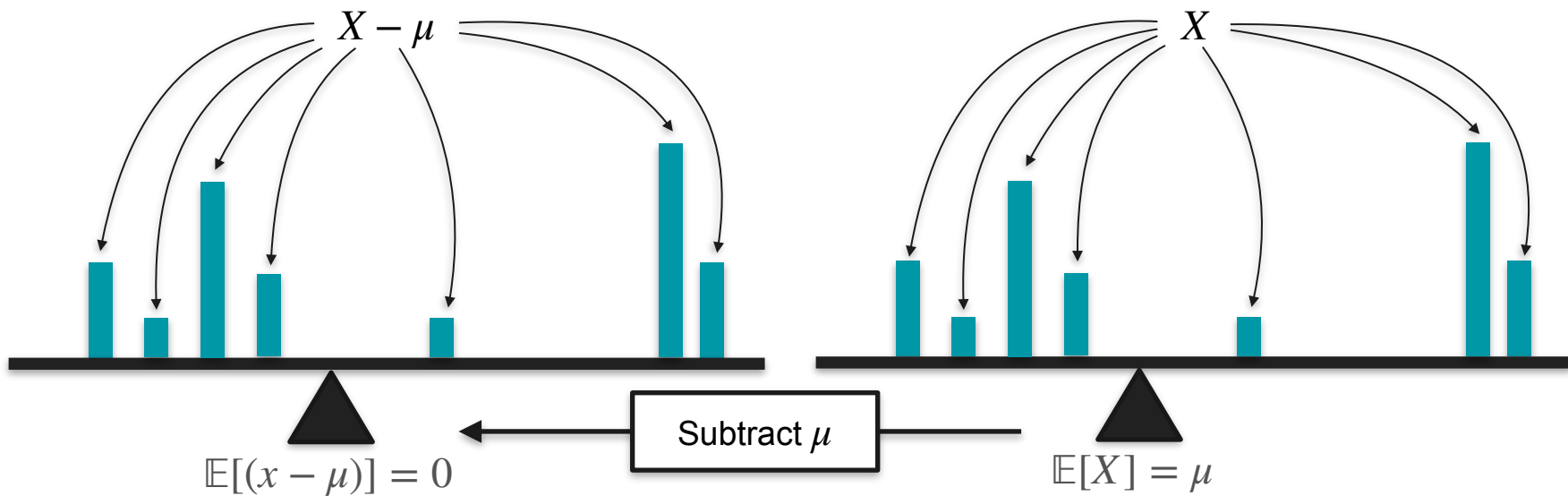
Variance Formula



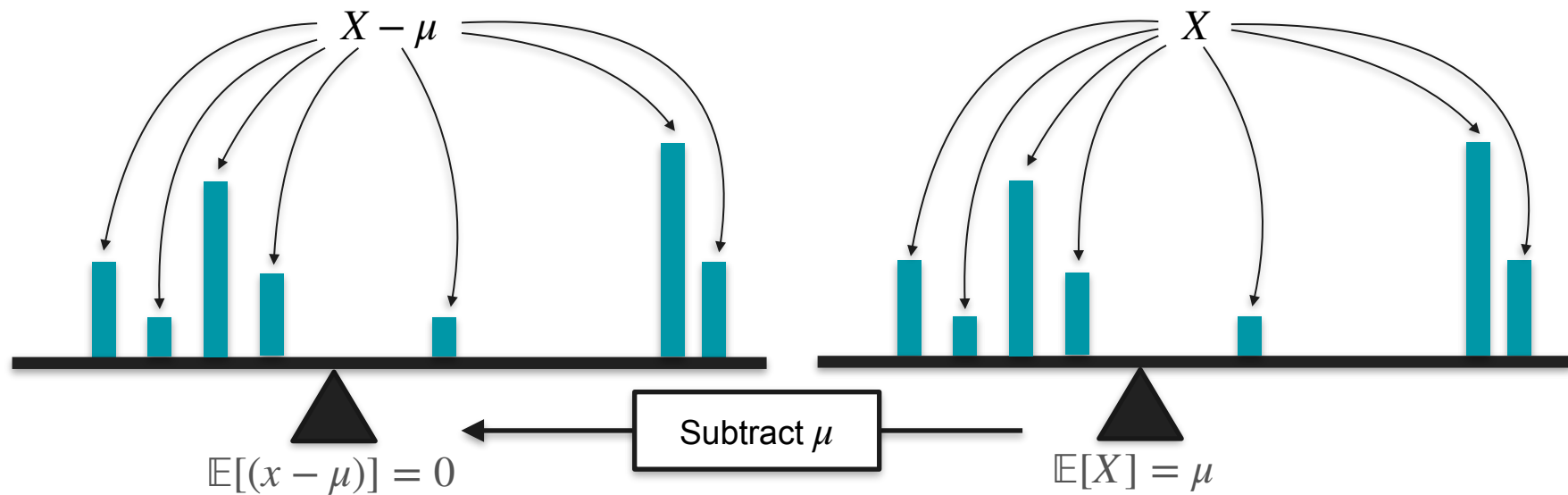
Variance Formula



Variance Formula



Variance Formula



$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$$

Variance Formula

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Variance Formula

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Variance Formula

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2]$$

Variance Formula

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \end{aligned}$$

$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

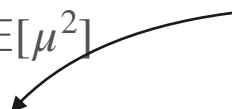
Variance Formula

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \quad \mathbb{E}[\text{constant} \cdot X] = \text{constant} \cdot \mathbb{E}[X] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \end{aligned}$$

Variance Formula

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \end{aligned}$$

$\mathbb{E}[\text{constant}] = \text{constant}$



Variance Formula

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\&= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \\&= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\&= \mathbb{E}[X^2] - 2\mathbb{E}[X] \cdot \mathbb{E}[X] + \mathbb{E}[X]^2\end{aligned}$$

$\mu = \mathbb{E}[X]$



Variance Formula

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X] \cdot \mathbb{E}[X] + \mathbb{E}[X]^2 \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]^2 + \mathbb{E}[X]^2\end{aligned}$$

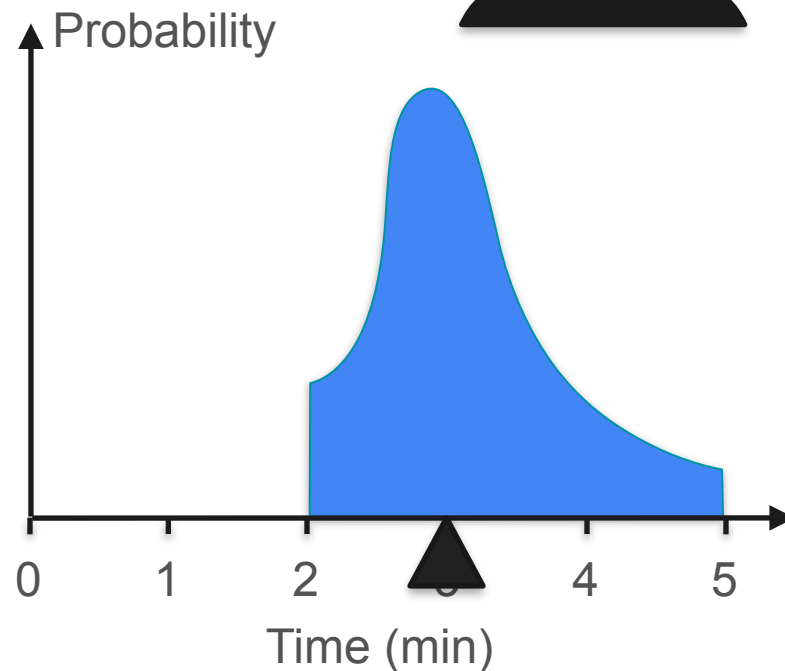
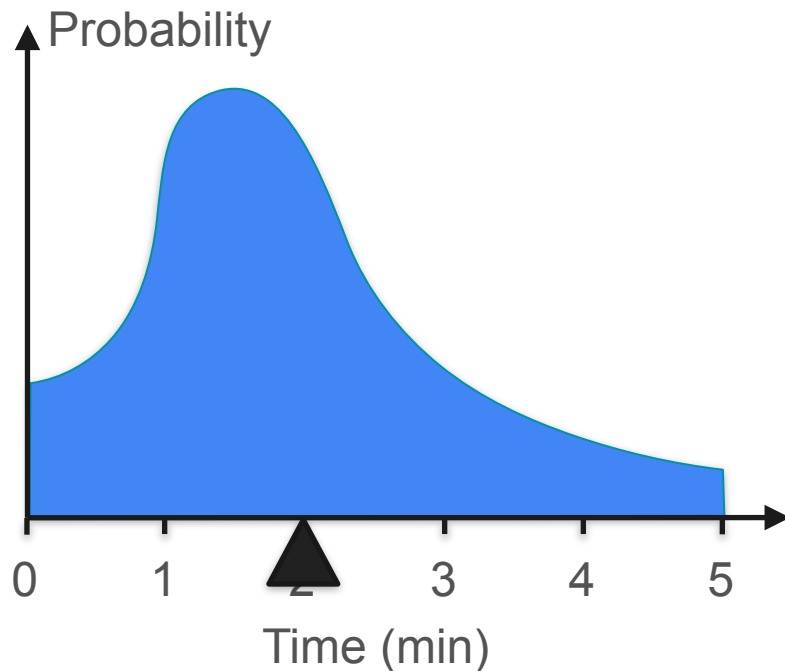
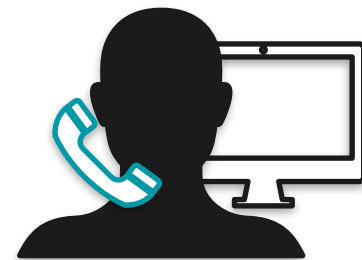
Variance Formula

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X] \cdot \mathbb{E}[X] + \mathbb{E}[X]^2 \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]^2 + \mathbb{E}[X]^2 \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \end{aligned}$$

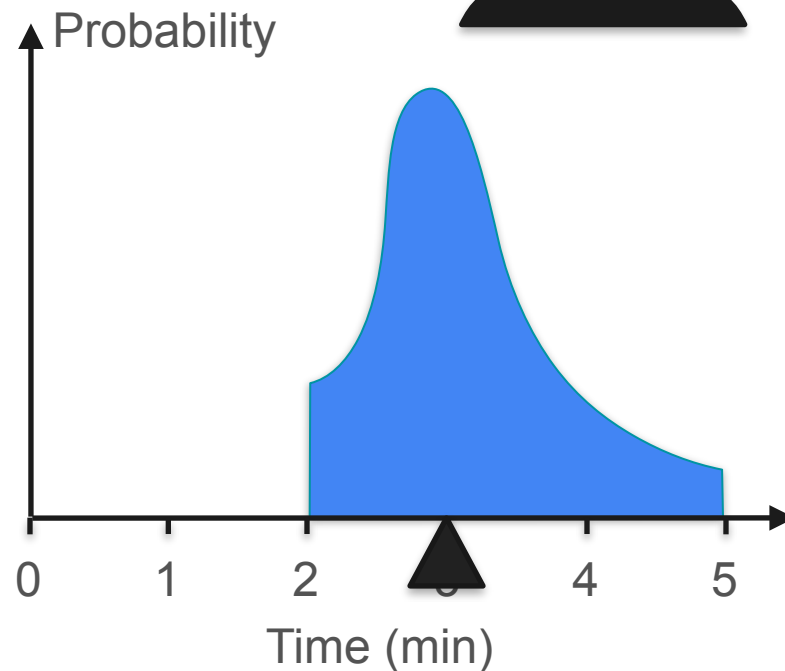
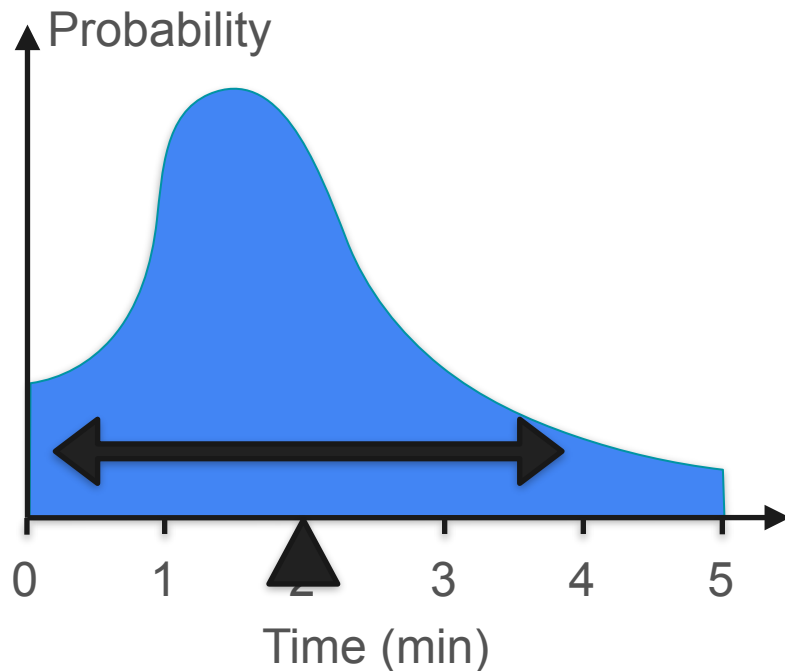
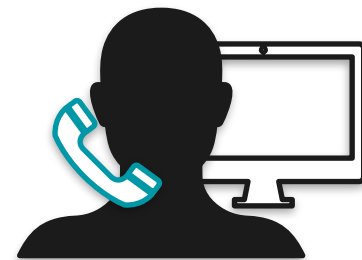
Variance Formula

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu)^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \end{aligned}$$

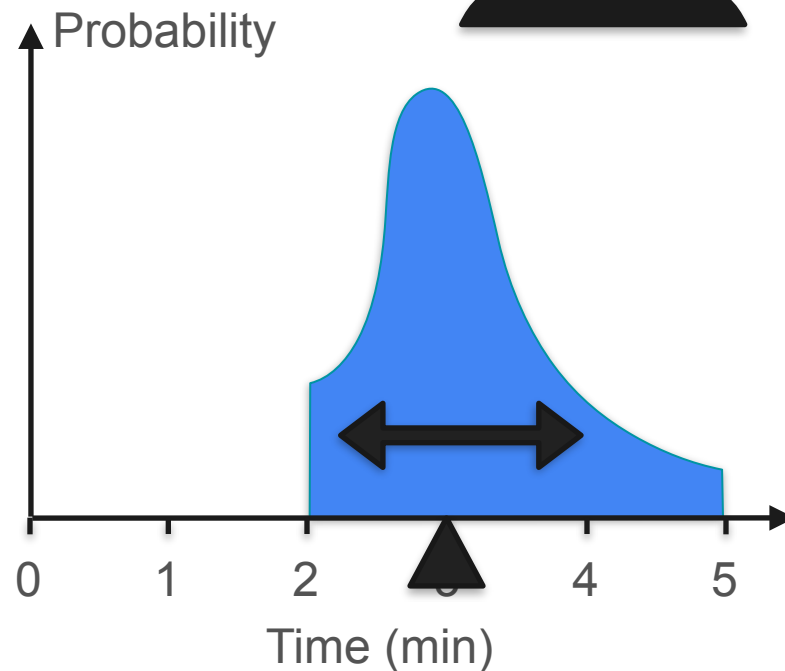
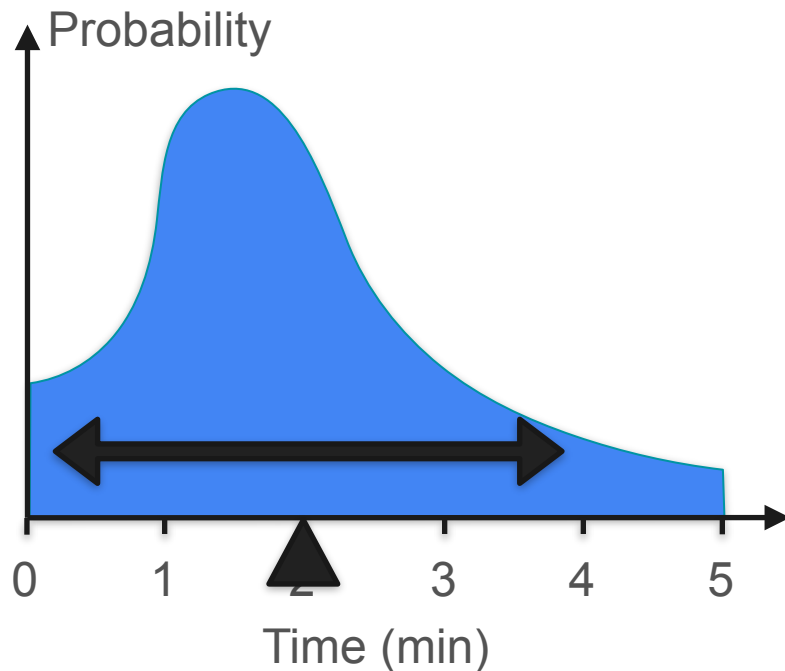
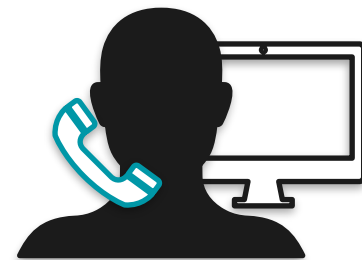
Variance for Continuous Distributions



Variance for Continuous Distributions

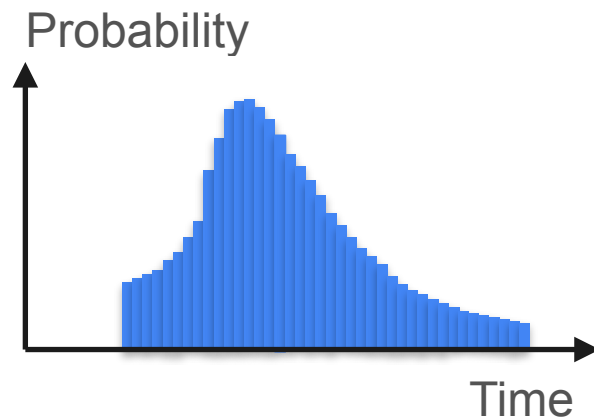


Variance for Continuous Distributions



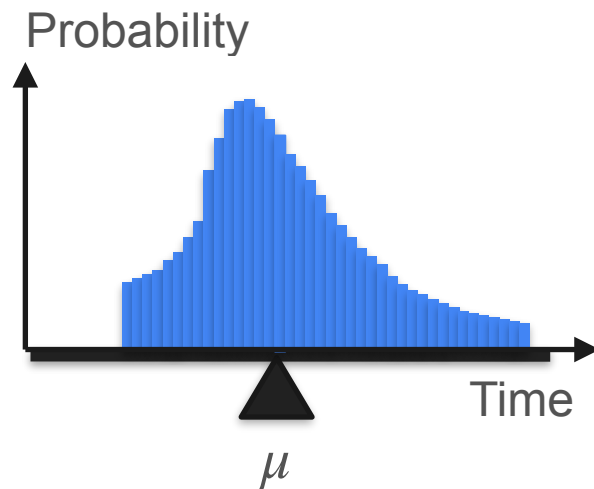
Variance

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$



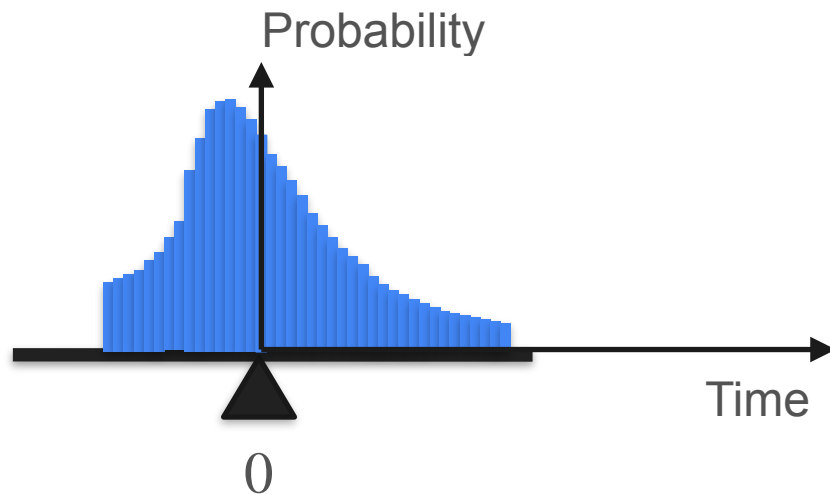
Variance

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$



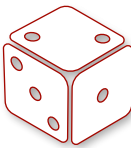
Variance

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$



Properties of the Variance

Properties of the Variance

Probability:	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$
Roll:	1	2	3	4	5	6
						
Double:	2	4	6	8	10	12
Wins	-3	-2	-1	0	1	2

Properties of the Variance

$$= 2.92$$

Properties of the Variance

$= 2.92$



Properties of the Variance

$= 2.92$

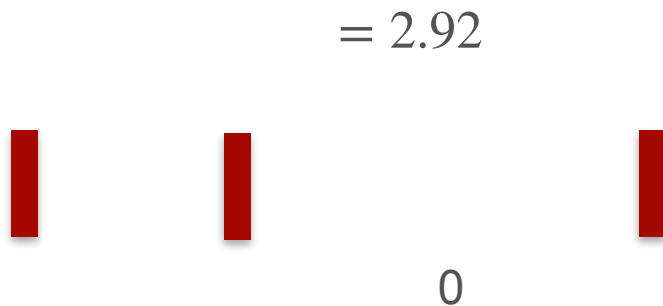


Properties of the Variance

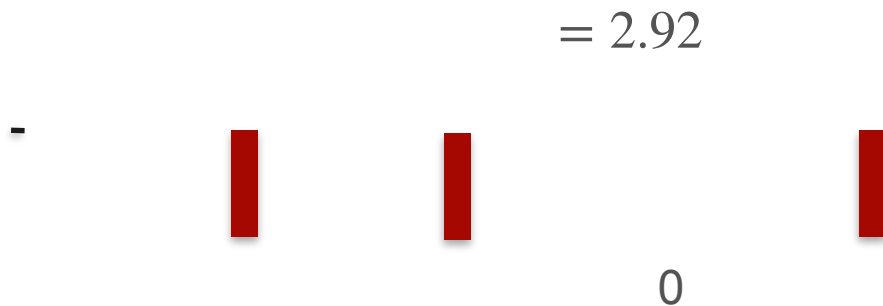

$$= 2.92$$

0

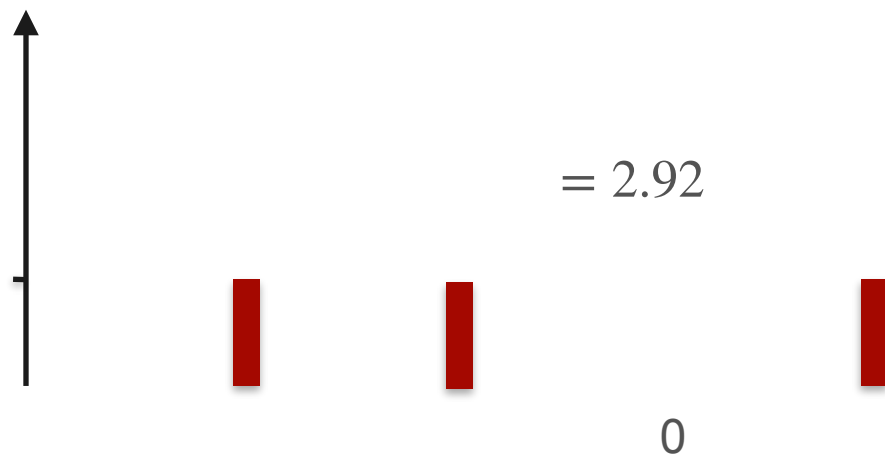
Properties of the Variance



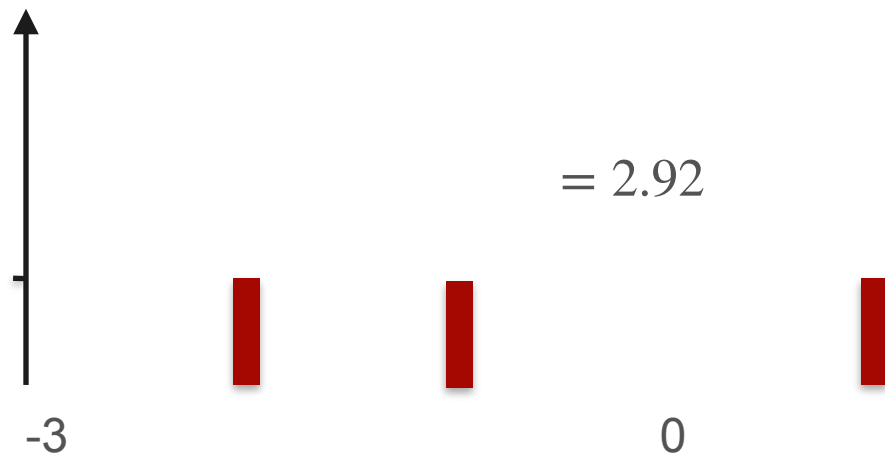
Properties of the Variance



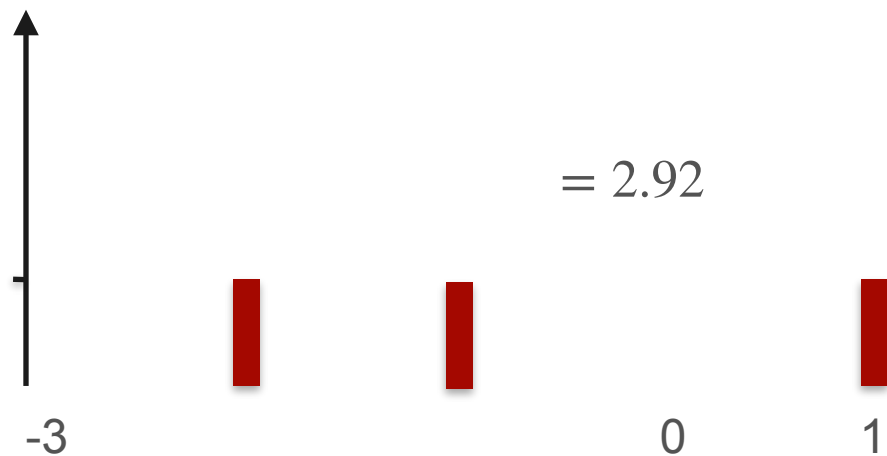
Properties of the Variance



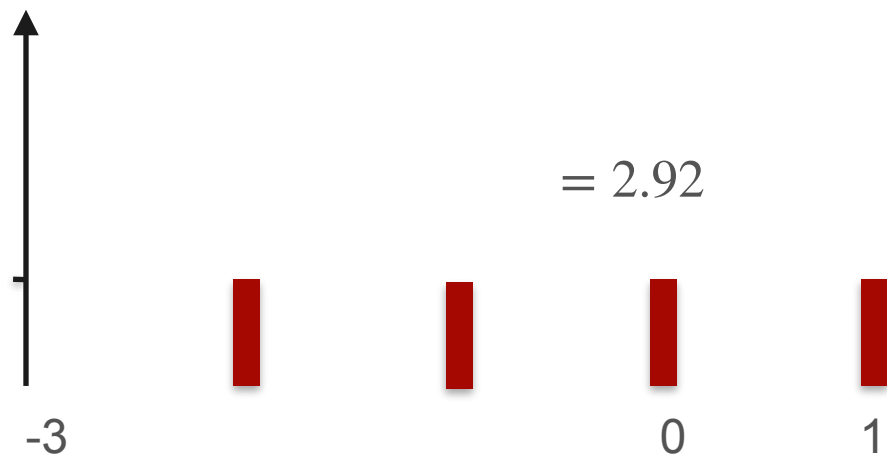
Properties of the Variance



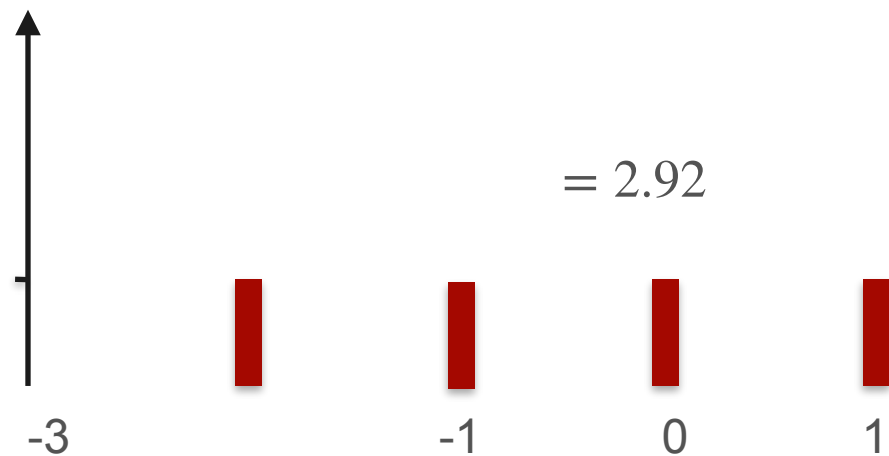
Properties of the Variance



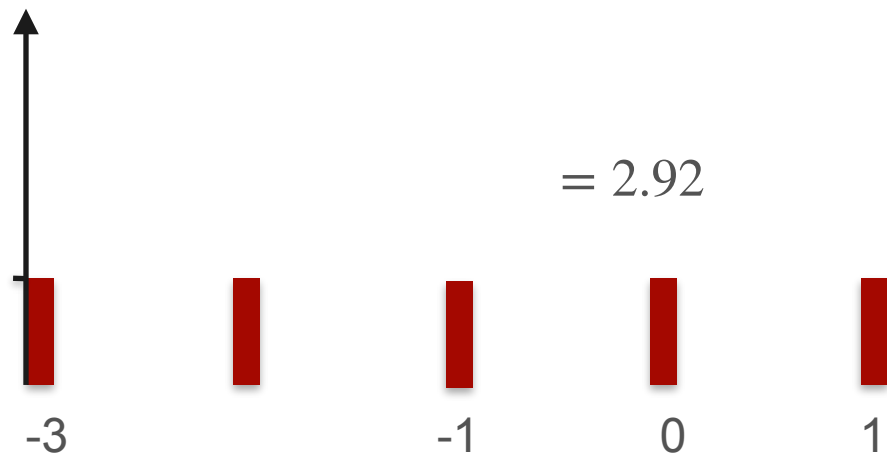
Properties of the Variance



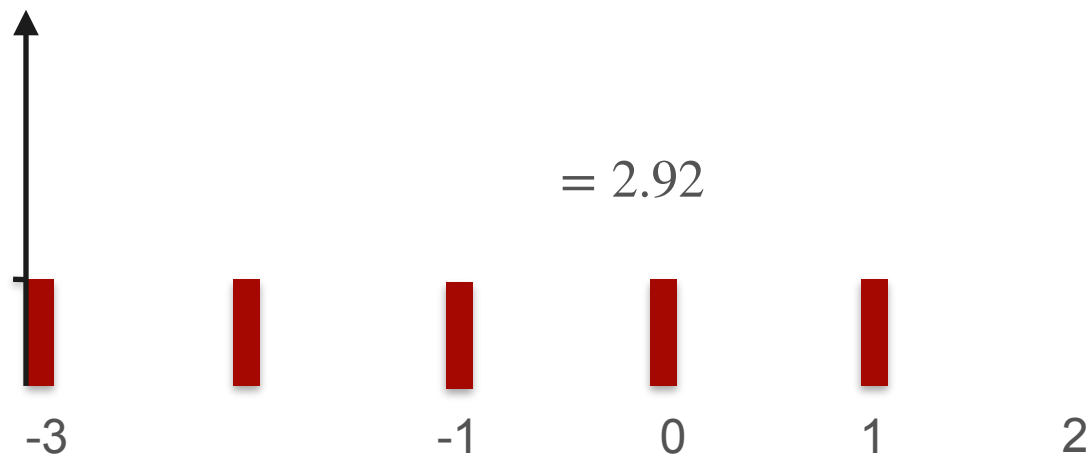
Properties of the Variance



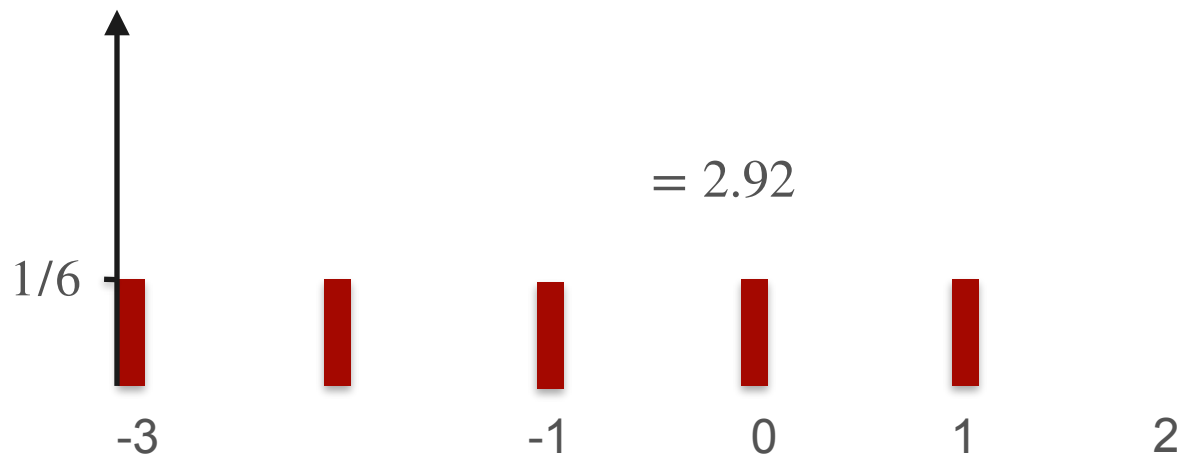
Properties of the Variance



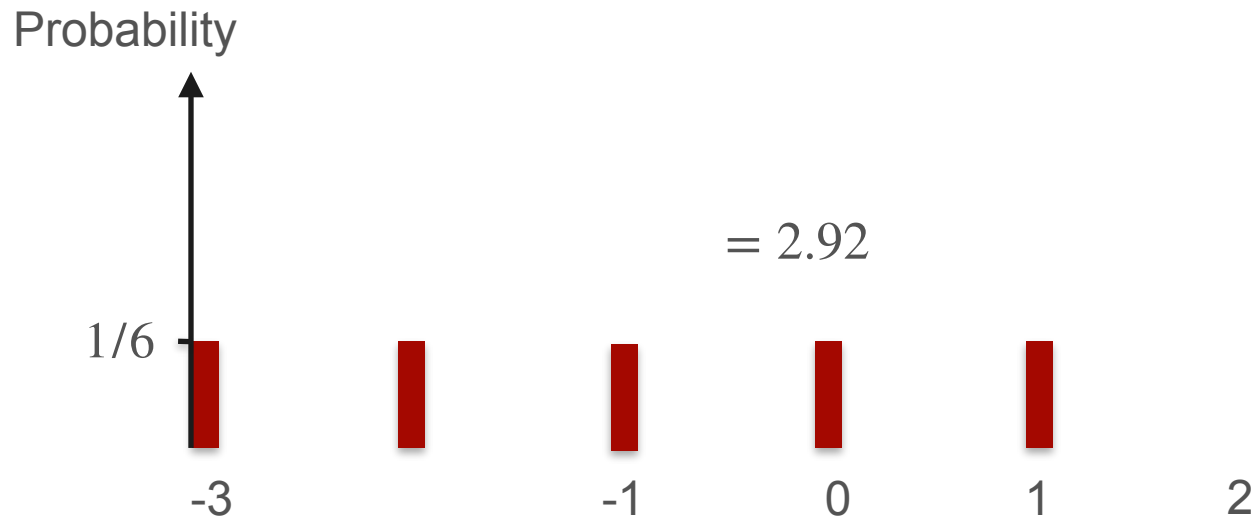
Properties of the Variance



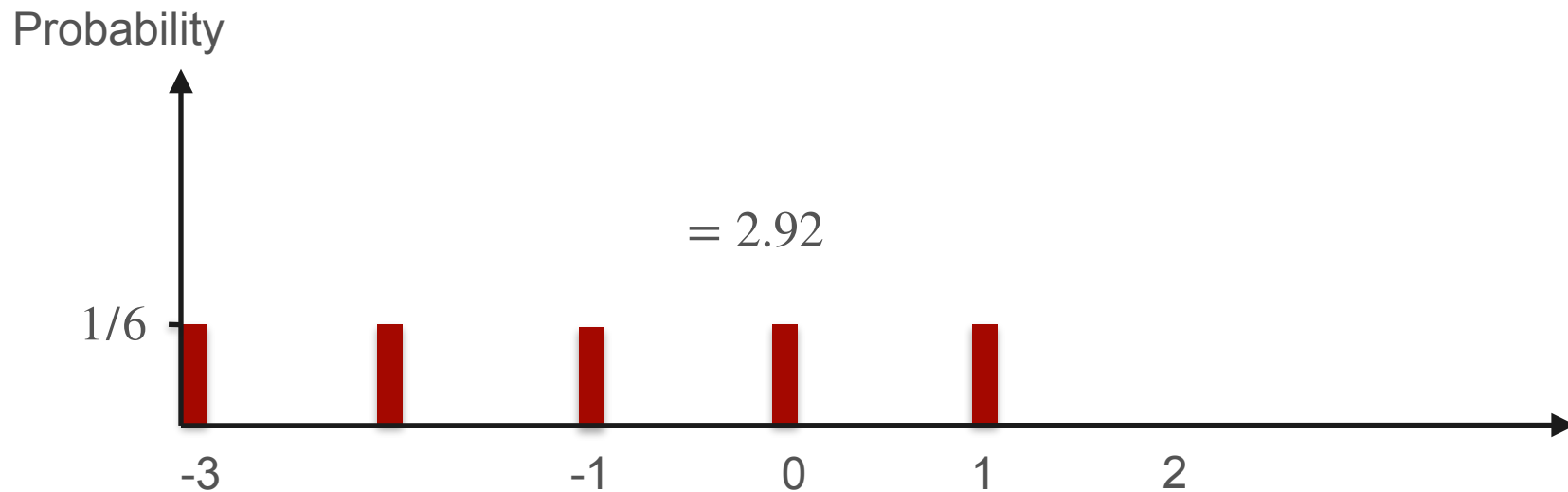
Properties of the Variance



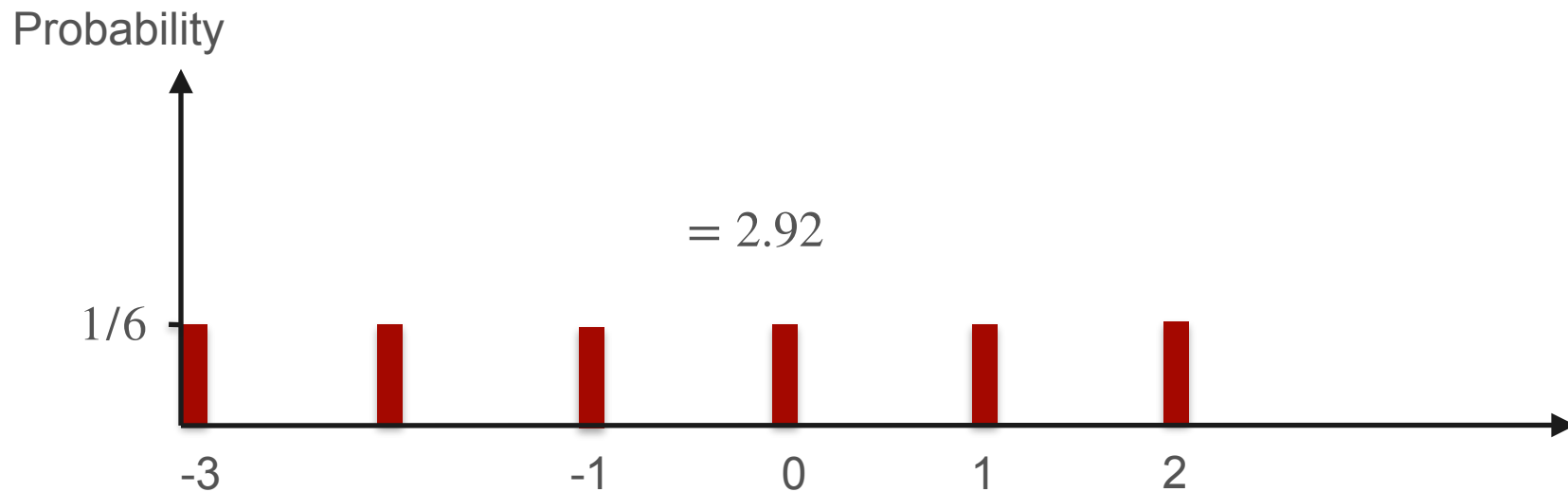
Properties of the Variance



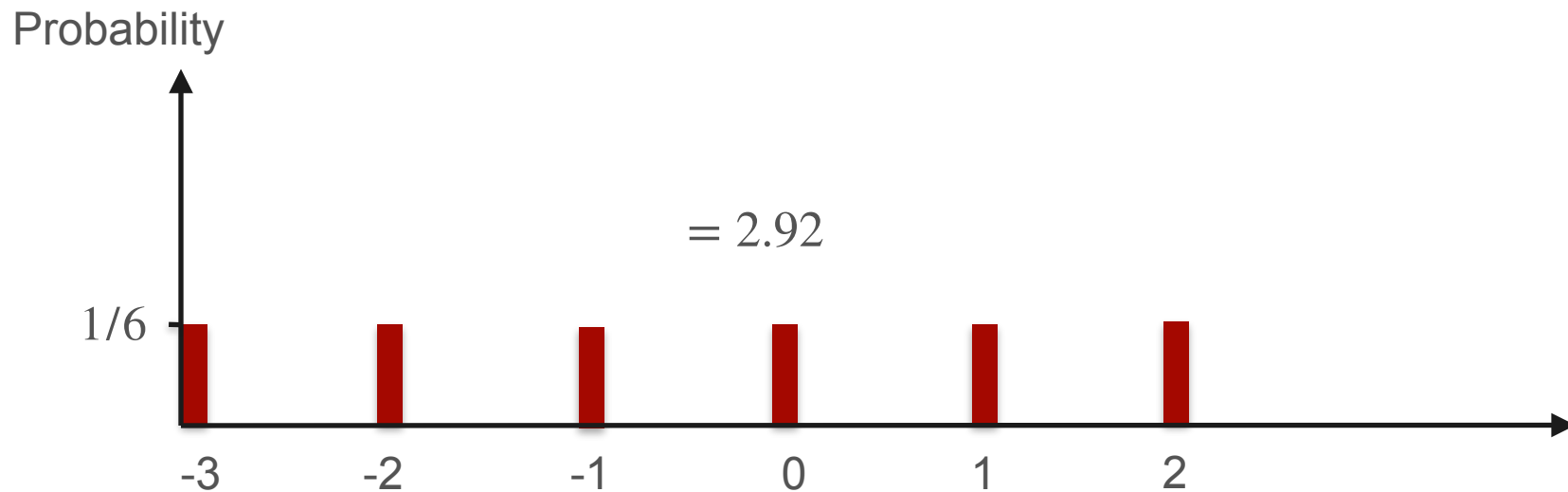
Properties of the Variance



Properties of the Variance

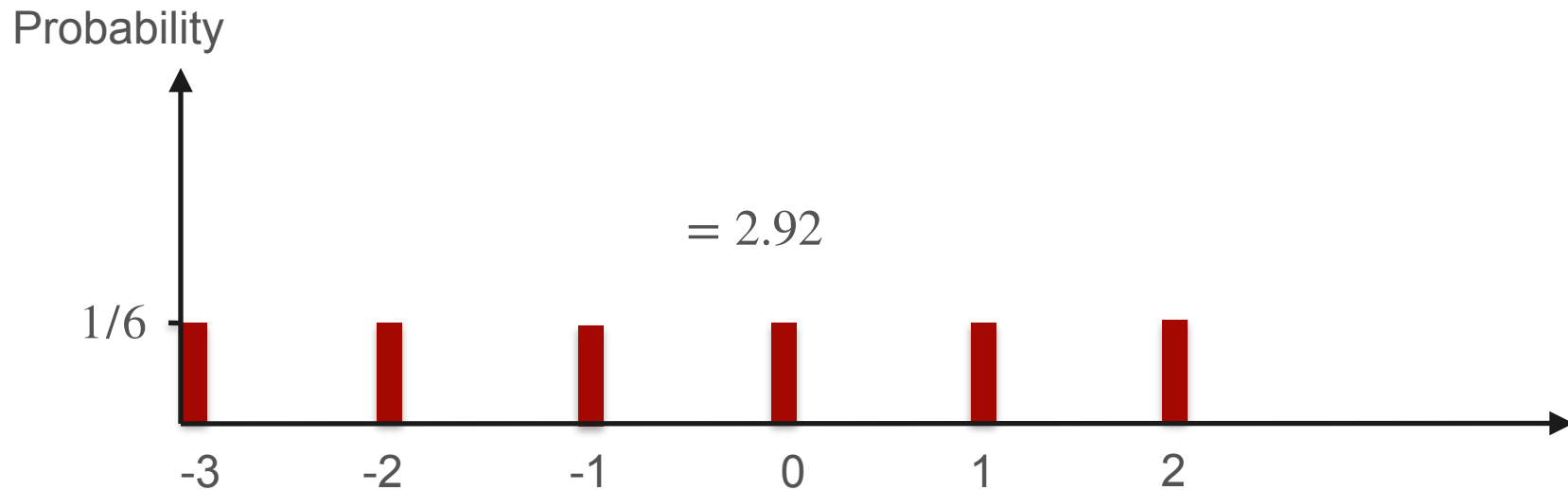


Properties of the Variance



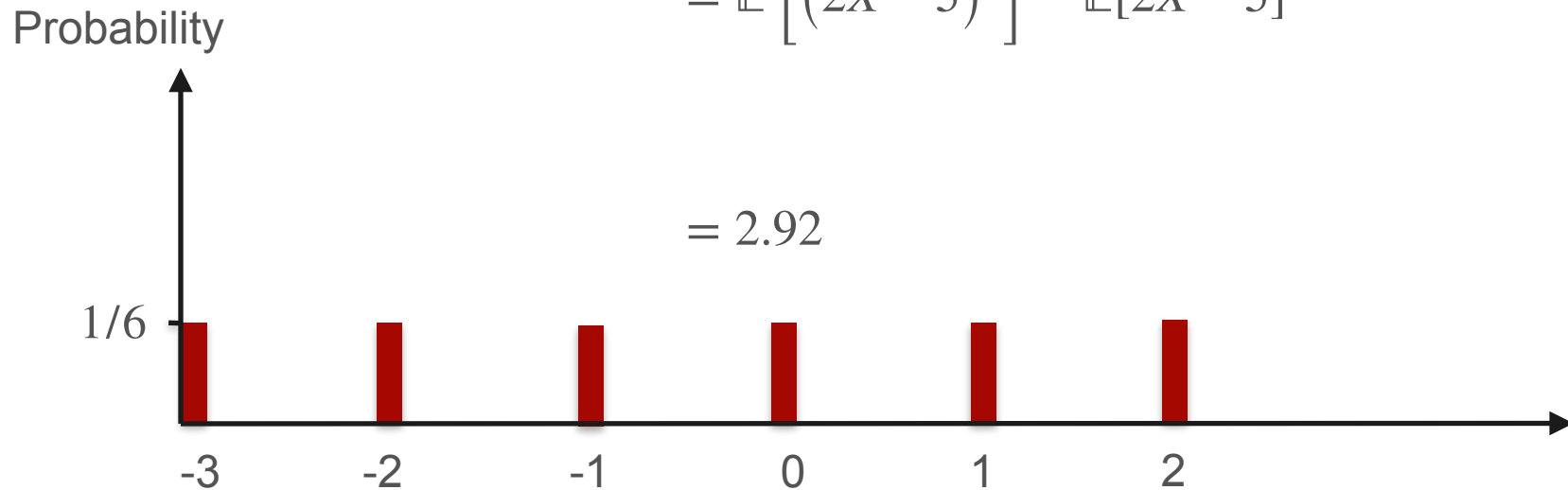
Properties of the Variance

$$\text{Var}(2X - 5) = \mathbb{E} \left[(2X - 5 - \mathbb{E}[2X - 5])^2 \right]$$



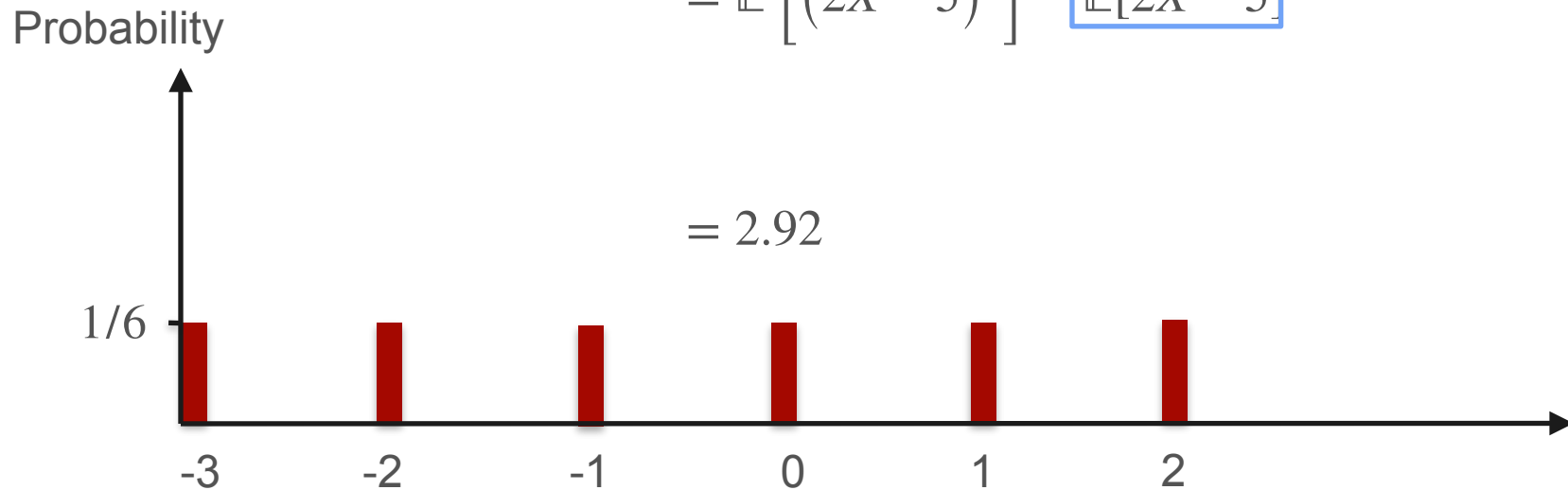
Properties of the Variance

$$\begin{aligned} \text{Var}(2X - 5) &= \mathbb{E} \left[(2X - 5 - \mathbb{E}[2X - 5])^2 \right] \\ &= \mathbb{E} \left[(2X - 5)^2 \right] - \mathbb{E}[2X - 5]^2 \end{aligned}$$



Properties of the Variance

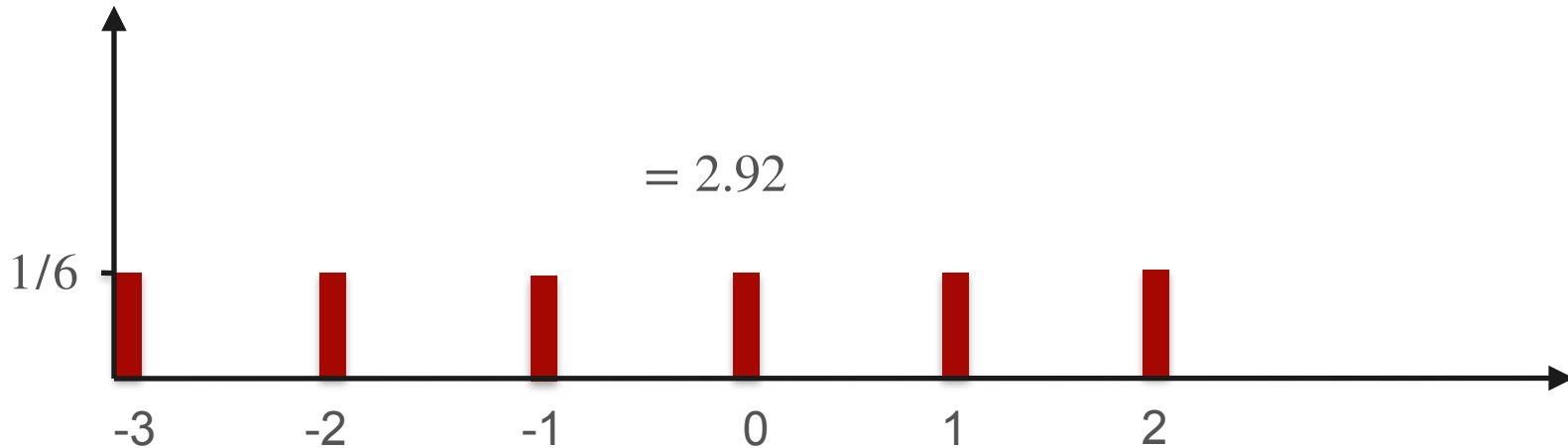
$$\begin{aligned} \text{Var}(2X - 5) &= \mathbb{E} \left[(2X - 5 - \mathbb{E}[2X - 5])^2 \right] \\ &= \mathbb{E} \left[(2X - 5)^2 \right] - \boxed{\mathbb{E}[2X - 5]^2} \end{aligned}$$



Properties of the Variance

$$\begin{aligned} \text{Var}(2X - 5) &= \mathbb{E} \left[(2X - 5 - \mathbb{E}[2X - 5])^2 \right] \\ &= \mathbb{E} \left[(2X - 5)^2 \right] - \overset{-0.5}{\boxed{\mathbb{E}[2X - 5]^2}} \end{aligned}$$

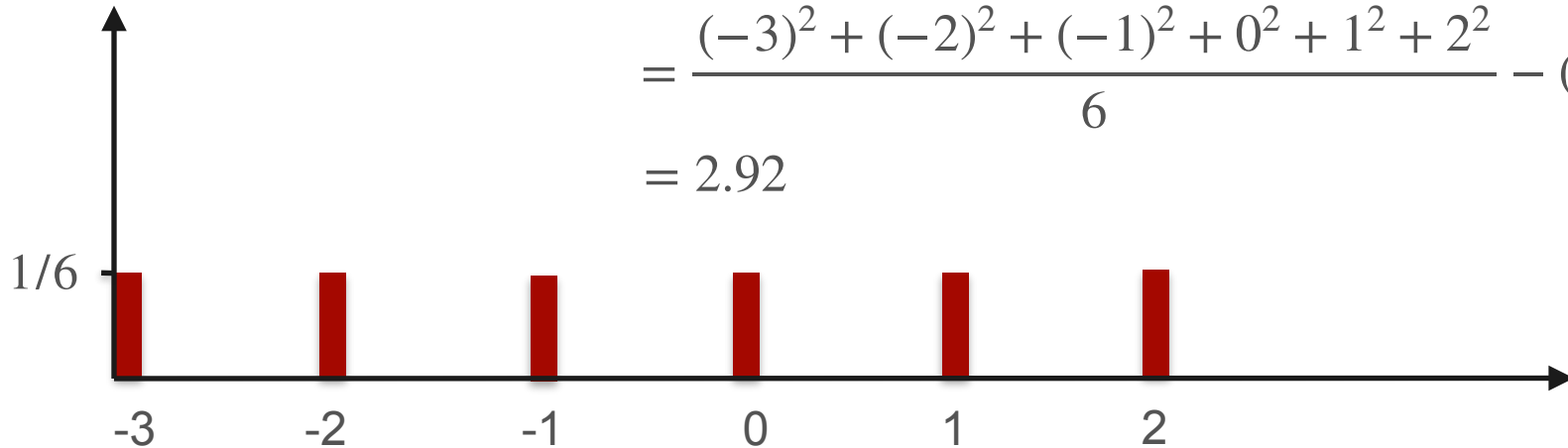
Probability



Properties of the Variance

$$\begin{aligned} \text{Var}(2X - 5) &= \mathbb{E} \left[(2X - 5 - \mathbb{E}[2X - 5])^2 \right] \\ &= \mathbb{E} \left[(2X - 5)^2 \right] - \overbrace{\mathbb{E}[2X - 5]^2}^{-0.5} \\ &= \frac{(-3)^2 + (-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2}{6} - (-0.5)^2 \\ &= 2.92 \end{aligned}$$

Probability



Properties of the Variance

Properties of the Variance

$$\mathbb{E} \left[(2X - 5)^2 \right]$$

Properties of the Variance

$$\mathbb{E} \left[(2X - 5)^2 \right] = \mathbb{E} \left[2^2 X^2 + (-5)^2 - 2(-5)X \right]$$

Properties of the Variance

$$\begin{aligned}\mathbb{E} \left[(2X - 5)^2 \right] &= \mathbb{E} \left[2^2 X^2 + (-5)^2 - 2(-5)X \right] \\ &= 2^2 \mathbb{E} \left[X^2 \right] + (-5)^2 - 2(-5) \mathbb{E}[X]\end{aligned}$$

Properties of the Variance

$$\begin{aligned}\mathbb{E} \left[(2X - 5)^2 \right] &= \mathbb{E} \left[2^2 X^2 + (-5)^2 - 2(-5)X \right] \\ &= 2^2 \mathbb{E} [X^2] + (-5)^2 - 2(-5) \mathbb{E}[X]\end{aligned}$$

$$\text{Var}(2X - 5) = \mathbb{E} \left[(2X - 5)^2 \right] - \mathbb{E}[2X - 5]^2$$

Properties of the Variance

$$\begin{aligned}\mathbb{E} \left[(2X - 5)^2 \right] &= \mathbb{E} \left[2^2 X^2 + (-5)^2 - 2(-5)X \right] \\ &= 2^2 \mathbb{E} [X^2] + (-5)^2 - 2(-5) \mathbb{E}[X]\end{aligned}$$

$$\text{Var}(2X - 5) = \mathbb{E} \left[(2X - 5)^2 \right] - \boxed{\mathbb{E}[2X - 5]^2}$$

$2\mathbb{E}[X] - 5$



Properties of the Variance

$$\begin{aligned}\mathbb{E} \left[(2X - 5)^2 \right] &= \mathbb{E} \left[2^2 X^2 + (-5)^2 - 2(-5)X \right] \\ &= 2^2 \mathbb{E} [X^2] + (-5)^2 - 2(-5) \mathbb{E}[X]\end{aligned}$$

$$\begin{aligned}\text{Var}(2X - 5) &= \mathbb{E} \left[(2X - 5)^2 \right] - \boxed{\mathbb{E}[2X - 5]^2} \\ &= 2^2 \mathbb{E} [X^2] + (-5)^2 - 2(-5) \mathbb{E}[X] - (2\mathbb{E}[X] - 5)^2\end{aligned}$$


Properties of the Variance

$$\begin{aligned}\mathbb{E} \left[(2X - 5)^2 \right] &= \mathbb{E} \left[2^2 X^2 + (-5)^2 - 2(-5)X \right] \\ &= 2^2 \mathbb{E} [X^2] + (-5)^2 - 2(-5) \mathbb{E}[X]\end{aligned}$$

$$\begin{aligned}\text{Var}(2X - 5) &= \mathbb{E} \left[(2X - 5)^2 \right] - \boxed{\mathbb{E}[2X - 5]^2} \\ &= 2^2 \mathbb{E} [X^2] + (-5)^2 - 2(-5) \mathbb{E}[X] - (2\mathbb{E}[X] - 5)^2 \\ &= 2^2 \mathbb{E} [X^2] + (-5)^2 - 2(-5) \mathbb{E}[X] - (2^2 \mathbb{E}[X]^2 + (-5)^2 - 2(-5) \mathbb{E}[X])\end{aligned}$$

Properties of the Variance

$$\text{Var}(2X - 5) = 2^2 \mathbb{E}[X^2] + (-5)^2 - 2(-5)\mathbb{E}[X] - (2^2 \mathbb{E}[X]^2 + (-5)^2 - 2(-5)\mathbb{E}[X])$$

Properties of the Variance

$$\text{Var}(2X - 5) = \boxed{2^2 \mathbb{E}[X^2]} + \cancel{(-5)^2} - 2(-5)\mathbb{E}[X] - (\boxed{2^2 \mathbb{E}[X]^2} + \cancel{(-5)^2} - 2(-5)\mathbb{E}[X])$$

Properties of the Variance

$$\text{Var}(2X - 5) = \boxed{2^2 \mathbb{E}[X^2]} + \cancel{(-5)^2} - \cancel{2(-5)\mathbb{E}[X]} - (\boxed{2^2 \mathbb{E}[X]^2} + \cancel{(-5)^2} - \cancel{2(-5)\mathbb{E}[X]})$$

Properties of the Variance

$$\begin{aligned} \text{Var}(2X - 5) &= \boxed{2^2 \mathbb{E}[X^2]} + \cancel{(-5)^2} - \cancel{2(-5)\mathbb{E}[X]} - (\boxed{2^2 \mathbb{E}[X]^2} + \cancel{(-5)^2} - \cancel{2(-5)\mathbb{E}[X]}) \\ &= 2^2 (\mathbb{E}[X^2] - \mathbb{E}[X]^2) \end{aligned}$$

Properties of the Variance

$$\begin{aligned} \text{Var}(2X - 5) &= \boxed{2^2 \mathbb{E}[X^2]} + \cancel{(-5)^2} - \cancel{2(-5)\mathbb{E}[X]} - (\boxed{2^2 \mathbb{E}[X]^2} + \cancel{(-5)^2} - \cancel{2(-5)\mathbb{E}[X]}) \\ &= 2^2 \underbrace{(\mathbb{E}[X^2] - \mathbb{E}[X]^2)}_{\text{Var}(X)} \end{aligned}$$

Properties of the Variance

$$\begin{aligned} \text{Var}(2X - 5) &= \boxed{2^2 \mathbb{E}[X^2]} + \cancel{(-5)^2} - \cancel{2(-5)\mathbb{E}[X]} - (\boxed{2^2 \mathbb{E}[X]^2} + \cancel{(-5)^2} - \cancel{2(-5)\mathbb{E}[X]}) \\ &= 2^2 \underbrace{(\mathbb{E}[X^2] - \mathbb{E}[X]^2)}_{\text{Var}(X)} = 2^2 \text{Var}(X) \end{aligned}$$

Properties of the Variance

$$\begin{aligned} \text{Var}(2X - 5) &= \boxed{2^2 \mathbb{E}[X^2]} + \cancel{(-5)^2} - \cancel{2(-5)\mathbb{E}[X]} - (\boxed{2^2 \mathbb{E}[X]^2} + \cancel{(-5)^2} - \cancel{2(-5)\mathbb{E}[X]}) \\ &= 2^2 \underbrace{(\mathbb{E}[X^2] - \mathbb{E}[X]^2)}_{\text{Var}(X)} = 2^2 \text{Var}(X) \end{aligned}$$

In general: $\text{Var}(aX + b) = a^2 \text{Var}(X)$



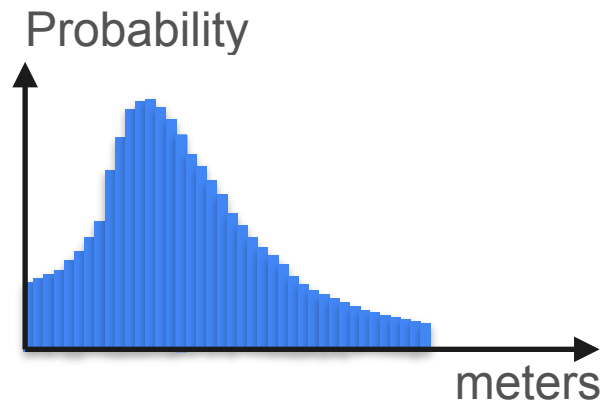
DeepLearning.AI

Describing Distributions

Standard deviation

Standard Deviation

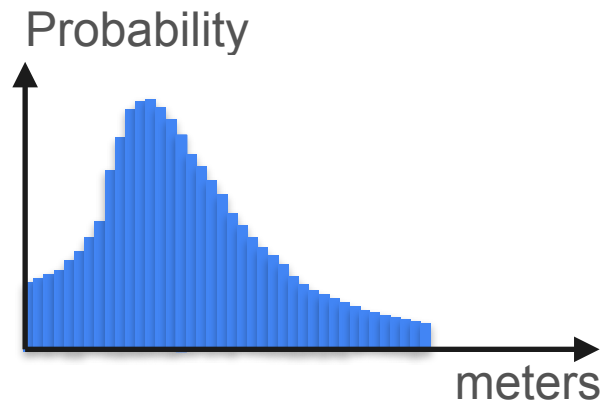
$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$



Standard Deviation

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Say X is measured in meters.

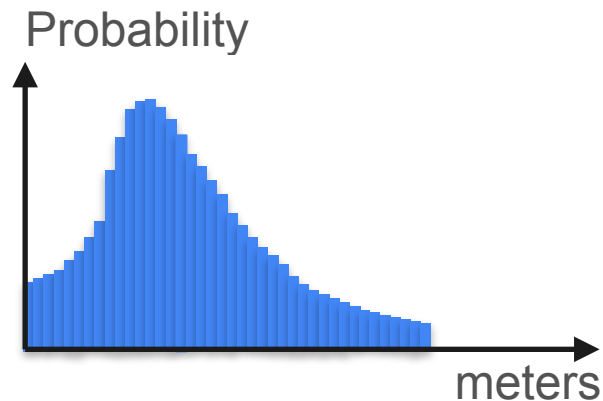


Standard Deviation

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Say X is measured in meters.

Then $\mathbb{E}[X]$ is measured in meters.

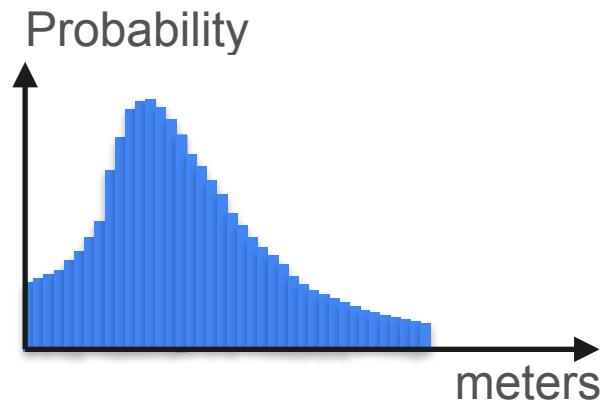


Standard Deviation

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \overset{m^2}{\mathbb{E}[X^2]} - \overset{m^2}{\mathbb{E}[X]^2}$$

Say X is measured in meters.

Then $\mathbb{E}[X]$ is measured in meters.

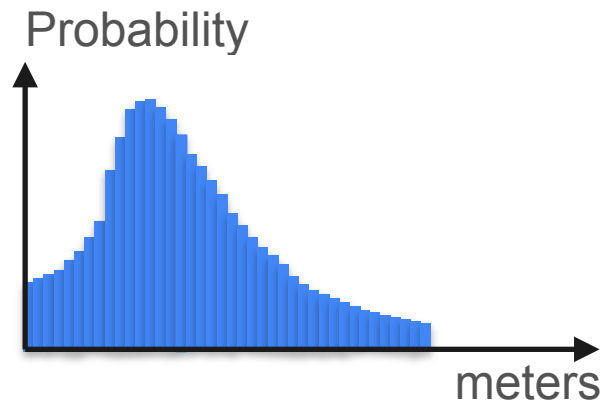


Standard Deviation

$$\overset{m^2}{\boxed{Var(X)}} = \mathbb{E}[(X - \mu)^2] = \overset{m^2}{\boxed{\mathbb{E}[X^2]}} - \overset{m^2}{\boxed{\mathbb{E}[X]^2}}$$

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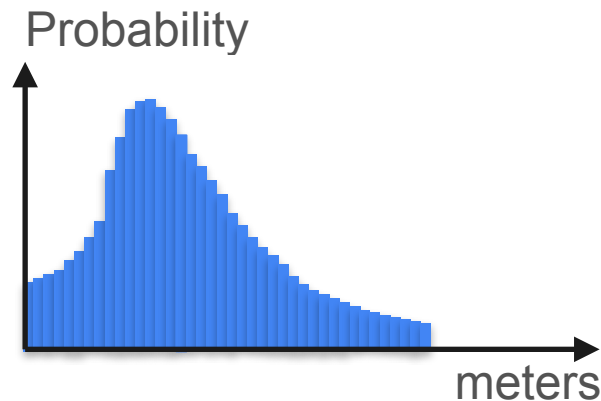
Standard Deviation

$$\overset{m^2}{\boxed{Var(X)}} = \mathbb{E}[(X - \mu)^2] = \overset{m^2}{\boxed{\mathbb{E}[X^2]}} - \overset{m^2}{\boxed{\mathbb{E}[X]^2}}$$

Say X is measured in meters.

Then $\mathbb{E}[X]$ is measured in meters.

Then $Var(X)$ is measured in meters².



Standard Deviation

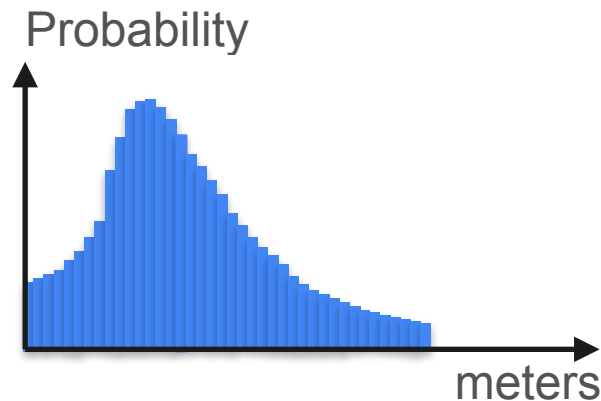
$$\overset{m^2}{\boxed{Var(X)}} = \mathbb{E}[(X - \mu)^2] = \overset{m^2}{\boxed{\mathbb{E}[X^2]}} - \overset{m^2}{\boxed{\mathbb{E}[X]^2}}$$

Say X is measured in meters.

Then $\mathbb{E}[X]$ is measured in meters.

Then $Var(X)$ is measured in meters².

Then $\sqrt{Var(X)}$ is measured in meters.



Standard Deviation

$$\overset{m^2}{\boxed{Var(X)}} = \mathbb{E}[(X - \mu)^2] = \overset{m^2}{\boxed{\mathbb{E}[X^2]}} - \overset{m^2}{\boxed{\mathbb{E}[X]^2}}$$

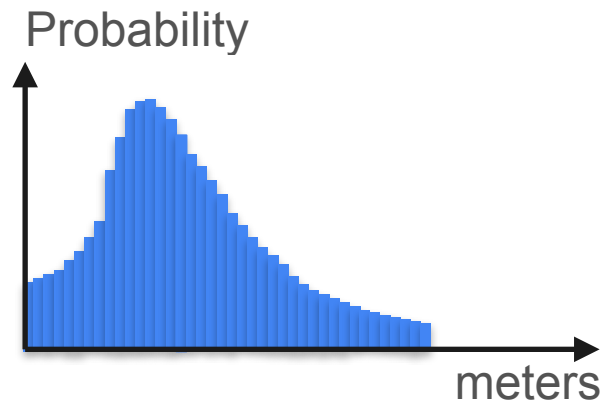
Say X is measured in meters.

Then $\mathbb{E}[X]$ is measured in meters.

Then $Var(X)$ is measured in meters^2 .

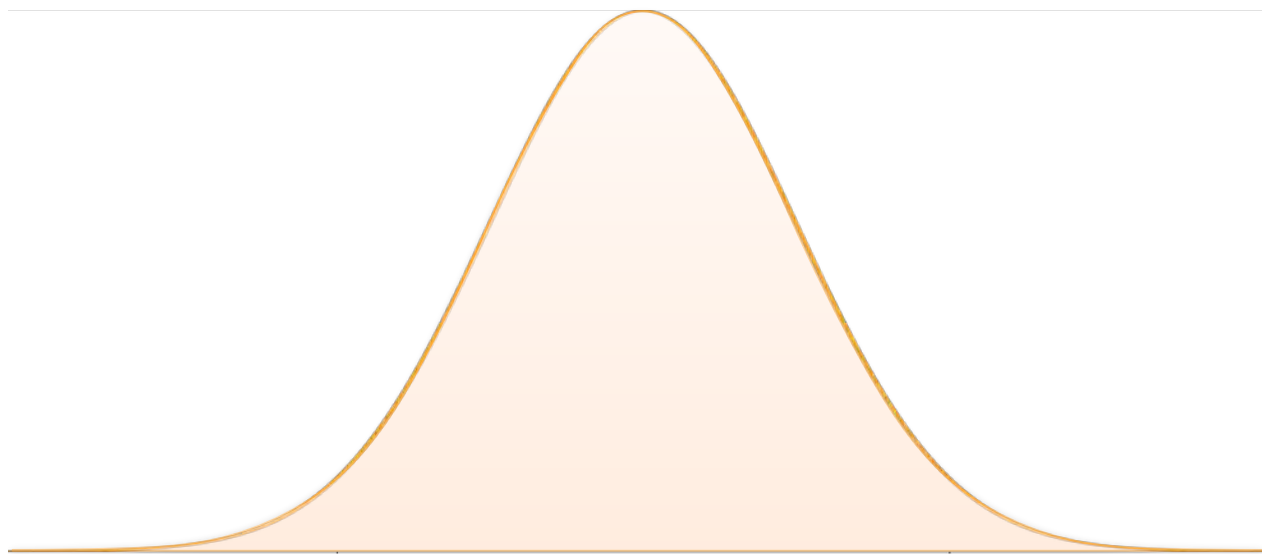
Then $\sqrt{Var(X)}$ is measured in meters.

Let's call $std(X) = \sqrt{Var(X)}$, the *standard deviation* of X



Normal Distribution: 68-95-99.7 Rule

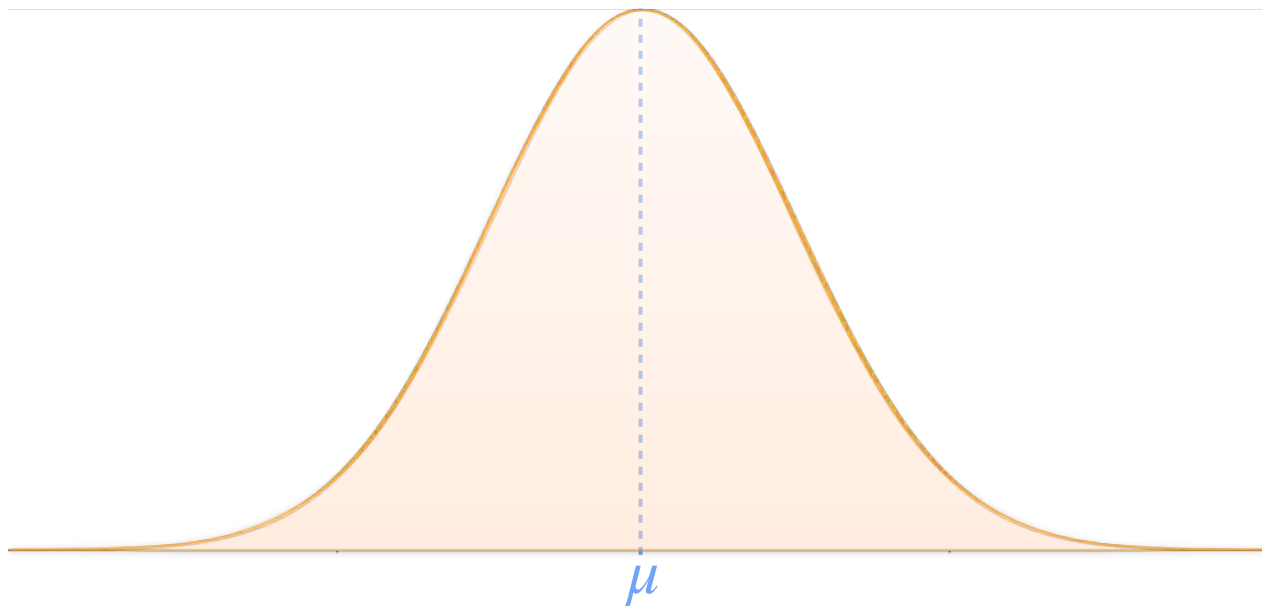
Normal Distribution: 68-95-99.7 Rule



Normal Distribution: 68-95-99.7 Rule

Parameters:

- μ : center of the bell

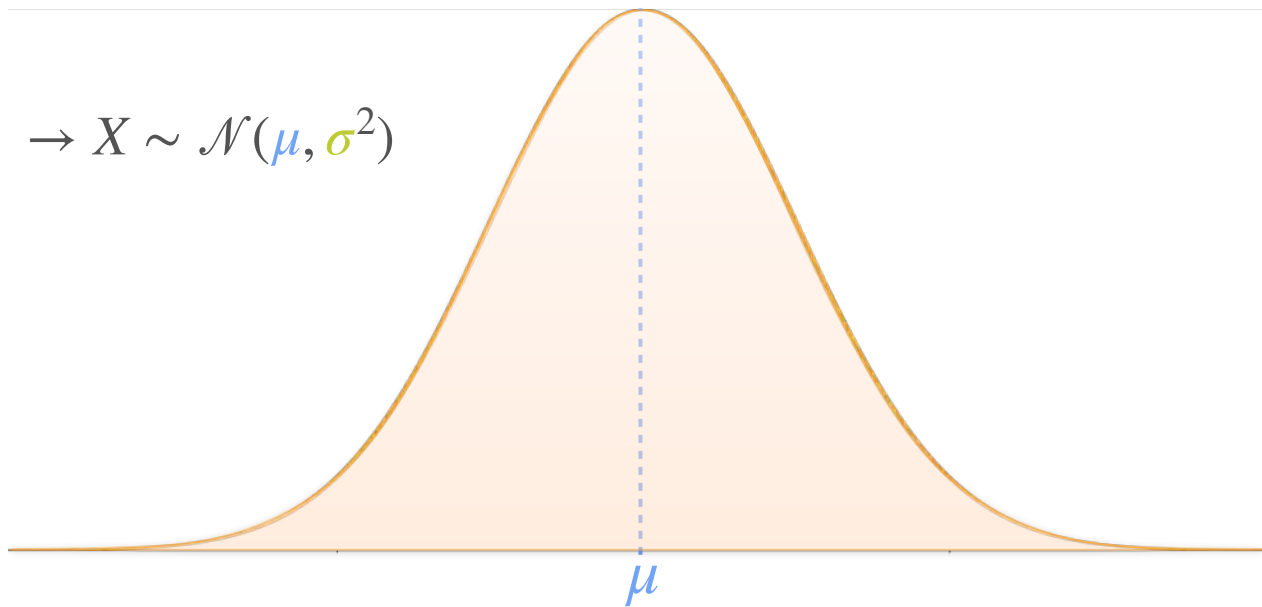


Normal Distribution: 68-95-99.7 Rule

Parameters:

- μ : center of the bell
- σ : spread of the bell

$$\rightarrow X \sim \mathcal{N}(\mu, \sigma^2)$$



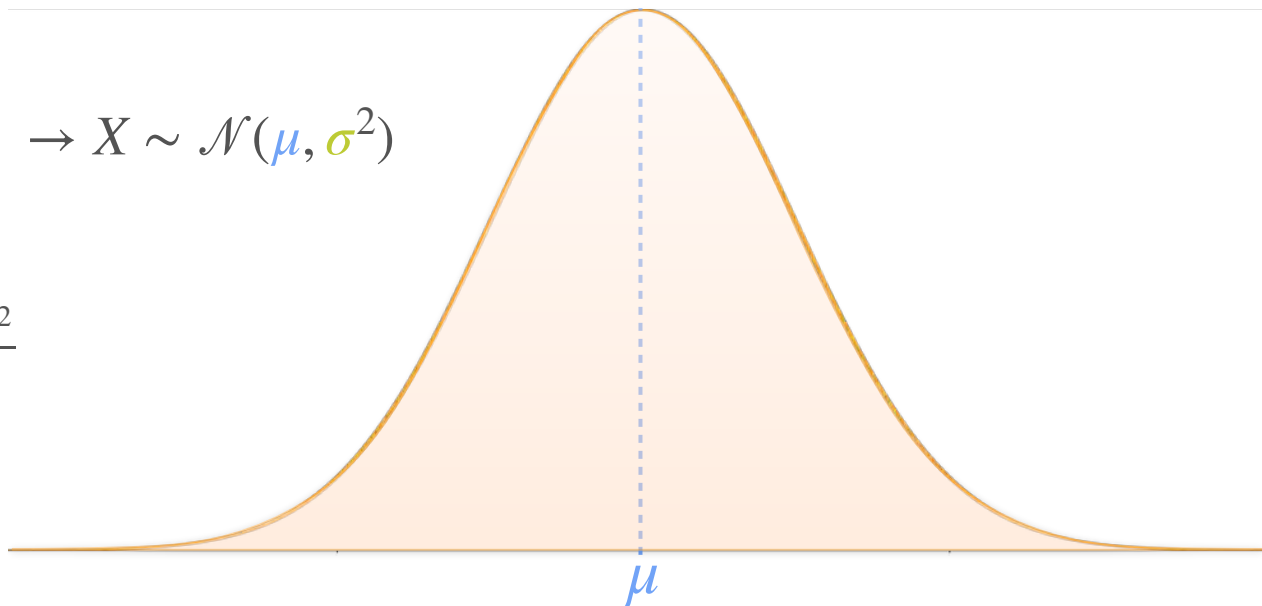
Normal Distribution: 68-95-99.7 Rule

Parameters:

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$$\rightarrow X \sim \mathcal{N}(\mu, \sigma^2)$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$



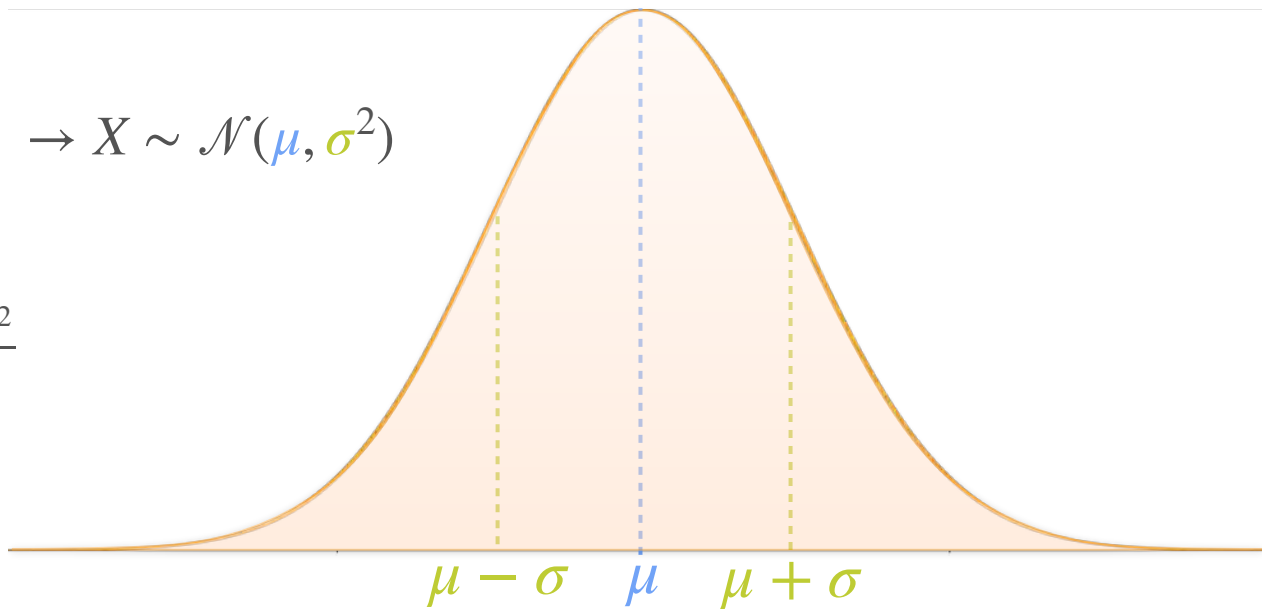
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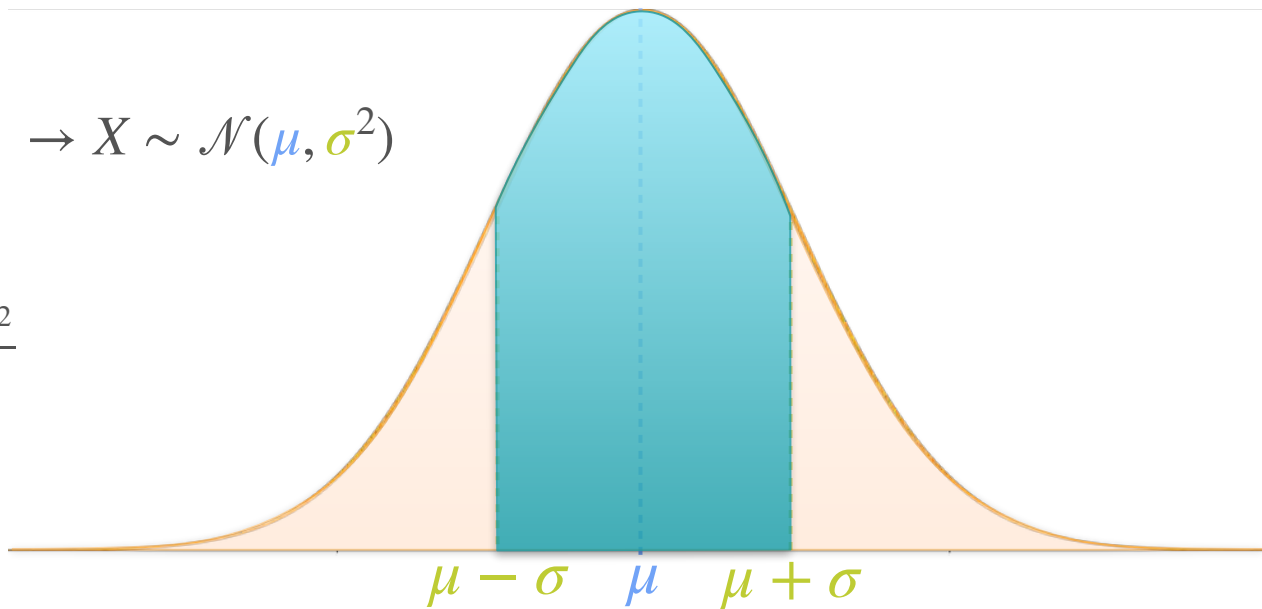
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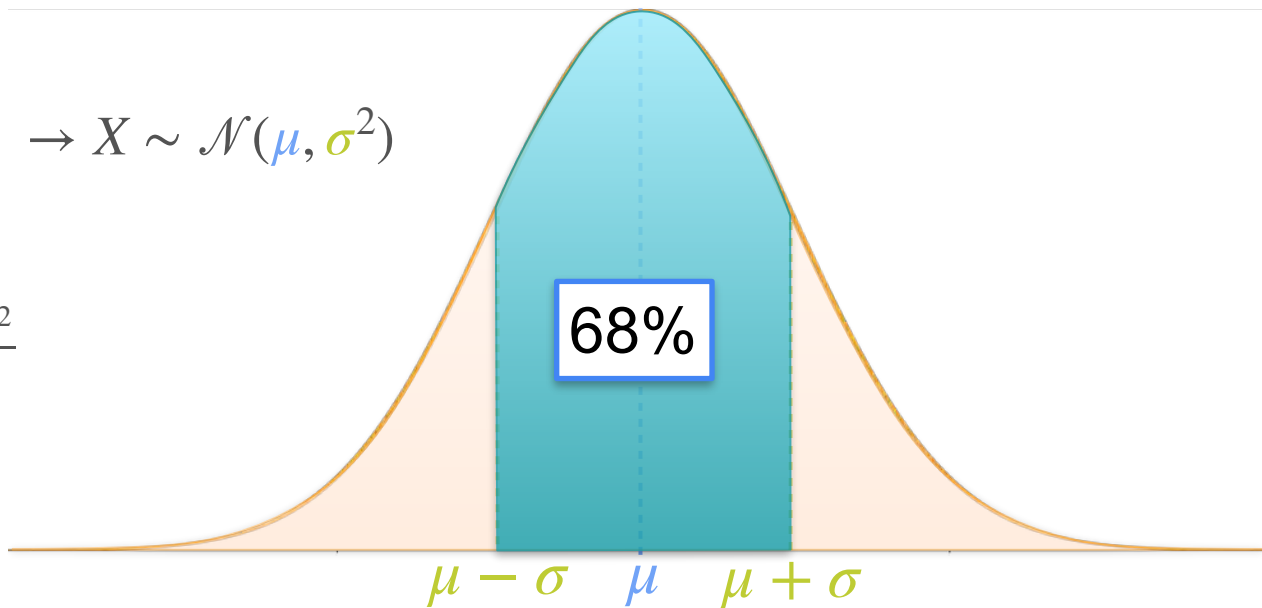
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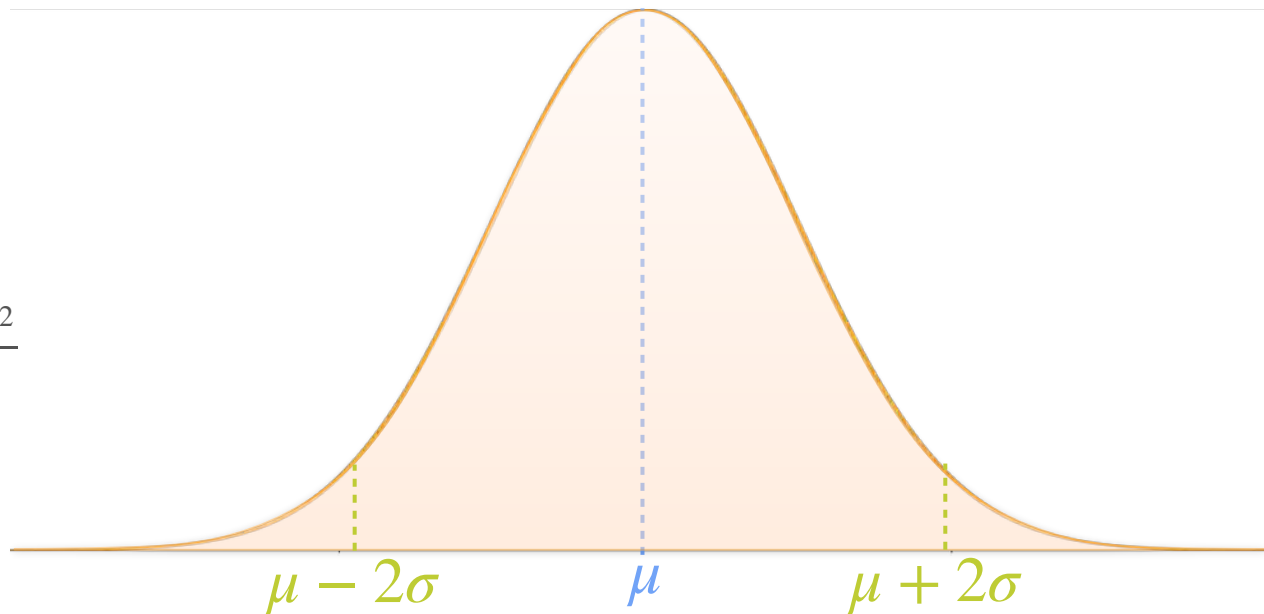


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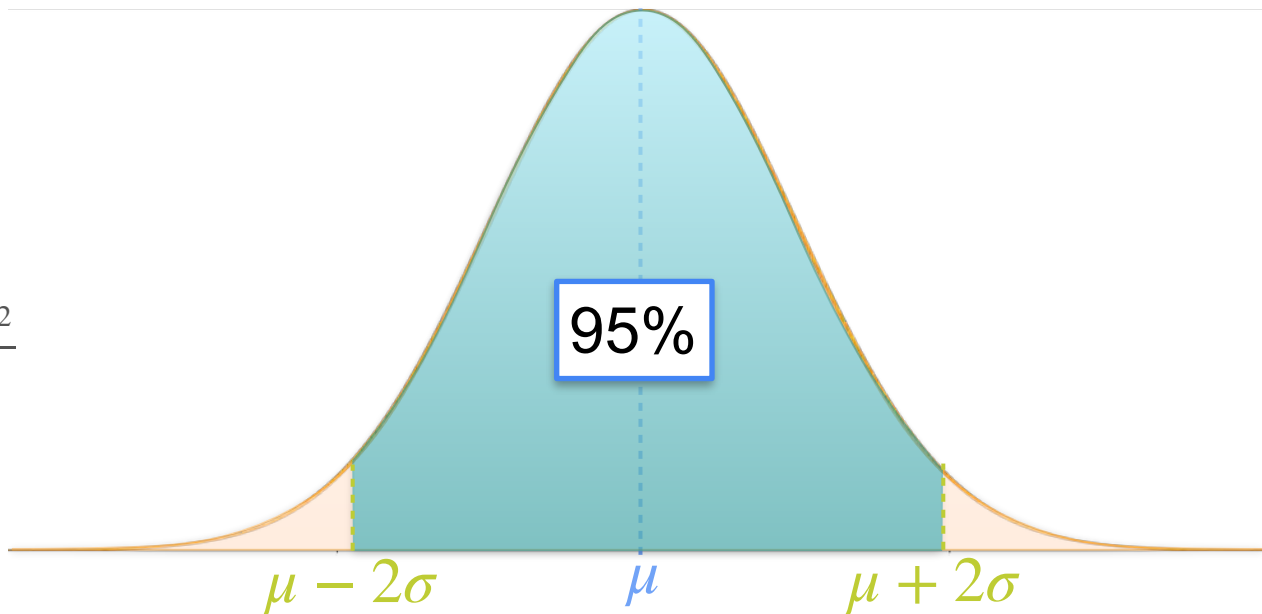


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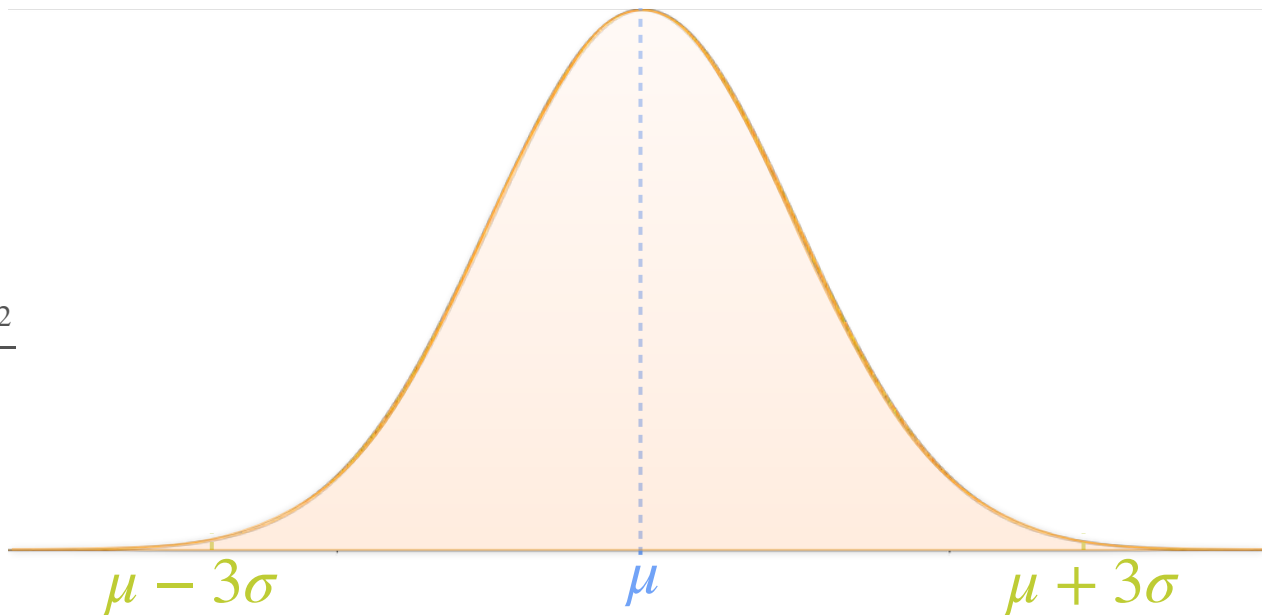


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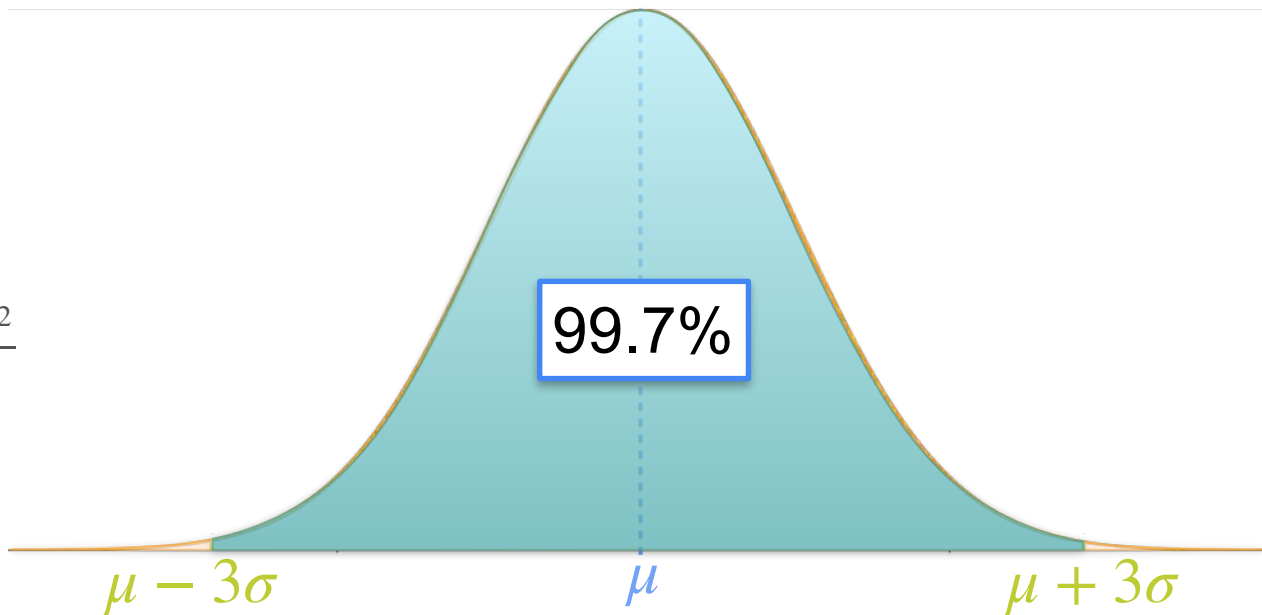


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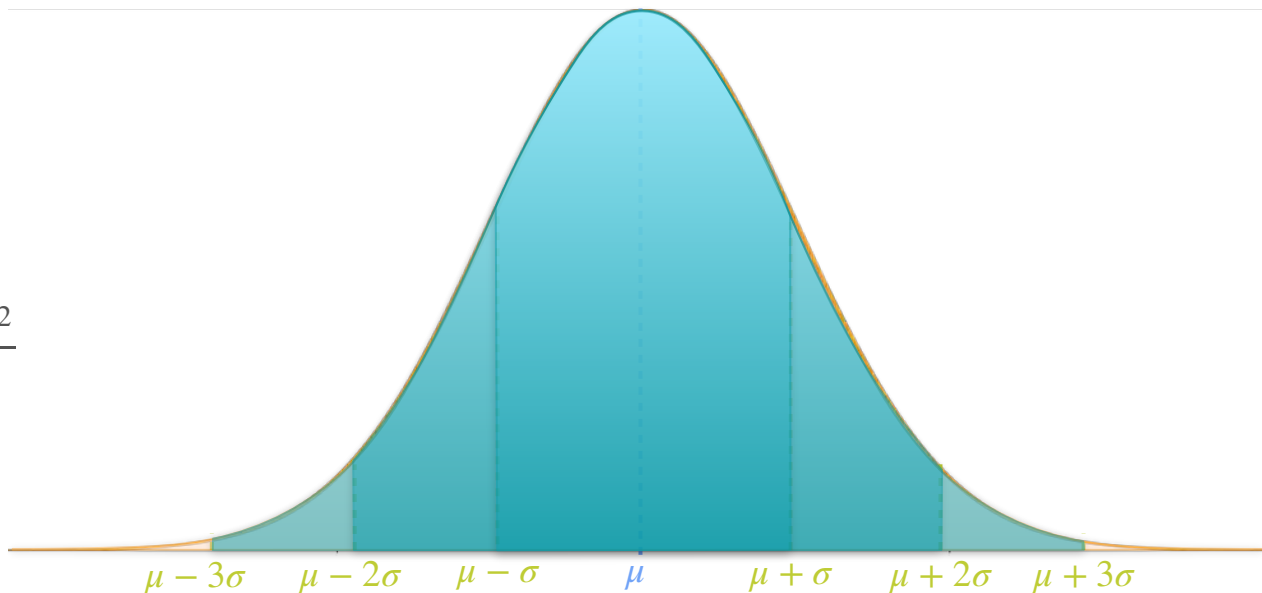


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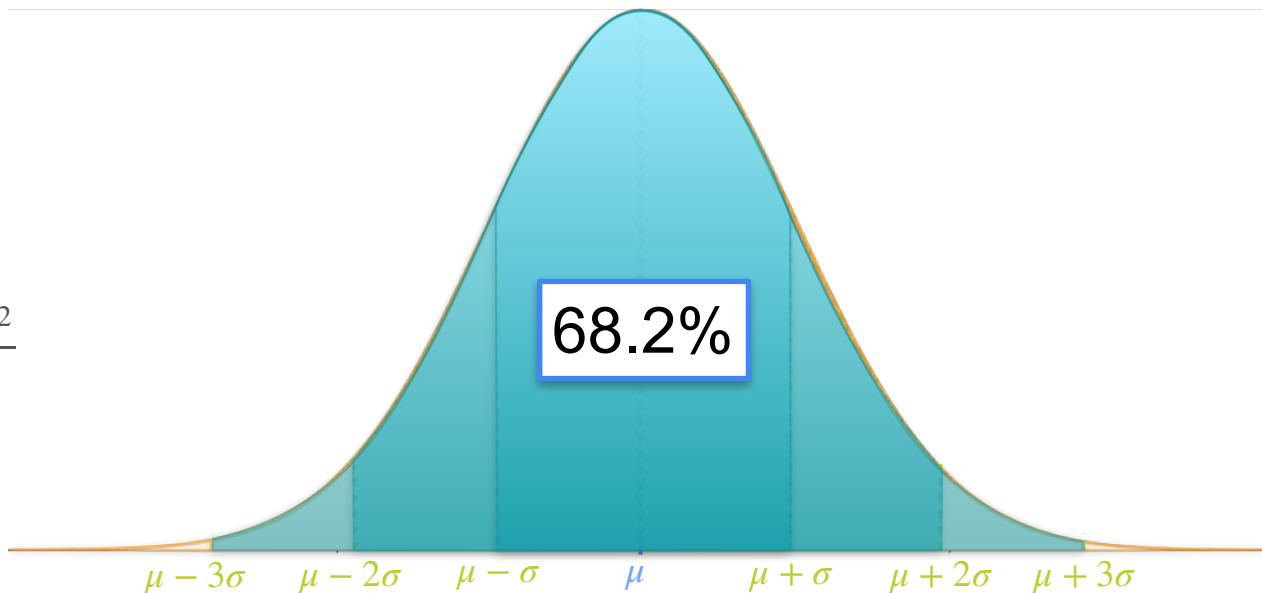


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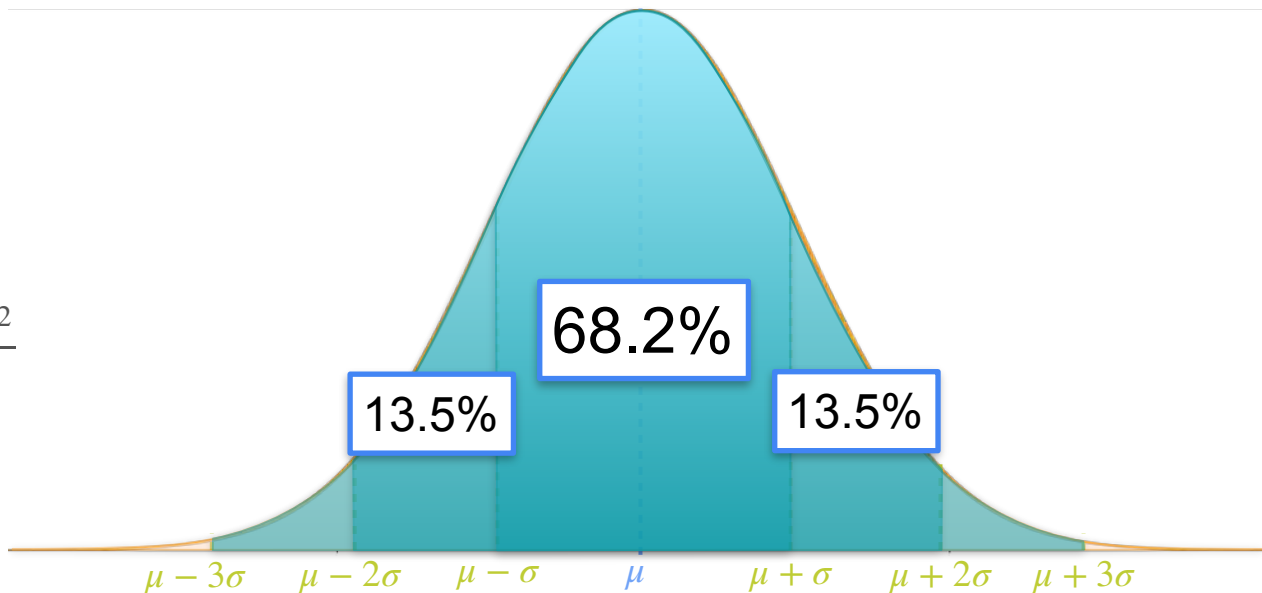


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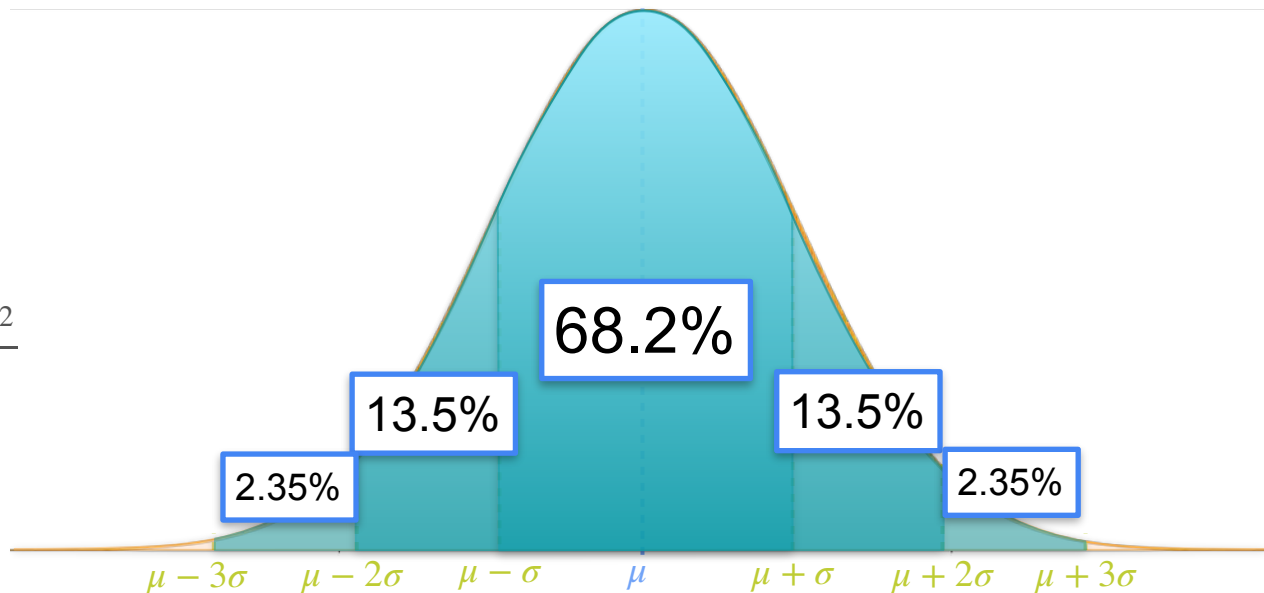


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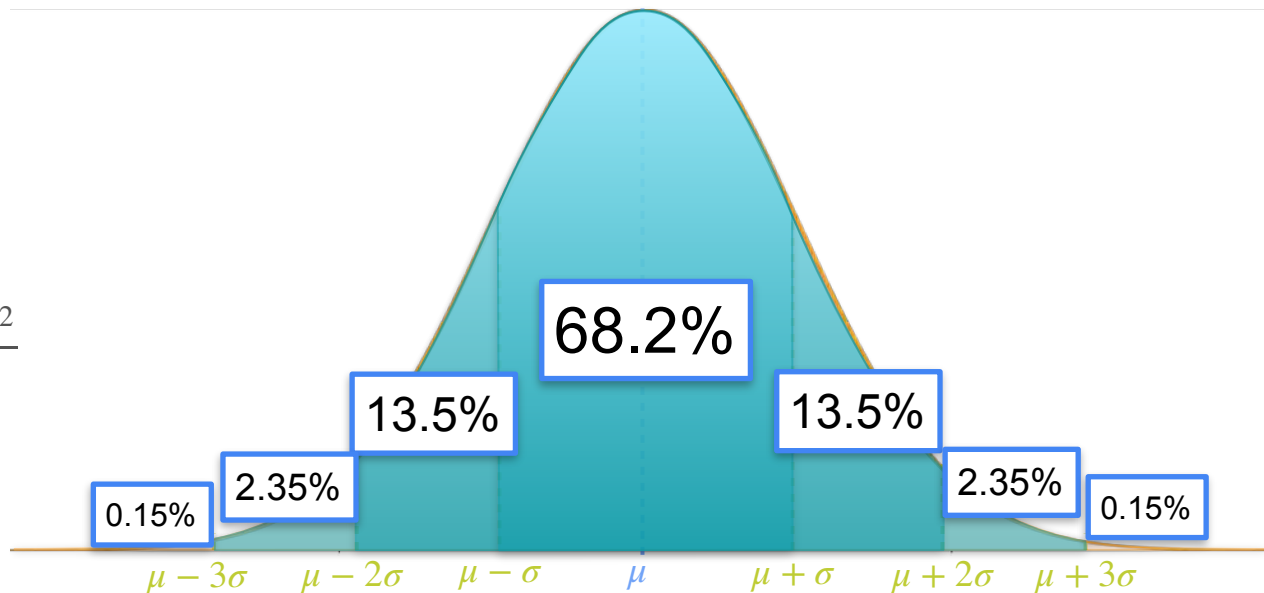


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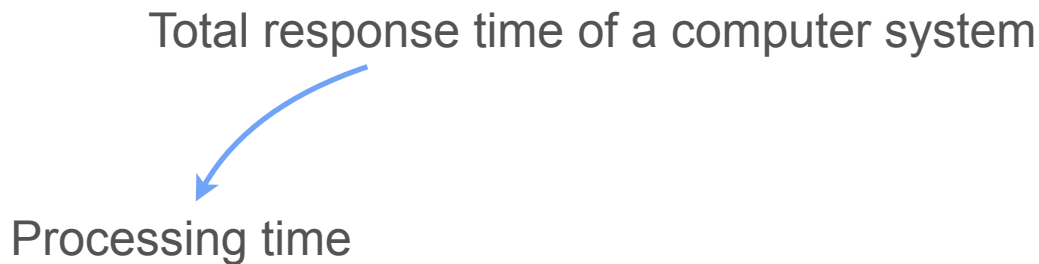


Sum of Gaussians: an Example

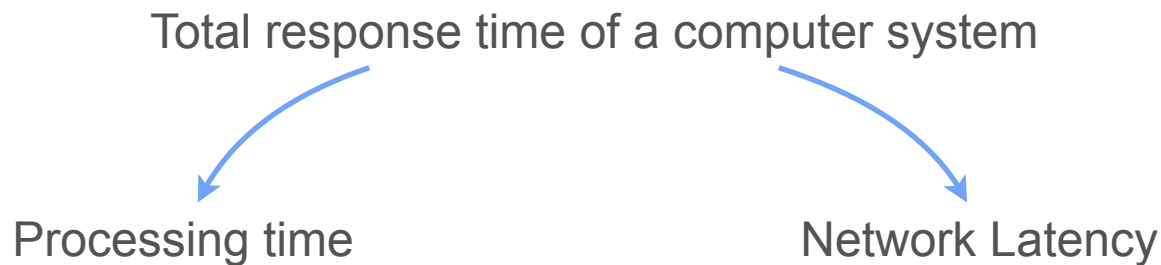
Sum of Gaussians: an Example

Total response time of a computer system

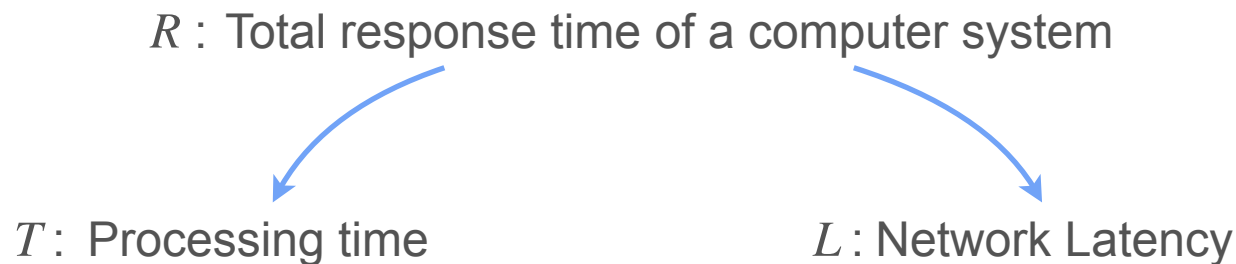
Sum of Gaussians: an Example



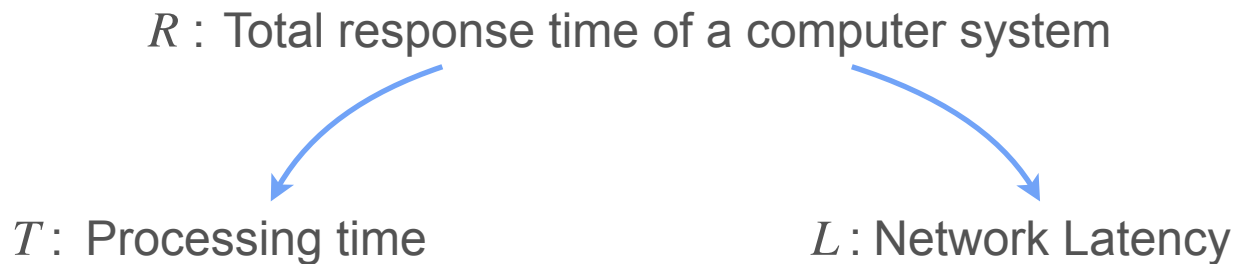
Sum of Gaussians: an Example



Sum of Gaussians: an Example

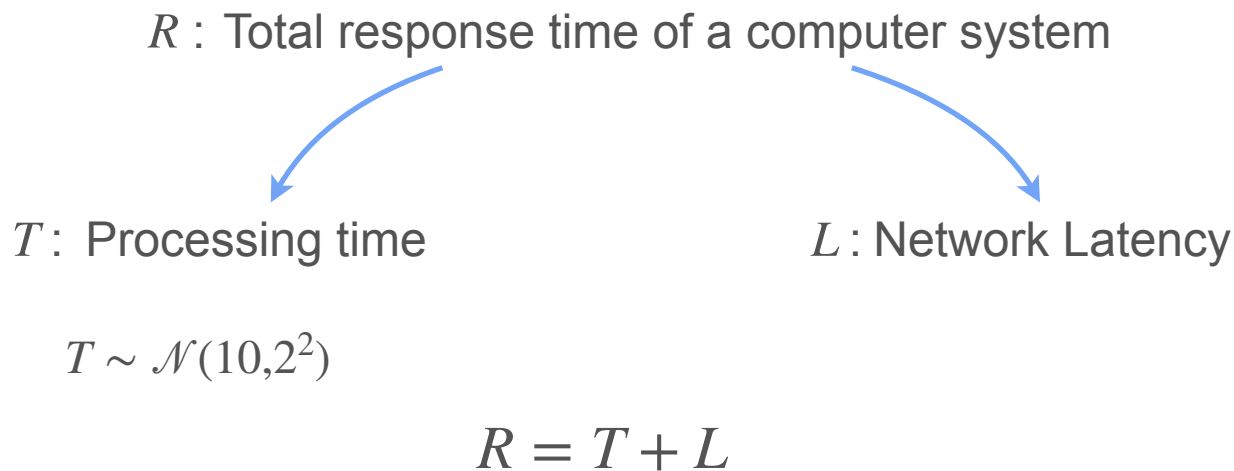


Sum of Gaussians: an Example

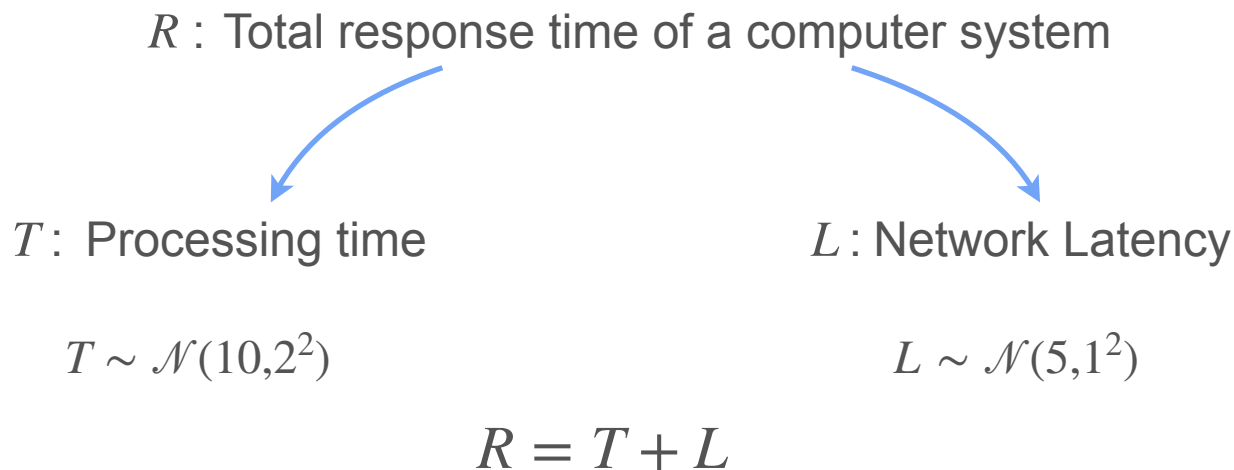


$$R = T + L$$

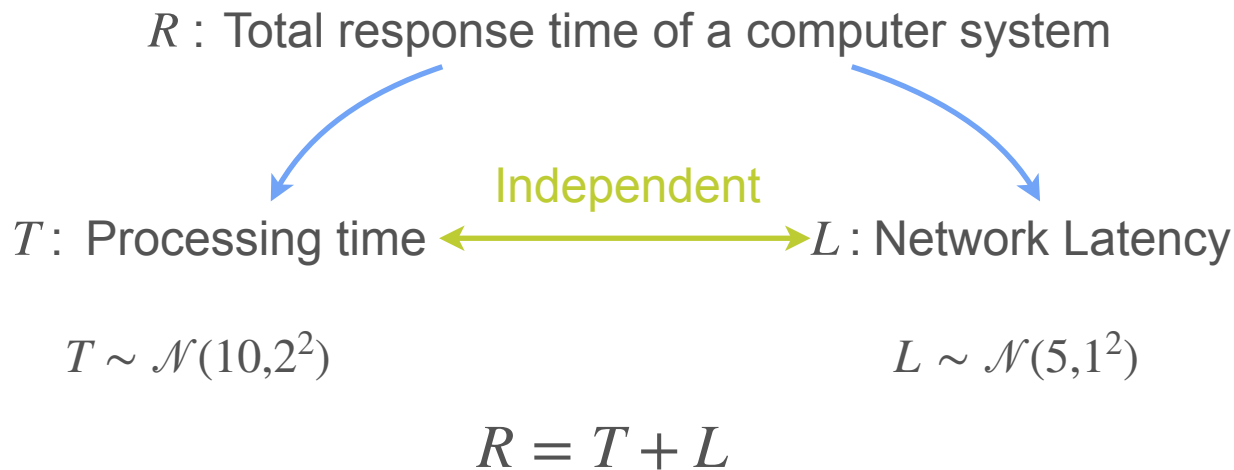
Sum of Gaussians: an Example



Sum of Gaussians: an Example



Sum of Gaussians: an Example



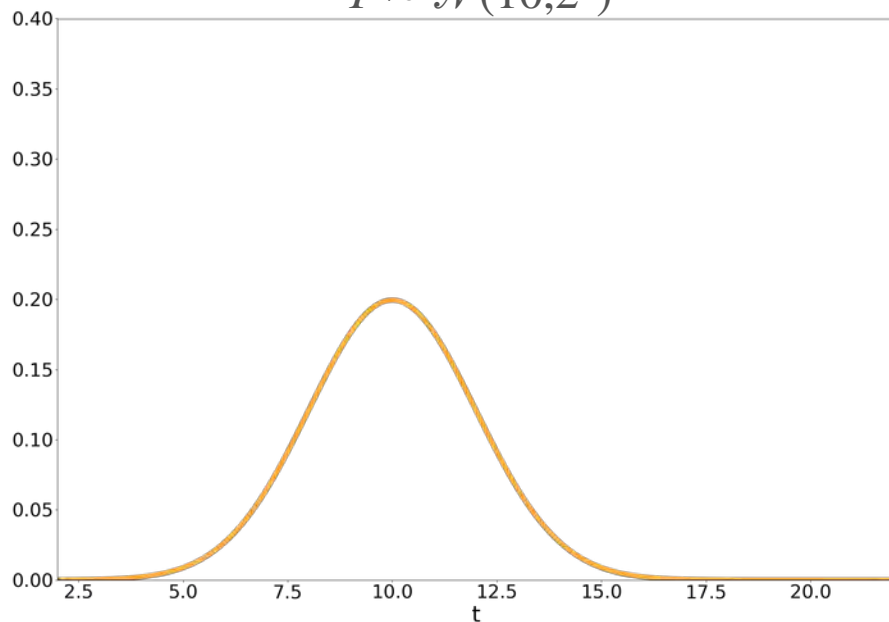
Sum Of Gaussians

$$T \sim \mathcal{N}(10, 2^2)$$

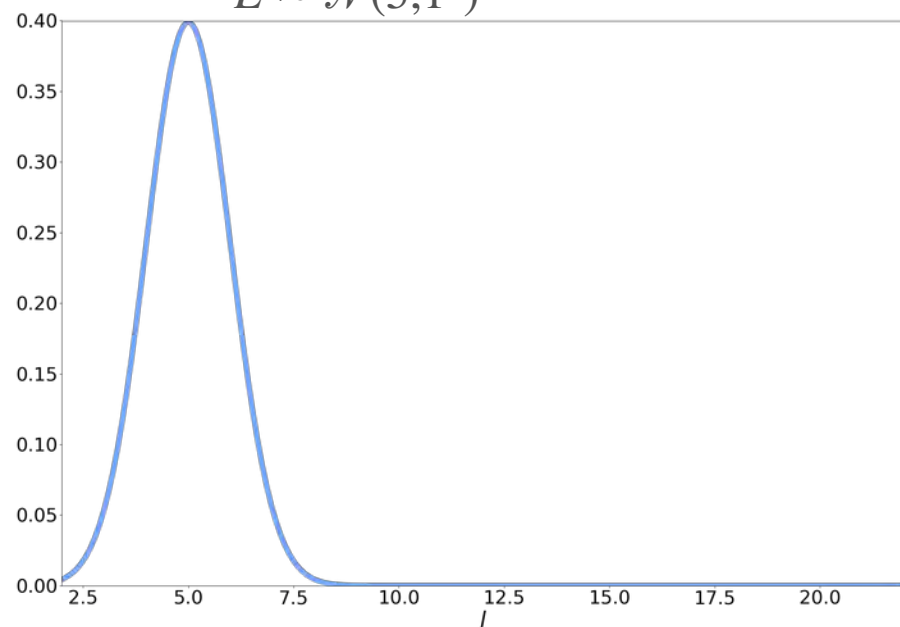
$$L \sim \mathcal{N}(5, 1^2)$$

Sum Of Gaussians

$$T \sim \mathcal{N}(10, 2^2)$$



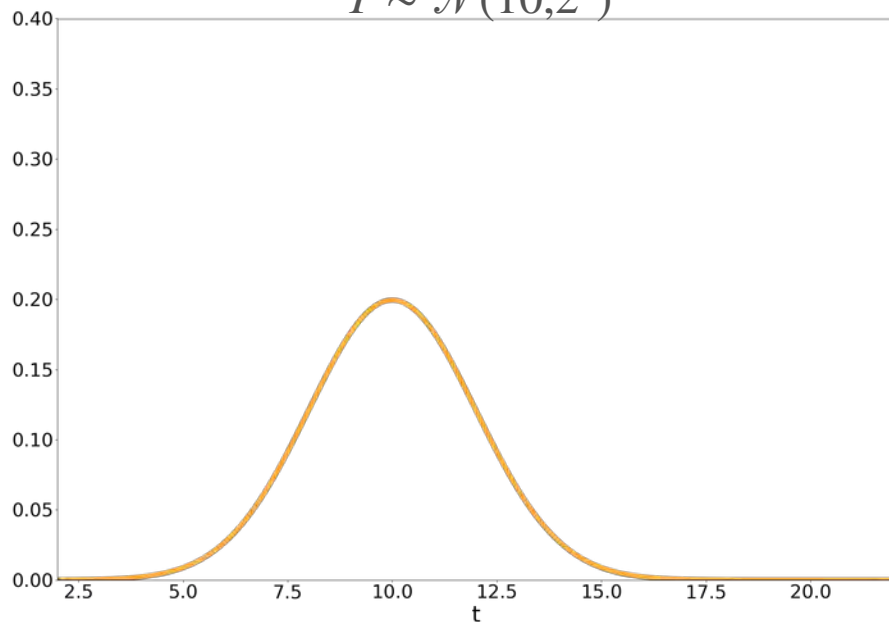
$$L \sim \mathcal{N}(5, 1^2)$$



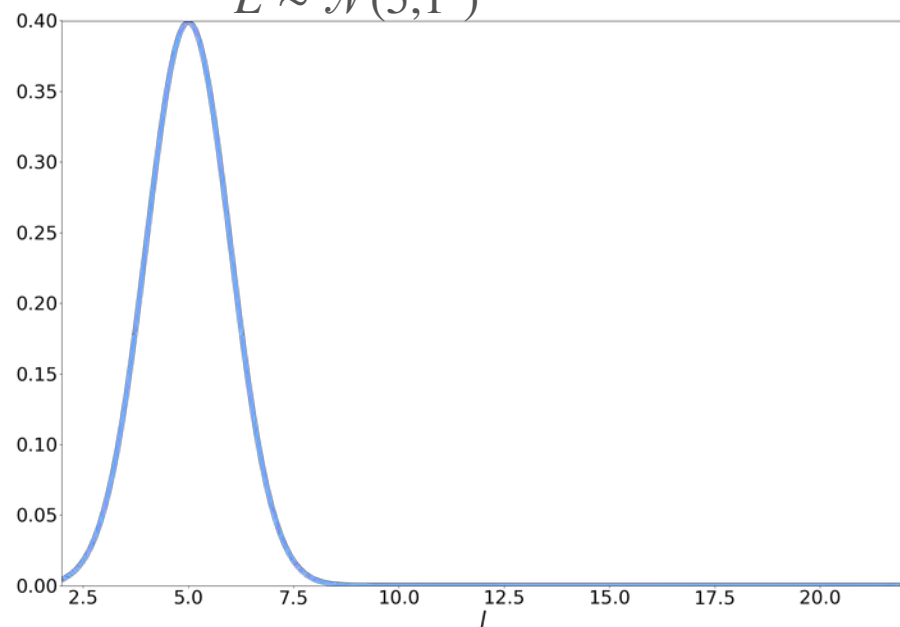
Sum Of Gaussians

Sample each variable 10000 times

$$T \sim \mathcal{N}(10, 2^2)$$



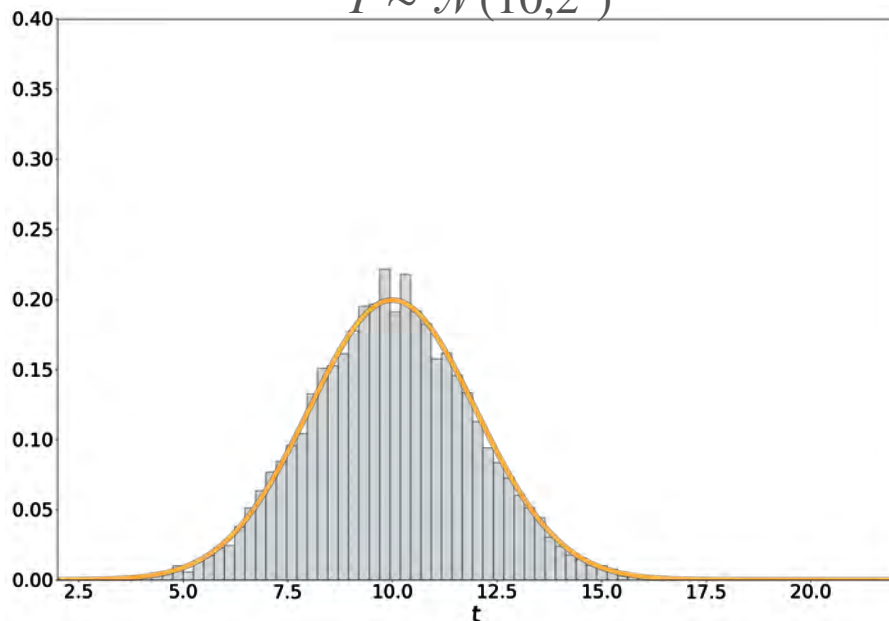
$$L \sim \mathcal{N}(5, 1^2)$$



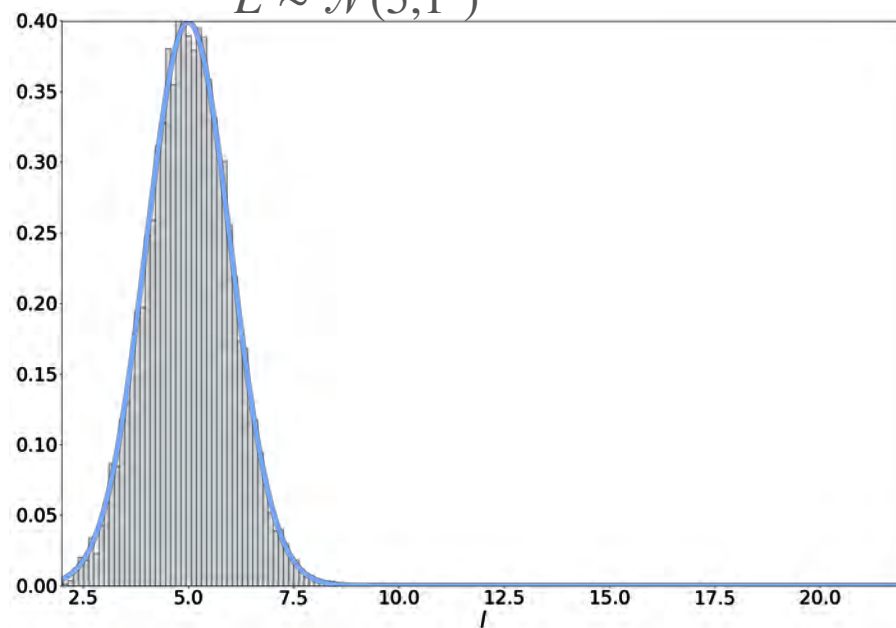
Sum Of Gaussians

Sample each variable 10000 times

$$T \sim \mathcal{N}(10, 2^2)$$



$$L \sim \mathcal{N}(5, 1^2)$$

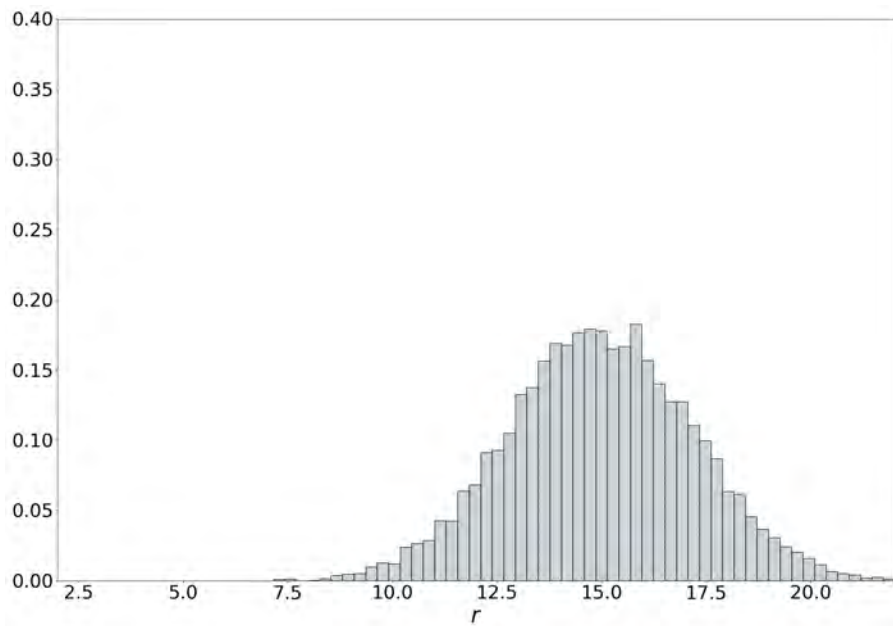


Sum Of Gaussians

$$R = T + L$$

Sum Of Gaussians

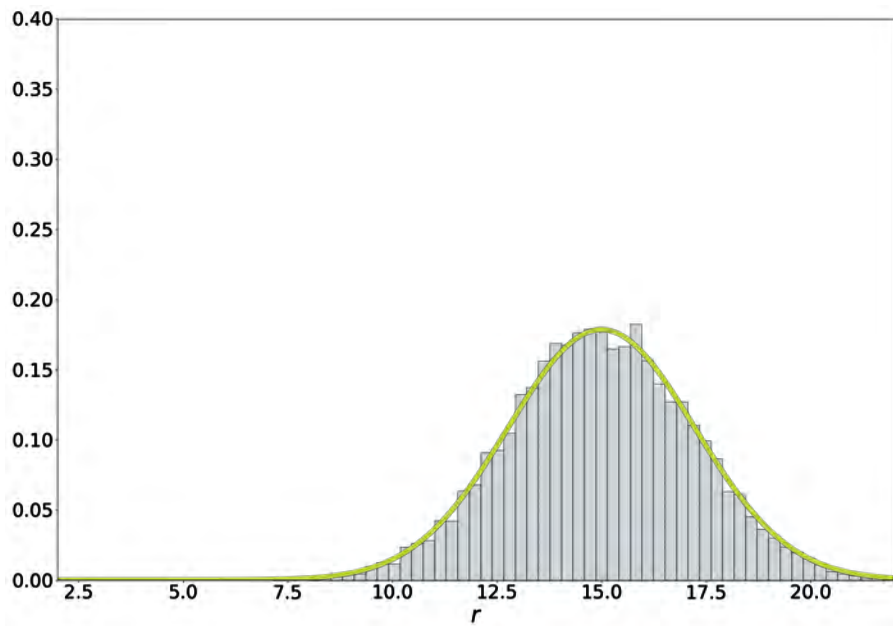
$$R = T + L$$



Sum Of Gaussians

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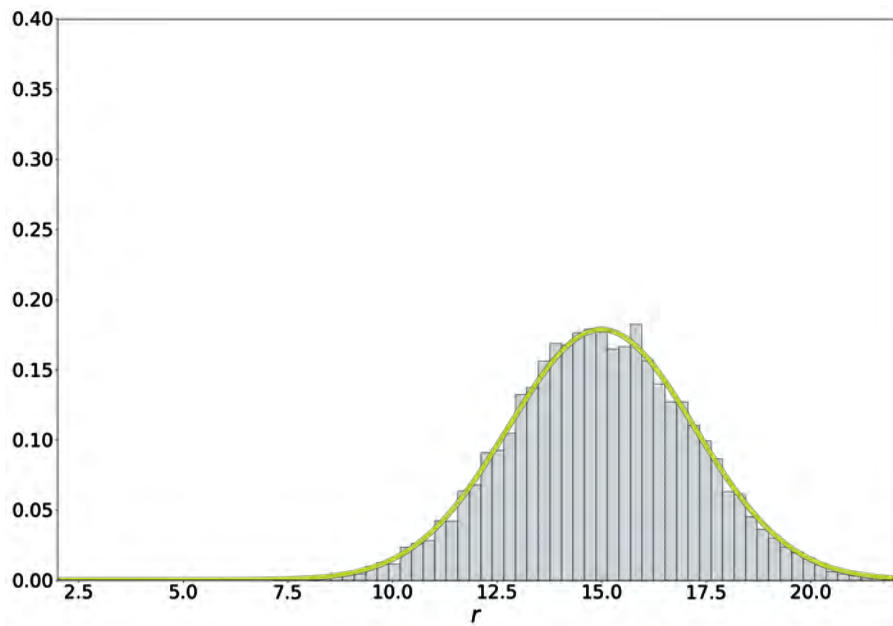
R is still Gaussian!



Sum Of Gaussians

$$R = T + L$$

R is still Gaussian!



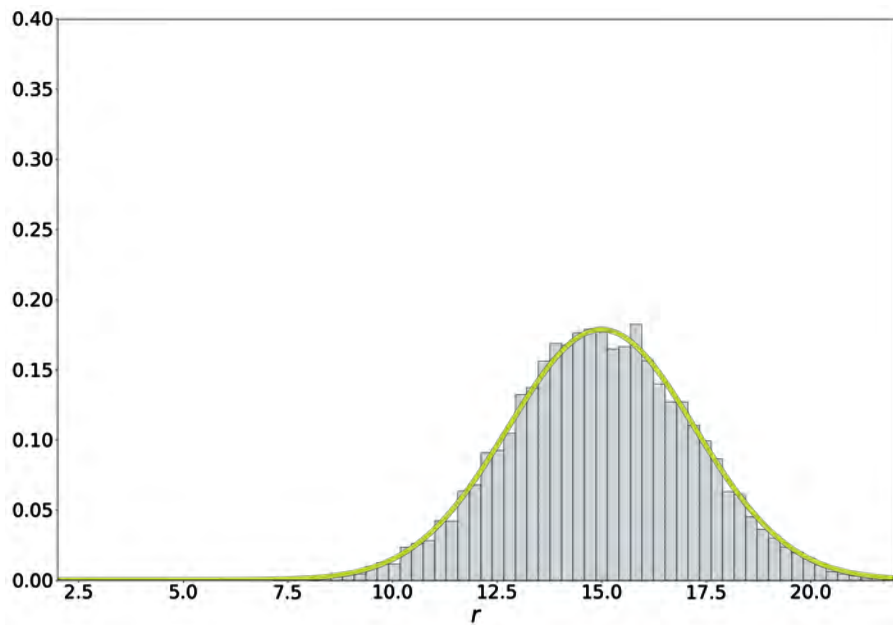
$$\mu_R = \mathbb{E}[R]$$

Sum Of Gaussians

$$R = T + L$$

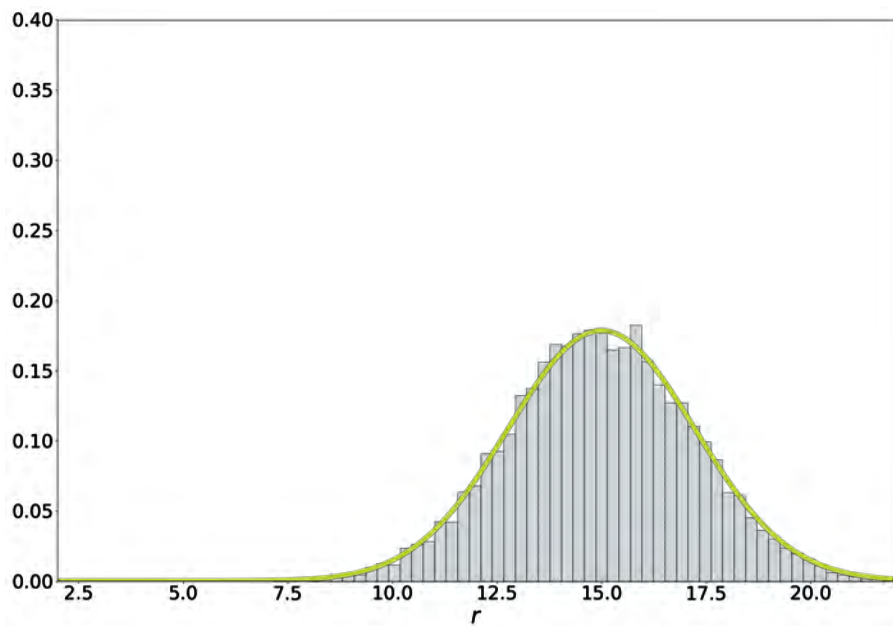
R is still Gaussian!

$$\mu_R = \mathbb{E}[R] = \mathbb{E}[T + L]$$



Sum Of Gaussians

$$R = T + L$$



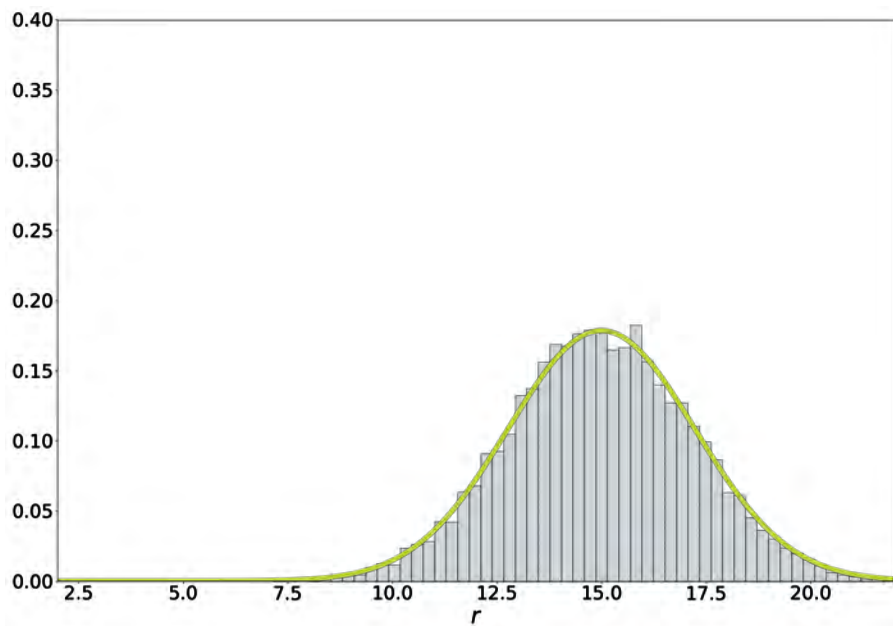
R is still Gaussian!

Expectation is linear

$$\mu_R = \mathbb{E}[R] = \mathbb{E}[T + L] = \mathbb{E}[T] + \mathbb{E}[L]$$

Sum Of Gaussians

$$R = T + L$$



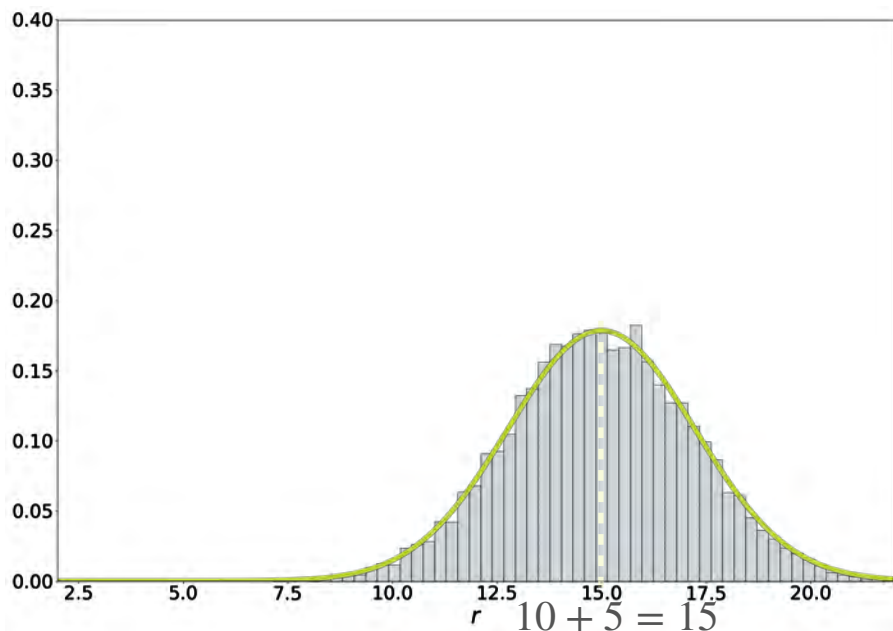
R is still Gaussian!

Expectation is linear

$$\begin{aligned}\mu_R = \mathbb{E}[R] &= \mathbb{E}[T + L] = \mathbb{E}[T] + \mathbb{E}[L] \\ &= \mu_T + \mu_L\end{aligned}$$

Sum Of Gaussians

$$R = T + L$$



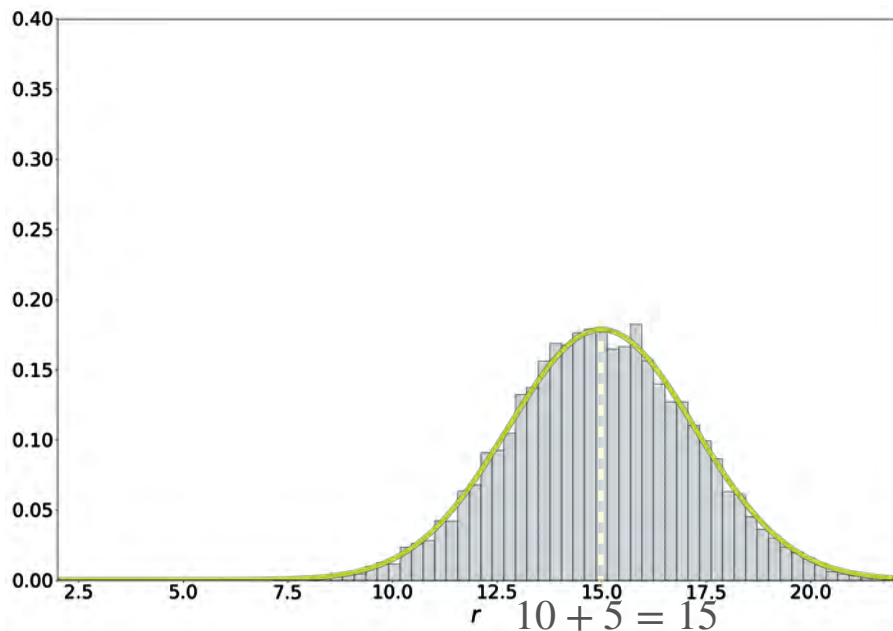
R is still Gaussian!

Expectation is linear

$$\begin{aligned}\mu_R = \mathbb{E}[R] &= \mathbb{E}[T + L] = \mathbb{E}[T] + \mathbb{E}[L] \\ &= \mu_T + \mu_L = 10 + 5 = 15\end{aligned}$$

Sum Of Gaussians

$$R = T + L$$



R is still Gaussian!

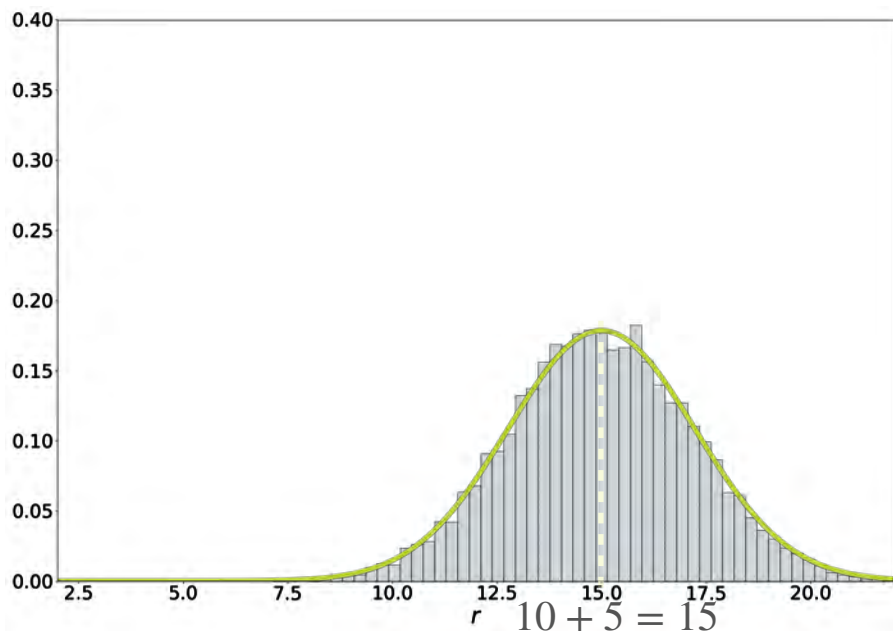
Expectation is linear

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$$\sigma_R^2 = \text{Var}(R)$$

Sum Of Gaussians

$$R = T + L$$



R is still Gaussian!

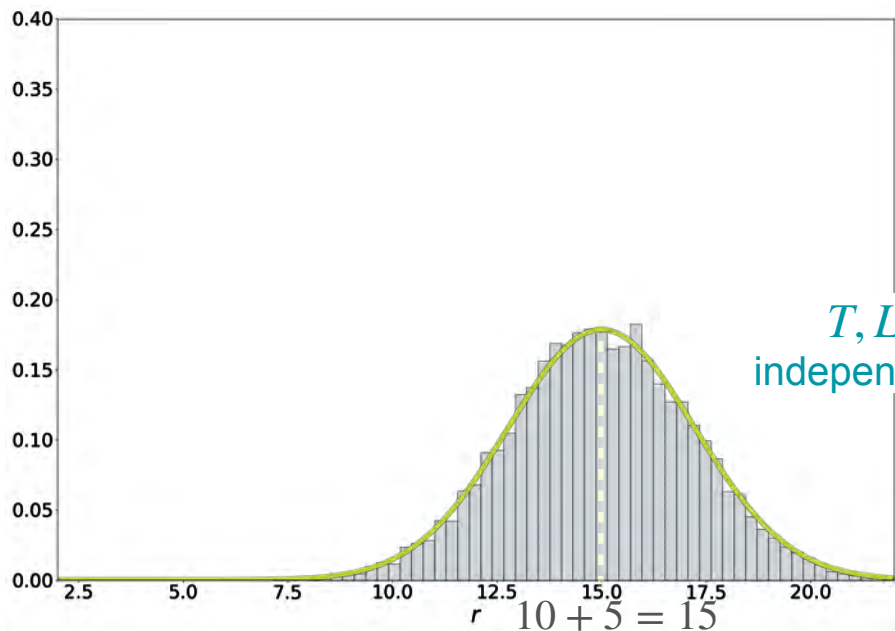
Expectation is linear

$$\begin{aligned}\mu_R = \mathbb{E}[R] &= \mathbb{E}[T + L] = \mathbb{E}[T] + \mathbb{E}[L] \\ &= \mu_T + \mu_L = 10 + 5 = 15\end{aligned}$$

$$\sigma_R^2 = \text{Var}(R) \quad \text{Var}(T + L)$$

Sum Of Gaussians

$$R = T + L$$



R is still Gaussian!

Expectation is linear

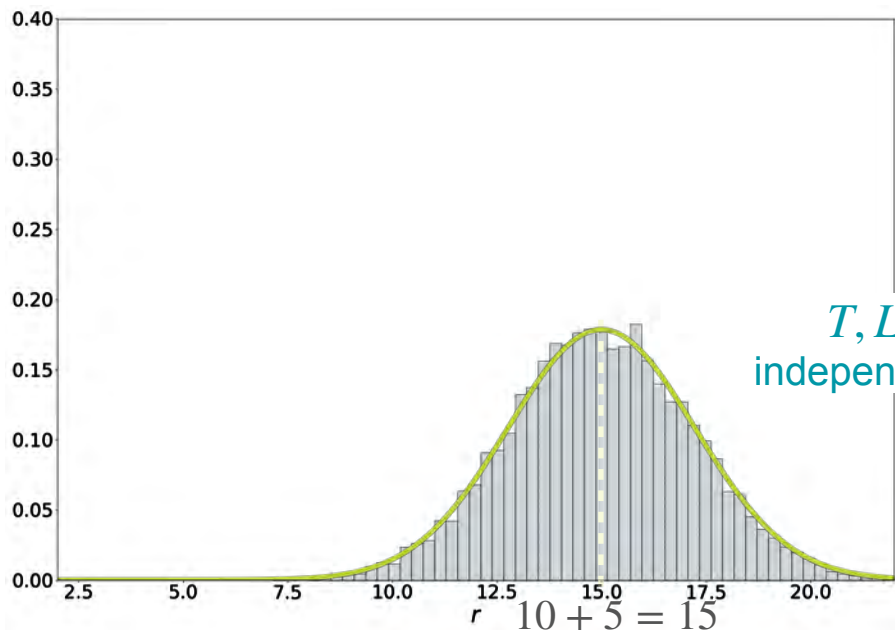
$$\begin{aligned}\mu_R = \mathbb{E}[R] &= \mathbb{E}[T + L] = \mathbb{E}[T] + \mathbb{E}[L] \\ &= \mu_T + \mu_L = 10 + 5 = 15\end{aligned}$$

T, L
independent

$$\begin{aligned}\sigma_R^2 &= \text{Var}(R) = \text{Var}(T + L) \\ &= \text{Var}(T) + \text{Var}(L)\end{aligned}$$

Sum Of Gaussians

$$R = T + L$$



R is still Gaussian!

Expectation is linear

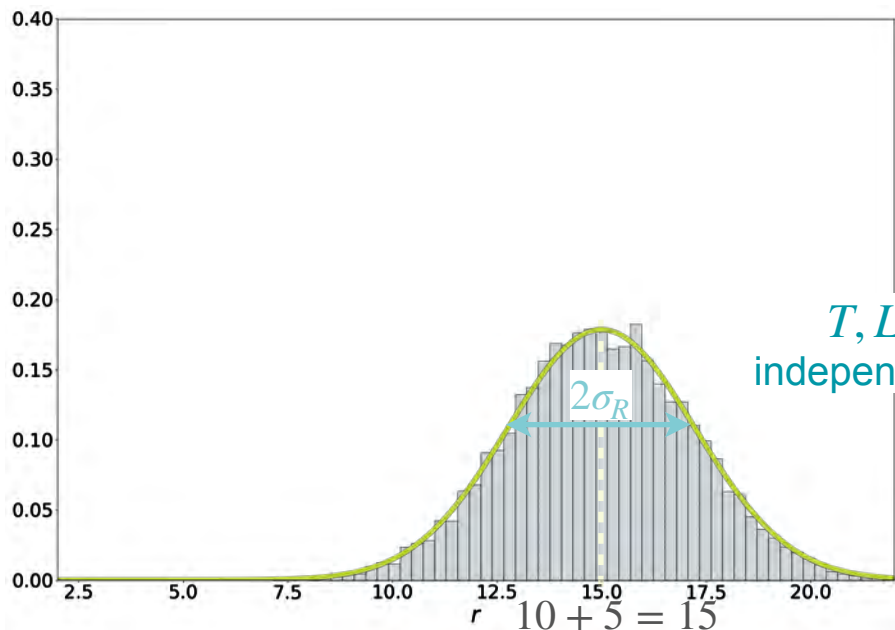
$$\begin{aligned}\mu_R &= \mathbb{E}[R] = \mathbb{E}[T + L] = \mathbb{E}[T] + \mathbb{E}[L] \\ &= \mu_T + \mu_L = 10 + 5 = 15\end{aligned}$$

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$$\begin{aligned}\sigma_R^2 &= \text{Var}(R) = \text{Var}(T + L) \\ &= \text{Var}(T) + \text{Var}(L) = \sigma_T^2 + \sigma_L^2\end{aligned}$$

Sum Of Gaussians

$$R = T + L$$



R is still Gaussian!

Expectation is linear

$$\begin{aligned}\mu_R &= \mathbb{E}[R] = \mathbb{E}[T + L] = \mathbb{E}[T] + \mathbb{E}[L] \\ &= \mu_T + \mu_L = 10 + 5 = 15\end{aligned}$$

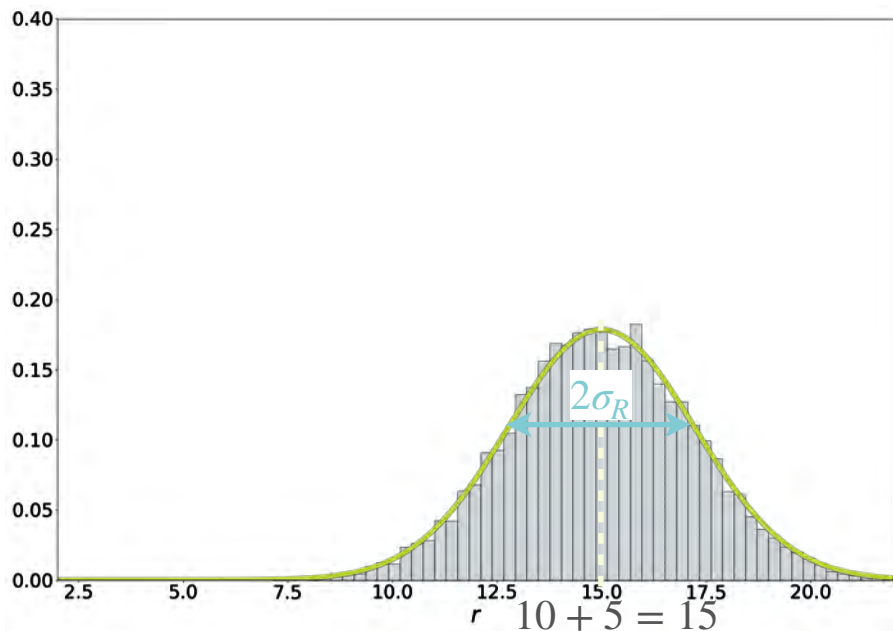
T, L
independent

$$\begin{aligned}\sigma_R^2 &= \text{Var}(R) = \text{Var}(T + L) \\ &= \text{Var}(T) + \text{Var}(L) = \sigma_T^2 + \sigma_L^2 \\ &= 4 + 1 = 5\end{aligned}$$

Sum Of Gaussians

$$R = T + L$$

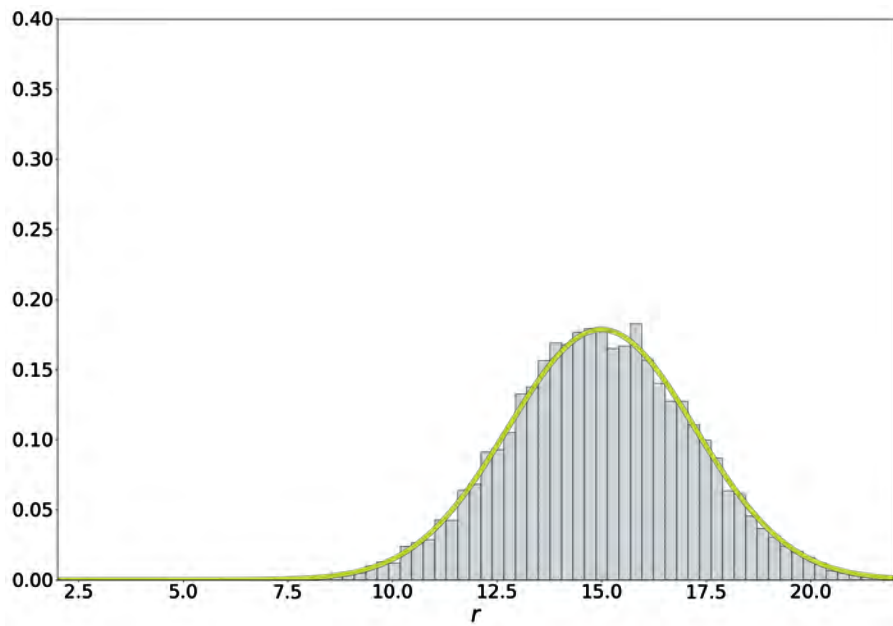
R is still Gaussian!



$$R = (T + L) \sim \mathcal{N} \left(10 + 5, \quad 4 + 1 \right)$$

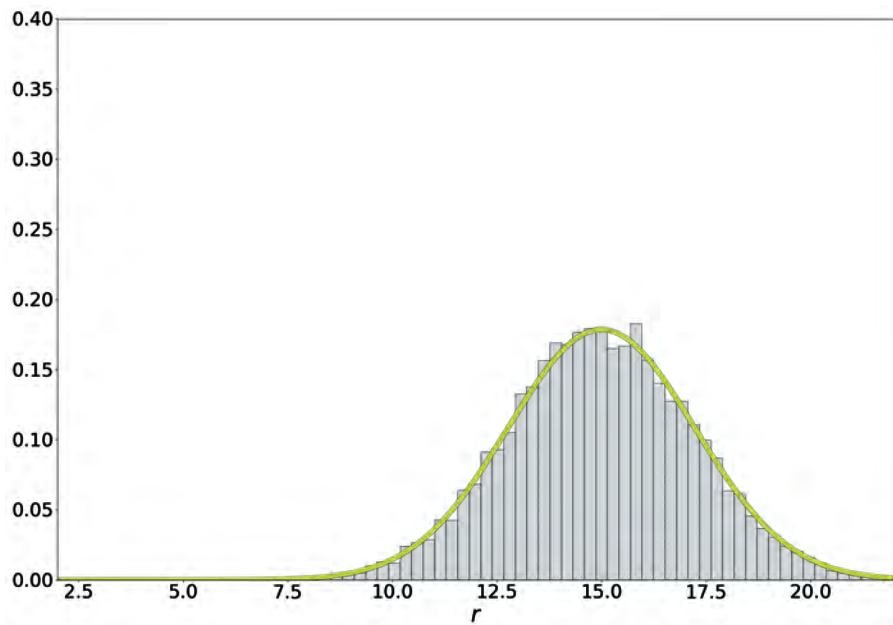
Sum Of Gaussians

$$R = T + L$$



Sum Of Gaussians

$$R = T + L$$

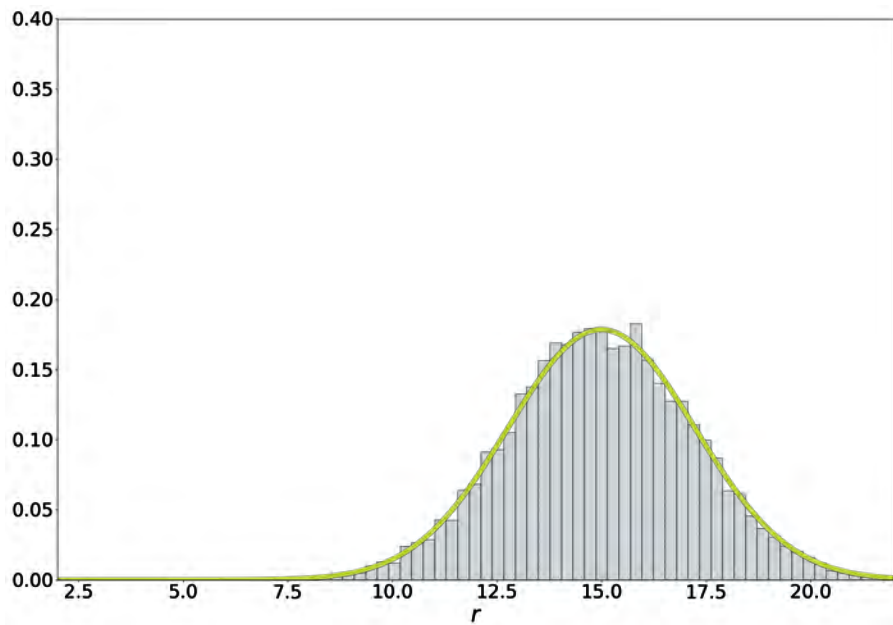


In general: $W = aX + bY$

Independent $\begin{cases} X \sim \mathcal{N}(\mu_X, \sigma_X^2) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2) \end{cases}$

Sum Of Gaussians

$$R = T + L$$



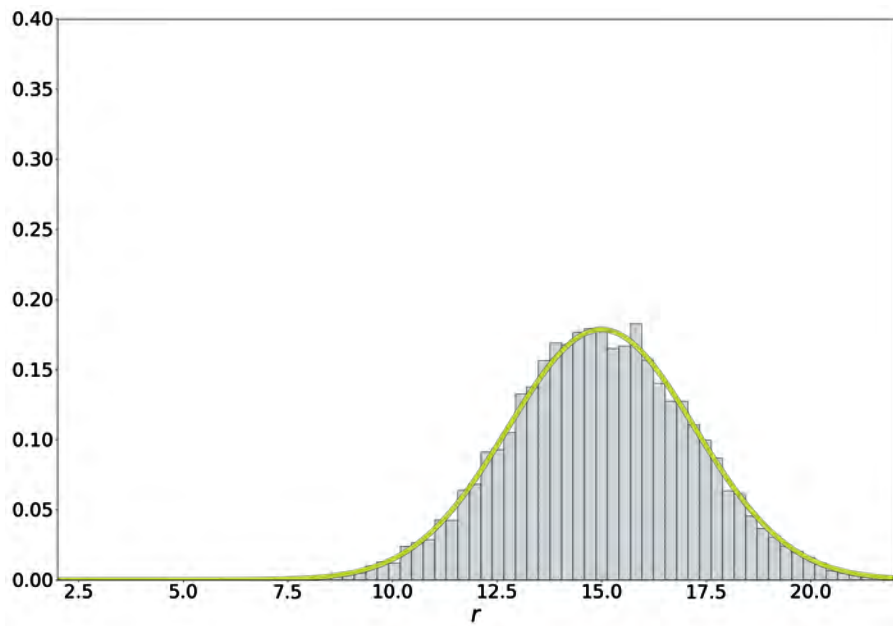
In general: $W = aX + bY$

Independent $\begin{cases} X \sim \mathcal{N}(\mu_X, \sigma_X^2) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2) \end{cases}$

$$\rightarrow W \sim \mathcal{N} \left(\quad , \quad \right)$$

Sum Of Gaussians

$$R = T + L$$



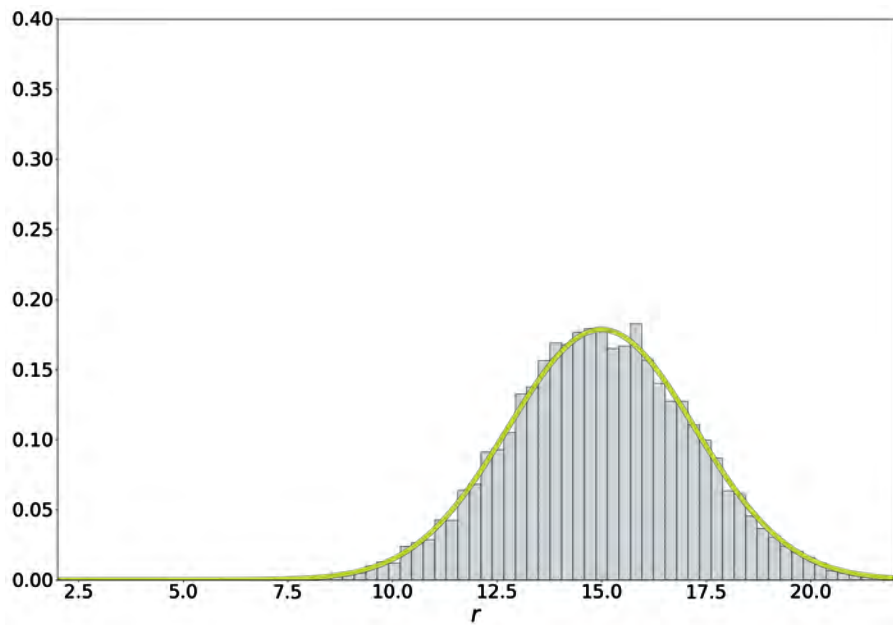
In general: $W = aX + bY$

Independent $\begin{cases} X \sim \mathcal{N}(\mu_X, \sigma_X^2) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2) \end{cases}$

$$\rightarrow W \sim \mathcal{N} \left(a\mu_x + b\mu_y, \right)$$

Sum Of Gaussians

$$R = T + L$$



In general: $W = aX + bY$

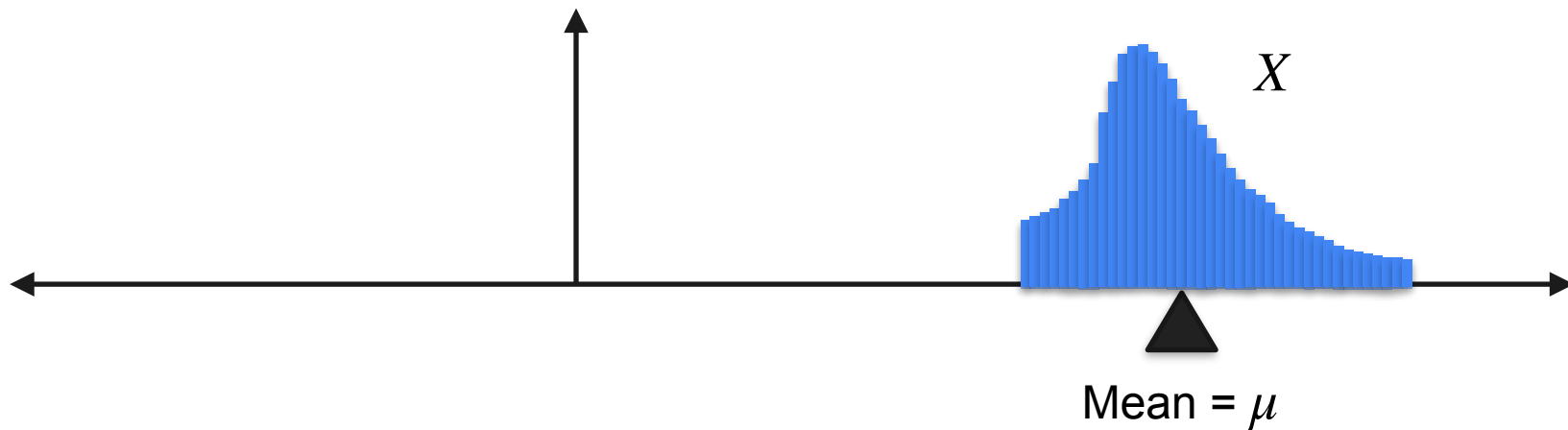
Independent $\begin{cases} X \sim \mathcal{N}(\mu_X, \sigma_X^2) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2) \end{cases}$

$$\rightarrow W \sim \mathcal{N}\left(a\mu_x + b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2\right)$$

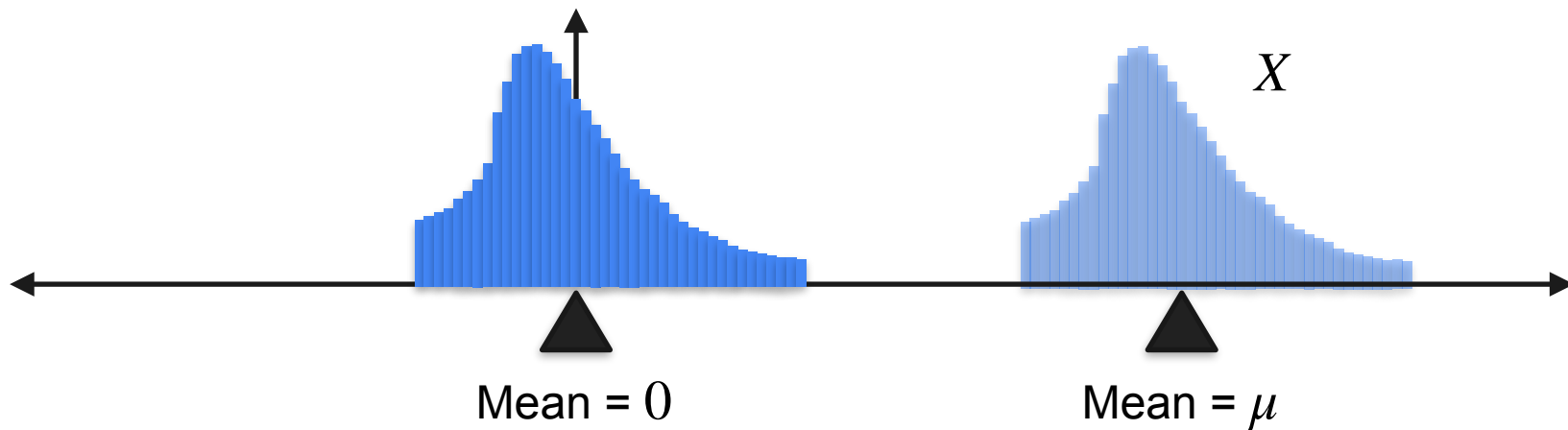
Everything Is Nicer When the Mean Is 0



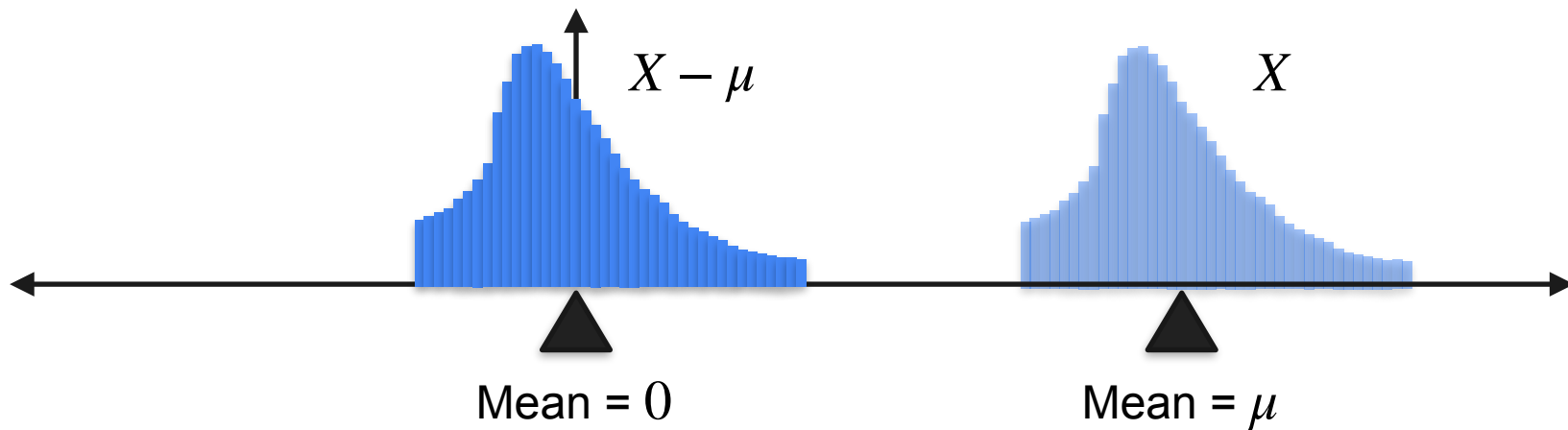
Everything Is Nicer When the Mean Is 0



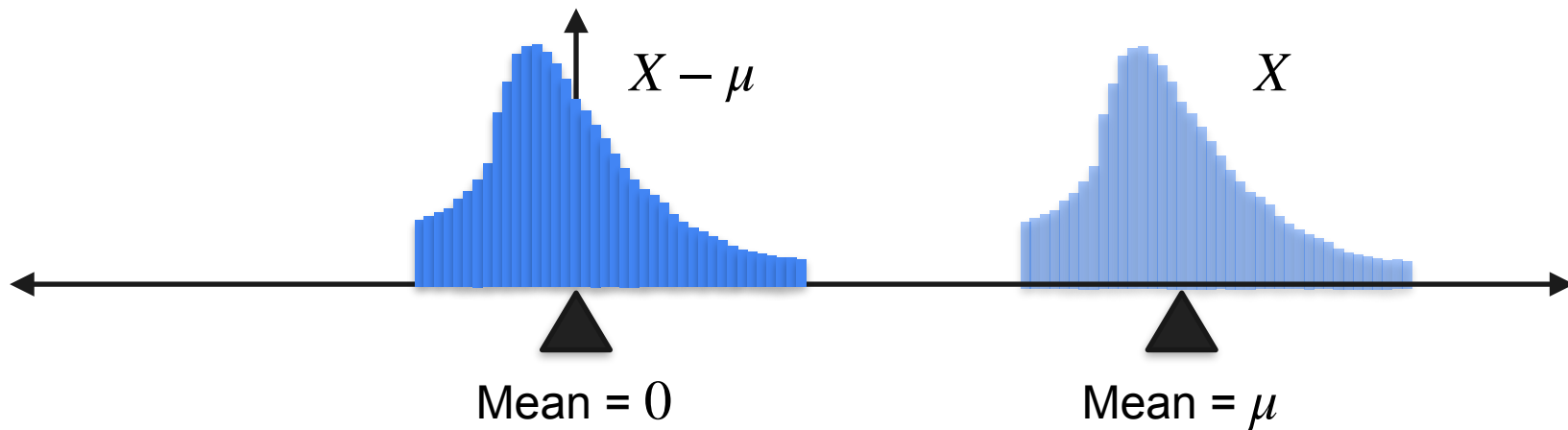
Everything Is Nicer When the Mean Is 0



Everything Is Nicer When the Mean Is 0



Everything Is Nicer When the Mean Is 0



$$X \rightarrow X - \mu$$

Everything Is Nicer When the Mean Is 0

Why?

Everything Is Nicer When the Mean Is 0

Why?

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

Everything Is Nicer When the Mean Is 0

Why?

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

$$\mathbb{E}[X - \mu]$$

Everything Is Nicer When the Mean Is 0

Why?

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

$$\mathbb{E}[X - \mu] = \mathbb{E}[X] - \mathbb{E}[\mu]$$

Everything Is Nicer When the Mean Is 0

Why?

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

$$\mathbb{E}[X - \mu] = \mathbb{E}[X] - \mathbb{E}[\mu]$$

$$= \mathbb{E}[X] - \mu$$

Everything Is Nicer When the Mean Is 0

Why?

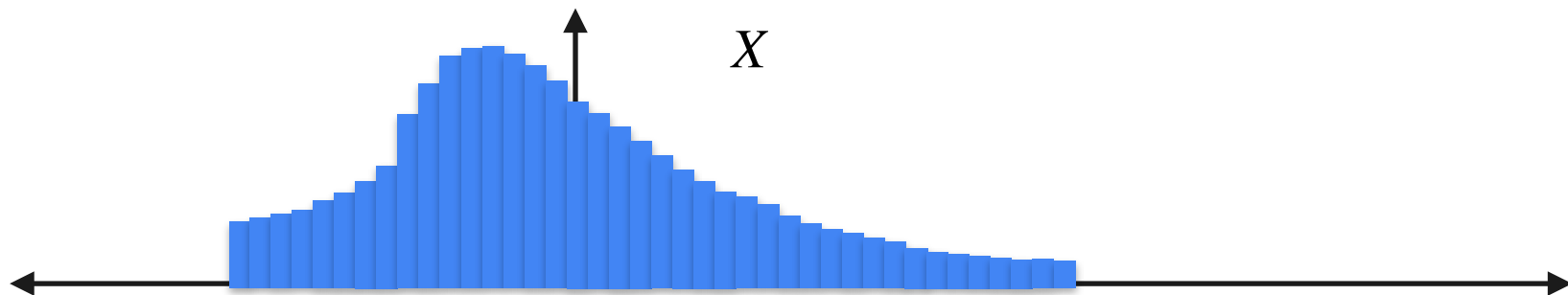
$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

$$\mathbb{E}[X - \mu] = \mathbb{E}[X] - \mathbb{E}[\mu]$$

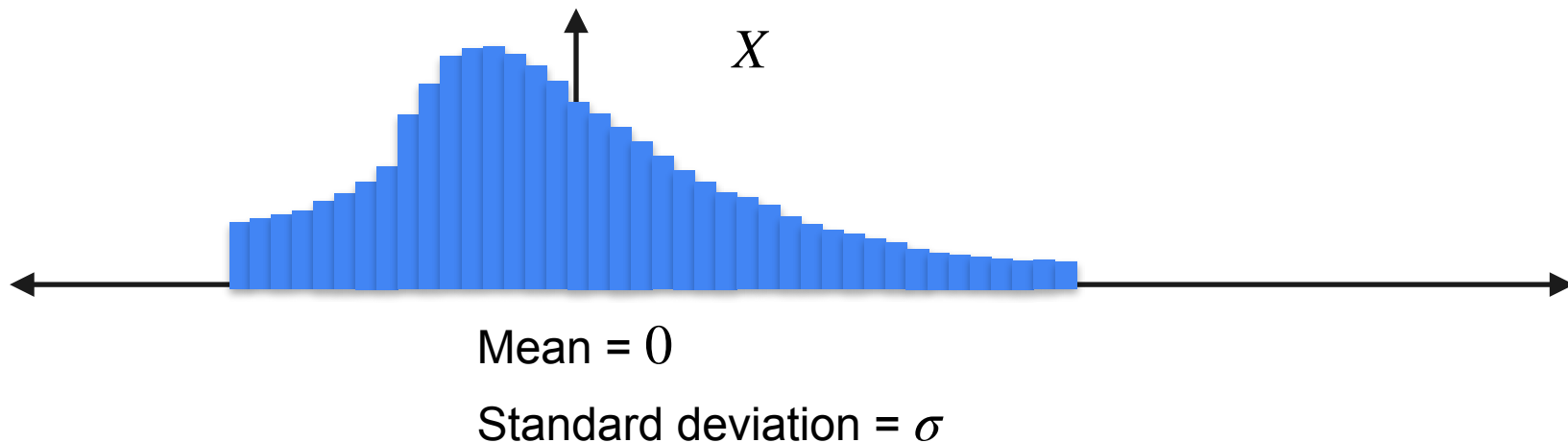
$$= \mathbb{E}[X] - \mu$$

$$= 0$$

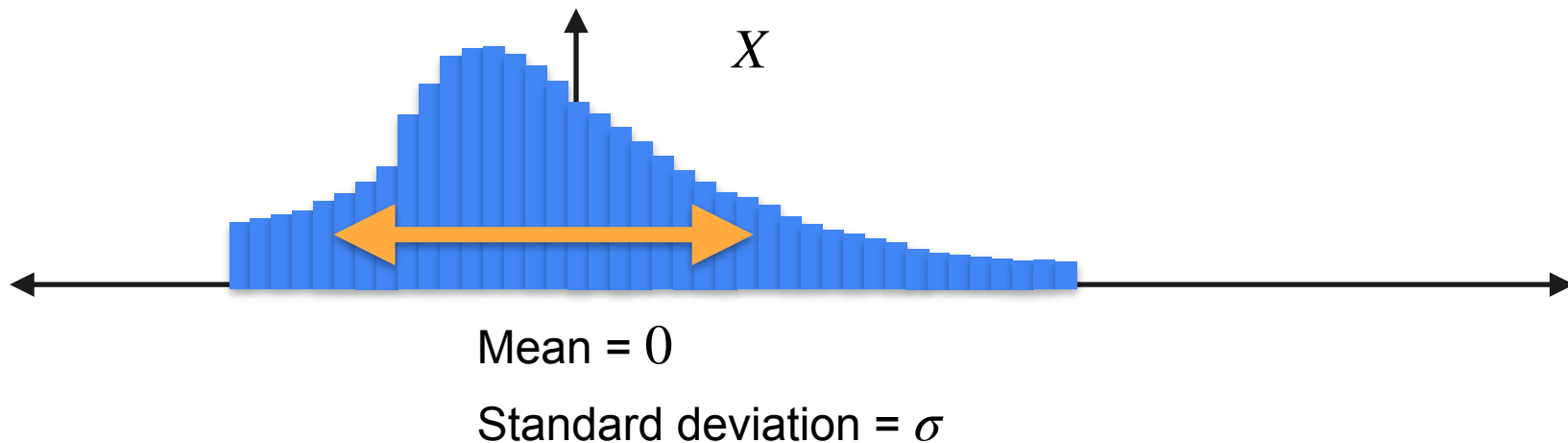
Everything Is Nicer When the Standard Deviation Is 1



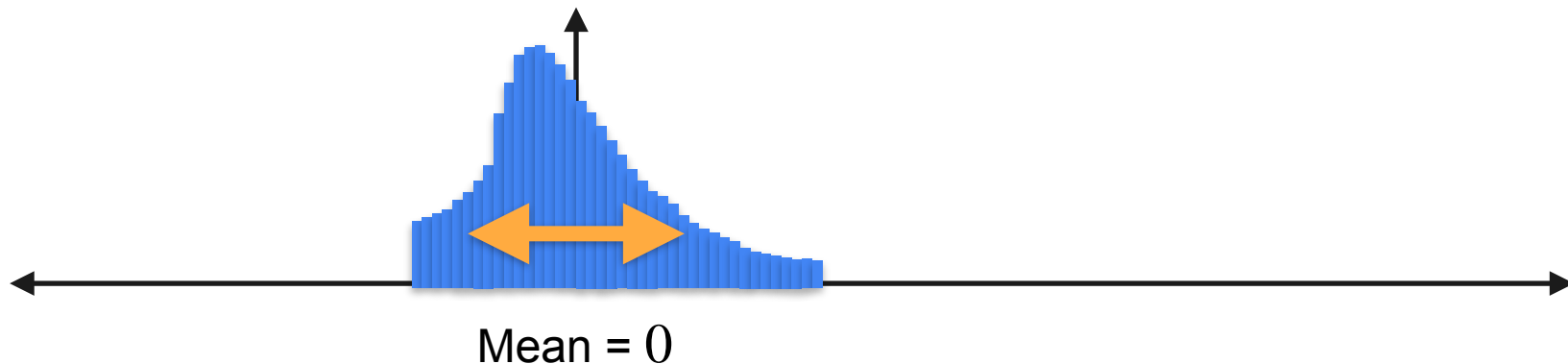
Everything Is Nicer When the Standard Deviation Is 1



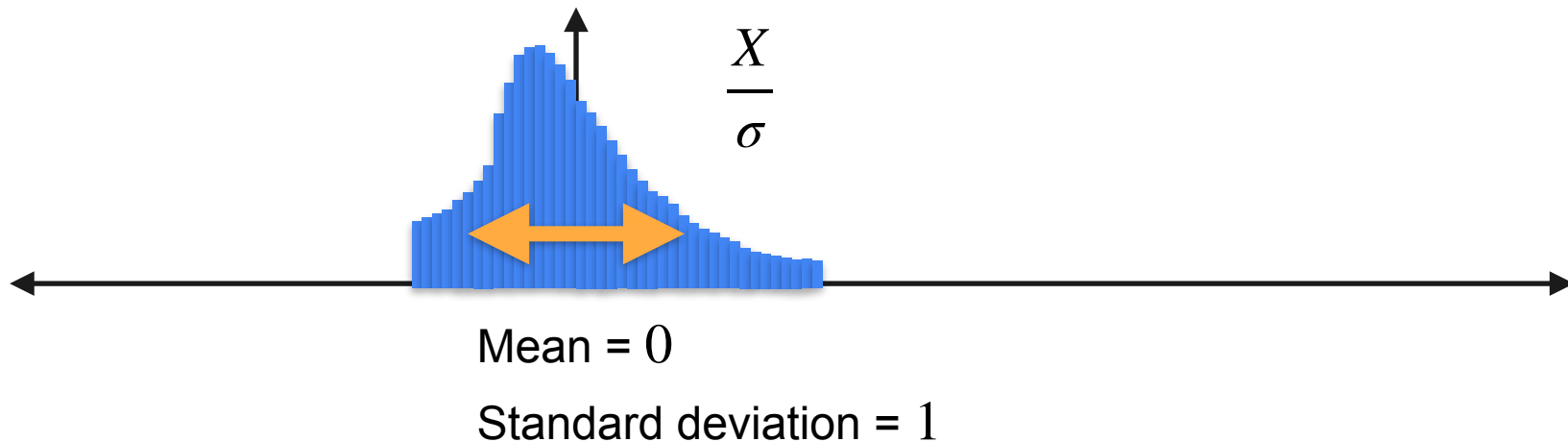
Everything Is Nicer When the Standard Deviation Is 1



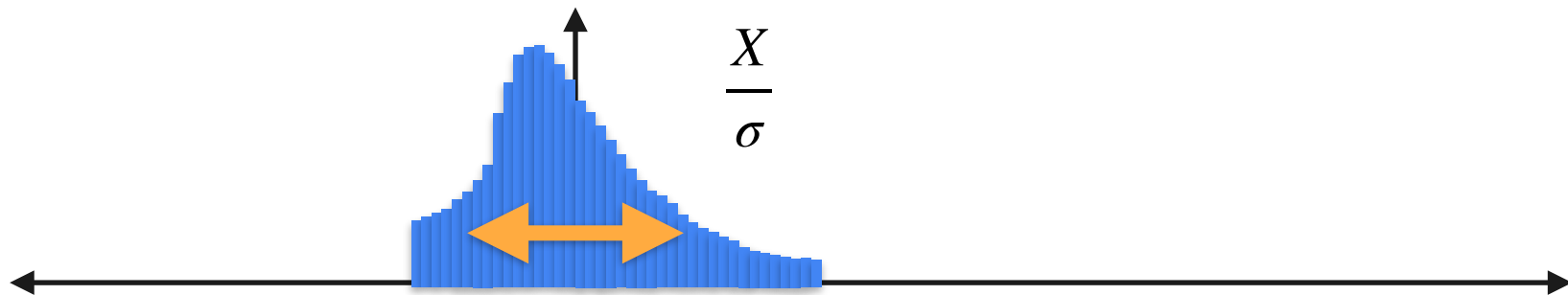
Everything Is Nicer When the Standard Deviation Is 1



Everything Is Nicer When the Standard Deviation Is 1



Everything Is Nicer When the Standard Deviation Is 1



Mean = 0

Standard deviation = 1

$$X \rightarrow \frac{X}{\sigma}$$

Everything Is Nicer When the Standard Deviation Is 1

Why?

$$\text{Var}\left(\frac{X}{\sigma}\right)$$

Everything Is Nicer When the Standard Deviation Is 1

Why?

$$\text{Var}(cX)$$

$$\text{Var}\left(\frac{X}{\sigma}\right)$$

Everything Is Nicer When the Standard Deviation Is 1

Why?

$$\text{Var}(cX) = \mathbb{E}[(cX)^2] - \mathbb{E}[cX]^2$$

$$\text{Var}\left(\frac{X}{\sigma}\right)$$

Everything Is Nicer When the Standard Deviation Is 1

Why?

$$\begin{aligned}\text{Var}(cX) &= \mathbb{E}[(cX)^2] - \mathbb{E}[cX]^2 \\ &= \mathbb{E}[c^2X^2] - (c\mathbb{E}[X])^2\end{aligned}$$

$$\text{Var}\left(\frac{X}{\sigma}\right)$$

Everything Is Nicer When the Standard Deviation Is 1

Why?

$$\text{Var}\left(\frac{X}{\sigma}\right)$$

$$\begin{aligned}\text{Var}(cX) &= \mathbb{E}[(cX)^2] - \mathbb{E}[cX]^2 \\ &= \mathbb{E}[c^2X^2] - (c\mathbb{E}[X])^2 \\ &= c^2\mathbb{E}[X^2] - c^2\mathbb{E}[X]^2\end{aligned}$$

Everything Is Nicer When the Standard Deviation Is 1

Why?

$$\text{Var}\left(\frac{X}{\sigma}\right)$$

$$\begin{aligned}\text{Var}(cX) &= \mathbb{E}[(cX)^2] - \mathbb{E}[cX]^2 \\ &= \mathbb{E}[c^2X^2] - (c\mathbb{E}[X])^2 \\ &= c^2\mathbb{E}[X^2] - c^2\mathbb{E}[X]^2 \\ &= c^2(\mathbb{E}[X^2] - \mathbb{E}[X]^2)\end{aligned}$$

Everything Is Nicer When the Standard Deviation Is 1

Why?

$$\text{Var}\left(\frac{X}{\sigma}\right)$$

$$\begin{aligned}\text{Var}(cX) &= \mathbb{E}[(cX)^2] - \mathbb{E}[cX]^2 \\ &= \mathbb{E}[c^2X^2] - (c\mathbb{E}[X])^2 \\ &= c^2\mathbb{E}[X^2] - c^2\mathbb{E}[X]^2 \\ &= c^2(\mathbb{E}[X^2] - \mathbb{E}[X]^2) \\ &= c^2\text{Var}(X)\end{aligned}$$

Everything Is Nicer When the Standard Deviation Is 1

Why?

$$\text{Var}\left(\frac{X}{\sigma}\right) = \frac{1}{\sigma^2} \text{Var}(X)$$

$$\begin{aligned}\text{Var}(cX) &= \mathbb{E}[(cX)^2] - \mathbb{E}[cX]^2 \\ &= \mathbb{E}[c^2X^2] - (c\mathbb{E}[X])^2 \\ &= c^2\mathbb{E}[X^2] - c^2\mathbb{E}[X]^2 \\ &= c^2(\mathbb{E}[X^2] - \mathbb{E}[X]^2) \\ &= c^2\text{Var}(X)\end{aligned}$$

Everything Is Nicer When the Standard Deviation Is 1

Why?

$$\begin{aligned} \text{Var}(cX) &= \mathbb{E}[(cX)^2] - \mathbb{E}[cX]^2 \\ &= \mathbb{E}[c^2X^2] - (c\mathbb{E}[X])^2 \\ &= c^2\mathbb{E}[X^2] - c^2\mathbb{E}[X]^2 \\ &= c^2(\mathbb{E}[X^2] - \mathbb{E}[X]^2) \\ &= c^2\text{Var}(X) \end{aligned}$$

$$\text{Var}\left(\frac{X}{\sigma}\right) = \frac{1}{\sigma^2}\text{Var}(X)$$

$$\text{std}\left(\frac{X}{\sigma}\right) = \frac{1}{\sigma}\text{std}(X)$$

Everything Is Nicer When the Standard Deviation Is 1

Why?

$$\begin{aligned} \text{Var}(cX) &= \mathbb{E}[(cX)^2] - \mathbb{E}[cX]^2 \\ &= \mathbb{E}[c^2X^2] - (c\mathbb{E}[X])^2 \\ &= c^2\mathbb{E}[X^2] - c^2\mathbb{E}[X]^2 \\ &= c^2(\mathbb{E}[X^2] - \mathbb{E}[X]^2) \\ &= c^2\text{Var}(X) \end{aligned}$$

$$\text{Var}\left(\frac{X}{\sigma}\right) = \frac{1}{\sigma^2}\text{Var}(X)$$

$$\begin{aligned} \text{std}\left(\frac{X}{\sigma}\right) &= \frac{1}{\sigma}\text{std}(X) \\ &= \frac{\sigma}{\sigma} \end{aligned}$$

Everything Is Nicer When the Standard Deviation Is 1

Why?

$$\begin{aligned} \text{Var}(cX) &= \mathbb{E}[(cX)^2] - \mathbb{E}[cX]^2 \\ &= \mathbb{E}[c^2X^2] - (c\mathbb{E}[X])^2 \\ &= c^2\mathbb{E}[X^2] - c^2\mathbb{E}[X]^2 \\ &= c^2(\mathbb{E}[X^2] - \mathbb{E}[X]^2) \\ &= c^2\text{Var}(X) \end{aligned}$$

$$\text{Var}\left(\frac{X}{\sigma}\right) = \frac{1}{\sigma^2}\text{Var}(X)$$

$$\text{std}\left(\frac{X}{\sigma}\right) = \frac{1}{\sigma}\text{std}(X)$$

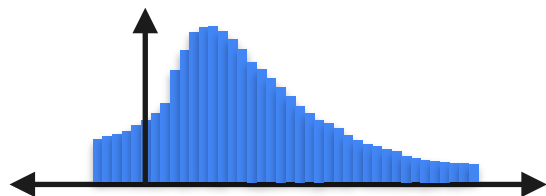
$$= \frac{\sigma}{\sigma}$$

$$= 1$$

Standardize a Distribution

Standardize a Distribution

X

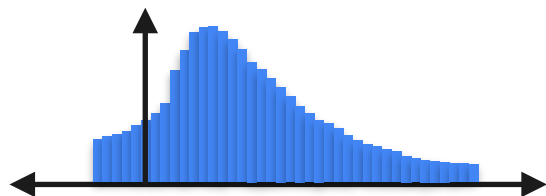


Mean = μ

std = σ

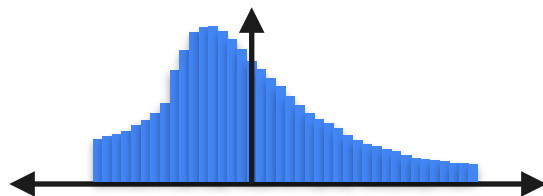
Standardize a Distribution

X



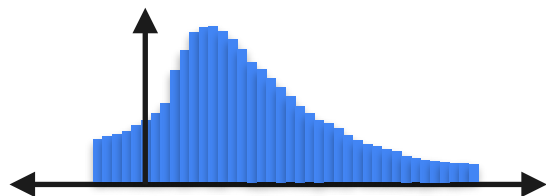
Mean = μ
std = σ

$X - \mu$



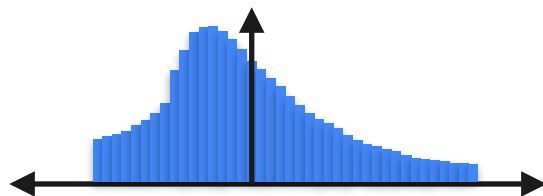
Mean = 0
std = σ

Standardize a Distribution

 X $X - \mu$ 

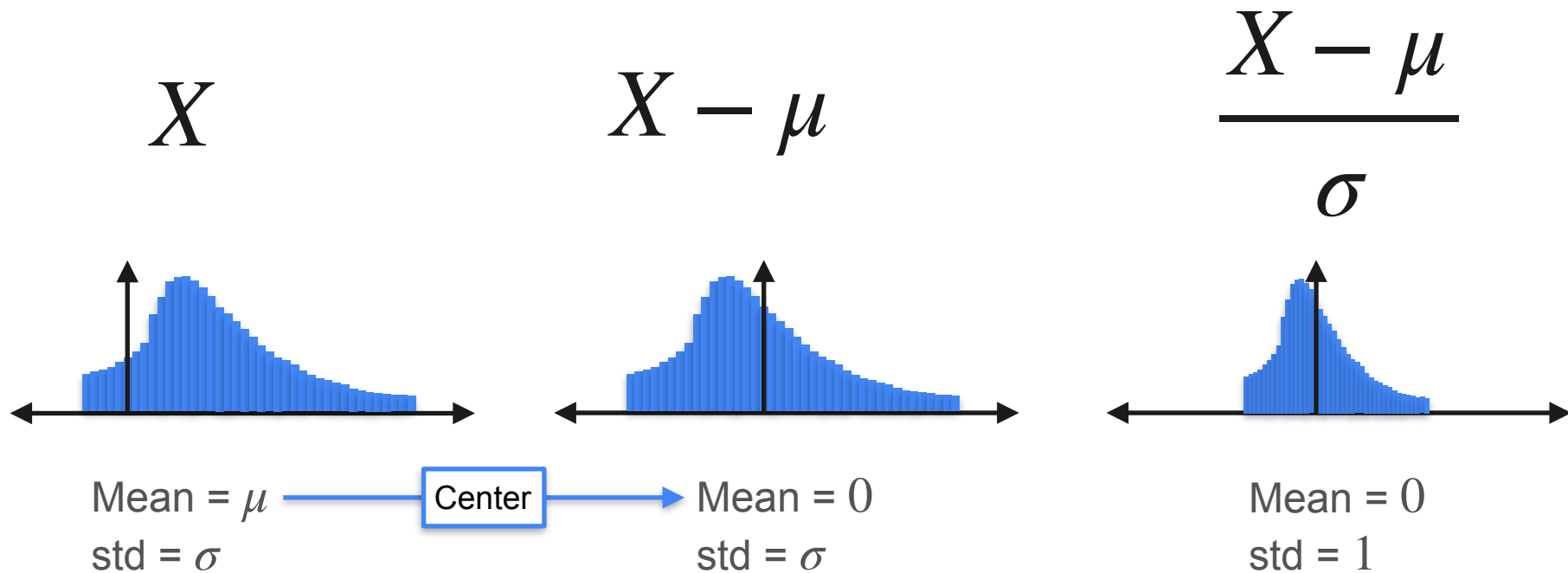
Mean = μ
std = σ

Center

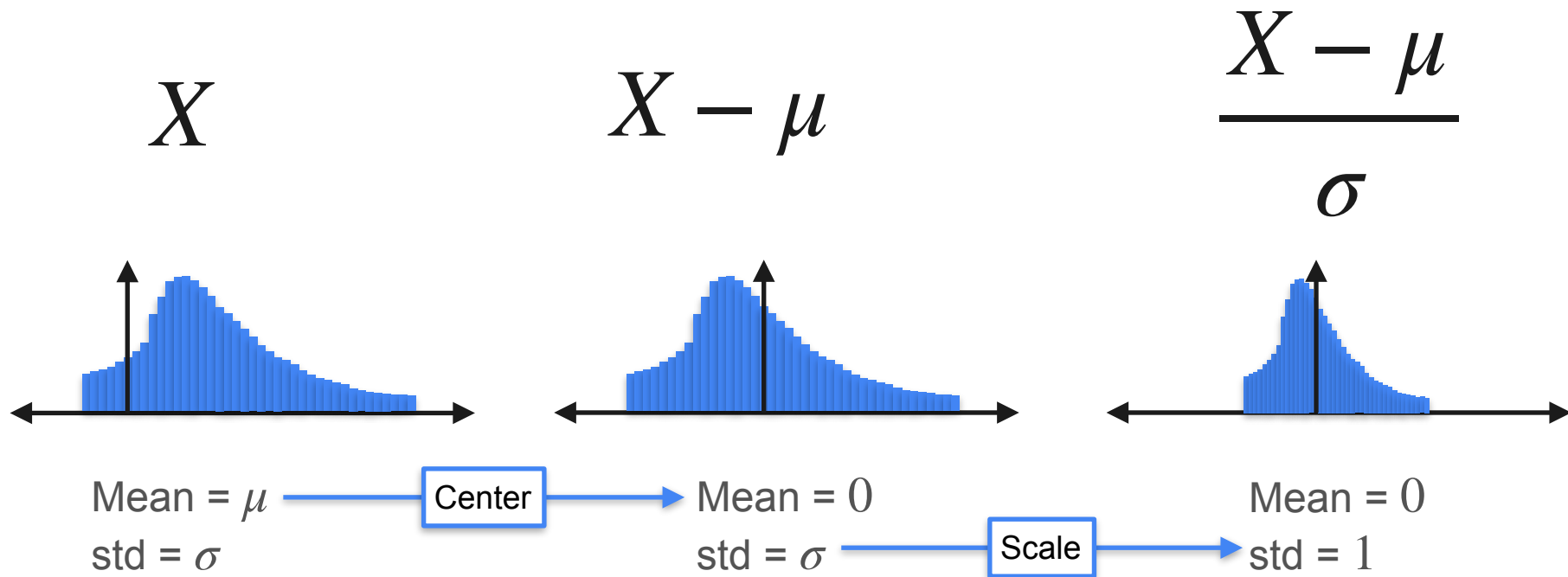


Mean = 0
std = σ

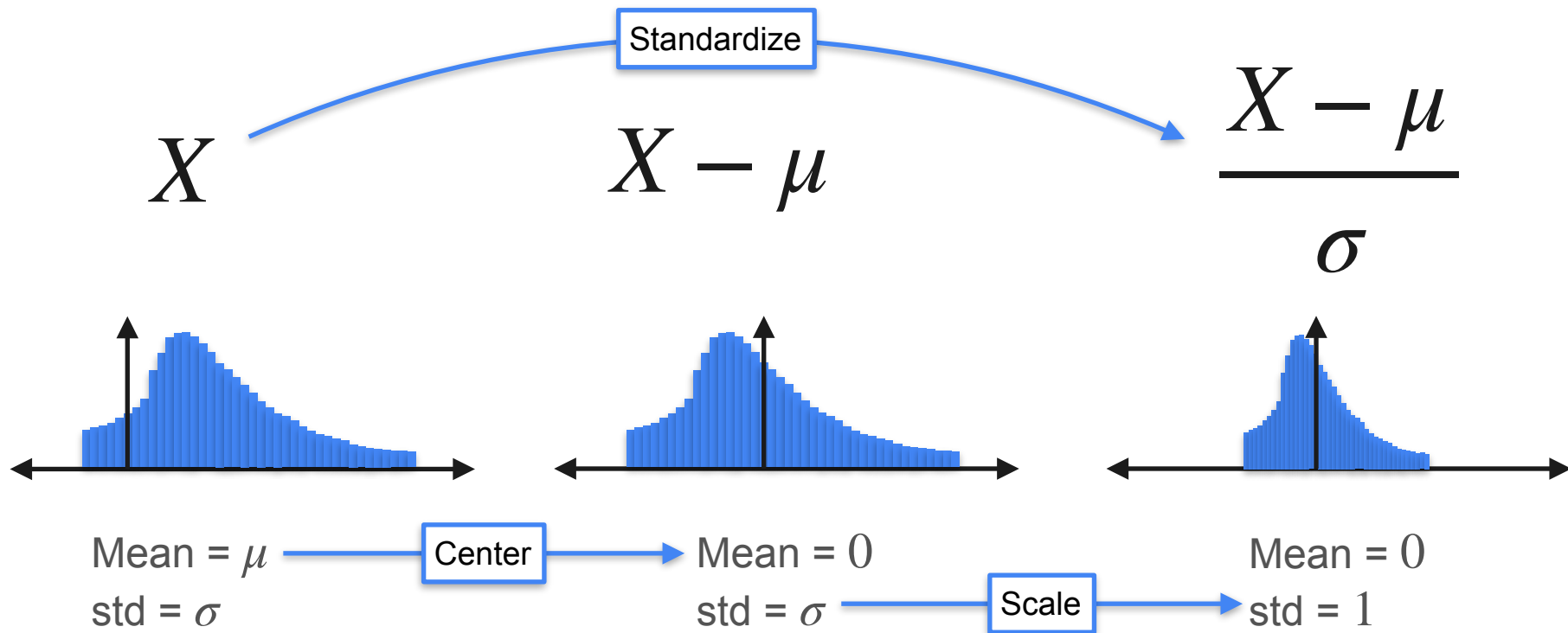
Standardize a Distribution



Standardize a Distribution



Standardize a Distribution





DeepLearning.AI

Describing Distributions

Skewness and Kurtosis

Moments of a Distribution

Random variable X



Moments of a Distribution

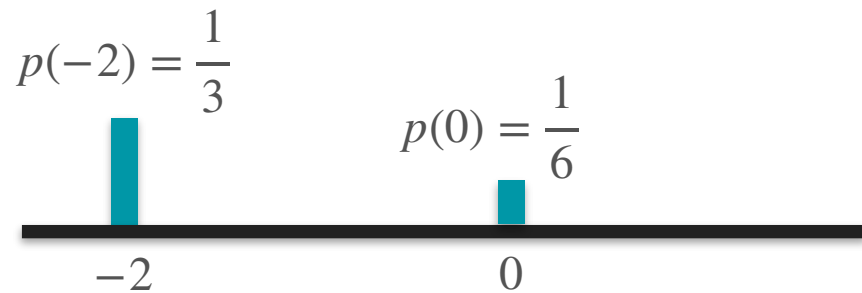
Random variable X

$$p(-2) = \frac{1}{3}$$

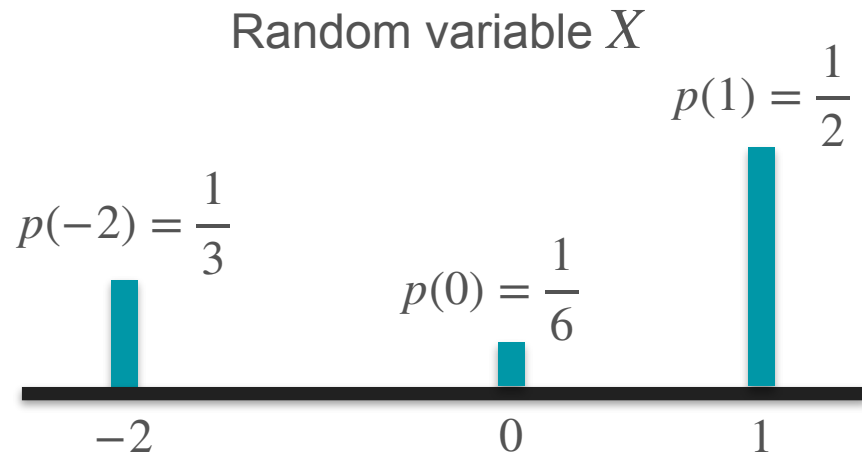


Moments of a Distribution

Random variable X

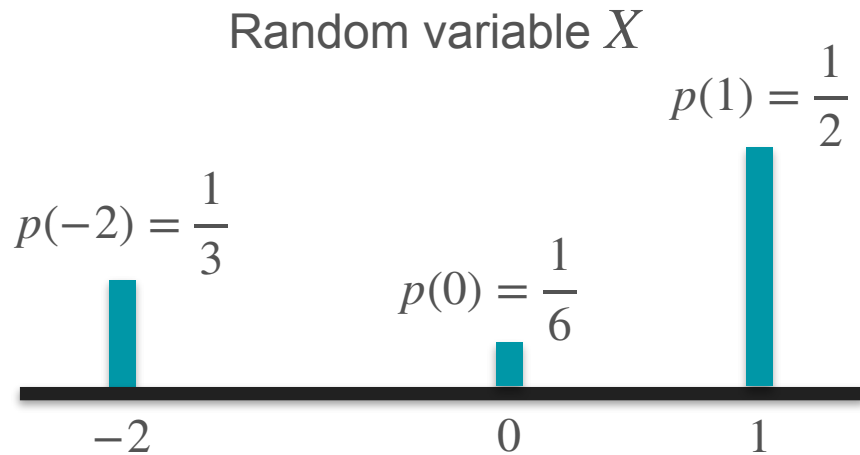


Moments of a Distribution



Moments of a Distribution

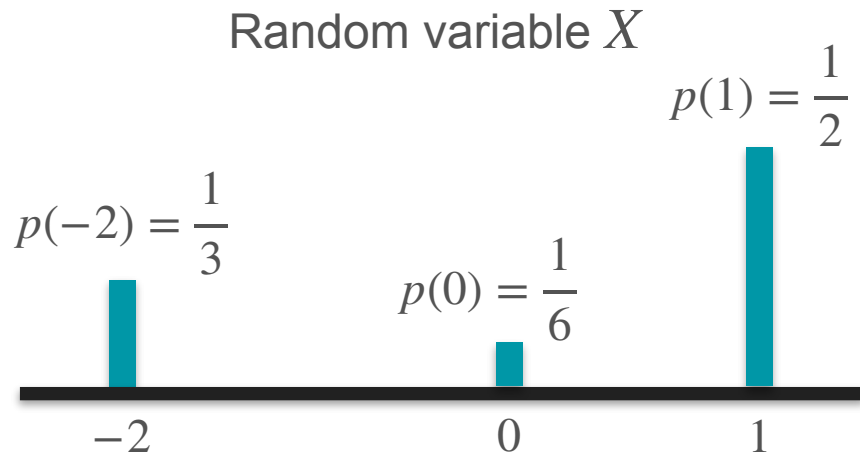
$$\mathbb{E}[X] = \frac{1}{3}(-2) + \frac{1}{6}(0) + \frac{1}{2}(1)$$



Moments of a Distribution

$$\mathbb{E}[X] = \frac{1}{3}(-2) + \frac{1}{6}(0) + \frac{1}{2}(1)$$

$$\mathbb{E}[X^2] = \frac{1}{3}(-2)^2 + \frac{1}{6}(0)^2 + \frac{1}{2}(1)^2$$

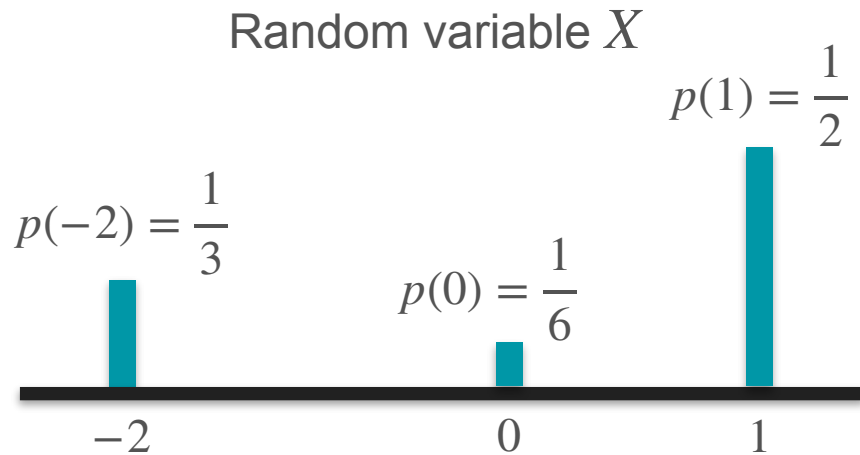


Moments of a Distribution

$$\mathbb{E}[X] = \frac{1}{3}(-2) + \frac{1}{6}(0) + \frac{1}{2}(1)$$

$$\mathbb{E}[X^2] = \frac{1}{3}(-2)^2 + \frac{1}{6}(0)^2 + \frac{1}{2}(1)^2$$

$$\mathbb{E}[X^3] = \frac{1}{3}(-2)^3 + \frac{1}{6}(0)^3 + \frac{1}{2}(1)^3$$



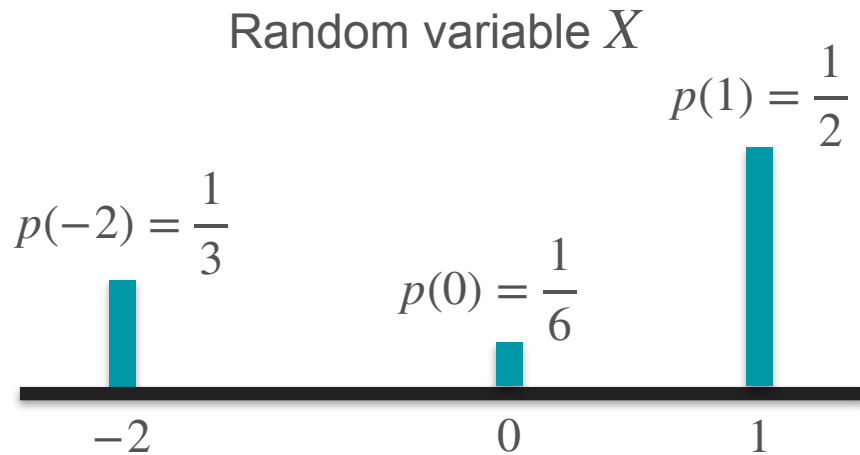
Moments of a Distribution

$$\mathbb{E}[X] = \frac{1}{3}(-2) + \frac{1}{6}(0) + \frac{1}{2}(1)$$

$$\mathbb{E}[X^2] = \frac{1}{3}(-2)^2 + \frac{1}{6}(0)^2 + \frac{1}{2}(1)^2$$

$$\mathbb{E}[X^3] = \frac{1}{3}(-2)^3 + \frac{1}{6}(0)^3 + \frac{1}{2}(1)^3$$

...



Moments of a Distribution

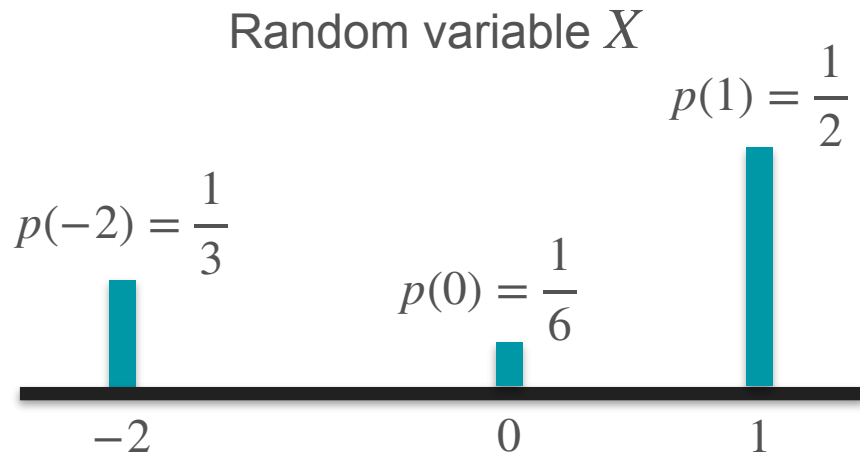
$$\mathbb{E}[X] = \frac{1}{3}(-2) + \frac{1}{6}(0) + \frac{1}{2}(1)$$

$$\mathbb{E}[X^2] = \frac{1}{3}(-2)^2 + \frac{1}{6}(0)^2 + \frac{1}{2}(1)^2$$

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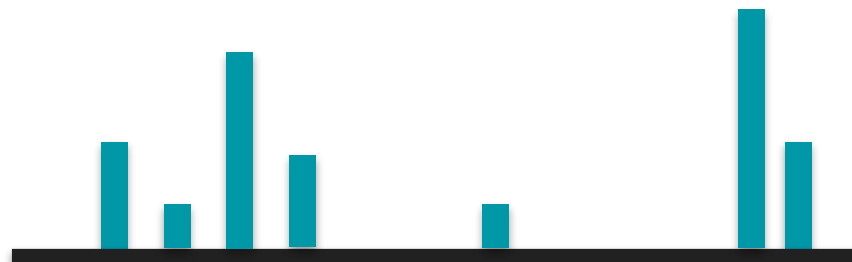
...

$$\mathbb{E}[X^k] = \frac{1}{3}(-2)^k + \frac{1}{6}(0)^k + \frac{1}{2}(1)^k$$



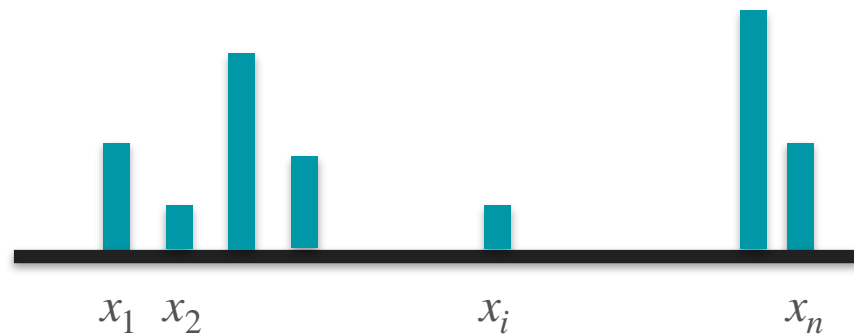
Moments of a Distribution

Random variable X



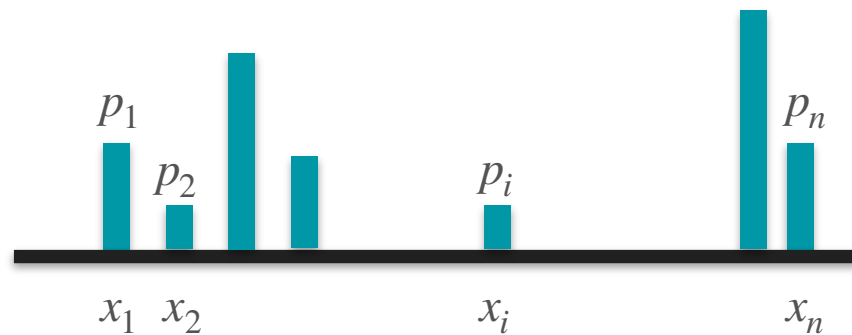
Moments of a Distribution

Random variable X



Moments of a Distribution

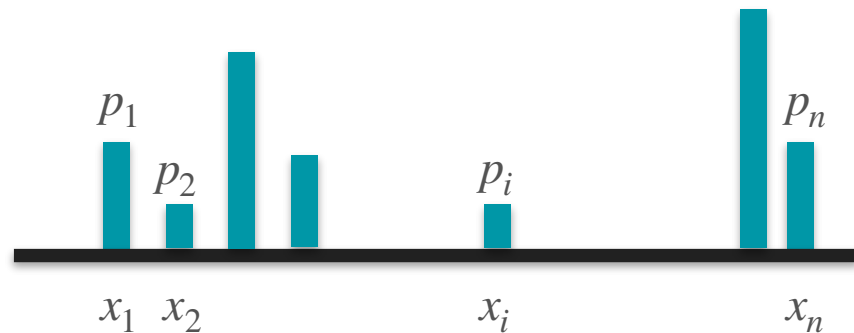
Random variable X



Moments of a Distribution

$$\mathbb{E}[X] = p_1x_1 + p_2x_2 + \cdots + p_nx_n$$

Random variable X

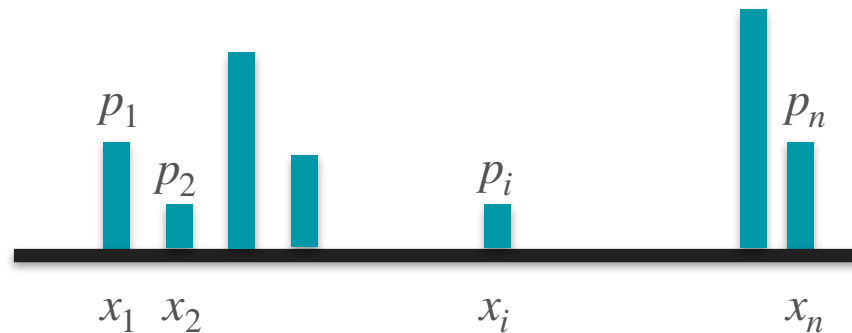


Moments of a Distribution

$$\mathbb{E}[X] = p_1x_1 + p_2x_2 + \cdots + p_nx_n$$

$$\mathbb{E}[X^2] = p_1x_1^2 + p_2x_2^2 + \cdots + p_nx_n^2$$

Random variable X



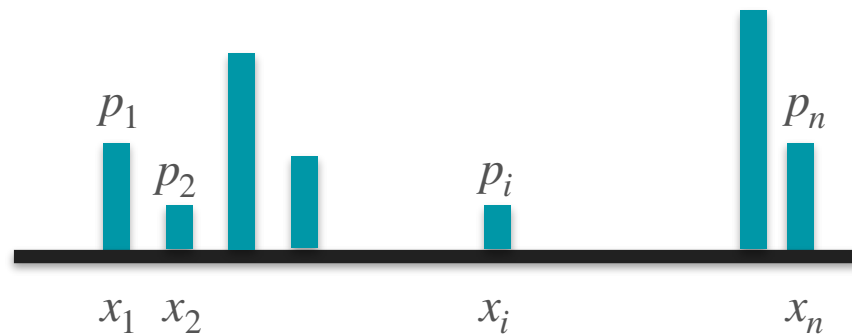
Moments of a Distribution

$$\mathbb{E}[X] = p_1x_1 + p_2x_2 + \cdots + p_nx_n$$

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$$\mathbb{E}[X^3] = p_1x_1^3 + p_2x_2^3 + \cdots + p_nx_n^3$$

Random variable X



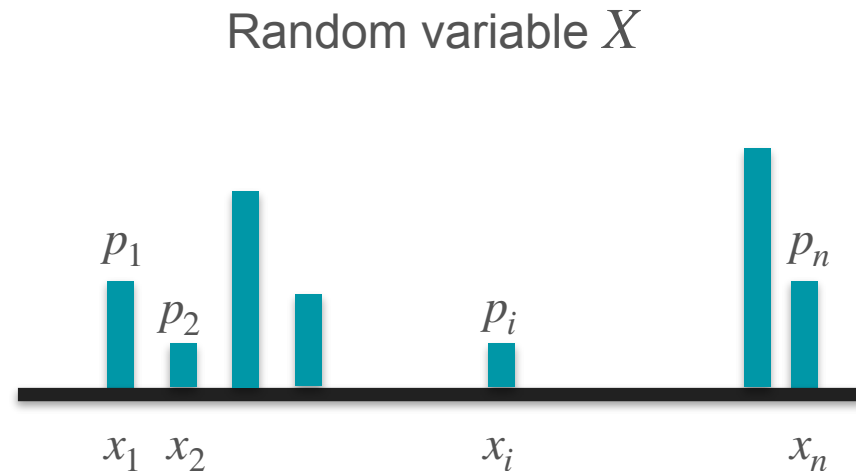
Moments of a Distribution

$$\mathbb{E}[X] = p_1x_1 + p_2x_2 + \cdots + p_nx_n$$

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$$\mathbb{E}[X^3] = p_1x_1^3 + p_2x_2^3 + \cdots + p_nx_n^3$$

$$\mathbb{E}[X^4] = p_1x_1^4 + p_2x_2^4 + \cdots + p_nx_n^4$$



Moments of a Distribution

$$\mathbb{E}[X] = p_1x_1 + p_2x_2 + \cdots + p_nx_n$$

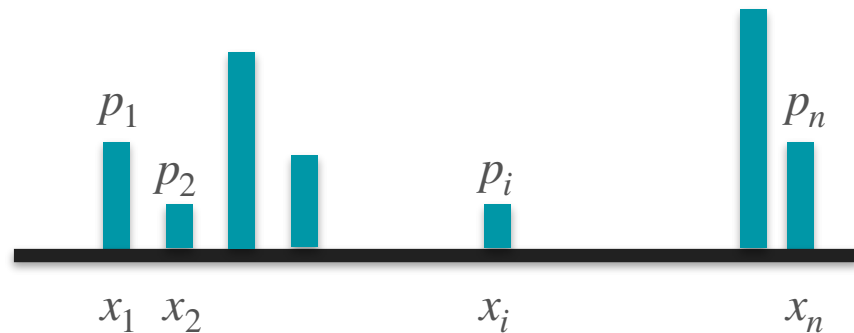
$$\mathbb{E}[X^2] = p_1x_1^2 + p_2x_2^2 + \cdots + p_nx_n^2$$

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$$\mathbb{E}[X^4] = p_1x_1^4 + p_2x_2^4 + \cdots + p_nx_n^4$$

...

Random variable X



Moments of a Distribution

$$\mathbb{E}[X] = p_1x_1 + p_2x_2 + \cdots + p_nx_n$$

$$\mathbb{E}[X^2] = p_1x_1^2 + p_2x_2^2 + \cdots + p_nx_n^2$$

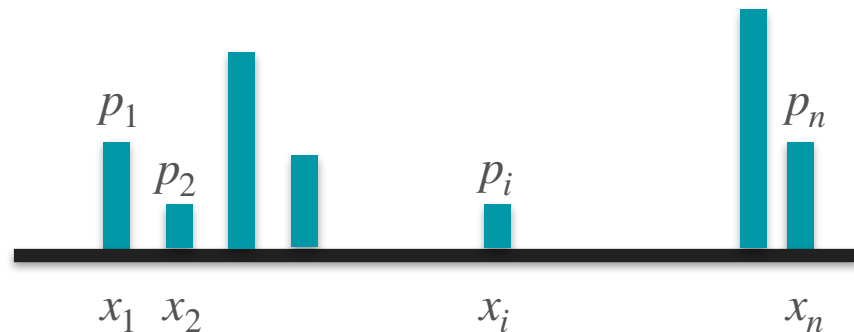
$$\mathbb{E}[X^3] = p_1x_1^3 + p_2x_2^3 + \cdots + p_nx_n^3$$

$$\mathbb{E}[X^4] = p_1x_1^4 + p_2x_2^4 + \cdots + p_nx_n^4$$

...

$$\mathbb{E}[X^k] = p_1x_1^k + p_2x_2^k + \cdots + p_nx_n^k$$

Random variable X



Video 7b:

- Skewness

Lottery vs Insurance

Lottery vs Insurance



Lottery

Lottery vs Insurance



Lottery



Car insurance

Lottery vs Insurance



Lottery

Ticket: \$1
Jackpot: \$100



Car insurance

Lottery vs Insurance



Lottery

Ticket: \$1
Jackpot: \$100



Car insurance

You **win** \$99 with 1% probability

You **lose** \$1 with 99% probability

Lottery vs Insurance



Lottery

Ticket: \$1
Jackpot: \$100



Car insurance

Cost: \$1
Crash Reparation: \$100

You **win** \$99 with 1% probability

You **lose** \$1 with 99% probability

Lottery vs Insurance



Lottery

Ticket: \$1
Jackpot: \$100

You **win** \$99 with 1% probability
You **lose** \$1 with 99% probability



Car insurance

Cost: \$1
Crash Reparation: \$100

You **win** \$1 with 99% probability
You **lose** \$99 with 1% probability

Lottery vs Insurance



Ticket: \$1
Jackpot: \$100

Lottery



Cost: \$1
Crash Reparation: \$100

Car insurance



Lottery vs Insurance



Ticket: \$1
Jackpot: \$100

Lottery



Cost: \$1
Crash Reparation: \$100

Car insurance



Lottery vs Insurance



Ticket: \$1
Jackpot: \$100

Lottery



Cost: \$1
Crash Reparation: \$100

Car insurance



Lottery vs Insurance



Ticket: \$1
Jackpot: \$100

Lottery



Cost: \$1
Crash Reparation: \$100

Car insurance



Lottery vs Insurance



Ticket: \$1
Jackpot: \$100

Lottery



Cost: \$1
Crash Reparation: \$100

Car insurance





Lottery

Ticket: \$1
Jackpot: \$100



Car insurance

Cost: \$1
Crash Reparation: \$100





Lottery

Ticket: \$1
Jackpot: \$100



Car insurance

Cost: \$1
Crash Reparation: \$100





Ticket: \$1
Jackpot: \$100

Lottery



Cost: \$1
Crash Reparation: \$100

Car insurance

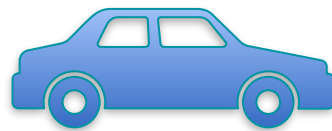
$$\mathbb{E}[X_1] = -1 \cdot 0.99 + 99 \cdot 0.01 = 0$$





Ticket: \$1
Jackpot: \$100

Lottery



Cost: \$1
Crash Reparation: \$100

Car insurance

$$\mathbb{E}[X_1] = -1 \cdot 0.99 + 99 \cdot 0.01 = 0$$

Lose 1
With probability 0.99

Win 99
With probability 0.01





Ticket: \$1
Jackpot: \$100

Lottery

$$\mathbb{E}[X_1] = -1 \cdot 0.99 + 99 \cdot 0.01 = 0$$

Lose 1
With probability 0.99

Win 99
With probability 0.01



Cost: \$1
Crash Reparation: \$100

Car insurance

$$\mathbb{E}[X_2] = -99 \cdot 0.01 + 1 \cdot 0.99 = 0$$





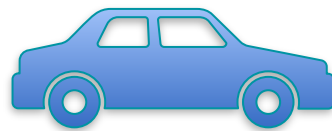
Ticket: \$1
Jackpot: \$100

Lottery

$$\mathbb{E}[X_1] = -1 \cdot 0.99 + 99 \cdot 0.01 = 0$$

Lose 1
With probability 0.99

Win 99
With probability 0.01



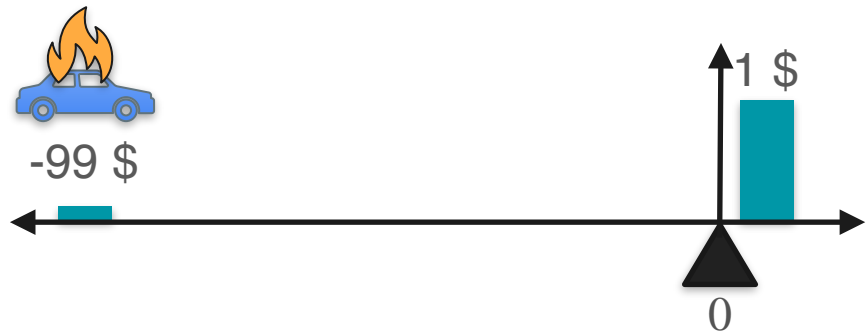
Cost: \$1
Crash Reparation: \$100

Car insurance

$$\mathbb{E}[X_2] = -99 \cdot 0.01 + 1 \cdot 0.99 = 0$$

Lose 99
With probability 0.01

Win 1
With probability 0.99





Ticket: \$1
Jackpot: \$100

Lottery

$$\mathbb{E}[X_1] = -1 \cdot 0.99 + 99 \cdot 0.01 = 0$$



Cost: \$1
Crash Reparation: \$100

Car insurance

$$\mathbb{E}[X_2] = -99 \cdot 0.01 + 1 \cdot 0.99 = 0$$

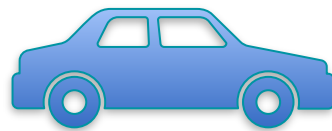




Ticket: \$1
Jackpot: \$100

Lottery

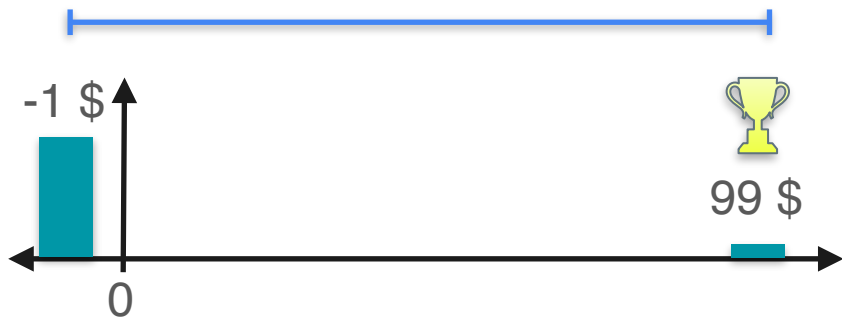
$$\mathbb{E}[X_1] = -1 \cdot 0.99 + 99 \cdot 0.01 = 0$$



Cost: \$1
Crash Reparation: \$100

Car insurance

$$\mathbb{E}[X_2] = -99 \cdot 0.01 + 1 \cdot 0.99 = 0$$

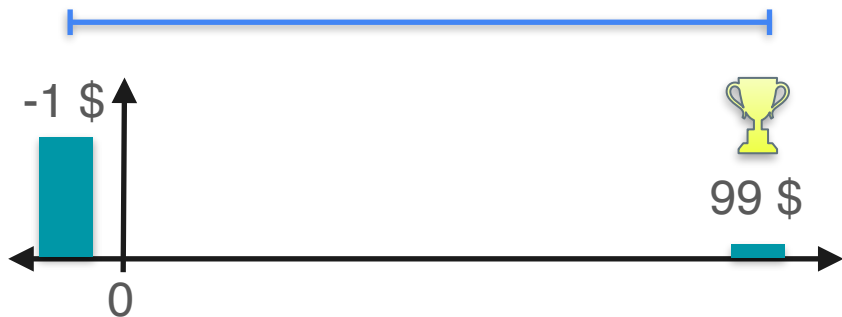




Ticket: \$1
Jackpot: \$100

Lottery

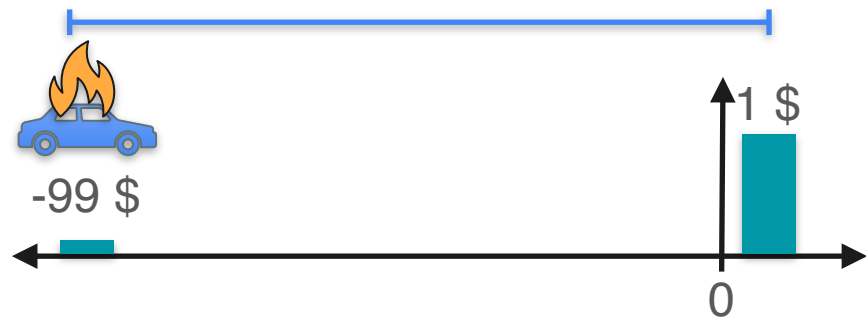
$$\mathbb{E}[X_1] = -1 \cdot 0.99 + 99 \cdot 0.01 = 0$$



Cost: \$1
Crash Reparation: \$100

Car insurance

$$\mathbb{E}[X_2] = -99 \cdot 0.01 + 1 \cdot 0.99 = 0$$



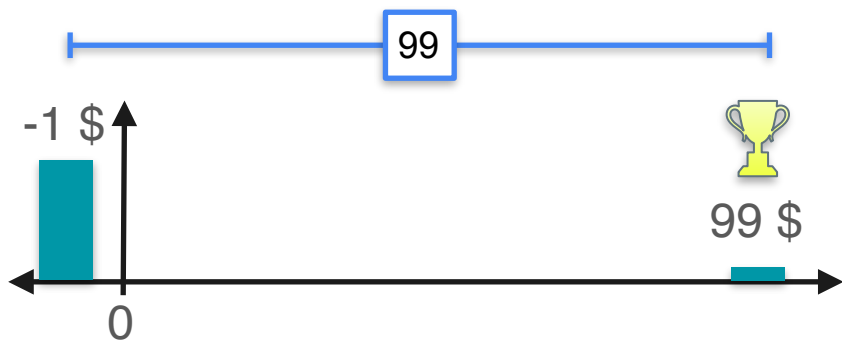


Ticket: \$1
Jackpot: \$100

Lottery

$$\mathbb{E}[X_1] = -1 \cdot 0.99 + 99 \cdot 0.01 = 0$$

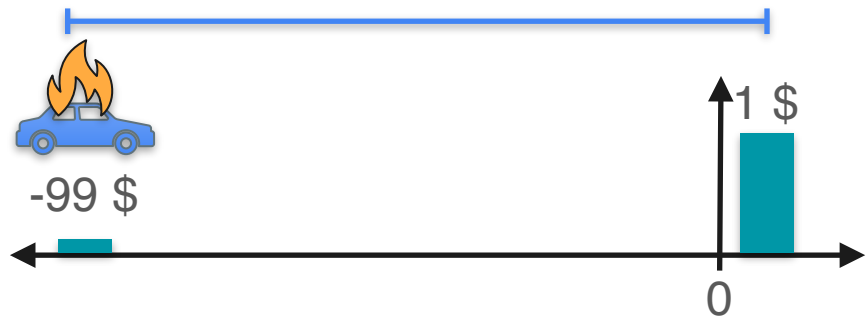
$$\text{Var}(X_1) = (-1)^2 \cdot 0.99 + (99)^2 \cdot 0.01 = 99$$



Cost: \$1
Crash Reparation: \$100

Car insurance

$$\mathbb{E}[X_2] = -99 \cdot 0.01 + 1 \cdot 0.99 = 0$$



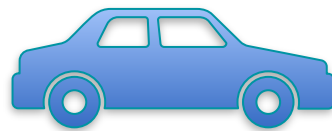
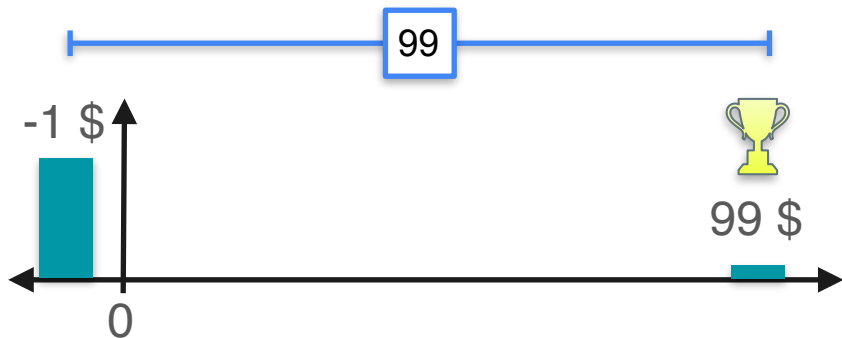


Ticket: \$1
Jackpot: \$100

Lottery

$$\mathbb{E}[X_1] = -1 \cdot 0.99 + 99 \cdot 0.01 = 0$$

$$\text{Var}(X_1) = (-1)^2 \cdot 0.99 + (99)^2 \cdot 0.01 = 99$$

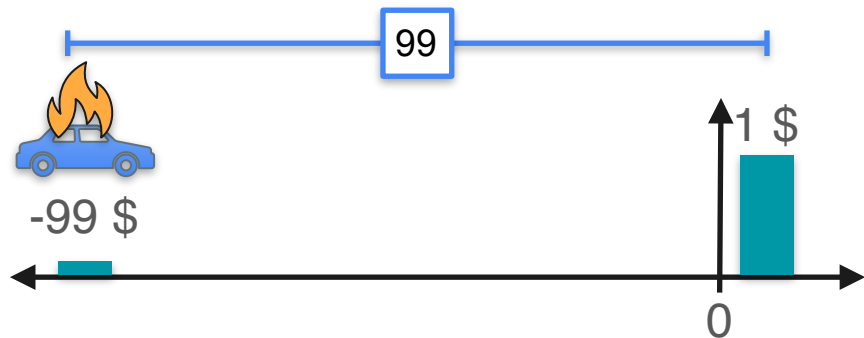


Cost: \$1
Crash Reparation: \$100

Car insurance

$$\mathbb{E}[X_2] = -99 \cdot 0.01 + 1 \cdot 0.99 = 0$$

$$\text{Var}(X_2) = (-99)^2 \cdot 0.01 + (1)^2 \cdot 0.99 = 99$$



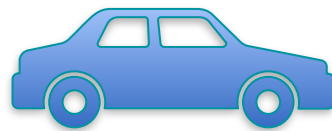
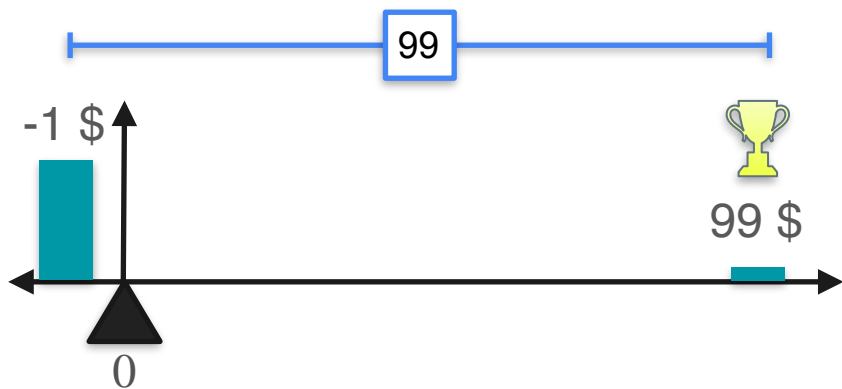


Ticket: \$1
Jackpot: \$100

Lottery

$$\mathbb{E}[X_1] = 0$$

$$\text{Var}(X_1) = 99$$

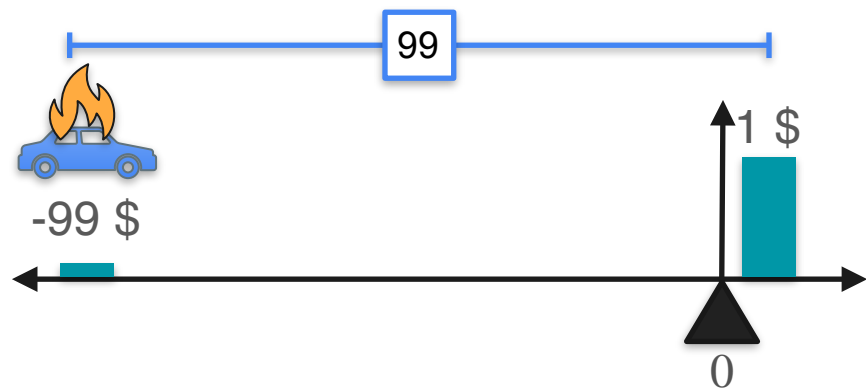


Cost: \$1
Crash Reparation: \$100

Car insurance

$$\mathbb{E}[X_2] = 0$$

$$\text{Var}(X_2) = 99$$



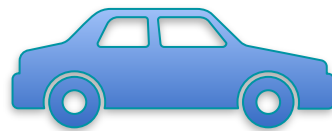


Lottery

Ticket: \$1
Jackpot: \$100

$$\mathbb{E}[X_1] = 0$$

$$\text{Var}(X_1) = 99$$



Car insurance

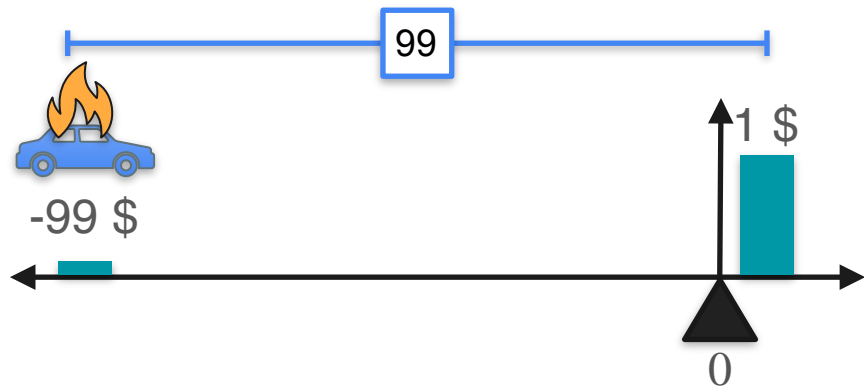
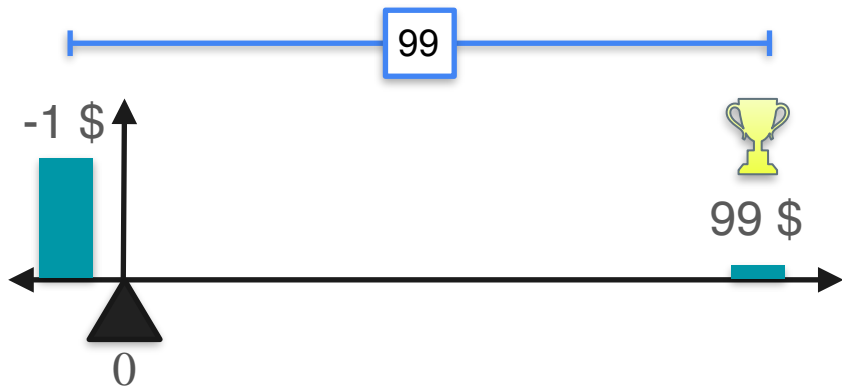
Cost: \$1
Crash Reparation: \$100

$$\mathbb{E}[X_2] = 0$$

$$\text{Var}(X_2) = 99$$

Same expectation
Same variance

How to tell them apart?



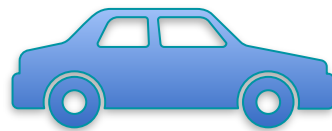


Lottery

Ticket: \$1
Jackpot: \$99

$$\mathbb{E}[X_1] = 0$$

$$\text{Var}(X_1) = 99$$



Car insurance

Cost: \$1
Crash Reparation: \$99

$$\mathbb{E}[X_2] = 0$$

$$\text{Var}(X_2) = 99$$

Same expectation
Same variance

How to tell them apart?





Lottery

Ticket: \$1
Jackpot: \$99

$$\mathbb{E}[X_1] = 0$$

$$\text{Var}(X_1) = 99$$

$$\mathbb{E}[X_1^2] - \mathbb{E}[X_1]^2 = 99$$



Car insurance

Cost: \$1
Crash Reparation: \$99

$$\mathbb{E}[X_2] = 0$$

$$\text{Var}(X_2) = 99$$

$$\mathbb{E}[X_2^2] - \mathbb{E}[X_2]^2 = 99$$



Same expectation
Same variance

How to tell them apart?



Lottery

Ticket: \$1
Jackpot: \$99

$$\mathbb{E}[X_1] = 0$$

$$\text{Var}(X_1) = 99$$

$$\mathbb{E}[X_1^2] - \cancel{\mathbb{E}[X_1]^2} = 99$$



Car insurance

Cost: \$1
Crash Reparation: \$99

$$\mathbb{E}[X_2] = 0$$

$$\text{Var}(X_2) = 99$$

$$\mathbb{E}[X_2^2] - \cancel{\mathbb{E}[X_2]^2} = 99$$



Same expectation
Same variance

How to tell them apart?



Ticket: \$1
Jackpot: \$99

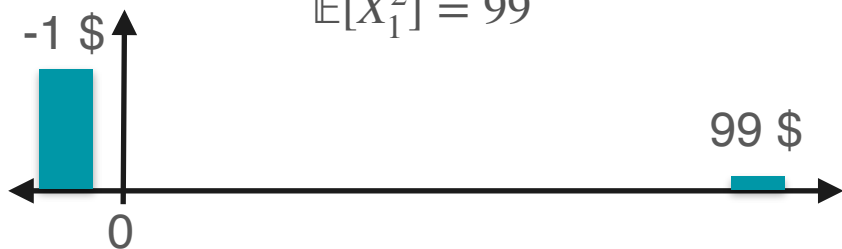
Lottery

$$\mathbb{E}[X_1] = 0$$

$$\text{Var}(X_1) = 99$$

$$\mathbb{E}[X_1^2] - \cancel{\mathbb{E}[X_1]^2} = 99$$

$$\mathbb{E}[X_1^2] = 99$$



Cost: \$1
Crash Reparation: \$99

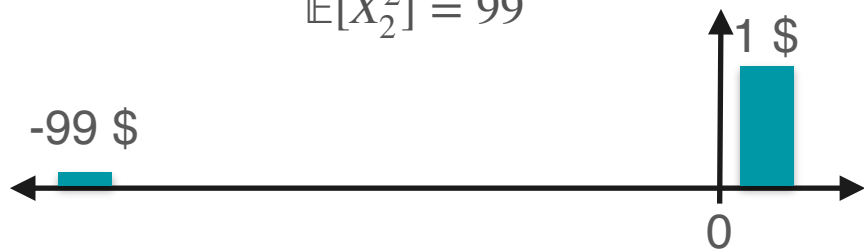
Car insurance

$$\mathbb{E}[X_2] = 0$$

$$\text{Var}(X_2) = 99$$

$$\mathbb{E}[X_2^2] - \cancel{\mathbb{E}[X_2]^2} = 99$$

$$\mathbb{E}[X_2^2] = 99$$



Same expectation
Same variance

How to tell them apart?



Ticket: \$1
Jackpot: \$99

Lottery

$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[X_1^2] = 99$$



Cost: \$1
Crash Reparation: \$99

Car insurance

Same expectation
Same variance

$$\mathbb{E}[X_2] = 0$$

How to tell them apart? $\mathbb{E}[X_2^2] = 99$





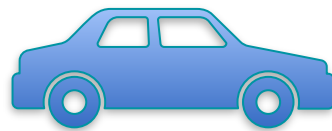
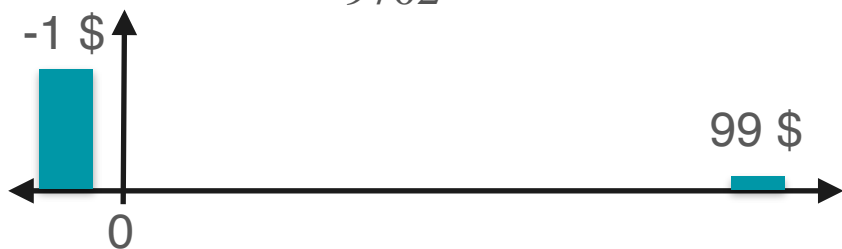
Ticket: \$1
Jackpot: \$99

Lottery

$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[X_1^2] = 99$$

$$\begin{aligned}\mathbb{E}[X_1^3] &= (-1)^3 \cdot 0.99 + (99)^3 \cdot 0.01 \\ &= 9702\end{aligned}$$



Cost: \$1
Crash Reparation: \$99

Car insurance

Same expectation
Same variance

$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[X_2^2] = 99$$

How to tell them apart?





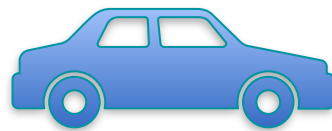
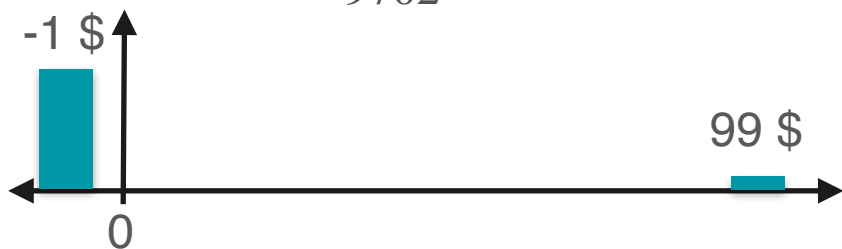
Ticket: \$1
Jackpot: \$99

Lottery

$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[X_1^2] = 99$$

$$\begin{aligned}\mathbb{E}[X_1^3] &= (-1)^3 \cdot 0.99 + (99)^3 \cdot 0.01 \\ &= 9702\end{aligned}$$



Cost: \$1
Crash Reparation: \$99

Car insurance

$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[X_2^2] = 99$$

$$\begin{aligned}\mathbb{E}[X_2^3] &= (1)^3 \cdot 0.99 + (-99)^3 \cdot 0.01 \\ &= -9702\end{aligned}$$



Same expectation
Same variance

How to tell them apart?



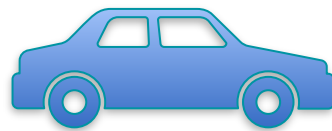
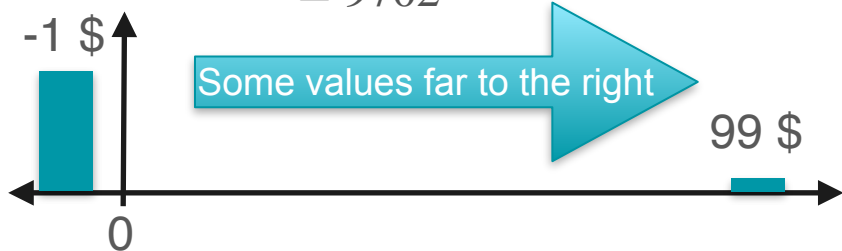
Ticket: \$1
Jackpot: \$99

Lottery

$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[X_1^2] = 99$$

$$\begin{aligned}\mathbb{E}[X_1^3] &= (-1)^3 \cdot 0.99 + (99)^3 \cdot 0.01 \\ &= 9702\end{aligned}$$



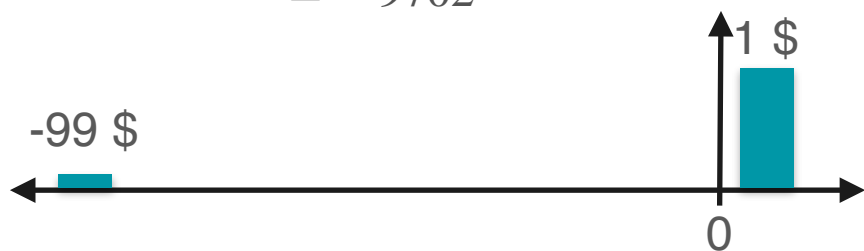
Cost: \$1
Crash Reparation: \$99

Car insurance

$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[X_2^2] = 99$$

$$\begin{aligned}\mathbb{E}[X_2^3] &= (1)^3 \cdot 0.99 + (-99)^3 \cdot 0.01 \\ &= -9702\end{aligned}$$



Same expectation
Same variance

How to tell them apart?



Ticket: \$1
Jackpot: \$99

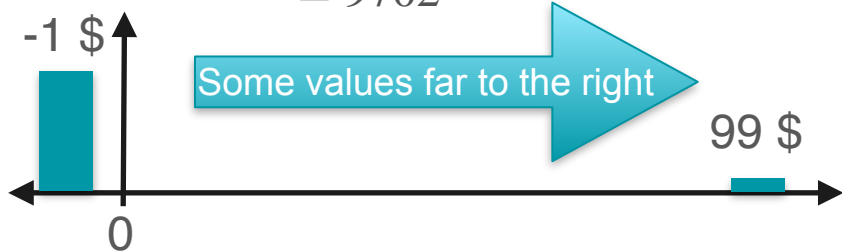
Lottery

$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[X_1^2] = 99$$

$$\mathbb{E}[X_1^3] = (-1)^3 \cdot 0.99 + (99)^3 \cdot 0.01$$

$$= 9702$$



Cost: \$1
Crash Reparation: \$99

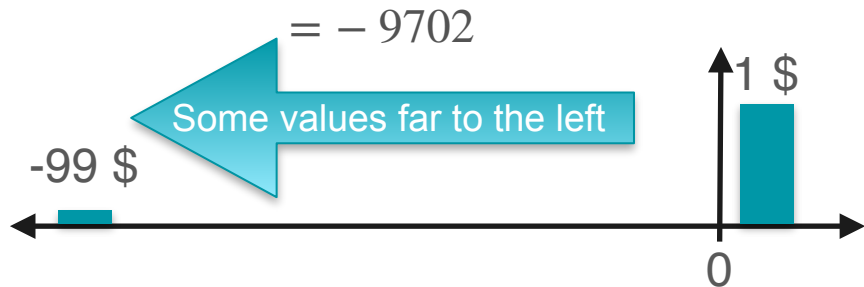
Car insurance

$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[X_2^2] = 99$$

$$\mathbb{E}[X_2^3] = (1)^3 \cdot 0.99 + (-99)^3 \cdot 0.01$$

$$= -9702$$



Same expectation
Same variance

How to tell them apart?



Ticket: \$1
Jackpot: \$99

Lottery

$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[X_1^2] = 99$$

$$\mathbb{E}[X_1^3] = \text{Large positive value}$$



Cost: \$1
Crash Reparation: \$99

Car insurance

Same expectation
Same variance

$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[X_1^2] = 99$$

$$\mathbb{E}[X_1^3] = \text{Large negative value}$$

How to tell them apart?





Ticket: \$1
Jackpot: \$99

Lottery

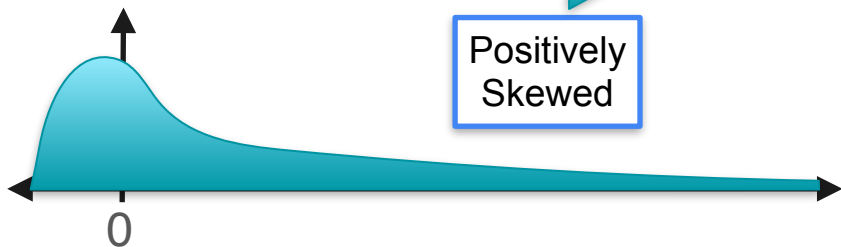
$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[X_1^2] = 99$$

$$\mathbb{E}[X_1^3] = \text{Large positive value}$$



Positively
Skewed



Cost: \$1
Crash Reparation: \$99

Car insurance

$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[X_1^2] = 99$$

$$\mathbb{E}[X_1^3] = \text{Large negative value}$$



Same expectation
Same variance

How to tell them apart?



Ticket: \$1
Jackpot: \$99

Lottery

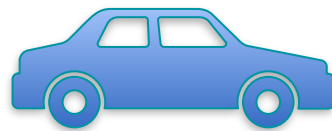
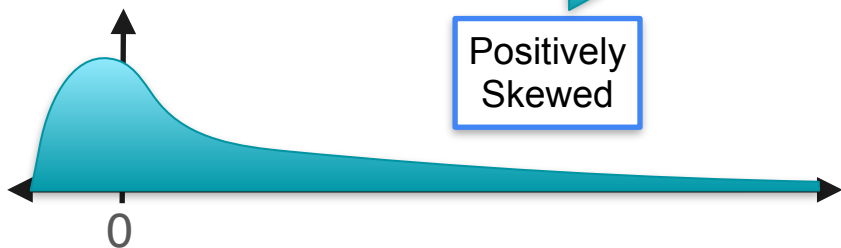
$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[X_1^2] = 99$$

$$\mathbb{E}[X_1^3] = \text{Large positive value}$$



Positively
Skewed



Cost: \$1
Crash Reparation: \$99

Car insurance

$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[X_1^2] = 99$$

$$\mathbb{E}[X_1^3] = \text{Large negative value}$$



Negatively
Skewed



Same expectation
Same variance

How to tell them apart?

Skewness

$$\mathbb{E}[X^3]$$

Skewness

$$\mathbb{E}[X^3]$$

Almost...

Skewness

$$\mathbb{E}[X^3]$$

Almost...

Need to standardize...

Skewness

Skewness

$$\text{Skewness} = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right]$$

Skewness

Skewness



Positively
Skewed



$$E \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right] > 0$$

Skewness



Positively
Skewed



$$E \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right] > 0$$



Not
Skewed



$$E \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right] = 0$$

Skewness



Positively
Skewed



$$E \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right] > 0$$



Not
Skewed



$$E \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right] = 0$$



Negatively
Skewed



$$E \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right] < 0$$

Video 7c:

- Kurtosis

Kurtosis: Example

Game 1

Game 2

Kurtosis: Example

Game 1

Game 2

probability $\frac{1}{2}$: You win 1 dollar

Kurtosis: Example

Game 1

Game 2

probability $\frac{1}{2}$: You win 1 dollar

probability $\frac{1}{2}$: You lose 1 dollar

Kurtosis: Example

Game 1

probability $\frac{1}{2}$: You win 1 dollar

probability $\frac{1}{2}$: You lose 1 dollar

Game 2

probability $\frac{100}{202}$: You win 10 cents

Kurtosis: Example

Game 1

probability $\frac{1}{2}$: You win 1 dollar

probability $\frac{1}{2}$: You lose 1 dollar

Game 2

probability $\frac{100}{202}$: You win 10 cents

probability $\frac{100}{202}$: You lose 10 cents

Kurtosis: Example

Game 1

probability $\frac{1}{2}$: You win 1 dollar

probability $\frac{1}{2}$: You lose 1 dollar

Game 2

probability $\frac{100}{202}$: You win 10 cents

probability $\frac{100}{202}$: You lose 10 cents

probability $\frac{1}{202}$: You win 10 dollars

Kurtosis: Example

Game 1

probability $\frac{1}{2}$: You win 1 dollar

probability $\frac{1}{2}$: You lose 1 dollar

Game 2

probability $\frac{100}{202}$: You win 10 cents

probability $\frac{100}{202}$: You lose 10 cents

probability $\frac{1}{202}$: You win 10 dollars

probability $\frac{1}{202}$: You lose 10 dollars

Kurtosis: Example

Game 1

probability $\frac{1}{2}$: You win 1 dollar

probability $\frac{1}{2}$: You lose 1 dollar

Which one
is riskier?

Game 2

probability $\frac{100}{202}$: You win 10 cents

probability $\frac{100}{202}$: You lose 10 cents

probability $\frac{1}{202}$: You win 10 dollars

probability $\frac{1}{202}$: You lose 10 dollars

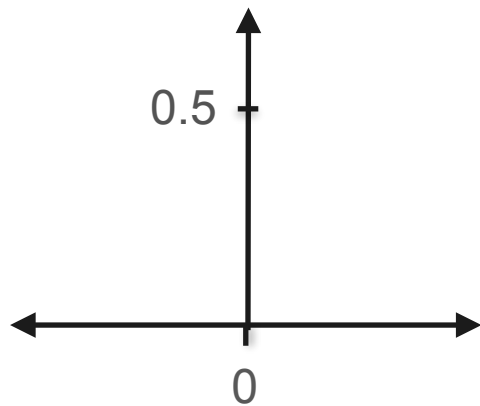
Kurtosis: Example

Game 1

Game 2

Kurtosis: Example

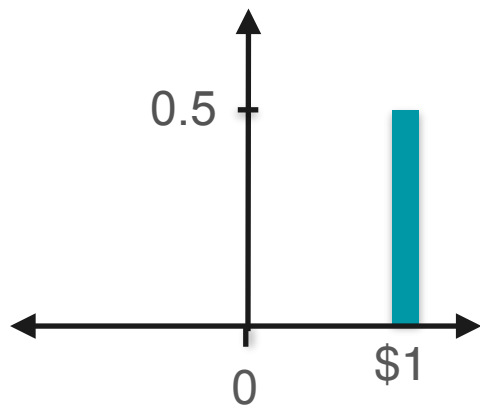
Game 1



Game 2

Kurtosis: Example

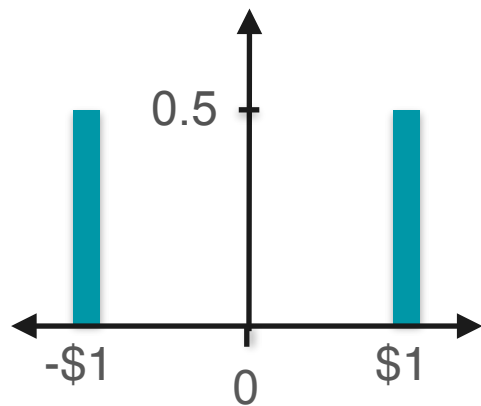
Game 1



Game 2

Kurtosis: Example

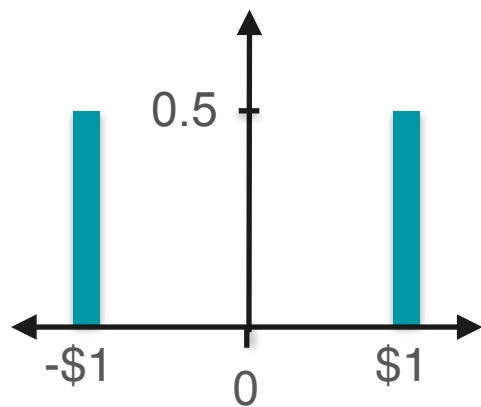
Game 1



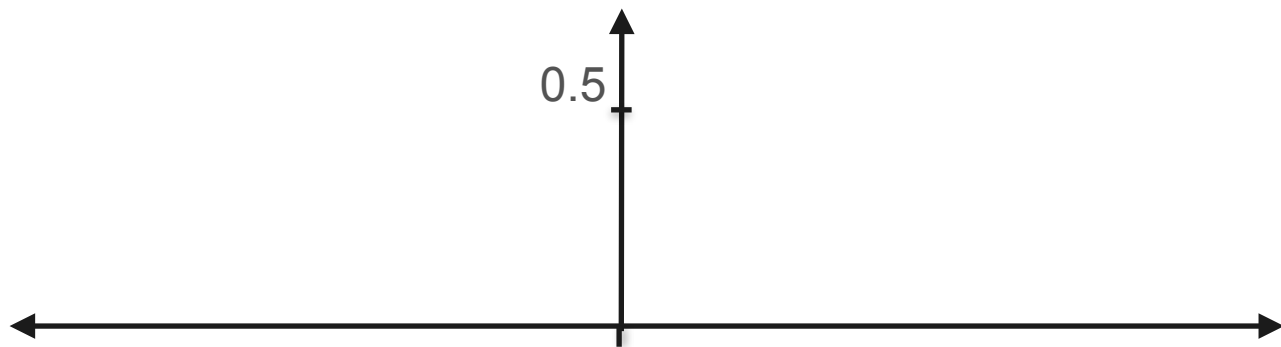
Game 2

Kurtosis: Example

Game 1

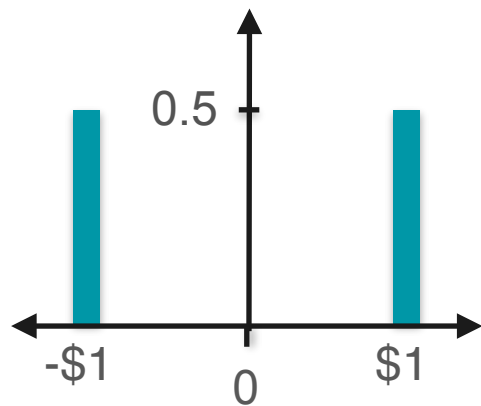


Game 2

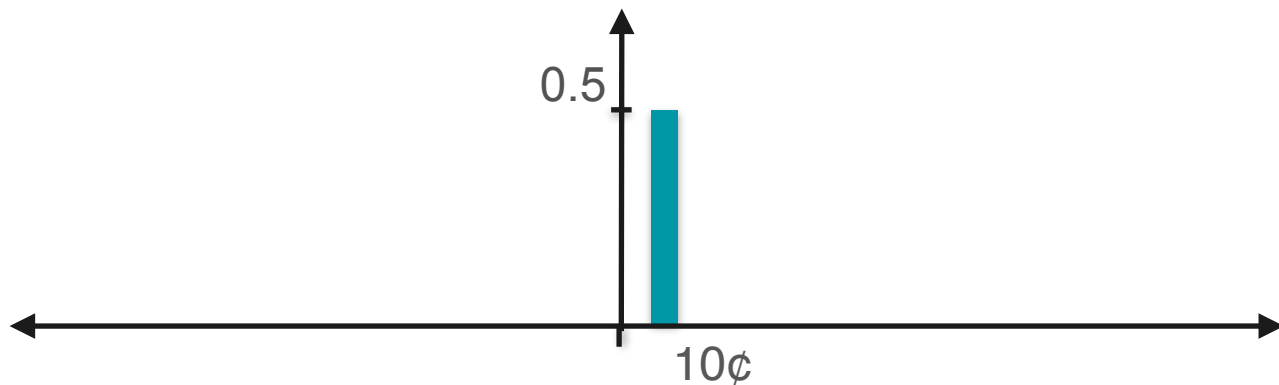


Kurtosis: Example

Game 1

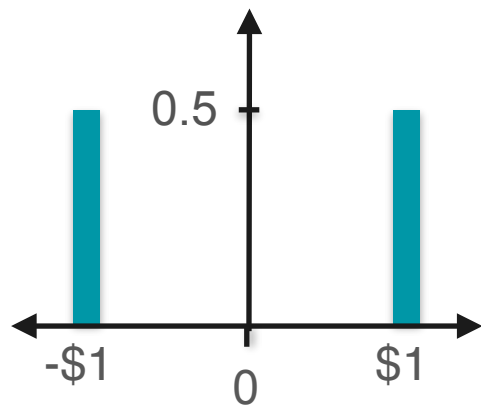


Game 2

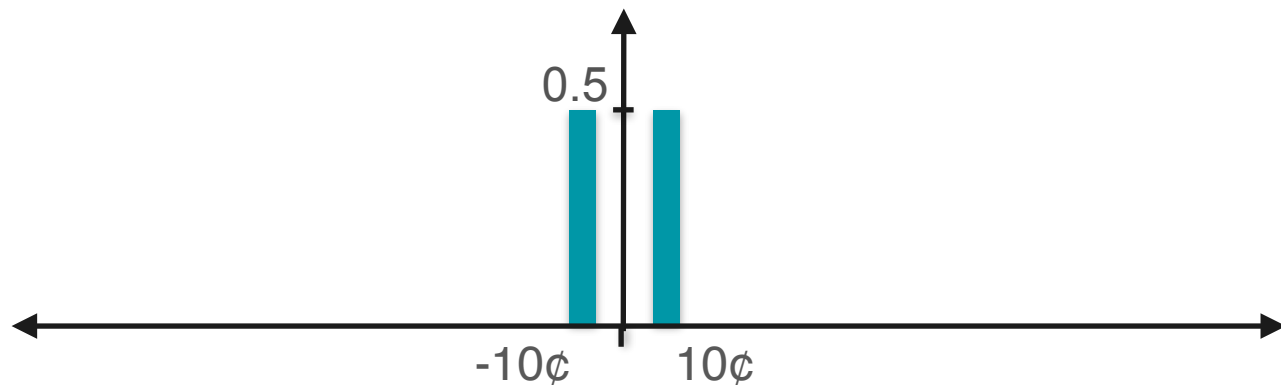


Kurtosis: Example

Game 1

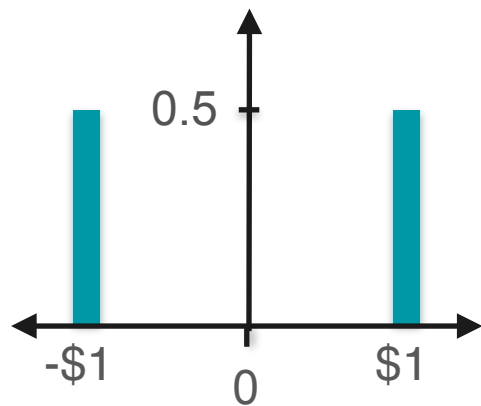


Game 2

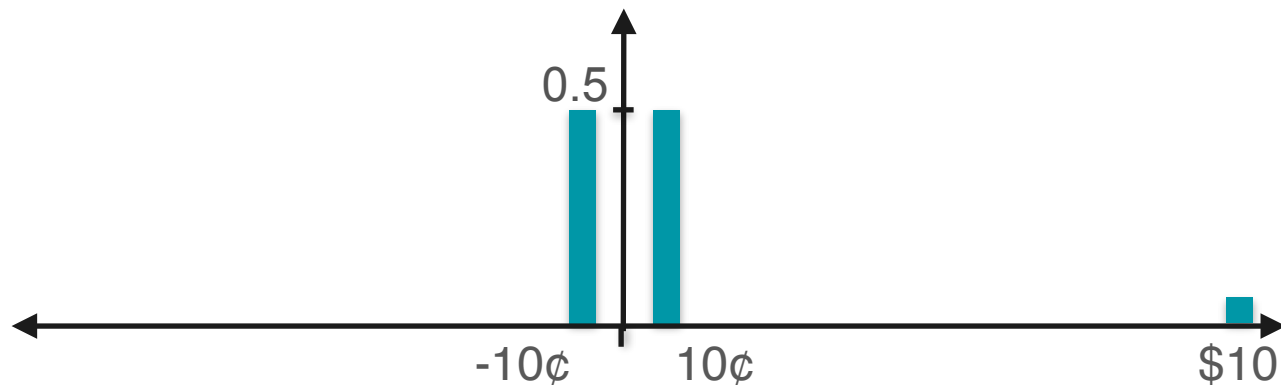


Kurtosis: Example

Game 1

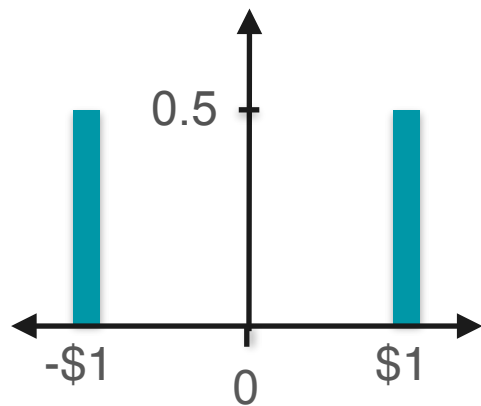


Game 2

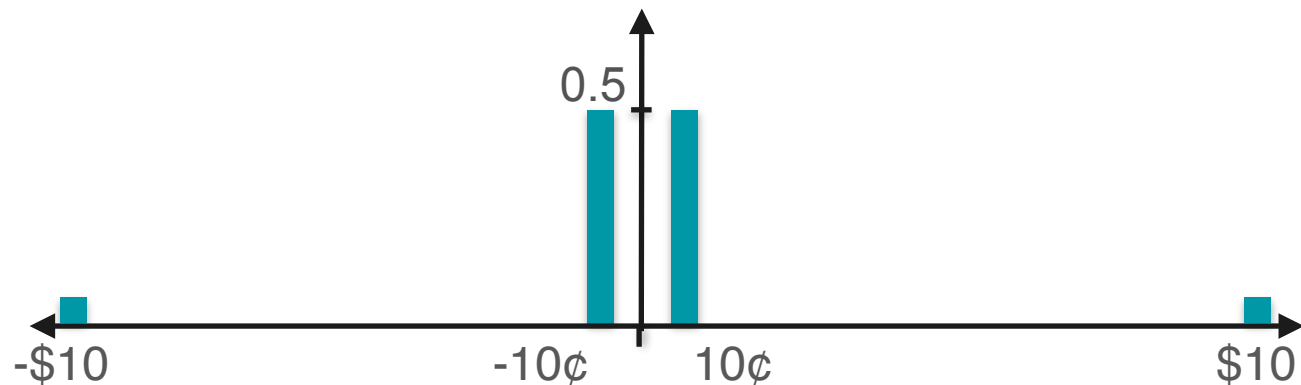


Kurtosis: Example

Game 1



Game 2

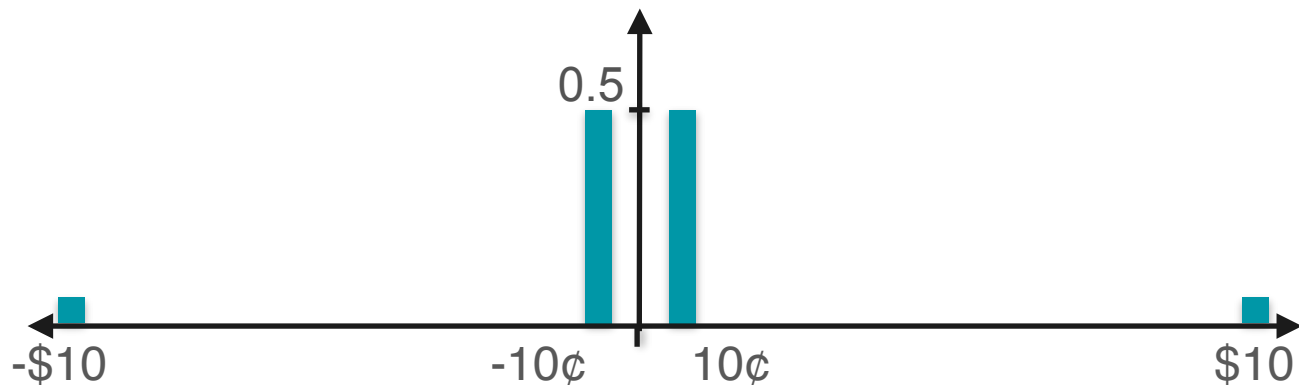
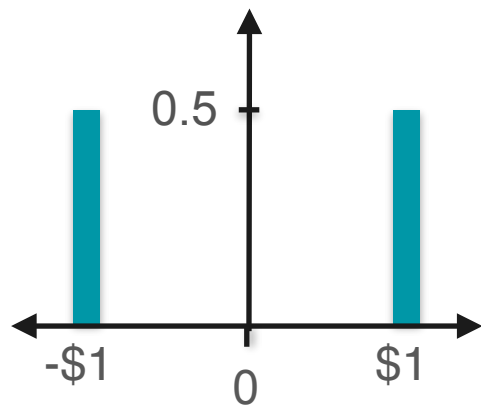


Kurtosis: Example

Game 1

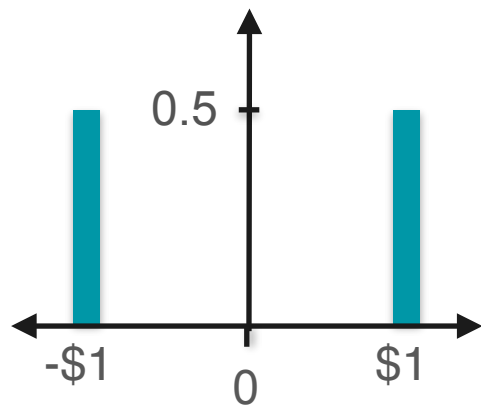
Expected value?

Game 2



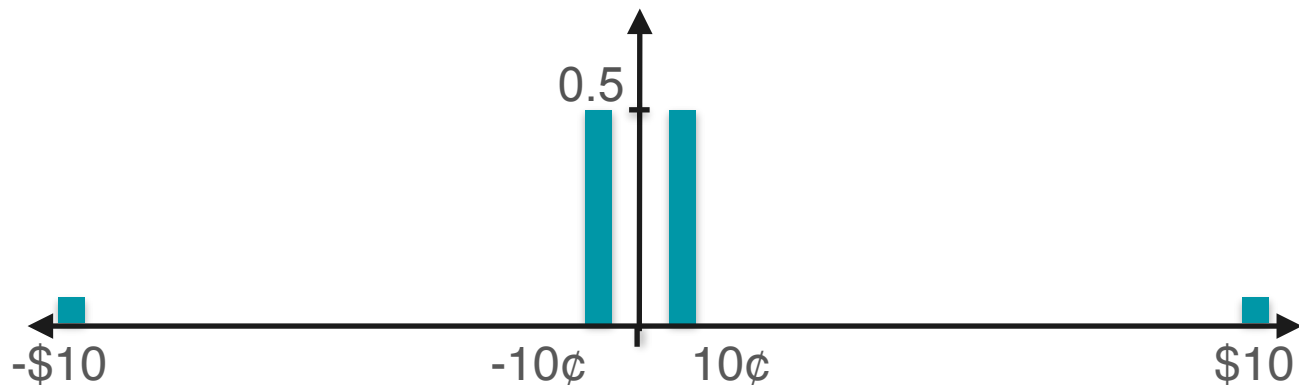
Kurtosis: Example

Game 1



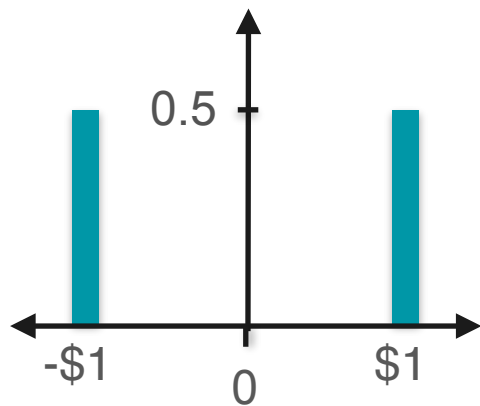
Expected value?
Standard deviation?

Game 2



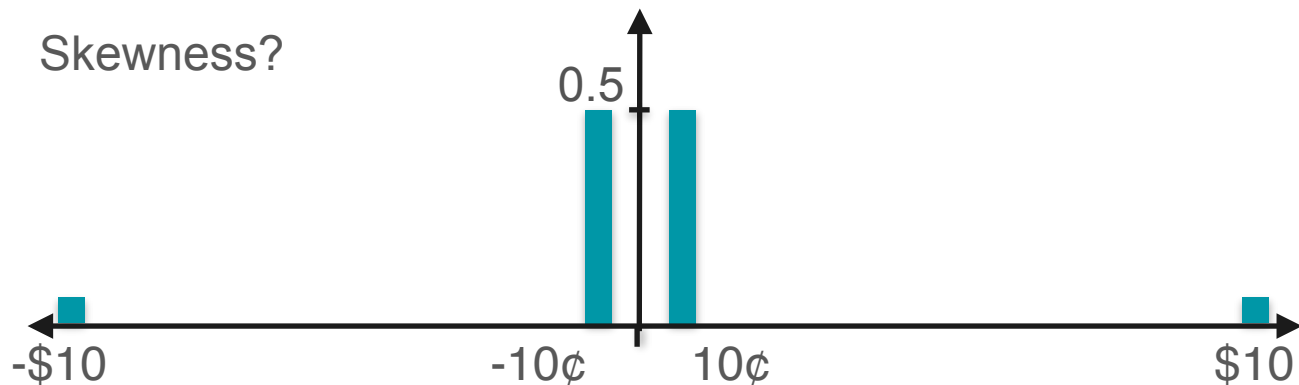
Kurtosis: Example

Game 1



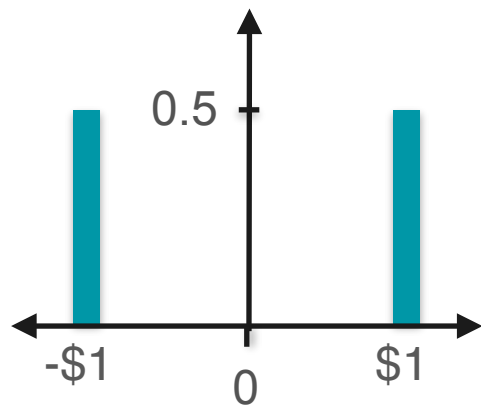
Expected value?
Standard deviation?
Skewness?

Game 2

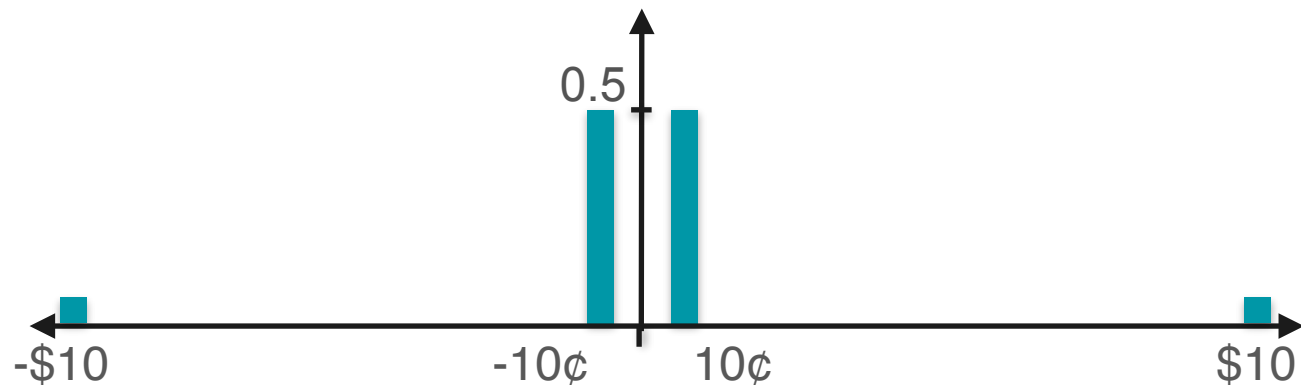


Kurtosis: Example Expected Value

Game 1

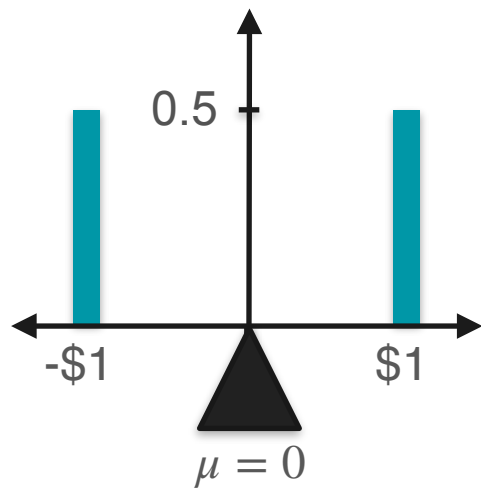


Game 2

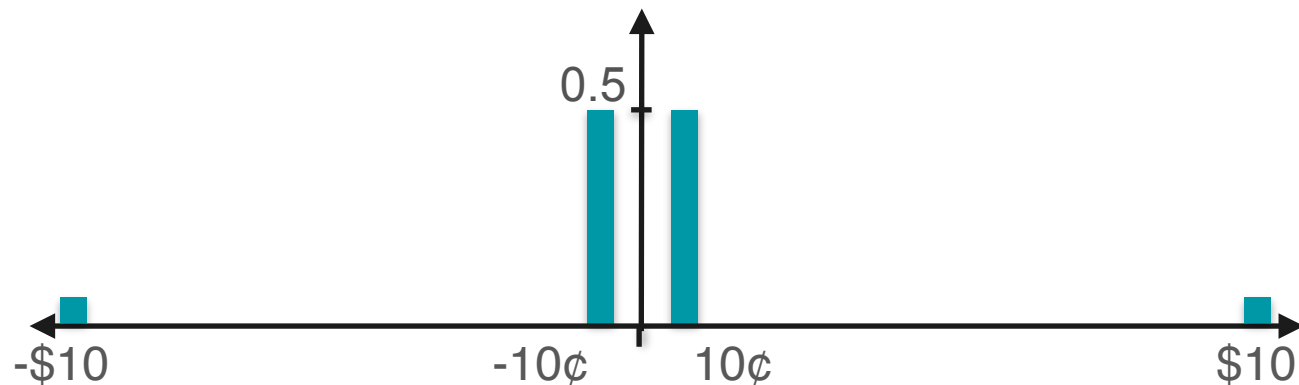


Kurtosis: Example Expected Value

Game 1

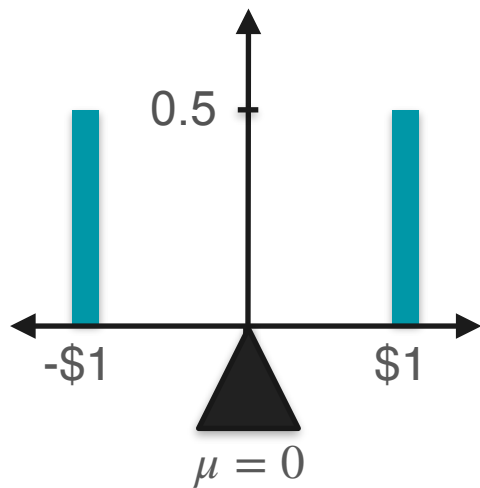


Game 2

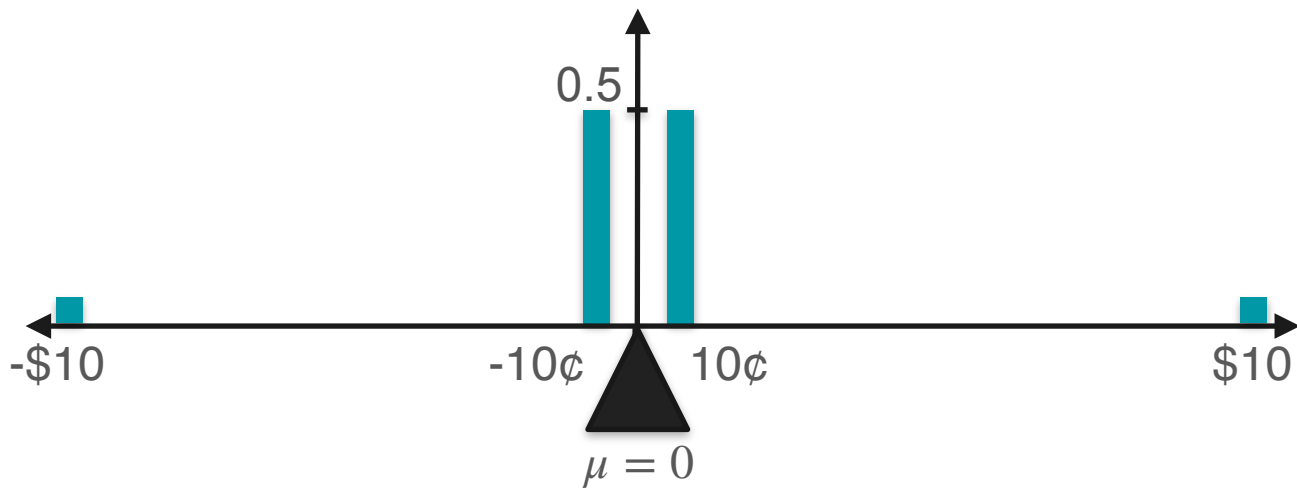


Kurtosis: Example Expected Value

Game 1



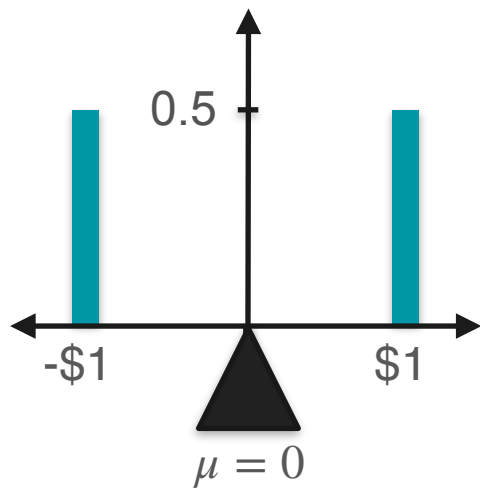
Game 2



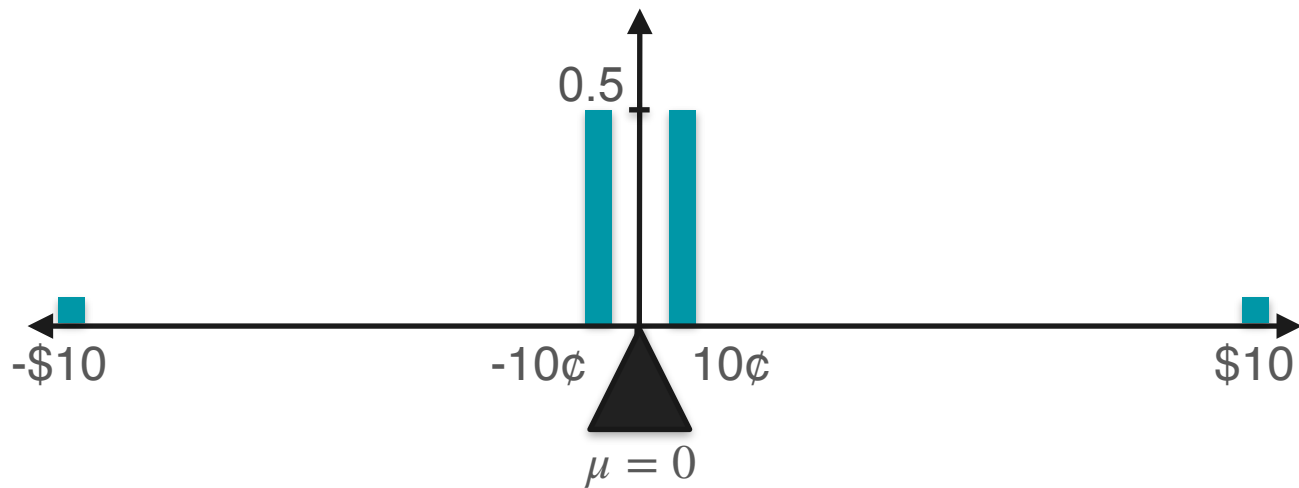
Kurtosis: Example Expected Value

Game 1

$$\mathbb{E}[X_1] = 0$$



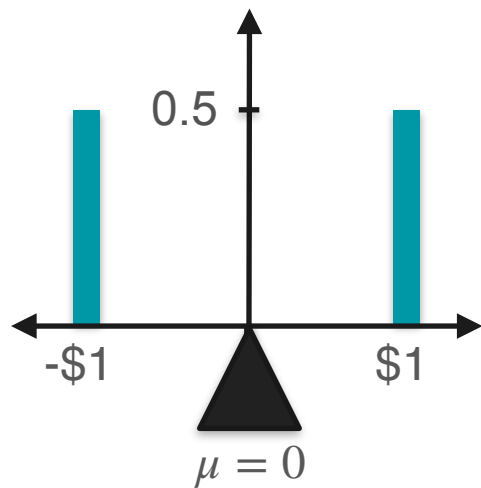
Game 2



Kurtosis: Example Expected Value

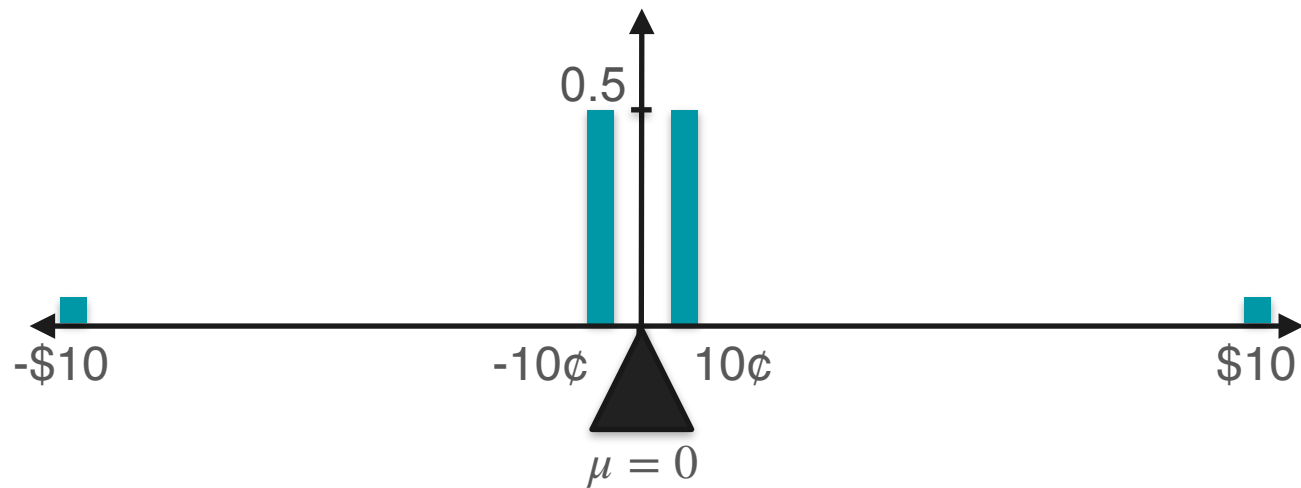
Game 1

$$\mathbb{E}[X_1] = 0$$

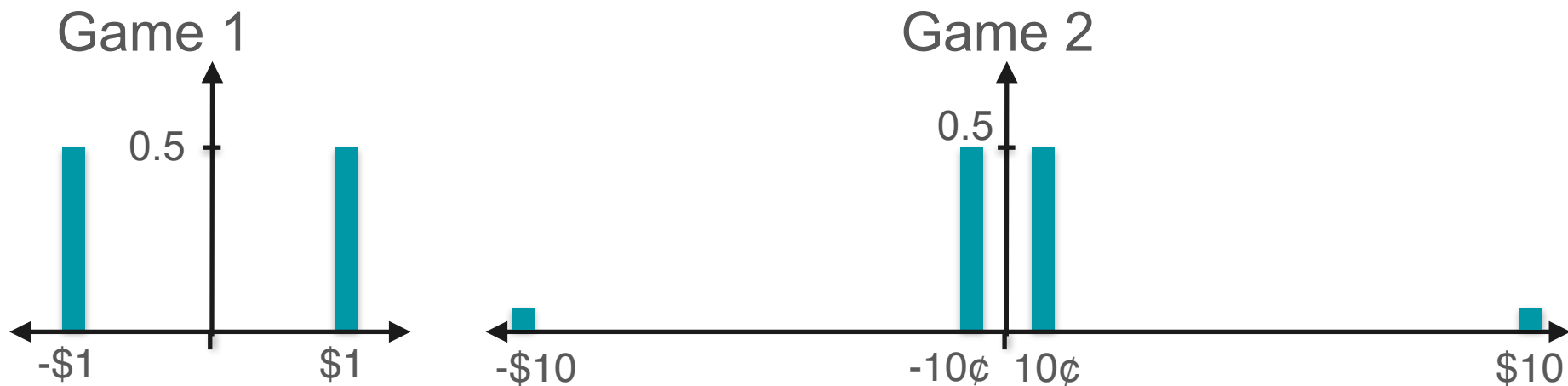


Game 2

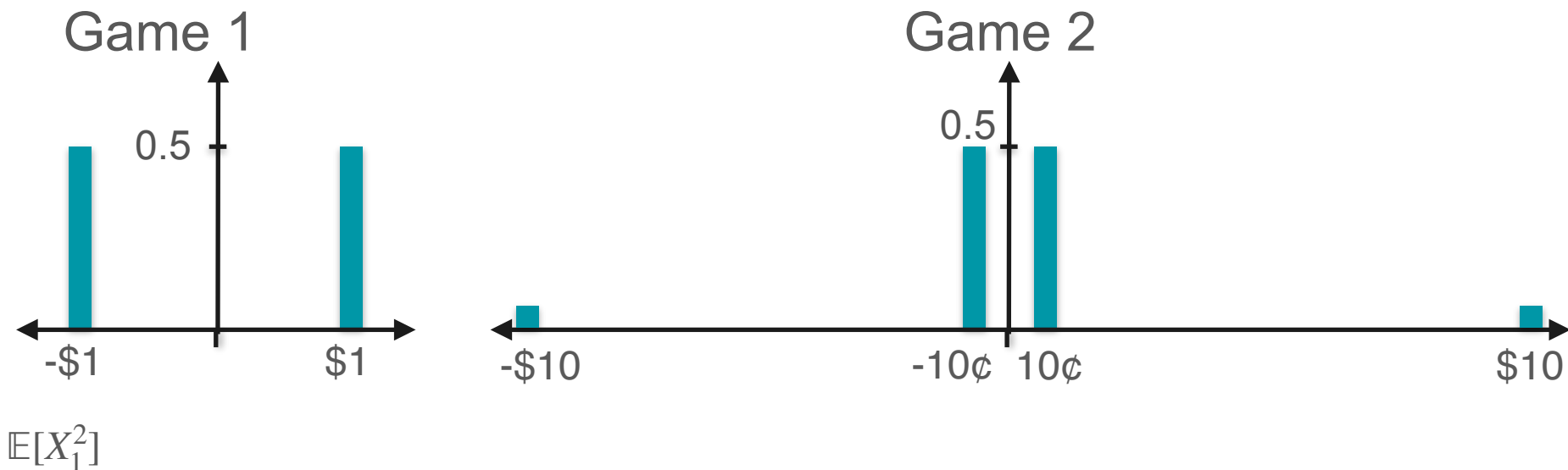
$$\mathbb{E}[X_2] = 0$$



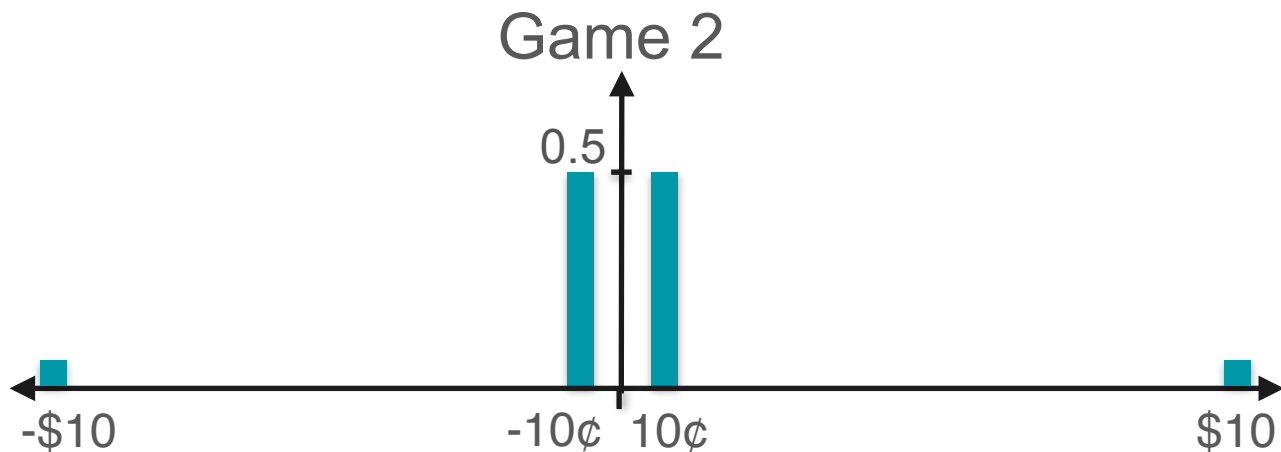
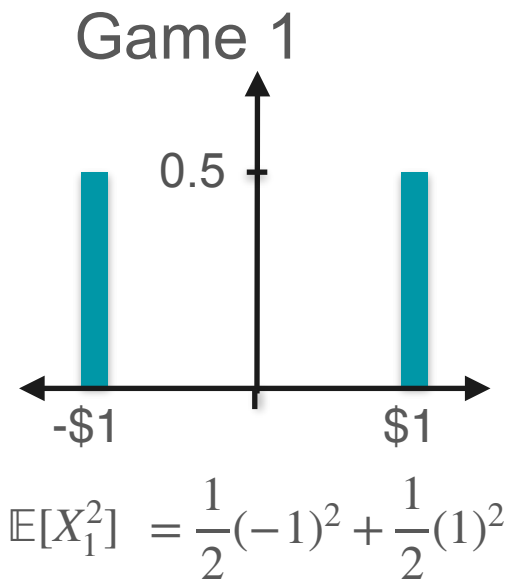
Kurtosis: Example Variance



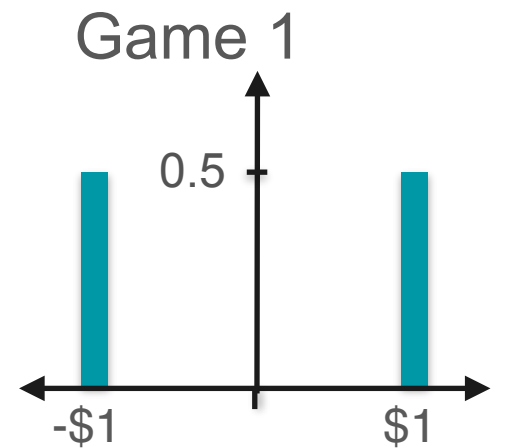
Kurtosis: Example Variance



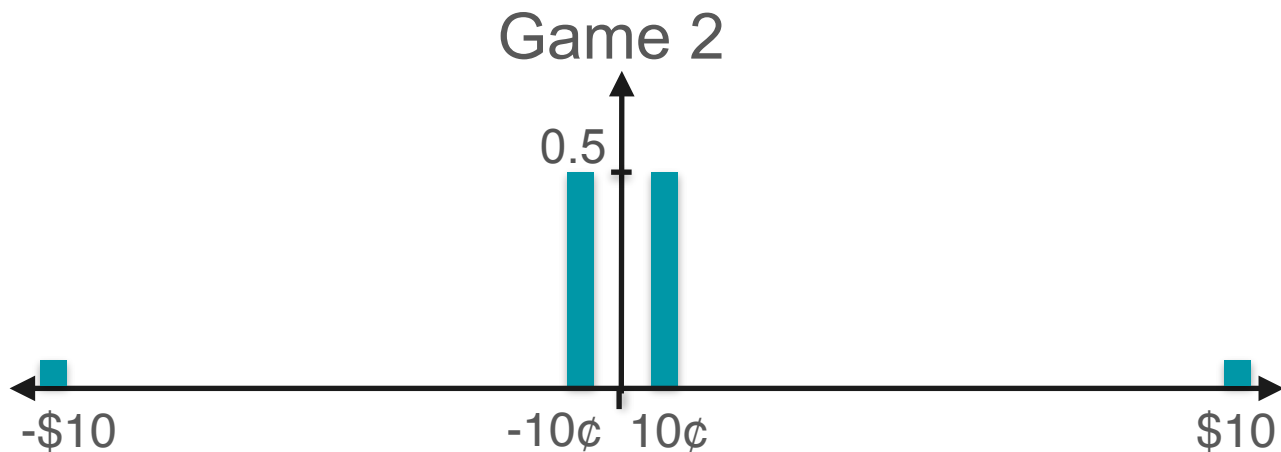
Kurtosis: Example Variance



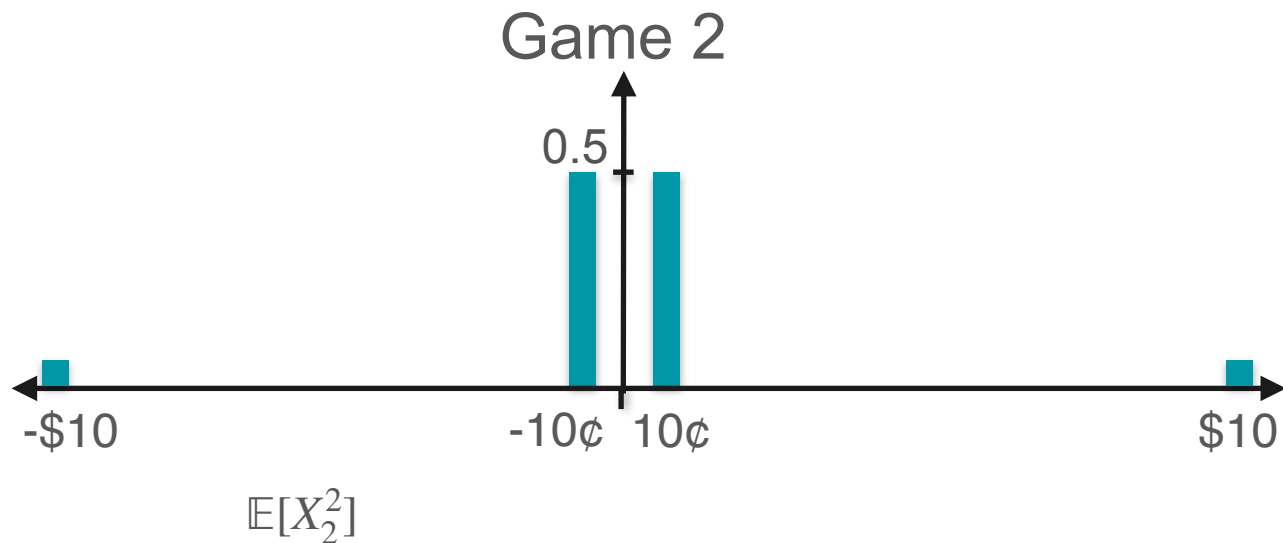
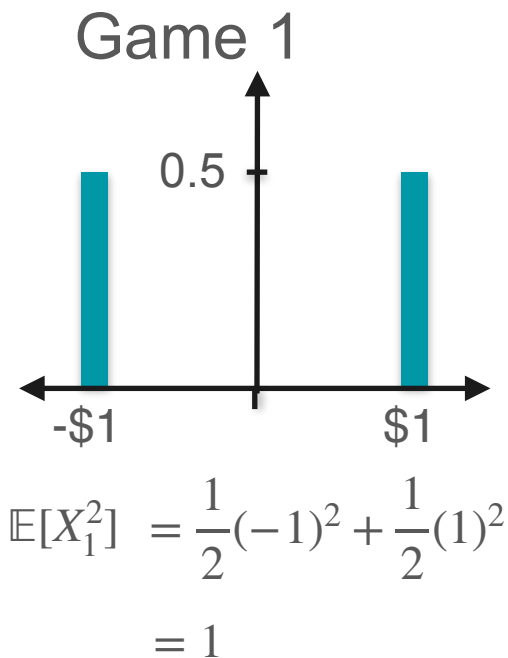
Kurtosis: Example Variance



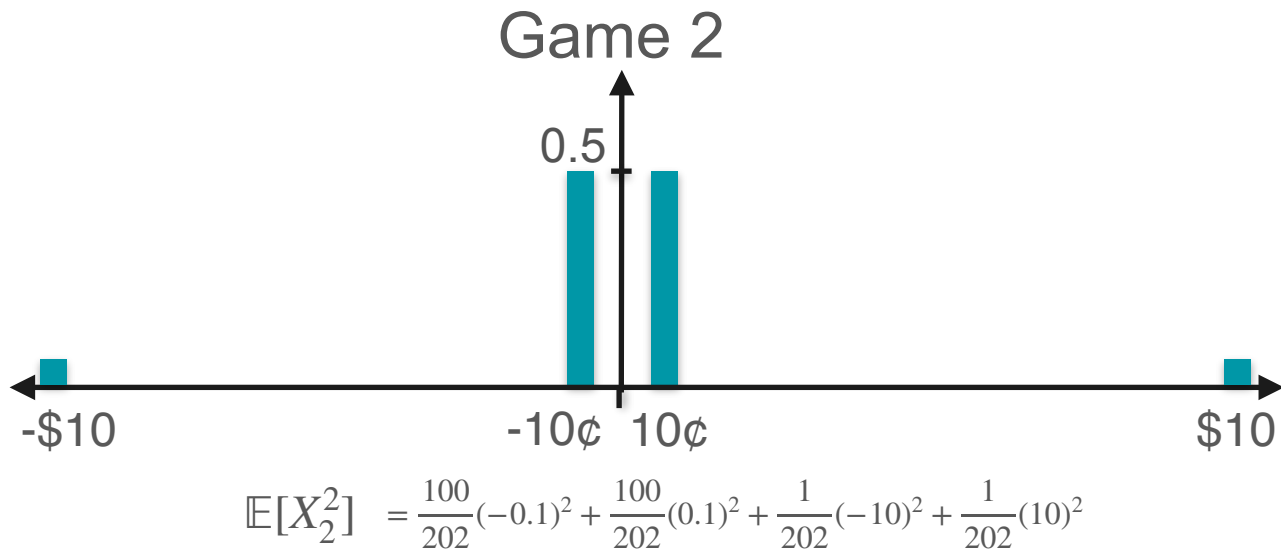
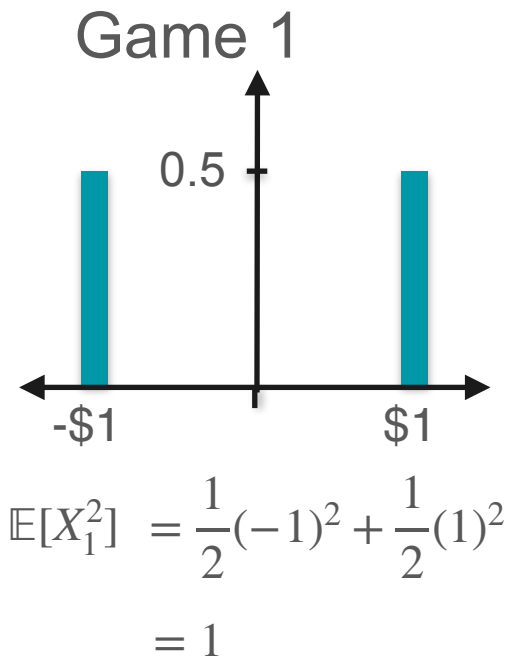
$$\begin{aligned}\mathbb{E}[X_1^2] &= \frac{1}{2}(-1)^2 + \frac{1}{2}(1)^2 \\ &= 1\end{aligned}$$



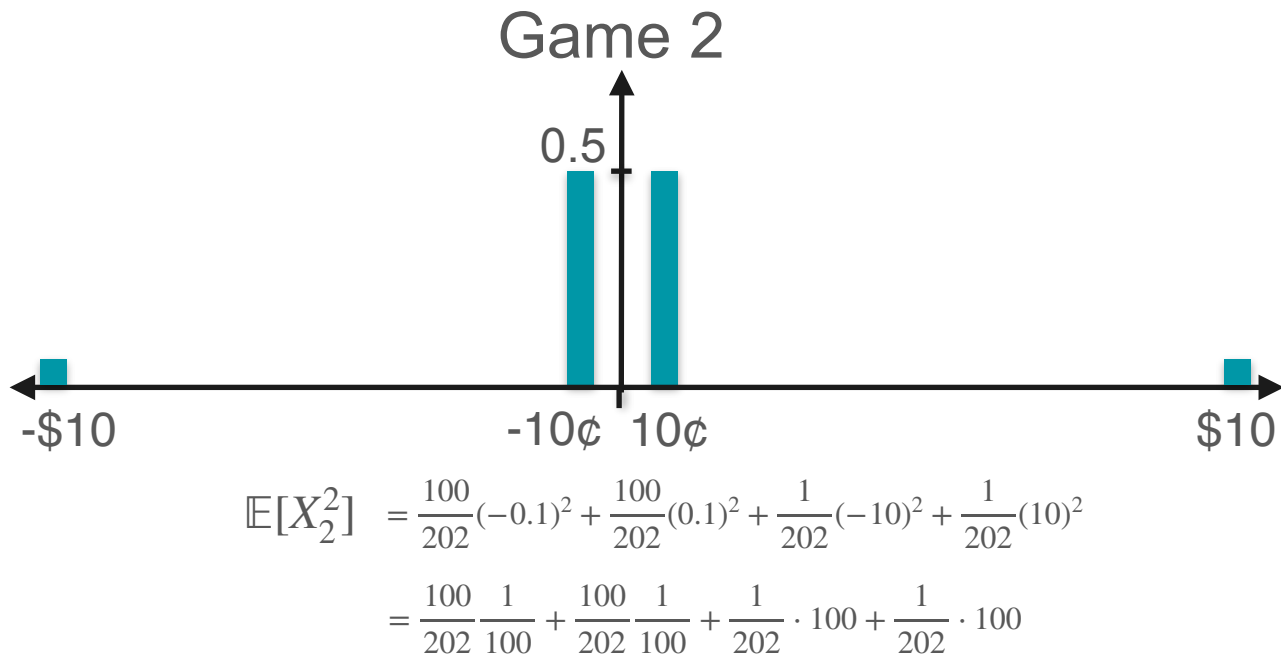
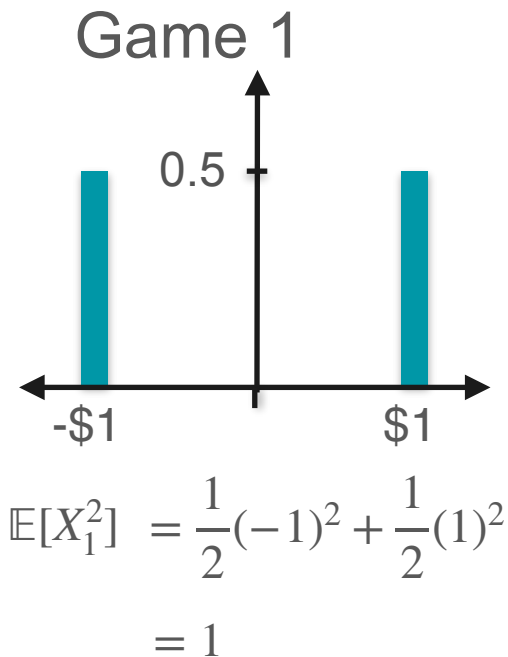
Kurtosis: Example Variance



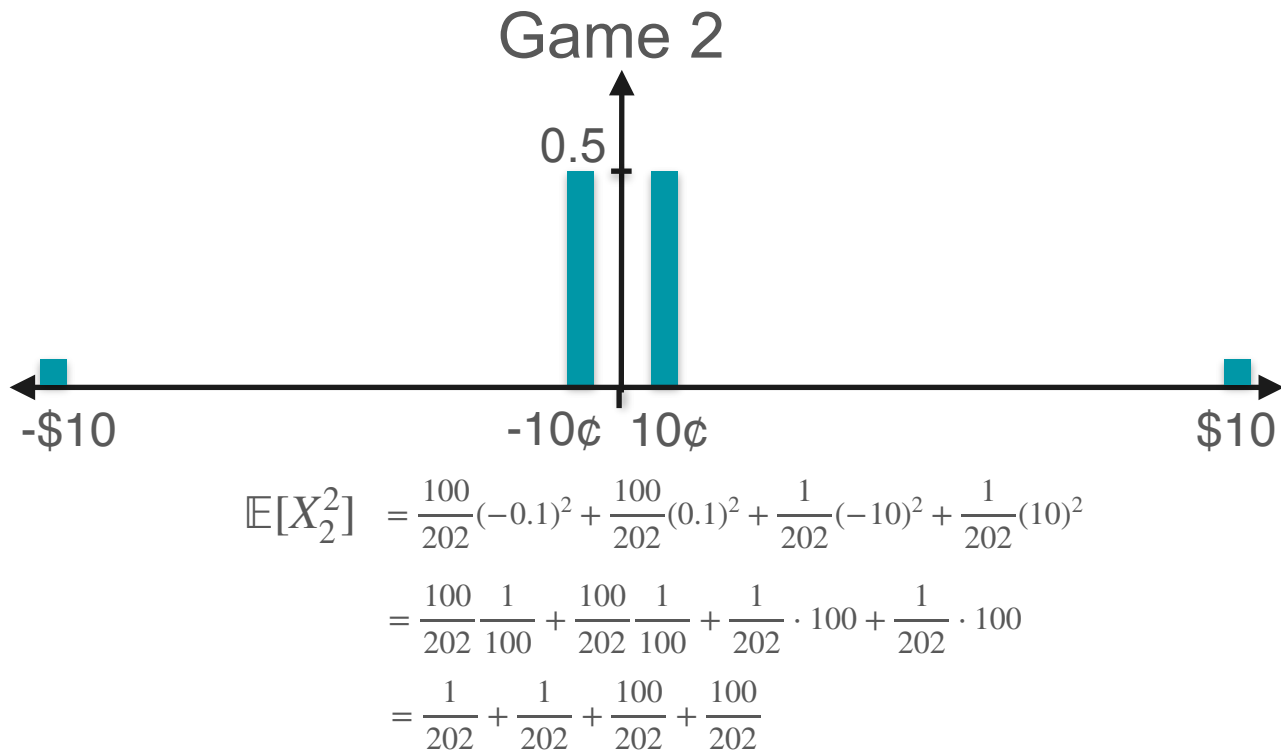
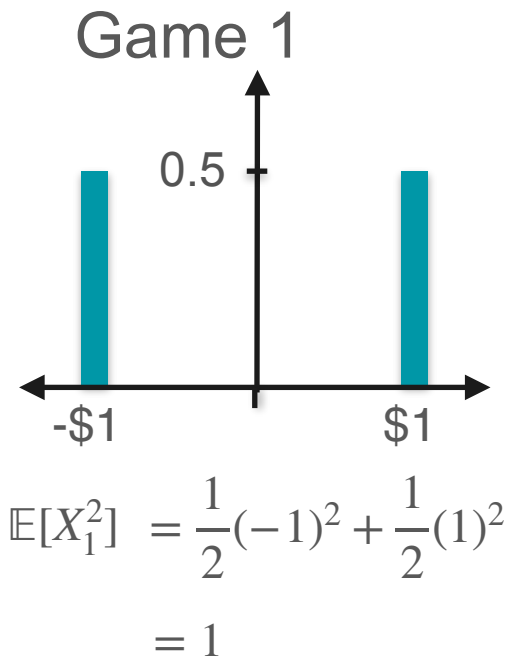
Kurtosis: Example Variance



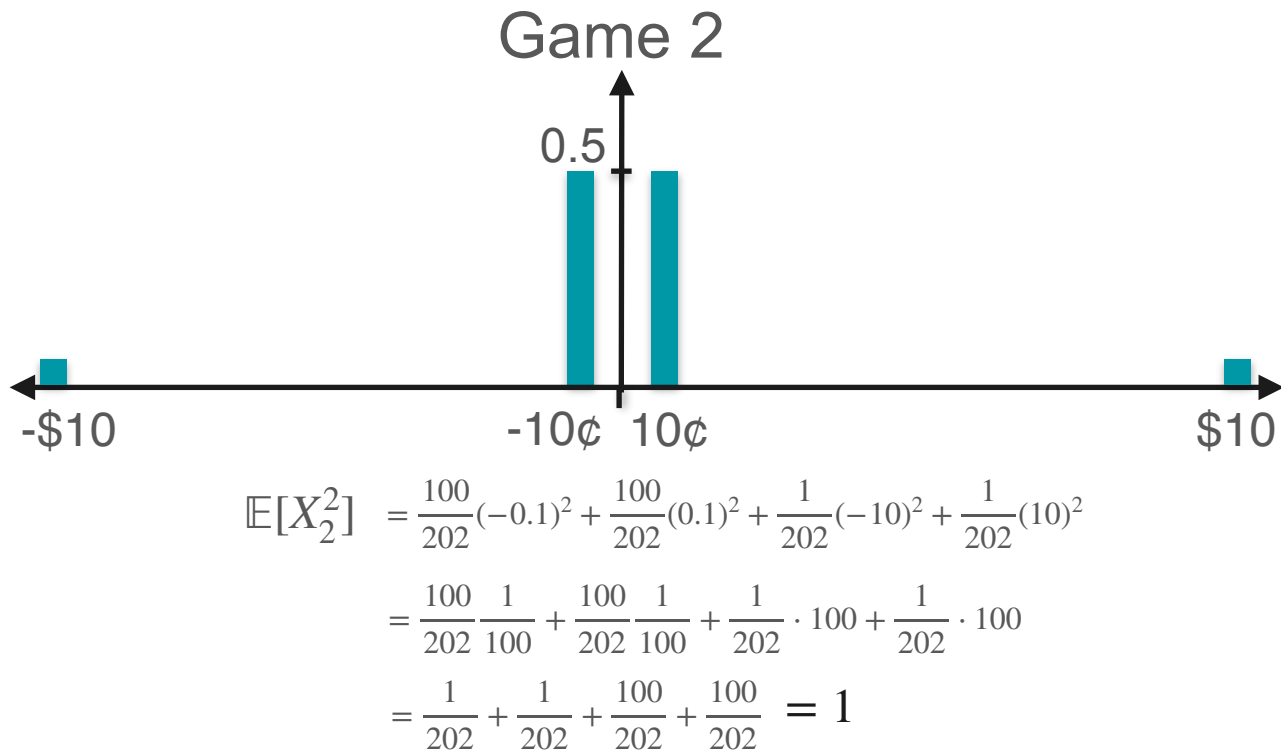
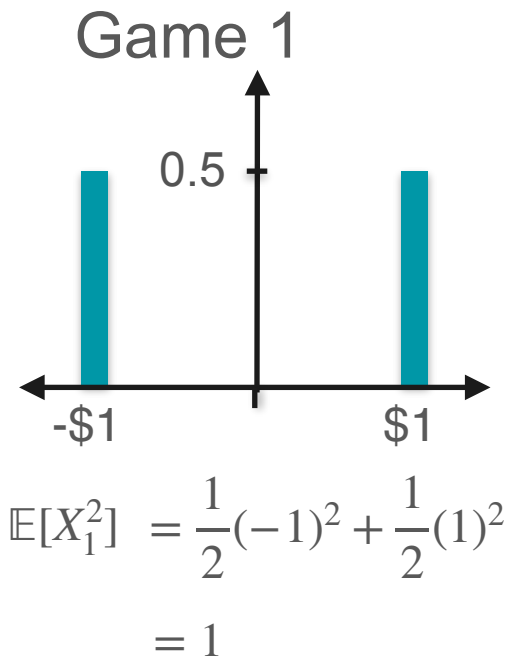
Kurtosis: Example Variance



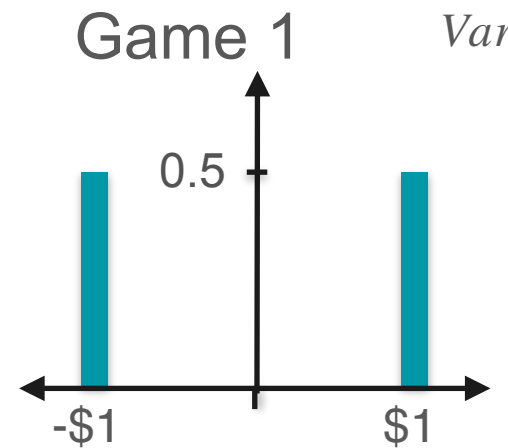
Kurtosis: Example Variance



Kurtosis: Example Variance

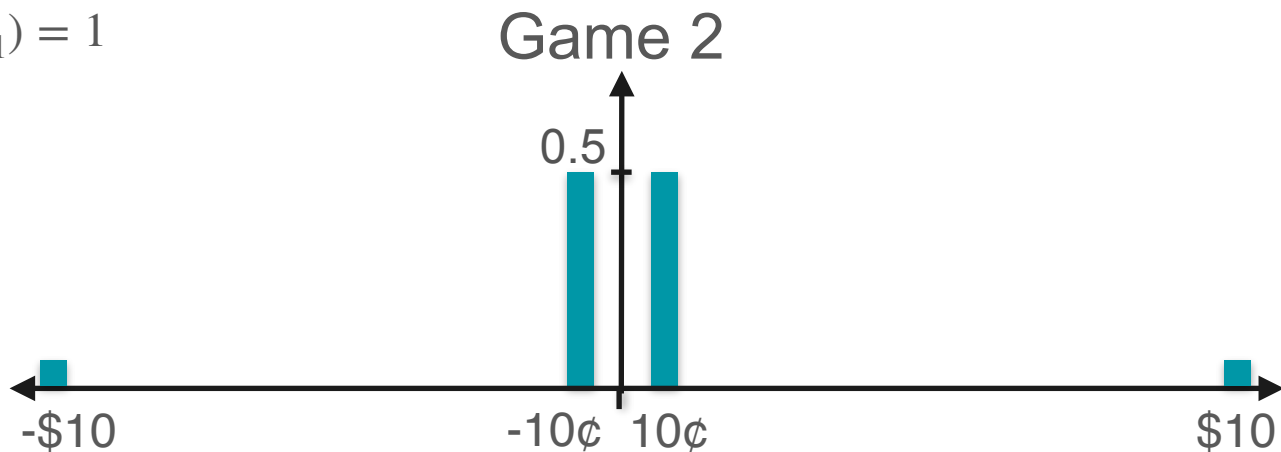


Kurtosis: Example Variance



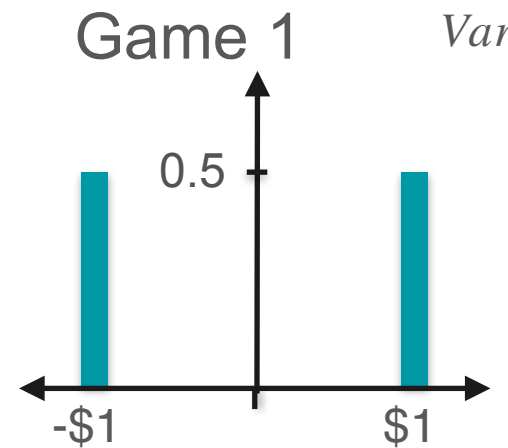
$$\text{Var}(X_1) = 1$$

$$\begin{aligned}\mathbb{E}[X_1^2] &= \frac{1}{2}(-1)^2 + \frac{1}{2}(1)^2 \\ &= 1\end{aligned}$$



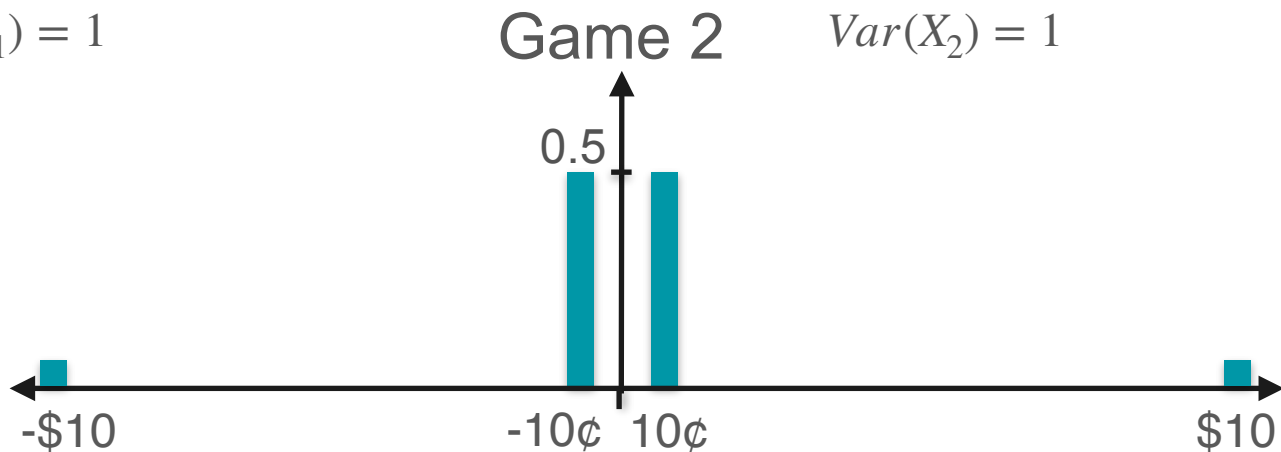
$$\begin{aligned}\mathbb{E}[X_2^2] &= \frac{100}{202}(-0.1)^2 + \frac{100}{202}(0.1)^2 + \frac{1}{202}(-10)^2 + \frac{1}{202}(10)^2 \\ &= \frac{100}{202} \frac{1}{100} + \frac{100}{202} \frac{1}{100} + \frac{1}{202} \cdot 100 + \frac{1}{202} \cdot 100 \\ &= \frac{1}{202} + \frac{1}{202} + \frac{100}{202} + \frac{100}{202} = 1\end{aligned}$$

Kurtosis: Example Variance



$$\text{Var}(X_1) = 1$$

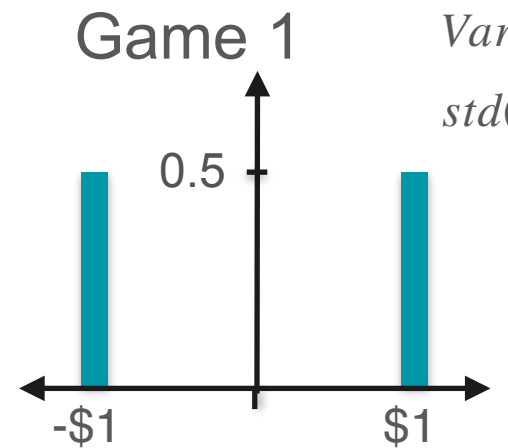
$$\begin{aligned}\mathbb{E}[X_1^2] &= \frac{1}{2}(-1)^2 + \frac{1}{2}(1)^2 \\ &= 1\end{aligned}$$



$$\text{Var}(X_2) = 1$$

$$\begin{aligned}\mathbb{E}[X_2^2] &= \frac{100}{202}(-0.1)^2 + \frac{100}{202}(0.1)^2 + \frac{1}{202}(-10)^2 + \frac{1}{202}(10)^2 \\ &= \frac{100}{202} \frac{1}{100} + \frac{100}{202} \frac{1}{100} + \frac{1}{202} \cdot 100 + \frac{1}{202} \cdot 100 \\ &= \frac{1}{202} + \frac{1}{202} + \frac{100}{202} + \frac{100}{202} = 1\end{aligned}$$

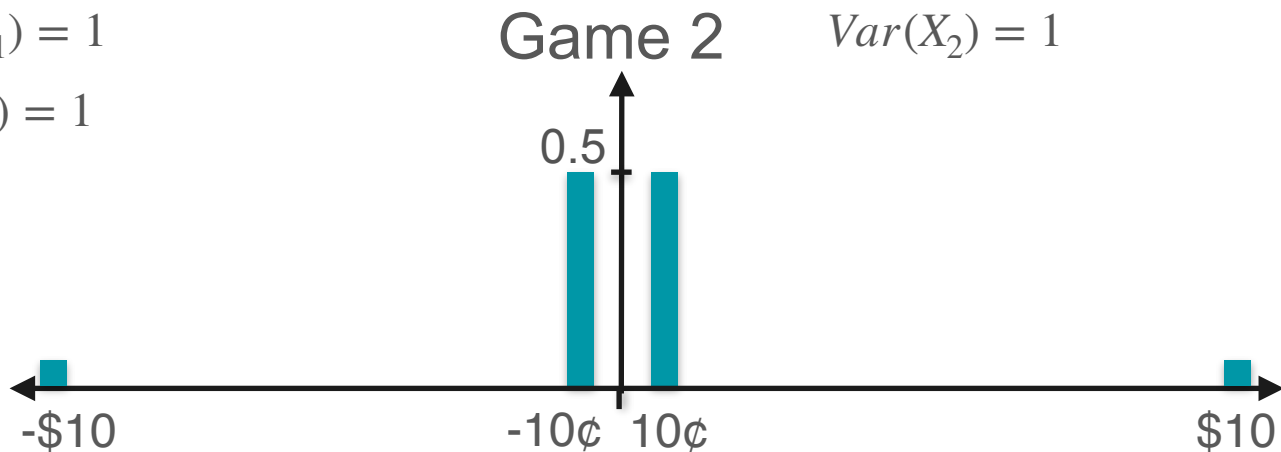
Kurtosis: Example Variance



$$\text{Var}(X_1) = 1$$

$$\text{std}(X_1) = 1$$

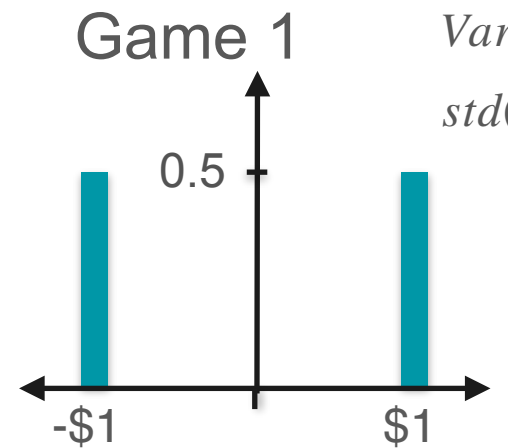
$$\begin{aligned}\mathbb{E}[X_1^2] &= \frac{1}{2}(-1)^2 + \frac{1}{2}(1)^2 \\ &= 1\end{aligned}$$



$$\text{Var}(X_2) = 1$$

$$\begin{aligned}\mathbb{E}[X_2^2] &= \frac{100}{202}(-0.1)^2 + \frac{100}{202}(0.1)^2 + \frac{1}{202}(-10)^2 + \frac{1}{202}(10)^2 \\ &= \frac{100}{202} \frac{1}{100} + \frac{100}{202} \frac{1}{100} + \frac{1}{202} \cdot 100 + \frac{1}{202} \cdot 100 \\ &= \frac{1}{202} + \frac{1}{202} + \frac{100}{202} + \frac{100}{202} = 1\end{aligned}$$

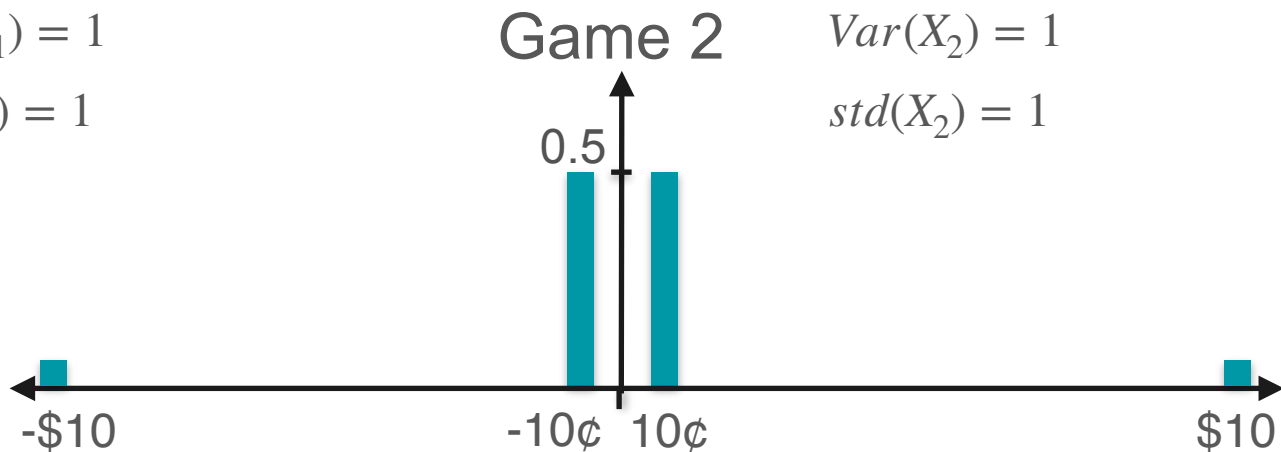
Kurtosis: Example Variance



$$\text{Var}(X_1) = 1$$

$$\text{std}(X_1) = 1$$

$$\begin{aligned}\mathbb{E}[X_1^2] &= \frac{1}{2}(-1)^2 + \frac{1}{2}(1)^2 \\ &= 1\end{aligned}$$

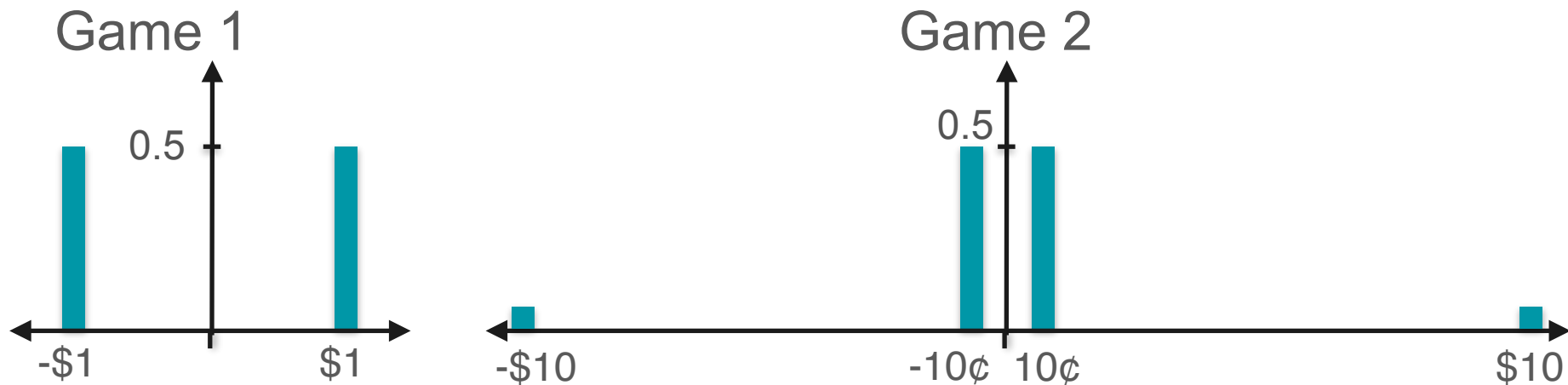


$$\text{Var}(X_2) = 1$$

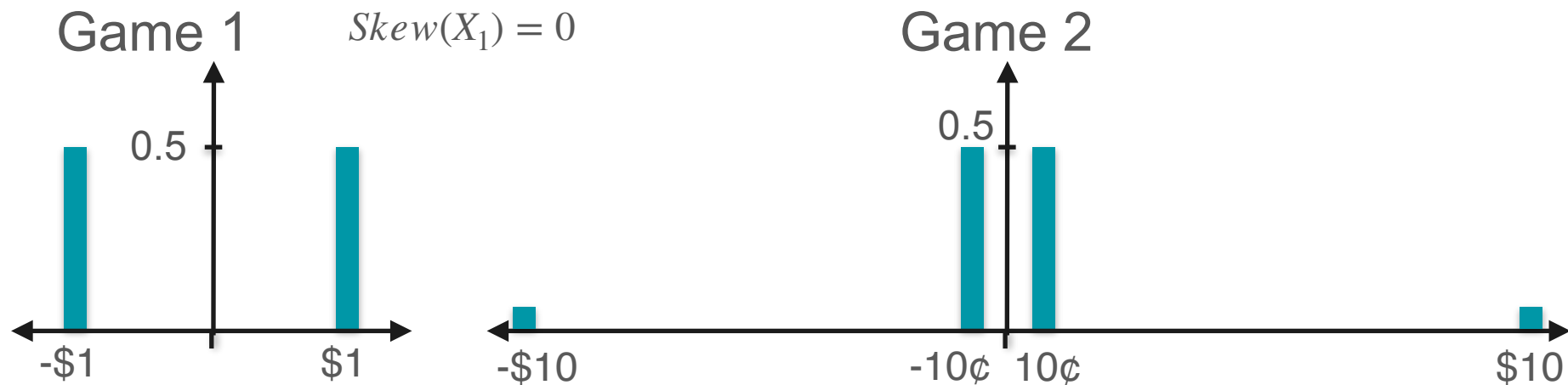
$$\text{std}(X_2) = 1$$

$$\begin{aligned}\mathbb{E}[X_2^2] &= \frac{100}{202}(-0.1)^2 + \frac{100}{202}(0.1)^2 + \frac{1}{202}(-10)^2 + \frac{1}{202}(10)^2 \\ &= \frac{100}{202} \frac{1}{100} + \frac{100}{202} \frac{1}{100} + \frac{1}{202} \cdot 100 + \frac{1}{202} \cdot 100 \\ &= \frac{1}{202} + \frac{1}{202} + \frac{100}{202} + \frac{100}{202} = 1\end{aligned}$$

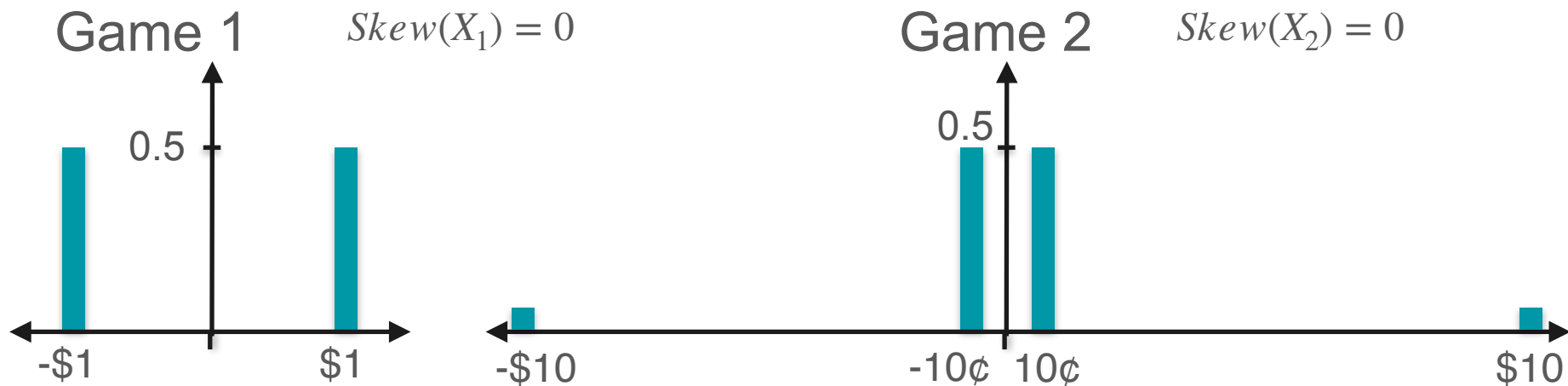
Kurtosis: Example Skewness



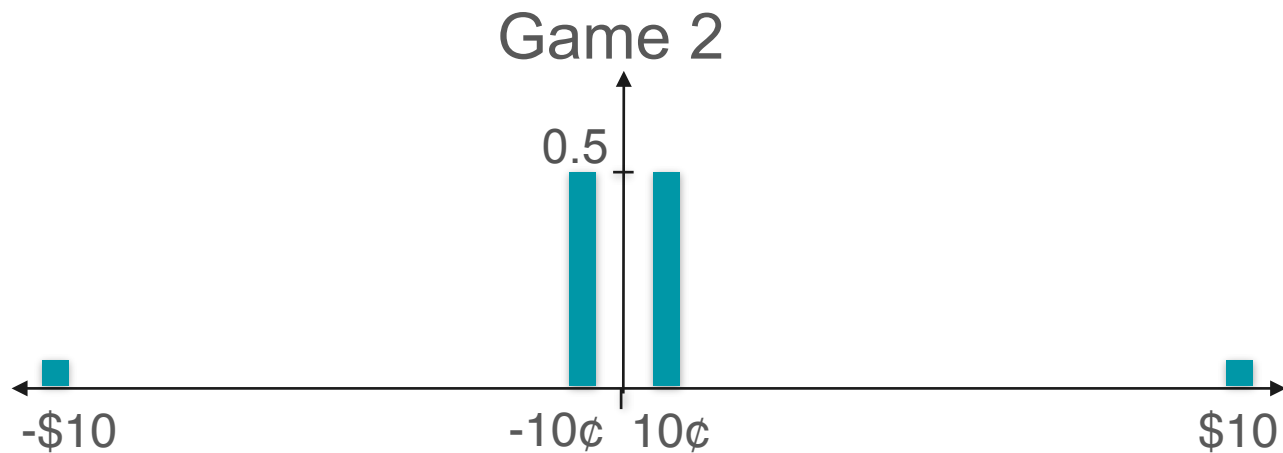
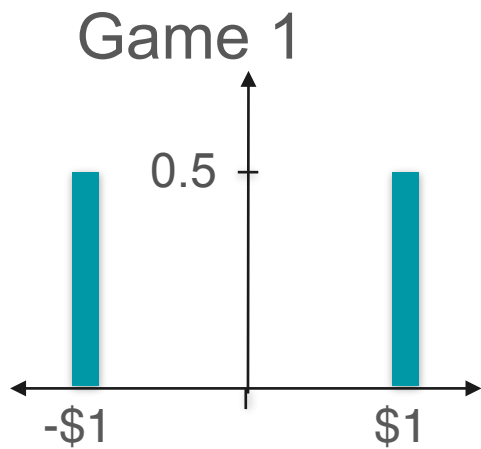
Kurtosis: Example Skewness



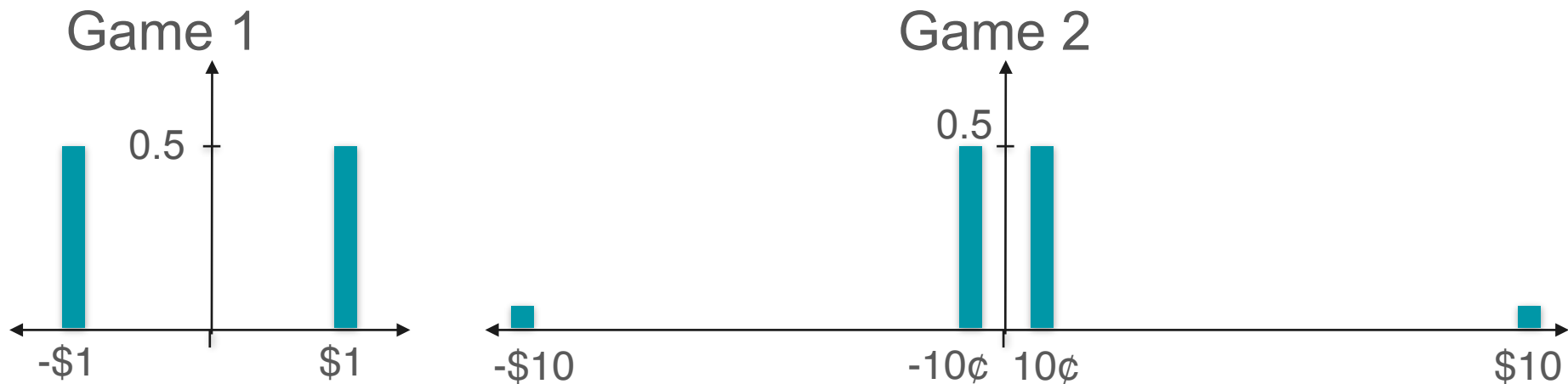
Kurtosis: Example Skewness



Kurtosis



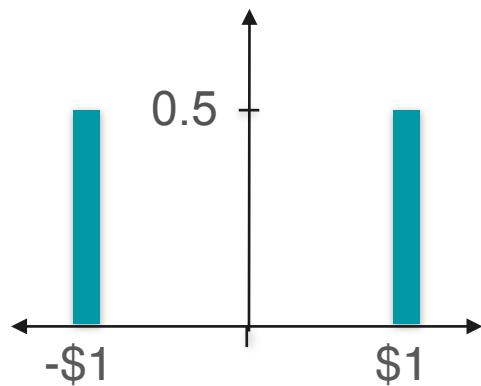
Kurtosis



$$\text{Skew}(X_1) = 0$$

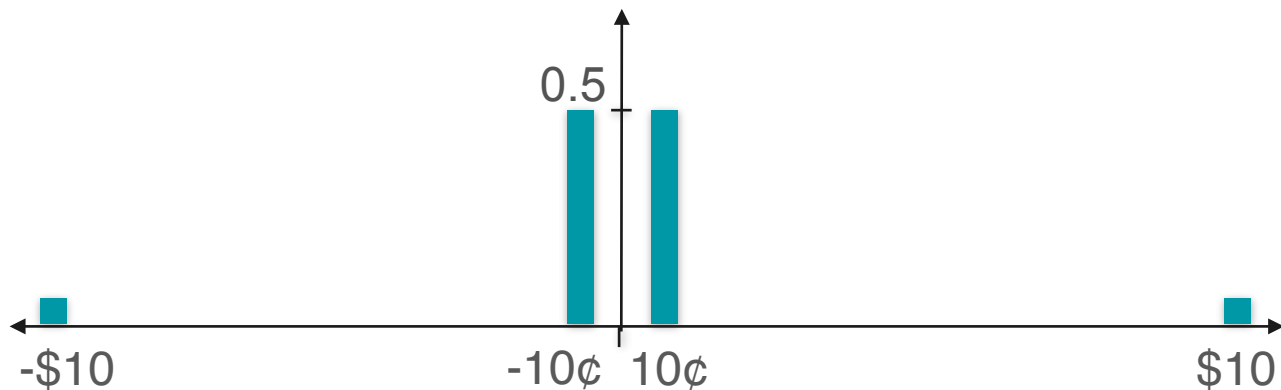
Kurtosis

Game 1



$$Skew(X_1) = 0$$

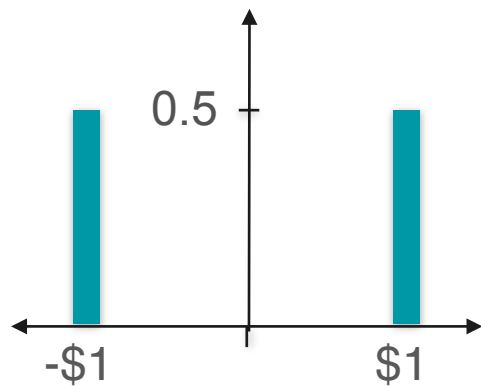
Game 2



$$Skew(X_2) = 0$$

Kurtosis

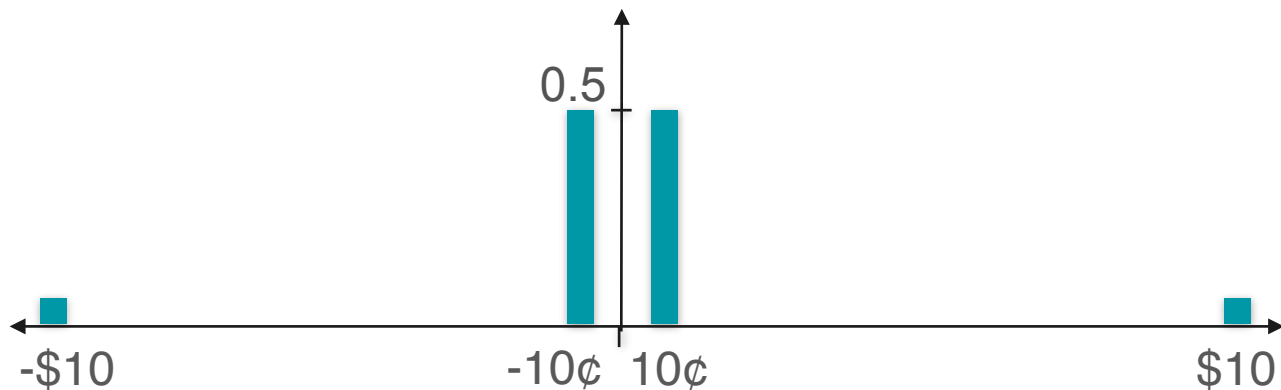
Game 1



$$\text{Var}(X_1) = 1$$

$$\text{Skew}(X_1) = 0$$

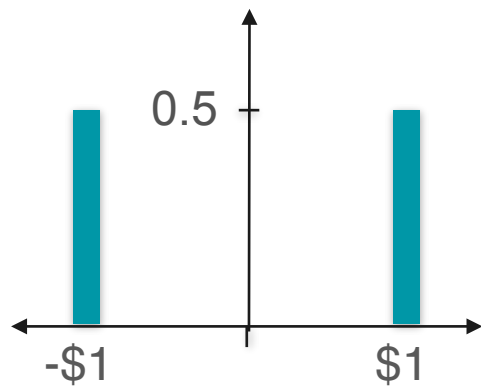
Game 2



$$\text{Skew}(X_2) = 0$$

Kurtosis

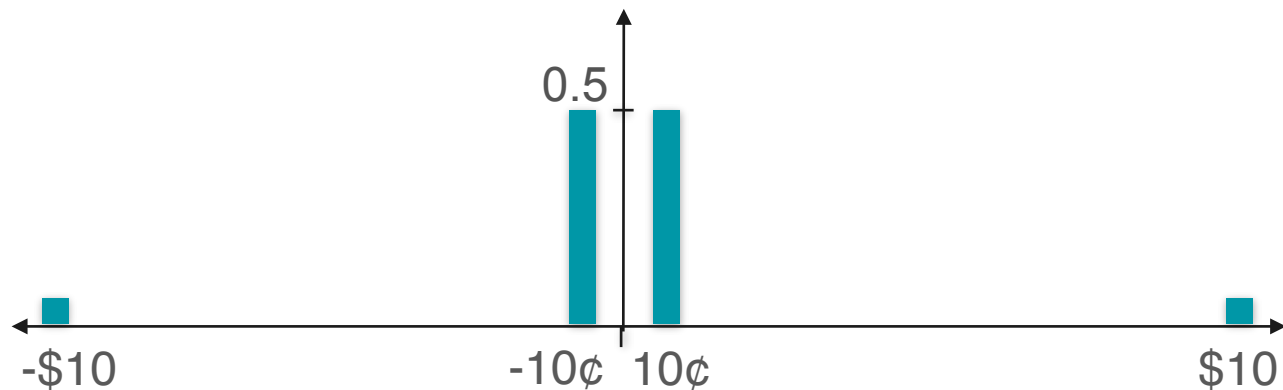
Game 1



$$\text{Var}(X_1) = 1$$

$$\text{Skew}(X_1) = 0$$

Game 2



$$\text{Var}(X_2) = 1$$

$$\text{Skew}(X_2) = 0$$

Kurtosis

Game 1

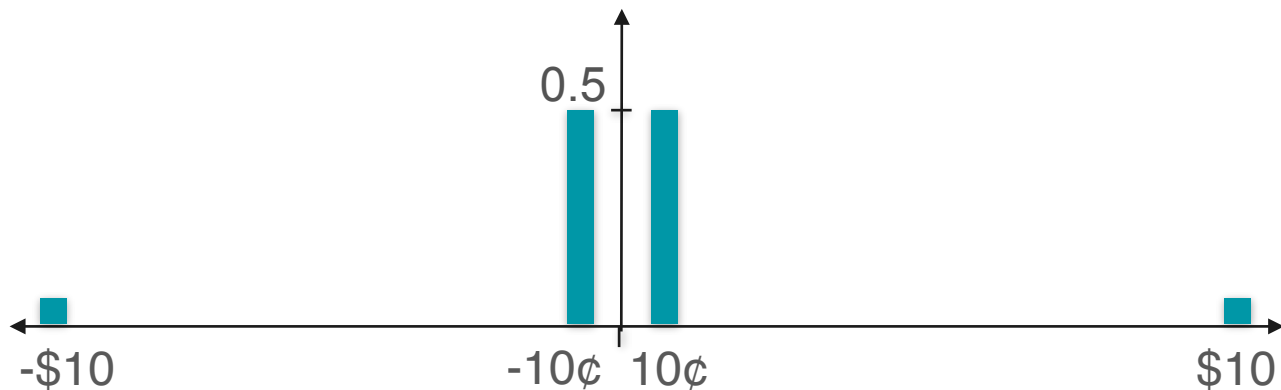


$$E[X_1] = 0$$

$$\text{Var}(X_1) = 1$$

$$\text{Skew}(X_1) = 0$$

Game 2

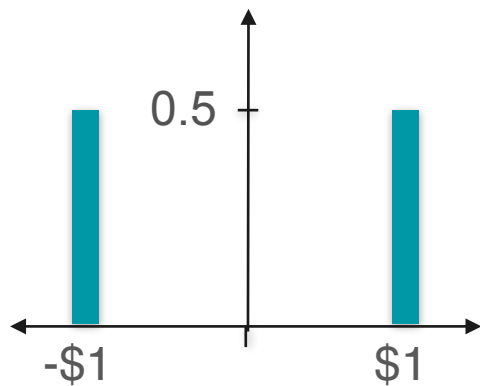


$$\text{Var}(X_2) = 1$$

$$\text{Skew}(X_2) = 0$$

Kurtosis

Game 1

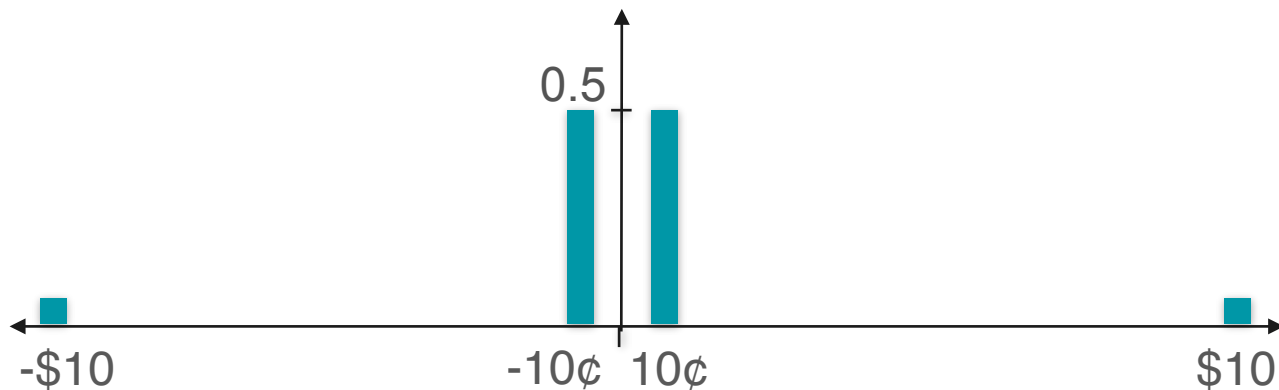


$$E[X_1] = 0$$

$$\text{Var}(X_1) = 1$$

$$\text{Skew}(X_1) = 0$$

Game 2



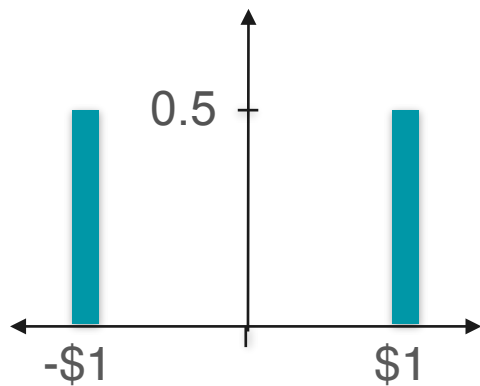
$$E[X_2] = 0$$

$$\text{Var}(X_2) = 1$$

$$\text{Skew}(X_2) = 0$$

Kurtosis

Game 1



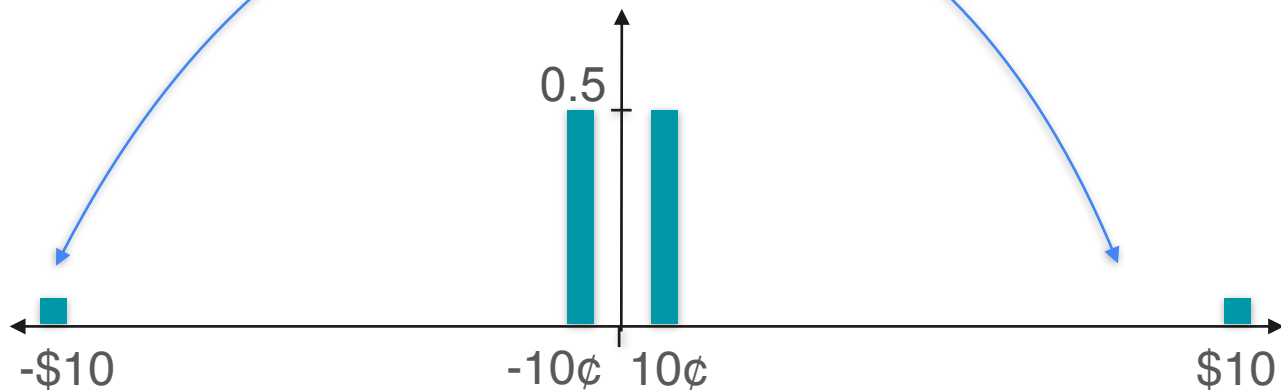
$$E[X_1] = 0$$

$$\text{Var}(X_1) = 1$$

$$\text{Skew}(X_1) = 0$$

Has values way
farther from 0

Game 2



$$E[X_2] = 0$$

$$\text{Var}(X_2) = 1$$

$$\text{Skew}(X_2) = 0$$

Kurtosis

Game 1



$$E[X_1] = 0$$

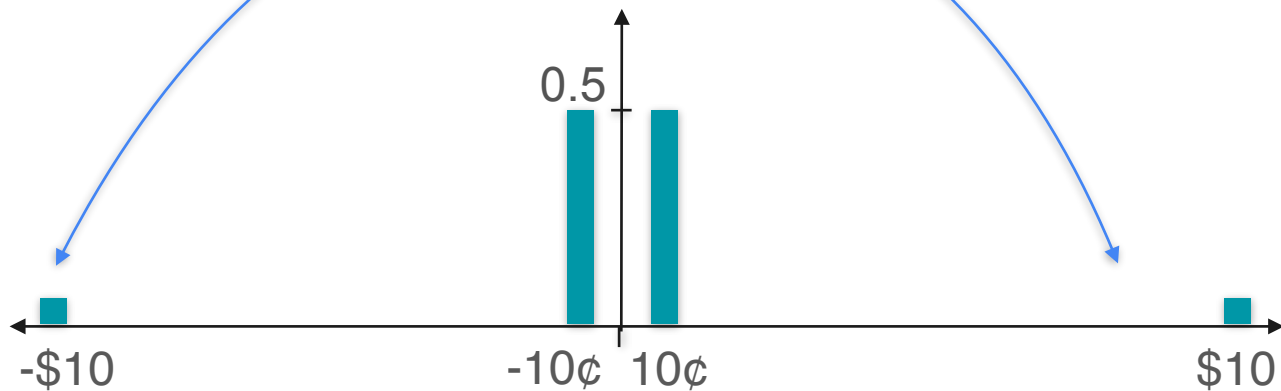
$$E[X_1] = 0$$

$$\text{Var}(X_1) = 1$$

$$\text{Skew}(X_1) = 0$$

Has values way
farther from 0

Game 2



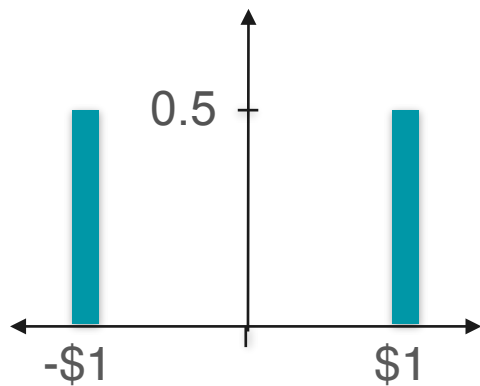
$$E[X_2] = 0$$

$$\text{Var}(X_2) = 1$$

$$\text{Skew}(X_2) = 0$$

Kurtosis

Game 1



$$E[X_1] = 0$$

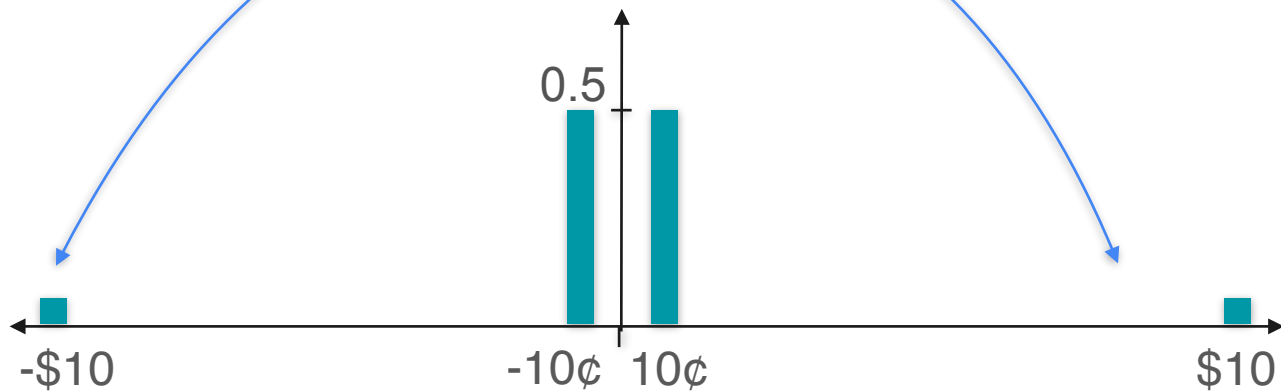
$$\text{Var}(X_1) = 1$$

$$\text{Skew}(X_1) = 0$$

$$E[X_1] = 0$$

$$E[X_1^2] = 1$$

Game 2



$$E[X_2] = 0$$

$$\text{Var}(X_2) = 1$$

$$\text{Skew}(X_2) = 0$$

Has values way
farther from 0

Kurtosis

Game 1



$$E[X_1] = 0$$

$$\text{Var}(X_1) = 1$$

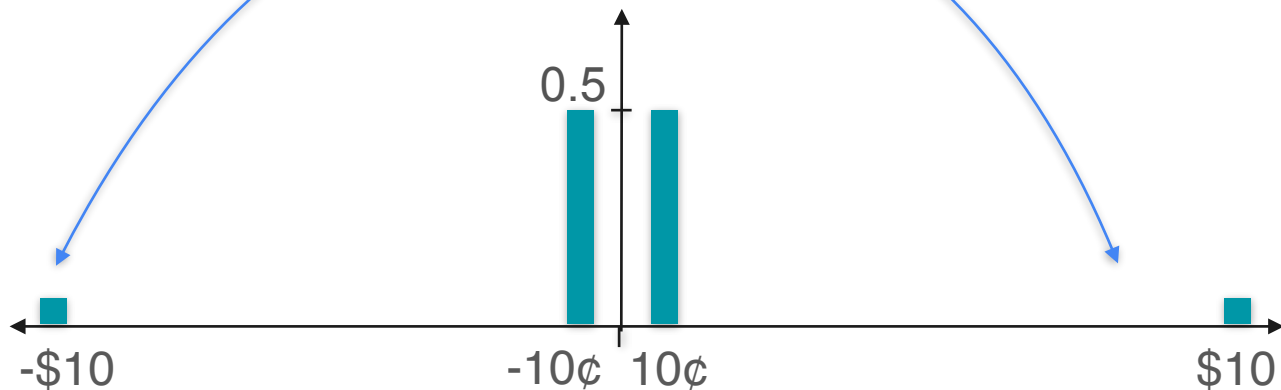
$$\text{Skew}(X_1) = 0$$

$$E[X_1] = 0$$

$$E[X_1^2] = 1$$

$$E[X_1^3] = 0$$

Game 2



$$E[X_2] = 0$$

$$\text{Var}(X_2) = 1$$

$$\text{Skew}(X_2) = 0$$

Has values way
farther from 0

Kurtosis

Game 1



$$E[X_1] = 0$$

$$E[X_1] = 0$$

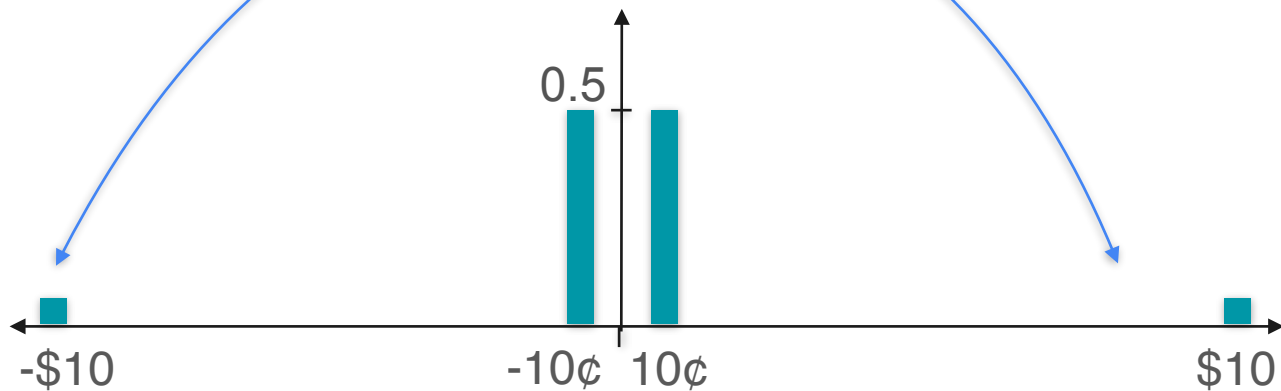
$$\text{Var}(X_1) = 1$$

$$E[X_1^2] = 1$$

$$\text{Skew}(X_1) = 0$$

$$E[X_1^3] = 0$$

Game 2



$$E[X_2] = 0$$

$$E[X_2] = 0$$

$$\text{Var}(X_2) = 1$$

$$\text{Skew}(X_2) = 0$$

Kurtosis

Game 1



$$E[X_1] = 0$$

$$E[X_1] = 0$$

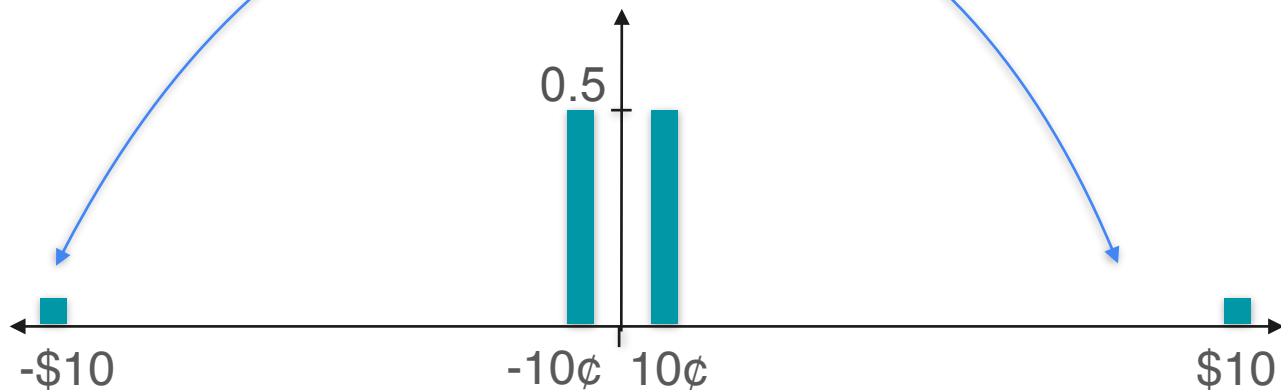
$$\text{Var}(X_1) = 1$$

$$E[X_1^2] = 1$$

$$\text{Skew}(X_1) = 0$$

$$E[X_1^3] = 0$$

Game 2



$$E[X_2] = 0$$

$$E[X_2] = 0$$

$$\text{Var}(X_2) = 1$$

$$E[X_2^2] = 1$$

$$\text{Skew}(X_2) = 0$$

Kurtosis

Game 1



$$E[X_1] = 0$$

$$E[X_1] = 0$$

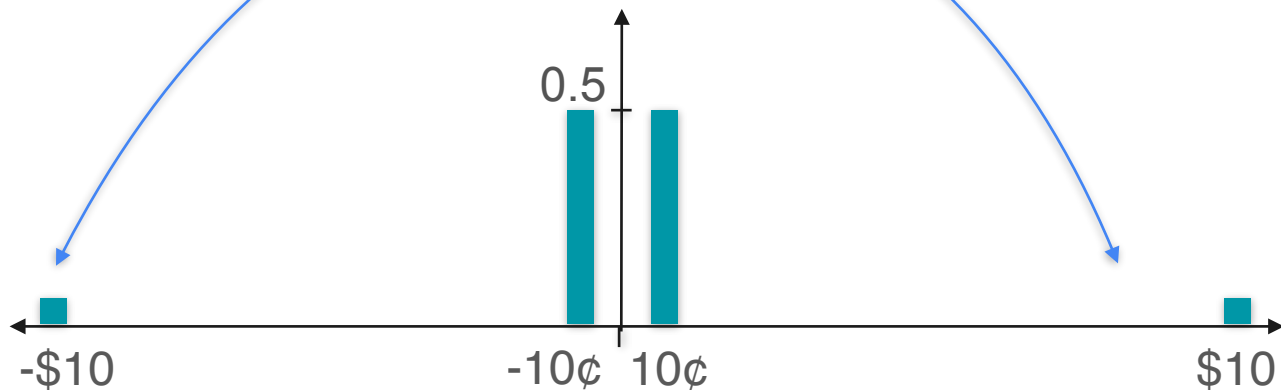
$$\text{Var}(X_1) = 1$$

$$E[X_1^2] = 1$$

$$\text{Skew}(X_1) = 0$$

$$E[X_1^3] = 0$$

Game 2



$$E[X_2] = 0$$

$$E[X_2] = 0$$

$$\text{Var}(X_2) = 1$$

$$E[X_2^2] = 1$$

$$\text{Skew}(X_2) = 0$$

$$E[X_2^3] = 0$$

Has values way
farther from 0

Kurtosis

Game 1



$$E[X_1] = 0$$

$$E[X_1] = 0$$

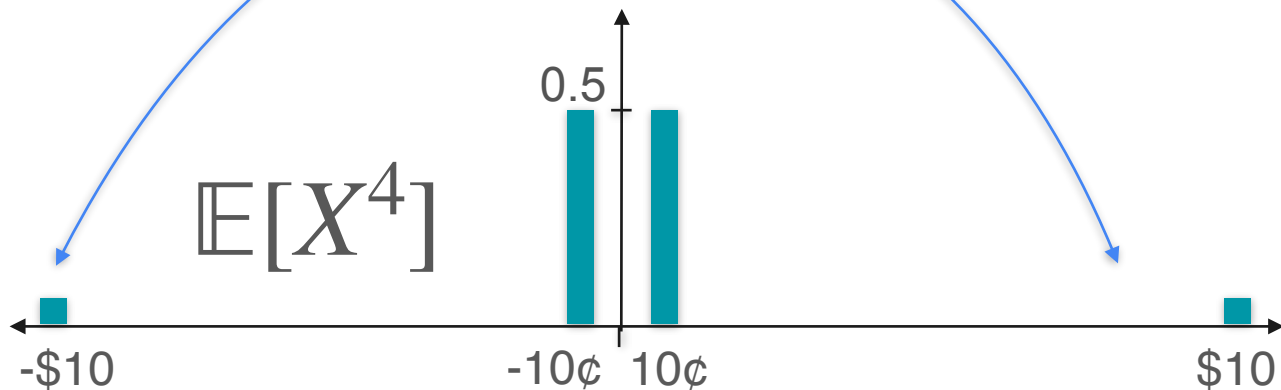
$$\text{Var}(X_1) = 1$$

$$E[X_1^2] = 1$$

$$\text{Skew}(X_1) = 0$$

$$E[X_1^3] = 0$$

Game 2



$$E[X_2] = 0$$

$$E[X_2] = 0$$

$$\text{Var}(X_2) = 1$$

$$E[X_2^2] = 1$$

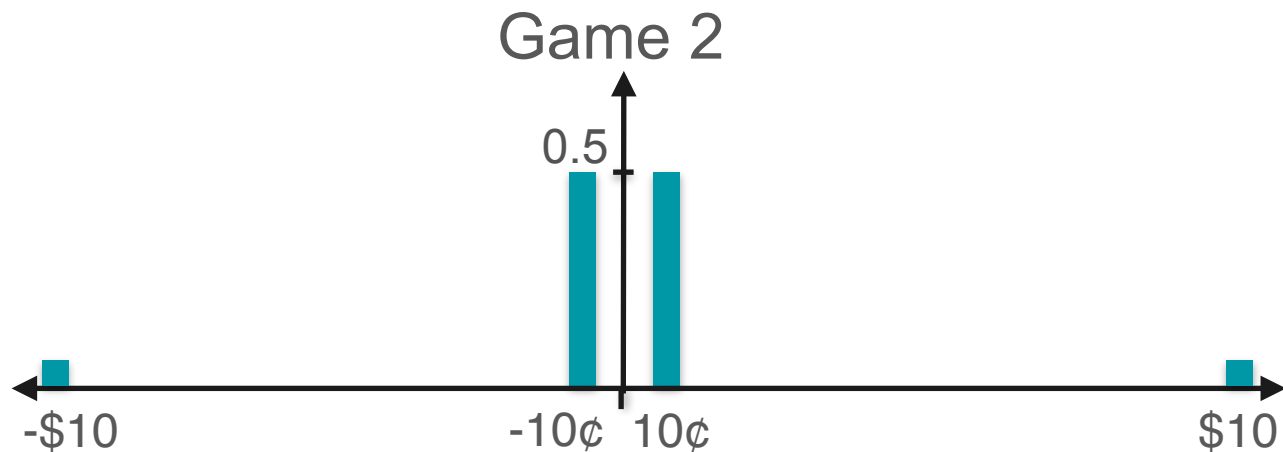
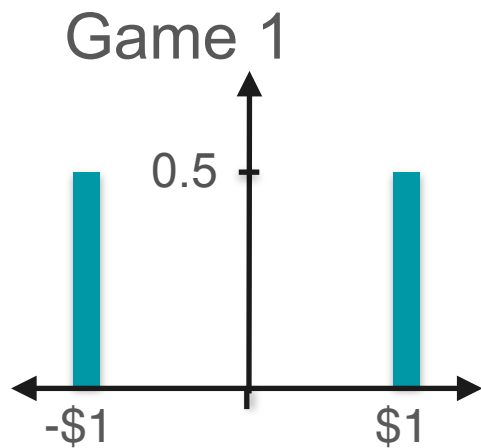
$$\text{Skew}(X_2) = 0$$

$$E[X_2^3] = 0$$

Has values way
farther from 0

$$E[X^4]$$

Kurtosis



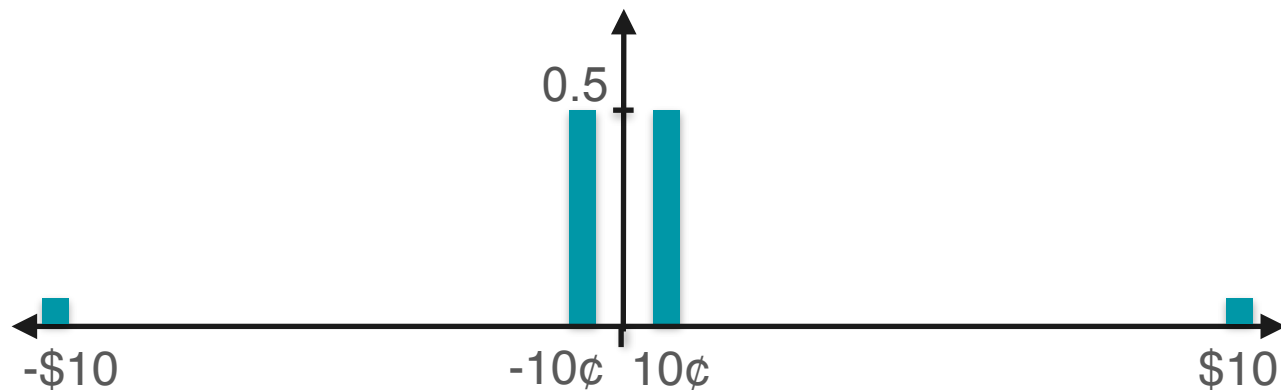
Kurtosis

Game 1



$\mathbb{E}[X_1^4]$

Game 2



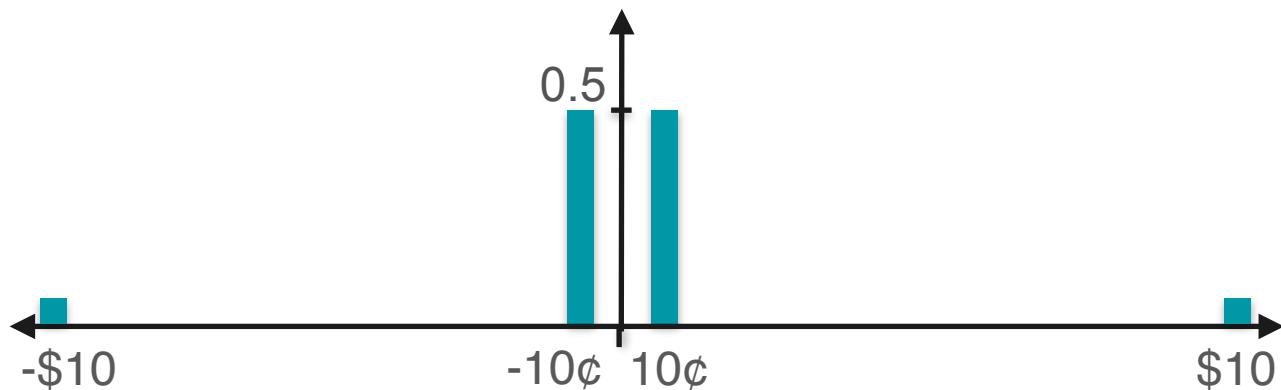
Kurtosis

Game 1



$$\mathbb{E}[X_1^4] = \frac{1}{2}(-1)^4 + \frac{1}{2}(1)^4$$

Game 2



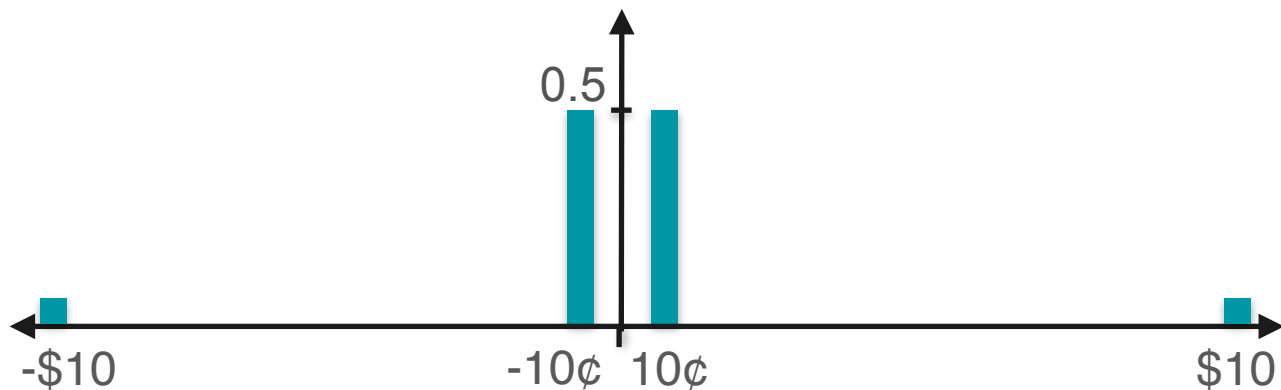
Kurtosis

Game 1



$$\begin{aligned}\mathbb{E}[X_1^4] &= \frac{1}{2}(-1)^4 + \frac{1}{2}(1)^4 \\ &= 1\end{aligned}$$

Game 2



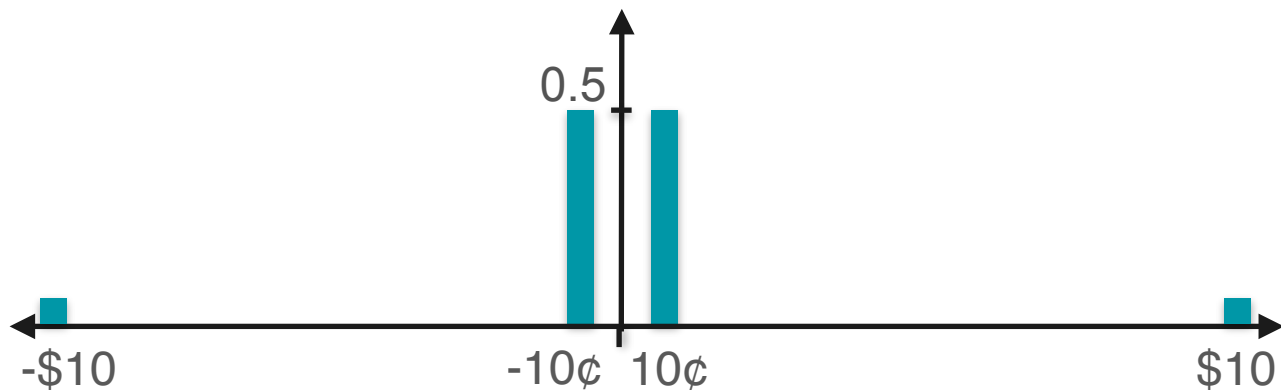
Kurtosis

Game 1



$$\begin{aligned}\mathbb{E}[X_1^4] &= \frac{1}{2}(-1)^4 + \frac{1}{2}(1)^4 \\ &= 1\end{aligned}$$

Game 2



$$\mathbb{E}[X_2^4]$$

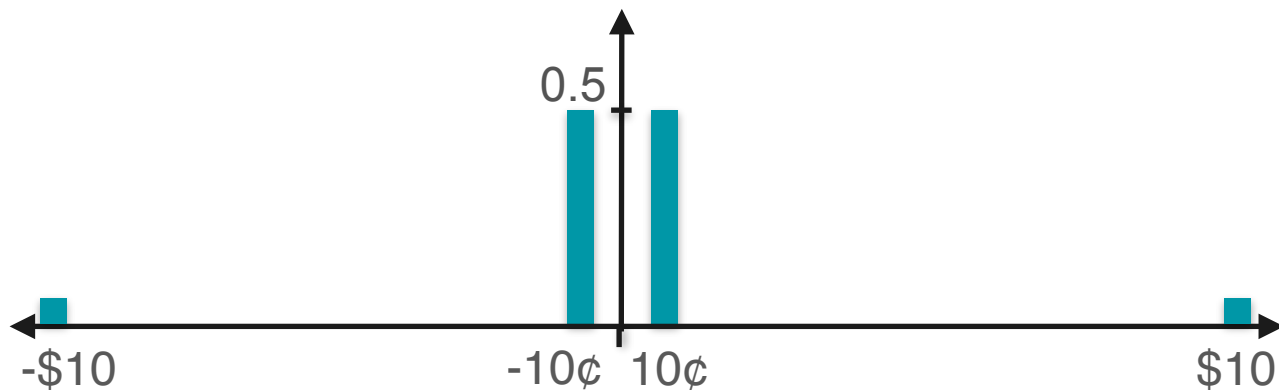
Kurtosis

Game 1



$$\begin{aligned}\mathbb{E}[X_1^4] &= \frac{1}{2}(-1)^4 + \frac{1}{2}(1)^4 \\ &= 1\end{aligned}$$

Game 2



$$\mathbb{E}[X_2^4] = \frac{100}{202}(-0.1)^4 + \frac{100}{202}(0.1)^4 + \frac{1}{202}(-10)^4 + \frac{1}{202}(10)^4$$

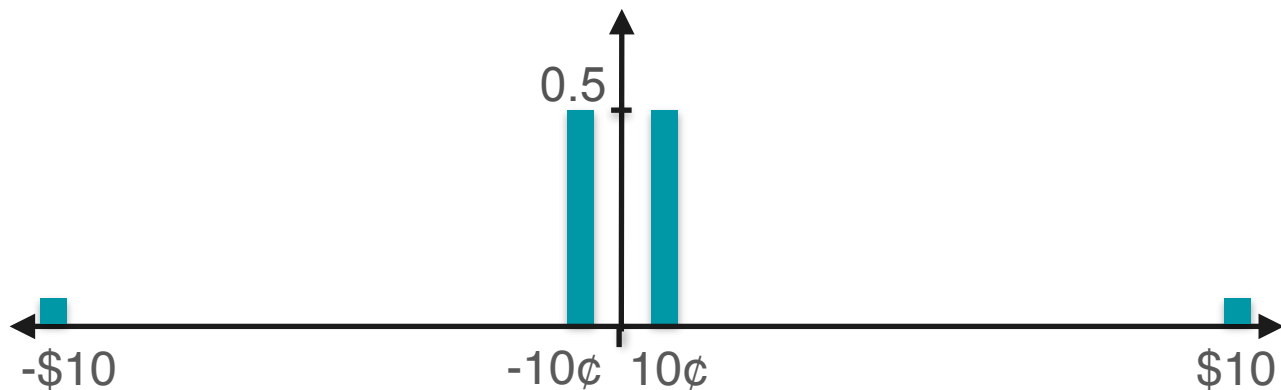
Kurtosis

Game 1



$$\begin{aligned}\mathbb{E}[X_1^4] &= \frac{1}{2}(-1)^4 + \frac{1}{2}(1)^4 \\ &= 1\end{aligned}$$

Game 2



$$\begin{aligned}\mathbb{E}[X_2^4] &= \frac{100}{202}(-0.1)^4 + \frac{100}{202}(0.1)^4 + \frac{1}{202}(-10)^4 + \frac{1}{202}(10)^4 \\ &= 99.01\end{aligned}$$

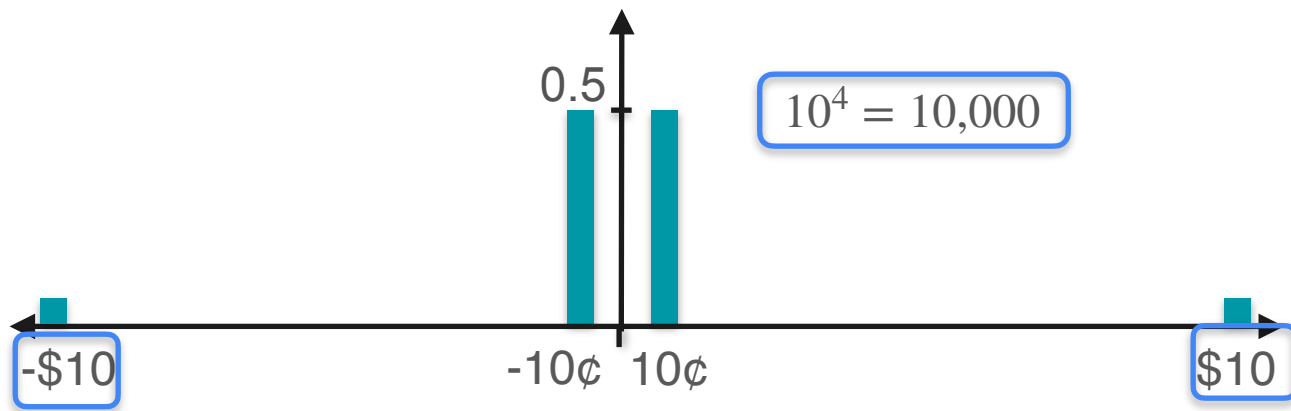
Kurtosis

Game 1



$$\begin{aligned}\mathbb{E}[X_1^4] &= \frac{1}{2}(-1)^4 + \frac{1}{2}(1)^4 \\ &= 1\end{aligned}$$

Game 2



$$\begin{aligned}\mathbb{E}[X_2^4] &= \frac{100}{202}(-0.1)^4 + \frac{100}{202}(0.1)^4 + \frac{1}{202}(-10)^4 + \frac{1}{202}(10)^4 \\ &= 99.01\end{aligned}$$

Kurtosis

$$\mathbb{E}[X^4]$$

Kurtosis

$$\mathbb{E}[X^4]$$

Almost...

Kurtosis

$$\mathbb{E}[X^4]$$

Almost...

Need to standardize...

Kurtosis

Kurtosis

$$\text{Kurtosis} = E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right]$$

Kurtosis: High and Low

Kurtosis: High and Low



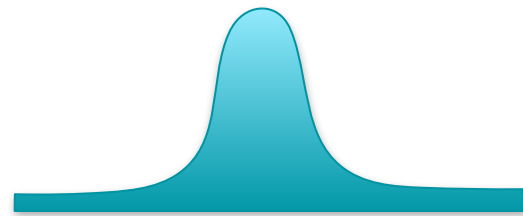
$$E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \text{small}$$

$$E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \text{large}$$

Kurtosis: High and Low



$$E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \text{small}$$

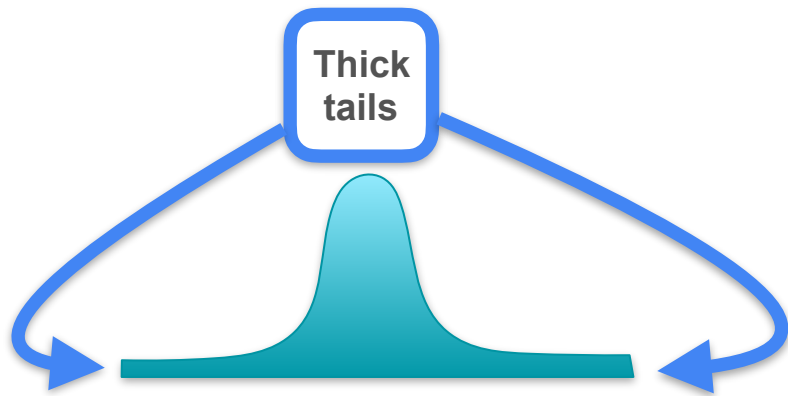


$$E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \text{large}$$

Kurtosis: High and Low

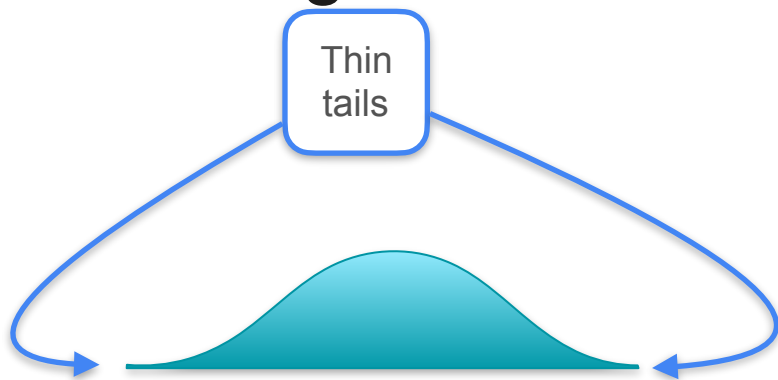


$$E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \text{small}$$

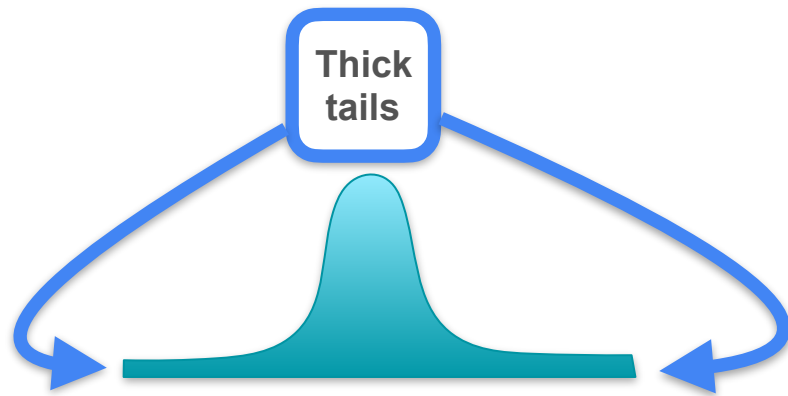


$$E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \text{large}$$

Kurtosis: High and Low

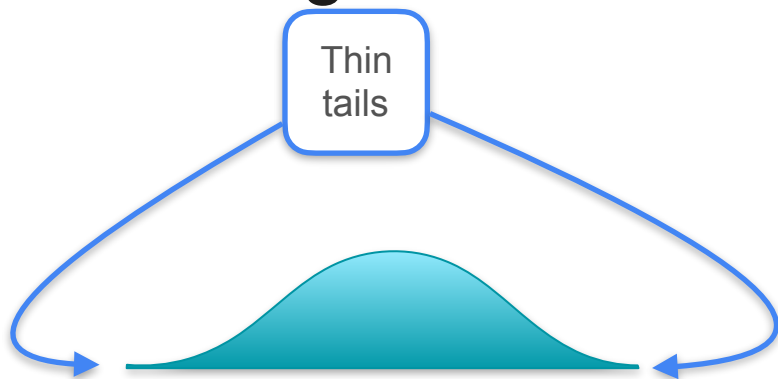


$$E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \text{small}$$

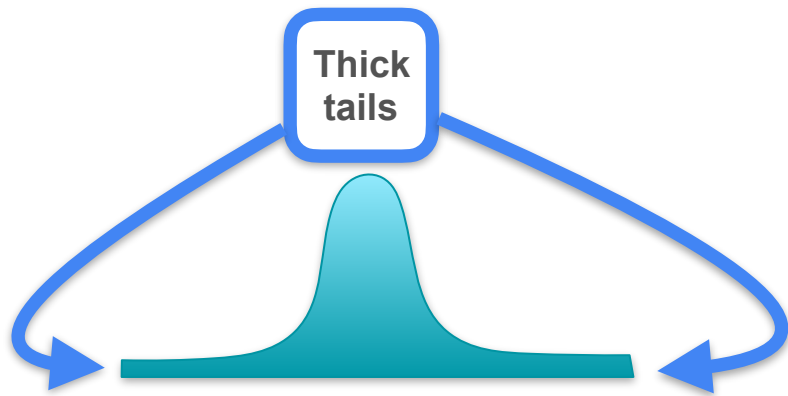


$$E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \text{large}$$

Kurtosis: High and Low



$$E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \text{small}$$



$$E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \text{large}$$

Even if they have the same variance!



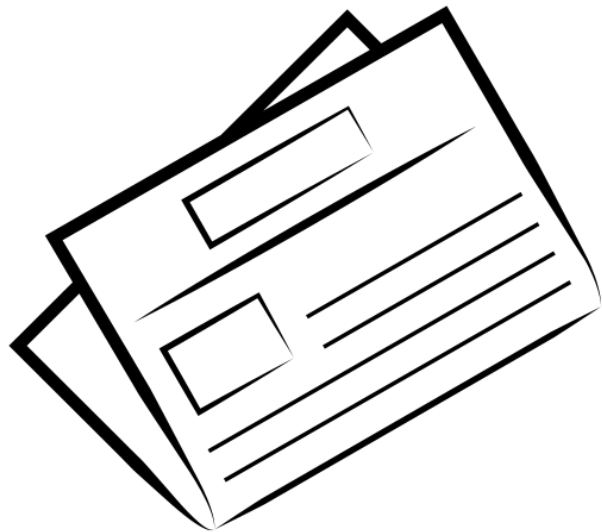
DeepLearning.AI

Describing Distributions

Quantiles and box-plots

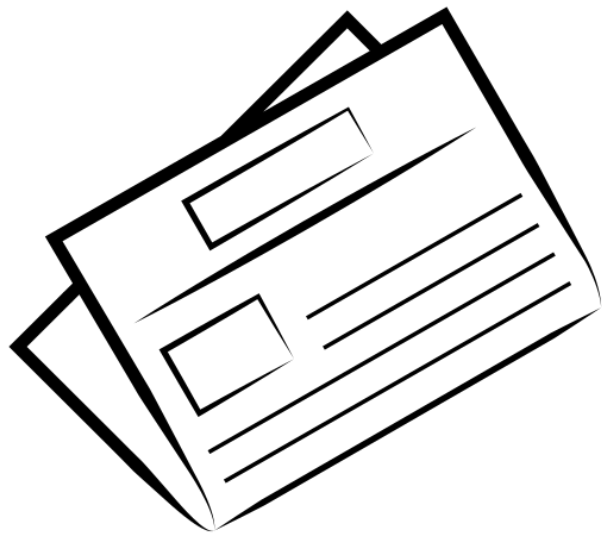
Quantiles: Example

Quantiles: Example



Newspaper advertisement

Quantiles: Example



Newspaper advertisement

Newspaper Advertisement (X)			
18.3	18.4	23.2	51.2
35.2	29.7	75	8.7
65.9	14.2	54.7	25.9

Quantiles: Example

Newspaper Advertisement (X)			
18.3	18.4	23.2	51.2
35.2	29.7	75	8.7
65.9	14.2	54.7	25.9

Quantiles: Example

What is the median here?

Newspaper Advertisement (X)			
18.3	18.4	23.2	51.2
35.2	29.7	75	8.7
65.9	14.2	54.7	25.9

Quantiles: Example

What is the median here?

The point that splits your data in half

Newspaper Advertisement (X)			
18.3	18.4	23.2	51.2
35.2	29.7	75	8.7
65.9	14.2	54.7	25.9

Quantiles: Example

What is the median here?

The point that splits your data in half

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Quantiles: Example

What is the median here?

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Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
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Quantiles: Example

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Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Quantiles: Example

What is the median here?

The point that splits your data in half

$$\text{Median} = \frac{25.9 + 29.7}{2} = 27.8$$

50% quantile

Second quartile

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Quantiles: Example

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Quantiles: Example

What about the point that leaves $1/4$ of your data to the left and $3/4$ to the right?

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Quantiles: Example

What about the point that leaves 1/4 of your data to the left and 3/4 to the right?

$$\frac{18.3 + 18.4}{2} = 18.35$$

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Quantiles: Example

What about the point that leaves 1/4 of your data to the left and 3/4 to the right?

$$\frac{18.3 + 18.4}{2} = 18.35$$

25% quantile

First quartile

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Quantiles: Example

What about the point that leaves 1/4 of your data to the left and 3/4 to the right?

$$q_{0.25} = Q1 = \frac{18.3 + 18.4}{2} = 18.35$$

25% quantile

First quartile

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Quantiles

Quantiles

In general:

The **k% quantile** ($q_{k/100}$) is the value that leaves k% of your data to the left and (100-k)% of your data to the right

Quantiles

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The **k% quantile** ($q_{k/100}$) is the value that leaves k% of your data to the left and (100-k)% of your data to the right

$k \%$

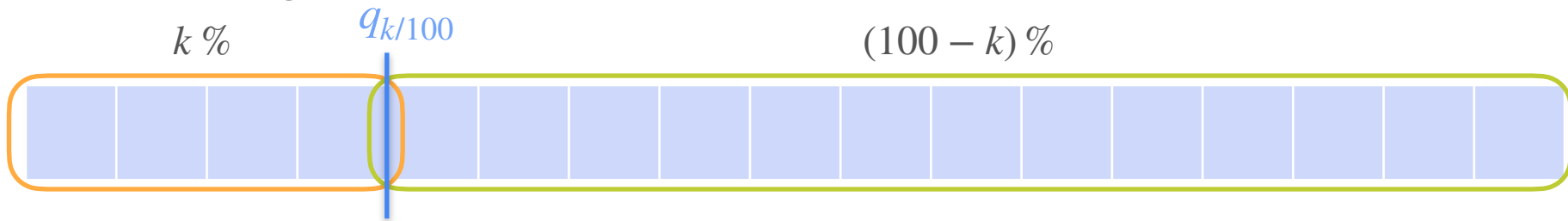
$(100 - k) \%$



Quantiles

In general:

The **k% quantile** ($q_{k/100}$) is the value that leaves k% of your data to the left and (100-k)% of your data to the right



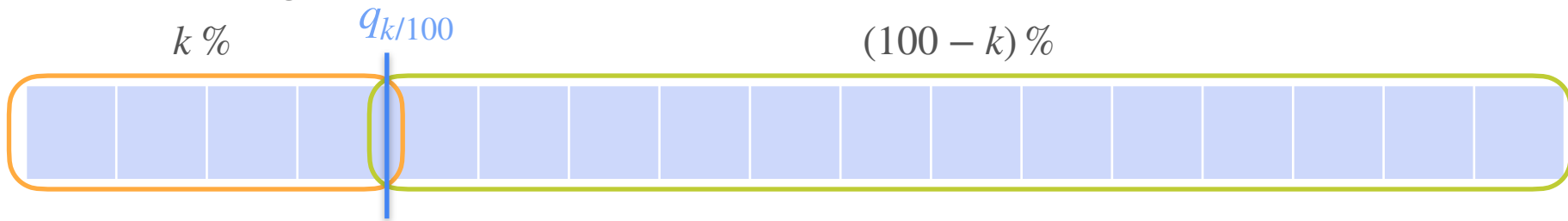
Quantiles

In general:

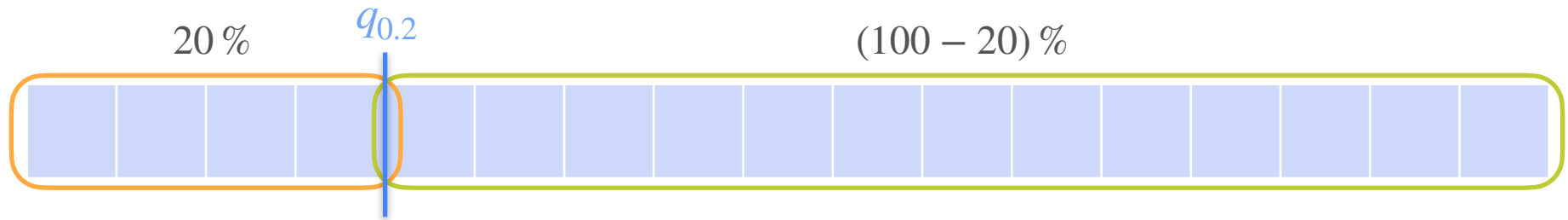
The **k% quantile** ($q_{k/100}$) is the value that leaves k% of your data to the left and (100-k)% of your data to the right

Some common quantiles:

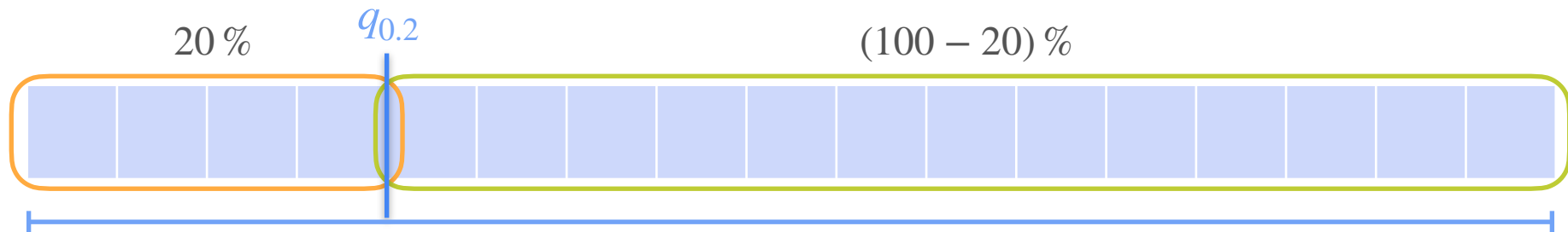
- 25% quantile (first quartile - Q1)
- 50% quantile (median - Q2)
- 75% quantile (third quartile - Q3)



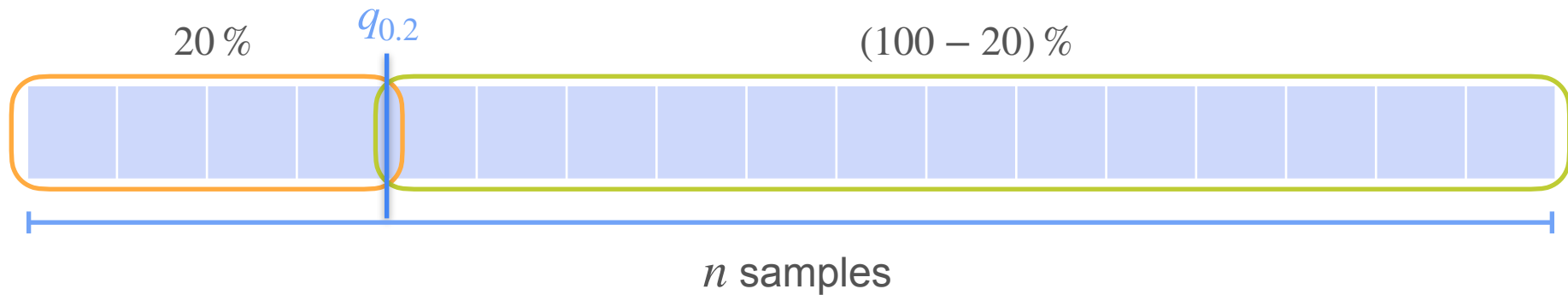
Quantiles



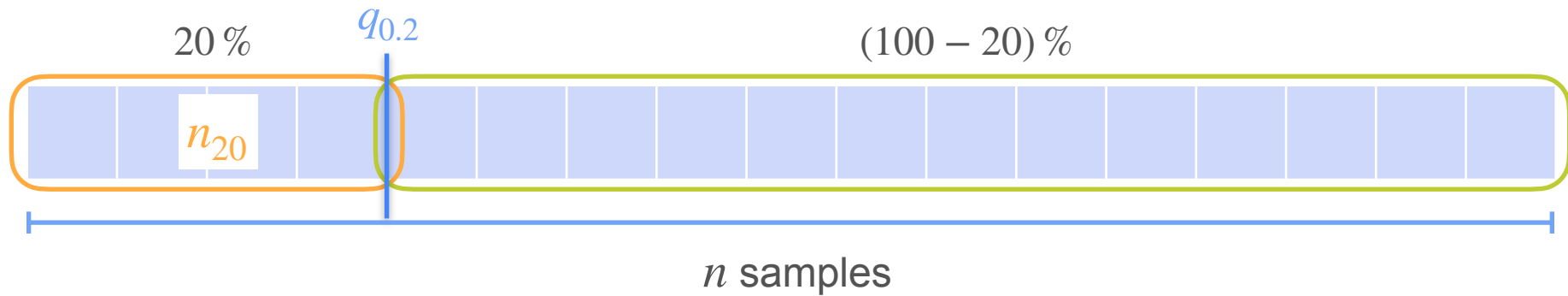
Quantiles



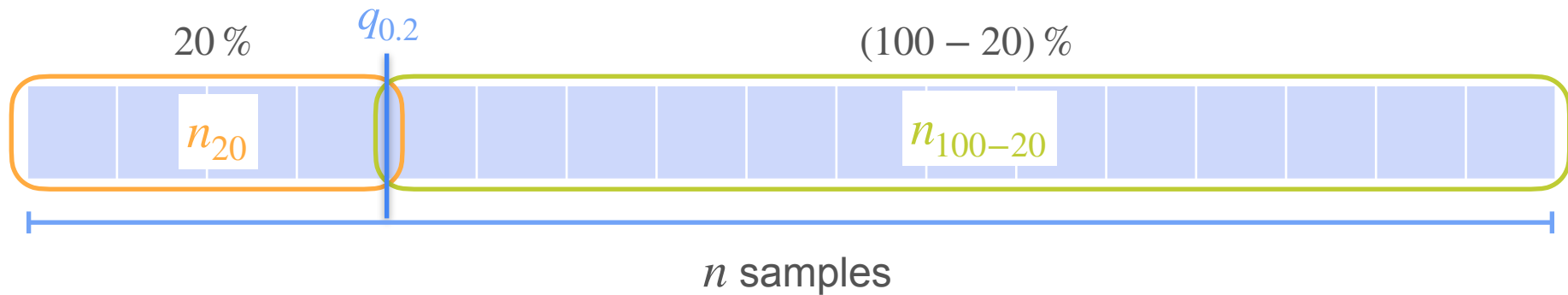
Quantiles



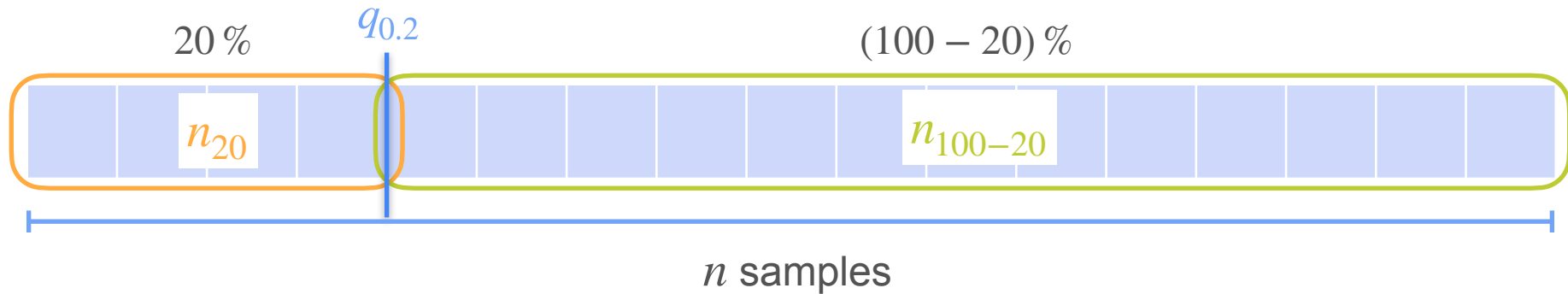
Quantiles



Quantiles

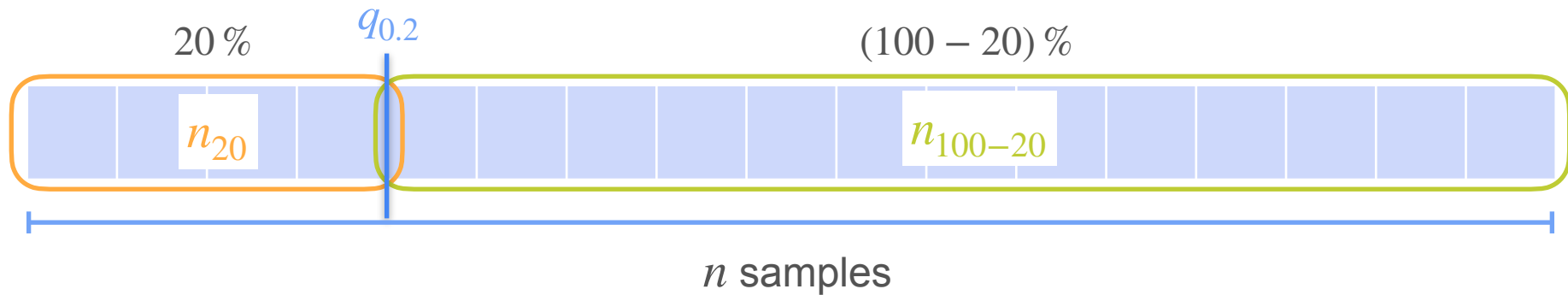


Quantiles



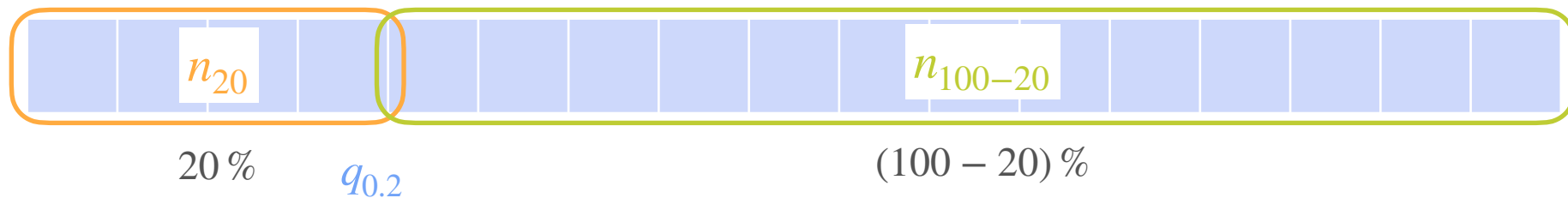
$$\frac{20}{100} = \frac{n_{20}}{n}$$

Quantiles

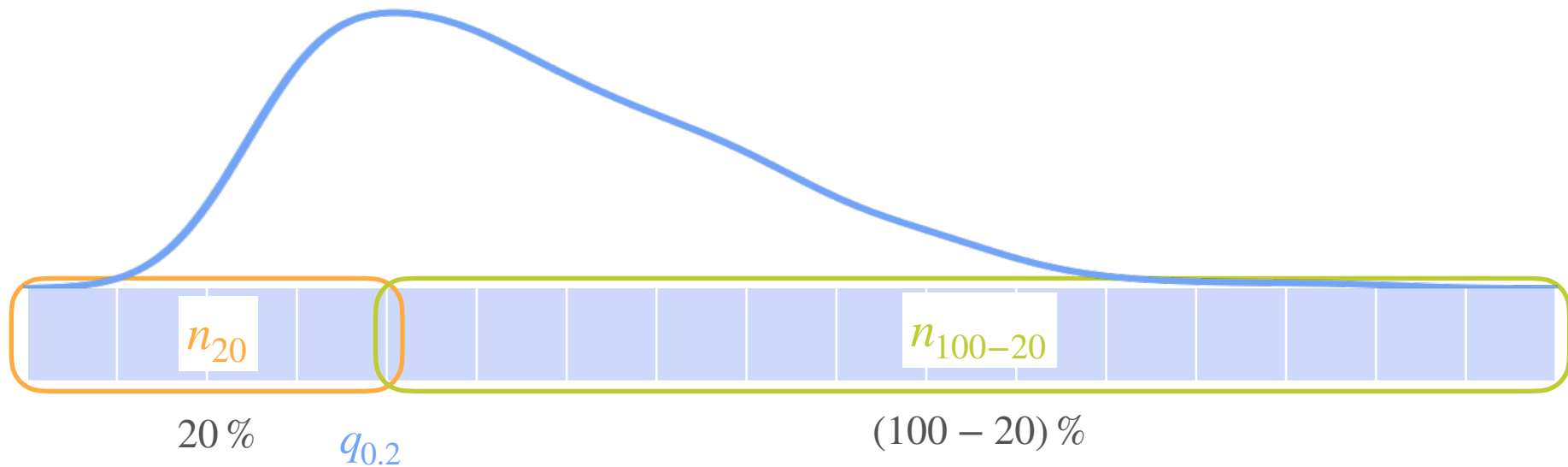


$$\frac{20}{100} = \frac{n_{20}}{n} \approx \mathbf{P}(X \leq q_{0.2})$$

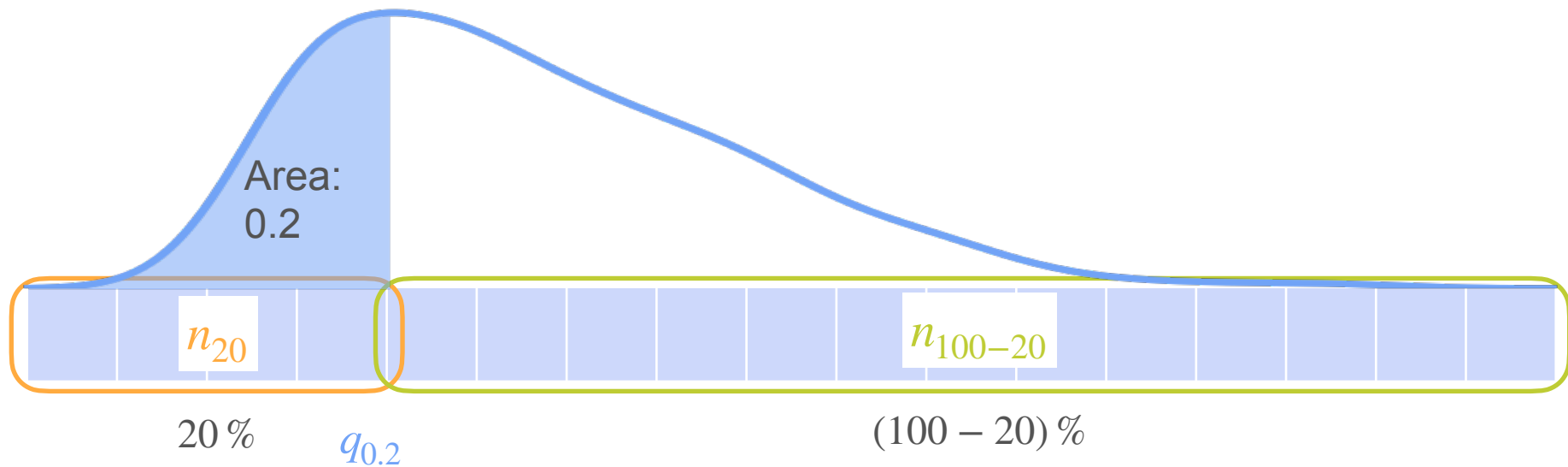
Quantiles



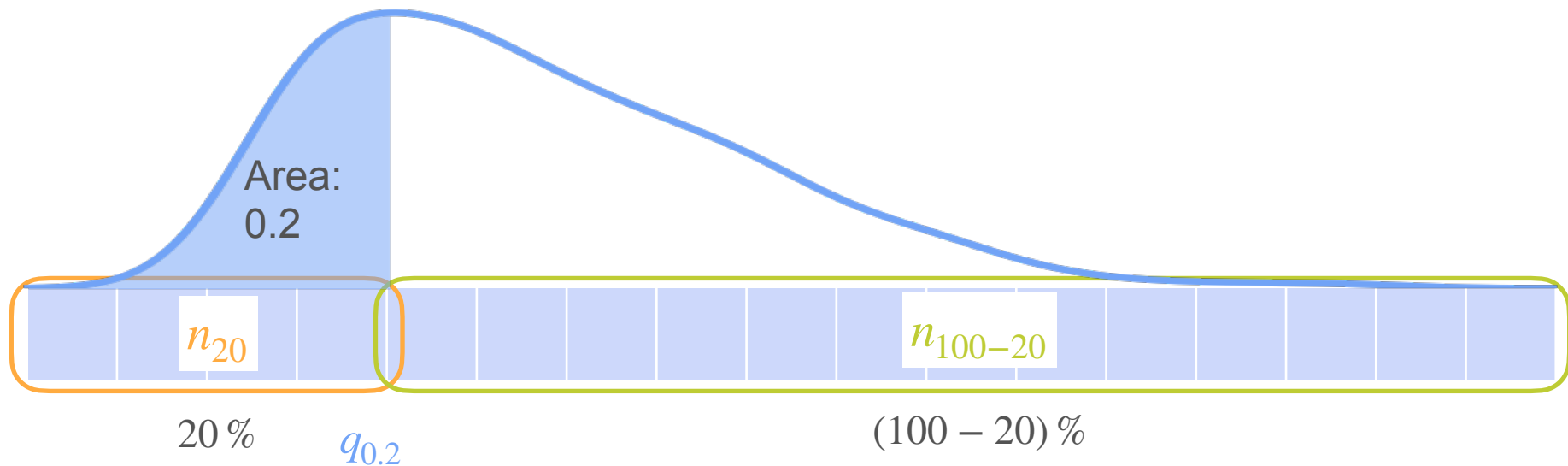
Quantiles



Quantiles



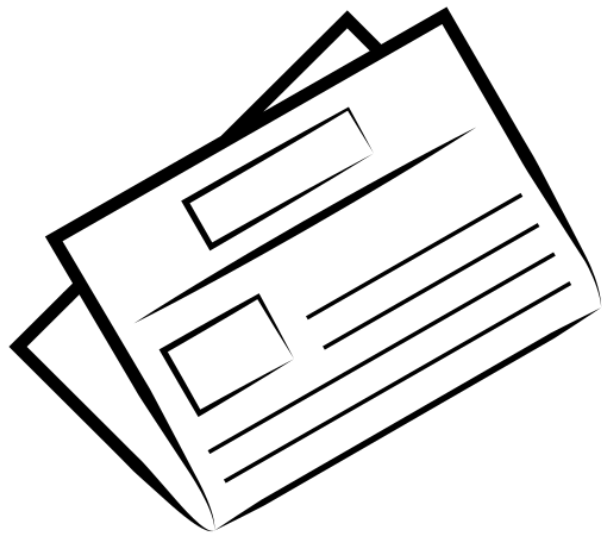
Quantiles



k% quantile ($q_{k/100}$) is the value such that $\mathbf{P} \left(X \leq q_{k/100} \right) = \frac{k}{100}$

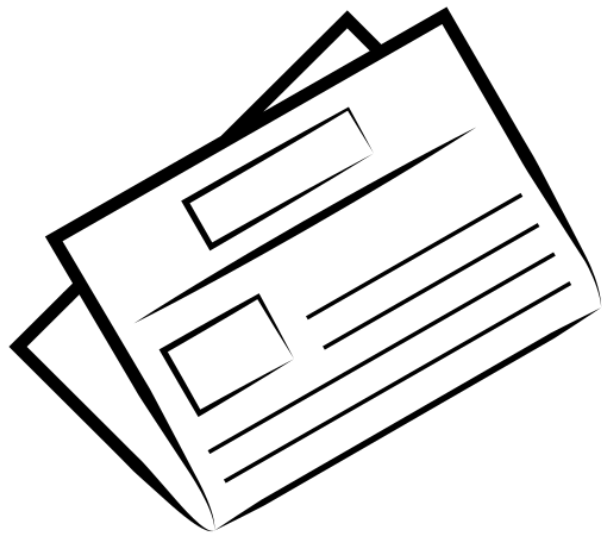
Box-Plots

Box-Plots



Newspaper advertisement

Box-Plots



Newspaper advertisement

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Box-Plots

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Box-Plots

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Box-Plots

$$Q1 = q_{0.25} = \frac{18.3 + 18.4}{2} = 18.35$$

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
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Box-Plots

$$Q1 = q_{0.25} = \frac{18.3 + 18.4}{2} = 18.35$$

$$Q2 = q_{0.50} = \frac{25.9 + 29.7}{2} = 27.8$$

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Box-Plots

$$Q1 = q_{0.25} = \frac{18.3 + 18.4}{2} = 18.35$$

$$Q2 = q_{0.50} = \frac{25.9 + 29.7}{2} = 27.8$$

$$Q3 = q_{0.75} = \frac{51.2 + 54.7}{2} = 52.95$$

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Box-Plots

$$Q1 = q_{0.25} = \frac{18.3 + 18.4}{2} = 18.35$$

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Box-Plots

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Interquartile range (IQR)

$$IQR = Q3 - Q1$$

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
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Box-Plots

$$Q1 = q_{0.25} = \frac{18.3 + 18.4}{2} = 18.35$$

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Interquartile range (IQR)

$$IQR = Q3 - Q1 = 52.95 - 18.35 = 34.6$$

Newspaper Advertisement (X)			
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Box-Plots

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Interquartile range (IQR)

$$IQR = Q3 - Q1 = 52.95 - 18.35 = 34.6$$

$$\min = 8.7$$

Newspaper Advertisement (X)			
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$$Q1 = q_{0.25} = \frac{18.3 + 18.4}{2} = 18.35$$

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Interquartile range (IQR)

$$IQR = Q3 - Q1 = 52.95 - 18.35 = 34.6$$

$$\min = 8.7 \quad \max = 75$$

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
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Box-Plots

$$Q1 = q_{0.25} = \frac{18.3 + 18.4}{2} = 18.35$$

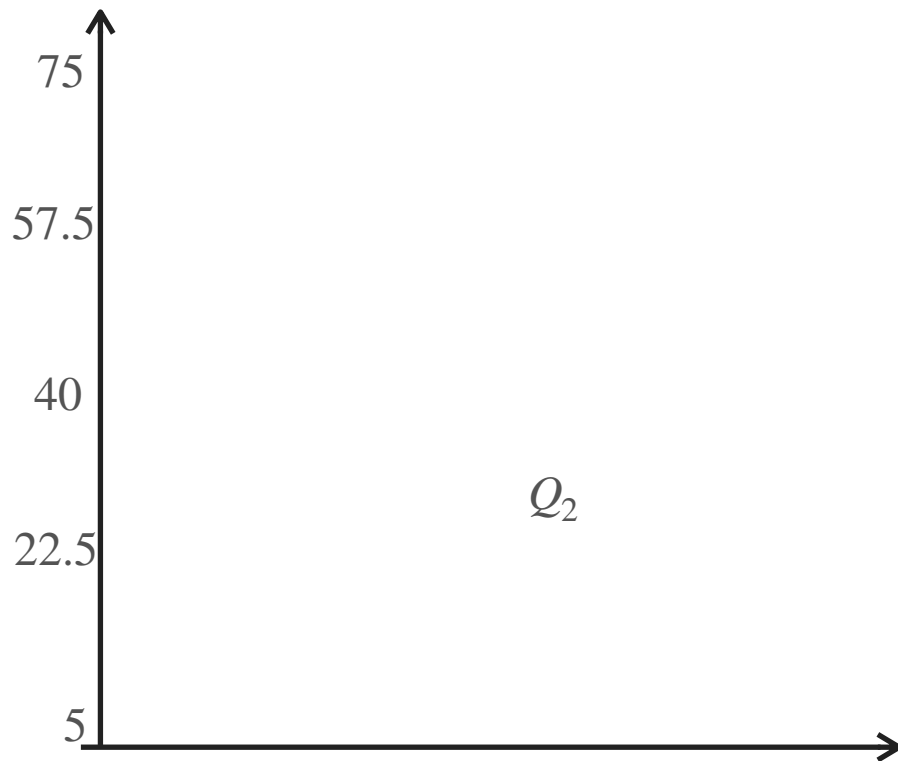
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Interquartile range (IQR)

$$IQR = Q3 - Q1 = 52.95 - 18.35 = 34.6$$

$$x_{\min} = 8.7 \quad x_{\max} = 75$$



Box-Plots

$$Q1 = q_{0.25} = \frac{18.3 + 18.4}{2} = 18.35$$

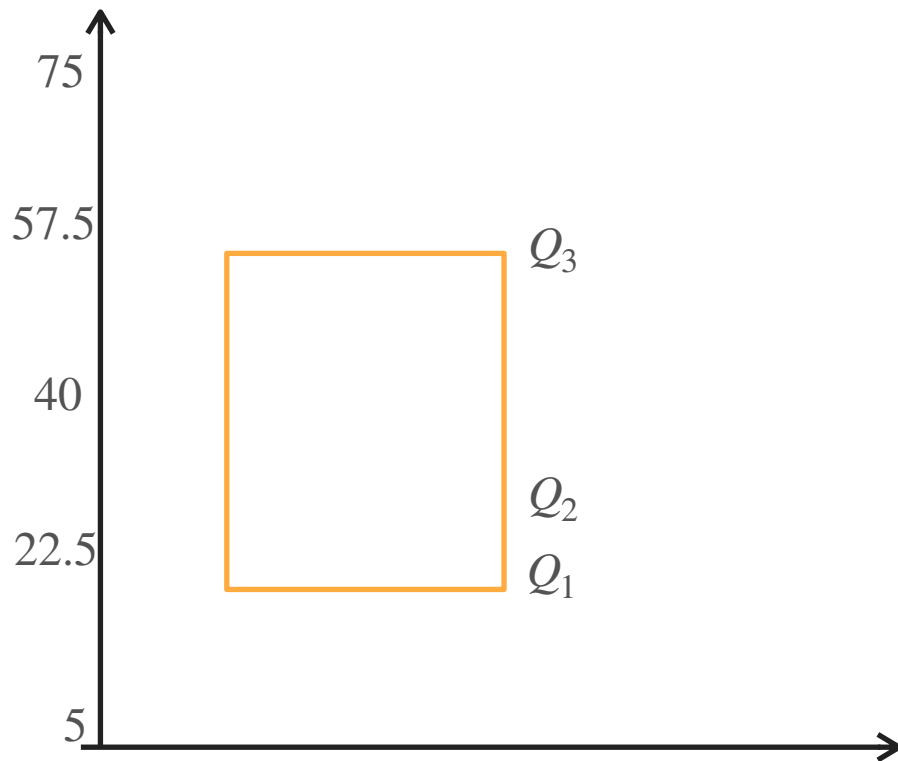
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Box-Plots

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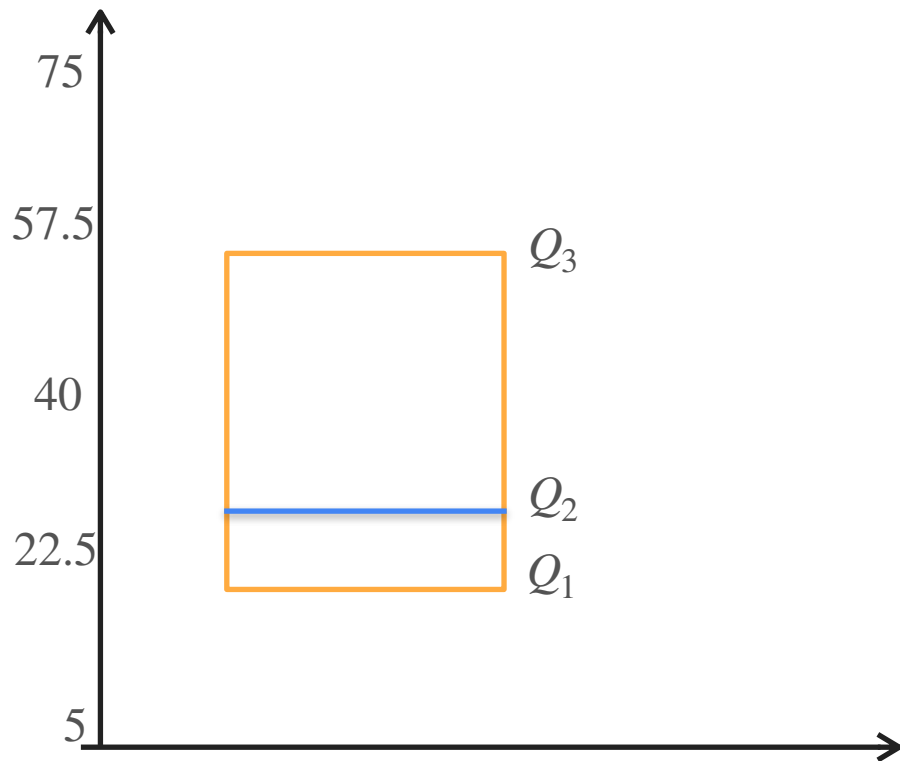
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Box-Plots

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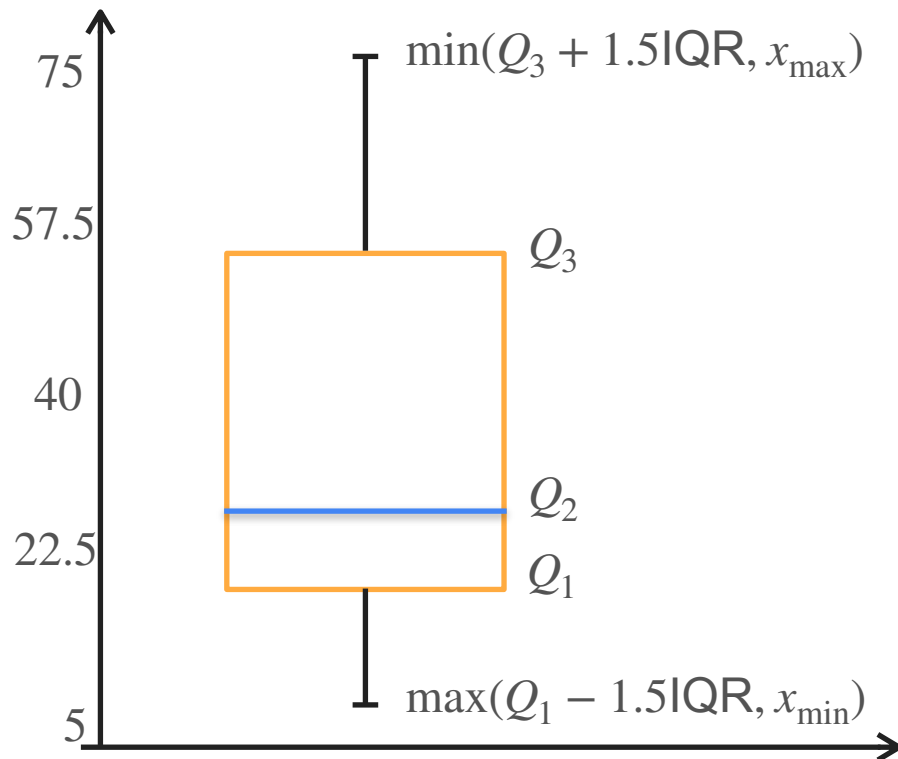
$$Q2 = q_{0.5} = \frac{25.9 + 29.7}{2} = 27.8$$

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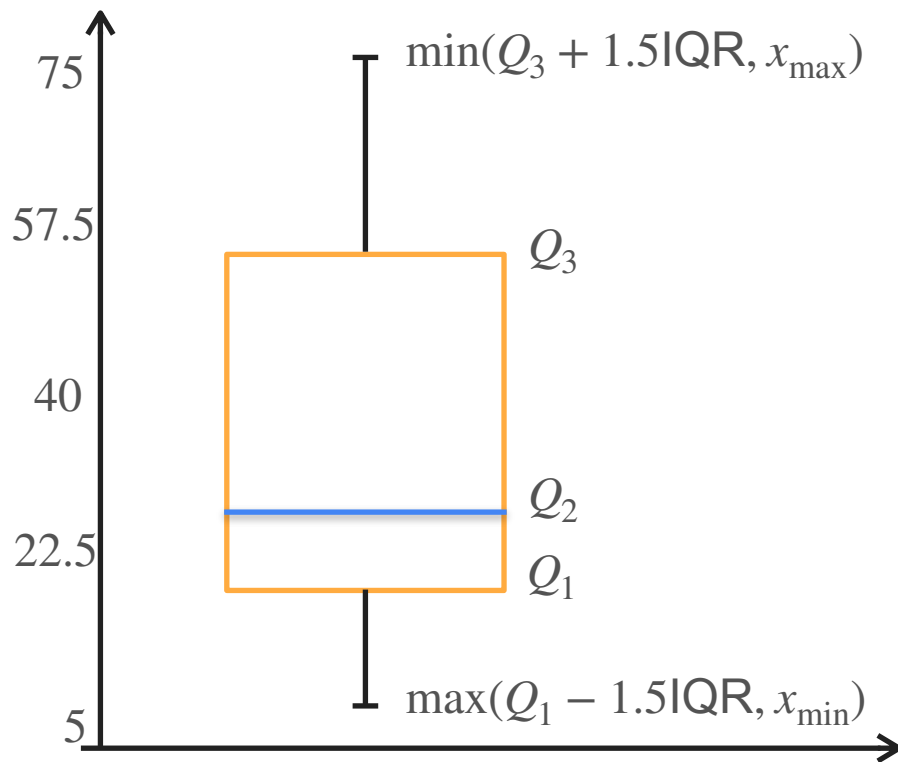
Interquartile range (IQR)

$$IQR = Q3 - Q1 = 52.95 - 18.35 = 34.6$$

$$x_{\min} = 8.7 \quad x_{\max} = 75$$



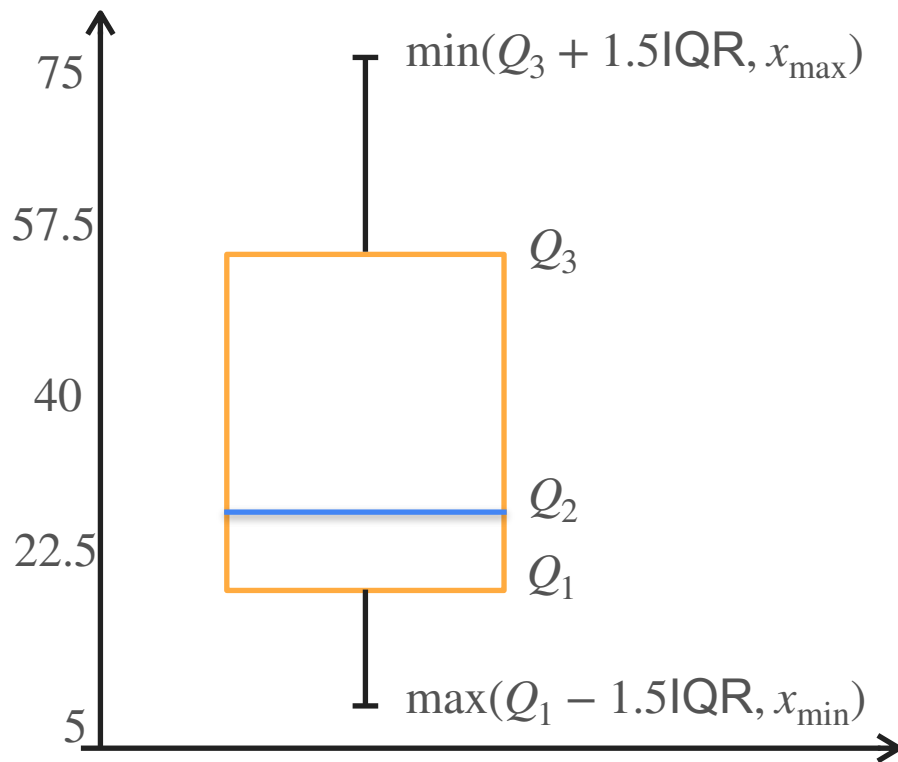
Box-Plots



Box-Plots

What can you tell from this plot?

- Data is skewed
- No outliers (whiskers were cut at max and min value)
- Analyze dispersion



Box-Plots

Box-Plots

Let's see how the box-plot
looks for the whole dataset

Box-Plots

Let's see how the box-plot
looks for the whole dataset

$$Q_1 = 12.75 \quad Q_2 = 25.75 \quad Q_3 = 45.1$$

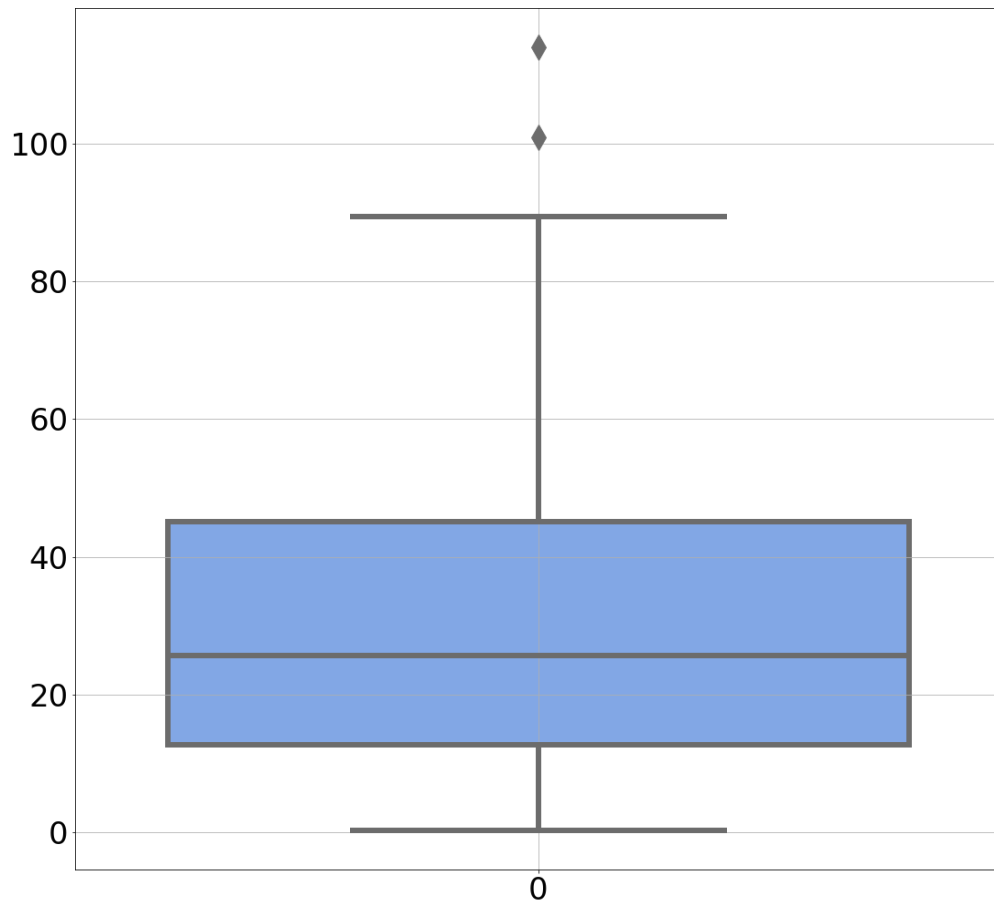
$$\text{IQR} = 32.35 \quad x_{\min} = 0.3 \quad x_{\max} = 114$$

Box-Plots

Let's see how the box-plot looks for the whole dataset

$$Q_1 = 12.75 \quad Q_2 = 25.75 \quad Q_3 = 45.1$$

$$\text{IQR} = 32.35 \quad x_{\min} = 0.3 \quad x_{\max} = 114$$

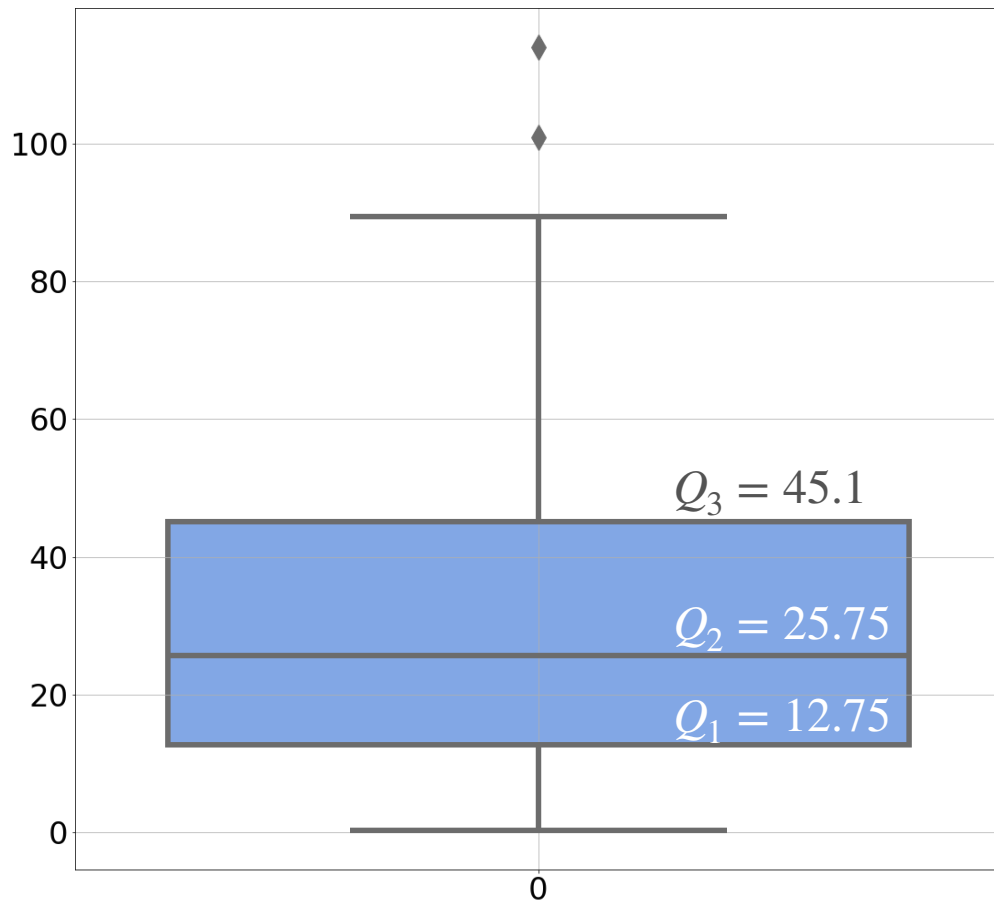


Box-Plots

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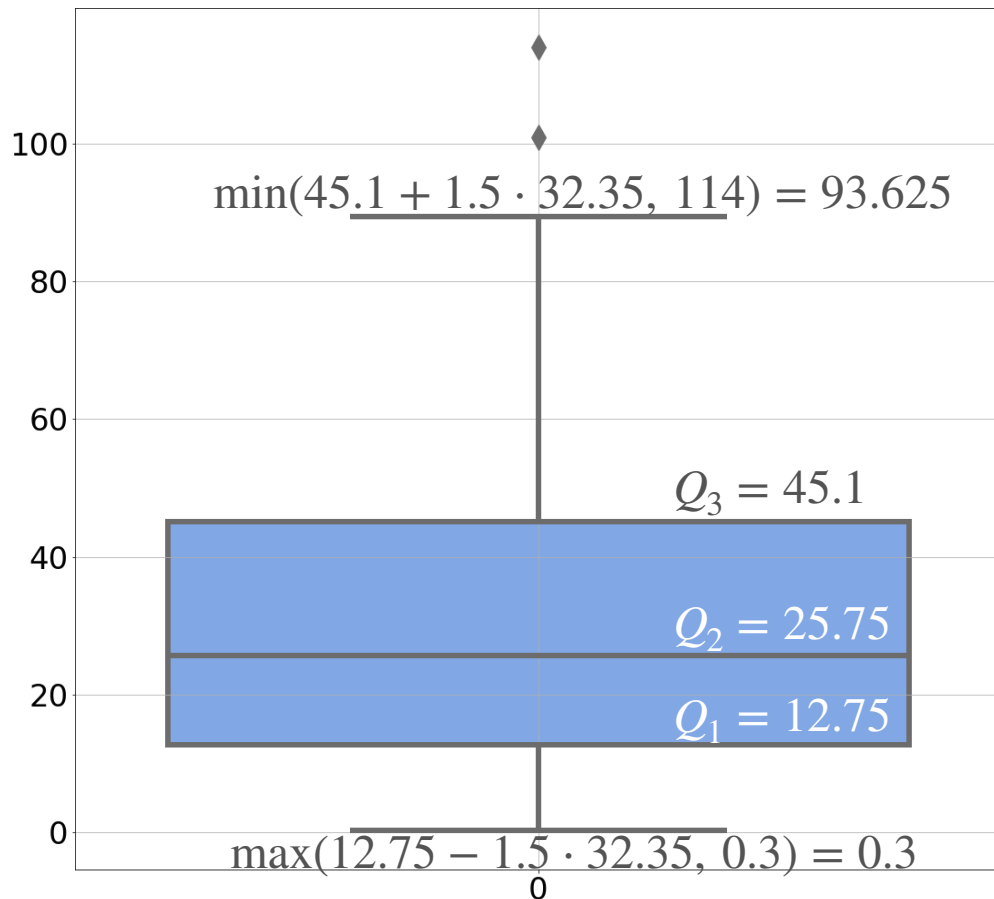


Box-Plots

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$$Q_1 = 12.75 \quad Q_2 = 25.75 \quad Q_3 = 45.1$$

$$\text{IQR} = 32.35 \quad x_{\min} = 0.3 \quad x_{\max} = 114$$



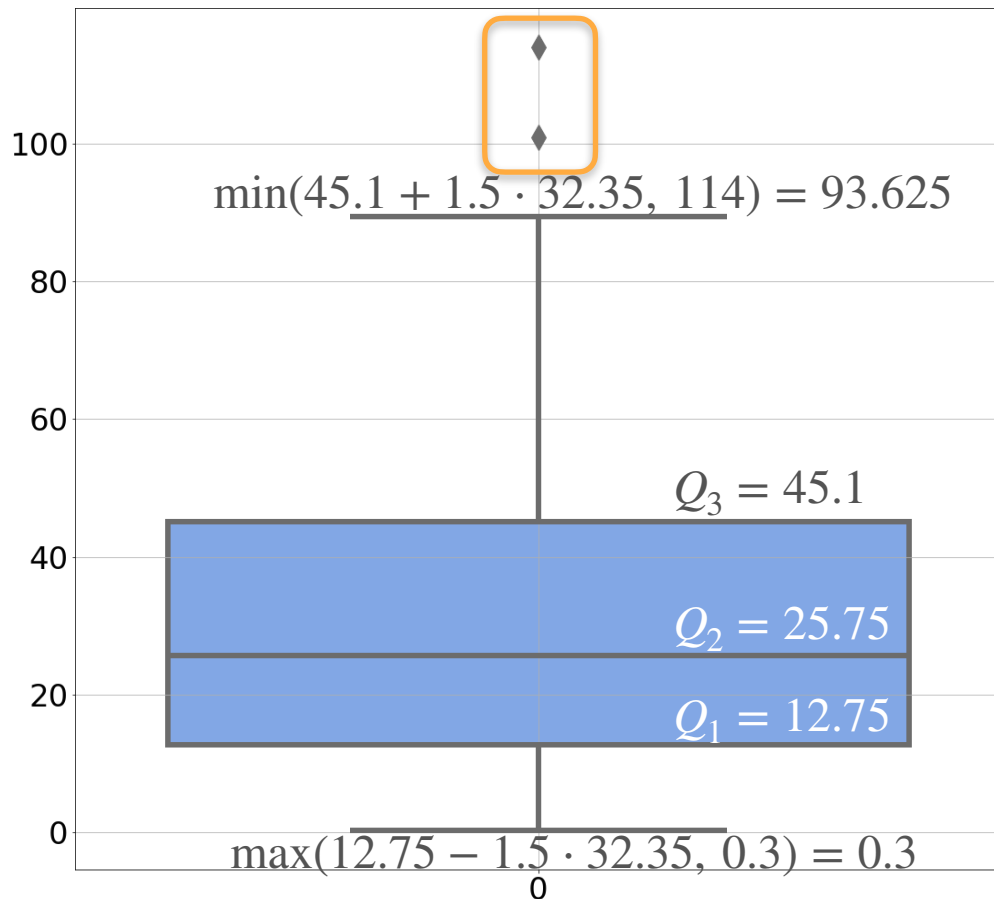
Box-Plots

Let's see how the box-plot looks for the whole dataset

$$Q_1 = 12.75 \quad Q_2 = 25.75 \quad Q_3 = 45.1$$

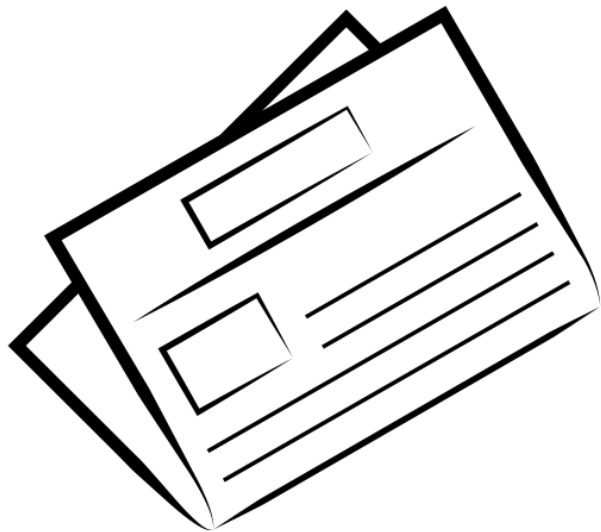
$$\text{IQR} = 32.35 \quad x_{\min} = 0.3 \quad x_{\max} = 114$$

Now you can see two outliers



Density Estimation

Density Estimation

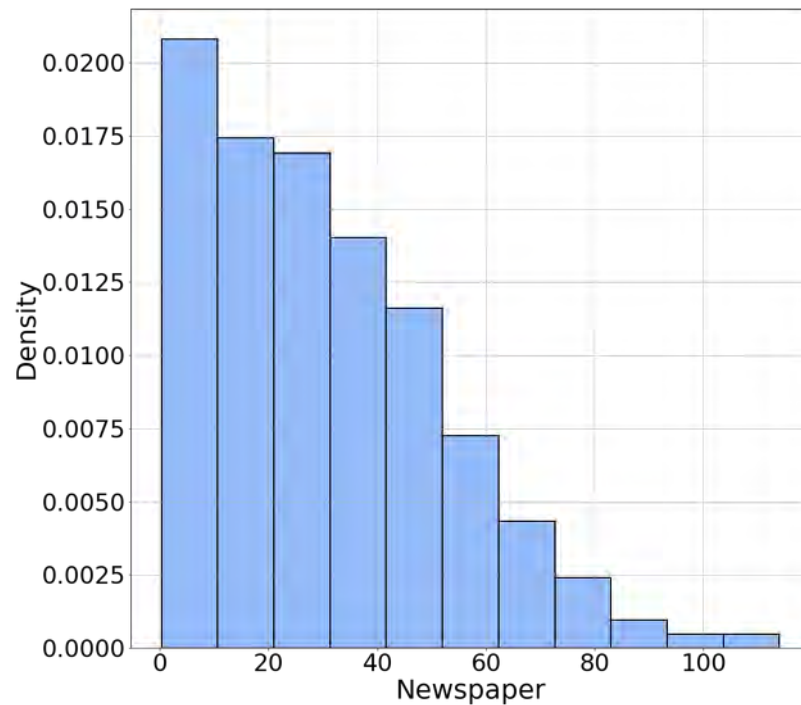


Newspaper advertisement

Newspaper Advertisement (X)			
8.7	14.2	18.3	18.4
23.2	25.9	29.7	35.2
51.2	54.7	65.9	75

Histograms

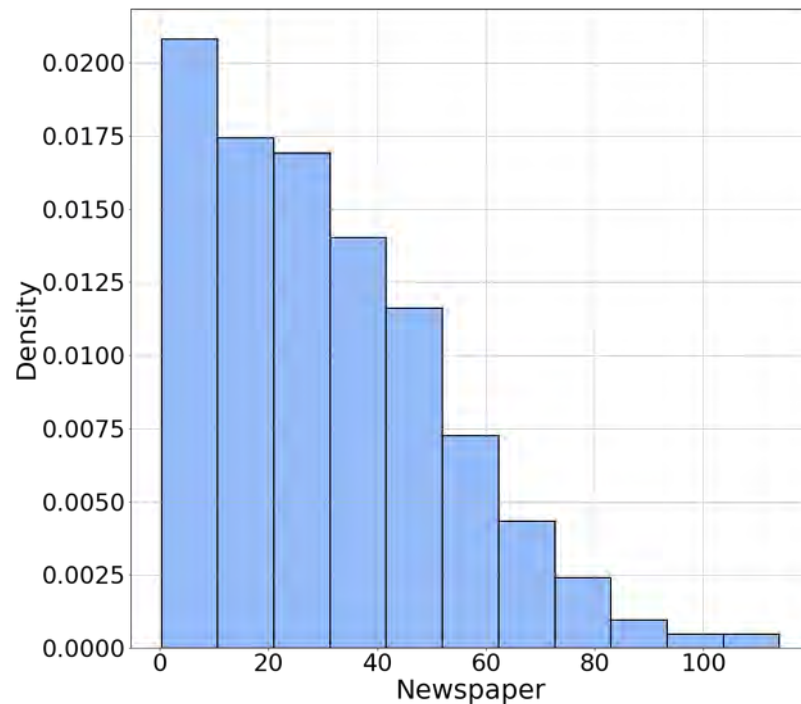
Histograms



Histograms

It represents a density function

- It is positive
- Area under the curve is 1



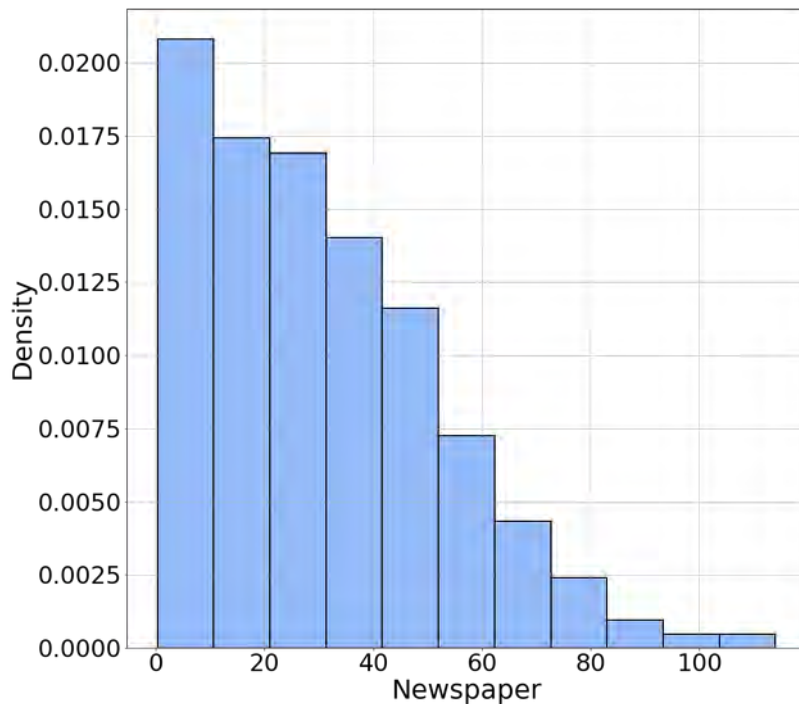
Histograms

It represents a density function

- It is positive
- Area under the curve is 1

But...

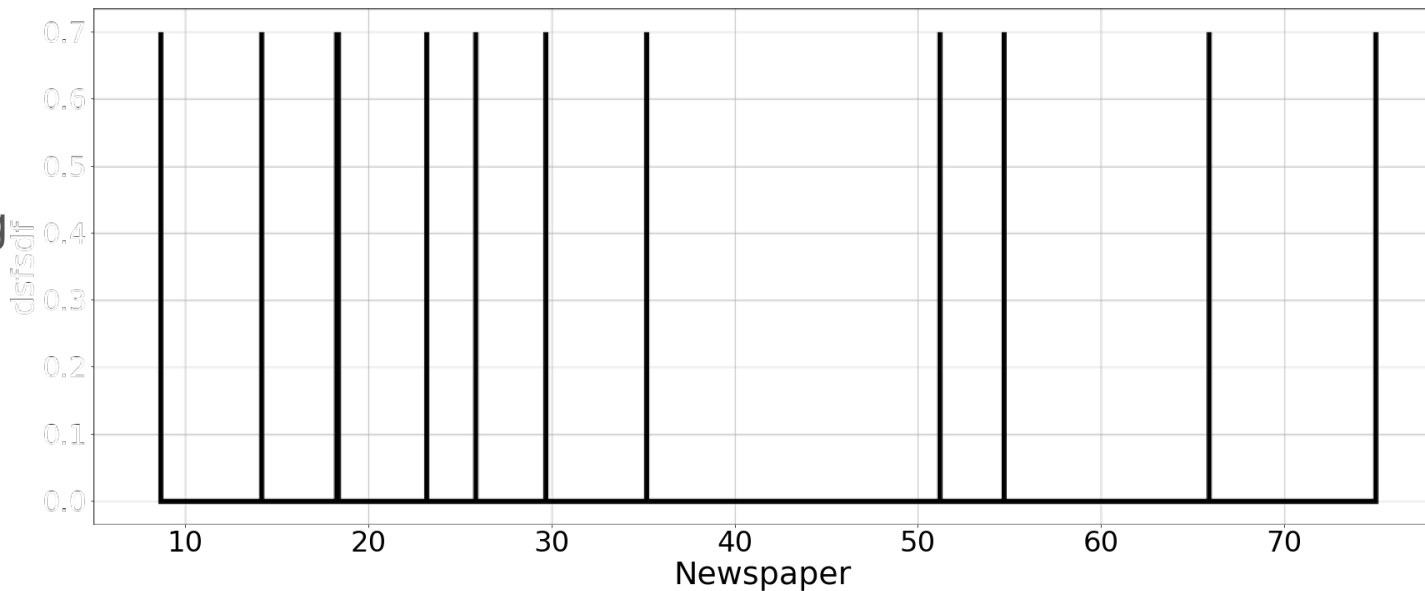
- PDFs are usually smooth function
- The discontinuities come from the method and not the data



Kernel Density Estimation

Kernel Density Estimation

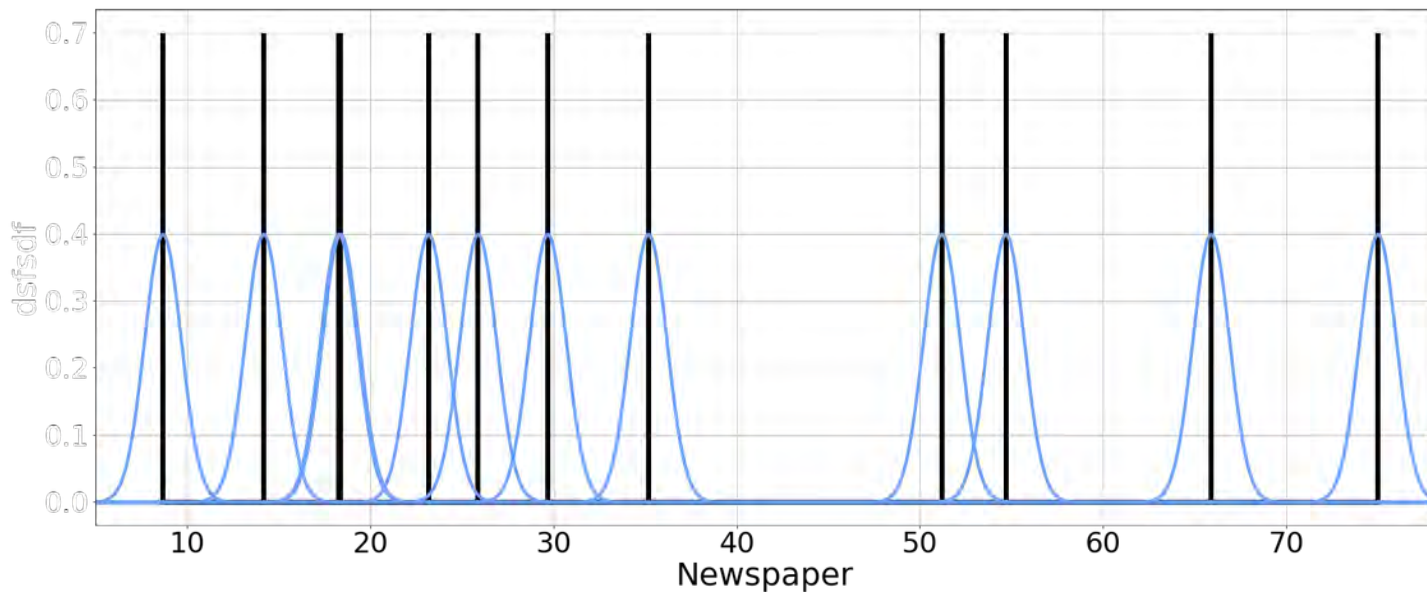
First: draw your observations along the x axis



Kernel Density Estimation

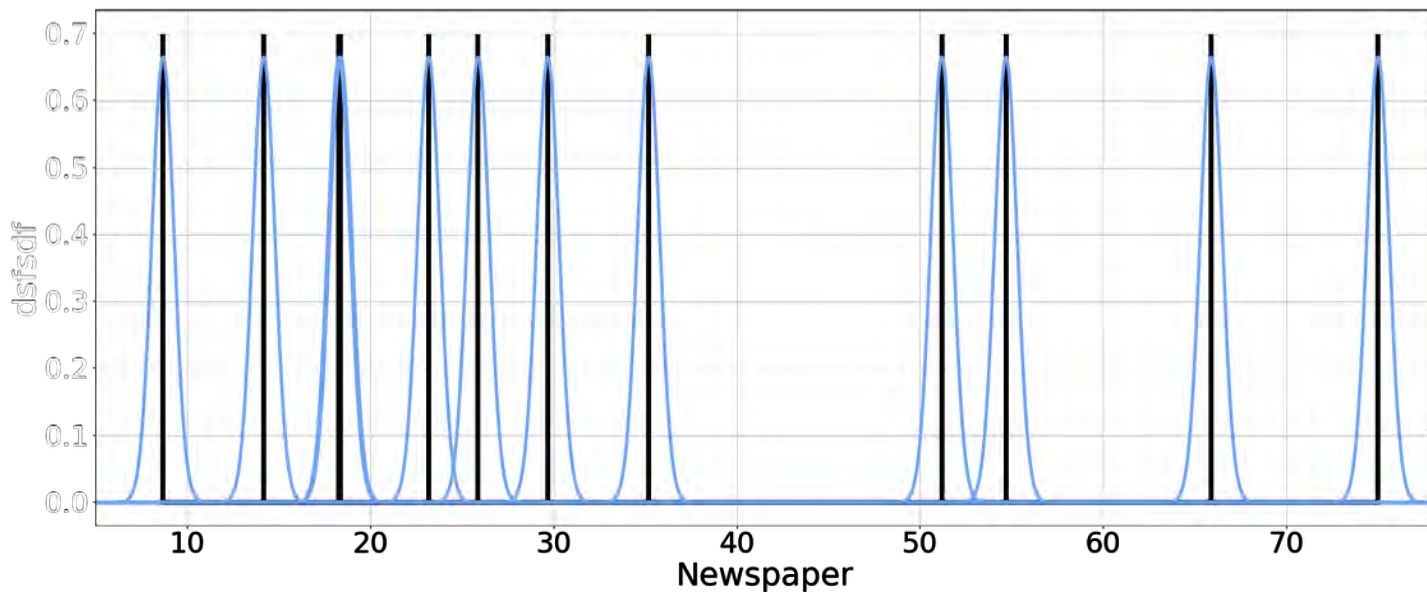
Kernel Density Estimation

Second: draw a gaussian centered at each observation



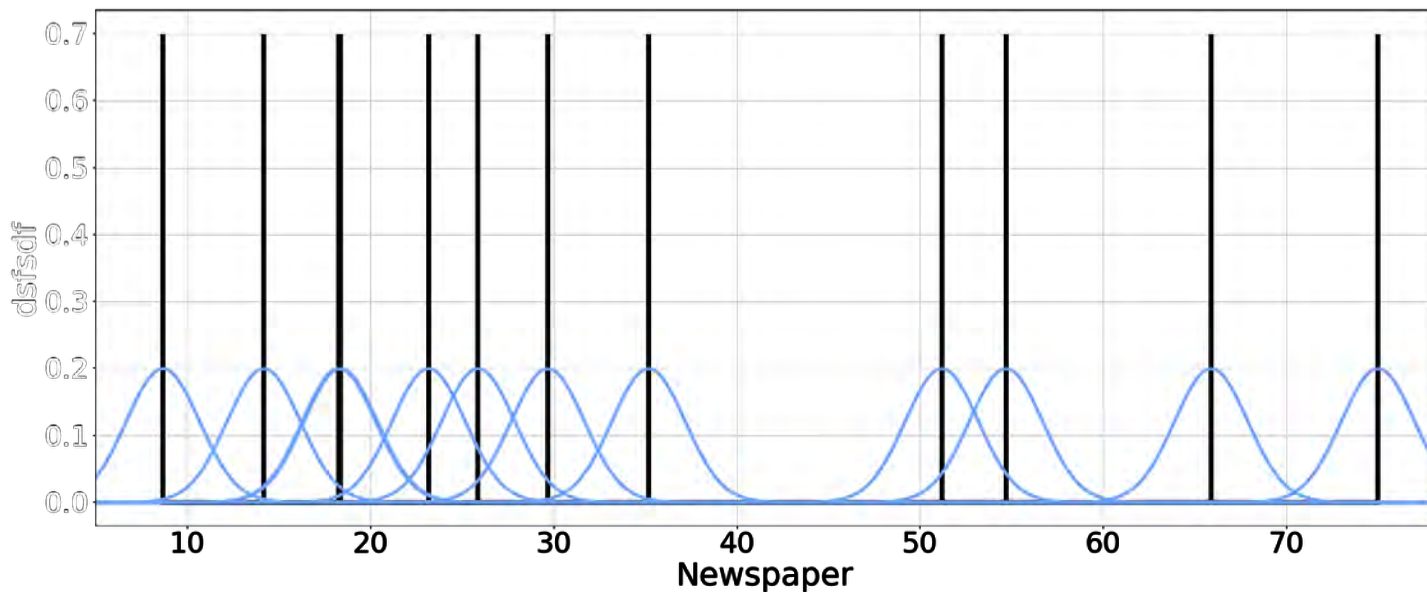
Kernel Density Estimation

Second: draw a gaussian centered at each observation



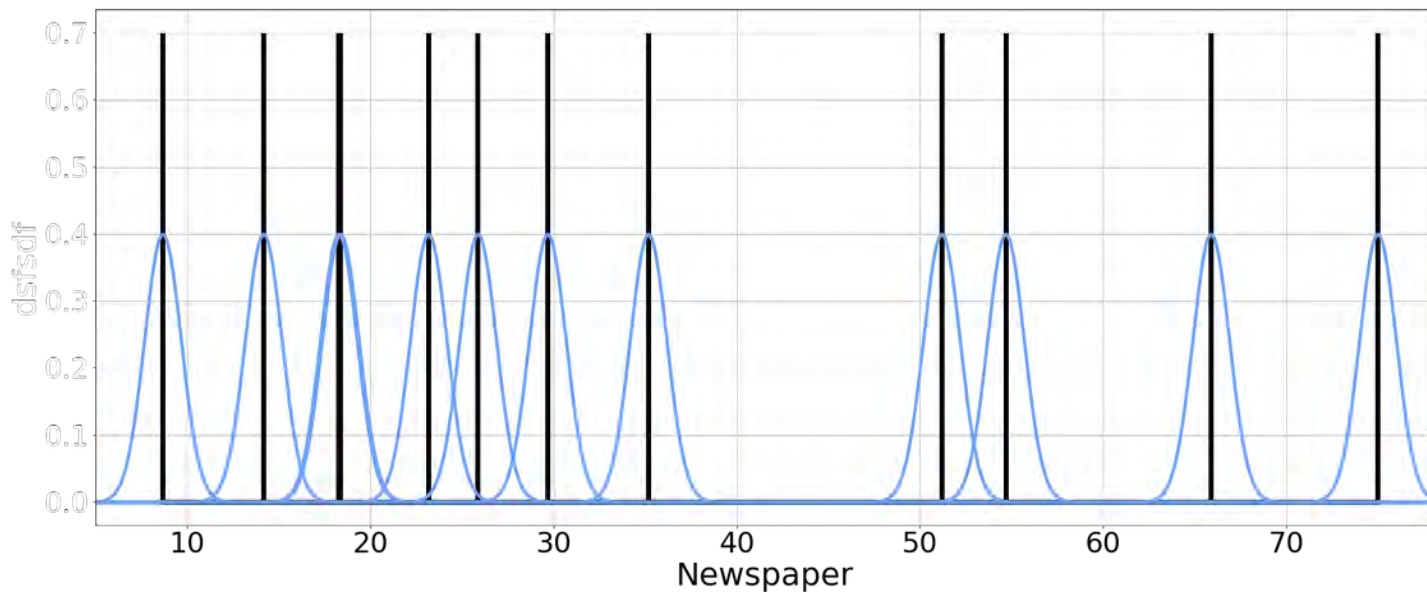
Kernel Density Estimation

Second: draw a gaussian centered at each observation

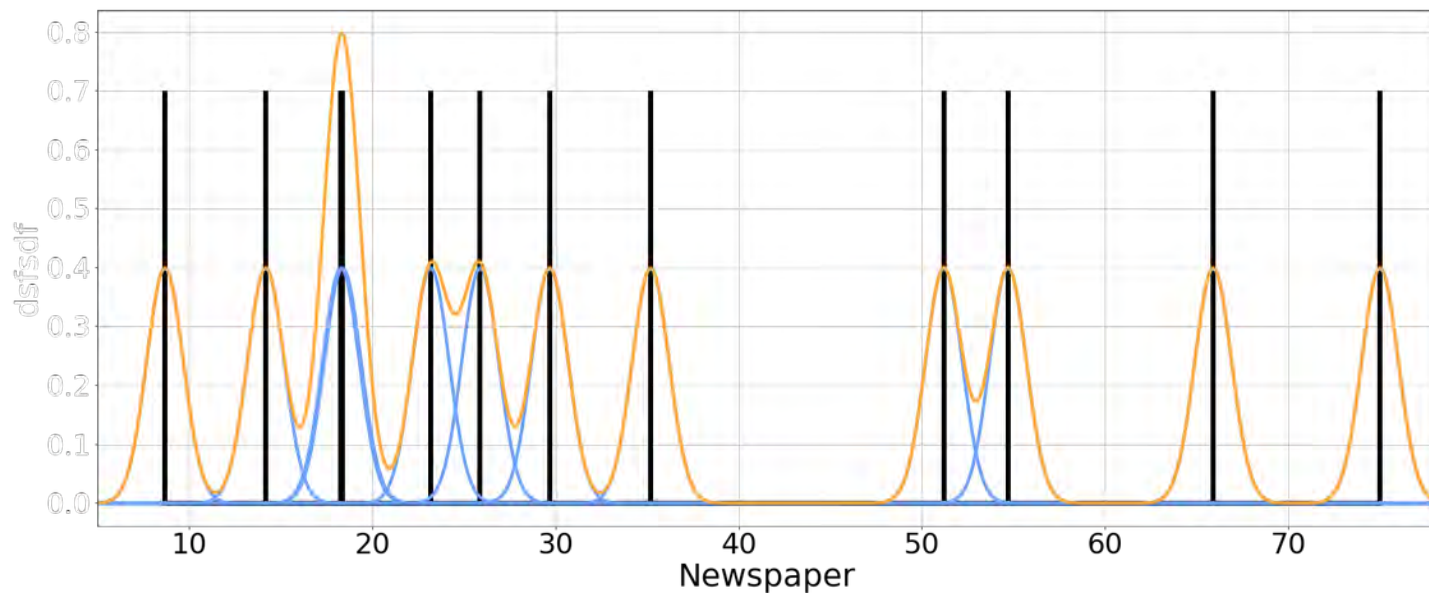


Kernel Density Estimation

Second: draw a gaussian centered at each observation

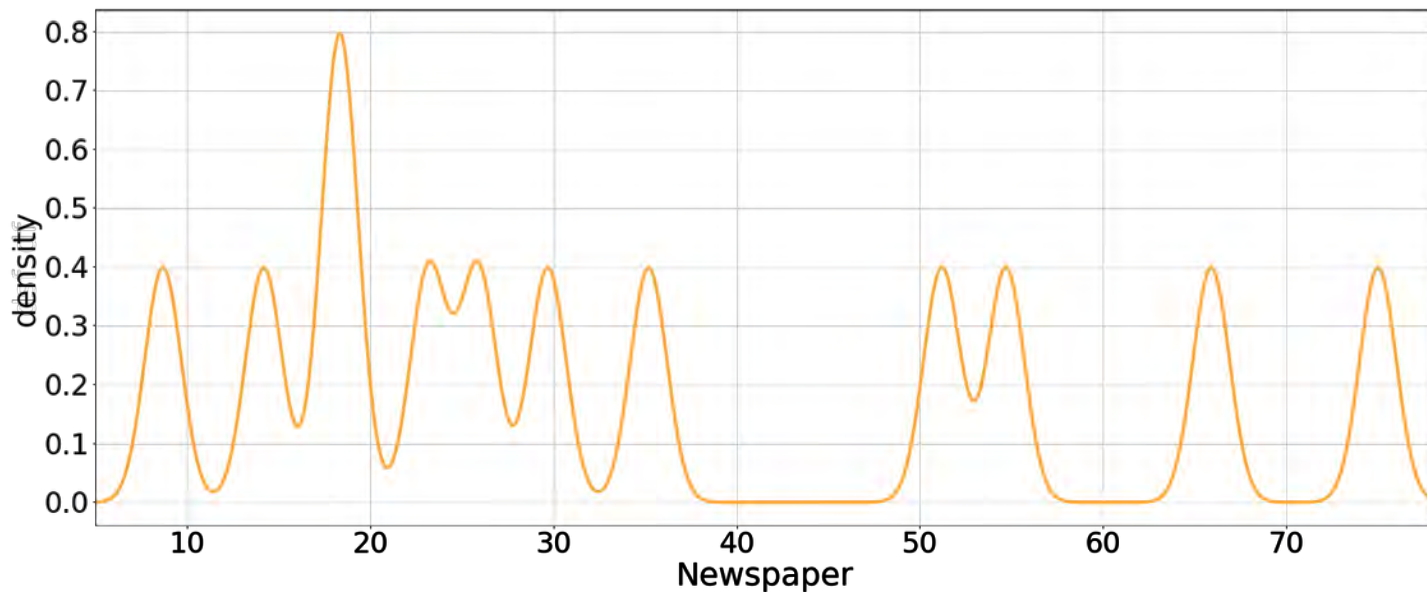


Kernel Density Estimation



Kernel Density Estimation

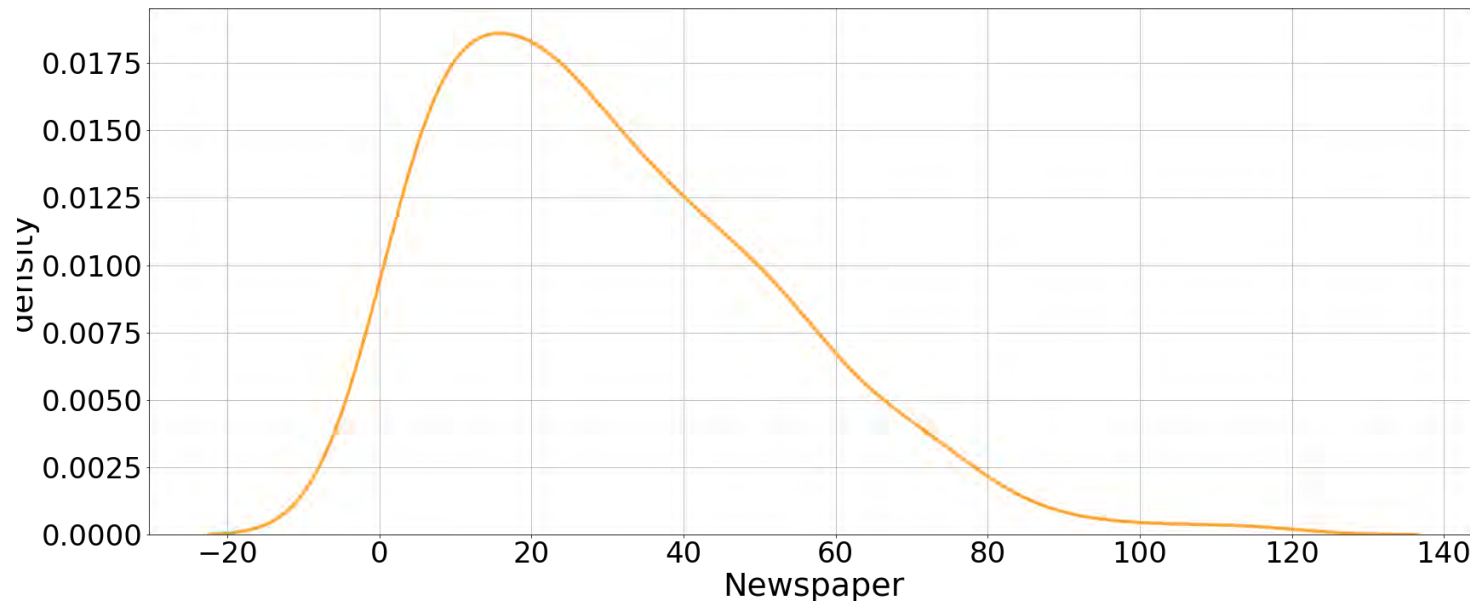
Third: multiply everything by $1/n$ and sum the curves



Kernel Density Estimation

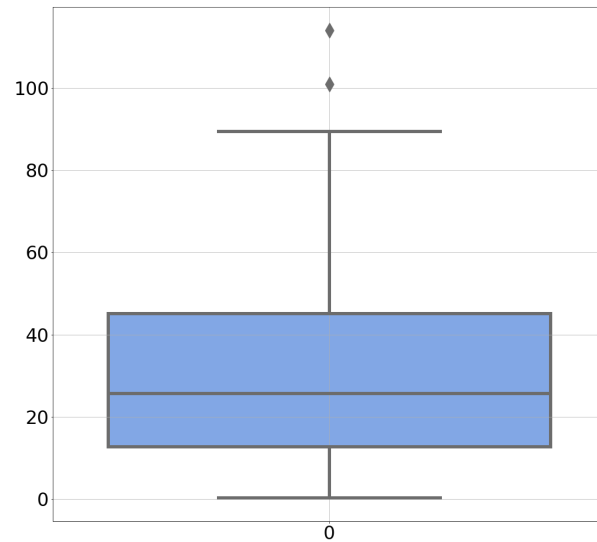
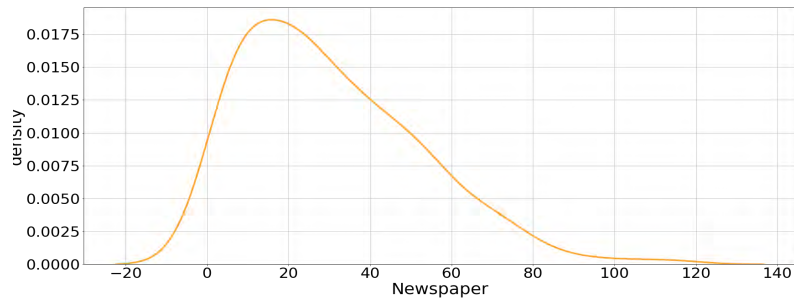
Kernel Density Estimation

What if
you used
all the
dataset?

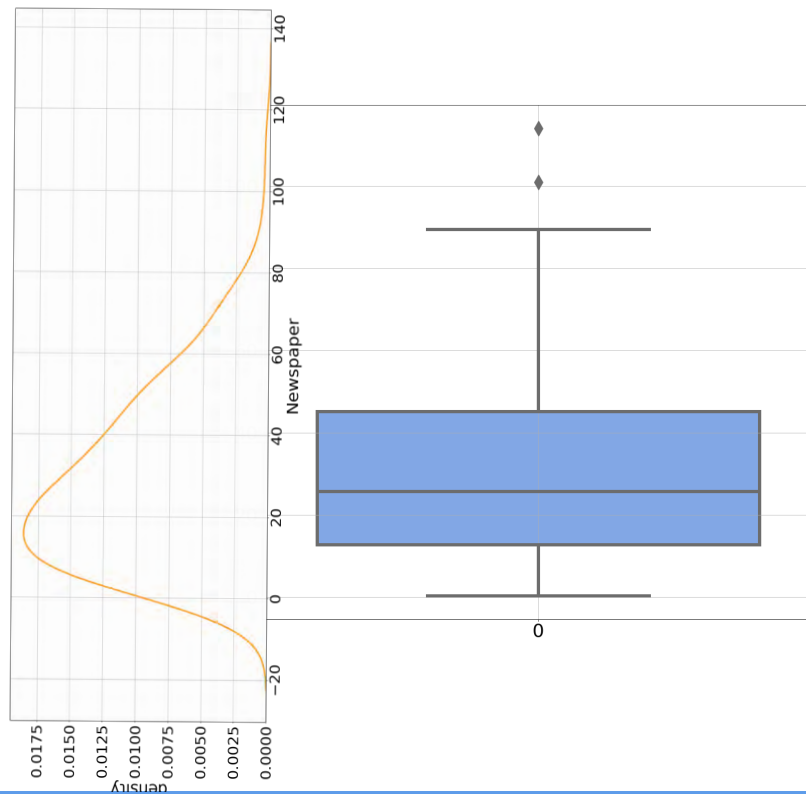


Violin Plots

Violin Plots

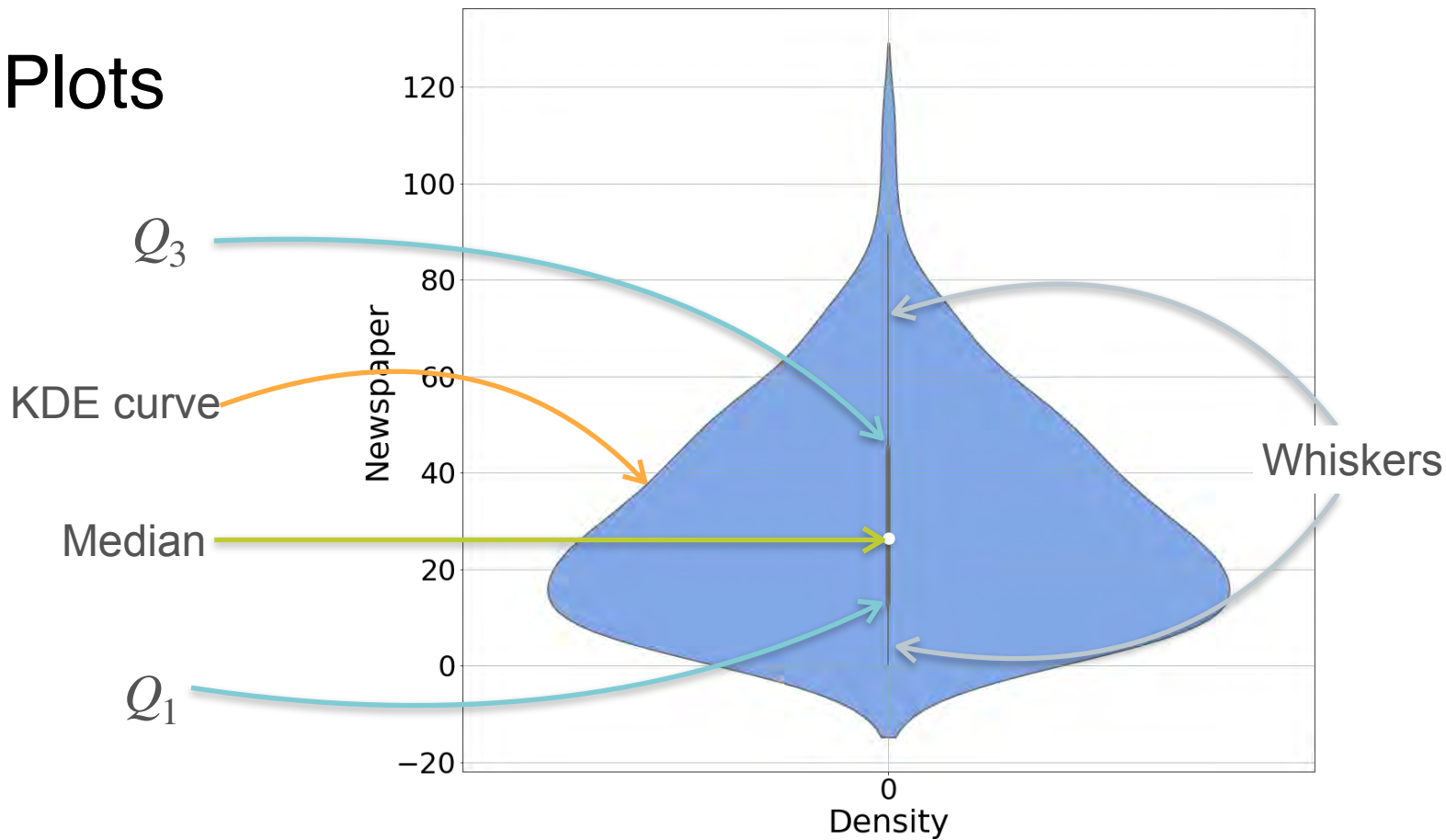


Violin Plots



Violin Plots

Violin Plots



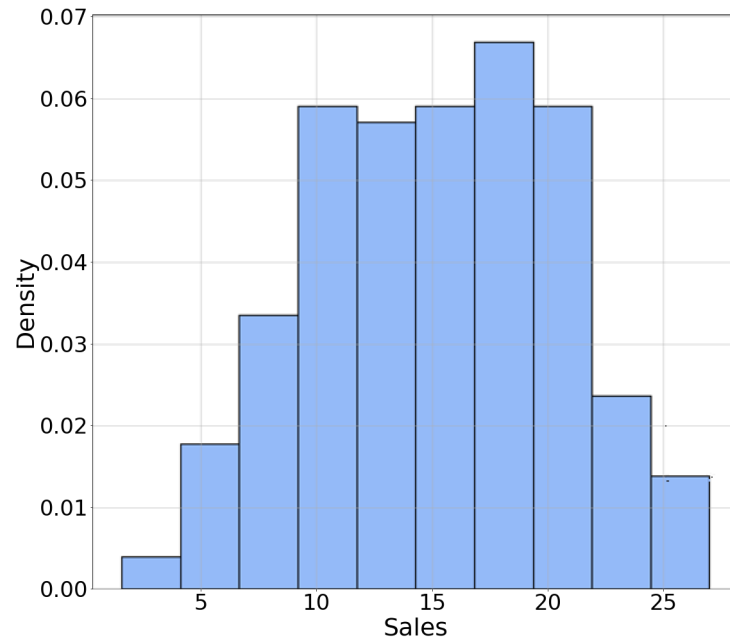
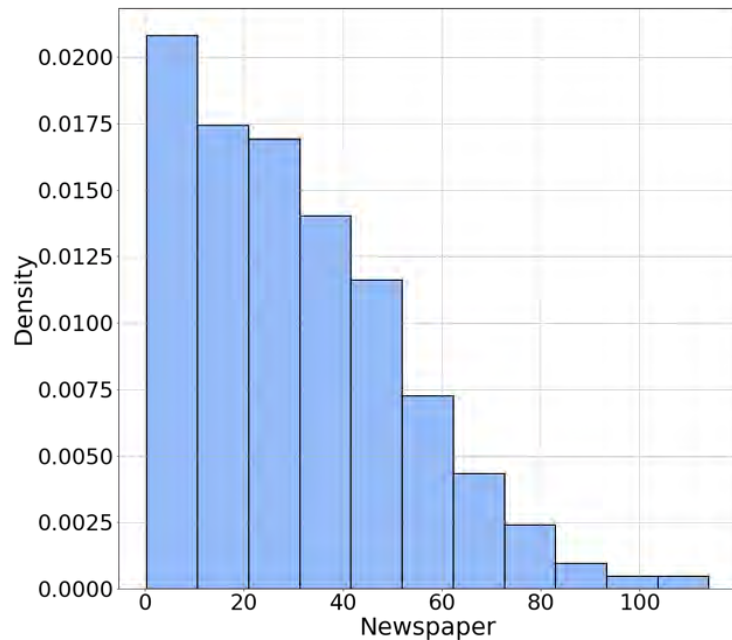
Assessing Normality of Data

Some models assume normally distributed data

- Linear regression
- Logistic regression
- Gaussian Naive Bayes
- Others

Some tests used in Data Science also assume normality.

Assessing Normality of Data



QQ Plots

Newspaper

QQ Plots

Newspaper

Quantile-Quantile plots (QQ Plots) compare quantiles

QQ Plots

Quantile-Quantile plots (QQ Plots) compare quantiles

- Standardize your data:

$$\left(\frac{x - \mu}{\sigma} \right)$$

QQ Plots

Quantile-Quantile plots (QQ Plots) compare quantiles

- Standardize your data:

$$\left(\frac{x - \mu}{\sigma} \right)$$

- Compute quantiles

QQ Plots

Quantile-Quantile plots (QQ Plots) compare quantiles

- Standardize your data:

$$\left(\frac{x - \mu}{\sigma} \right)$$

- Compute quantiles
- Compare to gaussian quantiles

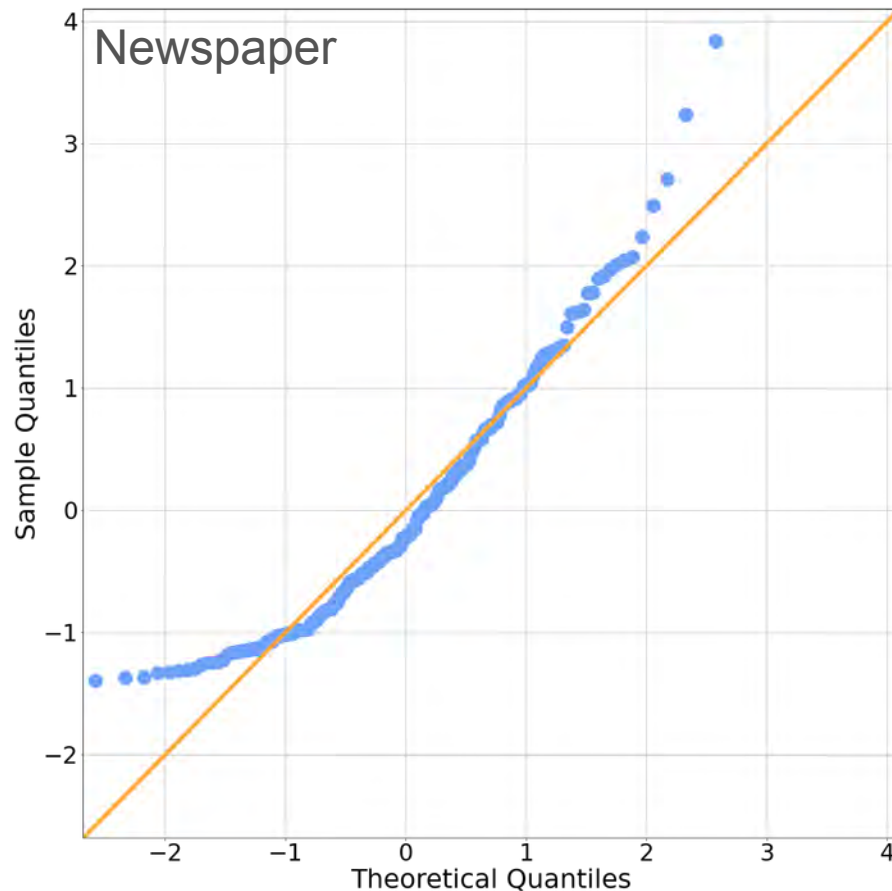
QQ Plots

Quantile-Quantile plots (QQ Plots) compare quantiles

- Standardize your data:

$$\left(\frac{x - \mu}{\sigma} \right)$$

- Compute quantiles
- Compare to gaussian quantiles



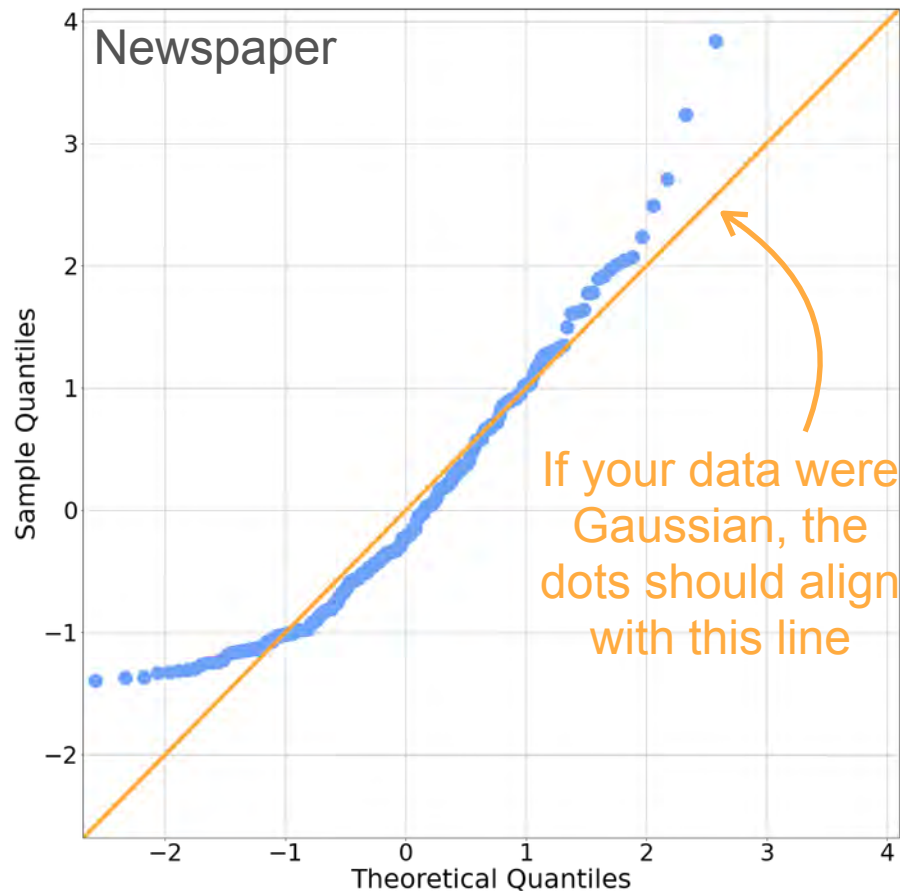
QQ Plots

Quantile-Quantile plots (QQ Plots) compare quantiles

- Standardize your data:

$$\left(\frac{x - \mu}{\sigma} \right)$$

- Compute quantiles
- Compare to gaussian quantiles



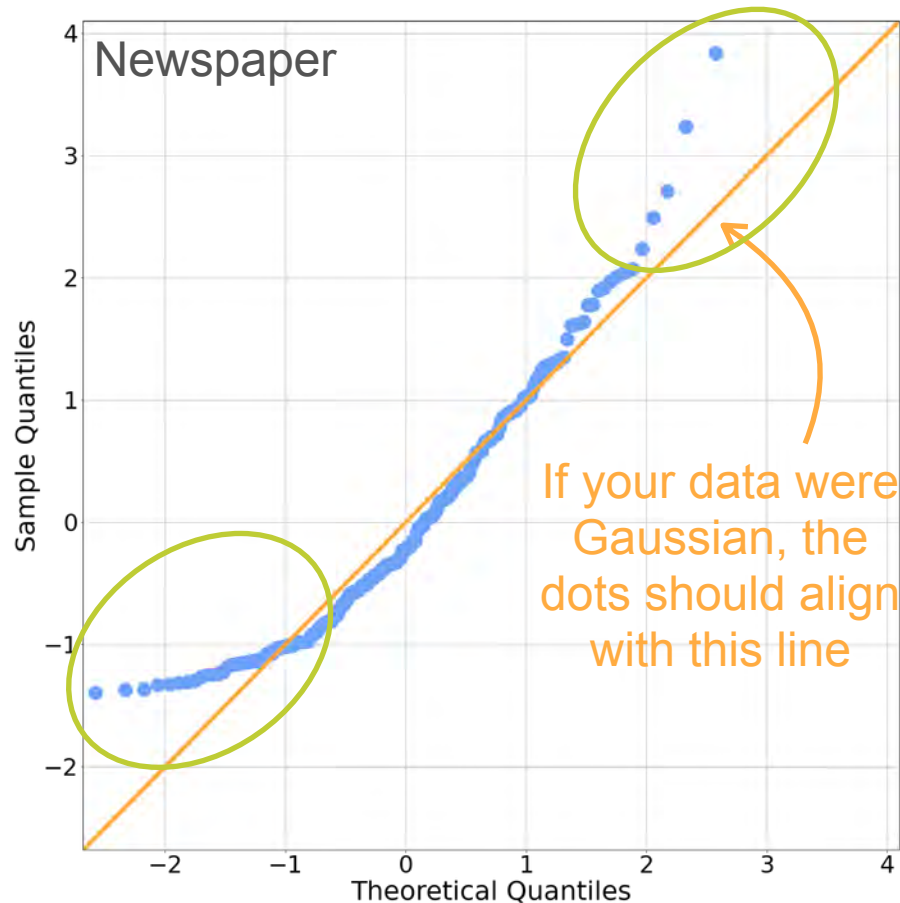
QQ Plots

Quantile-Quantile plots (QQ Plots) compare quantiles

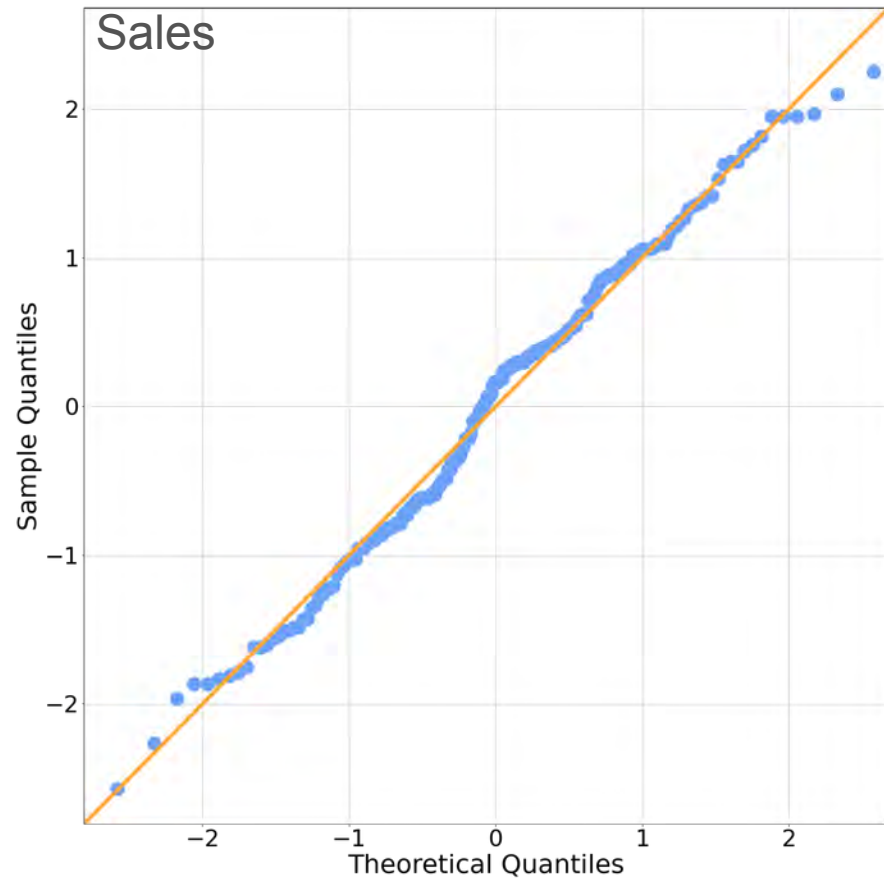
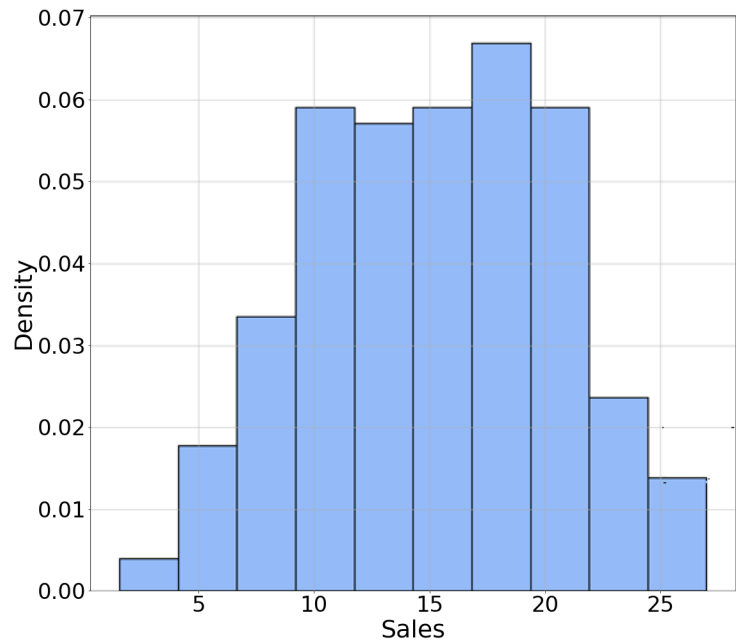
- Standardize your data:

$$\left(\frac{x - \mu}{\sigma} \right)$$

- Compute quantiles
- Compare to gaussian quantiles



QQ Plots



W2 Lesson 2



DeepLearning.AI

Probability Distributions with Multiple Variables

Joint Distribution (Discrete)

Joint Distributions (Discrete): Example 1

Joint Distributions (Discrete): Example 1

Age (Year)	Count
7	3
8	2
9	4
10	1

Joint Distributions (Discrete): Example 1

Age (Year)	Count
7	3
8	2
9	4
10	1



Joint Distributions (Discrete): Example 1

Age (Year)	Count
7	3
8	2
9	4
10	1



Joint Distributions (Discrete): Example 1

Age (Year)	Count
7	3
8	2
9	4
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Joint Distributions (Discrete): Example 1

Age (Year)	Count
7	3
8	2
9	4
10	1



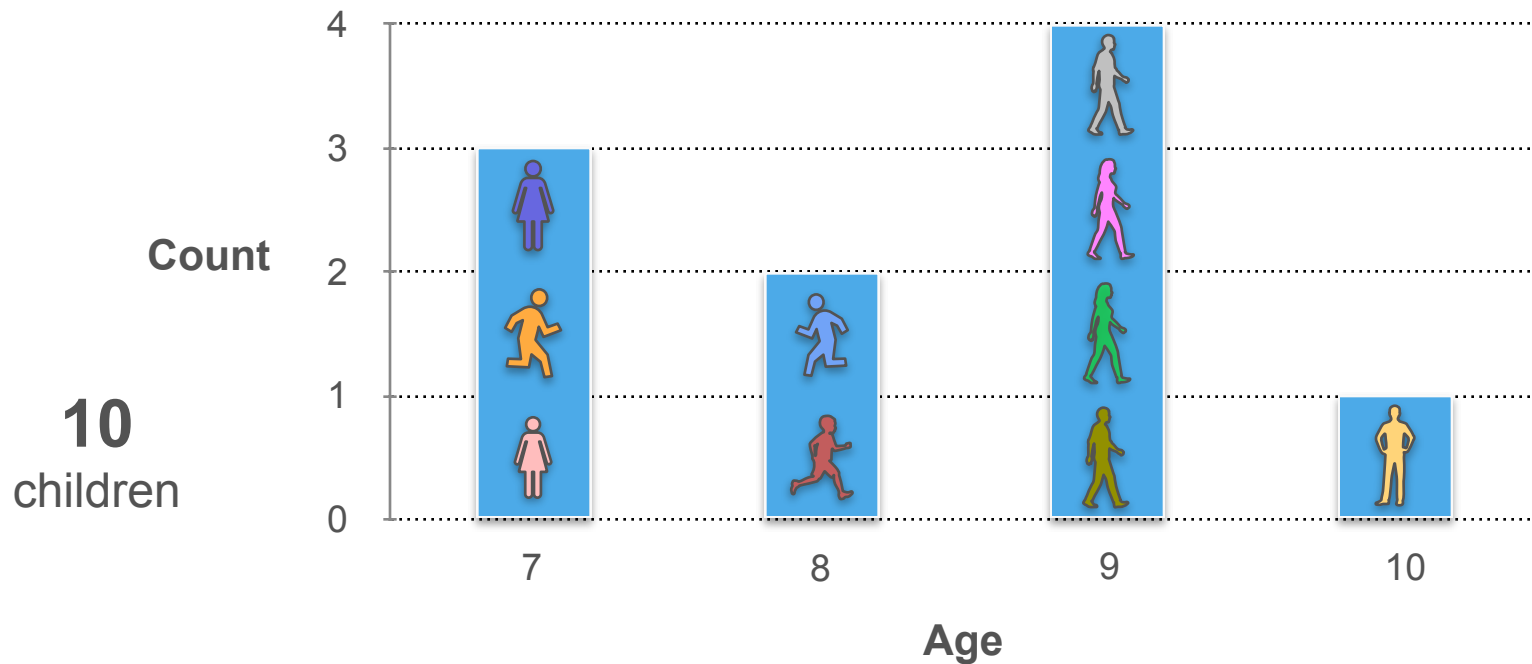
Joint Distributions (Discrete): Example 1

Age (Year)	Count
7	3
8	2
9	4
10	1

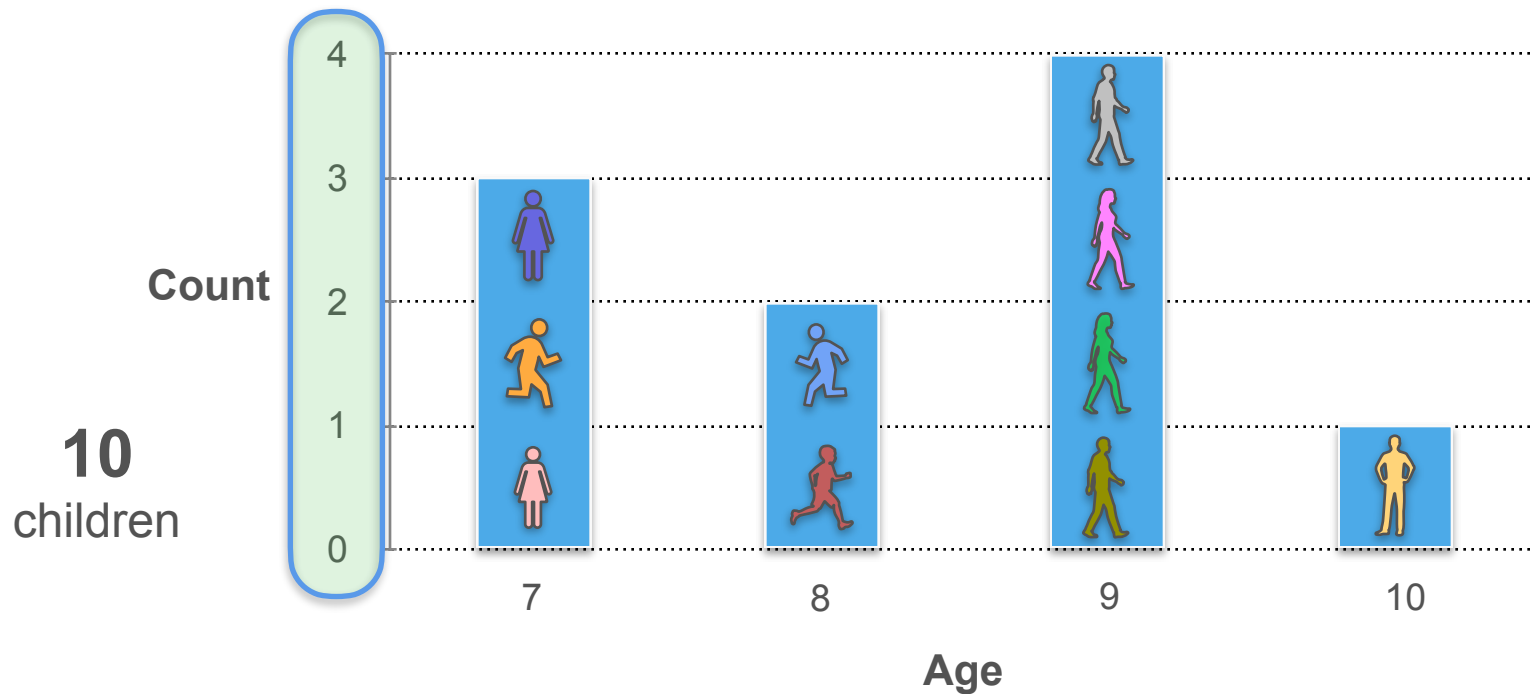


10
children

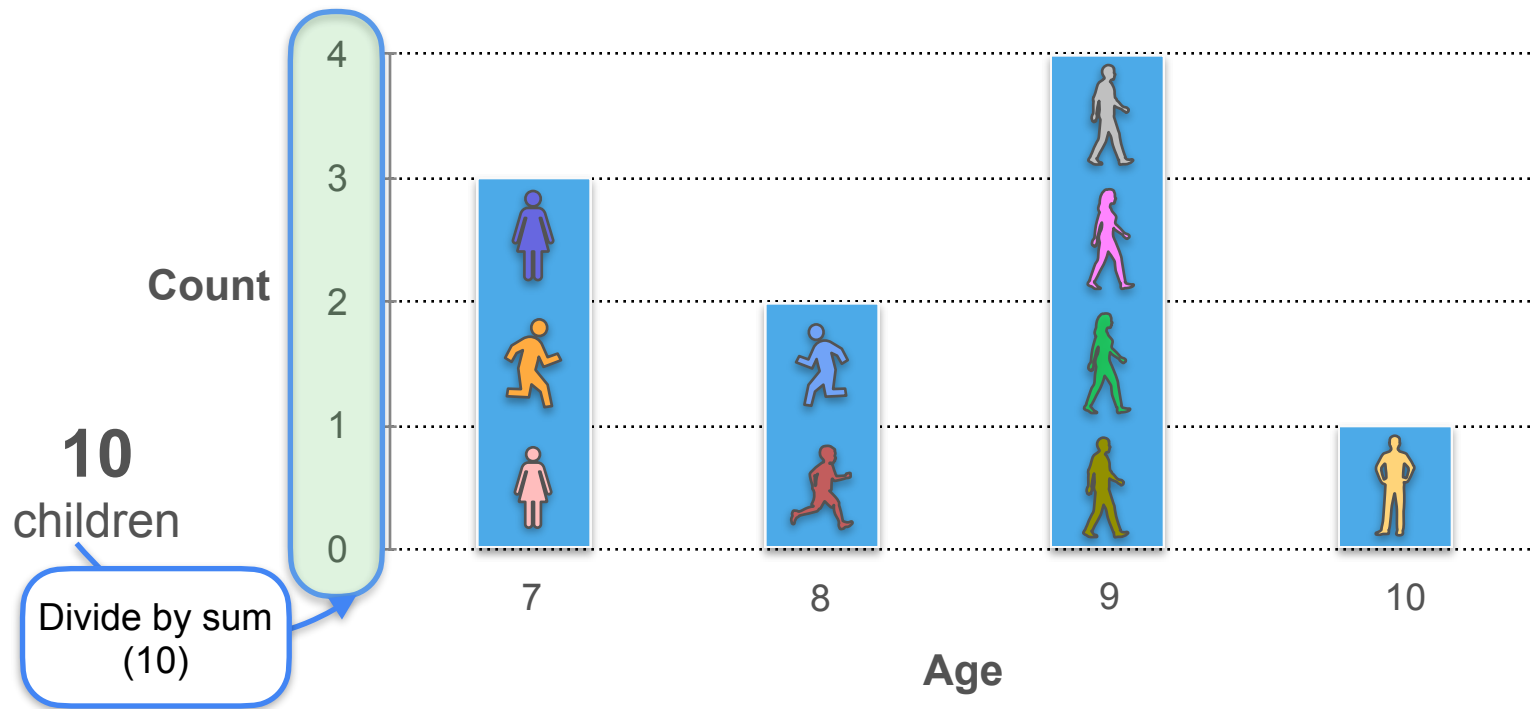
Joint Distributions (Discrete): Example 1



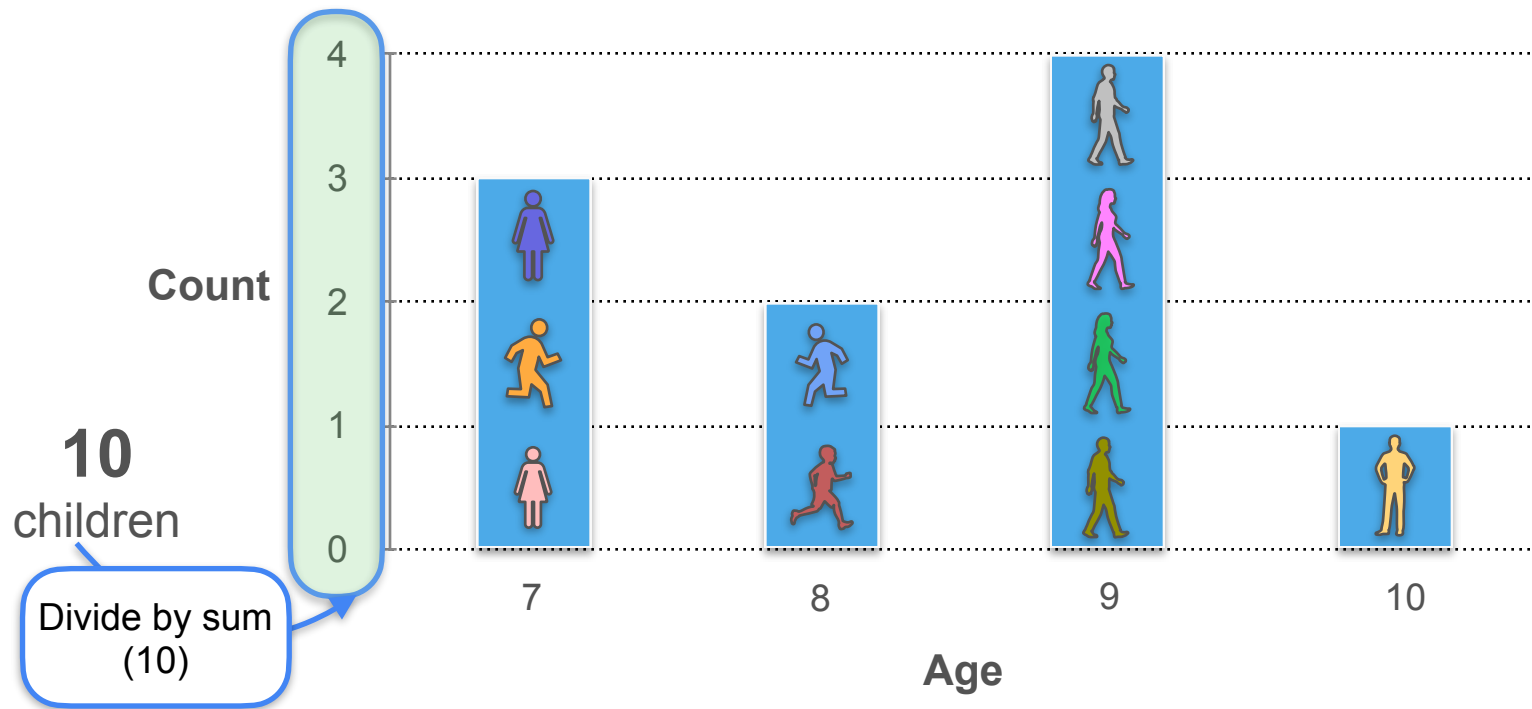
Joint Distributions (Discrete): Example 1



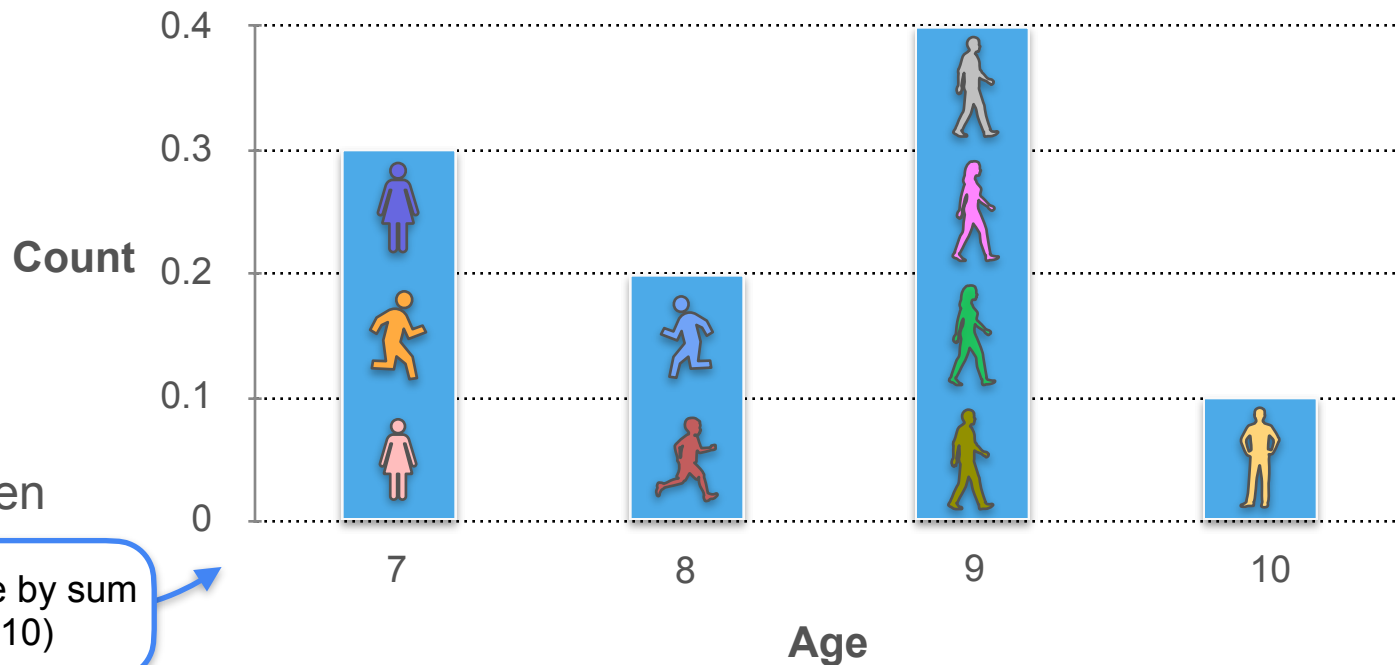
Joint Distributions (Discrete): Example 1



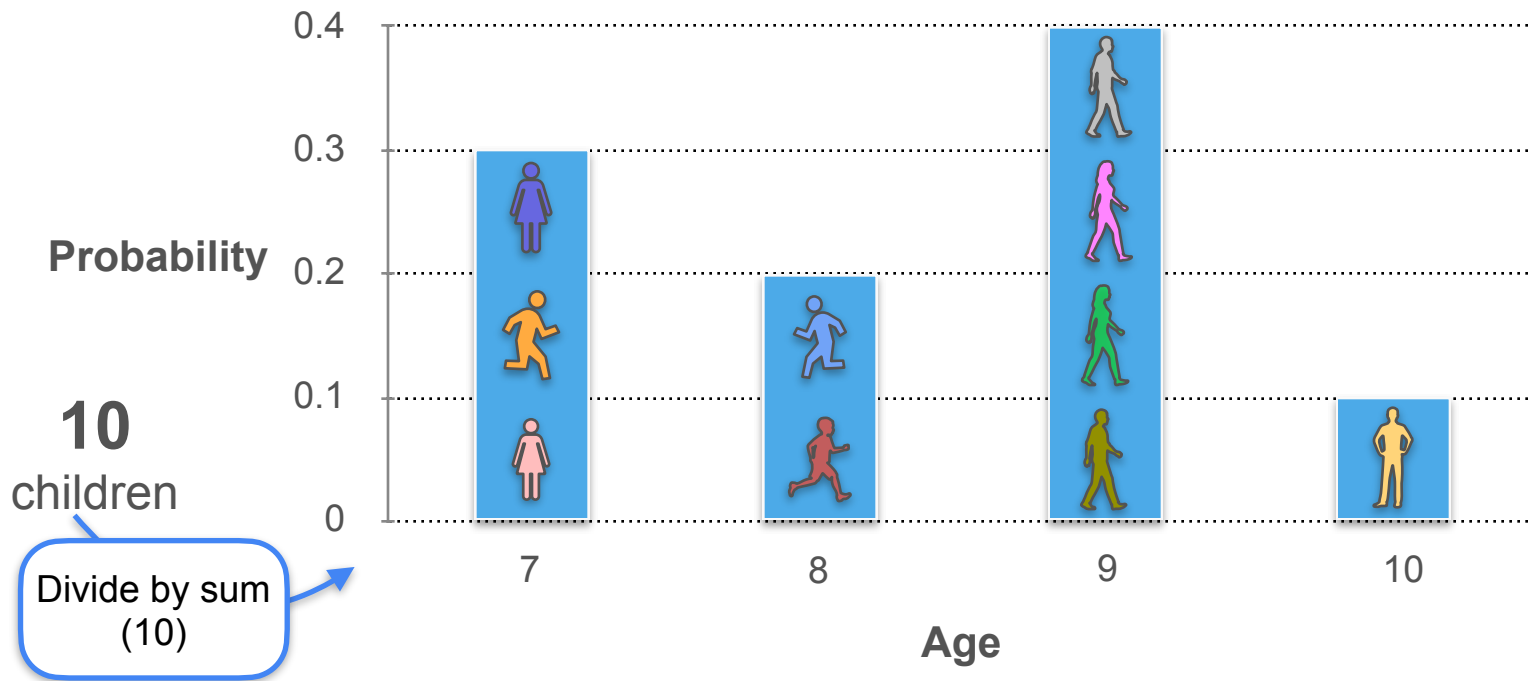
Joint Distributions (Discrete): Example 1



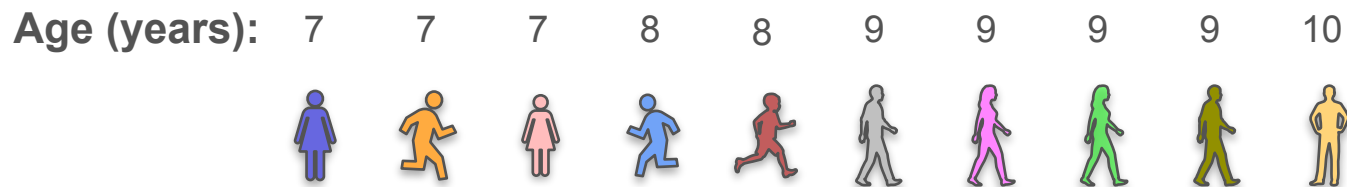
Joint Distributions (Discrete): Example 1



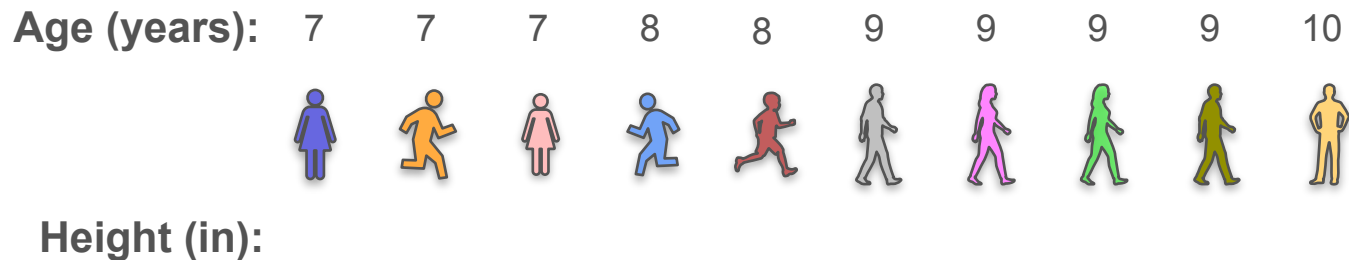
Joint Distributions (Discrete): Example 1



Joint Distributions (Discrete): Example 1



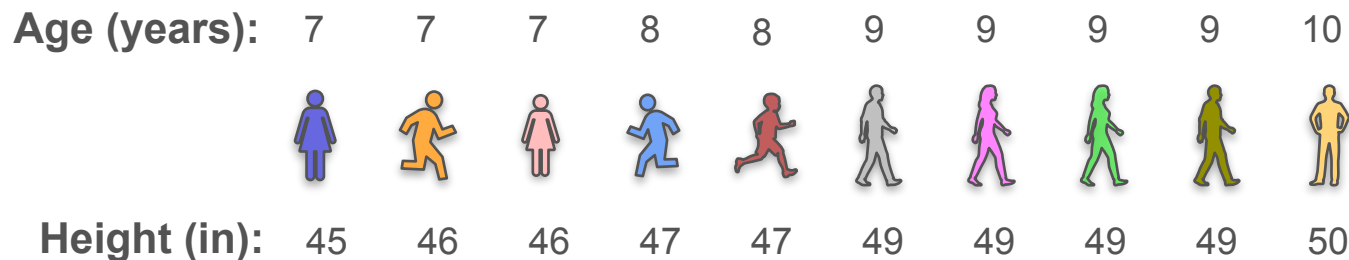
Joint Distributions (Discrete): Example 1



Joint Distributions (Discrete): Example 1

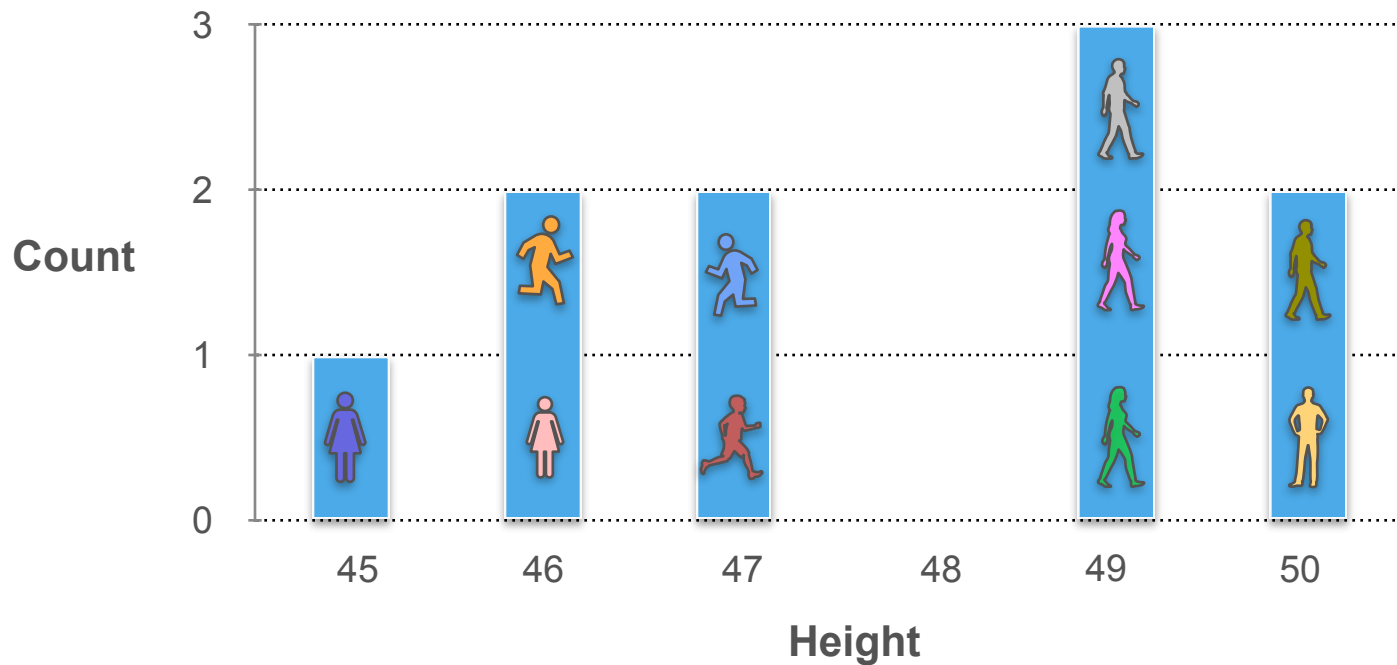
Age (years):	7	7	7	8	8	9	9	9	9	10
										
Height (in):	45	46	46	47	47	49	49	49	49	50

Joint Distributions (Discrete): Example 1

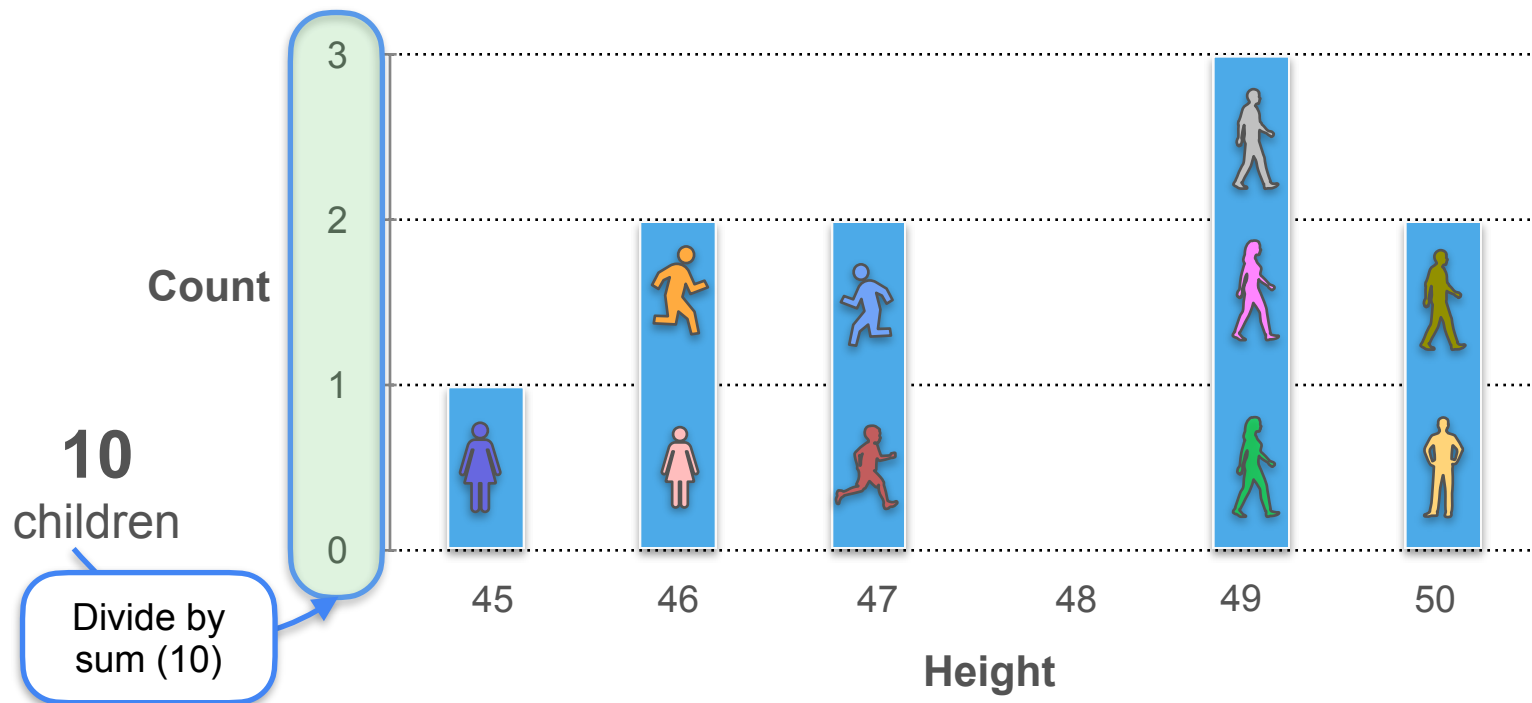


Height (in)	Count
45	1
46	2
47	2
48	0
49	4
50	1

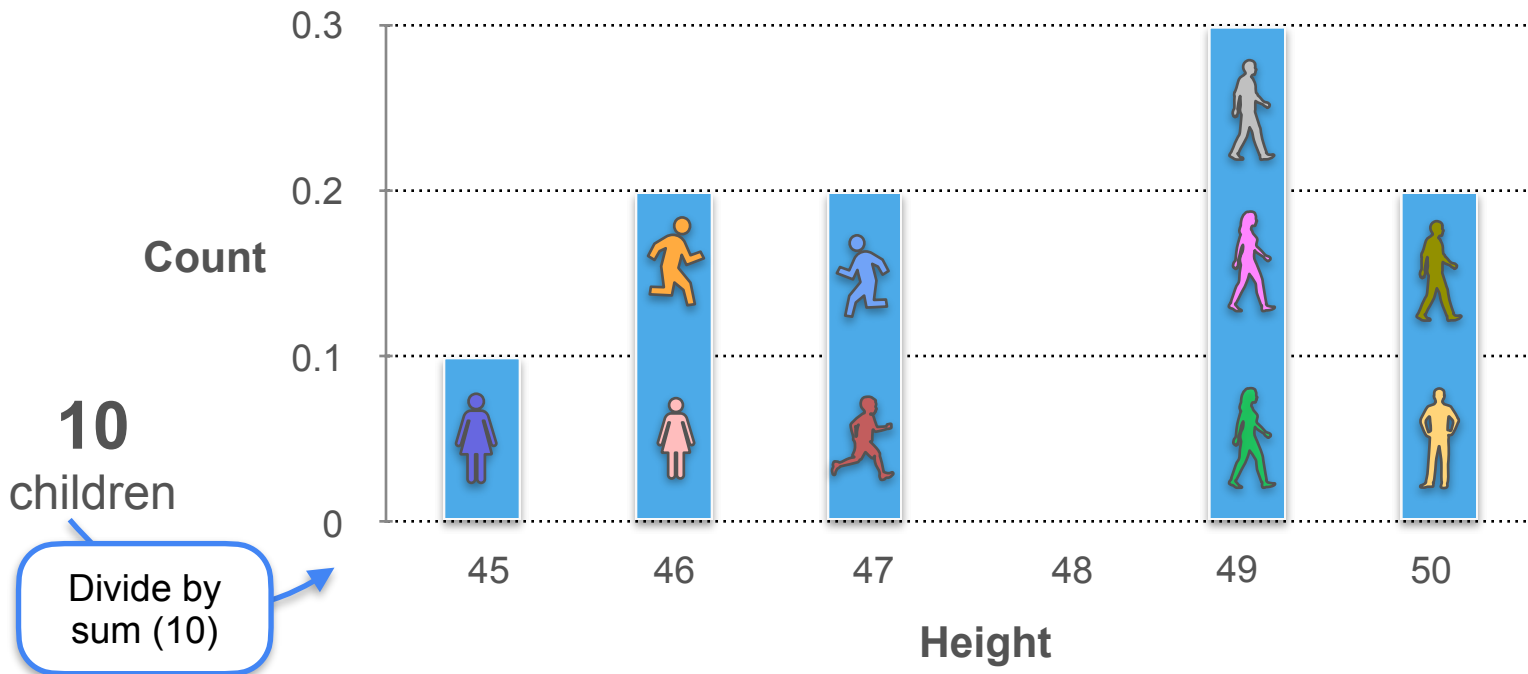
Joint Distributions (Discrete): Example 1



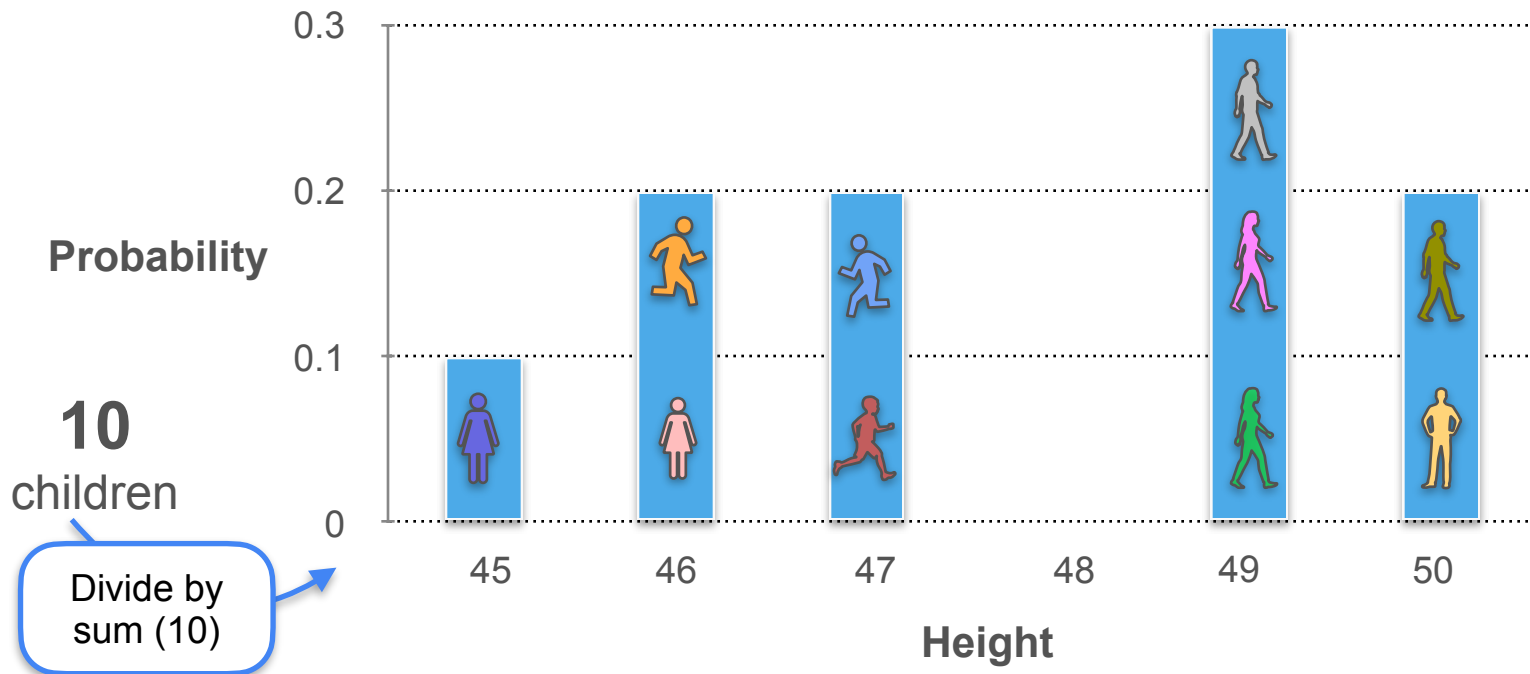
Joint Distributions (Discrete): Example 1



Joint Distributions (Discrete): Example 1

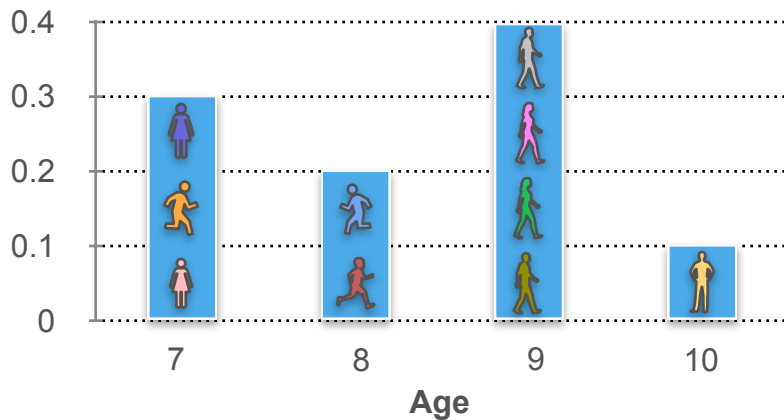


Joint Distributions (Discrete): Example 1

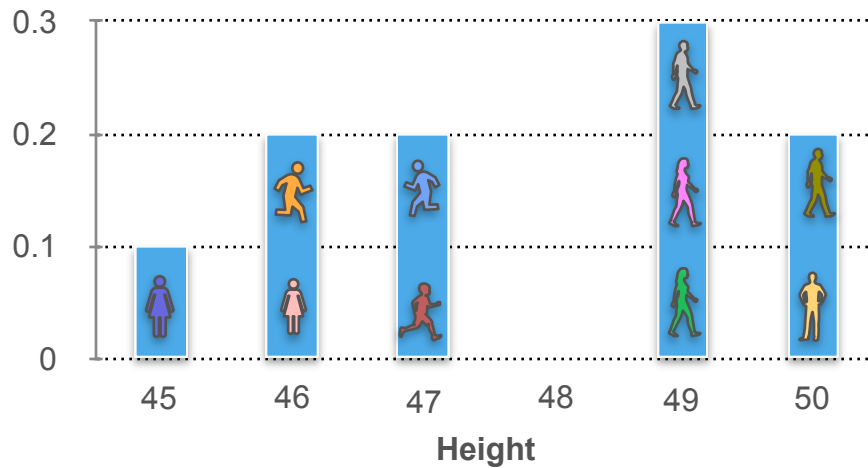
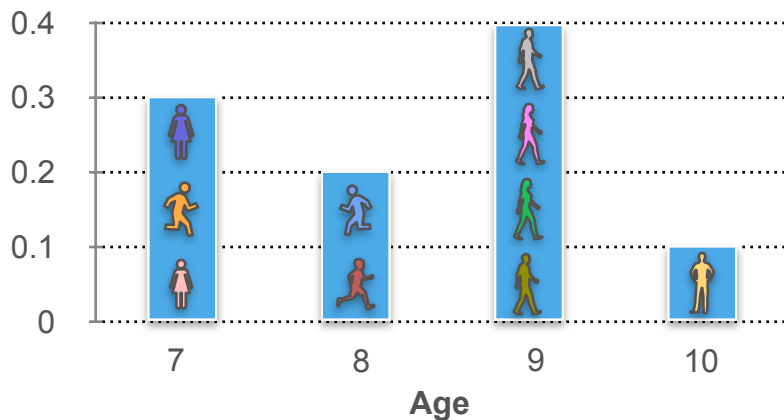


Joint Distributions (Discrete): Example 1

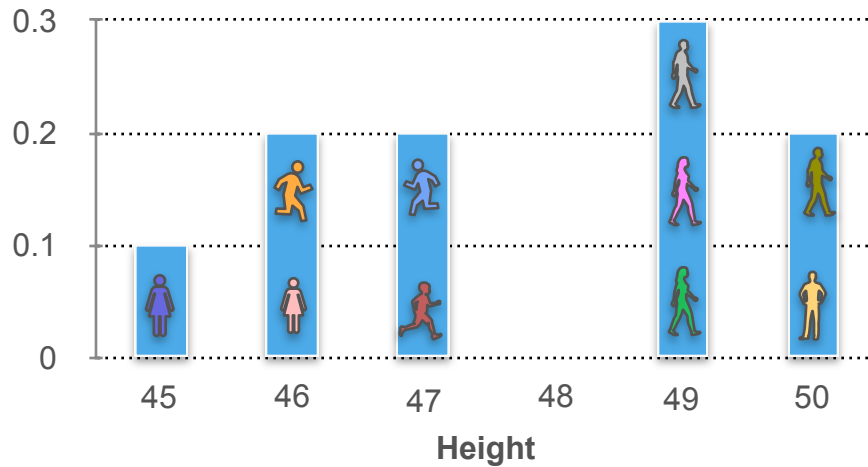
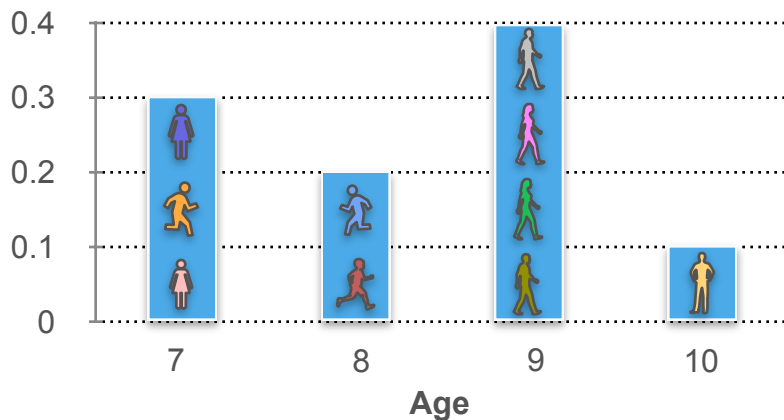
Joint Distributions (Discrete): Example 1



Joint Distributions (Discrete): Example 1

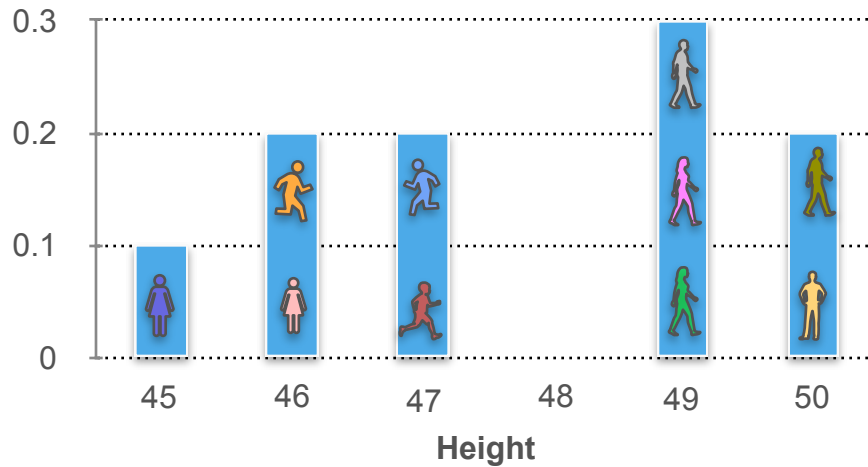
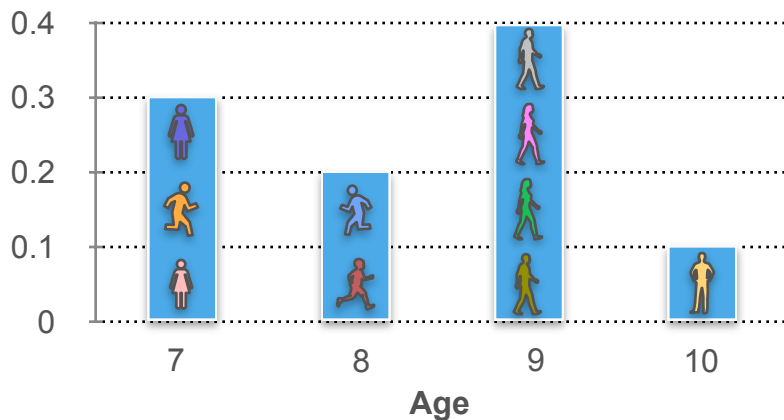


Joint Distributions (Discrete): Example 1



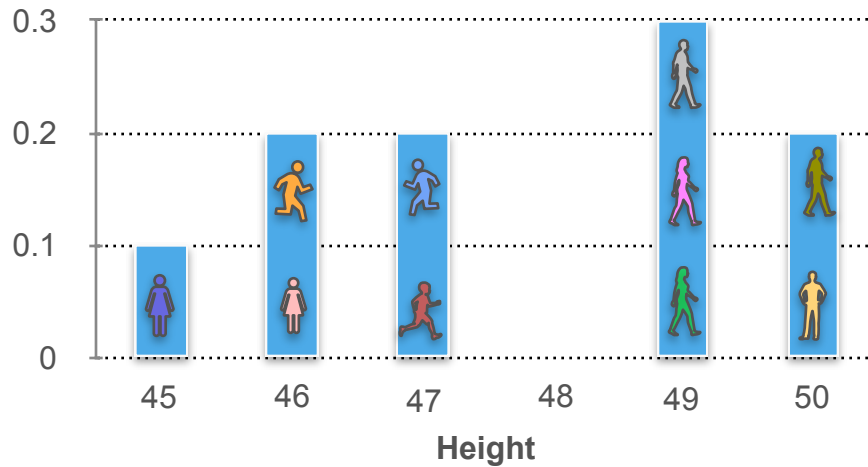
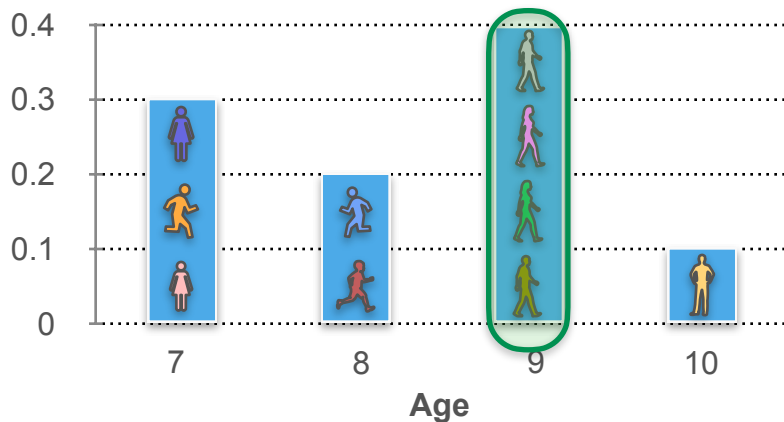
What is the probability that a child is **9 years** old and **49 inches** tall?

Joint Distributions (Discrete): Example 1



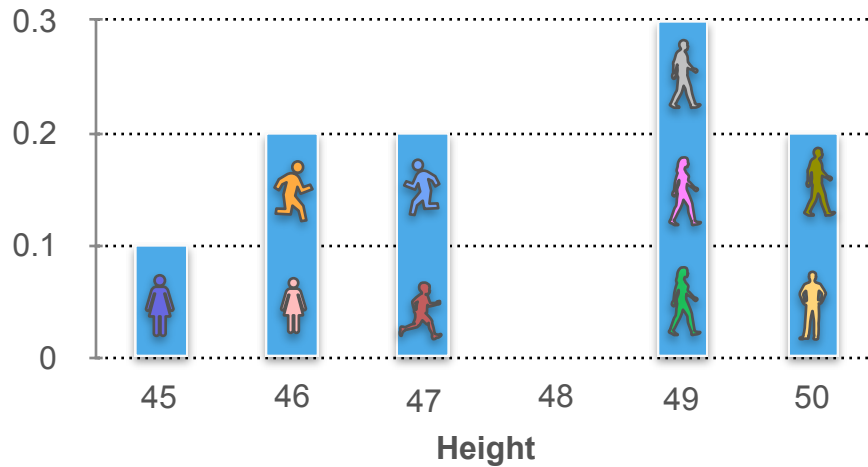
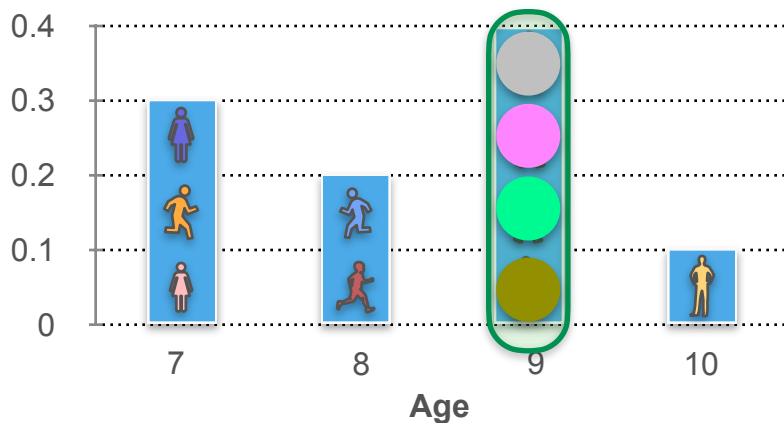
What is the probability that a child is **9 years** old and **49 inches** tall?

Joint Distributions (Discrete): Example 1



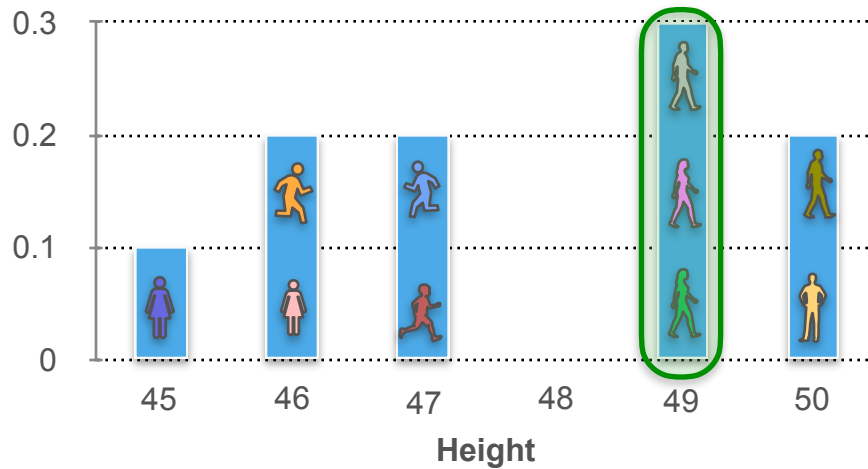
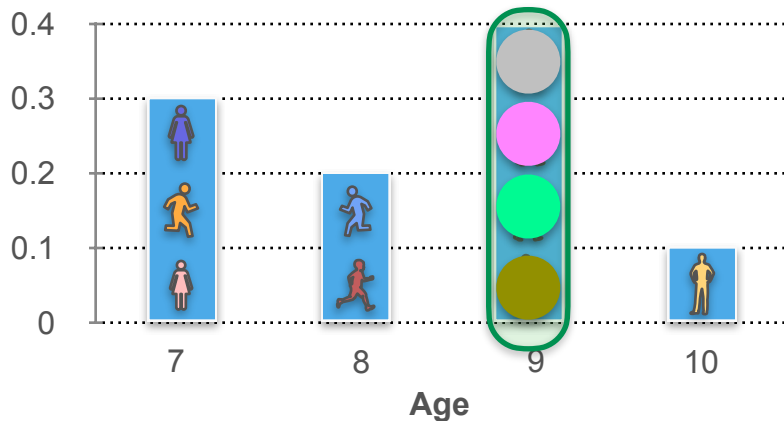
What is the probability that a child is **9 years** old and **49 inches** tall?

Joint Distributions (Discrete): Example 1



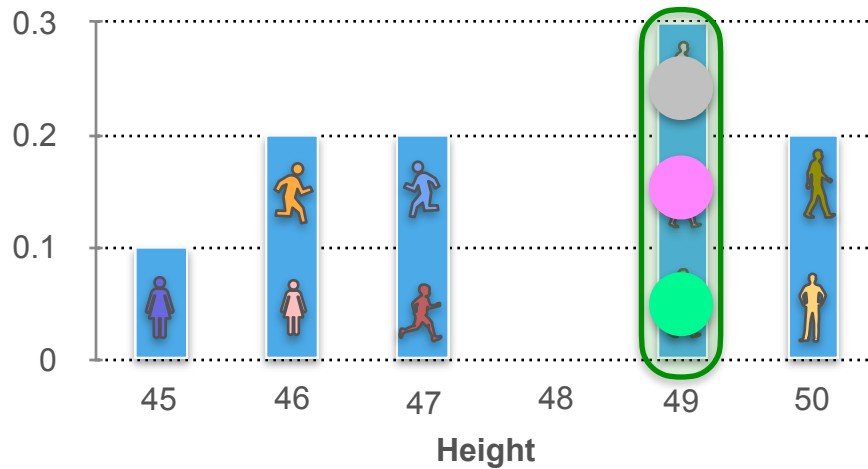
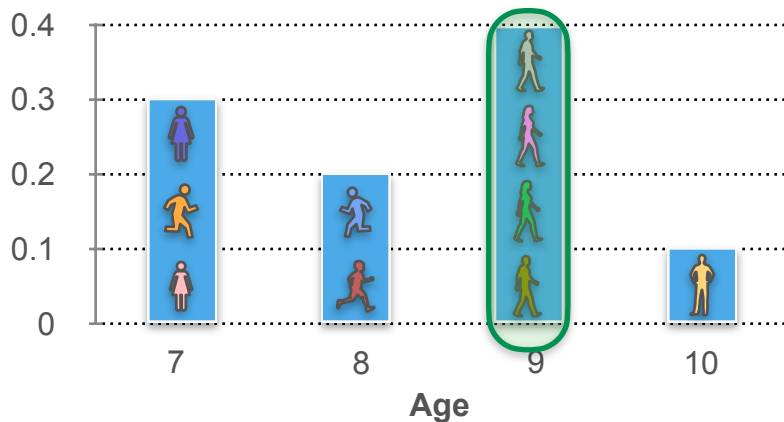
What is the probability that a child is **9 years** old and **49 inches** tall?

Joint Distributions (Discrete): Example 1



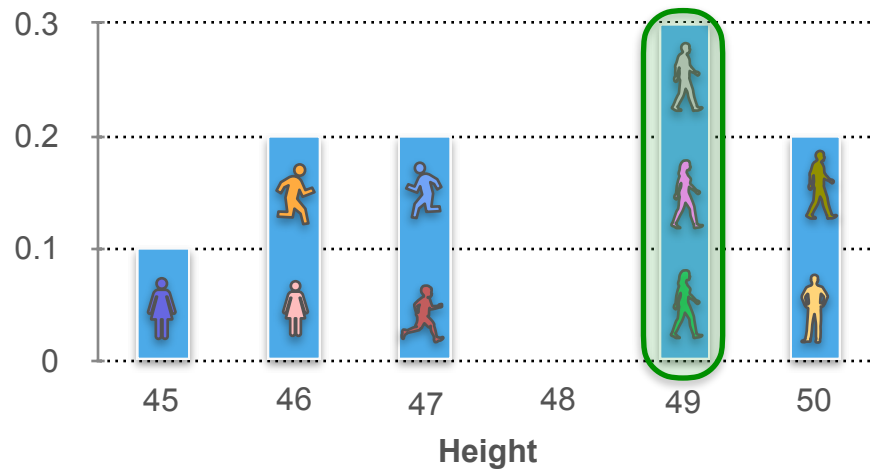
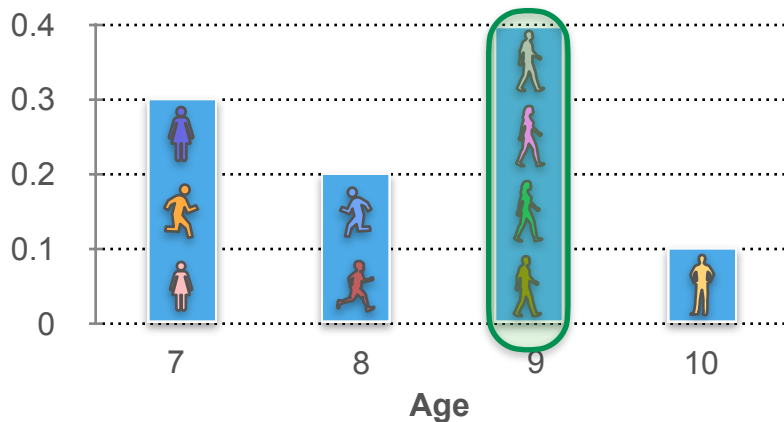
What is the probability that a child is **9 years** old and **49 inches** tall?

Joint Distributions (Discrete): Example 1



What is the probability that a child is **9 years** old and **49 inches** tall?

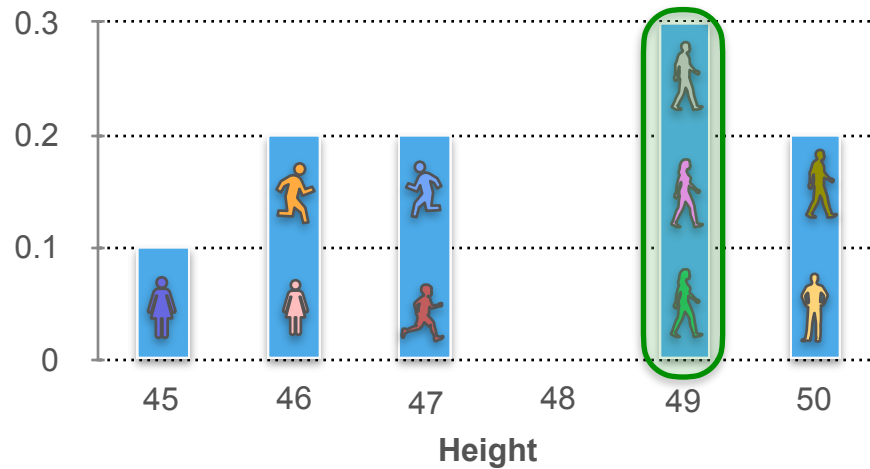
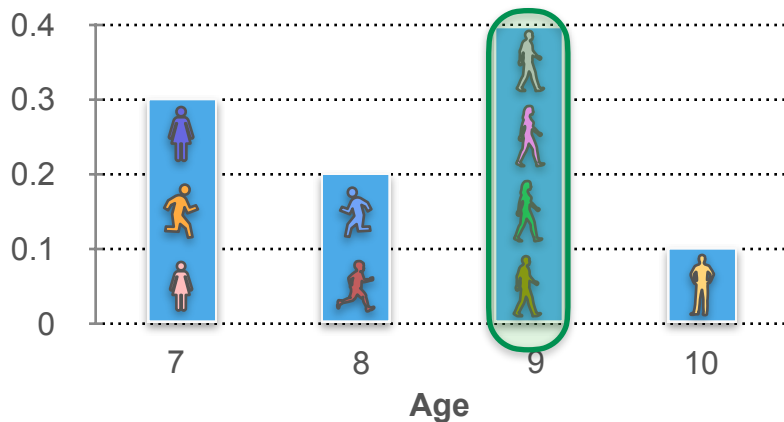
Joint Distributions (Discrete): Example 1



What is the probability that a child is 9 years old and 49 inches tall?



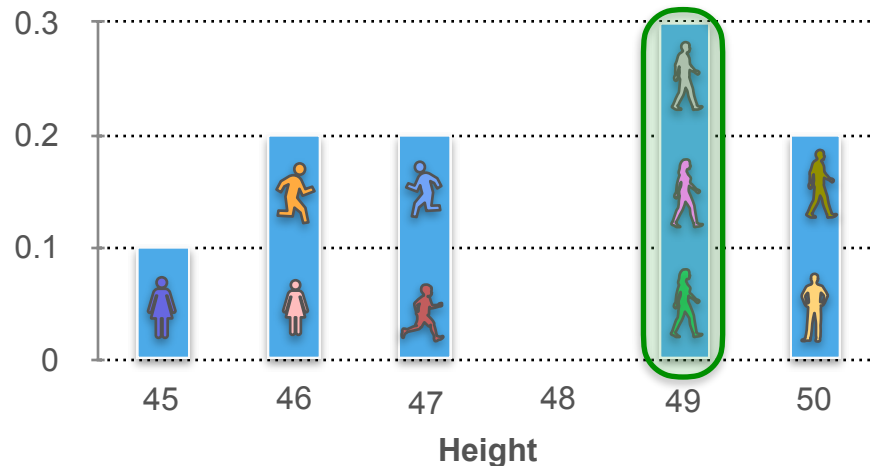
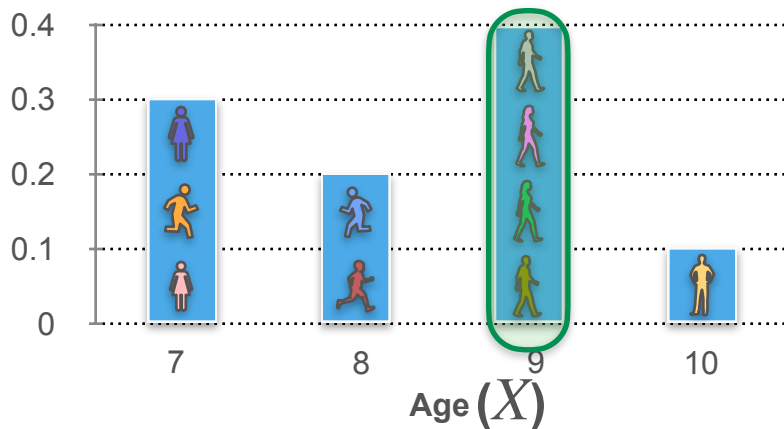
Joint Distributions (Discrete): Example 1



What is the probability that a child is **9 years** old and **49 inches** tall?

$$\frac{\text{Purple} + \text{Pink} + \text{Green}}{10} = \frac{3}{10}$$

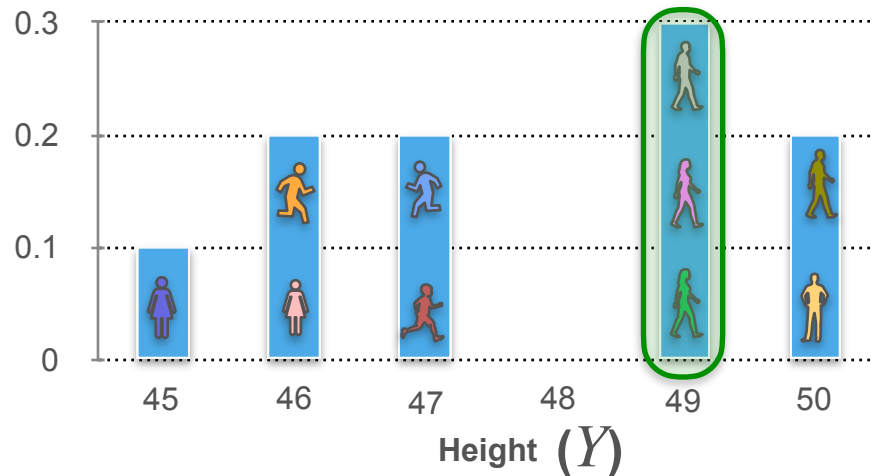
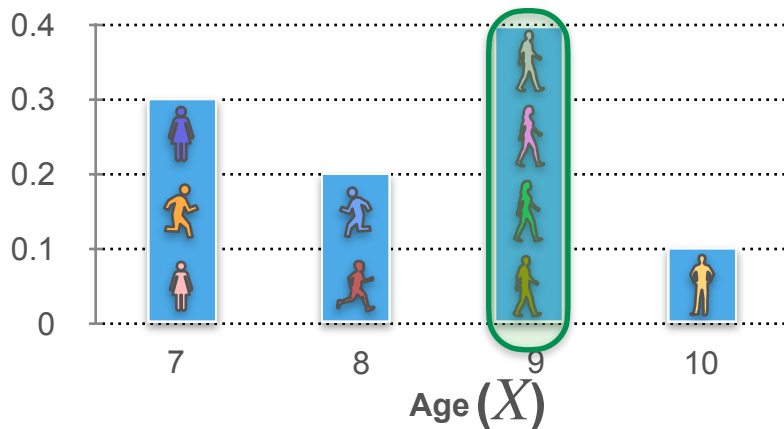
Joint Distributions (Discrete): Example 1



What is the probability that a child is **9 years** old and **49 inches** tall?

$$\frac{\text{Grey} \quad \text{Pink} \quad \text{Green}}{10} = \frac{3}{10}$$

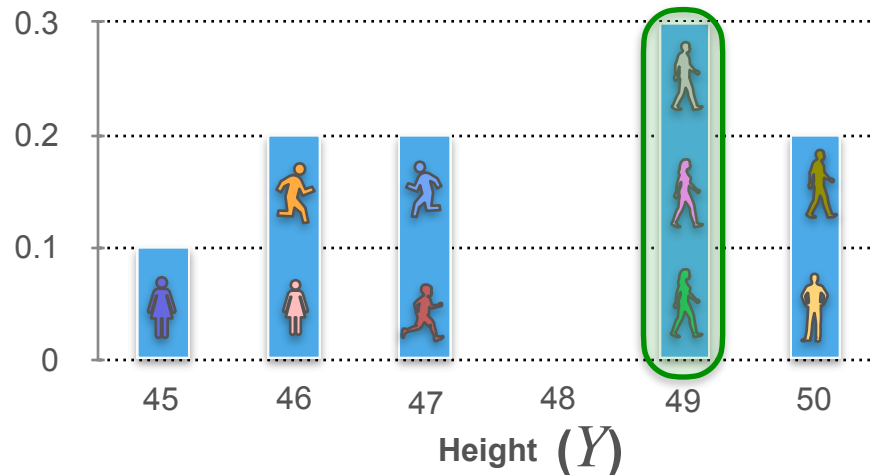
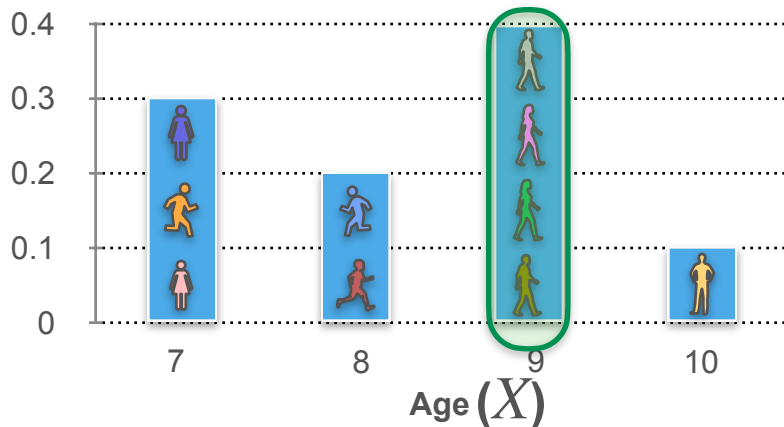
Joint Distributions (Discrete): Example 1



What is the probability that a child is **9 years** old and **49 inches** tall?

$$\frac{\text{Grey} \quad \text{Pink} \quad \text{Green}}{10} = \frac{3}{10}$$

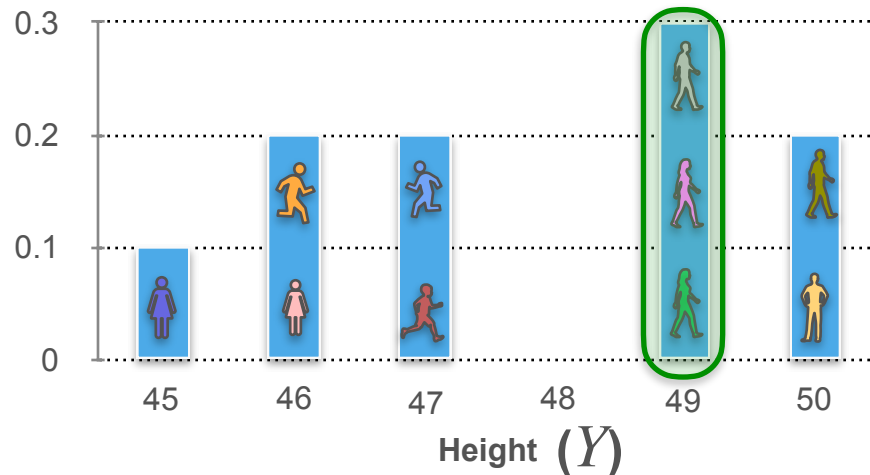
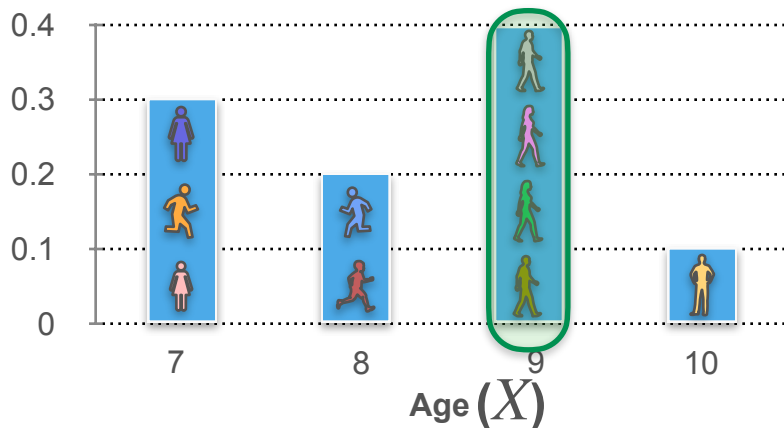
Joint Distributions (Discrete): Example 1



What is the probability that a child is 9 years old and 49 inches tall?

$$\frac{\text{3}}{10} = \frac{3}{10}$$

Joint Distributions (Discrete): Example 1

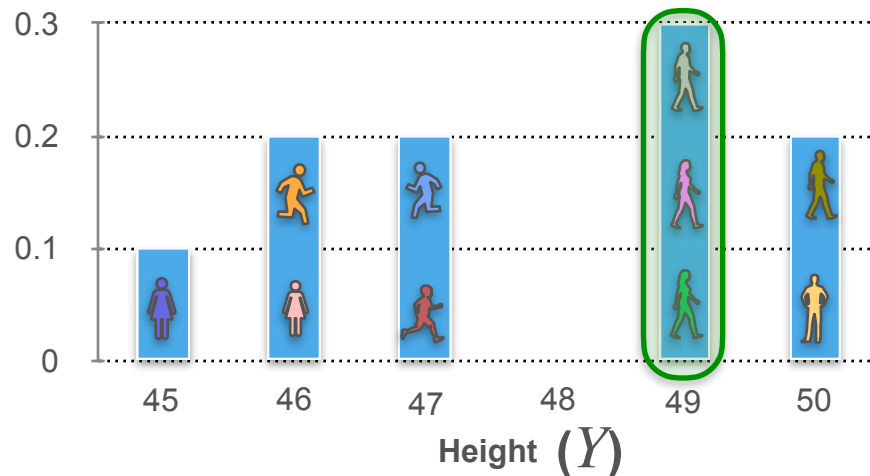
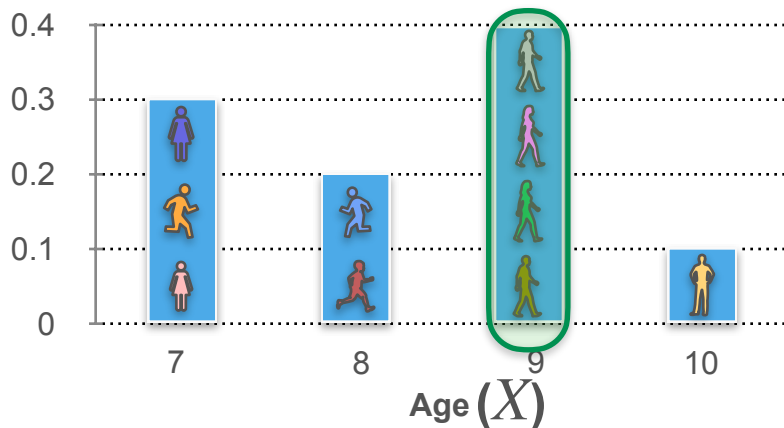


What is the probability that a child is 9 years old and 49 inches tall?

$$p_{XY}(9, 49)$$

$$\frac{3}{10} = \frac{3}{10}$$

Joint Distributions (Discrete): Example 1



What is the probability that a child is 9 years old and 49 inches tall?

$$p_{XY}(9, 49) = \mathbf{P}(X = 9, Y = 49) = \frac{\text{3}}{10} = \frac{3}{10}$$

Joint Distributions (Discrete): Example 1

What is the probability that a child is **9 years** old and **49 inches** tall?

$$p_{XY}(9, 49) = \mathbf{P}(X = 9, Y = 49) = \frac{\text{gray} \text{ magenta} \text{ green}}{10} = \frac{3}{10}$$

Joint Distributions (Discrete): Example 1

What is the probability that a child is 9 years old and 49 inches tall?

$$p_{XY}(9, 49) = \mathbf{P}(X = 9, Y = 49) = \frac{\text{gray} \text{ magenta} \text{ green}}{10} = \frac{3}{10}$$

$$p_{XY}(x, y)$$

Joint Distributions (Discrete): Example 1

What is the probability that a child is 9 years old and 49 inches tall?

$$p_{XY}(9, 49) = \mathbf{P}(X = 9, Y = 49) = \frac{\text{gray circle} \text{ pink circle} \text{ green circle}}{10} = \frac{3}{10}$$

$$p_{XY}(x, y) = \mathbf{P}(X = x, Y = y)$$

Joint Distributions: Example 1

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

Joint Distributions: Example 1



Age (years): 7 7 7 8 8 9 9 9 9 10

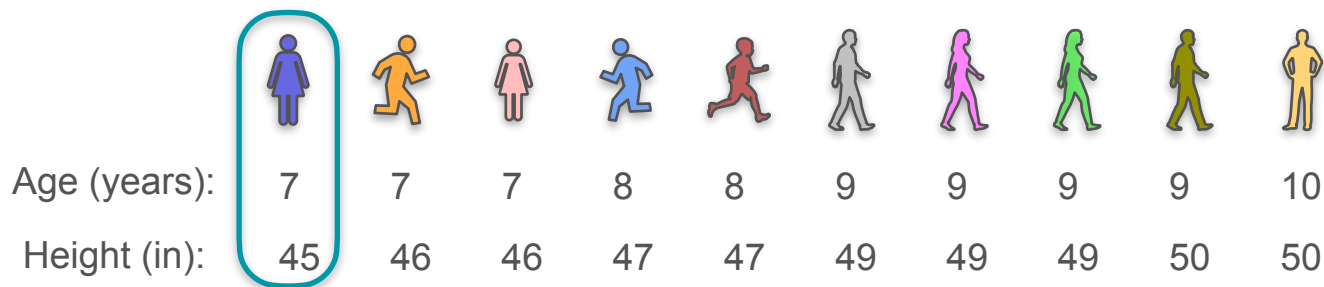
Height (in): 45 46 46 47 47 49 49 49 50 50

Height

Age

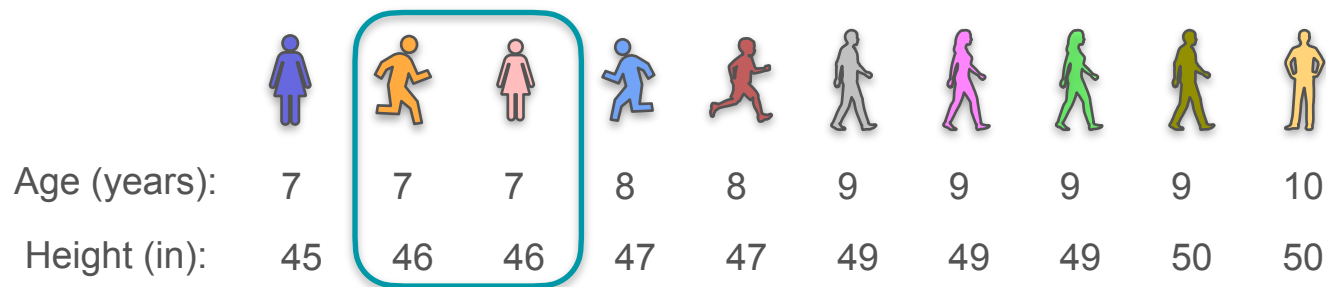
	45	46	47	48	49	50
7						
8						
9						
10						

Joint Distributions: Example 1



		Height					
Age		45	46	47	48	49	50
	7	1					
	8						
	9						
	10						

Joint Distributions: Example 1



		Height					
		45	46	47	48	49	50
Age	7	1	2				
	8						
	9						
	10						

Joint Distributions: Example 1



Age (years): 7 7 7 8 8 9 9 9 9 10

Height (in): 45 46 46 47 47 49 49 49 50 50

Height

Age

	45	46	47	48	49	50
7	1	2	0	0	0	0
8						
9						
10						

Joint Distributions: Example 1



Age (years): 7 7 7 8 8 9 9 9 9 10

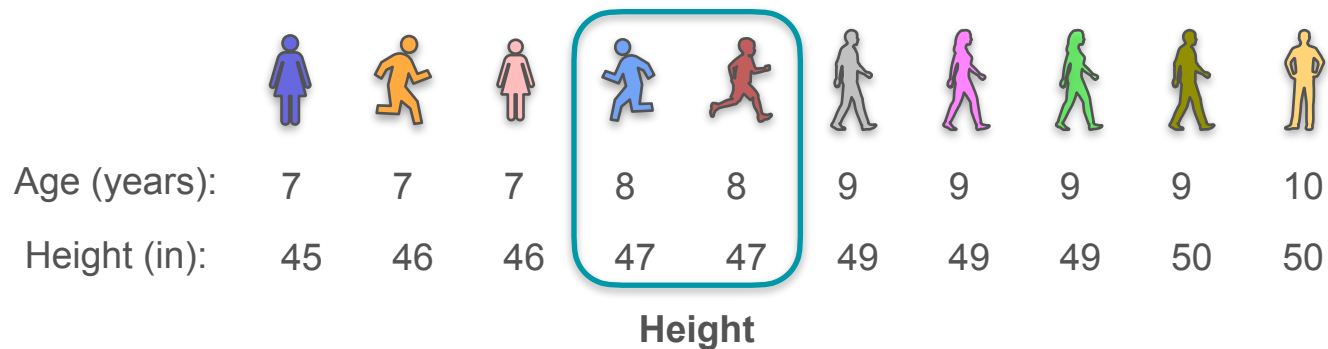
Height (in): 45 46 46 47 47 49 49 49 50 50

Height

Age

	45	46	47	48	49	50
7	1	2	0	0	0	0
8	0	0				
9						
10						

Joint Distributions: Example 1



Age		45	46	47	48	49	50
	7	1	2	0	0	0	0
	8	0	0	2			
	9						
	10						

Joint Distributions: Example 1



Age (years): 7 7 7 8 8 9 9 9 9 10

Height (in): 45 46 46 47 47 49 49 49 50 50

Height

Age

	45	46	47	48	49	50
7	1	2	0	0	0	0
8	0	0	2	0	0	0
9						
10						

Joint Distributions: Example 1



Age (years): 7 7 7 8 8 9 9 9 9 10

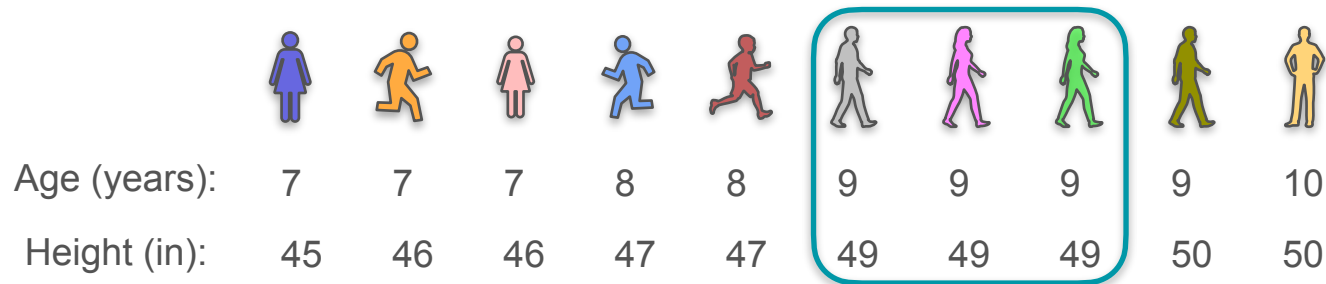
Height (in): 45 46 46 47 47 49 49 49 50 50

Height

Age

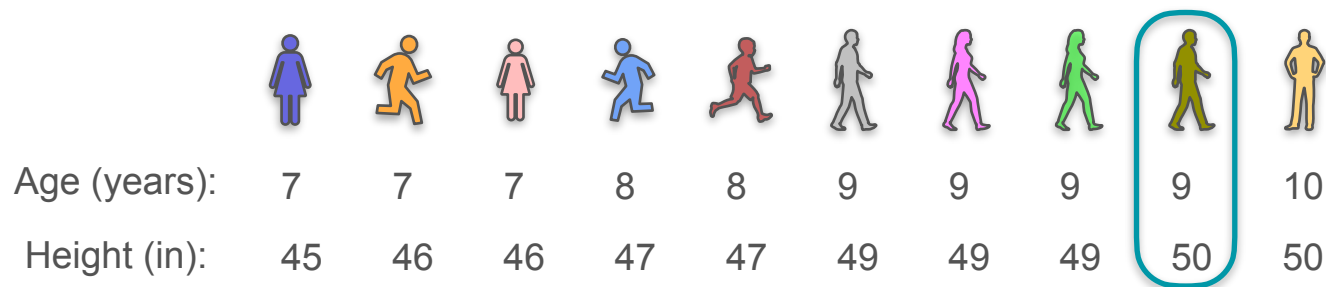
	45	46	47	48	49	50
7	1	2	0	0	0	0
8	0	0	2	0	0	0
9	0	0	0	0		
10						

Joint Distributions: Example 1



		Height					
Age		45	46	47	48	49	50
	7	1	2	0	0	0	0
	8	0	0	2	0	0	0
	9	0	0	0	0	3	
	10						

Joint Distributions: Example 1



		Height					
Age		45	46	47	48	49	50
	7	1	2	0	0	0	0
	8	0	0	2	0	0	0
	9	0	0	0	0	3	1
	10						

Joint Distributions: Example 1

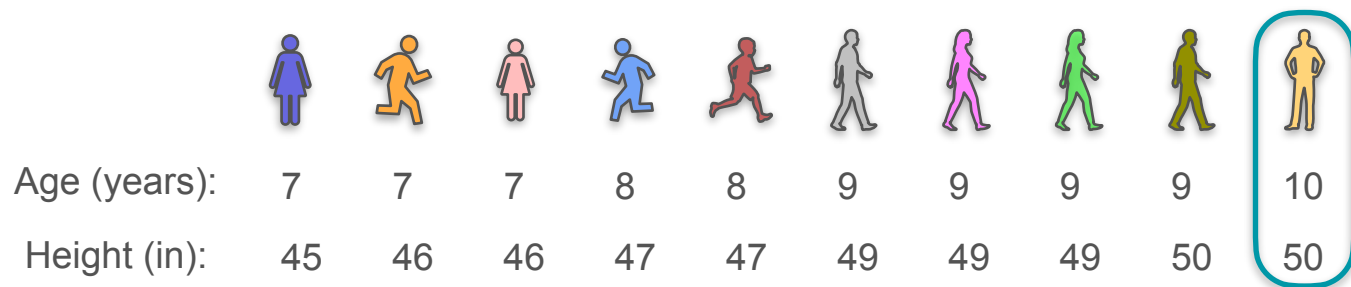


Age (years): 7 7 7 8 8 9 9 9 9 10

Height (in): 45 46 46 47 47 49 49 49 50 50

		Height					
Age		45	46	47	48	49	50
	7	1	2	0	0	0	0
	8	0	0	2	0	0	0
	9	0	0	0	0	3	1
	10	0	0	0	0	0	

Joint Distributions: Example 1



		Height					
Age		45	46	47	48	49	50
	7	1	2	0	0	0	0
	8	0	0	2	0	0	0
	9	0	0	0	0	3	1
	10	0	0	0	0	0	1

Joint Distributions: Example 1



Age (years): 7 7 7 8 8 9 9 9 9 10

Height (in): 45 46 46 47 47 49 49 49 50 50

		Height					
Age		45	46	47	48	49	50
	7	1	2	0	0	0	0
	8	0	0	2	0	0	0
	9	0	0	0	0	3	1
	10	0	0	0	0	0	1

Joint Distributions: Example 1

		Height					
		45	46	47	48	49	50
Age	7	1	2	0	0	0	0
	8	0	0	2	0	0	0
	9	0	0	0	0	3	1
	10	0	0	0	0	0	1
		Counts					

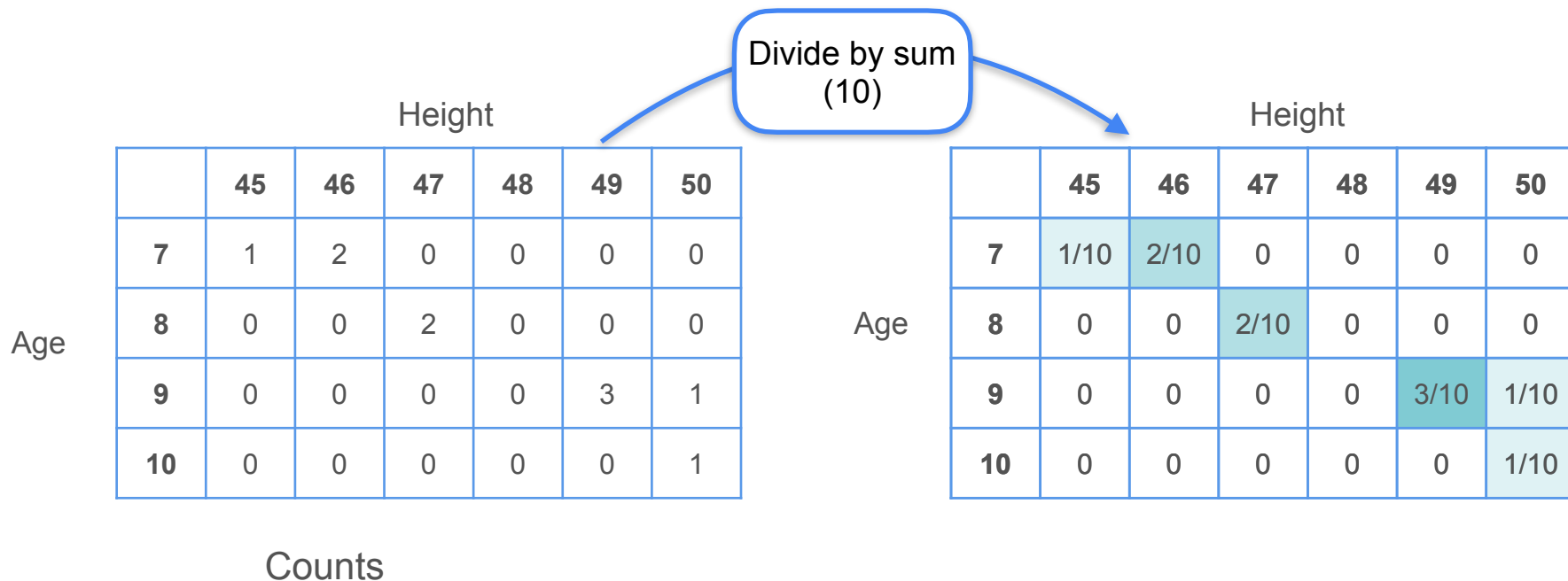
Joint Distributions: Example 1

Diagram illustrating the normalization of a joint distribution table. A blue box labeled "Divide by sum (10)" with arrows points from the left table to the right table.

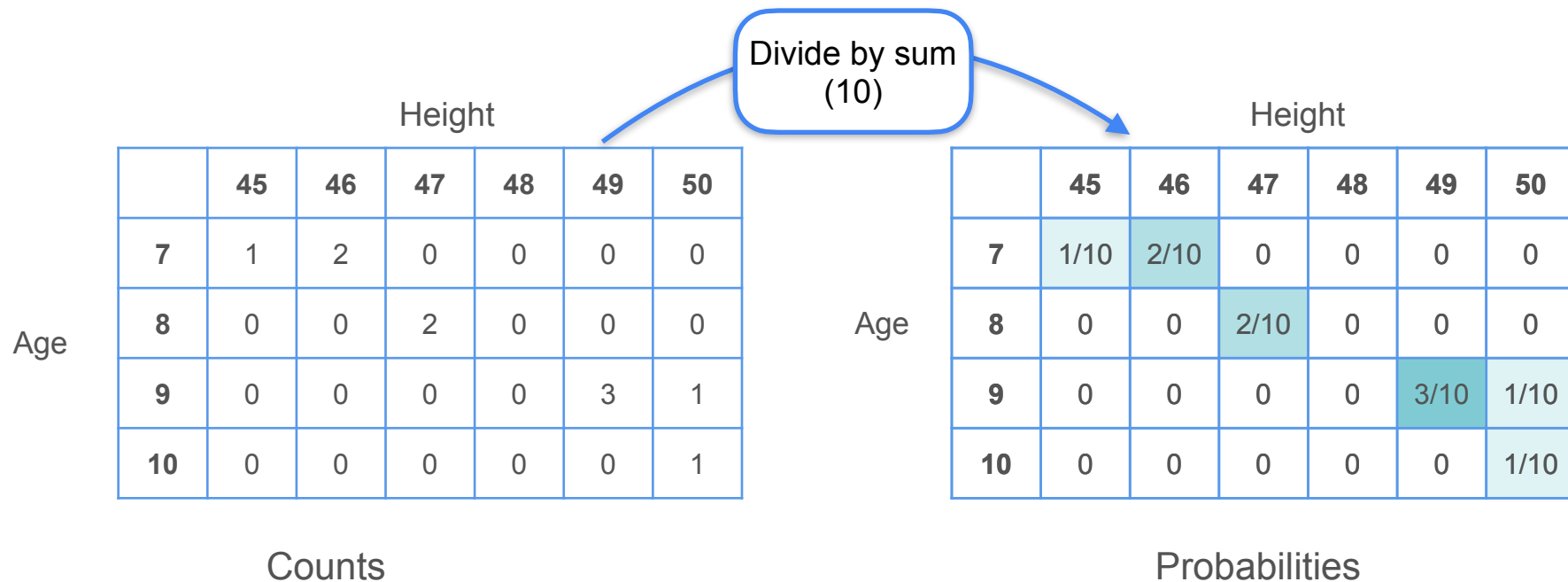
		Height					
Age		45	46	47	48	49	50
	7	1	2	0	0	0	0
	8	0	0	2	0	0	0
	9	0	0	0	0	3	1
	10	0	0	0	0	0	1
		Counts					

		Height					
Age		45	46	47	48	49	50
	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10

Joint Distributions: Example 1



Joint Distributions: Example 1



Joint Distributions: Example 1

		Height					
		45	46	47	48	49	50
Age	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
		Probabilities					

Joint Distributions: Example 1

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
		Probabilities					

Joint Distributions: Example 1

All probabilities for all possible combinations of X and Y

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
Probabilities							

Joint Distributions: Example 1

Joint Distribution

All probabilities for all possible combinations of X and Y

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10

Probabilities

Joint Distributions: Example 1

What is the probability that a child is 8 and 48 inches tall?

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10

Probabilities

Joint Distributions: Example 1

What is the probability that a child is 8 and 48 inches tall?

$$p_{XY}(x, y) = \mathbf{P}(X = x, Y = y)$$

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10

Probabilities

Joint Distributions: Example 1

What is the probability that a child is 8 and 48 inches tall?

$$p_{XY}(x, y) = \mathbf{P}(X = x, Y = y)$$

$$p_{XY}(8, 48) = \mathbf{P}(X = 8, Y = 48)$$

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10

Probabilities

Joint Distributions: Example 1

What is the probability that a child is 8 and 48 inches tall?

$$p_{XY}(x, y) = \mathbf{P}(X = x, Y = y)$$

$$p_{XY}(8, 48) = \mathbf{P}(X = 8, Y = 48)$$

Height (Y)

	45	46	47	48	49	50
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

Age (X)

Probabilities

Joint Distributions: Example 1

What is the probability that a child is 8 and 48 inches tall?

$$p_{XY}(x, y) = \mathbf{P}(X = x, Y = y)$$

$$p_{XY}(8, 48) = \mathbf{P}(X = 8, Y = 48)$$

$$p_{XY}(8, 48) = 0$$

Height (Y)

	45	46	47	48	49	50
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

Age (X)

Probabilities

Joint Distributions: Example 1

What is the probability that a child is 8 and 48 inches tall?

$$p_{XY}(x, y) = \mathbf{P}(X = x, Y = y)$$

$$p_{XY}(8, 48) = \mathbf{P}(X = 8, Y = 48)$$

$$p_{XY}(8, 48) = 0$$

$$p_{XY}(7, 46) =$$

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10

Probabilities

Joint Distributions: Example 1

What is the probability that a child is 8 and 48 inches tall?

$$p_{XY}(x, y) = \mathbf{P}(X = x, Y = y)$$

$$p_{XY}(8, 48) = \mathbf{P}(X = 8, Y = 48)$$

$$p_{XY}(8, 48) = 0$$

$$p_{XY}(7, 46) =$$

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
		Probabilities					

Joint Distributions: Example 1

What is the probability that a child is 8 and 48 inches tall?

$$p_{XY}(x, y) = \mathbf{P}(X = x, Y = y)$$

$$p_{XY}(8, 48) = \mathbf{P}(X = 8, Y = 48)$$

$$p_{XY}(8, 48) = 0$$

$$p_{XY}(7, 46) =$$

Height (Y)

	45	46	47	48	49	50
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

Age (X)

Probabilities

Joint Distributions: Example 1

What is the probability that a child is 8 and 48 inches tall?

$$p_{XY}(x, y) = \mathbf{P}(X = x, Y = y)$$

$$p_{XY}(8, 48) = \mathbf{P}(X = 8, Y = 48)$$

$$p_{XY}(8, 48) = 0$$

$$p_{XY}(7, 46) = \frac{2}{10}$$

Height (Y)

	45	46	47	48	49	50
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

Age (X)

Probabilities

Joint Distributions (Discrete): Example 2

Joint Distributions (Discrete): Example 2

X

Joint Distributions (Discrete): Example 2

X

the number rolled on the 1st dice

Joint Distributions (Discrete): Example 2

 X Y

the number rolled on the 1st dice

Joint Distributions (Discrete): Example 2

 X

the number rolled on the 1st dice

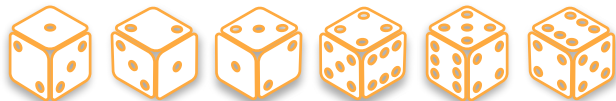
 Y

the number rolled on the 2nd dice

Joint Distributions (Discrete): Example 2

 X

the number rolled on the 1st dice

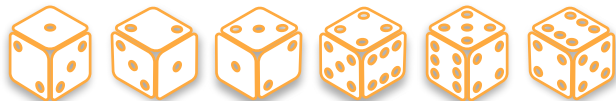
 Y

the number rolled on the 2nd dice

Joint Distributions (Discrete): Example 2

 X

the number rolled on the 1st dice



$$\frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6}$$

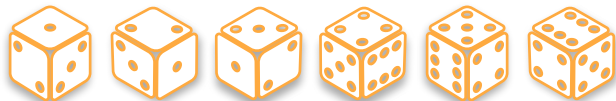
 Y

the number rolled on the 2nd dice

Joint Distributions (Discrete): Example 2

 X

the number rolled on the 1st dice



$$\frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6}$$

 Y

the number rolled on the 2nd dice



Joint Distributions (Discrete): Example 2

 X

the number rolled on the 1st dice



$$\frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6}$$

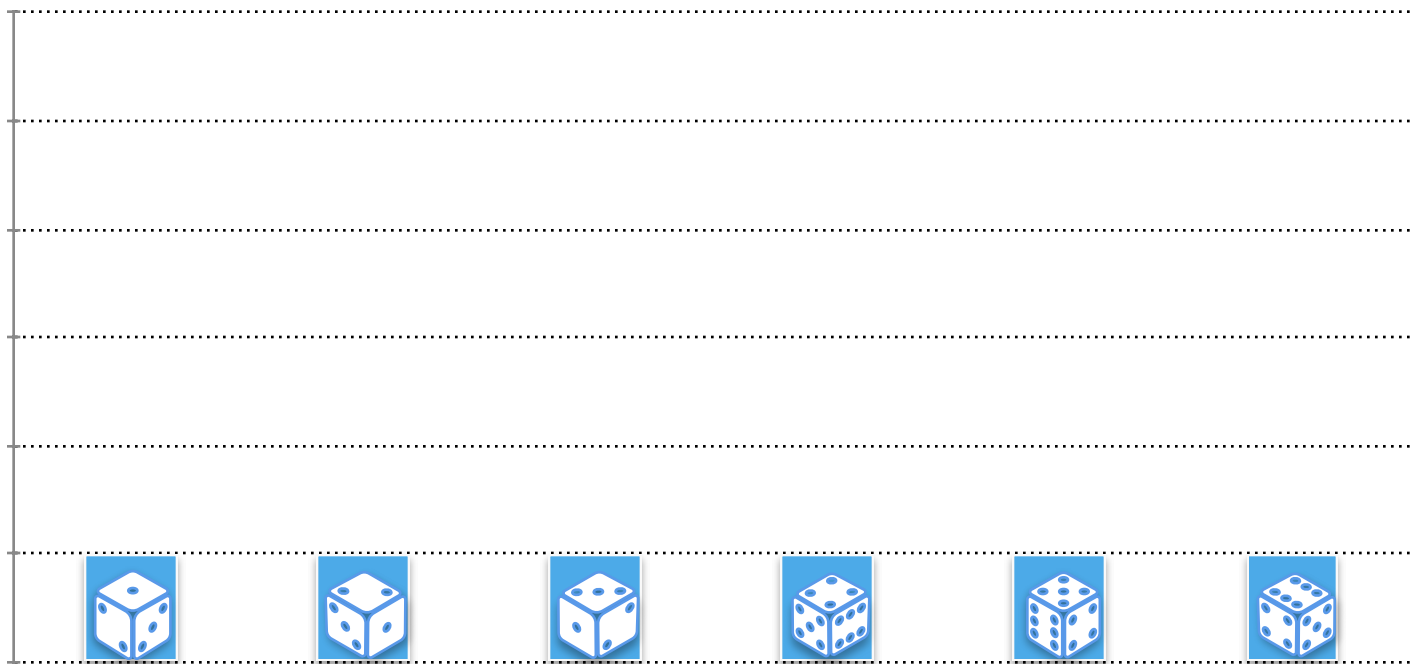
 Y

the number rolled on the 2nd dice

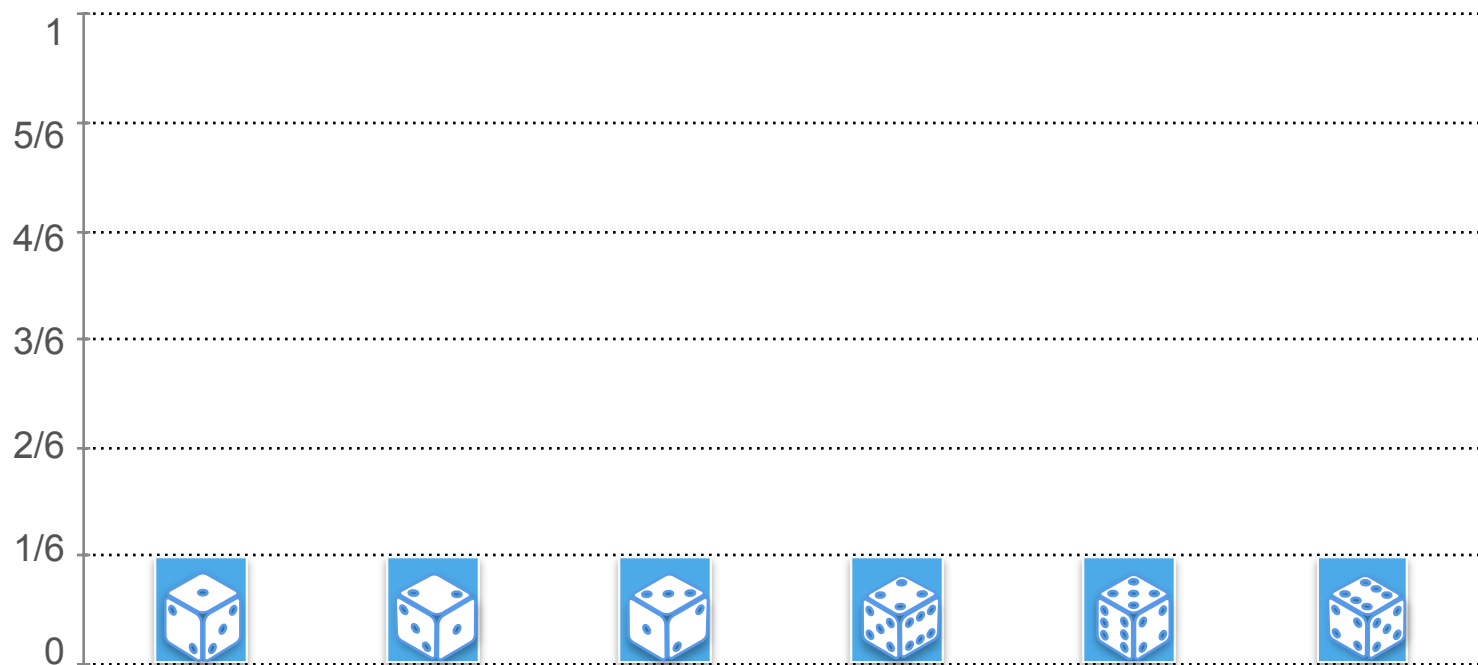


$$\frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6}$$

Joint Distributions: Example 2



Joint Distributions: Example 2



Joint Distributions: Example 2

Joint Distributions: Example 2

X

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

						
$P(x)$	1/6	1/6	1/6	1/6	1/6	1/6
$P(y)$	1/6	1/6	1/6	1/6	1/6	1/6

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

						
$P(x)$	1/6	1/6	1/6	1/6	1/6	1/6
$P(y)$	1/6	1/6	1/6	1/6	1/6	1/6

		Y					
		1	2	3	4	5	6
X	1						
	2						
	3						
	4						
	5						
	6						

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X and Y are independent



$P(x)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$P(y)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

		Y					
		1	2	3	4	5	6
X	1						
	2						
	3						
	4						
	5						
	6						

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X and Y are independent



$P(x)$ 1/6 1/6 1/6 1/6 1/6 1/6

$P(y)$ 1/6 1/6 1/6 1/6 1/6 1/6

$p_{XY}(\text{orange die}, \text{green die})$

		Y					
		1	2	3	4	5	6
X	1						
	2						
	3						
	4						
	5						
	6						

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X and Y are independent



$P(x)$ 1/6 1/6 1/6 1/6 1/6 1/6

$P(y)$ 1/6 1/6 1/6 1/6 1/6 1/6

$p_{XY}(\text{orange die}, \text{green die})$

		Y					
		1	2	3	4	5	6
X	1						
	2						
	3						
	4						
	5						
	6						

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X and Y are independent



$P(x)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$P(y)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$$p_{XY}(\text{orange die}, \text{green die}) = P(\text{orange die}) \cdot P(\text{green die})$$

		Y					
		1	2	3	4	5	6
X	1						
	2						
	3						
	4						
	5						
	6						

Joint Distributions: Example 2

X : the number rolled on the 1st die

Y : the number rolled on the 2nd die

X and Y are independent



$P(x)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$P(y)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$$p_{XY}(\text{orange die}, \text{green die}) = P(\text{orange die}) \cdot P(\text{green die}) = \frac{1}{6} \cdot \frac{1}{6}$$

		Y					
		1	2	3	4	5	6
X	1						
	2						
	3						
	4						
	5						
	6						

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X and Y are independent



$P(x)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$P(y)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$$p_{XY}(\text{orange die}, \text{green die}) = P(\text{orange die}) \cdot P(\text{green die}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

		Y					
		1	2	3	4	5	6
X	1						
	2						
	3						
	4						
	5						
	6						

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X and Y are independent



$P(x)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$P(y)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$$p_{XY}(\text{orange die}, \text{green die}) = P(\text{orange die}) \cdot P(\text{green die}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

		Y					
		1	2	3	4	5	6
X	1						
	2					1/36	
	3						
	4						
	5						
	6						

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X and Y are independent



$P(x)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$P(y)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$$p_{XY}(\text{orange die}, \text{green die}) = P(\text{orange die}) \cdot P(\text{green die}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

$$p_{XY}(x, y) = P(X = x, Y = y)$$

		Y					
		1	2	3	4	5	6
X	1						
	2					1/36	
	3						
	4						
	5						
	6						

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X and Y are independent



$P(x)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$P(y)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$$p_{XY}(\text{orange die}, \text{green die}) = P(\text{orange die}) \cdot P(\text{green die}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

$$p_{XY}(x, y) = P(X = x, Y = y) = \frac{1}{36}$$

		Y					
		1	2	3	4	5	6
X	1						
	2					1/36	
	3						
	4						
	5						
	6						

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X and Y are independent



$P(x)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$P(y)$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$$p_{XY}(\text{orange die}, \text{green die}) = P(\text{orange die}) \cdot P(\text{green die}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

$$p_{XY}(x, y) = P(X = x, Y = y) = \frac{1}{36}$$

		Y					
		1	2	3	4	5	6
X	1						
	2					1/36	
	3						
	4						
	5						
	6						

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X and Y are independent



$P(x)$ 1/6 1/6 1/6 1/6 1/6 1/6

$P(y)$ 1/6 1/6 1/6 1/6 1/6 1/6

$$p_{XY}(\text{orange die}, \text{green die}) = P(\text{orange die}) \cdot P(\text{green die}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

$$p_{XY}(x, y) = P(X = x, Y = y) = \frac{1}{36}$$

X	Y					
	1	2	3	4	5	6
1	1/36	1/36	1/36	1/36	1/36	1/36
2	1/36	1/36	1/36	1/36	1/36	1/36
3	1/36	1/36	1/36	1/36	1/36	1/36
4	1/36	1/36	1/36	1/36	1/36	1/36
5	1/36	1/36	1/36	1/36	1/36	1/36
6	1/36	1/36	1/36	1/36	1/36	1/36

Joint Distributions: Example 2

Joint Distributions: Example 2

$$p_{XY}(x, y) = \mathbf{P}(X = x, Y = y) = \mathbf{P}(x) \cdot \mathbf{P}(y)$$

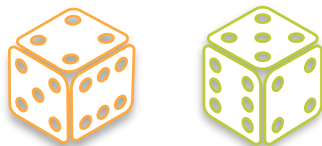
Joint Distributions: Example 2

Thus for independent discrete random variables:

$$p_{XY}(x, y) = \mathbf{P}(X = x, Y = y) = \mathbf{P}(x) \cdot \mathbf{P}(y)$$

Joint Distributions (Discrete): Example 2

Joint Distributions (Discrete): Example 2



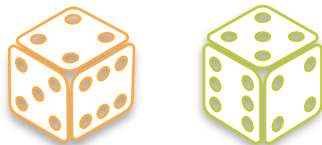
Joint Distributions (Discrete): Example 2

X



Joint Distributions (Discrete): Example 2

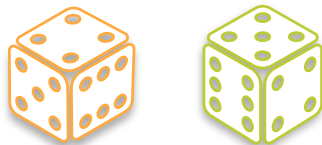
X



the number rolled on the 1st dice

Joint Distributions (Discrete): Example 2

X

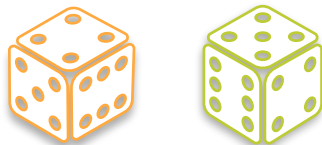


the number rolled on the 1st dice



Joint Distributions (Discrete): Example 2

X



the number rolled on the 1st dice



$X = 4$

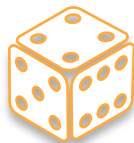
Joint Distributions (Discrete): Example 2

X



Y

the number rolled on the 1st dice



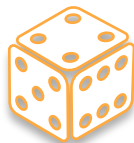
$$X = 4$$

Joint Distributions (Discrete): Example 2

 X  Y

the number rolled on the 1st dice

sum of the two dice



$$X = 4$$

Joint Distributions (Discrete): Example 2

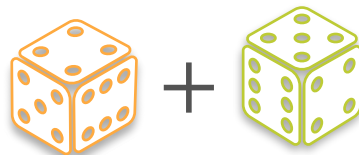
 X  Y

the number rolled on the 1st dice

sum of the two dice



$$X = 4$$



$$Y = 4 + 5$$

Joint Distributions: Example 2

Joint Distributions: Example 2

X

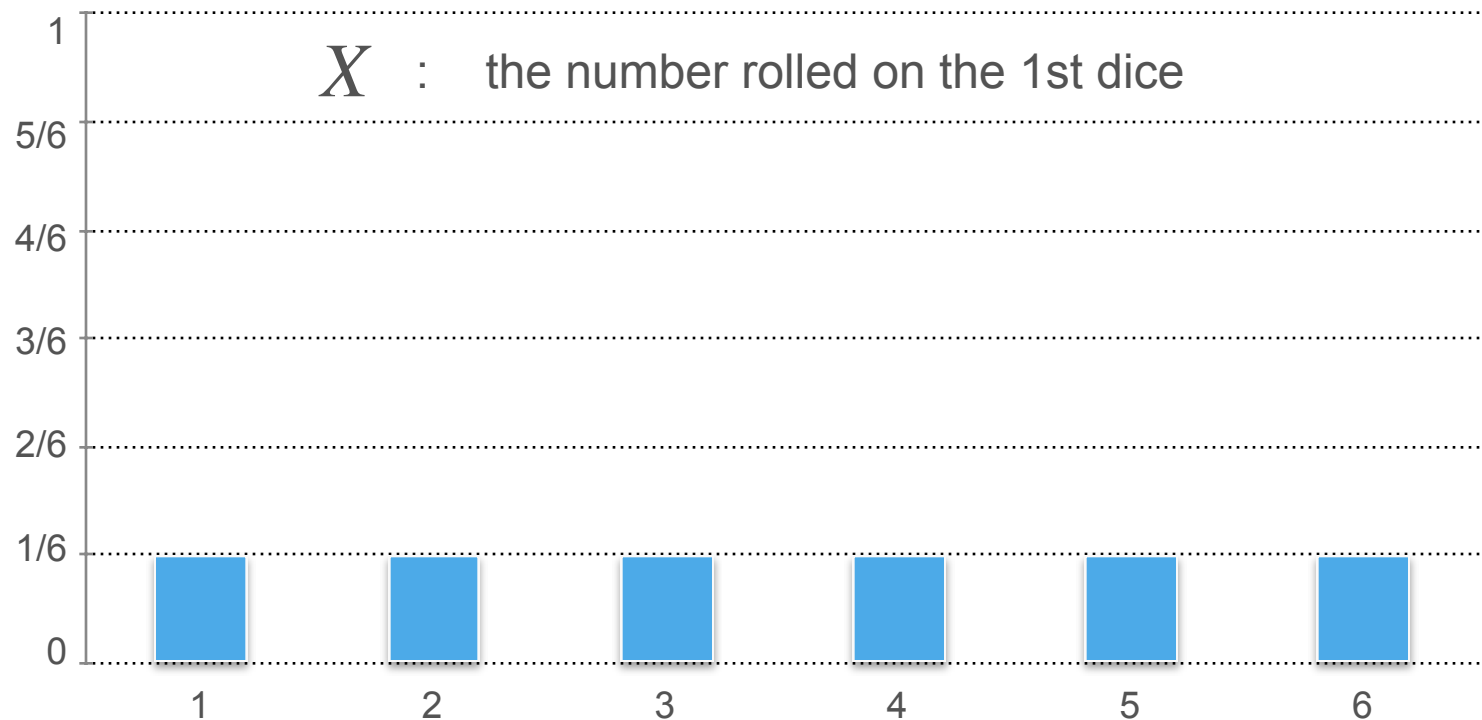
Joint Distributions: Example 2

X the number rolled on the 1st dice

Joint Distributions: Example 2

X : the number rolled on the 1st dice

Joint Distributions: Example 2



Joint Distributions - Example 3

Y : Sum of both dice

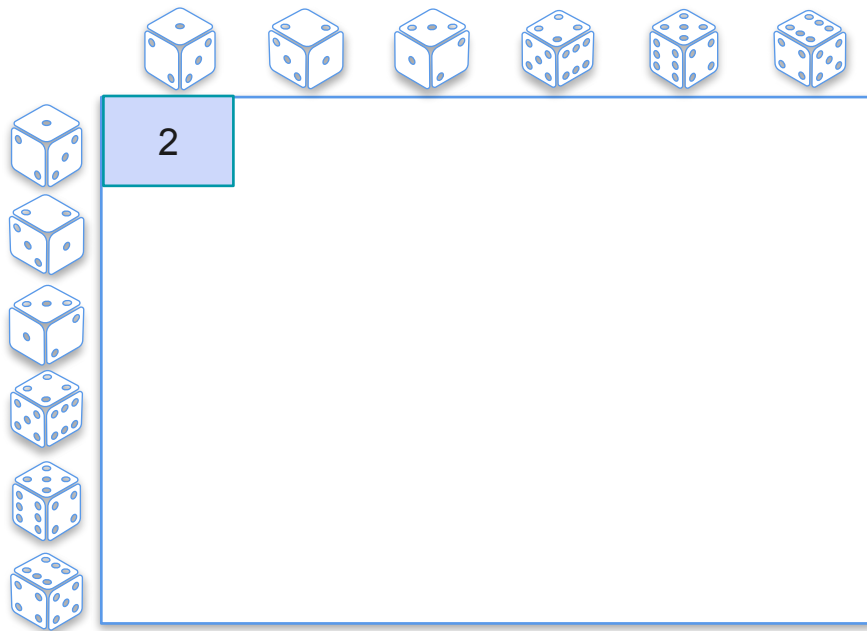
Joint Distributions - Example 3

Y : Sum of both dice



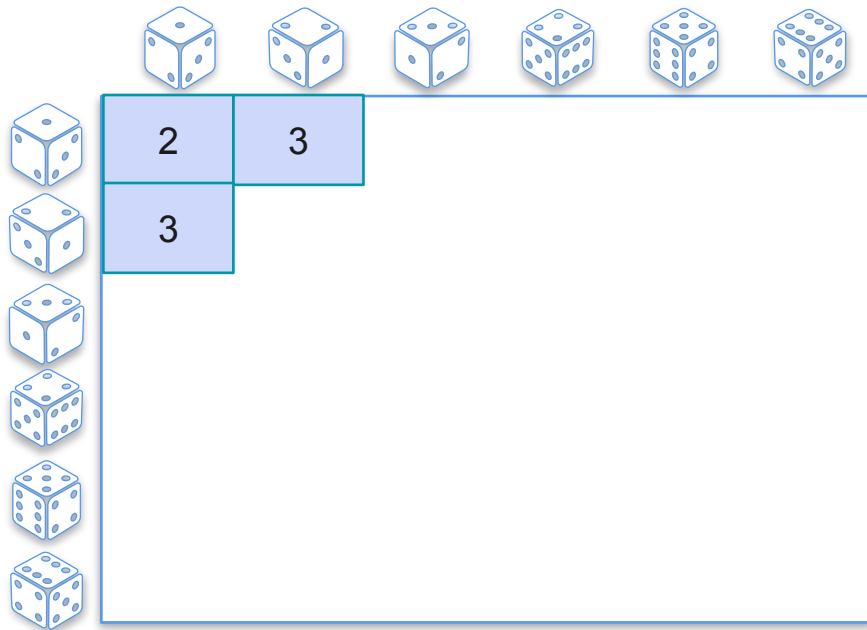
Joint Distributions - Example 3

Y : Sum of both dice



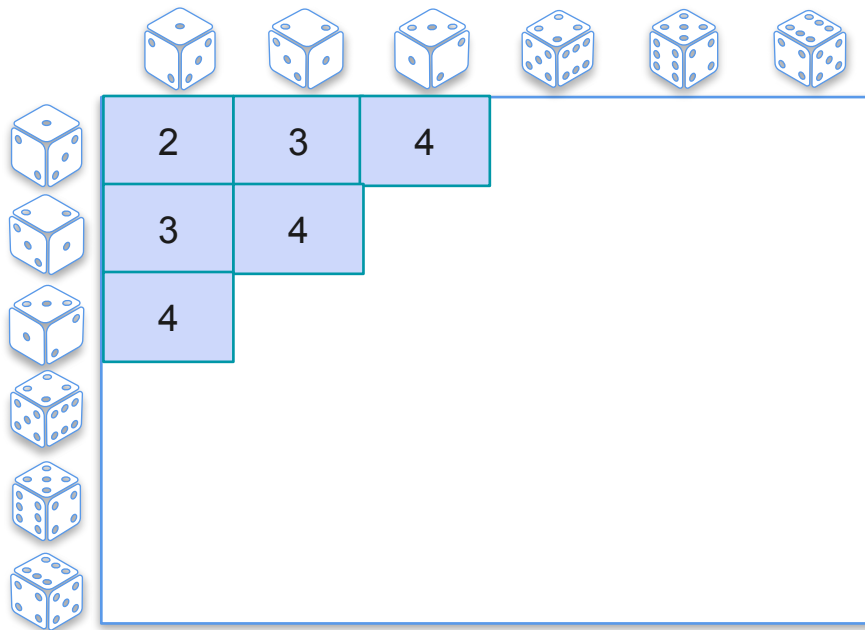
Joint Distributions - Example 3

Y : Sum of both dice



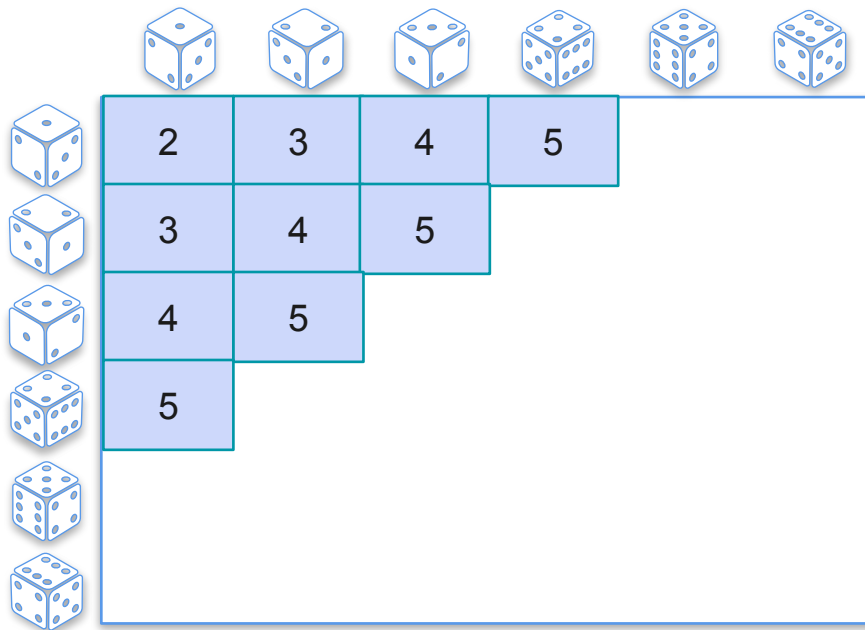
Joint Distributions - Example 3

Y : Sum of both dice



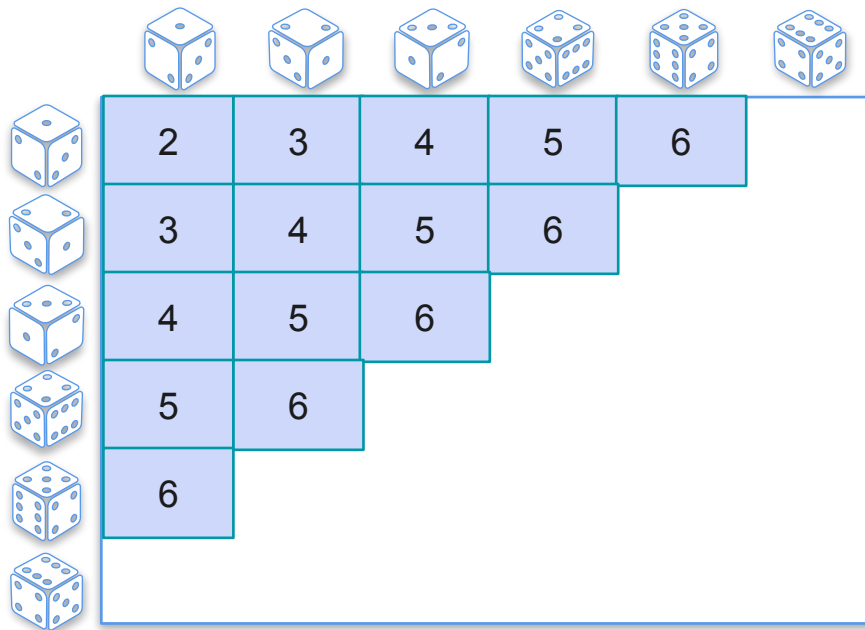
Joint Distributions - Example 3

Y : Sum of both dice



Joint Distributions - Example 3

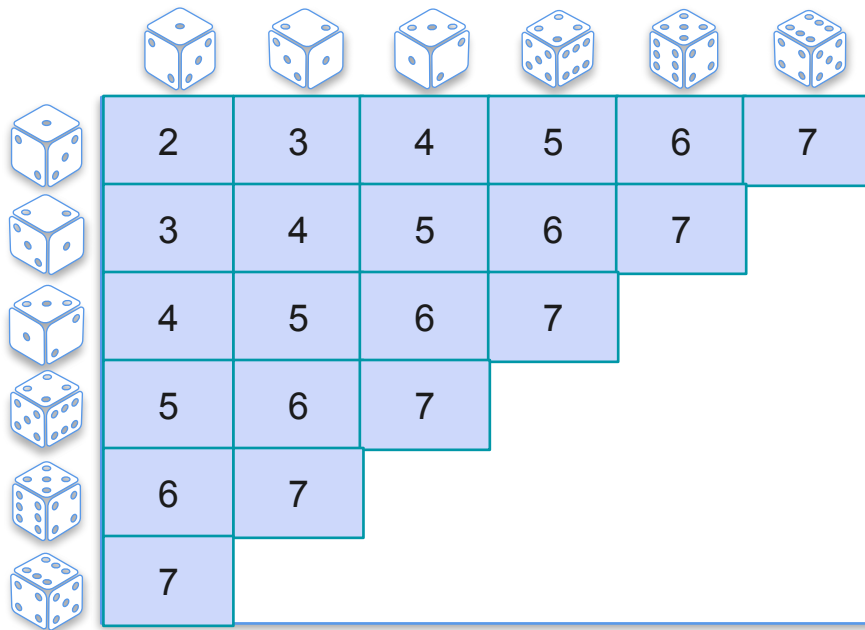
Y : Sum of both dice



						
	2	3	4	5	6	
	3	4	5	6		
	4	5	6			
	5	6				
	6					
						

Joint Distributions - Example 3

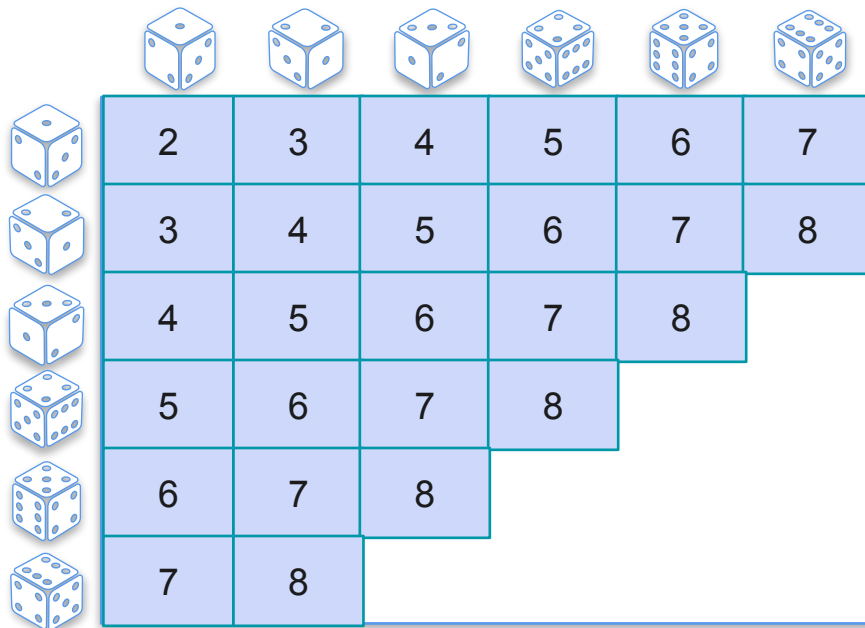
Y : Sum of both dice



						
	2	3	4	5	6	7
	3	4	5	6	7	
	4	5	6	7		
	5	6	7			
	6	7				
	7					

Joint Distributions - Example 3

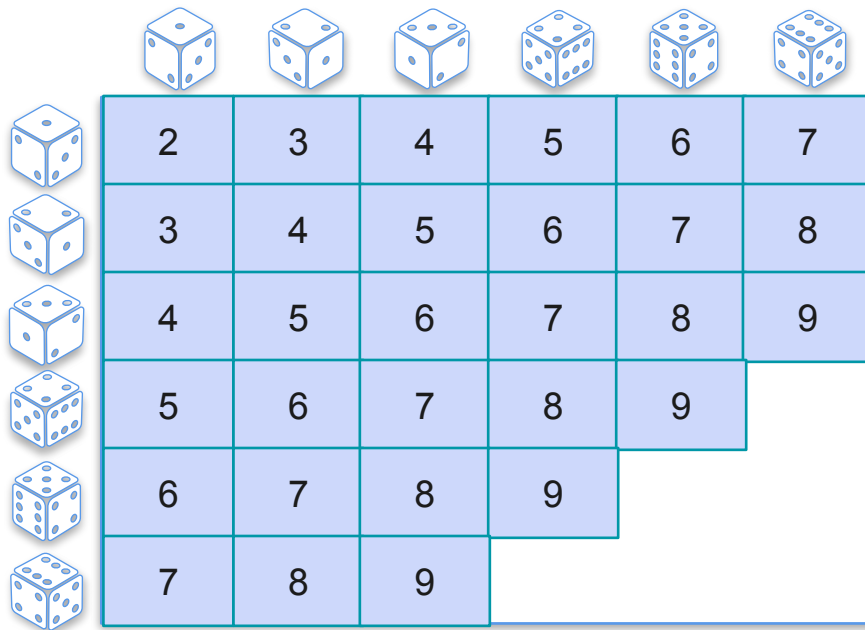
Y : Sum of both dice



						
	2	3	4	5	6	7
	3	4	5	6	7	8
	4	5	6	7	8	
	5	6	7	8		
	6	7	8			
	7	8				

Joint Distributions - Example 3

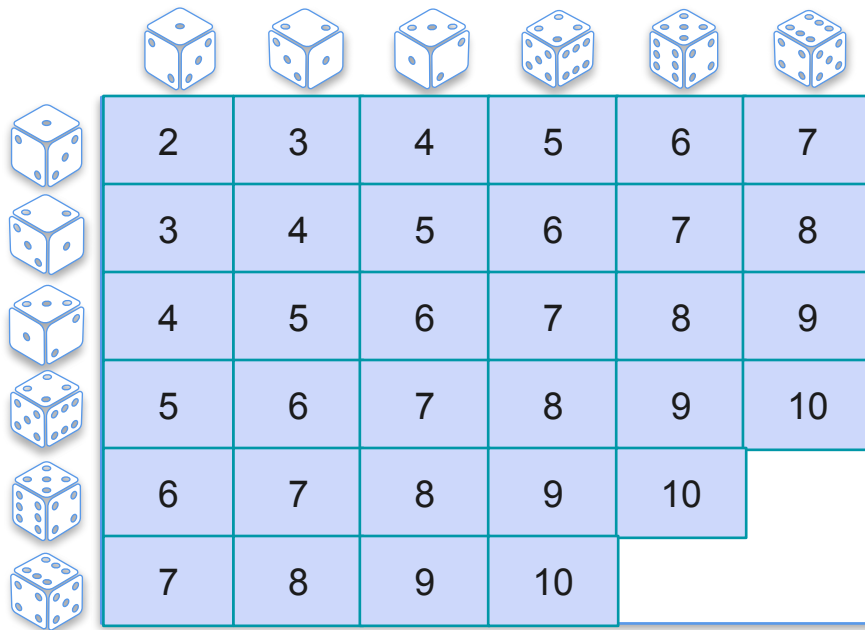
Y : Sum of both dice



						
	2	3	4	5	6	7
	3	4	5	6	7	8
	4	5	6	7	8	9
	5	6	7	8	9	
	6	7	8	9		
	7	8	9			

Joint Distributions - Example 3

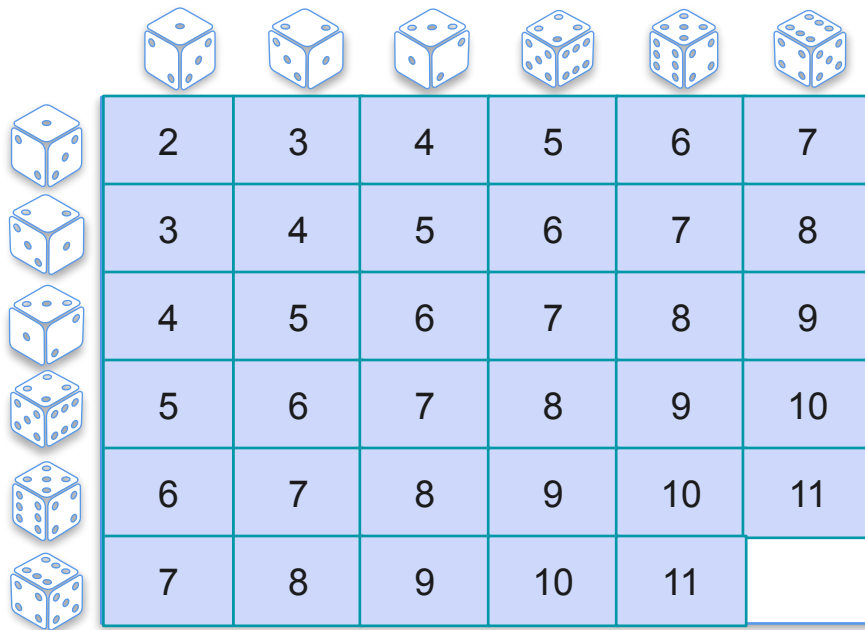
Y : Sum of both dice



						
	2	3	4	5	6	7
	3	4	5	6	7	8
	4	5	6	7	8	9
	5	6	7	8	9	10
	6	7	8	9	10	
	7	8	9	10		

Joint Distributions - Example 3

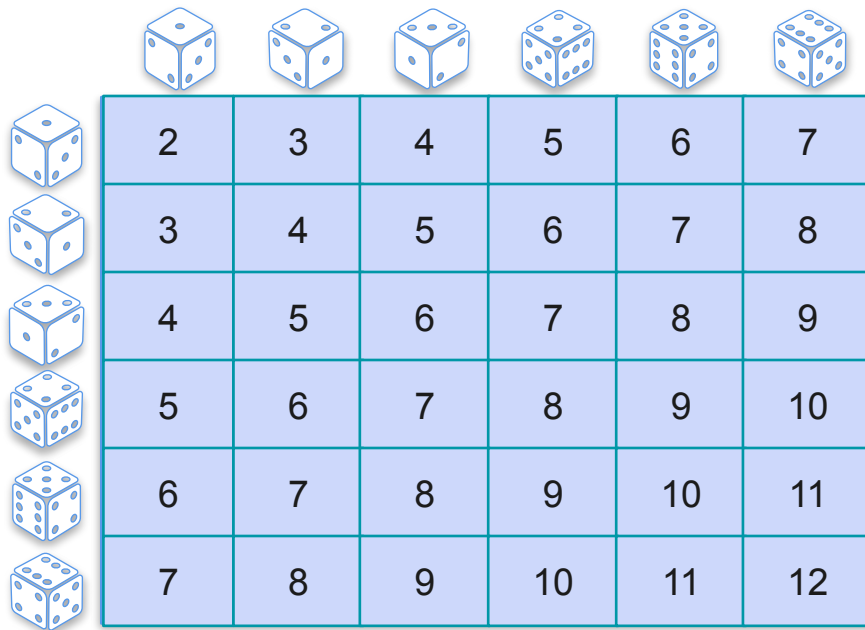
Y : Sum of both dice



						
	2	3	4	5	6	7
	3	4	5	6	7	8
	4	5	6	7	8	9
	5	6	7	8	9	10
	6	7	8	9	10	11
	7	8	9	10	11	

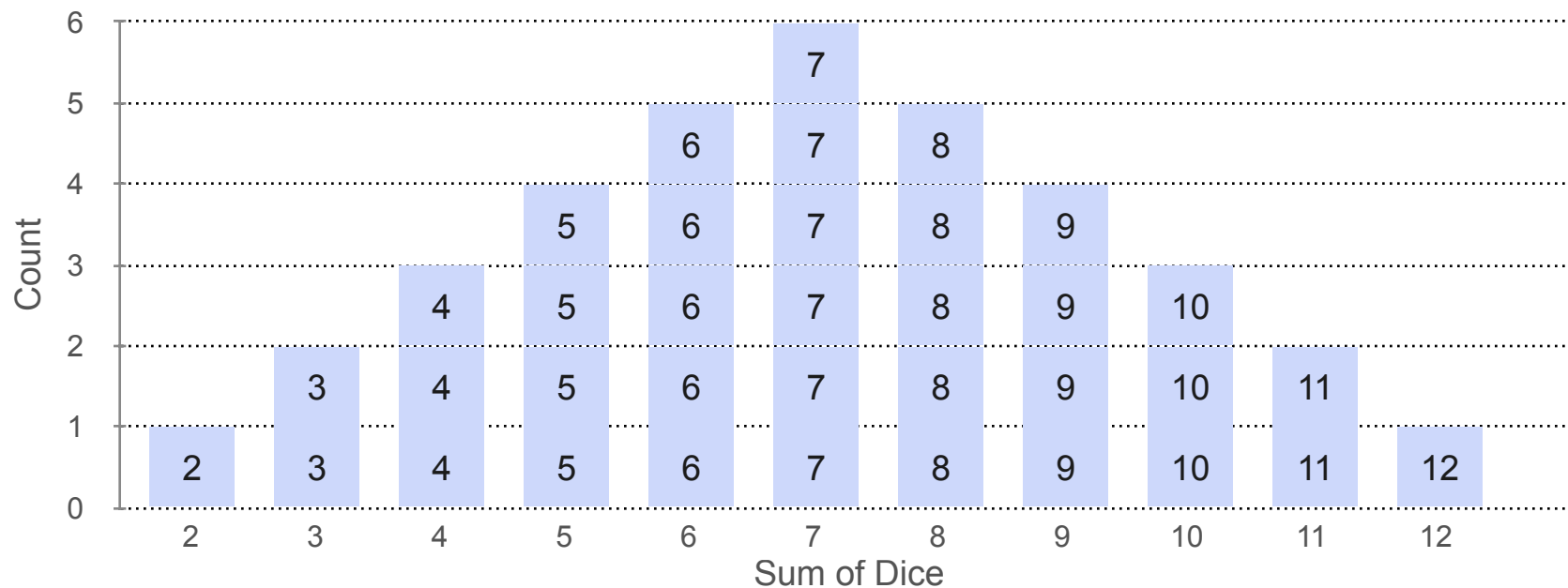
Joint Distributions - Example 3

Y : Sum of both dice

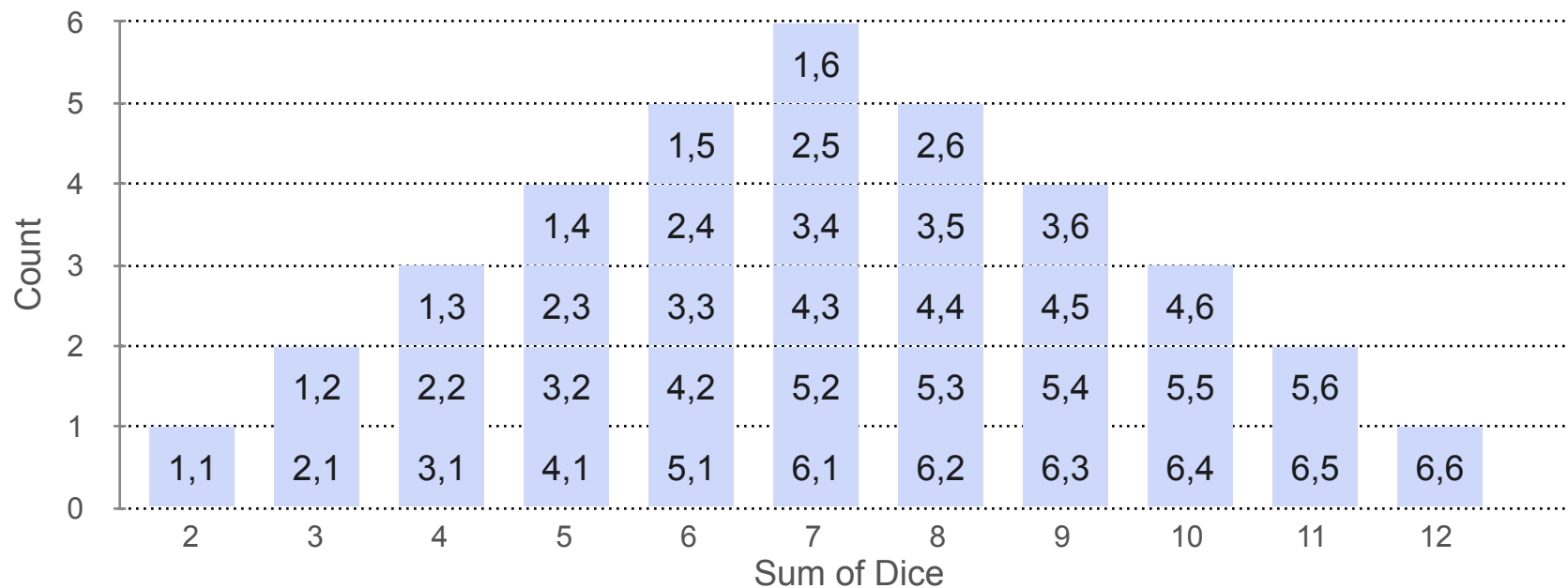


						
	2	3	4	5	6	7
	3	4	5	6	7	8
	4	5	6	7	8	9
	5	6	7	8	9	10
	6	7	8	9	10	11
	7	8	9	10	11	12

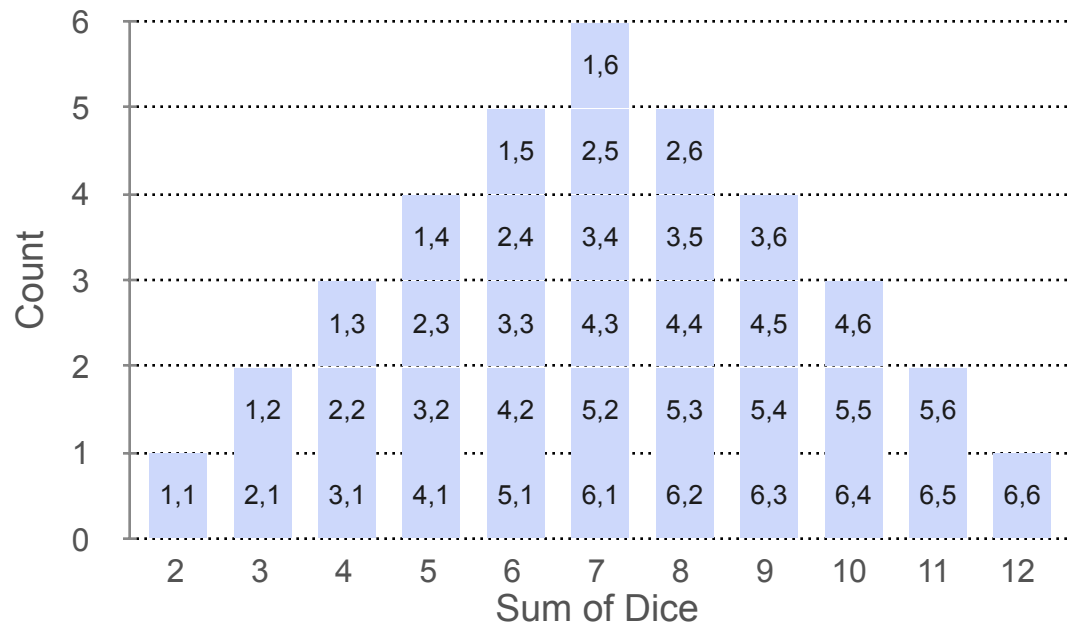
Joint Distributions: Example 3



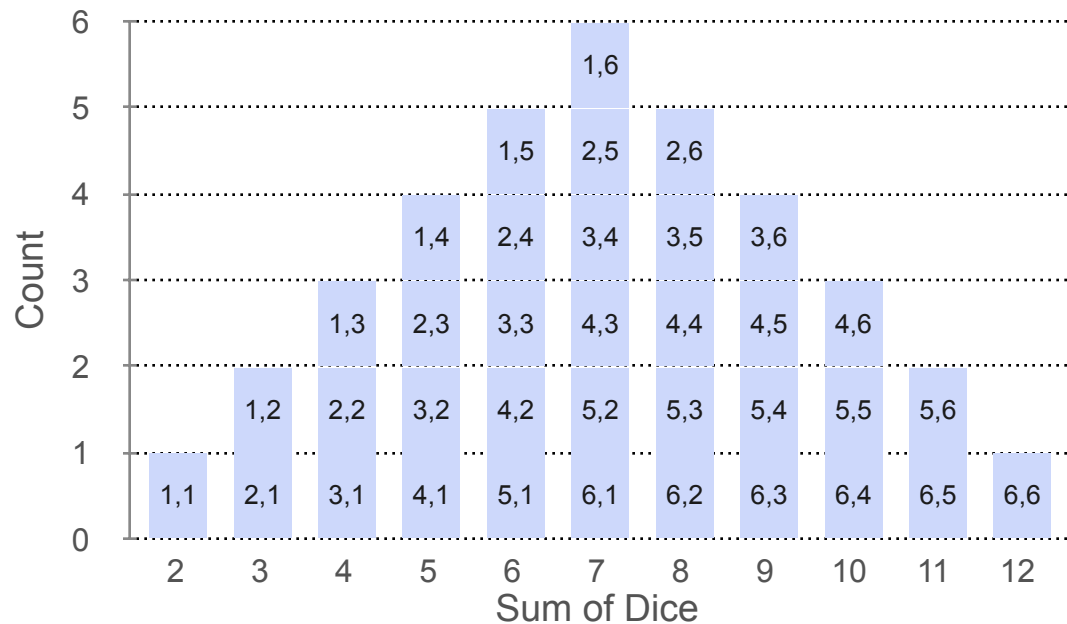
Joint Distributions: Example 3



Joint Distributions: Example 3

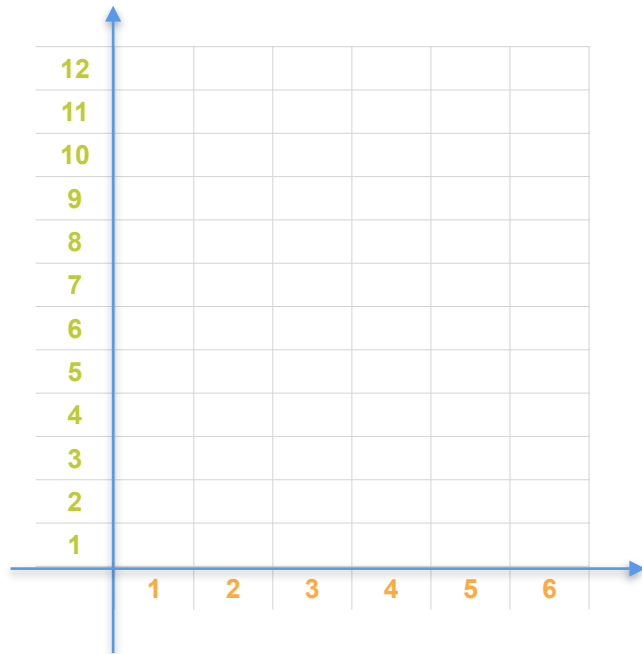
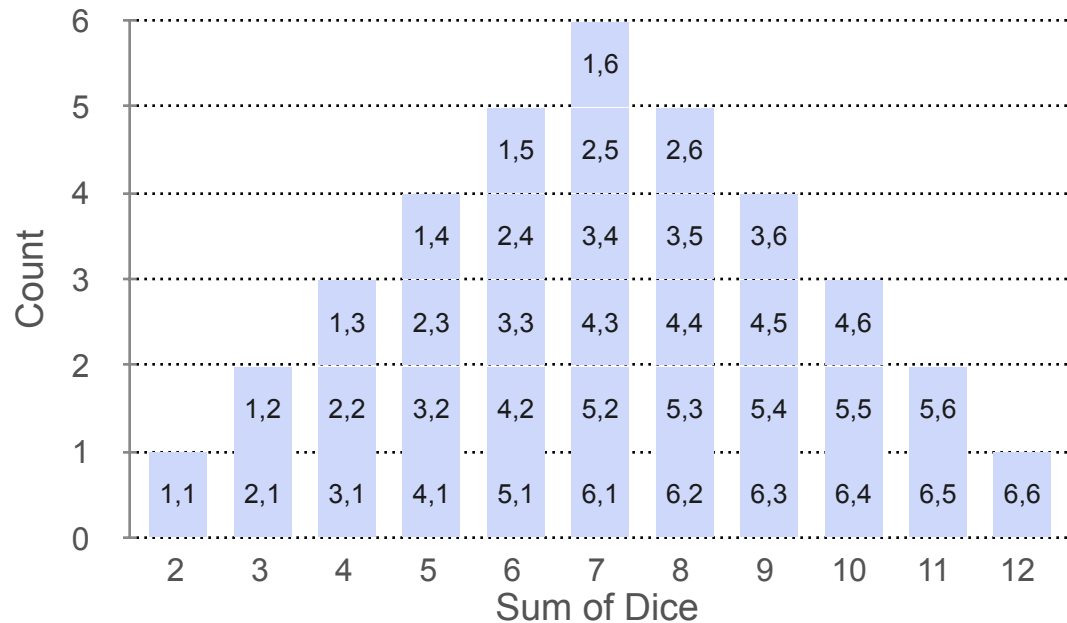


Joint Distributions: Example 3

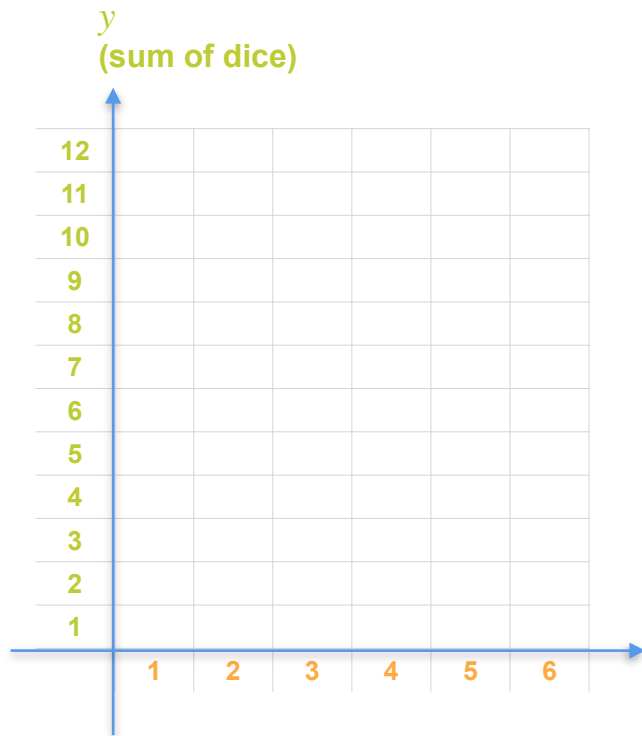
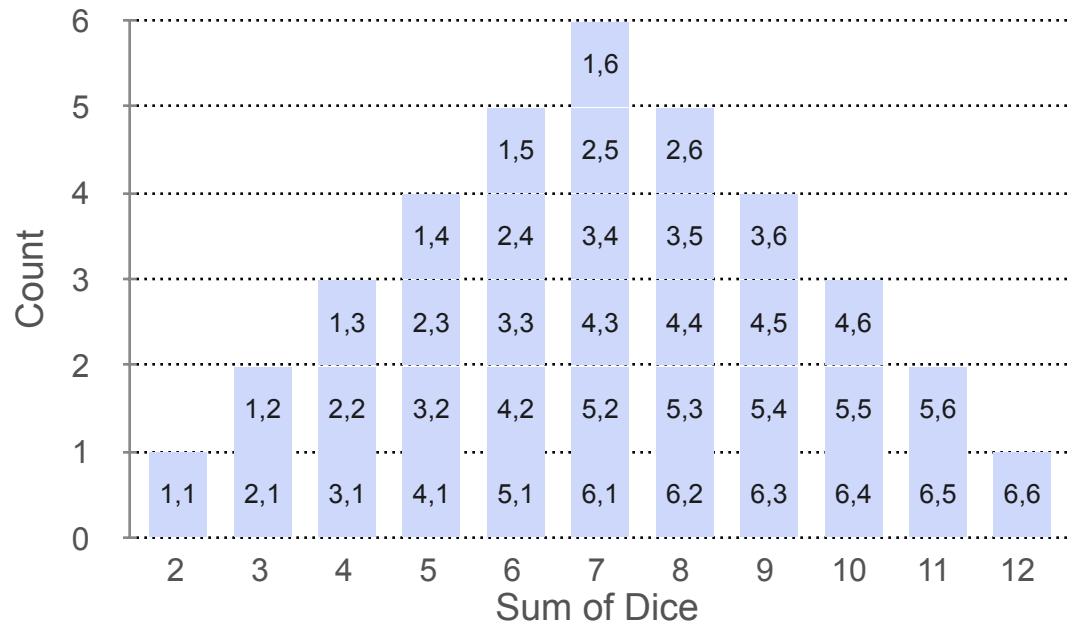


12						
11						
10						
9						
8						
7						
6						
5						
4						
3						
2						
1						
	1	2	3	4	5	6

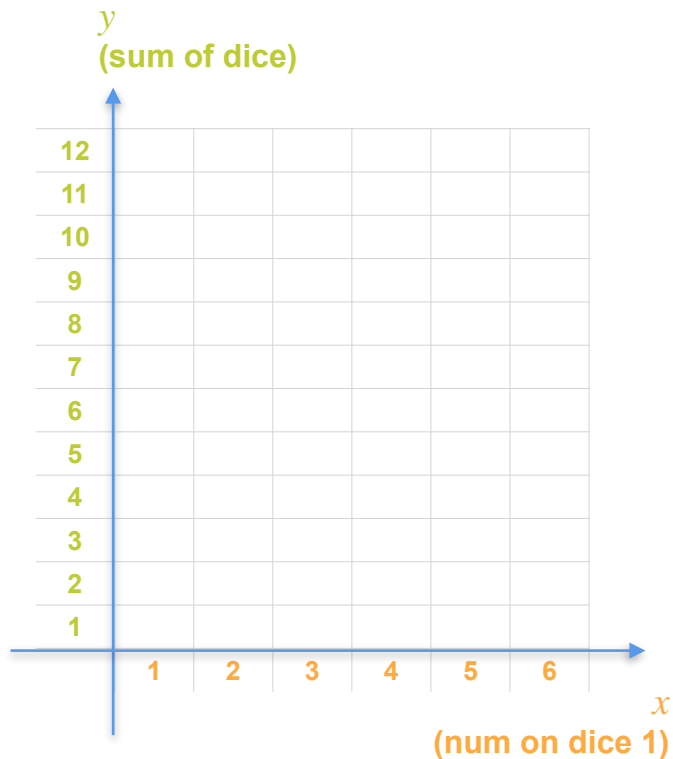
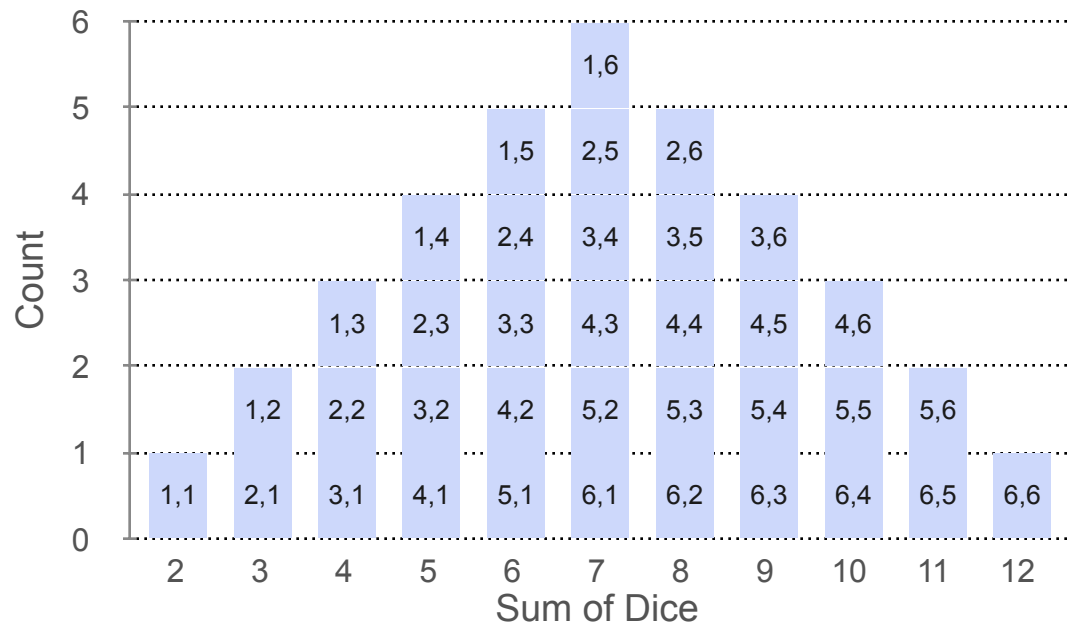
Joint Distributions: Example 3



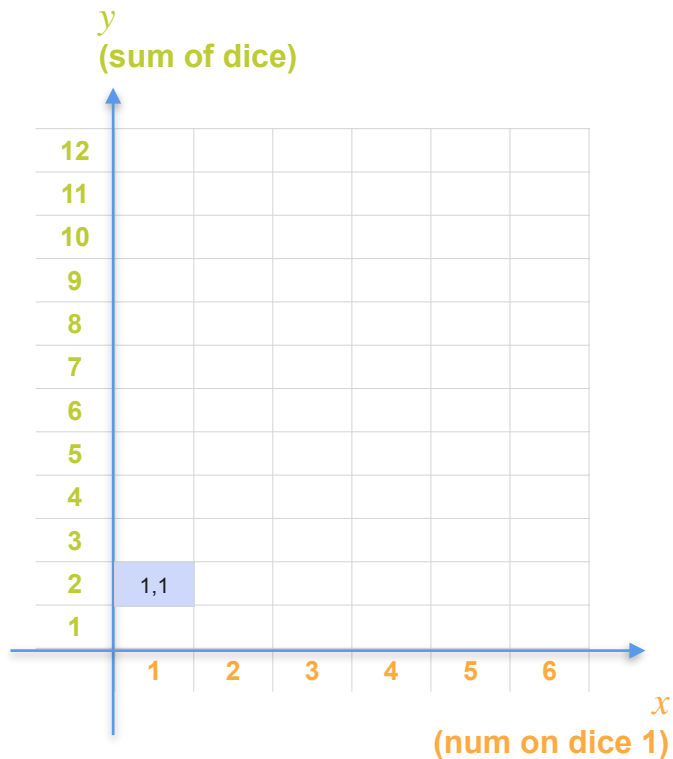
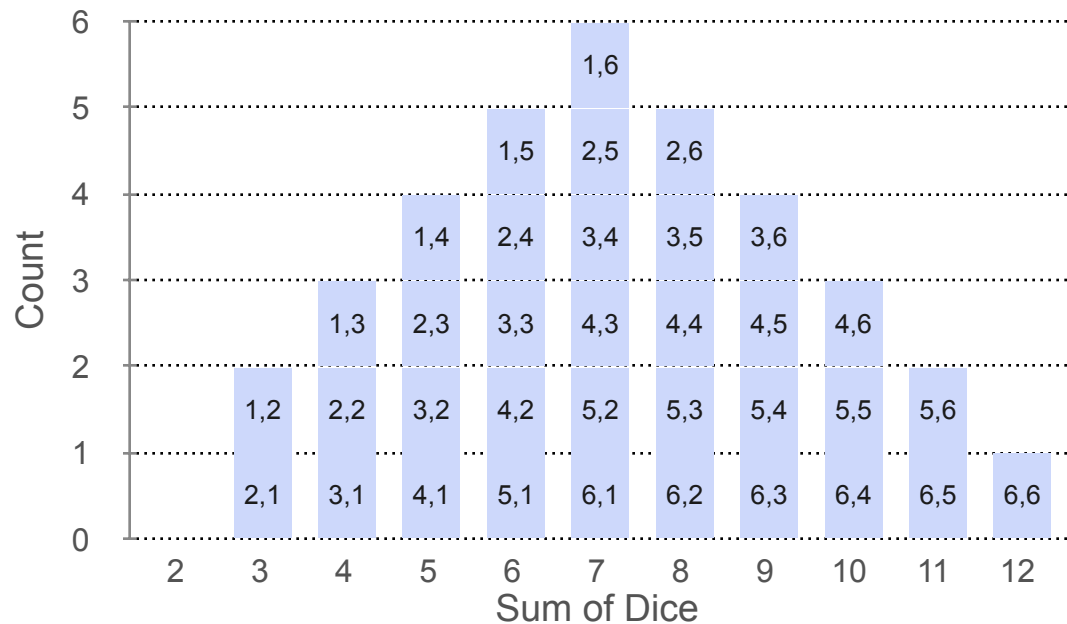
Joint Distributions: Example 3



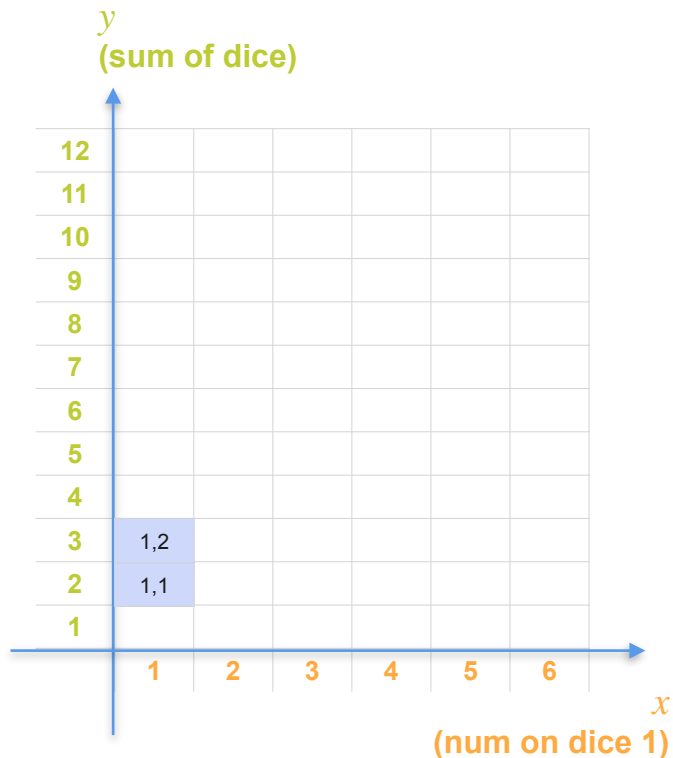
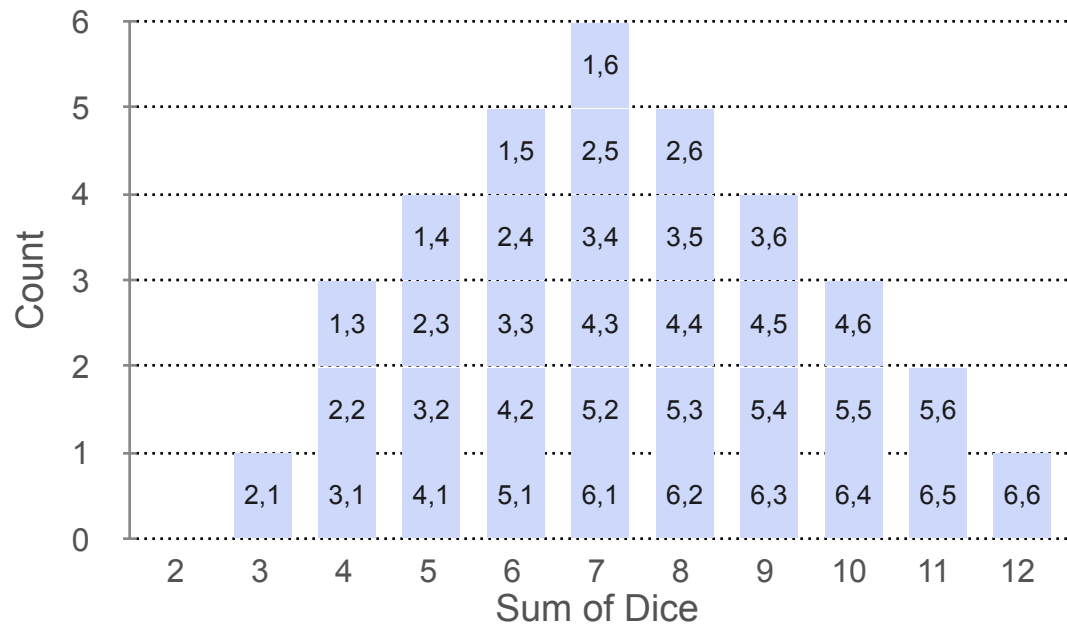
Joint Distributions: Example 3



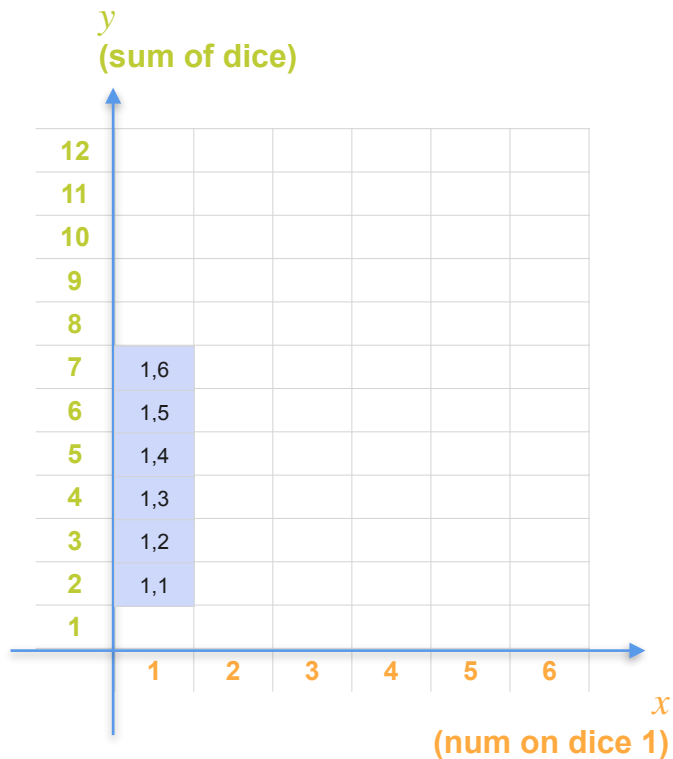
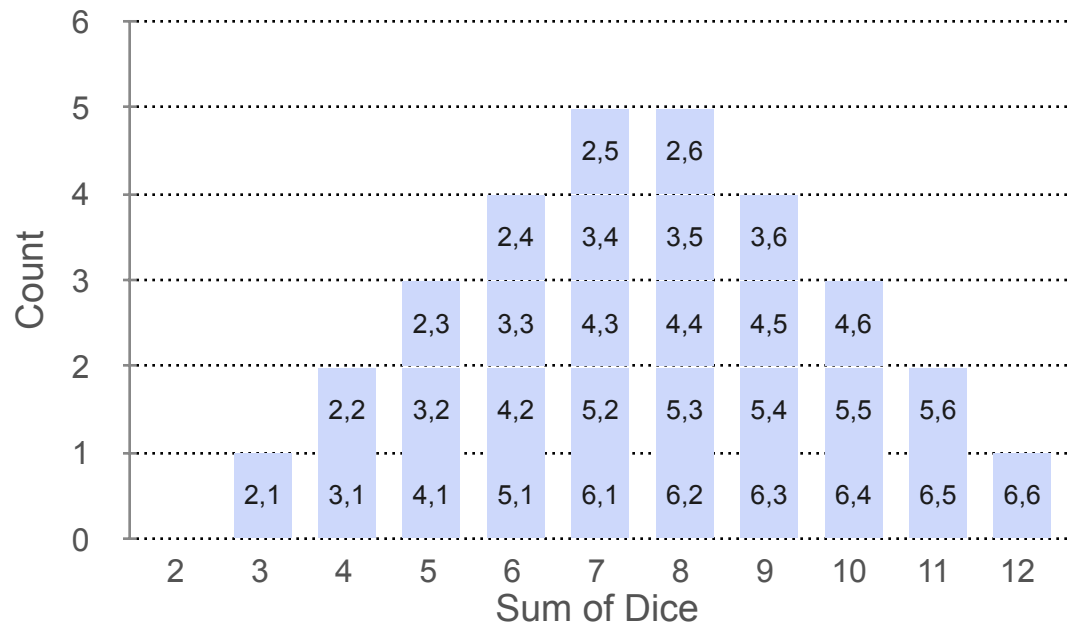
Joint Distributions: Example 3



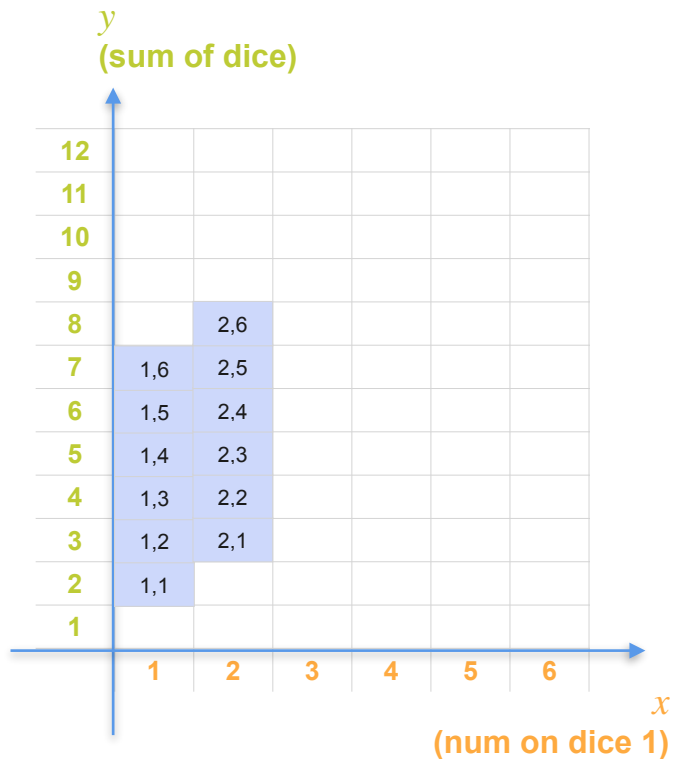
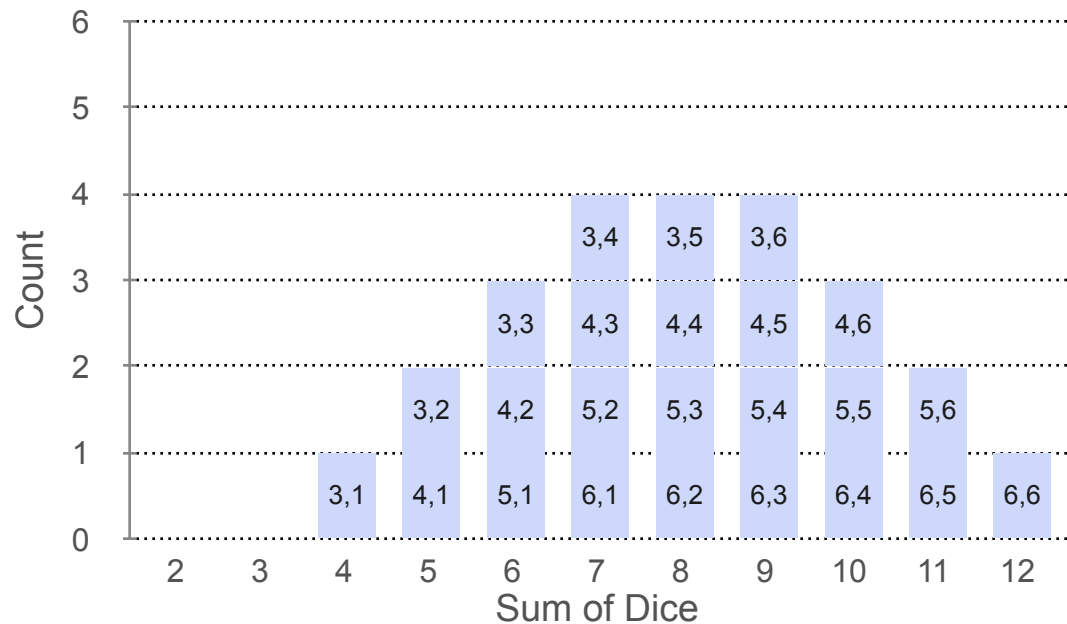
Joint Distributions: Example 3



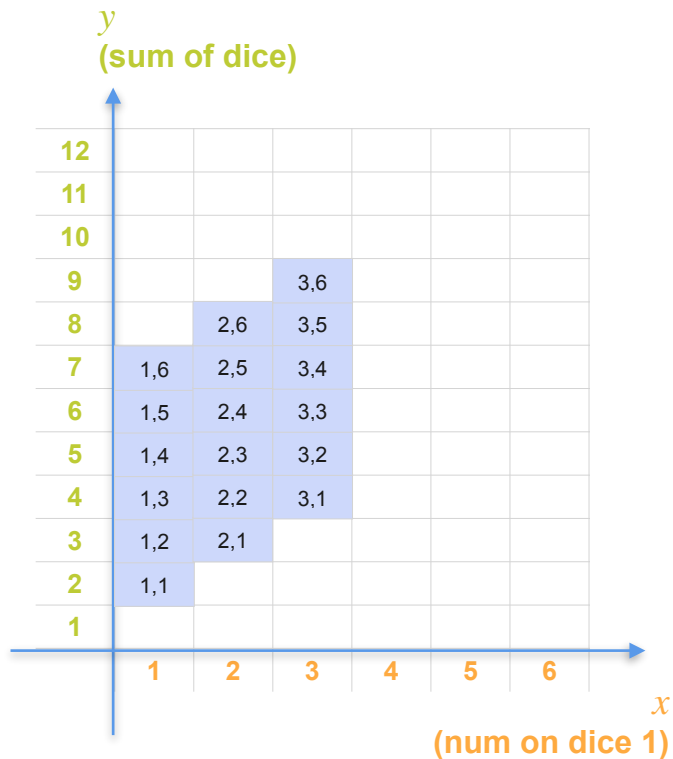
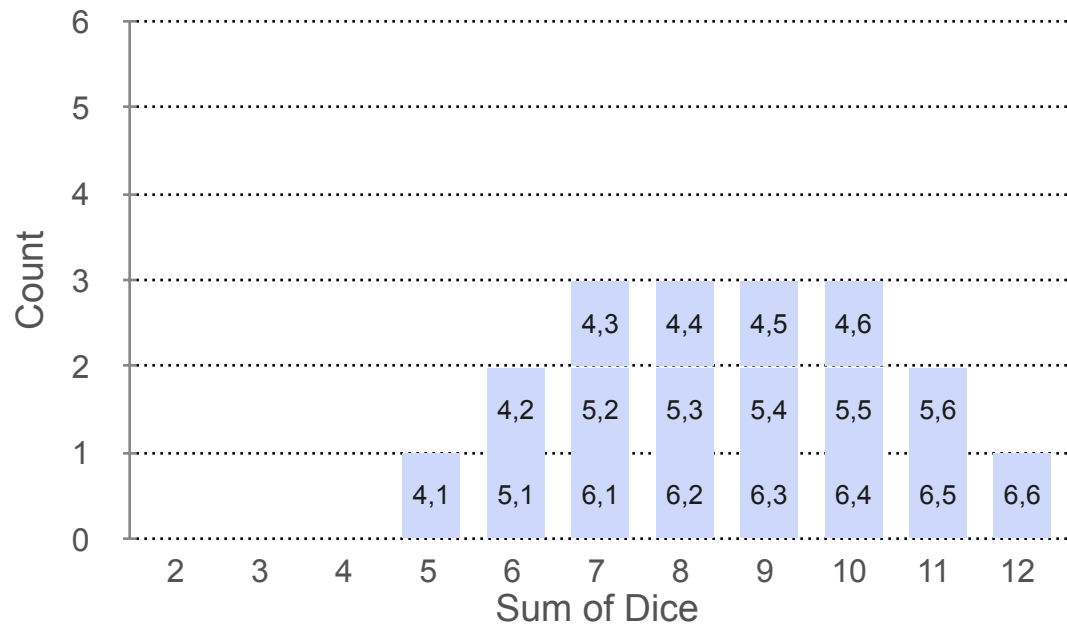
Joint Distributions: Example 3



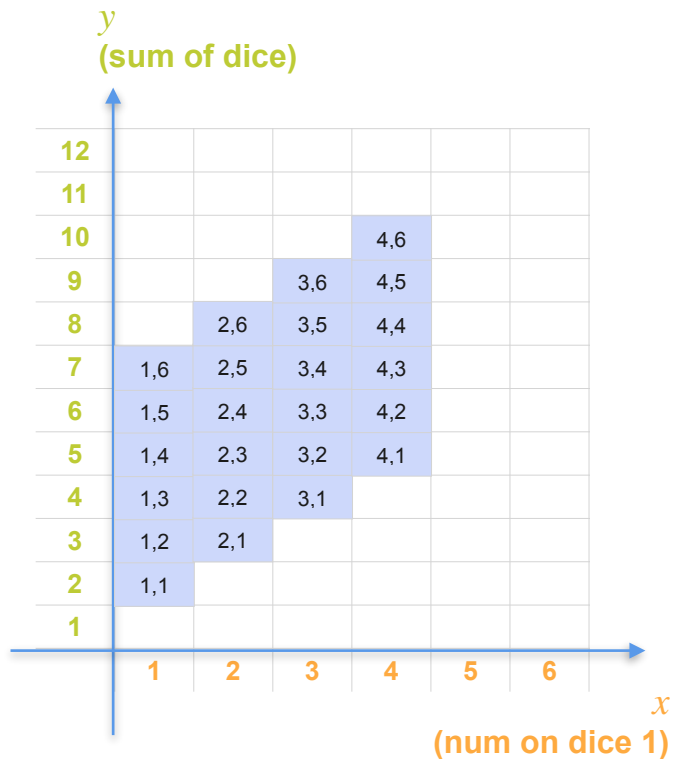
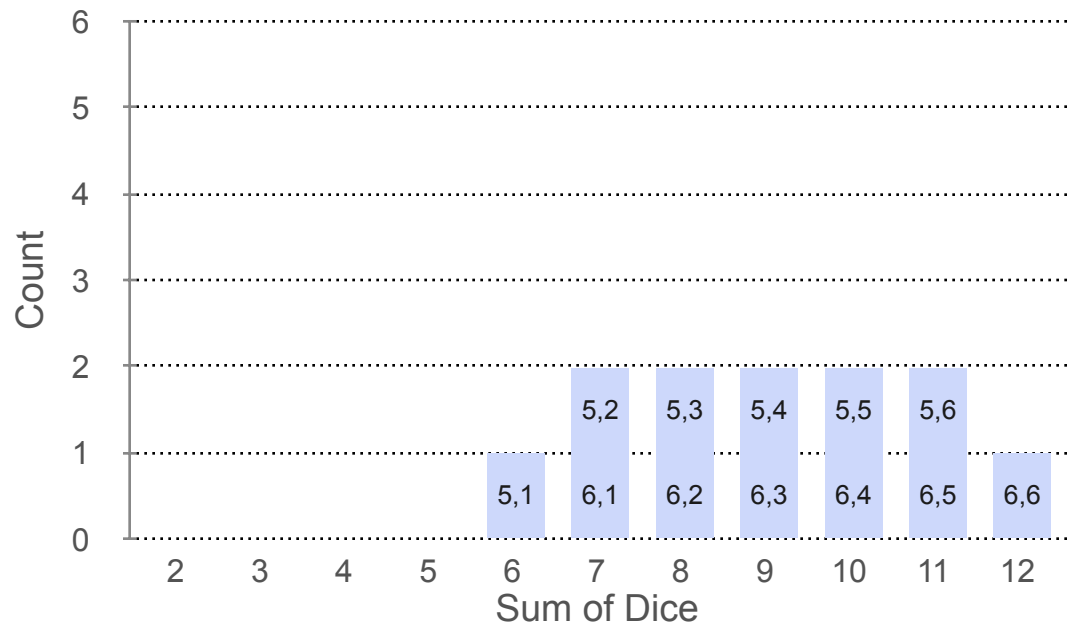
Joint Distributions: Example 3



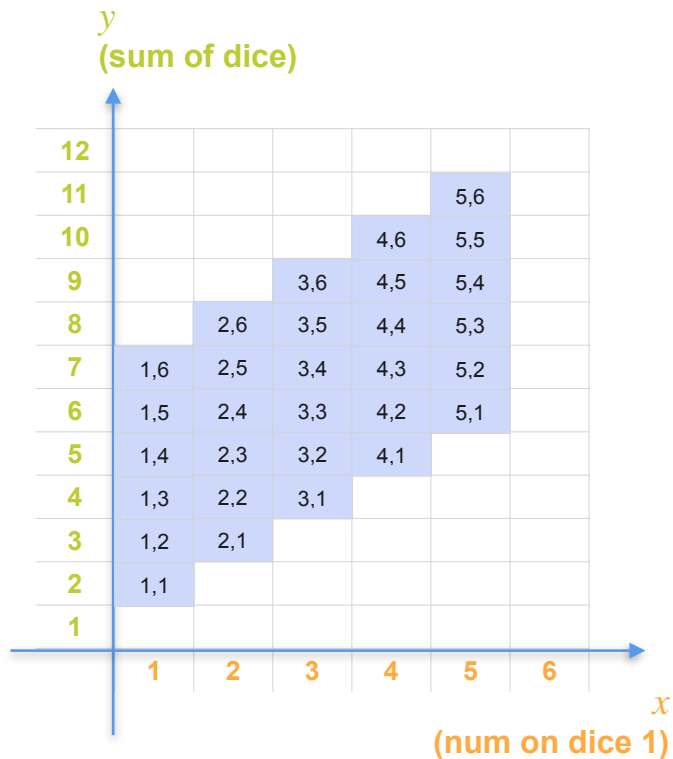
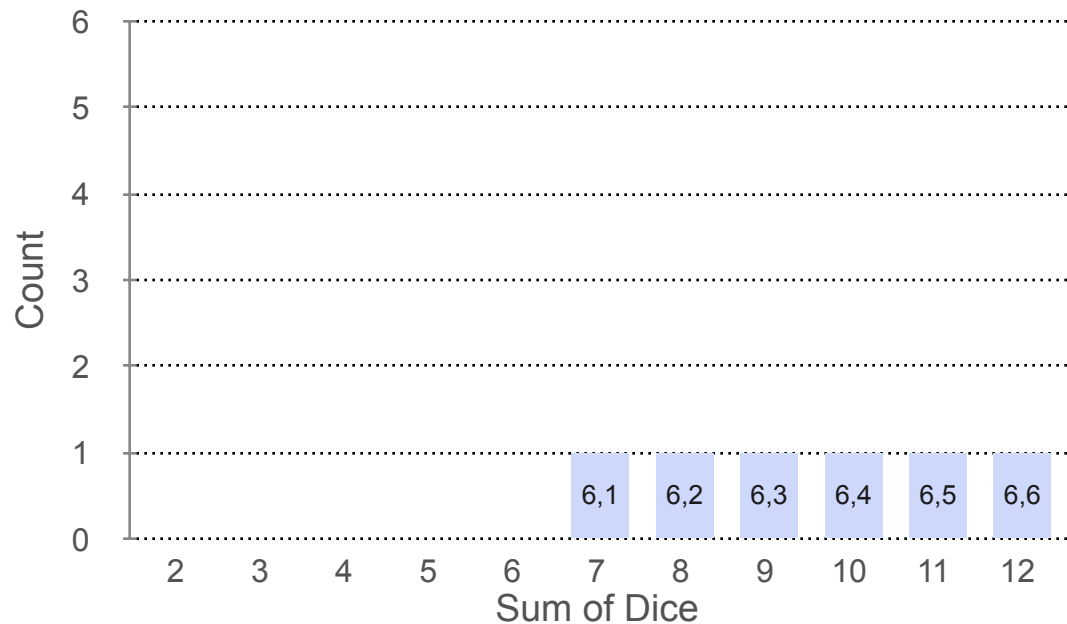
Joint Distributions: Example 3



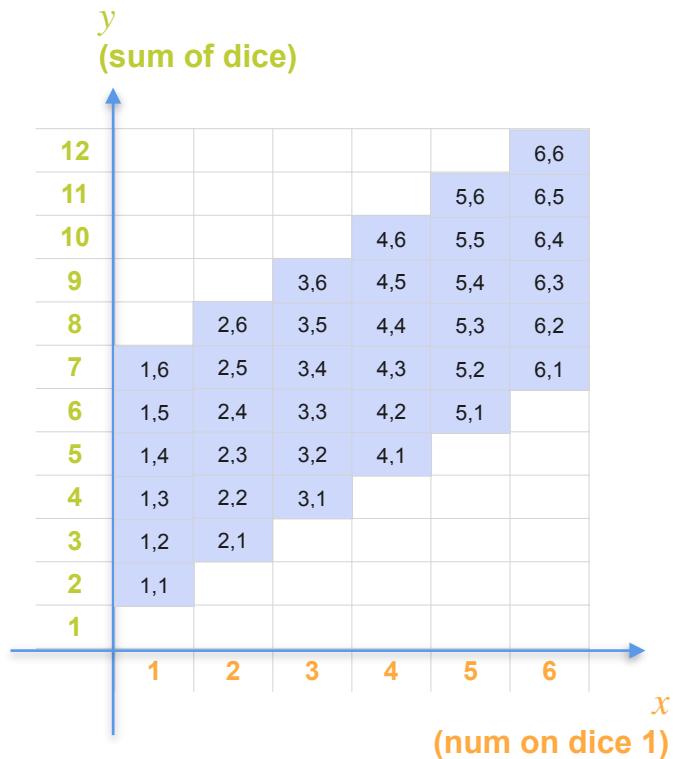
Joint Distributions: Example 3



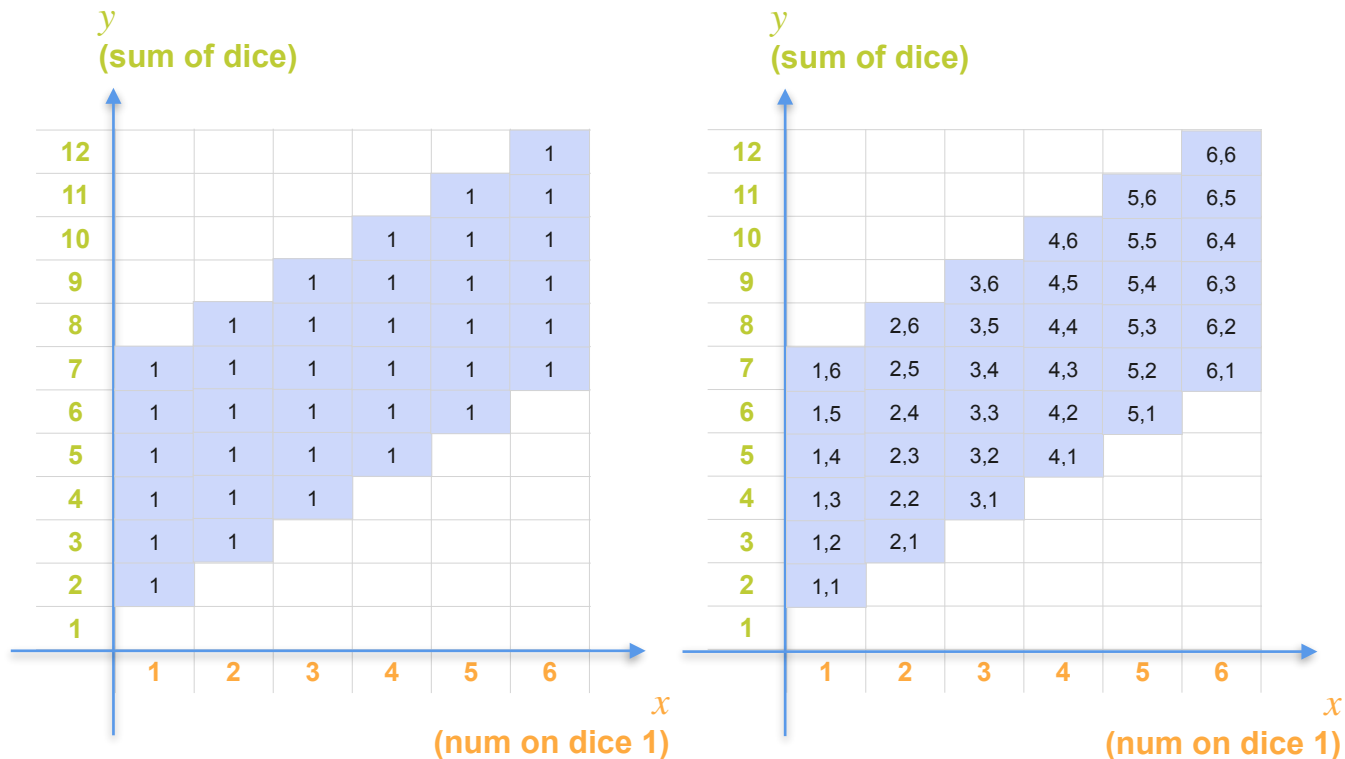
Joint Distributions: Example 3



Joint Distributions: Example 3



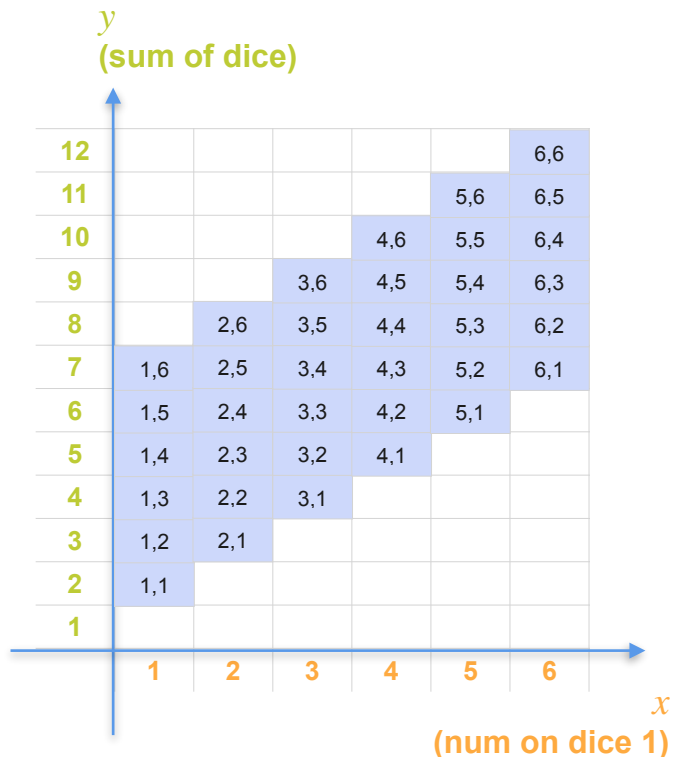
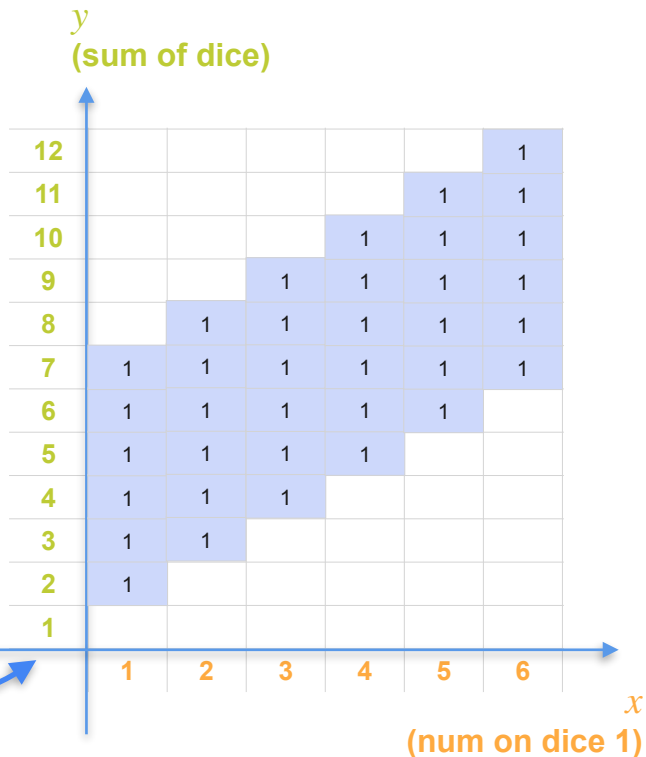
Joint Distributions: Example 3



Joint Distributions: Example 3

36
possible
outcomes

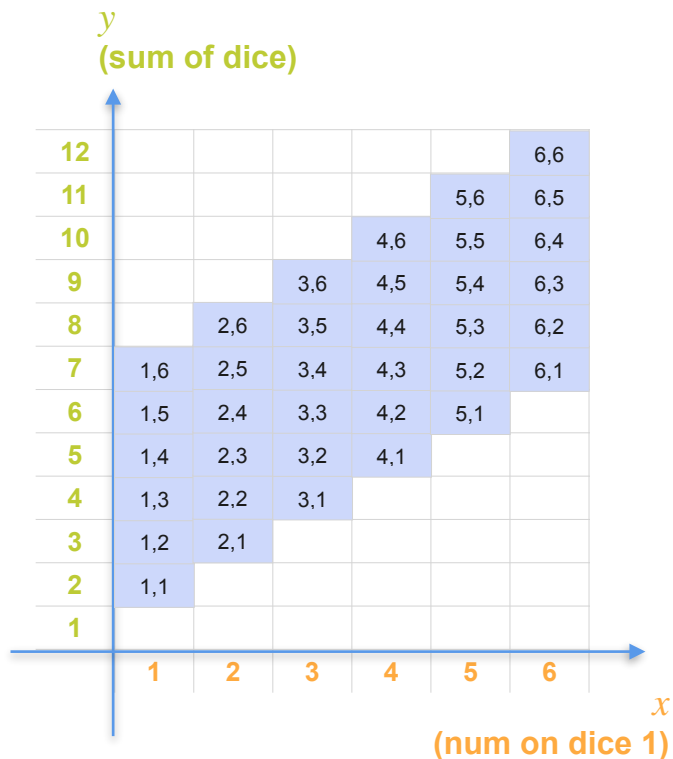
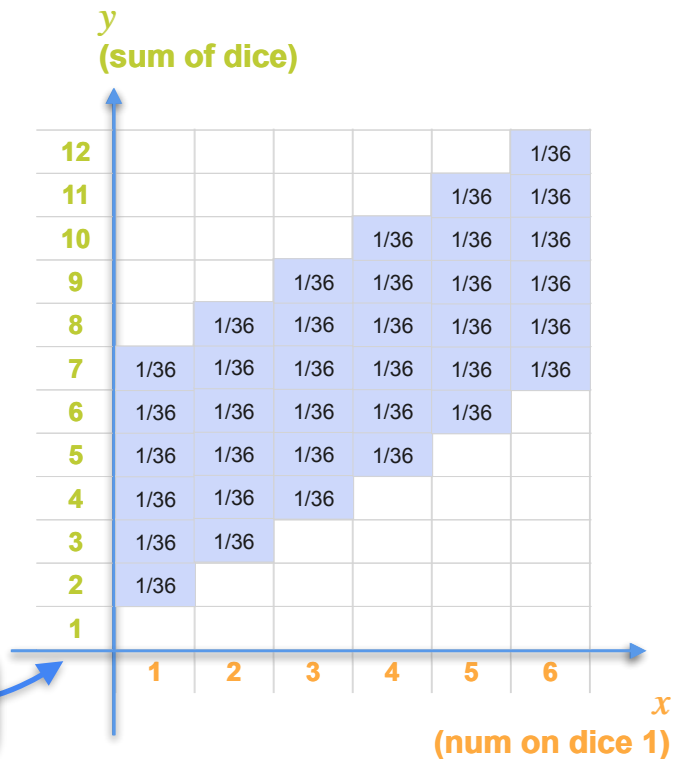
Divide by sum
(36)



Joint Distributions: Example 3

**36
possible
outcomes**

Divide by sum
(36)



Joint Distributions: Example 3

Joint Distribution for
 X and Y

36
possible
outcomes

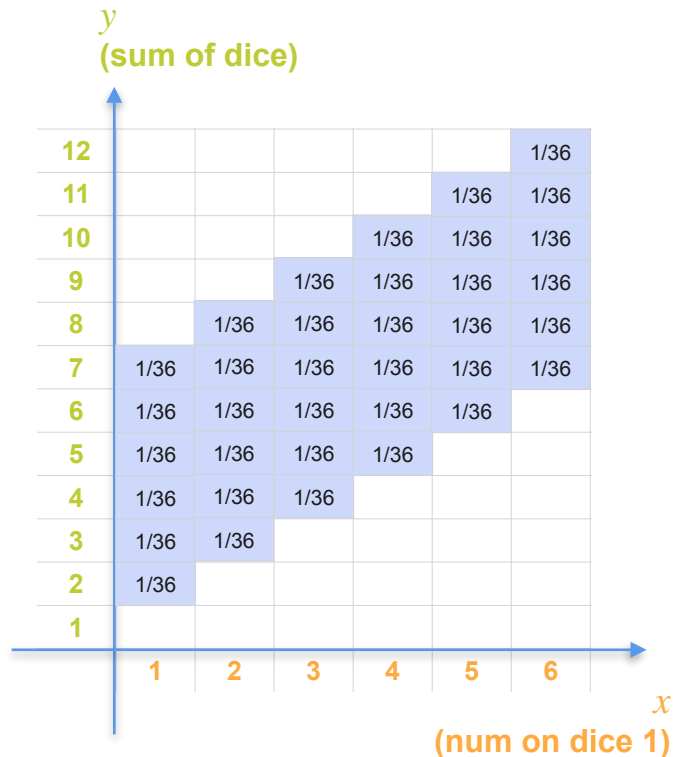
Divide by sum
(36)

y (sum of dice)	1	2	3	4	5	6
12						1/36
11					1/36	1/36
10				1/36	1/36	1/36
9			1/36	1/36	1/36	1/36
8		1/36	1/36	1/36	1/36	1/36
7	1/36	1/36	1/36	1/36	1/36	1/36
6	1/36	1/36	1/36	1/36	1/36	
5	1/36	1/36	1/36	1/36		
4	1/36	1/36	1/36			
3	1/36	1/36				
2	1/36					
1						
x (num on dice 1)						

y (sum of dice)	1	2	3	4	5	6
12						6,6
11					5,6	6,5
10				4,6	5,5	6,4
9			3,6	4,5	5,4	6,3
8		2,6	3,5	4,4	5,3	6,2
7	1,6	2,5	3,4	4,3	5,2	6,1
6	1,5	2,4	3,3	4,2	5,1	
5	1,4	2,3	3,2	4,1		
4	1,3	2,2	3,1			
3	1,2	2,1				
2	1,1					
1						
x (num on dice 1)						

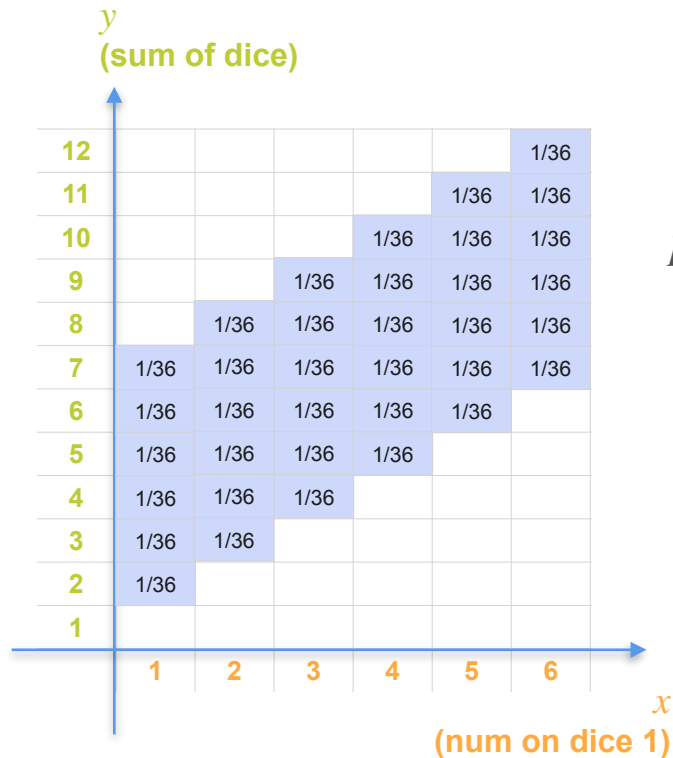
Joint Distributions: Example 3

Joint Distribution for
 X and Y



Joint Distributions: Example 3

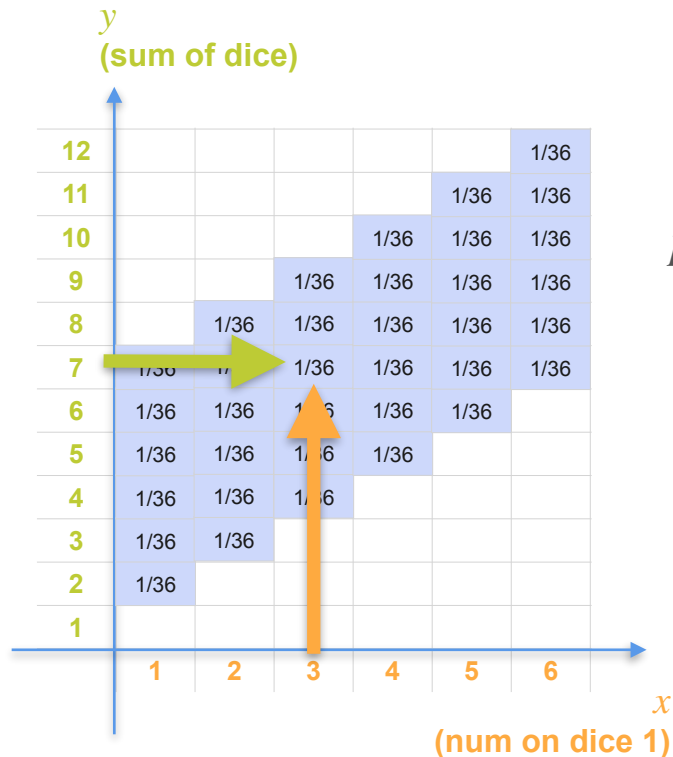
Joint Distribution for
 X and Y



$$p_{XY}(3, 7) = \mathbf{P}(X = 3, Y = 7)$$

Joint Distributions: Example 3

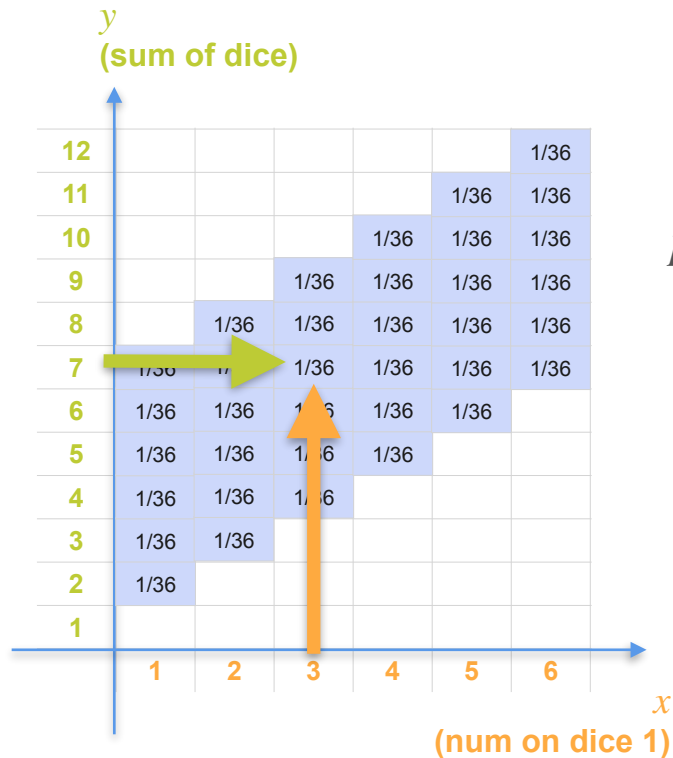
Joint Distribution for
 X and Y



$$p_{XY}(3, 7) = \mathbf{P}(X = 3, Y = 7)$$

Joint Distributions: Example 3

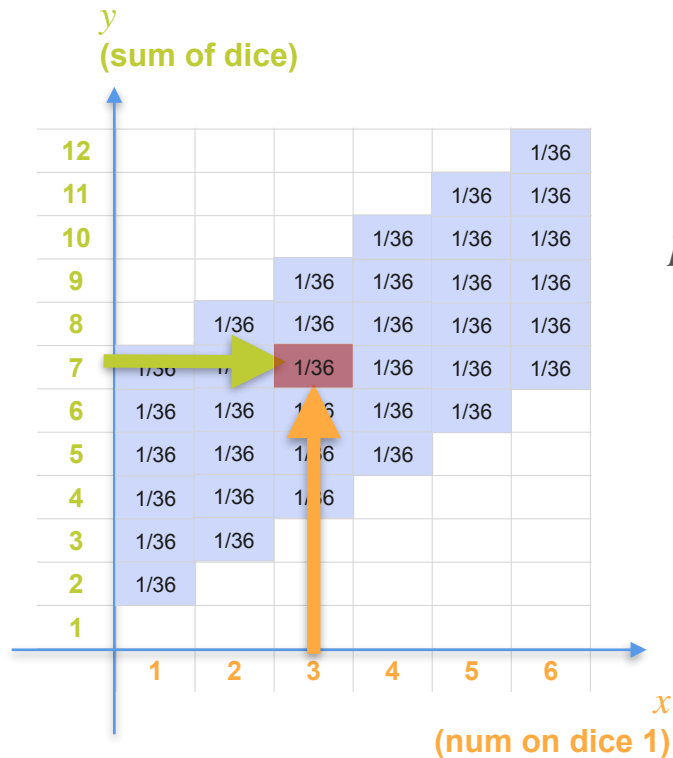
Joint Distribution for
 X and Y



$$p_{XY}(3, 7) = \mathbf{P}(X = 3, Y = 7) = \frac{1}{36}$$

Joint Distributions: Example 3

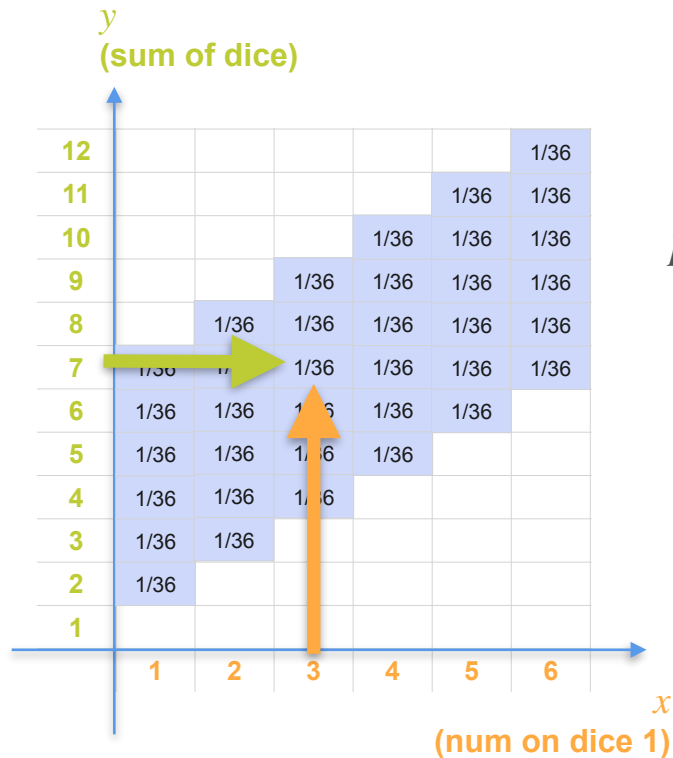
Joint Distribution for
 X and Y



$$p_{XY}(3, 7) = \mathbf{P}(X = 3, Y = 7) = \frac{1}{36}$$

Joint Distributions: Example 3

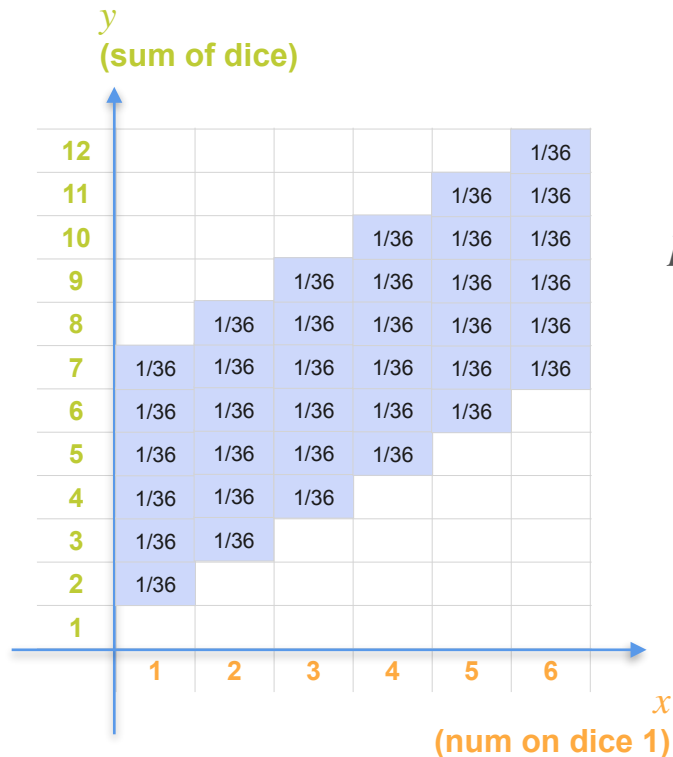
Joint Distribution for
 X and Y



$$p_{XY}(3, 7) = \mathbf{P}(X = 3, Y = 7) = \frac{1}{36}$$

Joint Distributions: Example 3

Joint Distribution for
 X and Y

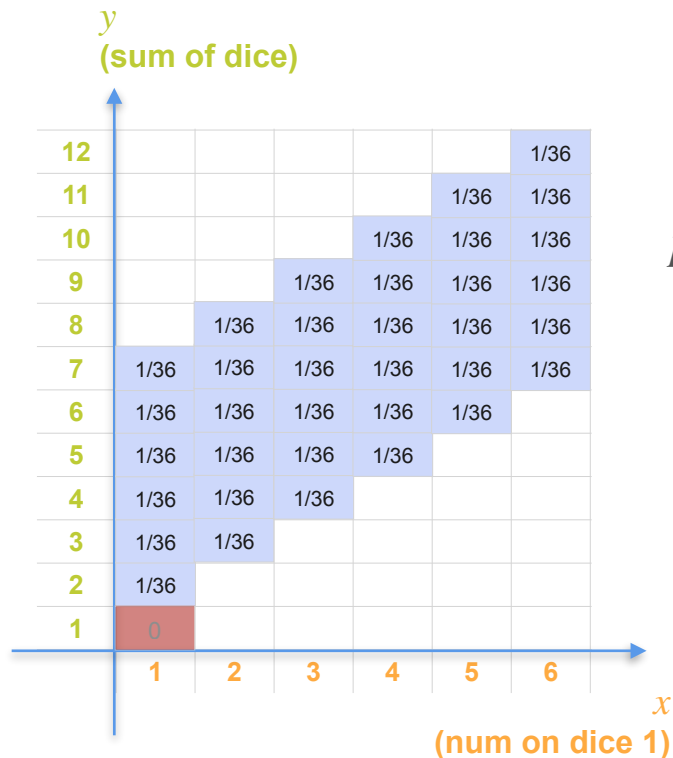


$$p_{XY}(3, 7) = \mathbf{P}(X = 3, Y = 7) = \frac{1}{36}$$

$$p_{XY}(1, 1) = \mathbf{P}(X = 1, Y = 1)$$

Joint Distributions: Example 3

Joint Distribution for
 X and Y

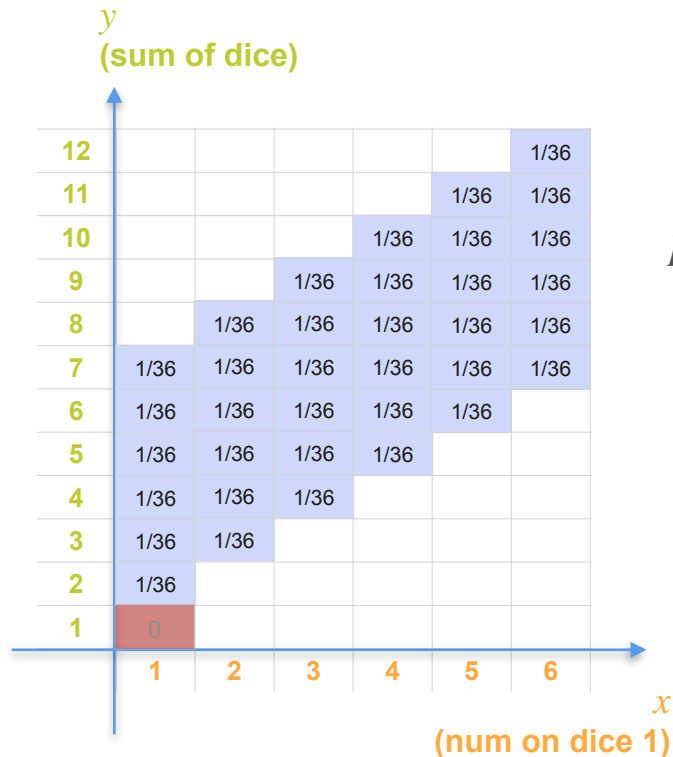


$$p_{XY}(3, 7) = \mathbf{P}(X = 3, Y = 7) = \frac{1}{36}$$

$$p_{XY}(1, 1) = \mathbf{P}(X = 1, Y = 1)$$

Joint Distributions: Example 3

Joint Distribution for
 X and Y



$$p_{XY}(3, 7) = \mathbf{P}(X = 3, Y = 7) = \frac{1}{36}$$

$$p_{XY}(1, 1) = \mathbf{P}(X = 1, Y = 1) = 0$$



DeepLearning.AI

Probability Distributions with Multiple Variables

Joint Distribution (Continuous)

Joint Continuous Distributions

Joint Continuous Distributions

X : age of a child in year

Y : discrete values of height of child in inches

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X : the number rolled on the 1st dice

Y : sum of both dice

Joint Continuous Distributions

X : age of a child in year

Y : discrete values of height of child in inches

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X : the number rolled on the 1st dice

Y : sum of both dice

X and Y are
Discrete Random Variables

Joint Continuous Distributions

X : age of a child in year

Y : discrete values of height of child in inches

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

X : the number rolled on the 1st dice

Y : sum of both dice

X and Y are
Discrete Random Variables

What about when X and Y are
Continuous Random Variables?

Joint Continuous Distributions

Joint Continuous Distributions



Joint Continuous Distributions

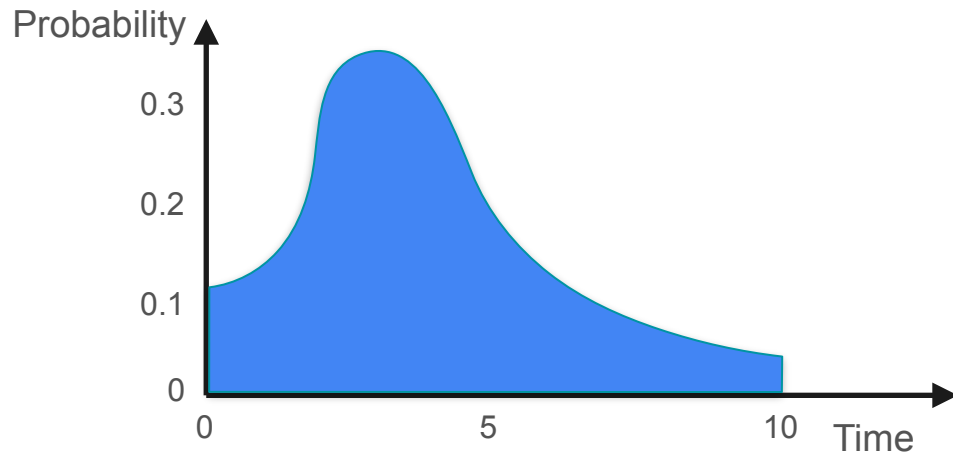


X variable: Waiting time

Joint Continuous Distributions



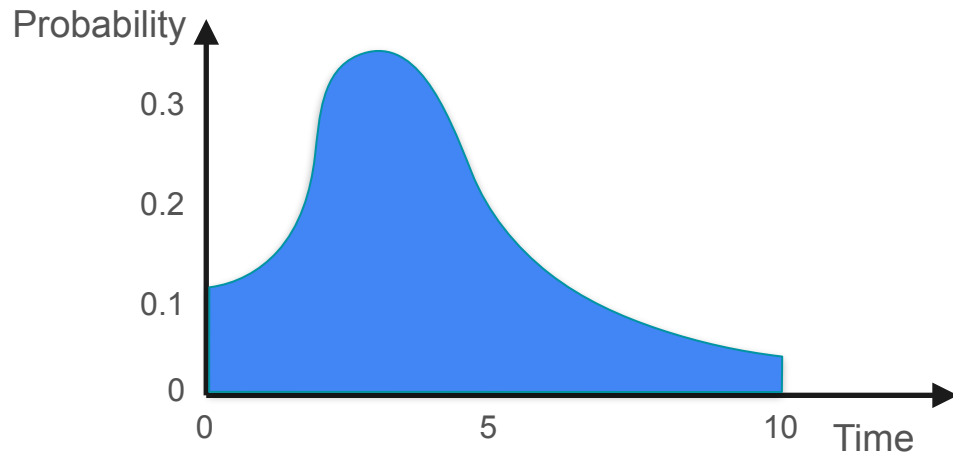
X variable: Waiting time



Joint Continuous Distributions



X variable: Waiting time

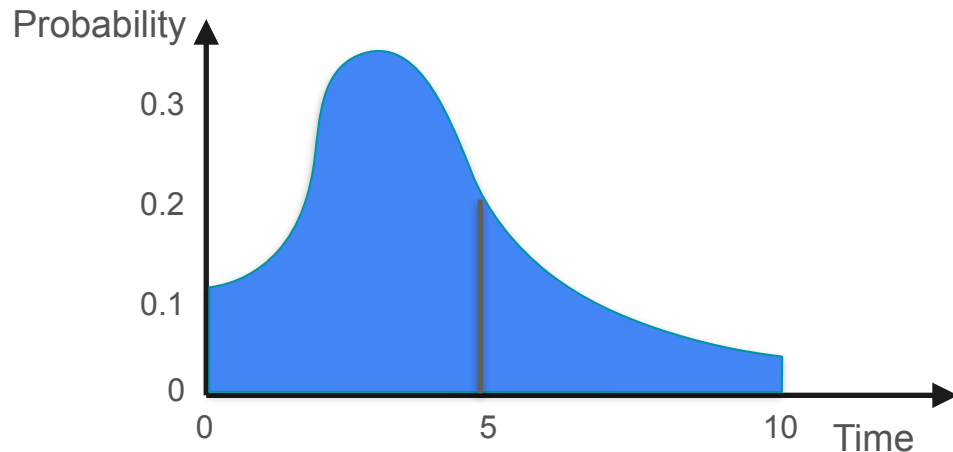


$P(X \text{ between } 0 \text{ and } 5 \text{ mins})$

Joint Continuous Distributions



X variable: Waiting time

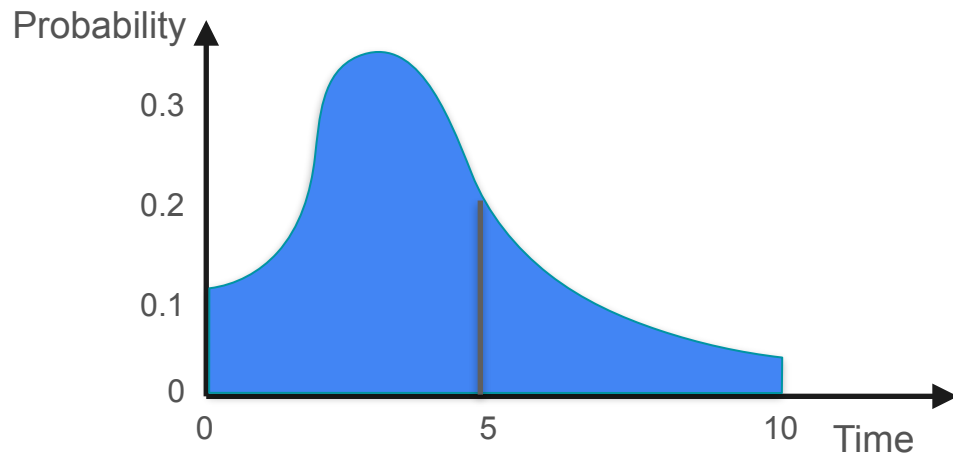


$P(X \text{ between } 0 \text{ and } 5 \text{ mins})$

Joint Continuous Distributions



X variable: Waiting time

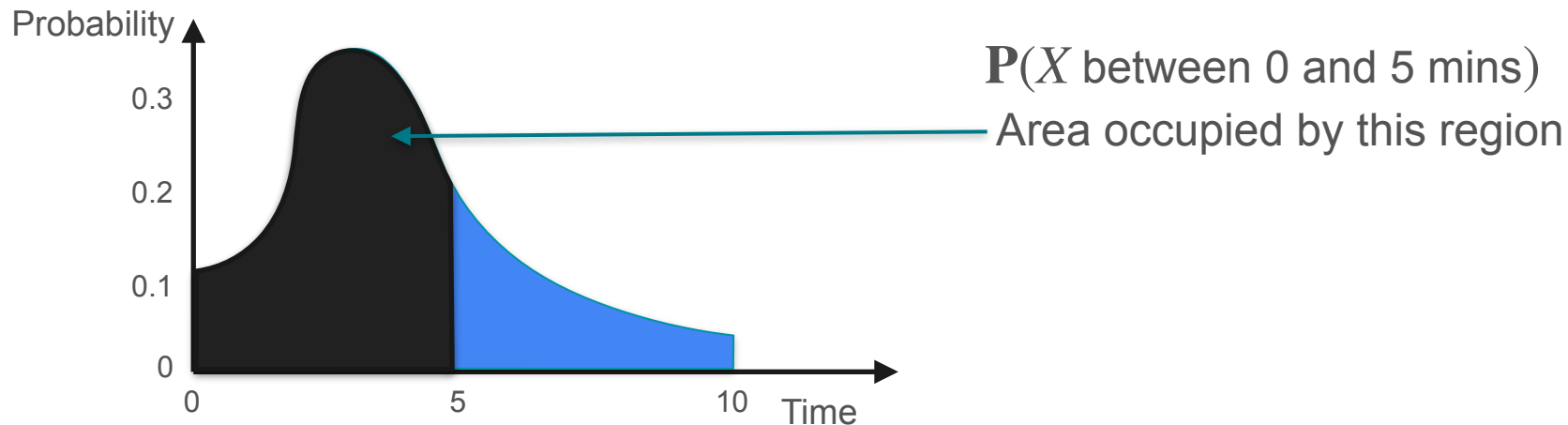


$P(X \text{ between } 0 \text{ and } 5 \text{ mins})$
Area occupied by this region

Joint Continuous Distributions



X variable: Waiting time



Joint Continuous Distributions

Joint Continuous Distributions

 X Y

Joint Continuous Distributions

 X

Waiting time
before a call is picked up
[0 - 10 minutes]

 Y 

Joint Continuous Distributions

 X

Waiting time
before a call is picked up
[0 - 10 minutes]

 Y

Customer
satisfaction rating
[0 - 10]



Joint Continuous Distributions

 X

Waiting time
before a call is picked up
[0 - 10 minutes]



**Both variables are
continuous**

 Y

Customer
satisfaction rating
[0 - 10]



Joint Continuous Distributions

 X

Waiting time
before a call is picked up
[0 - 10 minutes]



2.4 minutes

1.5 minutes

**Both variables are
continuous**

 Y

Customer
satisfaction rating
[0 - 10]



Joint Continuous Distributions

 X

Waiting time
before a call is picked up
[0 - 10 minutes]



2.4 minutes

1.5 minutes

**Both variables are
continuous**

 Y

Customer
satisfaction rating
[0 - 10]



0.0

5.7

Joint Continuous Distributions: Dataset

Joint Continuous Distributions: Dataset

X variable: Waiting time (mins)

0 - 10 mins

Joint Continuous Distributions: Dataset

X variable: Waiting time (mins)

0 - 10 mins

1000 customers

Joint Continuous Distributions: Dataset

X variable: Waiting time (mins)

0 - 10 mins

1000 customers



Joint Continuous Distributions: Dataset

Joint Continuous Distributions: Dataset

Y variable: Satisfaction rating

0 - 10

Joint Continuous Distributions: Dataset

Y variable: Satisfaction rating

0 - 10

1000 customers

Joint Continuous Distributions: Dataset

Y variable: Satisfaction rating
0 - 10

1000 customers



Joint Continuous Distributions: Dataset

Joint Continuous Distributions: Dataset

X variable: Waiting time (mins)
0 - 10 mins

Joint Continuous Distributions: Dataset

X variable: Waiting time (mins)
0 - 10 mins

Y variable: Satisfaction rating
0 - 10

Joint Continuous Distributions: Dataset

X variable: Waiting time (mins)

0 - 10 mins

Y variable: Satisfaction rating

0 - 10

1000 customers

Joint Continuous Distributions: Dataset

X variable: Waiting time (mins)
0 - 10 mins

Y variable: Satisfaction rating
0 - 10

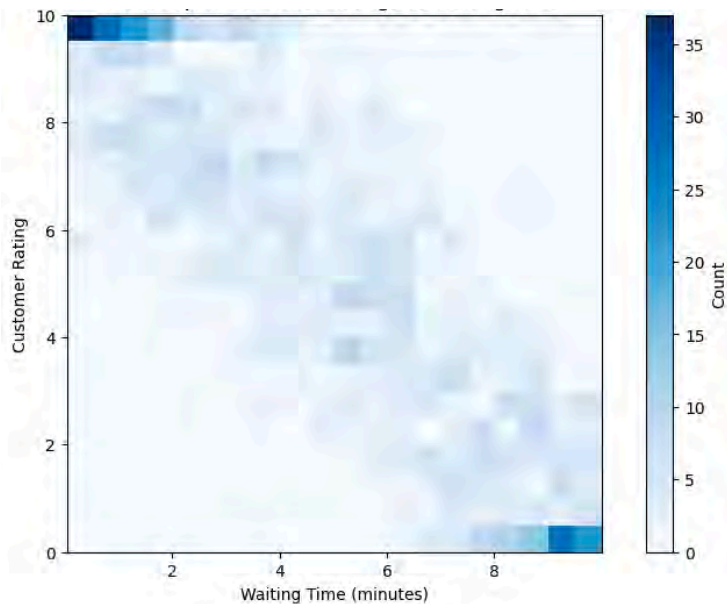
1000 customers



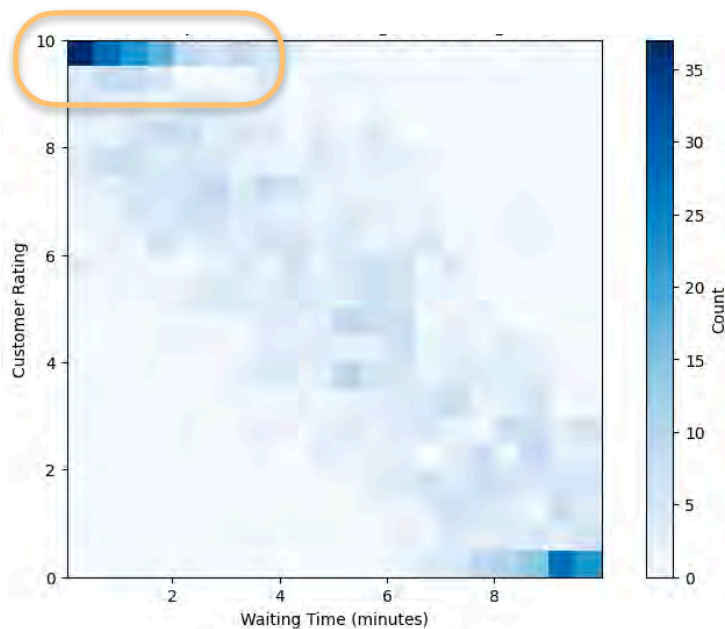
Joint Continuous Distributions: Dataset



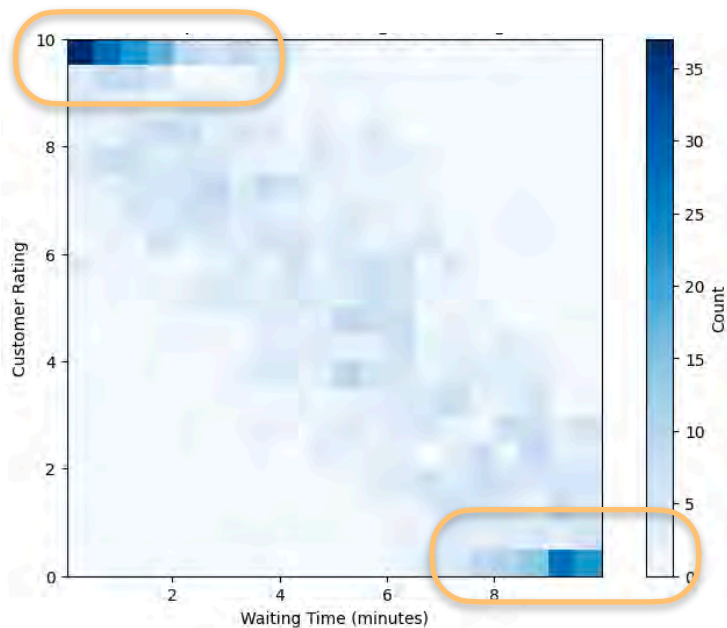
Joint Continuous Distributions: Dataset



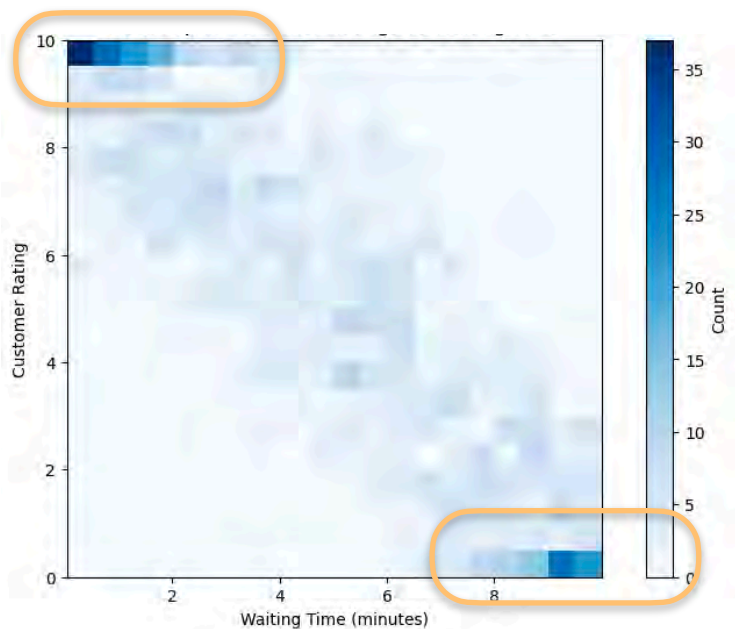
Joint Continuous Distributions: Dataset



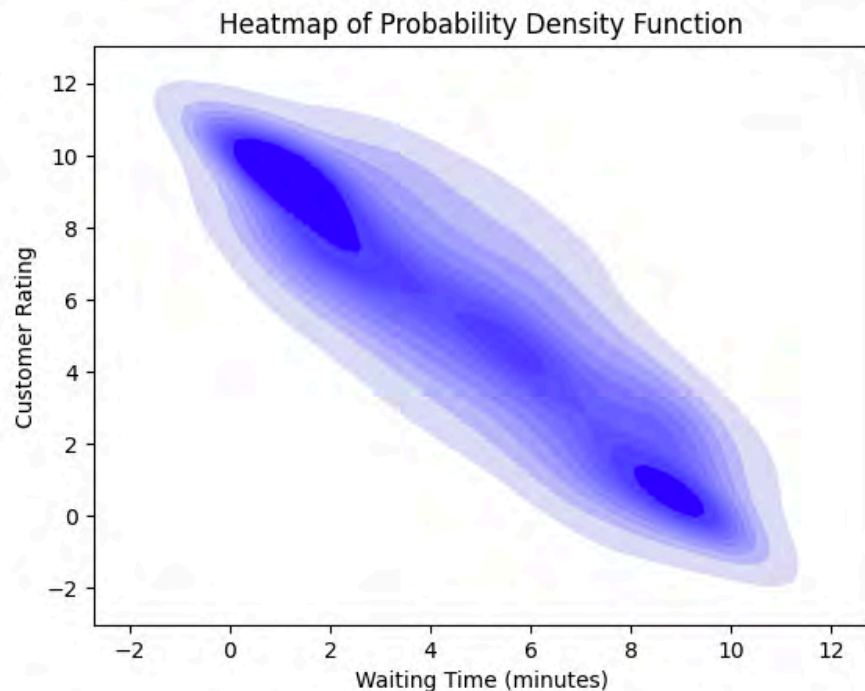
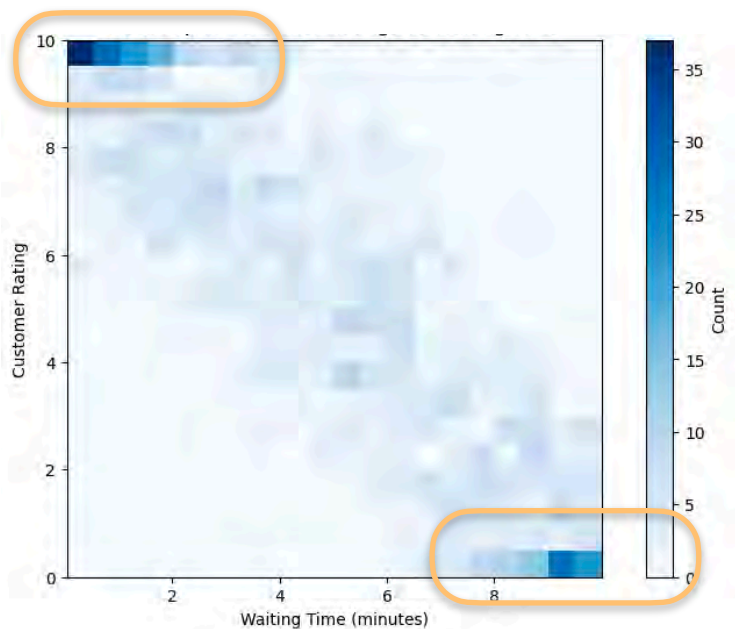
Joint Continuous Distributions: Dataset



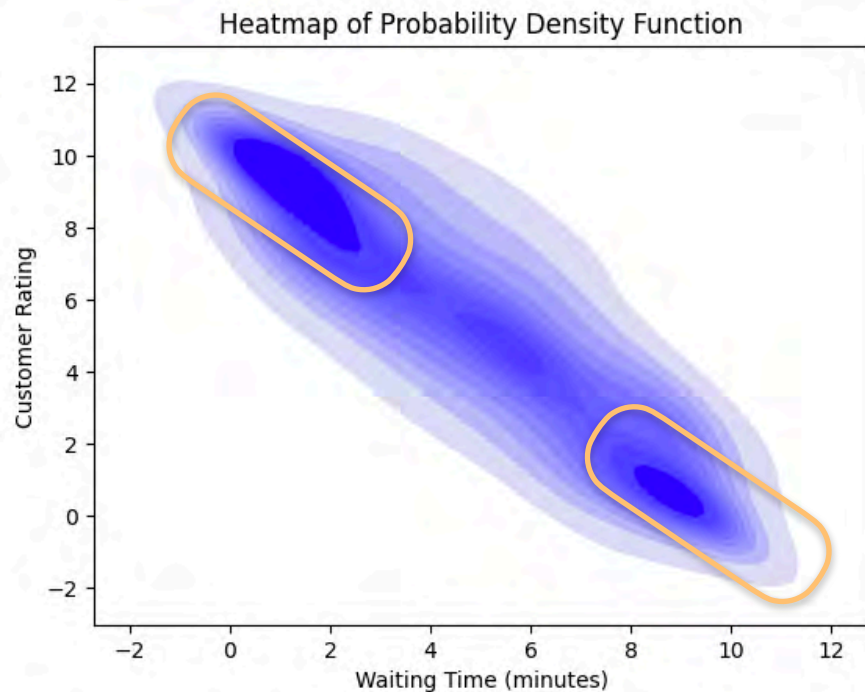
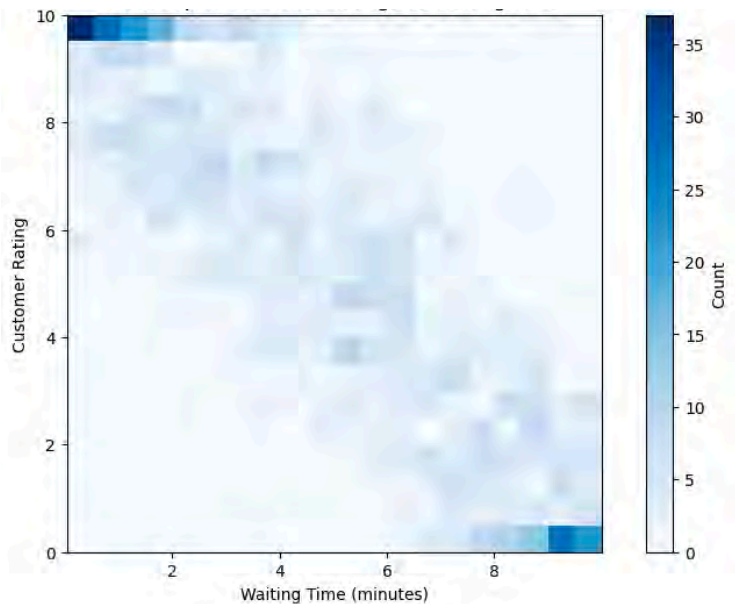
Joint Continuous Distributions: Dataset



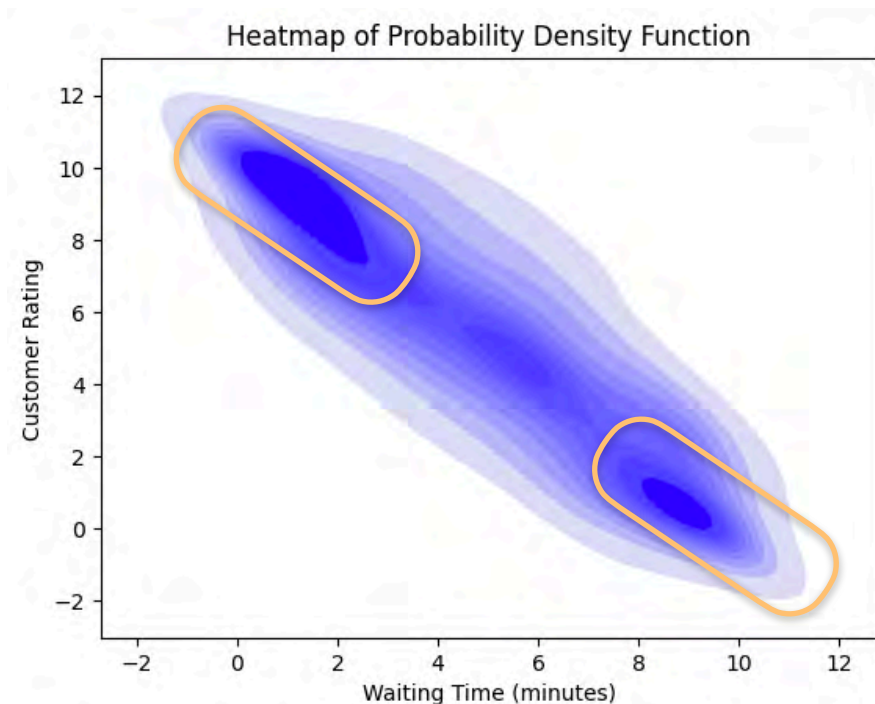
Joint Continuous Distributions: Dataset



Joint Continuous Distributions: Dataset

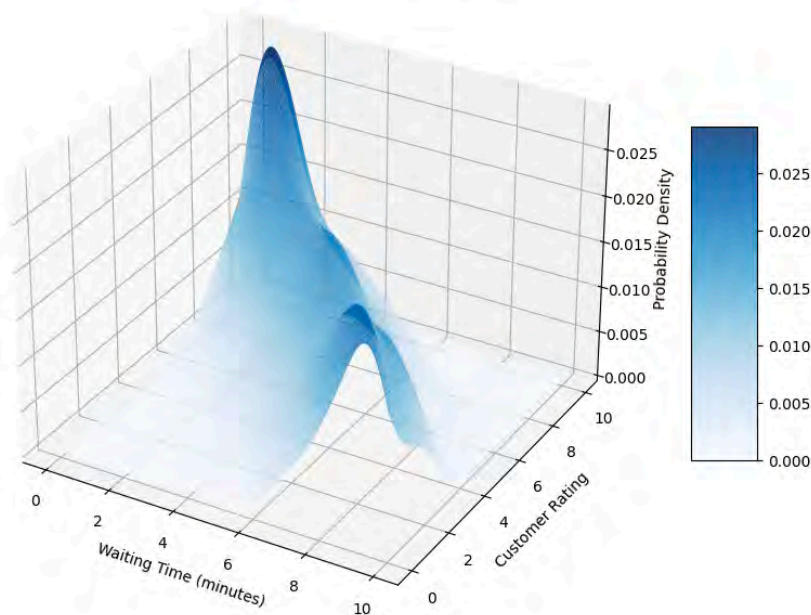


Joint Continuous Distributions: Dataset

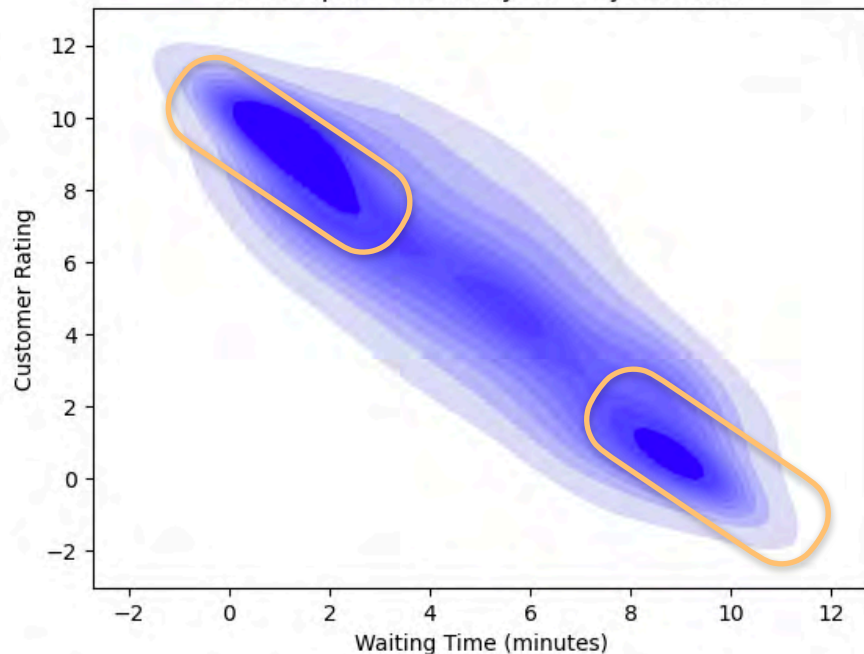


Joint Continuous Distributions: Dataset

3D Probability Density Distribution for Customer Ratings vs Waiting Time

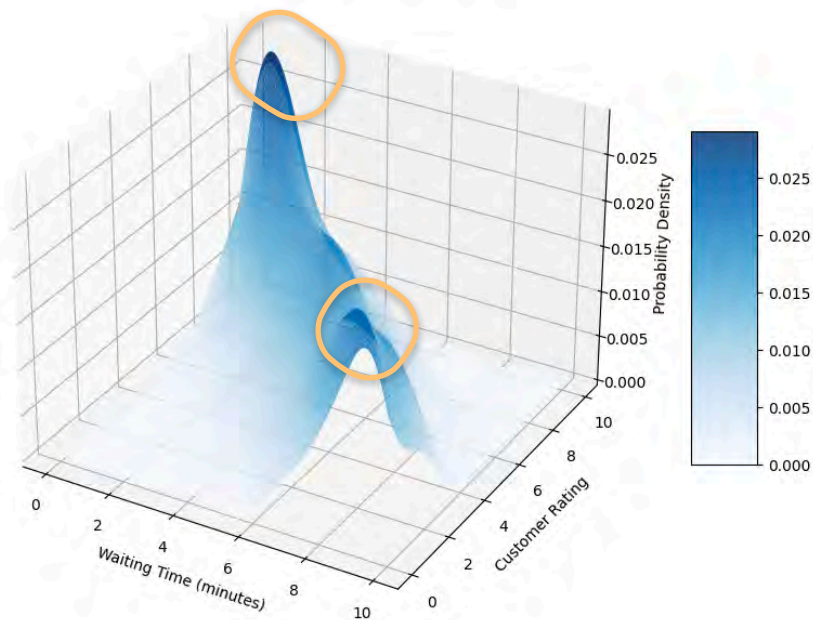


Heatmap of Probability Density Function

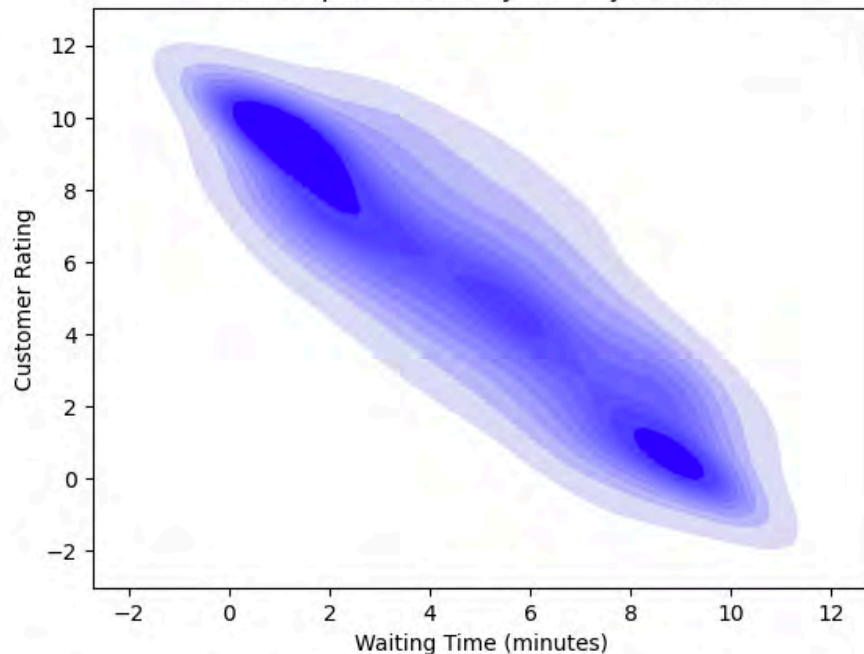


Joint Continuous Distributions: Dataset

3D Probability Density Distribution for Customer Ratings vs Waiting Time



Heatmap of Probability Density Function



Joint Continuous Distributions: Dataset

X variable: Waiting time (mins)
0 - 10 mins

Y variable: Satisfaction rating
0 - 10

1000 customers



Expected Value

X variable: Waiting time (mins)

0 - 10 mins

Y variable: Satisfaction rating

0 - 10

1000 customers



Expected Value

X variable: Waiting time (mins)

0 - 10 mins

Y variable: Satisfaction rating

0 - 10

1000 customers

$$\mathbb{E}[X] = 4.903 \text{ minutes}$$



Expected Value

X variable: Waiting time (mins)

0 - 10 mins

Y variable: Satisfaction rating

0 - 10

1000 customers

$$\mathbb{E}[X] = 4.903 \text{ minutes}$$

$$\mathbb{E}[Y] = 5.280$$



Expected Value

X variable: Waiting time (mins)

0 - 10 mins

Y variable: Satisfaction rating

0 - 10

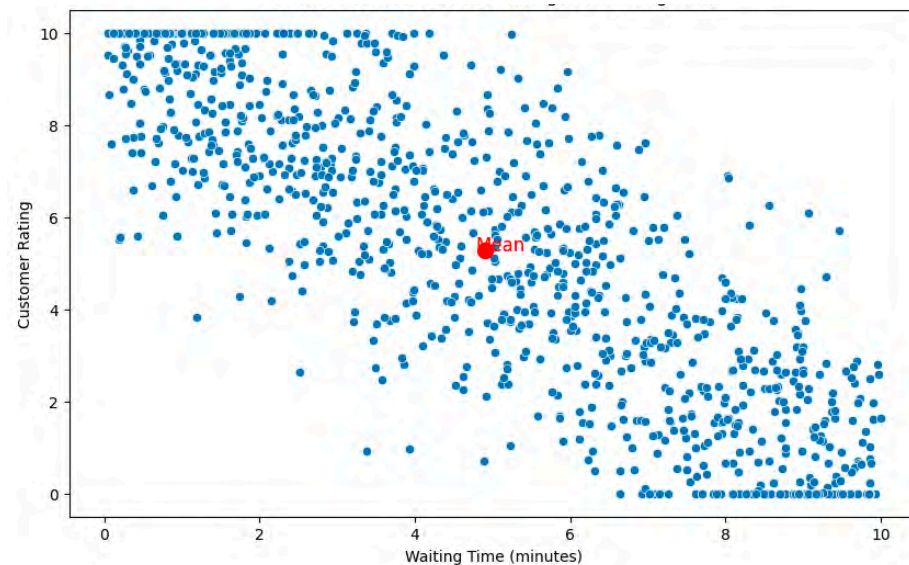
1000 customers

$$\mathbb{E}[X] = 4.903 \text{ minutes}$$

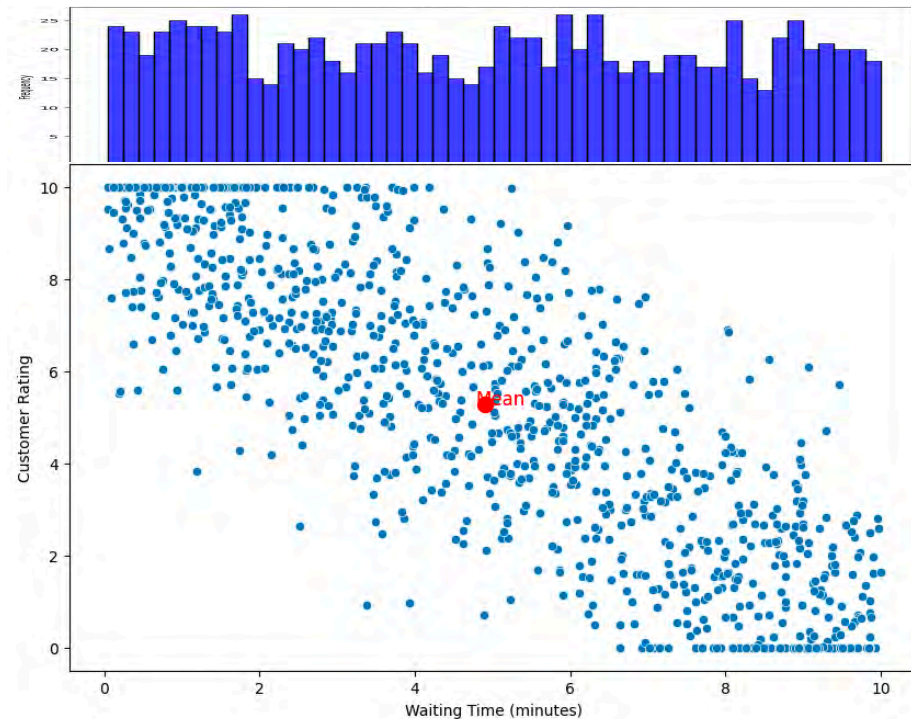
$$\mathbb{E}[Y] = 5.280$$



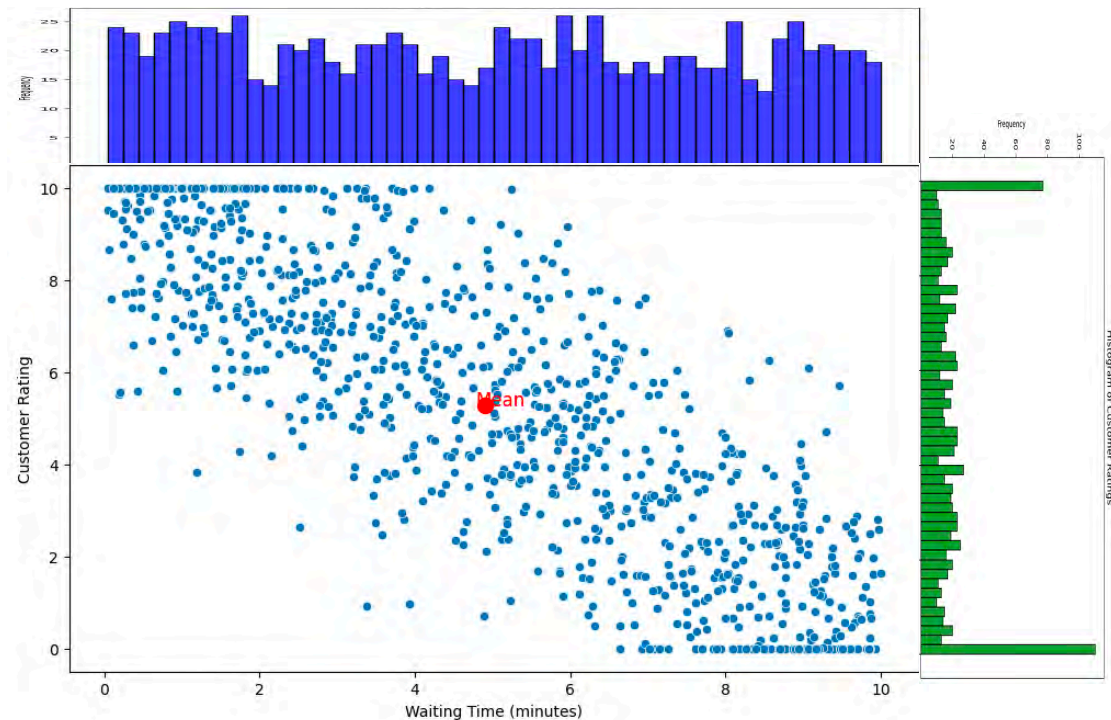
Variances



Variances

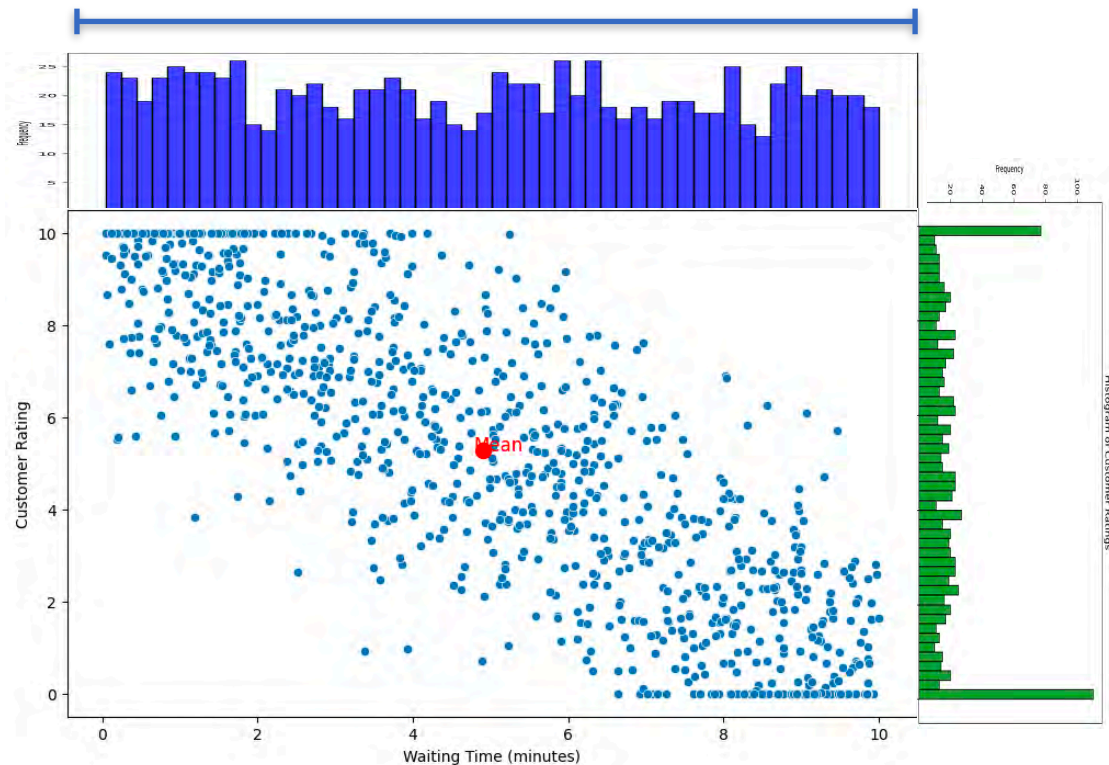


Variances

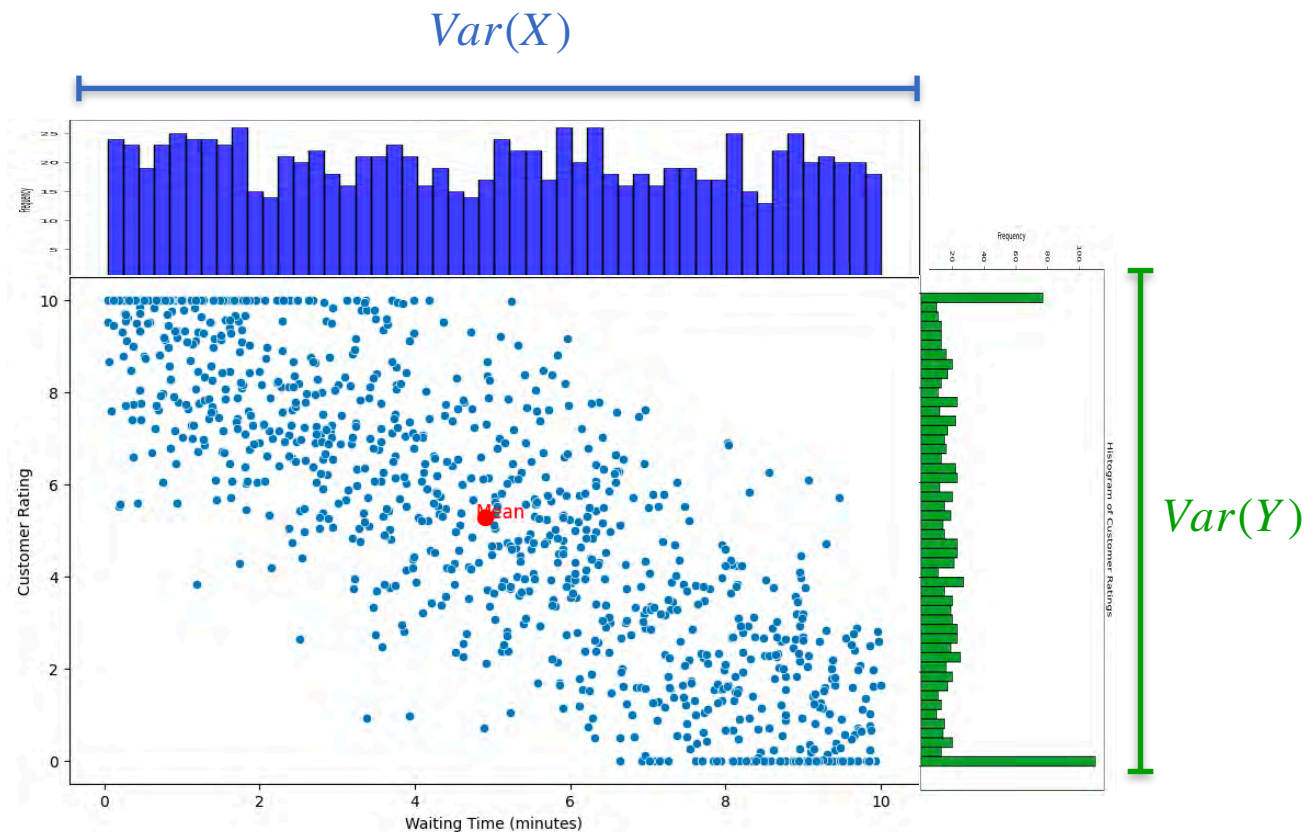


Variances

$$\text{Var}(X)$$

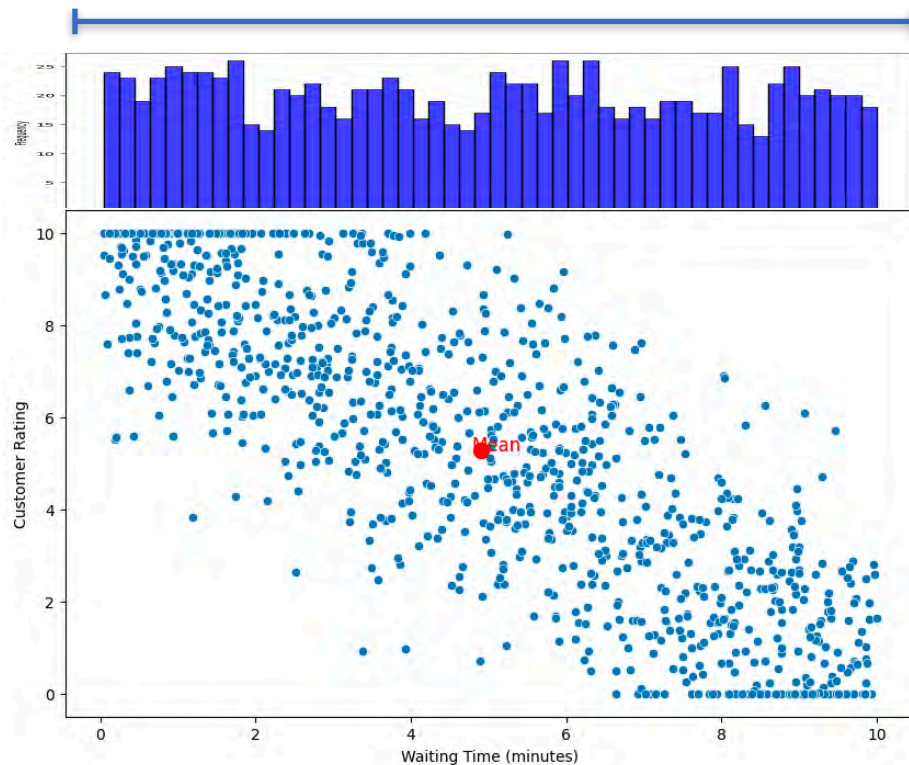


Variances



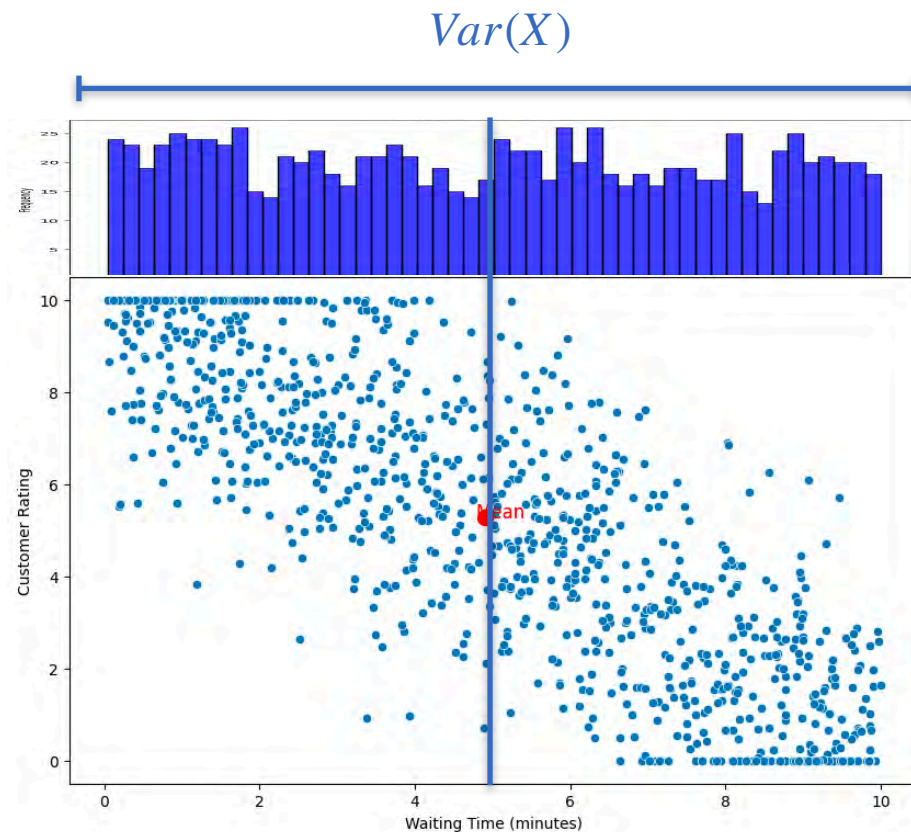
Variances

$$\text{Var}(X)$$



Variances

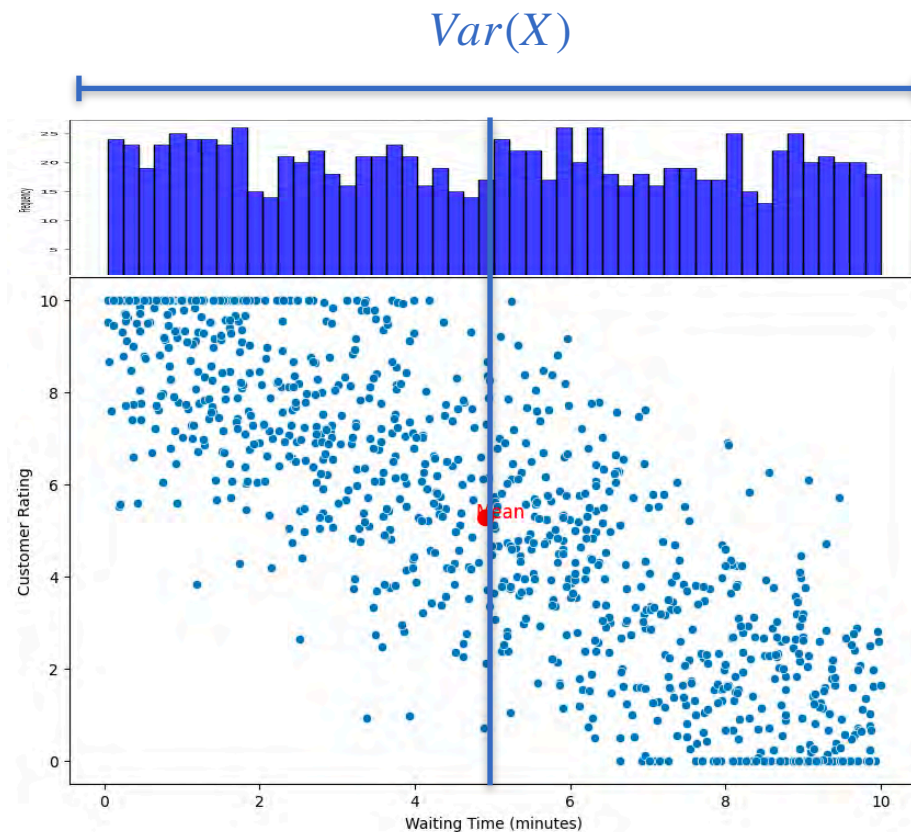
$$\mathbb{E}[X] = 4.903 \text{ minutes}$$



Variances

$$\mathbb{E}[X] = 4.903 \text{ minutes}$$

$$\mathbb{E}[X^2] = 32.561$$

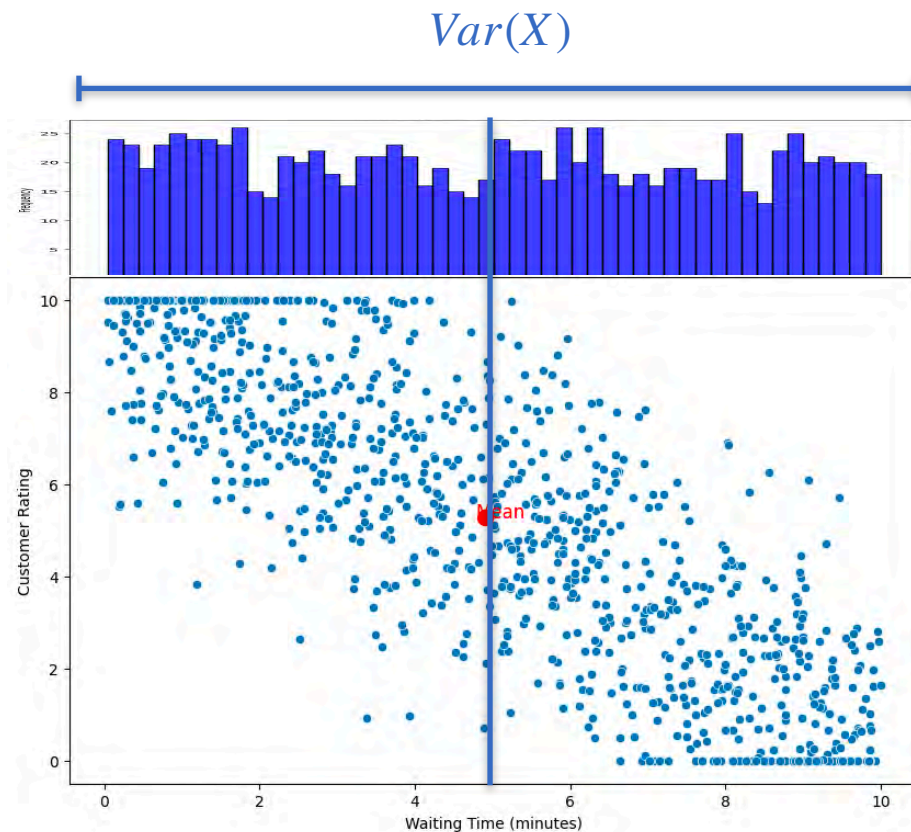


Variances

$$\mathbb{E}[X] = 4.903 \text{ minutes}$$

$$\mathbb{E}[X^2] = 32.561$$

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

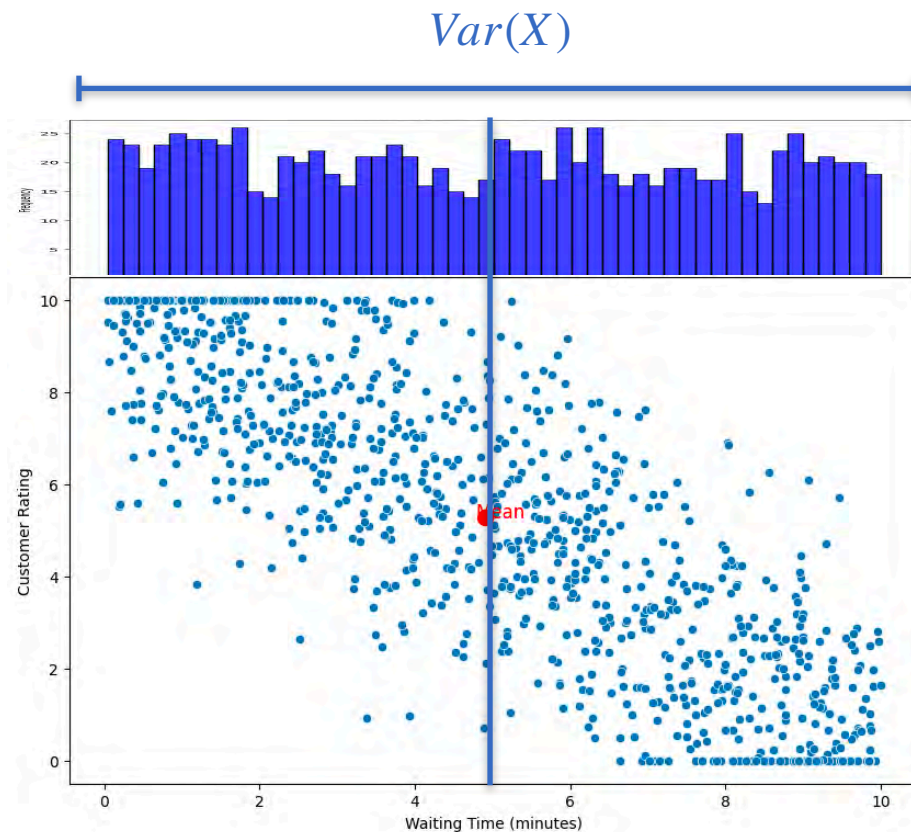


Variances

$$\mathbb{E}[X] = 4.903 \text{ minutes}$$

$$\mathbb{E}[X^2] = 32.561$$

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \\ &= 32.561 - 4.903^2 \end{aligned}$$

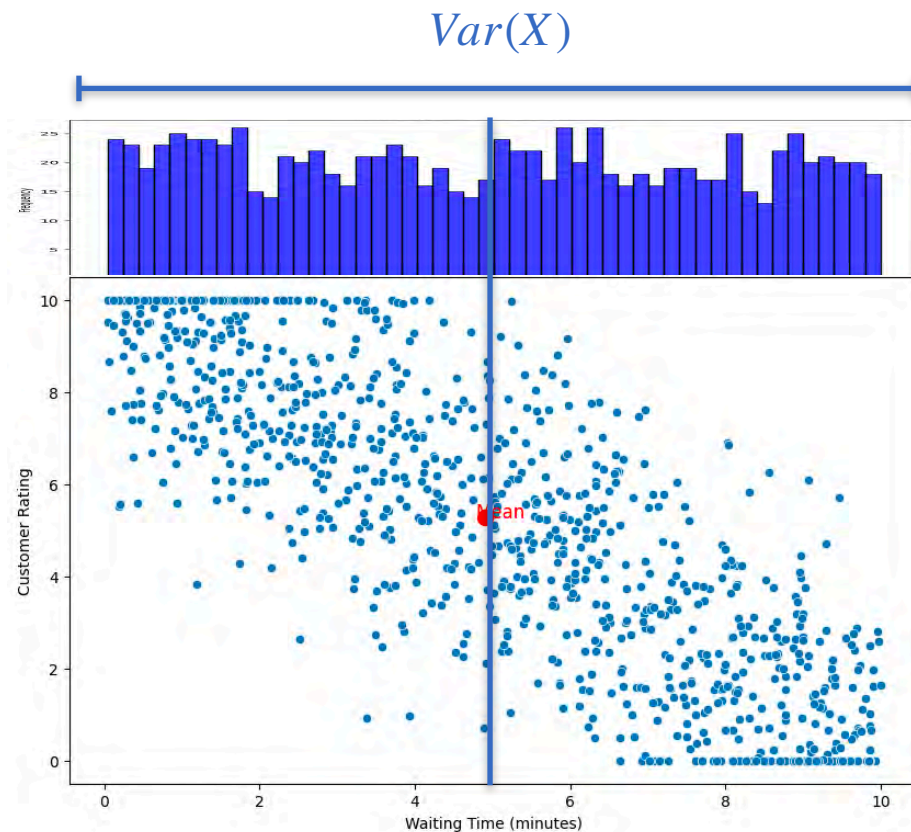


Variances

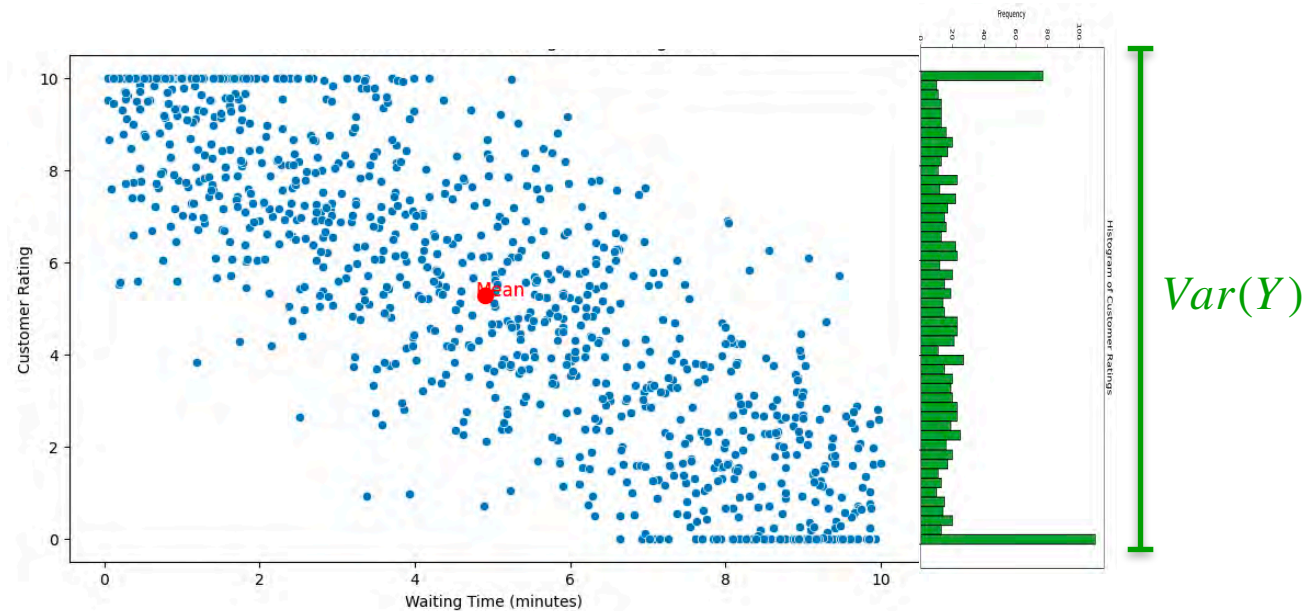
$$\mathbb{E}[X] = 4.903 \text{ minutes}$$

$$\mathbb{E}[X^2] = 32.561$$

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \\ &= 32.561 - 4.903^2 \\ &= 8.526 \end{aligned}$$

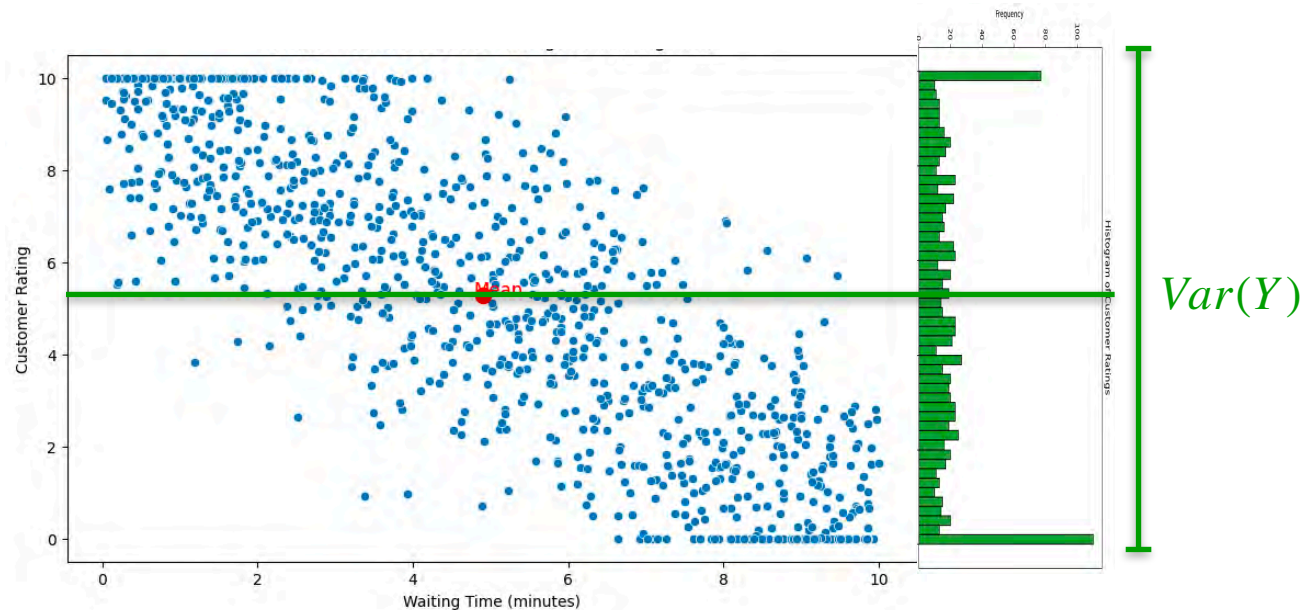


Variances



Variances

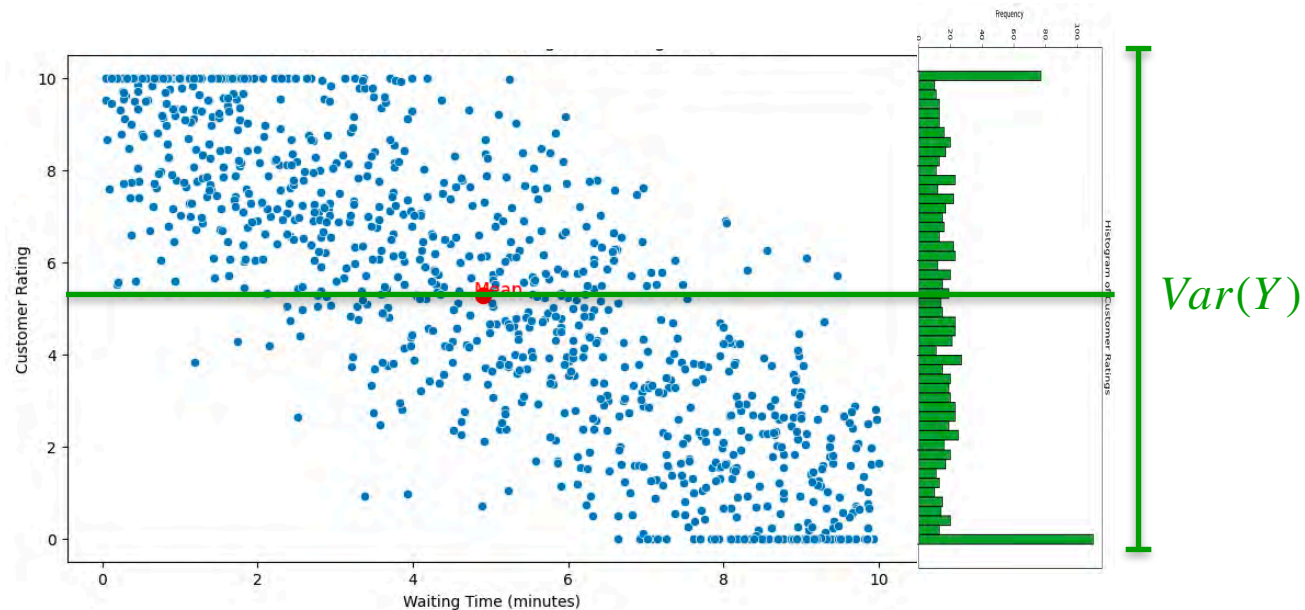
$$\mathbb{E}[Y] = 5.280$$



Variances

$$\mathbb{E}[Y] = 5.280$$

$$\mathbb{E}[Y^2] = 38.037$$

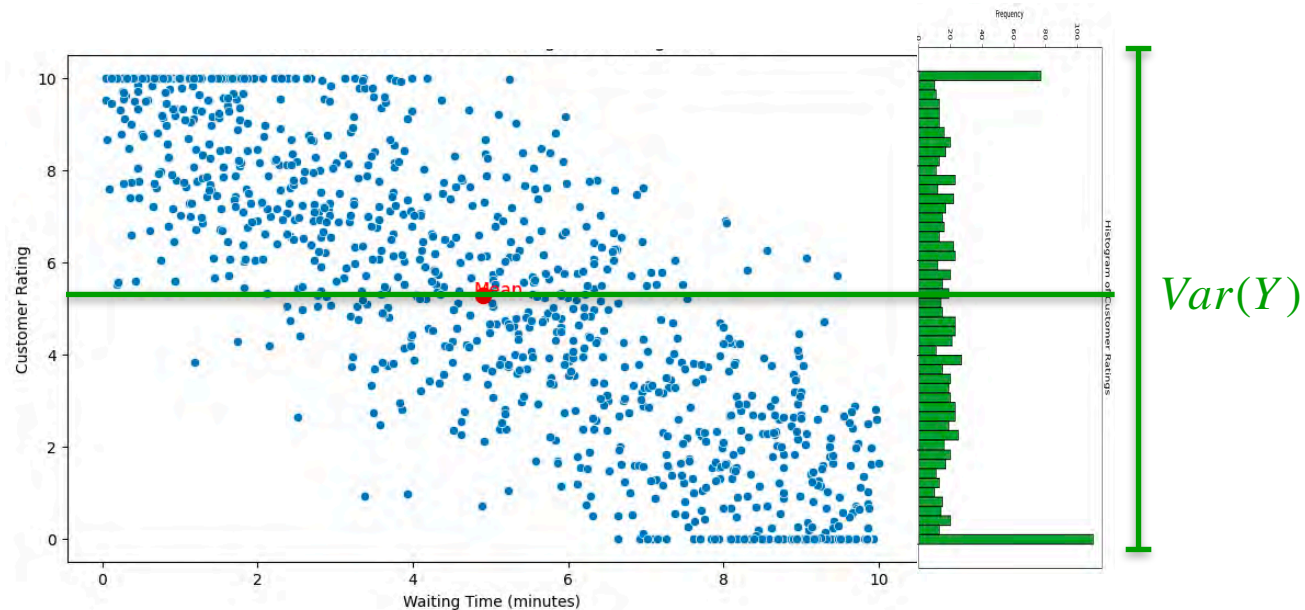


Variances

$$\mathbb{E}[Y] = 5.280$$

$$\mathbb{E}[Y^2] = 38.037$$

$$\text{Var}(Y) = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2$$

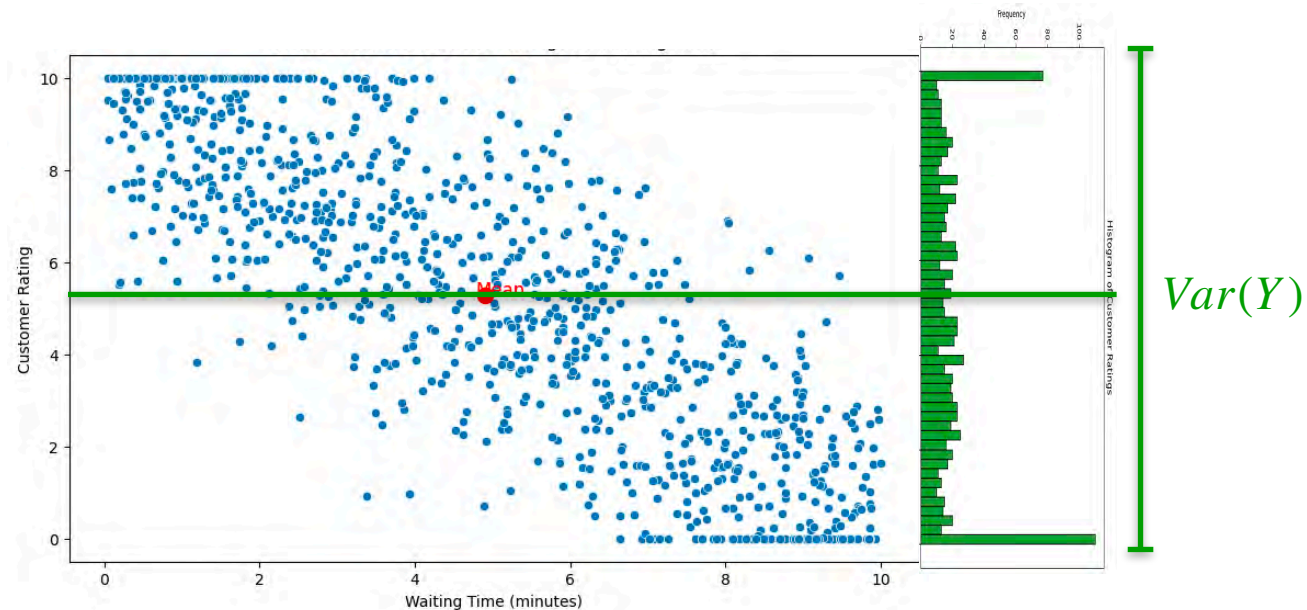


Variances

$$\mathbb{E}[Y] = 5.280$$

$$\mathbb{E}[Y^2] = 38.037$$

$$\begin{aligned} \text{Var}(Y) &= \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 \\ &= 38.037 - 5.280^2 \end{aligned}$$

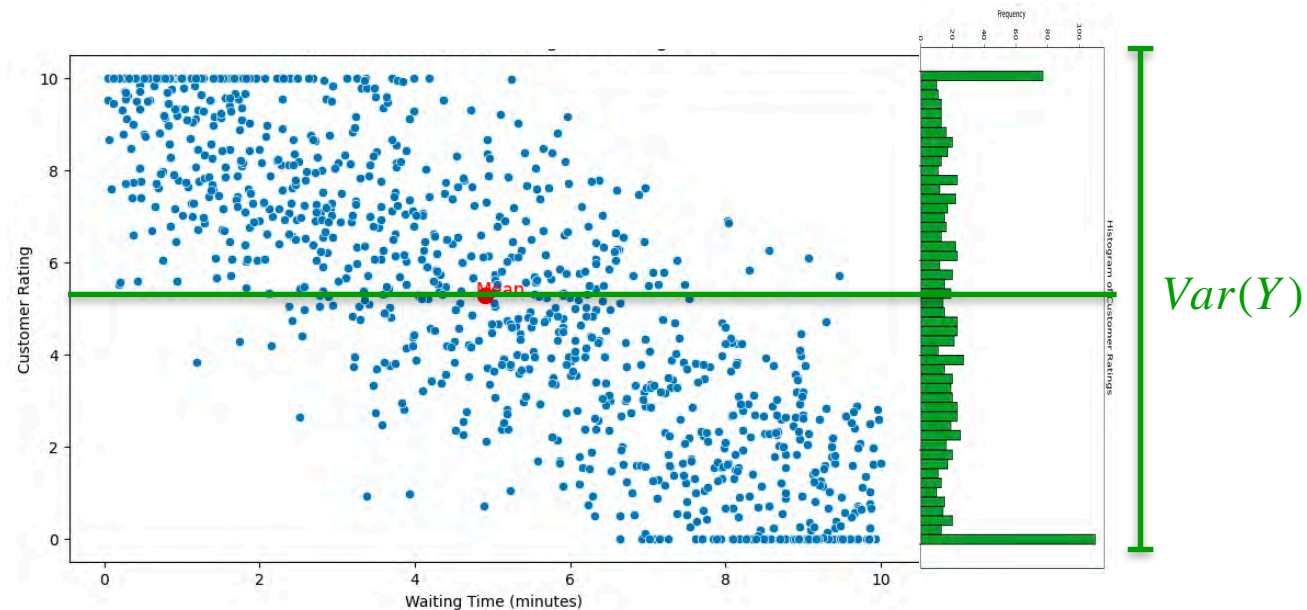


Variances

$$\mathbb{E}[Y] = 5.280$$

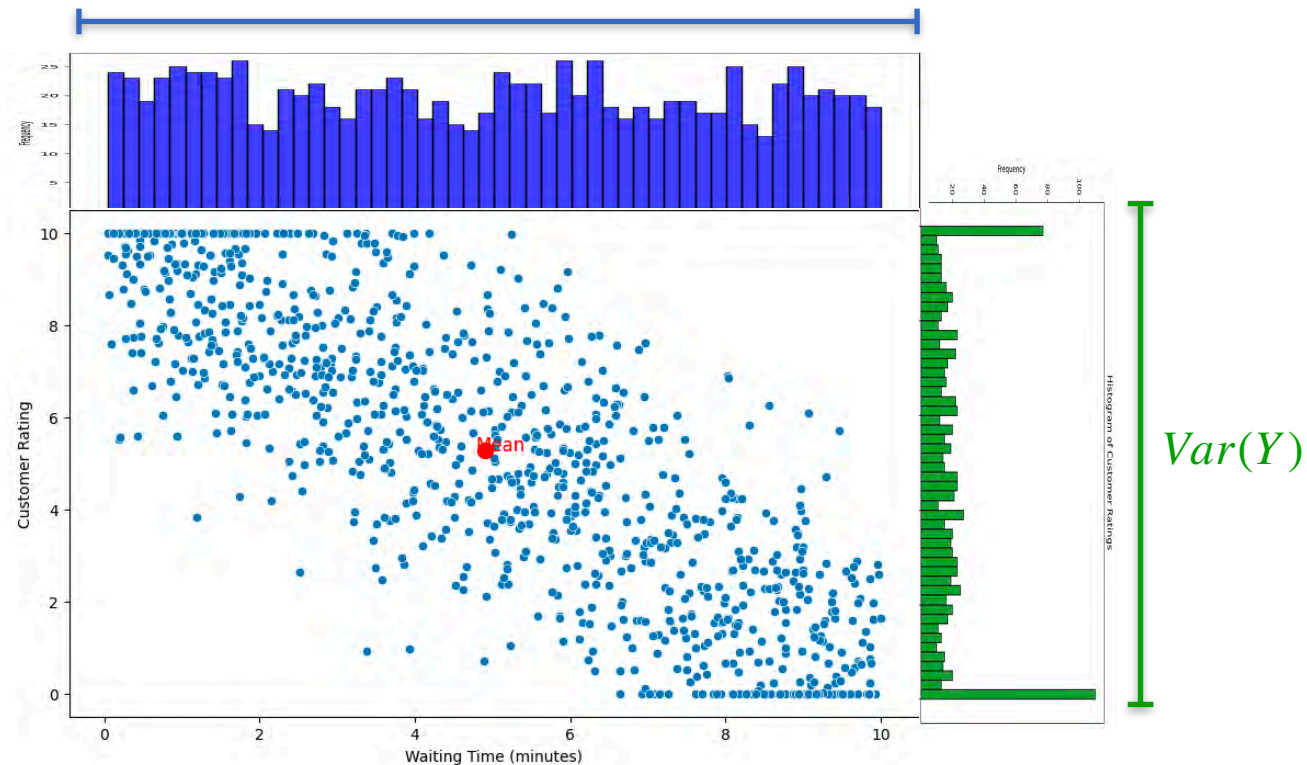
$$\mathbb{E}[Y^2] = 38.037$$

$$\begin{aligned} \text{Var}(Y) &= \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 \\ &= 38.037 - 5.280^2 \\ &= 10.163 \end{aligned}$$



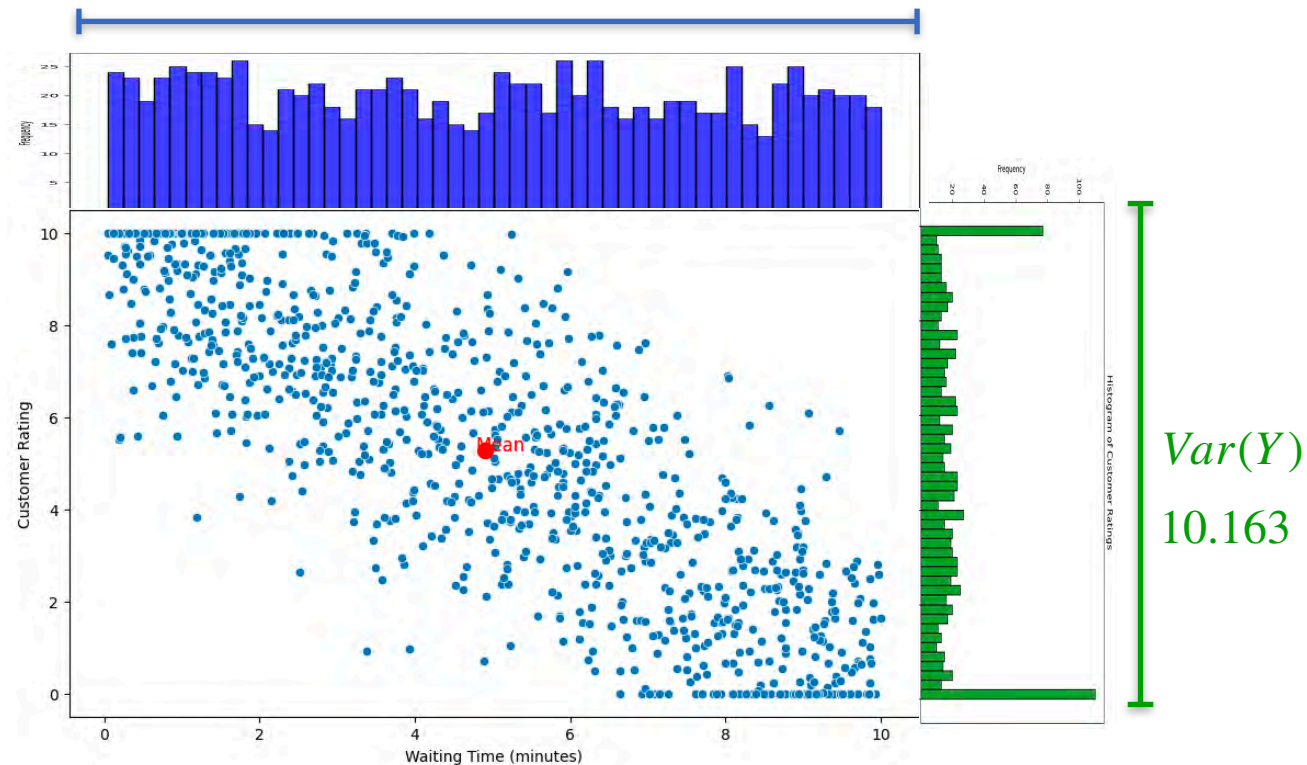
Variances

$Var(X)$

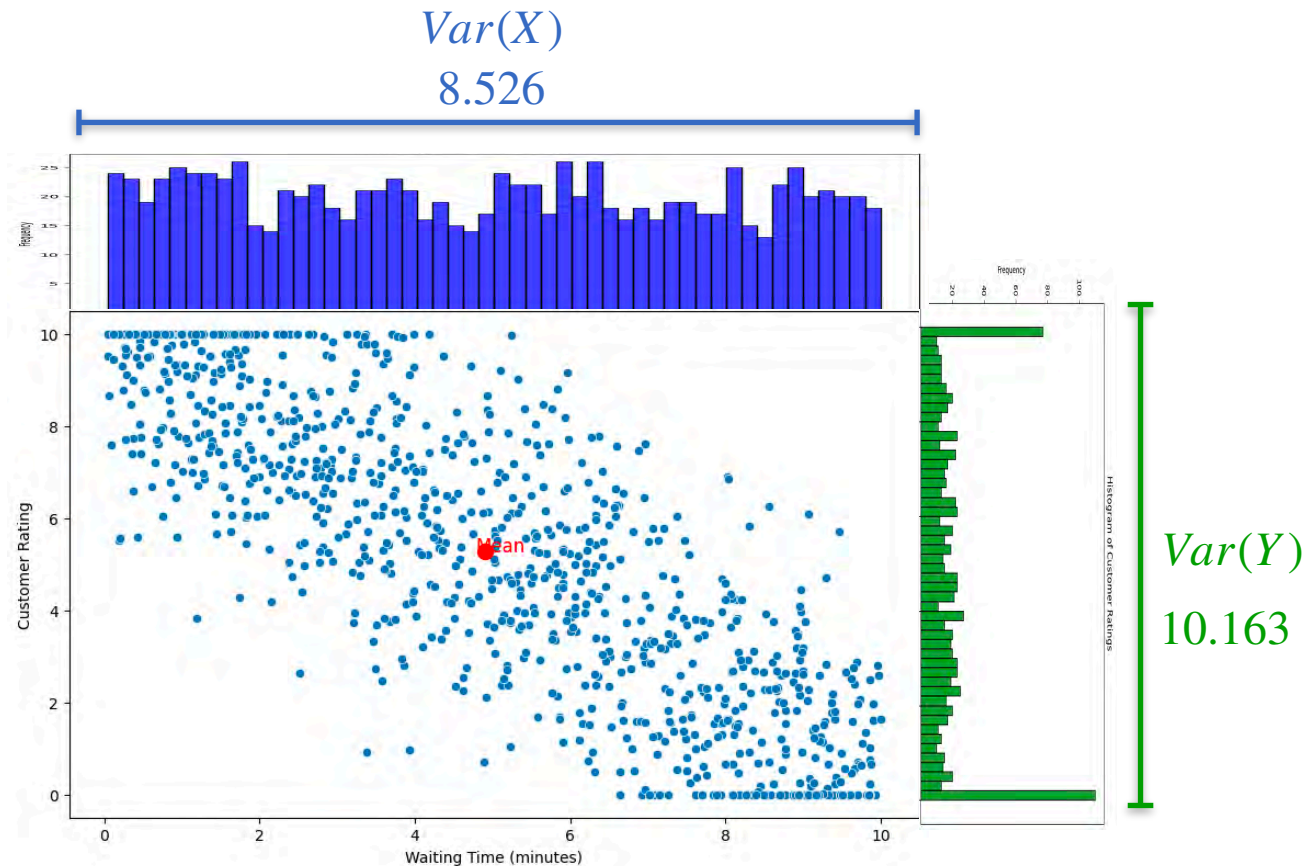


Variances

$Var(X)$



Variances





DeepLearning.AI

Probability Distributions with Multiple Variables

Marginal and Conditional Distribution

Marginal Distribution: Example 1

Marginal Distribution: Example 1

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

Marginal Distribution: Example 1

	Height (Y)					
	45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0
	8	0	0	2/10	0	0
	9	0	0	0	0	3/10
	10	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

Marginal Distribution: Example 1

	Height (Y)					
	45	46	47	48	49	50
Age (X) 7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

Marginal Distribution: Example 1

Age (X)	Height (Y)					
	45	46	47	48	49	50
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

Marginal Distribution

Marginal Distribution: Example 1

Age (X)	Height (Y)					
	45	46	47	48	49	50
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

Marginal Distribution

Distribution of one variable
while ignoring the other

Marginal Distribution: Example 1

Age (X)	Height (Y)					
	45	46	47	48	49	50
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

Marginal Distribution: Example 1

	Height (Y)					
	45	46	47	48	49	50
Age (X)						
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

To find the marginal distribution of height:

Marginal Distribution: Example 1

Age (X)	Height (Y)					
	45	46	47	48	49	50
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

To find the marginal distribution of height:

sum the joint probability distribution over all values of age

Marginal Distribution: Example 1

Age (X)	Height (Y)					
	45	46	47	48	49	50
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

Marginal Distribution: Example 1

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
		1/10	2/10	2/10	0	3/10	2/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

$$p_Y(y_j) = \sum_i p_{XY}(x_i, y_j)$$

Marginal Distribution: Example 1

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
		1/10	2/10	2/10	0	3/10	2/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

$$p_Y(y_j) = \sum_i p_{XY}(x_i, y_j)$$

Marginal Distribution: Example 1

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
		1/10	2/10	2/10	0	3/10	2/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

$$p_Y(y_j) = \sum_i p_{XY}(x_i, y_j)$$

$$p_Y(50) =$$

Marginal Distribution: Example 1

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
		1/10	2/10	2/10	0	3/10	2/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

$$p_Y(y_j) = \sum_i p_{XY}(x_i, y_j)$$

Marginal Distribution: Example 1

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
		1/10	2/10	2/10	0	3/10	2/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

$$p_Y(y_j) = \sum_i p_{XY}(x_i, y_j)$$

$$p_Y(50) = \sum_i p_{XY}(x_i, 50)$$

Marginal Distribution: Example 1

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
		1/10	2/10	2/10	0	3/10	2/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

$$p_Y(y_j) = \sum_i p_{XY}(x_i, y_j)$$

$$p_Y(50) = \sum_i p_{XY}(x_i, 50)$$

$$p_Y(50) = \frac{2}{10}$$

Marginal Distribution: Example 1

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
		1/10	2/10	2/10	0	3/10	2/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

Marginal Distribution: Example 1

		Height (Y)						Age (X)
		45	46	47	48	49	50	Height (Y)
Age (X)	7	1/10	2/10	0	0	0	0	3/10
	8	0	0	2/10	0	0	0	2/10
	9	0	0	0	0	3/10	1/10	4/10
	10	0	0	0	0	0	1/10	1/10
		1/10	2/10	2/10	0	3/10	2/10	

Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

$$p_X(x_i) = \sum_j p_{XY}(x_i, y_j)$$

Marginal Distribution: Example 1

		Height (Y)						
		45	46	47	48	49	50	
Age (X)	7	1/10	2/10	0	0	0	0	3/10
	8	0	0	2/10	0	0	0	2/10
	9	0	0	0	0	3/10	1/10	4/10
	10	0	0	0	0	0	1/10	1/10
		1/10	2/10	2/10	0	3/10	2/10	

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

$$p_X(x_i) = \sum_j p_{XY}(x_i, y_j)$$

Marginal Distribution: Example 1

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	2/10
	9	0	0	0	0	3/10	4/10
	10	0	0	0	0	0	1/10
		1/10	2/10	2/10	0	3/10	2/10

Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

$$p_X(x_i) = \sum_j p_{XY}(x_i, y_j)$$

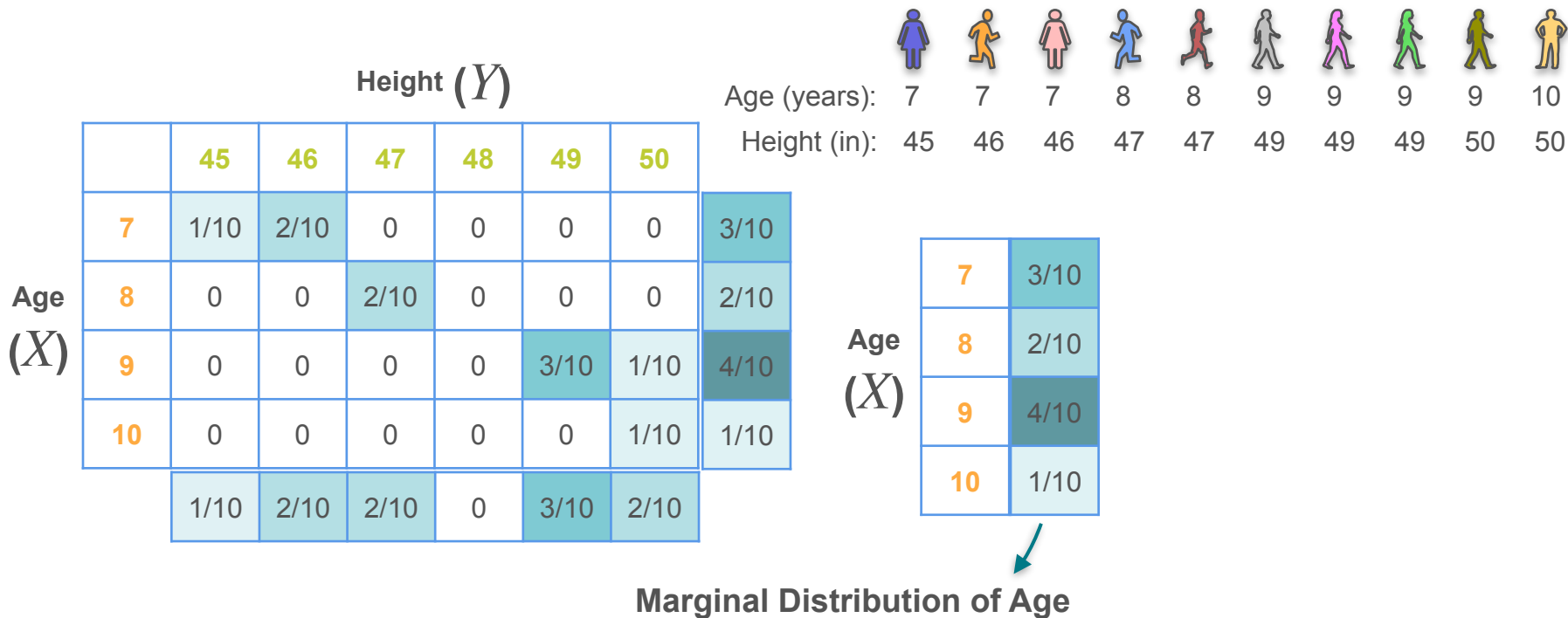
$$p_X(7) = \sum_j p_{XY}(7, y_j)$$

$$p_X(7) = \frac{3}{10}$$

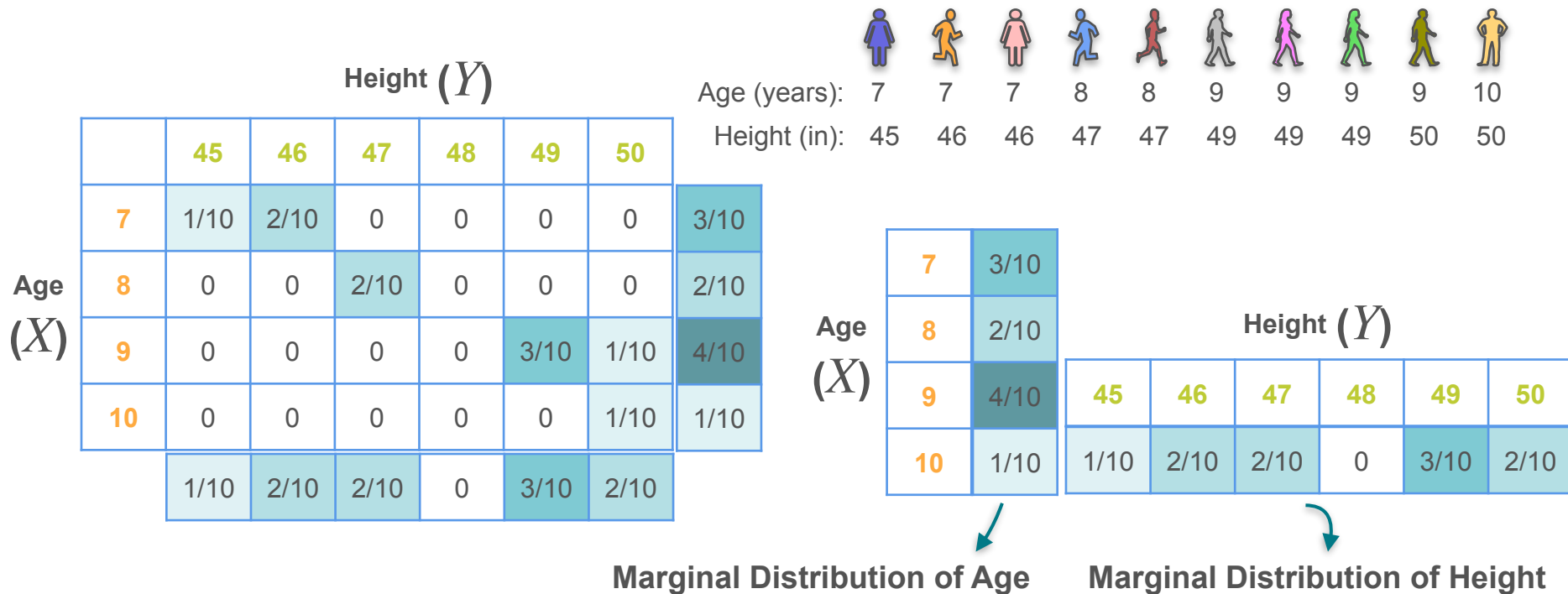
Marginal Distribution: Example 1

		Height (Y)						Age (years):									
		45	46	47	48	49	50	Height (in):									
Age (X)																	
	7	1/10	2/10	0	0	0	0	45	46	46	47	47	49	49	49	50	50
	8	0	0	2/10	0	0	0										
	9	0	0	0	0	3/10	1/10										
	10	0	0	0	0	0	1/10										
		1/10	2/10	2/10	0	3/10	2/10										

Marginal Distribution: Example 1



Marginal Distribution: Example 1



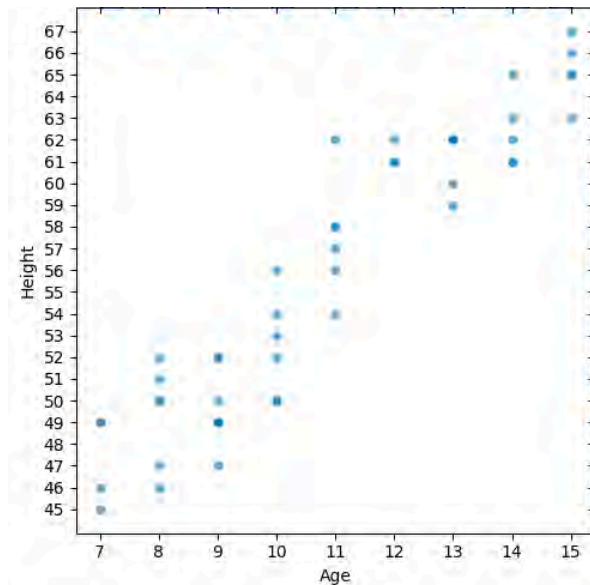
Marginal Distribution: Example 1

Marginal Distribution: Example 1

Age and Height Dataset
for 50 children

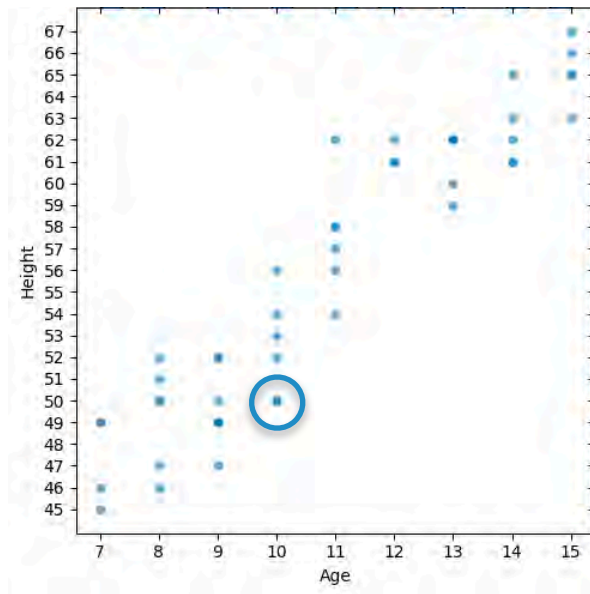
Marginal Distribution: Example 1

Age and Height Dataset
for 50 children



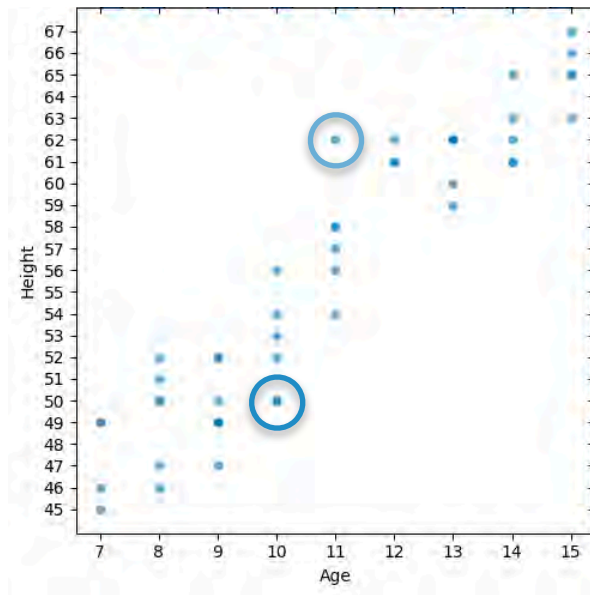
Marginal Distribution: Example 1

Age and Height Dataset
for 50 children



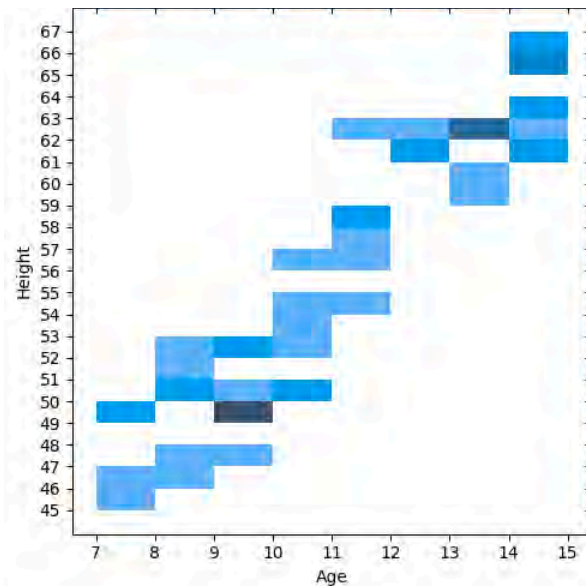
Marginal Distribution: Example 1

Age and Height Dataset
for 50 children



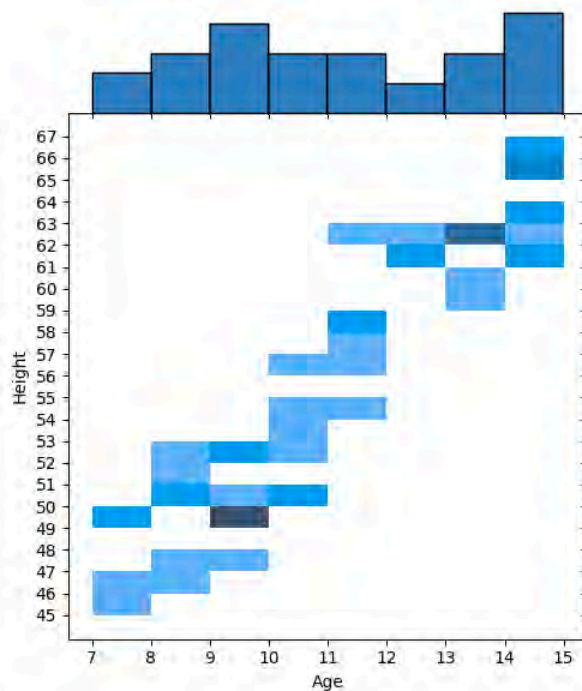
Marginal Distribution: Example 1

Age and Height Dataset
for 50 children



Marginal Distribution: Example 1

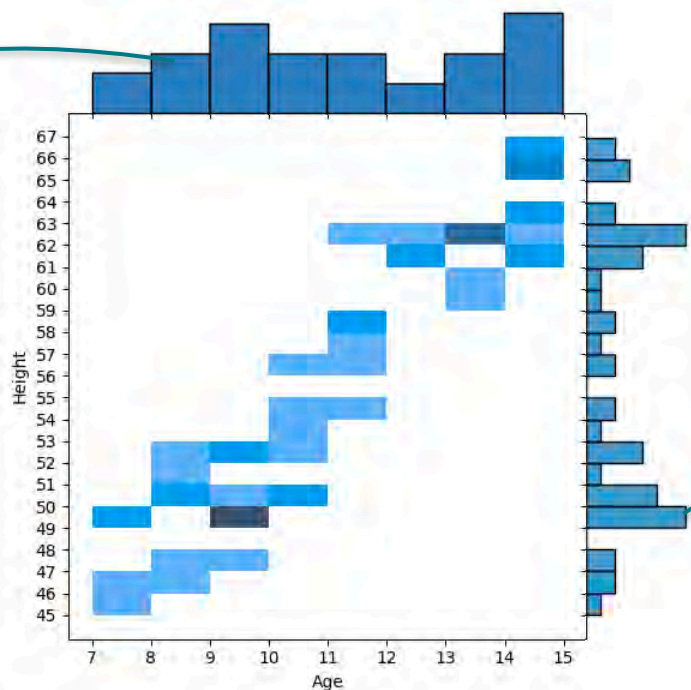
Age and Height Dataset
for 50 children



Marginal Distribution: Example 1

Marginal Distribution of Age

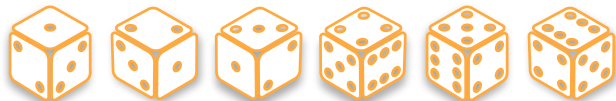
Age and Height Dataset
for 50 children



Marginal Distribution of Height

Marginal Distributions: Example 2

X : the number rolled on the 1st dice



$$\frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6}$$

Y : the number rolled on the 2nd dice



$$\frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6}$$

Marginal Distributions: Example 2

Y

X		1	2	3	4	5	6
	1	1/36	1/36	1/36	1/36	1/36	1/36
	2	1/36	1/36	1/36	1/36	1/36	1/36
	3	1/36	1/36	1/36	1/36	1/36	1/36
	4	1/36	1/36	1/36	1/36	1/36	1/36
	5	1/36	1/36	1/36	1/36	1/36	1/36
	6	1/36	1/36	1/36	1/36	1/36	1/36

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

Marginal Distributions: Example 2

Y

X	Y						
	1	2	3	4	5	6	
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	1/36	1/36	1/36	1/36	1/36	1/6
3	1/36	1/36	1/36	1/36	1/36	1/36	1/6
4	1/36	1/36	1/36	1/36	1/36	1/36	1/6
5	1/36	1/36	1/36	1/36	1/36	1/36	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

Marginal Distributions: Example 2

Y

X		1	2	3	4	5	6	
	1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	2	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	3	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	4	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	5	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
		1/6	1/6	1/6	1/6	1/6	1/6	

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

Marginal Distributions: Example 2

Y

X		1	2	3	4	5	6	
	1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	2	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	3	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	4	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	5	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
		1/6	1/6	1/6	1/6	1/6	1/6	

X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

$$p_X(x_i) = \frac{1}{6}$$

Marginal Distributions: Example 2

		Y						
		1	2	3	4	5	6	
X	1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	2	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	3	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	4	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	5	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	6	1/36	1/36	1/36	1/36	1/36	1/36	1/6
		1/6	1/6	1/6	1/6	1/6	1/6	

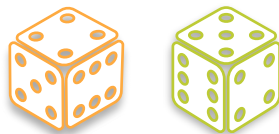
X : the number rolled on the 1st dice

Y : the number rolled on the 2nd dice

$$p_X(x_i) = \frac{1}{6}$$

$$p_Y(y_j) = \frac{1}{6}$$

Marginal Distributions: Example 3



X : the number rolled on the 1st dice

Y : sum of the two dice

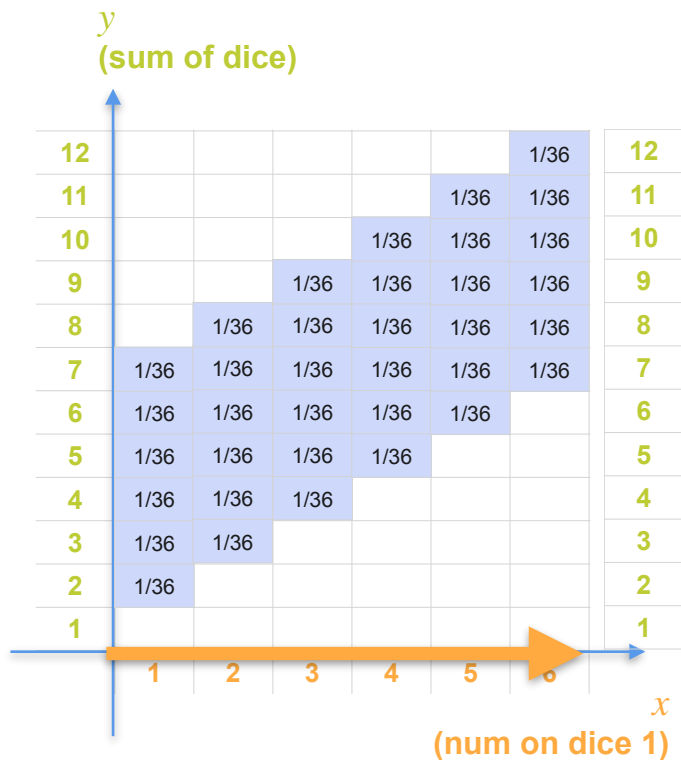
Marginal Distributions: Example 3



X : the number rolled on the 1st dice

Y : sum of the two dice

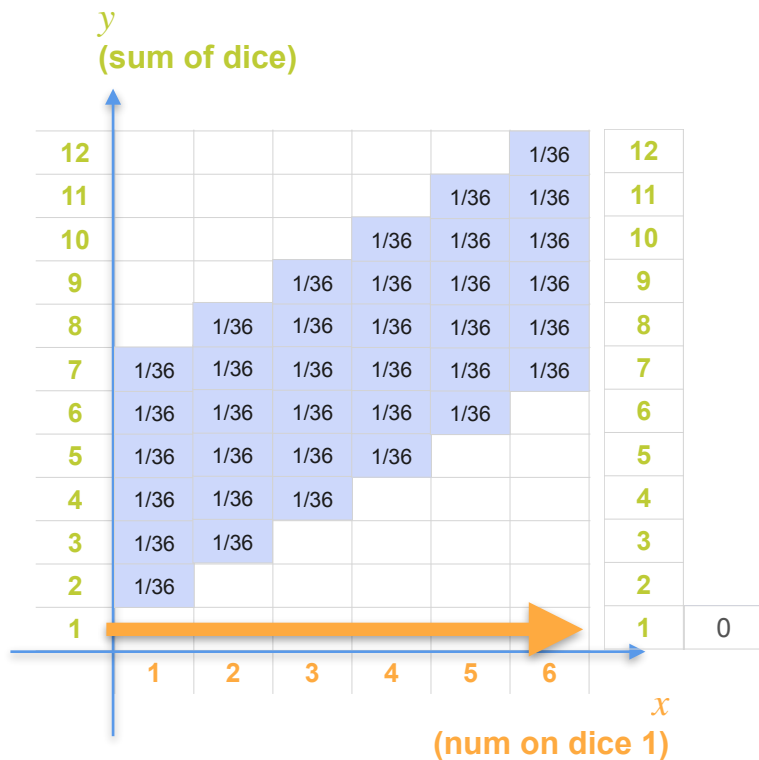
Marginal Distributions: Example 3



X : the number rolled on the 1st dice

Y : sum of the two dice

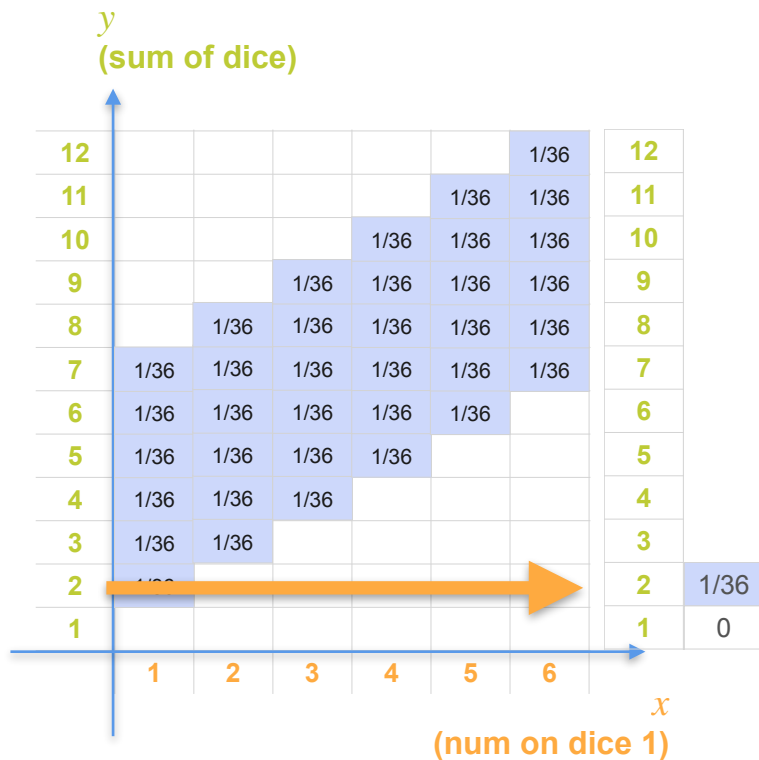
Marginal Distributions: Example 3



X : the number rolled on the 1st dice

Y : sum of the two dice

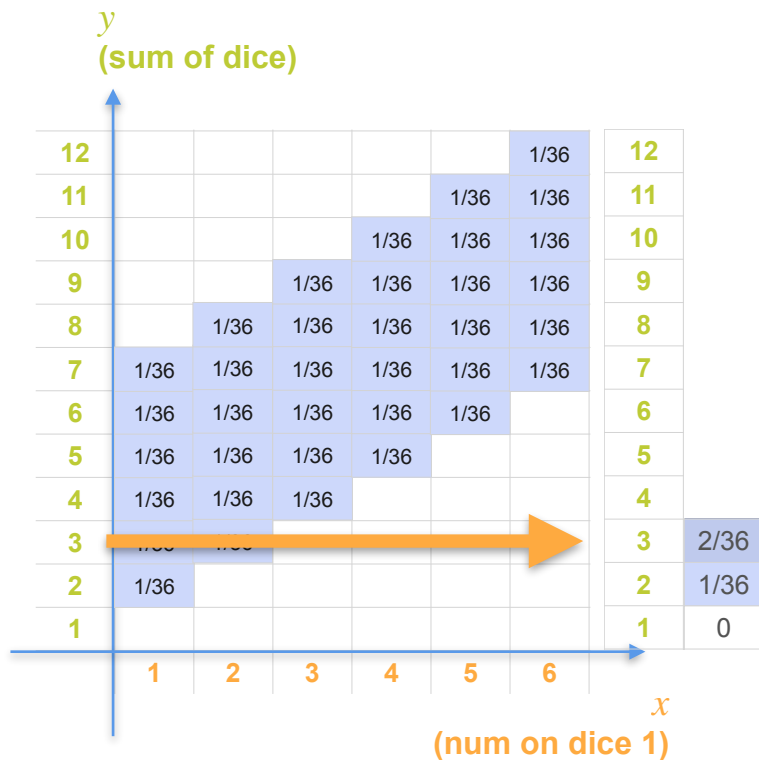
Marginal Distributions: Example 3



X : the number rolled on the 1st dice

Y : sum of the two dice

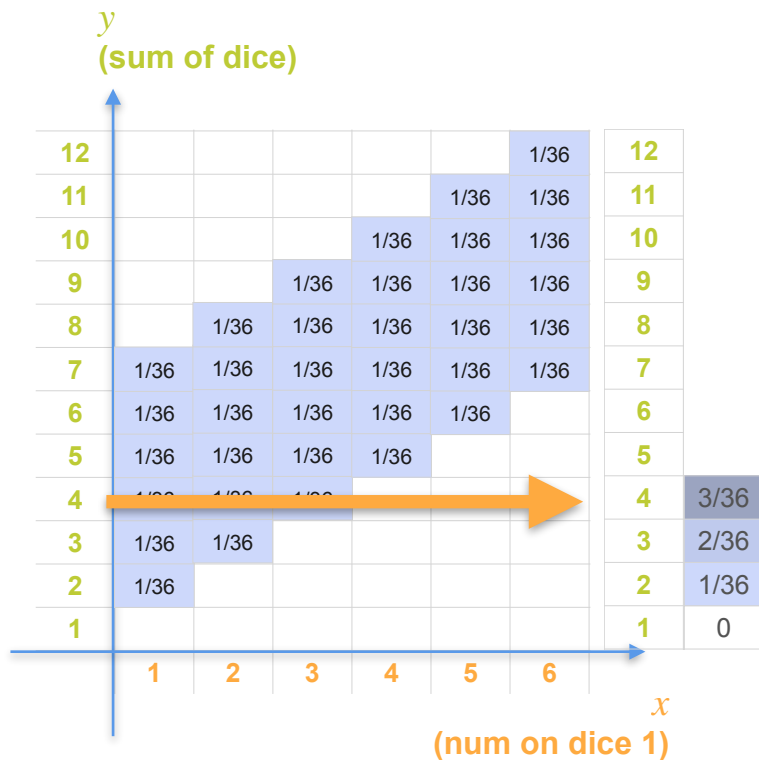
Marginal Distributions: Example 3



X : the number rolled on the 1st dice

Y : sum of the two dice

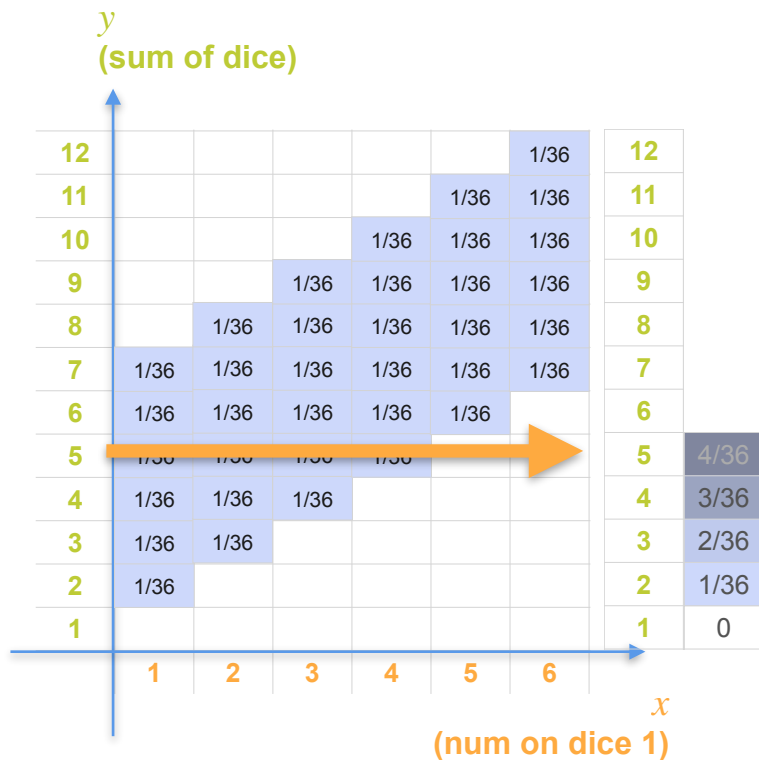
Marginal Distributions: Example 3



X : the number rolled on the 1st dice

Y : sum of the two dice

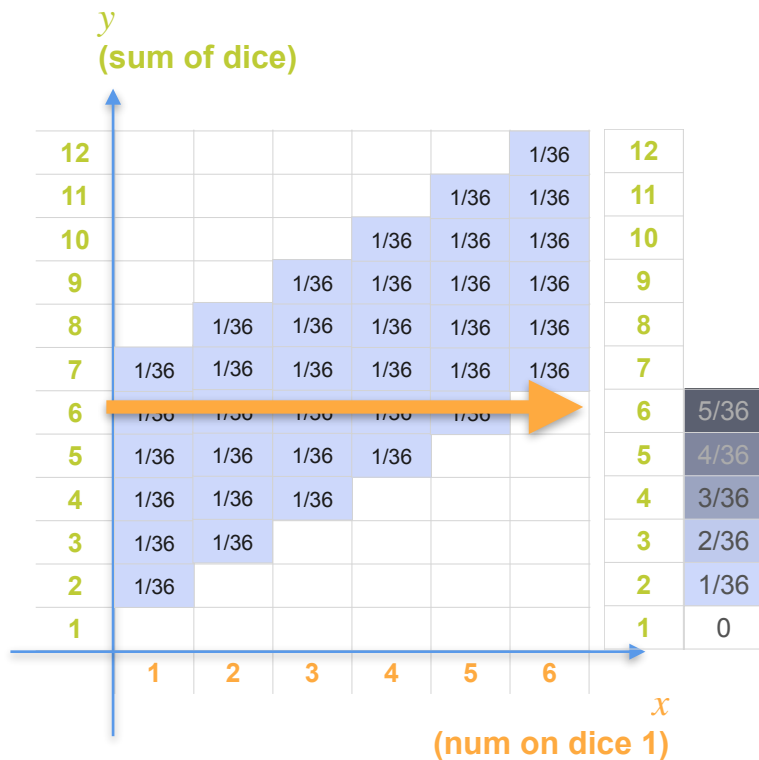
Marginal Distributions: Example 3



X : the number rolled on the 1st dice

Y : sum of the two dice

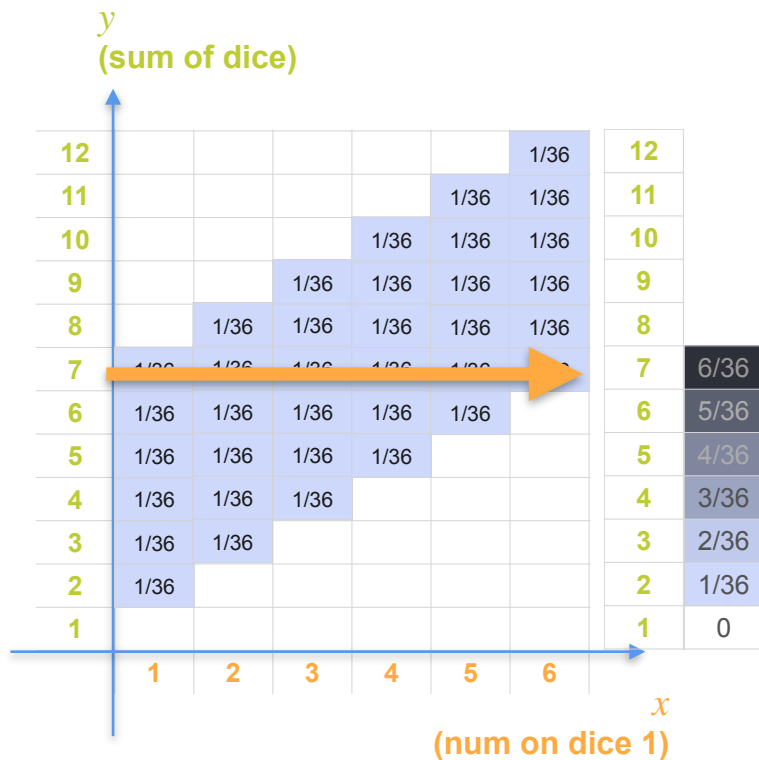
Marginal Distributions: Example 3



X : the number rolled on the 1st dice

Y : sum of the two dice

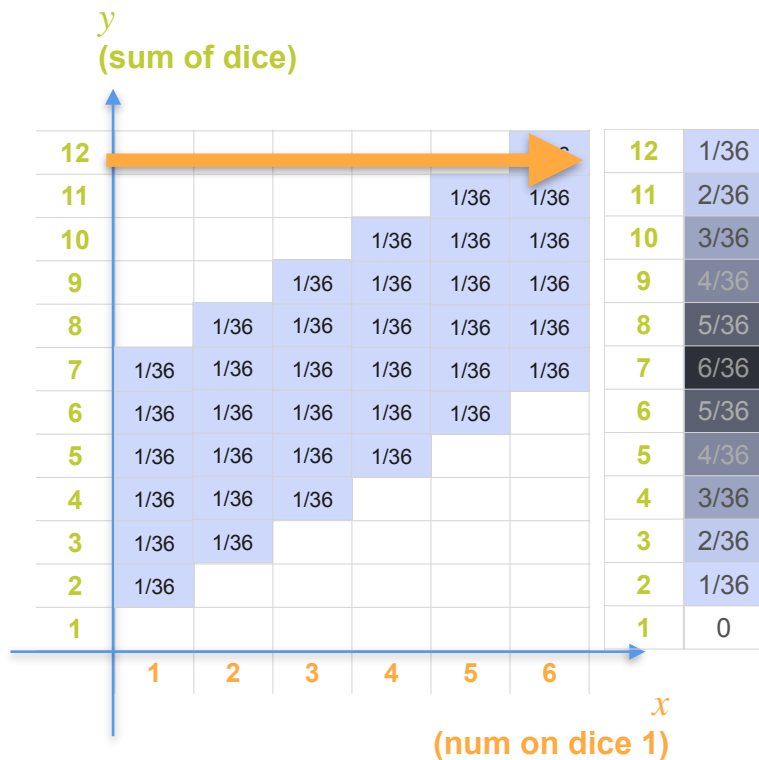
Marginal Distributions: Example 3



X : the number rolled on the 1st dice

Y : sum of the two dice

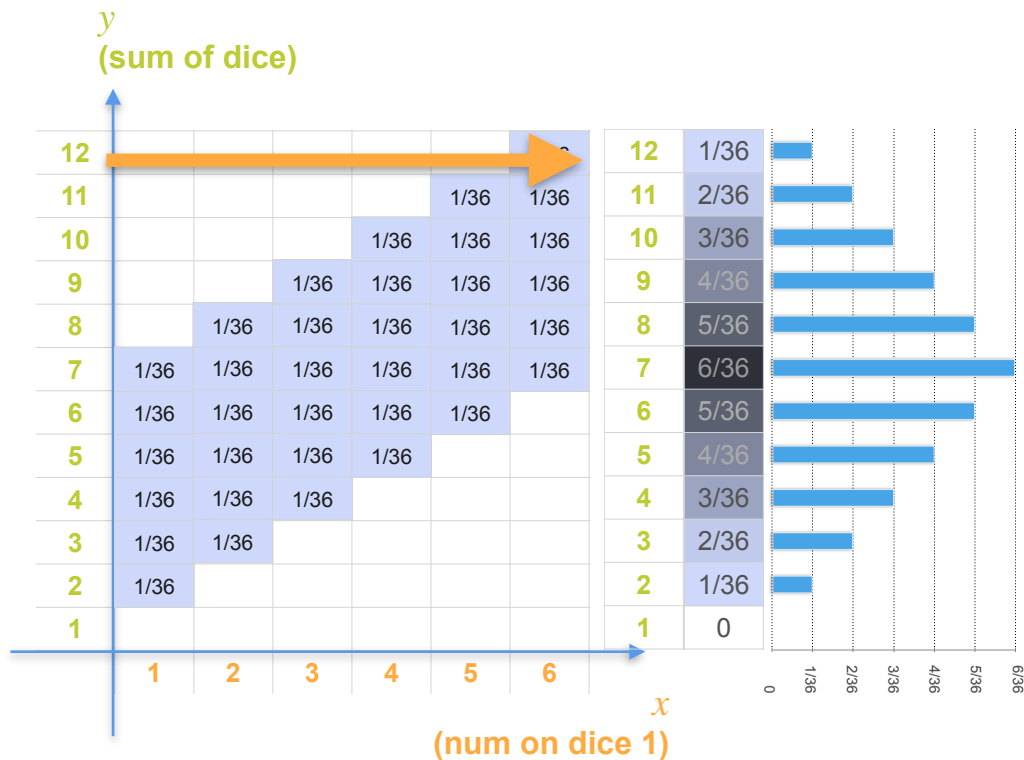
Marginal Distributions: Example 3



X : the number rolled on the 1st dice

Y : sum of the two dice

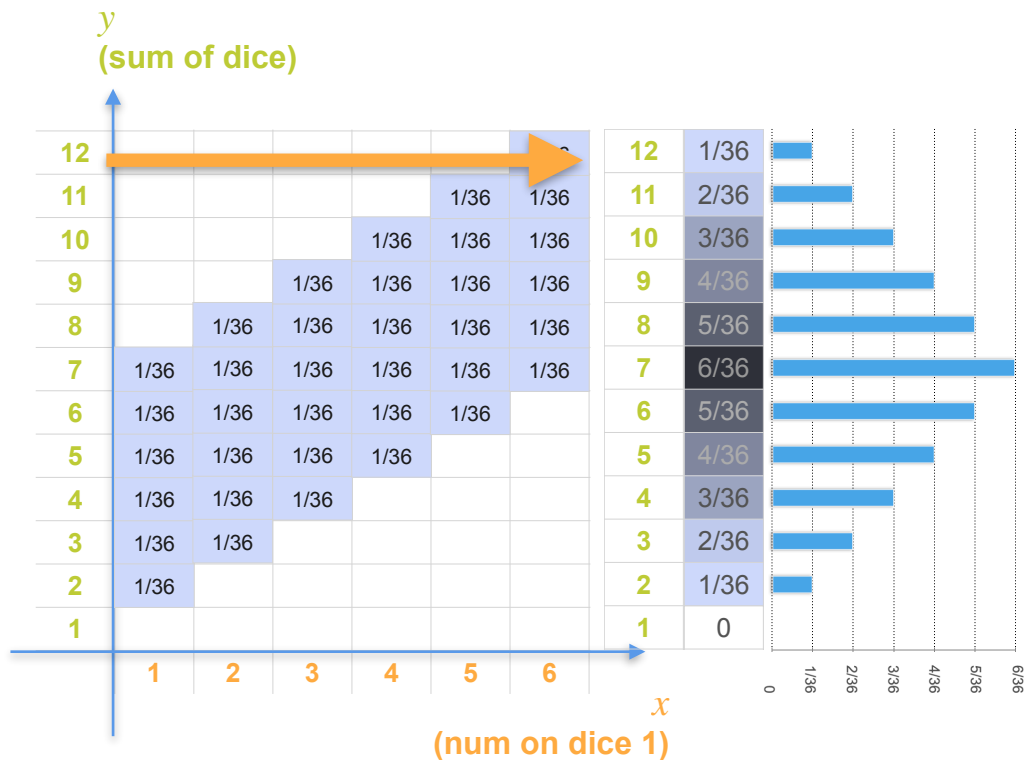
Marginal Distributions: Example 3



X : the number rolled on the 1st dice

Y : sum of the two dice

Marginal Distributions: Example 3

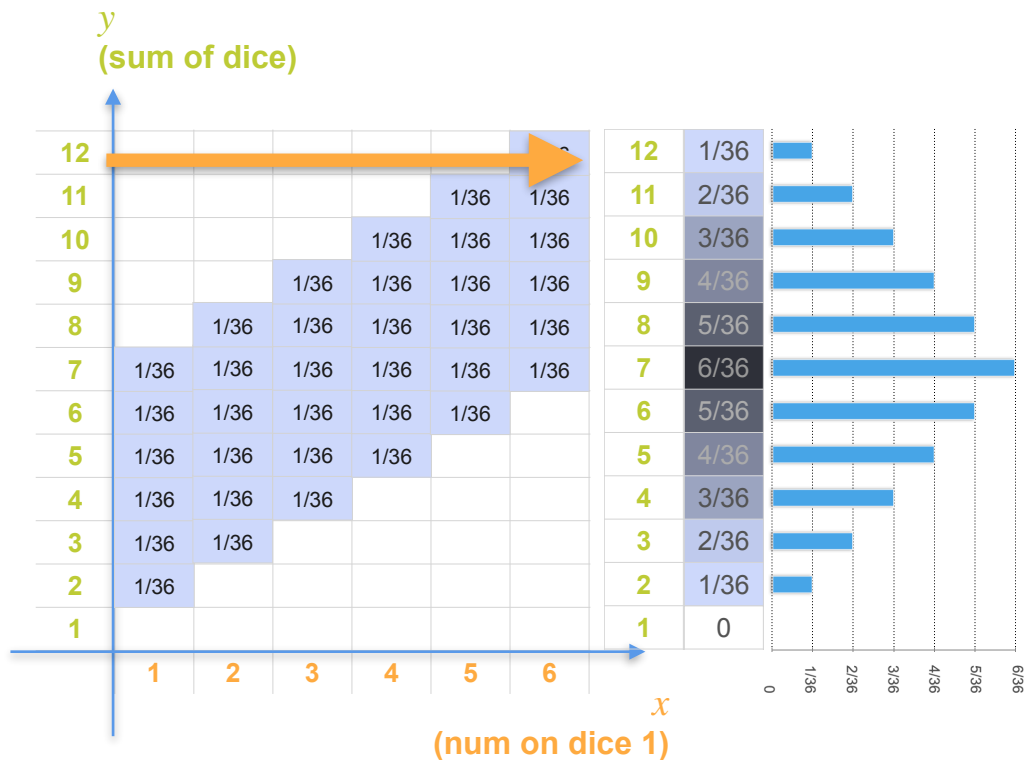


X : the number rolled on the 1st die

Y : sum of the two dice

$$p_Y(y_j) = \sum_i p_{XY}(x_i, y_j) = ?$$

Marginal Distributions: Example 3



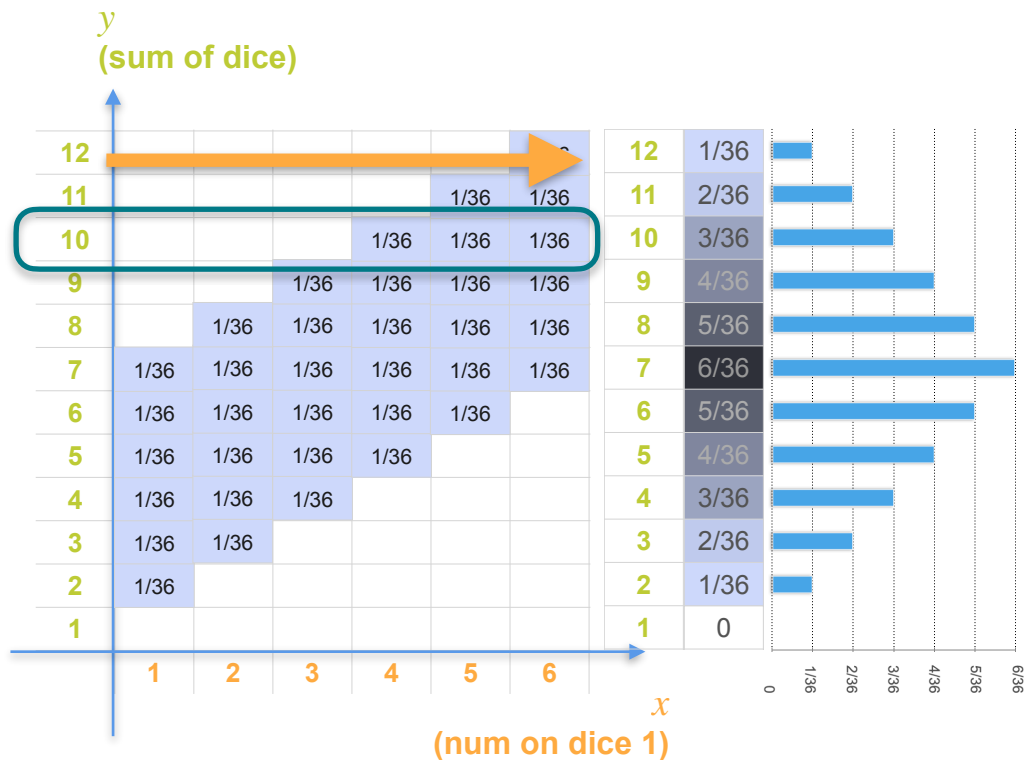
X : the number rolled on the 1st die

Y : sum of the two dice

$$p_Y(y_j) = \sum_i p_{XY}(x_i, y_j) = ?$$

$$p_Y(10) = ?$$

Marginal Distributions: Example 3



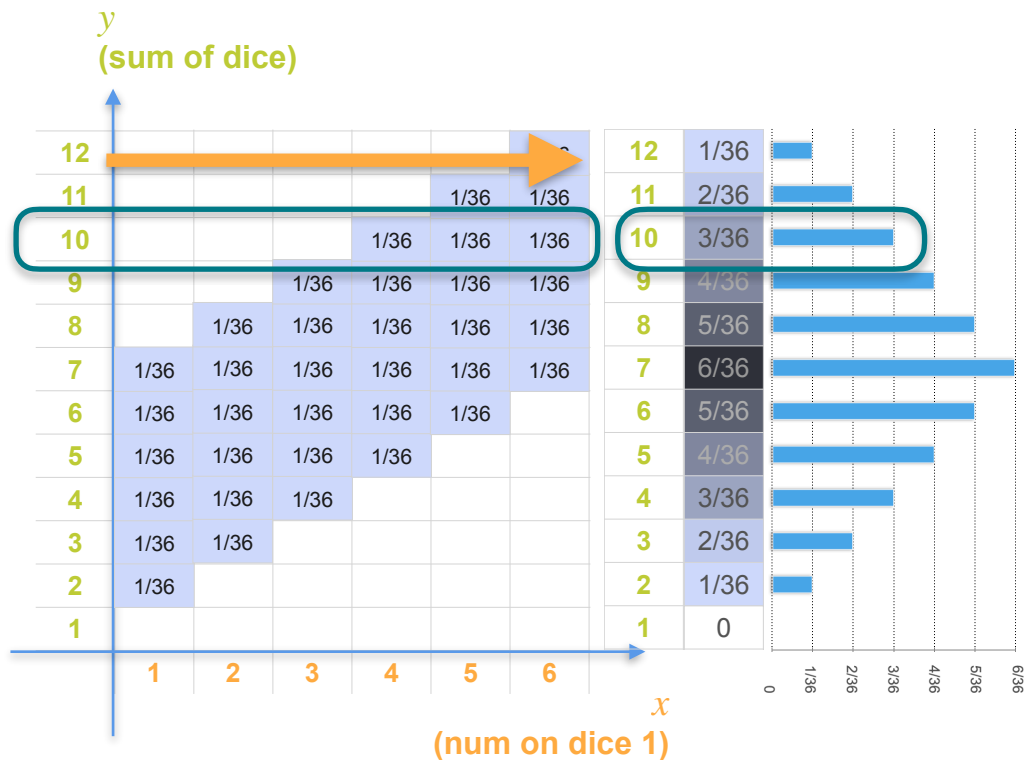
X : the number rolled on the 1st dice

Y : sum of the two dice

$$p_Y(y_j) = \sum_i p_{XY}(x_i, y_j) = ?$$

$$p_Y(10) = ?$$

Marginal Distributions: Example 3



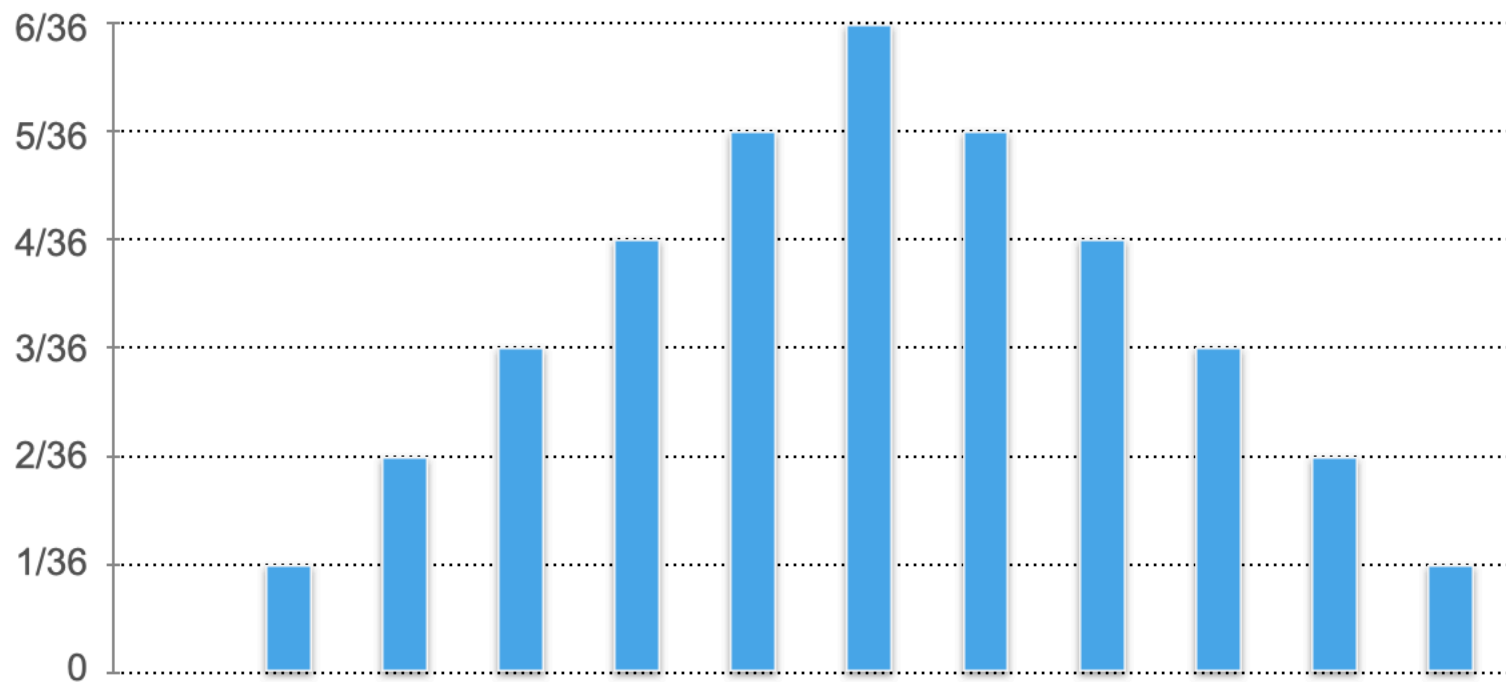
X : the number rolled on the 1st die

Y : sum of the two dice

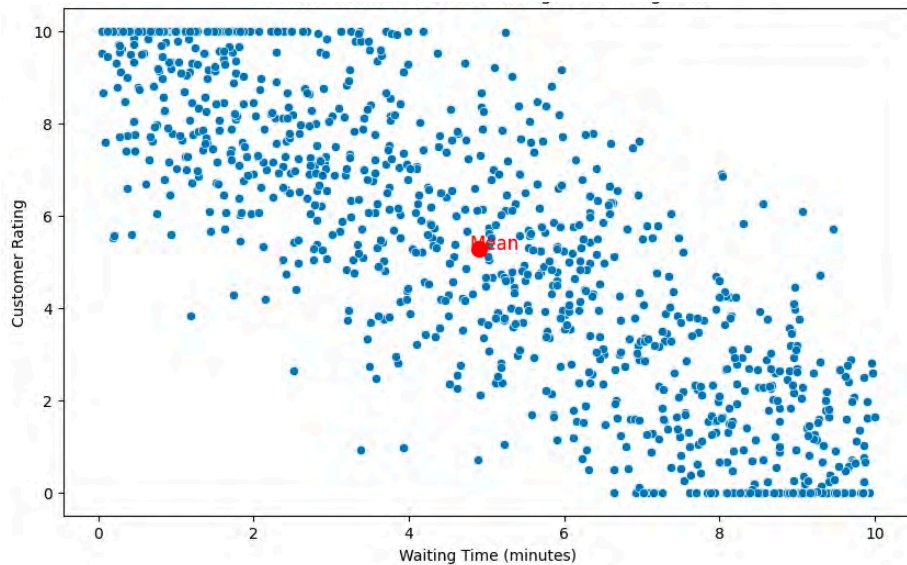
$$p_Y(y_j) = \sum_i p_{XY}(x_i, y_j) = ?$$

$$p_Y(10) = ? = \frac{3}{36}$$

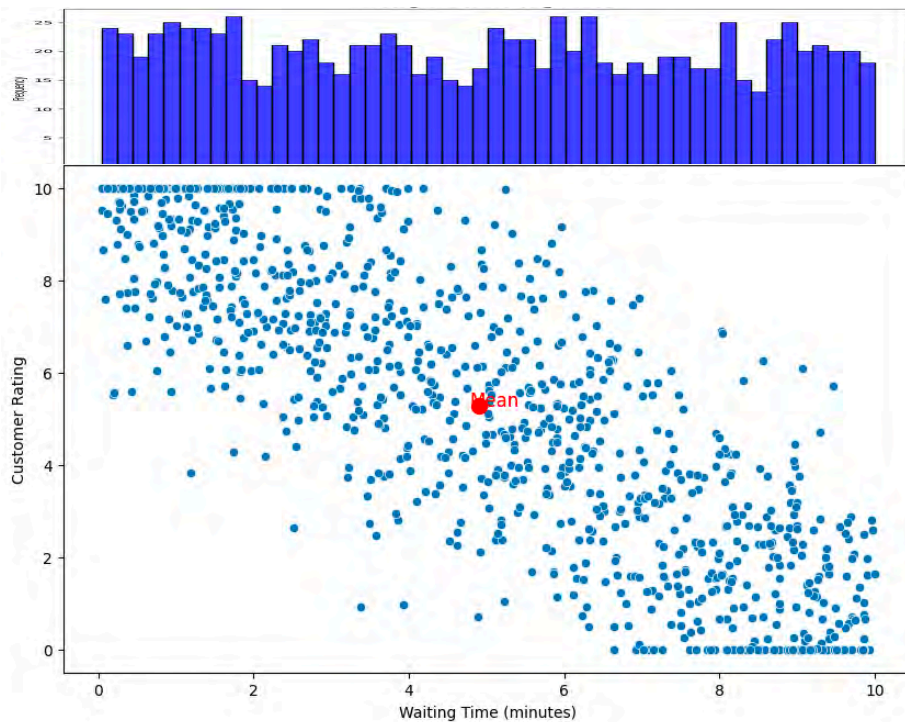
Marginal Distributions: Example 2



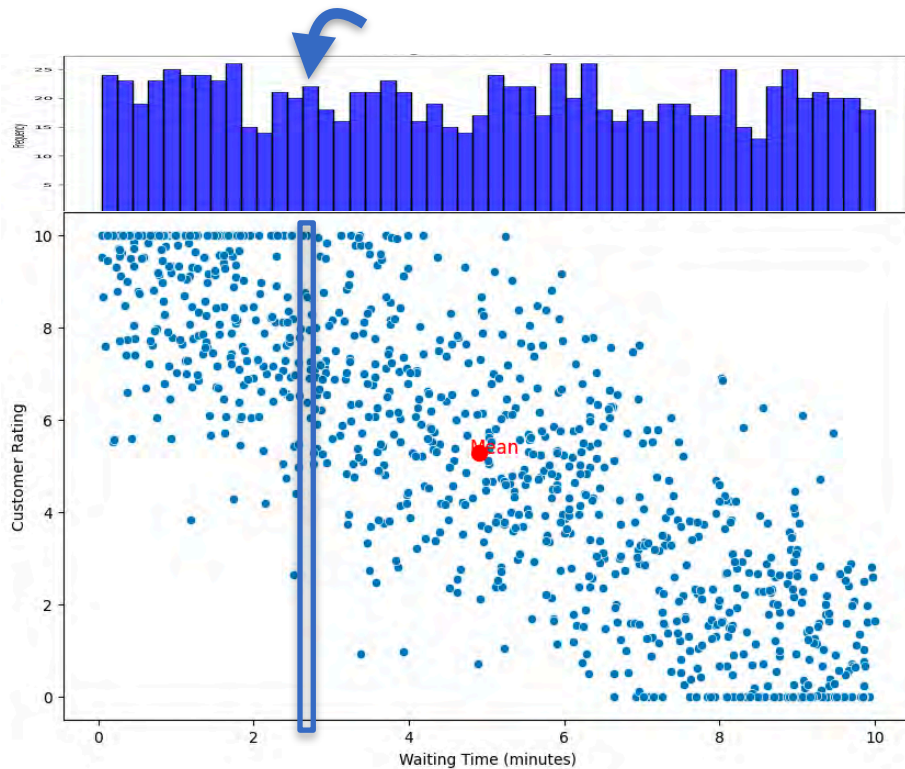
Marginal Distributions



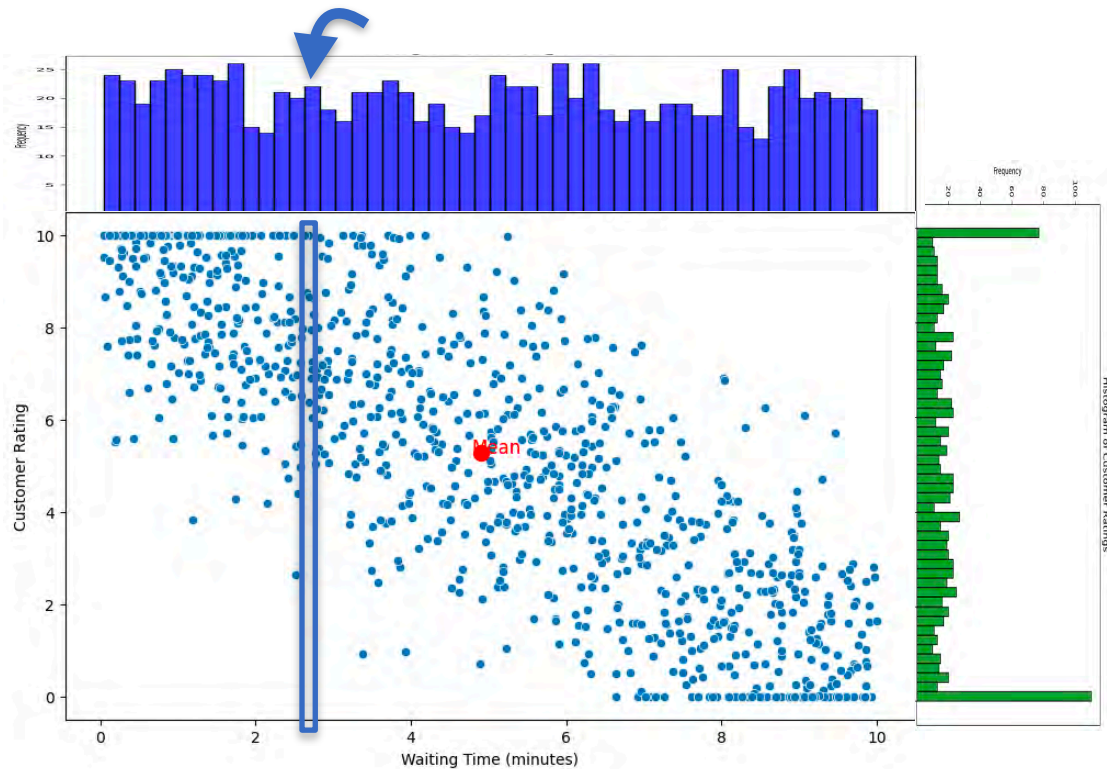
Marginal Distributions



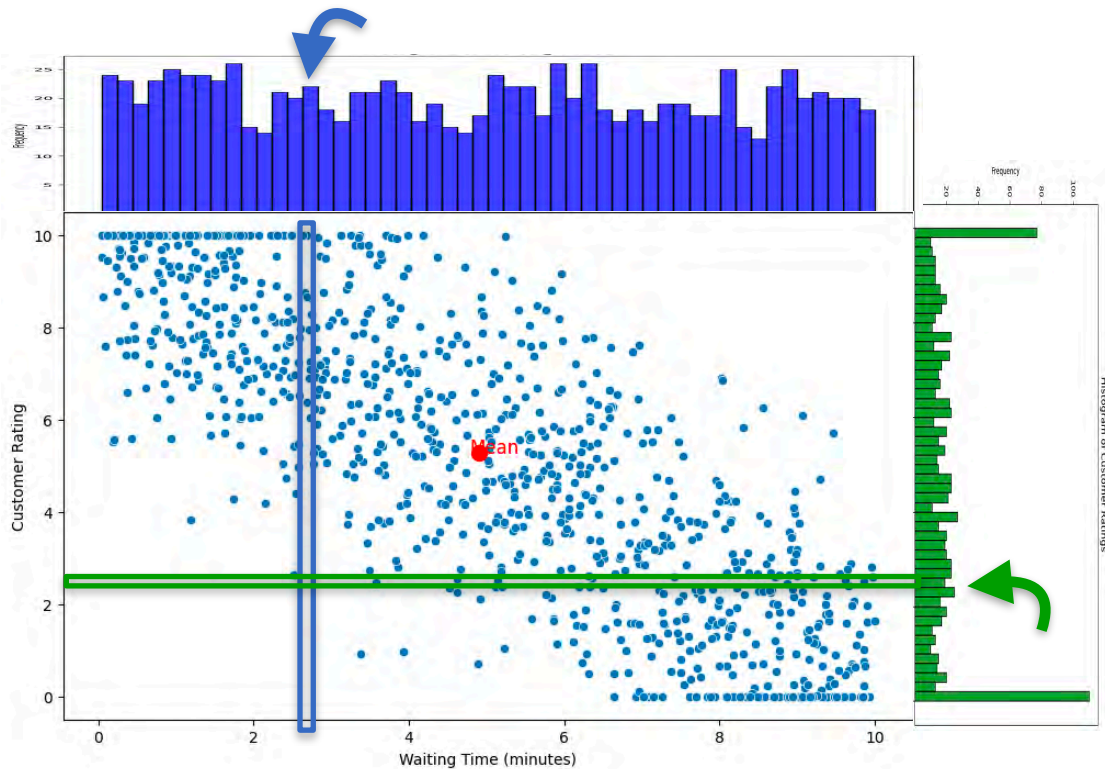
Marginal Distributions



Marginal Distributions



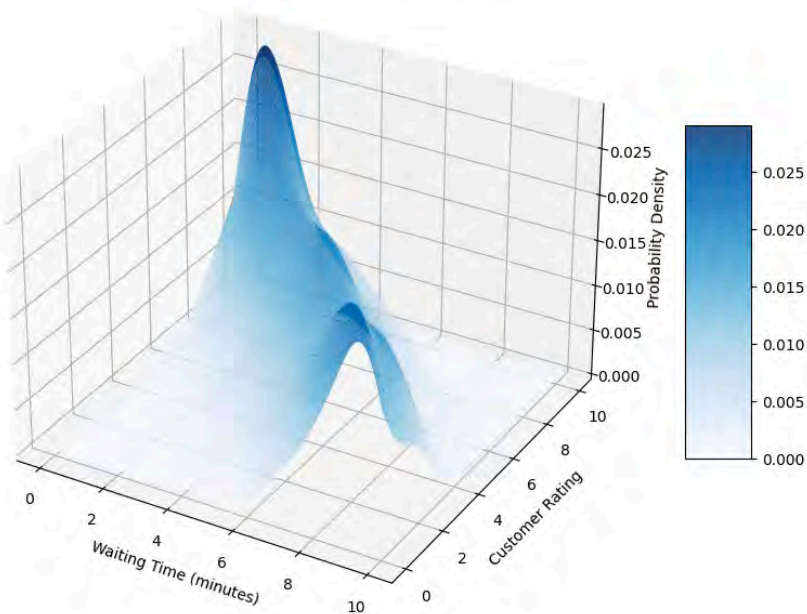
Marginal Distributions



Continuous Marginal Distribution

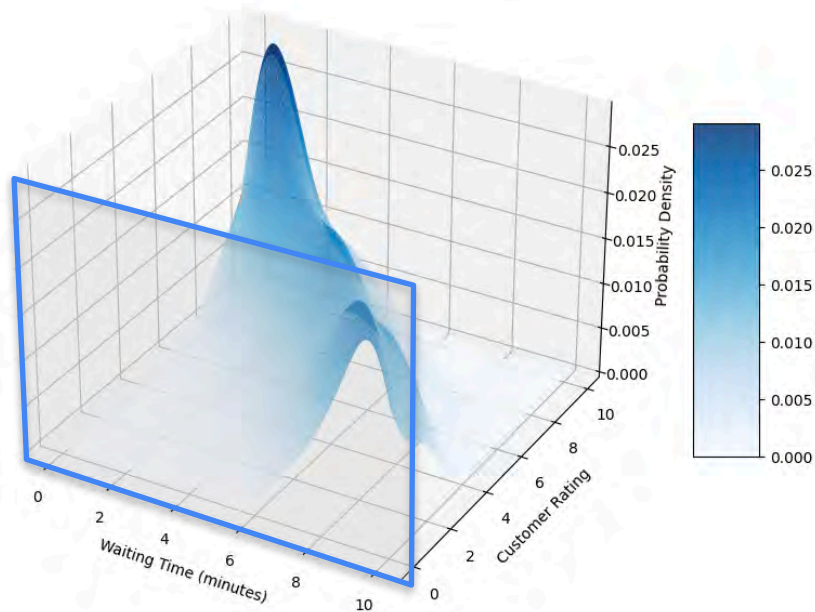
Continuous Marginal Distribution

3D Probability Density Distribution for Customer Ratings vs Waiting Time



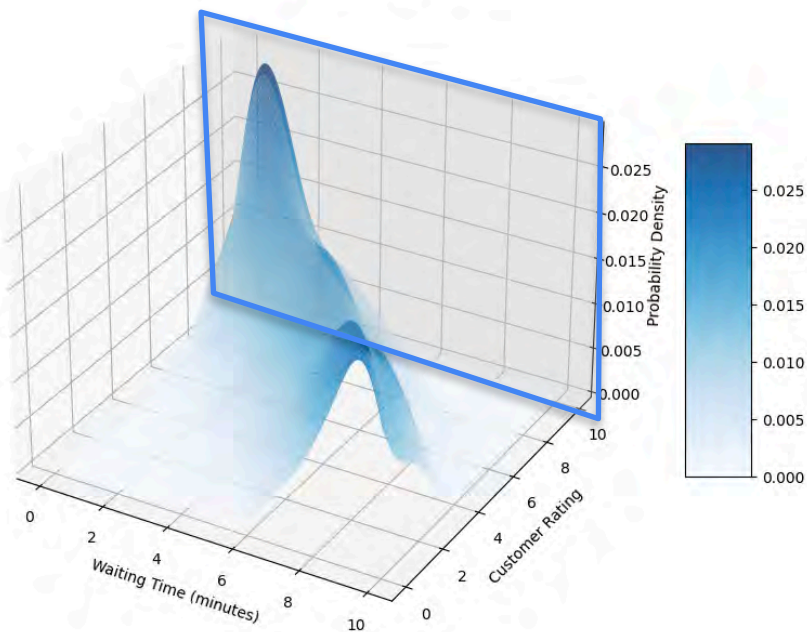
Continuous Marginal Distribution

3D Probability Density Distribution for Customer Ratings vs Waiting Time



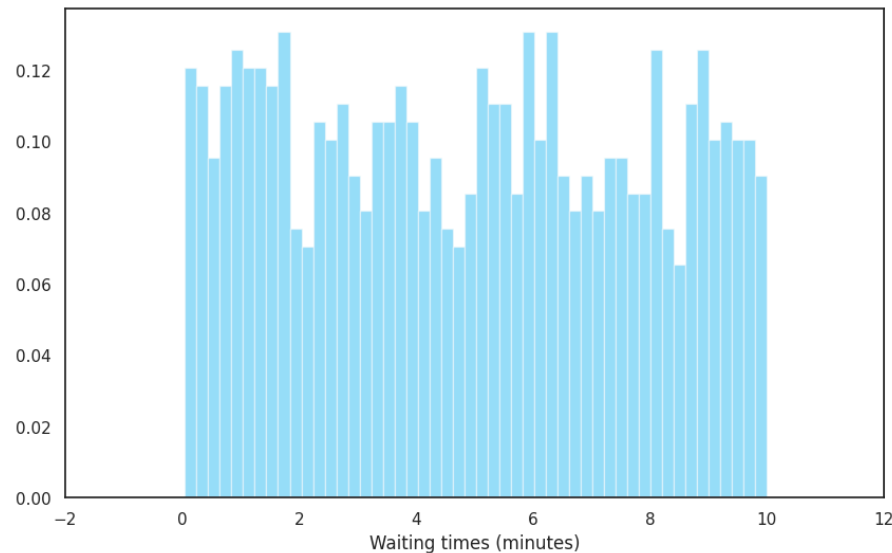
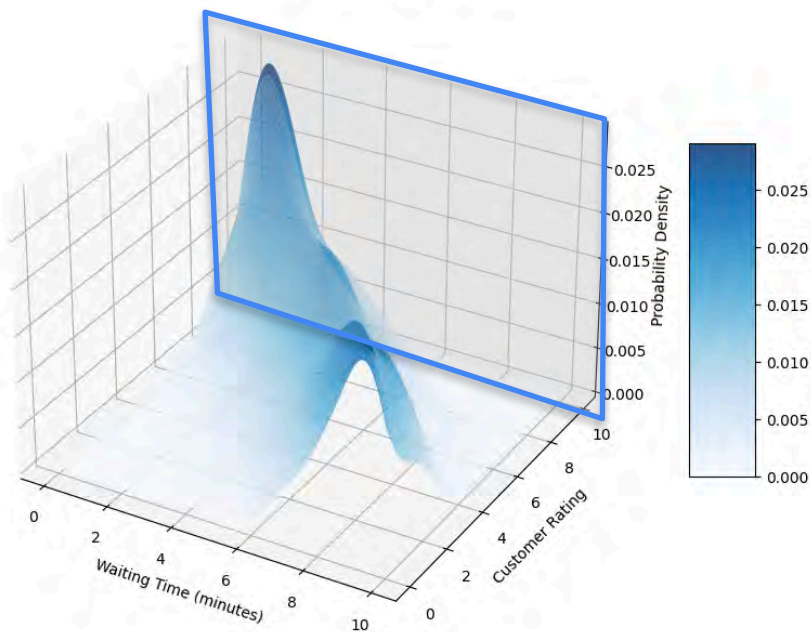
Continuous Marginal Distribution

3D Probability Density Distribution for Customer Ratings vs Waiting Time



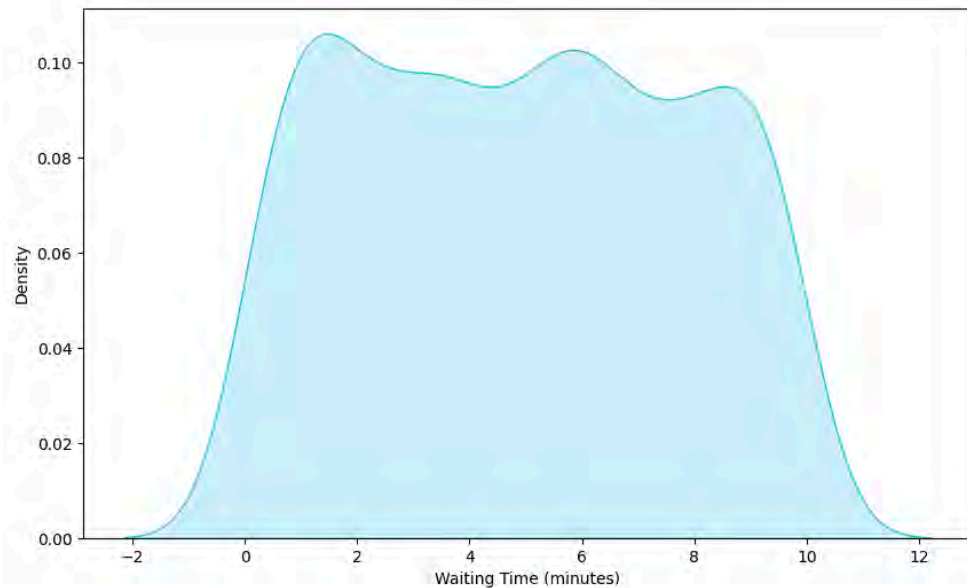
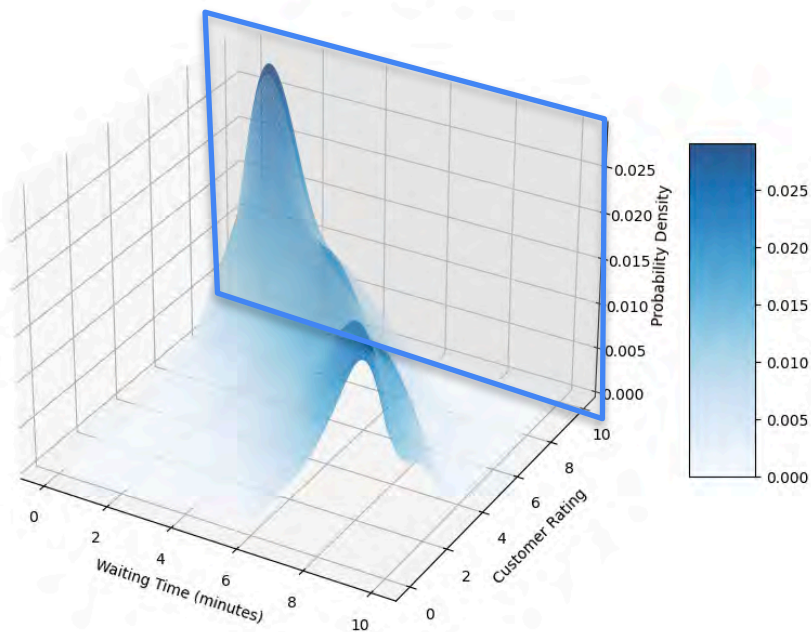
Continuous Marginal Distribution

3D Probability Density Distribution for Customer Ratings vs Waiting Time

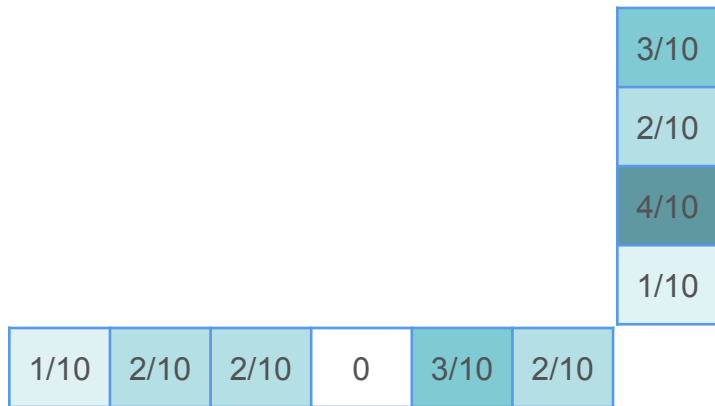


Continuous Marginal Distribution

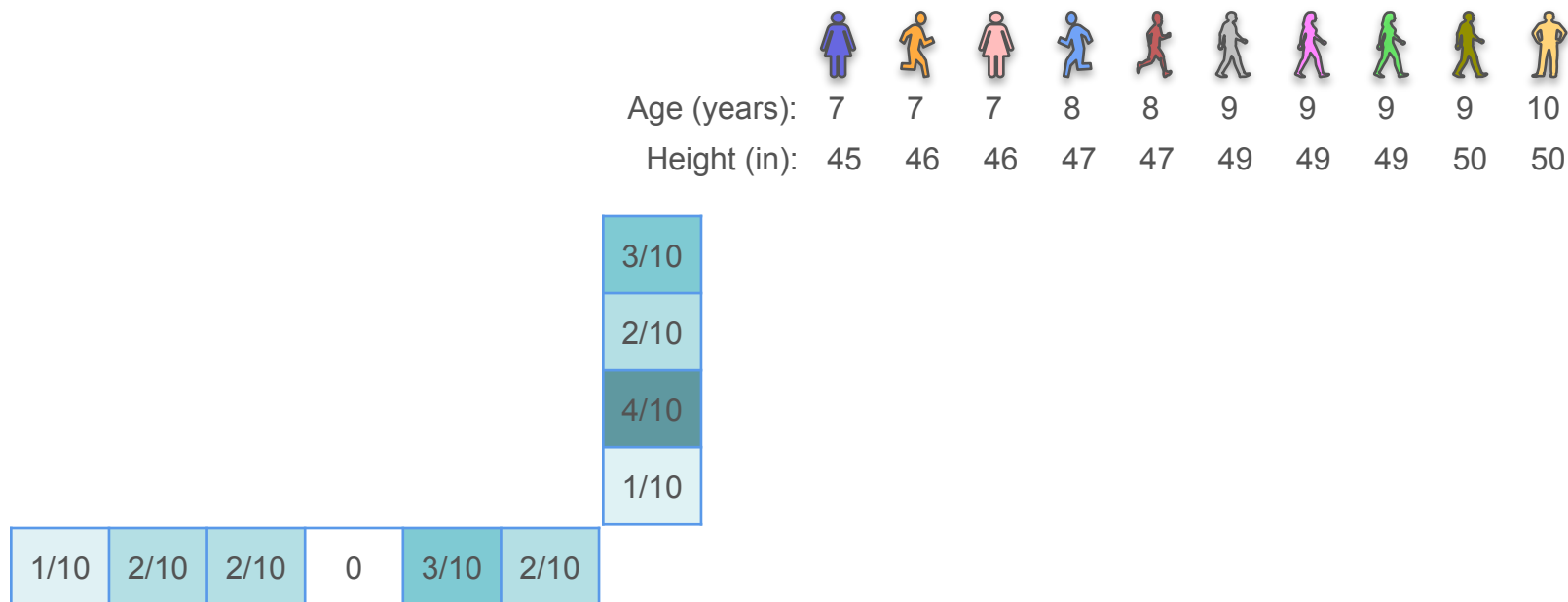
3D Probability Density Distribution for Customer Ratings vs Waiting Time



Conditional Distribution: Example 1



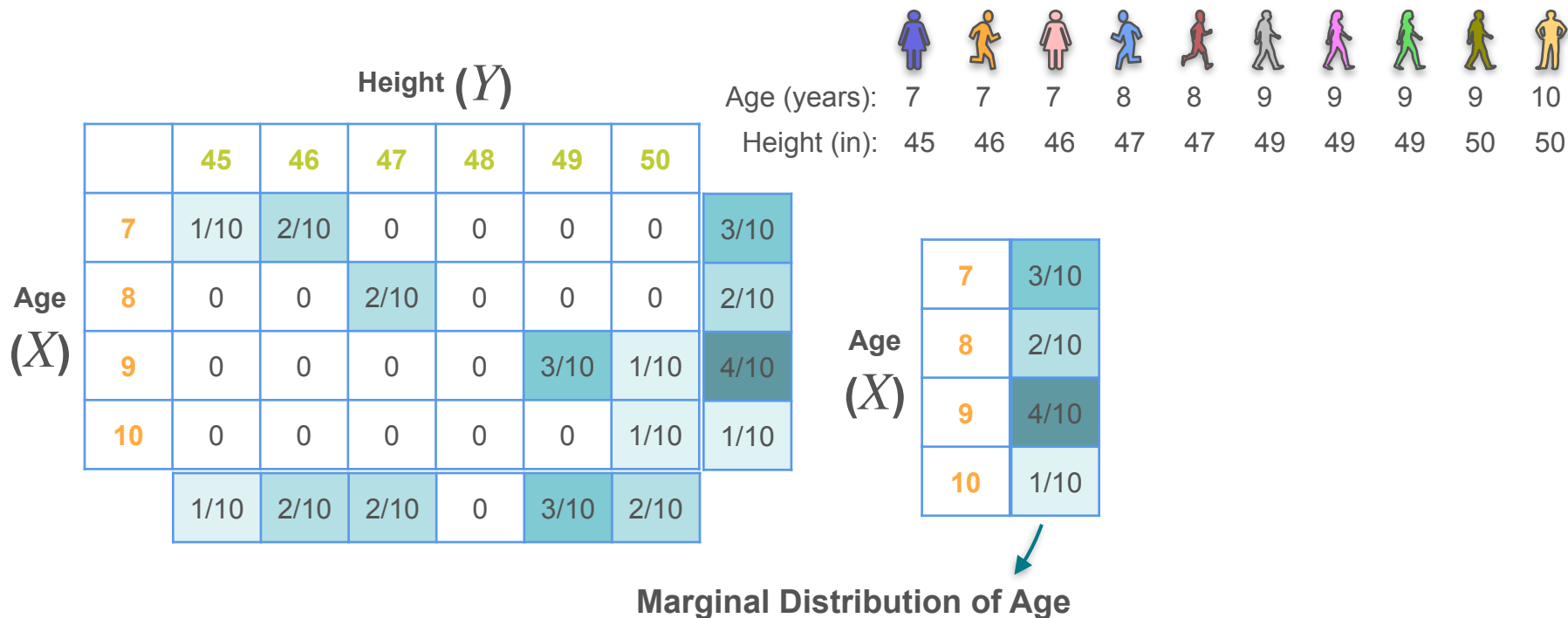
Conditional Distribution: Example 1



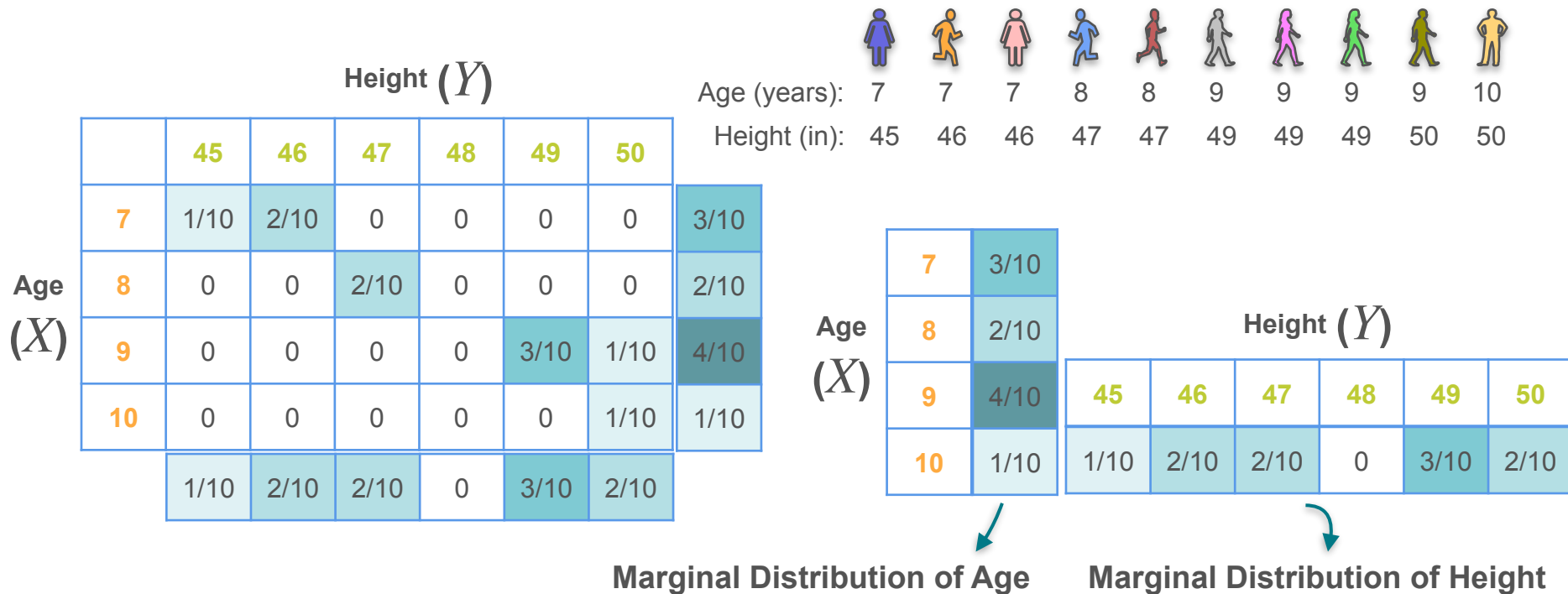
Conditional Distribution: Example 1

		Height (Y)						Age (years):											
		45	46	47	48	49	50												
		Height (in):						7	7	7	8	8	9	9	9	9	10	10	
Age (X)																			
	7	1/10	2/10	0	0	0	0	3/10											
	8	0	0	2/10	0	0	0	2/10											
	9	0	0	0	0	3/10	1/10	4/10											
	10	0	0	0	0	0	1/10	1/10											
		1/10	2/10	2/10	0	3/10	2/10												

Conditional Distribution: Example 1



Conditional Distribution: Example 1



Conditional Distribution: Example 1

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

Conditional Distribution: Example 1

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

If age = 9, what is the distribution across the height variable?

Conditional Distribution: Example 1

	Height (Y)					
	45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0
	8	0	0	2/10	0	0
	9	0	0	0	3/10	1/10
	10	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

If age = 9, what is the distribution across the height variable?

Conditional Distribution

Conditional Distribution: Example 1

Age (X)	Height (Y)					
	45	46	47	48	49	50
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

If age = 9, what is the distribution across the height variable?

Conditional Distribution: Example 1

Age (X)	Height (Y)					
	45	46	47	48	49	50
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

If age = 9, what is the distribution across the height variable?

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

Conditional Distribution: Example 1

Age (X)	Height (Y)					
	45	46	47	48	49	50
7	1/10	2/10	0	0	0	0
8	0	0	2/10	0	0	0
9	0	0	0	0	3/10	1/10
10	0	0	0	0	0	1/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

If age = 9, what is the distribution across the height variable?

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

Conditional Distribution: Example 1

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Conditional Distribution: Example 1

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Conditional Distribution: Example 1

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

↓

Normalize

↓

9	0	0	0	0	3/4	1/4
---	---	---	---	---	-----	-----

Conditional Distribution: Example 1

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

↓

Normalize

Divide by row sum

↓

	9	0	0	0	0	3/4	1/4
--	---	---	---	---	---	-----	-----

Conditional Distribution: Example 1

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Diagram illustrating the normalization process:

Initial joint distribution table (Row sum = 1/10):

Age (X)	9	0	0	0	0	3/10	1/10
-------------	---	---	---	---	---	------	------

Normalization process:

- Divide by row sum (1/10)
- Resulting conditional distribution table (Row sum = 1):

Age (X)	9	0	0	0	0	3/4	1/4
-------------	---	---	---	---	---	-----	-----

$$\mathbf{P}(Y = 49 | X = 9) = \frac{3}{4}$$

Conditional Distribution: Example 1

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

$$\mathbf{P}(A, B) = \mathbf{P}(A) \cdot \mathbf{P}(B | A)$$

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Diagram illustrating the normalization process:

Original joint distribution table (Row sum = 1/10):

Age (X)	9	0	0	0	0	3/10	1/10
-------------	---	---	---	---	---	------	------

Normalization process:

- Normalize (Divide by row sum)

Normalized conditional distribution table (Row sum = 1):

Age (X)	9	0	0	0	0	3/4	1/4
-------------	---	---	---	---	---	-----	-----

$$\mathbf{P}(Y = 49 | X = 9) = \frac{3}{4}$$

Conditional Distribution: Example 1

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Diagram illustrating the normalization process for the conditional distribution $P(Y = 49 | X = 9)$.

The initial joint probability table is shown above. The value $3/10$ is highlighted in the cell corresponding to $X = 9$ and $Y = 49$.

Two boxes indicate the steps:

- Normalize**: An arrow points from the initial table to the normalized table below.
- Divide by row sum**: A box indicating the operation performed on the row corresponding to $X = 9$.

The resulting normalized table (conditional distribution) is shown below:

	9	0	0	0	0	3/4	1/4
--	---	---	---	---	---	-----	-----

$$P(Y = 49 | X = 9) = \frac{3}{4}$$

$$p_{Y|X=9}(y) = P(Y = y | X = 9)$$

$$P(A, B) = P(A) \cdot P(B | A)$$

$$P(X = 9, Y = 49) = P(X = 9) \cdot P(Y = 49 | X = 9)$$

Conditional Distribution: Example 1

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Diagram illustrating the normalization process:

Original joint distribution table (top):

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Process: Normalize (Divide by row sum)

Normalized conditional distribution table (bottom):

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/4	1/4

$$\mathbf{P}(Y = 49 | X = 9) = \frac{3}{4}$$

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

$$\mathbf{P}(A, B) = \mathbf{P}(A) \cdot \mathbf{P}(B | A)$$

$$\mathbf{P}(X = 9, Y = 49) = P(X = 9) \cdot P(Y = 49 | X = 9)$$

$$P(Y = 49 | X = 9) = \frac{\mathbf{P}(X = 9, Y = 49)}{P(X = 9)}$$

Row sum

Conditional Distribution: Example 1

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Diagram illustrating the normalization process:

Original joint distribution table (top):

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Process: **Normalize** (Divide by row sum)

Normalized conditional distribution table (bottom):

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/4	1/4

$$\mathbf{P}(Y = 49 | X = 9) = \frac{3}{4}$$

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

$$\mathbf{P}(A, B) = \mathbf{P}(A) \cdot \mathbf{P}(B | A)$$

$$\mathbf{P}(X = 9, Y = 49) = \mathbf{P}(X = 9) \cdot \mathbf{P}(Y = 49 | X = 9)$$

$$\mathbf{P}(Y = 49 | X = 9) = \frac{\mathbf{P}(X = 9, Y = 49)}{\mathbf{P}(X = 9)}$$

Row sum

$$\mathbf{P}(Y = 49 | X = 9)$$

Conditional Distribution: Example 1

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Diagram illustrating the normalization process:

Original joint distribution table (highlighted cell is 3/10):

		45	46	47	48	49	50
9	0	0	0	0	0	3/10	1/10

Process: Normalize (Divide by row sum)

Normalized conditional distribution table (highlighted cell is 3/4):

		45	46	47	48	49	50
9	0	0	0	0	0	3/4	1/4

$$\mathbf{P}(Y = 49 | X = 9) = \frac{3}{4}$$

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

$$\mathbf{P}(A, B) = \mathbf{P}(A) \cdot \mathbf{P}(B | A)$$

$$\mathbf{P}(X = 9, Y = 49) = \mathbf{P}(X = 9) \cdot \mathbf{P}(Y = 49 | X = 9)$$

$$P(Y = 49 | X = 9) = \frac{\mathbf{P}(X = 9, Y = 49)}{\mathbf{P}(X = 9)}$$

Row sum

$$\mathbf{P}(Y = 49 | X = 9) = \frac{3/10}{4/10}$$

Conditional Distribution: Example 1

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Diagram illustrating the normalization process:

Original joint distribution table (top):

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Process: **Normalize** (Divide by row sum)

Normalized conditional distribution table (bottom):

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/4	1/4

$$\mathbf{P}(Y = 49 | X = 9) = \frac{3}{4}$$

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

$$\mathbf{P}(A, B) = \mathbf{P}(A) \cdot \mathbf{P}(B | A)$$

$$\mathbf{P}(X = 9, Y = 49) = \mathbf{P}(X = 9) \cdot \mathbf{P}(Y = 49 | X = 9)$$

$$P(Y = 49 | X = 9) = \frac{\mathbf{P}(X = 9, Y = 49)}{\mathbf{P}(X = 9)}$$

Row sum

$$\mathbf{P}(Y = 49 | X = 9) = \frac{3/10}{4/10} = \frac{3}{4}$$

Conditional Distribution: Example 1

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Diagram illustrating the normalization process:

Original joint distribution table (highlighted cell is 3/10):

		45	46	47	48	49	50
9	0	0	0	0	0	3/10	1/10

Normalization process:

1. Normalize (Divide by row sum):

		45	46	47	48	49	50
9	0	0	0	0	0	3/4	1/4

$$\mathbf{P}(Y = 49 | X = 9) = \frac{3}{4}$$

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

$$\mathbf{P}(A, B) = \mathbf{P}(A) \cdot \mathbf{P}(B | A)$$

$$\mathbf{P}(X = 9, Y = 49) = \mathbf{P}(X = 9) \cdot \mathbf{P}(Y = 49 | X = 9)$$

$$\mathbf{P}(Y = 49 | X = 9) = \frac{\mathbf{P}(X = 9, Y = 49)}{\mathbf{P}(X = 9)}$$

Row sum

$$\mathbf{P}(Y = 49 | X = 9) = \frac{3/10}{4/10} = \frac{3}{4}$$

Conditional Distribution: Example 1

		Sum					
		Height (Y)					
		P(X = 9)					
Age (X)		45	46	47	48	49	50
9		0	0	0	0	3/10	1/10

Normalize
↓

Divide by
row sum
↓

9	0	0	0	0	3/4	1/4
---	---	---	---	---	-----	-----

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

$$\mathbf{P}(A, B) = \mathbf{P}(A) \cdot \mathbf{P}(B | A)$$

$$\mathbf{P}(X = 9, Y = 49) = \mathbf{P}(X = 9) \cdot \mathbf{P}(Y = 49 | X = 9)$$

$$P(Y = 49 | X = 9) = \frac{\mathbf{P}(X = 9, Y = 49)}{\mathbf{P}(X = 9)}$$

Row sum

$$\mathbf{P}(Y = 49 | X = 9) = \frac{3}{4} \quad \longleftrightarrow \quad \mathbf{P}(Y = 49 | X = 9) = \frac{3/10}{4/10} = \frac{3}{4}$$

Conditional Distribution: Example 1

		Sum					
		Height (Y)					
		$P(X = 9)$					
(X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

$$p_{Y|X=9}(y) = \mathbf{P}(Y = y | X = 9)$$

$$\mathbf{P}(A, B) = \mathbf{P}(A) \cdot \mathbf{P}(B | A)$$

$$\mathbf{P}(X = 9, Y = 49) = \mathbf{P}(X = 9) \cdot \mathbf{P}(Y = 49 | X = 9)$$

$$P(Y = 49 | X = 9) = \frac{\mathbf{P}(X = 9, Y = 49)}{\mathbf{P}(X = 9)}$$

Row sum

$$\mathbf{P}(Y = 49 | X = 9) = \frac{3/10}{4/10} = \frac{3}{4}$$

Conditional Distribution: Example 1

		Sum					
		Height (Y)					
		$P(X = 9)$					
(X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

$$p_{Y|X=x}(y) = \frac{p_{XY}(x, y)}{p_X(x)}$$

$$p_{Y|X=9}(y) = P(Y = y | X = 9)$$

$$P(A, B) = P(A) \cdot P(B | A)$$

$$P(X = 9, Y = 49) = P(X = 9) \cdot P(Y = 49 | X = 9)$$

$$P(Y = 49 | X = 9) = \frac{P(X = 9, Y = 49)}{P(X = 9)}$$

Row sum

$$P(Y = 49 | X = 9) = \frac{3/10}{4/10} = \frac{3}{4}$$

Discrete Conditional Distribution: Formula

$$p_{Y|X=x}(y) = \frac{p_{XY}(x, y)}{p_X(x)}$$

Discrete Conditional Distribution: Formula

$$p_{Y|X=x}(y) = \frac{p_{XY}(x, y)}{p_X(x)}$$

Joint PDF of X and Y



Discrete Conditional Distribution: Formula

Conditional PDF of Y

Joint PDF of X and Y

$$p_{Y|X=x}(y) = \frac{p_{XY}(x, y)}{p_X(x)}$$


Discrete Conditional Distribution: Formula

Conditional PDF of Y

Joint PDF of X and Y

$$p_{Y|X=x}(y) = \frac{p_{XY}(x, y)}{p_X(x)}$$

Marginal distribution of X

Conditional Distributions: Example 2



Die 1: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

Die 2: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

X

Y

	Y					
	1	2	3	4	5	6
1	1/36	1/36	1/36	1/36	1/36	1/36
2	1/36	1/36	1/36	1/36	1/36	1/36
3	1/36	1/36	1/36	1/36	1/36	1/36
4	1/36	1/36	1/36	1/36	1/36	1/36
5	1/36	1/36	1/36	1/36	1/36	1/36
6	1/36	1/36	1/36	1/36	1/36	1/36

Conditional Distributions: Example 2



Die 1: 1/6 1/6 1/6 1/6 1/6 1/6

Die 2: 1/6 1/6 1/6 1/6 1/6 1/6

$$p_{Y|X=4}(y = 1)$$

		Y					
		1	2	3	4	5	6
X	1	1/36	1/36	1/36	1/36	1/36	1/36
	2	1/36	1/36	1/36	1/36	1/36	1/36
	3	1/36	1/36	1/36	1/36	1/36	1/36
	4	1/36	1/36	1/36	1/36	1/36	1/36
	5	1/36	1/36	1/36	1/36	1/36	1/36
	6	1/36	1/36	1/36	1/36	1/36	1/36

Conditional Distributions: Example 2



Die 1: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

Die 2: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$$p_{Y|X=4}(y = 1) = \frac{p_{XY}(x = 4, y = 1)}{p_X(x = 4)}$$

X

Y

	Y					
	1	2	3	4	5	6
1	1/36	1/36	1/36	1/36	1/36	1/36
2	1/36	1/36	1/36	1/36	1/36	1/36
3	1/36	1/36	1/36	1/36	1/36	1/36
4	1/36	1/36	1/36	1/36	1/36	1/36
5	1/36	1/36	1/36	1/36	1/36	1/36
6	1/36	1/36	1/36	1/36	1/36	1/36

Conditional Distributions: Example 2



Die 1: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

Die 2: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$$p_{Y|X=4}(y = 1) = \frac{p_{XY}(x = 4, y = 1)}{p_X(x = 4)}$$

X

Y

	Y						
	1	2	3	4	5	6	Sum
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	1/36	1/36	1/36	1/36	1/36	1/6
3	1/36	1/36	1/36	1/36	1/36	1/36	1/6
4	1/36	1/36	1/36	1/36	1/36	1/36	1/6
5	1/36	1/36	1/36	1/36	1/36	1/36	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6

Conditional Distributions: Example 2



Die 1: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

Die 2: $1/6$ $1/6$ $1/6$ $1/6$ $1/6$ $1/6$

$$\begin{aligned} p_{Y|X=4}(y = 1) &= \frac{p_{XY}(x = 4, y = 1)}{p_X(x = 4)} \\ &= \frac{1/36}{1/6} \end{aligned}$$

X

Y

	Y						
	1	2	3	4	5	6	Sum
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	1/36	1/36	1/36	1/36	1/36	1/6
3	1/36	1/36	1/36	1/36	1/36	1/36	1/6
4	1/36	1/36	1/36	1/36	1/36	1/36	1/6
5	1/36	1/36	1/36	1/36	1/36	1/36	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6

Conditional Distributions: Example 2



Die 1: 1/6 1/6 1/6 1/6 1/6 1/6

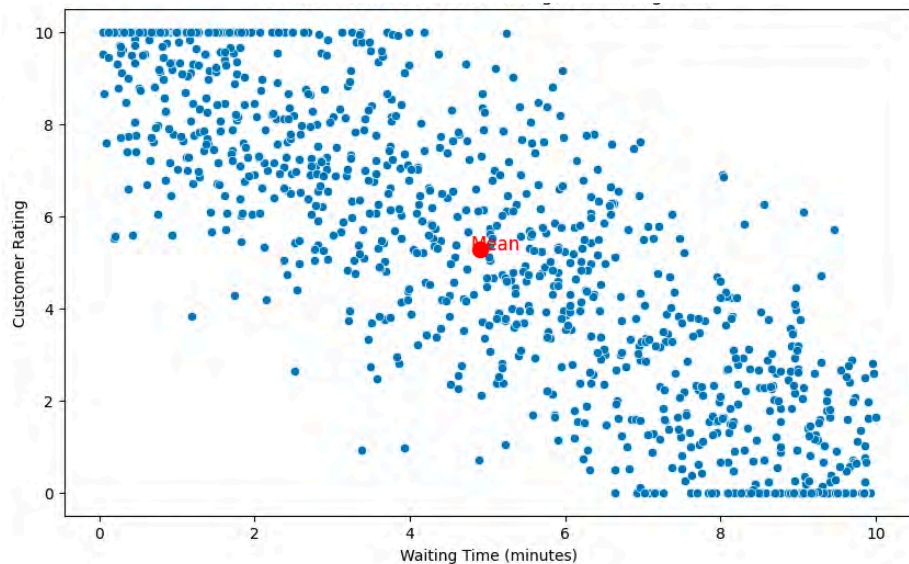
Die 2: 1/6 1/6 1/6 1/6 1/6 1/6

$$\begin{aligned} p_{Y|X=4}(y=1) &= \frac{p_{XY}(x=4, y=1)}{p_X(x=4)} \\ &= \frac{1/36}{1/6} \\ &= \frac{1}{6} \end{aligned}$$

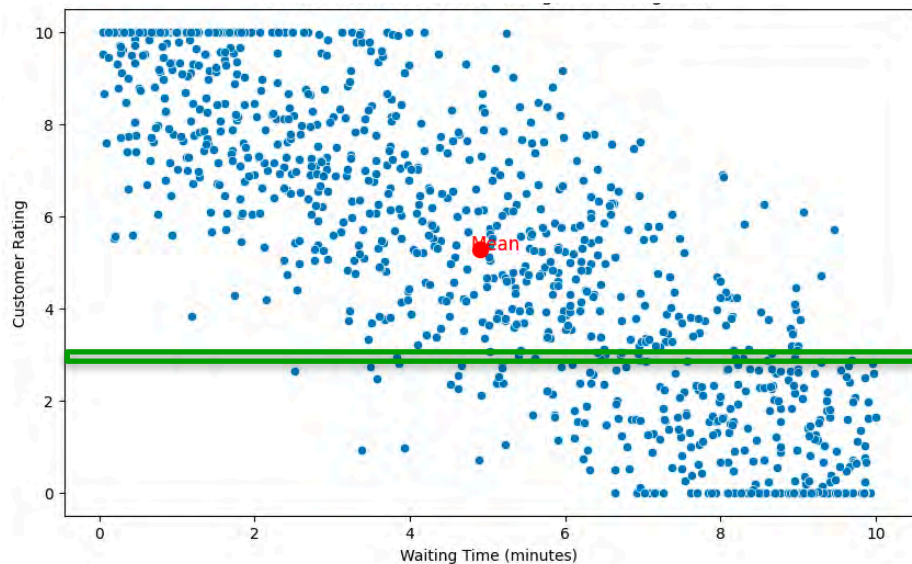
X

		Y						
		1	2	3	4	5	6	Sum
X	1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	2	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	3	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	4	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	5	1/36	1/36	1/36	1/36	1/36	1/36	1/6
	6	1/36	1/36	1/36	1/36	1/36	1/36	1/6

Conditional Distributions: Example 4



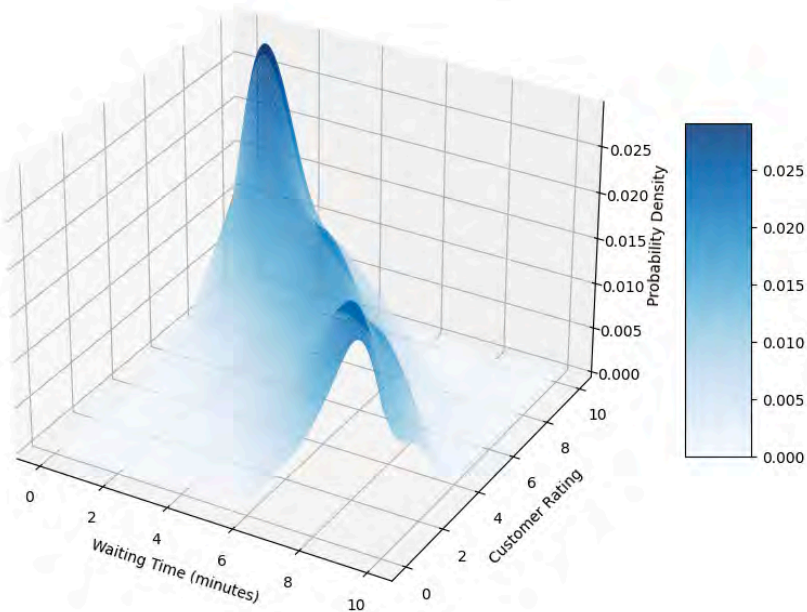
Conditional Distributions: Example 4



Continuous Conditional Distribution

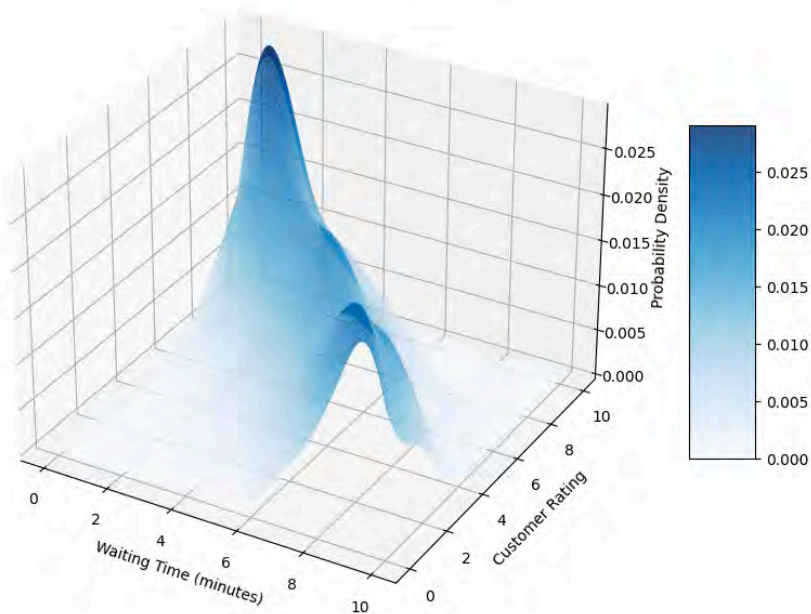
Continuous Conditional Distribution

3D Probability Density Distribution for Customer Ratings vs Waiting Time



Continuous Conditional Distribution

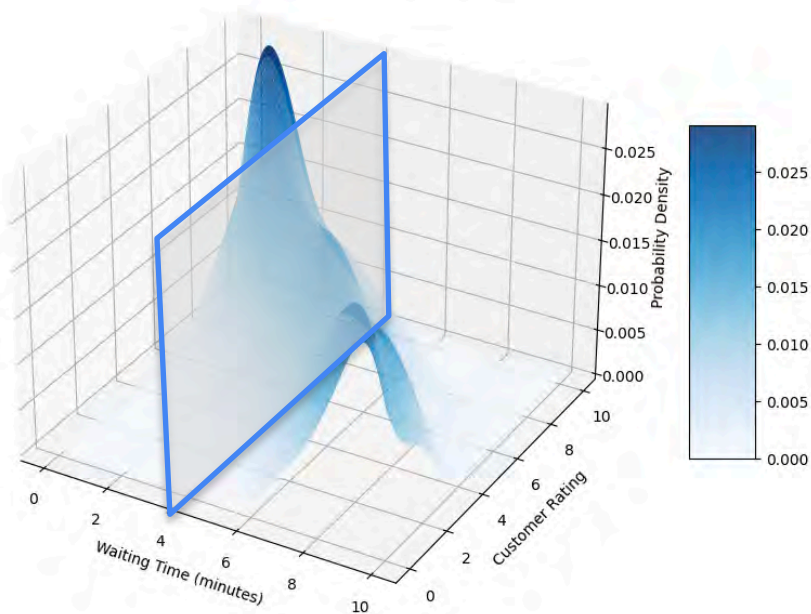
3D Probability Density Distribution for Customer Ratings vs Waiting Time



Probability distribution for rating given that waiting time was 4 minutes

Continuous Conditional Distribution

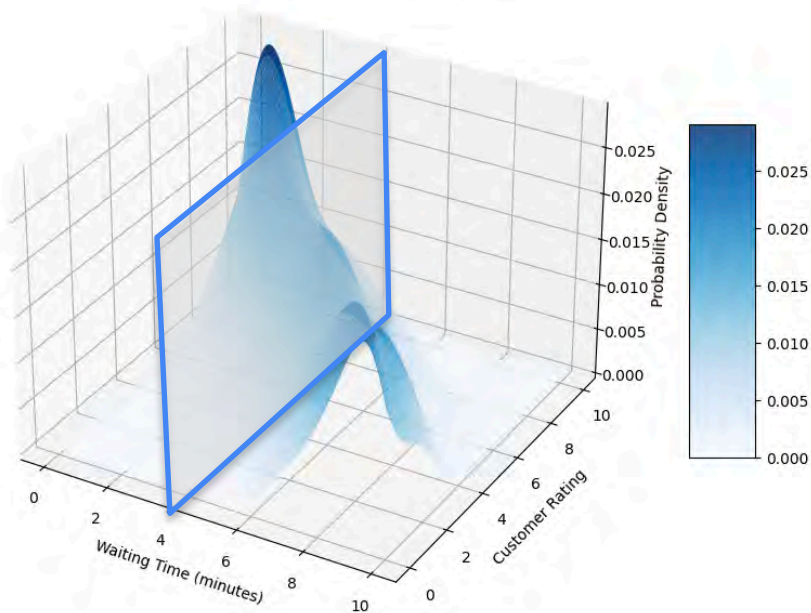
3D Probability Density Distribution for Customer Ratings vs Waiting Time



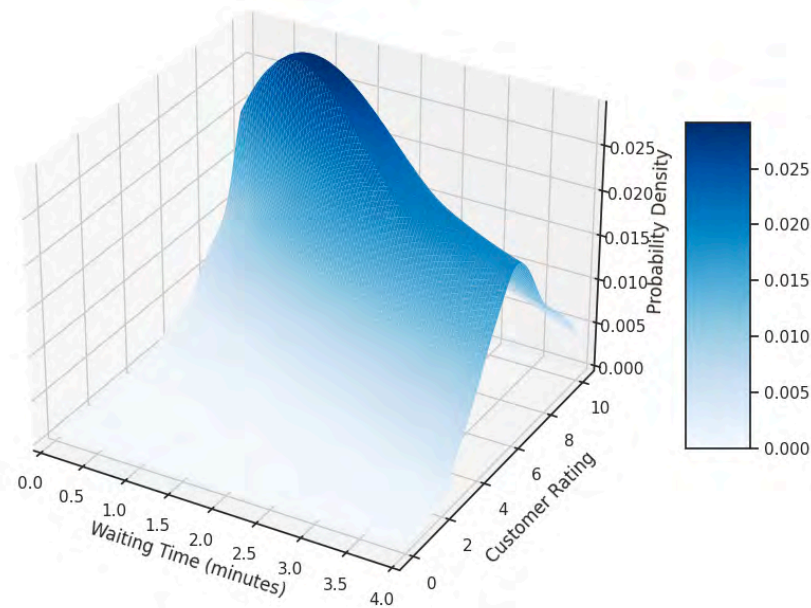
Probability distribution for rating given that waiting time was 4 minutes

Continuous Conditional Distribution

3D Probability Density Distribution for Customer Ratings vs Waiting Time

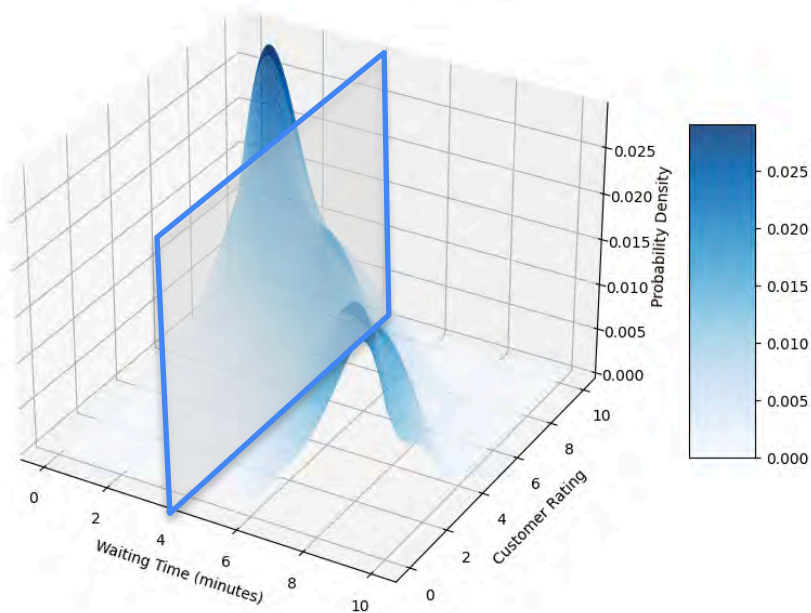


Probability distribution for rating given that waiting time was 4 minutes

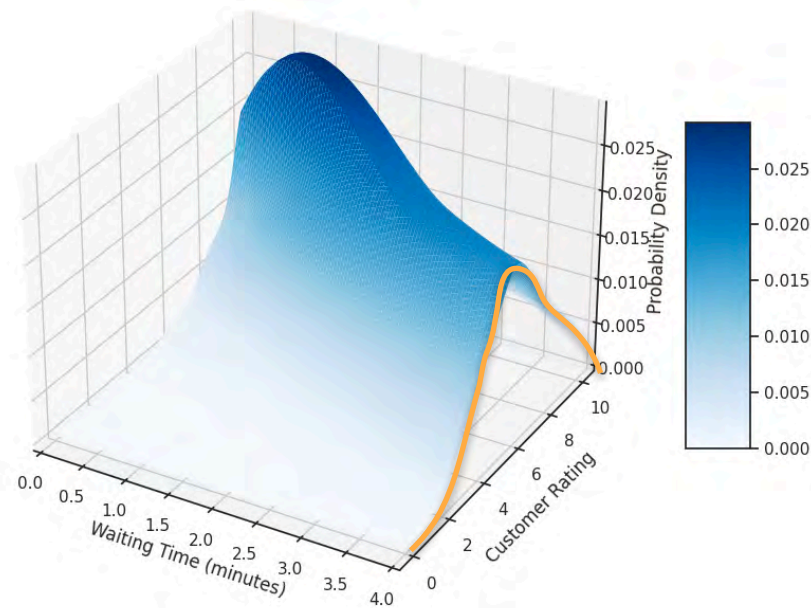


Continuous Conditional Distribution

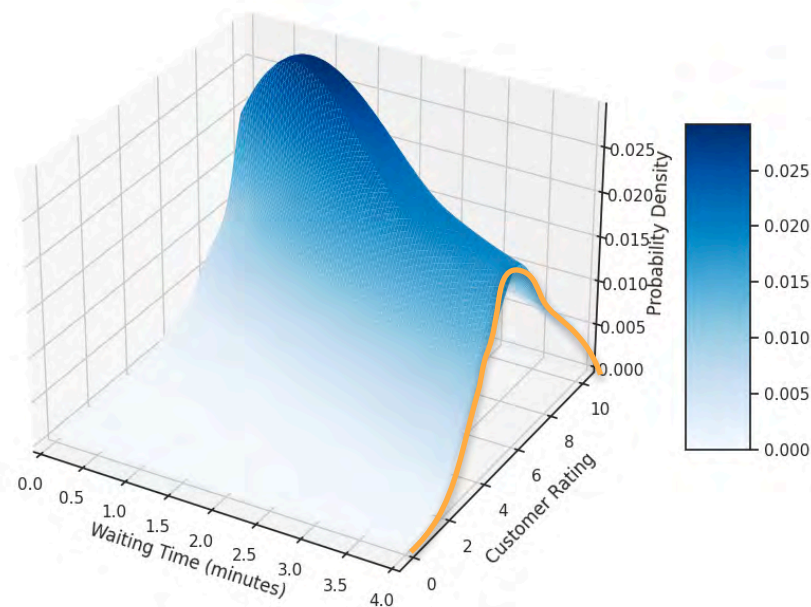
3D Probability Density Distribution for Customer Ratings vs Waiting Time



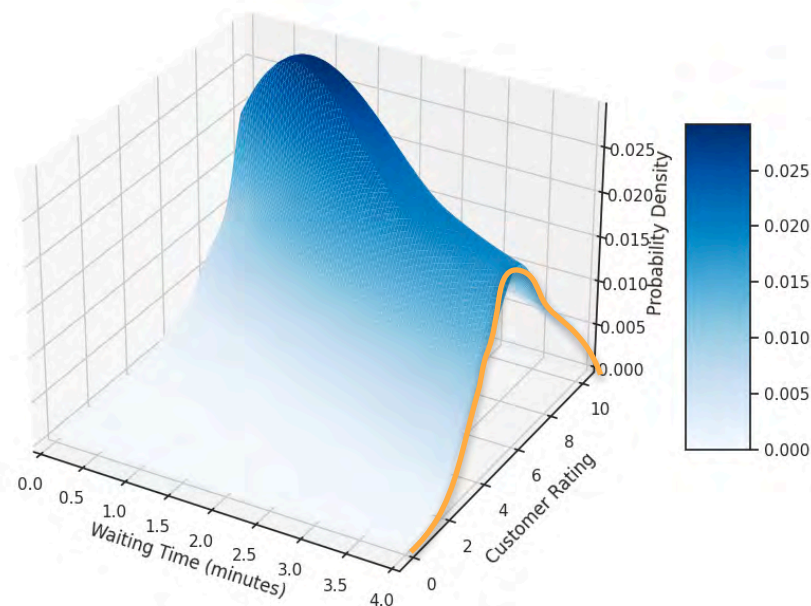
Probability distribution for rating given that waiting time was 4 minutes



Continuous Conditional Distribution



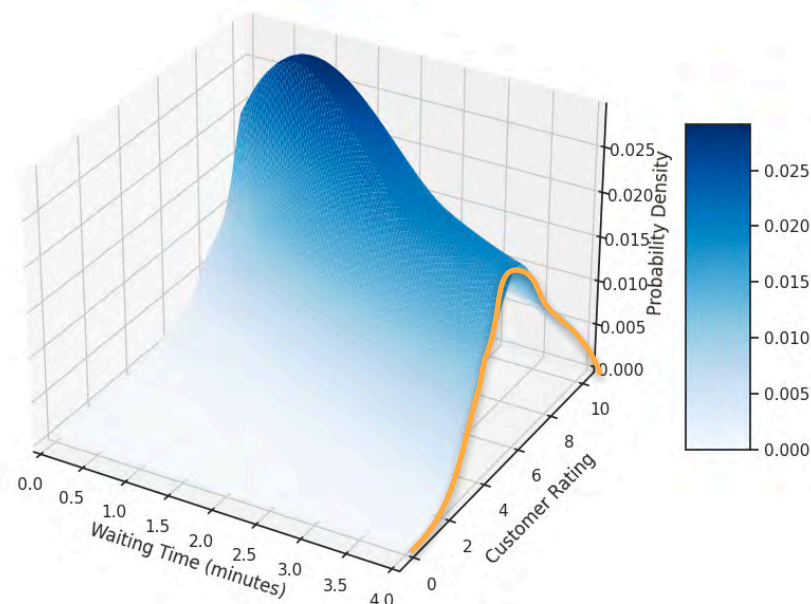
Continuous Conditional Distribution



Continuous Conditional Distribution



Conditional PDF of y given $x = 4$



Discrete Conditional Distribution

$$p_{Y|X=x}(y) = \frac{p_{XY}(x, y)}{p_X(x)}$$

Discrete Conditional Distribution

$$p_{Y|X=x}(y) = \frac{p_{XY}(x, y)}{p_X(x)}$$

Joint PDF of X and Y



Discrete Conditional Distribution

Conditional PDF of Y $p_{Y|X=x}(y) = \frac{p_{XY}(x, y)}{p_X(x)}$ Joint PDF of X and Y



Discrete Conditional Distribution

Conditional PDF of Y

Joint PDF of X and Y

$$p_{Y|X=x}(y) = \frac{p_{XY}(x, y)}{p_X(x)}$$

Marginal distribution of X

Continuous Conditional Distribution: Formula

Conditional PDF of Y

Joint PDF of X and Y

$$f_{Y|X=x}(y) = \frac{f_{XY}(x, y)}{f_X(x)}$$

Marginal distribution of X



DeepLearning.AI

Probability Distributions with Multiple Variables

Covariance

Introduction to Covariance

Introduction to Covariance

X : age of a child

Introduction to Covariance

X : age of a child

Y_1 : height of the child (in)

Introduction to Covariance

X : age of a child

Y_1 : height of the child (in)

Y_2 : grades in a test

Introduction to Covariance

X : age of a child

Y_1 : height of the child (in)

Y_2 : grades in a test

Y_3 : number of naps per day

Introduction to Covariance

X : age of a child

Y_1 : height of the child (in)

Y_2 : grades in a test

Y_3 : number of naps per day

Age (X)	Height (Y_1)
6	50
7	52
8	55
9	57
10	60
11	62
12	64
13	65
14	67
15	68

Introduction to Covariance

X : age of a child

Y_1 : height of the child (in)

Age (X)	Height (Y_1)
6	50
7	52
8	55
9	57
10	60
11	62
12	64
13	65
14	67
15	68

Y_2 : grades in a test

Age (X)	Grades (Y_2)
6	5
7	7
8	8
9	3
10	1
11	1
12	6
13	10
14	2
15	7

Y_3 : number of naps per day

Introduction to Covariance

X : age of a child

Y_1 : height of the child (in)

Age (X)	Height (Y_1)
6	50
7	52
8	55
9	57
10	60
11	62
12	64
13	65
14	67
15	68

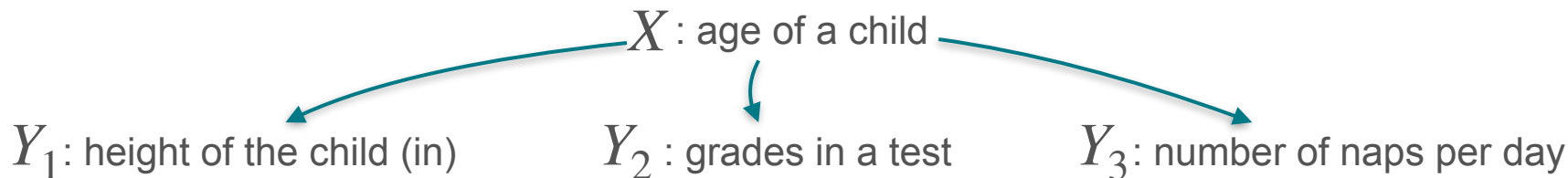
Y_2 : grades in a test

Age (X)	Grades (Y_2)
6	5
7	7
8	8
9	3
10	1
11	1
12	6
13	10
14	2
15	7

Y_3 : number of naps per day

Age (X)	Naps per day (Y_3)
6	8
7	7
8	6
9	5
10	4
11	3
12	2
13	1
14	1
15	0

Introduction to Covariance



Age (X)	Height (Y_1)
6	50
7	52
8	55
9	57
10	60
11	62
12	64
13	65
14	67
15	68

Age (X)	Grades (Y_2)
6	5
7	7
8	8
9	3
10	1
11	1
12	6
13	10
14	2
15	7

Age (X)	Naps per day (Y_3)
6	8
7	7
8	6
9	5
10	4
11	3
12	2
13	1
14	1
15	0

Introduction to Covariance

- How is variable X related to each of the Y variables?
- How do you compare these relations?

X : age of a child

Y_1 : height of the child (in)

Age (X)	Height (Y_1)
6	50
7	52
8	55
9	57
10	60
11	62
12	64
13	65
14	67
15	68

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6	5
7	7
8	8
9	3
10	1
11	1
12	6
13	10
14	2
15	7

Y_3 : number of naps per day

Age (X)	Naps per day (Y_3)
6	8
7	7
8	6
9	5
10	4
11	3
12	2
13	1
14	1
15	0

Introduction to Covariance

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6	50
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Y_2 : grades in a test

Age (X)	Grades (Y_2)
6	5
7	7
8	8
9	3
10	1
11	1
12	6
13	10
14	2
15	7

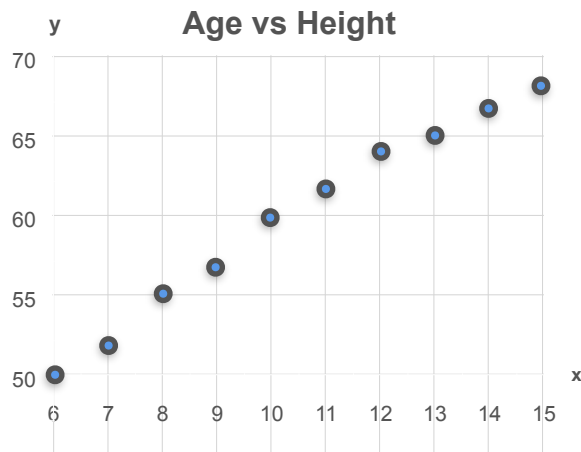
Y_3 : number of naps per day

Age (X)	Naps per day (Y_3)
6	8
7	7
8	6
9	5
10	4
11	3
12	2
13	1
14	1
15	0

Introduction to Covariance

X : age of a child

Y_1 : height of the child (in)



Y_2 : grades in a test

Age (X)	Grades (Y ₂)
6	5
7	7
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10	1
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12	6
13	10
14	2
15	7

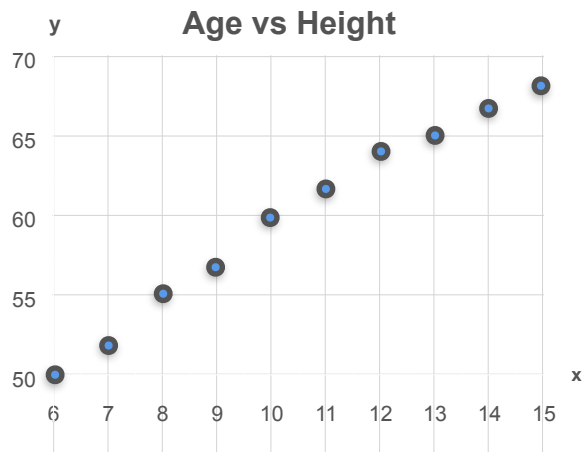
Y_3 : number of naps per day

Age (X)	Naps per day (Y ₃)
6	8
7	7
8	6
9	5
10	4
11	3
12	2
13	1
14	1
15	0

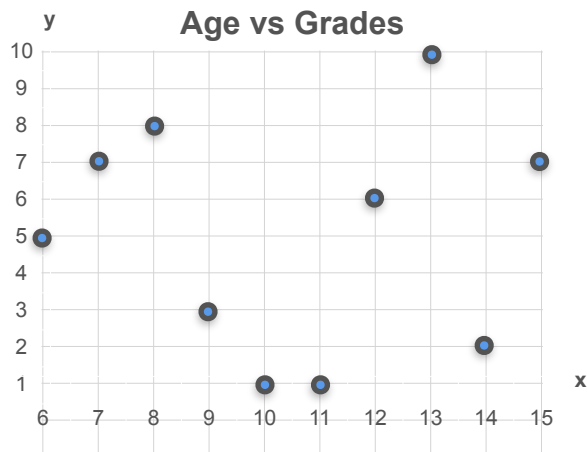
Introduction to Covariance

X : age of a child

Y_1 : height of the child (in)



Y_2 : grades in a test



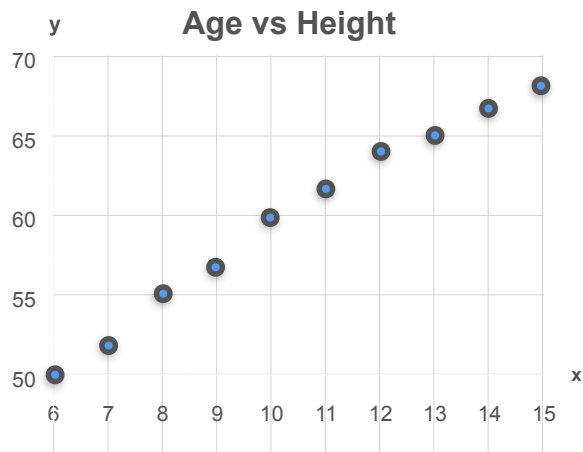
Y_3 : number of naps per day

Age (X)	Naps per day (Y_3)
6	8
7	7
8	6
9	5
10	4
11	3
12	2
13	1
14	1
15	0

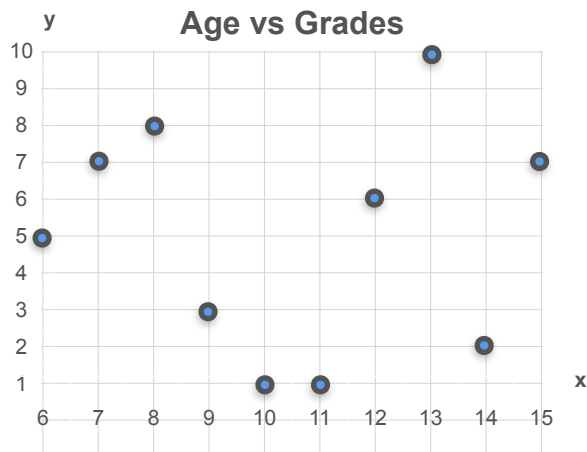
Introduction to Covariance

X : age of a child

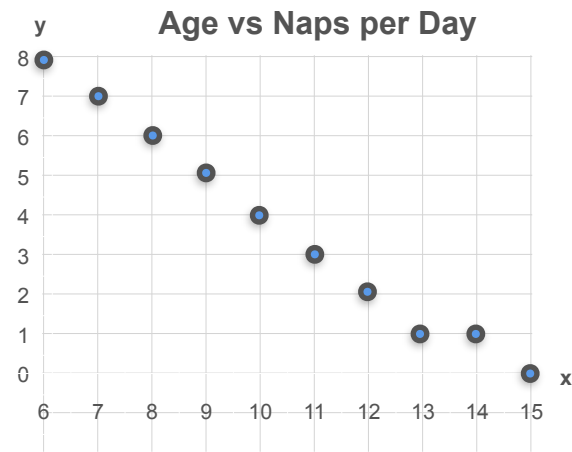
Y_1 : height of the child (in)



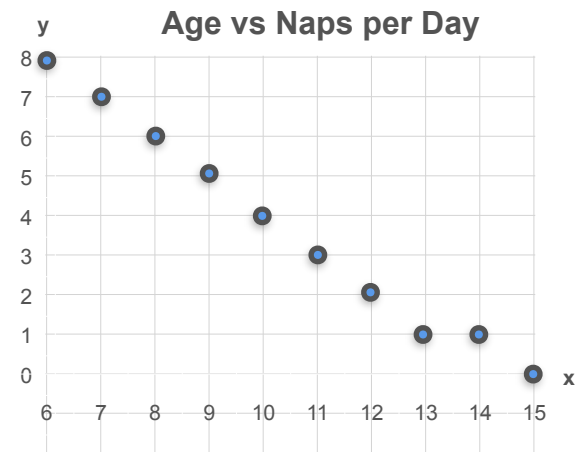
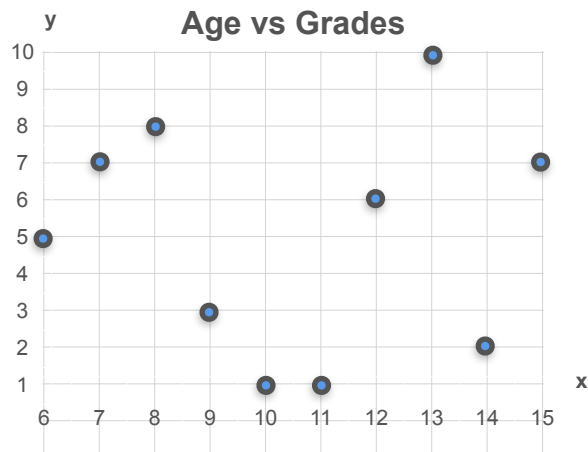
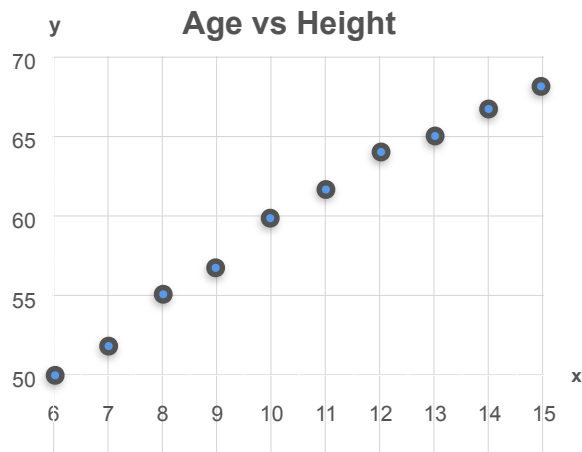
Y_2 : grades in a test



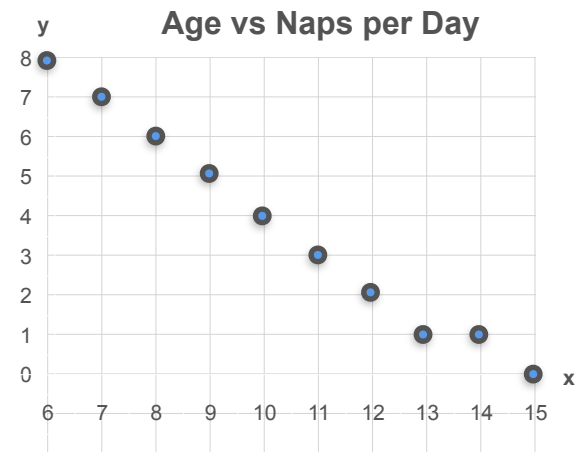
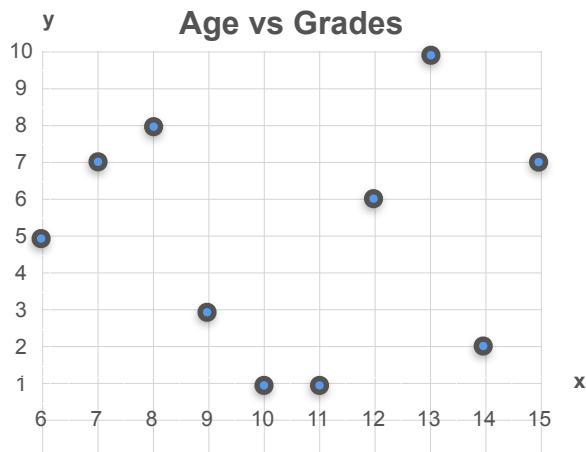
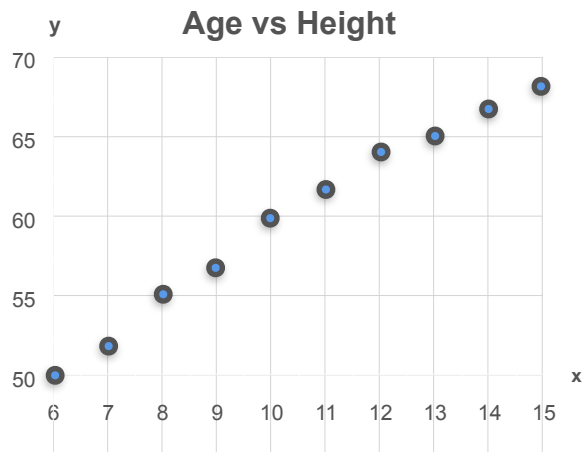
Y_3 : number of naps per day



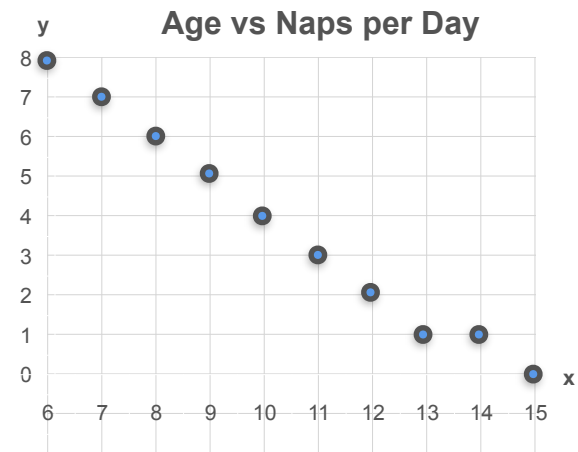
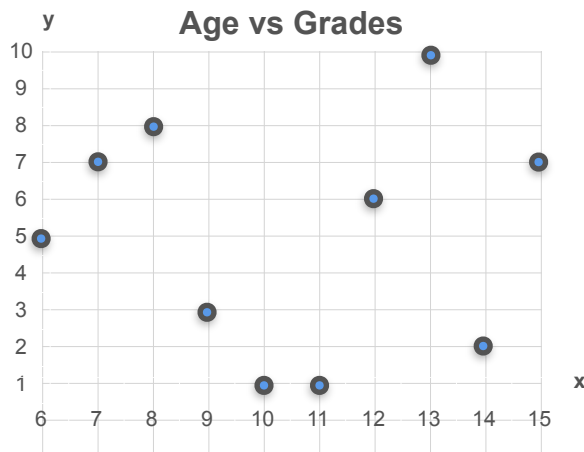
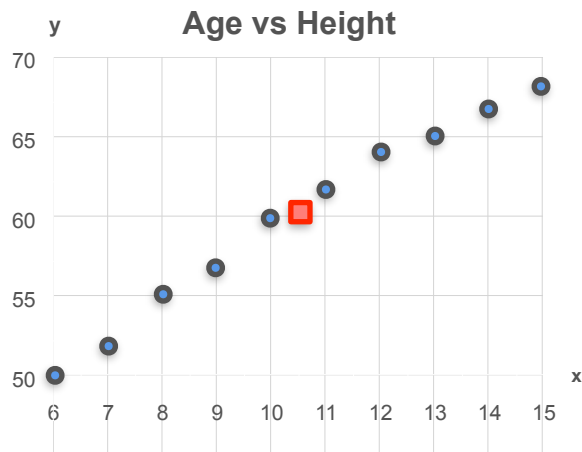
How To Compare These?



Mean?

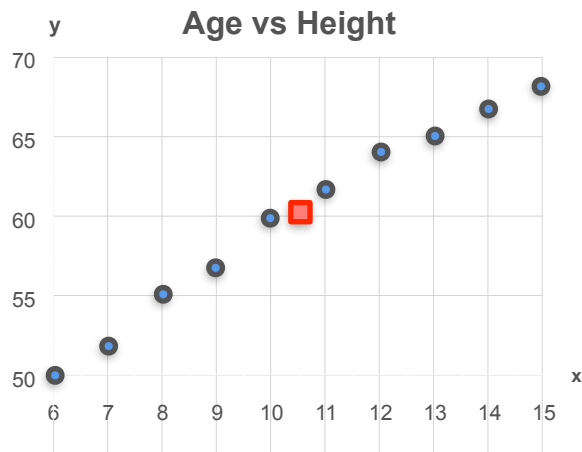


Mean?

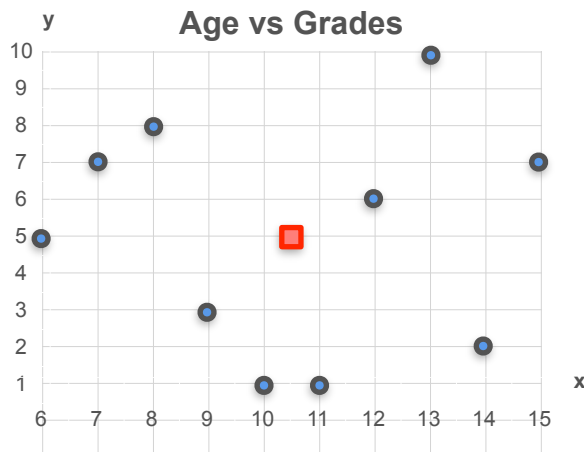


$$\mu_x = 10.5 \quad \mu_y = 60$$

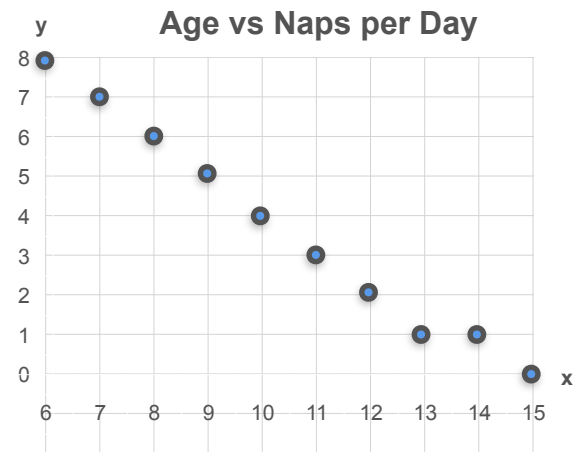
Mean?



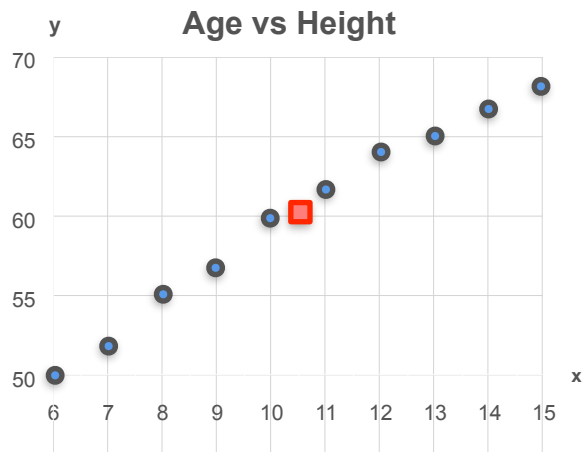
$$\mu_x = 10.5 \quad \mu_y = 60$$



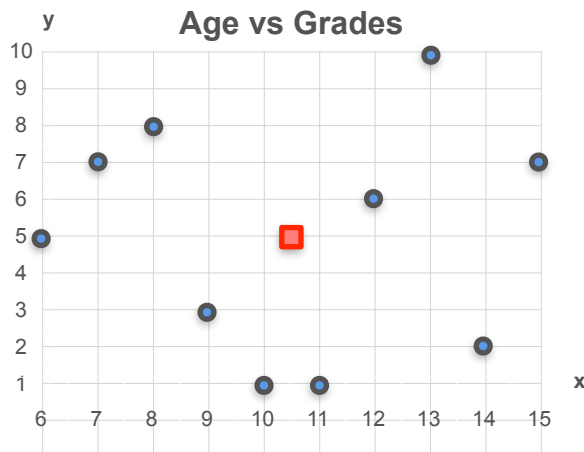
$$\mu_x = 10.5 \quad \mu_y = 5$$



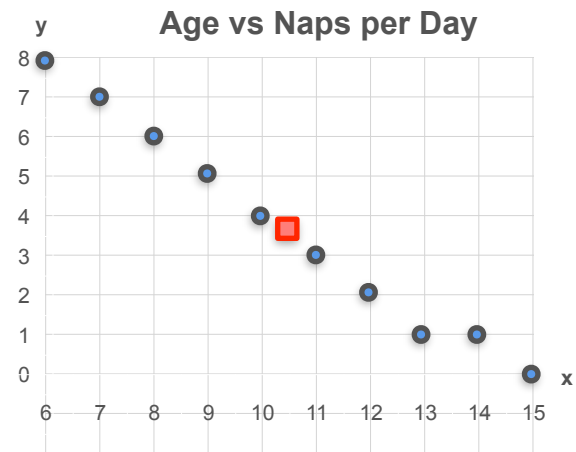
Mean?



$$\mu_x = 10.5 \quad \mu_y = 60$$

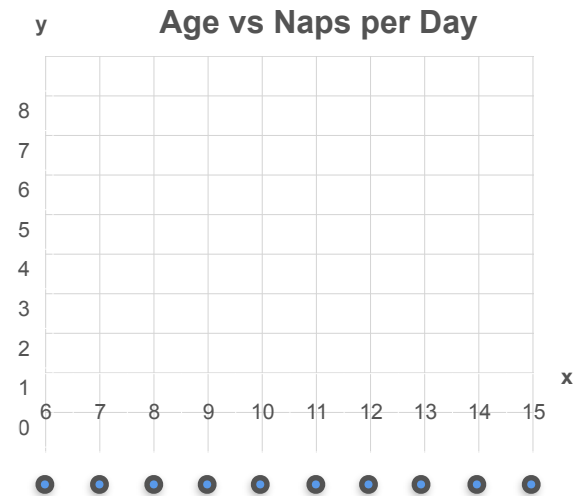
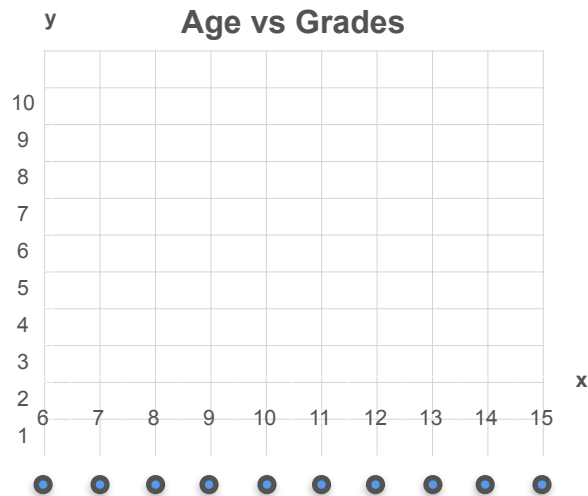
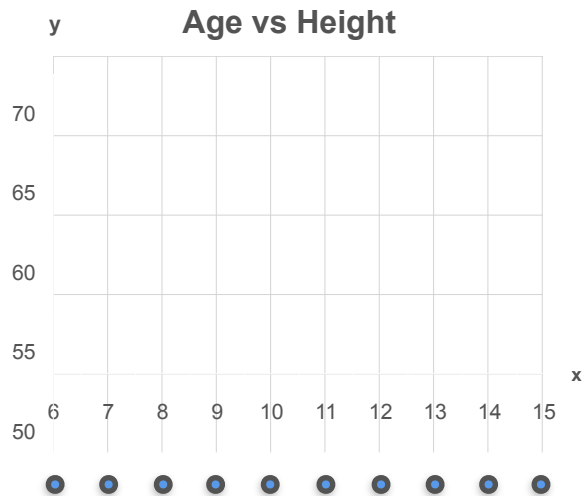


$$\mu_x = 10.5 \quad \mu_y = 5$$

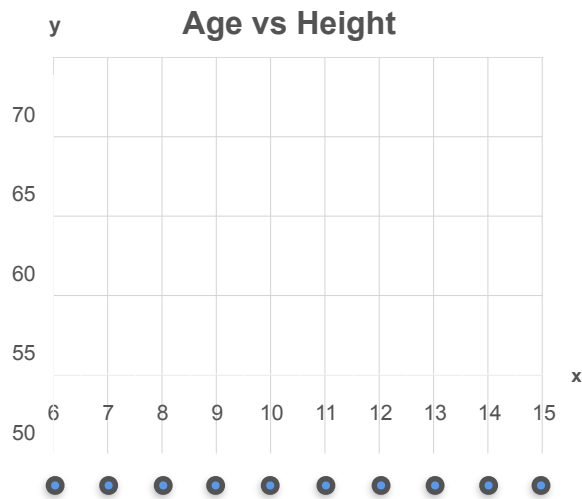


$$\mu_x = 10.5 \quad \mu_y = 3.7$$

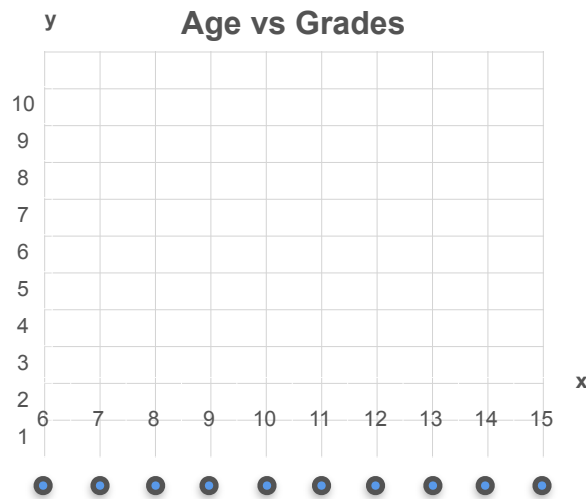
Horizontal (X) Variance



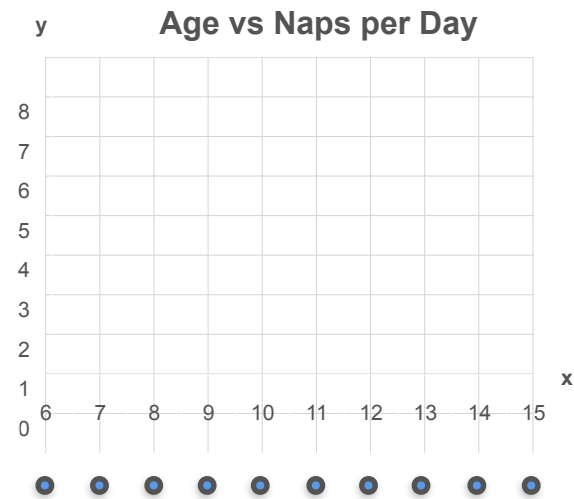
Horizontal (X) Variance



$$\text{Var}(X) = 9.17$$

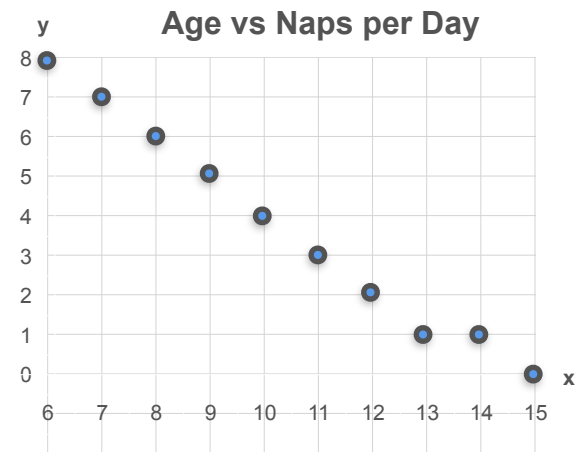
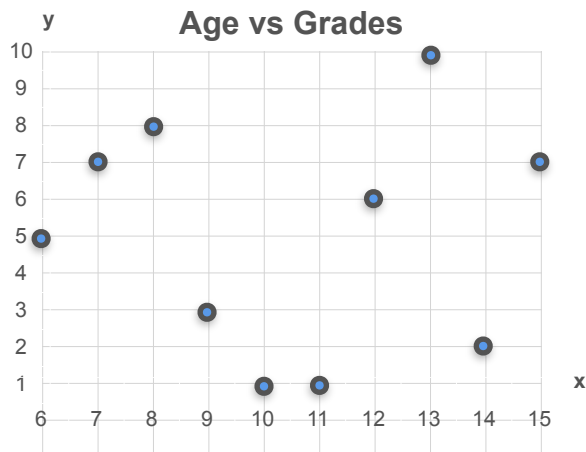
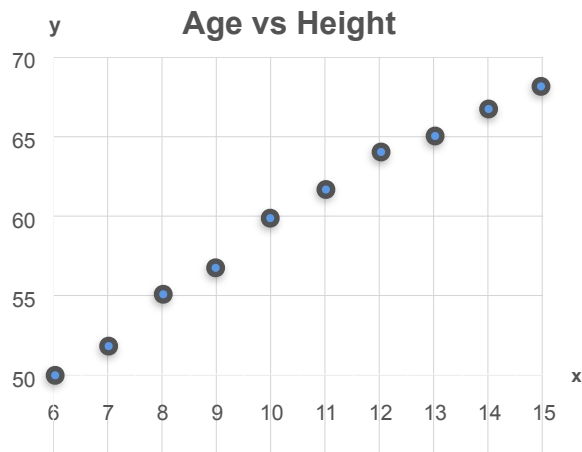


$$\text{Var}(X) = 9.17$$

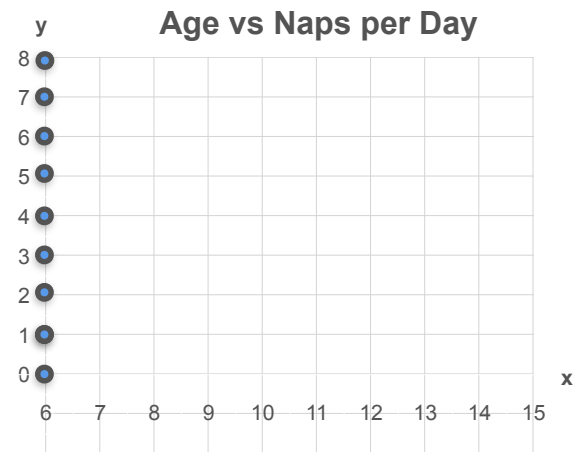
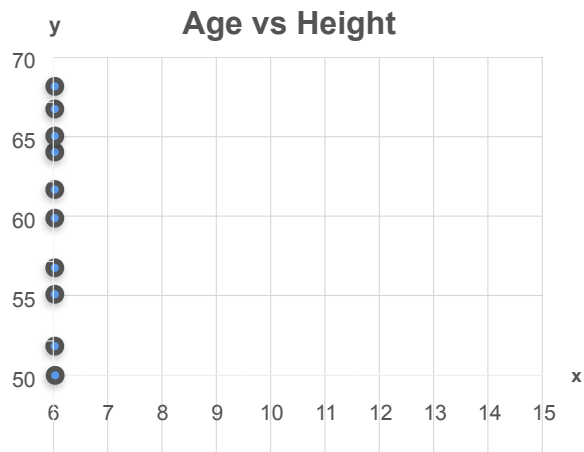


$$\text{Var}(X) = 9.17$$

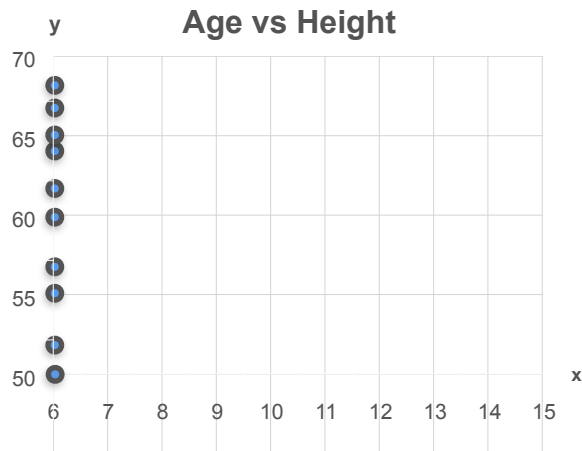
Anything Else?



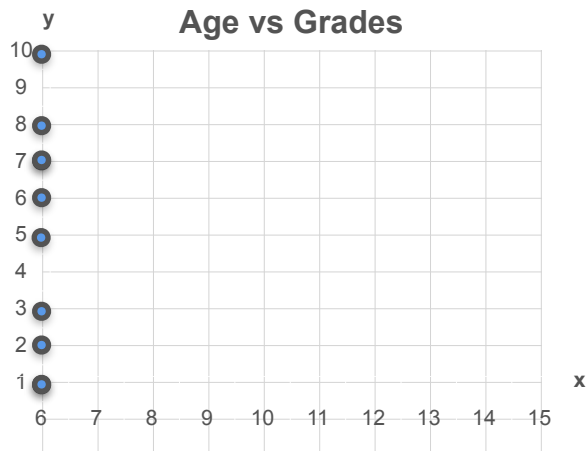
Vertical (Y) Variance



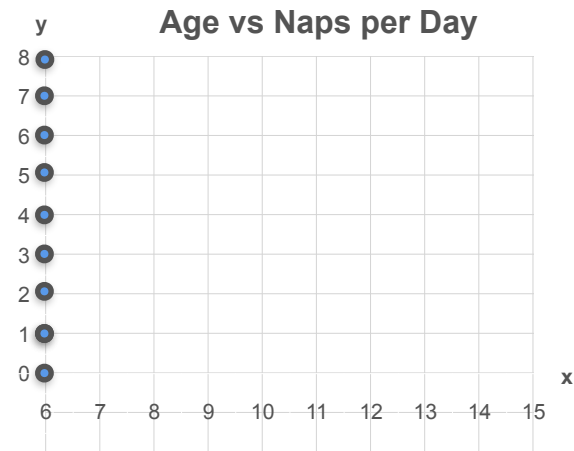
Vertical (Y) Variance



$$\text{Var}(Y) = 39.56$$

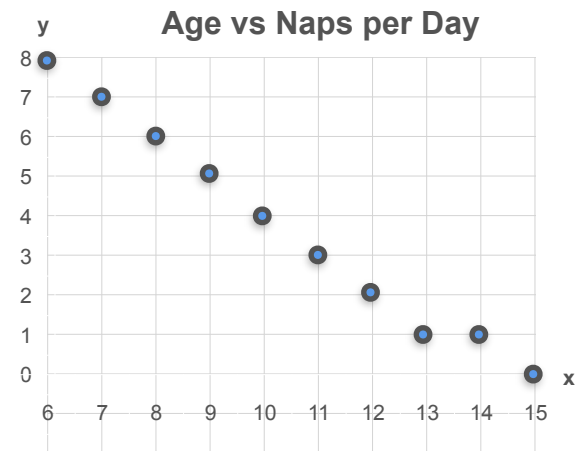
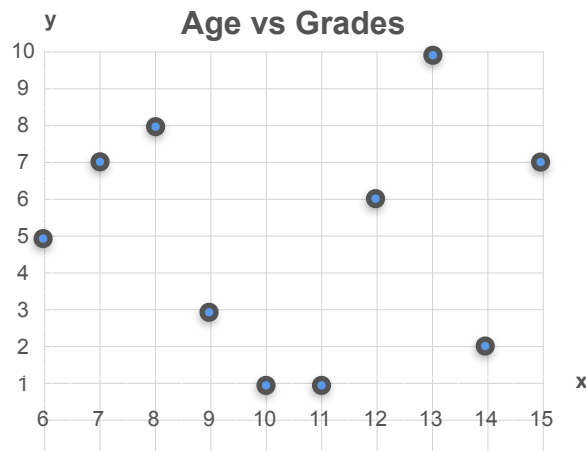
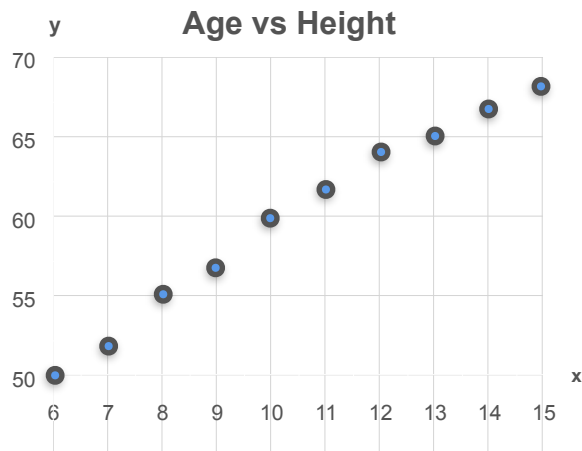


$$\text{Var}(Y) = 9.78$$

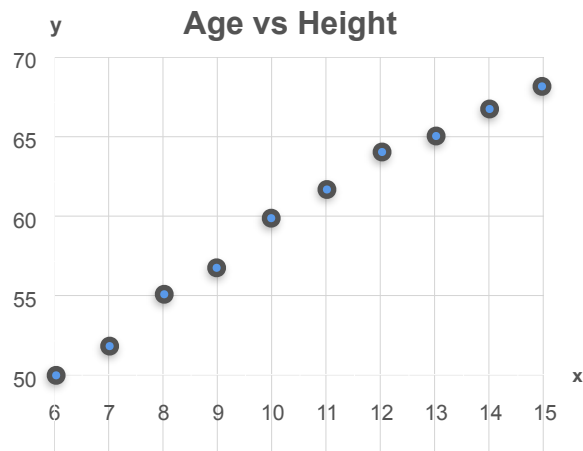


$$\text{Var}(Y) = 7.57$$

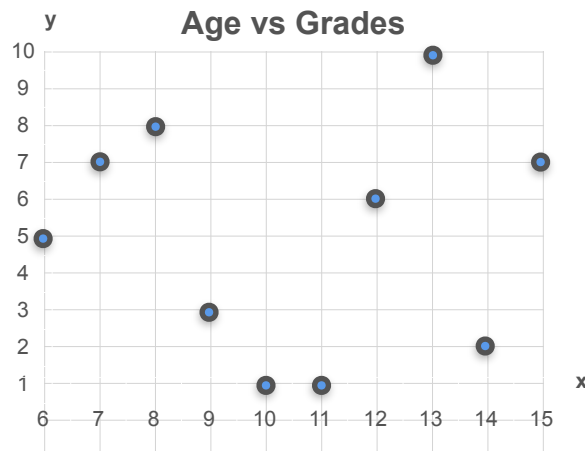
Still no Way To Compare Them



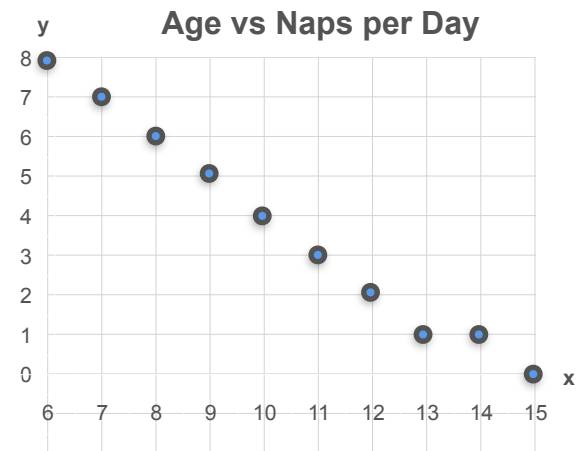
Still no Way To Compare Them



Covariance > 0



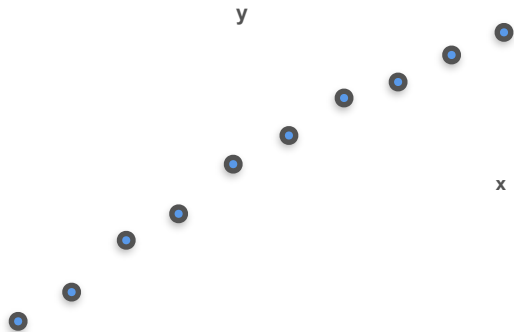
Covariance ≈ 0



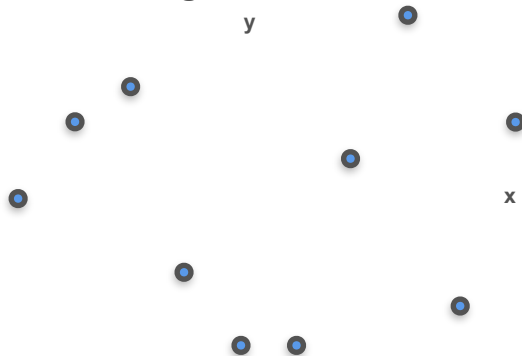
Covariance < 0

First Step: Center Them

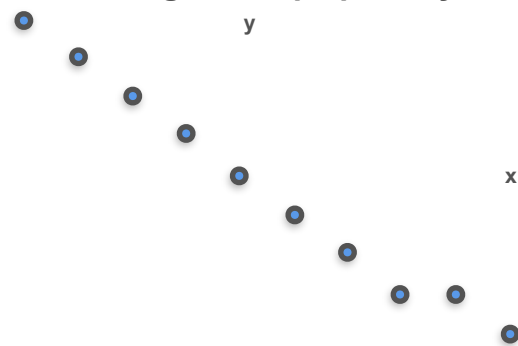
Age vs Height



Age vs Grades

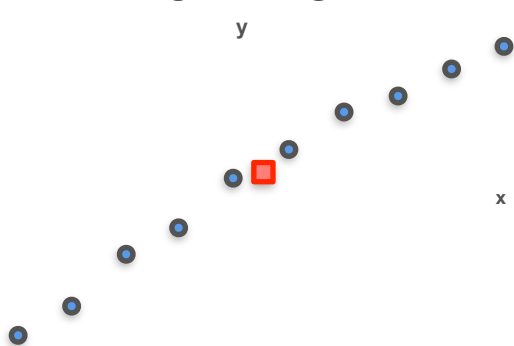


Age vs Naps per Day

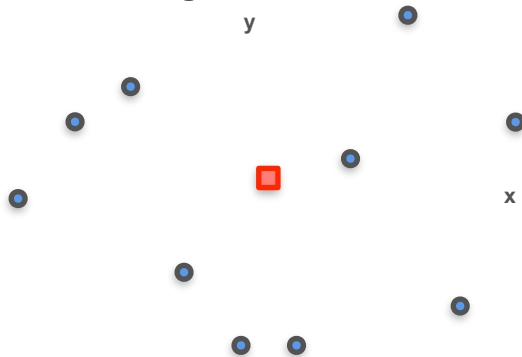


First Step: Center Them

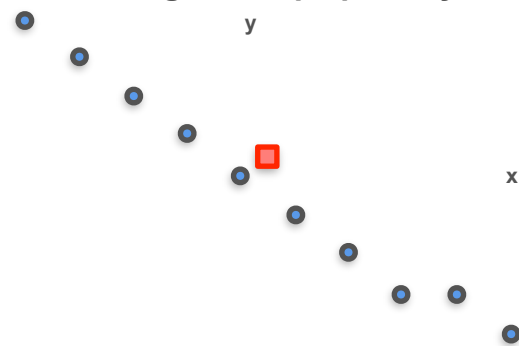
Age vs Height



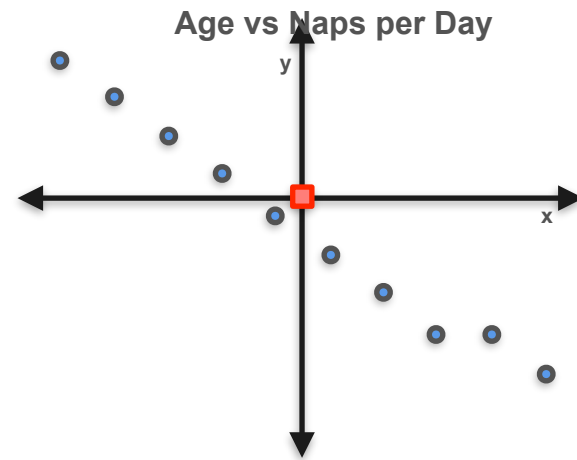
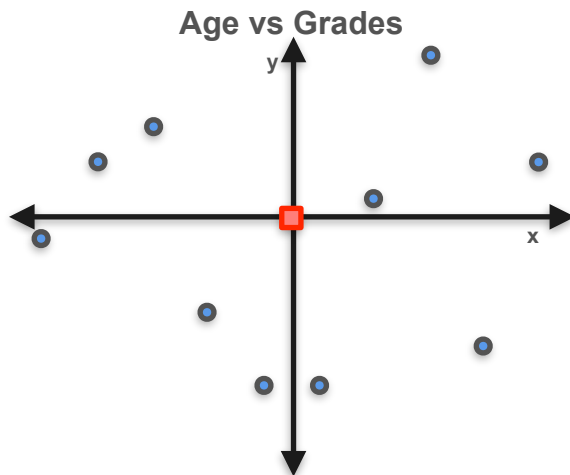
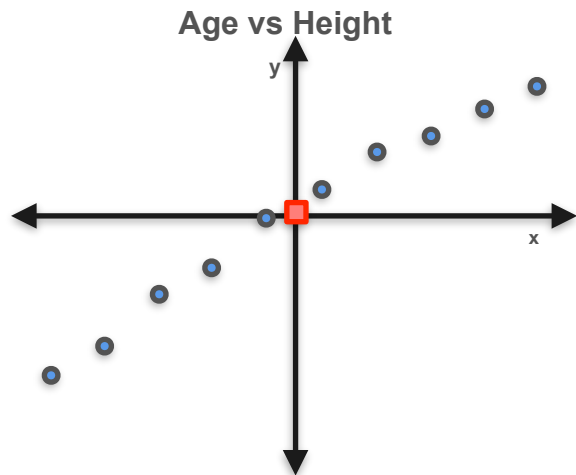
Age vs Grades



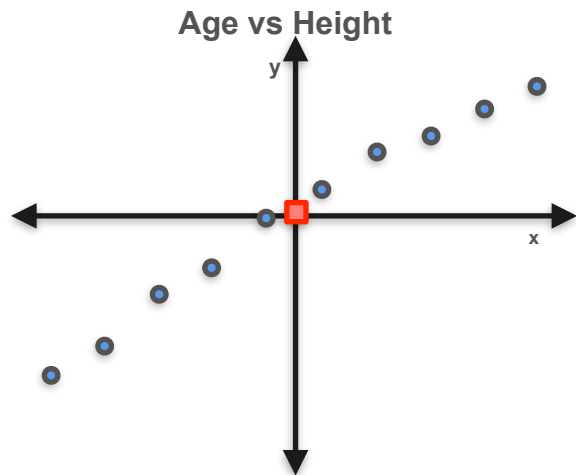
Age vs Naps per Day



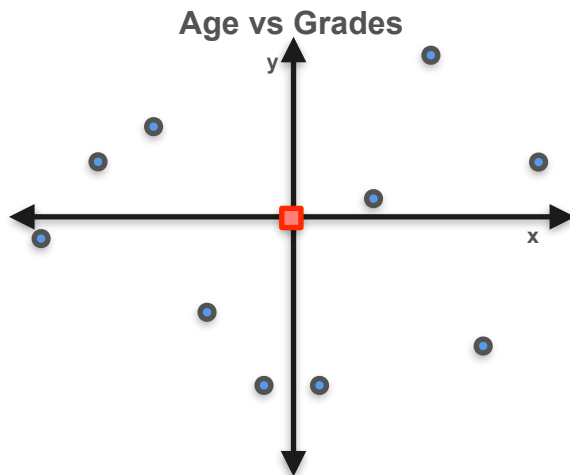
First Step: Center Them



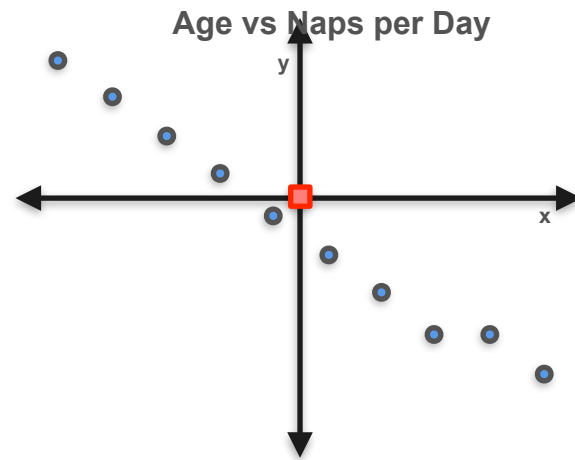
First Step: Center Them



$$\mu_x = 0$$
$$\mu_y = 0$$
$$\text{Var}(X) = 1$$
$$\text{Var}(Y) = 1$$

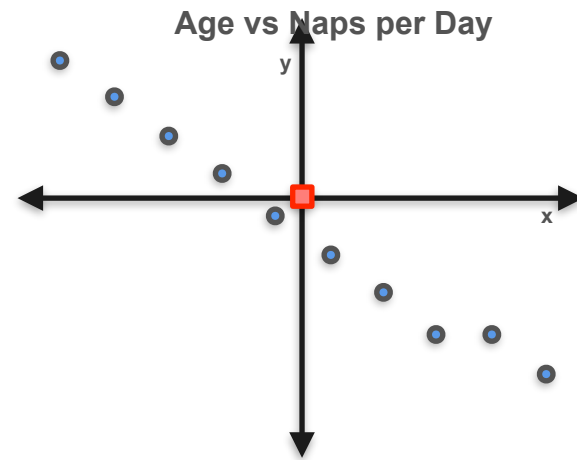
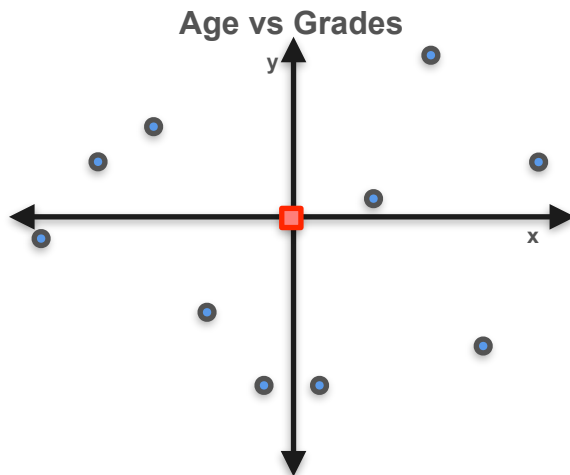
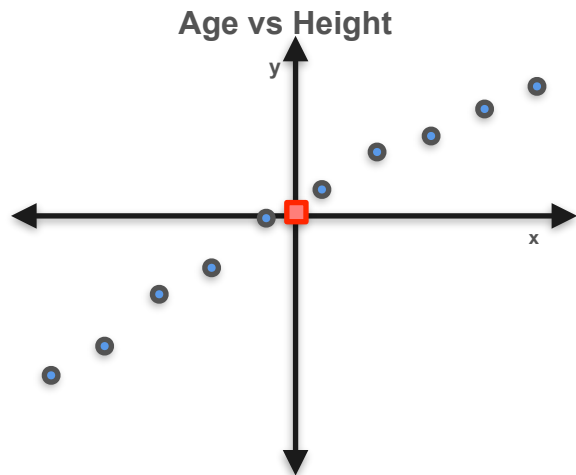


$$\mu_x = 0$$
$$\mu_y = 0$$
$$\text{Var}(X) = 1$$
$$\text{Var}(Y) = 1$$

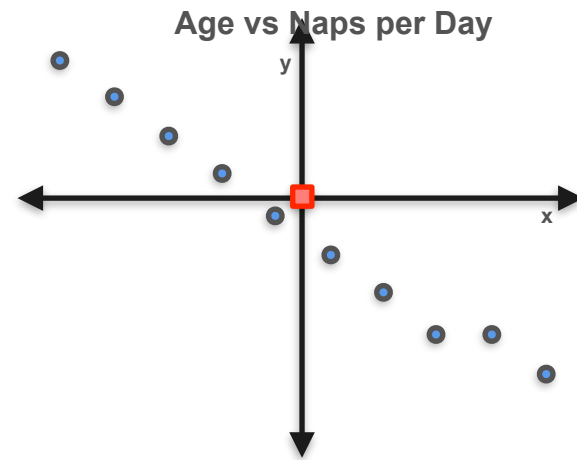
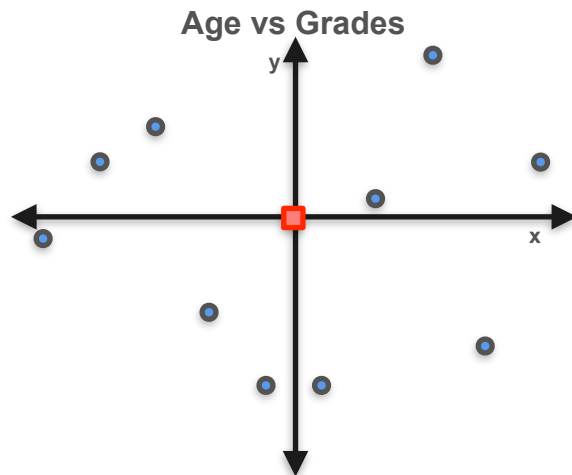
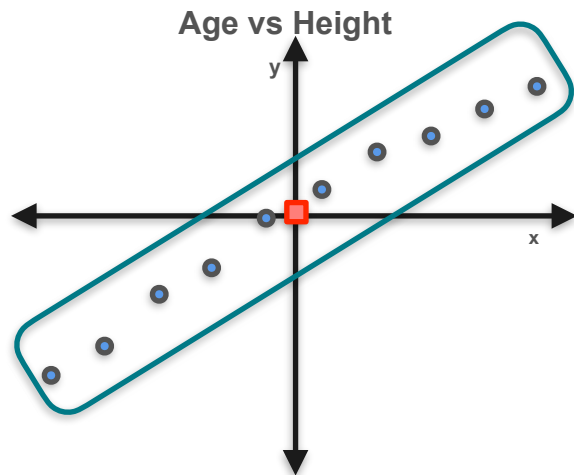


$$\mu_x = 0$$
$$\mu_y = 0$$
$$\text{Var}(X) = 1$$
$$\text{Var}(Y) = 1$$

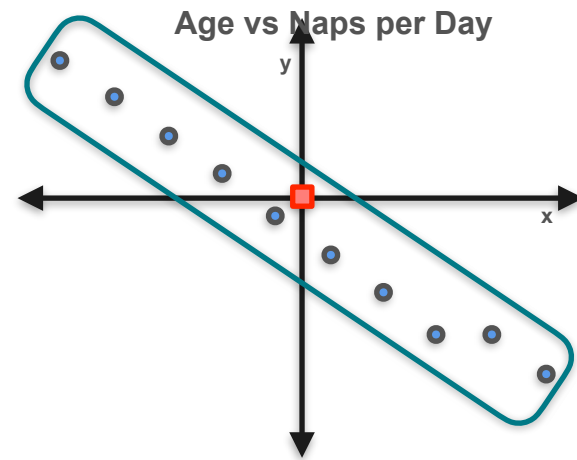
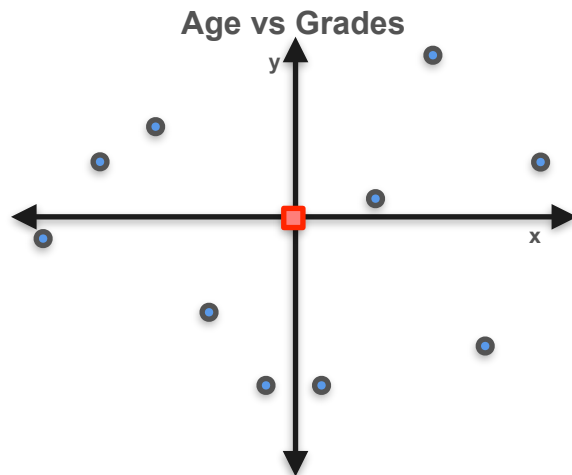
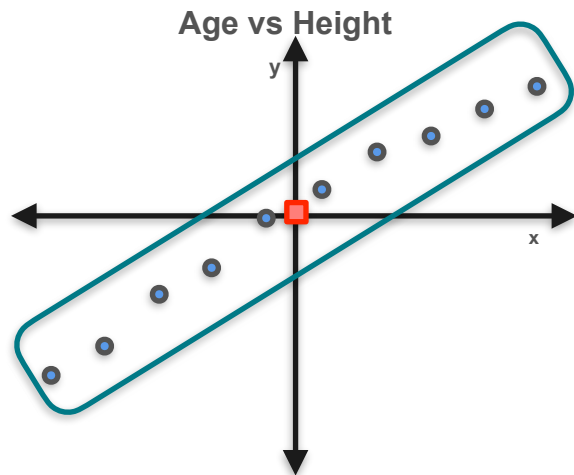
Second Step: Notice Trend



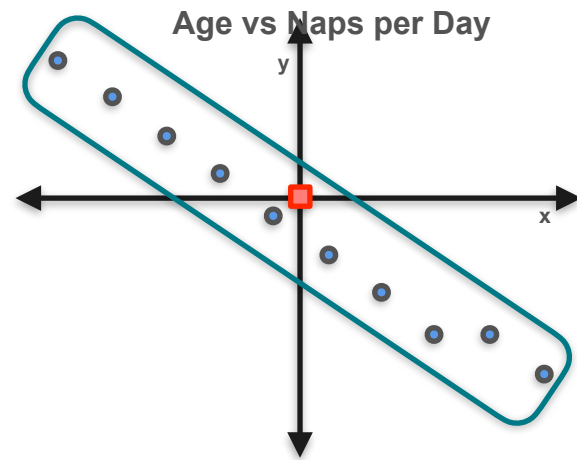
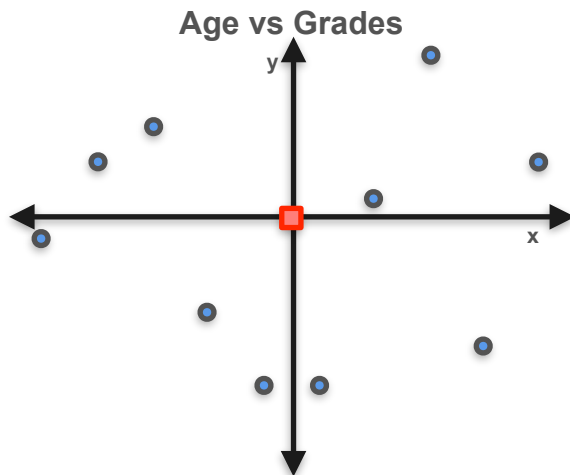
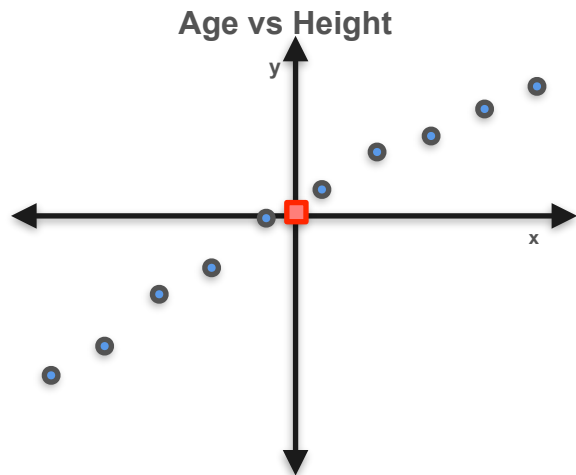
Second Step: Notice Trend



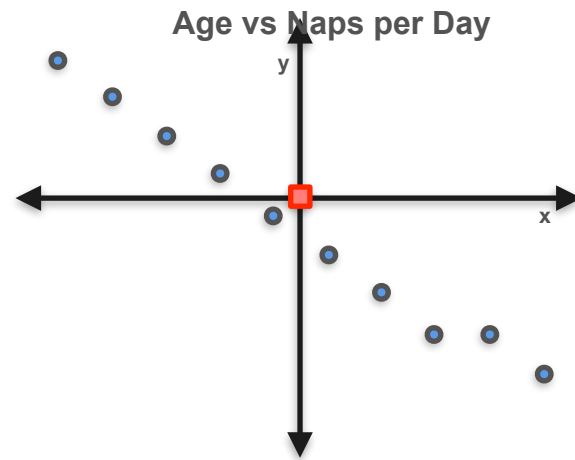
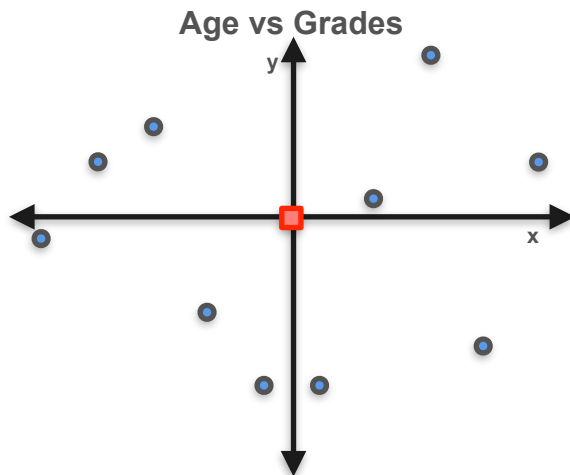
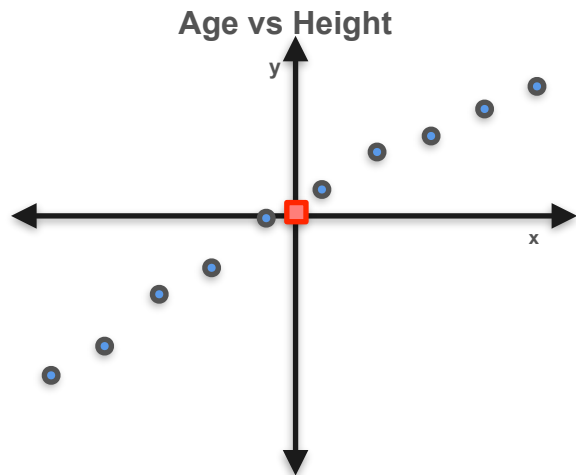
Second Step: Notice Trend



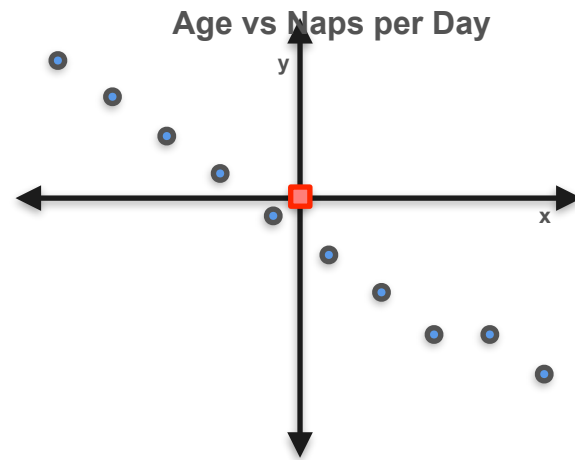
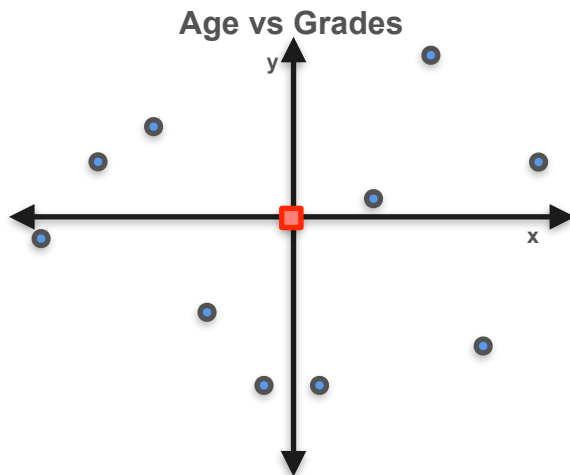
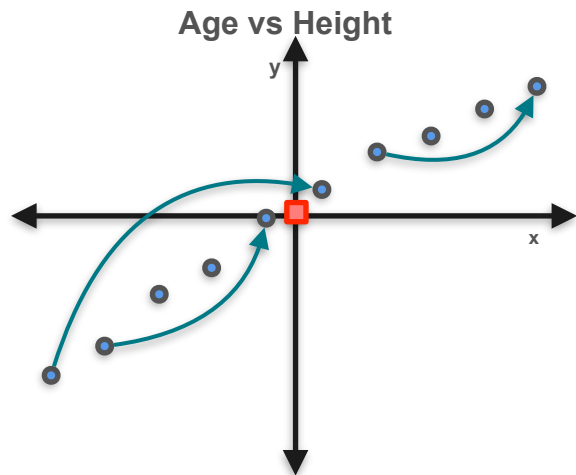
Second Step: Notice Trend



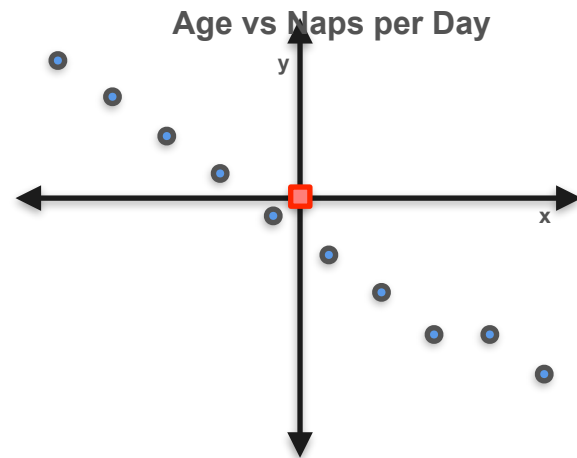
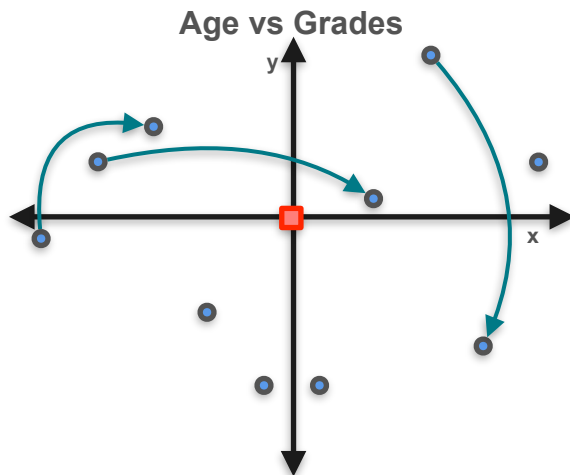
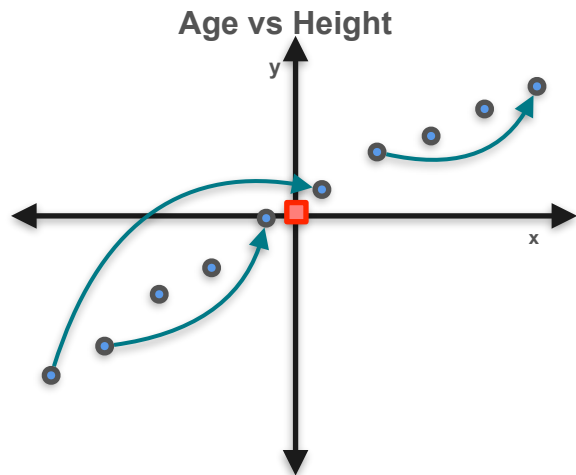
Second Step: Notice Trend



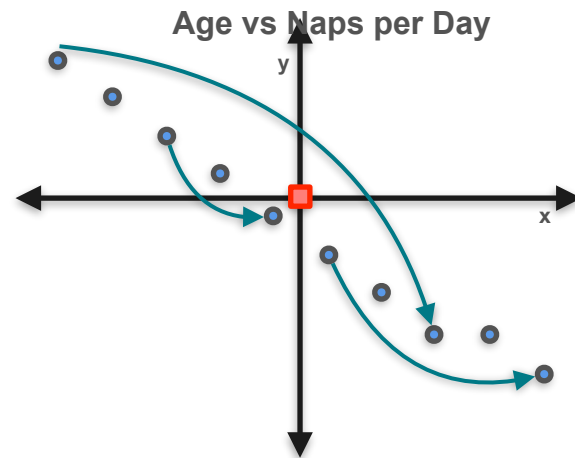
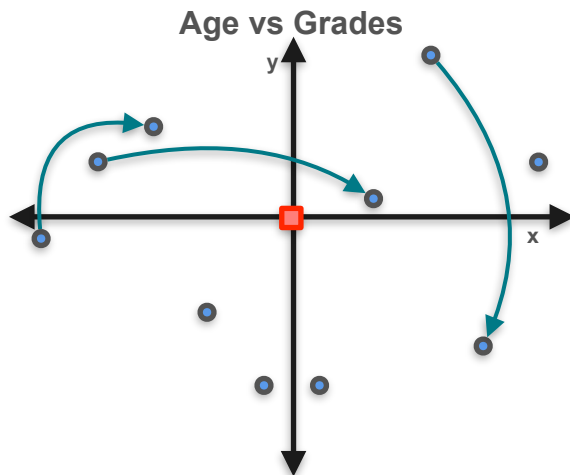
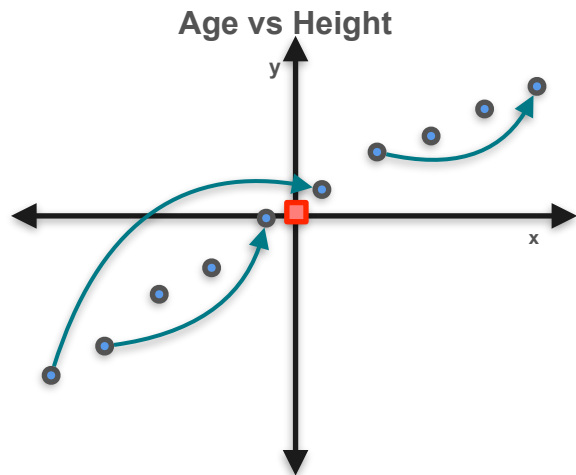
Second Step: Notice Trend



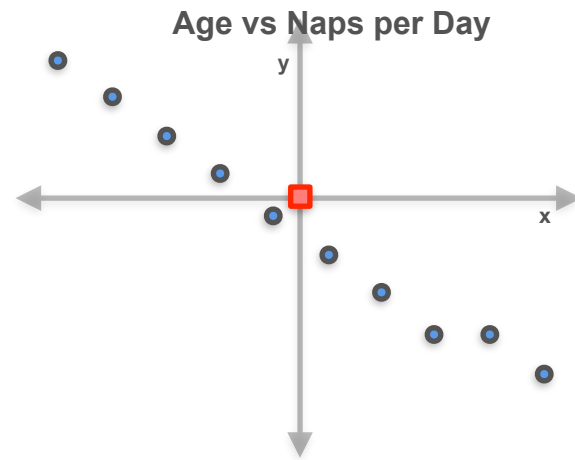
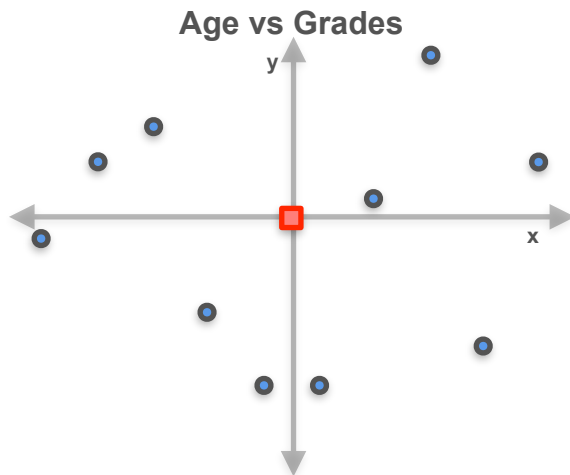
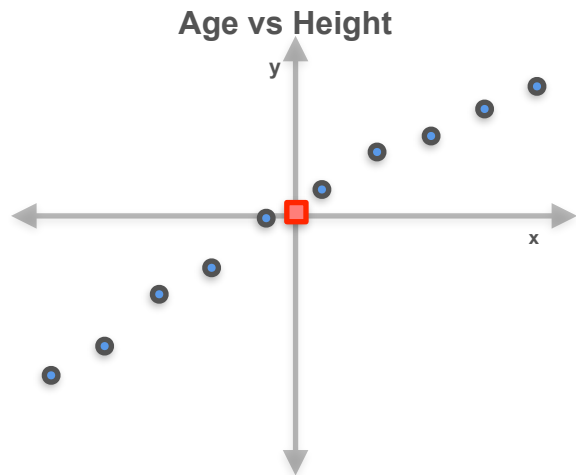
Second Step: Notice Trend



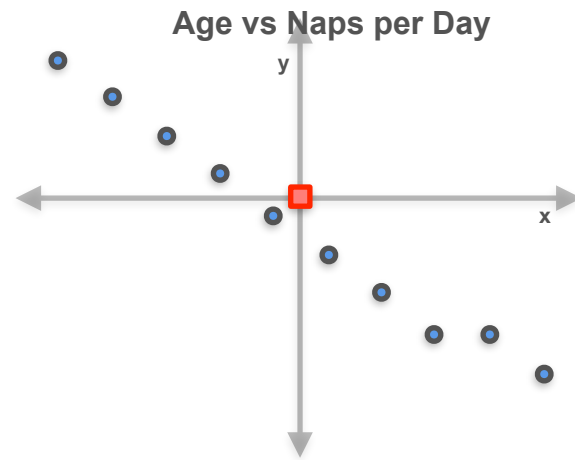
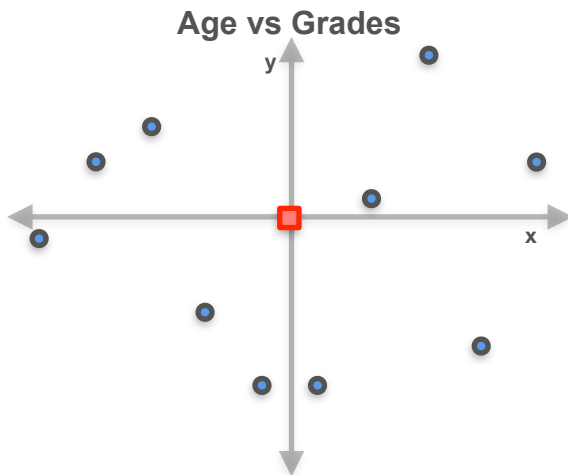
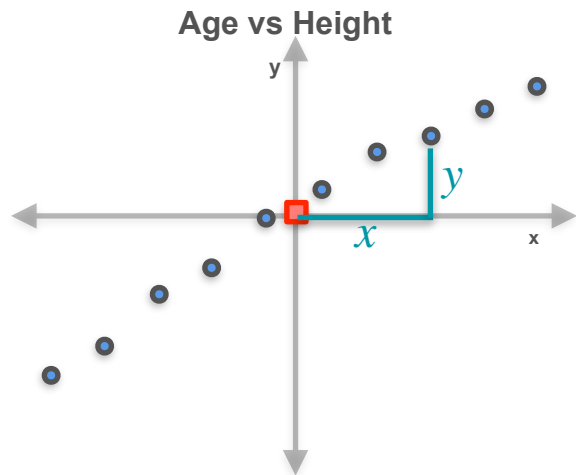
Second Step: Notice Trend



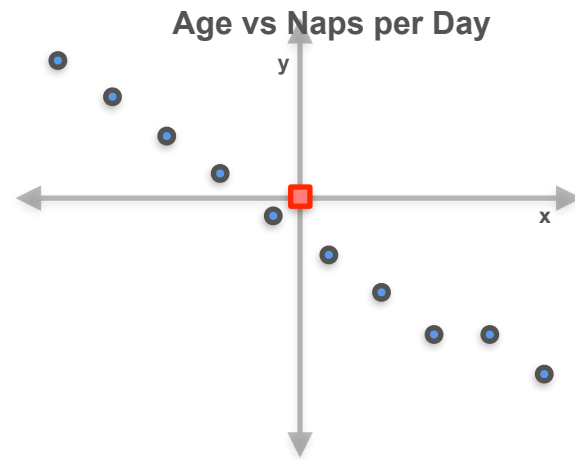
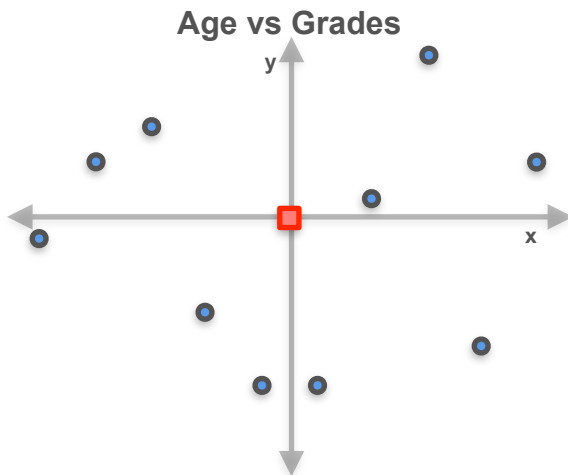
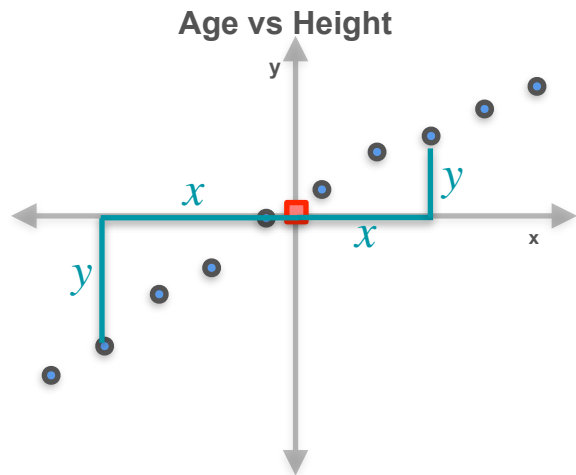
Positives and Negatives



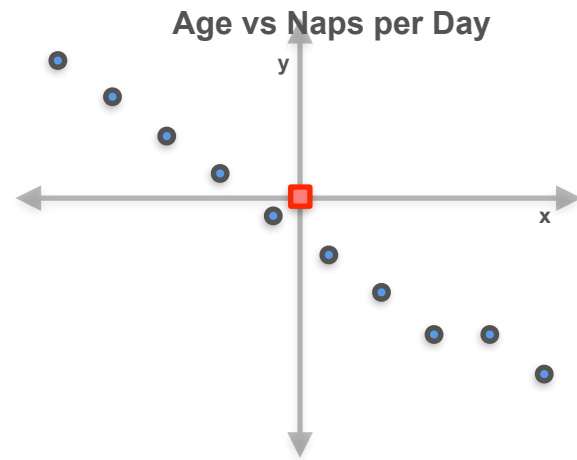
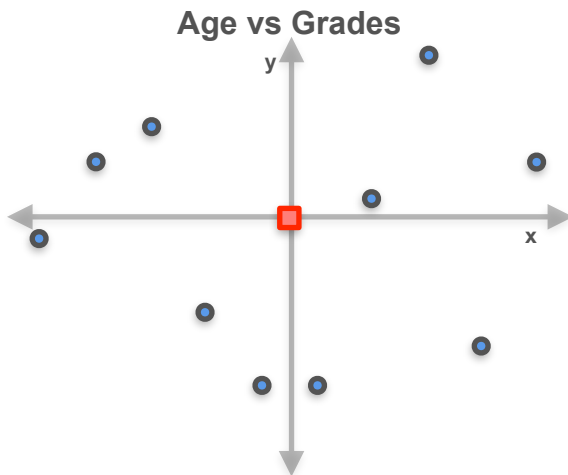
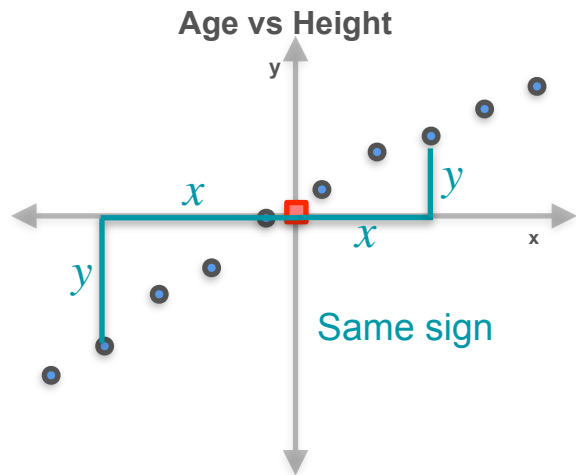
Positives and Negatives



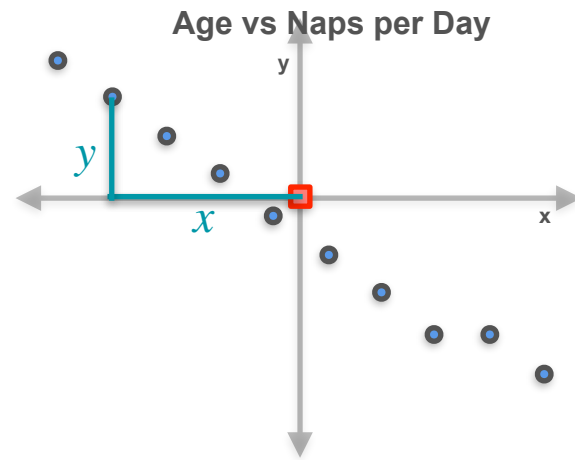
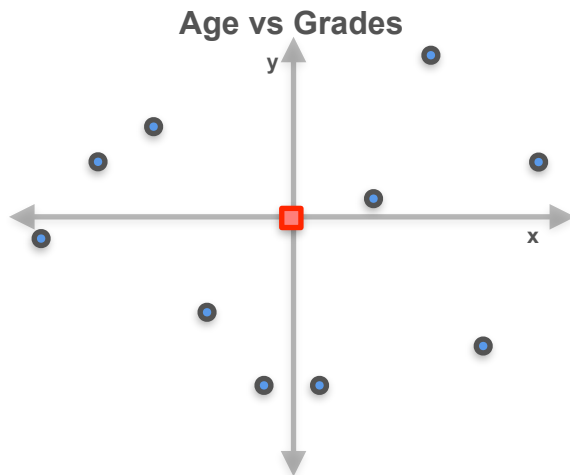
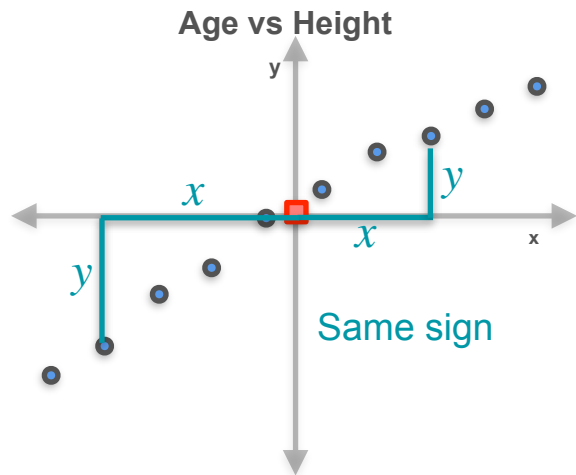
Positives and Negatives



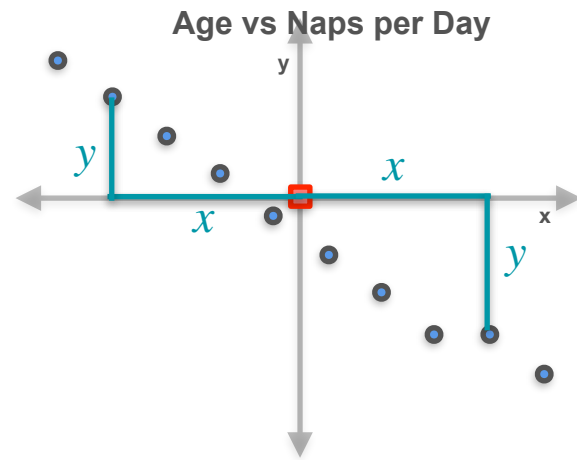
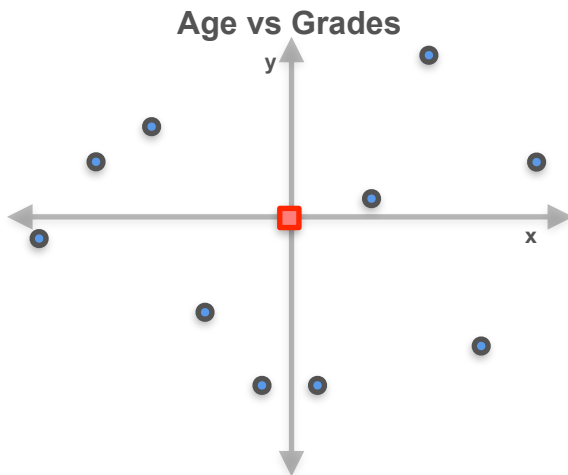
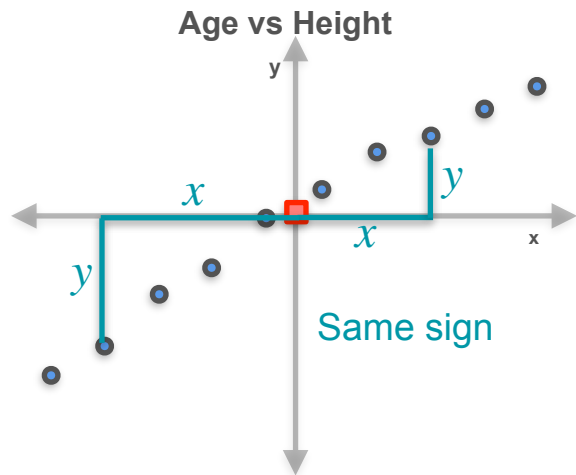
Positives and Negatives



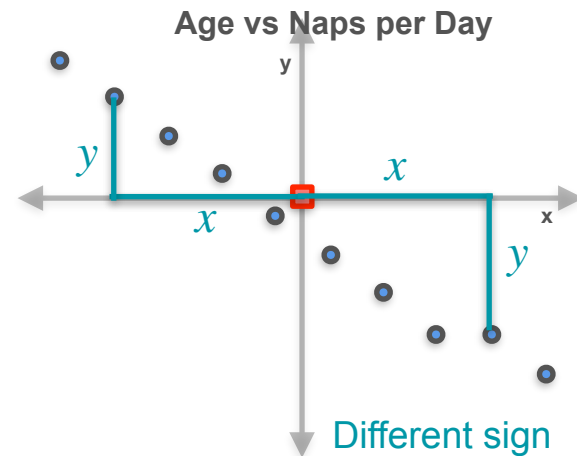
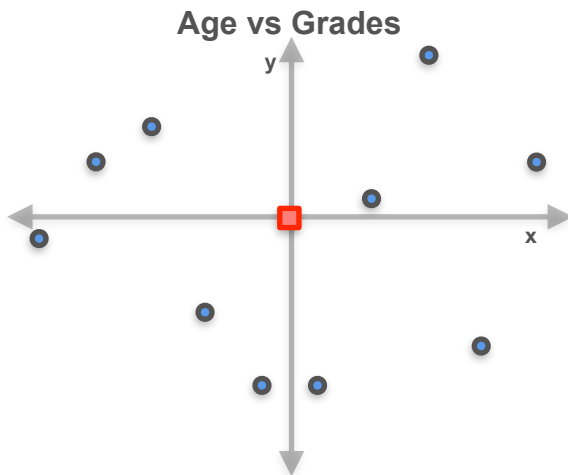
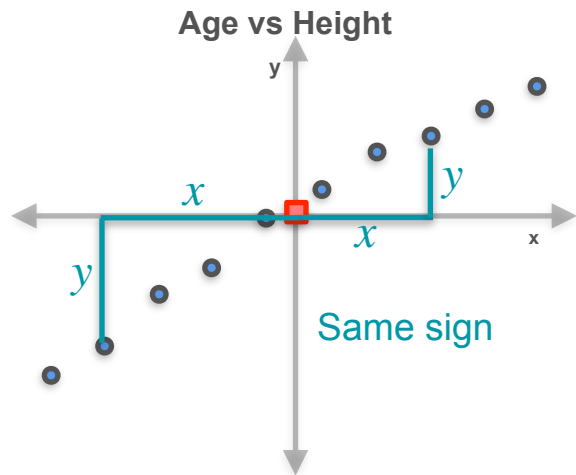
Positives and Negatives



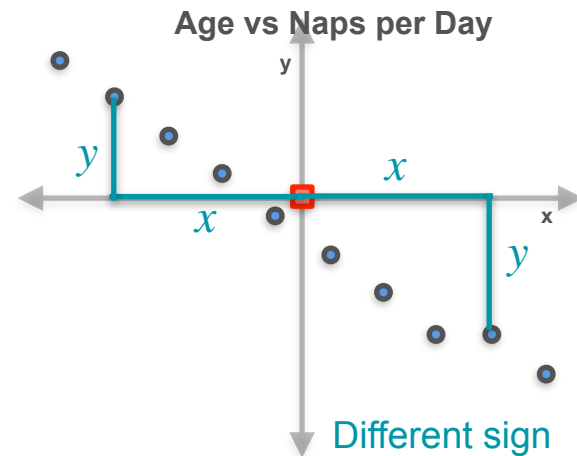
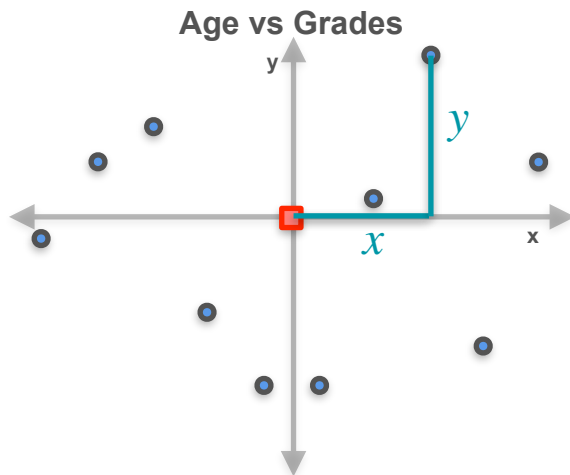
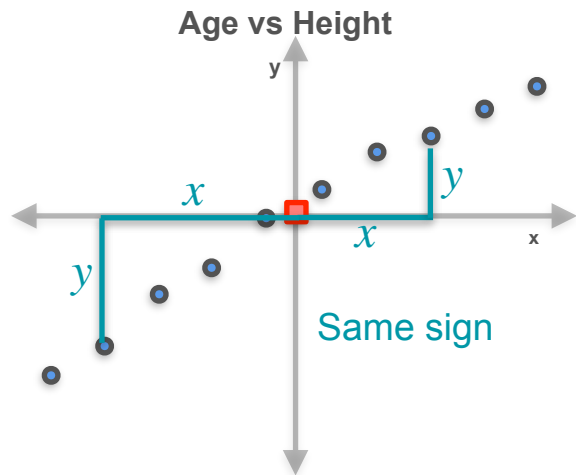
Positives and Negatives



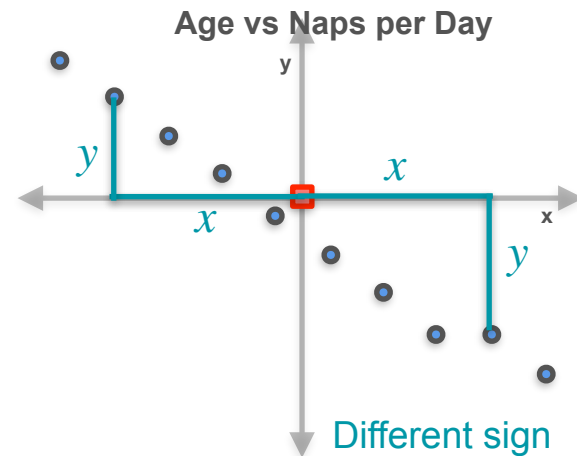
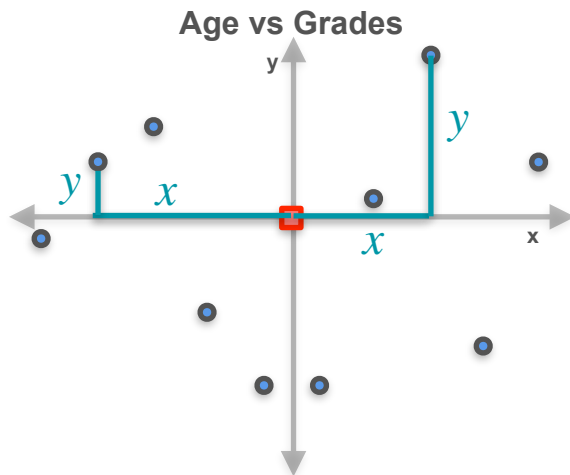
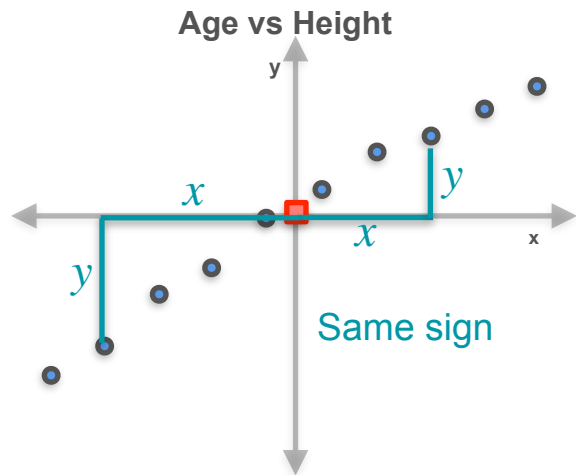
Positives and Negatives



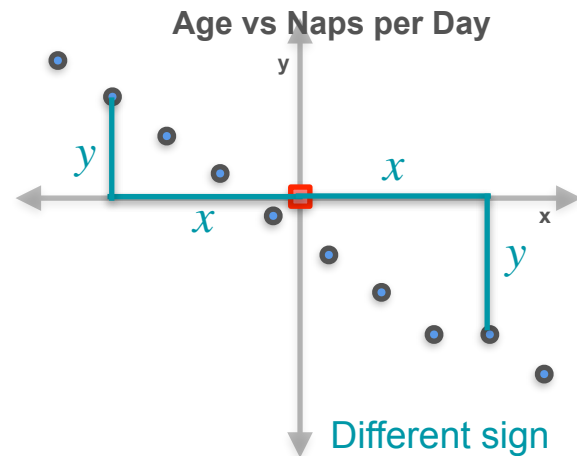
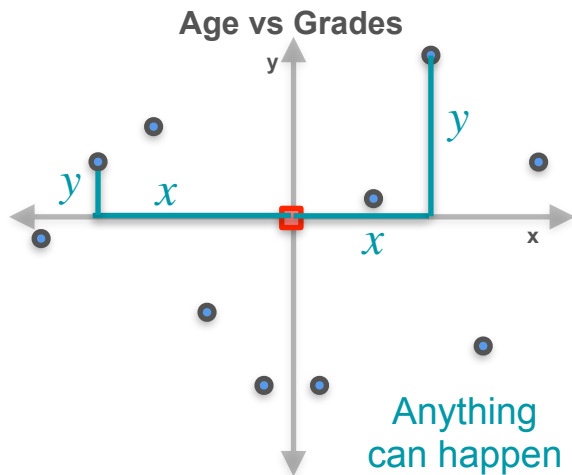
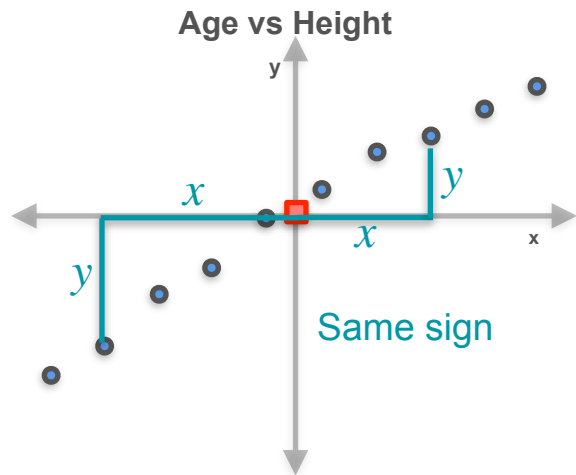
Positives and Negatives



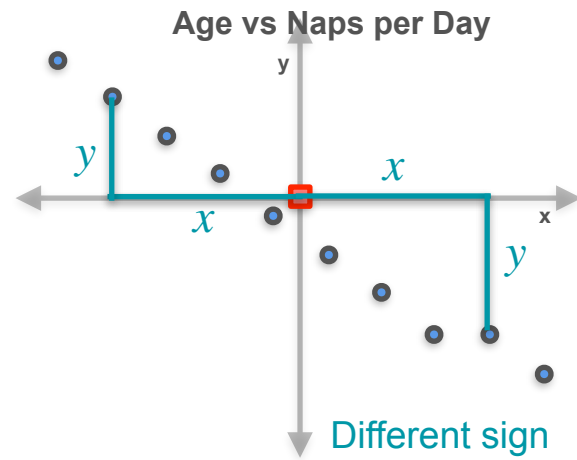
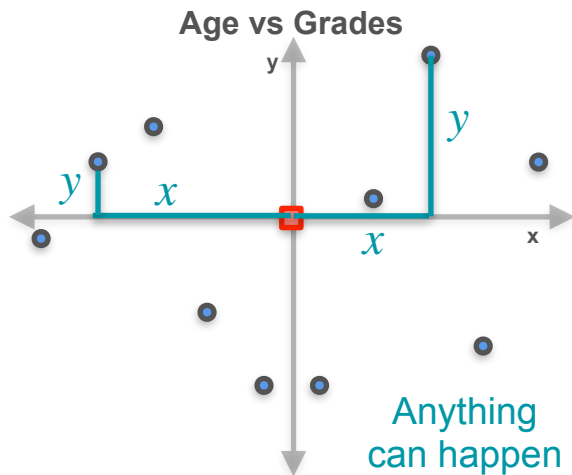
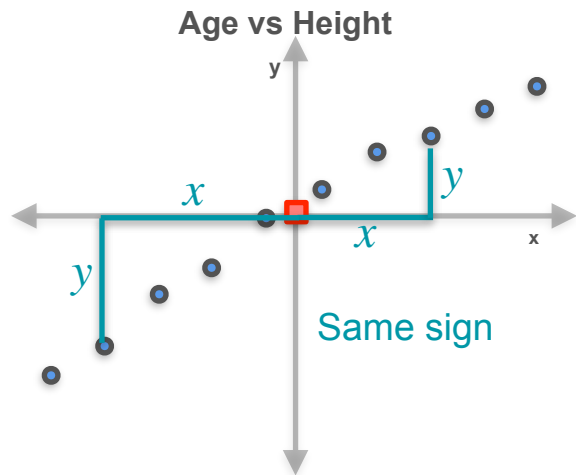
Positives and Negatives



Positives and Negatives



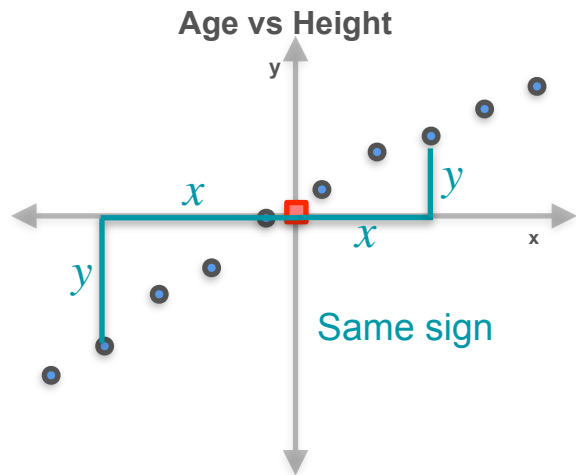
Positives and Negatives



xy

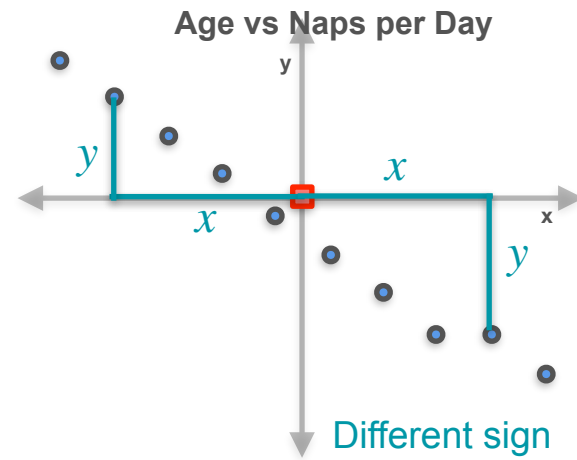
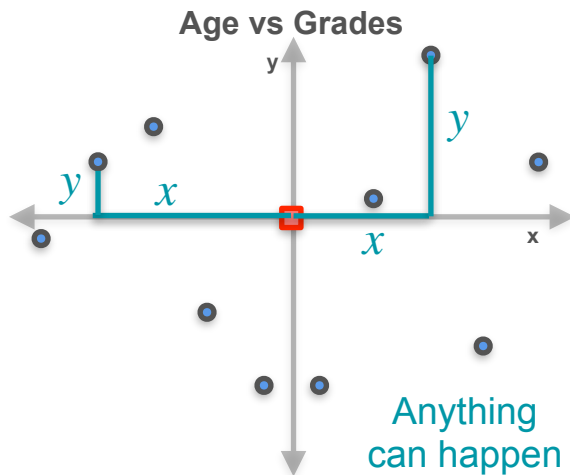
Positive

Positives and Negatives



xy

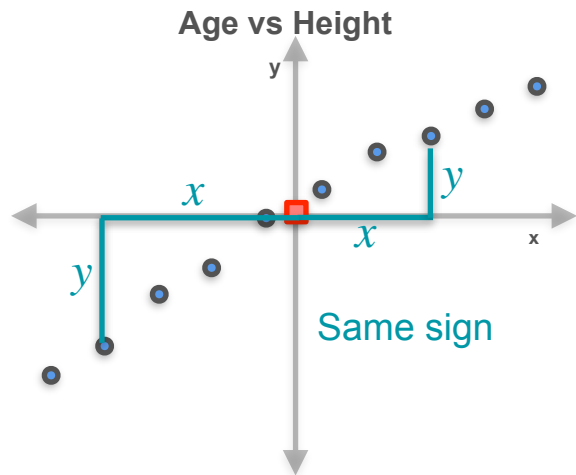
Positive



xy

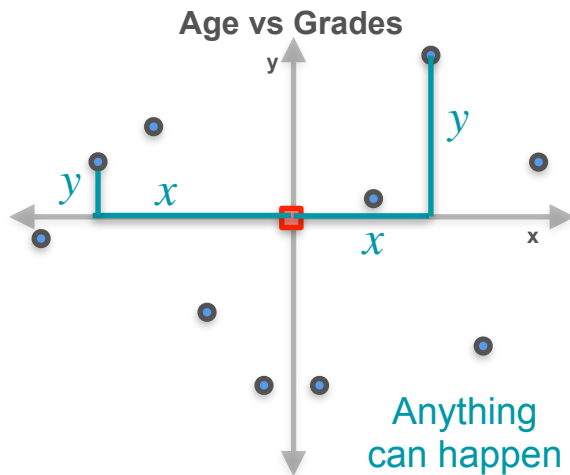
Negative

Positives and Negatives



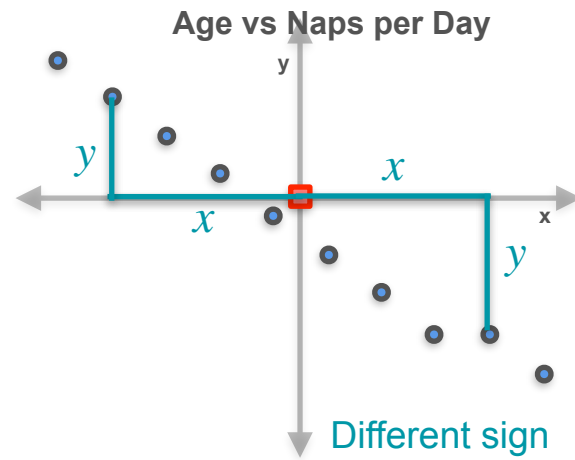
xy

Positive



xy

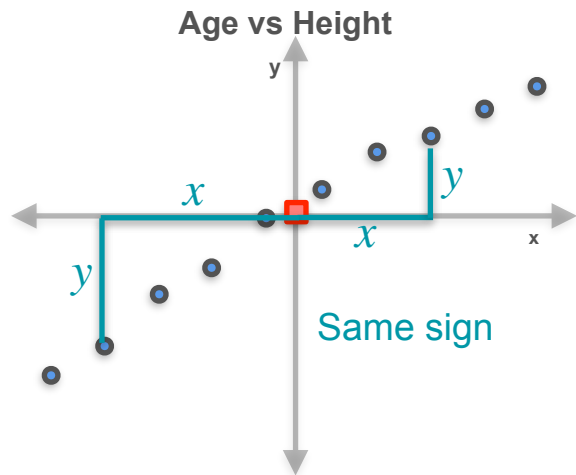
Both positive
and negative



xy

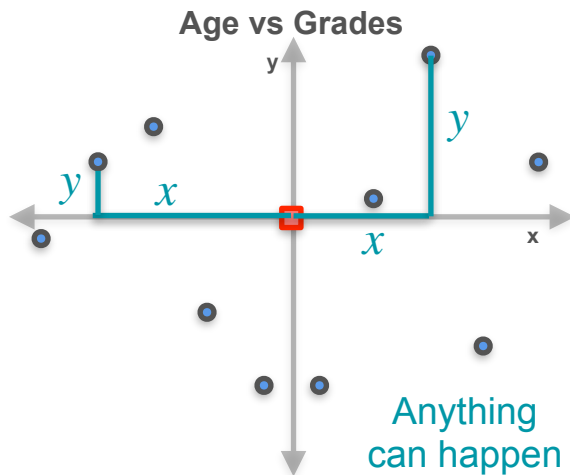
Negative

Positives and Negatives



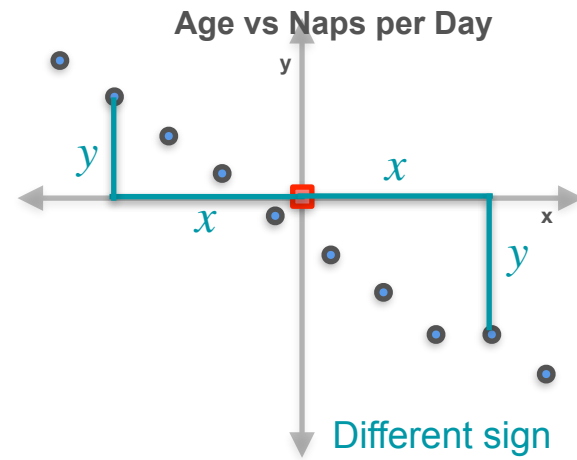
$$\sum xy$$

Positive



$$\sum xy$$

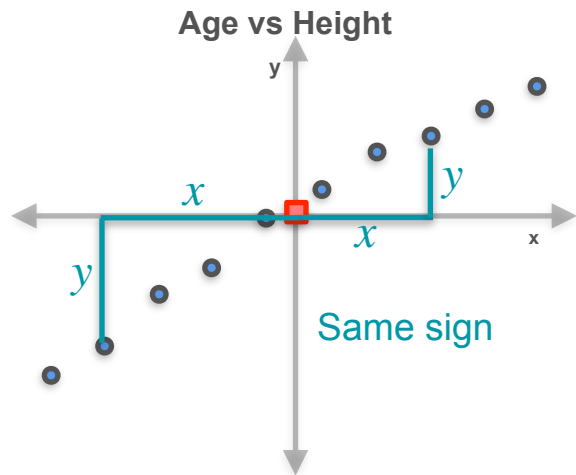
Both positive
and negative



$$\sum xy$$

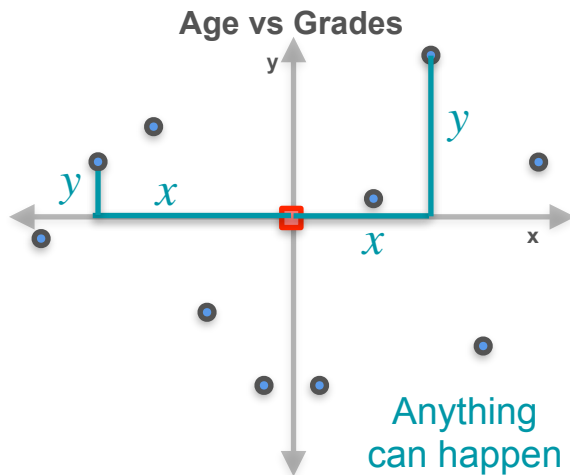
Negative

Positives and Negatives



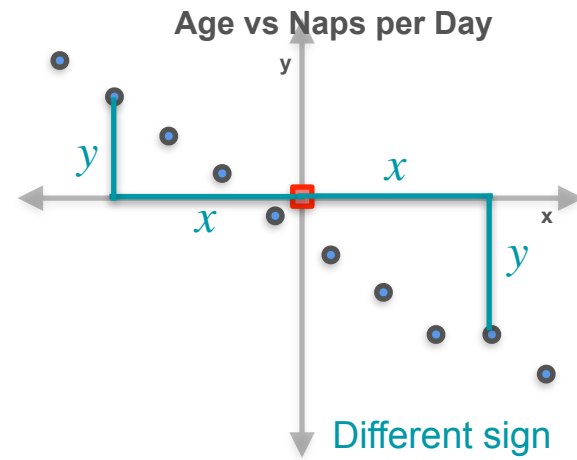
$$\sum xy > 0$$

Positive



$$\sum xy$$

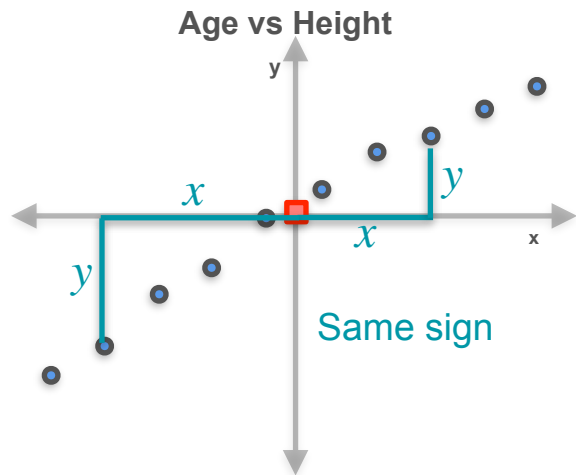
Both positive
and negative



$$\sum xy$$

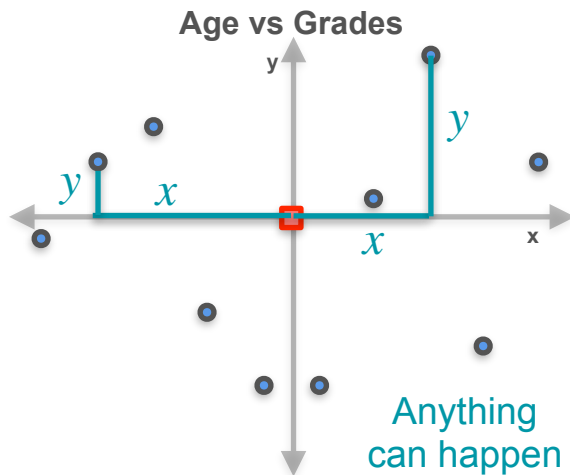
Negative

Positives and Negatives



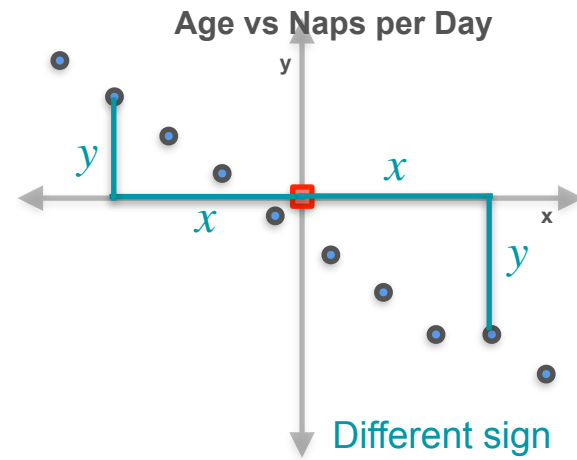
$$\sum xy > 0$$

Positive



$$\sum xy$$

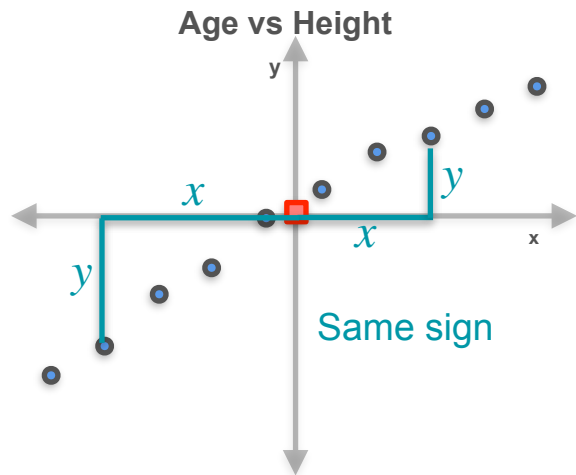
Both positive
and negative



$$\sum xy < 0$$

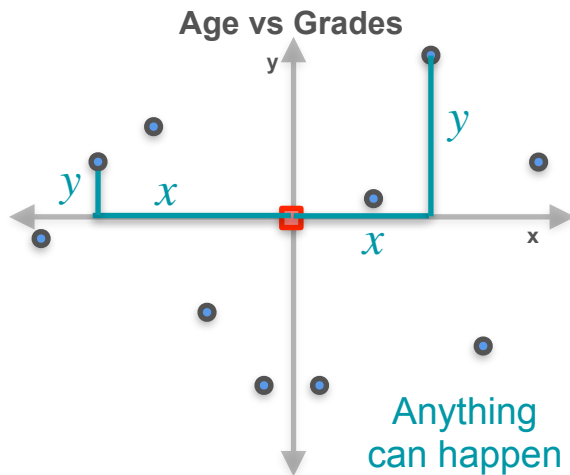
Negative

Positives and Negatives



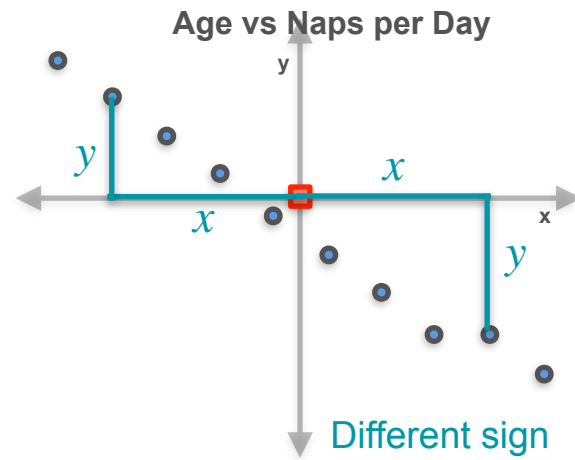
$$\sum xy > 0$$

Positive



$$\sum xy \approx 0$$

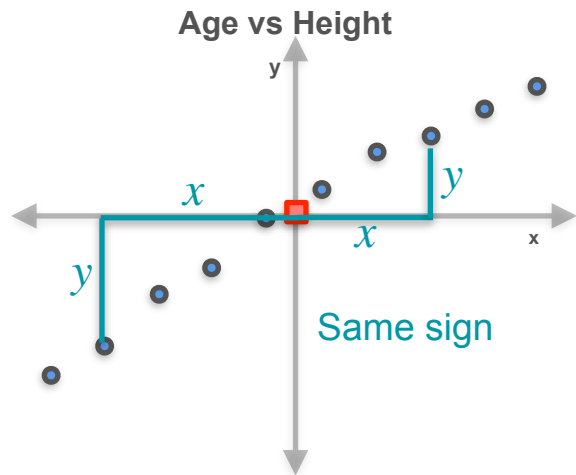
Both positive
and negative



$$\sum xy < 0$$

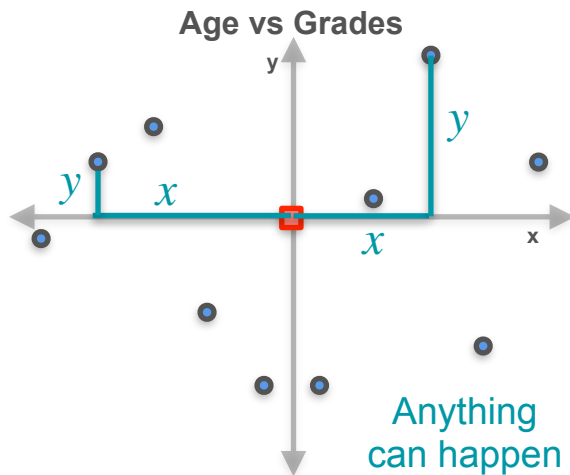
Negative

Covariance



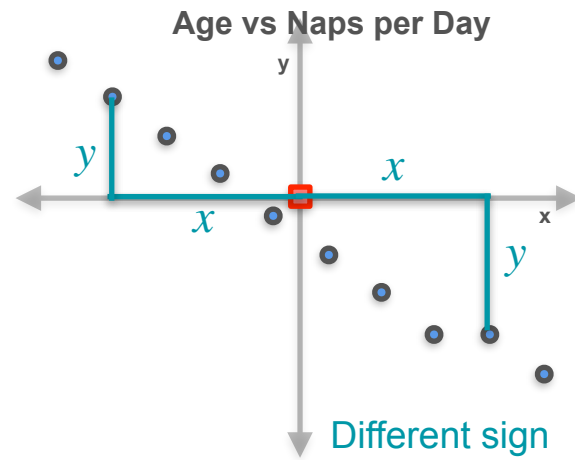
$$\sum xy > 0$$

Positive



$$\sum xy \approx 0$$

Both positive
and negative



$$\sum xy < 0$$

Negative

Covariance

Covariance

$$\text{Cov}(X, Y) = \sum xy$$

Covariance

$$\text{Cov}(X, Y) = \sum xy$$

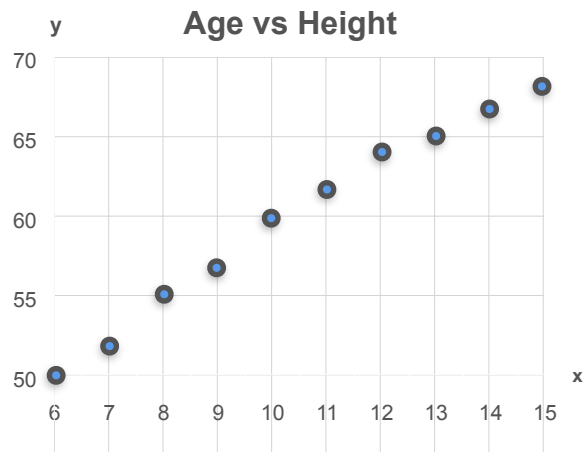
Almost...

Covariance

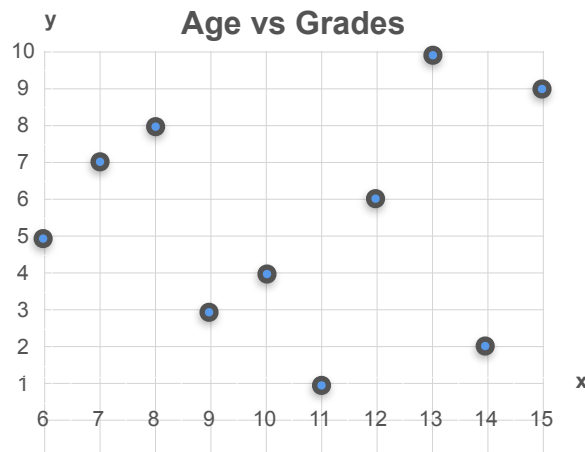
$$\text{Cov}(X, Y) = \sum xy \quad \text{Almost...}$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

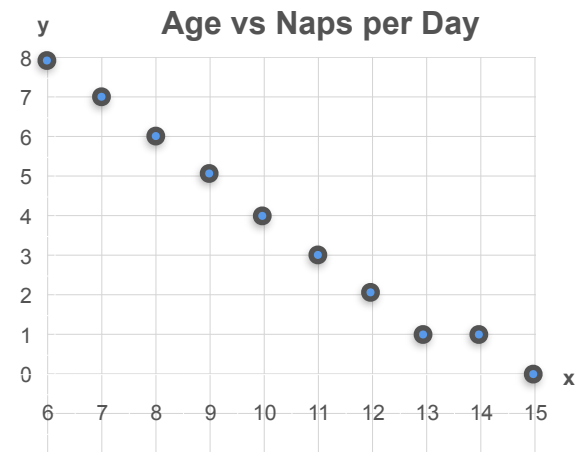
Covariance



$$\text{Cov}(X, Y) > 0$$



$$\text{Cov}(X, Y) \approx 0$$



$$\text{Cov}(X, Y) < 0$$

Covariance Formula

Covariance Formula

Age vs Height

Covariance Formula

Age vs Height

Covariance > 0

Covariance Formula

Age vs Height

Covariance > 0

$$\mu_x = 10.5$$

Covariance Formula

Age vs Height

Covariance > 0

$$\mu_x = 10.5 \quad \mu_y = 60$$

Covariance Formula

Age vs Height

Covariance > 0

$$\mu_x = 10.5 \quad \mu_y = 60$$

Age (x)	Height (y)
6	50
7	52
8	55
9	57
10	60
11	62
12	64
13	65
14	67
15	68

Covariance Formula

Age vs Height

Covariance > 0

$$\mu_x = 10.5 \quad \mu_y = 60$$

Age (x)	Height (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	50	-4.5	-10	
7	52	-3.5	-8	
8	55	-2.5	-5	
9	57	-1.5	-3	
10	60	-0.5	0	
11	62	0.5	2	
12	64	1.5	4	
13	65	2.5	5	
14	67	3.5	7	
15	68	4.5	8	

Covariance Formula

Age vs Height

Covariance > 0

$$\mu_x = 10.5 \quad \mu_y = 60$$

Age (x)	Height (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	50	-4.5	-10	45
7	52	-3.5	-8	28
8	55	-2.5	-5	12.5
9	57	-1.5	-3	4.5
10	60	-0.5	0	0
11	62	0.5	2	1
12	64	1.5	4	6
13	65	2.5	5	12.5
14	67	3.5	7	24.5
15	68	4.5	8	36

Covariance Formula

Age vs Height

Covariance > 0

$$\mu_x = 10.5 \quad \mu_y = 60$$

Age (x)	Height (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	50	-4.5	-10	45
7	52	-3.5	-8	28
8	55	-2.5	-5	12.5
9	57	-1.5	-3	4.5
10	60	-0.5	0	0
11	62	0.5	2	1
12	64	1.5	4	6
13	65	2.5	5	12.5
14	67	3.5	7	24.5
15	68	4.5	8	36

$$\Sigma = 170$$

Covariance Formula

Age vs Height

Covariance > 0

$$\mu_x = 10.5 \quad \mu_y = 60$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

Age (x)	Height (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	50	-4.5	-10	45
7	52	-3.5	-8	28
8	55	-2.5	-5	12.5
9	57	-1.5	-3	4.5
10	60	-0.5	0	0
11	62	0.5	2	1
12	64	1.5	4	6
13	65	2.5	5	12.5
14	67	3.5	7	24.5
15	68	4.5	8	36

$$\Sigma = 170$$

Covariance Formula

Age vs Height

Covariance > 0

$$\mu_x = 10.5 \quad \mu_y = 60$$

$$Cov(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

$$Cov(X, Y) = \frac{170}{10}$$

Age (x)	Height (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	50	-4.5	-10	45
7	52	-3.5	-8	28
8	55	-2.5	-5	12.5
9	57	-1.5	-3	4.5
10	60	-0.5	0	0
11	62	0.5	2	1
12	64	1.5	4	6
13	65	2.5	5	12.5
14	67	3.5	7	24.5
15	68	4.5	8	36

$$\Sigma = 170$$

Covariance Formula

Age vs Height

Covariance > 0

$$\mu_x = 10.5 \quad \mu_y = 60$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

$$\text{Cov}(X, Y) = \frac{170}{10} = 17$$

Age (x)	Height (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	50	-4.5	-10	45
7	52	-3.5	-8	28
8	55	-2.5	-5	12.5
9	57	-1.5	-3	4.5
10	60	-0.5	0	0
11	62	0.5	2	1
12	64	1.5	4	6
13	65	2.5	5	12.5
14	67	3.5	7	24.5
15	68	4.5	8	36

$$\Sigma = 170$$

Covariance Formula

Age vs Height

Covariance > 0

$$\mu_x = 10.5 \quad \mu_y = 60$$

$$Cov(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

$$Cov(X, Y) = \frac{170}{10} = 17 > 0$$

Age (x)	Height (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	50	-4.5	-10	45
7	52	-3.5	-8	28
8	55	-2.5	-5	12.5
9	57	-1.5	-3	4.5
10	60	-0.5	0	0
11	62	0.5	2	1
12	64	1.5	4	6
13	65	2.5	5	12.5
14	67	3.5	7	24.5
15	68	4.5	8	36

$$\Sigma = 170$$

Covariance Formula

Covariance Formula

Age vs Naps per Day

Covariance Formula

Age vs Naps per Day

Covariance < 0

Covariance Formula

Age vs Naps per Day

Covariance < 0

$$\mu_x = 10.5$$

Covariance Formula

Age vs Naps per Day

Covariance < 0

$$\mu_x = 10.5 \quad \mu_y = 3.7$$

Covariance Formula

Age vs Naps per Day

Covariance < 0

$$\mu_x = 10.5 \quad \mu_y = 3.7$$

Age (x)
6
7
8
9
10
11
12
13
14
15

Covariance Formula

Age vs Naps per Day

Covariance < 0

$$\mu_x = 10.5 \quad \mu_y = 3.7$$

Age (x)	Naps (y)
6	8
7	7
8	6
9	5
10	4
11	3
12	2
13	1
14	1
15	0

Covariance Formula

Age vs Naps per Day

Covariance < 0

$$\mu_x = 10.5 \quad \mu_y = 3.7$$

Age (x)	Naps (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	8	-4.5	4.3	-19.35
7	7	-3.5	3.3	-11.55
8	6	-2.5	2.3	-5.75
9	5	-1.5	1.3	-1.95
10	4	-0.5	0.3	-0.15
11	3	0.5	-0.7	-0.35
12	2	1.5	-1.7	-2.55
13	1	2.5	-2.7	-6.75
14	1	3.5	-2.7	-9.45
15	0	4.5	-3.7	-16.65

Covariance Formula

Age vs Naps per Day

Covariance < 0

$$\mu_x = 10.5 \quad \mu_y = 3.7$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

Age (x)	Naps (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	8	-4.5	4.3	-19.35
7	7	-3.5	3.3	-11.55
8	6	-2.5	2.3	-5.75
9	5	-1.5	1.3	-1.95
10	4	-0.5	0.3	-0.15
11	3	0.5	-0.7	-0.35
12	2	1.5	-1.7	-2.55
13	1	2.5	-2.7	-6.75
14	1	3.5	-2.7	-9.45
15	0	4.5	-3.7	-16.65

$$\Sigma = -74.5$$

Covariance Formula

Age vs Naps per Day

Covariance < 0

$$\mu_x = 10.5 \quad \mu_y = 3.7$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

$$\text{Cov}(X, Y) = \frac{-74.5}{10}$$

Age (x)	Naps (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	8	-4.5	4.3	-19.35
7	7	-3.5	3.3	-11.55
8	6	-2.5	2.3	-5.75
9	5	-1.5	1.3	-1.95
10	4	-0.5	0.3	-0.15
11	3	0.5	-0.7	-0.35
12	2	1.5	-1.7	-2.55
13	1	2.5	-2.7	-6.75
14	1	3.5	-2.7	-9.45
15	0	4.5	-3.7	-16.65

$$\sum = -74.5$$

Covariance Formula

Age vs Naps per Day

Covariance < 0

$$\mu_x = 10.5 \quad \mu_y = 3.7$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

$$\text{Cov}(X, Y) = \frac{-74.5}{10} = -7.45$$

Age (x)	Naps (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	8	-4.5	4.3	-19.35
7	7	-3.5	3.3	-11.55
8	6	-2.5	2.3	-5.75
9	5	-1.5	1.3	-1.95
10	4	-0.5	0.3	-0.15
11	3	0.5	-0.7	-0.35
12	2	1.5	-1.7	-2.55
13	1	2.5	-2.7	-6.75
14	1	3.5	-2.7	-9.45
15	0	4.5	-3.7	-16.65

$$\sum = -74.5$$

Covariance Formula

Age vs Naps per Day

Covariance < 0

$$\mu_x = 10.5 \quad \mu_y = 3.7$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

$$\text{Cov}(X, Y) = \frac{-74.5}{10} = -7.45 < 0$$

Age (x)	Naps (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	8	-4.5	4.3	-19.35
7	7	-3.5	3.3	-11.55
8	6	-2.5	2.3	-5.75
9	5	-1.5	1.3	-1.95
10	4	-0.5	0.3	-0.15
11	3	0.5	-0.7	-0.35
12	2	1.5	-1.7	-2.55
13	1	2.5	-2.7	-6.75
14	1	3.5	-2.7	-9.45
15	0	4.5	-3.7	-16.65

$$\sum = -74.5$$

Covariance Formula

$$(x_i - \mu_x)$$

Covariance Formula

Age vs Grades

Covariance ≈ 0

$$\mu_x = 10.5 \quad \mu_y = 5$$

Age (x)	Grades (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	5	-4.5	0	0
7	7	-3.5	2	-7
8	8	-2.5	3	-7.5
9	3	-1.5	-2	3
10	1	-0.5	-4	2
11	1	0.5	-4	-2
12	6	1.5	1	1.5
13	10	2.5	5	12.5
14	2	3.5	-3	-10.5
15	7	4.5	2	9

Covariance Formula

Age vs Grades

Covariance ≈ 0

$$\mu_x = 10.5 \quad \mu_y = 5$$

Age (x)	Grades (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	5	-4.5	0	0
7	7	-3.5	2	-7
8	8	-2.5	3	-7.5
9	3	-1.5	-2	3
10	1	-0.5	-4	2
11	1	0.5	-4	-2
12	6	1.5	1	1.5
13	10	2.5	5	12.5
14	2	3.5	-3	-10.5
15	7	4.5	2	9

$$\Sigma = 1$$

Covariance Formula

Age vs Grades

Covariance ≈ 0

$$\mu_x = 10.5 \quad \mu_y = 5$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

Age (x)	Grades (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	5	-4.5	0	0
7	7	-3.5	2	-7
8	8	-2.5	3	-7.5
9	3	-1.5	-2	3
10	1	-0.5	-4	2
11	1	0.5	-4	-2
12	6	1.5	1	1.5
13	10	2.5	5	12.5
14	2	3.5	-3	-10.5
15	7	4.5	2	9

$$\sum = 1$$

Covariance Formula

Age vs Grades

Covariance ≈ 0

$$\mu_x = 10.5 \quad \mu_y = 5$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

$$\text{Cov}(X, Y) = \frac{1}{10}$$

Age (x)	Grades (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	5	-4.5	0	0
7	7	-3.5	2	-7
8	8	-2.5	3	-7.5
9	3	-1.5	-2	3
10	1	-0.5	-4	2
11	1	0.5	-4	-2
12	6	1.5	1	1.5
13	10	2.5	5	12.5
14	2	3.5	-3	-10.5
15	7	4.5	2	9

$$\sum = 1$$

Covariance Formula

Age vs Grades

Covariance ≈ 0

$$\mu_x = 10.5 \quad \mu_y = 5$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

$$\text{Cov}(X, Y) = \frac{1}{10} = 0.1$$

Age (x)	Grades (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	5	-4.5	0	0
7	7	-3.5	2	-7
8	8	-2.5	3	-7.5
9	3	-1.5	-2	3
10	1	-0.5	-4	2
11	1	0.5	-4	-2
12	6	1.5	1	1.5
13	10	2.5	5	12.5
14	2	3.5	-3	-10.5
15	7	4.5	2	9

$$\Sigma = 1$$

Covariance Formula

Age vs Grades

Covariance ≈ 0

$$\mu_x = 10.5 \quad \mu_y = 5$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

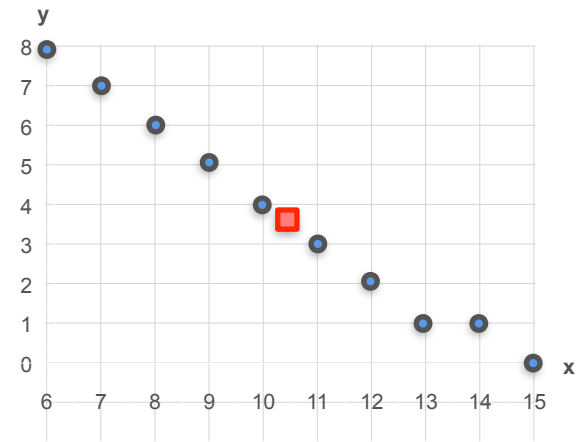
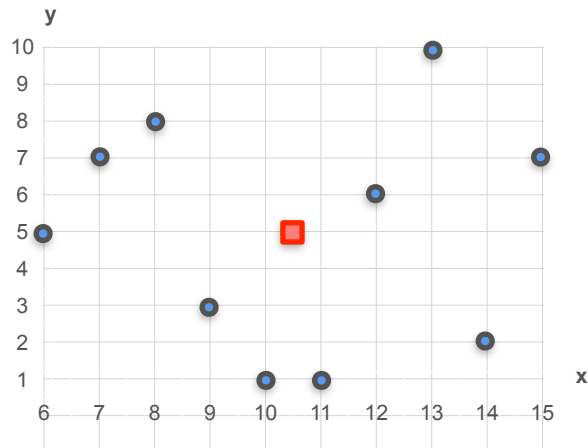
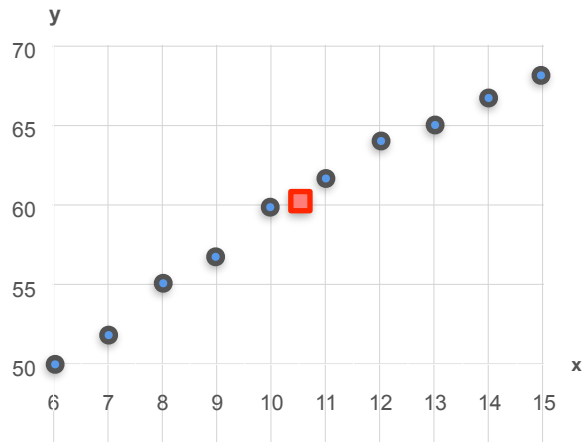
$$\text{Cov}(X, Y) = \frac{1}{10} = 0.1 \approx 0$$

Age (x)	Grades (y)	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
6	5	-4.5	0	0
7	7	-3.5	2	-7
8	8	-2.5	3	-7.5
9	3	-1.5	-2	3
10	1	-0.5	-4	2
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12	6	1.5	1	1.5
13	10	2.5	5	12.5
14	2	3.5	-3	-10.5
15	7	4.5	2	9

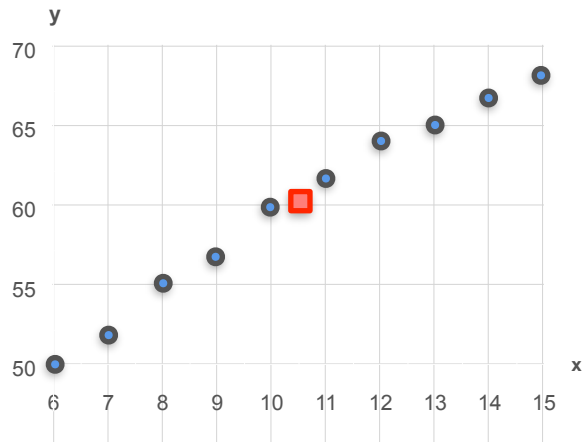
$$\sum = 1$$

Comparing Correlations

Comparing Correlations



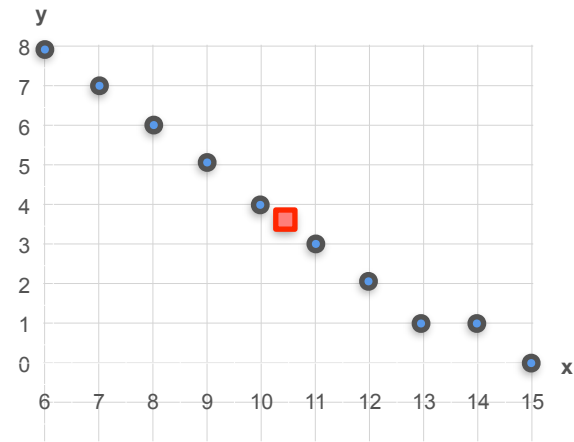
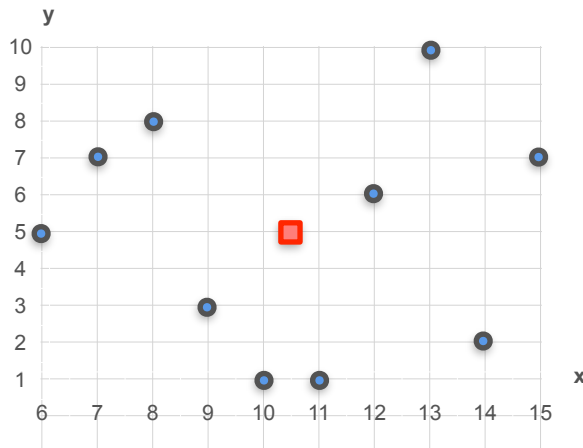
Comparing Correlations



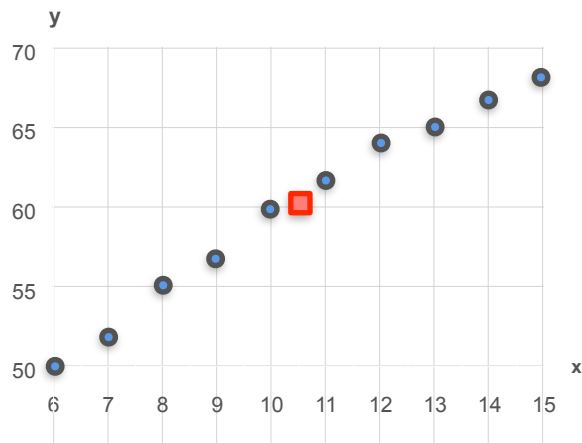
Age vs Height

Covariance > 0

$$\text{Cov}(x, y) = 17$$



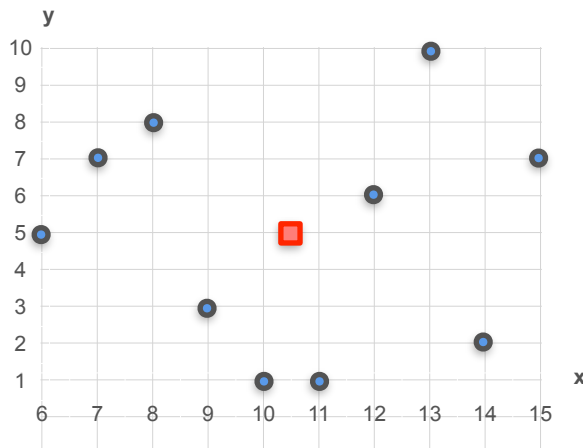
Comparing Correlations



Age vs Height

Covariance > 0

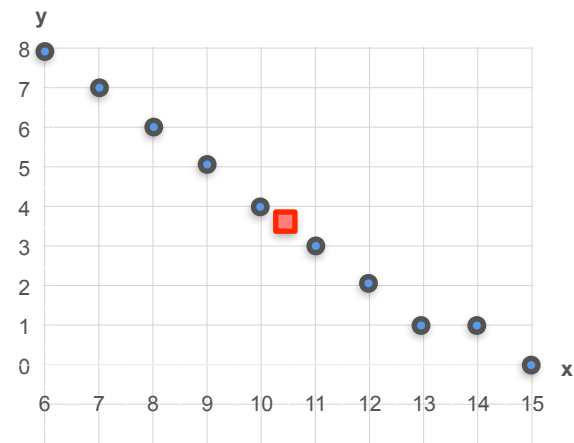
$$Cov(x, y) = 17$$



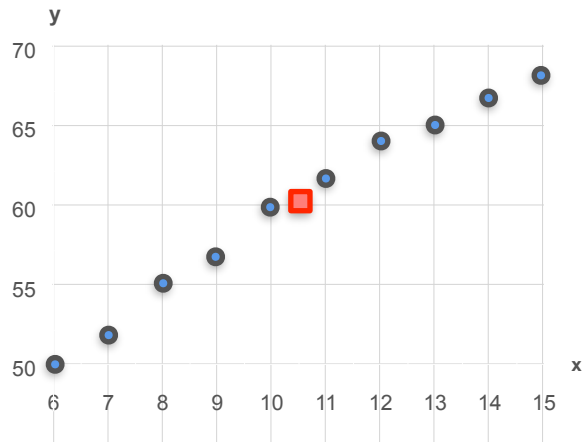
Age vs Grades

Covariance ≈ 0

$$Cov(x, y) = 0.1$$



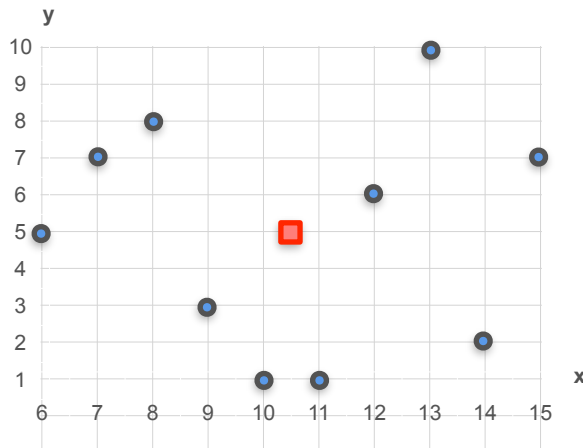
Comparing Correlations



Age vs Height

Covariance > 0

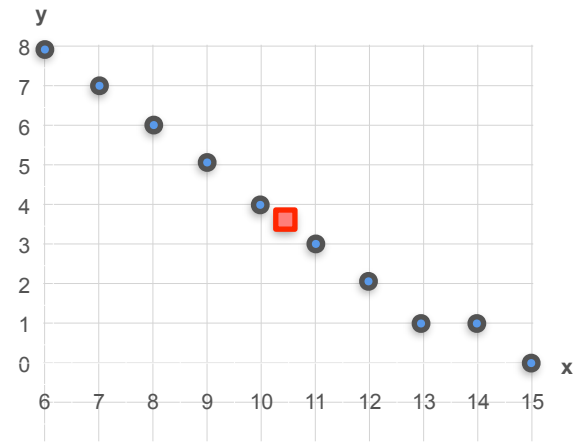
$$\text{Cov}(x, y) = 17$$



Age vs Grades

Covariance ≈ 0

$$\text{Cov}(x, y) = 0.1$$



Age vs Naps per Day

Covariance < 0

$$\text{Cov}(x, y) = -7.45$$

Covariance of a Probability Distribution: Motivation

Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

GAME 2

GAME 3

Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a : Both players win \$1 each

GAME 2

GAME 3

Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

b: Both players lose \$1 each

GAME 2

GAME 3

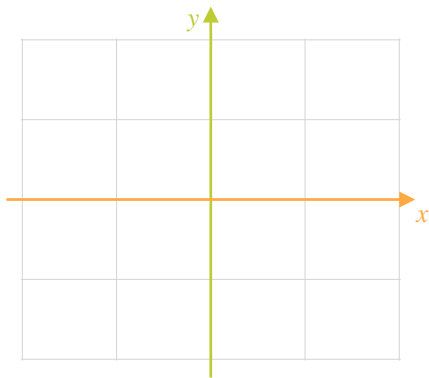
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

b: Both players lose \$1 each



GAME 2

GAME 3

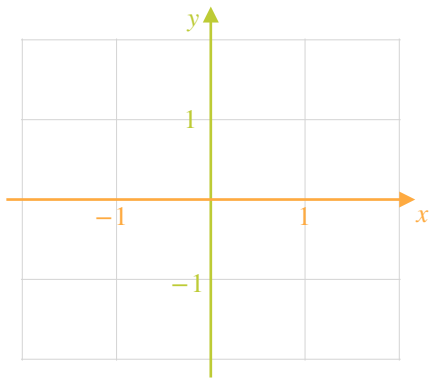
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

b: Both players lose \$1 each



GAME 2

GAME 3

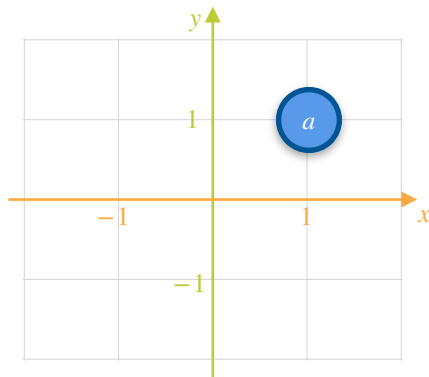
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a : Both players win \$1 each

b : Both players lose \$1 each



GAME 2

GAME 3

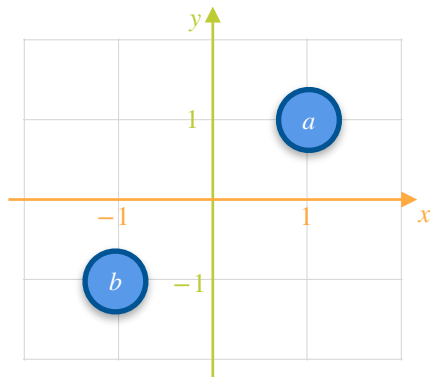
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a : Both players win \$1 each

b : Both players lose \$1 each



GAME 2

GAME 3

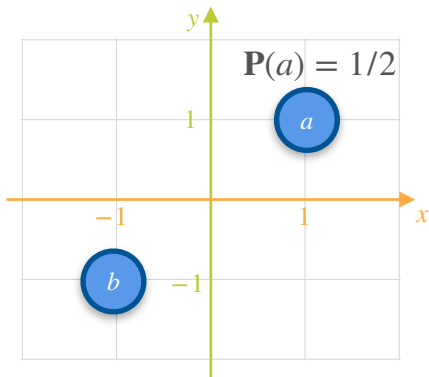
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a : Both players win \$1 each

b : Both players lose \$1 each



GAME 2

GAME 3

Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

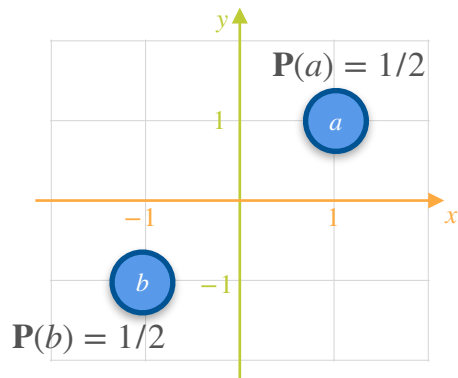
GAME 1

a : Both players win \$1 each

b : Both players lose \$1 each

GAME 2

GAME 3



Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

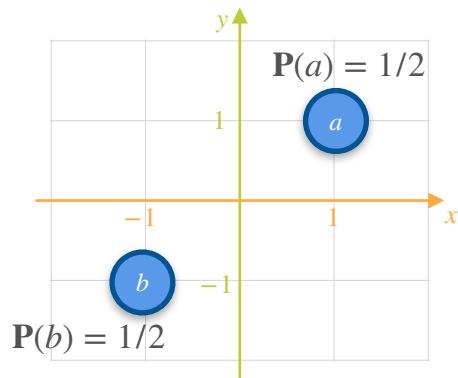
a: Both players win \$1 each

b: Both players lose \$1 each

GAME 2

c: X wins \$1 and Y loses \$1

GAME 3



Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

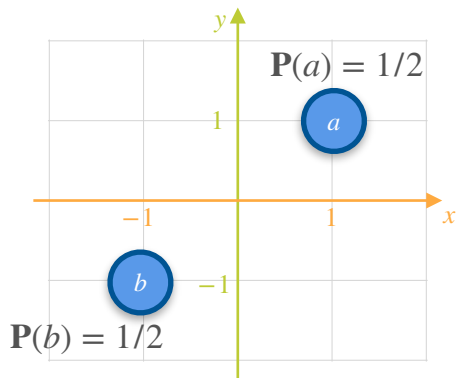
b: Both players lose \$1 each

GAME 2

c: X wins \$1 and Y loses \$1

d: X loses \$1 and Y win \$1

GAME 3



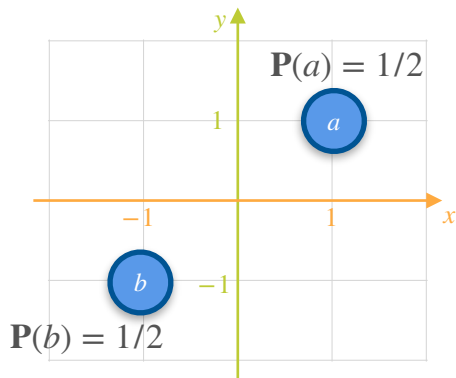
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

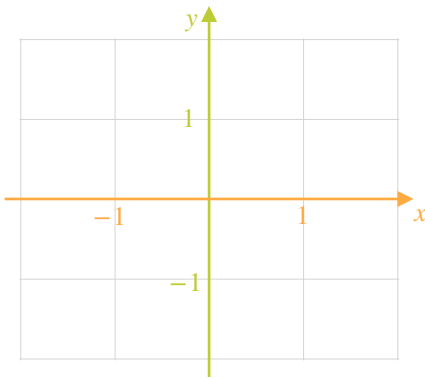
b: Both players lose \$1 each



GAME 2

c: X wins \$1 and Y loses \$1

d: X loses \$1 and Y win \$1



GAME 3

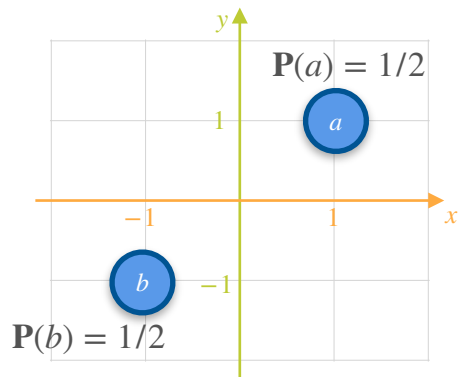
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

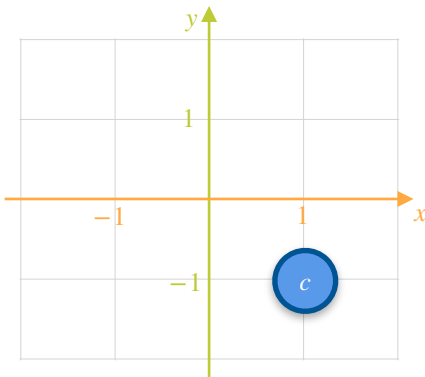
b: Both players lose \$1 each



GAME 2

c: X wins \$1 and Y loses \$1

d: X loses \$1 and Y win \$1



GAME 3

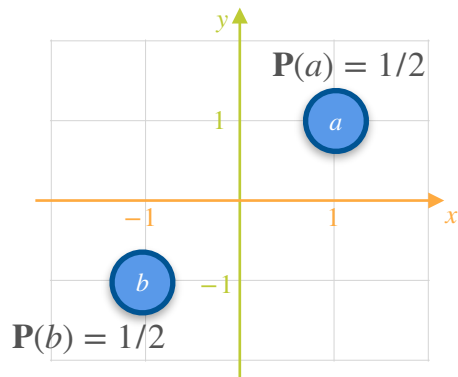
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

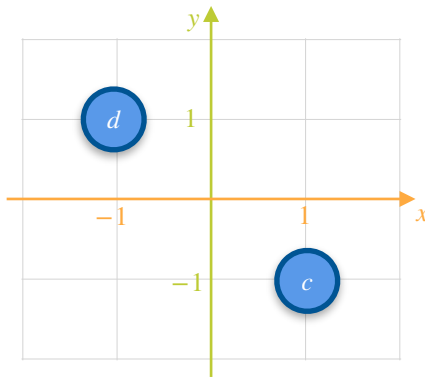
b: Both players lose \$1 each



GAME 2

c: X wins \$1 and Y loses \$1

d: X loses \$1 and Y win \$1



GAME 3

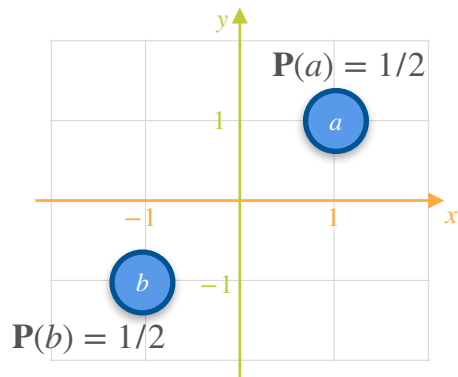
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

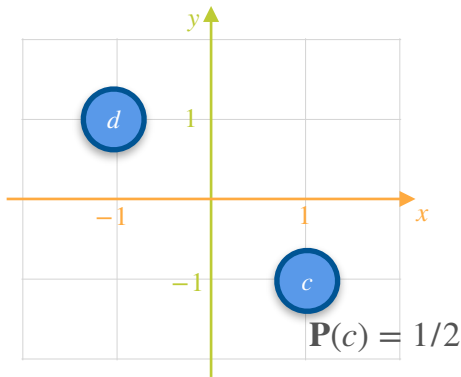
b: Both players lose \$1 each



GAME 2

c: X wins \$1 and Y loses \$1

d: X loses \$1 and Y win \$1



GAME 3

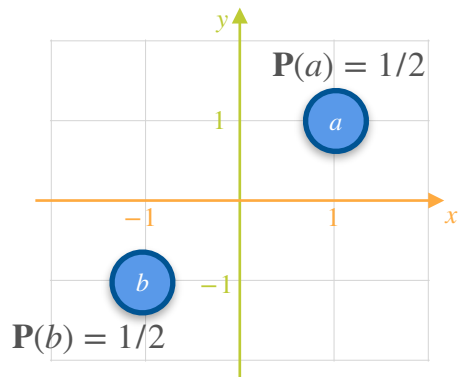
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

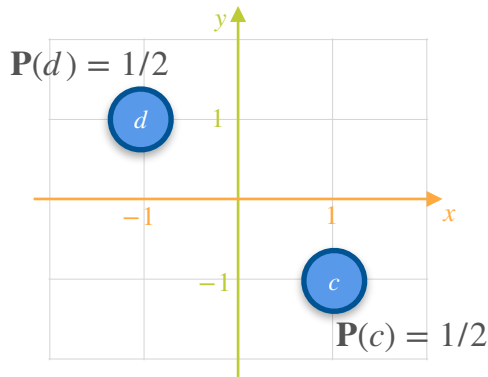
b: Both players lose \$1 each



GAME 2

c: X wins \$1 and Y loses \$1

d: X loses \$1 and Y win \$1



GAME 3

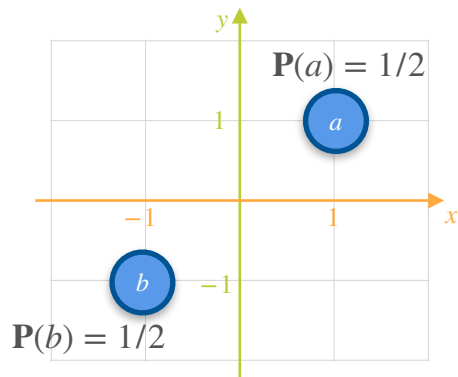
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

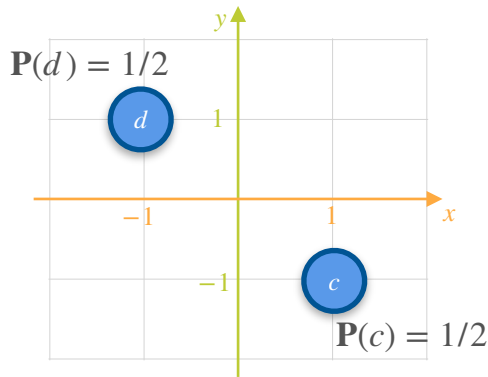
b: Both players lose \$1 each



GAME 2

c: X wins \$1 and Y loses \$1

d: X loses \$1 and Y win \$1



GAME 3

a, *b*, *c* or *d*

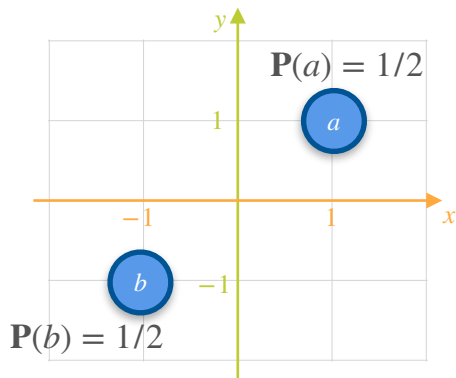
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

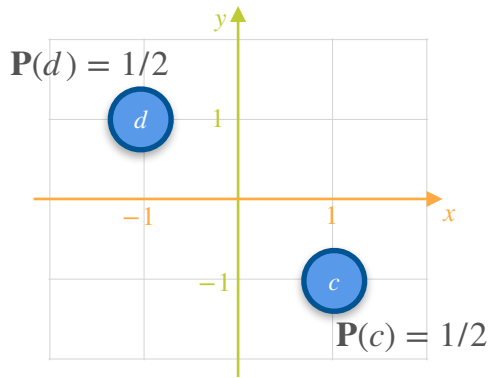
b: Both players lose \$1 each



GAME 2

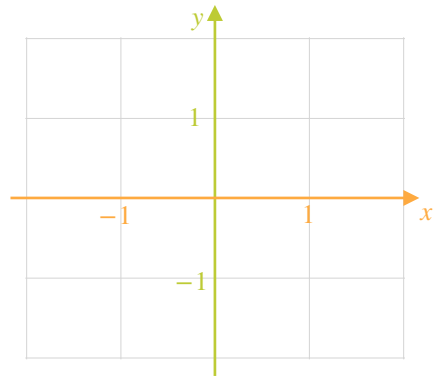
c: X wins \$1 and Y loses \$1

d: X loses \$1 and Y win \$1



GAME 3

a, *b*, *c* or *d*



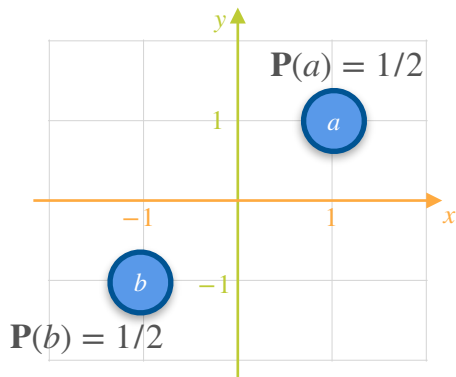
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

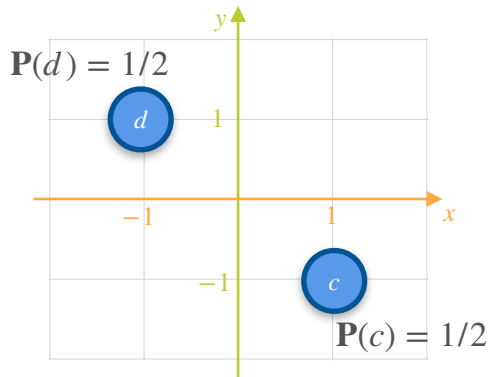
b: Both players lose \$1 each



GAME 2

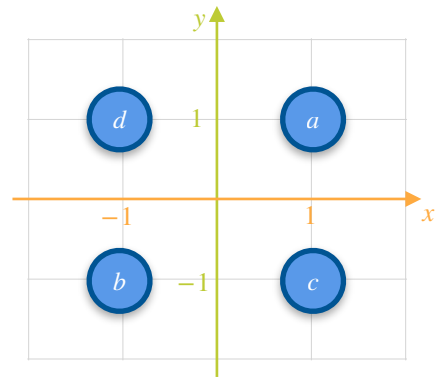
c: X wins \$1 and Y loses \$1

d: X loses \$1 and Y win \$1



GAME 3

a, *b*, *c* or *d*



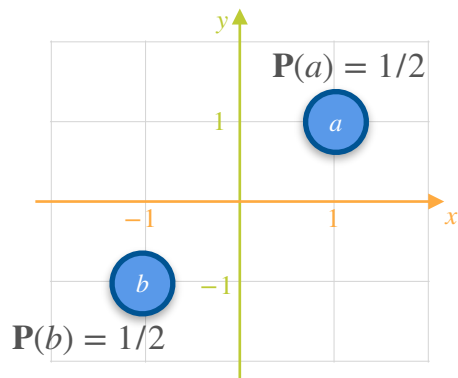
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

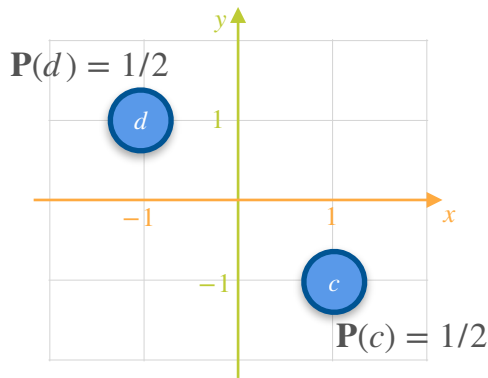
b: Both players lose \$1 each



GAME 2

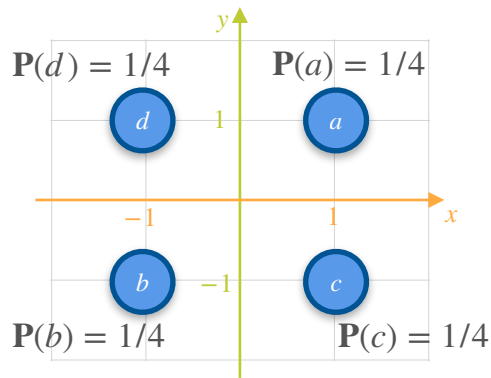
c: X wins \$1 and Y loses \$1

d: X loses \$1 and Y win \$1



GAME 3

a, *b*, *c* or *d*



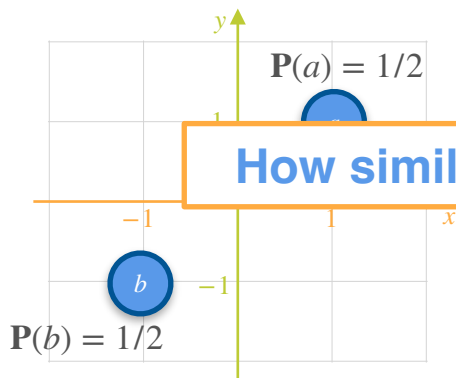
Covariance of a Probability Distribution: Motivation

X and Y are playing 3 games to either win or lose a dollar

GAME 1

a: Both players win \$1 each

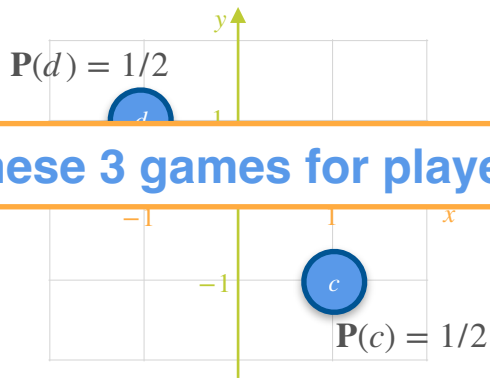
b: Both players lose \$1 each



GAME 2

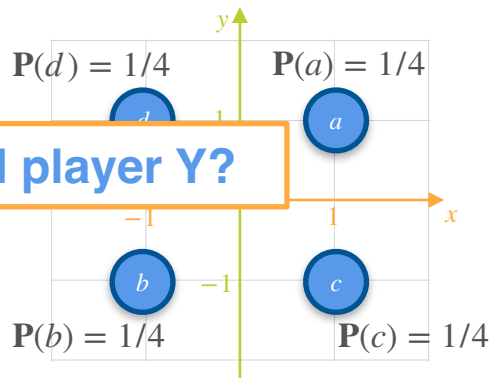
c: X wins \$1 and Y loses \$1

d: X loses \$1 and Y win \$1



GAME 3

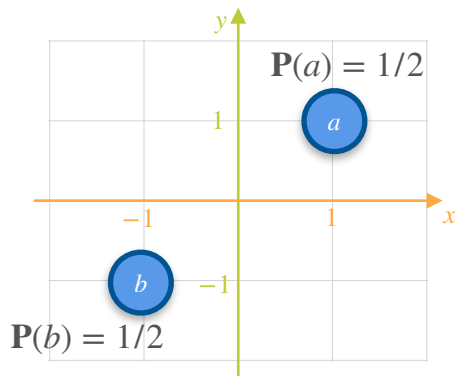
a, *b*, *c* or *d*



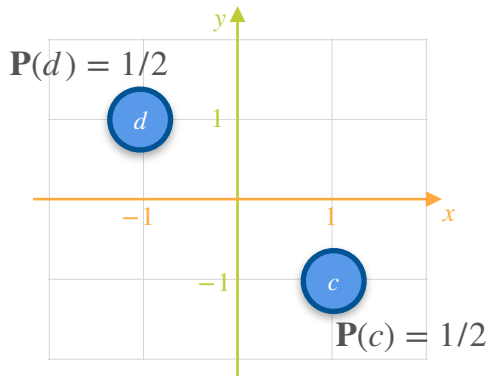
How similar are these 3 games for player X and player Y?

Covariance of a Probability Distribution: Motivation

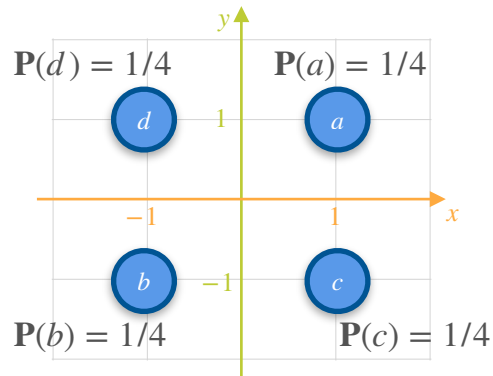
GAME 1



GAME 2



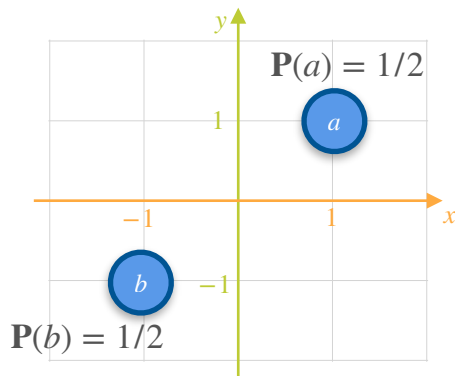
GAME 3



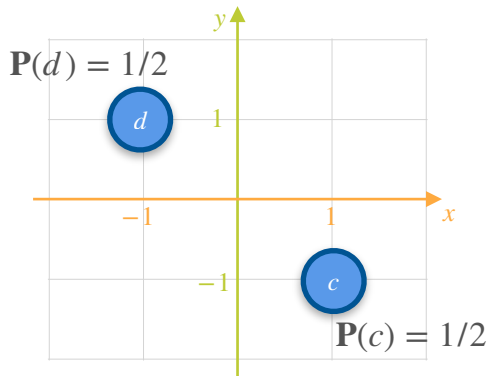
How similar are these 3 games for player X and player Y?

Covariance of a Probability Distribution: Motivation

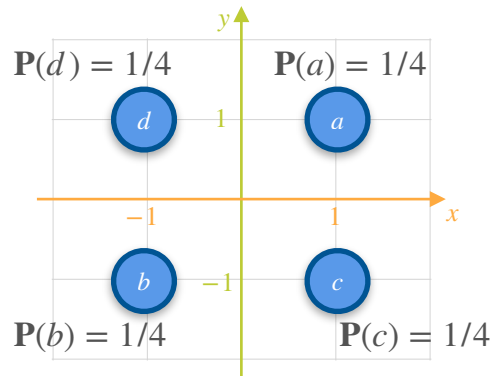
GAME 1



GAME 2



GAME 3

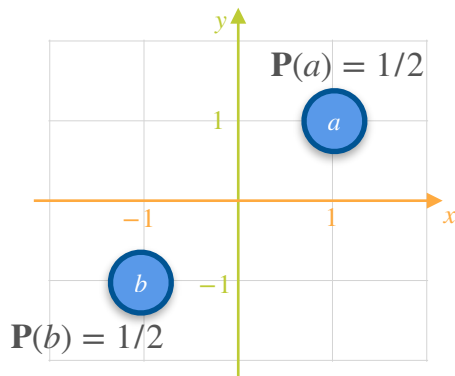


How similar are these 3 games for player X and player Y?

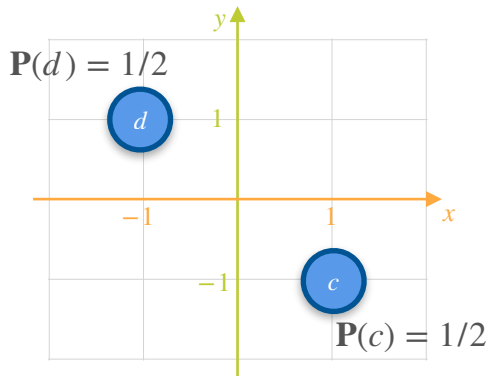
X : how much money in dollars player X wins

Covariance of a Probability Distribution: Motivation

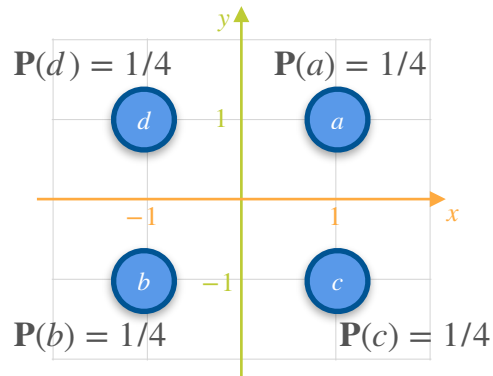
GAME 1



GAME 2



GAME 3



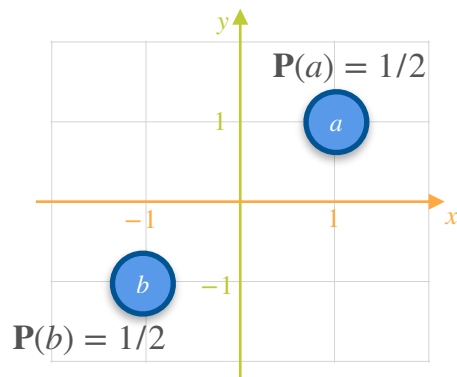
How similar are these 3 games for player X and player Y?

X : how much money in dollars player X wins

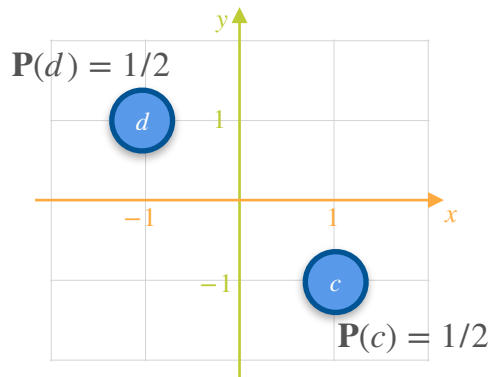
Y : how much money in dollars player Y wins

Covariance of a Probability Distribution: Motivation

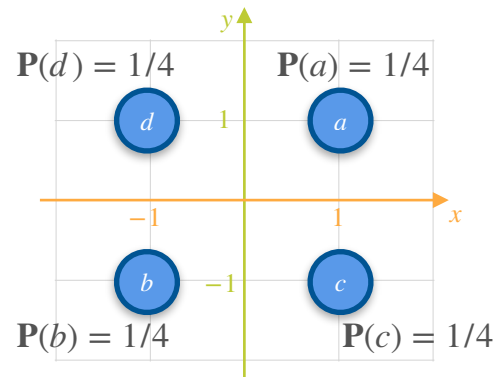
GAME 1



GAME 2

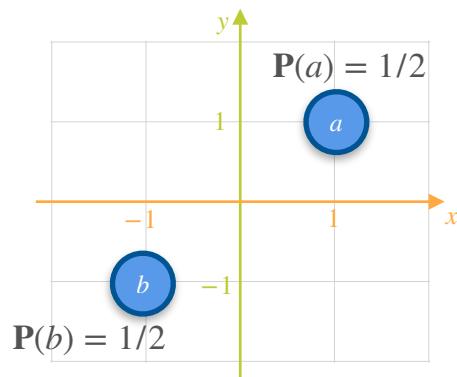


GAME 3

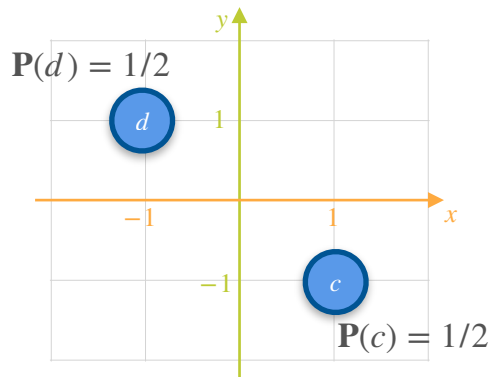


Covariance of a Probability Distribution: Motivation

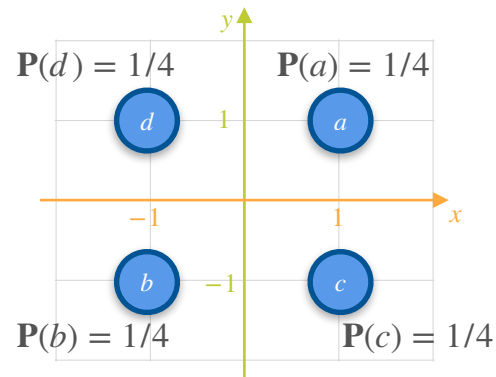
GAME 1



GAME 2



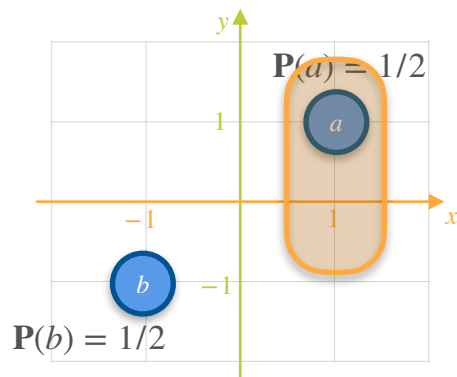
GAME 3



$$\mathbb{E}[X_1] =$$

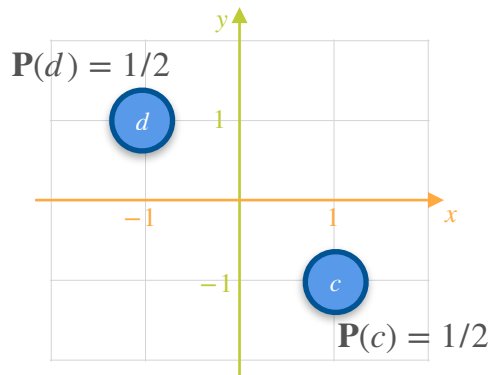
Covariance of a Probability Distribution: Motivation

GAME 1

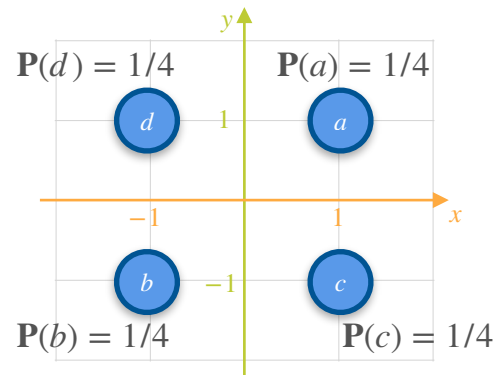


$$\mathbb{E}[X_1] = \frac{1}{2}(1)$$

GAME 2

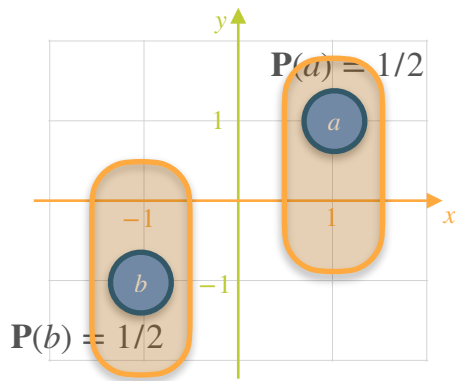


GAME 3



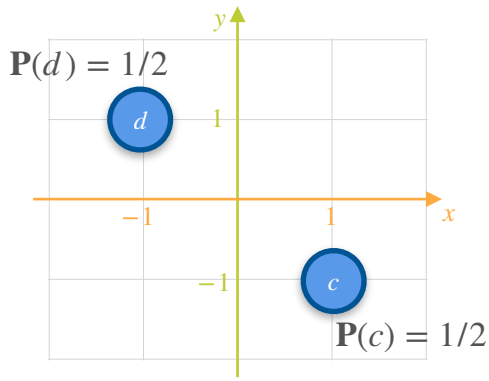
Covariance of a Probability Distribution: Motivation

GAME 1

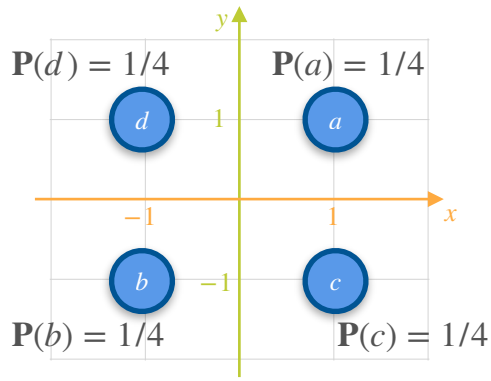


$$\mathbb{E}[X_1] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0$$

GAME 2

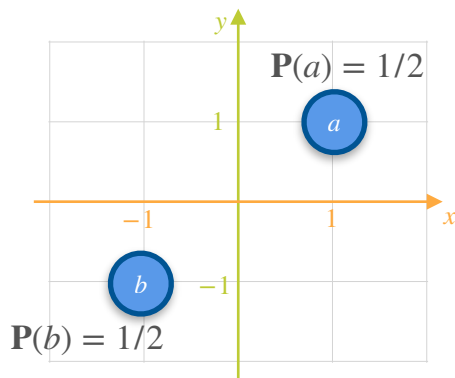


GAME 3



Covariance of a Probability Distribution: Motivation

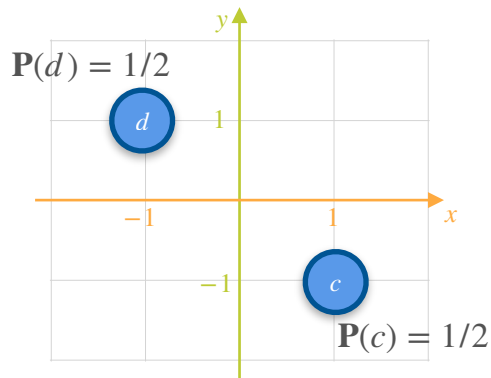
GAME 1



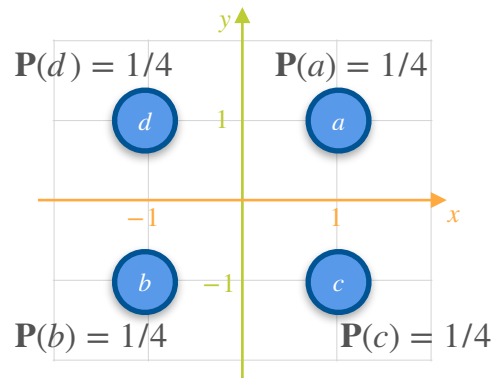
$$\mathbb{E}[X_1] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0$$

$$\mathbb{E}[Y_1] =$$

GAME 2

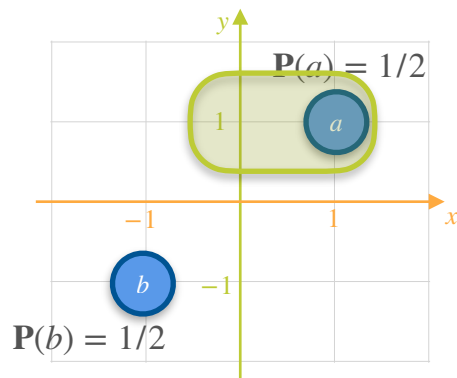


GAME 3

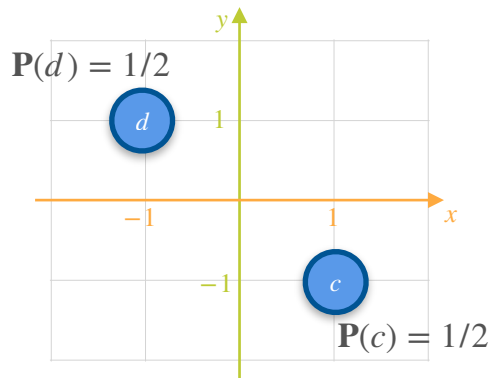


Covariance of a Probability Distribution: Motivation

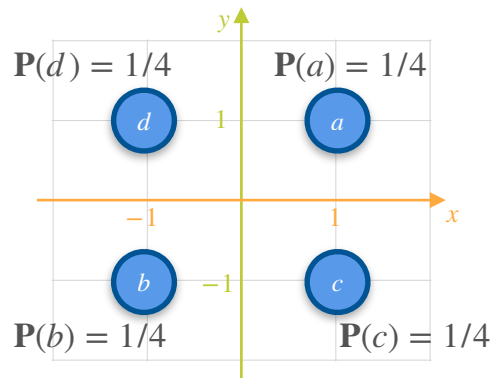
GAME 1



GAME 2



GAME 3

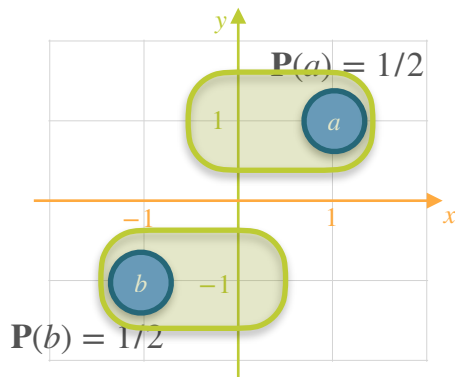


$$\mathbb{E}[X_1] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0$$

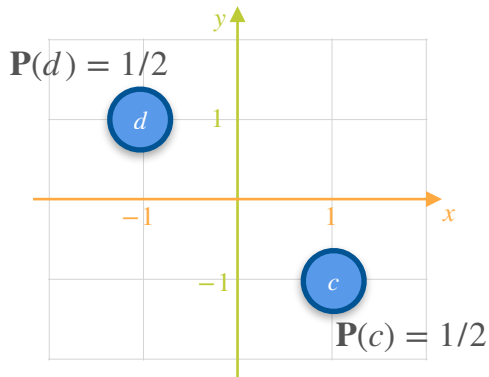
$$\mathbb{E}[Y_1] = \frac{1}{2}(1)$$

Covariance of a Probability Distribution: Motivation

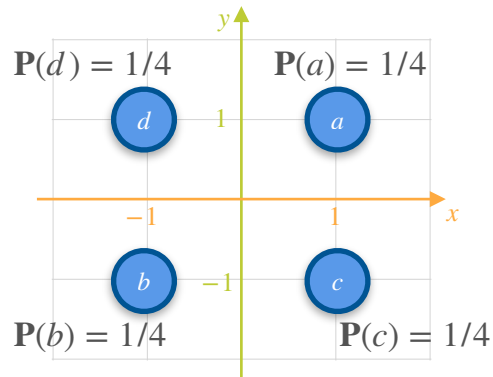
GAME 1



GAME 2



GAME 3

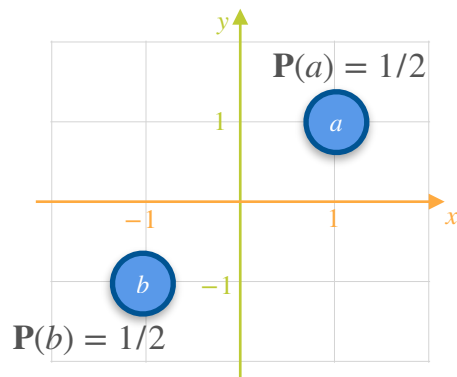


$$\mathbb{E}[X_1] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0$$

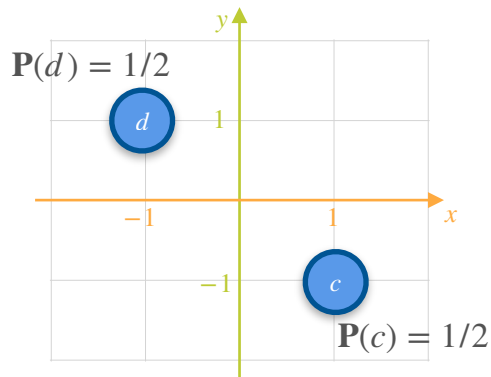
$$\mathbb{E}[Y_1] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0$$

Covariance of a Probability Distribution: Motivation

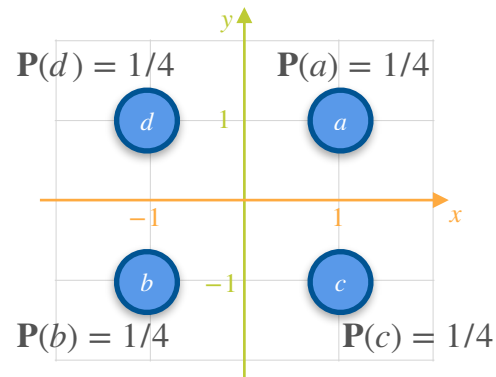
GAME 1



GAME 2



GAME 3

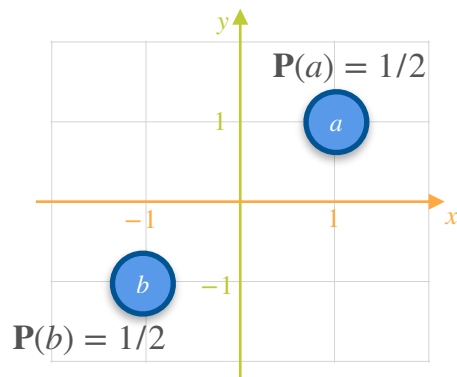


$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[Y_1] = 0$$

Covariance of a Probability Distribution: Motivation

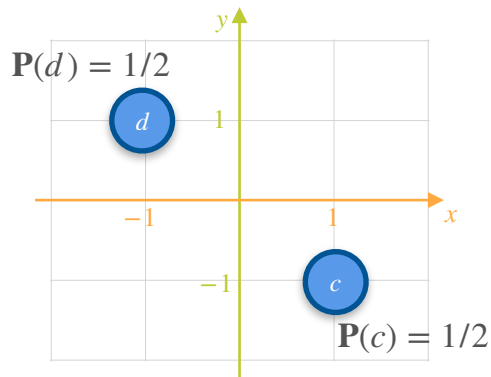
GAME 1



$$\mathbb{E}[X_1] = 0$$

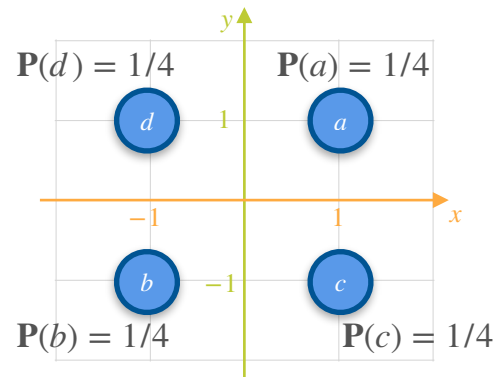
$$\mathbb{E}[Y_1] = 0$$

GAME 2



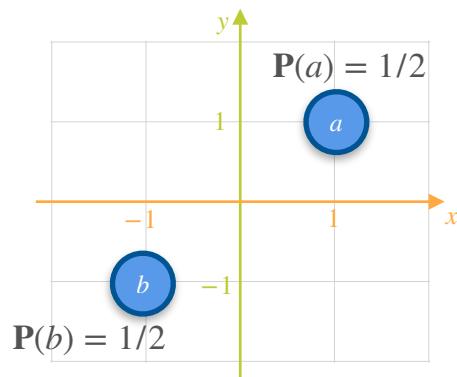
$$\mathbb{E}[X_2] =$$

GAME 3



Covariance of a Probability Distribution: Motivation

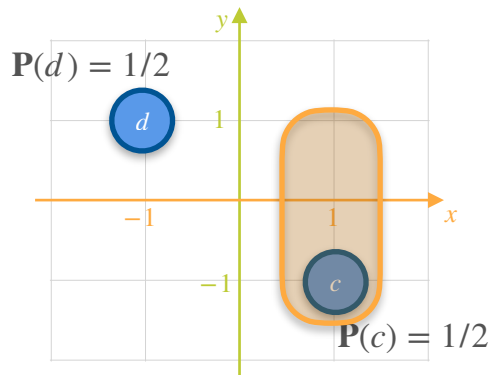
GAME 1



$$\mathbb{E}[X_1] = 0$$

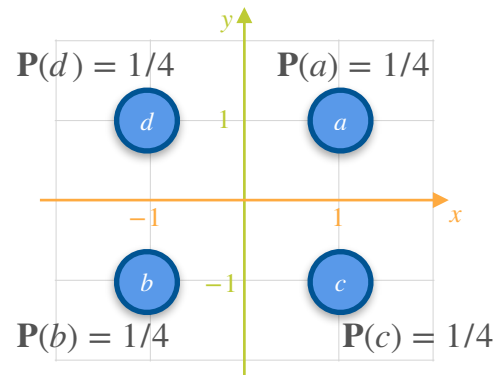
$$\mathbb{E}[Y_1] = 0$$

GAME 2



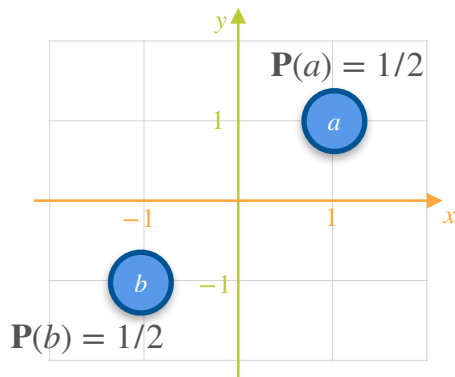
$$\mathbb{E}[X_2] = \frac{1}{2}(1)$$

GAME 3



Covariance of a Probability Distribution: Motivation

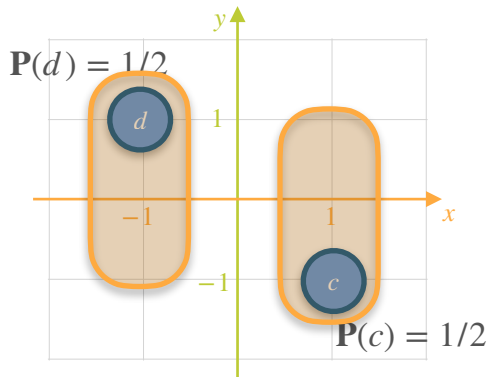
GAME 1



$$\mathbb{E}[X_1] = 0$$

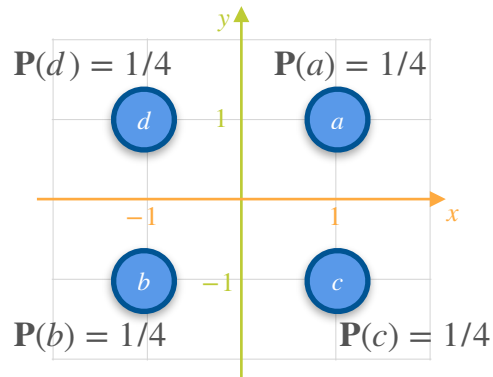
$$\mathbb{E}[Y_1] = 0$$

GAME 2



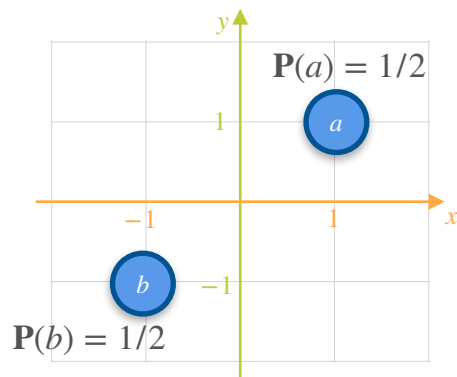
$$\mathbb{E}[X_2] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0$$

GAME 3



Covariance of a Probability Distribution: Motivation

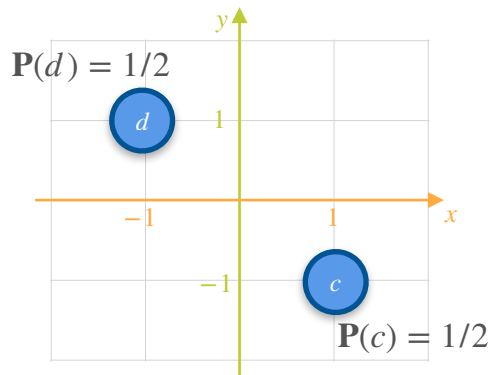
GAME 1



$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[Y_1] = 0$$

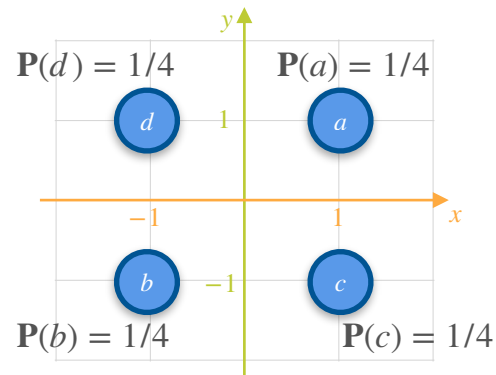
GAME 2



$$\mathbb{E}[X_2] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0$$

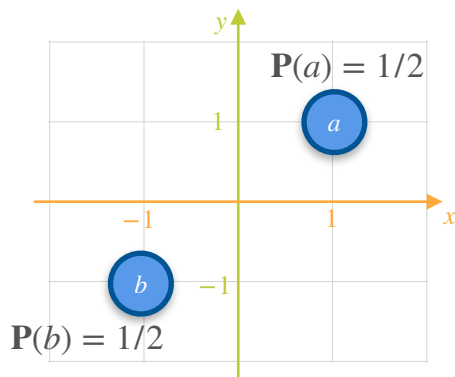
$$\mathbb{E}[Y_2] =$$

GAME 3



Covariance of a Probability Distribution: Motivation

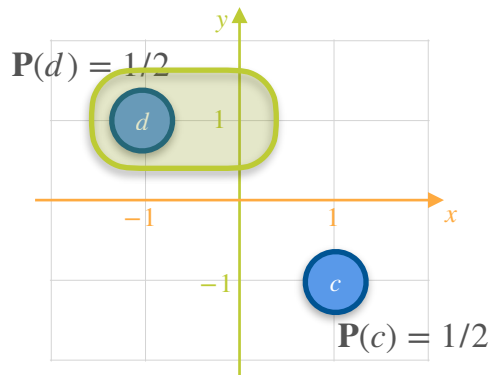
GAME 1



$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[Y_1] = 0$$

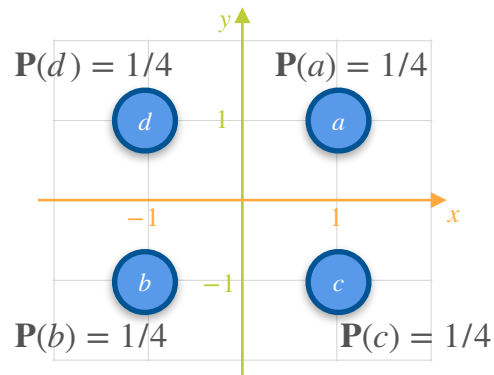
GAME 2



$$\mathbb{E}[X_2] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0$$

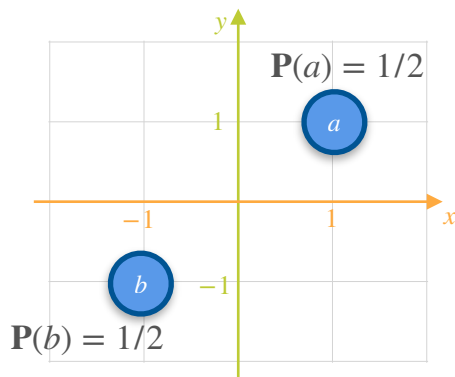
$$\mathbb{E}[Y_2] = \frac{1}{2}(1)$$

GAME 3



Covariance of a Probability Distribution: Motivation

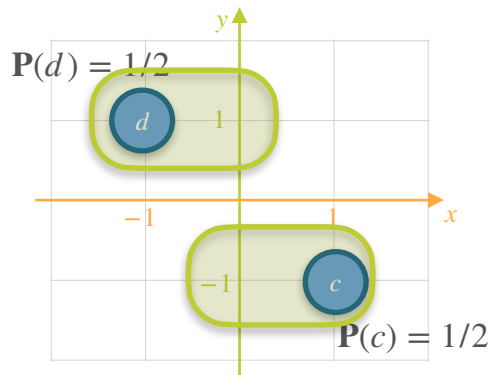
GAME 1



$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[Y_1] = 0$$

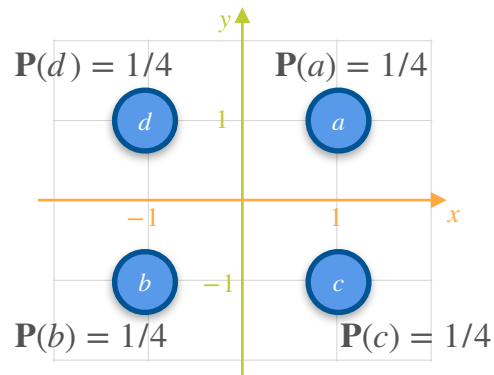
GAME 2



$$\mathbb{E}[X_2] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0$$

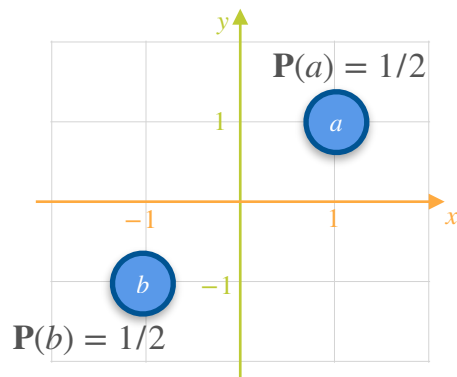
$$\mathbb{E}[Y_2] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0$$

GAME 3



Covariance of a Probability Distribution: Motivation

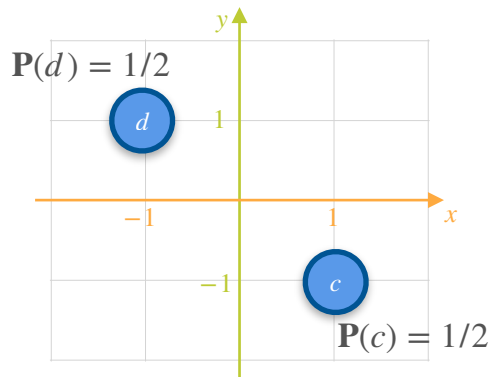
GAME 1



$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[Y_1] = 0$$

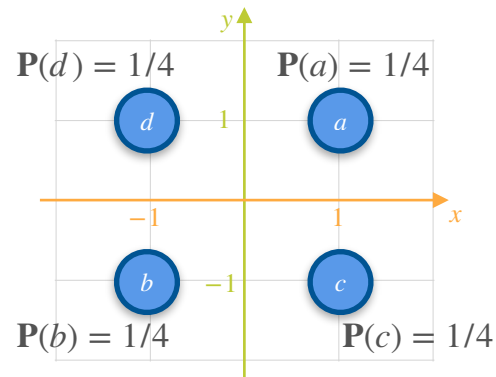
GAME 2



$$\mathbb{E}[X_2] = 0$$

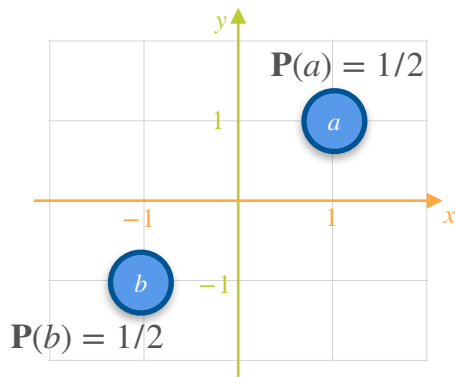
$$\mathbb{E}[Y_2] = 0$$

GAME 3



Covariance of a Probability Distribution: Motivation

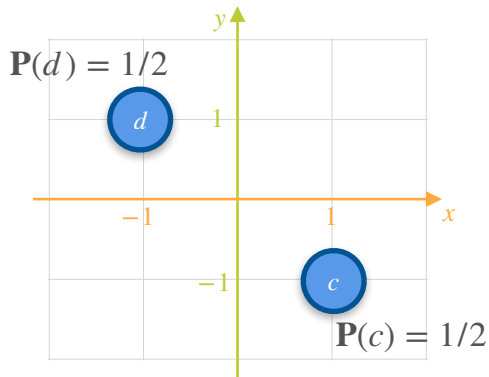
GAME 1



$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[Y_1] = 0$$

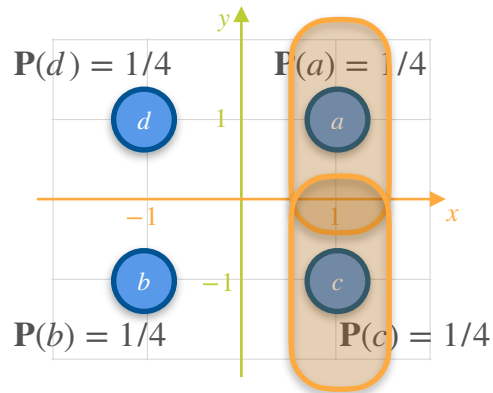
GAME 2



$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[Y_2] = 0$$

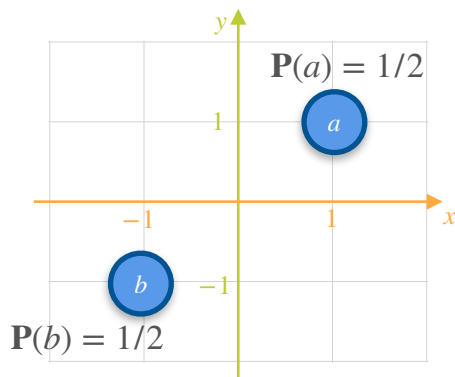
GAME 3



$$\mathbb{E}[X_3] = 2\left(\frac{1}{4}(1)\right)$$

Covariance of a Probability Distribution: Motivation

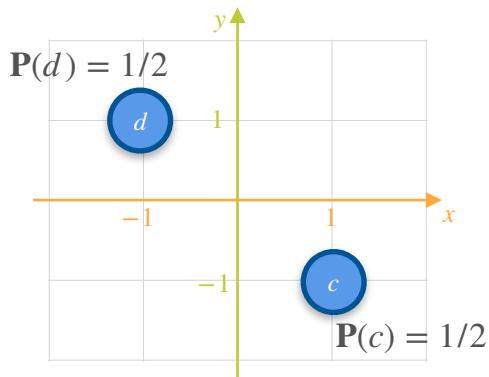
GAME 1



$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[Y_1] = 0$$

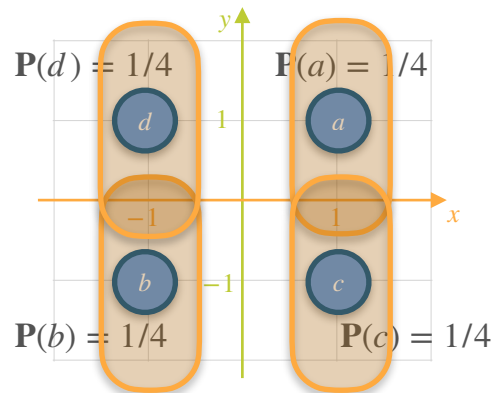
GAME 2



$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[Y_2] = 0$$

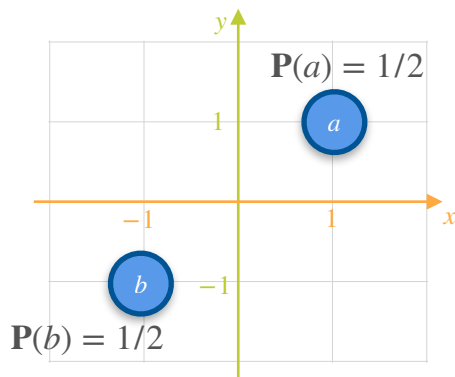
GAME 3



$$\mathbb{E}[X_3] = 2\left(\frac{1}{4}(1)\right) + 2\left(\frac{1}{4}(-1)\right) = 0$$

Covariance of a Probability Distribution: Motivation

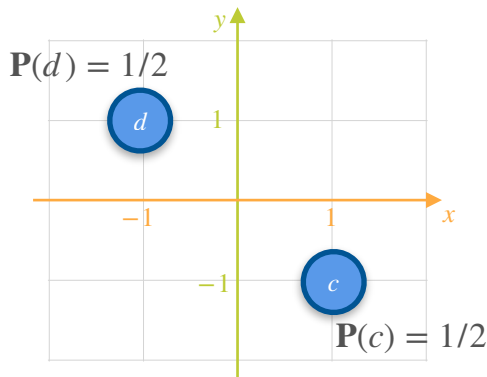
GAME 1



$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[Y_1] = 0$$

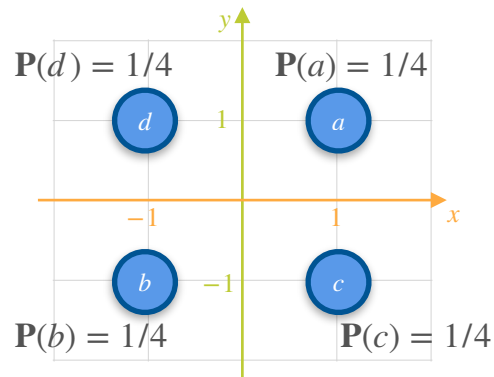
GAME 2



$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[Y_2] = 0$$

GAME 3

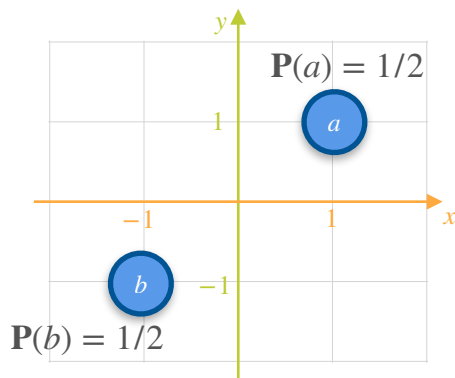


$$\mathbb{E}[X_3] = 2\left(\frac{1}{4}(1)\right) + 2\left(\frac{1}{4}(-1)\right) = 0$$

$$\mathbb{E}[Y_3] =$$

Covariance of a Probability Distribution: Motivation

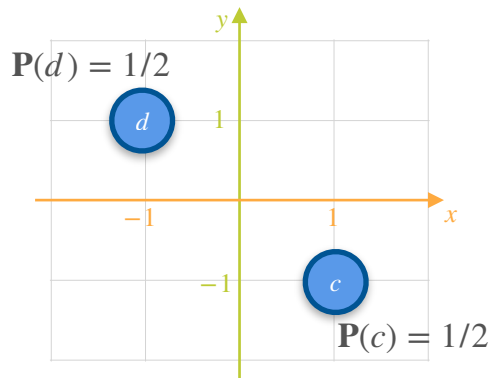
GAME 1



$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[Y_1] = 0$$

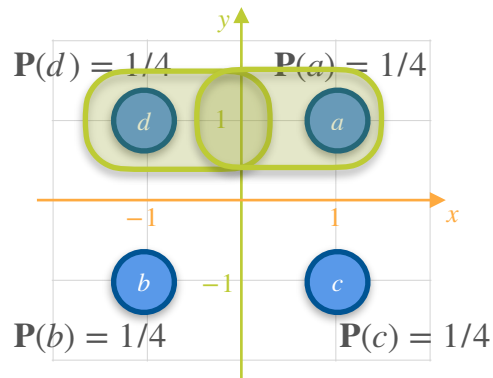
GAME 2



$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[Y_2] = 0$$

GAME 3

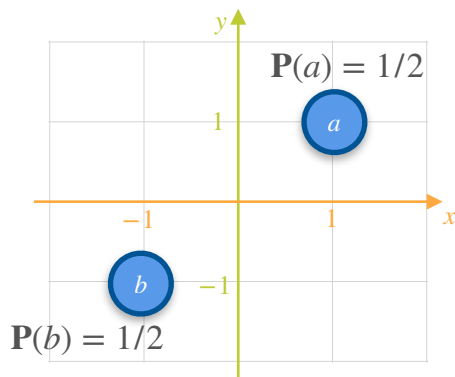


$$\mathbb{E}[X_3] = 2\left(\frac{1}{4}(1)\right) + 2\left(\frac{1}{4}(-1)\right) = 0$$

$$\mathbb{E}[Y_3] = 2\left(\frac{1}{4}(1)\right)$$

Covariance of a Probability Distribution: Motivation

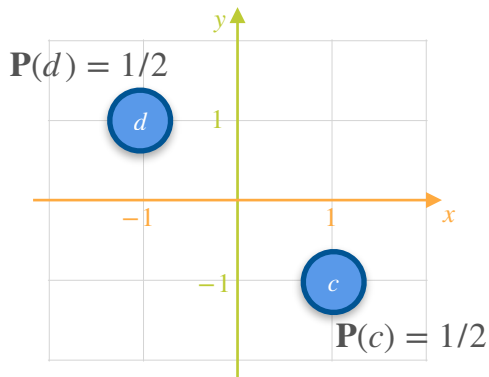
GAME 1



$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[Y_1] = 0$$

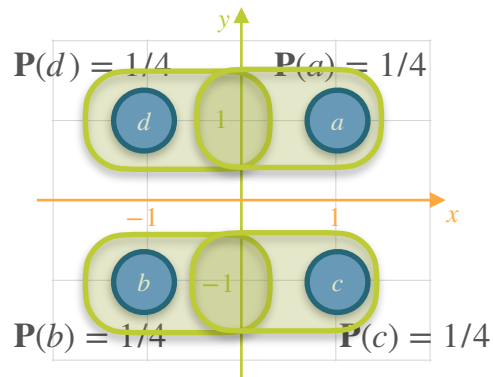
GAME 2



$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[Y_2] = 0$$

GAME 3

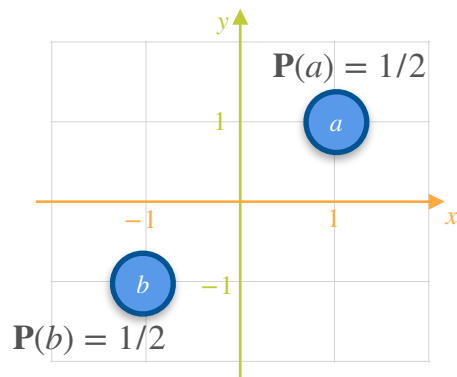


$$\mathbb{E}[X_3] = 2\left(\frac{1}{4}(1)\right) + 2\left(\frac{1}{4}(-1)\right) = 0$$

$$\mathbb{E}[Y_3] = 2\left(\frac{1}{4}(1)\right) + 2\left(\frac{1}{4}(-1)\right) = 0$$

Covariance of a Probability Distribution: Motivation

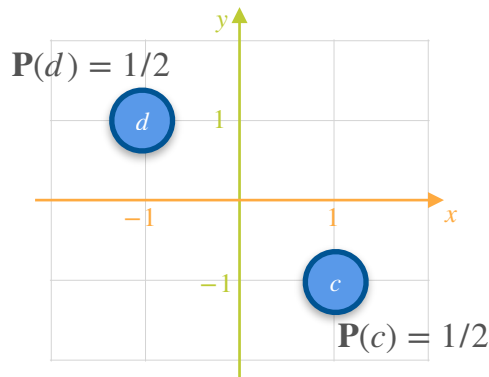
GAME 1



$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[Y_1] = 0$$

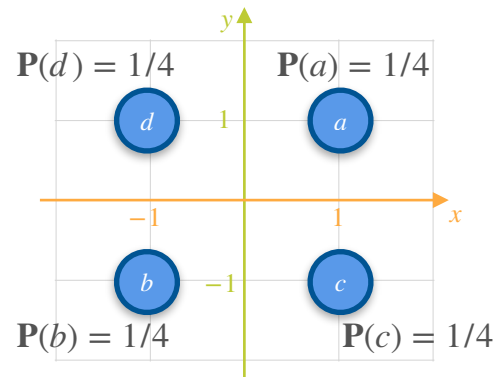
GAME 2



$$\mathbb{E}[X_2] = 0$$

$$\mathbb{E}[Y_2] = 0$$

GAME 3

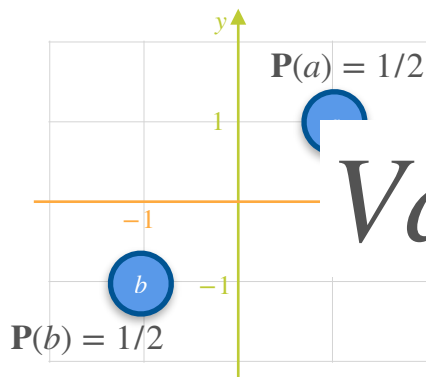


$$\mathbb{E}[X_3] = 0$$

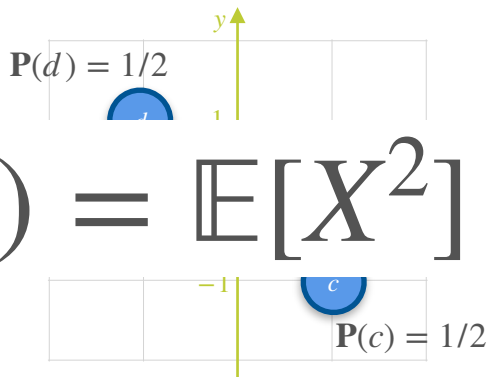
$$\mathbb{E}[Y_3] = 0$$

Covariance of a Probability Distribution: Motivation

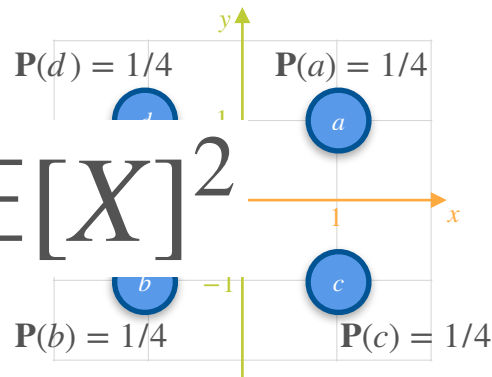
GAME 1



GAME 2



GAME 3



$$Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\mathbb{E}[X_1] = 0$$

$$\mathbb{E}[Y_1] = 0$$

$$\mathbb{E}[X_2] = 0$$

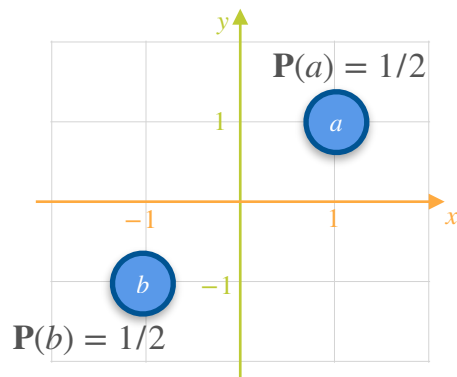
$$\mathbb{E}[Y_2] = 0$$

$$\mathbb{E}[X_3] = 0$$

$$\mathbb{E}[Y_3] = 0$$

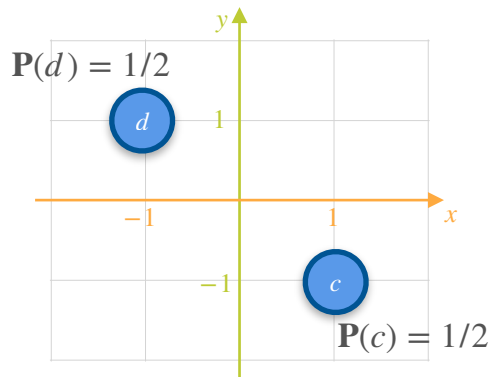
Covariance of a Probability Distribution: Motivation

GAME 1



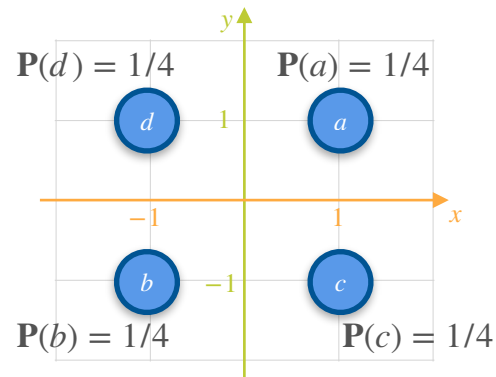
$$\mathbb{E}[X_1] = 0 \quad \mathbb{E}[Y_1] = 0$$

GAME 2



$$\mathbb{E}[X_2] = 0 \quad \mathbb{E}[Y_2] = 0$$

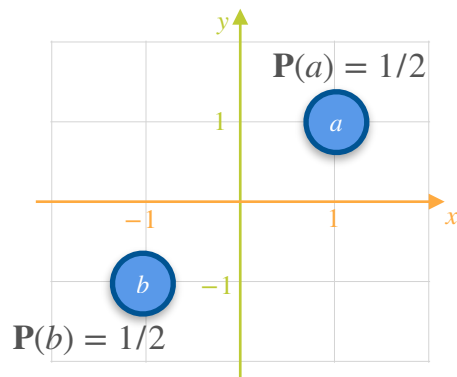
GAME 3



$$\mathbb{E}[X_3] = 0 \quad \mathbb{E}[Y_3] = 0$$

Covariance of a Probability Distribution: Motivation

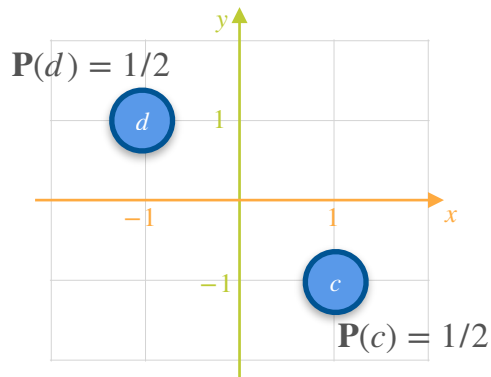
GAME 1



$$\mathbb{E}[X_1] = 0 \quad \mathbb{E}[Y_1] = 0$$

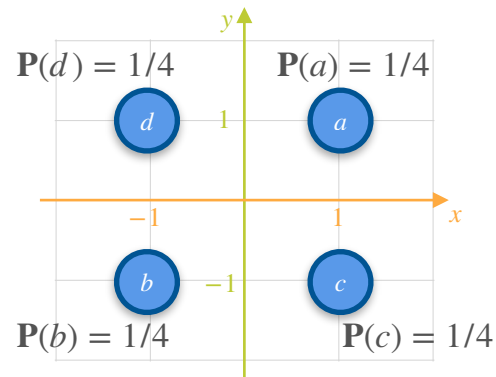
$$\text{Var}(X_1) =$$

GAME 2



$$\mathbb{E}[X_2] = 0 \quad \mathbb{E}[Y_2] = 0$$

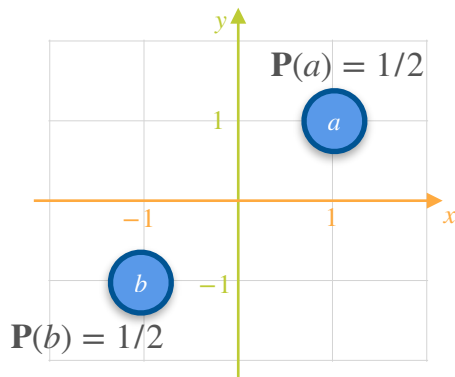
GAME 3



$$\mathbb{E}[X_3] = 0 \quad \mathbb{E}[Y_3] = 0$$

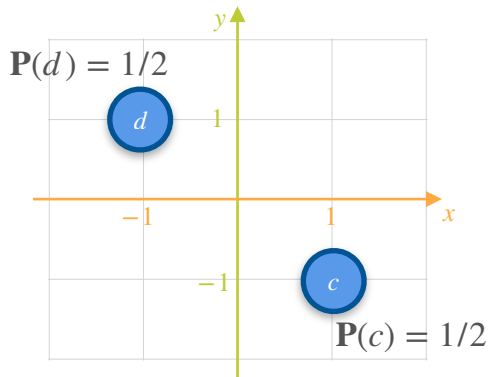
Covariance of a Probability Distribution: Motivation

GAME 1



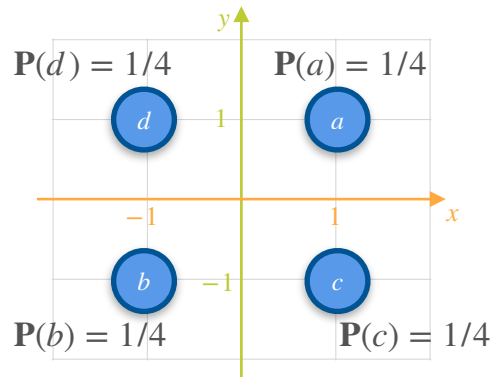
$$\mathbb{E}[X_1] = 0 \quad \mathbb{E}[Y_1] = 0$$

GAME 2



$$\mathbb{E}[X_2] = 0 \quad \mathbb{E}[Y_2] = 0$$

GAME 3

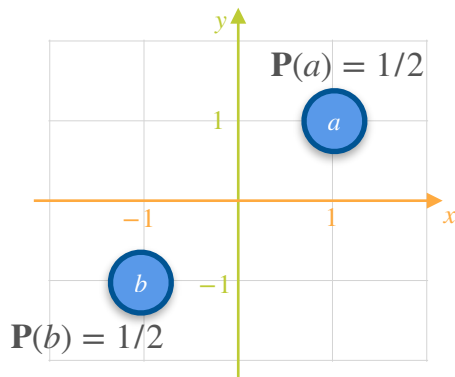


$$\mathbb{E}[X_3] = 0 \quad \mathbb{E}[Y_3] = 0$$

$$\text{Var}(X_1) = \mathbb{E}[X_1^2] - E[X_1]^2$$

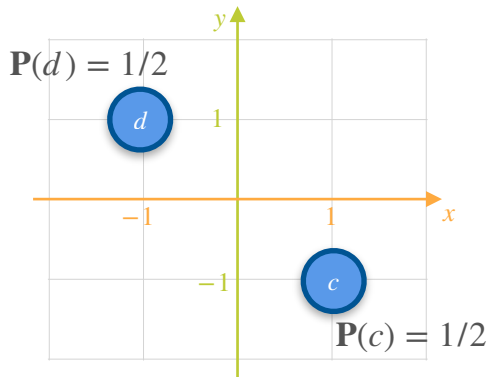
Covariance of a Probability Distribution: Motivation

GAME 1



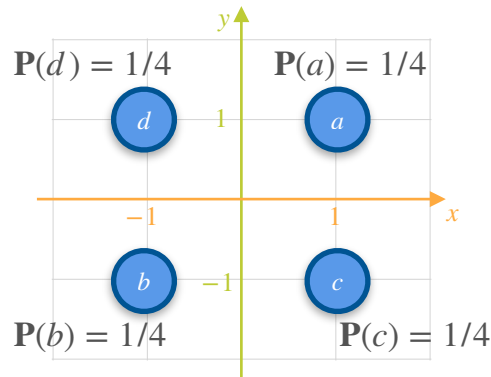
$$\mathbb{E}[X_1] = 0 \quad \mathbb{E}[Y_1] = 0$$

GAME 2



$$\mathbb{E}[X_2] = 0 \quad \mathbb{E}[Y_2] = 0$$

GAME 3

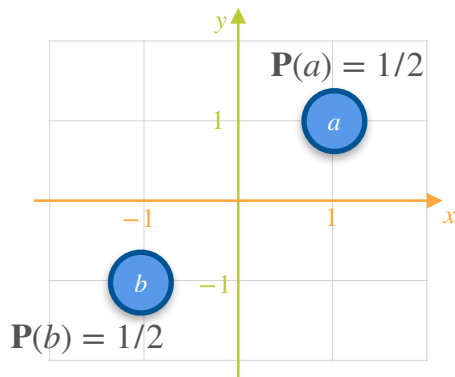


$$\mathbb{E}[X_3] = 0 \quad \mathbb{E}[Y_3] = 0$$

$$\text{Var}(X_1) = \mathbb{E}[X_1^2] - E[X_1]^2 = \frac{1}{2}(-1)^2 + \frac{1}{2}(1)^2 - 0^2 =$$

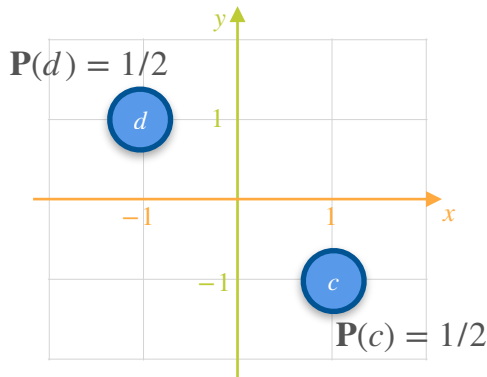
Covariance of a Probability Distribution: Motivation

GAME 1



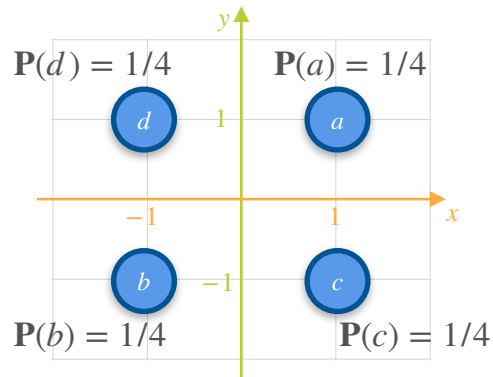
$$\mathbb{E}[X_1] = 0 \quad \mathbb{E}[Y_1] = 0$$

GAME 2



$$\mathbb{E}[X_2] = 0 \quad \mathbb{E}[Y_2] = 0$$

GAME 3

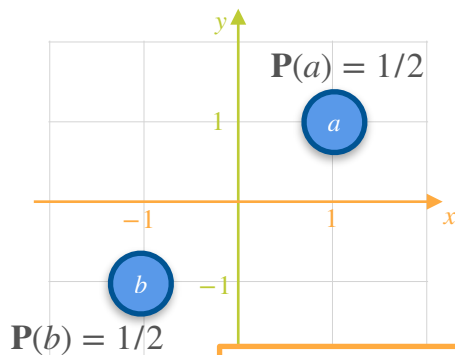


$$\mathbb{E}[X_3] = 0 \quad \mathbb{E}[Y_3] = 0$$

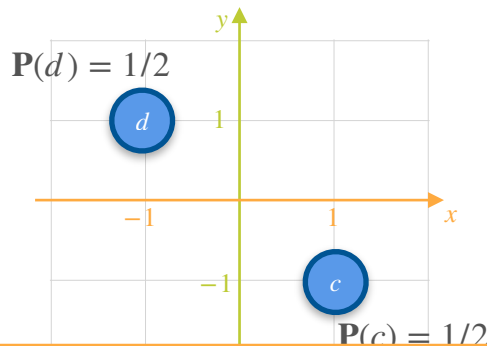
$$\text{Var}(X_1) = \mathbb{E}[X_1^2] - E[X_1]^2 = \frac{1}{2}(-1)^2 + \frac{1}{2}(1)^2 - 0^2 = 1$$

Covariance of a Probability Distribution: Motivation

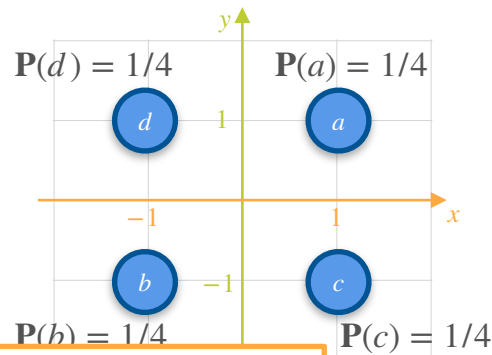
GAME 1



GAME 2



GAME 3



How similar are these 3 games for player X and player Y?

$$\mathbb{E}[X_1] = 0 \quad \mathbb{E}[Y_1] = 0$$

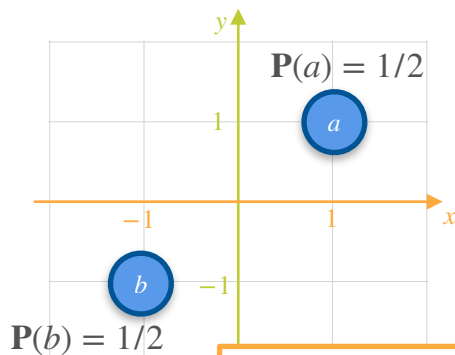
$$\mathbb{E}[X_2] = 0 \quad \mathbb{E}[Y_2] = 0$$

$$\mathbb{E}[X_3] = 0 \quad \mathbb{E}[Y_3] = 0$$

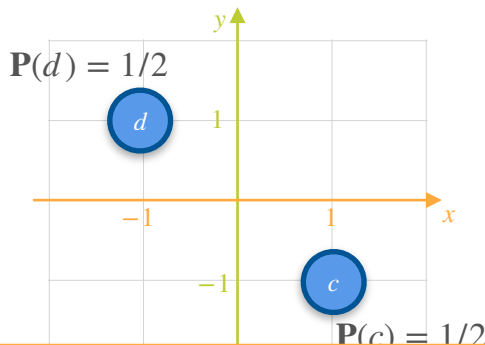
$$\text{Var}(X_1) = 1$$

Covariance of a Probability Distribution: Motivation

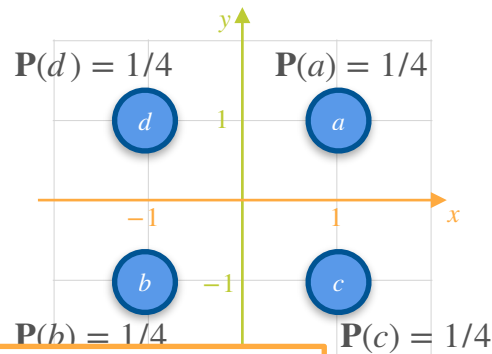
GAME 1



GAME 2



GAME 3



How similar are these 3 games for player X and player Y?

$$\mathbb{E}[X_1] = 0 \quad \mathbb{E}[Y_1] = 0$$

$$\text{Var}(X_1) = 1$$

$$\text{Var}(Y_1) = 1$$

$$\mathbb{E}[X_2] = 0 \quad \mathbb{E}[Y_2] = 0$$

$$\text{Var}(X_2) = 1$$

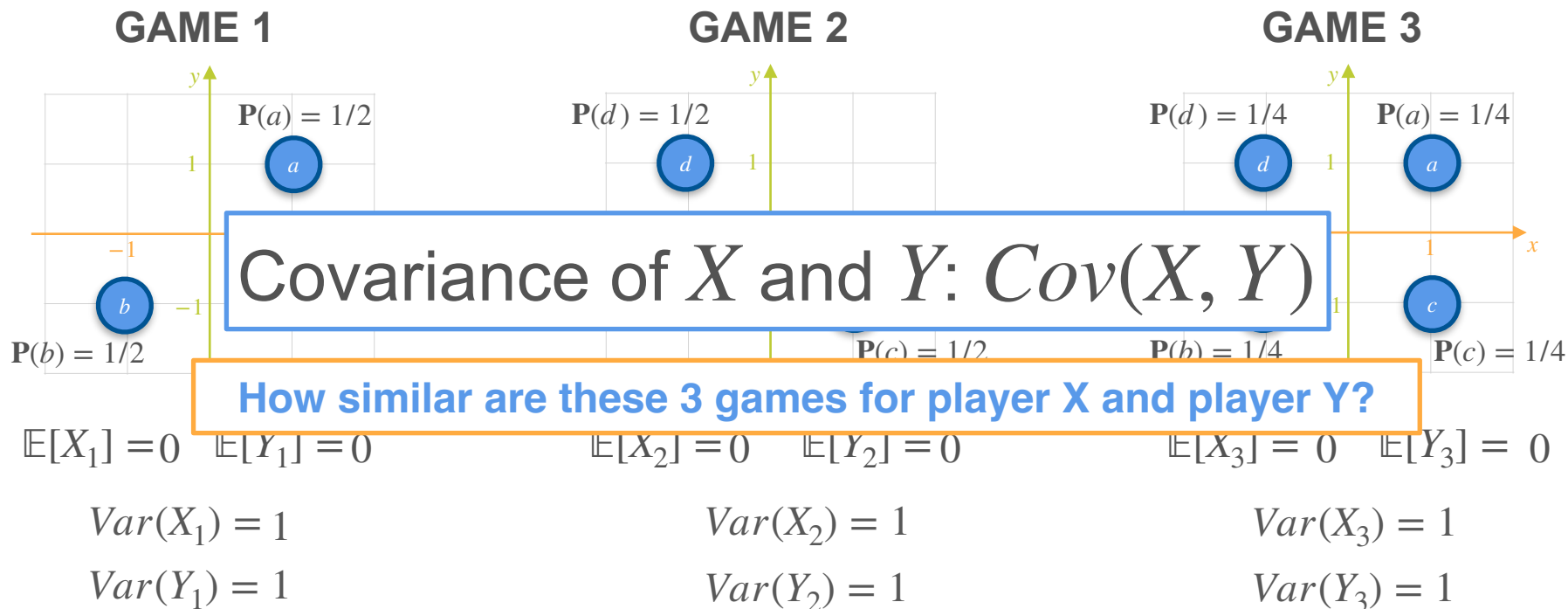
$$\text{Var}(Y_2) = 1$$

$$\mathbb{E}[X_3] = 0 \quad \mathbb{E}[Y_3] = 0$$

$$\text{Var}(X_3) = 1$$

$$\text{Var}(Y_3) = 1$$

Covariance of a Probability Distribution: Motivation



Covariance of a Probability Distribution: Motivation

Covariance of a Probability Distribution: Motivation

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

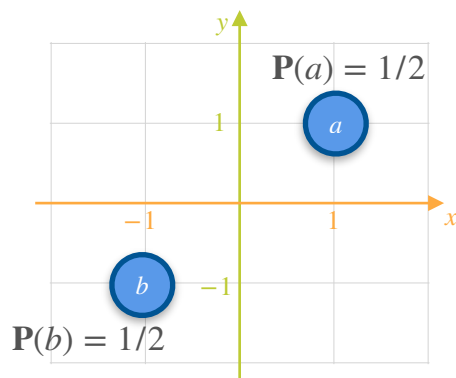
Covariance of a Probability Distribution: Motivation

Covariance of X and Y : $Cov(X, Y)$

$$Cov(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

Covariance of a Probability Distribution: Motivation

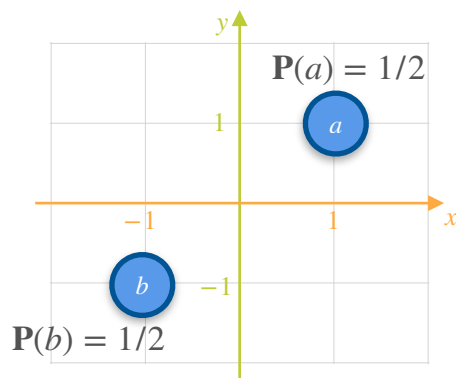
GAME 1



$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

Covariance of a Probability Distribution: Motivation

GAME 1

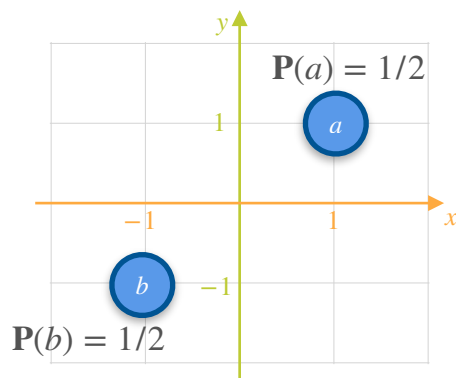


$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
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Covariance of a Probability Distribution: Motivation

GAME 1

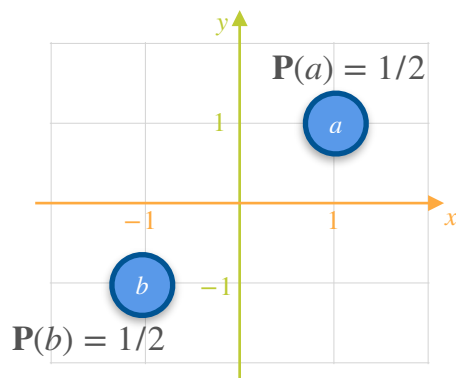


$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1				
-1				

Covariance of a Probability Distribution: Motivation

GAME 1

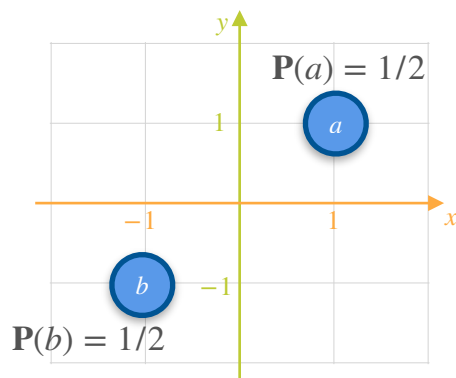


$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1			
-1	-1			

Covariance of a Probability Distribution: Motivation

GAME 1

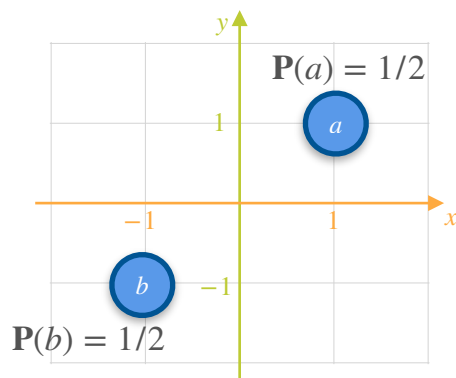


$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

Covariance of a Probability Distribution: Motivation

GAME 1

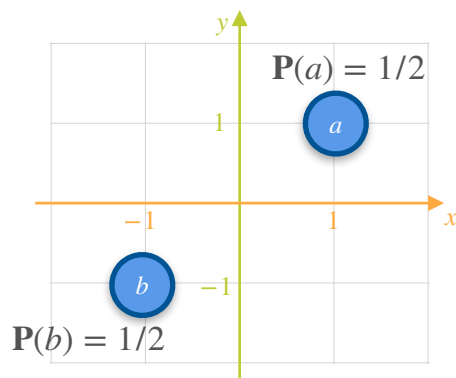


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1	1	1	1	1
-1	-1	-1	-1	1

Covariance of a Probability Distribution: Motivation

GAME 1

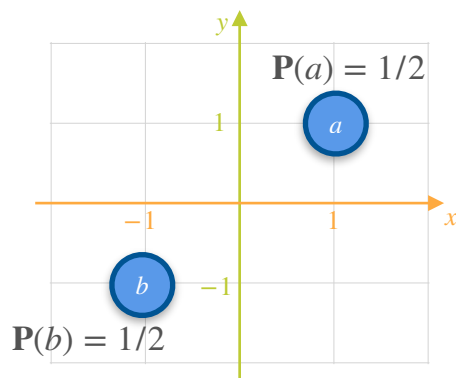


$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

Covariance of a Probability Distribution: Motivation

GAME 1



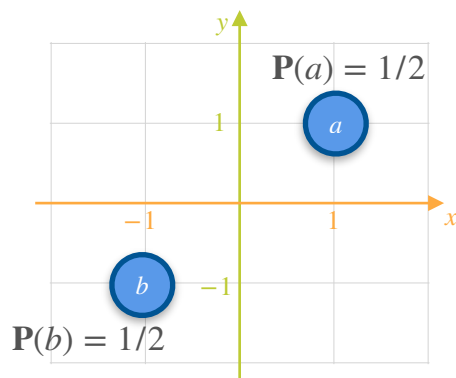
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x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

$$\sum (x_i - \mu_x)(y_i - \mu_y) =$$

Covariance of a Probability Distribution: Motivation

GAME 1



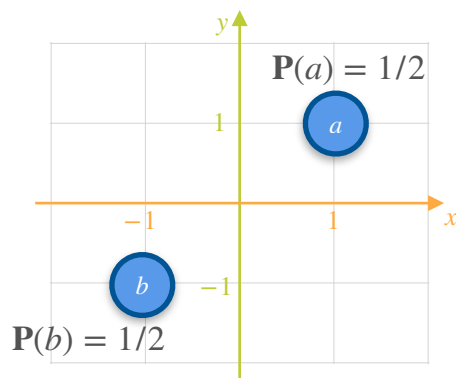
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

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1	1	1	1	1
-1	-1	-1	-1	1

$$\sum (x_i - \mu_x)(y_i - \mu_y) =$$

Covariance of a Probability Distribution: Motivation

GAME 1



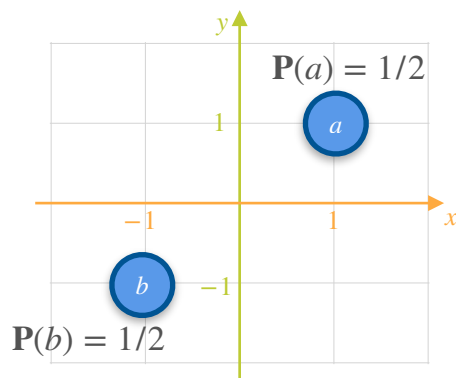
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

$$\sum (x_i - \mu_x)(y_i - \mu_y) = 2$$

Covariance of a Probability Distribution: Motivation

GAME 1



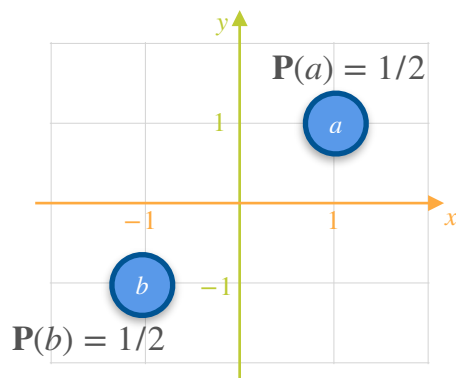
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

$$\text{Cov}(X, Y) = \sum (x_i - \mu_x)(y_i - \mu_y) = 2$$

Covariance of a Probability Distribution: Motivation

GAME 1



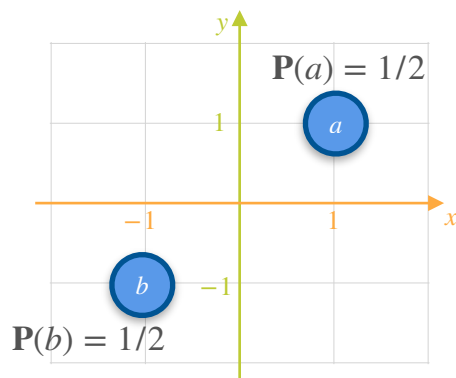
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = 2$$

Covariance of a Probability Distribution: Motivation

GAME 1



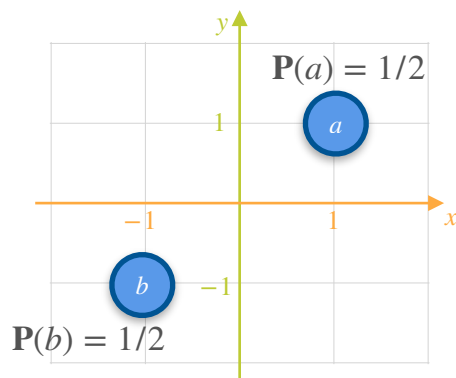
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \frac{2}{2}$$

Covariance of a Probability Distribution: Motivation

GAME 1



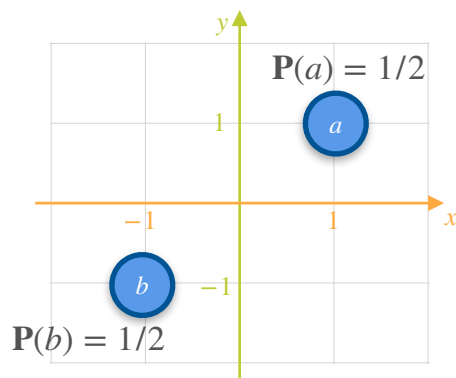
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \frac{2}{2} = 1$$

Covariance of a Probability Distribution: Motivation

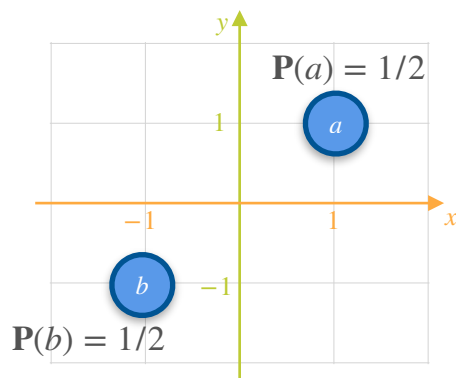
GAME 1



$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

Covariance of a Probability Distribution: Motivation

GAME 1

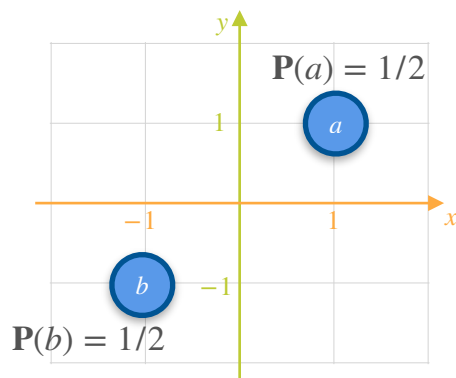


$$\mu_x = \mathbb{E}[X_1] = 0$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

Covariance of a Probability Distribution: Motivation

GAME 1



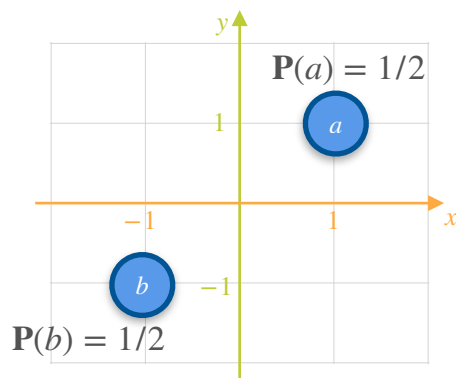
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$$\mu_x = \mathbb{E}[X_1] = 0$$

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Covariance of a Probability Distribution: Motivation

GAME 1



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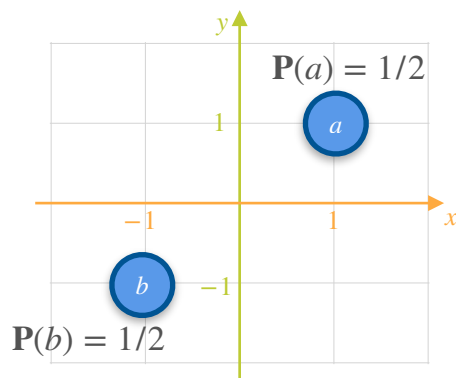
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
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$$\mu_x = \mathbb{E}[X_1] = 0$$

$$\mu_y = \mathbb{E}[Y_1] = 0$$

Covariance of a Probability Distribution: Motivation

GAME 1



$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

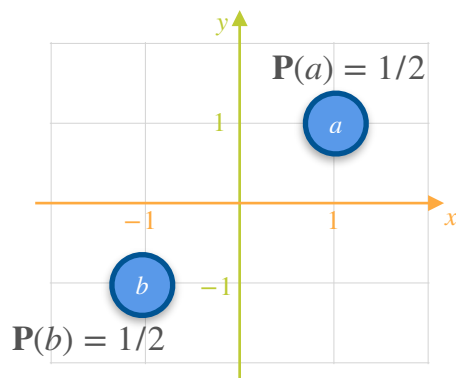
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1				
-1				

$$\mu_x = \mathbb{E}[X_1] = 0$$

$$\mu_y = \mathbb{E}[Y_1] = 0$$

Covariance of a Probability Distribution: Motivation

GAME 1



$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

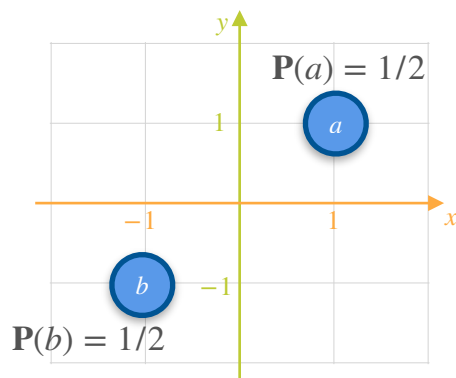
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1			
-1	-1			

$$\mu_x = \mathbb{E}[X_1] = 0$$

$$\mu_y = \mathbb{E}[Y_1] = 0$$

Covariance of a Probability Distribution: Motivation

GAME 1



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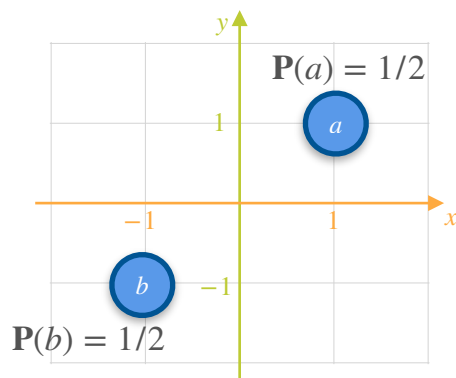
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

$$\mu_x = \mathbb{E}[X_1] = 0$$

$$\mu_y = \mathbb{E}[Y_1] = 0$$

Covariance of a Probability Distribution: Motivation

GAME 1



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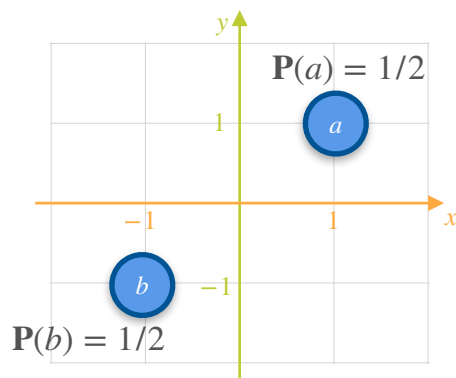
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

$$\mu_x = \mathbb{E}[X_1] = 0$$

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Covariance of a Probability Distribution: Motivation

GAME 1



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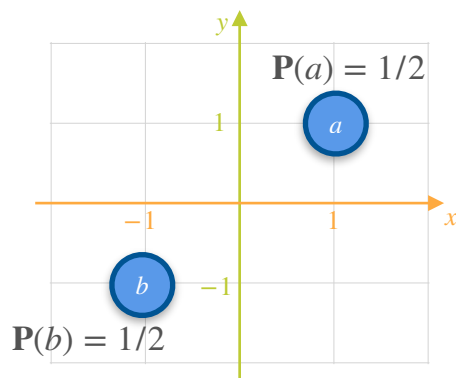
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

$$\mu_x = \mathbb{E}[X_1] = 0$$

$$\mu_y = \mathbb{E}[Y_1] = 0$$

Covariance of a Probability Distribution: Motivation

GAME 1



$$\mu_x = \mathbb{E}[X_1] = 0$$

$$\mu_y = \mathbb{E}[Y_1] = 0$$

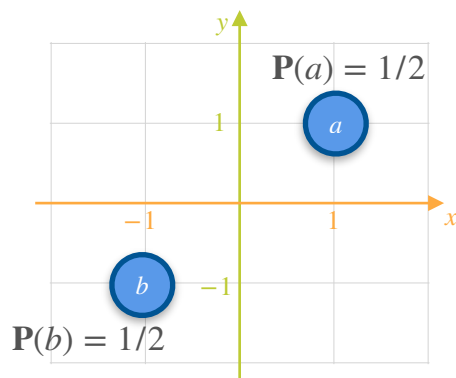
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x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

$$\sum (x_i - \mu_x)(y_i - \mu_y) =$$

Covariance of a Probability Distribution: Motivation

GAME 1



$$\mu_x = \mathbb{E}[X_1] = 0$$

$$\mu_y = \mathbb{E}[Y_1] = 0$$

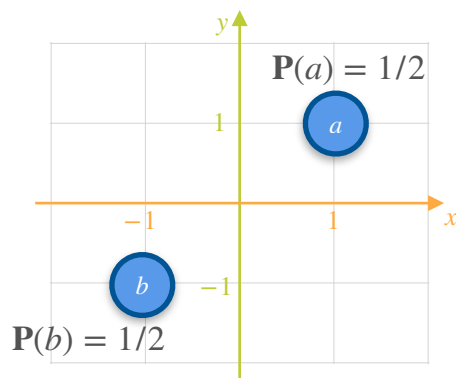
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
-1	-1	-1	-1	1

$$\sum (x_i - \mu_x)(y_i - \mu_y) =$$

Covariance of a Probability Distribution: Motivation

GAME 1



$$\mu_x = \mathbb{E}[X_1] = 0$$

$$\mu_y = \mathbb{E}[Y_1] = 0$$

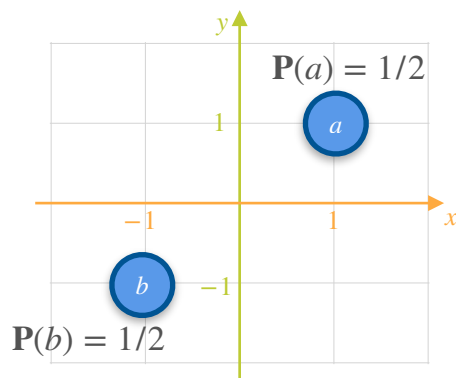
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-1	-1	-1	-1	1

$$\sum (x_i - \mu_x)(y_i - \mu_y) = 2$$

Covariance of a Probability Distribution: Motivation

GAME 1



$$\mu_x = \mathbb{E}[X_1] = 0$$

$$\mu_y = \mathbb{E}[Y_1] = 0$$

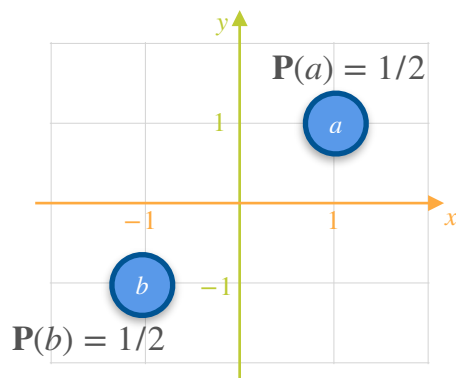
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Covariance of a Probability Distribution: Motivation

GAME 1



$$\mu_x = \mathbb{E}[X_1] = 0$$

$$\mu_y = \mathbb{E}[Y_1] = 0$$

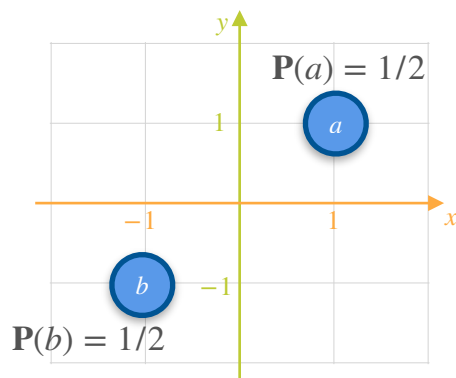
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

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$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = 2$$

Covariance of a Probability Distribution: Motivation

GAME 1



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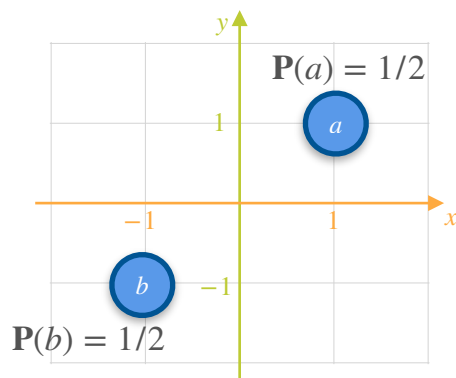
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1	1	1	1	1
-1	-1	-1	-1	1

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \frac{2}{2}$$

Covariance of a Probability Distribution: Motivation

GAME 1



$$\mu_x = \mathbb{E}[X_1] = 0$$

$$\mu_y = \mathbb{E}[Y_1] = 0$$

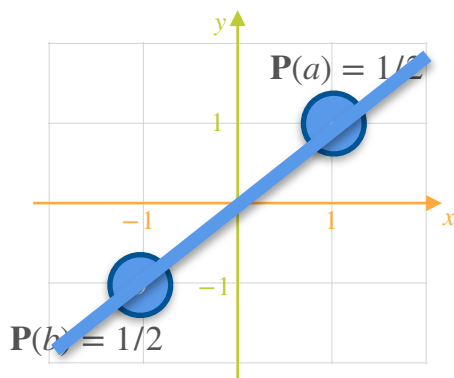
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
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$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \frac{2}{2} = 1$$

Covariance of a Probability Distribution: Motivation

GAME 1



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Covariance of a Probability Distribution: Motivation

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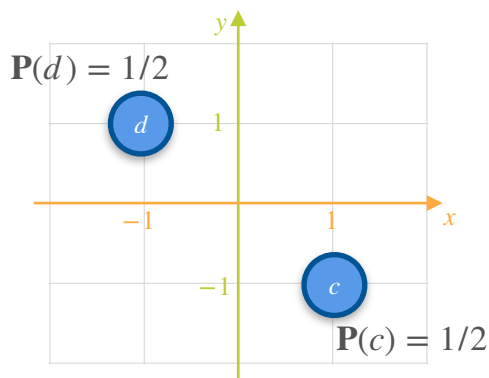
Covariance of a Probability Distribution: Motivation

GAME 2

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

Covariance of a Probability Distribution: Motivation

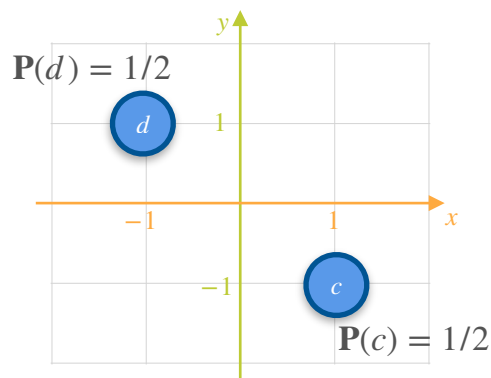
GAME 2



$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

Covariance of a Probability Distribution: Motivation

GAME 2

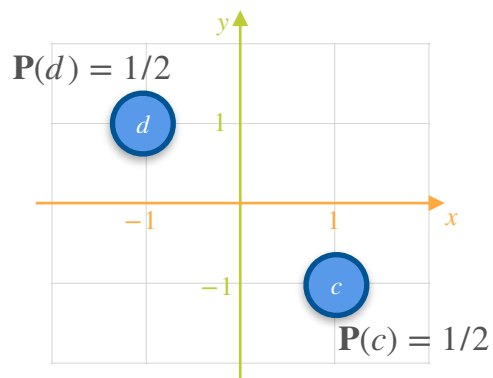


$$\mu_x = \mathbb{E}[X_2] = 0$$

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Covariance of a Probability Distribution: Motivation

GAME 2



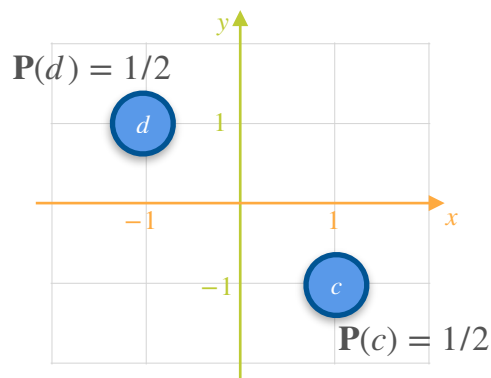
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Covariance of a Probability Distribution: Motivation

GAME 2



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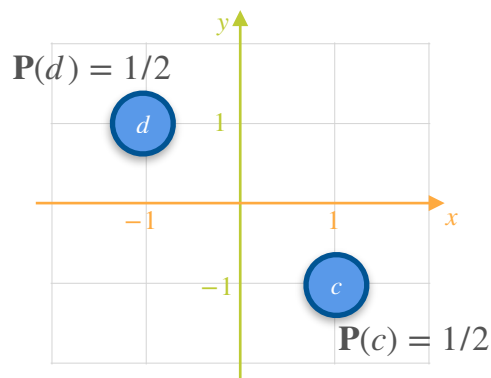
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
-----	-----	-----------------	-----------------	------------------------------

$$\mu_x = \mathbb{E}[X_2] = 0$$

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Covariance of a Probability Distribution: Motivation

GAME 2



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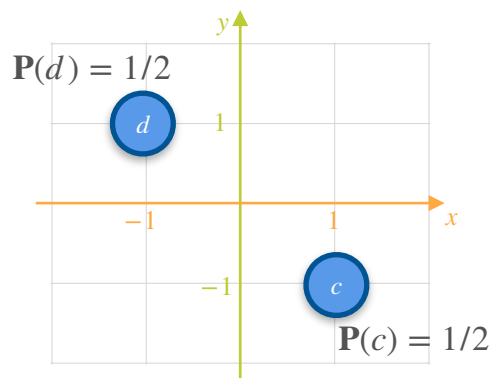
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1				
-1				

$$\mu_x = \mathbb{E}[X_2] = 0$$

$$\mu_y = \mathbb{E}[Y_2] = 0$$

Covariance of a Probability Distribution: Motivation

GAME 2



$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

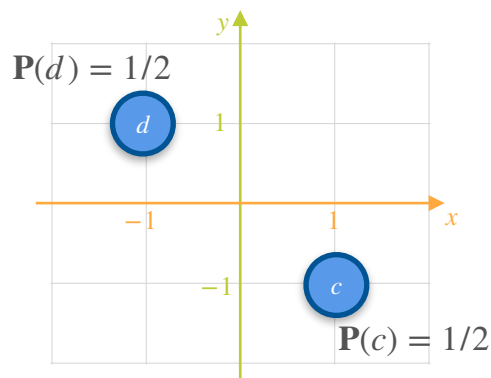
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	-1			
-1	1			

$$\mu_x = \mathbb{E}[X_2] = 0$$

$$\mu_y = \mathbb{E}[Y_2] = 0$$

Covariance of a Probability Distribution: Motivation

GAME 2



$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

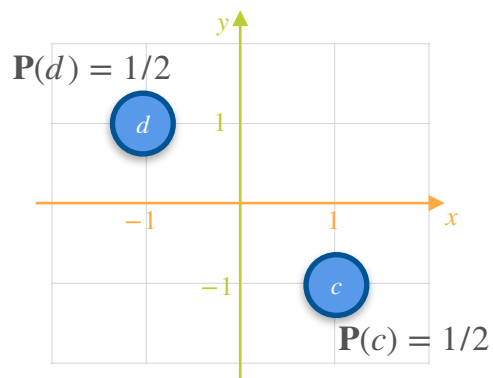
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	-1	1	-1	-1
-1	1	-1	1	-1

$$\mu_x = \mathbb{E}[X_2] = 0$$

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Covariance of a Probability Distribution: Motivation

GAME 2



$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

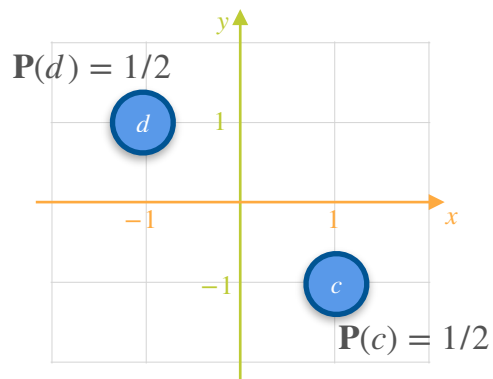
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	-1	1	-1	-1
-1	1	-1	1	-1

$$\mu_x = \mathbb{E}[X_2] = 0$$

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Covariance of a Probability Distribution: Motivation

GAME 2



$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

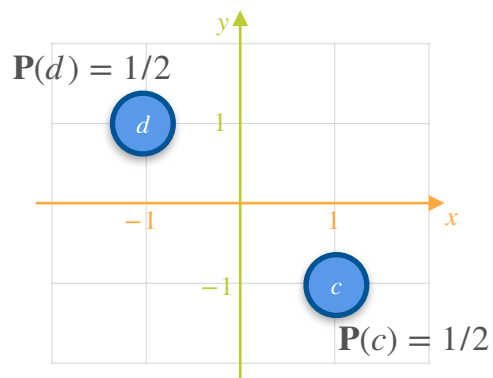
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-1	1	-1	1	-1

$$\mu_x = \mathbb{E}[X_2] = 0$$

$$\mu_y = \mathbb{E}[Y_2] = 0$$

Covariance of a Probability Distribution: Motivation

GAME 2



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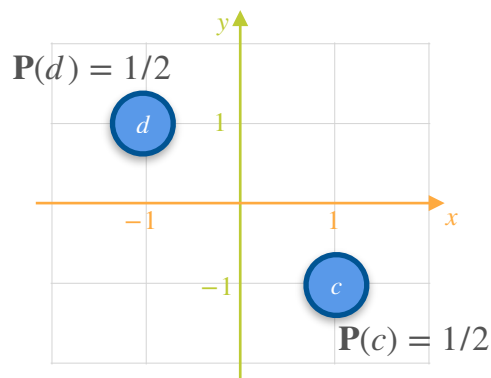
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1	-1	1	-1	-1
-1	1	-1	1	-1

$$\sum (x_i - \mu_x)(y_i - \mu_y) =$$

Covariance of a Probability Distribution: Motivation

GAME 2



$$\mu_x = \mathbb{E}[X_2] = 0$$

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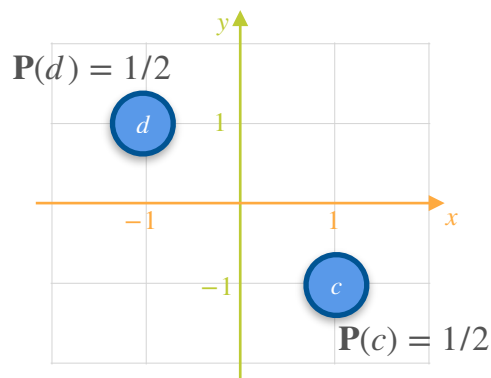
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-1	1	-1	1	-1

$$\sum (x_i - \mu_x)(y_i - \mu_y) =$$

Covariance of a Probability Distribution: Motivation

GAME 2



$$\mu_x = \mathbb{E}[X_2] = 0$$

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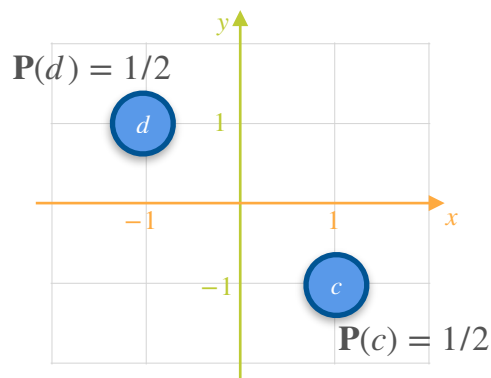
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x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	-1	1	-1	-1
-1	1	-1	1	-1

$$\sum (x_i - \mu_x)(y_i - \mu_y) = -2$$

Covariance of a Probability Distribution: Motivation

GAME 2



$$\mu_x = \mathbb{E}[X_2] = 0$$

$$\mu_y = \mathbb{E}[Y_2] = 0$$

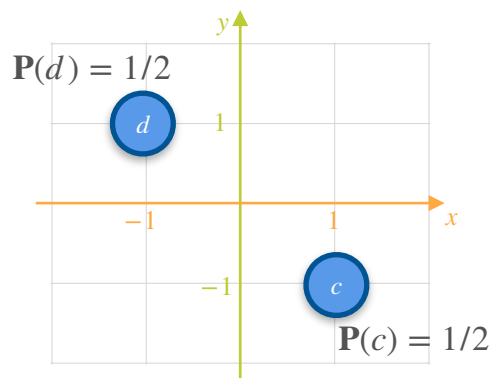
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x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	-1	1	-1	-1
-1	1	-1	1	-1

$$\text{Cov}(X, Y) = \sum (x_i - \mu_x)(y_i - \mu_y) = -2$$

Covariance of a Probability Distribution: Motivation

GAME 2



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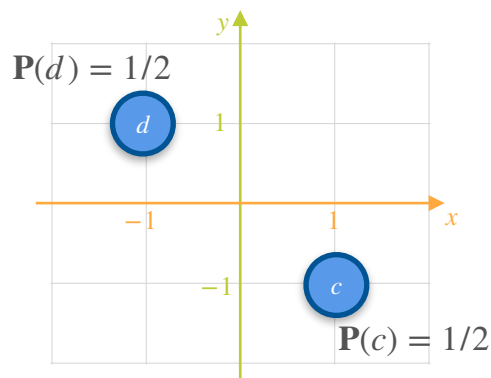
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-1	1	-1	1	-1



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Covariance of a Probability Distribution: Motivation

GAME 2



$$\mu_x = \mathbb{E}[X_2] = 0$$

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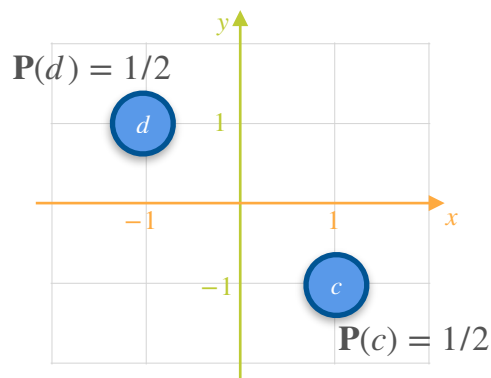
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1	-1	1	-1	-1
-1	1	-1	1	-1

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \frac{-2}{2}$$

Covariance of a Probability Distribution: Motivation

GAME 2



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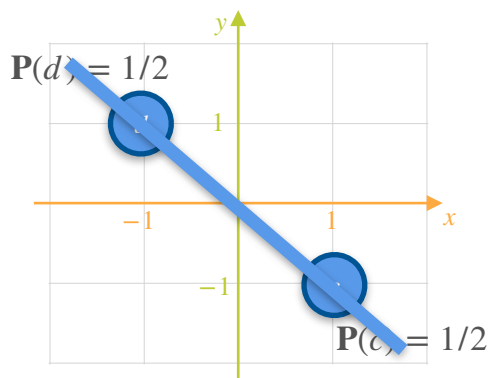
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-1	1	-1	1	-1

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Covariance of a Probability Distribution: Motivation

GAME 2



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Covariance of a Probability Distribution: Motivation

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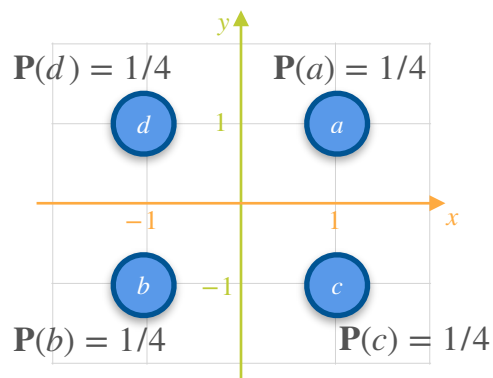
Covariance of a Probability Distribution: Motivation

GAME 3

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

Covariance of a Probability Distribution: Motivation

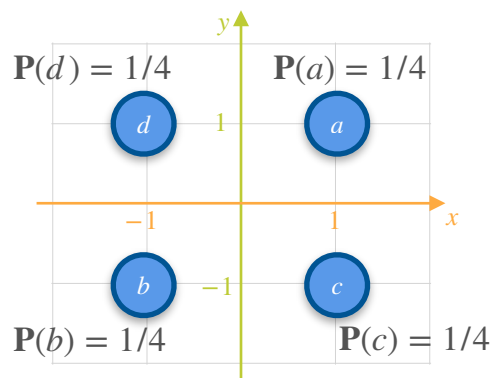
GAME 3



$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

Covariance of a Probability Distribution: Motivation

GAME 3

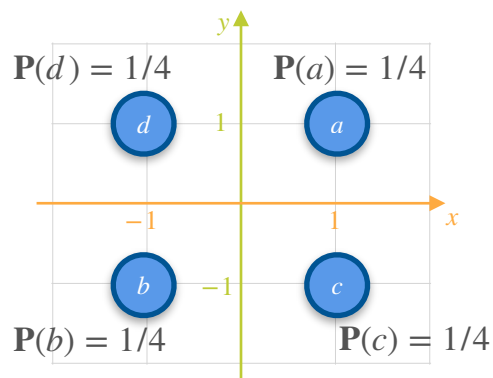


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$$\mu_x = \mathbb{E}[X_3] = 0$$

Covariance of a Probability Distribution: Motivation

GAME 3



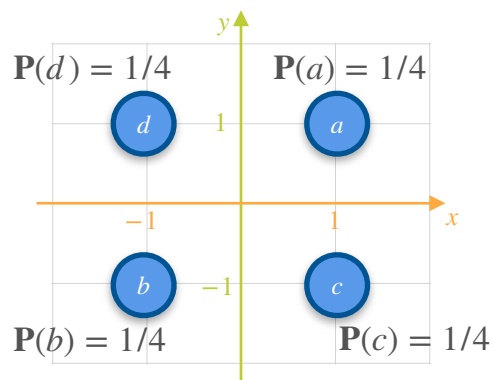
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Covariance of a Probability Distribution: Motivation

GAME 3



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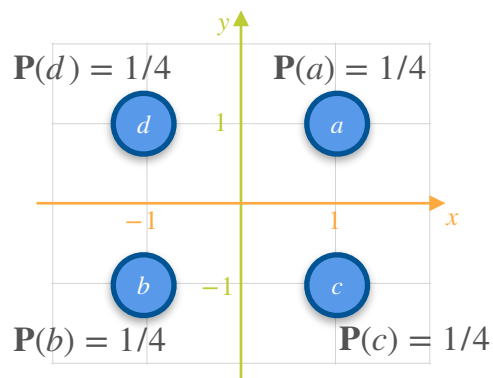
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
-----	-----	-----------------	-----------------	------------------------------

$$\mu_x = \mathbb{E}[X_3] = 0$$

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Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

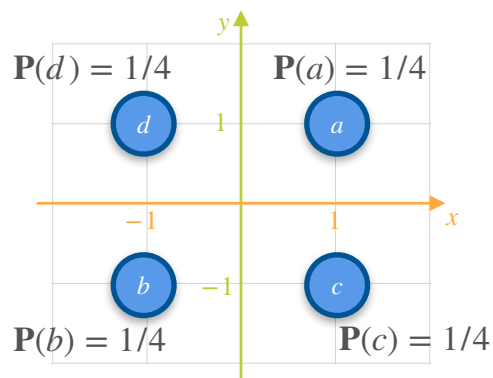
$$\mu_y = \mathbb{E}[Y_3] = 0$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1			
1	-1			
-1	1			
-1	-1			

Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

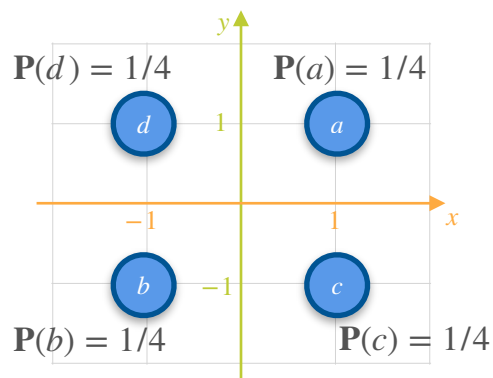
$$\mu_y = \mathbb{E}[Y_3] = 0$$

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x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1			
1	-1			
-1	1			
-1	-1			

Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

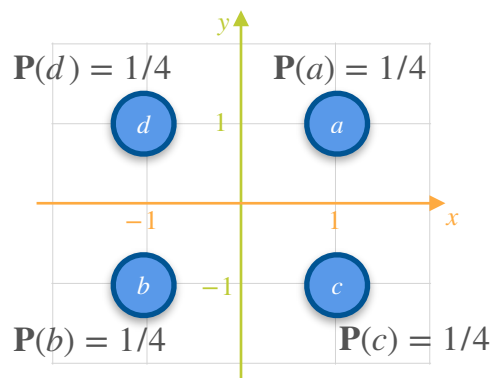
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x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
1	-1	-1	-1	1
-1	1	1	1	1
-1	-1	-1	-1	1

Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

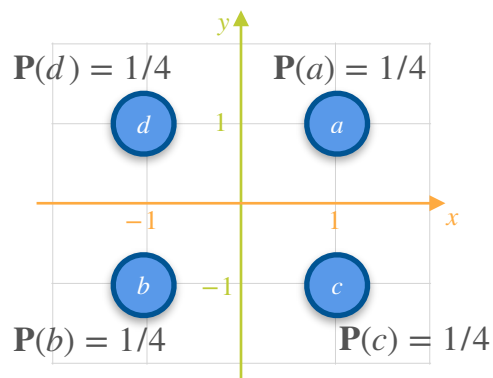
$$\mu_y = \mathbb{E}[Y_3] = 0$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
1	-1	-1	1	1
-1	1	1	-1	-1
-1	-1	-1	-1	-1

Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

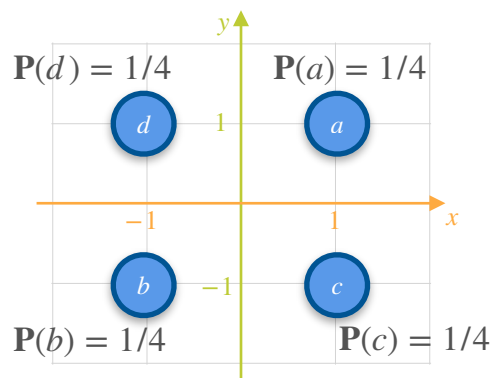
$$\mu_y = \mathbb{E}[Y_3] = 0$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
1	-1	-1	1	-1
-1	1	1	-1	-1
-1	-1	-1	-1	1

Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

$$\mu_y = \mathbb{E}[Y_3] = 0$$

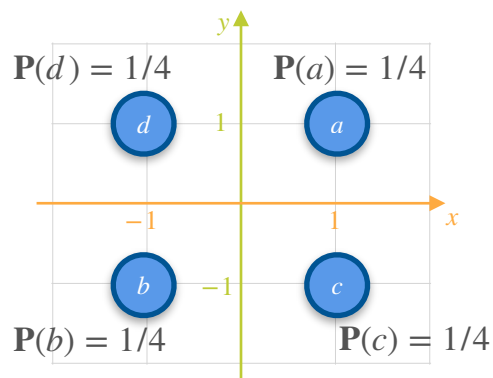
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
1	-1	-1	1	-1
-1	1	1	-1	-1
-1	-1	-1	-1	1

$$\sum (x_i - \mu_x)(y_i - \mu_y) =$$

Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

$$\mu_y = \mathbb{E}[Y_3] = 0$$

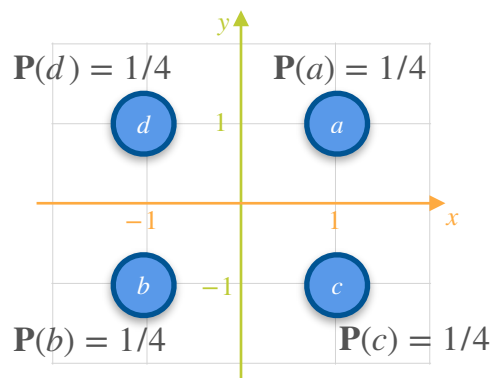
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
1	-1	-1	1	-1
-1	1	1	-1	-1
-1	-1	-1	-1	1

$$\sum (x_i - \mu_x)(y_i - \mu_y) =$$

Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

$$\mu_y = \mathbb{E}[Y_3] = 0$$

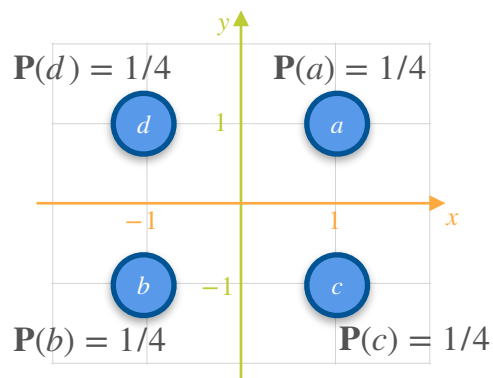
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
1	-1	-1	1	-1
-1	1	1	-1	-1
-1	-1	-1	-1	1

$$\sum (x_i - \mu_x)(y_i - \mu_y) = 0$$

Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

$$\mu_y = \mathbb{E}[Y_3] = 0$$

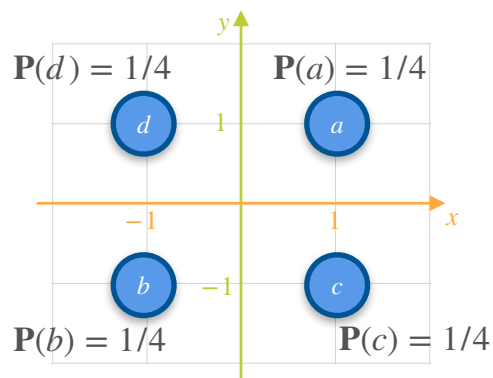
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
1	-1	-1	1	-1
-1	1	1	-1	-1
-1	-1	-1	-1	1

$$\text{Cov}(X, Y) = \sum (x_i - \mu_x)(y_i - \mu_y) = 0$$

Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

$$\mu_y = \mathbb{E}[Y_3] = 0$$

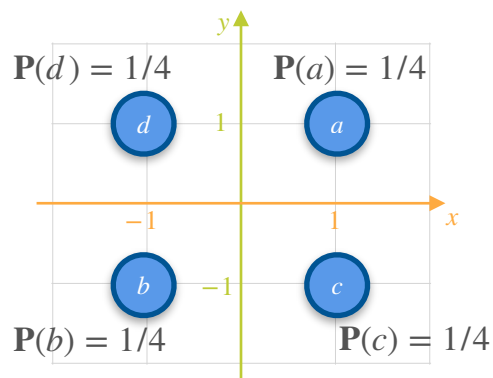
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
1	-1	-1	1	-1
-1	1	1	-1	-1
-1	-1	-1	-1	1

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = 0$$

Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

$$\mu_y = \mathbb{E}[Y_3] = 0$$

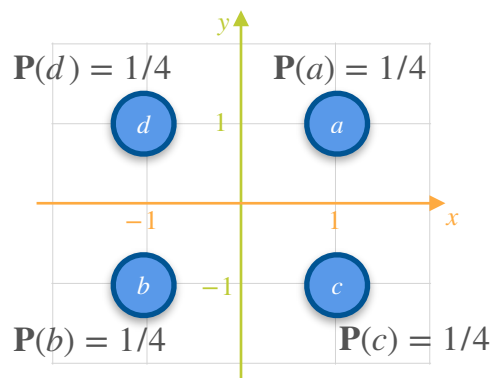
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
1	-1	-1	1	-1
-1	1	1	-1	-1
-1	-1	-1	-1	1

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \frac{0}{4}$$

Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

$$\mu_y = \mathbb{E}[Y_3] = 0$$

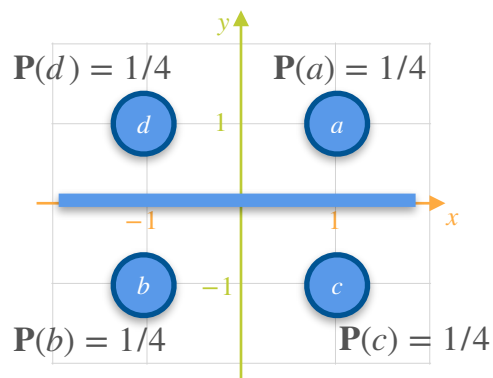
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
1	-1	-1	1	-1
-1	1	1	-1	-1
-1	-1	-1	-1	1

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \frac{0}{4} = 0$$

Covariance of a Probability Distribution: Motivation

GAME 3



$$\mu_x = \mathbb{E}[X_3] = 0$$

$$\mu_y = \mathbb{E}[Y_3] = 0$$

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

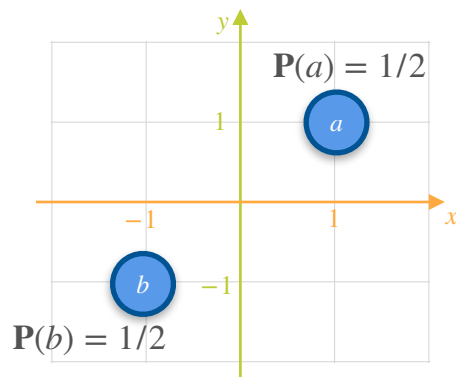
x	y	$(x_i - \mu_x)$	$(y_i - \mu_y)$	$(x_i - \mu_x)(y_i - \mu_y)$
1	1	1	1	1
1	-1	-1	1	-1
-1	1	1	-1	-1
-1	-1	-1	-1	1

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \frac{0}{4} = 0$$

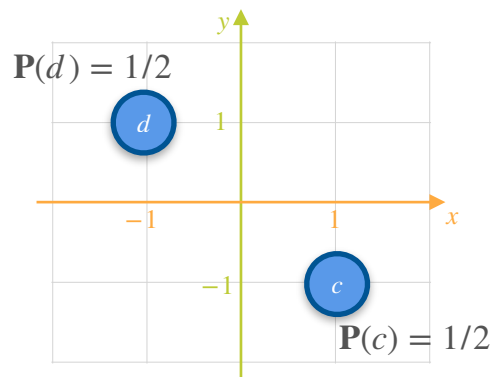
Covariance of a Probability Distribution: Motivation

Covariance of a Probability Distribution: Motivation

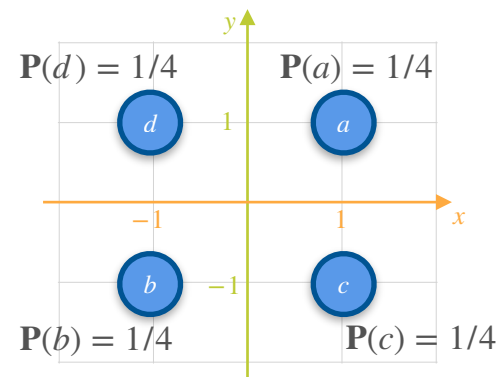
GAME 1



GAME 2

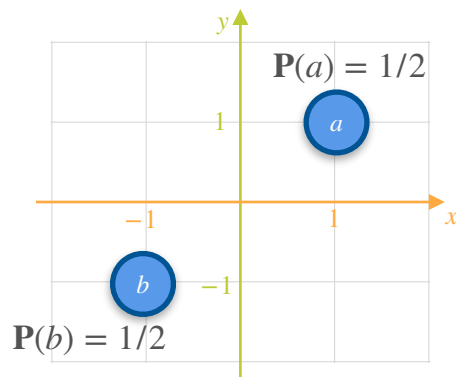


GAME 3



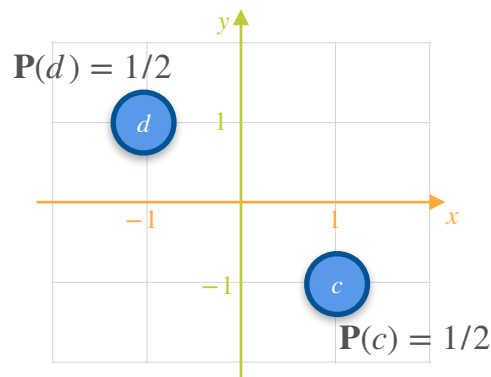
Covariance of a Probability Distribution: Motivation

GAME 1



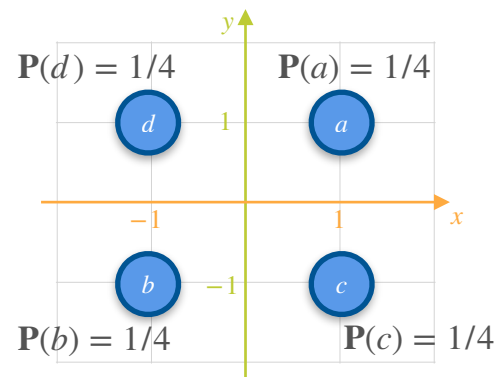
$$\text{Cov}(X, Y) = 1$$

GAME 2



$$\text{Cov}(X, Y) = -1$$

GAME 3



$$\text{Cov}(X, Y) = 0$$

Covariance of a Probability Distribution: Motivation

Covariance of a Probability Distribution: Motivation

GAME 4

Covariance of a Probability Distribution: Motivation

GAME 4

a: Both players win \$1 each

Covariance of a Probability Distribution: Motivation

GAME 4

a: Both players win \$1 each

b: Both players lose \$1 each

Covariance of a Probability Distribution: Motivation

GAME 4

a: Both players win \$1 each

b: Both players lose \$1 each

c: Neither players wins nor lose anything

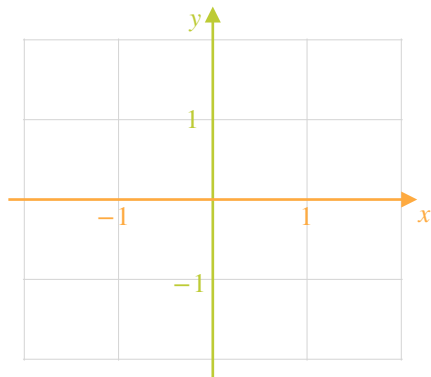
Covariance of a Probability Distribution: Motivation

GAME 4

a: Both players win \$1 each

b: Both players lose \$1 each

c: Neither players wins nor lose anything



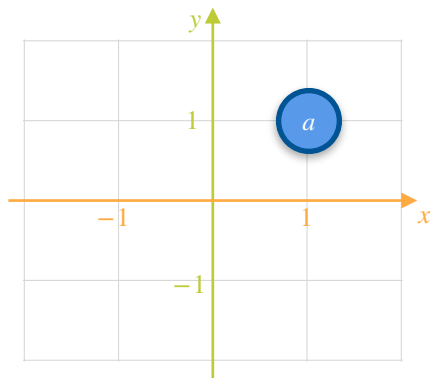
Covariance of a Probability Distribution: Motivation

GAME 4

a: Both players win \$1 each

b: Both players lose \$1 each

c: Neither players wins nor lose anything



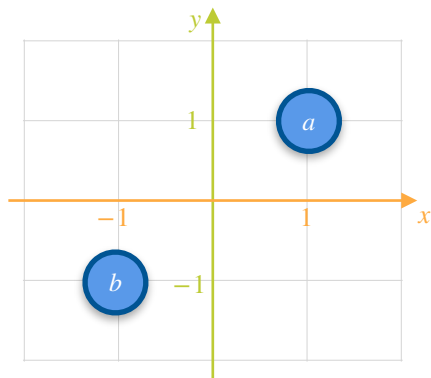
Covariance of a Probability Distribution: Motivation

GAME 4

a: Both players win \$1 each

b: Both players lose \$1 each

c: Neither players wins nor lose anything



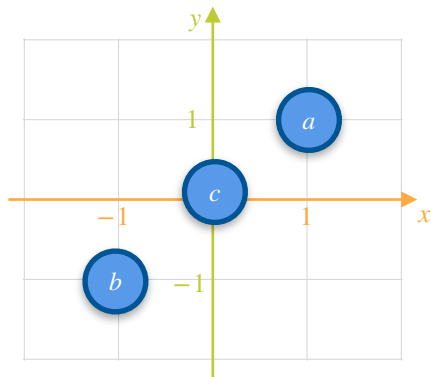
Covariance of a Probability Distribution: Motivation

GAME 4

a: Both players win \$1 each

b: Both players lose \$1 each

c: Neither players wins nor lose anything



Covariance of a Probability Distribution: Motivation

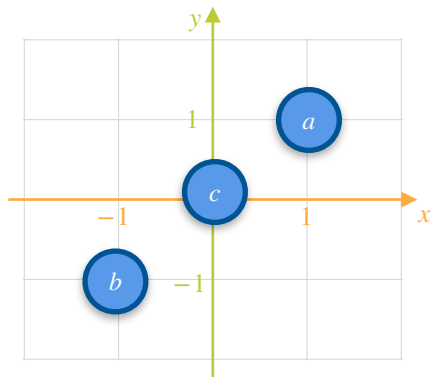
GAME 4

a: Both players win \$1 each

b: Both players lose \$1 each

c: Neither players wins nor lose anything

Unequal Probabilities



Covariance of a Probability Distribution: Motivation

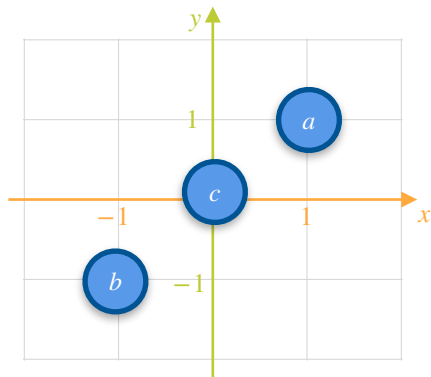
GAME 4

a : Both players win \$1 each $P(a) = 1/2$

b : Both players lose \$1 each

c : Neither players wins nor lose anything

Unequal Probabilities



Covariance of a Probability Distribution: Motivation

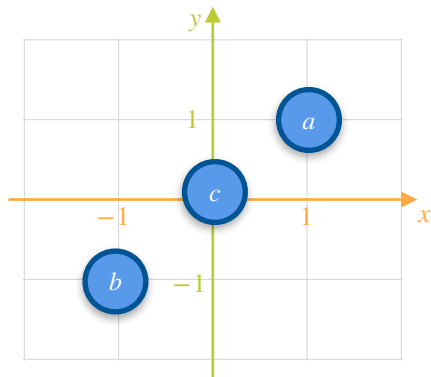
GAME 4

a : Both players win \$1 each $P(a) = 1/2$

b : Both players lose \$1 each $P(b) = 1/3$

c : Neither players wins nor lose anything

Unequal Probabilities



Covariance of a Probability Distribution: Motivation

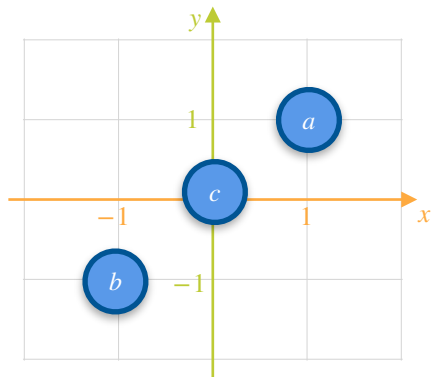
GAME 4

a : Both players win \$1 each $P(a) = 1/2$

b : Both players lose \$1 each $P(b) = 1/3$

c : Neither players wins nor lose anything $P(c) = 1/6$

Unequal Probabilities



Covariance of a Probability Distribution: Motivation

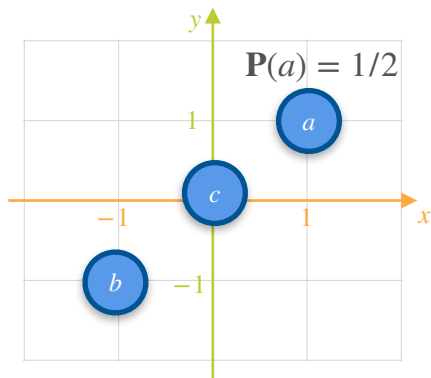
GAME 4

a : Both players win \$1 each $P(a) = 1/2$

b : Both players lose \$1 each $P(b) = 1/3$

c : Neither players wins nor lose anything $P(c) = 1/6$

Unequal Probabilities



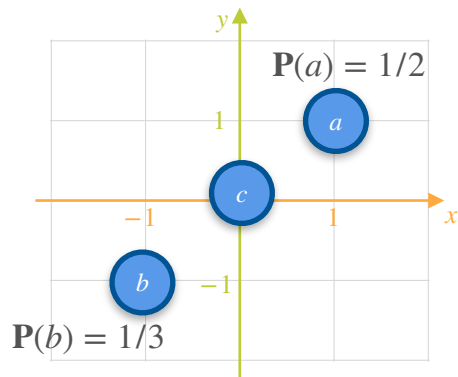
Covariance of a Probability Distribution: Motivation

GAME 4

a : Both players win \$1 each $P(a) = 1/2$

b : Both players lose \$1 each $P(b) = 1/3$

c : Neither players wins nor lose anything $P(c) = 1/6$



Unequal Probabilities

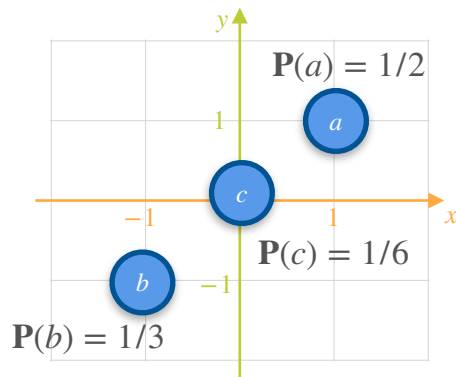
Covariance of a Probability Distribution: Motivation

GAME 4

a : Both players win \$1 each $P(a) = 1/2$

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Unequal Probabilities

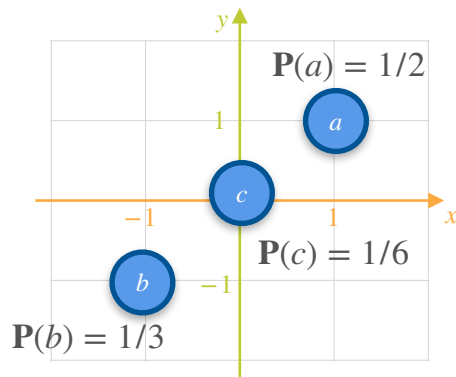
Covariance of a Probability Distribution: Motivation

GAME 4

a : Both players win \$1 each $P(a) = 1/2$

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Unequal Probabilities

$$\mathbb{E}[X_4] =$$

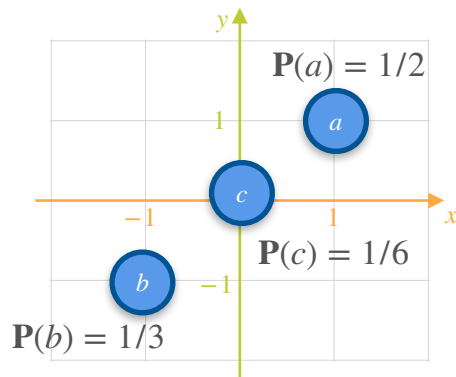
Covariance of a Probability Distribution: Motivation

GAME 4

a : Both players win \$1 each $\mathbf{P}(a) = 1/2$

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c : Neither players wins nor lose anything $\mathbf{P}(c) = 1/6$



Unequal Probabilities

$$\mathbb{E}[X_4] = \frac{1}{2}(1) + \frac{1}{6}(0) + \frac{1}{3}(-1) =$$

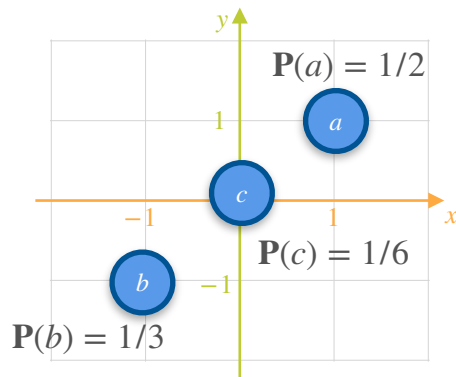
Covariance of a Probability Distribution: Motivation

GAME 4

a: Both players win \$1 each $P(a) = 1/2$

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c: Neither players wins nor lose anything $P(c) = 1/6$



Unequal Probabilities

$$\mathbb{E}[X_4] = \frac{1}{2}(1) + \frac{1}{6}(0) + \frac{1}{3}(-1) = \frac{1}{6}$$

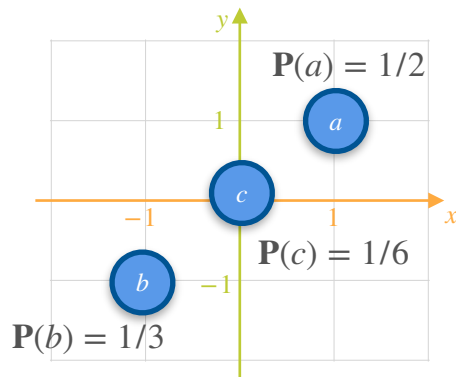
Covariance of a Probability Distribution: Motivation

GAME 4

a : Both players win \$1 each $P(a) = 1/2$

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c : Neither players wins nor lose anything $P(c) = 1/6$



Unequal Probabilities

$$\mathbb{E}[X_4] = \frac{1}{2}(1) + \frac{1}{6}(0) + \frac{1}{3}(-1) = \frac{1}{6}$$

$$\mathbb{E}[Y_4] =$$

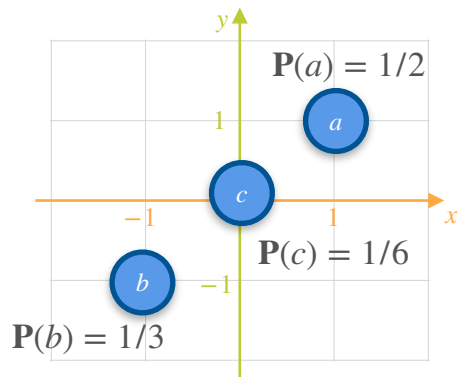
Covariance of a Probability Distribution: Motivation

GAME 4

a : Both players win \$1 each $P(a) = 1/2$

b : Both players lose \$1 each $P(b) = 1/3$

c : Neither players wins nor lose anything $P(c) = 1/6$



Unequal Probabilities

$$\mathbb{E}[X_4] = \frac{1}{2}(1) + \frac{1}{6}(0) + \frac{1}{3}(-1) = \frac{1}{6}$$

$$\mathbb{E}[Y_4] = \frac{1}{2}(1) + \frac{1}{6}(0) + \frac{1}{3}(-1) = \frac{1}{6}$$

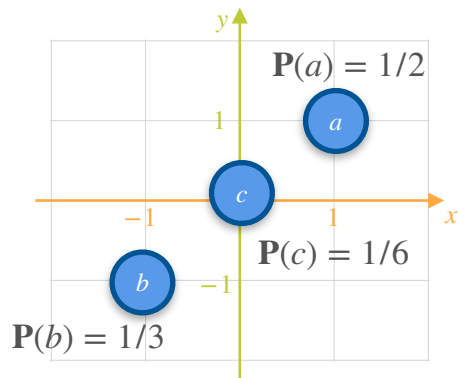
Covariance of a Probability Distribution: Motivation

GAME 4

a : Both players win \$1 each $P(a) = 1/2$

b : Both players lose \$1 each $P(b) = 1/3$

c : Neither players wins nor lose anything $P(c) = 1/6$



$$\mathbb{E}[X_4] = \frac{1}{6}$$

$$\mathbb{E}[Y_4] = \frac{1}{6}$$

Unequal Probabilities

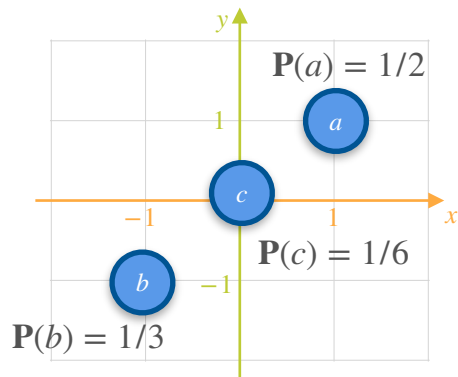
Covariance of a Probability Distribution: Motivation

GAME 4

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c : Neither players wins nor lose anything $P(c) = 1/6$



$$\mathbb{E}[X_4] = \frac{1}{6}$$

$$\mathbb{E}[Y_4] = \frac{1}{6}$$

Unequal Probabilities

$$\text{Var}(X_4) =$$

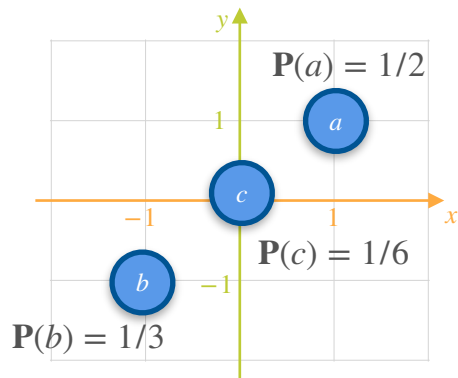
Covariance of a Probability Distribution: Motivation

GAME 4

a: Both players win \$1 each $\mathbf{P}(a) = 1/2$

b: Both players lose \$1 each $\mathbf{P}(b) = 1/3$

c: Neither players wins nor lose anything $\mathbf{P}(c) = 1/6$



$$\mathbb{E}[X_4] = \frac{1}{6}$$

$$\mathbb{E}[Y_4] = \frac{1}{6}$$

Unequal Probabilities

$$\text{Var}(X_4) = \sum_{n=1}^N (\mathbb{E}[X_4] - \mu_x)^2 \cdot \mathbf{P}(x_i)$$

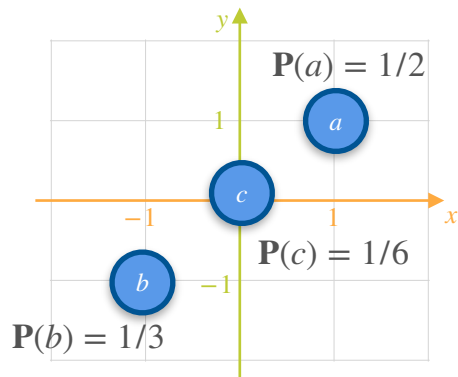
Covariance of a Probability Distribution: Motivation

GAME 4

a: Both players win \$1 each $\mathbf{P}(a) = 1/2$

b: Both players lose \$1 each $\mathbf{P}(b) = 1/3$

c: Neither players wins nor lose anything $\mathbf{P}(c) = 1/6$



$$\mathbb{E}[X_4] = \frac{1}{6} \quad \text{Var}(X_4) =$$

$$\mathbb{E}[Y_4] = \frac{1}{6}$$

Unequal Probabilities

$$\text{Var}(X_4) = \sum_{n=1}^N (\mathbb{E}[X_4] - \mu_x)^2 \cdot \mathbf{P}(x_i)$$

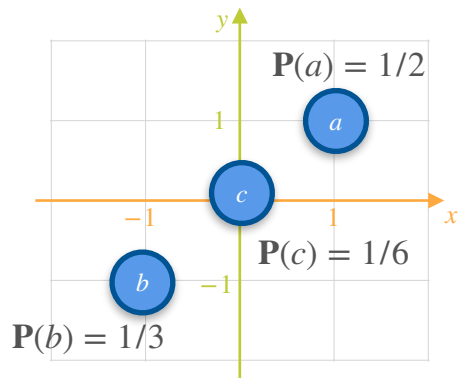
Covariance of a Probability Distribution: Motivation

GAME 4

a : Both players win \$1 each $\mathbf{P}(a) = 1/2$

b : Both players lose \$1 each $\mathbf{P}(b) = 1/3$

c : Neither players wins nor lose anything $\mathbf{P}(c) = 1/6$



$$\mathbb{E}[X_4] = \frac{1}{6}$$

$$\mathbb{E}[Y_4] = \frac{1}{6}$$

Unequal Probabilities

$$\text{Var}(X_4) = \sum_{n=1}^N (\mathbb{E}[X_4] - \mu_x)^2 \cdot \mathbf{P}(x_i)$$

$$\text{Var}(X_4) = \frac{1}{2} \left(1 - \frac{1}{6}\right)^2 + \frac{1}{6} \left(0 - \frac{1}{6}\right)^2 + \frac{1}{3} \left(-1 - \frac{1}{6}\right)^2$$

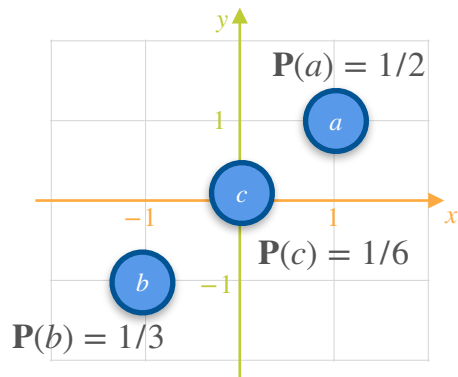
Covariance of a Probability Distribution: Motivation

GAME 4

a : Both players win \$1 each $\mathbf{P}(a) = 1/2$

b : Both players lose \$1 each $\mathbf{P}(b) = 1/3$

c : Neither players wins nor lose anything $\mathbf{P}(c) = 1/6$



$$\mathbb{E}[X_4] = \frac{1}{6}$$

$$\mathbb{E}[Y_4] = \frac{1}{6}$$

Unequal Probabilities

$$\text{Var}(X_4) = \sum_{n=1}^N (\mathbb{E}[X_4] - \mu_x)^2 \cdot \mathbf{P}(x_i)$$

$$\begin{aligned} \text{Var}(X_4) &= \frac{1}{2} \left(1 - \frac{1}{6}\right)^2 + \frac{1}{6} \left(0 - \frac{1}{6}\right)^2 + \frac{1}{3} \left(-1 - \frac{1}{6}\right)^2 \\ &= 0.806 \end{aligned}$$

Covariance of a Probability Distribution: Motivation

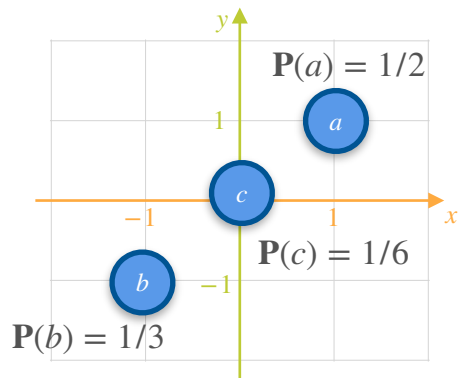
GAME 4

Unequal Probabilities

a : Both players win \$1 each $P(a) = 1/2$

b : Both players lose \$1 each $P(b) = 1/3$

c : Neither players wins nor lose anything $P(c) = 1/6$



$$\mathbb{E}[X_4] = \frac{1}{6}$$

$$\text{Var}(X_4) = 0.806$$

$$\mathbb{E}[Y_4] = \frac{1}{6}$$

Covariance of a Probability Distribution: Motivation

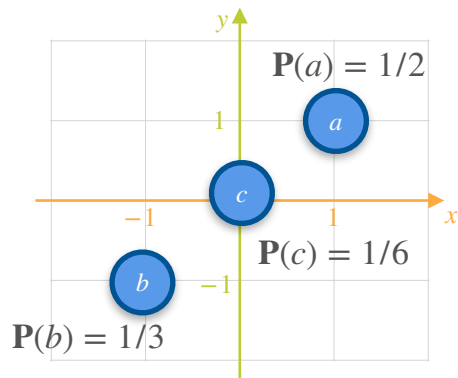
GAME 4

Unequal Probabilities

a : Both players win \$1 each $P(a) = 1/2$

b : Both players lose \$1 each $P(b) = 1/3$

c : Neither players wins nor lose anything $P(c) = 1/6$



$$\mathbb{E}[X_4] = \frac{1}{6} \quad \text{Var}(X_4) = 0.806$$

$$\mathbb{E}[Y_4] = \frac{1}{6} \quad \text{Var}(Y_4) = 0.806$$

Covariance of a Probability Distribution: Motivation

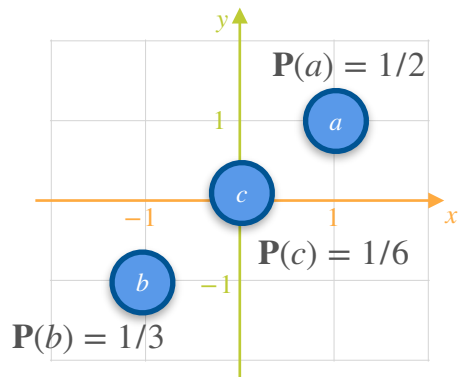
GAME 4

Unequal Probabilities

a : Both players win \$1 each $P(a) = 1/2$

b : Both players lose \$1 each $P(b) = 1/3$

c : Neither player wins nor loses anything $P(c) = 1/6$



$$\mathbb{E}[X_4] = \frac{1}{6}$$

$$\text{Var}(X_4) = 0.806$$

$$\mathbb{E}[Y_4] = \frac{1}{6}$$

$$\text{Var}(Y_4) = 0.806$$

$$\text{Cov}(X, Y) = ?$$

Covariance of Probability Distributions

Covariance of Probability Distributions

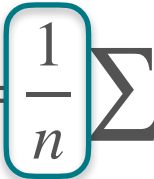
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

Covariance of Probability Distributions

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \frac{1}{n} \sum (x_i - \mu_x)(y_i - \mu_y)$$

Covariance of Probability Distributions

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \frac{1}{n} \sum (x_i - \mu_x)(y_i - \mu_y)$$

 **equal probabilities**

Covariance of Probability Distributions

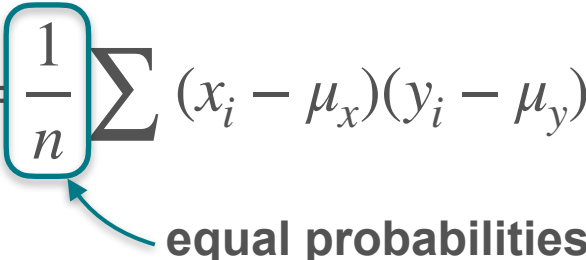
$$Cov(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \boxed{\frac{1}{n}} \sum (x_i - \mu_x)(y_i - \mu_y)$$

 **equal probabilities**

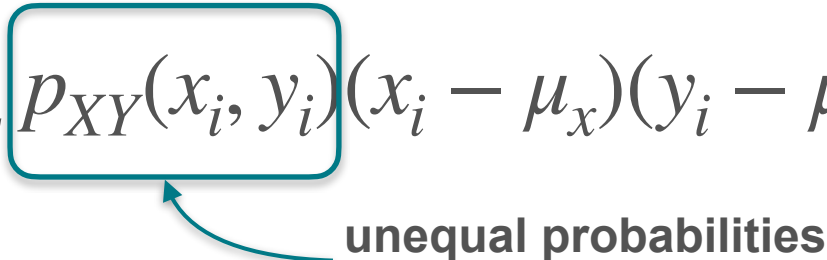
$$Cov(X, Y) = \sum p_{XY}(x_i, y_i)(x_i - \mu_x)(y_i - \mu_y)$$

Covariance of Probability Distributions

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \frac{1}{n} \sum (x_i - \mu_x)(y_i - \mu_y)$$

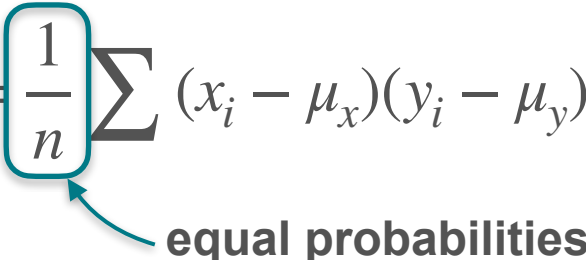
equal probabilities

$$\text{Cov}(X, Y) = \sum p_{XY}(x_i, y_i)(x_i - \mu_x)(y_i - \mu_y)$$

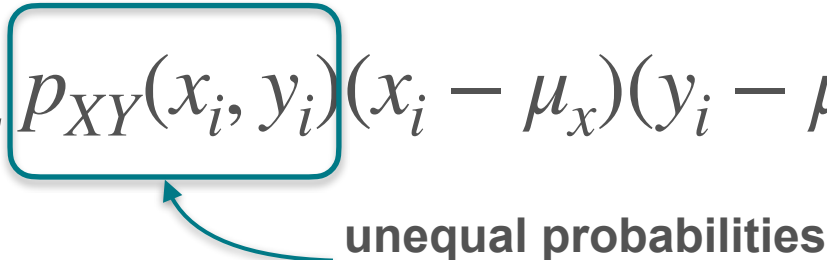
unequal probabilities

Covariance of Probability Distributions

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n} = \frac{1}{n} \sum (x_i - \mu_x)(y_i - \mu_y)$$

equal probabilities

$$\text{Cov}(X, Y) = \sum p_{XY}(x_i, y_i)(x_i - \mu_x)(y_i - \mu_y)$$

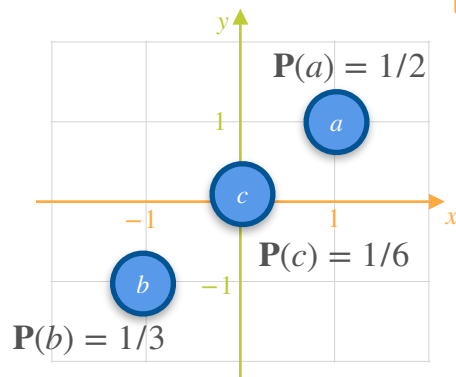
unequal probabilities

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Covariance of a Probability Distribution: Motivation

GAME 4

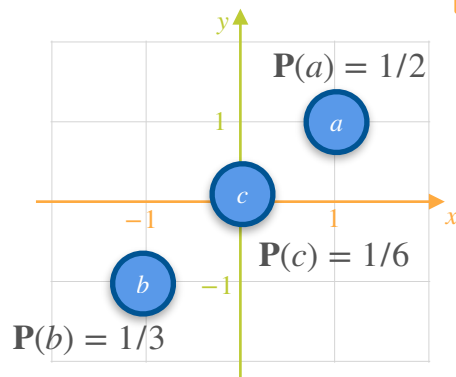
Unequal Probabilities



Covariance of a Probability Distribution: Motivation

GAME 4

Unequal Probabilities

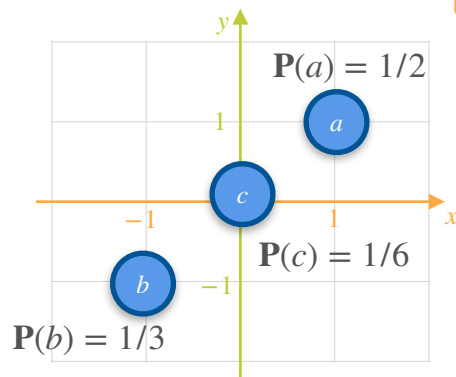


$$\text{Cov}(X, Y) = \sum p_{XY}(x_i, y_i)(x_i - \mu_x)(y_i - \mu_y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Covariance of a Probability Distribution: Motivation

GAME 4

Unequal Probabilities



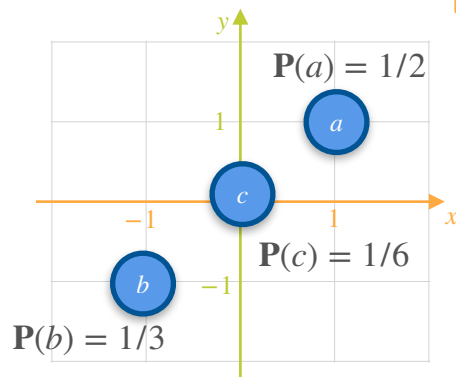
$$\text{Cov}(X, Y) = \sum p_{XY}(x_i, y_i)(x_i - \mu_x)(y_i - \mu_y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

$$\mu_x = \mu_y = \mathbb{E}[X_4] = \mathbb{E}[Y_4] = 1/6$$

Covariance of a Probability Distribution: Motivation

GAME 4

Unequal Probabilities



$$\text{Cov}(X, Y) = \sum p_{XY}(x_i, y_i)(x_i - \mu_x)(y_i - \mu_y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

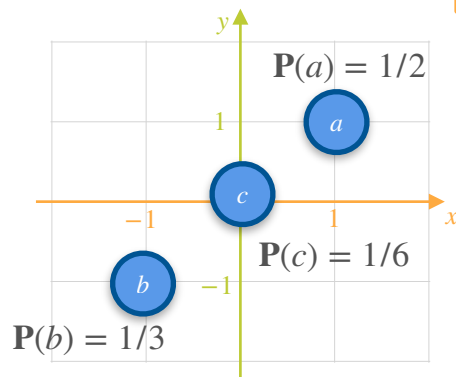
$$\mu_x = \mu_y = \mathbb{E}[X_4] = \mathbb{E}[Y_4] = 1/6$$

$$\text{Var}(X_4) = \text{Var}(Y_4) = 0.806$$

Covariance of a Probability Distribution: Motivation

GAME 4

Unequal Probabilities



$$\begin{aligned} \text{Cov}(X, Y) &= \sum p_{XY}(x_i, y_i)(x_i - \mu_x)(y_i - \mu_y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \\ &= \frac{1}{2}\left(1 - \frac{1}{6}\right)^2 + \frac{1}{6}\left(0 - \frac{1}{6}\right)^2 + \frac{1}{3}\left(-1 - \frac{1}{6}\right)^2 \end{aligned}$$

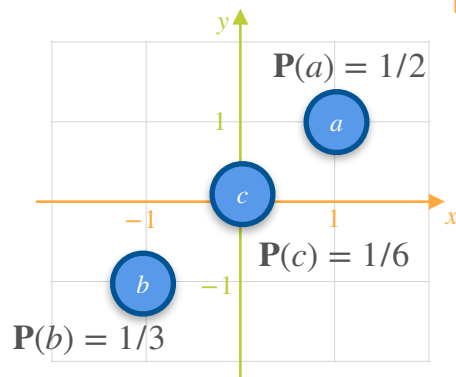
$$\mu_x = \mu_y = \mathbb{E}[X_4] = \mathbb{E}[Y_4] = 1/6$$

$$\text{Var}(X_4) = \text{Var}(Y_4) = 0.806$$

Covariance of a Probability Distribution: Motivation

GAME 4

Unequal Probabilities



$$\text{Cov}(X, Y) = \sum p_{XY}(x_i, y_i) (x_i - \mu_x)(y_i - \mu_y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

$$= \frac{1}{2} \left(1 - \frac{1}{6}\right)^2 + \frac{1}{6} \left(0 - \frac{1}{6}\right)^2 + \frac{1}{3} \left(-1 - \frac{1}{6}\right)^2$$

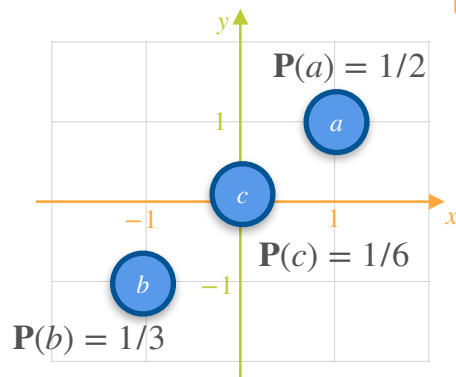
$$\mu_x = \mu_y = \mathbb{E}[X_4] = \mathbb{E}[Y_4] = 1/6$$

$$\text{Var}(X_4) = \text{Var}(Y_4) = 0.806$$

Covariance of a Probability Distribution: Motivation

GAME 4

Unequal Probabilities



$$\mu_x = \mu_y = \mathbb{E}[X_4] = \mathbb{E}[Y_4] = 1/6$$

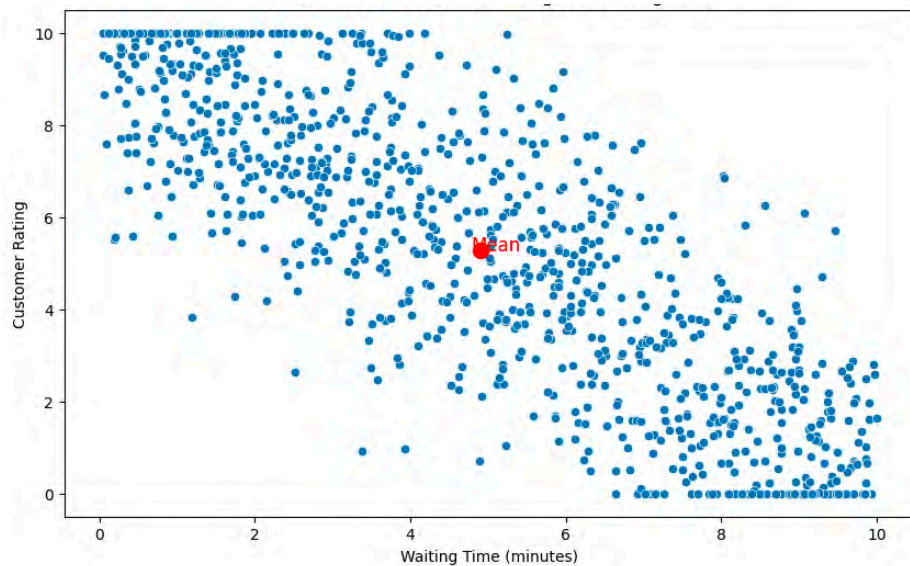
$$\text{Var}(X_4) = \text{Var}(Y_4) = 0.806$$

$$\text{Cov}(X, Y) = \sum p_{XY}(x_i, y_i) (x_i - \mu_x)(y_i - \mu_y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

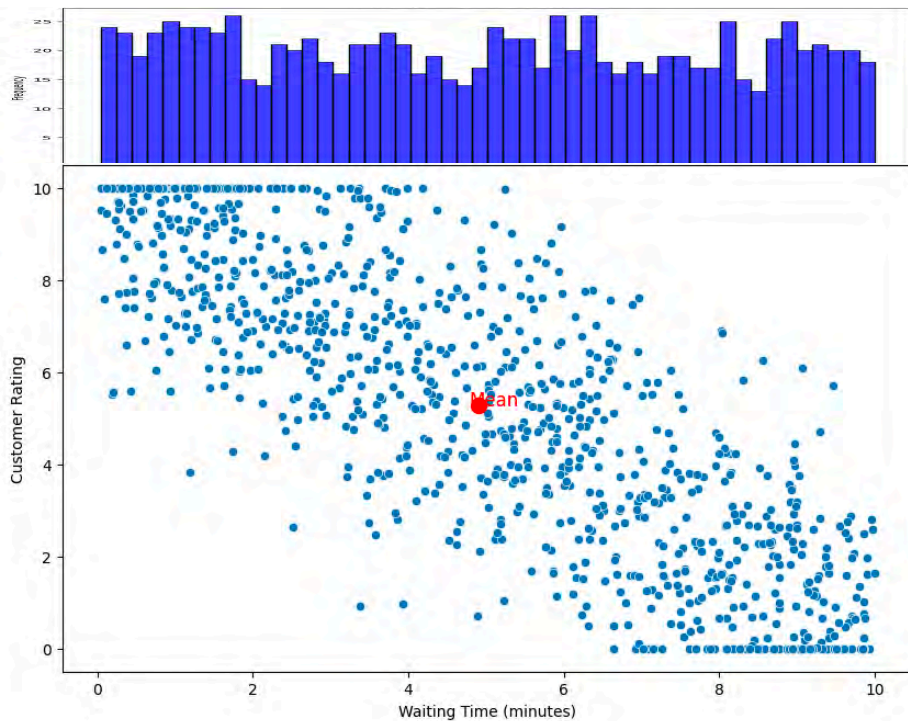
$$= \frac{1}{2} \left(1 - \frac{1}{6}\right)^2 + \frac{1}{6} \left(0 - \frac{1}{6}\right)^2 + \frac{1}{3} \left(-1 - \frac{1}{6}\right)^2$$

$$\text{Cov}(X, Y) = 0.806$$

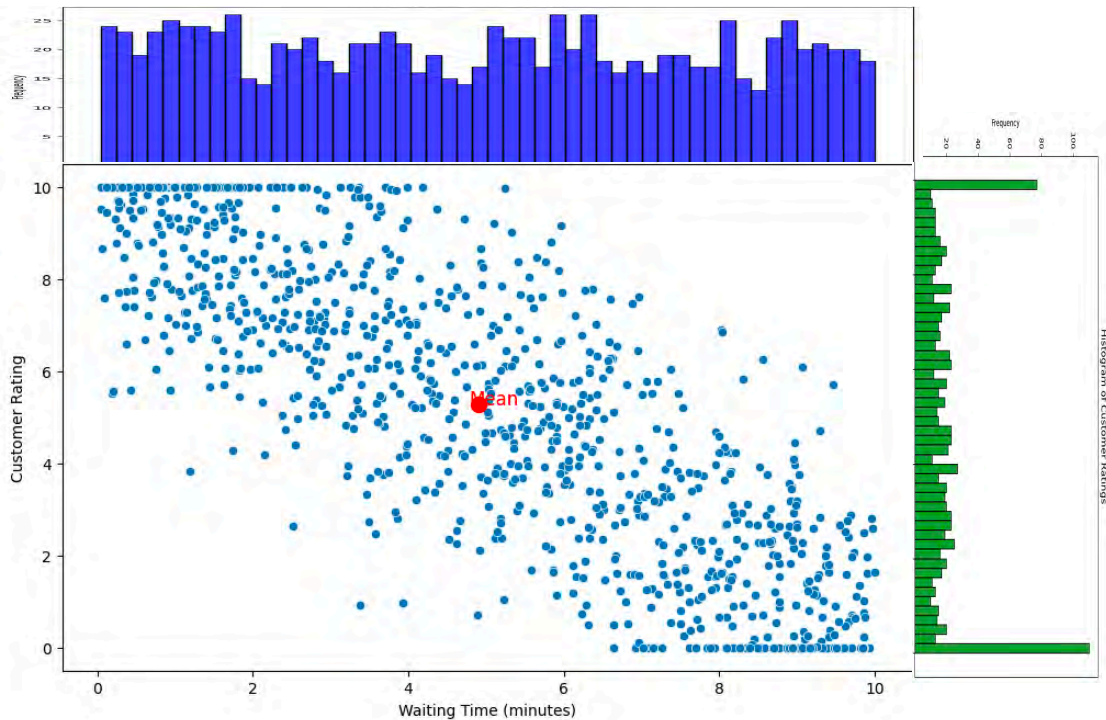
Covariance?



Covariance?

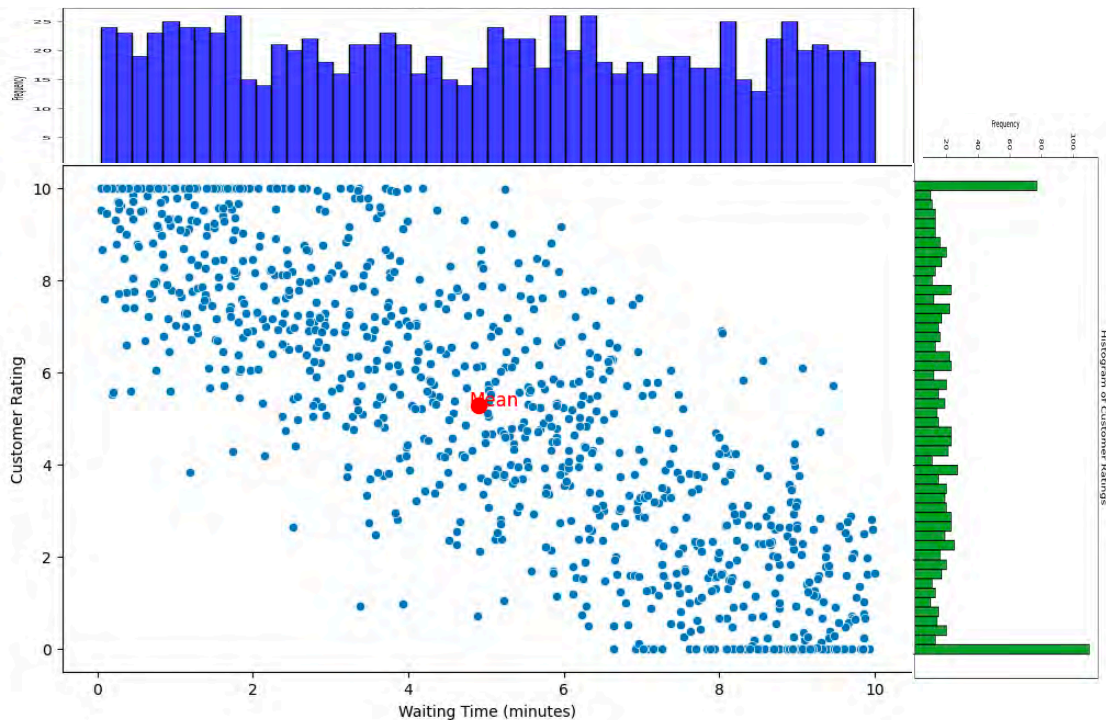


Covariance?



Covariance?

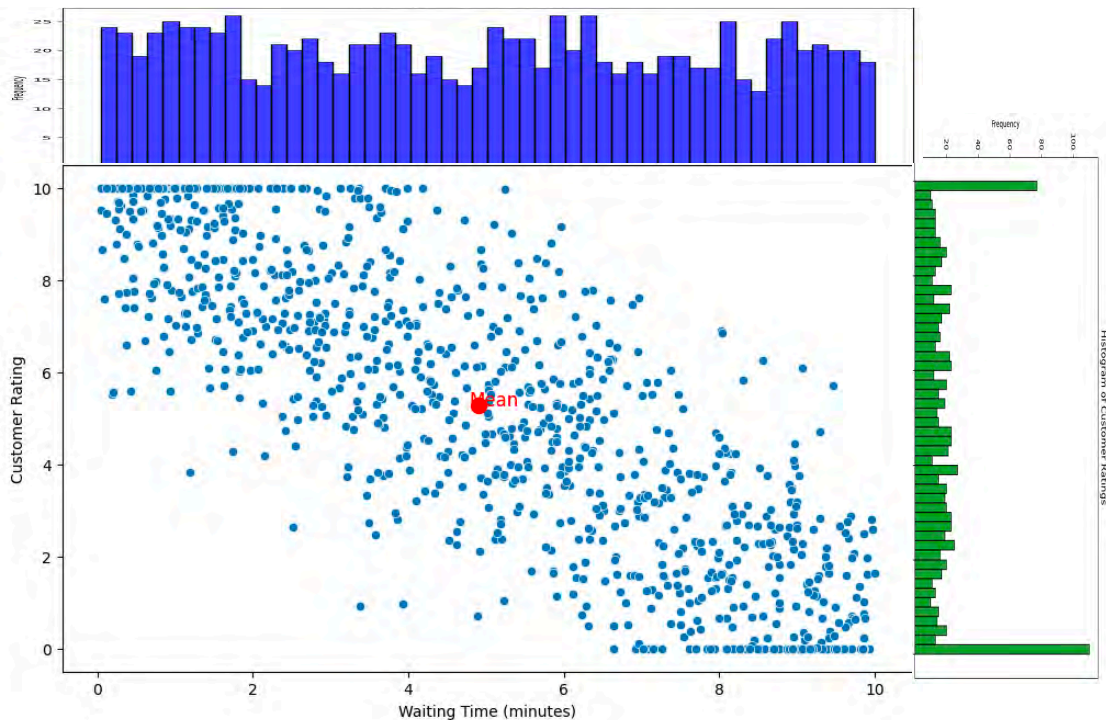
$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$



Covariance?

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

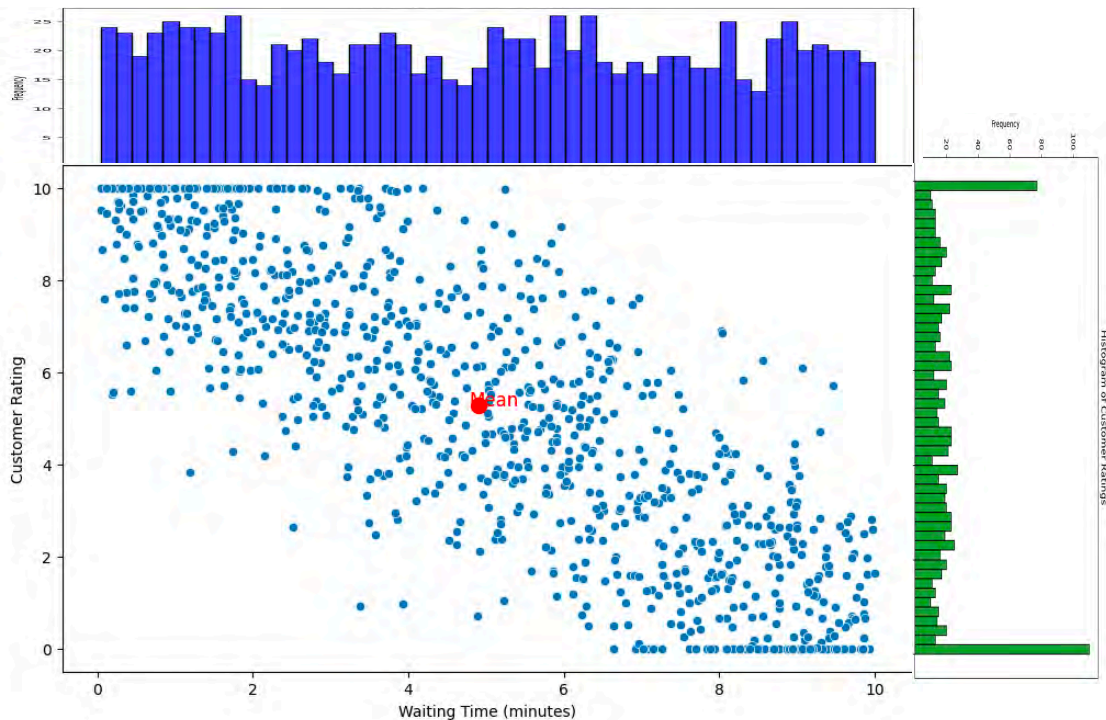


Covariance?

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

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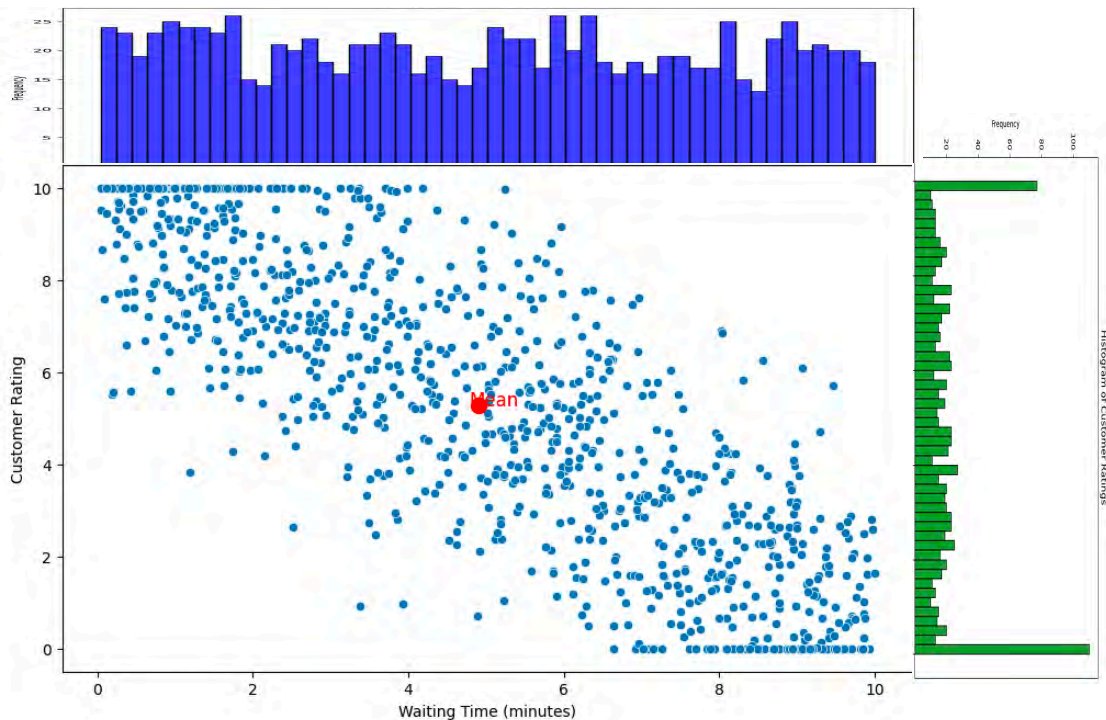
Covariance?

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\text{Cov}(X, Y) = -7.878$$



Covariance?

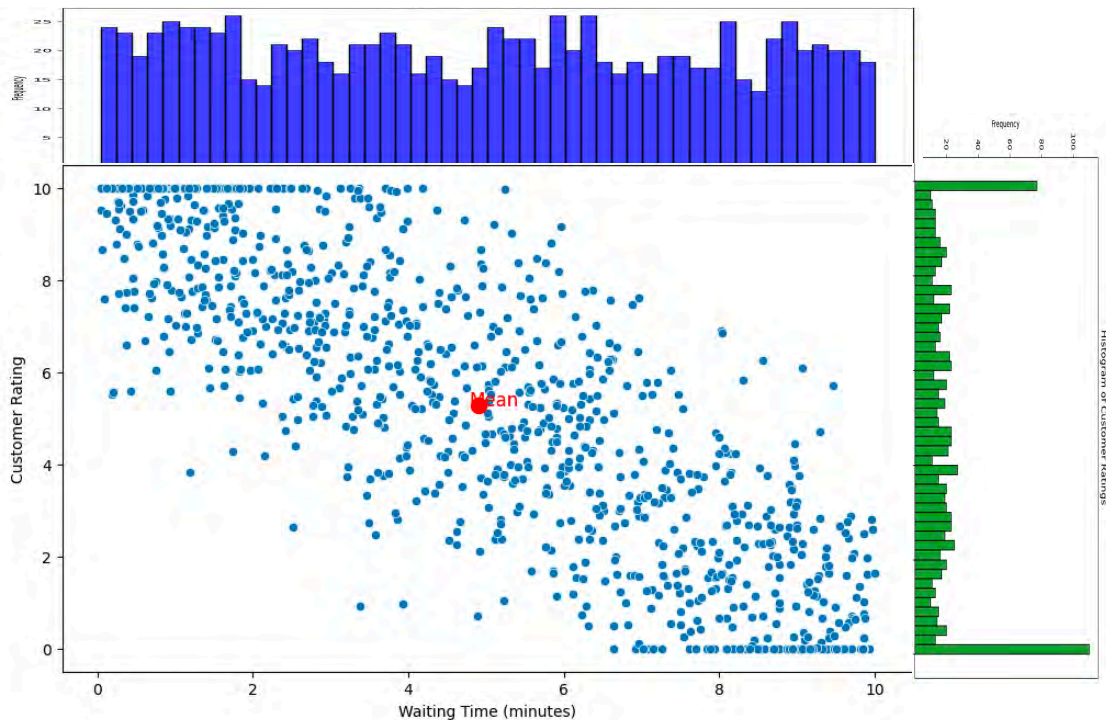
$$\mathbb{E}(X) = 4.903$$

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\text{Cov}(X, Y) = -7.878$$



Covariance?

$$\mathbb{E}(X) = 4.903$$

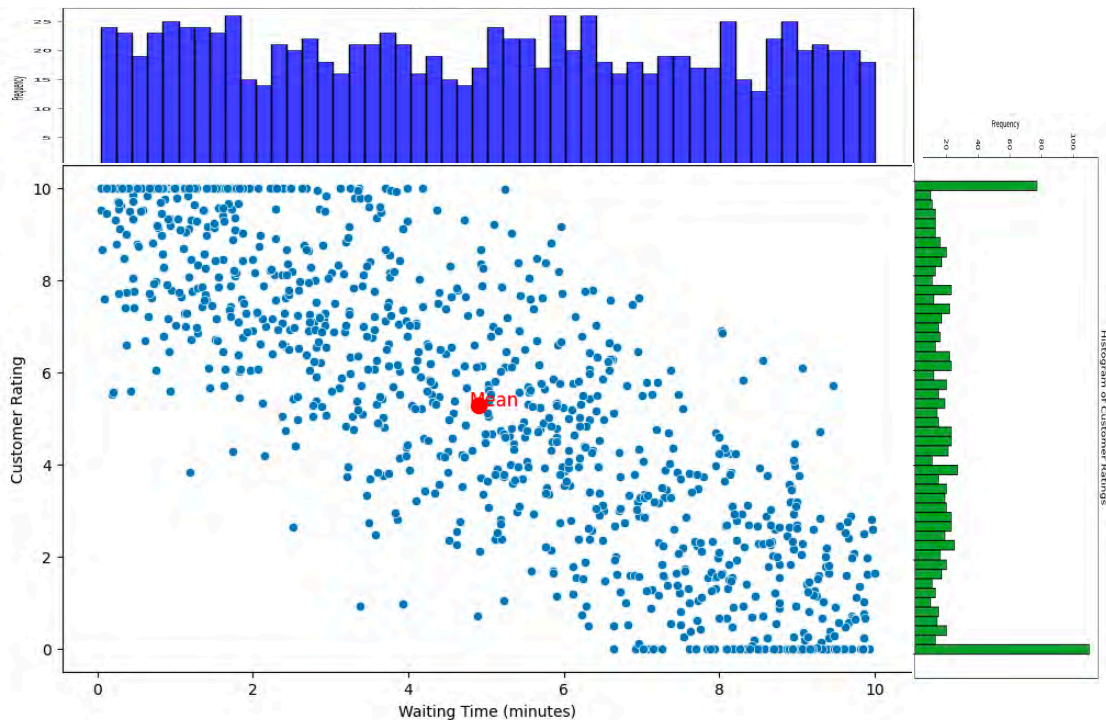
$$\mathbb{E}(Y) = 5.280$$

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\text{Cov}(X, Y) = -7.878$$



Covariance?

$$\mathbb{E}(X) = 4.903$$

$$\mathbb{E}(Y) = 5.280$$

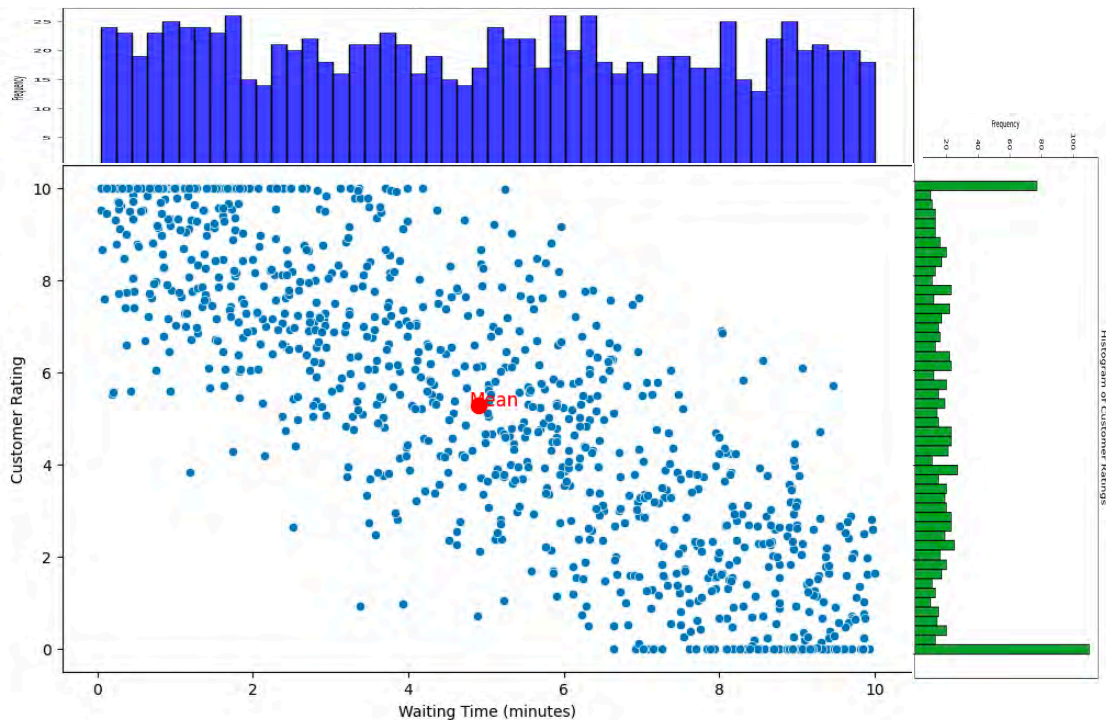
$$\mathbb{E}(XY) = 18.014$$

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\text{Cov}(X, Y) = -7.878$$



Covariance of a Joint Continuous Distribution

Covariance of a Joint Continuous Distribution

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

Covariance of a Joint Continuous Distribution

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

$$\mathbb{E}(X) = 4.903$$

Covariance of a Joint Continuous Distribution

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

$$\mathbb{E}(X) = 4.903$$

$$\mathbb{E}(Y) = 5.280$$

Covariance of a Joint Continuous Distribution

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

$$\mathbb{E}(X) = 4.903$$

$$\mathbb{E}(Y) = 5.280$$

$$\mathbb{E}(XY) = 18.014$$

Covariance of a Joint Continuous Distribution

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

$$\mathbb{E}(X) = 4.903$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\mathbb{E}(Y) = 5.280$$

$$\mathbb{E}(XY) = 18.014$$

Covariance of a Joint Continuous Distribution

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

$$\mathbb{E}(X) = 4.903$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\mathbb{E}(Y) = 5.280$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\mathbb{E}(XY) = 18.014$$

Covariance of a Joint Continuous Distribution

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

$$\mathbb{E}(X) = 4.903$$

$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\mathbb{E}(Y) = 5.280$$

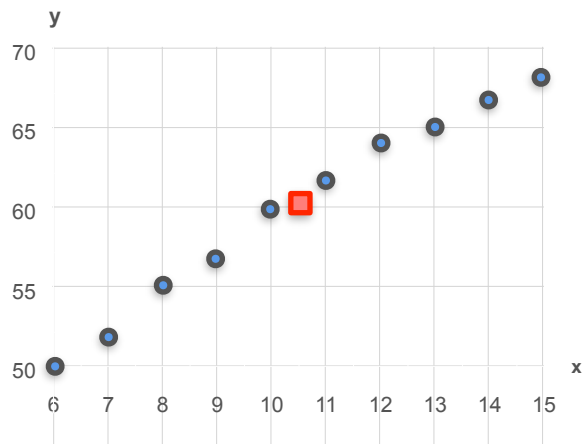
$$\text{Cov}(X, Y) = 18.014 - (4.903)(5.280)$$

$$\mathbb{E}(XY) = 18.014$$

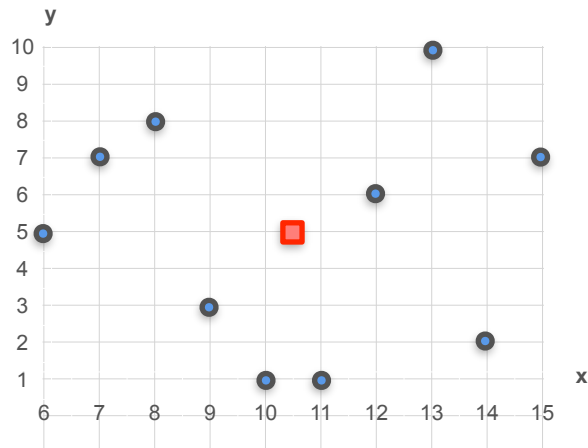
$$\text{Cov}(X, Y) = -7.878$$

Covariance Matrix

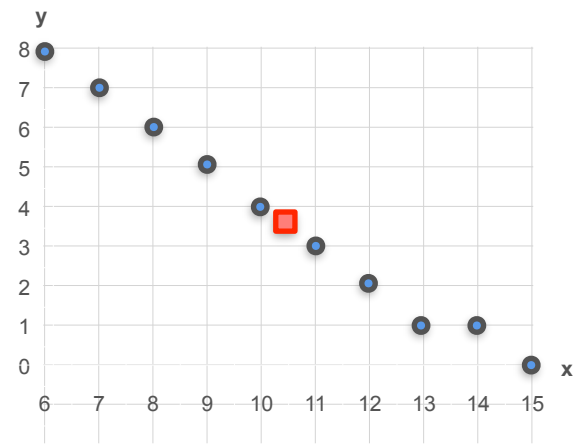
Covariance Matrix



Age vs Height

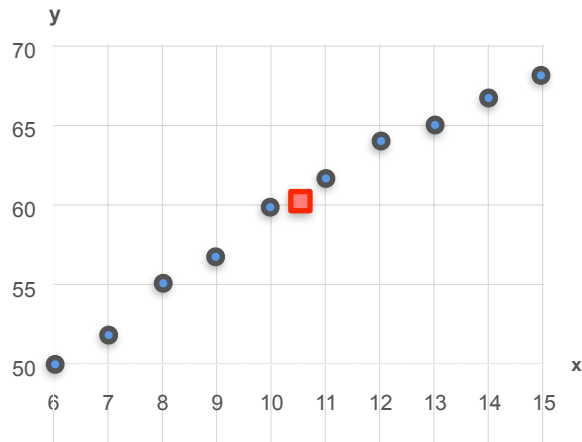


Age vs Grades



Age vs Naps per Day

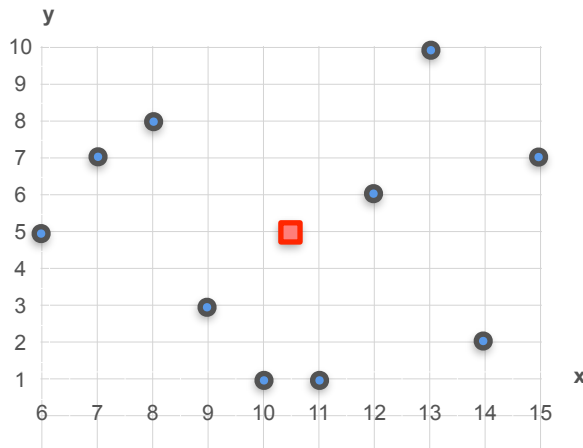
Covariance Matrix



Age vs Height

$$\text{Var}(X) = 9.17$$

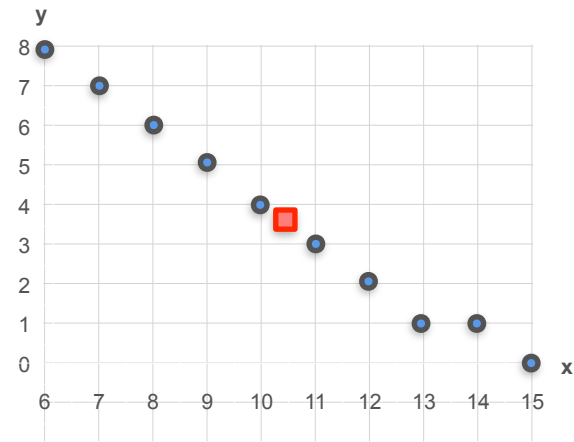
$$\text{Var}(Y) = 39.56$$



Age vs Grades

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 9.78$$

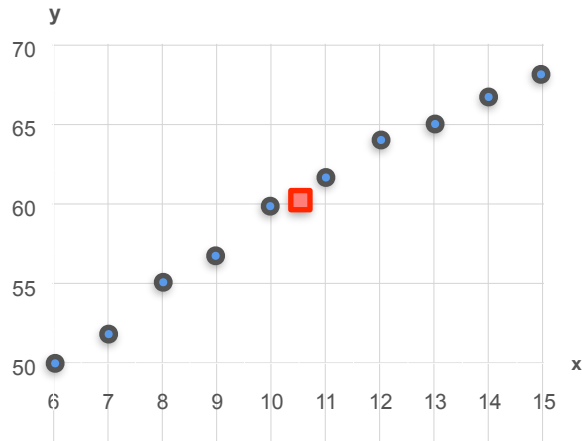


Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

Covariance Matrix

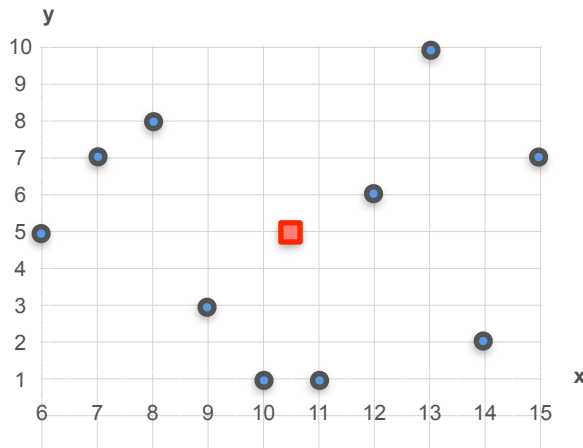


Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$

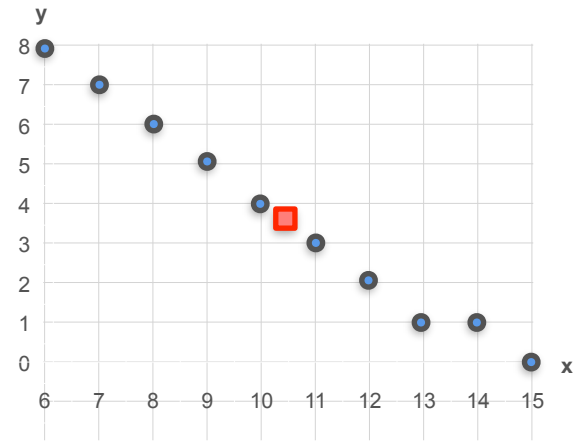


Age vs Grades

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 9.78$$

$$\text{Cov}(X, Y) = 0.1$$



Age vs Naps per Day

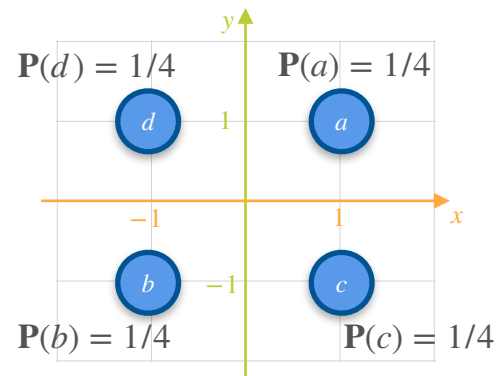
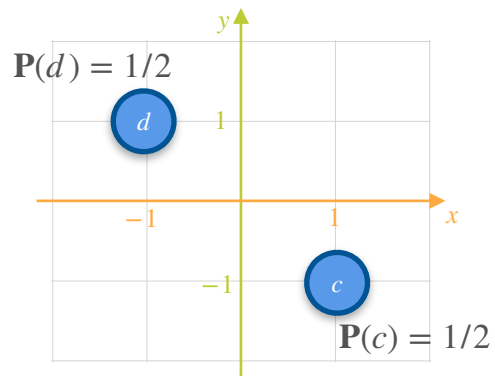
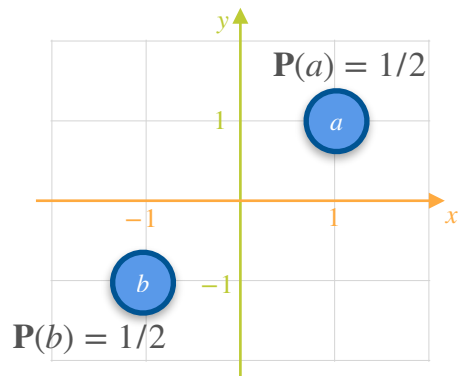
$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

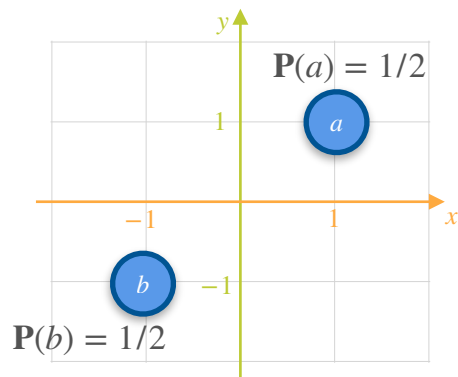
$$\text{Cov}(X, Y) = -7.45$$

Covariance Matrix

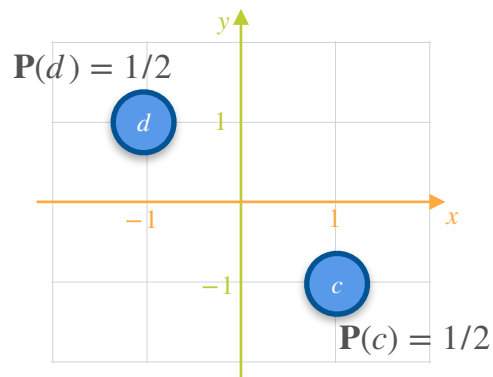
Covariance Matrix



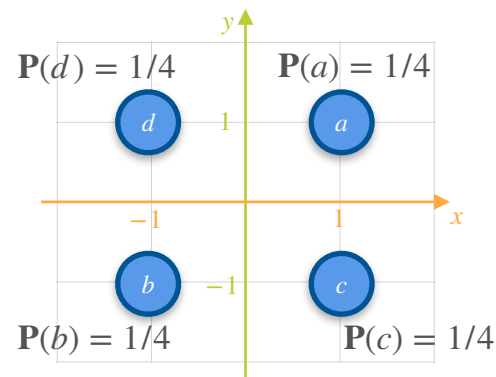
Covariance Matrix



$$\text{Var}(X) = \text{Var}(Y) = 1$$

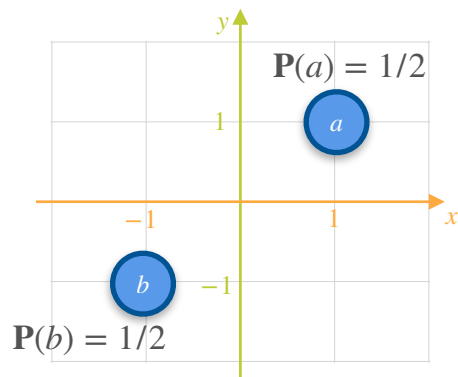


$$\text{Var}(X) = \text{Var}(Y) = 1$$



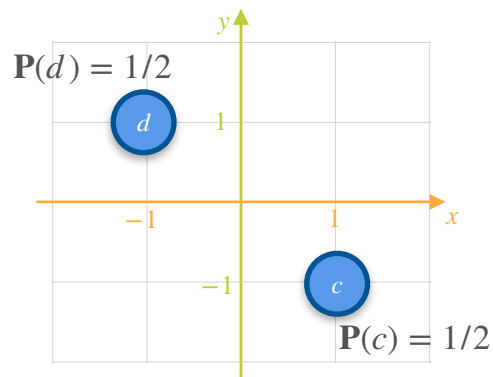
$$\text{Var}(X) = \text{Var}(Y) = 1$$

Covariance Matrix



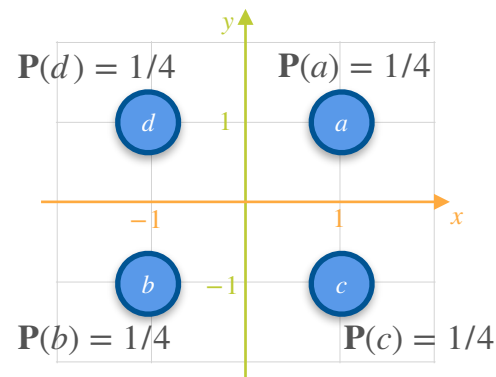
$$\text{Var}(X) = \text{Var}(Y) = 1$$

$$\text{Cov}(X, Y) = 1$$



$$\text{Var}(X) = \text{Var}(Y) = 1$$

$$\text{Cov}(X, Y) = -1$$

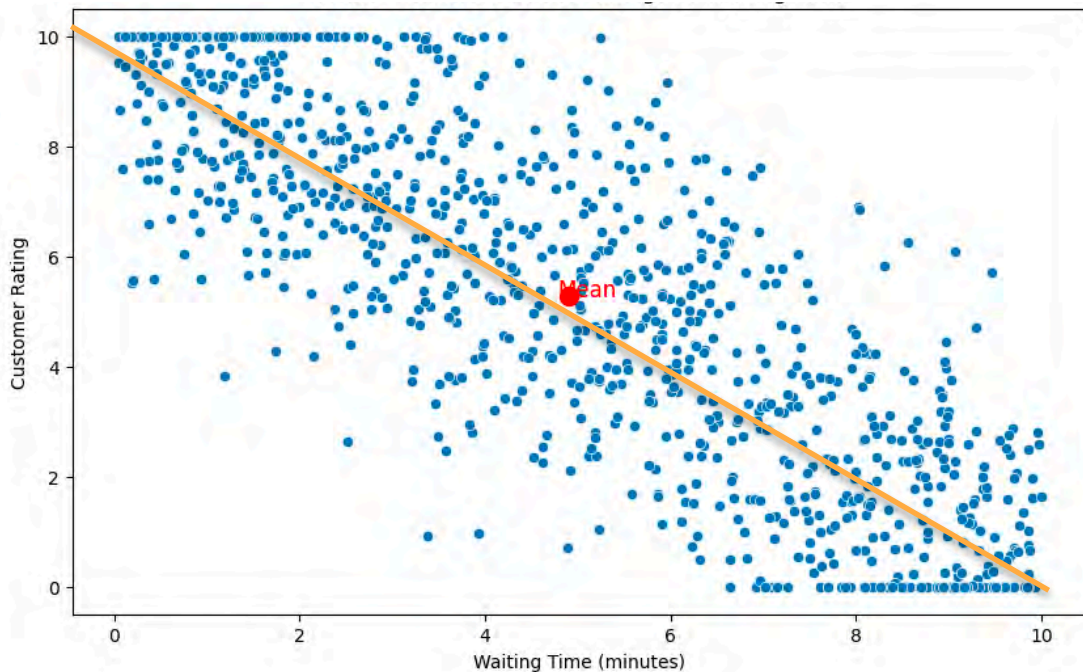


$$\text{Var}(X) = \text{Var}(Y) = 1$$

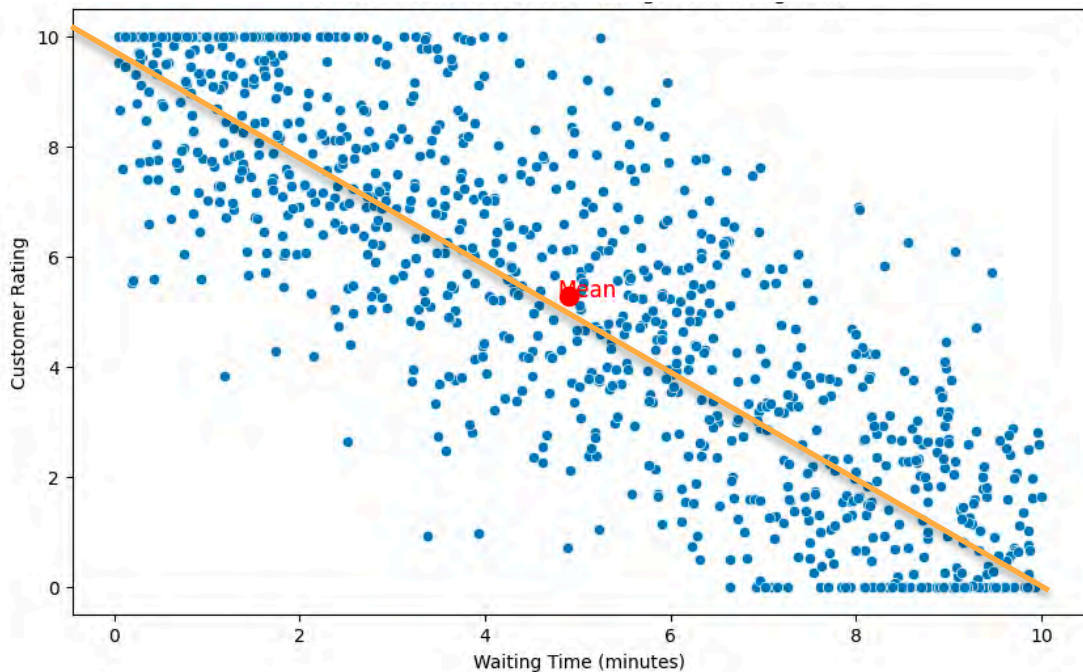
$$\text{Cov}(X, Y) = 0$$

Covariance Matrix

Covariance Matrix



Covariance Matrix

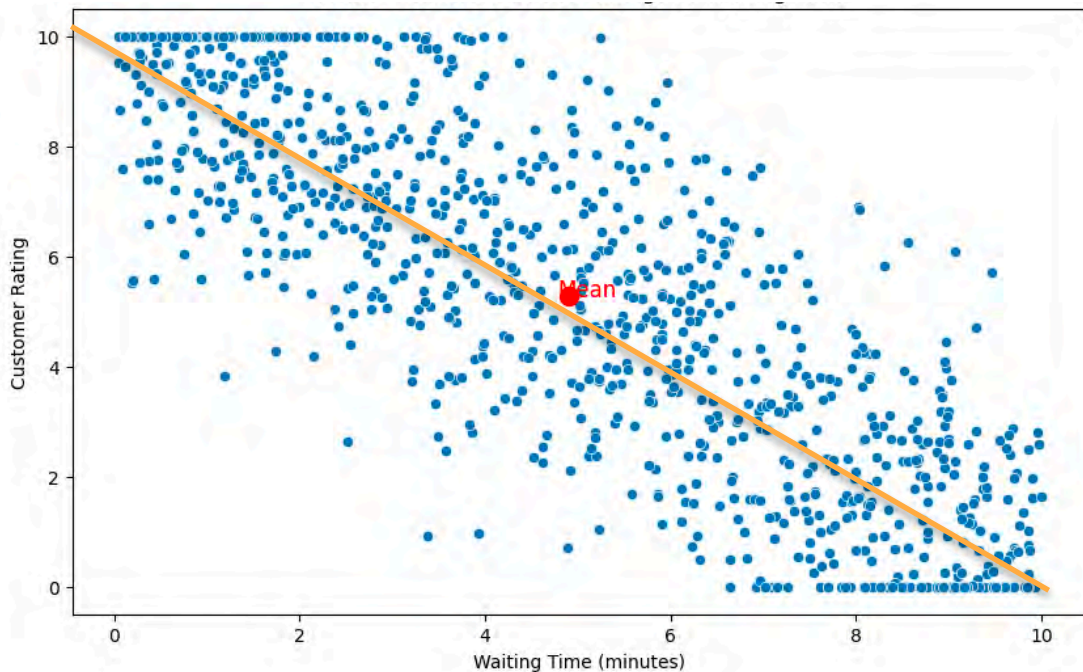


$$\text{Var}(X) = 8.526$$

$$\text{Var}(Y) = 10.163$$

$$\text{Cov}(X, Y) = -7.878$$

Covariance Matrix

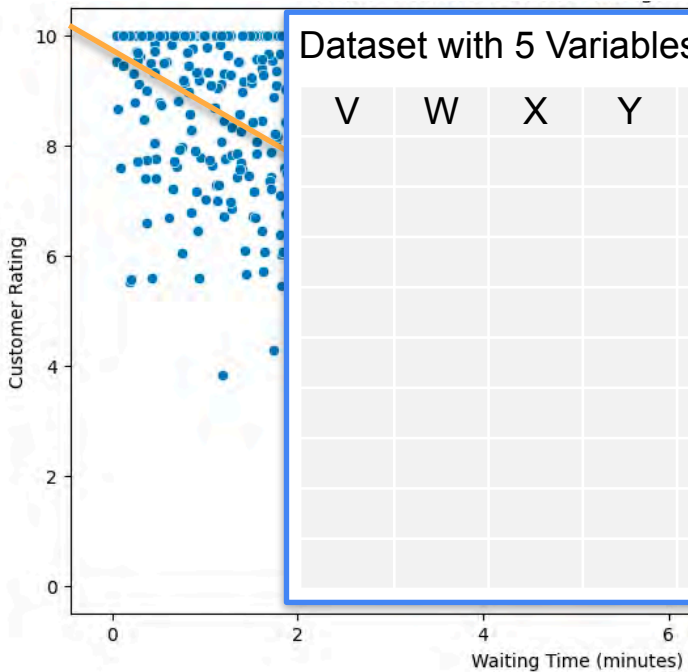


$$\text{Var}(X) = 8.526$$

$$\text{Var}(Y) = 10.163$$

$$\text{Cov}(X, Y) = -7.878$$

Covariance Matrix



Dataset with 5 Variables:

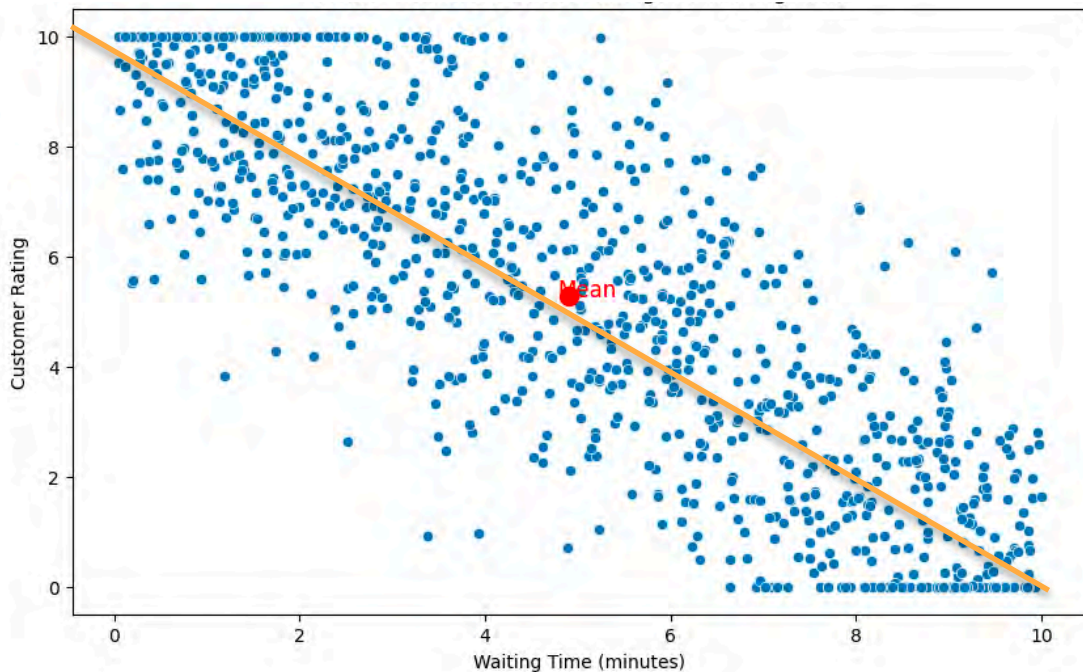
V	W	X	Y	Z	$Var(V)$	$Cov(V, W)$
					$Var(W)$	$Cov(V, X)$
					$Var(X)$	$Cov(V, Y)$
					$Var(Y)$	$Cov(V, Z)$
					$Var(Z)$	$Cov(W, X)$
						$\vdots$
						$Cov(Y, Z)$

$$ar(X) = 8.526$$

$$\bar{ar}(Y) = 10.163$$

$$\text{cov}(X, Y) = -7.878$$

Covariance Matrix

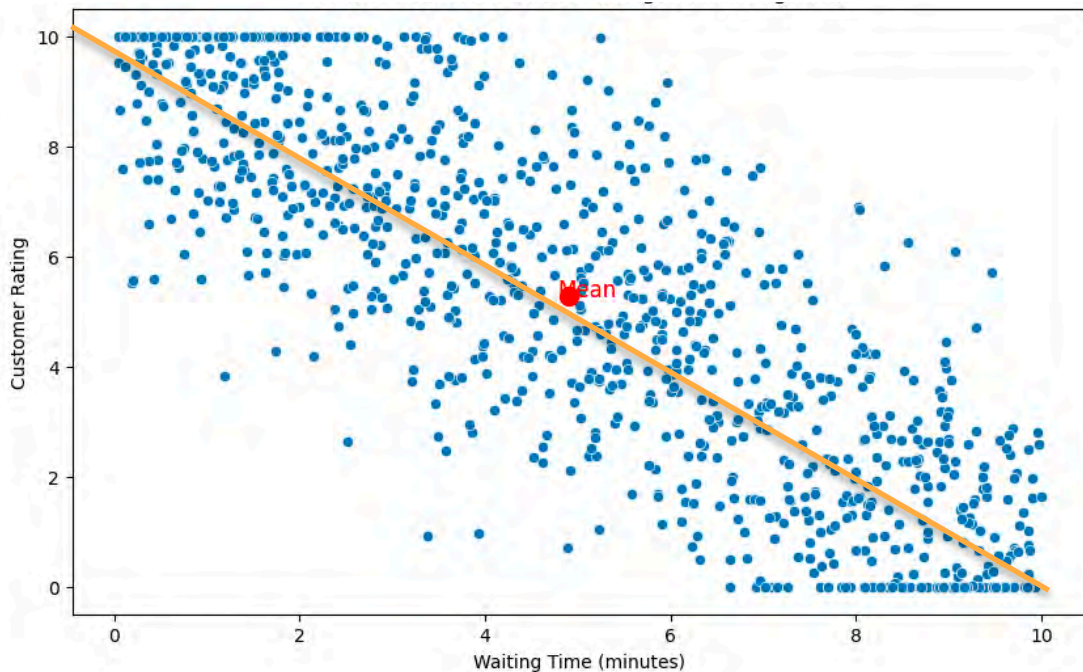


$$\text{Var}(X) = 8.526$$

$$\text{Var}(Y) = 10.163$$

$$\text{Cov}(X, Y) = -7.878$$

Covariance Matrix



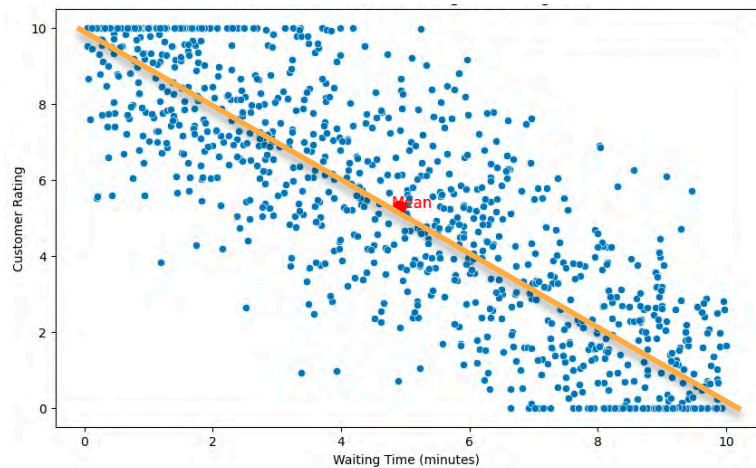
$$\text{Var}(X) = 8.526$$

$$\text{Var}(Y) = 10.163$$

$$\text{Cov}(X, Y) = -7.878$$

Covariance Matrix

Covariance Matrix

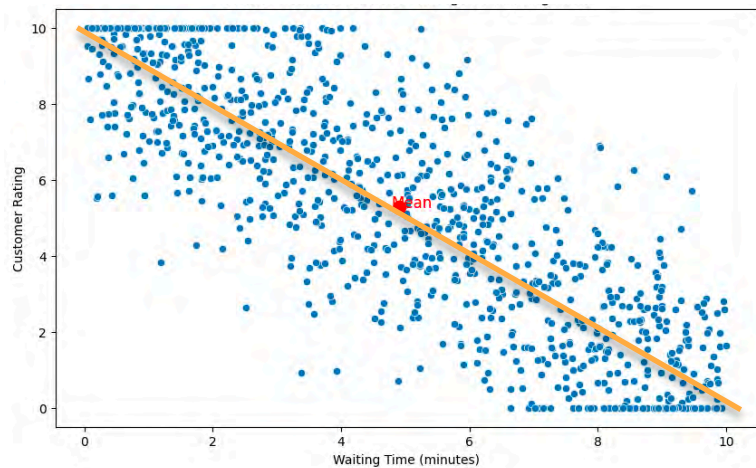


$$\text{Var}(X) = 8.526$$

$$\text{Var}(Y) = 10.163$$

$$\text{Cov}(X, Y) = -7.878$$

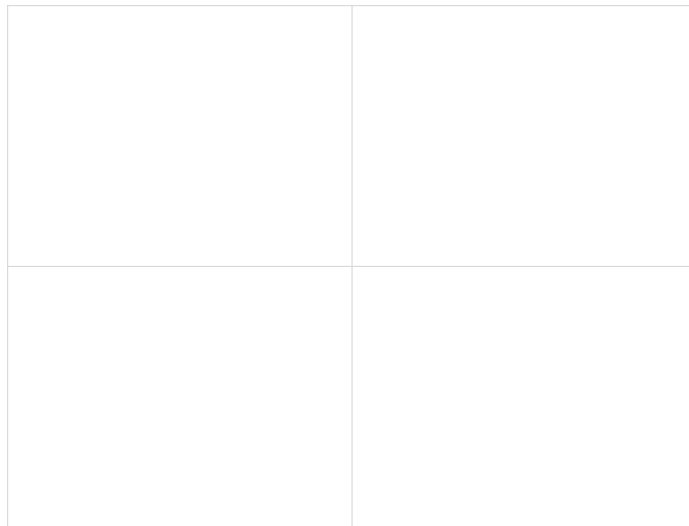
Covariance Matrix



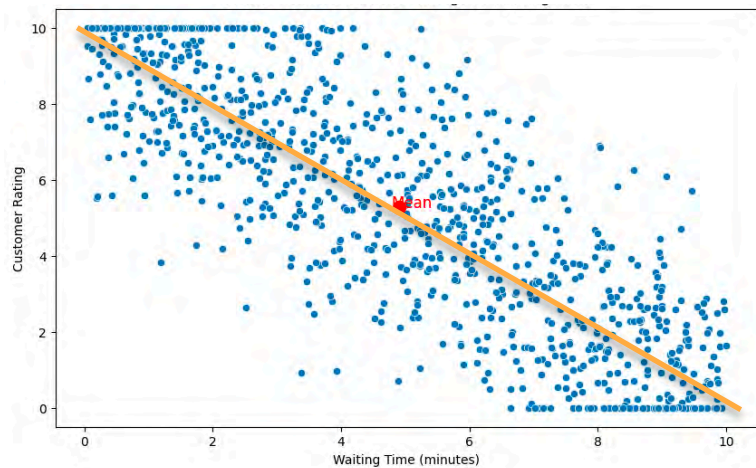
$$\text{Var}(X) = 8.526$$

$$\text{Var}(Y) = 10.163$$

$$\text{Cov}(X, Y) = -7.878$$



Covariance Matrix



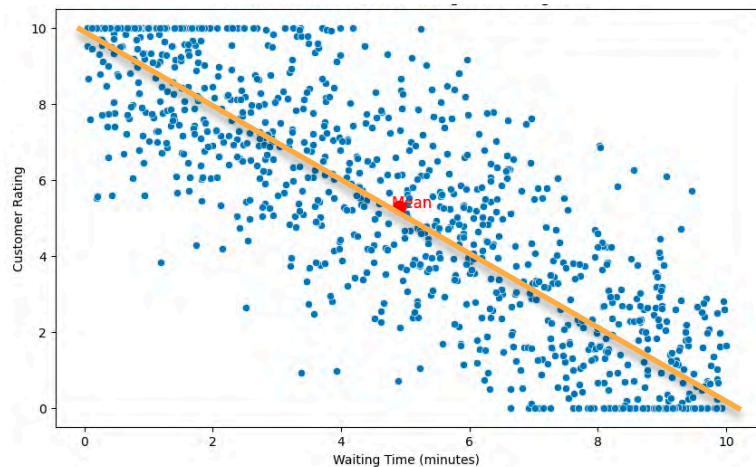
$$\text{Var}(X) = 8.526$$

$$\text{Var}(Y) = 10.163$$

$$\text{Cov}(X, Y) = -7.878$$

	X	Y
X		
Y		

Covariance Matrix



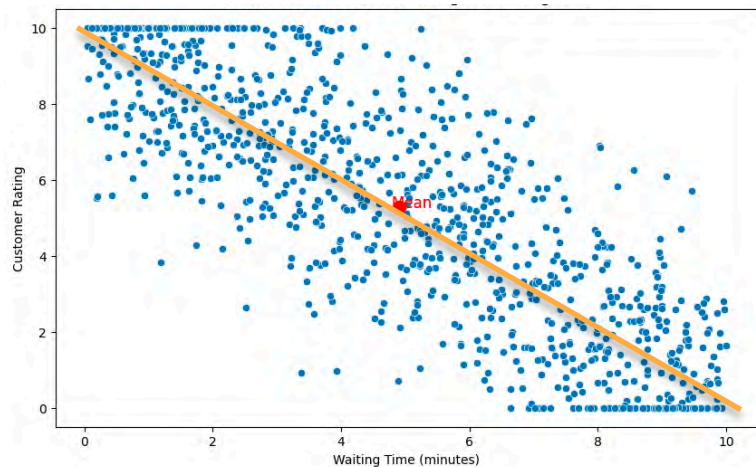
$$Var(X) = 8.526$$

$$Var(Y) = 10.163$$

$$Cov(X, Y) = -7.878$$

	X	Y
X	$Var(X)$	
Y		$Var(Y)$

Covariance Matrix



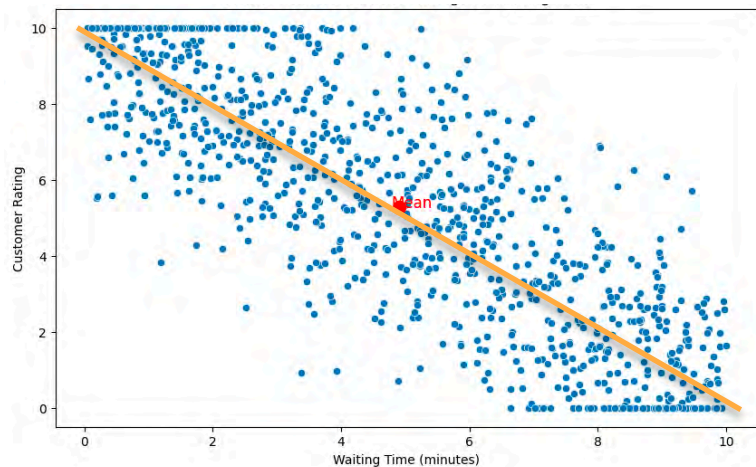
$$Var(X) = 8.526$$

$$Var(Y) = 10.163$$

$$Cov(X, Y) = -7.878$$

	X	Y
X	$Var(X)$	$Cov(X, Y)$
Y	$Cov(X, Y)$	$Var(Y)$

Covariance Matrix



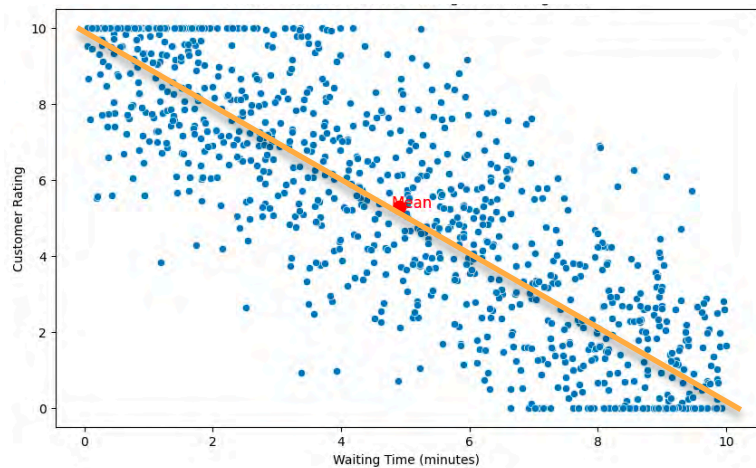
$$Var(X) = 8.526$$

$$Var(Y) = 10.163$$

$$Cov(X, Y) = -7.878$$

	X	Y
X	$Var(X)$	$Cov(X, Y)$
Y	$Cov(X, Y)$	$Var(Y)$

Covariance Matrix



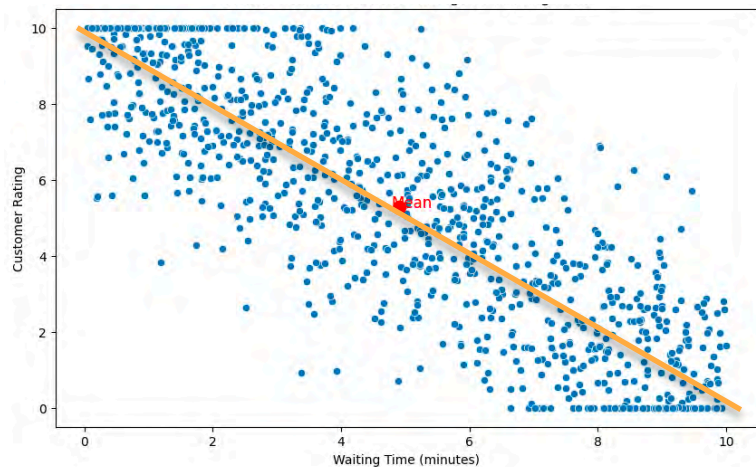
	X	Y
X	$Var(X)$	$Cov(X, Y)$
Y	$Cov(X, Y)$	$Var(Y)$

$$Var(X) = 8.526$$

$$Var(Y) = 10.163$$

$$Cov(X, Y) = -7.878$$

Covariance Matrix



$$Var(X) = 8.526$$

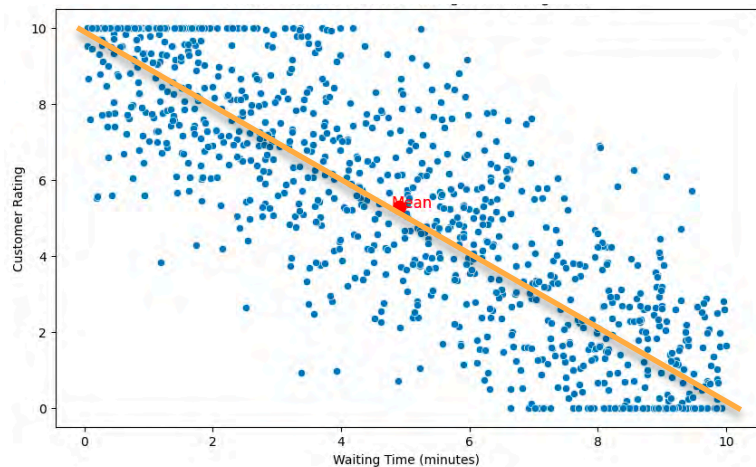
$$Var(Y) = 10.163$$

$$Cov(X, Y) = -7.878$$

	X	Y
X	$Var(X)$	$Cov(X, Y)$
Y	$Cov(X, Y)$	$Var(Y)$

$$\begin{bmatrix} Var(X) & Cov(X, Y) \\ Cov(X, Y) & Var(Y) \end{bmatrix}$$

Covariance Matrix



$$Var(X) = 8.526$$

$$Var(Y) = 10.163$$

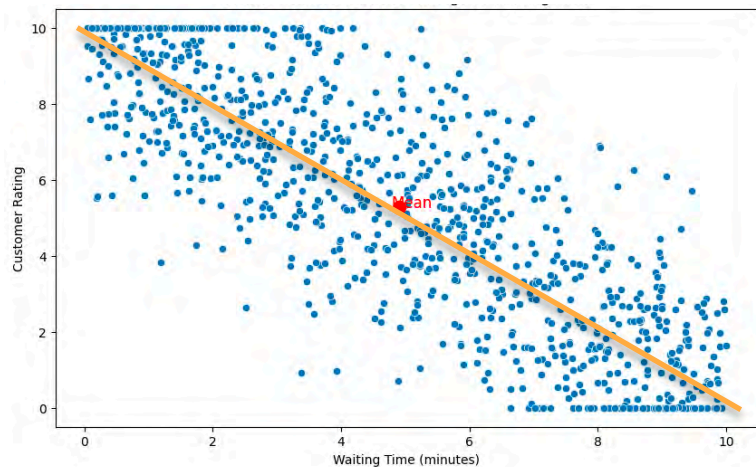
$$Cov(X, Y) = -7.878$$

	X	Y
X	$Var(X)$	$Cov(X, Y)$
Y	$Cov(X, Y)$	$Var(Y)$

$$\begin{bmatrix} Var(X) & Cov(X, Y) \\ Cov(X, Y) & Var(Y) \end{bmatrix}$$

8.526	-7.878
-7.878	10.163

Covariance Matrix



$$Var(X) = 8.526$$

$$Var(Y) = 10.163$$

$$Cov(X, Y) = -7.878$$

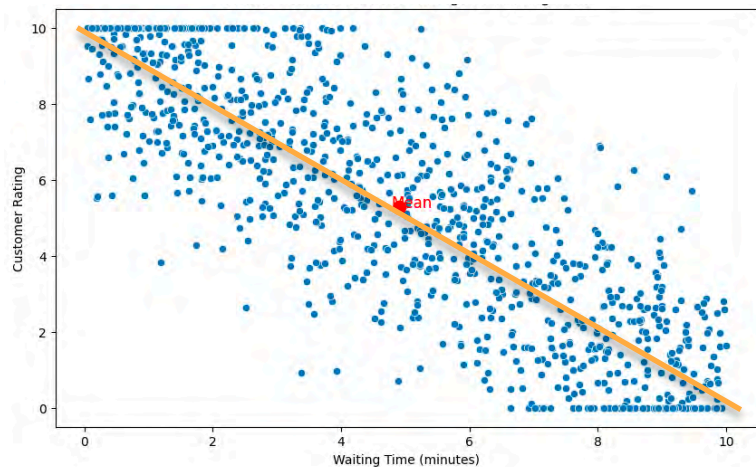
	X	Y
X	$Var(X)$	$Cov(X, Y)$
Y	$Cov(X, Y)$	$Var(Y)$

8.526	-7.878
-7.878	10.163

$$\begin{bmatrix} Var(X) & Cov(X, Y) \\ Cov(X, Y) & Var(Y) \end{bmatrix}$$

$$\begin{bmatrix} 8.534 & -7.878 \\ -7.878 & 10.173 \end{bmatrix}$$

Covariance Matrix



$$Var(X) = 8.526$$

$$Var(Y) = 10.163$$

$$Cov(X, Y) = -7.878$$

	X	Y
X	$Var(X)$	$Cov(X, Y)$
Y	$Cov(X, Y)$	$Var(Y)$

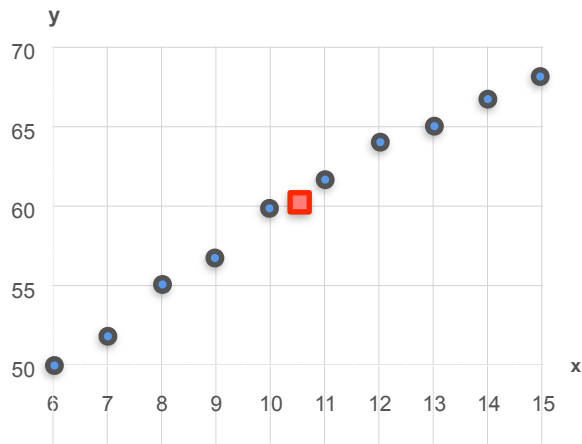
8.526	-7.878
-7.878	10.163

$$\begin{bmatrix} Var(X) & Cov(X, Y) \\ Cov(X, Y) & Var(Y) \end{bmatrix}$$

$$\begin{bmatrix} 8.534 & -7.878 \\ -7.878 & 10.173 \end{bmatrix}$$

Covariance Matrix

Covariance Matrix



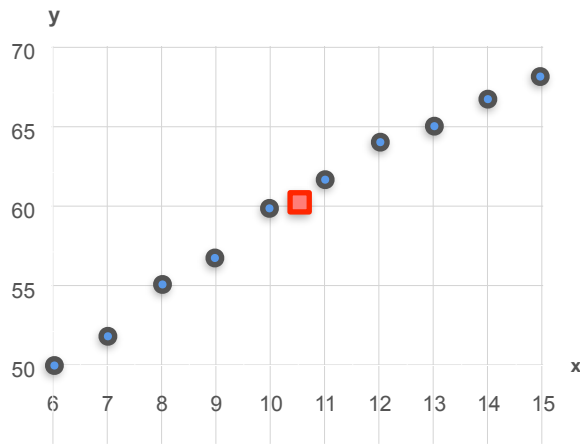
Age vs Height

$$Var(X) = 9.17$$

$$Var(Y) = 39.56$$

$$Cov(X, Y) = 17$$

Covariance Matrix



Age vs Height

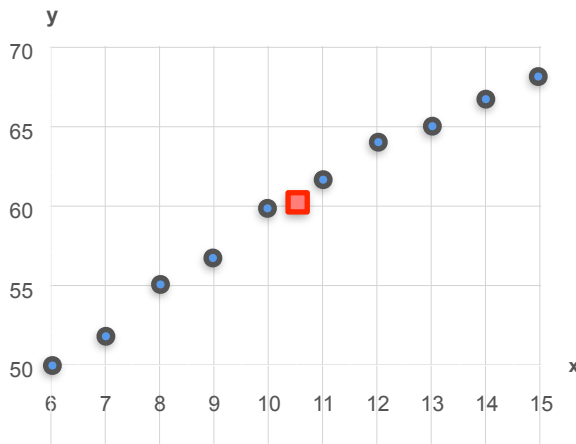
$$Var(X) = 9.17$$

$$Var(Y) = 39.56$$

$$Cov(X, Y) = 17$$

$$\begin{bmatrix} Var(X) & Cov(X, Y) \\ Cov(X, Y) & Var(Y) \end{bmatrix}$$

Covariance Matrix



Age vs Height

$$Var(X) = 9.17$$

$$Var(Y) = 39.56$$

$$Cov(X, Y) = 17$$

$$\begin{bmatrix} Var(X) & Cov(X, Y) \\ Cov(X, Y) & Var(Y) \end{bmatrix}$$

$$\begin{bmatrix} 9.17 & 17 \\ 17 & 39.56 \end{bmatrix}$$

Covariance Matrix

$$\text{Var}(X) = 1$$

$$\text{Var}(Y) = 1$$

$$\text{Cov}(X, Y) = 1$$

$$\text{Var}(X) = 1$$

$$\text{Var}(Y) = 1$$

$$\text{Cov}(X, Y) = -1$$

Covariance Matrix

$$\begin{bmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{bmatrix}$$

$$\text{Var}(X) = 1$$

$$\text{Var}(Y) = 1$$

$$\text{Cov}(X, Y) = 1$$

$$\text{Var}(X) = 1$$

$$\text{Var}(Y) = 1$$

$$\text{Cov}(X, Y) = -1$$

Covariance Matrix

$$\begin{bmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{bmatrix}$$

$$\begin{array}{l} \text{Var}(X) = 1 \\ \text{Var}(Y) = 1 \\ \text{Cov}(X, Y) = 1 \end{array} \quad \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{array}{l} \text{Var}(X) = 1 \\ \text{Var}(Y) = 1 \\ \text{Cov}(X, Y) = -1 \end{array}$$

Covariance Matrix

$$\begin{bmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{bmatrix}$$

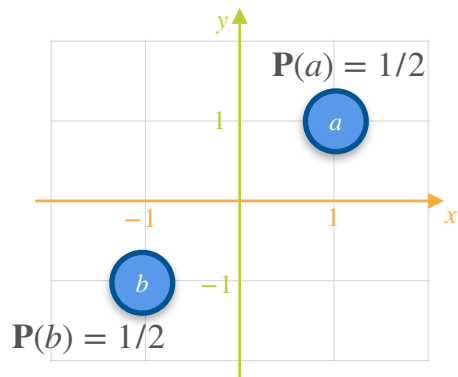
$$\begin{aligned} \text{Var}(X) &= 1 \\ \text{Var}(Y) &= 1 \\ \text{Cov}(X, Y) &= 1 \end{aligned}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

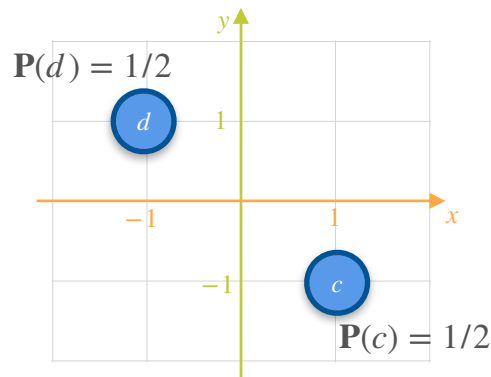
$$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\begin{aligned} \text{Var}(X) &= 1 \\ \text{Var}(Y) &= 1 \\ \text{Cov}(X, Y) &= -1 \end{aligned}$$

Covariance Matrix



$$\begin{bmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{bmatrix}$$



$$\begin{aligned} \text{Var}(X) &= 1 \\ \text{Var}(Y) &= 1 \\ \text{Cov}(X, Y) &= 1 \end{aligned} \quad \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\begin{aligned} \text{Var}(X) &= 1 \\ \text{Var}(Y) &= 1 \\ \text{Cov}(X, Y) &= -1 \end{aligned}$$

Covariance of a Joint Continuous Distribution

Covariance of a Joint Continuous Distribution

Dataset with 3 Variables:

X	Y	Z	$Var(X)$	$Cov(X, Y)$
			$Var(Y)$	$Cov(X, Z)$
			$Var(Z)$	$Cov(Y, Z)$

Covariance of a Joint Continuous Distribution

Dataset with 3 Variables:

X	Y	Z	$Var(X)$	$Cov(X, Y)$
			$Var(Y)$	$Cov(X, Z)$
			$Var(Z)$	$Cov(Y, Z)$

X Y Z

X

Y

Z

Covariance of a Joint Continuous Distribution

Dataset with 3 Variables:

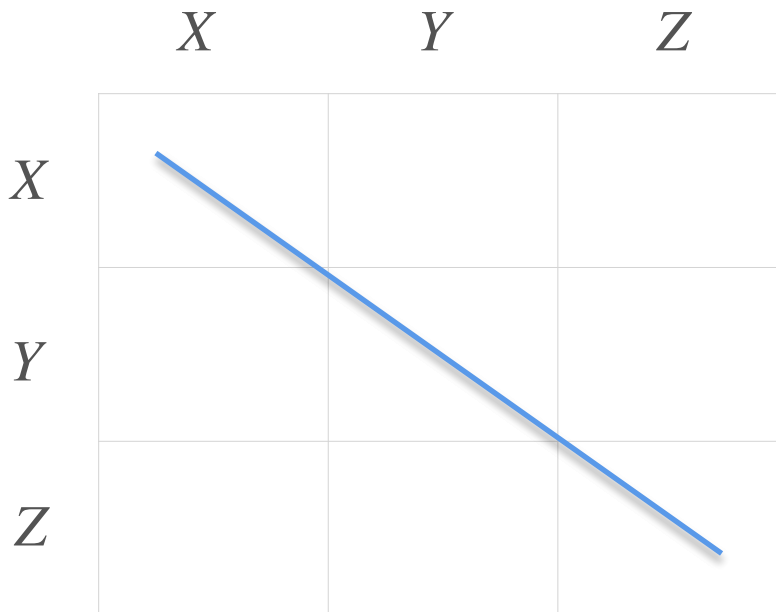
X	Y	Z	$Var(X)$	$Cov(X, Y)$
			$Var(Y)$	$Cov(X, Z)$
			$Var(Z)$	$Cov(Y, Z)$

	X	Y	Z
X			
Y			
Z			

Covariance of a Joint Continuous Distribution

Dataset with 3 Variables:

X	Y	Z	$Var(X)$	$Cov(X, Y)$
			$Var(Y)$	$Cov(X, Z)$
			$Var(Z)$	$Cov(Y, Z)$



Covariance of a Joint Continuous Distribution

Dataset with 3 Variables:

X	Y	Z	$Var(X)$	$Cov(X, Y)$
			$Var(Y)$	$Cov(X, Z)$
			$Var(Z)$	$Cov(Y, Z)$

	X	Y	Z
X	$Var(X)$		
Y		$Var(Y)$	
Z			$Var(Z)$

Covariance of a Joint Continuous Distribution

Dataset with 3 Variables:

X	Y	Z	$Var(X)$	$Cov(X, Y)$
			$Var(Y)$	$Cov(X, Z)$
			$Var(Z)$	$Cov(Y, Z)$

	X	Y	Z
X	$Var(X)$	$Cov(X, Y)$	$Cov(X, Z)$
Y	$Cov(X, Y)$	$Var(Y)$	$Cov(Y, Z)$
Z	$Cov(X, Z)$	$Cov(Y, Z)$	$Var(Z)$

Covariance of a Joint Continuous Distribution

Covariance of a Joint Continuous Distribution

Dataset with 5 Variables:

V	W	X	Y	Z	$Var(V)$	$Cov(V, W)$
					$Var(W)$	$Cov(V, X)$
					$Var(X)$	$Cov(V, Y)$
					$Var(Y)$	$Cov(V, Z)$
					$Var(Z)$	$Cov(W, X)$
						$\vdots$
						$Cov(Y, Z)$

Covariance of a Joint Continuous Distribution

Dataset with 5 Variables:

V	W	X	Y	Z	$Var(V)$	$Cov(V, W)$
					$Var(W)$	$Cov(V, X)$
					$Var(X)$	$Cov(V, Y)$
					$Var(Y)$	$Cov(V, Z)$
					$Var(Z)$	$Cov(W, X)$
						$\vdots$
						$Cov(Y, Z)$

	V	W	X	Y	Z
V					
W					
X					
Y					
Z					

Covariance of a Joint Continuous Distribution

Dataset with 5 Variables:

V	W	X	Y	Z

$Var(V)$ $Cov(V, W)$

$Var(W)$ $Cov(V, X)$

$Var(X)$ $Cov(V, Y)$

$Var(Y)$ $Cov(V, Z)$

$Var(Z)$ $Cov(W, X)$

$\vdots$

$Cov(Y, Z)$

	V	W	X	Y	Z
V					
W					
X					
Y					
Z					

Covariance of a Joint Continuous Distribution

Dataset with 5 Variables:

V	W	X	Y	Z	$Var(V)$	$Cov(V, W)$
					$Var(W)$	$Cov(V, X)$
					$Var(X)$	$Cov(V, Y)$
					$Var(Y)$	$Cov(V, Z)$
					$Var(Z)$	$Cov(W, X)$
						$\vdots$
						$Cov(Y, Z)$

	V	W	X	Y	Z
V	$Var(V)$				
W		$Var(W)$			
X			$Var(X)$		
Y				$Var(Y)$	
Z					$Var(Z)$

Covariance of a Joint Continuous Distribution

Dataset with 5 Variables:

V	W	X	Y	Z	$Var(V)$	$Cov(V, W)$
					$Var(W)$	$Cov(V, X)$
					$Var(X)$	$Cov(V, Y)$
					$Var(Y)$	$Cov(V, Z)$
					$Var(Z)$	$Cov(W, X)$
						$\vdots$
						$Cov(Y, Z)$

	V	W	X	Y	Z
V	$Var(V)$	$Cov(V, W)$	$Cov(V, X)$	$Cov(V, Y)$	$Cov(V, Z)$
W	$Cov(V, W)$	$Var(W)$	$Cov(W, X)$	$Cov(W, Y)$	$Cov(W, Z)$
X	$Cov(V, X)$	$Cov(W, X)$	$Var(X)$	$Cov(X, Y)$	$Cov(X, Z)$
Y	$Cov(V, Y)$	$Cov(W, Y)$	$Cov(X, Y)$	$Var(Y)$	$Cov(Y, Z)$
Z	$Cov(V, Z)$	$Cov(W, Z)$	$Cov(X, Z)$	$Cov(Y, Z)$	$Var(Z)$

Covariance of a Joint Continuous Distribution

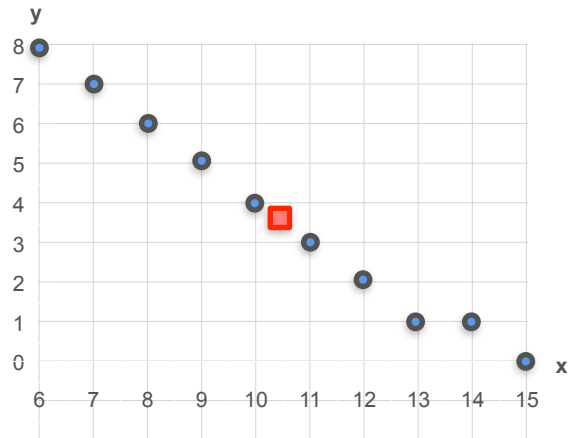
$$\Sigma =$$

Covariance Matrix

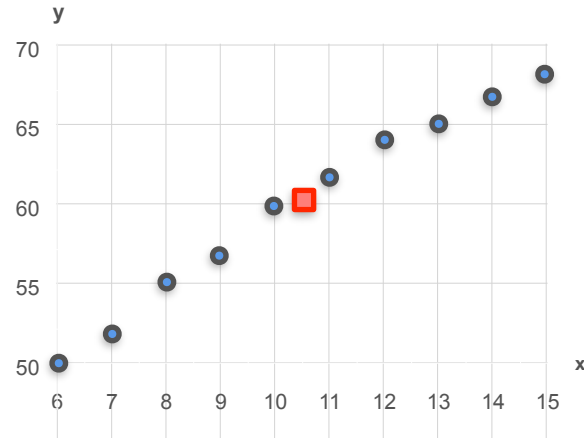
	<i>V</i>	<i>W</i>	<i>X</i>	<i>Y</i>	<i>Z</i>
<i>V</i>	$Var(V)$	$Cov(V, W)$	$Cov(V, X)$	$Cov(V, Y)$	$Cov(V, Z)$
<i>W</i>	$Cov(V, W)$	$Var(W)$	$Cov(W, X)$	$Cov(W, Y)$	$Cov(W, Z)$
<i>X</i>	$Cov(V, X)$	$Cov(W, X)$	$Var(X)$	$Cov(X, Y)$	$Cov(X, Z)$
<i>Y</i>	$Cov(V, Y)$	$Cov(W, Y)$	$Cov(X, Y)$	$Var(Y)$	$Cov(Y, Z)$
<i>Z</i>	$Cov(V, Z)$	$Cov(W, Z)$	$Cov(X, Z)$	$Cov(Y, Z)$	$Var(Z)$

Correlation Coefficient

Correlation Coefficient

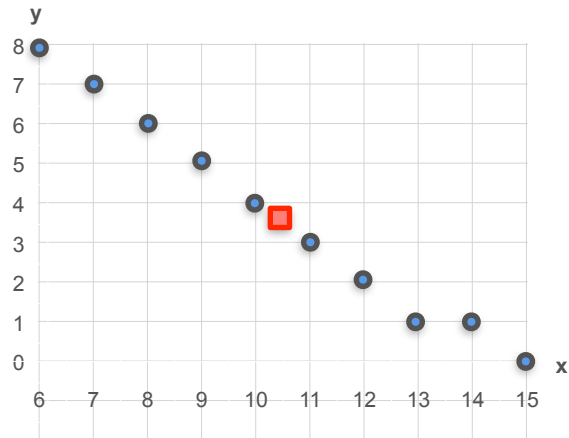


Age vs Naps per Day



Age vs Height

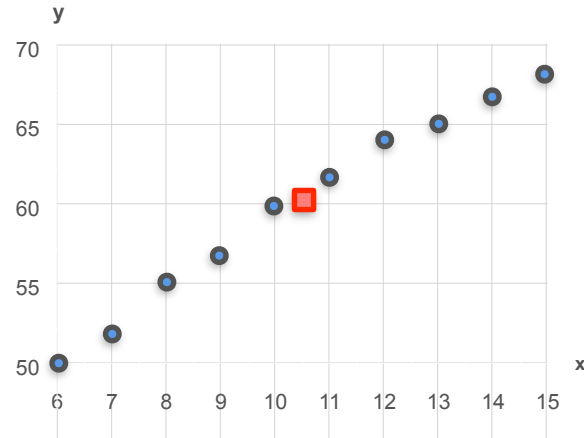
Correlation Coefficient



Age vs Naps per Day

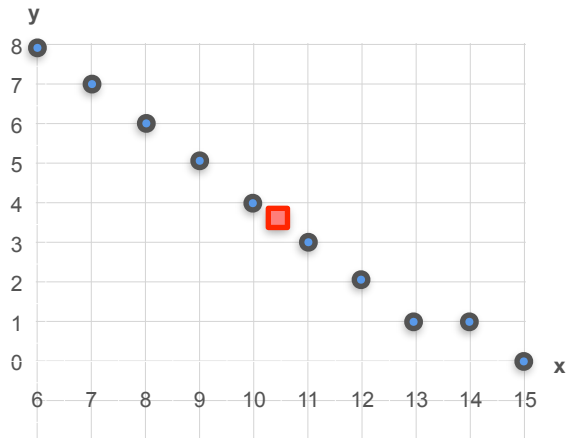
$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$



Age vs Height

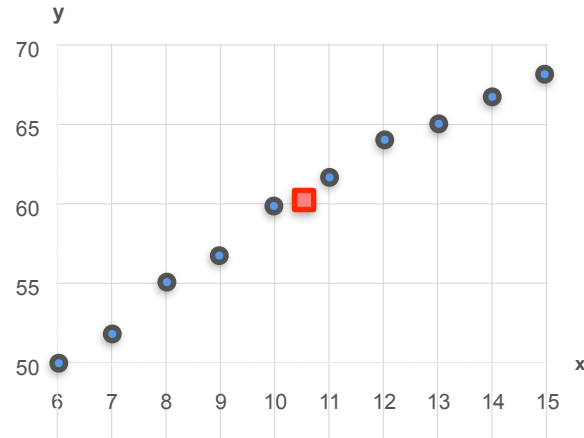
Correlation Coefficient



Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

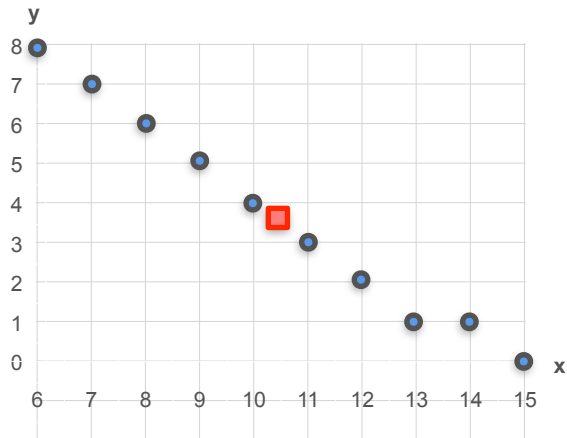


Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

Correlation Coefficient

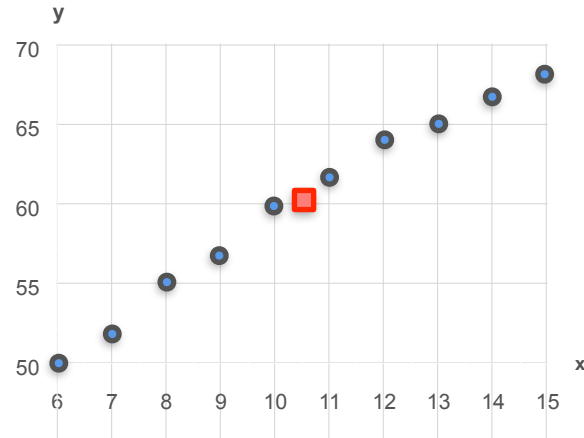


Age vs Naps per Day

$$Var(X) = 9.17$$

$$Var(Y) = 7.57$$

$$Cov(X, Y) = -7.45$$



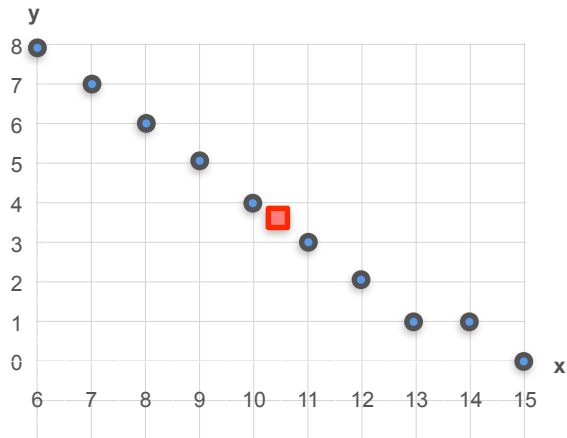
Age vs Height

$$Var(X) = 9.17$$

$$Var(Y) = 39.56$$

$$Cov(X, Y) = 17$$

Correlation Coefficient

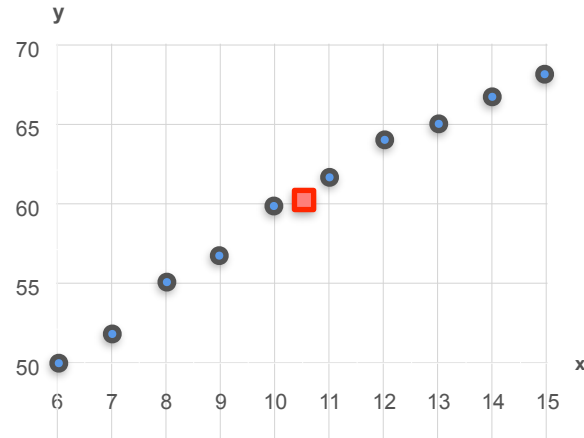


Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$



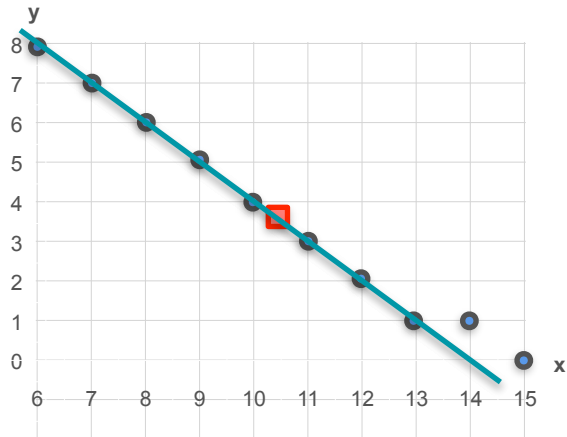
Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$

Correlation Coefficient

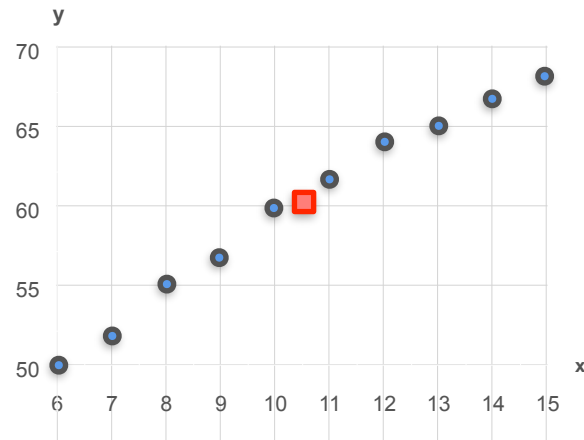


Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$



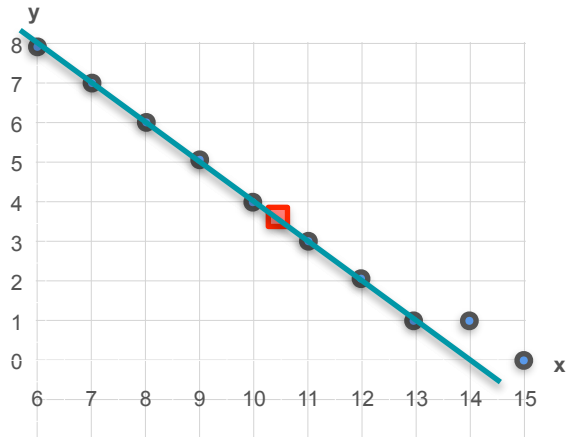
Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$

Correlation Coefficient

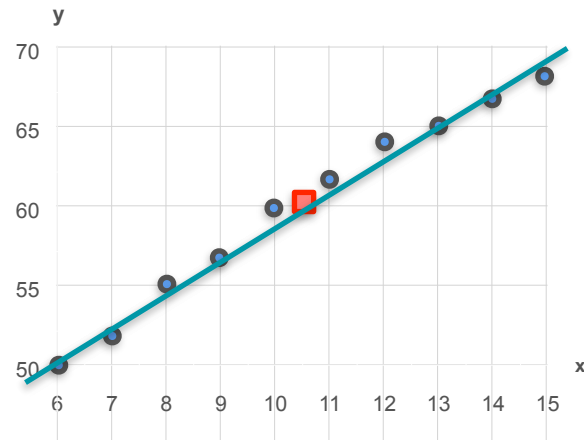


Age vs Naps per Day

$$Var(X) = 9.17$$

$$Var(Y) = 7.57$$

$$Cov(X, Y) = -7.45$$



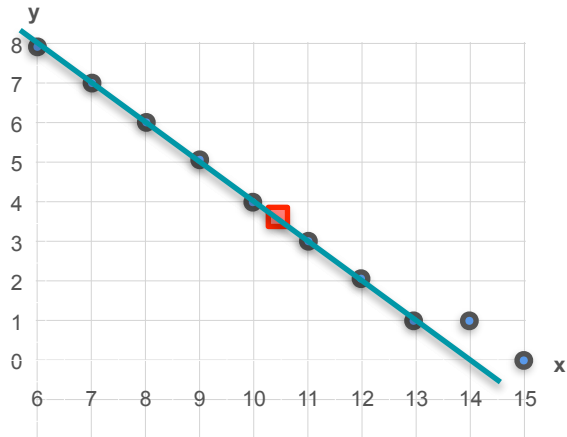
Age vs Height

$$Var(X) = 9.17$$

$$Var(Y) = 39.56$$

$$Cov(X, Y) = 17$$

Correlation Coefficient

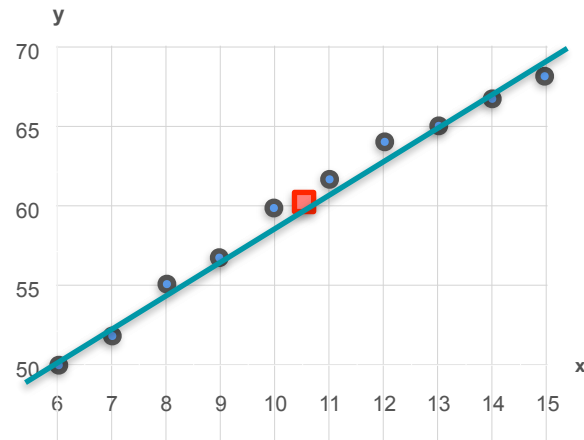


Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$



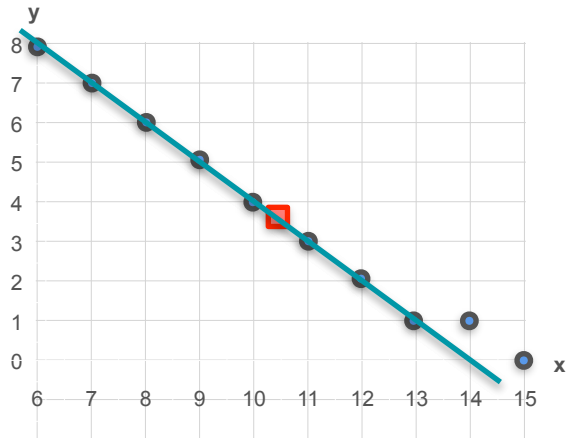
Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$

Correlation Coefficient



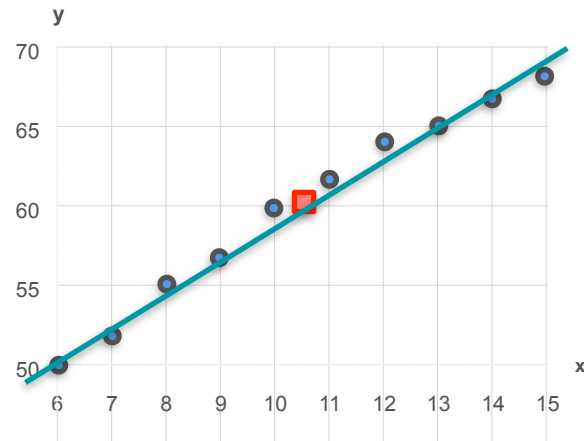
Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

Is the correlation here stronger?



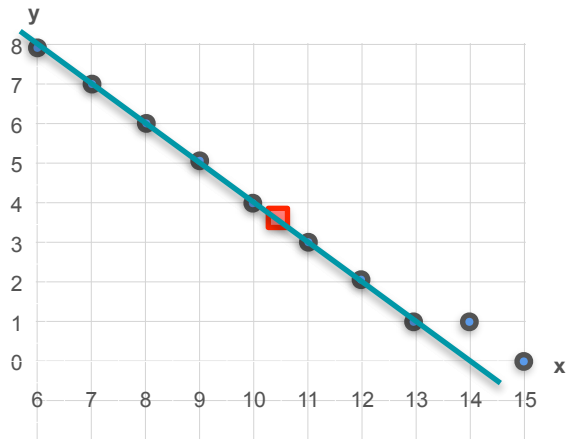
Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$

Correlation Coefficient



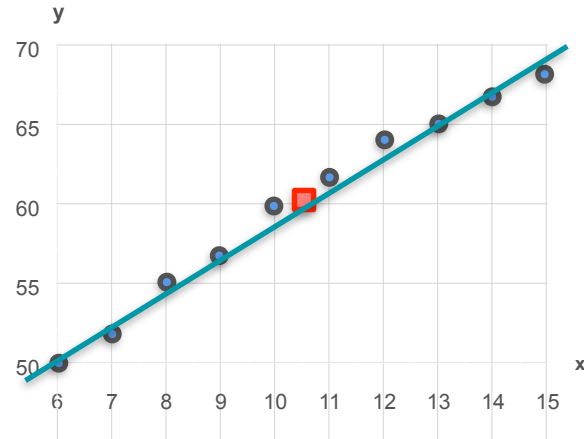
Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

Is the correlation here strong?



Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$

Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$

Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

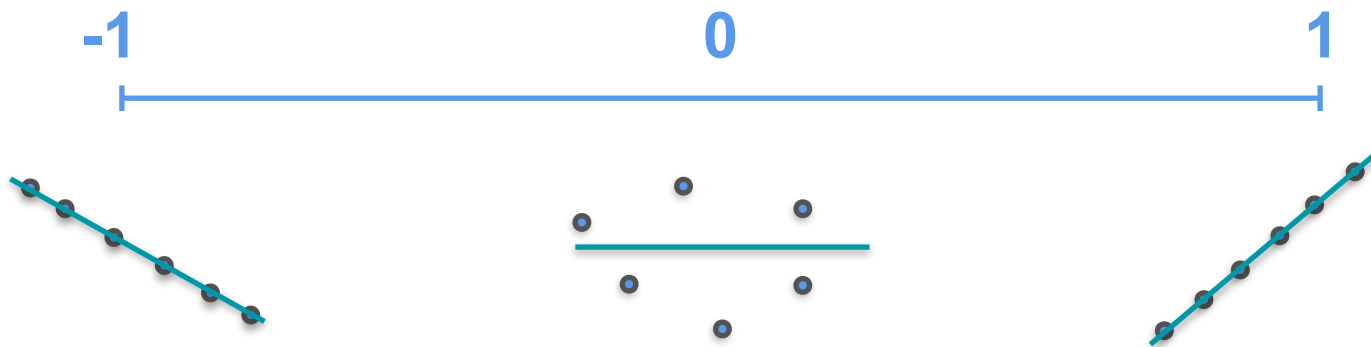
$$\text{Cov}(X, Y) = -7.45$$

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

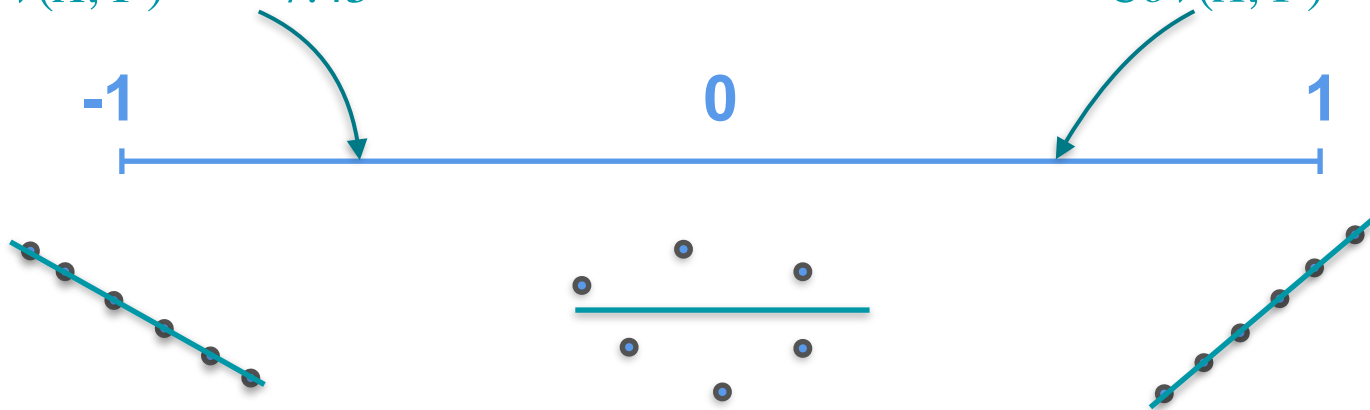
$$\text{Cov}(X, Y) = -7.45$$

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

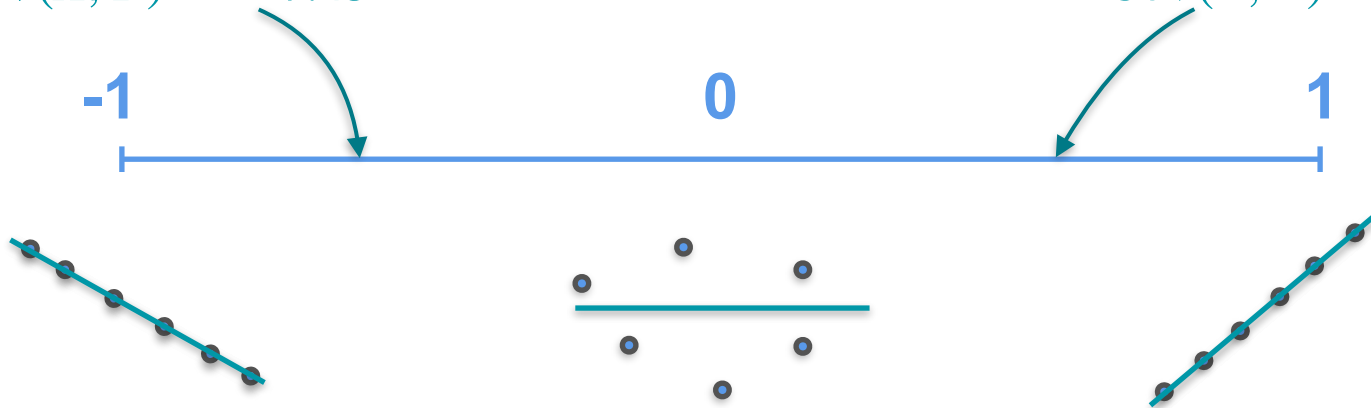
Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$

Correlation Coefficient



Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



Correlation Coefficient

Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



**Correlation
Coefficient =**

Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



$$\text{Correlation Coefficient} = \text{Cov}(X, Y)$$

Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



$$\text{Correlation Coefficient} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



$$\text{Correlation Coefficient} = \frac{\text{Cov}(X, Y)}{\sigma_x}$$

Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



$$\text{Correlation Coefficient} = \frac{\text{Cov}(X, Y)}{\sigma_x \cdot \sigma_y}$$

Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



$$\text{Correlation Coefficient} = \frac{\text{Cov}(X, Y)}{\sigma_x \cdot \sigma_y}$$

Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



$$\text{Correlation Coefficient} = \frac{\text{Cov}(X, Y)}{\sigma_x \cdot \sigma_y} =$$

Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$

-1

0

1

**Correlation
Coefficient**

$$= \frac{\text{Cov}(X, Y)}{\sigma_x \cdot \sigma_y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

Correlation Coefficient

Age vs Naps per Day

$$Var(X) = 9.17$$

$$Var(Y) = 7.57$$

$$Cov(X, Y) = -7.45$$

$$\text{Correlation Coefficient} = \frac{Cov(X, Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}}$$

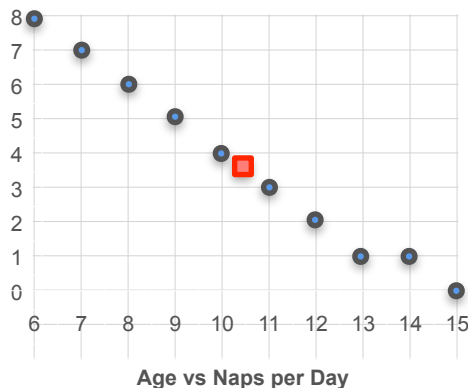
Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$



$$\text{Correlation Coefficient} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

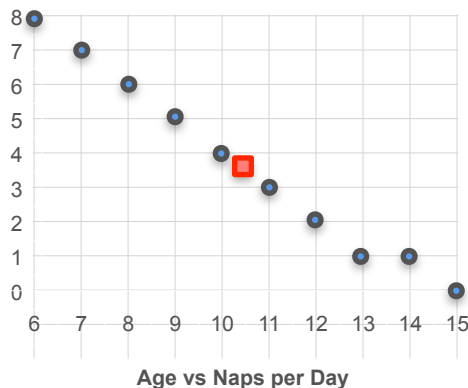
Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$



**Correlation
Coefficient**

$$= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

=

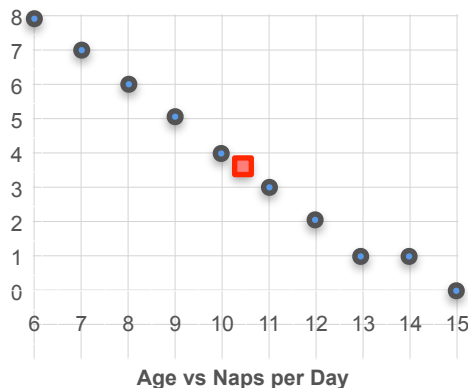
Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$



Correlation Coefficient

$$= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

$$= \frac{-7.45}{\sqrt{9.17} \cdot \sqrt{7.57}}$$

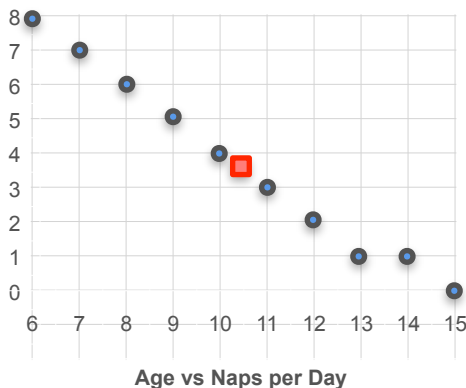
Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$



**Correlation
Coefficient**

$$= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

$$= \frac{-7.45}{\sqrt{9.17} \cdot \sqrt{7.57}}$$

$$\approx -0.894$$

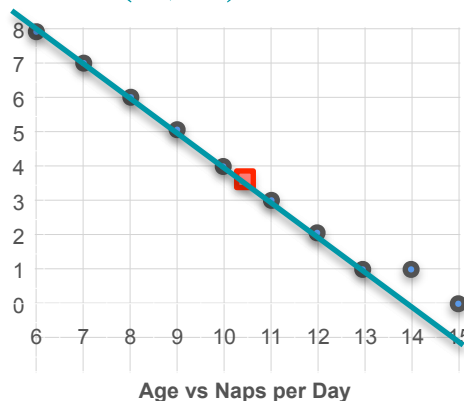
Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$



Correlation Coefficient

$$= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

$$= \frac{-7.45}{\sqrt{9.17} \cdot \sqrt{7.57}}$$

$$\approx -0.894$$

Correlation Coefficient

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$

$$\text{Correlation Coefficient} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

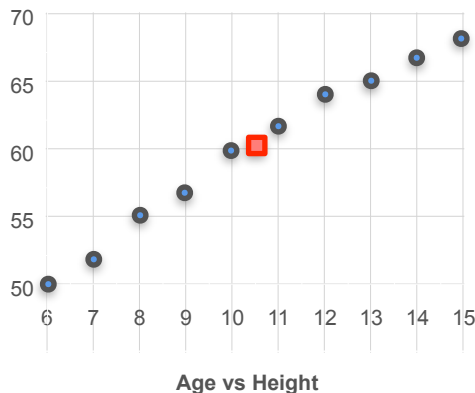
Correlation Coefficient

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



**Correlation
Coefficient**

$$= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

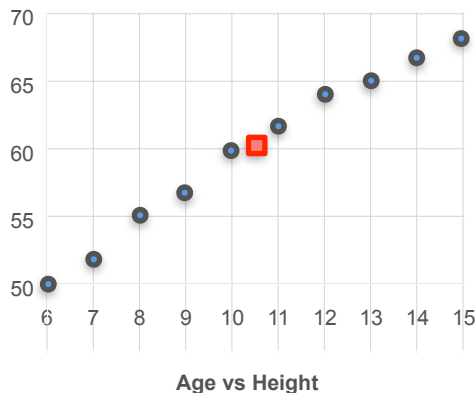
Correlation Coefficient

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



**Correlation
Coefficient**

$$= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

=

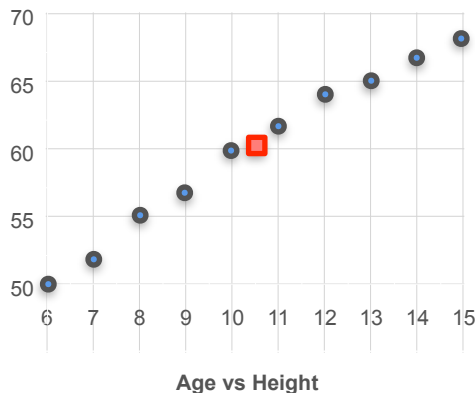
Correlation Coefficient

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



Correlation Coefficient

$$= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

$$= \frac{17}{\sqrt{9.17} \cdot \sqrt{39.56}}$$

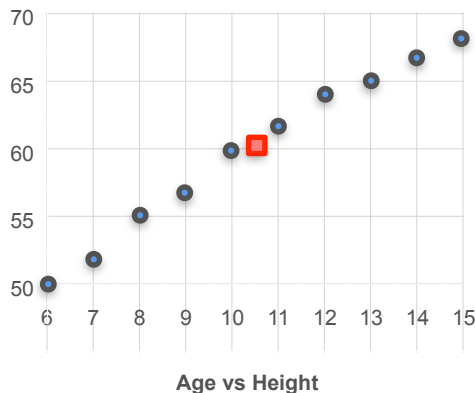
Correlation Coefficient

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



Correlation Coefficient

$$= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

$$= \frac{17}{\sqrt{9.17} \cdot \sqrt{39.56}}$$

$$\approx 0.893$$

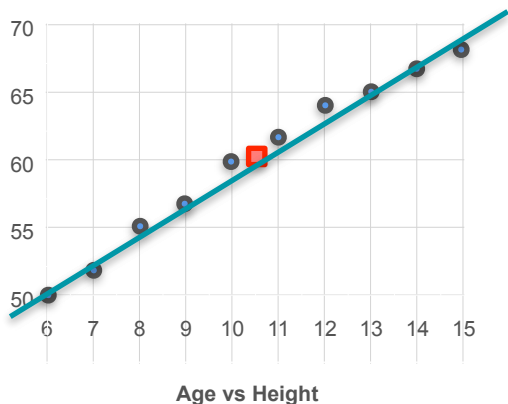
Correlation Coefficient

Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



**Correlation
Coefficient**

$$= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

$$= \frac{17}{\sqrt{9.17} \cdot \sqrt{39.56}}$$

$$\approx 0.893$$

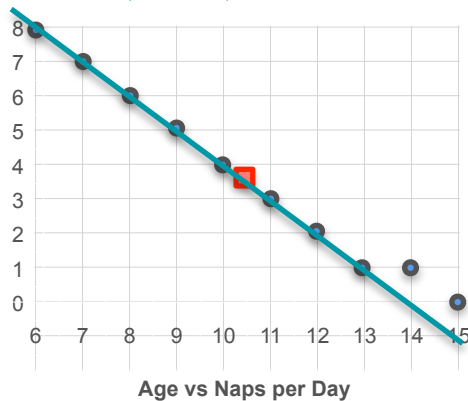
Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$



Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$

$$\approx 0.893$$

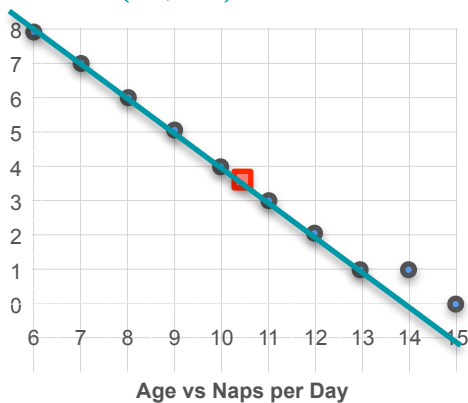
Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

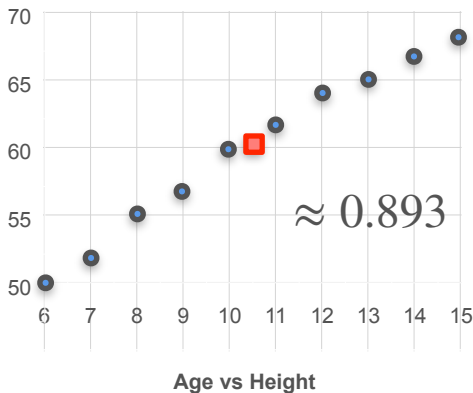


Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



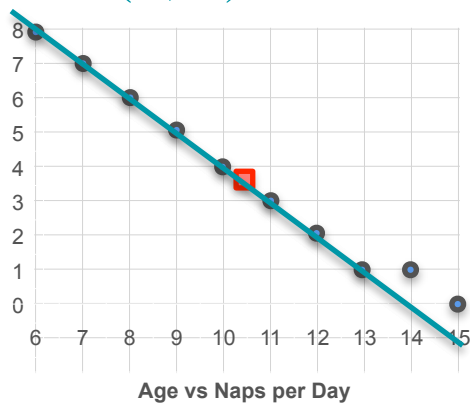
Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

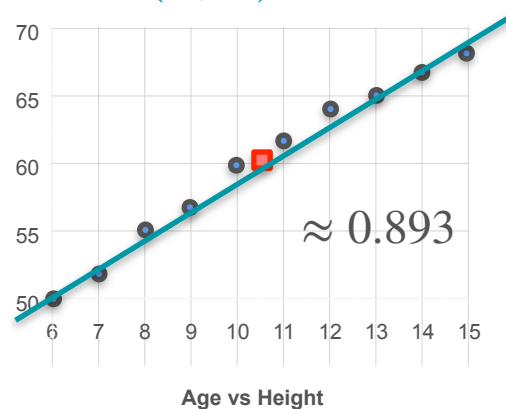


Age vs Height

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



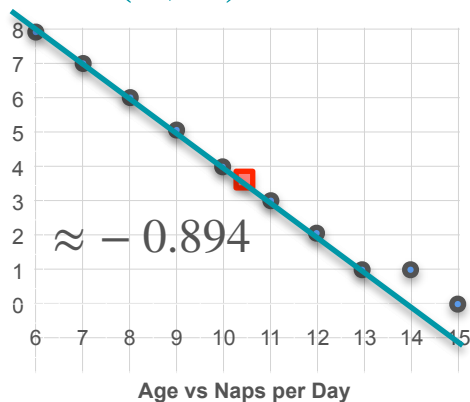
Correlation Coefficient

Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$

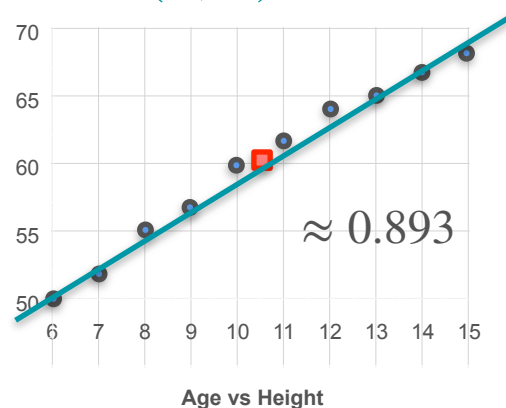


Age vs Height

$$\text{Var}(X) = 9.17$$

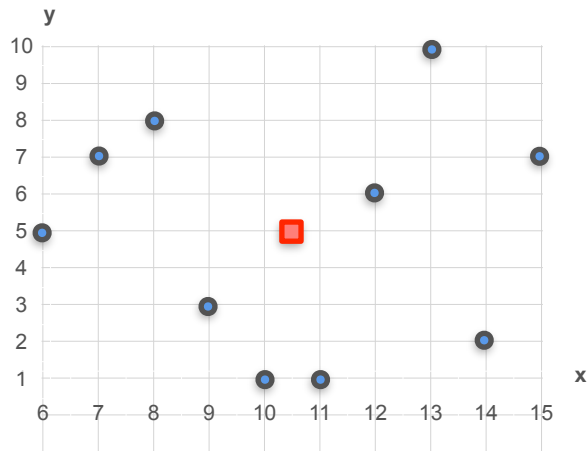
$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



Correlation Coefficient

Correlation Coefficient



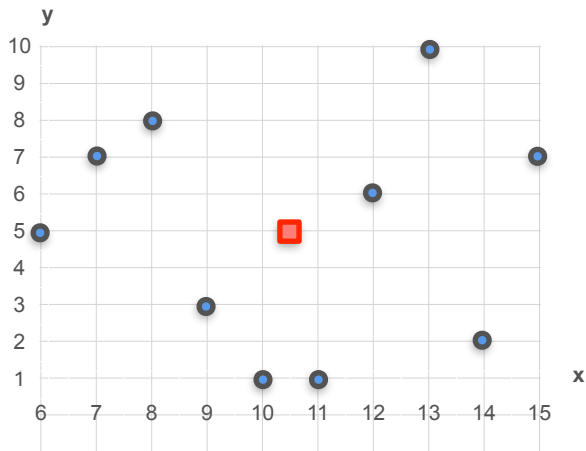
Age vs Grades

$$Var(X) = 9.17$$

$$Var(Y) = 9.78$$

$$Cov(X, Y) = 0.1$$

Correlation Coefficient



Age vs Grades

$$Var(X) = 9.17$$

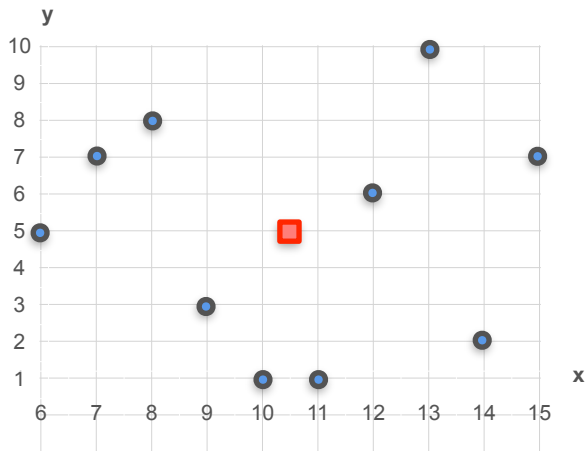
$$Var(Y) = 9.78$$

$$Cov(X, Y) = 0.1$$

**Correlation
Coefficient**

$$= \frac{Cov(X, Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}}$$

Correlation Coefficient



Age vs Grades

$$Var(X) = 9.17$$

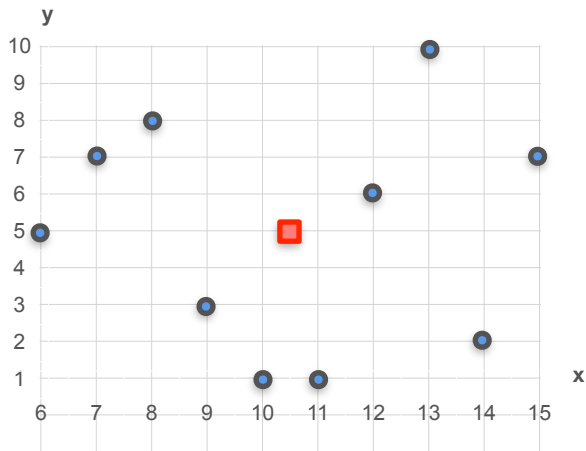
$$Var(Y) = 9.78$$

$$Cov(X, Y) = 0.1$$

**Correlation
Coefficient**

$$= \frac{Cov(X, Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}}$$
$$=$$

Correlation Coefficient



Age vs Grades

$$Var(X) = 9.17$$

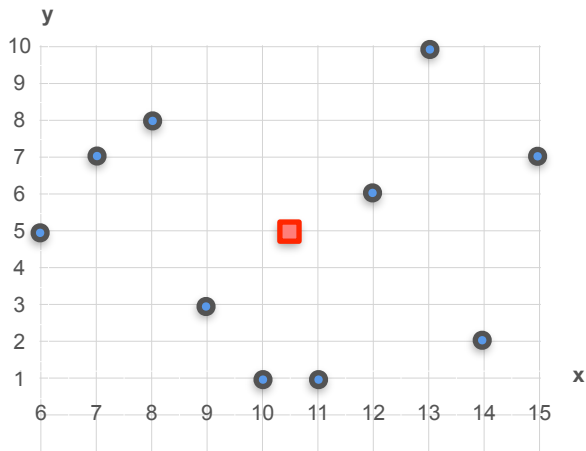
$$Var(Y) = 9.78$$

$$Cov(X, Y) = 0.1$$

**Correlation
Coefficient**

$$\begin{aligned} &= \frac{Cov(X, Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}} \\ &= \frac{0.1}{\sqrt{9.17} \cdot \sqrt{9.78}} \end{aligned}$$

Correlation Coefficient



Age vs Grades

$$Var(X) = 9.17$$

$$Var(Y) = 9.78$$

$$Cov(X, Y) = 0.1$$

**Correlation
Coefficient**

$$= \frac{Cov(X, Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}}$$

$$= \frac{0.1}{\sqrt{9.17} \cdot \sqrt{9.78}}$$

$$\approx 0.01$$

Correlation Coefficient

Correlation Coefficient

$$\text{Correlation Coefficient} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

Correlation Coefficient

$$\begin{aligned}\text{Correlation Coefficient} &= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}} \\ &= \end{aligned}$$

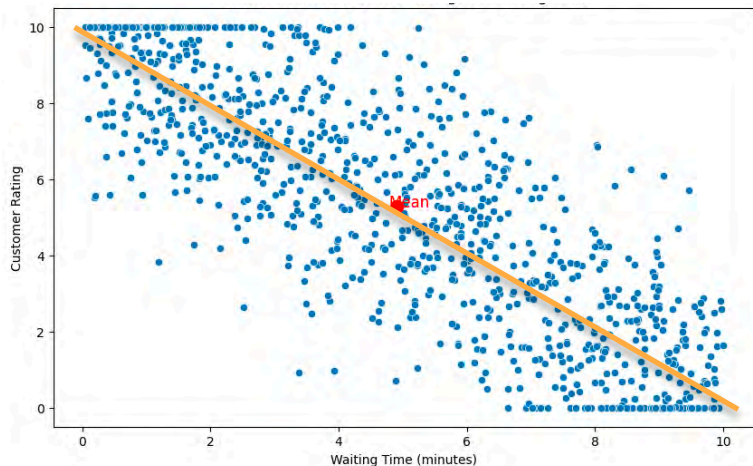
Correlation Coefficient

$$\begin{aligned}\text{Correlation Coefficient} &= \frac{Cov(X, Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}} \\ &= \frac{-7.878}{\sqrt{8.562} \cdot \sqrt{10.163}}\end{aligned}$$

Correlation Coefficient

$$\begin{aligned}\text{Correlation Coefficient} &= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}} \\ &= \frac{-7.878}{\sqrt{8.562} \cdot \sqrt{10.163}} \\ &\approx -0.845\end{aligned}$$

Correlation Coefficient



$$Var(X) = 8.526$$

$$Var(Y) = 10.163$$

$$Cov(X, Y) = -7.878$$

**Correlation
Coefficient**

$$\begin{aligned} &= \frac{Cov(X, Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}} \\ &= \frac{-7.878}{\sqrt{8.526} \cdot \sqrt{10.163}} \\ &\approx -0.845 \end{aligned}$$

Correlation Coefficient

Correlation Coefficient

**Correlation
Coefficient =**

Correlation Coefficient

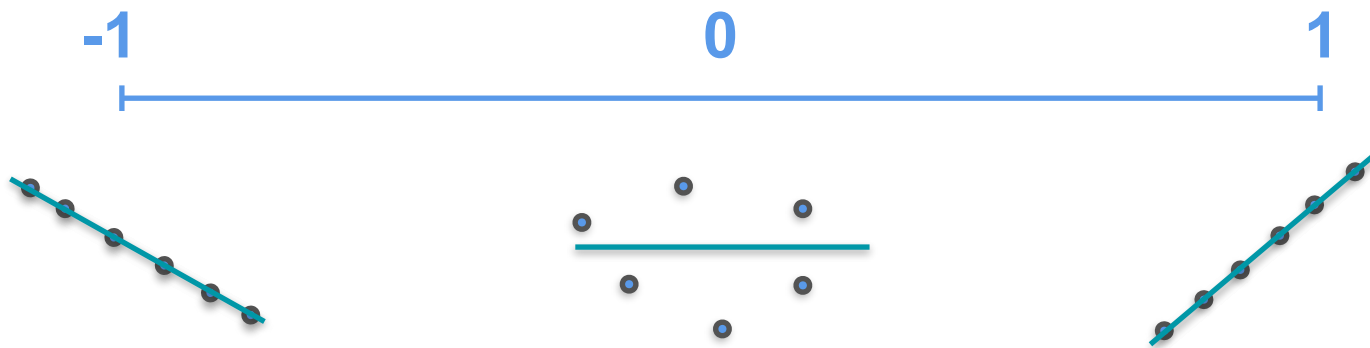
$$\text{Correlation Coefficient} = \frac{\text{Cov}(X, Y)}{\sigma_x \cdot \sigma_y}$$

Correlation Coefficient

$$\text{Correlation Coefficient} = \frac{Cov(X, Y)}{\sigma_x \cdot \sigma_y} = \frac{Cov(X, Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}}$$

Correlation Coefficient

$$\text{Correlation Coefficient} = \frac{\text{Cov}(X, Y)}{\sigma_x \cdot \sigma_y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$





DeepLearning.AI

Probability Distributions with Multiple Variables

Multivariate Gaussian Distribution

Multivariate Gaussian Distribution

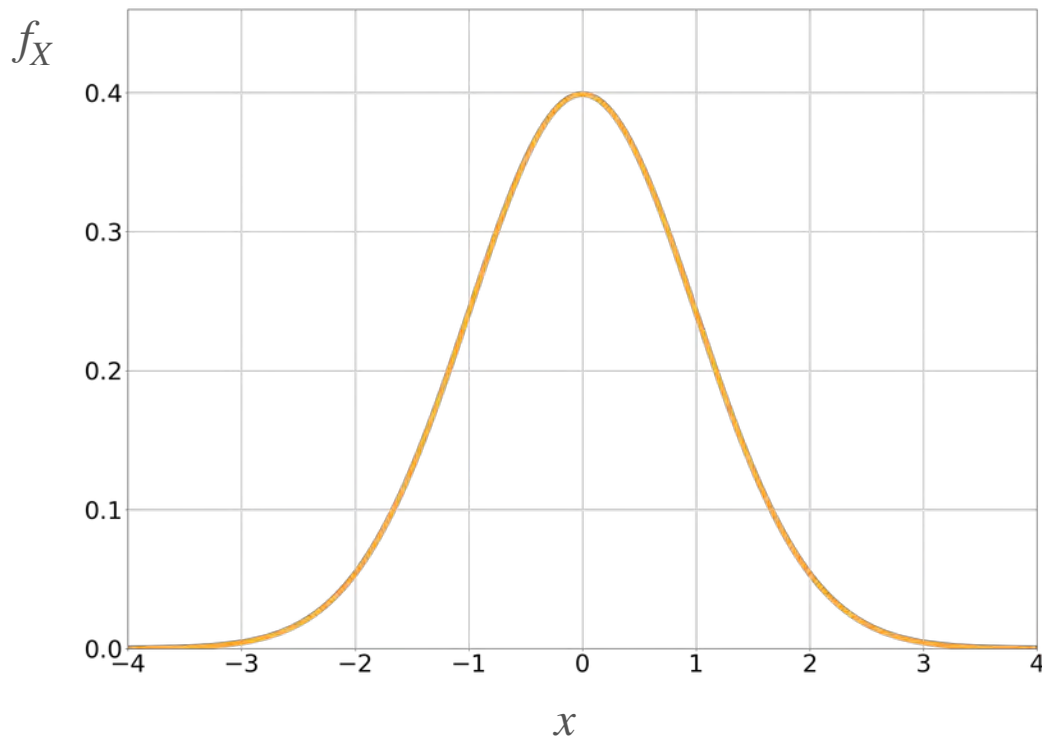
Multivariate Gaussian Distribution

For a single variable, X

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

Parameters:

- μ : center of the bell
- σ : spread of the bell



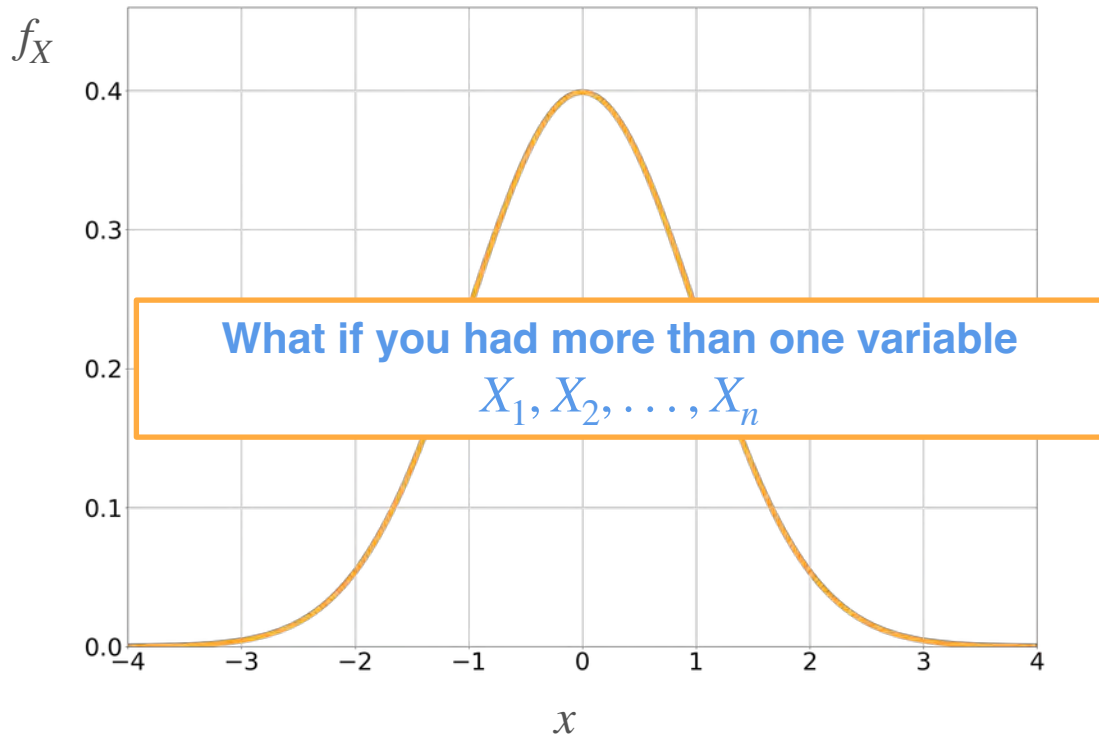
Multivariate Gaussian Distribution

For a single variable, X

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

Parameters:

- μ : center of the bell
- σ : spread of the bell



Multivariate Gaussian Distribution: An Example

Multivariate Gaussian Distribution: An Example

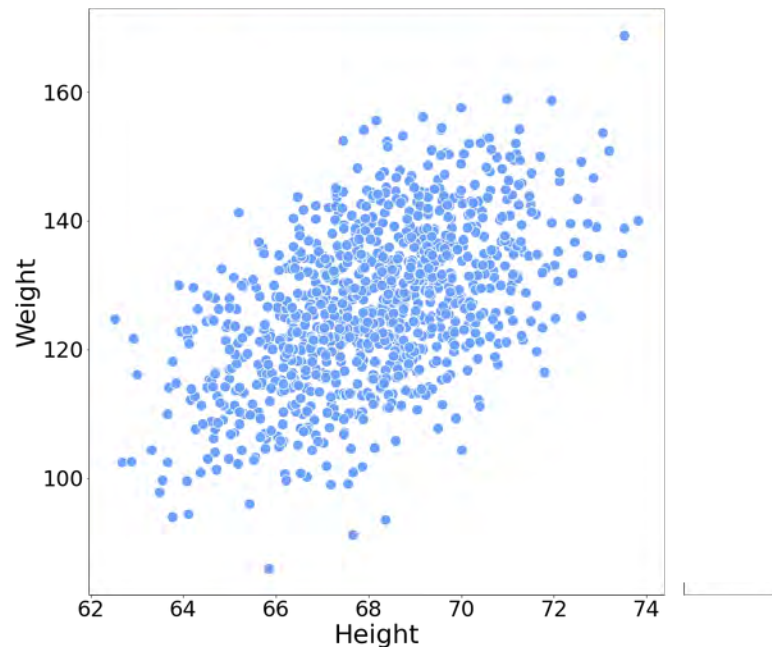
Two variables

Multivariate Gaussian Distribution: An Example

Two variables

H : Height of an adult in inches

W : Weight of an adult in pounds

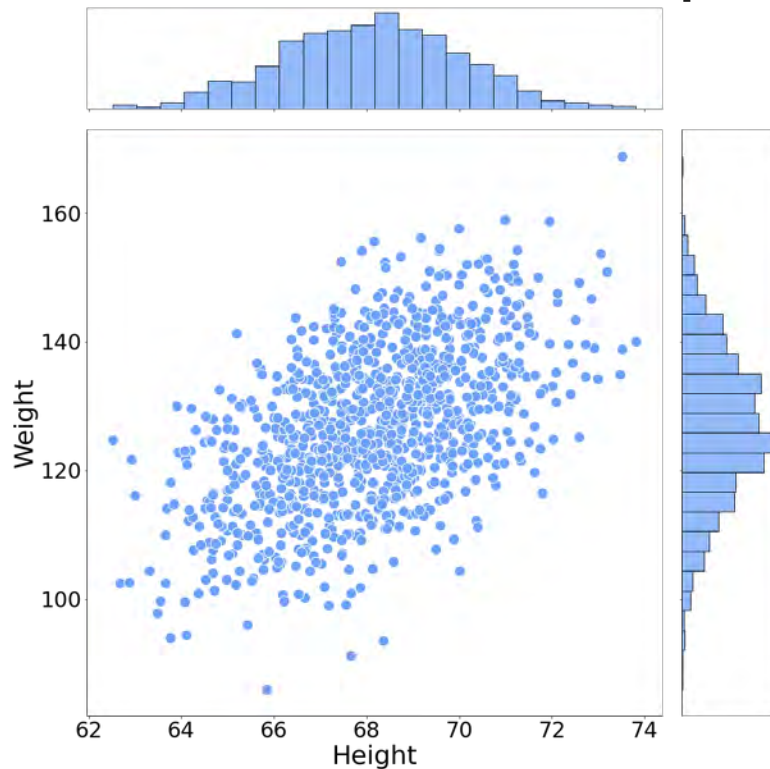


Multivariate Gaussian Distribution: An Example

Two variables

H : Height of an adult in inches

W : Weight of an adult in pounds



Multivariate Gaussian Distribution: An Example

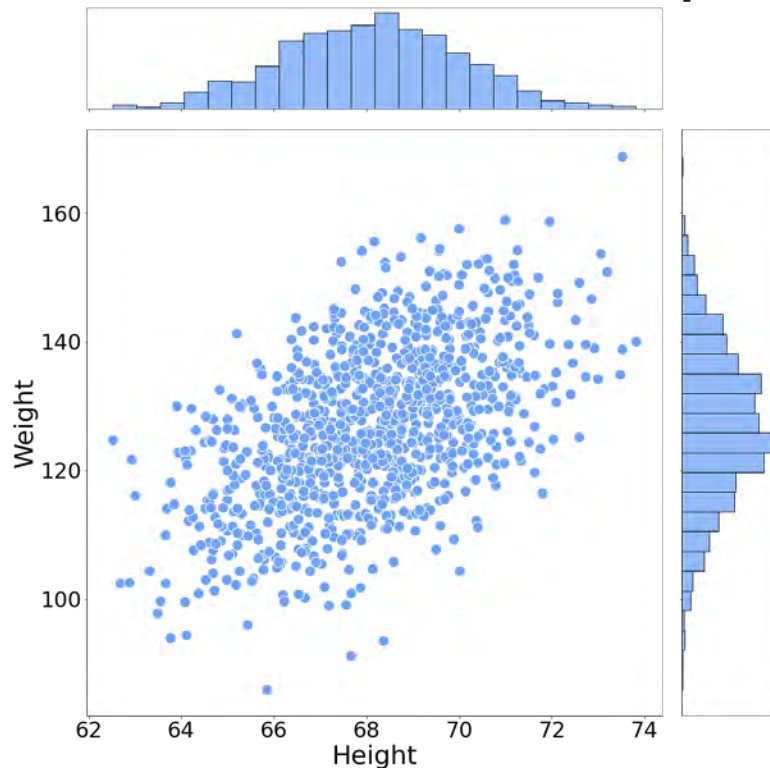
Two variables

H : Height of an adult in inches

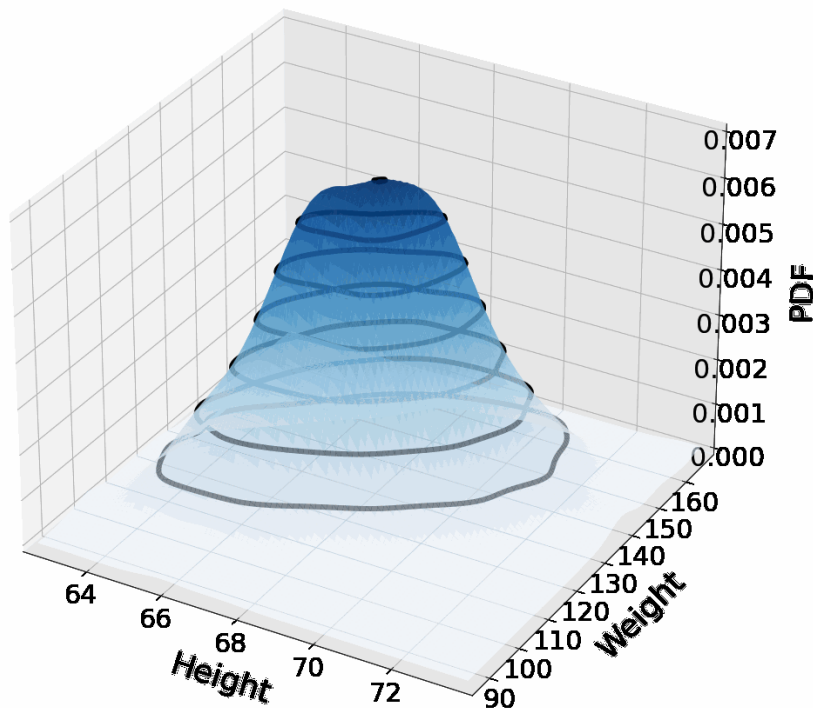
$$H \sim \mathcal{N}(\mu_H, \sigma_H)$$

W : Weight of an adult in pounds

$$W \sim \mathcal{N}(\mu_W, \sigma_W)$$



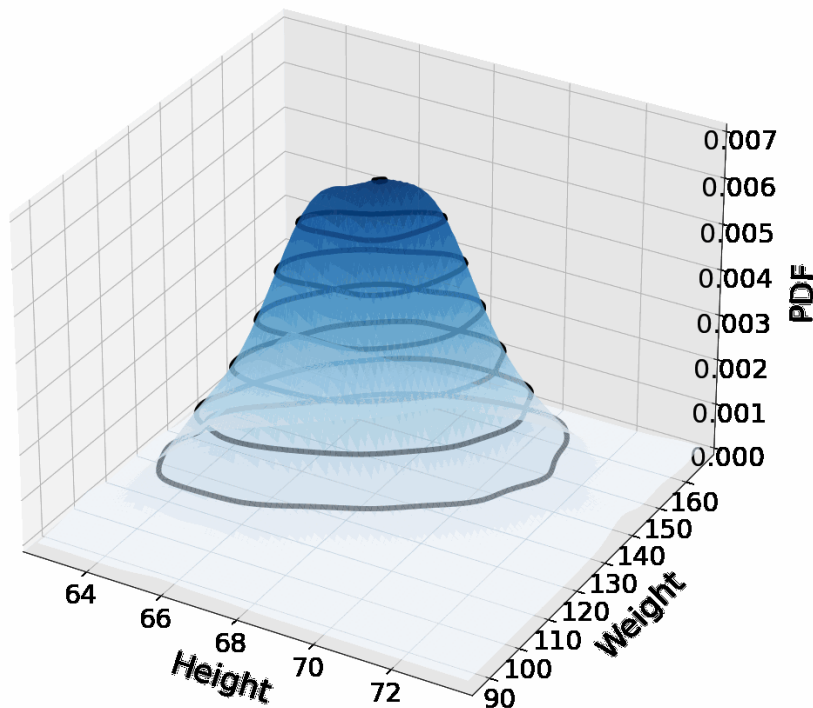
Multivariate Gaussian Distribution: An Example



If W, H were independent

$$f_{HW}(h, w) = f_H(h)f_W(w)$$

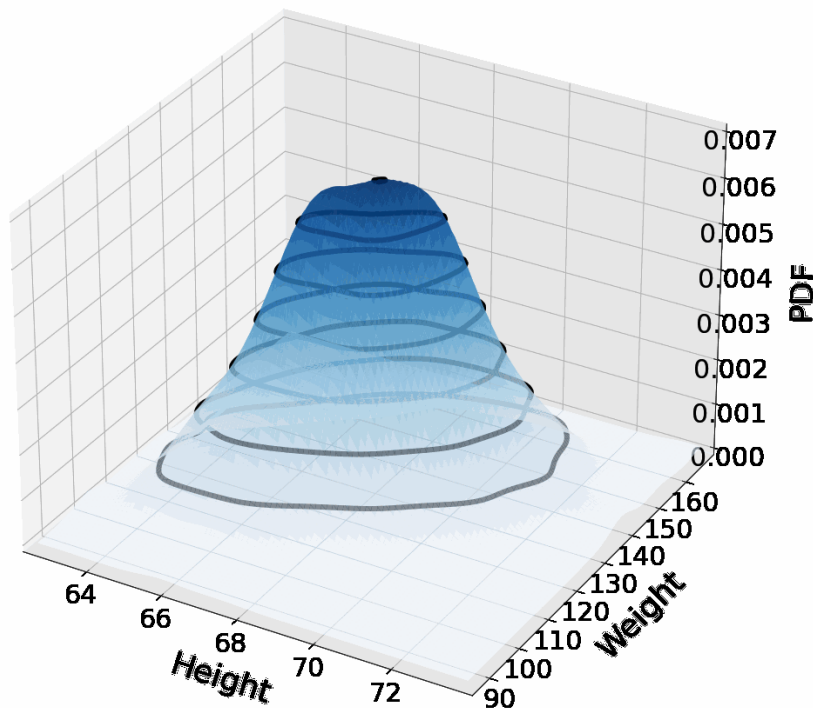
Multivariate Gaussian Distribution: An Example



If W, H were independent

$$f_{HW}(h, w) = f_H(h)f_W(w)$$

Multivariate Gaussian Distribution: An Example

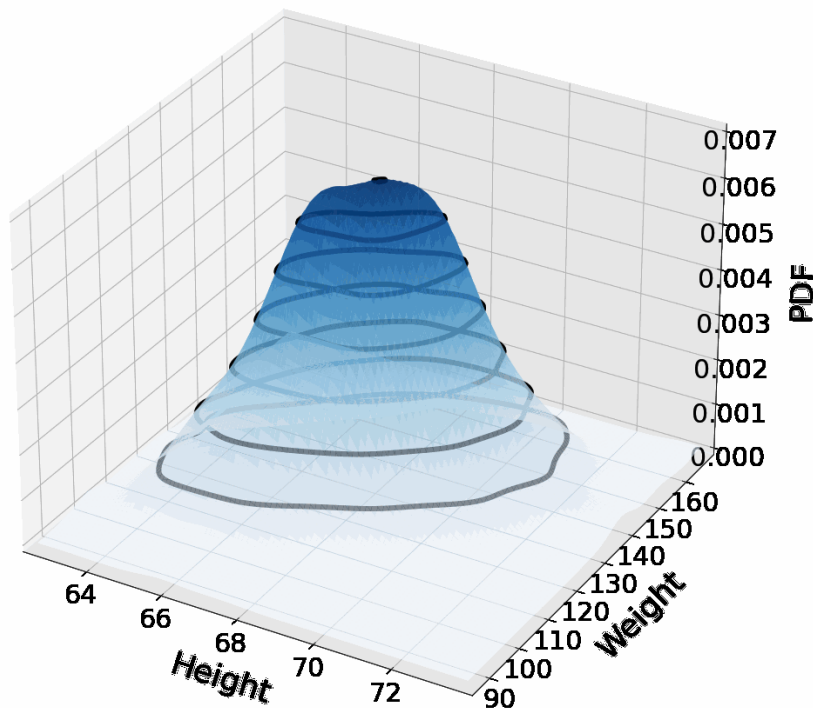


If W, H were independent

$$f_{HW}(h, w) = f_H(h)f_W(w)$$

$$= \frac{1}{\sqrt{2\pi}\sigma_H} e^{-\frac{1}{2}\frac{(h-\mu_H)^2}{\sigma_H^2}} \cdot \frac{1}{\sqrt{2\pi}\sigma_W} e^{-\frac{1}{2}\frac{(w-\mu_W)^2}{\sigma_W^2}}$$

Multivariate Gaussian Distribution: An Example

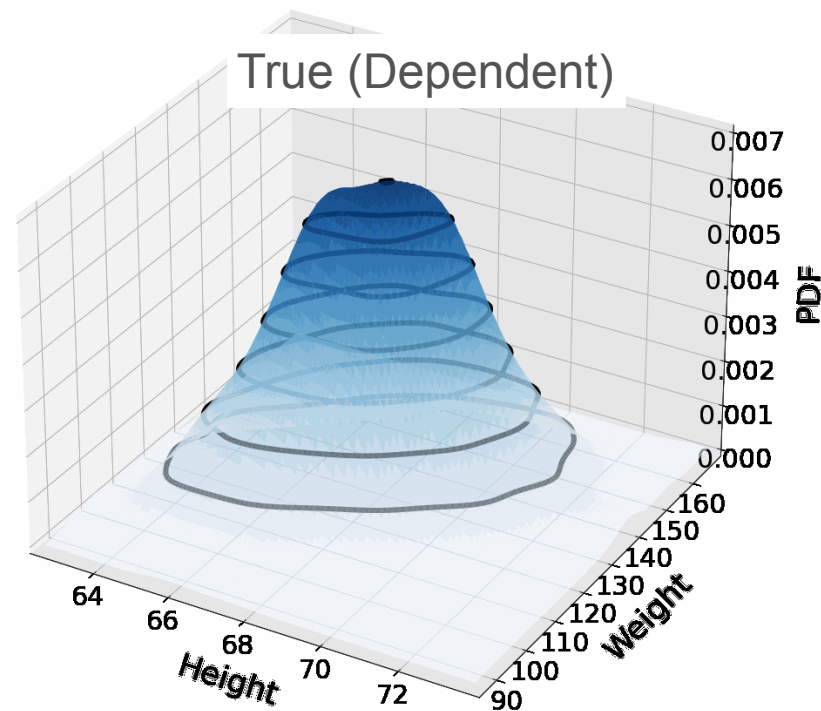
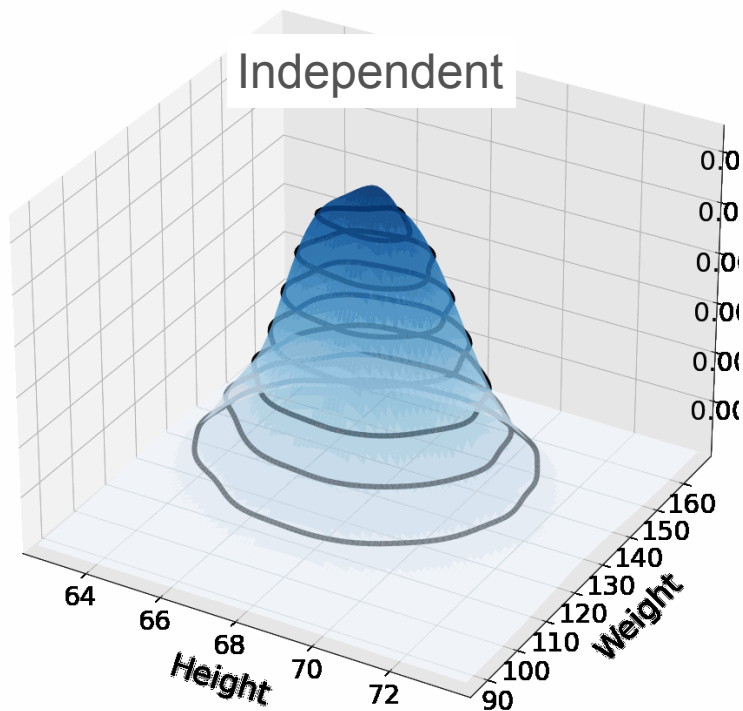


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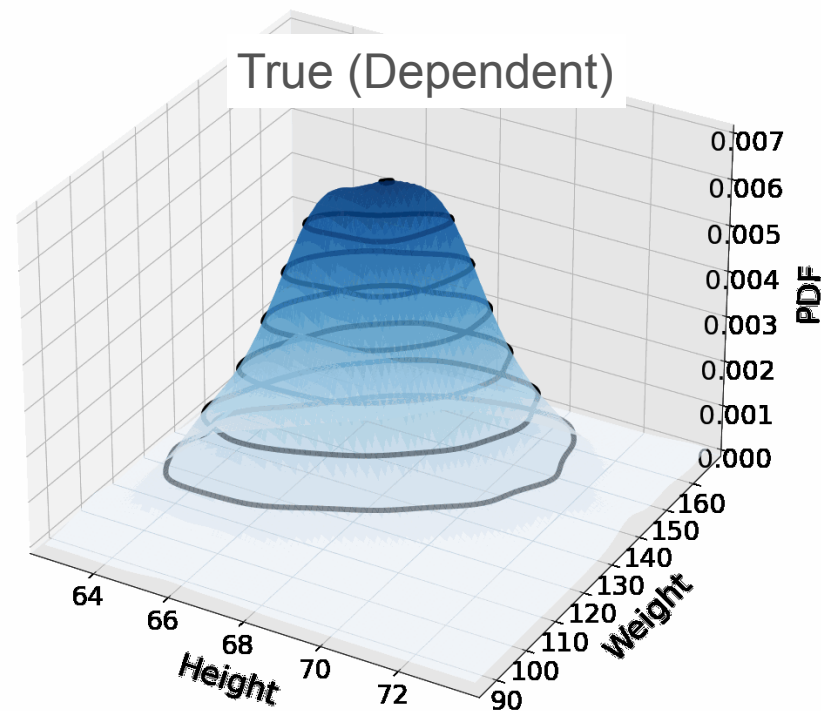
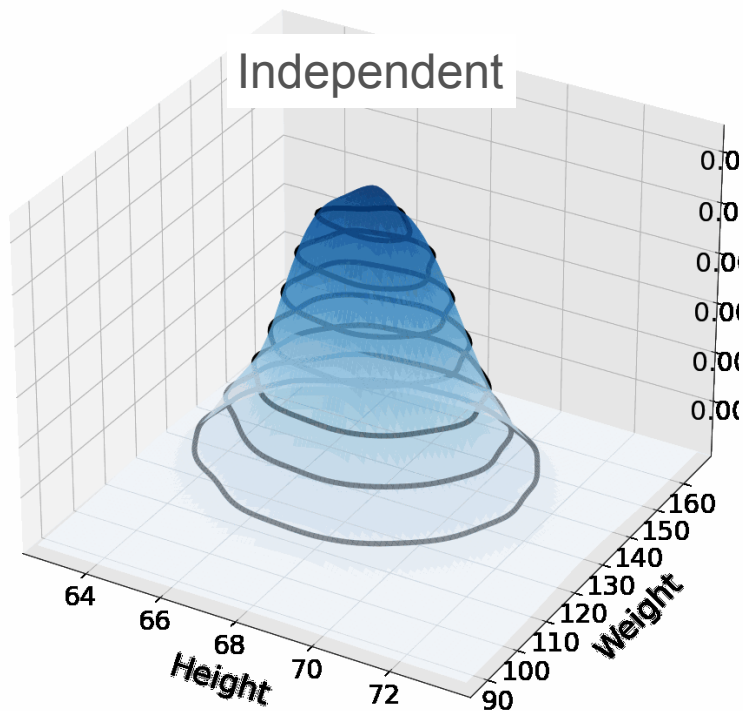
$$f_{HW}(h, w) = f_H(h)f_W(w)$$

$$\begin{aligned} &= \frac{1}{\sqrt{2\pi}\sigma_H} e^{-\frac{1}{2}\frac{(h-\mu_H)^2}{\sigma_H^2}} \cdot \frac{1}{\sqrt{2\pi}\sigma_W} e^{-\frac{1}{2}\frac{(w-\mu_W)^2}{\sigma_W^2}} \\ &= \frac{1}{2\pi\sigma_H\sigma_W} e^{-\frac{1}{2}\left(\frac{(h-\mu_H)^2}{\sigma_H^2} + \frac{(w-\mu_W)^2}{\sigma_W^2}\right)} \end{aligned}$$

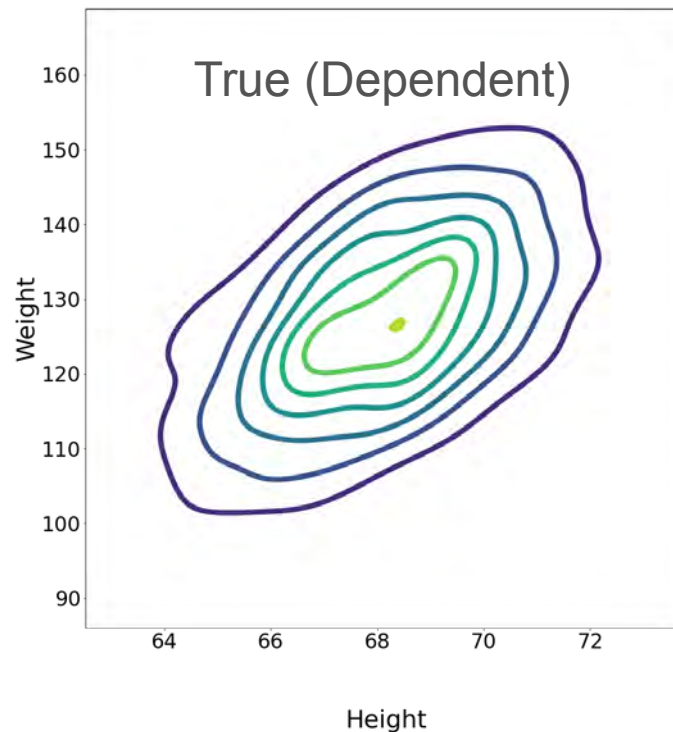
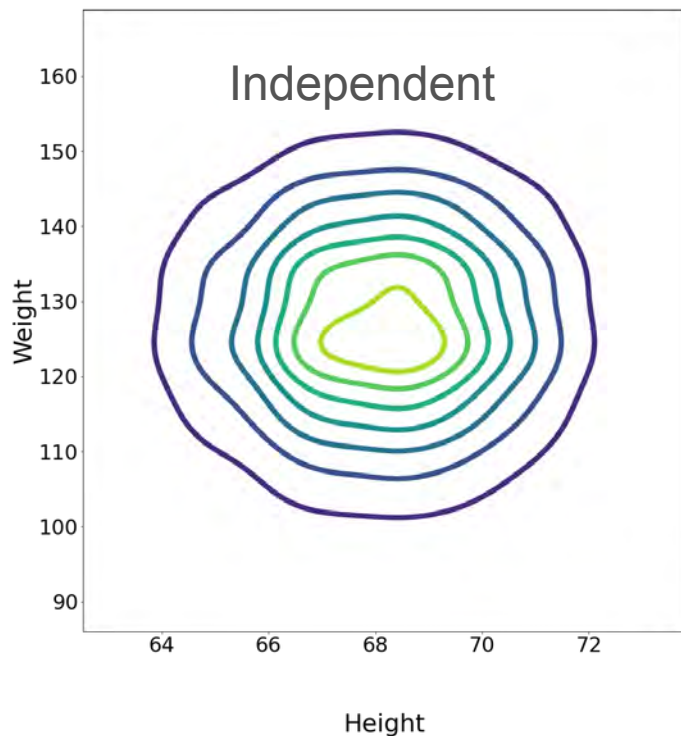
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Multivariate Gaussian Distribution: An Example

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$$\left\| \begin{bmatrix} \frac{h - \mu_H}{\sigma_H} \\ \frac{w - \mu_W}{\sigma_W} \end{bmatrix} \right\|_2^2$$

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$$\left\| \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \right\|_2^2 = \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} & \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix}$$

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$$\left\| \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \right\|^2 = \begin{bmatrix} h-\mu_H & w-\mu_W \\ \sigma_H & \sigma_W \end{bmatrix} \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix}$$

$$\begin{bmatrix} h-\mu_h \\ w-\mu_w \end{bmatrix} = \begin{bmatrix} h \\ w \end{bmatrix} - \begin{bmatrix} \mu_h \\ \mu_w \end{bmatrix}$$

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$$\left\| \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \right\|^2$$

$$\Rightarrow \begin{bmatrix} h-\mu_H & w-\mu_W \\ \sigma_H & \sigma_W \end{bmatrix} \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix}$$

$$\begin{bmatrix} h-\mu_h \\ w-\mu_w \end{bmatrix} = \begin{bmatrix} h \\ w \end{bmatrix} - \begin{bmatrix} \mu_h \\ \mu_w \end{bmatrix}$$

Multiply by
diagonal matrix

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$$= \left\| \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \right\|_2^2 = \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} & \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix}$$
$$= ([h \ w] - [\mu_H \ \mu_W]) \begin{bmatrix} \frac{1}{\sigma_H^2} & 0 \\ 0 & \frac{1}{\sigma_W^2} \end{bmatrix} \left(\begin{bmatrix} h \\ w \end{bmatrix} - \begin{bmatrix} \mu_H \\ \mu_W \end{bmatrix} \right)$$

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$$\begin{aligned} &= \left\| \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \right\|_2^2 = \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} & \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \\ &= ([h \ w] - [\mu_H \ \mu_W]) \begin{bmatrix} \frac{1}{\sigma_H^2} & 0 \\ 0 & \frac{1}{\sigma_W^2} \end{bmatrix} \left(\begin{bmatrix} h \\ w \end{bmatrix} - \begin{bmatrix} \mu_H \\ \mu_W \end{bmatrix} \right) \\ &= \left(\begin{bmatrix} h \\ w \end{bmatrix} - \begin{bmatrix} \mu_H \\ \mu_W \end{bmatrix} \right)^T \begin{bmatrix} \sigma_H^2 & 0 \\ 0 & \sigma_W^2 \end{bmatrix}^{-1} \left(\begin{bmatrix} h \\ w \end{bmatrix} - \begin{bmatrix} \mu_H \\ \mu_W \end{bmatrix} \right) \end{aligned}$$

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$$= \left\| \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \right\|_2^2 = \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} & \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix}$$
$$= ([h \ w] - [\mu_H \ \mu_W]) \begin{bmatrix} \frac{1}{\sigma_H^2} & 0 \\ 0 & \frac{1}{\sigma_W^2} \end{bmatrix} \left(\begin{bmatrix} h \\ w \end{bmatrix} - \begin{bmatrix} \mu_H \\ \mu_W \end{bmatrix} \right)$$

Covariance matrix!
(Σ)

$$= \left(\begin{bmatrix} h \\ w \end{bmatrix} - \begin{bmatrix} \mu_H \\ \mu_W \end{bmatrix} \right)^T \begin{bmatrix} \sigma_H^2 & 0 \\ 0 & \sigma_W^2 \end{bmatrix}^{-1} \left(\begin{bmatrix} h \\ w \end{bmatrix} - \begin{bmatrix} \mu_H \\ \mu_W \end{bmatrix} \right)$$

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$\det(\Sigma)^{1/2}$

Covariance matrix!
(Σ)

$$= \left\| \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \right\|_2^2 = \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} & \frac{w-\mu_W}{\sigma_W} \end{bmatrix} \begin{bmatrix} \frac{h-\mu_H}{\sigma_H} \\ \frac{w-\mu_W}{\sigma_W} \end{bmatrix}$$

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$$= \frac{1}{2\pi \det \Sigma^{1/2}} \exp \left(-\frac{1}{2} \left(\begin{bmatrix} h \\ w \end{bmatrix} - \mu \right)^T \Sigma^{-1} \left(\begin{bmatrix} h \\ w \end{bmatrix} - \mu \right) \right)$$

Multivariate Gaussian Distribution: An Example

Dependent case:

$$f_{HW}(h, w) = f_H(h)f_W(w)$$

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$$\Sigma = \begin{bmatrix} \sigma_H^2 & \text{Cov}(H, W) \\ \text{Cov}(H, W) & \sigma_W^2 \end{bmatrix}$$

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covariance matrix
spread of the bell

$|\Sigma|$ determinant of the covariance matrix



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$\mathbf{x} = [x_1 \ x_2 \ \dots \ x_n]^T$

Mean vector
 $\boldsymbol{\mu} = [\mu_1, \mu_2, \dots, \mu_n]^T$

$$f_X(x_1, x_2, \dots, x_n) = \frac{1}{(2\pi)^{n/2} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu})}$$

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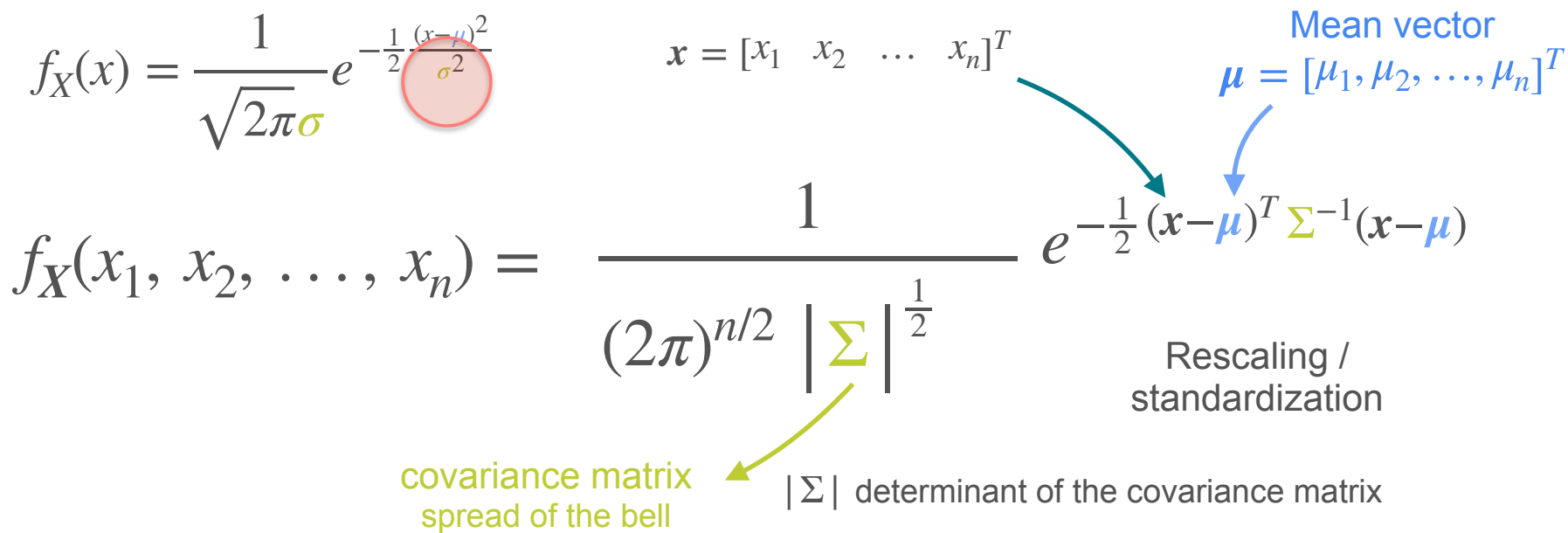
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$$f_X(x_1, x_2, \dots, x_n) = \frac{1}{(2\pi)^{n/2} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$

Rescaling / standardization

covariance matrix
spread of the bell

$|\boldsymbol{\Sigma}|$ determinant of the covariance matrix



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$\mathbf{x} = [x_1 \ x_2 \ \dots \ x_n]^T$

Mean vector
 $\boldsymbol{\mu} = [\mu_1, \mu_2, \dots, \mu_n]^T$

$f_X(x_1, x_2, \dots, x_n)$

random
 $\mathbf{X} = [X_1 \ X_2 \ \dots \ X_n]$

- For univariate, we work with scalar values and variances
- For multivariate, we work with vectors and the covariance matrix

covariance matrix
spread of the bell

$|\Sigma|$ determinant of the covariance matrix

$\frac{1}{|\Sigma|^{1/2}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})}$

Multivariate Gaussian Distribution: Conditionals



DeepLearning.AI

Probability Distributions with Multiple Variables

Conclusion